

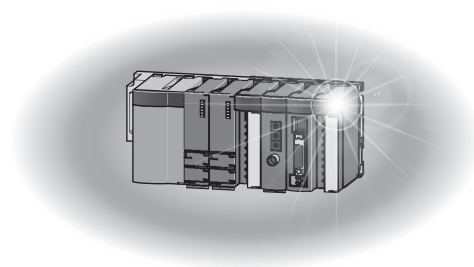
Programmable Controller

MELSEC **Q** series

## Type QD75P/QD75D Positioning Module User's Manual

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-QD75P1N  
-QD75P2N  
-QD75P4N  
-QD75D1N  
-QD75D2N  
-QD75D4N  
-QD75P1  
-QD75P2  
-QD75P4  
-QD75D1  
-QD75D2  
-QD75D4





## • SAFETY PRECAUTIONS •

(Read these precautions before using this product.)

Before using this product, please read this manual and the relevant manuals carefully and pay full attention to safety to handle the product correctly.

The precautions given in this manual are concerned with this product only. For the safety precautions of the programmable controller system, refer to the user's manual for the CPU module used.

In this manual, the safety precautions are classified into two levels: "⚠WARNING" and "⚠CAUTION".



**WARNING**

Indicates that incorrect handling may cause hazardous conditions, resulting in death or severe injury.



**CAUTION**

Indicates that incorrect handling may cause hazardous conditions, resulting in minor or moderate injury or property damage.

Under some circumstances, failure to observe the precautions given under "⚠CAUTION" may lead to serious consequences.

Observe the precautions of both levels because they are important for personal and system safety.

Make sure that the end users read this manual and then keep the manual in a safe place for future reference.

### [Design Precautions]



#### **WARNING**

- Provide a safety circuit outside the programmable controller so that the entire system will operate safely even when an external power supply error or programmable controller fault occurs.

Failure to observe this could lead to accidents for incorrect outputs or malfunctioning.

- (1) Configure an emergency stop circuit and interlock circuit such as a positioning upper limit/lower limit to prevent mechanical damage outside the programmable controller.
- (2) The machine OPR operation is controlled by the OPR direction and OPR speed data. Deceleration starts when the near-point dog turns ON. Thus, if the OPR direction is incorrectly set, deceleration will not start and the machine will continue to travel. Configure an interlock circuit to prevent mechanical damage outside the programmable controller.
- (3) When the module detects an error, normally deceleration stop or sudden stop will take place according to the parameter stop group settings.  
Set the parameters to the positioning system specifications.  
Make sure that the OPR parameter and positioning data are within the parameter setting values.

## [Design Precautions]



### CAUTION

- Do not bundle or adjacently lay the connection cable connected to the module external I/O signals or drive unit with the main circuit line, power line, or the load line other than that for the programmable controller. Separate these by 100mm as a guide. Failure to observe this could lead to malfunctioning caused by noise, surge, or induction.

## [Security Precautions]



### WARNING

- To maintain the security (confidentiality, integrity, and availability) of the programmable controller and the system against unauthorized access, denial-of-service (DoS) attacks, computer viruses, and other cyberattacks from external devices via the network, take appropriate measures such as firewalls, virtual private networks (VPNs), and antivirus solutions.

## [Mounting Precautions]



### CAUTION

- Use the programmable controller in an environment that meets the general specifications contained in QCPU User's Manual(Hardware Design, Maintenance and Inspection) to use. Using this programmable controller in an environment outside the range of the general specifications may cause electric shock, fire, malfunction, and damage to or deterioration of the product.
- While pressing the installation lever located at the bottom of module, insert the module fixing tab into the fixing hole in the base unit until it stops. Then, securely mount the module with the fixing hole as a supporting point.  
Incorrect loading of the module can cause a malfunction, failure or drop.  
When using the programmable controller in the environment of much vibration, tighten the module with a screw.  
Tighten the screw in the specified torque range.  
Undertightening can cause a drop, short circuit or malfunction.  
Overtightening can cause a drop, short circuit or malfunction due to damage to the screw or module.
- Completely turn off the externally supplied power used in the system before mounting or removing the module.  
Not doing so may damage the product.



## [Wiring Precautions]



### WARNING

- Always confirm the terminal layout before connecting the wires to the module.

## [Wiring Precautions]



### CAUTION

- Use applicable solderless terminals and tighten them within the specified torque range. If any spade solderless terminal is used, it may be disconnected when the terminal screw comes loose, resulting in failure.
- Tighten the connector screws within the specified torque range.  
Undertightening can cause short circuit, fire, or malfunction.  
Overtightening can damage the screw and/or module, resulting in drop, short circuit, fire, or malfunction.
- Connectors for external devices must be crimped with the tool specified by the manufacturer, or must be correctly soldered. Incomplete connections may cause short circuit, fire, or malfunction.
- When disconnecting the cable from the module, do not pull the cable by the cable part.  
For the cable with connector, hold the connector part of the cable.  
Pulling the cable connected to the module may result in malfunction or damage to the module or cable.
- Prevent foreign matter such as dust or wire chips from entering the module. Such foreign matter can cause a fire, failure, or malfunction.
- A protective film is attached to the top of the module to prevent foreign matter, such as wire chips, from entering the module during wiring. Do not remove the film during wiring. Remove it for heat dissipation before system operation.

## [Startup/Maintenance Precautions]



### WARNING

- Completely turn off the externally supplied power used in the system before cleaning or tightening the screws.  
Failure to turn all phases OFF could lead to electric shocks.

## [Startup/Maintenance Precautions]

### CAUTION

- Never disassemble or modify the module.  
Failure to observe this could lead to trouble, malfunctioning, injuries or fires.
- Completely turn off the externally supplied power used in the system before installing or removing the module.  
Failure to turn all phases OFF could lead to module trouble or malfunctioning.
- Do not install/remove the module to/from the base unit, or the terminal block to/from the module more than 50 times after the first use of the product. (IEC 61131-2 compliant)  
Failure to do so may cause malfunction.
- Before starting test operation, set the parameter speed limit value to the slowest value, and make sure that operation can be stopped immediately if a hazardous state occurs.
- Always make sure to touch the grounded metal to discharge the electricity charged in the body, etc., before touching the module.  
Failure to do so may cause a failure or malfunctions of the module.

## [Precautions for use]

### CAUTION

- Note that when the reference axis speed is designated for interpolation operation, the speed of the partner axis (2nd axis, 3rd axis and 4th axis) may be larger than the set speed (larger than the speed limit value).

## [Disposal Precautions]

### CAUTION

- When disposing of the product, handle it as industrial waste.

## • CONDITIONS OF USE FOR THE PRODUCT •

(1) MELSEC programmable controller ("the PRODUCT") shall be used in conditions;

i) where any problem, fault or failure occurring in the PRODUCT, if any, shall not lead to any major or serious accident; and

ii) where the backup and fail-safe function are systematically or automatically provided outside of the PRODUCT for the case of any problem, fault or failure occurring in the PRODUCT.

(2) The PRODUCT has been designed and manufactured for the purpose of being used in general industries.

MITSUBISHI ELECTRIC SHALL HAVE NO RESPONSIBILITY OR LIABILITY (INCLUDING, BUT NOT LIMITED TO ANY AND ALL RESPONSIBILITY OR LIABILITY BASED ON CONTRACT, WARRANTY, TORT, PRODUCT LIABILITY) FOR ANY INJURY OR DEATH TO PERSONS OR LOSS OR DAMAGE TO PROPERTY CAUSED BY the PRODUCT THAT ARE OPERATED OR USED IN APPLICATION NOT INTENDED OR EXCLUDED BY INSTRUCTIONS, PRECAUTIONS, OR WARNING CONTAINED IN MITSUBISHI ELECTRIC USER'S, INSTRUCTION AND/OR SAFETY MANUALS, TECHNICAL BULLETINS AND GUIDELINES FOR the PRODUCT.

("Prohibited Application")

Prohibited Applications include, but not limited to, the use of the PRODUCT in;

- Nuclear Power Plants and any other power plants operated by Power companies, and/or any other cases in which the public could be affected if any problem or fault occurs in the PRODUCT.
- Railway companies or Public service purposes, and/or any other cases in which establishment of a special quality assurance system is required by the Purchaser or End User.
- Aircraft or Aerospace, Medical applications, Train equipment, transport equipment such as Elevator and Escalator, Incineration and Fuel devices, Vehicles, Manned transportation, Equipment for Recreation and Amusement, and Safety devices, handling of Nuclear or Hazardous Materials or Chemicals, Mining and Drilling, and/or other applications where there is a significant risk of injury to the public or property.

Notwithstanding the above restrictions, Mitsubishi Electric may in its sole discretion, authorize use of the PRODUCT in one or more of the Prohibited Applications, provided that the usage of the PRODUCT is limited only for the specific applications agreed to by Mitsubishi Electric and provided further that no special quality assurance or fail-safe, redundant or other safety features which exceed the general specifications of the PRODUCTS are required. For details, please contact the Mitsubishi Electric representative in your region.

(3) Mitsubishi Electric shall have no responsibility or liability for any problems involving programmable controller trouble and system trouble caused by DoS attacks, unauthorized access, computer viruses, and other cyberattacks.

# REVISIONS

\* The manual number is given on the bottom left of the back cover.

| Print Date | * Manual Number  | Revision                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dec., 1999 | SH (NA)-080058-A | First edition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Oct., 2000 | SH (NA)-080058-B | Addition of function version B<br>(Overall revisions based on the Japanese Manual Version SH-080047-E)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Jun., 2001 | SH (NA)-080058-C | <ul style="list-style-type: none"> <li>The software package names (GPP function software package, QD75 software package) have been replaced by the product names (GX Developer, GX Configurator-QP) for standardization.</li> </ul> <div>Partial corrections and additions</div> <p>CONTENTS, About Manuals, Generic Terms and Abbreviations, Section 1.4, Section 2.2, Section 2.3, Section 3.2.2 to Section 3.2.4, Section 3.3.2, Section 3.3.3, Section 3.4.1, Section 3.4.3, Section 3.4.4, Section 4.1.2, Section 4.3, Section 5.1.2, Section 5.1.3, Section 5.2.3, Section 5.2.5, Section 5.6.2, Section 5.7.1, Section 6.2 to Section 6.4, Section 6.5.3, Section 7.2, Section 8.2.2, Section 8.2.5, Section 8.2.6, Section 9.1.2, Section 9.2.1, Section 9.2.16, Section 9.2.17, Section 10.3.2, Section 10.6.2, Section 11.2.3, Section 11.3.3, Section 11.3.4, Section 11.4.3, Section 12.1.1, Section 12.5 to Section 12.7, Section 13.1, Section 13.3, Section 13.4, Section 14.2 to Section 14.7, Section 15.1, Section 15.2, Section 15.4, Appendix 1, Appendix 9.2, Appendix 11, INDEX</p> |
| Apr., 2003 | SH (NA)-080058-D | <div>Partial corrections and additions</div> <p>SAFETY INSTRUCTIONS, CONTENTS, Component List, Section 1.2.3, Section 1.4, Section 2.3, Section 2.4, Section 3.1, Section 3.2.1, Section 3.2.3, Section 3.2.4, Section 3.4.1, Section 3.4.4, Section 4.1.2, Section 4.3.1, Section 4.3.2, Section 5.1.1, Section 5.1.7, Section 5.1.8, Section 5.2.1, Section 5.2.4, Section 5.6.2, Section 5.7.1, Section 6.4, Section 6.5.4, Section 6.5.6, Section 8.2.3 to Section 8.2.8, Section 9.2.17, Section 9.2.19, Section 11.2.1, Section 11.3.1, Section 11.4.1, Section 12.1.1, Section 12.5.1, Section 12.5.2, Section 12.7.3, Section 12.7.5, Section 12.7.9, Section 14.4, Section 15.2, Appendix 1.1, Appendix 4.1 to Appendix 4.3, Appendix 7.1, Appendix 9.2, Appendix 10 to Appendix 13, INDEX</p>                                                                                                                                                                                                                                                                                                   |
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| Jun., 2005 | SH (NA)-080058-H | <p>Partial corrections and additions</p> <p>Section 5.1.2, Section 9.1.2, Section 9.2.10, Section 9.2.21, Section 10.3.8, Section 11.4.1, Section 12.5.2, Section 12.7.1, Section 12.7.6, Section 15.1, Section 15.2</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| Jul., 2008 | SH (NA)-080058-J | <p>Partial corrections and additions</p> <p>SAFETY INSTRUCTIONS, ABOUT MANUALS, Compliance with the EMC and Low Voltage Directives, Section 1.3, Section 2.3, 2.4, Section 3.1, 3.4.1, Section 4.1.2, 4.2.1, 4.3.1, Section 5.1.2, 5.2.1, 5.2.4, 5.2.5, Section 6.2, Section 6.4, Section 9.2.16, 9.2.17, Section 12.6, Section 12.7.2, Section 14.3 to 14.7, Appendix 12, 13</p>                                                                                                                                                                                                                                                                                                                                                                  |
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| Dec., 2019 | SH (NA)-080058-U | Partial corrections and additions<br>Appendix 9.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| May, 2024  | SH (NA)-080058-V | Additions<br>LIST OF DISCONTINUED MODELS<br>Partial corrections and additions<br>Section 2.2, Section 12.4.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|            |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

Japanese Manual Version SH-080047-AB

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## INTRODUCTION

Thank you for purchasing the Mitsubishi Electric general-purpose programmable controller MELSEC-Q Series. Always read through this manual, and fully comprehend the functions and performance of the Q Series programmable controller before starting use to ensure correct usage of this product. When applying the program examples provided in this manual to an actual system, ensure the applicability and confirm that it will not cause system control problems. Please make sure that the end users read this manual.

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## ABOUT MANUALS

The following manuals are also related to this product.  
In necessary, order them by quoting the details in the tables below.

### **Related Manuals**

| Manual Name                                                                                                                                                                                                                                            | Manual Number<br>(Model Code) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| GX Configurator-QP Version 2 Operating Manual<br>Data creation (such as parameters and positioning data) and operations of transferring data to modules, positioning monitor, and tests using GX Configurator-QP ..... (sold separately)* <sup>1</sup> | SH-080172<br>(13JU19)         |
| GX Developer Version 8 Operating Manual<br>Operating methods of GX Developer, such as programming, printing, monitoring, and debugging<br>(sold separately)                                                                                            | SH-080373E<br>(13JU41)        |
| GX Works2 Version1 Operating Manual<br>(Common)<br>System configuration, parameter settings, and online operations (common to Simple project and Structured project) of GX Works2..... (sold separately)                                               | SH-080779ENG<br>(13JU63)      |
| GX Works2 Version1 Operating Manual<br>(Intelligent Function Module)<br>Parameter settings, monitoring, and operations of the pre-defined protocol support function of intelligent function modules, using GX Works2 ..... (sold separately)           | SH-080921ENG<br>(13JU69)      |

\*1: The manual is included in the CD-ROM of the software package in a PDF-format file.  
For users interested in buying the manual separately, a printed version is available. Please contact us with the manual number (model code) in the list above.

## USING THIS MANUAL

■ The symbols used in this manual are shown below.

|                   |                                                                               |
|-------------------|-------------------------------------------------------------------------------|
| <b>Pr.*</b> ..... | Symbol indicating positioning parameter and OPR parameter item.               |
| <b>Da.*</b> ..... | Symbol indicating positioning data, block start data and condition data item. |
| <b>Md.*</b> ..... | Symbol indicating monitor data item.                                          |
| <b>Cd.*</b> ..... | Symbol indicating control data item.                                          |

(A serial No. is inserted in the \* mark.)

■ Representation of numerical values used in this manual.

- Buffer memory addresses, error codes and warning codes are represented in decimal.
- X/Y devices are represented in hexadecimal.
- Setting data and monitor data are represented in decimal or hexadecimal.

Data ended by "H" are represented in hexadecimal.

(Example) 10.....Decimal

10H .....Hexadecimal

## COMPLIANCE WITH EMC AND LOW VOLTAGE DIRECTIVES

### (1) Method of ensuring compliance

To ensure that Mitsubishi Electric programmable controllers maintain EMC and Low Voltage Directives when incorporated into other machinery or equipment, certain measures may be necessary. Please refer to one of the following manuals.

- QCPU User's Manual (Hardware Design, Maintenance and Inspection)
- Safety Guidelines (this manual is included with the CPU module or base unit)

The CE mark on the side of the programmable controller indicates compliance with EMC and Low Voltage Directives.

### (2) Additional measures

To ensure that this product maintains EMC and Low Voltage Directives, please refer to Section 4.3.1.

## GENERIC TERMS AND ABBREVIATIONS

Unless specially noted, the following generic terms and abbreviations are used in this manual.

| Generic term/abbreviation      | Details of generic term/abbreviation                                                                                                                                                                         |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CPU module                     | Generic term for CPU module on which QD75 can be mounted.                                                                                                                                                    |
| QD75                           | Generic term for positioning module QD75P1N, QD75P2N, QD75P4N, QD75D1N, QD75D2N, QD75D4N, QD75P1, QD75P2, QD75P4, QD75D1, QD75D2, and QD75D4.<br>The module type is described to indicate a specific module. |
| QD75P□N                        | Generic term for positioning module QD75P1N, QD75P2N, QD75P4N.                                                                                                                                               |
| QD75D□N                        | Generic term for positioning module QD75D1N, QD75D2N, QD75D4N.                                                                                                                                               |
| QD75P□                         | Generic term for positioning module QD75P1, QD75P2, QD75P4.                                                                                                                                                  |
| QD75D□                         | Generic term for positioning module QD75D1, QD75D2, QD75D4.                                                                                                                                                  |
| Peripheral device              | Generic term for DOS/V personal computer that can run the following "GX Developer" and "GX Configurator-QP".                                                                                                 |
| GX Configurator-QP             | Abbreviation for GX Configurator-QP (SW2D5C-QD75P-E or later).                                                                                                                                               |
| GX Developer                   | Abbreviation for GX Developer (SW4D5C-GPPW-E or later).                                                                                                                                                      |
| GX Works2                      | Product name of the software package for the MELSEC programmable controllers.                                                                                                                                |
| Drive unit (servo amplifier)   | Abbreviation for pulse input compatible drive unit (servo amplifier).                                                                                                                                        |
| Manual pulse generator         | Abbreviation for manual pulse generator (prepared by user).                                                                                                                                                  |
| DOS/V personal computer        | IBM PC/AT® and compatible DOS/V compliant personal computer.                                                                                                                                                 |
| Personal computer              | Generic term for personal computer which supports Windows®.                                                                                                                                                  |
| Workpiece                      | Generic term for moving body such as workpiece and tool, and for various control targets.                                                                                                                    |
| Axis 1, axis 2, axis 3, axis 4 | Indicates each axis connected to QD75.                                                                                                                                                                       |
| 1-axis, 2-axis, 3-axis, 4-axis | Indicates the number of axes. (Example: 2-axis = Indicates two axes such as axis 1 and axis 2, axis 2 and axis 3, and axis 3 and axis 1.)                                                                    |



## LIST OF DISCONTINUED MODELS

The following table lists discontinued models described in this manual. For onerous repair term after discontinuation of production, refer to WARRANTY in this manual.

| Model  | Production discontinuation |
|--------|----------------------------|
| QD75P1 | September 2021             |
| QD75P2 | September 2021             |
| QD75P4 | September 2021             |
| QD75D1 | September 2021             |
| QD75D2 | September 2021             |
| QD75D4 | September 2021             |

## COMPONENT LIST

The table below shows the component included in respective positioning modules:

| Module name | Description                                                          | Quantity |
|-------------|----------------------------------------------------------------------|----------|
| QD75P1N     | QD75P1N Positioning Module(1-axis open collector output system)      | 1        |
| QD75P2N     | QD75P2N Positioning Module(2-axes open collector output system)      | 1        |
| QD75P4N     | QD75P4N Positioning Module(4-axes open collector output system)      | 1        |
| QD75D1N     | QD75D1N Positioning Module(1-axis differential driver output system) | 1        |
|             | Differential driver common terminal                                  | 1        |
| QD75D2N     | QD75D2N Positioning Module(2-axes differential driver output system) | 1        |
|             | Differential driver common terminal                                  | 1        |
| QD75D4N     | QD75D4N Positioning Module(4-axes differential driver output system) | 1        |
|             | Differential driver common terminal                                  | 1        |
| QD75P1      | QD75P1 Positioning Module(1-axis open collector output system)       | 1        |
| QD75P2      | QD75P2 Positioning Module(2-axes open collector output system)       | 1        |
| QD75P4      | QD75P4 Positioning Module(4-axes open collector output system)       | 1        |
| QD75D1      | QD75D1 Positioning Module(1-axis differential driver output system)  | 1        |
|             | Differential driver common terminal                                  | 1        |
| QD75D2      | QD75D2 Positioning Module(2-axes differential driver output system)  | 1        |
|             | Differential driver common terminal                                  | 1        |
| QD75D4      | QD75D4 Positioning Module(4-axes differential driver output system)  | 1        |
|             | Differential driver common terminal                                  | 1        |

# MEMO

[illegible]

## PART 1 PRODUCT SPECIFICATIONS AND HANDLING

PART 1 is configured for the following purposes (1) to (5).

- (1) To understand the outline of positioning control, and the QD75 specifications and functions
- (2) To carry out actual work such as installation and wiring
- (3) To set parameters and data required for positioning control
- (4) To create a sequence program required for positioning control
- (5) To understand the memory configuration and data transmission process

Read PART 2 for details on each control.

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| CHAPTER 7 | MEMORY CONFIGURATION AND DATA PROCESS .....             | 7- 1 to 7- | 12  |

# MEMO

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# CHAPTER 1 PRODUCT OUTLINE

The purpose and outline of positioning control using QD75 are explained in this chapter. Reading this chapter will help you understand what can be done using the positioning system and which procedure to use for a specific purpose.

By understanding "What can be done", and "Which procedure to use" beforehand, the positioning system can be structured smoothly.

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## 1.1 Positioning control

### 1.1.1 Features of QD75

The features of the QD75 are shown below.

#### (1) Availability of one, two, and four axis modules

- (a) The pulse output types of the available modules are either the open collector output system or the differential driver output system. A module can be selected from the following depending on the drive unit type and the number of axes. (Refer to Section 2.2.)
  - Open collector output system:  
QD75P1N/QD75P2N/QD75P4N (QD75P1/QD75P2/QD75P4)
  - Differential driver output system:  
QD75D1N/QD75D2N/QD75D4N (QD75D1/QD75D2/QD75D4)
- (b) For connecting any of the QD75 modules to the base unit, a single slot and 32 dedicated I/O channels are required.  
Within the limit imposed by the maximum number of inputs and outputs supported by the CPU module, up to 64 modules can be used. (Refer to Section 3.1.)

#### (2) Wide variety of positioning control functions

- (a) A wide variety of positioning control functions essential to any positioning system are supported: positioning to an arbitrary position, fixed-feed control, equal-speed control, and so on. (Refer to Section 5.3 and 9.2.)
  - 1) Up to 600 positioning data items, including such information as positioning addresses, control systems, and operation patterns, can be prepared for each axis.  
Using the prepared positioning data, the positioning control is performed independently for each axis. (In addition, such controls as interpolation involving two to four axes and simultaneous startup of multiple axes are possible.)
  - 2) Independent control of each axis can be achieved in linear control mode (executable simultaneously over four axes).  
Such control can either be the independent positioning control using a single positioning data or the continuous positioning control enabled by the continuous processing of multiple positioning data.
  - 3) Coordinated control over multiple axes can take the form of either the linear interpolation through the speed or position control of two to four axes or the circular interpolation involving two axes.  
Such control can either be the independent positioning control using a single positioning data or the continuous positioning control enabled by the continuous processing of multiple positioning data.
- (b) For each positioning data, the user can specify any of the following control systems: position control, speed control, speed-position switching control, position-speed switching control, and so on. (Refer to Section 5.3 and 9.2.)

- (c) Continuous positioning control using multiple positioning data can be executed in accordance with the operation patterns the user assigned to the positioning data. (Refer to Section 5.3 and 9.1.2)  
Continuous positioning control can be executed over multiple blocks, where each block consists of multiple positioning data. (Refer to Section 10.3.2.)
  - (d) OPR control is given additional features (Refer to Section 8.2.)
    - 1) Six different machine OPR methods are provided: near point dog method (one method), stopper methods (three methods), and count methods (two methods).
    - 2) OPR retry function facilitates the machine OPR control from an arbitrary position.  
(The machine OP a premier reference position for positioning control. The machine is set to the machine OP through one of the machine OPR methods mentioned in 1) above.)
  - (e) Two acceleration/deceleration control methods are provided: trapezoidal acceleration/deceleration and S-curve acceleration/deceleration. (Refer to Section 12.7.6.)  
(Note that there is a restriction on executing the S-curve acceleration/deceleration for the system using stepping motors. Refer to Section 1.3.)
- (3) Quick startup (Refer to Section 3.1.)  
The processing time to start the positioning operation is shortened.  
QD75P□N/QD75D□N: 1.5ms (QD75P□/QD75D□: 6ms)  
When operation using simultaneous start function or interpolation operation is executed, the axes start without delay.
- |           |                                                                  |   |                                     |
|-----------|------------------------------------------------------------------|---|-------------------------------------|
| (Example) | Axis 1 and Axis 3 are started by the simultaneous start function | : | No delay in Axis 1 and Axis 3 start |
|           | Axis 2 and Axis 4 are started by the interpolation operation     | : | No delay in Axis 2 and Axis 4 start |
- (4) Faster pulse output and allowance of longer distance to drive unit (Refer to Section 3.1.)  
The modules with a differential driver (QD75D□N (QD75D□)) incorporate the improvements in pulse output speed and maximum distance to the drive unit.
- QD75D□N: 4Mpulse/s, 10m max. (QD75D□: 1Mpulse/s, 10m max.)
  - QD75P□N: 200kpulse/s, 2m max. (QD75P□: 200kpulse/s, 2m max.)
- (5) Easy maintenance  
Each QD75 positioning module incorporates the following improvements in maintainability:
- (a) Data such as the positioning data and parameters can be stored on a flash ROM inside the QD75, eliminating the need of a battery for retaining data. (Refer to Section 7.1.1.)
  - (b) Error messages are classified in more detail to facilitate the initial troubleshooting procedure. (Refer to Section 15.1.)
  - (c) The module retains 16 error messages and 16 warning messages recently output, offering more complete error and warning histories. (Refer to Section 5.6.1.)

(6) Support of intelligent function module dedicated instructions

Dedicated instructions such as the absolute position restoration instruction, positioning start instruction, and teaching instruction are provided.

The use of such dedicated instruction simplifies sequence programs. (Refer to CHAPTER 14.)

(7) Setups, monitoring, and testing through GX Configurator-QP

Using GX Configurator-QP, the user can control the QD75 parameters and positioning data without having to be conscious of the buffer memory addresses. Moreover, GX Configurator-QP has a test function which allows the user to check the wiring before creating a sequence program for positioning control, or test operation the QD75 using created parameters and positioning data for checking their integrity.

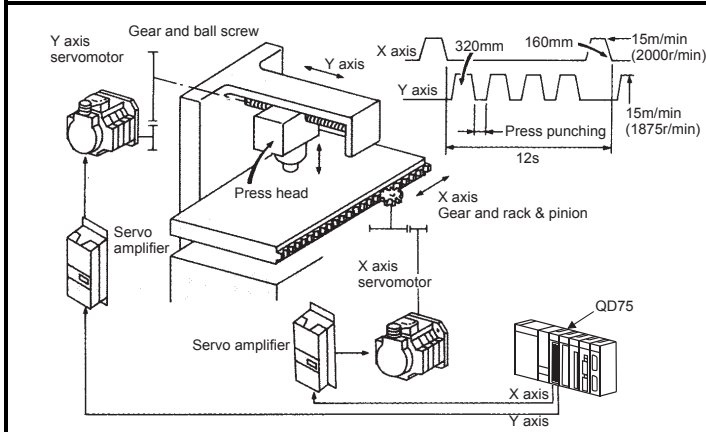
The control monitor function of GX Configurator-QP allows the user to debug programs efficiently.



## 1.1.2 Purpose and applications of positioning control

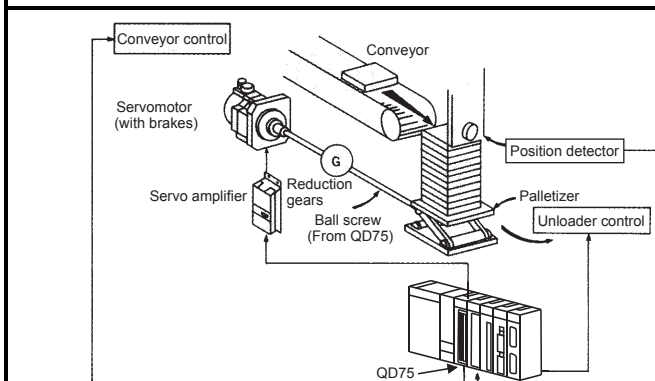
"Positioning" refers to moving a moving body, such as a workpiece or tool (hereinafter, generically called "workpiece") at a designated speed, and accurately stopping it at the target position. The main application examples are shown below.

### ■ Punch press (X, Y feed positioning)



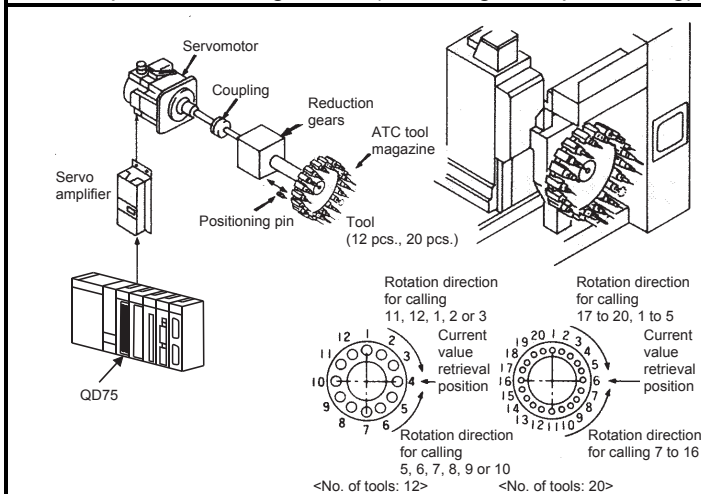
- To punch insulation material or leather, etc., as the same shape at a high yield, positioning is carried out with the X axis and Y axis servos.
- After positioning the table with the X axis servo, the press head is positioned with the Y axis servo, and is then punched with the press.
- When the material type or shape changes, the press head die is changed, and the positioning pattern is changed.

### ■ Palletizer



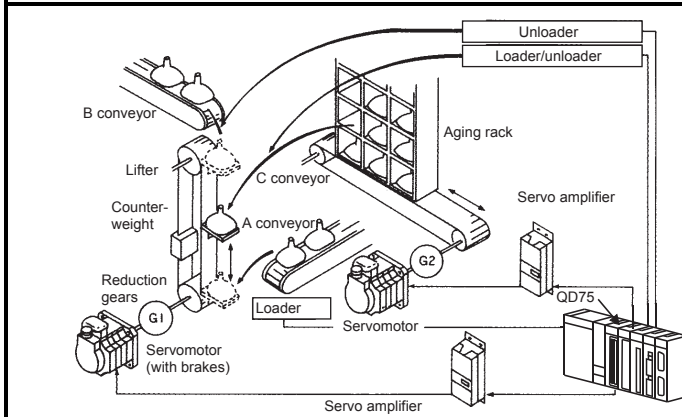
- Using the servo for one axis, the palletizer is positioned at a high accuracy.
- The amount to lower the palletizer according to the material thickness is saved.

### ■ Compact machining center (ATC magazine positioning)



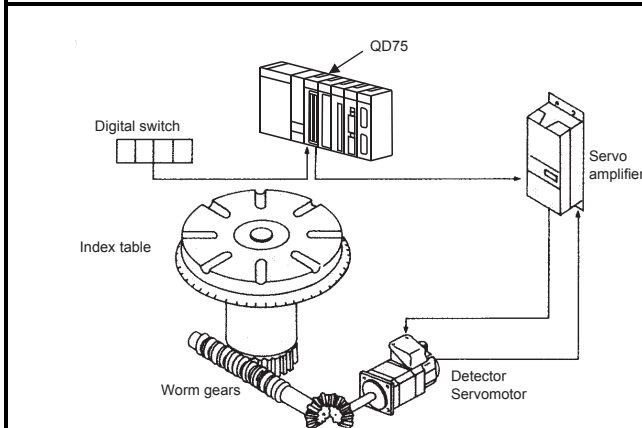
- The ATC tool magazine for a compact machining center is positioned.
- The relation of the magazine's current value and target value is calculated, and positioning is carried out with forward run or reverse run to achieve the shortest access time.

### ■ Lifter (Storage of Braun tubes onto aging rack)



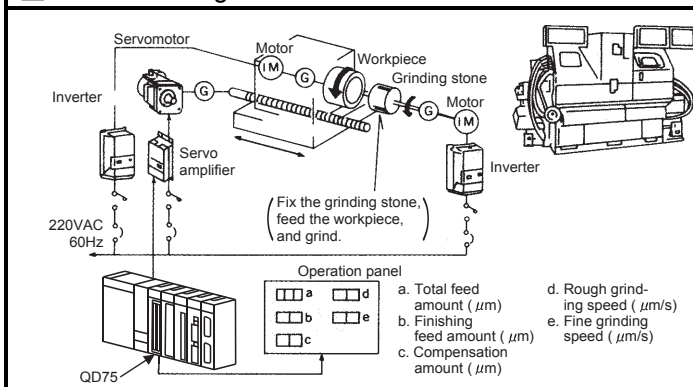
- During the aging process of braun tubes, storage onto the rack is carried out by positioning with the AC servo.
- The up/down positioning of the lifter is carried out with the 1-axis servo, and the horizontal position of the aging rack is positioned with the 2-axis servo.

### ■ Index table (High-accuracy indexing of angle)



- The index table is positioned at a high accuracy using the 1-axis servo.

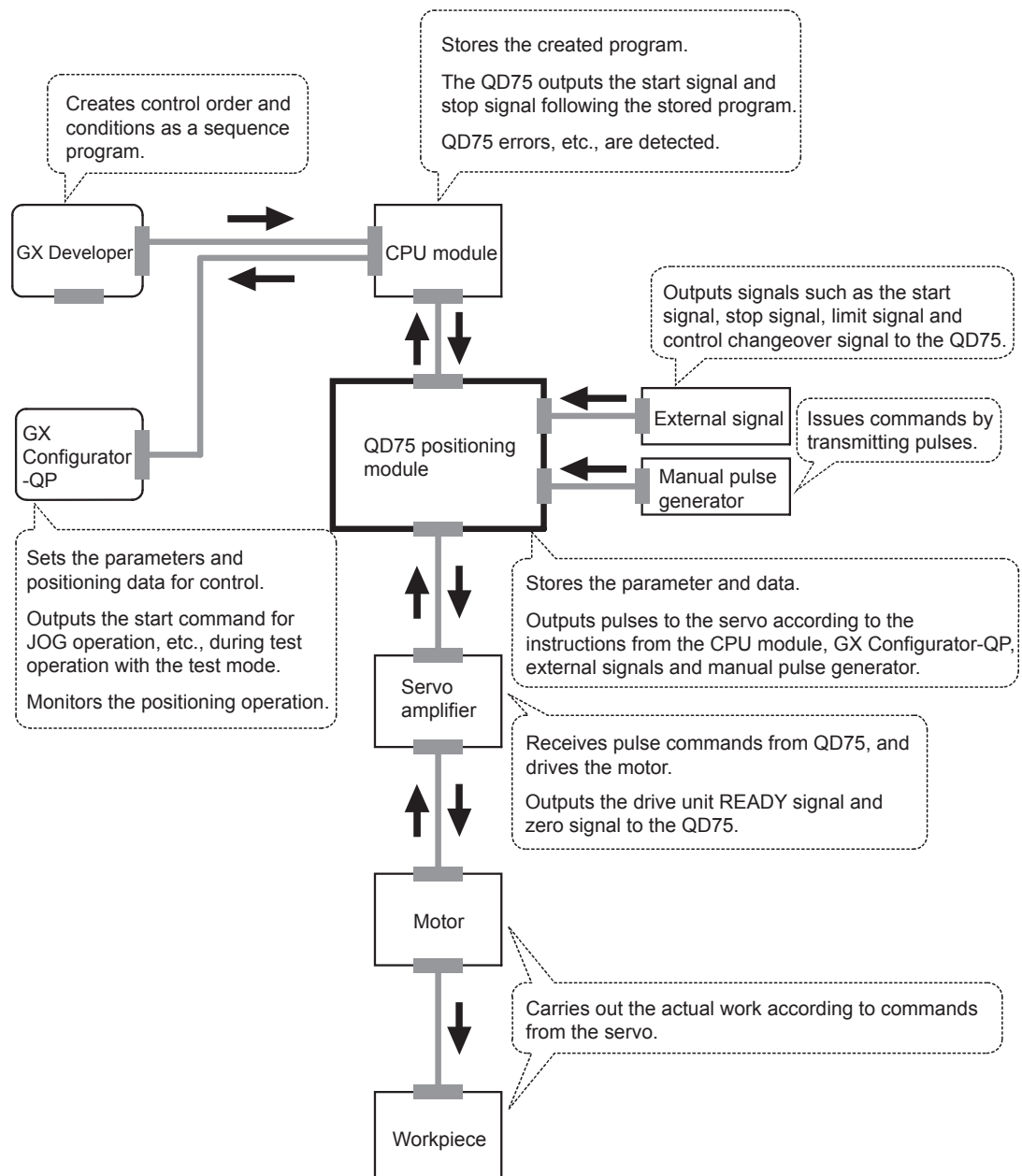
### ■ Inner surface grinder



- The grinding of the workpiece's inner surface is controlled with the servo and inverter.
- The rotation of the workpiece is controlled with the 1-axis inverter, and the rotation of the grinding stone is controlled with the 2-axis inverter. The workpiece is fed and ground with the 3-axis servo.

## 1.1.3 Mechanism of positioning control

Positioning control using the QD75 is carried out with "pulse signals". (The QD75 is a module that generates pulses). In the positioning system using the QD75, various software and devices are used for the following roles. The QD75 realizes complicated positioning control when it reads in various signals, parameters and data and is controlled with the CPU module.



The principle of "position control" and "speed control" operation is shown below.

### ■ Position control

The total No. of pulses required to move the designated distance is obtained in the following manner.

$$\left[ \begin{array}{l} \text{Total No. of pulses} \\ \text{required to move} \\ \text{designated distance} \end{array} \right] = \left[ \begin{array}{l} \text{Designated distance} \\ \text{Movement amount of machine (load)} \\ \text{side when motor rotates once} \end{array} \right] \times \left[ \begin{array}{l} \text{No. of pulses} \\ \text{required for motor to} \\ \text{rotate once} \end{array} \right]$$

\*The No. of pulses required for the motor to rotate once is the "encoder resolution" described in the motor catalog specification list.

When this total No. of pulses is issued from the QD75 to the servo amplifier, control to move the designated distance can be executed.

The machine side movement amount when one pulse is issued to the servo amplifier is called the "movement amount per pulse". This value is the min. value for the workpiece to move, and is also the electrical positioning precision.

### ■ Speed control

The "Total No. of pulses" mentioned above is invariably required for controlling the distance. For positioning or speed control, the speed must be controlled as well.

The speed is determined by the frequency of pulses sent from the QD75 to the drive unit.

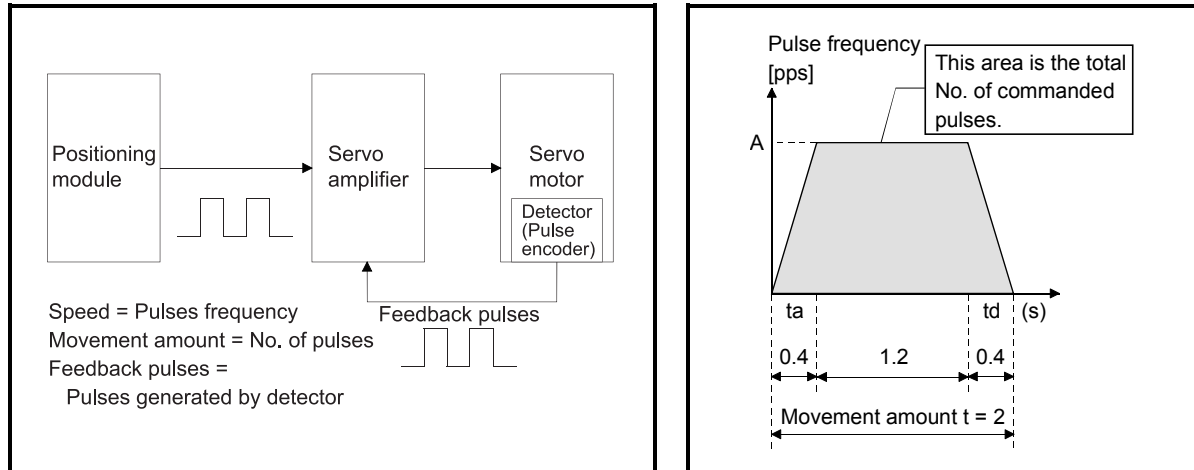


Fig. 1.1 Relationship between position control and speed control

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The QD75 controls the position with the "total No. of pulses", and the speed with the "pulse frequency".

## 1.1.4 Outline design of positioning system

The outline of the positioning system operation and design, using the QD75, is shown below.

## (1) Positioning system using QD75

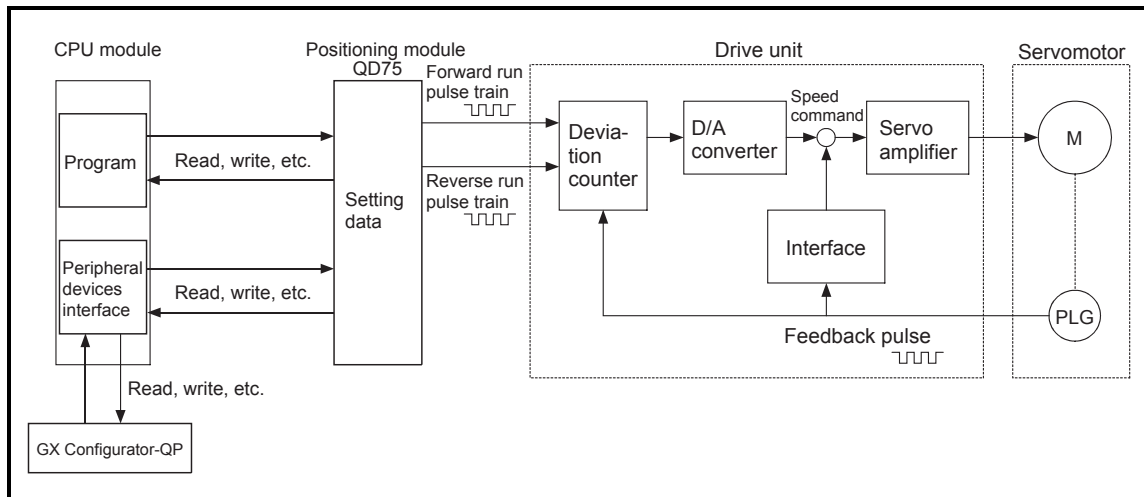


Fig. 1.2 Outline of the operation of positioning system using QD75

## (a) Positioning operation by the QD75

## 1) The QD75 output is a pulse train.

The pulse train output by the QD75 is counted by and stored in the deviation counter in the drive unit.

The D/A converter outputs an analog DC current proportionate to the count maintained by the deviation counter (called "pulse droop"). The analog DC current serves as the servomotor speed control signal.

## 2) The motor rotation is controlled by the speed control signal from the drive unit.

As the motor rotates, the pulse encoder (PLG) attached to the motor generates feedback pulses, the frequency of which is proportionate to the rotation speed.

The feedback pulses are fed back to the drive unit and decrements the pulse droop, the pulse count maintained by the deviation counter.

The motor keeps on rotating as the pulse droop is maintained at a certain level.

## 3) When the QD75 terminates the output of a pulse train, the motor decelerates as the pulse droop decreases and stops when the count drops to zero.

Thus, the motor rotation speed is proportionate to the pulse frequency, while the overall motor rotation angle is proportionate to the total number of pulses output by the QD75.

Therefore, when a movement amount per pulse is given, the overall movement amount can be determined by the number of pulses in the pulse train.

The pulse frequency, on the other hand, determines the motor rotation speed (feed speed).

## (b) Pulse train output from the QD75

- 1) As shown in Fig. 1.3, the pulse frequency increases as the motor accelerates. The pulses are sparse when the motor starts and more frequent when the motor speed comes close to the target speed.
- 2) The pulse frequency stabilizes when the motor speed equals the target speed.
- 3) The QD75 decreases the pulse frequency (sparser pulses) to decelerate the motor before it finally stops the output. There will be a little difference in timing between the decrease in the pulse frequency and the actual deceleration and stopping of the motor. This difference, called "the stop settling time", is required for gaining a stopping accuracy.

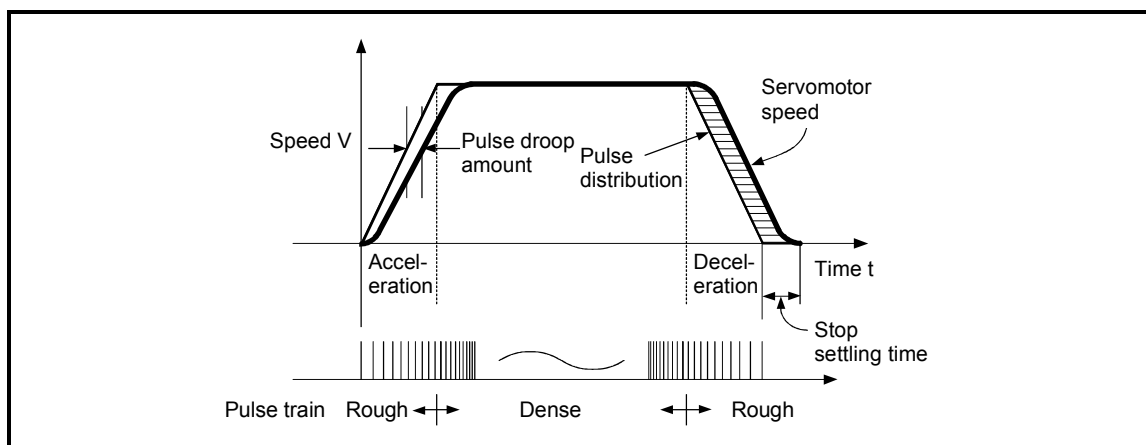


Fig. 1.3 QD75 output pulses

## (2) Movement amount and speed in a system using worm gears

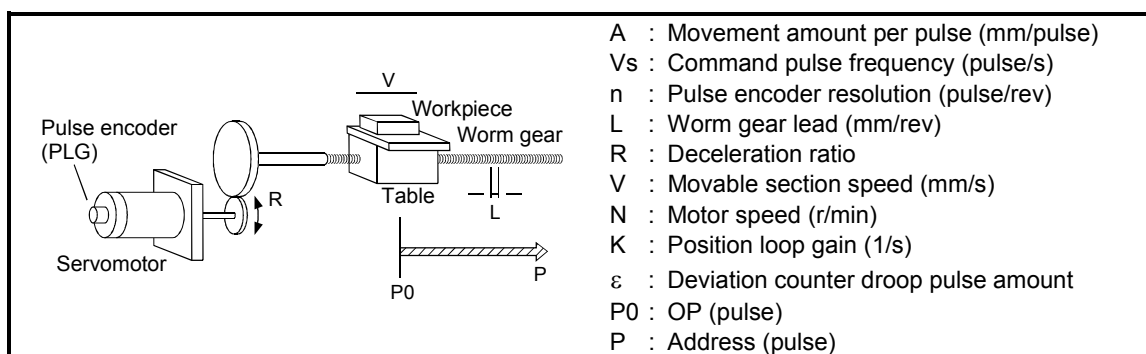


Fig. 1.4 System using worm gears

- (a) In the system shown in Fig. 1.4, the movement amount per pulse, command pulse frequency, and the deviation counter droop pulse amount are determined as follows:
- 1) Movement amount per pulse  
 The movement amount per pulse is determined by the worm gear lead, deceleration ratio, and the pulse encoder resolution.  
 The movement amount, therefore, is given as follows: (Number of pulses output) × (Movement amount per pulse).  

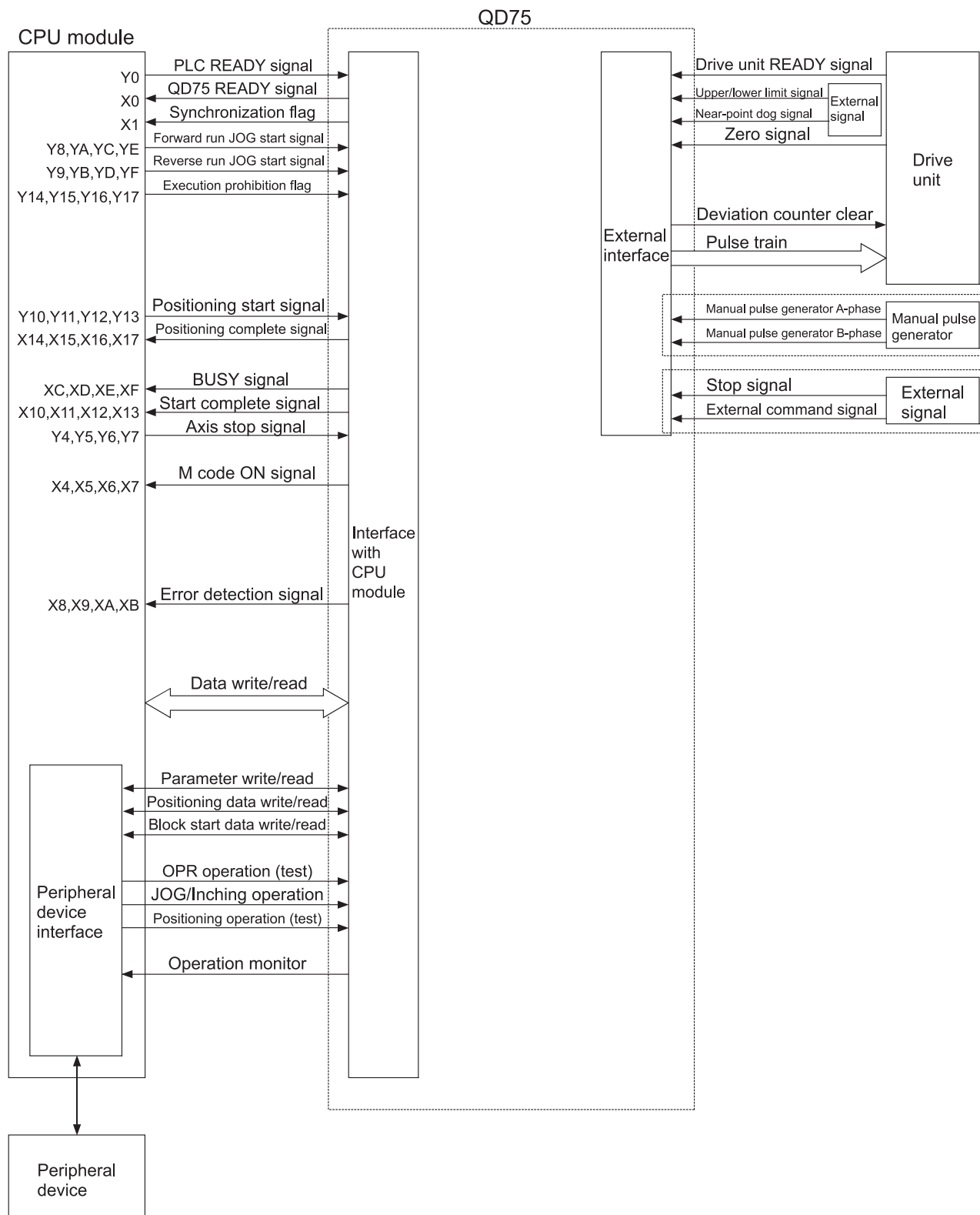
$$A = \frac{L}{R \times n} \text{ [mm/pulse]}$$
  - 2) Command pulse frequency  
 The command pulse frequency is determined by the speed of the moving part and movement amount per pulse:  

$$V_s = \frac{V}{A} \text{ [pulse/s]}$$
  - 3) Deviation counter droop pulse amount.  
 The deviation counter droop pulse amount is determined by the command pulse frequency and position loop gain.  

$$\varepsilon = \frac{V_s}{K} \text{ [pulse]}$$
- (b) The QD75 allows the user to select from the following four units as the unit used by positioning commands to any of the axes (1 to 4, if the module supports four axes): mm, inch, degree, and pulse.  
 The unit selected for one axis may differ from the unit selected for another axis.  
 When such data as the acceleration/deceleration time, positioning speed, and positioning address are correctly set in consideration of the chosen unit, the QD75 can calculate the number of pulses required for a movement amount to the target positioning address and execute the positioning by outputting a pulse train that includes the calculated number of pulses.

## 1.1.5 Communicating signals between QD75 and each module

The outline of the signal communication between the QD75 and CPU module, peripheral device and drive unit, etc., is shown below.  
(A peripheral device communicates with the QD75 via the CPU module to which it is connected)





### ■ QD75 ↔ CPU module

The QD75 and CPU module communicate the following data via the base unit.

| Direction<br>Communication | QD75 → CPU module                                                                                                                                                     | CPU module → QD75                                                                                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Control signal *           | Signal indicating QD75 state, such as QD75 READY signal, BUSY signal.                                                                                                 | Signal related to commands such as PLC READY signal, various start signals, stop signals                                                      |
| Data (read/write)          | <ul style="list-style-type: none"> <li>• Parameter</li> <li>• Positioning data</li> <li>• Block start data</li> <li>• Control data</li> <li>• Monitor data</li> </ul> | <ul style="list-style-type: none"> <li>• Parameter</li> <li>• Positioning data</li> <li>• Block start data</li> <li>• Control data</li> </ul> |

\* Refer to Section 3.3 "Specifications of input/output signals with CPU module " for details.

### ■ QD75 ↔ Peripheral device

The QD75 and peripheral device communicate the following data via the CPU module:

| Direction<br>Communication | QD75 → Peripheral device                                                                                              | Peripheral device → QD75                                                                                                                                                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data (read/write)          | <ul style="list-style-type: none"> <li>• Parameter</li> <li>• Positioning data</li> <li>• Block start data</li> </ul> | <ul style="list-style-type: none"> <li>• Parameter</li> <li>• Positioning data</li> <li>• Block start data</li> </ul>                                                                                                                                                          |
| Test operation             | —                                                                                                                     | <ul style="list-style-type: none"> <li>• OPR control start command</li> <li>• Positioning control start command</li> <li>• JOG/Inching operation start command</li> <li>• Teaching start command</li> <li>• Manual pulse generator operation enable/disable command</li> </ul> |
| Operation monitor          | • Monitor data                                                                                                        | —                                                                                                                                                                                                                                                                              |

### ■ QD75 ↔ Drive unit

The QD75 and drive unit communicate the following data via the external device connection connector.

| Direction<br>Communication | QD75 → Drive unit                                                  | Drive unit → QD75                                                   |
|----------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------|
| Control signal             | Signals related to commands such as deviation counter clear signal | Signals indicating drive unit state such as drive unit READY signal |
| Pulse train                | • Pulse train output                                               | —                                                                   |

### ■ QD75 ↔ Manual pulse generator

The QD75 and manual pulse generator communicate the following data via the external device connection connector.

(The manual pulse generator should be connected to an external device connection connector for axis 1 or for axes 1 and 2.)

| Direction<br>Communication | QD75 → Manual pulse generator | Manual pulse generator → QD75                                  |
|----------------------------|-------------------------------|----------------------------------------------------------------|
| Pulse signal               | —                             | Manual pulse generator A-phase, manual pulse generator B-phase |

## ■ QD75 ↔ External signal

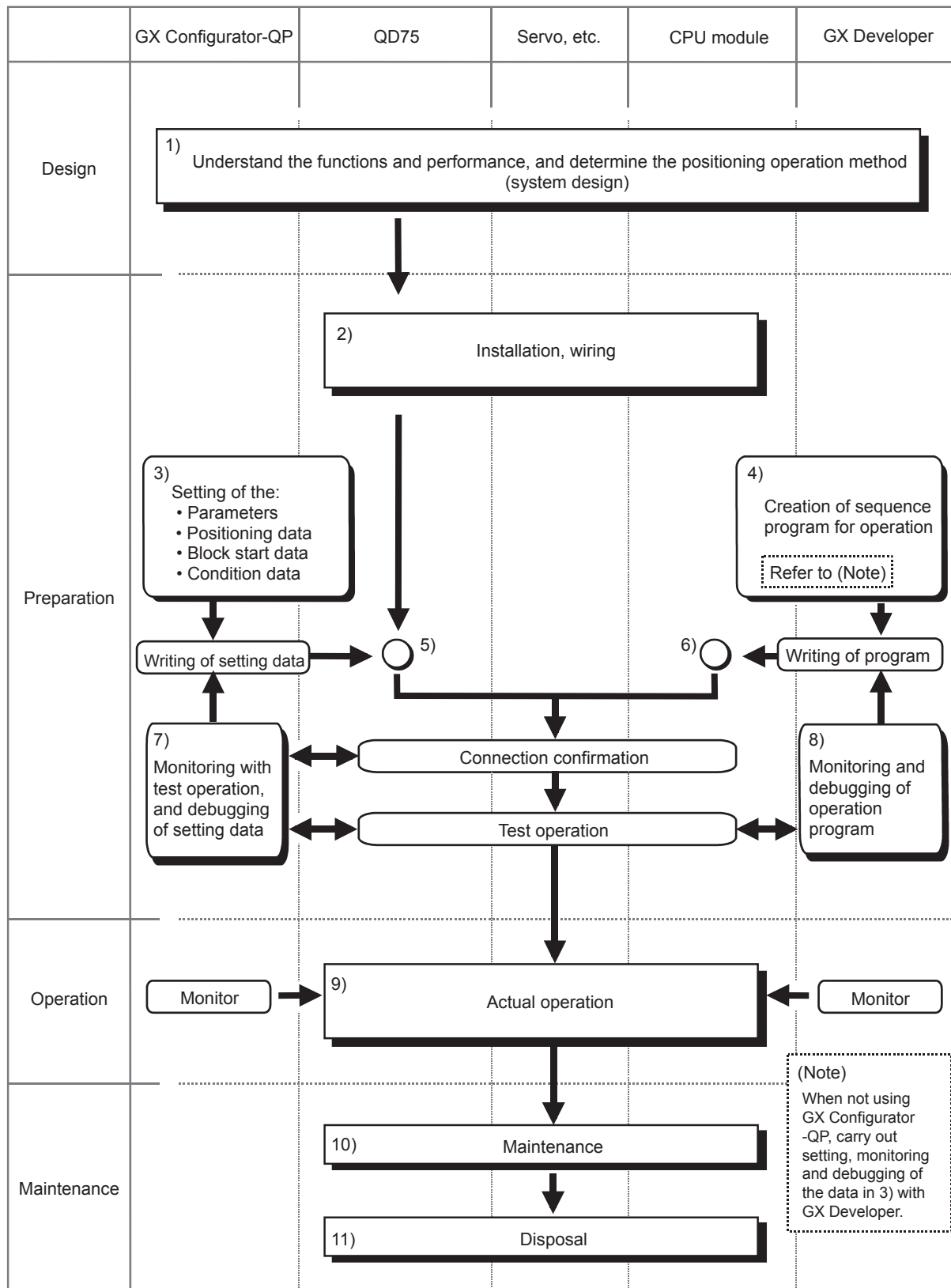
The QD75 and external signal communicate the following data via the external device connection connector.

| Direction<br>Communication | QD75 → External signal | External signal → QD75                                                                                                                                                                                                                   |
|----------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Control signal             | —                      | <ul style="list-style-type: none"><li>• Signals from detector such as near-point dog signal, upper/lower limit signal, zero signal</li><li>• Control signals from external device such as stop signal, external command signal</li></ul> |

## 1.2 Flow of system operation

## 1.2.1 Flow of all processes

The positioning control processes, using the QD75, are shown below.



The following work is carried out with the processes shown on the previous page.

|     | Details                                                                                                                                                                                                                                                                   | Reference                                                                                                                                                                   |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1)  | Understand the product functions and usage methods, the configuration devices and specifications required for positioning control, and design the system.                                                                                                                 | <ul style="list-style-type: none"> <li>• CHAPTER 1</li> <li>• CHAPTER 2</li> <li>• CHAPTER 3</li> <li>• CHAPTER 8 to CHAPTER 13</li> </ul>                                  |
| 2)  | Install the QD75 onto the base unit, wire the QD75 and external connection devices (drive unit, etc.).                                                                                                                                                                    | <ul style="list-style-type: none"> <li>• CHAPTER 4</li> </ul>                                                                                                               |
| 3)  | Using GX Configurator-QP, set the parameters, positioning data, block start data and condition data required for the positioning control to be executed. *1                                                                                                               | <ul style="list-style-type: none"> <li>• CHAPTER 5</li> <li>• CHAPTER 8 to CHAPTER 13</li> <li>• GX Configurator-QP Operating Manual</li> </ul>                             |
| 4)  | Using GX Developer, create the sequence program required for positioning operation. (When not using GX Configurator-QP, also create the sequence program for setting data.)                                                                                               | <ul style="list-style-type: none"> <li>• CHAPTER 6</li> <li>• GX Developer Operating Manual</li> </ul>                                                                      |
| 5)  | Write the parameters and positioning data, etc., created with GX Configurator-QP into the QD75.                                                                                                                                                                           | <ul style="list-style-type: none"> <li>• CHAPTER 7</li> <li>• GX Configurator-QP Operating Manual</li> </ul>                                                                |
| 6)  | Using GX Developer, write the created sequence program into the CPU module. (When not using GX Configurator-QP, also write in the sequence program for setting data.)                                                                                                     | <ul style="list-style-type: none"> <li>• CHAPTER 7</li> <li>• GX Developer Operating Manual</li> </ul>                                                                      |
| 7)  | Carry out test operation and adjustments in the test mode to check the connection with the QD75 and external connection device, and to confirm that the designated positioning operation is executed correctly. (Debug the set "parameters" and "positioning data", etc.) | <ul style="list-style-type: none"> <li>• GX Configurator-QP Operating Manual</li> <li>• CHAPTER 13</li> <li>• GX Developer Operating Manual</li> </ul>                      |
| 8)  | Carry out test operation and adjustment to confirm that the designated positioning operation is executed correctly. (Debug the created sequence program. When not using GX Configurator-QP, also debug the set data.)                                                     | <ul style="list-style-type: none"> <li>• GX Developer Operating Manual</li> </ul>                                                                                           |
| 9)  | Actually operate the positioning operation. At this time, monitor the operation state as required. If an error or warning occurs, remedy.                                                                                                                                 | <ul style="list-style-type: none"> <li>• CHAPTER 5</li> <li>• CHAPTER 15</li> <li>• GX Configurator-QP Operating Manual</li> <li>• GX Developer Operating Manual</li> </ul> |
| 10) | Service the QD75 as required.                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>• CHAPTER 4</li> </ul>                                                                                                               |
| 11) | Dispose of the QD75.                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>• CHAPTER 4</li> </ul>                                                                                                               |

\*1: When setting the QD75P□N/QD75D□N using GX Configurator-QP, there are restrictions on the setting ranges of some items. (Refer to Appendix 1.2.)

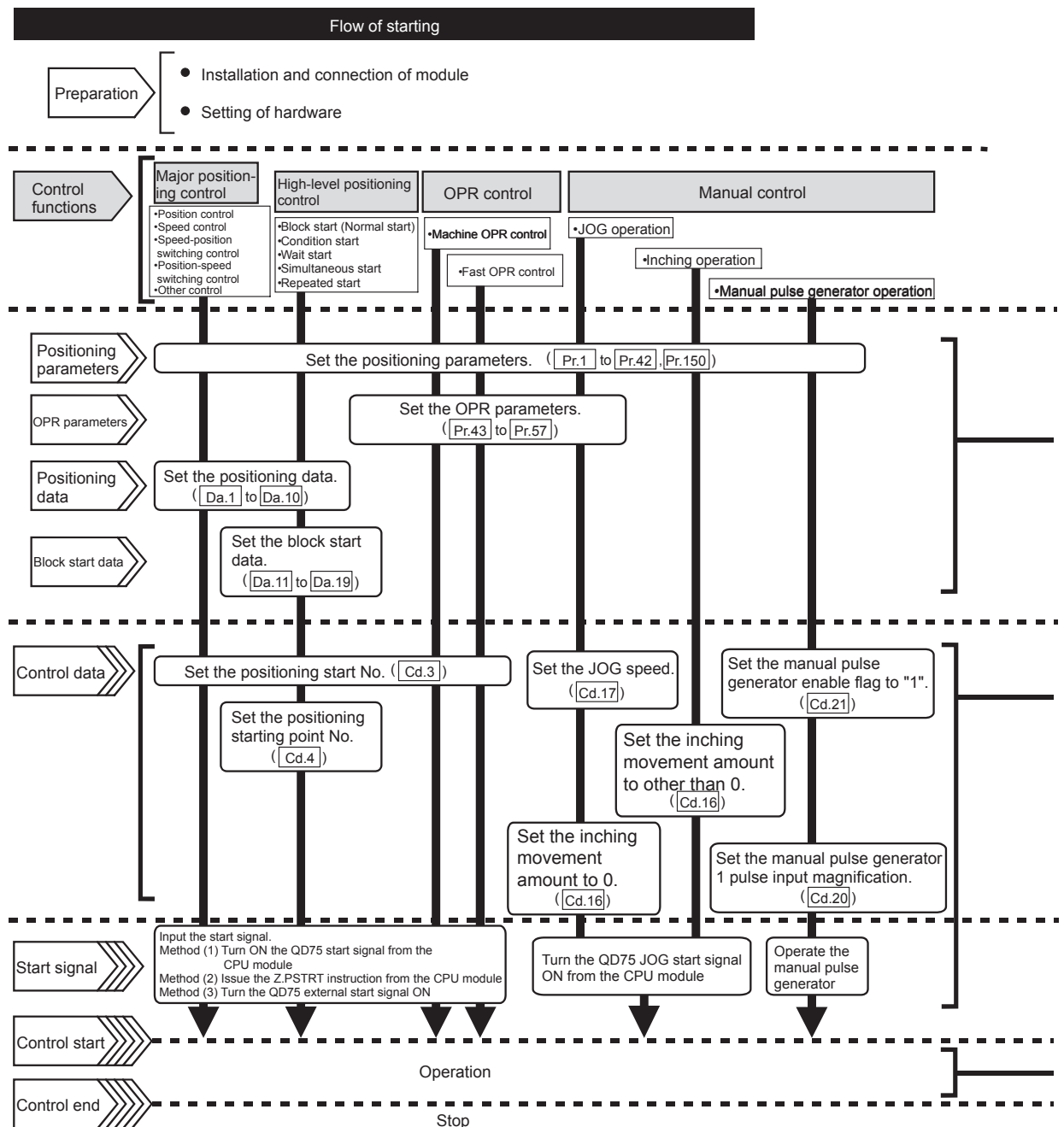
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## 1.2.2 Outline of starting

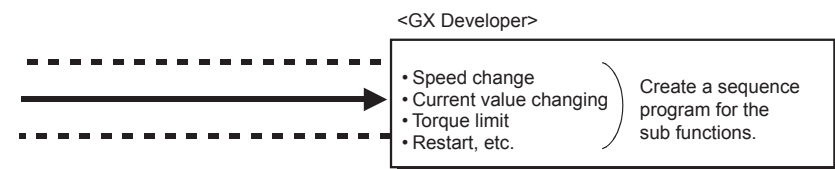
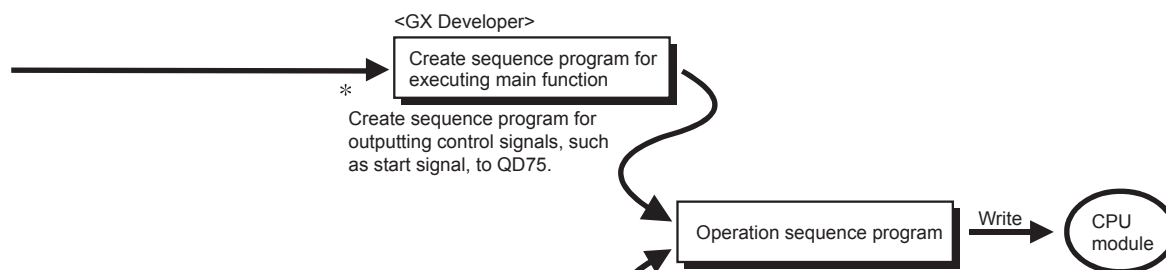
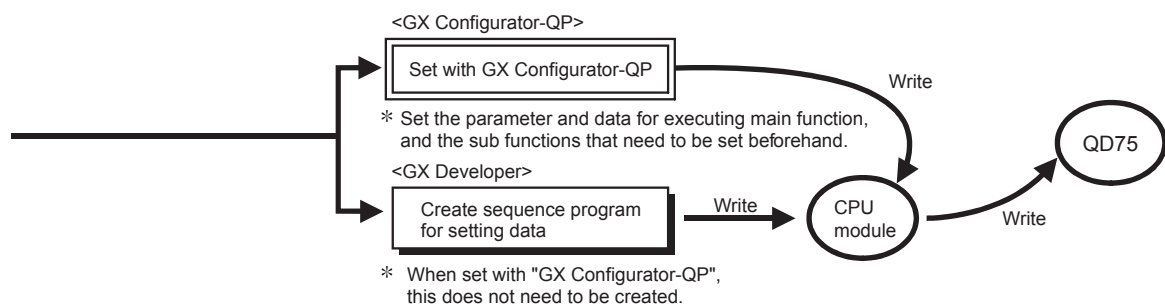
The outline for starting each control is shown with the following flowchart.

\*It is assumed that each module is installed, and the required system configuration, etc., has been prepared.



## Setting method

: Indicates the sequence program that must be created.



## 1.2.3 Outline of stopping

Each control is stopped in the following cases.

- (1) When each control is completed normally.
- (2) When the drive unit READY signal is turned OFF.
- (3) When a CPU module error occurs
- (4) When the PLC READY signal is turned OFF.
- (5) When an error occurs in the QD75.
- (6) When control is intentionally stopped (Stop signal from CPU module turned ON, stop signal from an external device, etc.)

The outline for the stopping process in these cases is shown below. (Excluding (1) for normal stopping.)

| Stop cause                             |                                                              | Stop axis | M code ON signal after stop | Axis operation status after stopping<br>(Md.26) | Stop process                                                                               |                  |                           |                                |                       |                                  |
|----------------------------------------|--------------------------------------------------------------|-----------|-----------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------|------------------|---------------------------|--------------------------------|-----------------------|----------------------------------|
|                                        |                                                              |           |                             |                                                 | OPR control                                                                                |                  | Major positioning control | High-level positioning control | Manual control        |                                  |
|                                        |                                                              |           |                             |                                                 | Machine OPR control                                                                        | Fast OPR control |                           |                                | JOG/Inching operation | Manual pulse generator operation |
| Forced stop                            | Drive unit READY signal OFF                                  | Each axis | No change                   | Error                                           | Immediate stop                                                                             |                  |                           |                                | Deceleration stop     |                                  |
| Fatal stop<br>(Stop group 1)           | Hardware stroke limit upper/lower limit error occurrence     | Each axis | No change                   | Error                                           | Deceleration stop/sudden stop<br>(Select with " Pr.37 Stop group 1 sudden stop selection") |                  |                           |                                | Deceleration stop     |                                  |
| Emergency stop<br>(Stop group 2)       | CPU module error occurrence                                  | All axes  | No change                   | Error                                           | Deceleration stop/sudden stop<br>(Select with " Pr.38 Stop group 2 sudden stop selection") |                  |                           |                                | Deceleration stop     |                                  |
|                                        | PLC READY signal OFF                                         |           | Turns OFF                   |                                                 |                                                                                            |                  |                           |                                |                       |                                  |
|                                        | Error in test mode                                           |           | No change                   |                                                 |                                                                                            |                  |                           |                                |                       |                                  |
| Relatively safe stop<br>(Stop group 3) | Axis error detection (Error other than stop group 1 or 2) *1 | Each axis | No change                   | Error                                           | Deceleration stop/sudden stop<br>(Select with " Pr.39 Stop group 3 sudden stop selection") |                  |                           |                                | Deceleration stop     |                                  |
| Intentional stop<br>(Stop group 3)     | "Stop signal" ON from external source                        | Each axis | No change                   | Stopped (Standby)                               |                                                                                            |                  |                           |                                |                       |                                  |
|                                        | "Stop signal" input from peripheral device*2                 |           |                             |                                                 |                                                                                            |                  |                           |                                |                       |                                  |



\*1: When multiple positioning data is executed by the continuous positioning control and there is invalid setting value in a positioning data, an error occurs and deceleration is performed at the previous positioning data. In this case, sudden stop is not performed even when it is set for the stop group 3. If any of the following error occurs, the operation is immediately stopped after executing up to previous positioning data of the positioning data where an error occurred.

- No command speed (error code: 503)
- Outside linear movement amount range (error code: 504)
- Large arc error deviation (error code: 506)
- Software stroke limit + (error code: 507)
- Software stroke limit - (error code: 508)
- Sub point setting error (error code: 525)
- End point setting error (error code: 526)
- Center point setting error (error code: 527)
- Outside radius range (error code: 544)
- Illegal setting of ABS direction in unit of degree (error code: 546)

\*2: This means the stop using a stop button on the test dialog box when using the positioning test function of GX Works2 or test function of GX Configurator-QP.

## 1.2.4 Outline of restarting

When a stop cause has occurred during operation with position control causing the axis to stop, positioning to the end point of the positioning data can be restarted from the stopped position by using the " **Cd.6** Restart command".

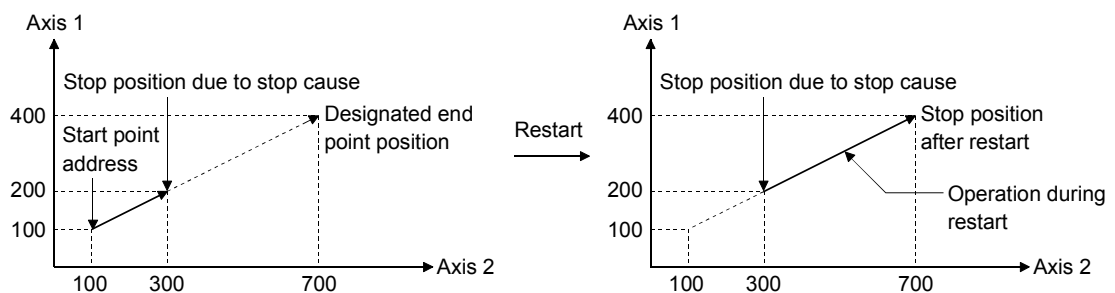
If issued during a continuous positioning or continuous path control operation, the restart command will cause the positioning to be re-executed using the current position (pointed by the positioning data No. associated with the moment when the movement was interrupted) as the start point.

■ When " **Cd.6** Restart command" is ON

- (1) If the " **Md.26** Axis operation status" is "Stopped", positioning to the end point of the positioning data will be restarted from the stopped position regardless of the absolute system or incremental system.
- (2) When " **Md.26** Axis operation status" is not "Stopped", the warning "Restart not possible" (warning code: 104) will be applied, and the restart command will be ignored.

[Example for incremental system]

- (a) The restart operation when the axis 1 movement amount is 300, and the axis 2 movement amount is 600 is shown below.

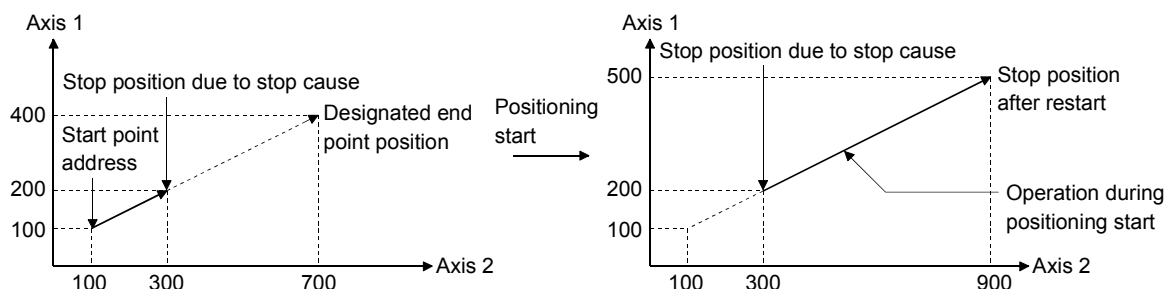


■ Reference

If the positioning start signal [Y10 to Y13]/external command signal\* is turned ON while the " **Md.26** Axis operation status" is "Standby" or "Stopped", positioning will be restarted from the start of the positioning start data regardless of the absolute system or incremental system. (\*: When the external command signal is set to "External positioning start")  
(Same as normal positioning.)

[Example for incremental system]

- (a) The positioning start operation when the axis 1 movement amount is 300 and the axis 2 movement amount is 600 is shown below.



### 1.3 Restrictions on using a stepping motor

Note the following restrictions on using a stepping motor:

- (1) For an axis where a stepping motor is connected, executing the S-curve acceleration/deceleration may cause step-out.  
Before using the S-curve acceleration/deceleration, confirm that step-out does not occur.
- (2) The circular interpolation control and 3-axis helical interpolation control cannot be used for an axis where a stepping motor is connected.  
Although setting the bias speed at start is required by the characteristics of the stepping motor, the setting of the bias speed at start is disabled for circular interpolation control.  
Ensure to use a servomotor for all axes where interpolation controls are carried out when the circular interpolation control or 3-axis helical interpolation control is carried out.
- (3) The backlash compensation function cannot be used for an axis where a stepping motor is connected. If the function is used, the motor may lose steps.

### 1.4 Function additions/modifications according to function version B

| POINT                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| When it is desired to check the following items, refer to the corresponding explanation sections.<br>How to check the function version and SERIAL No. (Refer to Section 2.4) |

The following function additions/modifications according to the function version B of the QD75:

| Function                                                  | Outline                                                                                                                                                                                              | Reference      |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Multiple CPU compatible function                          | Refer to QCPU User's Manual (Multiple CPU System).                                                                                                                                                   |                |
| Speed-position switching control (ABS mode)               | Function which starts the axis under speed control, switches to position control at the external command signal input, then positions the axis to a stop at the designated address.                  | Section 9.2.19 |
| Pre-reading start function                                | Function which shortens virtual start time.                                                                                                                                                          | Section 12.7.7 |
| External I/O signal monitor function                      | Function which monitors the external I/O signal monitor information in the module's detailed information which can be displayed on the system monitor of GX Developer *1.                            | Section 13.5   |
| Deceleration start flag function *2                       | Function that turns ON the flag when the constant speed status or acceleration status switches to the deceleration status during position control whose operation pattern is "Positioning complete". | Section 12.7.8 |
| Stop command processing for deceleration stop function *3 | Function that selects a deceleration curve when a stop cause occurs during deceleration stop processing to speed 0.                                                                                  | Section 12.7.9 |

\*1: For the QD75P□/QD75D□, this function is available in GX Developer (SW6D5C-GPPW-E or later). For details, refer to GX Developer Operating Manual.

For the QD75P□N/QD75D□N, external I/O signals cannot be monitored on GX Developer. Use the system monitor of GX Works2. For details on the system monitor of GX Works2, refer to GX Works2 Version 1 Operating Manual (Common).

\*2: For the QD75P□/QD75D□, this function is available for modules whose serial No. (first five digits) is "03042" or later.

\*3: For the QD75P□/QD75D□, this function is available for modules whose serial No. (first five digits) is "05072" or later.

# MEMO

[illegible]

# CHAPTER 2 SYSTEM CONFIGURATION

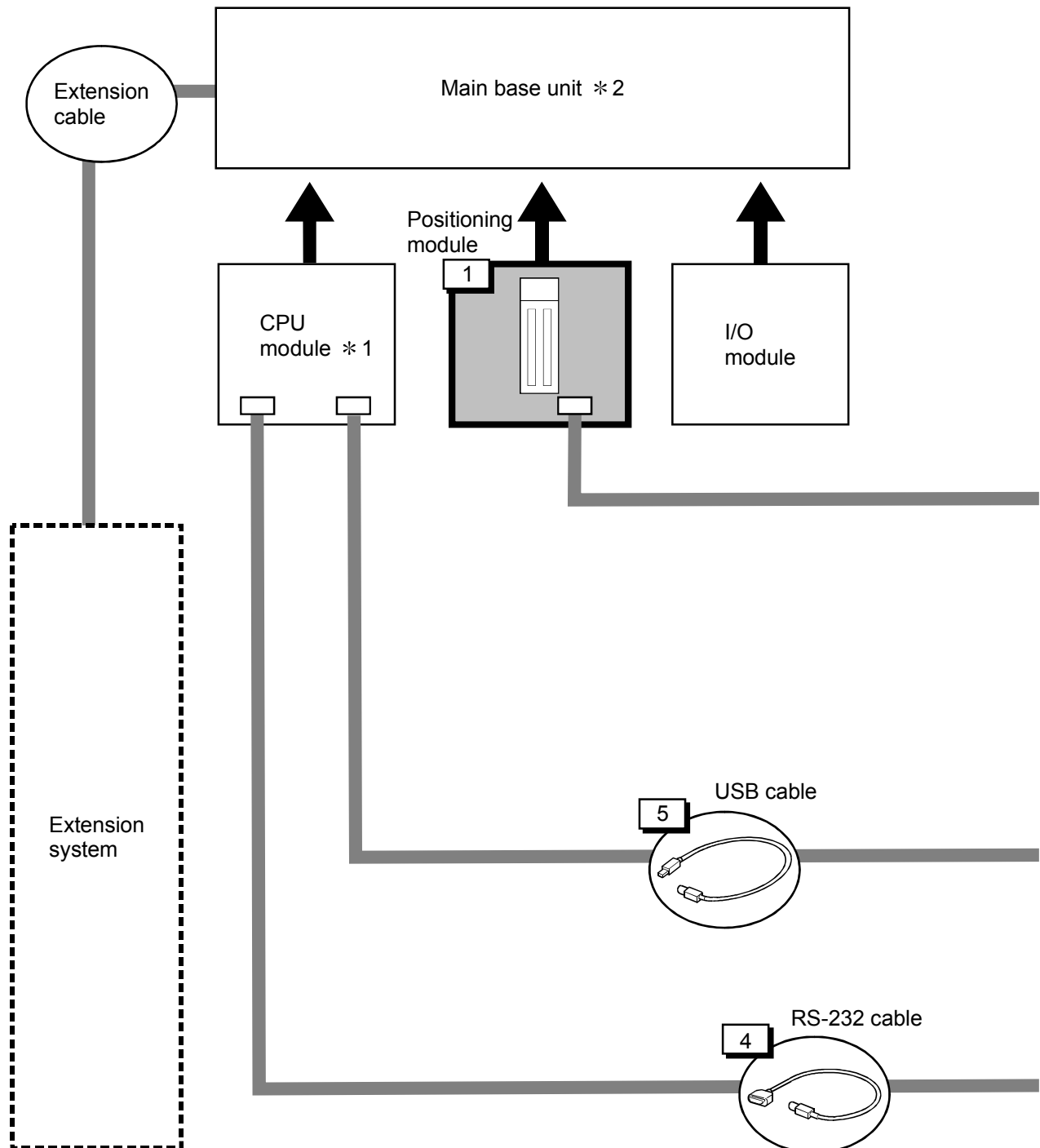
In this chapter, the general image of the system configuration of the positioning control using QD75, the configuration devices, applicable CPU and the precautions of configuring the system are explained.  
Prepare the required configuration devices to match the positioning control system.

|     |                                                       |    |   |
|-----|-------------------------------------------------------|----|---|
| 2.1 | General image of system .....                         | 2- | 2 |
| 2.2 | Configuration list .....                              | 2- | 4 |
| 2.3 | Applicable system .....                               | 2- | 6 |
| 2.4 | How to check the function version and SERIAL No. .... | 2- | 8 |

## 2.1 General image of system

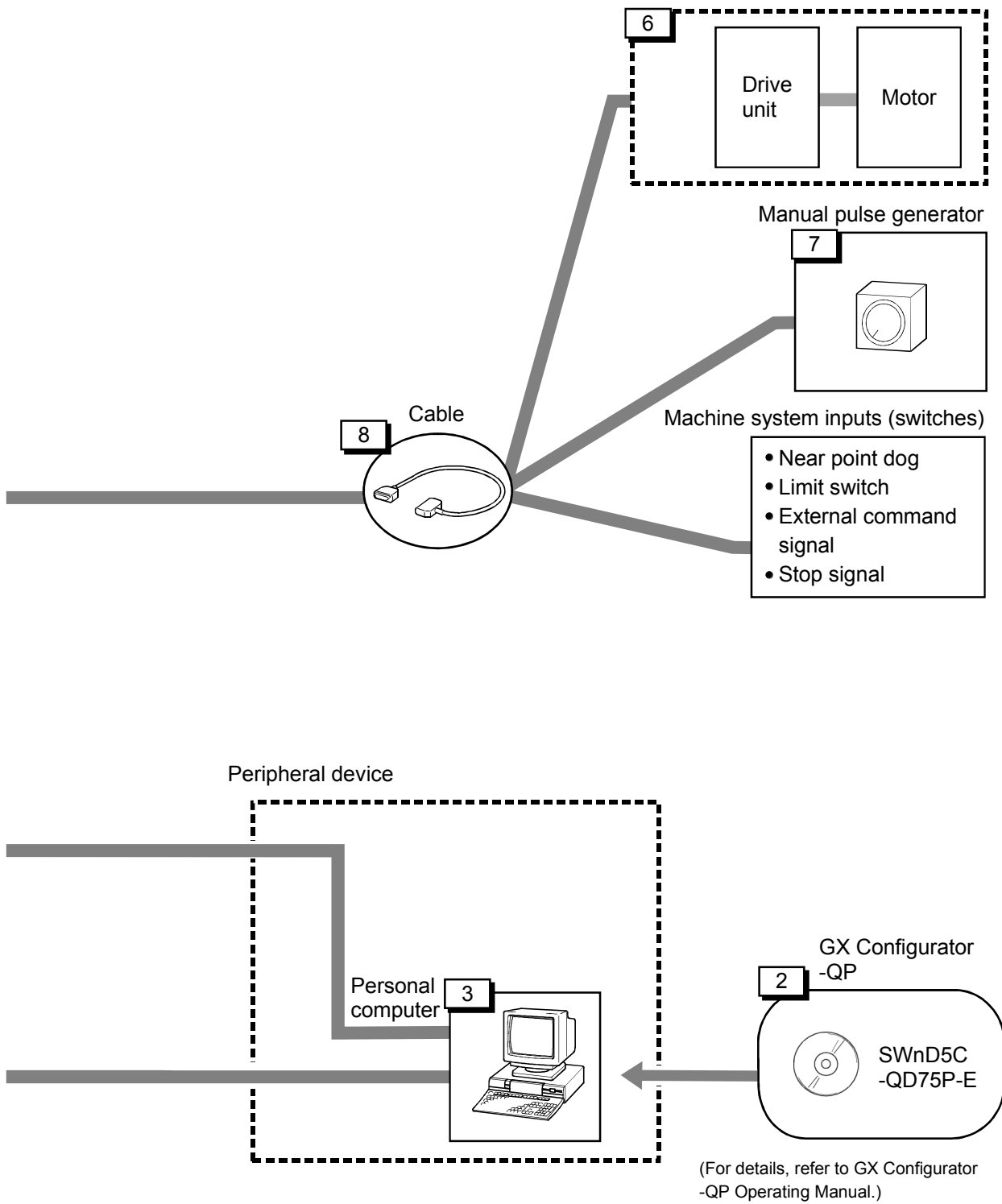
The general image of the system, including the QD75, CPU module and peripheral devices is shown below.

(The Nos. in the illustration refer to the "No." in Section 2.2 "Component list".)

**REMARK**

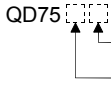
\*1 Refer to Section 2.3 "Applicable system" for the CPU modules that can be used.

\*2 Refer to the CPU module User's Manual for the base units that can be used.



## 2.2 Configuration list

The positioning system using the QD75 is configured of the following devices.

| No. | Part name                                                                | Type                                                                                                                       | Remarks                                                                                                                                                                                                                                                                  |
|-----|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Positioning module                                                       | QD75P1N<br>QD75P2N<br>QD75P4N<br>QD75D1N<br>QD75D2N<br>QD75D4N<br>QD75P1<br>QD75P2<br>QD75P4<br>QD75D1<br>QD75D2<br>QD75D4 |  No. of control axes<br>• P: Open collector output system<br>• D: Differential driver output system<br>Refer to Appendix 1.2 for differences between QD75P□N/QD75D□N and QD75P□/QD75D□. |
| 2   | GX Configurator-QP                                                       | SW□D5C-QD75P-E                                                                                                             | Refer to GX Configurator-QP Operating Manual for details.                                                                                                                                                                                                                |
| 3   | Personal computer                                                        | Personal computer which supports Windows®.                                                                                 | (Prepared by user)<br>Refer to GX Configurator-QP Operating Manual for details.                                                                                                                                                                                          |
| 4   | RS-232 cable                                                             | QC30R2                                                                                                                     | (Prepared by user)<br>An RS-232 cable is needed for connecting the CPU module with a personal computer.<br>For details, refer to GX Configurator-QP Operating Manual.                                                                                                    |
| 5   | USB cable                                                                | —                                                                                                                          | (Prepared by user)<br>A USB cable is needed for connecting the CPU module with a personal computer.<br>For details, refer to GX Configurator-QP Operating Manual.                                                                                                        |
| 6   | Drive unit                                                               | —                                                                                                                          | (Prepared by user)                                                                                                                                                                                                                                                       |
| 7   | Manual pulse generator                                                   | —                                                                                                                          | (Prepared by user)<br>Recommended: MR-HDP01 (Mitsubishi Electric)                                                                                                                                                                                                        |
| 8   | Connection cable<br>(For connecting between the QD75 and the drive unit) | —                                                                                                                          | (Prepared by user)<br>Cables are needed to connect the QD75 with the drive unit, manual pulse generator, and input devices in the machine system.<br>(Prepare them referring to the manuals for the connected devices and information given in 3.4.2 of this manual.)    |



■ Specifications of recommended manual pulse generator

| Item                      | Specifications                                                                                                              |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Model name                | MR-HDP01                                                                                                                    |
| Pulse resolution          | 25pulse/rev (100 pulse/rev after magnification by 4)                                                                        |
| Output method             | Voltage-output (power supply voltage -1V or more),<br>Output current = Max. 20mA                                            |
| Power supply voltage      | 4.5 to 13.2VDC <sup>*1</sup>                                                                                                |
| Current consumption       | 60mA                                                                                                                        |
| Output level              | "H" level: power supply voltage <sup>*1</sup> -1V or more (in no-load)<br>"L" level: 0.5V or less (with maximum leading-in) |
| Life time                 | 100 revolutions (at 200r/min)                                                                                               |
| Permitted axial loads     | Radial load: Max. 19.6N                                                                                                     |
|                           | Thrust load: Max. 9.8N                                                                                                      |
| Operation temperature     | -10 to 60°C (14 to 140°F)                                                                                                   |
| Weight                    | 0.4 (0.88) [kg(lb)]                                                                                                         |
| Number of max. revolution | Instantaneous Max. 600r/min. normal 200r/min                                                                                |
| Pulse signal status       | 2 signals: A phase, B phase, 90° phase difference                                                                           |
| Start friction torque     | 0.06N·m (at 20°C (68°F))                                                                                                    |

<sup>\*1</sup>: Use the stabilized power supply of 4.5 to 6.1 VDC for the power supply of the manual pulse generator.

## 2.3 Applicable system

The QD75 can be used in the following system.

### (1) Applicable modules and base units, and No. of modules

#### (a) When mounted with a CPU module

For the CPU modules, the number of modules, and base units applicable to the QD75, refer to the user's manual for the CPU modules used.

Check the following when the QD75 is mounted with the CPU module.

- Depending on the combination with other modules or the number of mounted modules, power supply capacity may be insufficient. Pay attention to the power supply capacity before mounting modules, and if the power supply capacity is insufficient, change the combination of the modules.
- Mount a module within the number of I/O points for the CPU module. If the number of slots is within the available range, the module can be mounted on any slot.

|               |
|---------------|
| <b>REMARK</b> |
|---------------|

To use a C Controller module with the QD75, refer to C Controller Module User's Manual.

#### (b) Mounting to a MELSECNET/H remote I/O station

For the MELSECNET/H remote I/O station, the number of modules, and base units applicable to the QD75, refer to the Q Corresponding MELSECNET/H Network System Reference Manual (Remote I/O network).

### (2) Compatibility with multiple CPU system

When using the QD75 in a multiple CPU system, refer to the QCPU User's Manual (Multiple CPU System).

## (3) Supported software packages

The following table lists the compatibility between the systems using the QD75 and the software packages. GX Developer or GX Works2 is required for use of the QD75.

|                                                                     |                     | Software version       |                                  |                                                         |
|---------------------------------------------------------------------|---------------------|------------------------|----------------------------------|---------------------------------------------------------|
|                                                                     |                     | GX Developer           | GX Configurator-QP <sup>*1</sup> | GX Works2                                               |
| Q00J/Q00/Q01CPU                                                     | Single CPU system   | Version 7 or later     | Version 2.10L or later           | Refer to GX Works2 Version 1 Operating Manual (Common). |
|                                                                     | Multiple CPU system | Version 8 or later     |                                  |                                                         |
| Q02/Q02H/Q06H/<br>Q12H/Q25HCPU                                      | Single CPU system   | Version 4 or later     | Version 2.00A or later           |                                                         |
|                                                                     | Multiple CPU system | Version 6 or later     |                                  |                                                         |
| Q02PH/Q06PHCPU                                                      | Single CPU system   | Version 8.68W or later | Version 2.29F or later           |                                                         |
|                                                                     | Multiple CPU system |                        |                                  |                                                         |
| Q12PH/Q25PHCPU                                                      | Single CPU system   | Version 7.10L or later | Version 2.13P or later           |                                                         |
|                                                                     | Multiple CPU system |                        |                                  |                                                         |
| Q02U/Q03UD/<br>Q04UDH/<br>Q06UDHCPU                                 | Single CPU system   | Version 8.48A or later | Version 2.25B or later           |                                                         |
|                                                                     | Multiple CPU system |                        |                                  |                                                         |
| Q13UDH/Q26UDH/<br>Q03UDE/Q04UDEH/<br>Q06UDEH/Q13UDEH/<br>Q26UDEHCPU | Single CPU system   | Version 8.68W or later | Version 2.29F or later           |                                                         |
|                                                                     | Multiple CPU system |                        |                                  |                                                         |
| Q00U/Q00UJ/Q01U/Q<br>10UDH/Q20UDH/<br>Q10UDEH/<br>Q20UDEHCPU        | Single CPU system   | Version 8.76E or later | Version 2.32J or later           |                                                         |
|                                                                     | Multiple CPU system |                        |                                  |                                                         |
| CPU modules other than above                                        |                     | Not supported          | Not supported                    |                                                         |
| For use on MELSECNET/H remote I/O station                           |                     | Version 6 or later     | Version 2.00A or later           |                                                         |

\*1: To use the QD75P□N/QD75D□N with GX Configurator-QP, select the QD75P□/QD75D□ as the model name. The QD75P□N/QD75D□N can be used in the same manner as the QD75P□/QD75D□; the setting ranges are the same. Therefore, a speed exceeding 1000000pulse/s cannot be set in some items. (Refer to Appendix 1.2)

**POINT**

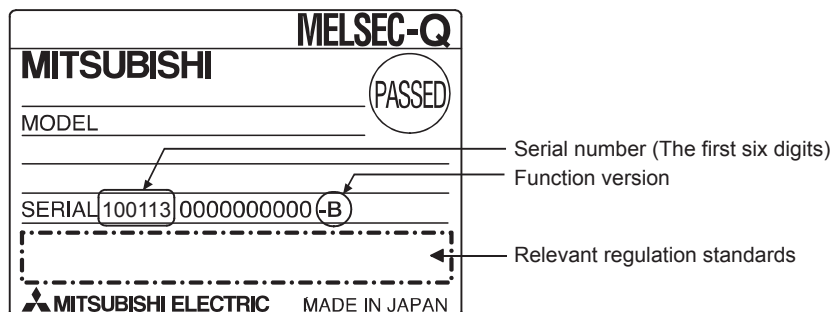
When using GX Works2, refer to the following:

- GX Works2 Version 1 Operating Manual (Common)
- GX Works2 Version 1 Operating Manual (Intelligent Function Module)

## 2.4 How to check the function version and SERIAL No.

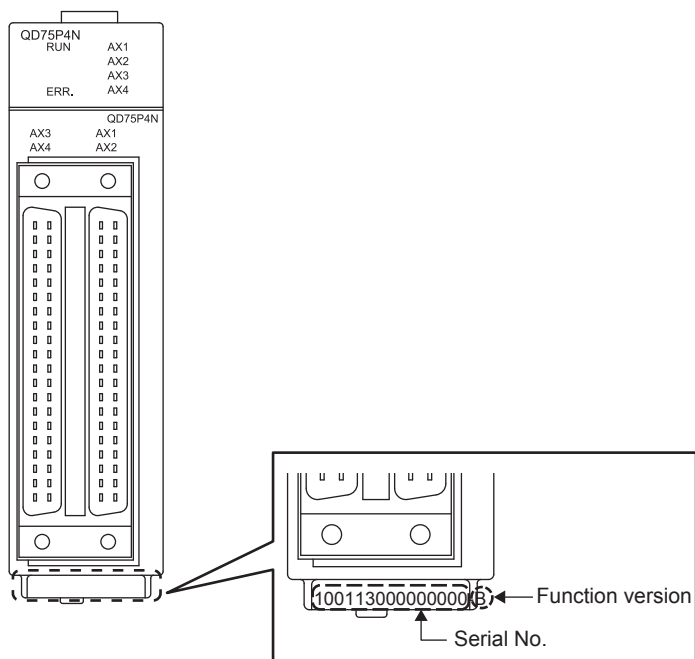
## (1) Confirming the serial number on the rating plate

The rating plate is situated on the side face of the QD75.



## (2) Checking on the front of the module

The serial No. on the rating plate is also indicated on the front of the module (lower part).



## (3) Confirming the serial number with software

Check the function version and SERIAL No. in "Product information" displayed on System monitor "Module's Detailed Information" of GX Developer or on "OS information" of GX Configurator-QP <sup>\*1</sup>

## (a) Checking on the System monitor (Product Information List) screen

To open the screen, select [Diagnostics] → [System monitor] and click the Product Information List button in GX Developer.

<GX Developer display screen>

| Product Information List |          |        |            |        |         |            |                  |      |                   |
|--------------------------|----------|--------|------------|--------|---------|------------|------------------|------|-------------------|
| Slot                     | Type     | Series | Model name | Points | I/O No. | Master PLC | Serial No.       | Ver. | Product No.       |
| PLC                      | PLC      | Q      | Q02UCPU    | -      | -       | -          | 1002200000000000 | B    | 091012092915091-B |
| 0-0                      | Intelli. | Q      | QD75P4     | 32pt   | 0010    | PLC No.1   | 0711200000000000 | B    | -                 |
| 0-1                      | -        | -      | None       | -      | -       | -          | -                | -    | -                 |
| 0-2                      | -        | -      | None       | -      | -       | -          | -                | -    | -                 |
| 0-3                      | -        | -      | None       | -      | -       | -          | -                | -    | -                 |
| 0-4                      | -        | -      | None       | -      | -       | -          | -                | -    | -                 |

[Production No. display]

Since the QD75 does not support the production number display, "-" is displayed.

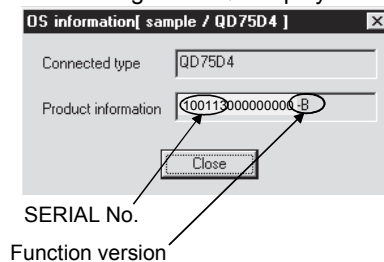
**POINT**

The serial No. on the rating plate may be different from the serial No. displayed on the product information screen of GX Developer.

- The serial No. on the rating plate indicates the management information of the product.
- The serial No. displayed on the product information screen of GX Developer indicates the function information of the product. The function information of the product is updated when a new function is added.

## (b) Checking with GX Configurator-QP

<GX Configurator-QP display screen>



\*1: For details, refer to GX Configurator-QP Operating Manual.

[illegible]

# CHAPTER 3 SPECIFICATIONS AND FUNCTIONS

The various specifications of the QD75 are explained in this chapter.

The "Performance specifications", "List of functions", "Specifications of input/output signals with CPU module", and the "Specifications of input/output interfaces with external devices", etc., are described as information required when designing the positioning system.

Confirm each specification before designing the positioning system.

|       |                                                                       |       |
|-------|-----------------------------------------------------------------------|-------|
| 3.1   | Performance specifications .....                                      | 3- 2  |
| 3.2   | List of functions .....                                               | 3- 6  |
| 3.2.1 | QD75 control functions .....                                          | 3- 6  |
| 3.2.2 | QD75 main functions .....                                             | 3- 8  |
| 3.2.3 | QD75 sub functions and common functions .....                         | 3- 10 |
| 3.2.4 | Combination of QD75 main functions and sub functions .....            | 3- 14 |
| 3.3   | Specifications of input/output signals with CPU module .....          | 3- 16 |
| 3.3.1 | List of input/output signals with CPU module .....                    | 3- 16 |
| 3.3.2 | Details of input signals (QD75 → CPU module) .....                    | 3- 17 |
| 3.3.3 | Details of output signals (CPU module → QD75) .....                   | 3- 18 |
| 3.4   | Specifications of input/output interfaces with external devices ..... | 3- 19 |
| 3.4.1 | Electrical specifications of input/output signals .....               | 3- 19 |
| 3.4.2 | Signal layout for external device connection connector .....          | 3- 25 |
| 3.4.3 | List of input/output signal details .....                             | 3- 26 |
| 3.4.4 | Input/output interface internal circuit .....                         | 3- 28 |

## 3.1 Performance specifications

## ■ QD75P□N/QD75D□N

| Model                  |                                       | QD75P1N *1<br>QD75D1N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | QD75P2N *1<br>QD75D2N                                        | QD75P4N *1<br>QD75D4N                                                                                   |
|------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Item                   |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                              |                                                                                                         |
| No. of control axes    |                                       | 1 axis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2 axes                                                       | 4 axes                                                                                                  |
| Interpolation function |                                       | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2-axis linear interpolation<br>2-axis circular interpolation | 2-, 3-, or 4-axis linear interpolation<br>2-axis circular interpolation<br>3-axis helical interpolation |
| Control system         |                                       | PTP (Point To Point) control, path control (all of linear, circular, and helical can be set), speed control, speed-position switching control, position-speed switching control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                              |                                                                                                         |
| Control unit           |                                       | mm, inch, degree, pulse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                              |                                                                                                         |
| Positioning data       |                                       | 600 data /axis<br>(Can be set with peripheral device or sequence program.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                              |                                                                                                         |
| Backup                 |                                       | Parameters, positioning data, and block start data can be saved on flash ROM (battery-less backup)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                              |                                                                                                         |
| Positioning            | Positioning system                    | PTP control: Incremental system/absolute system<br>Speed-position switching control: Incremental system/absolute system *2<br>Position-speed switching control: Incremental system<br>Path control: Incremental system/absolute system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                              |                                                                                                         |
|                        | Positioning range                     | <div>In absolute system</div> <ul style="list-style-type: none"> <li>• -214748364.8 to 214748364.7 (μm)</li> <li>• -21474.83648 to 21474.83647 (inch)</li> <li>• 0 to 359.99999 (degree)</li> <li>• -2147483648 to 2147483647 (pulse)</li> </ul> <div>In incremental system</div> <ul style="list-style-type: none"> <li>• -214748364.8 to 214748364.7 (μm)</li> <li>• -21474.83648 to 21474.83647 (inch)</li> <li>• -21474.83648 to 21474.83647 (degree)</li> <li>• -2147483648 to 2147483647 (pulse)</li> </ul> <div>In speed-position switching control (INC mode) / position-speed switching control</div> <ul style="list-style-type: none"> <li>• 0 to 214748364.7 (μm)</li> <li>• 0 to 21474.83647 (inch)</li> <li>• 0 to 21474.83647 (degree)</li> <li>• 0 to 2147483647 (pulse)</li> </ul> <div>In speed-position switching control (ABS mode) *2</div> <ul style="list-style-type: none"> <li>• 0 to 359.99999 (degree)</li> </ul> |                                                              |                                                                                                         |
|                        | Speed command                         | 0.01 to 20000000.00 (mm/min)<br>0.001 to 2000000.000 (inch/min)<br>0.001 to 2000000.000 (degree/min)<br>1 to 4000000 (pulse/s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                              |                                                                                                         |
|                        | Acceleration/<br>deceleration process | Trapezoidal acceleration/deceleration, S-curve acceleration/deceleration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                              |                                                                                                         |
|                        | Acceleration/<br>deceleration time    | 1 to 8388608 (ms)<br>Four patterns can be set for each of acceleration time and deceleration time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                              |                                                                                                         |
|                        | Sudden stop<br>deceleration time      | 1 to 8388608 (ms)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                              |                                                                                                         |

\*1: QD75P□N represents the open collector output system, and QD75D□N represents the differential driver output system.

\*2: In speed-position switching control (ABS mode), the control unit available is "degree" only. (For details, refer to Section 9.2.19 "Speed-position switching control (ABS mode)".)



| Model                                    | QD75P1N *1<br>QD75D1N                                                                                         | QD75P2N *1<br>QD75D2N              | QD75P4N *1<br>QD75D4N                                                                                                                                                                                                                                                                                                    |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Item                                     |                                                                                                               |                                    |                                                                                                                                                                                                                                                                                                                          |
| Starting time *3                         | 1-axis linear control                                                                                         | 1.5ms                              | Factors in starting time extension<br>The following times will be added to the starting time in the described conditions:<br>• S-curve acceleration/ deceleration is selected: 0.1ms<br>• Other axis is in operation: 0.5ms<br>• During continuous positioning control: 0.3ms<br>• During continuous path control: 0.3ms |
|                                          | 1-axis speed control                                                                                          | 1.5ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 2-axis linear interpolation control (Composite speed)                                                         | 1.5ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 2-axis linear interpolation control (Reference axis speed)                                                    | 1.5ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 2-axis circular interpolation control                                                                         | 2.0ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 2-axis speed control                                                                                          | 1.5ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 3-axis linear interpolation control (Composite speed)                                                         | 1.7ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 3-axis linear interpolation control (Reference axis speed)                                                    | 1.7ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 3-axis helical interpolation control                                                                          | 2.6ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 3-axis speed control                                                                                          | 1.7ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 4-axis linear interpolation control                                                                           | 1.8ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 4-axis speed control                                                                                          | 1.8ms                              |                                                                                                                                                                                                                                                                                                                          |
| External wiring connection system        | 40-pin connector                                                                                              |                                    |                                                                                                                                                                                                                                                                                                                          |
| Applicable wire size                     | 0.3mm <sup>2</sup> (22AWG) (for A6CON1 or A6CON4),<br>0.088 to 0.24mm <sup>2</sup> (28 to 24AWG) (for A6CON2) |                                    |                                                                                                                                                                                                                                                                                                                          |
| Applicable connector for external device | A6CON1, A6CON2, A6CON4 (sold separately)                                                                      |                                    |                                                                                                                                                                                                                                                                                                                          |
| Max. output pulse                        | QD75P1N, QD75P2N, QD75P4N: 200kpps<br>QD75D1N, QD75D2N, QD75D4N: 4Mpps                                        |                                    |                                                                                                                                                                                                                                                                                                                          |
| Max. connection distance between servos  | QD75P1N, QD75P2N, QD75P4N: 2m<br>QD75D1N, QD75D2N, QD75D4N: 10m                                               |                                    |                                                                                                                                                                                                                                                                                                                          |
| Internal current consumption (5VDC)      | QD75P1N: 0.29A<br>QD75D1N: 0.43A                                                                              | QD75P2N: 0.30A<br>QD75D2N: 0.45A   | QD75P4N: 0.36A<br>QD75D4N: 0.66A                                                                                                                                                                                                                                                                                         |
| Flash ROM write count                    | Max. 100000 times                                                                                             |                                    |                                                                                                                                                                                                                                                                                                                          |
| No. of occupied I/O points               | 32 points (I/O assignment: intelligent 32 points)                                                             |                                    |                                                                                                                                                                                                                                                                                                                          |
| Outline dimensions                       | 98 (H) × 27.4 (W) × 90 (D) mm                                                                                 |                                    |                                                                                                                                                                                                                                                                                                                          |
| Weight                                   | QD75P1N: 0.14kg<br>QD75D1N: 0.15kg                                                                            | QD75P2N: 0.14kg<br>QD75D2N: 0.15kg | QD75P4N: 0.16kg<br>QD75D4N: 0.16kg                                                                                                                                                                                                                                                                                       |

\*1: QD75P□N represents the open collector output system, and QD75D□N represents the differential driver output system.

\*3: Using the "Pre-reading start function", the virtual start time can be shortened. (For details, refer to Section 12.7.7 "Pre-reading start function".

## ■ QD75P□/QD75D□

| Model                  |                                       | QD75P1 *1<br>QD75D1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | QD75P2 *1<br>QD75D2                                          | QD75P4 *1<br>QD75D4                                                     |
|------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------|
| Item                   |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                              |                                                                         |
| No. of control axes    |                                       | 1 axis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2 axes                                                       | 4 axes                                                                  |
| Interpolation function |                                       | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2-axis linear interpolation<br>2-axis circular interpolation | 2-, 3-, or 4-axis linear interpolation<br>2-axis circular interpolation |
| Control system         |                                       | PTP (Point To Point) control, path control (both linear and arc can be set), speed control, speed-position switching control, position-speed switching control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                              |                                                                         |
| Control unit           |                                       | mm, inch, degree, pulse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                              |                                                                         |
| Positioning data       |                                       | 600 data /axis<br>(Can be set with peripheral device or sequence program.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                              |                                                                         |
| Backup                 |                                       | Parameters, positioning data, and block start data can be saved on flash ROM (battery-less backup)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                              |                                                                         |
| Positioning            | Positioning system                    | PTP control: Incremental system/absolute system<br>Speed-position switching control: Incremental system/absolute system *2<br>Position-speed switching control: Incremental system<br>Path control: Incremental system/absolute system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                              |                                                                         |
|                        | Positioning range                     | <div>In absolute system</div> <ul style="list-style-type: none"> <li>• -214748364.8 to 214748364.7 (μm)</li> <li>• -21474.83648 to 21474.83647 (inch)</li> <li>• 0 to 359.99999 (degree)</li> <li>• -2147483648 to 2147483647 (pulse)</li> </ul> <div>In incremental system</div> <ul style="list-style-type: none"> <li>• -214748364.8 to 214748364.7 (μm)</li> <li>• -21474.83648 to 21474.83647 (inch)</li> <li>• -21474.83648 to 21474.83647 (degree)</li> <li>• -2147483648 to 2147483647 (pulse)</li> </ul> <div>In speed-position switching control (INC mode) / position-speed switching control</div> <ul style="list-style-type: none"> <li>• 0 to 214748364.7 (μm)</li> <li>• 0 to 21474.83647 (inch)</li> <li>• 0 to 21474.83647 (degree)</li> <li>• 0 to 2147483647 (pulse)</li> </ul> <div>In speed-position switching control (ABS mode) *2</div> <ul style="list-style-type: none"> <li>• 0 to 359.99999 (degree)</li> </ul> |                                                              |                                                                         |
|                        | Speed command                         | 0.01 to 20000000.00 (mm/min)<br>0.001 to 2000000.000 (inch/min)<br>0.001 to 2000000.000 (degree/min)<br>1 to 1000000 (pulse/s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                              |                                                                         |
|                        | Acceleration/<br>deceleration process | Trapezoidal acceleration/deceleration, S-curve acceleration/deceleration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                              |                                                                         |
|                        | Acceleration/<br>deceleration time    | 1 to 8388608 (ms)<br>Four patterns can be set for each of acceleration time and deceleration time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                              |                                                                         |
|                        | Sudden stop<br>deceleration time      | 1 to 8388608 (ms)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                              |                                                                         |

\*1: QD75P □ represents the open collector output system, and QD75D □ represents the differential driver output system.

\*2: In speed-position switching control (ABS mode), the control unit available is "degree" only. (For details, refer to Section 9.2.19 "Speed-position switching control (ABS mode)".)

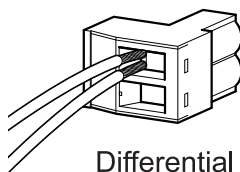
| Model                                    | QD75P1 *1<br>QD75D1                                                                                           | QD75P2 *1<br>QD75D2              | QD75P4 *1<br>QD75D4                                                                                                                                                                                                                                                                                                      |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Item                                     |                                                                                                               |                                  |                                                                                                                                                                                                                                                                                                                          |
| Starting time *3                         | 1-axis linear control                                                                                         | 6ms                              | Factors in starting time extension<br>The following times will be added to the starting time in the described conditions:<br>• S-curve acceleration/ deceleration is selected: 0.5ms<br>• Other axis is in operation: 1.5ms<br>• During continuous positioning control: 0.2ms<br>• During continuous path control: 1.0ms |
|                                          | 1-axis speed control                                                                                          | 6ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 2-axis linear interpolation control (Composite speed)                                                         | 7ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 2-axis linear interpolation control (Reference axis speed)                                                    | 7ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 2-axis circular interpolation control                                                                         | 7ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 2-axis speed control                                                                                          | 6ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 3-axis linear interpolation control (Composite speed)                                                         | 7ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 3-axis linear interpolation control (Reference axis speed)                                                    | 7ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 3-axis speed control                                                                                          | 6ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 4-axis linear interpolation control                                                                           | 7ms                              |                                                                                                                                                                                                                                                                                                                          |
|                                          | 4-axis speed control                                                                                          | 7ms                              |                                                                                                                                                                                                                                                                                                                          |
| External wiring connection system        | 40-pin connector                                                                                              |                                  |                                                                                                                                                                                                                                                                                                                          |
| Applicable wire size                     | 0.3mm <sup>2</sup> (22AWG) (for A6CON1 or A6CON4),<br>0.088 to 0.24mm <sup>2</sup> (28 to 24AWG) (for A6CON2) |                                  |                                                                                                                                                                                                                                                                                                                          |
| Applicable connector for external device | A6CON1, A6CON2, A6CON4 (sold separately)                                                                      |                                  |                                                                                                                                                                                                                                                                                                                          |
| Max. output pulse                        | QD75P1, QD75P2, QD75P4 : 200kpps<br>QD75D1, QD75D2, QD75D4 : 1Mpps                                            |                                  |                                                                                                                                                                                                                                                                                                                          |
| Max. connection distance between servos  | QD75P1, QD75P2, QD75P4 : 2m<br>QD75D1, QD75D2, QD75D4 : 10m                                                   |                                  |                                                                                                                                                                                                                                                                                                                          |
| Internal current consumption (5VDC)      | QD75P1 : 0.40A<br>QD75D1 : 0.52A                                                                              | QD75P2 : 0.46A<br>QD75D2 : 0.56A | QD75P4 : 0.58A<br>QD75D4 : 0.82A                                                                                                                                                                                                                                                                                         |
| Flash ROM write count                    | Max. 100000 times                                                                                             |                                  |                                                                                                                                                                                                                                                                                                                          |
| No. of occupied I/O points               | 32 points (I/O assignment: intelligent 32 points)                                                             |                                  |                                                                                                                                                                                                                                                                                                                          |
| Outline dimensions                       | 98 (H) × 27.4 (W) × 90 (D) mm                                                                                 |                                  |                                                                                                                                                                                                                                                                                                                          |
| Weight                                   | 0.15 kg                                                                                                       | 0.15 kg                          | 0.16 kg                                                                                                                                                                                                                                                                                                                  |

\*1: QD75P □ represents the open collector output system, and QD75D □ represents the differential driver output system.

\*3: Using the "Pre-reading start function", the virtual start time can be shortened. (For details, refer to Section 12.7.7 "Pre-reading start function".

#### ■ Differential driver common terminal specifications(QD75D□N/QD75D□ only)

|                                                                   |                                                   |
|-------------------------------------------------------------------|---------------------------------------------------|
| Applicable wire size                                              | 0.24 to 2.5mm <sup>2</sup> (24 to 12AWG)          |
| Applicable sizes of two wires inserted into same spot of terminal | Solid wire: 0.2 to 0.8mm <sup>2</sup> × 2 pcs.    |
|                                                                   | Stranded wire: 0.2 to 0.8mm <sup>2</sup> × 2 pcs. |
| Screw tightening torque                                           | 0.5 to 0.6N · m                                   |



Differential driver common terminal (accessory)

Fig. 3.1 Two wires inserted into same spot of terminal

## 3.2 List of functions

### 3.2.1 QD75 control functions

The QD75 has several functions. In this manual, the QD75 functions are categorized and explained as follows.

#### ■ Main functions

##### (1) OPR control

"OPR control" is a function that established the start point for carrying out positioning control, and carries out positioning toward that start point. This is used to return a workpiece, located at a position other than the OP when the power is turned ON or after positioning stop, to the OP. The "OPR control" is preregistered in the QD75 as the "Positioning start data No. 9001 (Machine OPR)", and "Positioning start data No. 9002 (Fast OPR)". (Refer to CHAPTER 8 "OPR CONTROL".)

##### (2) Major positioning control

This control is carried out using the "Positioning data" stored in the QD75. Positioning control, such as position control and speed control, is executed by setting the required items in this "positioning data" and starting that positioning data. An "operation pattern" can be set in this "positioning data", and with this whether to carry out control with continuous positioning data (ex.: positioning data No. 1, No. 2, No. 3, ...) can be set. (Refer to CHAPTER 9 "MAJOR POSITIONING CONTROL".)

##### (3) High-level positioning control

This control executes the "positioning data" stored in the QD75 using the "block start data". The following types of applied positioning control can be carried out.

- Random blocks, handling several continuing positioning data items as "blocks", can be executed in the designated order.
- "Condition judgment" can be added to position control and speed control.
- The operation of the designated positioning data No. that is set for multiple axes can be started simultaneously. (Pulses are output simultaneously to multiple servos.)
- The designated positioning data can be executed repeatedly, etc., (Refer to CHAPTER 10 "HIGH-LEVEL POSITIONING CONTROL".)

##### (4) Manual control

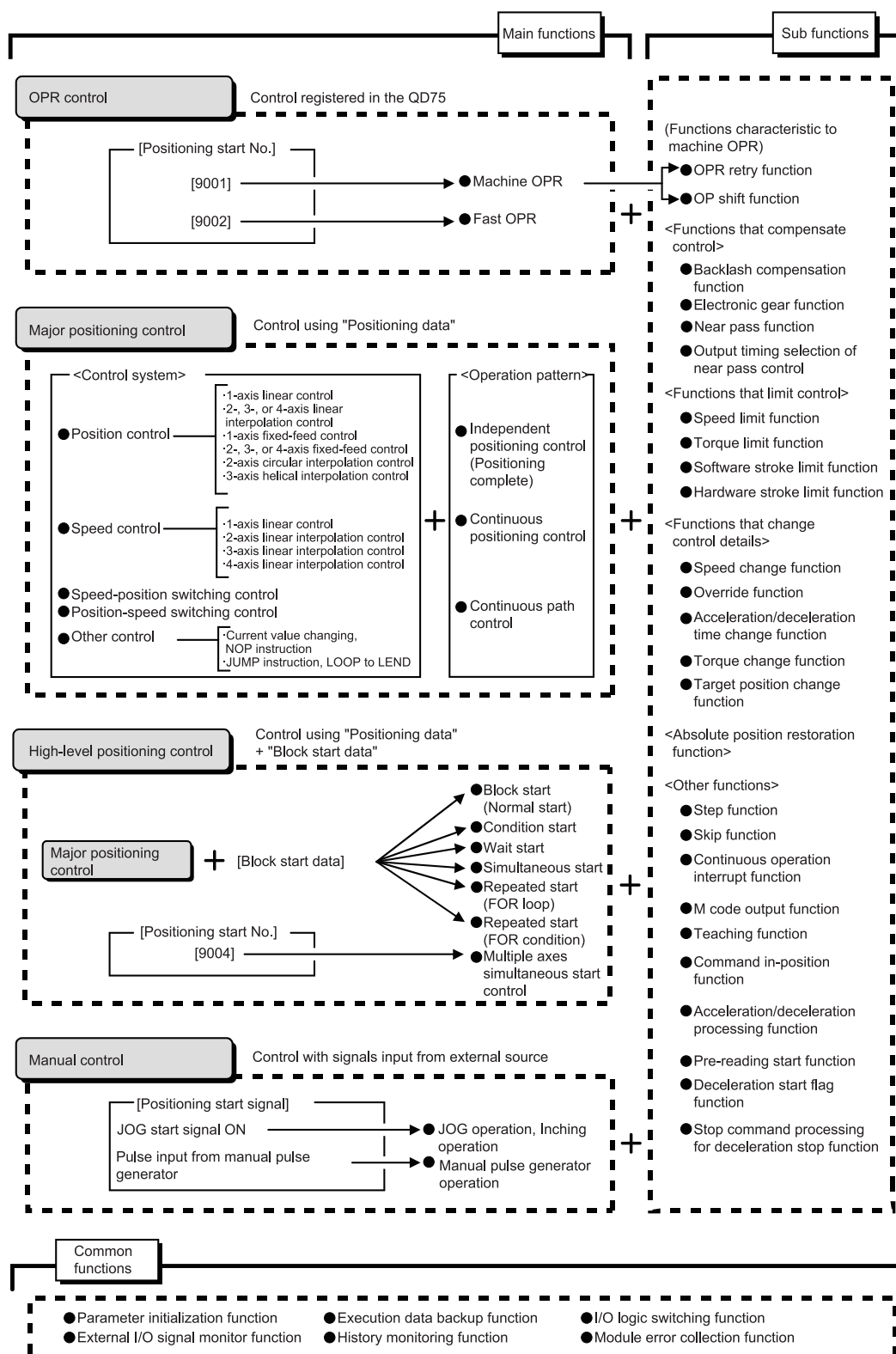
By inputting a signal into the QD75 from an external source, the QD75 will output a random pulse train and carry out control. Use this manual control to move the workpiece to a random position (JOG operation), and to finely adjust the positioning (inching operation, manual pulse generator operation), etc. (Refer to CHAPTER 11 "MANUAL CONTROL".)

#### ■ Sub functions

When executing the main functions, control compensation, limits and functions can be added. (Refer to CHAPTER 12 "CONTROL SUB FUNCTIONS".)

#### ■ Common functions

Common control using the QD75 for "parameter initialization" or "backup of execution data" can be carried out. (Refer to CHAPTER 13 "COMMON FUNCTIONS".)



## 3.2.2 QD75 main functions

The outline of the main functions for positioning control with the QD75 is described below. (Refer to PART 2 for details on each function.)

| Main functions            |                                  | Details                                                                                                                                                              | Reference section                    |
|---------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| OPR control               | Machine OPR control              | Mechanically establishes the positioning start point using a near-point dog or stopper. (Positioning start No. 9001)                                                 | 8.2                                  |
|                           | Fast OPR control                 | Positions a target to the OP address ( <u>Md.21</u> Machine feed value) stored in the QD75 using machine OPR. (Positioning start No. 9002)                           | 8.3                                  |
| Major positioning control | Position control                 | Linear control<br>(1-axis linear control)<br>(2-axis linear interpolation control)<br>(3-axis linear interpolation control)<br>(4-axis linear interpolation control) | 9.2.2<br>9.2.3<br>9.2.4<br>9.2.5     |
|                           |                                  | Fixed-feed control<br>(1-axis fixed-feed control)<br>(2-axis fixed-feed control)<br>(3-axis fixed-feed control)<br>(4-axis fixed-feed control)                       | 9.2.6<br>9.2.7<br>9.2.8<br>9.2.9     |
|                           |                                  | 2-axis circular interpolation control                                                                                                                                | 9.2.10<br>9.2.11                     |
|                           |                                  | 3-axis helical interpolation control                                                                                                                                 | 9.2.12<br>9.2.13                     |
|                           |                                  |                                                                                                                                                                      |                                      |
|                           | Speed control                    | Linear control<br>(1-axis speed control)<br>(2-axis speed control)<br>(3-axis speed control)<br>(4-axis speed control)                                               | 9.2.14<br>9.2.15<br>9.2.16<br>9.2.17 |
|                           |                                  |                                                                                                                                                                      |                                      |
|                           | Speed-position switching control |                                                                                                                                                                      | 9.2.18<br>9.2.19                     |
|                           | Position-speed switching control |                                                                                                                                                                      | 9.2.20                               |
|                           | Other control                    | Current value changing                                                                                                                                               | 9.2.21                               |
|                           |                                  | NOP instruction                                                                                                                                                      | 9.2.22                               |
|                           |                                  | JUMP instruction                                                                                                                                                     | 9.2.23                               |
|                           |                                  | LOOP                                                                                                                                                                 | 9.2.24                               |
|                           |                                  | LEND                                                                                                                                                                 | 9.2.25                               |

| Main functions                 |                                          | Details                                                                                                                                                                                                                                                                                                                            | Reference section |
|--------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| High-level positioning control | Block start (Normal start)               | With one start, executes the positioning data in a random block with the set order.                                                                                                                                                                                                                                                | 10.3.2            |
|                                | Condition start                          | Carries out condition judgment set in the "condition data" for the designated positioning data, and then executes the "block start data".<br>When the condition is established, the "block start data" is executed. When not established, that "block start data" is ignored, and the next point's "block start data" is executed. | 10.3.3            |
|                                | Wait start                               | Carries out condition judgment set in the "condition data" for the designated positioning data, and then executes the "block start data".<br>When the condition is established, the "block start data" is executed. When not established, stops the control until the condition is established. (Waits.)                           | 10.3.4            |
|                                | Simultaneous start                       | Simultaneously executes the positioning data having the No. for the axis designated with the "condition data". (Outputs pulses at the same timing.)                                                                                                                                                                                | 10.3.5            |
|                                | Repeated start (FOR loop)                | Repeats the program from the block start data set with the "FOR loop" to the block start data set in "NEXT" for the designated No. of times.                                                                                                                                                                                       | 10.3.6            |
|                                | Repeated start (FOR condition)           | Repeats the program from the block start data set with the "FOR condition" to the block start data set in "NEXT" until the conditions set in the "condition data" are established.                                                                                                                                                 | 10.3.7            |
|                                | Multiple axes simultaneous start control | Starts the operation of multiple axes simultaneously according to the pulse output level.<br>(Positioning start No. 9004, same as the "simultaneous start" above)                                                                                                                                                                  | 10.5              |
| Manual control                 | JOG operation                            | Outputs a pulse to drive unit while the JOG start signal is ON.                                                                                                                                                                                                                                                                    | 11.2              |
|                                | Inching operation                        | Outputs pulses corresponding to minute movement amount by manual operation to drive unit.<br>(Performs fine adjustment with the JOG start signal.)                                                                                                                                                                                 | 11.3              |
|                                | Manual pulse generator operation         | Outputs pulses commanded with the manual pulse generator to drive unit. (Carry out fine adjustment, etc., at the pulse level.)                                                                                                                                                                                                     | 11.4              |

With the "major positioning control" ("high-level positioning control"), whether or not to continuously execute the positioning data can be set with the "operation pattern".

Outlines of the "operation patterns" are given below.

| Da.1 Operation pattern                                    | Details                                                                                                                                                                                                                                               | Reference section |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Independent positioning control<br>(Positioning complete) | When "independent positioning control" is set for the operation pattern of the started positioning data, only the designated positioning data will be executed, and then the positioning will end.                                                    | 9.1.2             |
| Continuous positioning control                            | When "continuous positioning control" is set for the operation pattern of the started positioning data, after the designated positioning data is executed, the program will stop once, and then the next following positioning data will be executed. |                   |
| Continuous path control                                   | When "continuous path control" is set for the operation pattern of the started positioning data, the designated positioning data will be executed, and then without decelerating, the next following positioning data will be executed.               |                   |

## 3.2.3 QD75 sub functions and common functions

## ■ Sub functions

The functions that assist positioning control using the QD75 are described below.  
(Refer to PART 2 for details on each function.)

| Sub function                            |                                                | Details                                                                                                                                                                                                                                                                                   | Reference section |
|-----------------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Functions characteristic to machine OPR | OPR retry function                             | This function retries the machine OPR with the upper/lower limit switches during machine OPR. This allows machine OPR to be carried out even if the axis is not returned to before the near-point dog with JOG operation, etc.                                                            | 12.2.1            |
|                                         | OP shift function                              | After returning to the machine OP, this function compensates the position by the designated distance from the machine OP position and sets that position as the OP address.                                                                                                               | 12.2.2            |
| Functions that compensate control       | Backlash compensation function                 | This function compensates the mechanical backlash. Feed pulses equivalent to the set backlash amount are output each time the movement direction changes.                                                                                                                                 | 12.3.1            |
|                                         | Electronic gear function                       | By setting the movement amount per pulse, this function can freely change the machine movement amount per commanded pulse.<br>A flexible positioning system that matches the machine system can be structured with this function.                                                         | 12.3.2            |
|                                         | Near pass function *1                          | This function suppresses the machine vibration when the positioning data is switched during continuous path control in the interpolation control.                                                                                                                                         | 12.3.3            |
|                                         | Output timing selection of near pass control   | This function allows the user to select the timing to output the difference ( $\Delta \ell$ ) between the actual and the set positioning end addresses in continuous path control, in which the difference ( $\Delta \ell$ ) is output during the execution of the next positioning data. | 12.3.4            |
| Functions that limit control            | Speed limit function                           | If the command speed exceeds " [Pr.8] Speed limit value" during control, this function limits the commanded speed to within the " [Pr.8] Speed limit value" setting range.                                                                                                                | 12.4.1            |
|                                         | Torque limit function *2                       | If the torque generated by the servomotor exceeds " [Pr.17] Torque limit setting value" during control, this function limits the generated torque to within the " [Pr.17] Torque limit setting value" setting range.                                                                      | 12.4.2            |
|                                         | Software stroke limit function                 | If a command outside of the upper/lower limit stroke limit setting range, set in the parameters, is issued, this function will not execute positioning for that command.                                                                                                                  | 12.4.3            |
|                                         | Hardware stroke limit function                 | This function carries out deceleration stop with the limit switch connected to the QD75 external device connector.                                                                                                                                                                        | 12.4.4            |
| Functions that change control details   | Speed change function                          | This function changes the speed during positioning.<br>Set the new speed in the speed change buffer memory ([Cd.14] New speed value), and change the speed with the Speed change request ([Cd.15]).                                                                                       | 12.5.1            |
|                                         | Override function                              | This function changes the speed within a percentage of 1 to 300% during positioning. This is executed using " [Cd.13] Positioning operation speed override".                                                                                                                              | 12.5.2            |
|                                         | Acceleration/deceleration time change function | This function changes the acceleration/deceleration time during speed change. (Functions added to the speed change function and override function)                                                                                                                                        | 12.5.3            |
|                                         | Torque change function                         | This function changes the "torque limit value" during control.                                                                                                                                                                                                                            | 12.5.4            |
|                                         | Target position change function                | This function changes the target position during positioning. Position and speed can be changed simultaneously.                                                                                                                                                                           | 12.5.5            |



\* 1: The near pass function is featured as standard and is valid only for position control. It cannot be set to be invalid with parameters.

\* 2: Using "Torque limit function" requires a "D/A conversion module" and a "drive unit capable of torque limit command with analog voltage".

| Sub function                              | Details                                                                                                                                                                                           | Reference section |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Absolute position restoration function *3 | This function restores the absolute position of designated axis. By this function, the OPR after power ON from OFF is not required once the OPR is executed when the system operation is started. | 12.6              |

\* 3: To execute the "absolute position restoration function", the following are required: "16-point input module"; "16-point output module"; and "drive unit capable of configuring an absolute position detection system, which is a Mitsubishi Electric General-Purpose AC Servo having an absolute position detection function (absolute position data transference protocol) equivalent to that of MR-J3-□A".

| Sub function    |                                                        | Details                                                                                                                                                                                                                                                                                                                                     | Reference section |
|-----------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Other functions | Step function                                          | This function temporarily stops the operation to confirm the positioning operation during debugging, etc. The operation can be stopped at each "automatic deceleration" or "positioning data".                                                                                                                                              | 12.7.1            |
|                 | Skip function                                          | This function stops (decelerates to a stop) the positioning being executed when the skip signal is input, and carries out the next positioning.                                                                                                                                                                                             | 12.7.2            |
|                 | M code output function                                 | This function issues a command for a sub work (clamp or drill stop, tool change, etc.) corresponding to the M code No. (0 to 65535) that can be set for each positioning data.                                                                                                                                                              | 12.7.3            |
|                 | Teaching function                                      | This function stores the address positioned with manual control into the positioning address having the designated positioning data No. ( <u>Cd.39</u> ).                                                                                                                                                                                   | 12.7.4            |
|                 | Command in-position function                           | At each automatic deceleration, this function calculates the remaining distance for the QD75 to reach the positioning stop position. When the value is less than the set value, the "command in-position flag" is set to "1". When using another auxiliary work before ending the control, use this function as a trigger for the sub work. | 12.7.5            |
|                 | Acceleration/deceleration process function             | This function adjusts the control acceleration/deceleration. (acceleration/deceleration time and curve).                                                                                                                                                                                                                                    | 12.7.6            |
|                 | Continuous operation interrupt function                | This function interrupts continuous operation. When this request is accepted, the operation stops when the execution of the current positioning data is completed.                                                                                                                                                                          | 6.5.4             |
|                 | Pre-reading start function                             | This function shortens the virtual start time.                                                                                                                                                                                                                                                                                              | 12.7.7            |
|                 | Deceleration start flag function                       | Function that turns ON the flag when the constant speed status or acceleration status switches to the deceleration status during position control, whose operation pattern is "Positioning complete", to make the stop timing known.                                                                                                        | 12.7.8            |
|                 | Stop command processing for deceleration stop function | Function that selects a deceleration curve when a stop cause occurs during deceleration stop processing to speed 0.                                                                                                                                                                                                                         | 12.7.9            |

### ■ Common functions

The outline of the functions executed as necessary are described below.  
(Refer to PART 2 for details on each function.)

| Common functions                             | Details                                                                                                                                                                                                                                                                      | Reference section |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Parameter initialization function            | This function returns the "parameters" stored in the QD75 buffer memory and flash ROM to the default values.<br>The following two methods can be used.<br>1) Method using sequence program<br>2) Method using GX Configurator-QP                                             | 13.2              |
| Execution data backup function               | This functions stores the "setting data", currently being executed, into the flash ROM.<br>1) Method using sequence program<br>2) Method using GX Configurator-QP                                                                                                            | 13.3              |
| External I/O signal logic switching function | This function switches I/O signal logic according to externally connected devices.<br>This function enables the use of the system that does not use b (N.C.)-contact signals, such as Drive unit READY or Upper/lower limit signal, by setting parameters to positive logic. | 13.4              |
| External I/O signal monitor function         | This function monitors the external I/O signal monitor information in the module's detailed information which can be displayed on the system monitor of GX Developer*4.                                                                                                      | 13.5              |

\*4: For the QD75P□/QD75D□, this function is available in GX Developer (SW6D5C-GPPW-E or later).

For the QD75P□N/QD75D□N, external I/O signals cannot be monitored on GX Developer. Use the system monitor of GX Works2.

## 3.2.4 Combination of QD75 main functions and sub functions

With positioning control using the QD75, the main functions and sub functions can be combined and used as necessary. A list of the main function and sub function combinations is given below.

| Main functions            |                                  |                                                      | Sub functions                                       | Functions characteristic to machine OPR |                   |
|---------------------------|----------------------------------|------------------------------------------------------|-----------------------------------------------------|-----------------------------------------|-------------------|
|                           |                                  |                                                      |                                                     | OPR retry function                      | OP shift function |
|                           |                                  |                                                      | Combination with operation pattern. ※1              |                                         |                   |
| OPR control               | Machine OPR control              |                                                      | ×                                                   | ○                                       | ○                 |
|                           | Fast OPR control                 |                                                      | ×                                                   | ×                                       | ×                 |
| Major positioning control | Position control                 | 1-axis linear control                                | ○                                                   | ×                                       | ×                 |
|                           |                                  | 2-, 3-, or 4-axis linear interpolation control       | ○                                                   | ×                                       | ×                 |
|                           |                                  | 1-axis fixed-feed control                            | △ (Continuous path control cannot be set)           | ×                                       | ×                 |
|                           |                                  | 2-, 3-, or 4-axis fixed-feed control (interpolation) | △ (Continuous path control cannot be set)           | ×                                       | ×                 |
|                           |                                  | 2-axis circular interpolation control                | ○                                                   | ×                                       | ×                 |
|                           |                                  | 3-axis helical interpolation control                 | ○                                                   | ×                                       | ×                 |
|                           |                                  | Speed control (1- to 4-axis)                         | △ (Only independent positioning control can be set) | ×                                       | ×                 |
|                           | Other control                    | Speed-position switching control                     | △ (Continuous path control cannot be set)           | ×                                       | ×                 |
|                           |                                  | Position-speed switching control                     | △ (Only independent positioning control can be set) |                                         |                   |
|                           |                                  | Current value changing                               | △ (Continuous path control cannot be set)           | ×                                       | ×                 |
|                           |                                  | NOP instruction                                      | ×                                                   |                                         |                   |
|                           |                                  | JUMP instruction                                     | ×                                                   | ×                                       | ×                 |
|                           |                                  | LOOP to LEND                                         |                                                     |                                         |                   |
|                           |                                  |                                                      |                                                     |                                         |                   |
| Manual control            | JOG operation, inching operation |                                                      | ×                                                   | ×                                       | ×                 |
|                           | Manual pulse generator operation |                                                      | ×                                                   | ×                                       | ×                 |

◎: Always combine, ○: Combination possible, △: Combination limited, ×: Combination not possible

※1 The operation pattern is one of the "positioning data" setting items.

※2 The near pass function is featured as standard and is valid only for setting continuous path control for position control.

※3 Invalid during creep speed.

※4 Invalid during continuous path control.

※5 Combination with the inching operation is not available. (Inching operation does not perform acceleration/deceleration processing.)

※6 Valid for the reference axis only.

※7 Valid for only the case where a deceleration start is made during position control.

※8 Valid when using the positioning data but invalid when using the positioning start No. 9003.

|  | Functions that compensate control |                          |                    | Functions that limit control                 |                      |                       |                                | Functions that change control details |                       |                   |                                                 |                        | Other functions                 |               |               |                        |                   |                              |                                            |                            |                                  |                                                        |
|--|-----------------------------------|--------------------------|--------------------|----------------------------------------------|----------------------|-----------------------|--------------------------------|---------------------------------------|-----------------------|-------------------|-------------------------------------------------|------------------------|---------------------------------|---------------|---------------|------------------------|-------------------|------------------------------|--------------------------------------------|----------------------------|----------------------------------|--------------------------------------------------------|
|  | Backlash compensation function    | Electronic gear function | Near pass function | Output timing selection of near pass control | Speed limit function | Torque limit function | Software stroke limit function | Hardware stroke limit function        | Speed change function | Override function | Acceleration/ deceleration time change function | Torque change function | Target position change function | Step function | Skip function | M code output function | Teaching function | Command in-position function | Acceleration/deceleration process function | Pre-reading start function | Deceleration start flag function | Stop command processing for deceleration stop function |
|  | ○                                 | ○                        | *2                 | ×                                            | ○                    | ○                     | ×                              | ◎                                     | △*3                   | △*3               | △*3                                             | ○                      | ×                               | ×             | ×             | ×                      | ×                 | ×                            | ○                                          | ×                          | ×                                | ○                                                      |
|  | ○                                 | ○                        |                    | ×                                            | ○                    | ○                     | ×                              | ◎                                     | ○                     | ○                 | ○                                               | ○                      | ×                               | ×             | ×             | ×                      | ×                 | ×                            | ○                                          | ×                          | ×                                | ○                                                      |
|  | ○                                 | ○                        |                    | ○                                            | ○                    | ○                     | ○                              | ◎                                     | ○                     | ○                 | ○                                               | ○                      | △*4                             | ○             | ○             | ○                      | ×                 | ○                            | ○                                          | ○                          | ○                                | ○                                                      |
|  | ○                                 | ○                        |                    | ○                                            | ○                    | ○                     | ○                              | ◎                                     | ○                     | ○                 | ○                                               | ○                      | ×                               | ○             | ○             | ○                      | ×                 | ○                            | ○                                          | ○                          | △*6                              | ○                                                      |
|  | ○                                 | ○                        |                    | ×                                            | ○                    | ○                     | ○                              | ◎                                     | ○                     | ○                 | ○                                               | ○                      | ×                               | ○             | ○             | ○                      | ×                 | ○                            | ○                                          | ○                          | ○                                | ○                                                      |
|  | ○                                 | ○                        |                    | ×                                            | ○                    | ○                     | ○                              | ◎                                     | ○                     | ○                 | ○                                               | ○                      | ×                               | ○             | ○             | ○                      | ×                 | ○                            | ○                                          | ○                          | △*6                              | ○                                                      |
|  | ○                                 | ○                        |                    | ○                                            | ○                    | ○                     | ○                              | ◎                                     | ○                     | ○                 | ○                                               | ○                      | ×                               | ○             | ○             | ○                      | ×                 | ○                            | ○                                          | ○                          | ×                                | ○                                                      |
|  | ○                                 | ○                        |                    | ○                                            | ○                    | ○                     | ○                              | ◎                                     | ○                     | ○                 | ○                                               | ○                      | ×                               | ○             | ○             | ○                      | ×                 | ○                            | ○                                          | ○                          | ×                                | ○                                                      |
|  | ○                                 | ○                        |                    | ×                                            | ○                    | ○                     | ○                              | ◎                                     | ○                     | ○                 | ○                                               | ○                      | ×                               | ×             | ×             | ○                      | ×                 | ×                            | ○                                          | ○                          | ×                                | ○                                                      |
|  | ○                                 | ○                        |                    | ×                                            | ○                    | ○                     | ○                              | ◎                                     | ○                     | ○                 | ○                                               | ○                      | ×                               | ○             | ○             | ○                      | ×                 | ○                            | ○                                          | ○                          | △*7                              | ○                                                      |
|  | ×                                 | ×                        |                    | ×                                            | ×                    | ×                     | ×                              | ◎                                     | ×                     | ×                 | ×                                               | ×                      | ×                               | ○             | ×             | △*8                    | ×                 | ×                            | ×                                          | ×                          | ×                                | ×                                                      |
|  | ×                                 | ×                        |                    | ×                                            | ×                    | ×                     | ×                              | ◎                                     | ×                     | ×                 | ×                                               | ×                      | ×                               | ×             | ×             | ×                      | ×                 | ×                            | ×                                          | ×                          | ×                                | ×                                                      |
|  | ○                                 | ○                        |                    | ×                                            | ○                    | ○                     | ○                              | ◎                                     | △*5                   | △*5               | △*5                                             | ○                      | ×                               | ×             | ×             | ×                      | ○                 | ×                            | △*5                                        | ×                          | ×                                | ×                                                      |
|  | ○                                 | ○                        |                    | ×                                            | ×                    | ○                     | ○                              | ◎                                     | ×                     | ×                 | ×                                               | ○                      | ×                               | ×             | ×             | ×                      | ○                 | ×                            | ×                                          | ×                          | ×                                | ×                                                      |

## 3.3 Specifications of input/output signals with CPU module

## 3.3.1 List of input/output signals with CPU module

The QD75 uses 32 input points and 32 output points for exchanging data with the CPU module.

The input/output signals when the QD75 is mounted in slot No. 0 of the main base unit are shown below.

Device X refers to the signals input from the QD75 to the CPU module, and device Y refers to the signals output from the CPU module to the QD75.

| Signal direction: QD75 → CPU module |                      |                      | Signal direction: CPU module → QD75 |                |                            |
|-------------------------------------|----------------------|----------------------|-------------------------------------|----------------|----------------------------|
| Device No.                          | Signal name          |                      | Device No.                          | Signal name    |                            |
| X0                                  | QD75 READY           |                      | Y0                                  | PLC READY      |                            |
| X1                                  | Synchronization flag |                      | Y1                                  | Use prohibited |                            |
| X2                                  | Use prohibited       |                      | Y2                                  |                |                            |
| X3                                  |                      |                      | Y3                                  |                |                            |
| X4                                  | Axis 1               | M code ON            | Y4                                  | Axis 1         | Axis stop                  |
| X5                                  | Axis 2               |                      | Y5                                  | Axis 2         |                            |
| X6                                  | Axis 3               |                      | Y6                                  | Axis 3         |                            |
| X7                                  | Axis 4               |                      | Y7                                  | Axis 4         |                            |
| X8                                  | Axis 1               | Error detection      | Y8                                  | Axis 1         | Forward run JOG start      |
| X9                                  | Axis 2               |                      | Y9                                  | Axis 1         | Reverse run JOG start      |
| XA                                  | Axis 3               |                      | YA                                  | Axis 2         | Forward run JOG start      |
| XB                                  | Axis 4               |                      | YB                                  | Axis 2         | Reverse run JOG start      |
| XC                                  | Axis 1               | BUSY                 | YC                                  | Axis 3         | Forward run JOG start      |
| XD                                  | Axis 2               |                      | YD                                  | Axis 3         | Reverse run JOG start      |
| XE                                  | Axis 3               |                      | YE                                  | Axis 4         | Forward run JOG start      |
| XF                                  | Axis 4               |                      | YF                                  | Axis 4         | Reverse run JOG start      |
| X10                                 | Axis 1               | Start complete       | Y10                                 | Axis 1         | Positioning start          |
| X11                                 | Axis 2               |                      | Y11                                 | Axis 2         |                            |
| X12                                 | Axis 3               |                      | Y12                                 | Axis 3         |                            |
| X13                                 | Axis 4               |                      | Y13                                 | Axis 4         |                            |
| X14                                 | Axis 1               | Positioning complete | Y14                                 | Axis 1         | Execution prohibition flag |
| X15                                 | Axis 2               |                      | Y15                                 | Axis 2         |                            |
| X16                                 | Axis 3               |                      | Y16                                 | Axis 3         |                            |
| X17                                 | Axis 4               |                      | Y17                                 | Axis 4         |                            |
| X18                                 | Use prohibited       |                      | Y18                                 | Use prohibited |                            |
| X19                                 |                      |                      | Y19                                 |                |                            |
| X1A                                 |                      |                      | Y1A                                 |                |                            |
| X1B                                 |                      |                      | Y1B                                 |                |                            |
| X1C                                 |                      |                      | Y1C                                 |                |                            |
| X1D                                 |                      |                      | Y1D                                 |                |                            |
| X1E                                 |                      |                      | Y1E                                 |                |                            |
| X1F                                 |                      |                      | Y1F                                 |                |                            |

**Important**

[Y1 to Y3], [Y18 to Y1F], [X2, X3], and [X18 to X1F] are used by the system, and cannot be used by the user.

If these devices are used, the operation of the QD75 will not be guaranteed.

## 3.3.2 Details of input signals (QD75 → CPU module)

The ON/OFF timing and conditions of the input signals are shown below.

| Device No.               | Signal name                          |                         | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|--------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| X0                       | QD75 READY                           |                         | <p>ON: READY<br/>OFF: Not READY/<br/>Watch dog timer error</p> <ul style="list-style-type: none"> <li>When the PLC READY signal [Y0] turns from OFF to ON, the parameter setting range is checked. If no error is found, this signal turns ON.</li> <li>When the PLC READY signal [Y0] turns OFF, this signal turns OFF.</li> <li>When watch dog timer error occurs, this signal turns OFF.</li> <li>This signal is used for interlock in a sequence program, etc.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| X1                       | Synchronization flag                 |                         | <p>OFF: Module access disabled<br/>ON: Module access enabled</p> <ul style="list-style-type: none"> <li>After the programmable controller is turned ON or the CPU module is reset, this signal turns ON if the access from the CPU module to the QD75 is possible.</li> <li>When "Asynchronous" is selected in the module synchronization setting of the CPU module, this signal can be used as interlock for the access from a sequence program to the QD75.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| X4<br>X5<br>X6<br>X7     | Axis 1<br>Axis 2<br>Axis 3<br>Axis 4 | M code ON               | <p>OFF: M code is not set<br/>ON: M code is set</p> <ul style="list-style-type: none"> <li>In the WITH mode, this signal turns ON when the positioning data operation is started. In the AFTER mode, this signal turns ON when the positioning data operation is completed.</li> <li>This signal turns OFF with the "Cd.7 M code OFF request".</li> <li>When M code is not designated (when "Da.10 M code" is "0"), this signal will remain OFF.</li> <li>With using continuous path control for the positioning operation, the positioning will continue even when this signal does not turn OFF. However, the warning "M CODE ON SIGNAL START" (warning code: 503) will occur.</li> <li>When the PLC READY signal [Y0] turns OFF, the M code ON signal will also turn OFF.</li> <li>If the operation is started while the M code is ON, the error "M CODE ON SIGNAL START" (error code: 536) will occur.</li> </ul> |
| X8<br>X9<br>XA<br>XB     | Axis 1<br>Axis 2<br>Axis 3<br>Axis 4 | Error detection         | <p>OFF: No error<br/>ON: Error occurrence</p> <ul style="list-style-type: none"> <li>This signal turns ON when an error listed in Section 15.2 occurs, and turns OFF when the error is reset on "Cd.5 Axis error rest".</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| XC<br>XD<br>XE<br>XF     | Axis 1<br>Axis 2<br>Axis 3<br>Axis 4 | BUSY *1                 | <p>OFF: Not BUSY<br/>ON: BUSY</p> <ul style="list-style-type: none"> <li>This signal turns ON at the start of positioning, OPR or JOG operation. It turns OFF when the "Da.9 Dwell time" has passed after positioning stops. (This signal remains ON during positioning.) This signal turns OFF when the positioning is stopped with step operation.</li> <li>During manual pulse generator operation, this signal turns ON while the "Cd.21 Manual pulse generator enable flag" is ON.</li> <li>This signal turns OFF at error completion or positioning stop.</li> </ul>                                                                                                                                                                                                                                                                                                                                            |
| X10<br>X11<br>X12<br>X13 | Axis 1<br>Axis 2<br>Axis 3<br>Axis 4 | Start complete          | <p>OFF: Start incomplete<br/>ON: Start complete</p> <ul style="list-style-type: none"> <li>This signal turns ON when the positioning start signal turns ON and the QD75 starts the positioning process.<br/>(The start complete signal also turns ON during OPR control.)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| X14<br>X15<br>X16<br>X17 | Axis 1<br>Axis 2<br>Axis 3<br>Axis 4 | Positioning complete *2 | <p>OFF: Positioning incomplete<br/>ON: Positioning complete</p> <ul style="list-style-type: none"> <li>This signal turns ON for the time set in "Pr.40 Positioning complete signal output time" from the instant when the positioning control for each positioning data No. is completed. For the interpolation control, the positioning completed signal of interpolation axis turns ON during the time set to the reference axis.<br/>(It does not turn ON when "Pr.40 Positioning complete signal output time" is "0".)</li> <li>If positioning (including OPR), JOG/Inching operation, or manual pulse generator operation is started while this signal is ON, the signal will turn OFF.</li> <li>This signal will not turn ON when speed control or positioning is canceled midway.</li> </ul>                                                                                                                   |

**Important**

- \*1: The BUSY signal turns ON even when position control of movement amount 0 is executed. However, since the ON time is short, the ON status may not be detected in the sequence program.
- \*2: "Positioning complete" of the QD75 refers to the point when the pulse output from QD75 is completed.  
Thus, even if the QD75's positioning complete signal turns ON, the system may continue operation.

## 3.3.3 Detail of output signals (CPU module → QD75)

The ON/OFF timing and conditions of the output signals are shown below.

| Device No.                                   | Signal name                                                                  |                                                                                                                                                                                                      | Details                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y0                                           | PLC READY                                                                    |                                                                                                                                                                                                      | OFF:<br>PLC READY OFF<br>ON:<br>PLC READY ON                                    | <ul style="list-style-type: none"> <li>(a) This signal notifies the QD75 that the CPU module is normal. <ul style="list-style-type: none"> <li>• It is turned ON/OFF with the sequence program.</li> <li>• The PLC READY signal is turned ON during positioning control, OPR control, JOG operation, inching operation, and manual pulse generator operation, unless the system is in the peripheral device test mode.</li> </ul> </li> <li>(b) When parameters are changed, the PLC READY signal is turned OFF depending on the parameter (Refer to CHAPTER 7.).</li> <li>(c) The following processes are carried out when the PLC READY signal turns from OFF to ON. <ul style="list-style-type: none"> <li>• The parameter setting range is checked.</li> <li>• The QD75 READY signal [X0] turns ON.</li> </ul> </li> <li>(d) The following processes are carried out when the PLC READY signal turns from ON to OFF. <ul style="list-style-type: none"> <li>• In these cases, the OFF time should be set to 100ms or more.</li> <li>• The QD75 READY signal [X0] turns OFF.</li> <li>• The operating axis stops.</li> <li>• The M code ON signal [X4 to X7] for each axis turns OFF, and "0" is stored in " [Md.25] Valid M code".</li> </ul> </li> <li>(e) When parameters or positioning data (No. 1 to 600) are written from the peripheral device or CPU module to the flash ROM, the PLC READY signal will turn OFF.</li> </ul> |
| Y4<br>Y5<br>Y6<br>Y7                         | Axis 1<br>Axis 2<br>Axis 3<br>Axis 4                                         | Axis stop                                                                                                                                                                                            | OFF:<br>Axis stop not requested<br>ON:<br>Axis stop requested                   | <ul style="list-style-type: none"> <li>• When the axis stop signal turns ON, the OPR control, positioning control, JOG operation, inching operation and manual pulse generator operation will stop.</li> <li>• By turning the axis stop signal ON during positioning operation, the positioning operation will be "stopped".</li> <li>• Whether to decelerate or suddenly stop can be selected with " [Pr.39] Stop group 3 sudden stop selection".</li> <li>• During interpolation control of the positioning operation, if the axis stop signal of any axis turns ON, all axes in the interpolation control will decelerate and stop.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Y8<br>Y9<br>YA<br>YB<br>YC<br>YD<br>YE<br>YF | Axis 1<br>Axis 1<br>Axis 2<br>Axis 2<br>Axis 3<br>Axis 3<br>Axis 4<br>Axis 4 | Forward run JOG start<br>Reverse run JOG start<br>Forward run JOG start<br>Reverse run JOG start<br>Forward run JOG start<br>Reverse run JOG start<br>Forward run JOG start<br>Reverse run JOG start | OFF:<br>JOG not started<br>ON:<br>JOG started                                   | <ul style="list-style-type: none"> <li>• When the JOG start signal is ON, JOG operation will be carried out at the " [Cd.17] JOG speed". When the JOG start signal turns OFF, the operation will decelerate and stop.</li> <li>• When inching movement amount is set, the designated movement amount is output for one control cycle and then the operation stops.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Y10<br>Y11<br>Y12<br>Y13                     | Axis 1<br>Axis 2<br>Axis 3<br>Axis 4                                         | Positioning start                                                                                                                                                                                    | OFF:<br>Positioning start not requested<br>ON:<br>Positioning start requested   | <ul style="list-style-type: none"> <li>• OPR operation or positioning operation is started.</li> <li>• The positioning start signal is valid at the rising edge, and the operation is started.</li> <li>• When the positioning start signal turns ON during BUSY, the warning "Start during operation" (warning code: 100) will occur.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Y14<br>Y15<br>Y16<br>Y17                     | Axis 1<br>Axis 2<br>Axis 3<br>Axis 4                                         | Execution prohibition flag                                                                                                                                                                           | OFF:<br>Not during execution prohibition<br>ON:<br>During execution prohibition | <ul style="list-style-type: none"> <li>• If the execution prohibition flag is ON when the positioning start signal turns ON, positioning control does not start until the execution prohibition flag turns OFF. (Pulse output not provided) Used with the "Pre-reading start function". (Refer to Section 12.7.7)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

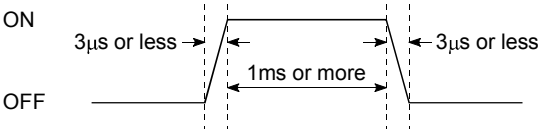
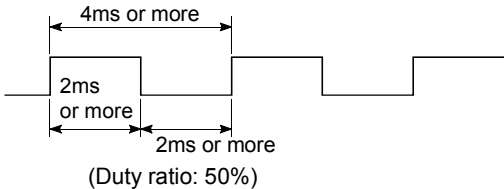
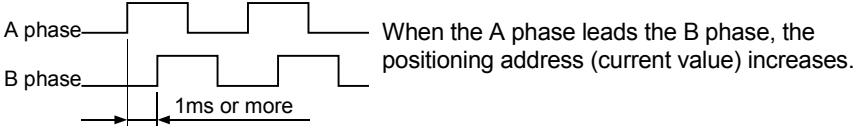


## 3.4 Specifications of input/output interfaces with external devices

## 3.4.1 Electrical specifications of input/output signals

## ■ QD75P□N/QD75D□N

## (1) Input specifications

| Signal name                                                                                              | Rated input voltage/current                                                                                                                                                                                                                       | Working voltage range | ON voltage/current                | OFF voltage/current              | Input resistance | Response time |
|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------|----------------------------------|------------------|---------------|
| Drive unit READY (READY)<br>Stop signal (STOP)<br>Upper limit signal (FLS)<br>Lower limit signal (RLS)   | 24VDC/5mA                                                                                                                                                                                                                                         | 19.2 to 26.4VDC       | 17.5VDC or more/<br>3.5mA or more | 7VDC or less/<br>1.7mA or less   | Approx. 4.7kΩ    | 4ms or less   |
| Zero signal<br>(PG05/PG024)                                                                              | 5VDC/5mA                                                                                                                                                                                                                                          | 4.5 to 6.1VDC         | 2VDC or more/<br>2mA or more      | 0.5VDC or less/<br>0.5mA or less | Approx. 620Ω     | 1ms or less   |
|                                                                                                          | 24VDC/5mA                                                                                                                                                                                                                                         | 12 to 26.4VDC         | 10VDC or more/<br>3mA or more     | 3VDC or less/<br>0.2mA or less   | Approx. 4.7kΩ    | 1ms or less   |
|                                                                                                          |                                                                                                                                                                |                       |                                   |                                  |                  |               |
|                                                                                                          | Differential receiver equivalent to AM26LS32<br>(ON/OFF level ON: 1.8V or more, OFF: 0.6V or less)                                                                                                                                                |                       |                                   |                                  |                  |               |
| Manual pulse generator A phase (PULSE GENERATOR A)<br>Manual pulse generator B phase (PULSE GENERATOR B) | 5VDC/5mA                                                                                                                                                                                                                                          | 4.5 to 6.1VDC         | 2.5VDC or more/<br>2mA or more    | 1VDC or less/<br>0.1mA or less   | Approx. 1.1kΩ    | 1ms or less   |
|                                                                                                          | <p>① Pulse width</p>  <p>(Duty ratio: 50%)</p> <p>② Phase difference</p>  |                       |                                   |                                  |                  |               |
| Near-point dog signal (DOG)                                                                              | 24VDC/5mA                                                                                                                                                                                                                                         | 19.2 to 26.4VDC       | 17.5VDC or more/<br>3.5mA or more | 7VDC or less/<br>1.7mA or less   | Approx. 4.3kΩ    | 1ms or less   |
| External command signal (CHG)                                                                            | 24VDC/5mA                                                                                                                                                                                                                                         | 19.2 to 26.4VDC       | 19VDC or more/<br>2.7mA or more   | 7VDC or less/<br>0.8mA or less   | Approx. 7.7kΩ    | 1ms or less   |

## (2) Output specifications

| Signal name                                                                       | Rated load voltage                        | Operating load voltage range | Max. load current/inrush current    | Max. voltage drop at ON    | Leakage current at OFF | Response time                    |
|-----------------------------------------------------------------------------------|-------------------------------------------|------------------------------|-------------------------------------|----------------------------|------------------------|----------------------------------|
| Deviation counter clear (CLEAR)                                                   | 5 to 24VDC                                | 4.75 to 30VDC                | 0.1A/1 point/0.4A<br>10ms or less   | 1VDC (TYP)<br>2.5VDC (MAX) | 0.1mA or less          | 2ms or less<br>(resistance load) |
| QD75P□N<br>Pulse output F (PULSE F)<br>Pulse output R (PULSE R)                   | 5 to 24VDC                                | 4.75 to 30VDC                | 50mA/1 point/<br>200mA 10ms or less | 0.5VDC (TYP)               | 0.1mA or less          | —                                |
| QD75D□N<br>Pulse output F (+/-) (PULSE F+/-)<br>Pulse output R (+/-) (PULSE R+/-) | Differential driver equivalent to AM26C31 |                              |                                     |                            |                        |                                  |

- Select the PULSE/SIGN type, CW/CCW type, and A phase/B phase type using the parameter (Pr.5 Pulse output mode) according to the drive unit specifications.
- The relation of the pulse output with the "Pr.5 Pulse output mode" and "Pr.23 Output signal logic selection" is shown below:

| Pr.5 Pulse output mode <sup>*1</sup>                                                                                                                                                                                                                              | Terminal name        | Pr.23 Output signal logic selection (bit0) |             |                      |             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------|-------------|----------------------|-------------|
|                                                                                                                                                                                                                                                                   |                      | Positive logic                             |             | Negative logic       |             |
|                                                                                                                                                                                                                                                                   |                      | Forward run                                | Reverse run | Forward run          | Reverse run |
| Open collector connection (QD75P□N)                                                                                                                                                                                                                               |                      |                                            |             |                      |             |
| • The voltage of a terminal having the PULSE COM terminal as a reference is shown. (Refer to Section 3.4.4, (2).)<br>(The transistor output becomes OFF to High and ON to Low.)                                                                                   |                      |                                            |             |                      |             |
| PULSE                                                                                                                                                                                                                                                             | PULSE F              | High Low                                   |             | High Low             |             |
| SIGN                                                                                                                                                                                                                                                              | PULSE R              | High Low                                   |             | High Low             |             |
| CW                                                                                                                                                                                                                                                                | PULSE F              | High Low                                   |             | High Low             |             |
| CCW                                                                                                                                                                                                                                                               | PULSE R              | High Low                                   |             | High Low             |             |
| A phase                                                                                                                                                                                                                                                           | PULSE F              | High Low                                   |             | High Low             |             |
| B phase                                                                                                                                                                                                                                                           | PULSE R              | High Low                                   |             | High Low             |             |
| Differential driver connection ( QD75D□N )                                                                                                                                                                                                                        |                      |                                            |             |                      |             |
| • The voltage of a terminal having the differential driver common terminal as a reference is shown. (Refer to Section 3.4.4, (3).)<br>• For the differential driver common terminal, refer to "Section 4.3.2, Wiring of the differential driver common terminal". |                      |                                            |             |                      |             |
| PULSE                                                                                                                                                                                                                                                             | PULSE F+<br>PULSE F- | High Low<br>High Low                       |             | High Low<br>High Low |             |
| SIGN                                                                                                                                                                                                                                                              | PULSE R+<br>PULSE R- | High Low<br>High Low                       |             | High Low<br>High Low |             |
| CW                                                                                                                                                                                                                                                                | PULSE F+<br>PULSE F- | High Low<br>High Low                       |             | High Low<br>High Low |             |
| CCW                                                                                                                                                                                                                                                               | PULSE R+<br>PULSE R- | High Low<br>High Low                       |             | High Low<br>High Low |             |
| A phase                                                                                                                                                                                                                                                           | PULSE F+<br>PULSE F- | High Low<br>High Low                       |             | High Low<br>High Low |             |
| B phase                                                                                                                                                                                                                                                           | PULSE R+<br>PULSE R- | High Low<br>High Low                       |             | High Low<br>High Low |             |

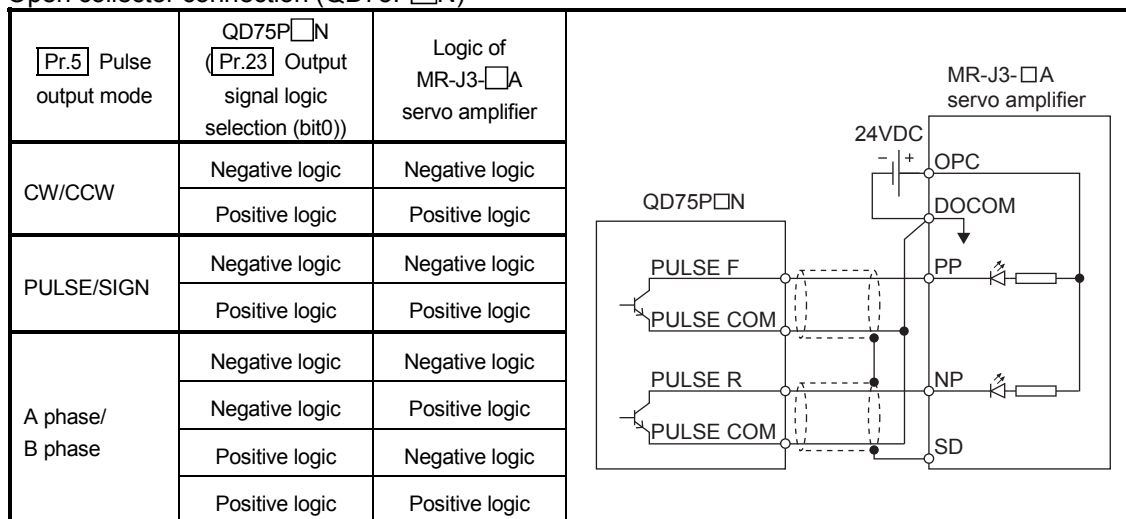
\*1: For details on "Pr.5 Pulse output mode", refer to "Section 5.2.1, Basic parameters 1".

## POINT

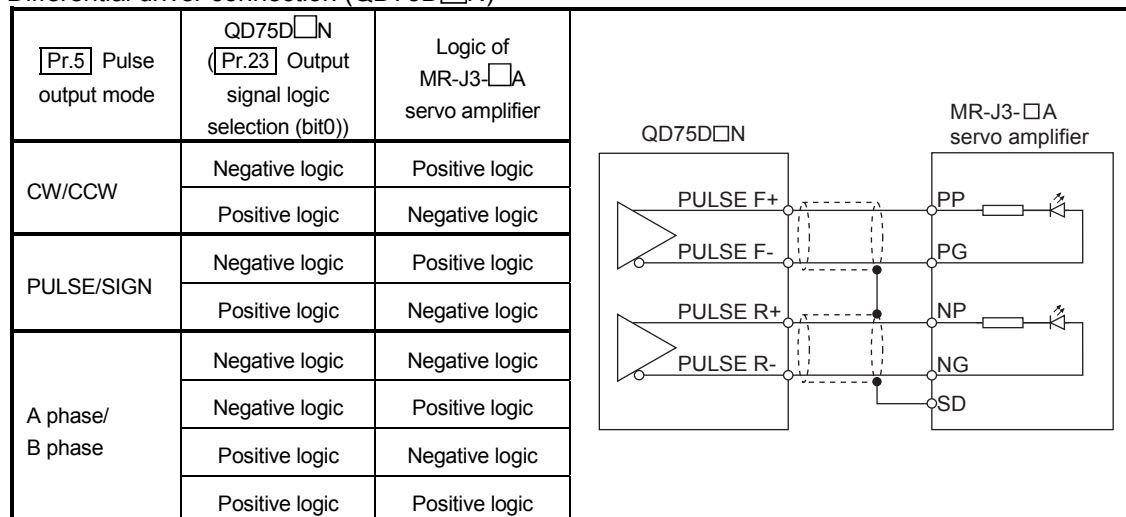
Set the parameters, " [Pr.5] Pulse output mode" and " [Pr.23] Output signal logic selection", in accordance with the specifications of a connected servo amplifier. If not, the motor may rotate in the opposite direction or may not rotate at all.

Connection examples with a MELSERVO-J3 series servo amplifier are shown below.

## Open collector connection (QD75P□N)



## Differential driver connection (QD75D□N)



## ■ QD75P□/QD75D□

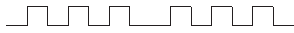
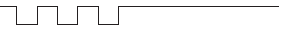
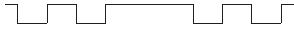
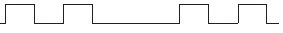
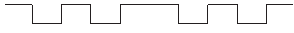
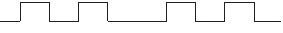
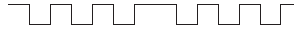
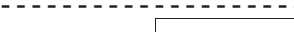
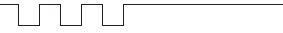
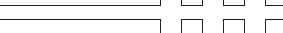
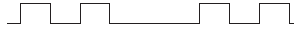
## (1) Input specifications

| Signal name                                                                                              | Rated input voltage/current                                                                        | Working voltage range | ON voltage/current                | OFF voltage/current              | Input resistance      | Response time |
|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------|----------------------------------|-----------------------|---------------|
| Drive unit READY (READY)<br>Stop signal (STOP)<br>Upper limit signal (FLS)<br>Lower limit signal (RLS)   | 24VDC/5mA                                                                                          | 19.2 to 26.4VDC       | 17.5VDC or more/<br>3.5mA or more | 7VDC or less/<br>1.7mA or less   | Approx. 4.7k $\Omega$ | 4ms or less   |
| Zero signal (PG05/PG024)                                                                                 | 5VDC/5mA                                                                                           | 4.5 to 6.1VDC         | 2VDC or more/<br>2mA or more      | 0.5VDC or less/<br>0.5mA or less | Approx. 0.3k $\Omega$ | 1ms or less   |
|                                                                                                          | 24VDC/5mA                                                                                          | 12 to 26.4VDC         | 10VDC or more/<br>3mA or more     | 3VDC or less/<br>0.2mA or less   | Approx. 4.7k $\Omega$ | 1ms or less   |
|                                                                                                          |                                                                                                    |                       |                                   |                                  |                       |               |
|                                                                                                          | Differential receiver equivalent to AM26LS32<br>(ON/OFF level ON: 1.8V or more, OFF: 0.6V or less) |                       |                                   |                                  |                       |               |
| Manual pulse generator A phase (PULSE GENERATOR A)<br>Manual pulse generator B phase (PULSE GENERATOR B) | 5VDC/5mA                                                                                           | 4.5 to 6.1VDC         | 2.5VDC or more/<br>2mA or more    | 1VDC or less/<br>0.1mA or less   | Approx. 1.5k $\Omega$ | 1ms or less   |
|                                                                                                          | <p>① Pulse width</p> <p>(Duty ratio: 50%)</p> <p>② Phase difference</p>                            |                       |                                   |                                  |                       |               |
| Near-point dog signal (DOG)<br>External command signal (CHG)                                             | 24VDC/5mA                                                                                          | 19.2 to 26.4VDC       | 17.5VDC or more/<br>3.5mA or more | 7VDC or less/<br>1.7mA or less   | Approx. 4.3k $\Omega$ | 1ms or less   |

## (2) Output specifications

| Signal name                                                                      | Rated load voltage                        | Operating load voltage range | Max. load current/inrush current    | Max. voltage drop at ON    | Leakage current at OFF | Response time                    |
|----------------------------------------------------------------------------------|-------------------------------------------|------------------------------|-------------------------------------|----------------------------|------------------------|----------------------------------|
| Deviation counter clear (CLEAR)                                                  | 5 to 24VDC                                | 4.75 to 30VDC                | 0.1A/1 point/0.4A<br>10ms or less   | 1VDC (TYP)<br>2.5VDC (MAX) | 0.1mA or less          | 2ms or less<br>(resistance load) |
| QD75P□<br>Pulse output F (PULSE F)<br>Pulse output R (PULSE R)                   | 5 to 24VDC                                | 4.75 to 30VDC                | 50mA/1 point/<br>200mA 10ms or less | 0.5VDC (TYP)               | 0.1mA or less          | —                                |
| QD75D□<br>Pulse output F (+/-) (PULSE F+/-)<br>Pulse output R (+/-) (PULSE R+/-) | Differential driver equivalent to AM26C31 |                              |                                     |                            |                        |                                  |

- Select the PULSE/SIGN type, CW/CCW type, and A phase/B phase type using the parameter ( [Pr.5] Pulse output mode) according to the drive unit specifications.
- The relation of the pulse output with the " [Pr.5] Pulse output mode" and " [Pr.23] Output signal logic selection" is shown below:

| [Pr.5] Pulse output mode <sup>*1</sup>                                                                                                                                                                                                                            | Terminal name        | [Pr.23] Output signal logic selection (bit0) |                                                                                     |                      |                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------|-------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                   |                      | Positive logic                               |                                                                                     | Negative logic       |                                                                                       |
|                                                                                                                                                                                                                                                                   |                      | Forward run                                  | Reverse run                                                                         | Forward run          | Reverse run                                                                           |
| Open collector connection ( QD75P□ )                                                                                                                                                                                                                              |                      |                                              |                                                                                     |                      |                                                                                       |
| • The voltage of a terminal having the PULSE COM terminal as a reference is shown. (Refer to Section 3.4.4, (2).)<br>(The transistor output becomes OFF to High and ON to Low.)                                                                                   |                      |                                              |                                                                                     |                      |                                                                                       |
| PULSE                                                                                                                                                                                                                                                             | PULSE F              | High Low                                     |  | High Low             |  |
| SIGN                                                                                                                                                                                                                                                              | PULSE R              | High Low                                     |  | High Low             |  |
| CW                                                                                                                                                                                                                                                                | PULSE F              | High Low                                     |  | High Low             |  |
| CCW                                                                                                                                                                                                                                                               | PULSE R              | High Low                                     |  | High Low             |  |
| A phase                                                                                                                                                                                                                                                           | PULSE F              | High Low                                     |  | High Low             |  |
| B phase                                                                                                                                                                                                                                                           | PULSE R              | High Low                                     |  | High Low             |  |
| Differential driver connection (QD75P□)                                                                                                                                                                                                                           |                      |                                              |                                                                                     |                      |                                                                                       |
| • The voltage of a terminal having the differential driver common terminal as a reference is shown. (Refer to Section 3.4.4, (3).)<br>• For the differential driver common terminal, refer to "Section 4.3.2, Wiring of the differential driver common terminal". |                      |                                              |                                                                                     |                      |                                                                                       |
| PULSE                                                                                                                                                                                                                                                             | PULSE F+<br>PULSE F- | High Low<br>High Low                         |  | High Low<br>High Low |  |
| SIGN                                                                                                                                                                                                                                                              | PULSE R+<br>PULSE R- | High Low<br>High Low                         |  | High Low<br>High Low |  |
| CW                                                                                                                                                                                                                                                                | PULSE F+<br>PULSE F- | High Low<br>High Low                         |  | High Low<br>High Low |  |
| CCW                                                                                                                                                                                                                                                               | PULSE R+<br>PULSE R- | High Low<br>High Low                         |  | High Low<br>High Low |  |
| A phase                                                                                                                                                                                                                                                           | PULSE F+<br>PULSE F- | High Low<br>High Low                         |  | High Low<br>High Low |  |
| B phase                                                                                                                                                                                                                                                           | PULSE R+<br>PULSE R- | High Low<br>High Low                         |  | High Low<br>High Low |  |

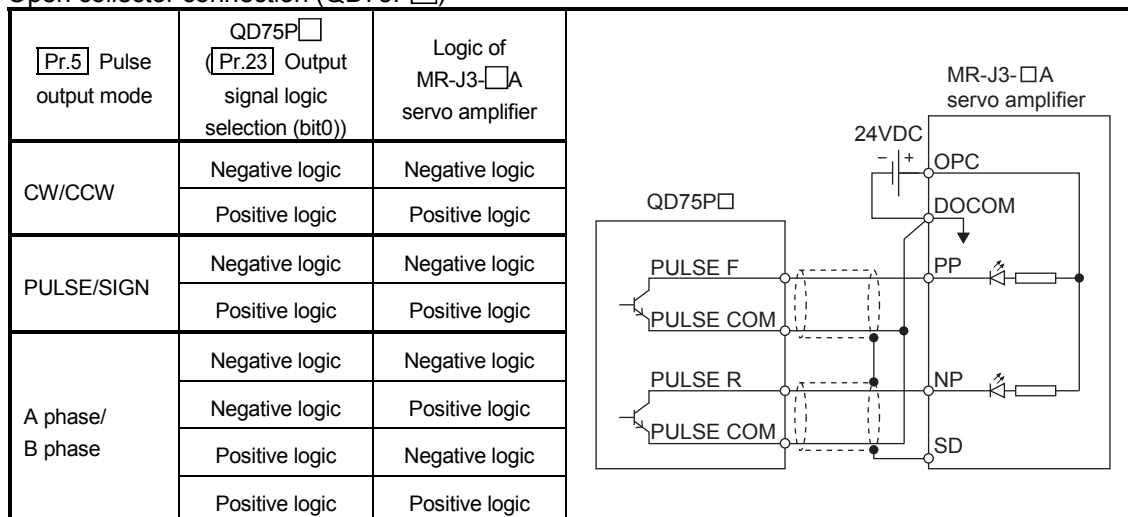
\* 1: For details on " [Pr.5] Pulse output mode", refer to "Section 5.2.1, Basic parameters 1".

## POINT

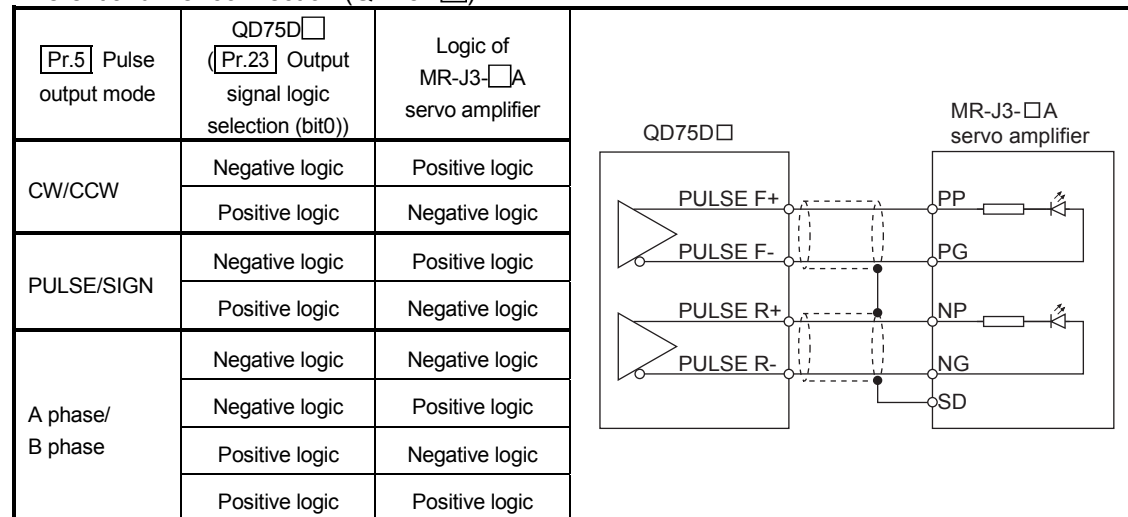
Set the parameters, " [Pr.5] Pulse output mode" and " [Pr.23] Output signal logic selection", in accordance with the specifications of a connected servo amplifier. If not, the motor may rotate in the opposite direction or may not rotate at all.

Connection examples with a MELSERVO-J3 series servo amplifier are shown below.

## Open collector connection (QD75P□)



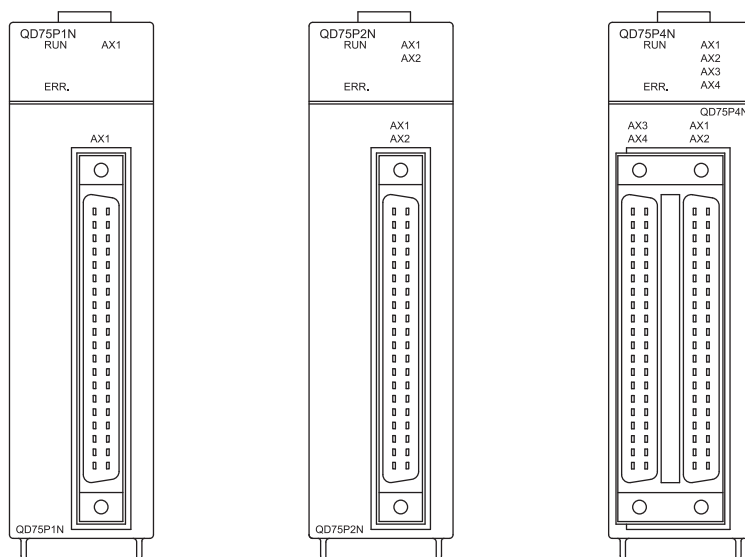
## Differential driver connection (QD75D□)



## 3.4.2 Signal layout for external device connection connector

The specifications of the connector section, which is the input/output interface for the QD75 and external device, are shown below.

The signal layout for the QD75 external device connection connector is shown.



| Pin layout |     |     | Axis 4(AX4) |             | Axis 3(AX3) |             | Axis 2(AX2)   |             | Axis 1(AX1) |             |
|------------|-----|-----|-------------|-------------|-------------|-------------|---------------|-------------|-------------|-------------|
|            |     |     | Pin No.     | Signal name | Pin No.     | Signal name | Pin No.<br>*2 | Signal name | Pin No.     | Signal name |
| B20        | □ □ | A20 | 2B20        | Vacant      | 2A20        | Vacant      | 1B20          | PULSER B-   | 1A20        | PULSER B+   |
| B19        | □ □ | A19 | 2B19        | Vacant      | 2A19        | Vacant      | 1B19          | PULSER A-   | 1A19        | PULSER A+   |
| B18        | □ □ | A18 | 2B18        | PULSE COM   | 2A18        | PULSE COM   | 1B18          | PULSE COM   | 1A18        | PULSE COM   |
| B17        | □ □ | A17 | *3          | PULSE R-    | *3          | PULSE R-    | *3            | PULSE R-    | *3          | PULSE R-    |
| B16        | □ □ | A16 | 2B17        | PULSE R     | 2A17        | PULSE R     | 1B17          | PULSE R     | 1A17        | PULSE R     |
| B15        | □ □ | A15 | *3          | PULSE R+    | *3          | PULSE R+    | *3            | PULSE R+    | *3          | PULSE R+    |
| B14        | □ □ | A14 | 2B16        | PULSE COM   | 2A16        | PULSE COM   | 1B16          | PULSE COM   | 1A16        | PULSE COM   |
| B13        | □ □ | A13 | *3          | PULSE F-    | *3          | PULSE F-    | *3            | PULSE F-    | *3          | PULSE F-    |
| B12        | □ □ | A12 | 2B15        | PULSE F     | 2A15        | PULSE F     | 1B15          | PULSE F     | 1A15        | PULSE F     |
| B11        | □ □ | A11 | *3          | PULSE F+    | *3          | PULSE F+    | *3            | PULSE F+    | *3          | PULSE F+    |
| B10        | □ □ | A10 | 2B14        | CLRCOM      | 2A14        | CLRCOM      | 1B14          | CLRCOM      | 1A14        | CLRCOM      |
| B9         | □ □ | A9  | 2B13        | CLEAR       | 2A13        | CLEAR       | 1B13          | CLEAR       | 1A13        | CLEAR       |
| B8         | □ □ | A8  | 2B12        | RDYCOM      | 2A12        | RDYCOM      | 1B12          | RDYCOM      | 1A12        | RDYCOM      |
| B7         | □ □ | A7  | 2B11        | READY       | 2A11        | READY       | 1B11          | READY       | 1A11        | READY       |
| B6         | □ □ | A6  | 2B10        | PG0COM      | 2A10        | PG0COM      | 1B10          | PG0COM      | 1A10        | PG0COM      |
| B5         | □ □ | A5  | 2B9         | PG05        | 2A9         | PG05        | 1B9           | PG05        | 1A9         | PG05        |
| B4         | □ □ | A4  | 2B8         | PG024       | 2A8         | PG024       | 1B8           | PG024       | 1A8         | PG024       |
| B3         | □ □ | A3  | 2B7         | COM         | 2A7         | COM         | 1B7           | COM         | 1A7         | COM         |
| B2         | □ □ | A2  | 2B6         | COM         | 2A6         | COM         | 1B6           | COM         | 1A6         | COM         |
| B1         | □ □ | A1  | 2B5         | CHG         | 2A5         | CHG         | 1B5           | CHG         | 1A5         | CHG         |
|            |     |     | 2B4         | STOP        | 2A4         | STOP        | 1B4           | STOP        | 1A4         | STOP        |
|            |     |     | 2B3         | DOG         | 2A3         | DOG         | 1B3           | DOG         | 1A3         | DOG         |
|            |     |     | 2B2         | RLS         | 2A2         | RLS         | 1B2           | RLS         | 1A2         | RLS         |
|            |     |     | 2B1         | FLS         | 2A1         | FLS         | 1B1           | FLS         | 1A1         | FLS         |

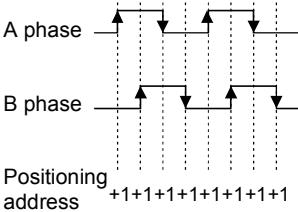
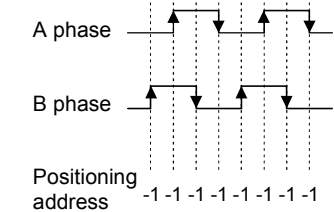
\* 1: Pin No. "1□ □ □" indicates the pin No. for the right connector. Pin No. "2 □ □ □" indicates the pin No. for the left connector.

\* 2: When a 1-axis module is used, pin Nos. 1B1 to 1B18 are "vacant".

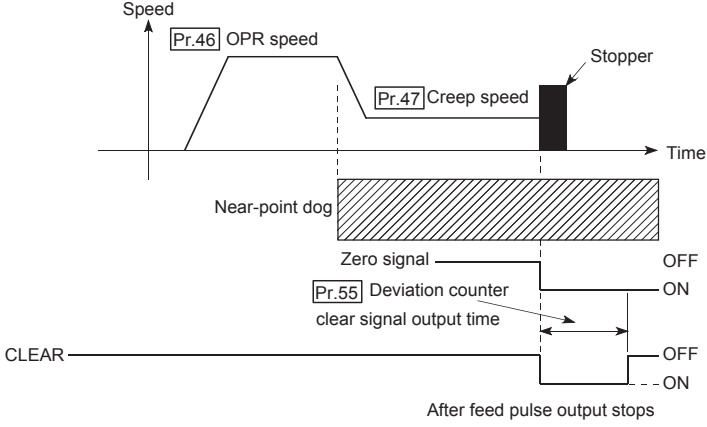
\* 3: The upper line indicates the signal name for the QD75P□N/QD75P□, and the lower line indicates the signal name for the QD75D□N/QD75D□.

## 3.4.3 List of input/output signal details

The details of each QD75 external device connection connector are shown below:

| Signal name                                                                                | Pin No.      |      |      |      | Signal details<br>(Negative logic is selected by external I/O signal logic selection)                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------|--------------|------|------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                            | AX1          | AX2  | AX3  | AX4  |                                                                                                                                                                                                                                                                                                                                                                                               |
| Manual pulse generator A phase (PULSER A+)<br>Manual pulse generator B phase (PULSER B+)   | 1A19<br>1A20 |      | —    |      | <ul style="list-style-type: none"> <li>Input the pulse signal from the manual pulse generator A phase and B phase.</li> <li>If the A phase leads the B phase, the positioning address will increase at the rising and falling edges of each phase.</li> <li>If the B phase leads the A phase, the positioning address will decrease at the rising and falling edges of each phase.</li> </ul> |
| Manual pulse generator A common (PULSER A-)<br>Manual pulse generator B common (PULSER B-) | 1B19<br>1B20 |      | —    |      | <p>[When increased]</p>  <p>A phase</p> <p>B phase</p> <p>Positioning address +1 +1 +1 +1 +1 +1 +1</p> <p>[When decreased]</p>  <p>A phase</p> <p>B phase</p> <p>Positioning address -1 -1 -1 -1 -1 -1 -1</p>            |
| Zero signal (+24V) (PG024)                                                                 | 1A8          | 1B8  | 2A8  | 2B8  | <ul style="list-style-type: none"> <li>Input the zero signal for machine OPR. Use the pulse encoder's zero signal and so on.</li> <li>Also use this signal when the machine OPR method is the stopper method and the OPR complete is input from an external source.</li> <li>The zero signal is detected at turning from OFF to ON.</li> </ul>                                                |
| Zero signal (+5V) (PG05)                                                                   | 1A9          | 1B9  | 2A9  | 2B9  |                                                                                                                                                                                                                                                                                                                                                                                               |
| Zero signal common (PG0COM)                                                                | 1A10         | 1B10 | 2A10 | 2B10 | <ul style="list-style-type: none"> <li>Common for zero signal (+5V) and zero signal (+24V).</li> </ul>                                                                                                                                                                                                                                                                                        |
| Pulse output F (+) (PULSE F+)                                                              | 1A15         | 1B15 | 2A15 | 2B15 | <ul style="list-style-type: none"> <li>Output the positioning pulses and pulse sign for the differential driver output system compatible drive unit. (QD75D□N/QD75D□ only)</li> </ul>                                                                                                                                                                                                         |
| Pulse output F (–) (PULSE F–)                                                              | 1A16         | 1B16 | 2A16 | 2B16 |                                                                                                                                                                                                                                                                                                                                                                                               |
| Pulse output R (+) (PULSE R+)                                                              | 1A17         | 1B17 | 2A17 | 2B17 |                                                                                                                                                                                                                                                                                                                                                                                               |
| Pulse output R (–) (PULSE R–)                                                              | 1A18         | 1B18 | 2A18 | 2B18 |                                                                                                                                                                                                                                                                                                                                                                                               |
| Pulse output F (PULSE F)                                                                   | 1A15         | 1B15 | 2A15 | 2B15 | <ul style="list-style-type: none"> <li>Output the positioning pulses and pulse sign for the open collector output system compatible drive unit. (QD75P□N/QD75P□ only)</li> </ul>                                                                                                                                                                                                              |
| Pulse output F common (PULSE COM)                                                          | 1A16         | 1B16 | 2A16 | 2B16 |                                                                                                                                                                                                                                                                                                                                                                                               |
| Pulse output R (PULSE R)                                                                   | 1A17         | 1B17 | 2A17 | 2B17 |                                                                                                                                                                                                                                                                                                                                                                                               |
| Pulse output R common (PULSE COM)                                                          | 1A18         | 1B18 | 2A18 | 2B18 |                                                                                                                                                                                                                                                                                                                                                                                               |
| Upper limit signal (FLS)                                                                   | 1A1          | 1B1  | 2A1  | 2B1  | <ul style="list-style-type: none"> <li>This signal is input from the limit switch installed at the upper limit position of the stroke.</li> <li>Positioning will stop when this signal turns OFF.</li> <li>When OPR retry function is valid, this will be the upper limit for finding the near-point dog signal.</li> </ul>                                                                   |
| Lower limit signal (RLS)                                                                   | 1A2          | 1B2  | 2A2  | 2B2  | <ul style="list-style-type: none"> <li>This signal is input from the limit switch installed at the lower limit position of the stroke.</li> <li>Positioning will stop when this signal turns OFF.</li> <li>When OPR retry function is valid, this will be the lower limit for finding the near-point dog signal.</li> </ul>                                                                   |



| Signal name                             | Pin No.    |            |            |            | Signal details<br>(Negative logic is selected by external I/O signal logic selection)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------|------------|------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | AX1        | AX2        | AX3        | AX4        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Near-point dog signal (DOG)             | 1A3        | 1B3        | 2A3        | 2B3        | <ul style="list-style-type: none"> <li>This signal is used for detecting the near-point dog during OPR.</li> <li>The near-point dog signal is detected at turning from OFF to ON.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Stop signal (STOP)                      | 1A4        | 1B4        | 2A4        | 2B4        | <ul style="list-style-type: none"> <li>Input this signal to stop positioning.</li> <li>When this signal turns ON, the QD75 will stop the positioning being executed.</li> <li>After that, even if this signal is turned from ON to OFF, the system will not start.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| External command signal (CHG)           | 1A5        | 1B5        | 2A5        | 2B5        | <ul style="list-style-type: none"> <li>Input a control switching signal during speed-position or position-speed switching control.</li> <li>Use this signal as the input signal of positioning start, speed change request, and skip request from an external source.</li> <li>Set the function to use this signal in " <a href="#">Pr.42</a> External command function selection".</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                               |
| Common (COM)                            | 1A6<br>1A7 | 1B6<br>1B7 | 2A6<br>2A7 | 2B6<br>2B7 | <ul style="list-style-type: none"> <li>Common for upper/lower limit, near-point dog, stop, and external command signals.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Drive unit READY (READY)                | 1A11       | 1B11       | 2A11       | 2B11       | <ul style="list-style-type: none"> <li>This signal turns ON when the drive unit is normal and can accept the feed pulse.</li> <li>The QD75 checks the drive unit READY signal, and outputs the OPR request if the system is not in the READY state.</li> <li>When the drive unit is inoperable, such as if an error occurs in the drive unit's control power supply, this signal will turn OFF.</li> <li>If this signal is turned OFF during positioning, the system will stop. The system will not start even if this signal is turned ON again.</li> <li>When this signal turns OFF, the OPR complete signal will also turn OFF.</li> </ul>                                                                                                                                                |
| Drive unit READY common (RDYCOM)        | 1A12       | 1B12       | 2A12       | 2B12       | <ul style="list-style-type: none"> <li>Common for drive unit READY signal.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Deviation counter clear (CLEAR)         | 1A13       | 1B13       | 2A13       | 2B13       | <ul style="list-style-type: none"> <li>This signal is output during machine OPR. (Note that it is not output during the count method ②.)</li> <li>(Example) When machine OPR is carried out in the stopper ② method.</li> </ul>  <ul style="list-style-type: none"> <li>The output time of the deviation counter clear signal is set in " <a href="#">Pr.55</a> Deviation counter clear signal output time".</li> <li>Use the drive unit that can reset the droop pulse amount in the internal deviation counter when the QD75 turns this signal ON.</li> <li>(Note) The deviation counter clear is a signal output by the QD75 during machine OPR. It cannot be output randomly by the user.</li> </ul> |
| Deviation counter clear common (CLRCOM) | 1A14       | 1B14       | 2A14       | 2B14       | <ul style="list-style-type: none"> <li>Common for deviation counter clear signal</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## 3.4.4 Input/output interface internal circuit

The outline diagrams of the internal circuits for the QD75P1N/QD75D1N/QD75P1/QD75D1 external device connection interface are shown below.

## (1) Input (Common to QD75P1N, QD75D1N, QD75P1, and QD75D1)

| External wiring | Pin No.  | Internal circuit | Signal name                    |           | Need for wiring<br>*1 |
|-----------------|----------|------------------|--------------------------------|-----------|-----------------------|
|                 | 1A3      |                  | Near-point dog signal          | DOG       | △                     |
|                 | 1A1      |                  | Upper limit signal             | FLS       | ○                     |
|                 | 1A2      |                  | Lower limit signal             | RLS       | ○                     |
|                 | 1A4      |                  | Stop signal                    | STOP      | △                     |
|                 | 1A5      |                  | External command signal        | CHG       | △                     |
|                 | 1A6      |                  | Common                         | COM       | ○                     |
|                 | 1A7      |                  |                                |           |                       |
|                 | (+) 1A19 |                  | Manual pulse generator A phase | PULSER A+ | △                     |
|                 | (-) 1B19 |                  |                                | PULSER A- |                       |
|                 | (+) 1A20 |                  | Manual pulse generator B phase | PULSER B+ |                       |
|                 | (-) 1B20 |                  |                                | PULSER B- |                       |
|                 | 1A11     |                  | Drive unit READY               | READY     | ○                     |
|                 | 1A12     |                  | Drive unit READY common        | RDY COM   | ○                     |
|                 | 1A8      |                  | Zero signal                    | PG024     | △                     |
|                 | 1A9      |                  |                                | PG05      |                       |
|                 | 1A10     |                  | Zero signal common             | PG0 COM   |                       |

\*1: The symbols in Need for wiring column indicate the following meanings:

- : Wiring is necessary for positioning.
- △ : Wiring is necessary depending on the situation.

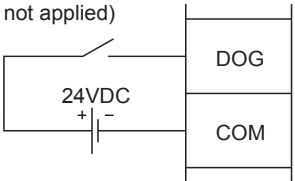
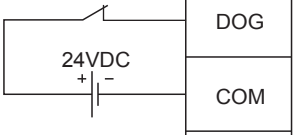
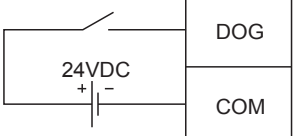
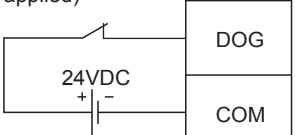
\*2: Either polarity can be connected to the common (COM).

## (a) Input signal ON/OFF status

The input signal ON/OFF status is defined by the external wiring and logic setting.

This is explained below with the example of near-point dog signal (DOG).

(The other input signals also perform the same operations as the near-point dog signal (DOG).)

| Logic setting<br>*3, *4           | External wiring *4                                                                                           | ON/OFF status of near-point dog<br>signal (DOG) as seen from QD75 |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Negative logic<br>(Initial value) | (Voltage not applied)<br>   | OFF                                                               |
|                                   | (Voltage applied)<br>       | ON                                                                |
| Positive logic                    | (Voltage not applied)<br> | ON                                                                |
|                                   | (Voltage applied)<br>     | OFF                                                               |

\*3: Set the logic setting using " [Pr.22] Input signal logic selection". For details of the settings, refer to "Section 5.2.3 Detailed parameters 1" and "Section 13.4 External I/O signal logic switching function".

\*4: When using the upper limit signal (FLS) or lower limit signal (RLS), always wire it as a "b" (normally closed) contact in the negative logic setting. The signal will turn OFF to stop positioning.

## (b) About logic setting and internal circuit

In the QD75, the case where the internal circuit (photocoupler) is OFF in the negative logic setting is defined as "input signal OFF".

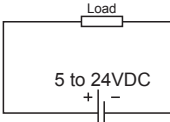
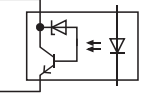
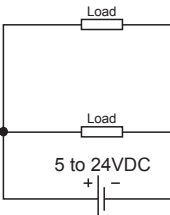
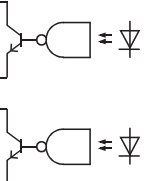
Reversely, the case where the internal circuit (photocoupler) is OFF in the positive logic setting is defined as "input signal ON".

## &lt;Photocoupler ON/OFF status&gt;

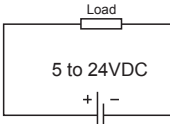
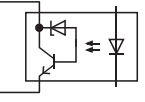
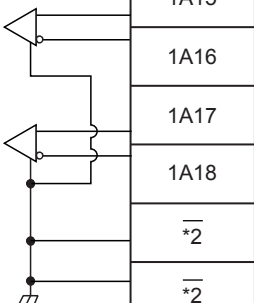
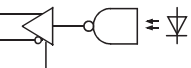
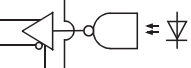
When voltage is not applied : Photocoupler OFF

When voltage is applied : Photocoupler ON

## (2) Output (For QD75P1N and QD75P1)

| External wiring                                                                   | Pin No. | Internal circuit                                                                  | Signal name             |           | Need for wiring <sup>*1</sup> |
|-----------------------------------------------------------------------------------|---------|-----------------------------------------------------------------------------------|-------------------------|-----------|-------------------------------|
|  | 1A13    |  | Deviation counter clear | CLEAR     | △                             |
|                                                                                   | 1A14    |                                                                                   | Common                  | CLEAR COM |                               |
|  | 1A15    |  | CW A phase PULSE        | PULSE F   | ○                             |
|                                                                                   | 1A16    |                                                                                   |                         | PULSE COM |                               |
|                                                                                   | 1A17    |                                                                                   | CCW B phase SIGN        | PULSE R   |                               |
|                                                                                   | 1A18    |                                                                                   |                         | PULSE COM |                               |

## (3) Output (For QD75D1N and QD75D1)

| External wiring                                                                     | Pin No. | Internal circuit                                                                    | Signal name                         |           | Need for wiring <sup>*1</sup> |
|-------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------|-------------------------------------|-----------|-------------------------------|
|   | 1A13    |   | Deviation counter clear             | CLEAR     | △                             |
|                                                                                     | 1A14    |                                                                                     | Common                              | CLEAR COM |                               |
|  | 1A15    |  | CW A phase PULSE                    | PULSE F+  | ○                             |
|                                                                                     | 1A16    |                                                                                     |                                     | PULSE F-  |                               |
|                                                                                     | 1A17    |  | CCW B phase SIGN                    | PULSE R+  |                               |
|                                                                                     | 1A18    |                                                                                     |                                     | PULSE R-  |                               |
|                                                                                     | *2      |  | Differential driver common terminal | PULSE COM | △                             |
|                                                                                     | *2      |                                                                                     |                                     |           |                               |

\*1: The symbols in Need for wiring column indicate the following meanings:

- : Wiring is necessary for positioning.
- △: Wiring is necessary depending on the situation.

\*2: A terminal block at the bottom of the module. (Refer to Section 4.1.2)

# CHAPTER 4 INSTALLATION, WIRING AND MAINTENANCE OF THE PRODUCT

The installation, wiring and maintenance of the QD75 are explained in this chapter.

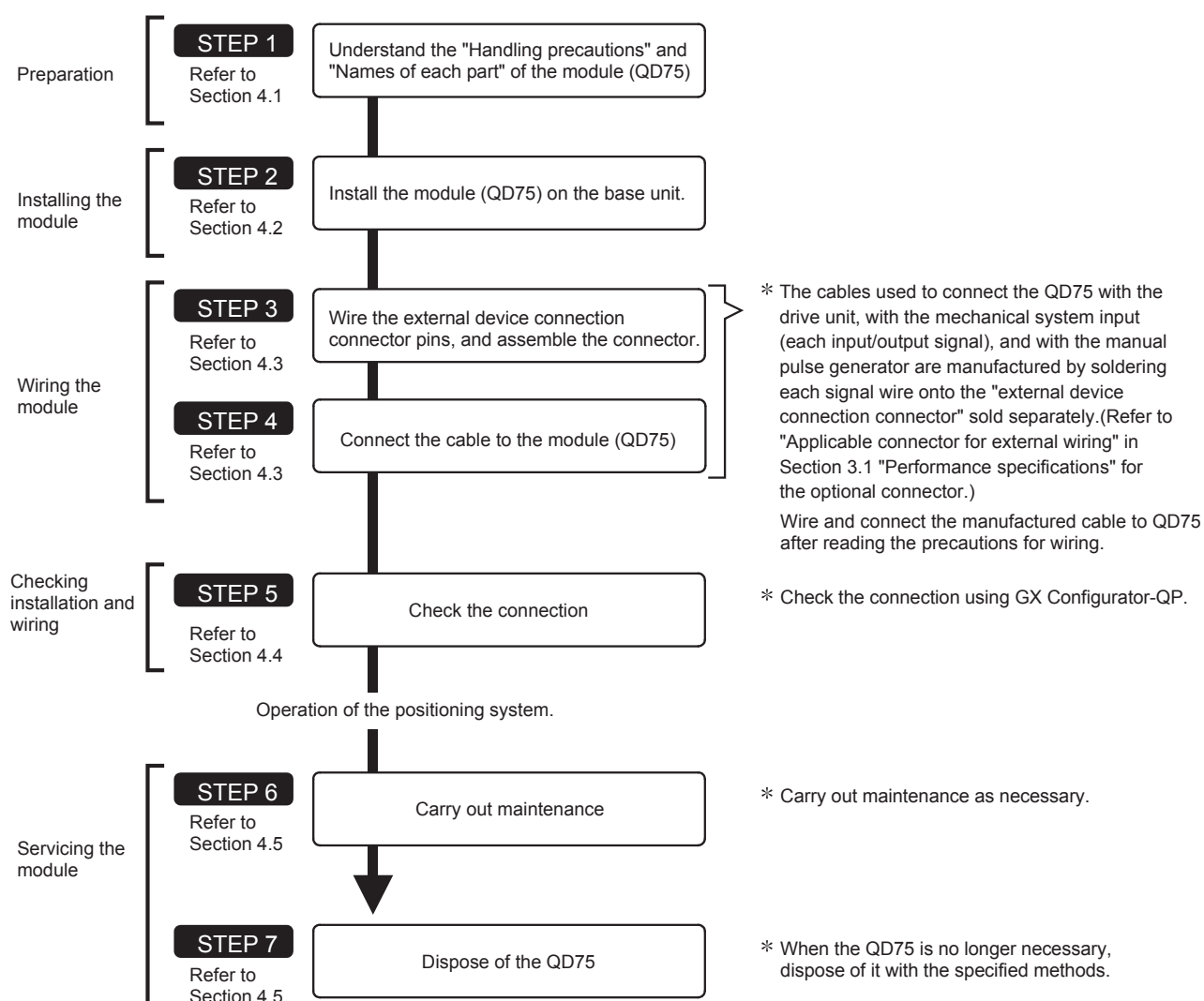
Important information such as precautions to prevent malfunctioning of the QD75, accidents and injuries as well as the proper work methods are described. Read this chapter thoroughly before starting installation, wiring or maintenance, and always following the precautions.

|       |                                                                 |       |
|-------|-----------------------------------------------------------------|-------|
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## 4.1 Outline of installation, wiring and maintenance

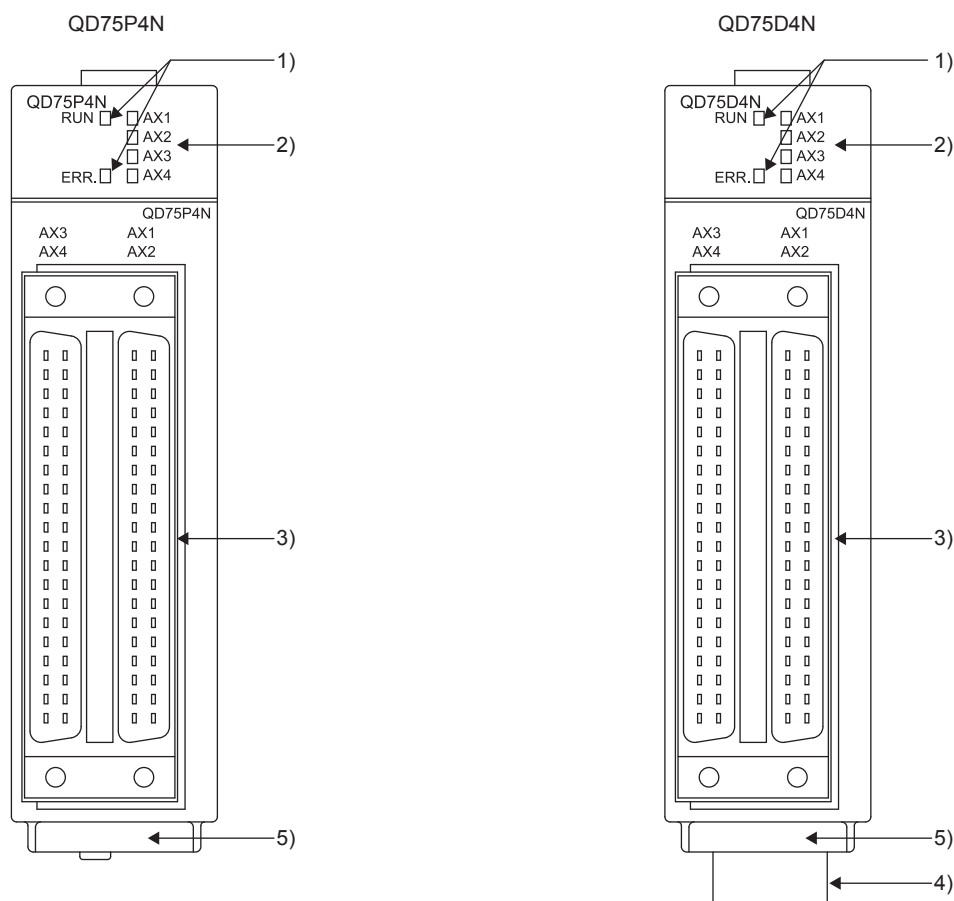
## 4.1.1 Installation, wiring and maintenance procedures

The outline and procedures for QD75 installation, wiring and maintenance are shown below.



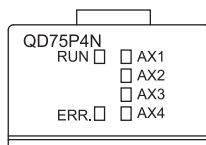
## 4.1.2 Names of each part

- (1) The part names of the QD75 are described with the QD75P□N/QD75D□N as an example. The QD75P□N/QD75D□N and QD75P□/QD75D□ are different in the indication of their model names and serial numbers



| No. | Name                                                                                              | Details                                                                                                                                                                                                                                                                    |
|-----|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1)  | RUN indicator LED, ERR indicator LED                                                              | Refer to the next page.                                                                                                                                                                                                                                                    |
| 2)  | Axis display LED (AX1 to AX4)                                                                     |                                                                                                                                                                                                                                                                            |
| 3)  | External device connector                                                                         | Connector for connection with the drive unit, mechanical system input or manual pulse generator. (40-pin connector)<br>AX1: Axis 1, AX2: Axis 2, AX3: Axis 3, AX4: Axis 4<br>For details, refer to Section 3.4.2 "Signal layout for external device connection connector". |
| 4)  | Differential driver common terminal (Differential driver output system (the QD75D□N/QD75D□) only) | Terminal connected to the differential receiver common of the drive unit. For details, refer to Section 4.3.2 "Wiring of the differential driver common terminal".                                                                                                         |
| 5)  | Serial number plate                                                                               | Indicates the serial number of the QD75                                                                                                                                                                                                                                    |

- (2) The LED display indicates the following operation statuses of the QD75 and axes.



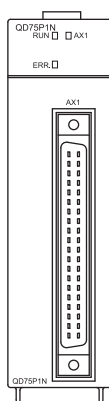
| Display                                                                                                                                                                                                      | Attention point              | Description                             | Display                                                                                                                                                                                                                                                  | Attention point                           | Description                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------|
| RUN <input type="checkbox"/> <input type="checkbox"/> AX1<br><input type="checkbox"/> AX2<br><input type="checkbox"/> AX3<br>ERR <input type="checkbox"/> <input type="checkbox"/> AX4                       | RUN is OFF.                  | Hardware failure, watch dog timer error | RUN <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> AX1<br><input type="checkbox"/> AX2<br><input type="checkbox"/> AX3<br>ERR <input type="checkbox"/> <input type="checkbox"/> AX4                                             | AX1 (or other axis) illuminates.          | The corresponding axis is in operation.    |
| RUN <input checked="" type="checkbox"/> <input type="checkbox"/> AX1<br><input type="checkbox"/> AX2<br><input type="checkbox"/> AX3<br>ERR <input type="checkbox"/> <input type="checkbox"/> AX4            | RUN illuminates. ERR is OFF. | The module operates normally.           | RUN <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> AX1<br><input type="checkbox"/> AX2<br><input type="checkbox"/> AX3<br>ERR <input checked="" type="checkbox"/> <input type="checkbox"/> AX4                                  | ERR flashes. AX1 (or other axis) flashes. | An error occurs on the corresponding axis. |
| RUN <input checked="" type="checkbox"/> <input type="checkbox"/> AX1<br><input type="checkbox"/> AX2<br><input type="checkbox"/> AX3<br>ERR <input checked="" type="checkbox"/> <input type="checkbox"/> AX4 | ERR illuminates.             | System error                            | RUN <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> AX1<br><input checked="" type="checkbox"/> AX2<br><input checked="" type="checkbox"/> AX3<br>ERR <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> AX4 | All LEDs illuminate.                      | Hardware failure                           |
| RUN <input checked="" type="checkbox"/> <input type="checkbox"/> AX1<br><input type="checkbox"/> AX2<br><input type="checkbox"/> AX3<br>ERR <input type="checkbox"/> <input type="checkbox"/> AX4            | AX1 to AX4 are OFF.          | The axes are stopped or on standby.     |                                                                                                                                                                                                                                                          |                                           |                                            |

The symbols in the Display column indicate the following statuses:

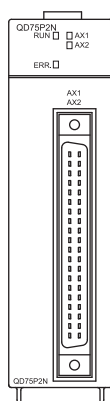
☐: Turns OFF. ☒: Illuminates. ☒: Flashes.

- (3) The interface of each QD75 is as shown below.

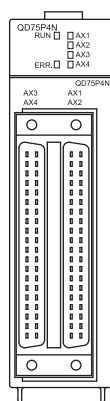
QD75P1N



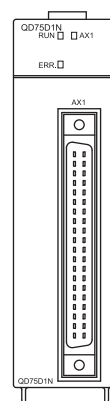
QD75P2N



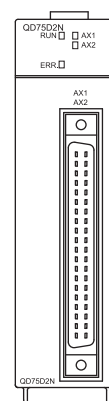
QD75P4N



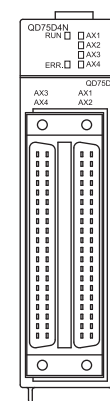
QD75D1N



QD75D2N



QD75D4N





### 4.1.3 Handling precautions

Handle the QD75 and cable while observing the following precautions.

#### [1] Handling precautions

#### CAUTION

- Use the programmable controller in an environment that meets the general specifications contained in QCPU User's Manual(Hardware Design, Maintenance and Inspection) to use. Using this programmable controller in an environment outside the range of the general specifications may cause electric shock, fire, malfunction, and damage to or deterioration of the product.
- Do not directly touch the conductive section and electronic parts of the module. Failure to observe this could lead to module malfunctioning or trouble.
- Make sure that foreign matter, such as cutting chips or wire scraps, do not enter the module. Failure to observe this could lead to fires, trouble or malfunctioning.
- Never disassemble or modify the module. Failure to observe this could lead to trouble, malfunctioning, injuries or fires.

## [2] Other precautions

### (1) Main body

- The main body case is made of plastic. Take care not to drop or apply strong impacts onto the case.
- Do not remove the QD75 PCB from the case. Failure to observe this could lead to faults.

### (2) Cable

- Do not press on the cable with a sharp object.
- Do not twist the cable with force.
- Do not forcibly pull on the cable.
- Do not step on the cable.
- Do not place objects on the cable.
- Do not damage the cable sheath.

### (3) Installation environment

Do not install the module in the following type of environment.

- Where the ambient temperature exceeds the 0 to 55°C range.
- Where the ambient humidity exceeds the 5 to 95%RH range.
- Where there is sudden temperature changes, or where dew condenses.
- Where there is corrosive gas or flammable gas.
- Where there are high levels of dust, conductive powder, such as iron chips, oil mist, salt or organic solvents.
- Where the module will be subject to direct sunlight.
- Where there are strong electric fields or magnetic fields.
- Where vibration or impact could be directly applied onto the main body.

## 4.2 Installation

### 4.2.1 Installation precautions

The precautions for installing the QD75 are given below. Refer to this section as well as "4.1.3 Handling precautions" when carrying out the work.

#### Installation precautions



#### WARNING

- Completely turn off the externally supplied power used in the system before cleaning or tightening the screws.  
Failure to turn all phases OFF could lead to electric shocks.



#### CAUTION

- Never disassemble or modify the module.  
Failure to observe this could lead to trouble, malfunctioning, injuries or fires.
- Completely turn off the externally supplied power used in the system before installing or removing the module.  
Failure to turn all phases OFF could lead to module trouble or malfunctioning.
- Do not install/remove the module to/from the base unit, or the terminal block to/from the module more than 50 times after the first use of the product (IEC 61131-2 compliant). Failure to do so may cause the module to malfunction due to poor contact of connector.
- Use the programmable controller in an environment that meets the general specifications contained in QCPU User's Manual(Hardware Design, Maintenance and Inspection) to use.  
Using this programmable controller in an environment outside the range of the general specifications may cause electric shock, fire, malfunction, and damage to or deterioration of the product.
- Hold down the module mounting lever at the bottom of the module and securely insert the module fixing projection into the fixing holes of the base module. Improper mounting of the module may lead to malfunctioning, faults, or dropping.  
When using the module in the environment subject to much vibration, secure the module with a screw.  
Tighten the screw within the range of the specified tightening torque.  
Insufficient tightening may lead to dropping, short-circuit, or malfunctioning.  
Excessive tightening may damage the screw or module, leading to dropping, short-circuit, or malfunctioning.

### 4.3 Wiring

The precautions for wiring the QD75 are given below. Refer to this section as well as "4.1.3 Handling precautions" when carrying out the work.

#### 4.3.1 Wiring precautions

- (1) Always confirm the terminal layout before connecting the wires to the QD75.  
(For the terminal layout, refer to Section 3.4.2 "Signal layout for external device connection connector".)
- (2) Correctly solder the external wiring connector. An incomplete soldering could lead to malfunctioning.
- (3) Make sure that foreign matter such as cutting chips and wire scraps does not enter the QD75. Failure to observe this could lead to fires, faults or malfunctioning.
- (4) A protective label is attached on the top of the QD75 to avoid foreign matter such as wire scraps from entering inside during wiring process. Do not remove the label until the wiring is completed. Before starting the system, however, be sure to remove the label to ensure heat radiation.
- (5) Tighten the connector screws within the specified torque range.  
Undertightening can cause short circuit, fire, or malfunction.  
Overtightening can damage the screw and/or module, resulting in drop, short circuit, fire, or malfunction.
- (6) When disconnecting the cable from the QD75 or the drive unit, do not pull the cable by the cable part. Hold the connector part of the cable. Pulling the cable connected to the QD75 or the drive unit may result in malfunction or damage to the module, drive unit, or cable.
- (7) Do not bundle or adjacently lay the connection cable connected to the QD75 external I/O signals or drive unit with the main circuit line, power line, or the load line other than that for the programmable controller. Separate these by 100mm as a guide. Failure to observe this could lead to malfunctioning caused by noise, surge, or induction.
- (8) The shielded cable for connecting QD75 can be secured in place.  
If the shielded cable is not secured, unevenness or movement of the shielded cable or careless pulling on it could result in damage to the QD75 or drive unit or shielded cable or defective cable connections could cause mis-operation of the unit.
- (9) If the cable connected to the QD75 and the power line must be adjacently laid (less than 100mm), use a shielded cable. Ground the shield of the cable securely to the control panel on the QD75 side. (A wiring example is given on the next page.)

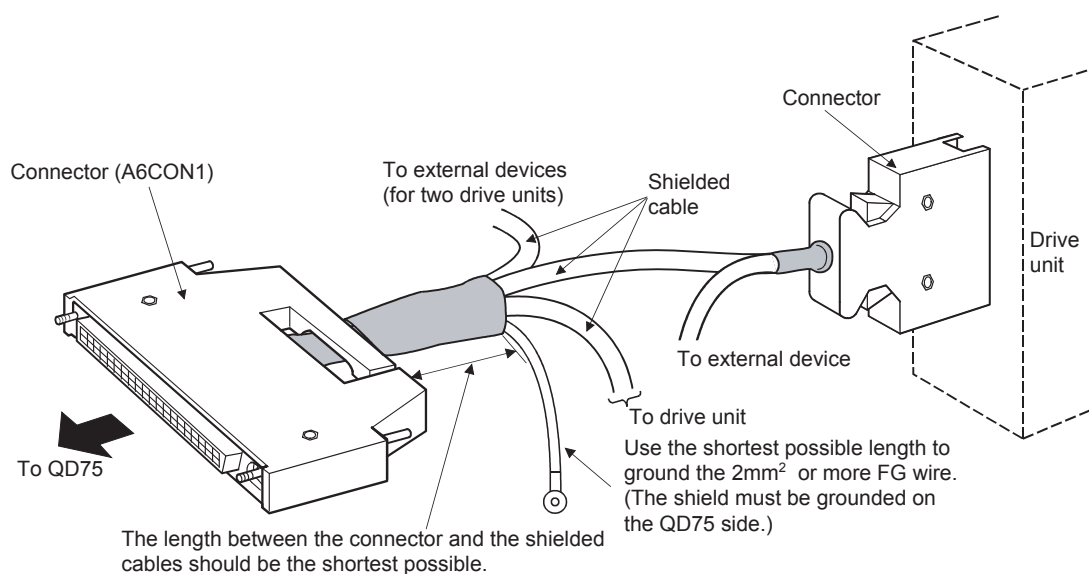
## [Applicable connectors]

The table below shows applicable connectors for external devices. When wiring, use applicable wires and an appropriate tightening torque.

| Mitsubishi Electric 40-pin connector |                   | Wire                                          |          |          |                    |
|--------------------------------------|-------------------|-----------------------------------------------|----------|----------|--------------------|
| Model                                | Tightening torque | Diameter                                      | Type     | Material | Temperature rating |
| A6CON1                               | 0.20 to 0.29N • m | 0.3mm <sup>2</sup> (22AWG)                    | Stranded | Copper   | 75°C or more       |
| A6CON2                               |                   | 0.088 to 0.24mm <sup>2</sup><br>(28 to 24AWG) |          |          |                    |
| A6CON4                               |                   | 0.3mm <sup>2</sup> (22AWG)                    |          |          |                    |

## [Wiring example of shielded cable]

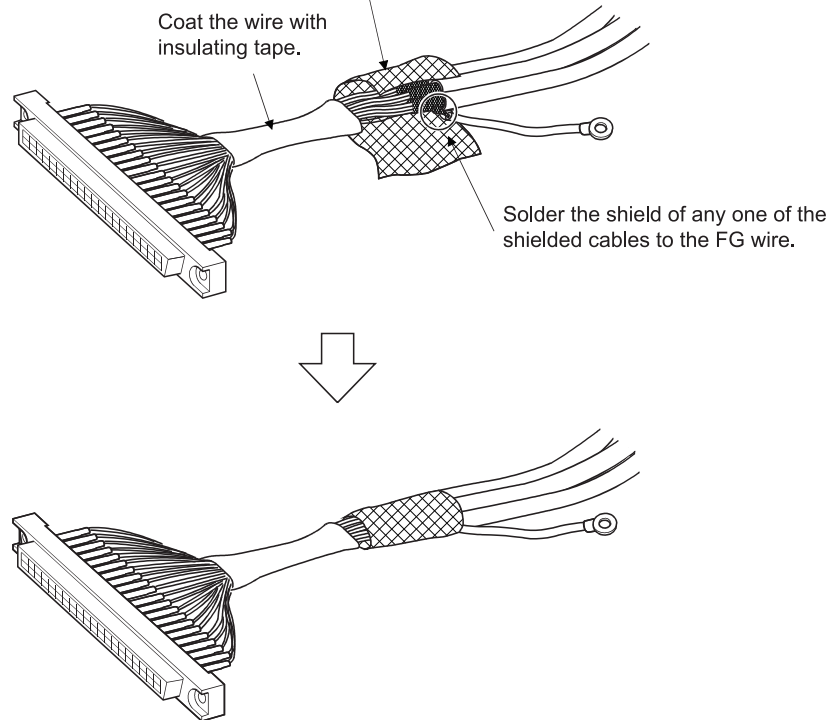
The following shows a wiring example for noise reduction in the case where the connector A6CON1 is used.



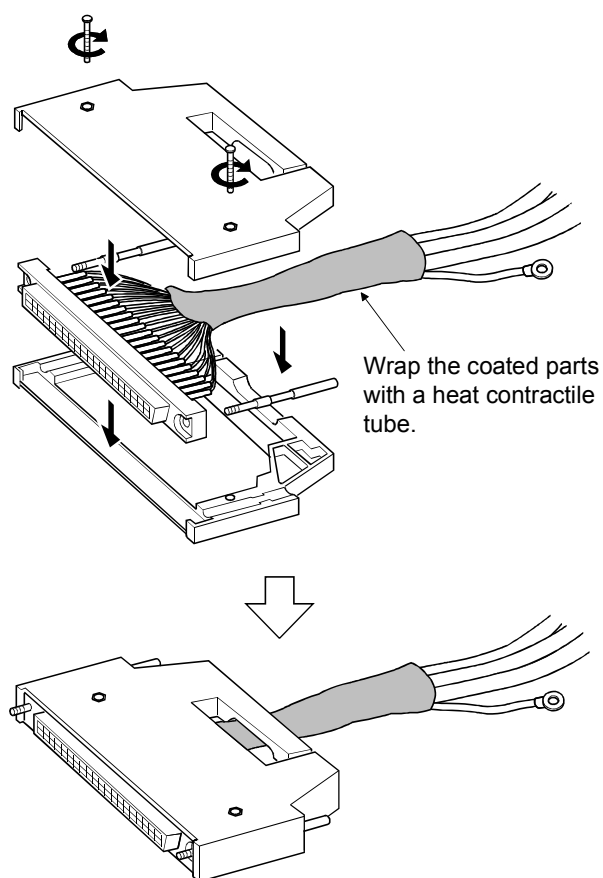
**[Processing example of shielded cables]**

Connect a cable with the FG wire and bind all shielded cables as shown below.

Remove the covering from all shielded cables and bind the appeared shield with a conductive tape.



## Assembling of connector (A6CON1)



(10) For compliance with the EMC and Low Voltage Directives, satisfy the following requirements.

- The cable connecting the drive unit and the QD75 must be the length below.

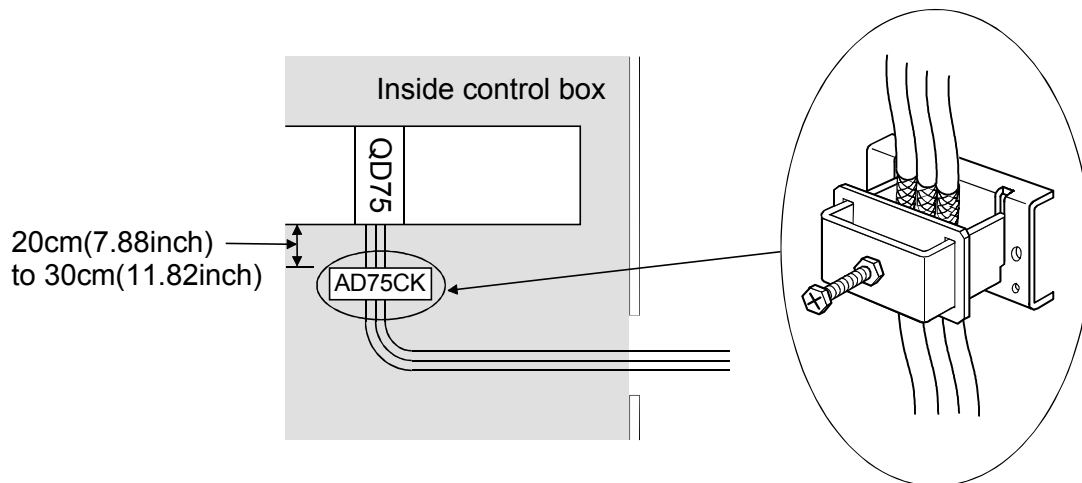
QD75P□N/QD75P□: 2m or shorter

QD75D□N/QD75D□: 10m or shorter

- The cables connected to the external device for the QD75 must be 30m or shorter. (except the one for pulse output)

- Use shielded twisted pair cables and an AD75CK type cable clamp (manufactured by Mitsubishi Electric) to ground the cables to the control box.

Even when compliance with the EMC Directive is not required, attaching an AD75CK type cable clamp to the cable connected to the QD75 may reduce the influence of external noise.

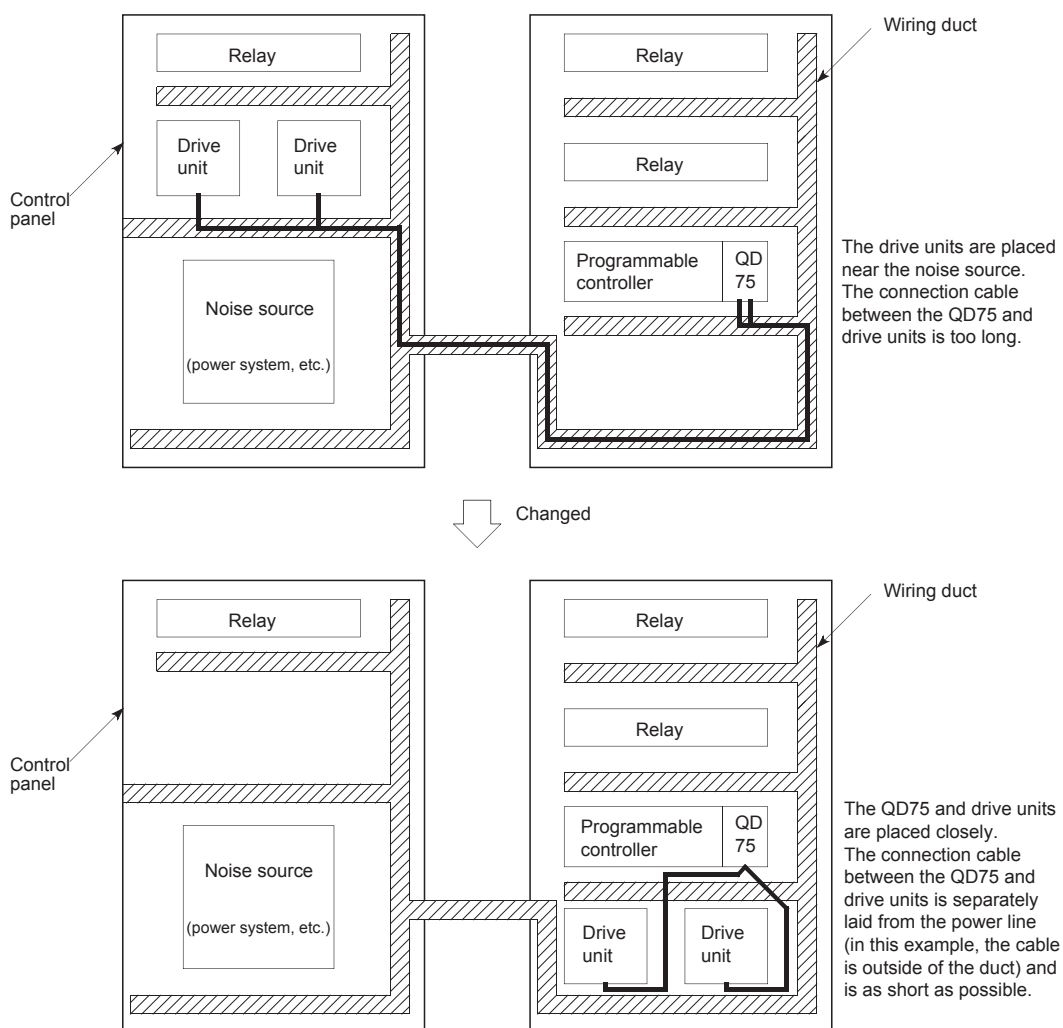


For details on AD75CK, refer to the following.

AD75CK-type Cable Clamping Instruction Manual



[Wiring examples using duct (incorrect example and corrected example)]



- (11) The influence of noise may be reduced by installing ferrite cores to the cable connected to the QD75 as a noise reduction technique.  
For the noise reduction techniques related to connection with the servo amplifier, also refer to the instruction manual of the servo amplifier.

### 4.3.2 Wiring of the differential driver common terminal

When the differential driver output system (QD75D□N/QD75D□) is used, a potential difference between commons may occur between the differential driver common terminal and the differential receiver common terminal of the drive unit. To remove the potential difference between commons, connect the differential driver common terminal of the QD75D□N/QD75D□ and the differential receiver common terminal of the drive unit.

When the common terminal of the drive unit is photocoupler-connected, the wiring to the differential driver common terminal of the QD75D□N/QD75D□ is not needed since a potential difference between commons does not exist.

(For the drive unit specifications, refer to the manual of the used drive unit.)

#### ■ Applicable connectors

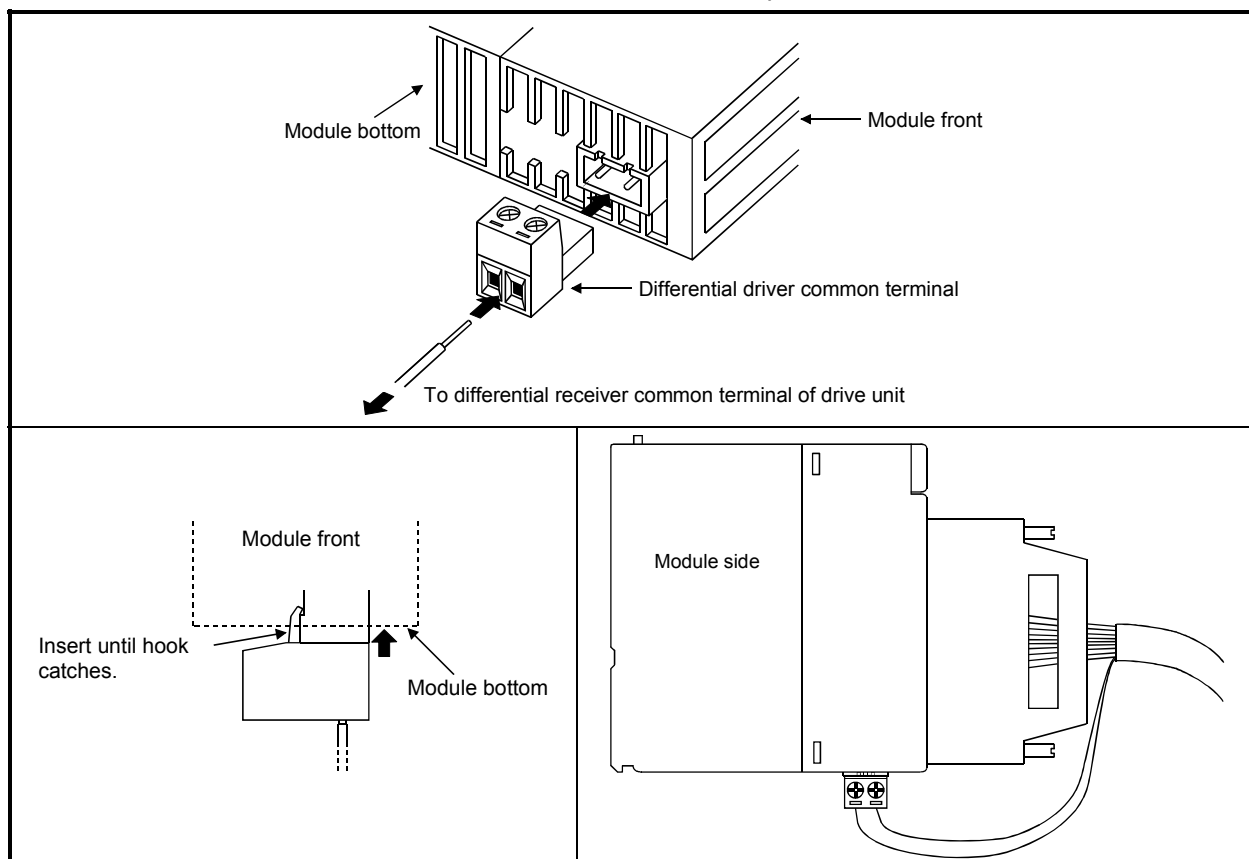
To wire the differential driver common terminal, use the wire applied to the following table.

| Wire diameter                               | Type                      | Material | Temperature rating |
|---------------------------------------------|---------------------------|----------|--------------------|
| 0.24 to 2.5mm <sup>2</sup><br>(24 to 12AWG) | Stranded wire/single wire | Copper   | 75°C or more       |

#### ■ Wiring example

The following shows an example of wiring to the differential driver common terminal of the QD75D□N/QD75D□.

Up to two cables can be connected to one differential driver common terminal. (For details, refer to Section 3.1 "Performance specifications".)



## 4.4 Checking installation and wiring

### 4.4.1 Items to check when installation and wiring are completed

Check the following points when completed with the QD75 installation and wiring.

- Is the module correctly wired? ... "Connection confirmation"

With "connection confirmation", the following three points are confirmed using GX Configurator-QP's connection confirmation function. (GX Configurator-QP is required for this "connection confirmation".)

- Are the QD75 and servo amplifier correctly connected?
- Are the servo amplifier and servomotor correctly connected?
- Are the QD75 and external device (input/output signal) correctly connected?

With this "connection confirmation", "whether the direction that the QD75 recognizes as forward run matches the address increment direction in the actual positioning work", and "whether the QD75 recognizes the external input/output signals such as the near-point dog signal and stop signal" can be checked.

Refer to GX Configurator-QP Operating Manual for details on "Connection confirmation".

Note that GX Developer (SW6D5C-GPPW-E or later)\*1 may also be used to "confirm the connection between the QD75 and external device (I/O signals)". For details, refer to Section 13.5 "External I/O signal monitor function" and GX Developer Operating Manual.

\*1: For the QD75P□N/QD75D□N, external I/O signals cannot be monitored on GX Developer. Use the system monitor of GX Works2. For details on the system monitor of GX Works2, refer to GX Works2 Version 1 Operating Manual (Common).

| Important                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If the QD75 is faulty, or when the required signals such as the near-point dog signal and stop signal are not recognized, unexpected accidents such as "not decelerating at the near-point dog during machine OPR and colliding with the stopper", or "not being able to stop with the stop signal" may occur. The "connection confirmation" must be carried out not only when structuring the positioning system, but also when the system has been changed with module replacement or rewiring, etc. |

## 4.5 Maintenance

### 4.5.1 Maintenance precautions

The precautions for servicing the QD75 are given below. Refer to this section as well as "4.1.3 Handling precautions" when carrying out the work.

#### WARNING

- Always turn all phases of the power supply OFF externally before cleaning or tightening the screws.  
Failure to turn all phases OFF could lead to electric shocks.

#### CAUTION

- Never disassemble or modify the module.  
Failure to observe this could lead to trouble, malfunctioning, injuries or fires.
- Completely turn off the externally supplied power used in the system before installing or removing the module.  
Failure to turn all phases OFF could lead to module trouble or malfunctioning.

### 4.5.2 Disposal precautions

#### CAUTION

- When disposing of the product, handle it as industrial waste.

# CHAPTER 5 DATA USED FOR POSITIONING CONTROL

The parameters and data used to carry out positioning control with the QD75 are explained in this chapter.

With the positioning system using the QD75, the various parameters and data explained in this chapter are used for control. The parameters and data include parameters set according to the device configuration, such as the system configuration, and parameters and data set according to each control. Read this chapter thoroughly and make settings according to each control or application.

\* Read PART 2 for details on each control.

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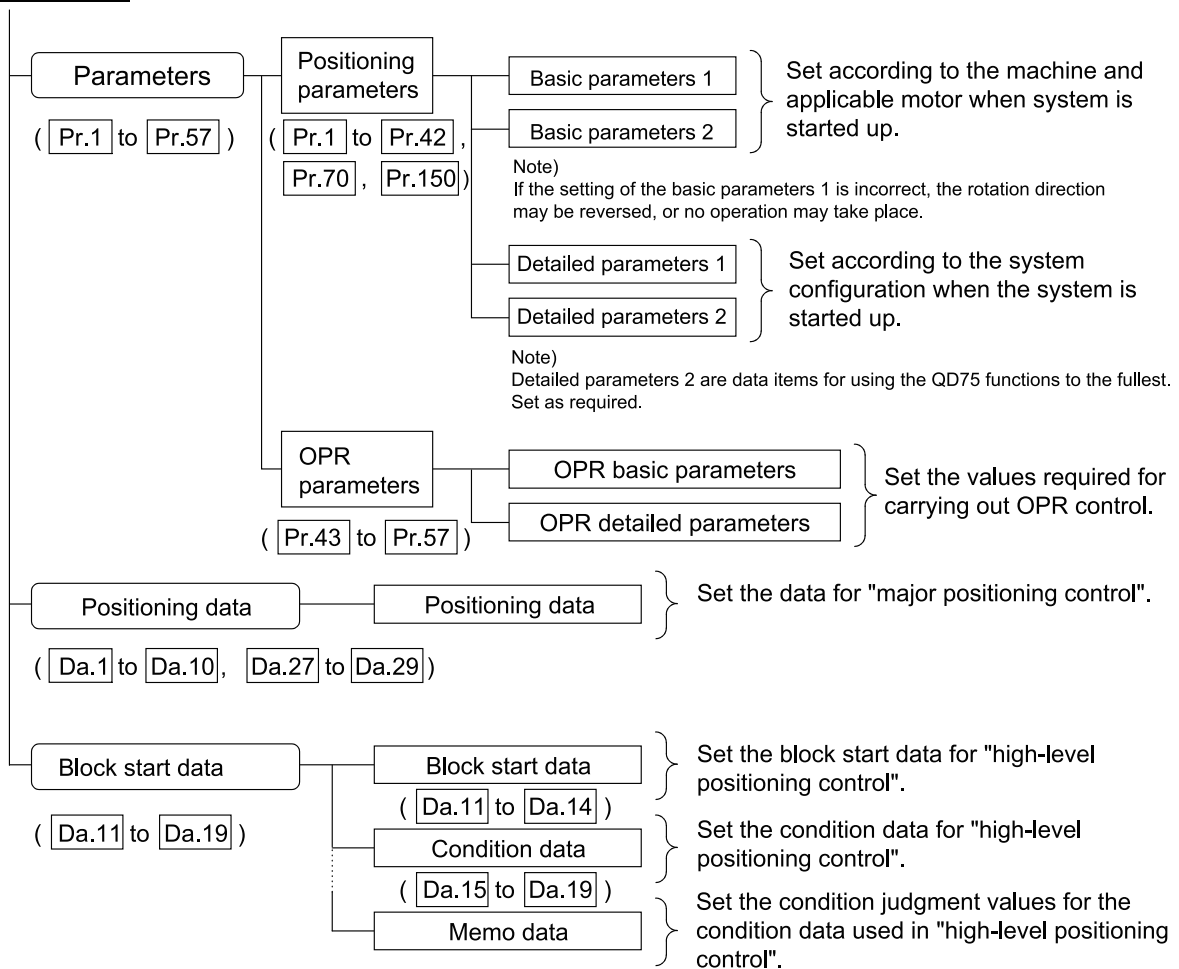
## 5.1 Types of data

## 5.1.1 Parameters and data required for control

The parameters and data required to carry out control with the QD75 include the "setting data", "monitor data" and "control data" shown below.

**Setting data**

(Data set beforehand according to the machine and application, and stored in the flash ROM.)



- ◇ The data is set with the sequence program or peripheral device.  
In this chapter, the method using the peripheral device will be explained.  
(Refer to "Point" on the next page.)
- ◇ The basic parameters 1, detailed parameters 1, and OPR parameters become valid when the PLC READY signal [Y0] turns from OFF to ON. Note, however, that the only valid value of the " Pr.5 Pulse output mode" is the value at the moment when the PLC READY signal [Y0] turns from OFF to ON for the first time after the power is switched ON or the CPU module is reset. Once the PLC READY signal [Y0] has been turned ON, the value will not be reset even if another value is set to the parameter and the PLC READY signal [Y0] is turned from OFF to ON.
- ◇ Even when the PLC READY signal [Y0] is ON, the values or contents of the following can be changed: basic parameters 2, detailed parameters 2, positioning data, and block start data.

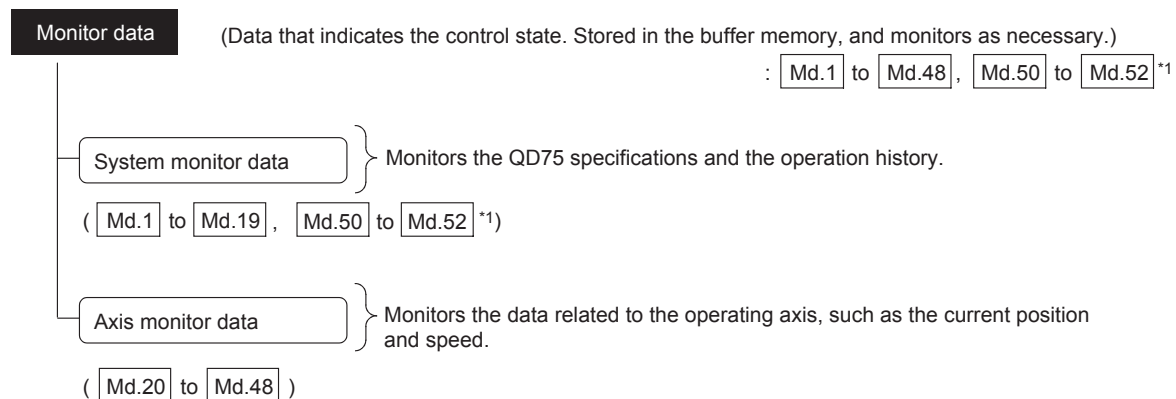
- ◇ The only valid data assigned to basic parameters 2, detailed parameters 2, positioning data, or block start data are the data read at the moment when a positioning or JOG operation is started. Once the operation has started, any modification to the data is ignored.

Exceptionally, however, modifications to the following are valid even when they are made during a positioning operation: acceleration time 0 to 3, deceleration time 0 to 3, and external start command.

- Acceleration time 0 to 3 and deceleration time 0 to 3:

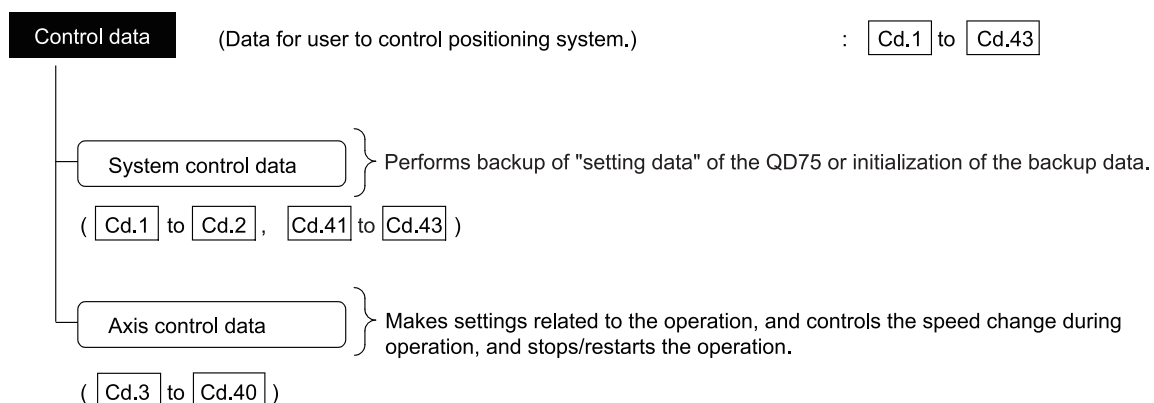
Positioning data are pre-read and pre-analyzed. Modifications to the data four or more steps after the current step are valid.

- External command function selection: The value at the time of detection is valid.



\*1: Md.50 to Md.52 are only for the QD75P□N/QD75D□N.

- ◇ The data is monitored with the sequence program or peripheral device. In this chapter, the method using the peripheral device will be explained.



- ◇ Control using the control data is carried out with the sequence program.  
" Cd.41 Deceleration start flag valid" is valid for only the value at the time when the PLC READY signal [Y0] turns from OFF to ON.

| POINT |
|-------|
|-------|

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>(1) The "setting data" is created for each axis.</li><li>(2) The "setting data" parameters have determined default values, and are set to the default values before shipment from the factory. (Parameters related to axes that are not used are left at the default value.)</li><li>(3) The "setting data" can be initialized with GX Configurator-QP or the sequence program.</li><li>(4) It is recommended to set the "setting data" with GX Configurator-QP. When executed with the sequence program, many sequence programs and devices must be used. This will not only complicate the program, but will also increase the scan time.</li></ul> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



## 5.1.2 Setting items for positioning parameters

The table below lists items set to the positioning parameters. Setting of positioning parameters is similarly done for individual axes for all controls achieved by the QD75. For details of controls, refer to PART 2. For details of setting items, refer to Section 5.2 "List of parameters".

| Positioning parameter |        |                                                  | Control | OPR control | Major positioning control                                        |                                                            |                                                                               |                        |                           |                                          |                                                    | Manual control |                                  | Related sub function |                                    |
|-----------------------|--------|--------------------------------------------------|---------|-------------|------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------|---------------------------|------------------------------------------|----------------------------------------------------|----------------|----------------------------------|----------------------|------------------------------------|
|                       |        |                                                  |         |             | Position control                                                 |                                                            |                                                                               |                        | 1 to 4 axis speed control | Speed-position or position-speed control | Other control                                      |                | Manual pulse generator operation |                      | JOG operation<br>Inching operation |
|                       |        |                                                  |         |             | 1-axis linear control<br>2/3/4-axis linear interpolation control | 1-axis fixed-feed control<br>2/3/4-axis fixed-feed control | 2-axis circular interpolation control<br>3-axis helical interpolation control | Current value changing |                           |                                          | JUMP instruction, NOP instruction,<br>LOOP to LEND |                |                                  |                      |                                    |
|                       |        |                                                  |         |             |                                                                  |                                                            |                                                                               |                        |                           |                                          |                                                    |                |                                  |                      |                                    |
| Basic parameters 1    | Pr.1   | Unit setting                                     | ⊙       | ⊙           | ⊙                                                                | △                                                          | ⊙                                                                             | ⊙                      | ⊙                         | ⊙                                        | ⊙                                                  | ⊙              | —                                |                      |                                    |
|                       | Pr.2   | No. of pulses per rotation (Ap)<br>(Unit: pulse) | ⊙       | ⊙           | ⊙                                                                | ⊙                                                          | ⊙                                                                             | ⊙                      | ⊙                         | ⊙                                        | ⊙                                                  | ⊙              | 12.3.2                           |                      |                                    |
|                       | Pr.3   | Movement amount per rotation (Al)                | ⊙       | ⊙           | ⊙                                                                | ⊙                                                          | ⊙                                                                             | ⊙                      | ⊙                         | ⊙                                        | ⊙                                                  | ⊙              |                                  |                      |                                    |
|                       | Pr.4   | Unit magnification (Am)                          | ⊙       | ⊙           | ⊙                                                                | ⊙                                                          | ⊙                                                                             | ⊙                      | ⊙                         | ⊙                                        | ⊙                                                  | ⊙              |                                  |                      |                                    |
|                       | Pr.5   | Pulse output mode                                | ⊙       | ⊙           | ⊙                                                                | ⊙                                                          | ⊙                                                                             | ⊙                      | ⊙                         | ⊙                                        | ⊙                                                  | ⊙              | —                                |                      |                                    |
|                       | Pr.6   | Rotation direction setting                       | ⊙       | ⊙           | ⊙                                                                | ⊙                                                          | ⊙                                                                             | ⊙                      | ⊙                         | ⊙                                        | ⊙                                                  | ⊙              | —                                |                      |                                    |
|                       | Pr.7   | Bias speed at start                              | ○       | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                      | —                         | —                                        | —                                                  | ○              | —                                |                      |                                    |
| Basic parameters 2    | Pr.8   | Speed limit value                                | ⊙       | ⊙           | ⊙                                                                | ⊙                                                          | ⊙                                                                             | ⊙                      | —                         | —                                        | —                                                  | ⊙              | 12.4.1                           |                      |                                    |
|                       | Pr.9   | Acceleration time 0                              | ⊙       | ⊙           | ⊙                                                                | ⊙                                                          | ⊙                                                                             | ⊙                      | —                         | —                                        | —                                                  | ⊙              | 12.7.6                           |                      |                                    |
|                       | Pr.10  | Deceleration time 0                              | ⊙       | ⊙           | ⊙                                                                | ⊙                                                          | ⊙                                                                             | ⊙                      | —                         | —                                        | —                                                  | ⊙              |                                  |                      |                                    |
| Detailed parameters 1 | Pr.11  | Backlash compensation amount                     | ○       | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                      | —                         | —                                        | ○                                                  | ○              | 12.3.1                           |                      |                                    |
|                       | Pr.12  | Software stroke limit upper limit value          | —       | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                      | —                         | —                                        | ○                                                  | ○              | 12.4.3                           |                      |                                    |
|                       | Pr.13  | Software stroke limit lower limit value          | —       | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                      | —                         | —                                        | ○                                                  | ○              |                                  |                      |                                    |
|                       | Pr.14  | Software stroke limit selection                  | —       | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                      | —                         | —                                        | ○                                                  | ○              |                                  |                      |                                    |
|                       | Pr.15  | Software stroke limit valid/invalid setting      | —       | —           | —                                                                | —                                                          | —                                                                             | —                      | ○                         | ○                                        | ○                                                  | ○              |                                  |                      |                                    |
|                       | Pr.16  | Command in-position width                        | —       | ○           | ○                                                                | ○                                                          | —                                                                             | ○                      | —                         | —                                        | —                                                  | —              | 12.7.5                           |                      |                                    |
|                       | Pr.17  | Torque limit setting value                       | △       | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                      | —                         | —                                        | △                                                  | △              | 12.4.2                           |                      |                                    |
|                       | Pr.18  | M code ON signal output timing                   | —       | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                      | ○                         | —                                        | —                                                  | —              | 12.7.3                           |                      |                                    |
|                       | Pr.19  | Speed switching mode                             | —       | ○           | ○                                                                | ○                                                          | —                                                                             | —                      | —                         | —                                        | —                                                  | —              | —                                |                      |                                    |
|                       | Pr.20  | Interpolation speed designation method           | —       | △           | △                                                                | △                                                          | △                                                                             | —                      | —                         | —                                        | —                                                  | —              | —                                |                      |                                    |
|                       | Pr.21  | Current feed value during speed control          | —       | —           | —                                                                | —                                                          | ○                                                                             | ○                      | —                         | —                                        | —                                                  | —              | —                                |                      |                                    |
|                       | Pr.22  | Input signal logic selection                     | ⊙       | ⊙           | ⊙                                                                | ⊙                                                          | ⊙                                                                             | ⊙                      | ⊙                         | ⊙                                        | ⊙                                                  | ⊙              | —                                |                      |                                    |
|                       | Pr.23  | Output signal logic selection                    | ⊙       | ⊙           | ⊙                                                                | ⊙                                                          | ⊙                                                                             | ⊙                      | ⊙                         | ⊙                                        | ⊙                                                  | ⊙              | —                                |                      |                                    |
|                       | Pr.24  | Manual pulse generator input selection           | —       | —           | —                                                                | —                                                          | —                                                                             | —                      | —                         | —                                        | ⊙                                                  | —              | —                                |                      |                                    |
|                       | Pr.150 | Speed-position function selection                | —       | —           | —                                                                | —                                                          | —                                                                             | ⊙                      | —                         | —                                        | —                                                  | —              | —                                |                      |                                    |
|                       | Pr.70  | Positioning option valid/invalid setting         | —       | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                      | ○                         | —                                        | —                                                  | —              | —                                |                      |                                    |

⊙ : Always set, ○ : Set as required, × : Setting not possible, △ : Setting restricted

— : Setting not required (This is an irrelevant item, so the set value will be ignored. If the value is the default value or within the setting range, there is no problem.)

| Positioning parameter |       | Control                                      | OPR control | Major positioning control                                        |                                                            |                                                                               |                                          |                        |                                                    | Manual control                   |                                    | Related sub function |                  |
|-----------------------|-------|----------------------------------------------|-------------|------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------|------------------------|----------------------------------------------------|----------------------------------|------------------------------------|----------------------|------------------|
|                       |       |                                              |             | Position control                                                 |                                                            |                                                                               | Speed-position or position-speed control | Other control          |                                                    | Manual pulse generator operation | JOG operation<br>Inching operation |                      |                  |
|                       |       |                                              |             | 1-axis linear control<br>2/3/4-axis linear interpolation control | 1-axis fixed-feed control<br>2/3/4-axis fixed-feed control | 2-axis circular interpolation control<br>3-axis helical interpolation control |                                          | Current value changing | JUMP instruction, NOP instruction,<br>LOOP to LEND |                                  |                                    |                      |                  |
| Detailed parameters 2 | Pr.25 | Acceleration time 1                          | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                                        | ○                      | -                                                  | -                                | -                                  | ○                    | 12.7.6           |
|                       | Pr.26 | Acceleration time 2                          | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                                        | ○                      | -                                                  | -                                | -                                  | ○                    |                  |
|                       | Pr.27 | Acceleration time 3                          | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                                        | ○                      | -                                                  | -                                | -                                  | ○                    |                  |
|                       | Pr.28 | Deceleration time 1                          | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                                        | ○                      | -                                                  | -                                | -                                  | ○                    |                  |
|                       | Pr.29 | Deceleration time 2                          | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                                        | ○                      | -                                                  | -                                | -                                  | ○                    |                  |
|                       | Pr.30 | Deceleration time 3                          | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                                        | ○                      | -                                                  | -                                | -                                  | ○                    |                  |
|                       | Pr.31 | JOG speed limit value                        | -           | -                                                                | -                                                          | -                                                                             | -                                        | -                      | -                                                  | -                                | -                                  | ◎                    | 12.4.1           |
|                       | Pr.32 | JOG operation acceleration time selection    | -           | -                                                                | -                                                          | -                                                                             | -                                        | -                      | -                                                  | -                                | -                                  | ◎                    | -                |
|                       | Pr.33 | JOG operation deceleration time selection    | -           | -                                                                | -                                                          | -                                                                             | -                                        | -                      | -                                                  | -                                | -                                  | ◎                    | -                |
|                       | Pr.34 | Acceleration/deceleration process selection  | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                                        | ○                      | -                                                  | -                                | -                                  | ○                    | 12.7.6           |
|                       | Pr.35 | S-curve ratio                                | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                                        | ○                      | -                                                  | -                                | -                                  | ○                    |                  |
|                       | Pr.36 | Sudden stop deceleration time                | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                                        | ○                      | -                                                  | -                                | -                                  | ○                    |                  |
|                       | Pr.37 | Stop group 1 sudden stop selection           | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                                        | ○                      | -                                                  | -                                | -                                  | ○                    | -                |
|                       | Pr.38 | Stop group 2 sudden stop selection           | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                                        | ○                      | -                                                  | -                                | -                                  | ○                    | -                |
|                       | Pr.39 | Stop group 3 sudden stop selection           | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                                        | ○                      | -                                                  | -                                | -                                  | ○                    | -                |
|                       | Pr.40 | Positioning complete signal output time      | -           | ○                                                                | ○                                                          | ○                                                                             | ○                                        | ○                      | ○                                                  | -                                | -                                  | -                    | -                |
|                       | Pr.41 | Allowable circular interpolation error width | -           | -                                                                | -                                                          | ○                                                                             | -                                        | -                      | -                                                  | -                                | -                                  | -                    | -                |
|                       | Pr.42 | External command function selection          | ○           | ○                                                                | ○                                                          | ○                                                                             | ○                                        | ◎                      | ○                                                  | -                                | -                                  | ○                    | 12.5.1<br>12.7.2 |

◎ : Always set

○ : Set as required

— : Setting not required (This is an irrelevant item, so the setting value will be ignored. If the value is the default value or within the setting range, there is no problem.)

### ■ Checking the positioning parameters

Pr.1 to Pr.42 are checked with the following timing.

- (1) When the "PLC READY signal [Y0]" output from the CPU module to the QD75 changes from OFF to ON
- (2) When the test operation button is turned ON in the test mode using GX Configurator-QP
- (3) When an error check is carried out with GX Configurator-QP

### REMARK

- "High-level positioning control" is carried out in combination with the "major positioning control".  
Refer to the "major positioning control" parameter settings for details on the parameters required for "high-level positioning control".

## 5.1.3 Setting items for OPR parameters

When carrying out "OPR control", the "OPR parameters" must be set. The setting items for the "OPR parameters" are shown below.

The "OPR parameters" are set commonly for each axis.

Refer to CHAPTER 8 "OPR CONTROL" for details on the "OPR control", and to Section 5.2 "List of parameters" for details on each setting item.

| OPR control             |       |                                                         | Machine OPR control   |                   |                   |                   |                 |                 | Fast OPR control |
|-------------------------|-------|---------------------------------------------------------|-----------------------|-------------------|-------------------|-------------------|-----------------|-----------------|------------------|
| OPR parameters          |       |                                                         |                       |                   |                   |                   |                 |                 |                  |
| OPR basic parameters    | Pr.43 | OPR method                                              | Near-point dog method | Stopper method 1) | Stopper method 2) | Stopper method 3) | Count method 1) | Count method 2) | ○                |
|                         | Pr.44 | OPR direction                                           | ○                     | ○                 | ○                 | ○                 | ○               | ○               |                  |
|                         | Pr.45 | OP address                                              | ○                     | ○                 | ○                 | ○                 | ○               | ○               |                  |
|                         | Pr.46 | OPR speed                                               | ○                     | ○                 | ○                 | ○                 | ○               | ○               |                  |
|                         | Pr.47 | Creep speed                                             | ○                     | ○                 | ○                 | ○                 | ○               | ○               |                  |
|                         | Pr.48 | OPR retry                                               | R                     | R                 | R                 | —                 | R               | R               |                  |
| OPR detailed parameters | Pr.49 | OPR dwell time                                          | —                     | ○                 | —                 | —                 | —               | —               |                  |
|                         | Pr.50 | Setting for the movement amount after near-point dog ON | —                     | —                 | —                 | —                 | ○               | ○               |                  |
|                         | Pr.51 | OPR acceleration time selection                         | ○                     | ○                 | ○                 | ○                 | ○               | ○               |                  |
|                         | Pr.52 | OPR deceleration time selection                         | ○                     | ○                 | ○                 | ○                 | ○               | ○               |                  |
|                         | Pr.53 | OP shift amount                                         | S                     | S                 | S                 | S                 | S               | S               |                  |
|                         | Pr.54 | OPR torque limit value                                  | —                     | ○                 | ○                 | ○                 | —               | —               |                  |
|                         | Pr.55 | Deviation counter clear signal output time              | C                     | C                 | C                 | C                 | C               | —               |                  |
|                         | Pr.56 | Speed designation during OP shift                       | S                     | S                 | S                 | S                 | S               | S               |                  |
|                         | Pr.57 | Dwell time during OPR retry                             | R                     | R                 | R                 | —                 | R               | R               |                  |

○ : Always set

○ : Preset parameters are used for machine OPR control.

— : Setting not required (This is an irrelevant item, so the setting value will be ignored. If the value is the default value or within the setting range, there is no problem.)

R : Set when using the "12.2.1 OPR retry function".

S : Set when using the "12.2.2 OP shift function".

C : Set the deviation counter clear signal output time.

#### ■ Checking the OPR parameters.

Pr.43 to Pr.57 are checked with the following timing.

- (1) When the "PLC READY signal [Y0]" output from the CPU module to the QD75 changes from OFF to ON
- (2) When the test operation button is turned ON in the test mode using GX Configurator-QP
- (3) When an error check is carried out with GX Configurator-QP

## 5.1.4 Setting items for positioning data

Positioning data must be set for carrying out any "major positioning control". The table below lists the items to be set for producing the positioning data.

One to 600 positioning data items can be set for each axis.

For details of the major positioning controls, refer to CHAPTER 9 "MAJOR POSITIONING CONTROL". For details of the individual setting items, refer to Section 5.3 "List of positioning data".

| Major positioning control<br><br>Positioning data |                                        |                                 | Position control                                                                                                                                       |                                                              |                                                        |                                                     | 1 to 4 axis speed control                                                                                                                                                            | Speed-position switching control                         | Position-speed switching control                         | Other control   |                        |                                       |                    |      |
|---------------------------------------------------|----------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|-----------------|------------------------|---------------------------------------|--------------------|------|
|                                                   |                                        |                                 | 1-axis linear control<br>2/3/4-axis linear interpolation control                                                                                       | 1-axis fixed-feed control<br>2/3/4-axis fixed-feed control   | 2-axis circular interpolation control                  | 3-axis helical interpolation control                |                                                                                                                                                                                      |                                                          |                                                          | NOP instruction | Current value changing | JUMP instruction                      | LOOP               | LEND |
| Da.1                                              | Operation pattern                      | Independent positioning control | ◎                                                                                                                                                      | ◎                                                            | ◎                                                      | ◎                                                   | ◎                                                                                                                                                                                    | ◎                                                        | ◎                                                        | -               | ◎                      | -                                     | -                  | -    |
|                                                   |                                        | Continuous positioning control  | ◎                                                                                                                                                      | ◎                                                            | ◎                                                      | ◎                                                   | ×                                                                                                                                                                                    | ◎                                                        | ×                                                        | -               | ◎                      | -                                     | -                  | -    |
|                                                   |                                        | Continuous path control         | ◎                                                                                                                                                      | ×                                                            | ◎                                                      | ◎                                                   | ×                                                                                                                                                                                    | ×                                                        | ×                                                        | -               | ×                      | -                                     | -                  | -    |
| Da.2                                              | Control system                         |                                 | Linear 1<br>Linear 2<br>Linear 3<br>Linear 4<br>* 1                                                                                                    | Fixed-feed 1<br>Fixed-feed 2<br>Fixed-feed 3<br>Fixed-feed 4 | Circular sub<br>Circular right<br>Circular left<br>* 1 | Helical sub<br>Helical right<br>Helical left<br>* 1 | Forward run speed 1<br>Reverse run speed 1<br>Forward run speed 2<br>Reverse run speed 2<br>Forward run speed 3<br>Reverse run speed 3<br>Forward run speed 4<br>Reverse run speed 4 | Forward run speed/position<br>Reverse run speed/position | Forward run position/speed<br>Reverse run position/speed | NOP instruction | Current value changing | JUMP instruction                      | LOOP               | LEND |
| Da.3                                              | Acceleration time No.                  |                                 | ○                                                                                                                                                      | ○                                                            | ○                                                      | ○                                                   | ○                                                                                                                                                                                    | ○                                                        | ○                                                        | -               | -                      | -                                     | -                  | -    |
| Da.4                                              | Deceleration time No.                  |                                 | ○                                                                                                                                                      | ○                                                            | ○                                                      | ○                                                   | ○                                                                                                                                                                                    | ○                                                        | ○                                                        | -               | -                      | -                                     | -                  | -    |
| Da.5                                              | Axis to be interpolated                |                                 | ◎: 2-axis interpolation control, 3-axis helical interpolation control<br>—: 1-axis control, 3-axis interpolation control, 4-axis interpolation control |                                                              |                                                        |                                                     |                                                                                                                                                                                      | -                                                        | -                                                        | -               | -                      | -                                     | -                  | -    |
| Da.6                                              | Positioning address/movement amount    |                                 | ◎                                                                                                                                                      | ◎                                                            | ◎                                                      | ◎                                                   | -                                                                                                                                                                                    | ◎                                                        | ◎                                                        | -               | New address            | -                                     | -                  | -    |
| Da.7                                              | Arc address                            |                                 | -                                                                                                                                                      | -                                                            | ◎                                                      | ◎                                                   | -                                                                                                                                                                                    | -                                                        | -                                                        | -               | -                      | -                                     | -                  | -    |
| Da.8                                              | Command speed                          |                                 | ◎                                                                                                                                                      | ◎                                                            | ◎                                                      | ◎                                                   | ◎                                                                                                                                                                                    | ◎                                                        | ◎                                                        | -               | -                      | -                                     | -                  | -    |
| Da.9                                              | Dwell time                             |                                 | ○                                                                                                                                                      | ○                                                            | ○                                                      | ○                                                   | ○                                                                                                                                                                                    | ○                                                        | ○                                                        | -               | -                      | JUMP destination positioning data No. | -                  | -    |
| Da.10                                             | M code                                 |                                 | ○                                                                                                                                                      | ○                                                            | ○                                                      | ○<br>* 2                                            | ○                                                                                                                                                                                    | ○                                                        | ○                                                        | -               | ○                      | JUMP condition data No.               | No. of repetitions | -    |
| Da.27                                             | M code ON signal output timing         |                                 | ○                                                                                                                                                      | ○                                                            | ○                                                      | ○                                                   | ○                                                                                                                                                                                    | ○                                                        | ○                                                        | -               | ○                      | -                                     | -                  | -    |
| Da.28                                             | ABS direction in degrees               |                                 | ○                                                                                                                                                      | ○                                                            | ○                                                      | ○                                                   | ○                                                                                                                                                                                    | ○                                                        | ○                                                        | -               | -                      | -                                     | -                  | -    |
| Da.29                                             | Interpolation speed designation method |                                 | —: 1-axis control<br>○: 2-axis interpolation control, 3-axis interpolation control, 4-axis interpolation control                                       |                                                              |                                                        |                                                     |                                                                                                                                                                                      | -                                                        | -                                                        | -               | -                      | -                                     | -                  | -    |

◎ : Always set, ○ : Set as required, × : Setting not possible

—: Setting not required (This is an irrelevant item, so the set value will be ignored. If the value is the default value or within the setting range, there is no problem.)

\* 1: Two control systems are available: the absolute (ABS) system and incremental (INC) system.

\* 2: Set the M code for the reference axis and the number of pitch for the linear interpolation axis.

### ■ Checking the positioning data

The items Da.1 to Da.10 are checked at the following timings:

- (1) Startup of a positioning operation
- (2) Error check performed by GX Configurator-QP

## 5.1.5 Setting items for block start data

The "block start data" must be set when carrying out "high-level positioning control".

The setting items for the "block start data" are shown below.

Up to 50 points of "block start data" can be set for each axis.

Refer to CHAPTER 10 "HIGH-LEVEL POSITIONING CONTROL" for details on the "high-level positioning control", and to Section 5.4 "List of block start data" for details on each setting item.

| High-level positioning control<br>Block start data |                           | Block start<br>(Normal start) | Condition<br>start | Wait start | Simultaneous<br>start | Repeated<br>start<br>(FOR loop) | Repeated<br>start<br>(FOR<br>condition) |
|----------------------------------------------------|---------------------------|-------------------------------|--------------------|------------|-----------------------|---------------------------------|-----------------------------------------|
| Da.11                                              | Shape (end/continue)      | ○                             | ○                  | ○          | ○                     | ○                               | ○                                       |
| Da.12                                              | Start data No.            | ○                             | ○                  | ○          | ○                     | ○                               | ○                                       |
| Da.13                                              | Special start instruction | —                             | ○                  | ○          | ○                     | ○                               | ○                                       |
| Da.14                                              | Parameter                 | —                             | ○                  | ○          | ○                     | ○                               | ○                                       |

○ : Set as required

— : Setting not required (This is an irrelevant item, so the setting value will be ignored. If the value is the default value or within the setting range, there is no problem.)

### ■ Checking the block start data

Da.11 to Da.14 are checked with the following timing.

- (1) When the "Block start data" starts
- (2) When an error check is carried out with GX Configurator-QP

## 5.1.6 Setting items for condition data

When carrying out "high-level positioning control" or using the JUMP instruction in the "major positioning control", the "condition data" must be set as required. The setting items for the "condition data" are shown below.

Up to 10 "condition data" items can be set for each axis.

For the "high-level positioning control", JUMP instruction, and setting items, refer to the following.

- "High-level positioning control" Refer to CHAPTER 10 "HIGH-LEVEL POSITIONING CONTROL"
- JUMP instruction.....Refer to Section 9.2.23 "JUMP instruction".
- Setting items .....Refer to Section 5.5 "List of condition data".

| Control<br>Condition data |                    | Major positioning control   |                  | High-level positioning control |                 |            |                     |                           |                                |
|---------------------------|--------------------|-----------------------------|------------------|--------------------------------|-----------------|------------|---------------------|---------------------------|--------------------------------|
|                           |                    | Other than JUMP instruction | JUMP instruction | Block start (Normal start )    | Condition start | Wait start | Simul-taneous start | Repeated start (FOR loop) | Repeated start (FOR condition) |
| Da.15                     | Condition target   | —                           | ○                | —                              | ○               | ○          | ○                   | —                         | ○                              |
| Da.16                     | Condition operator | —                           | ○                | —                              | ○               | ○          | ○                   | —                         | ○                              |
| Da.17                     | Address            | —                           | △                | —                              | △               | △          | —                   | —                         | △                              |
| Da.18                     | Parameter 1        | —                           | ○                | —                              | ○               | ○          | △                   | —                         | ○                              |
| Da.19                     | Parameter 2        | —                           | △                | —                              | △               | △          | △                   | —                         | △                              |

○ : Set as required

△ : Setting limited

— : Setting not required (This is an irrelevant item, so the setting value will be ignored. If the value is the default value or within the setting range, there is no problem.)

### ■ Checking the condition data

Da.15 to Da.19 are checked with the following timing.

- (1) When the "Block start data" starts
- (2) When "JUMP instruction" starts
- (3) When an error check is carried out with GX Configurator-QP

## 5.1.7 Types and roles of monitor data

The monitor data area in the buffer memory stores data relating to the operating state of the positioning system, which are monitored as required while the positioning system is operating.

The following data are available for monitoring.

- System monitoring:

Monitoring of the QD75 configuration and operation history (through the system monitor data [Md.1] through, [Md.19] [Md.50] through [Md.52] \*1)

\*1: [Md.50] through [Md.52] are for the QD75P□N/QD75D□N only.

- Axis operation monitoring:

Monitoring of the current position and speed, and other data related to the movements of axes (through the axis monitor data [Md.20] through [Md.48] \*2)

\*2: The axis monitor data are refreshed every 0.9ms (QD75P□N/QD75D□N) or 1.8ms (QD75P□/QD75D□). Note that " [Md.21] Machine feed value", " [Md.22] Feedrate", " [Md.28] Axis feedrate" and " [Md.30] External input/output signal" of QD75P□/QD75D□ are refreshed every 56.8ms. Also, " [Md.23] Valid M code", is updated when the "M code ON signal [X4, X5, X6, X7]" turns ON.

## [1] Monitoring the system

## ■ Monitoring the positioning system operation history

| Monitoring details                            |                                                                        |                                | Corresponding item |                                       |
|-----------------------------------------------|------------------------------------------------------------------------|--------------------------------|--------------------|---------------------------------------|
| Whether the system is in the test mode or not |                                                                        |                                | Md.1               | In test mode flag                     |
| History of data that started an operation     | Start information                                                      |                                | Md.3               | Start information                     |
|                                               | Start No.                                                              |                                | Md.4               | Start No.                             |
|                                               | Start                                                                  | Year:month<br>(QD75P□N/QD75D□N | Md.50              | Start (Year:month)                    |
|                                               |                                                                        | Day:hour<br>(QD75P□N/QD75D□N   | Md.5               | Start (Day:hour)                      |
|                                               |                                                                        | Hour<br>(QD75P□/QD75D□)        |                    |                                       |
|                                               |                                                                        | Minute:second                  | Md.6               | Start (Minute:second)                 |
|                                               | Error upon starting                                                    |                                | Md.7               | Error judgment                        |
|                                               | Pointer No. next to the pointer No. where the latest history is stored |                                | Md.8               | Start history pointer                 |
| History of all errors                         | Axis in which the error occurred                                       |                                | Md.9               | Axis in which the error occurred      |
|                                               | Axis error No.                                                         |                                | Md.10              | Axis error No.                        |
|                                               | Axis error occurrence                                                  | Year:month<br>(QD75P□N/QD75D□N | Md.51              | Axis error occurrence (Year:month)    |
|                                               |                                                                        | Day:hour<br>(QD75P□N/QD75D□N   | Md.11              | Axis error occurrence (Day:hour)      |
|                                               |                                                                        | Hour<br>(QD75P□/QD75D□)        |                    |                                       |
|                                               |                                                                        | Minute:second                  | Md.12              | Axis error occurrence (Minute:second) |
|                                               | Pointer No. next to the pointer No. where the latest history is stored |                                | Md.13              | Error history pointer                 |



| Monitoring details                                                       |                                                                        |                                | Corresponding item                                   |
|--------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------|------------------------------------------------------|
| History of all warnings                                                  | Axis in which the warning occurred                                     |                                | <b>Md.14</b> Axis in which the warning occurred      |
|                                                                          | Axis warning No.                                                       |                                | <b>Md.15</b> Axis warning No.                        |
|                                                                          | Axis warning occurrence                                                | Year:month<br>(QD75P□/QD75D□N) | <b>Md.52</b> Axis warning occurrence (Year:month)    |
|                                                                          |                                                                        | Day:hour<br>(QD75P□/QD75D□N)   | <b>Md.16</b> Axis warning occurrence (Day:hour)      |
|                                                                          |                                                                        | Hour<br>(QD75P□/QD75D□)        |                                                      |
|                                                                          |                                                                        | Minute:second                  | <b>Md.17</b> Axis warning occurrence (Minute:second) |
| Number of write accesses to the flash ROM after the power is switched ON | Pointer No. next to the pointer No. where the latest history is stored |                                | <b>Md.18</b> Warning history pointer                 |
|                                                                          | Number of write accesses to flash ROM                                  |                                | <b>Md.19</b> No. of write accesses to flash ROM      |

## [2] Monitoring the axis operation state

## ■ Monitoring the position

| Monitor details                          | Corresponding item              |
|------------------------------------------|---------------------------------|
| Monitor the current machine feed value   | <b>Md.21</b> Machine feed value |
| Monitor the current "current feed value" | <b>Md.20</b> Current feed value |
| Monitor the current target value         | <b>Md.32</b> Target value       |

## ■ Monitoring the speed

| Monitor details                  |                                             |                                                                                                      |                                    | Corresponding item        |               |
|----------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------|---------------------------|---------------|
| Monitor the current speed        | During independent axis control             |                                                                                                      | Indicates the speed of each axis   | <div>Md.22</div> Feedrate |               |
|                                  | During interpolation control                | When "0: Composite speed" is set for " <div>Pr.20</div> Interpolation speed designation method"      | Indicates the composite speed      |                           |               |
|                                  |                                             | When "1: Reference axis speed" is set for " <div>Pr.20</div> Interpolation speed designation method" | Indicates the reference axis speed |                           |               |
|                                  | Constantly indicates the speed of each axis |                                                                                                      |                                    | <div>Md.28</div>          | Axis feedrate |
| Monitor the current target speed |                                             |                                                                                                      |                                    | <div>Md.33</div>          | Target speed  |

■ Monitoring the state

| Monitor details                                                                                                                                                              | Corresponding item                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Monitor the axis operation state                                                                                                                                             | <a href="#">Md.26</a> Axis operation status                                  |
| Monitor the latest error code that occurred with the axis                                                                                                                    | <a href="#">Md.23</a> Axis error No.                                         |
| Monitor the latest warning code that occurred with the axis                                                                                                                  | <a href="#">Md.24</a> Axis warning No.                                       |
| Monitor the external input/output signal and flag                                                                                                                            | <a href="#">Md.30</a> External input/output signal                           |
|                                                                                                                                                                              | <a href="#">Md.31</a> Status                                                 |
| Monitor the valid M codes                                                                                                                                                    | <a href="#">Md.25</a> Valid M code                                           |
| Monitor whether the speed is being limited                                                                                                                                   | <a href="#">Md.39</a> In speed limit flag                                    |
| Monitor whether the speed is being changed                                                                                                                                   | <a href="#">Md.40</a> In speed change processing flag                        |
| Monitor the "start data" point currently being executed                                                                                                                      | <a href="#">Md.43</a> Start data pointer being executed                      |
| Monitor the "positioning data No." currently being executed                                                                                                                  | <a href="#">Md.44</a> Positioning data No. being executed                    |
| Monitor the remaining No. of repetitions (special start)                                                                                                                     | <a href="#">Md.41</a> Special start repetition counter                       |
| Monitor the remaining No. of repetitions (control system)                                                                                                                    | <a href="#">Md.42</a> Control system repetition counter                      |
| Monitor the block No.                                                                                                                                                        | <a href="#">Md.45</a> Block No. being executed                               |
| Monitor the current torque limit value                                                                                                                                       | <a href="#">Md.35</a> Torque limit stored value                              |
| Monitor the "instruction code" of the special start data when using special start                                                                                            | <a href="#">Md.36</a> Special start data instruction code setting value      |
| Monitor the "instruction parameter" of the special start data when using special start                                                                                       | <a href="#">Md.37</a> Special start data instruction parameter setting value |
| Monitor the "start data No." of the special start data when using special start                                                                                              | <a href="#">Md.38</a> Start positioning data No. setting value               |
| Monitor the "positioning data No." executed last                                                                                                                             | <a href="#">Md.46</a> Last executed positioning data No.                     |
| Monitor the positioning data currently being executed                                                                                                                        | <a href="#">Md.47</a> Positioning data being executed                        |
| Monitor the movement amount after the current position control switching when using "speed-position switching control (INC mode)"                                            | <a href="#">Md.29</a> Speed-position switching control positioning amount    |
| Monitor switching from the constant speed status or acceleration status to the deceleration status during position control whose operation pattern is "Positioning complete" | <a href="#">Md.48</a> Deceleration start flag                                |

## 5.1.8 Types and roles of control data

Operation of the positioning system is achieved through the execution of necessary controls. (Data required for controls are given through the default values when the power is switched ON, which can be modified as required by the sequence program.) Controls are performed over system data or machine operation.

- Controlling the system data :

Performs write/initialization, etc. of the QD75 "setting data". (the system control data Cd.1 and Cd.2)

- Controlling the operation :

Setting operation parameters, changing speed during operation, interrupting or restarting operation (the system control data Cd.41 and Cd.42, and the axis control data Cd.3 to Cd.40)

## [1] Controlling the system data

## ■ Setting and resetting the setting data

| Control details                                    | Controlled data item                         |
|----------------------------------------------------|----------------------------------------------|
| Write setting data from buffer memory to flash ROM | <u>Cd.1</u> Flash ROM write request          |
| Reset (initialize) parameters                      | <u>Cd.2</u> Parameter initialization request |

## [2] Controlling the operation

## ■ Controlling the operation

| Control details                                                                              | Corresponding item                                                                      |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Set which positioning to execute (start No.)                                                 | <a href="#">Cd.3</a> Positioning start No.                                              |
| Clear (reset) the axis error ( <a href="#">Md.23</a> ) and warning ( <a href="#">Md.24</a> ) | <a href="#">Cd.5</a> Axis error reset                                                   |
| Issue instruction to restart (When axis operation is stopped)                                | <a href="#">Cd.6</a> Restart command                                                    |
| End current positioning (deceleration stop), and start next positioning                      | <a href="#">Cd.37</a> Skip command                                                      |
| Set start point No. for executing block start                                                | <a href="#">Cd.4</a> Positioning starting point No.                                     |
| Stop continuous control                                                                      | <a href="#">Cd.18</a> Continuous operation interrupt request                            |
| Set start data Nos. for axes that start up simultaneously                                    | <a href="#">Cd.30</a> Simultaneous starting axis start data No. (axis 1 start data No.) |
|                                                                                              | <a href="#">Cd.31</a> Simultaneous starting axis start data No. (axis 2 start data No.) |
|                                                                                              | <a href="#">Cd.32</a> Simultaneous starting axis start data No. (axis 3 start data No.) |
|                                                                                              | <a href="#">Cd.33</a> Simultaneous starting axis start data No. (axis 4 start data No.) |
| Specify write destination for teaching results                                               | <a href="#">Cd.38</a> Teaching data selection                                           |
| Specify data to be taught                                                                    | <a href="#">Cd.39</a> Teaching positioning data No.                                     |

## ■ Controlling operation per step

| Control details                                                  | Corresponding item                           |
|------------------------------------------------------------------|----------------------------------------------|
| Stop positioning operation after each operation                  | <a href="#">Cd.35</a> Step valid flag        |
| Set unit to carry out step                                       | <a href="#">Cd.34</a> Step mode              |
| Issue instruction to continue operation after a step is finished | <a href="#">Cd.36</a> Step start information |

## ■ Controlling the speed

| Control details                                                                                                                     | Corresponding item                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Set new speed when changing speed during operation                                                                                  | <a href="#">Cd.14</a> New speed value                                                                     |
| Issue instruction to change speed in operation to <a href="#">Cd.14</a> value (Only during positioning operation and JOG operation) | <a href="#">Cd.15</a> Speed change request                                                                |
| Change positioning operation speed between 1 and 300% range                                                                         | <a href="#">Cd.13</a> Positioning operation speed override                                                |
| Set inching movement amount                                                                                                         | <a href="#">Cd.16</a> Inching movement amount                                                             |
| Set JOG speed                                                                                                                       | <a href="#">Cd.17</a> JOG speed                                                                           |
| When changing acceleration time during speed change, set new acceleration time                                                      | <a href="#">Cd.10</a> New acceleration time value                                                         |
| When changing deceleration time during speed change, set new deceleration time                                                      | <a href="#">Cd.11</a> New deceleration time value                                                         |
| Set acceleration/deceleration time validity during speed change                                                                     | <a href="#">Cd.12</a> Acceleration/deceleration time change during speed change, enable/disable selection |

■ Making settings related to operation

| Control details                                                                                                                   | Corresponding item                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Turn M code ON signal OFF                                                                                                         | <a href="#">Cd.7</a> M code OFF request                                                |
| Set new value when changing current value                                                                                         | <a href="#">Cd.9</a> New current value                                                 |
| Validate speed-position switching signal from external source                                                                     | <a href="#">Cd.24</a> Speed-position switching enable flag                             |
| Change movement amount for position control during speed-position switching control (INC mode)                                    | <a href="#">Cd.23</a> Speed-position switching control movement amount change register |
| Validate external position-speed switching signal                                                                                 | <a href="#">Cd.26</a> Position-speed switching enable flag                             |
| Change speed for speed control during position-speed switching control                                                            | <a href="#">Cd.25</a> Position-speed switching control speed change register           |
| Set up a flag when target position is changed during positioning                                                                  | <a href="#">Cd.29</a> Target position change request flag                              |
| Set new positioning address when changing target position during positioning                                                      | <a href="#">Cd.27</a> Target position change value(new address)                        |
| Set new speed when changing target position during positioning                                                                    | <a href="#">Cd.28</a> Target position change value(new speed)                          |
| Set absolute (ABS) moving direction in degrees                                                                                    | <a href="#">Cd.40</a> ABS direction in degrees                                         |
| Set manual pulse generator operation validity                                                                                     | <a href="#">Cd.21</a> Manual pulse generator enable flag                               |
| Set scale per pulse of No. of input pulses from manual pulse generator                                                            | <a href="#">Cd.20</a> Manual pulse generator 1 pulse input magnification               |
| Change OPR request flag from "ON to OFF"                                                                                          | <a href="#">Cd.19</a> OPR request flag OFF request                                     |
| Validate external command signal                                                                                                  | <a href="#">Cd.8</a> External command valid                                            |
| Change " <a href="#">Md.35</a> Torque limit stored value"                                                                         | <a href="#">Cd.22</a> New torque value                                                 |
| Set whether " <a href="#">Md.48</a> Deceleration start flag" is valid or invalid                                                  | <a href="#">Cd.41</a> Deceleration start flag valid                                    |
| Set the stop command processing for deceleration stop function (deceleration curve re-processing/deceleration curve continuation) | <a href="#">Cd.42</a> Stop command processing for deceleration stop selection          |

## 5.2 List of parameters

## 5.2.1 Basic parameters 1

| Item                      | Setting value, setting range                                     |                                                                                                                                                                                                                                                 | Default value | Setting value buffer memory address |        |        |        |
|---------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------|--------|--------|--------|
|                           | Value set with peripheral device                                 | Value set with sequence program                                                                                                                                                                                                                 |               | Axis 1                              | Axis 2 | Axis 3 | Axis 4 |
| <b>Pr.1</b> Unit setting  | 0 : mm                                                           | 0                                                                                                                                                                                                                                               | 3             | 0                                   | 150    | 300    | 450    |
|                           | 1 : inch                                                         | 1                                                                                                                                                                                                                                               |               |                                     |        |        |        |
|                           | 2 : degree                                                       | 2                                                                                                                                                                                                                                               |               |                                     |        |        |        |
|                           | 3 : pulse                                                        | 3                                                                                                                                                                                                                                               |               |                                     |        |        |        |
| Movement amount per pulse | <b>Pr.2</b><br>No. of pulses per rotation (Ap)<br>(Unit : pulse) | 1 to 65535<br>1 to 32767<br>:Set as a decimal<br>32768 to 65535<br>:Convert into hexadecimal<br>and set                                                                                                                                         | 20000         | 1                                   | 151    | 301    | 451    |
|                           | <b>Pr.3</b><br>Movement amount per rotation (Al)                 | The setting value range differs according to the " <b>Pr.1</b> Unit setting".<br>Here, the value within the [Table 1] range is set.<br> [Table 1] on next page | 20000         | 2                                   | 152    | 302    | 452    |
|                           | <b>Pr.4</b><br>Unit magnification (Am)                           | 1 : 1-fold                                                                                                                                                                                                                                      | 1             | 3                                   | 153    | 303    | 453    |
|                           |                                                                  | 10 : 10-fold                                                                                                                                                                                                                                    |               |                                     |        |        |        |
|                           |                                                                  | 100 : 100-fold                                                                                                                                                                                                                                  |               |                                     |        |        |        |
|                           |                                                                  | 1000 : 1000-fold                                                                                                                                                                                                                                |               |                                     |        |        |        |

**Pr.1** Unit setting

Set the unit used for defining positioning operations. Choose from the following units depending on the type of the control target: mm, inch, degree, or pulse. Different units can be defined for different axes (axis 1 to 4).

(Example) Different units (mm, inch, degree, and pulse) are applicable to different systems:

mm or inch..... X-Y table, conveyor (Select mm or inch depending on the machine specifications.)

degree ..... Rotating body (360 degrees/rotation)

pulse ..... X-Y table, conveyor

\* When you change the unit, note that the values of other parameters and data will not be changed automatically.

After changing the unit, check if the parameter and data values are within the allowable range.

Set "degree" to exercise speed-position switching control (ABS mode).

**Pr.2** to **Pr.4** Movement amount per pulse

These parameters define the amount of movement achieved by each single pulse within a pulse train output by the QD75.

The following paragraphs explain how to set the individual parameters **Pr.2**, **Pr.3**, and **Pr.4** assuming that the unit "mm" is selected with **Pr.1**.

The movement amount per pulse is given by the following expression:

$$\text{Movement amount per pulse} = \frac{\text{Movement amount per rotation (Al)}}{\text{No. of pulses per rotation (Ap)}}$$

- \* Due to the mechanical tolerance, the actual movement amount may differ slightly from the instructed movement amount. The error can be compensated by adjusting the movement amount per pulse defined here.  
(Refer to Section 12.3.2 "Electronic gear function".)

**POINT**

If the movement amount per pulse is less than 1, command frequency variations will occur. Smaller setting will increase variations and may cause machine vibration. If the movement amount per pulse becomes less than 1, also use the electronic gear function of the drive unit and make setting so that the movement amount per pulse is 1 or greater.

[Table 1]

| <b>Pr.1</b> setting value | Value set with peripheral device<br>(unit) | Value set with sequence program<br>(unit) * |
|---------------------------|--------------------------------------------|---------------------------------------------|
| 0 : mm                    | 0.1 to 6553.5 (μm)                         | 1 to 65535 (×10 <sup>-1</sup> μm)           |
| 1 : inch                  | 0.00001 to 0.65535 (inch)                  | 1 to 65535 (×10 <sup>-5</sup> inch)         |
| 2 : degree                | 0.00001 to 0.65535 (degree)                | 1 to 65535 (×10 <sup>-5</sup> degree)       |
| 3 : pulse                 | 1 to 65535 (pulse)                         | 1 to 65535 (pulse)                          |

- \* 1 to 32767 : Set as a decimal  
32768 to 65535 : Convert into hexadecimal and set

**Pr.2** No. of pulses per rotation (Ap)

Set the number of pulses required for a complete rotation of the motor shaft.

If you are using the Mitsubishi servo amplifier, set the value given as the "resolution per servomotor rotation" in the speed/position detector specifications.

No. of pulses per rotation (Ap) = Resolution per servomotor rotation

- \* When the "Resolution per servomotor revolution" of Mitsubishi servo amplifier exceeds 65535 pulses, make setting after referring to the Servo Amplifier Instruction Manual.

**Pr.3** Movement amount per rotation (Al), **Pr.4** Unit magnification (Am)

The amount how the workpiece moves with one motor rotation is determined by the mechanical structure.

If the worm gear lead (mm/rev) is PB and the deceleration rate is 1/n, then

$$\text{Movement amount per rotation (AL)} = \text{PB} \times 1/n$$

However, the maximum value that can be set for this "movement amount per rotation (AL)" parameter is 6553.5 $\mu$ m (approx. 6.5mm). Set the "movement amount per rotation (AL)" as shown below so that the "movement amount per rotation (AL)" does not exceed this maximum value.

$$\begin{aligned} \text{Movement amount per rotation (AL)} \\ &= \text{PB} \times 1/n \\ &= \text{Movement amount per rotation (AL)} \times \text{Unit magnification (Am)} \end{aligned}$$

Note) The unit magnification (Am) is a value of 1, 10, 100 or 1000. If the "PB  $\times$  1/n" value exceeds 6553.5 $\mu$ m, adjust with the unit magnification so that the "movement amount per rotation (AL)" does not exceed 6553.5 $\mu$ m.

#### Example 1)

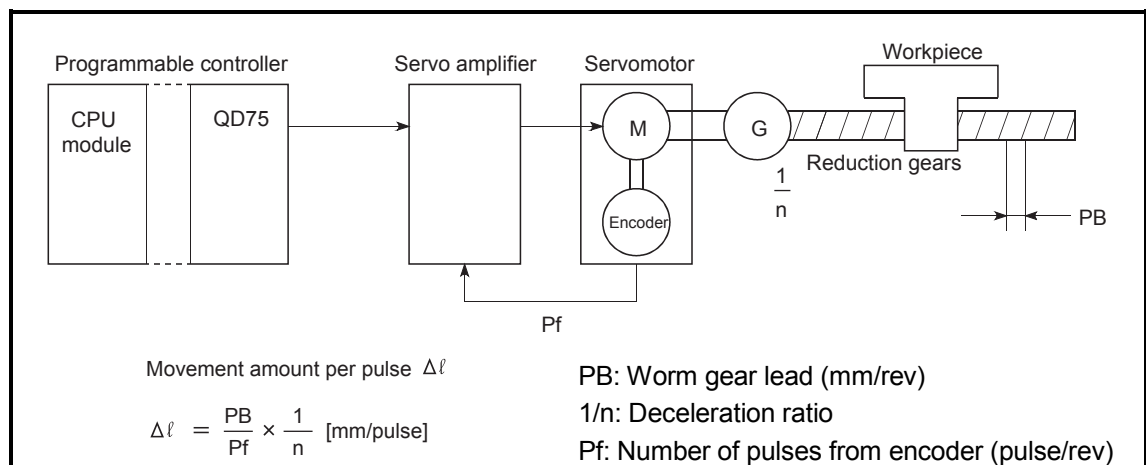
When movement amount per rotation (AL) = PB  $\times$  1/n = 6000.0 $\mu$ m (= 6mm)

$$\begin{aligned} \text{Movement amount per rotation (AL)} \\ &= \text{Movement amount per rotation (AL)} \times \text{Unit magnification (Am)} \\ &= 6000.0\mu\text{m} \times 1 \text{ time} \end{aligned}$$

#### Example 2)

When movement amount per rotation (AL) = PB  $\times$  1/n = 60000.0 $\mu$ m (= 60mm)

$$\begin{aligned} \text{Movement amount per rotation (AL)} \\ &= \text{Movement amount per rotation (AL)} \times \text{Unit magnification (Am)} \\ &= 6000.0\mu\text{m} \times 10 \text{ times} \end{aligned}$$



| Item                               | Setting value, setting range                              |                                 | Default value | Setting value buffer memory address |        |        |        |
|------------------------------------|-----------------------------------------------------------|---------------------------------|---------------|-------------------------------------|--------|--------|--------|
|                                    | Value set with peripheral device                          | Value set with sequence program |               | Axis 1                              | Axis 2 | Axis 3 | Axis 4 |
| Pr.5<br>Pulse output mode          | 0 : PULSE/SIGN mode                                       | 0                               | 1             | 4                                   | 154    | 304    | 454    |
|                                    | 1 : CW/CCW mode                                           | 1                               |               |                                     |        |        |        |
|                                    | 2 : A phase/B phase (multiple of 4)                       | 2                               |               |                                     |        |        |        |
|                                    | 3 : A phase/B phase (multiple of 1)                       | 3                               |               |                                     |        |        |        |
| Pr.6<br>Rotation direction setting | 0 : Current value increment with forward run pulse output | 0                               | 0             | 5                                   | 155    | 305    | 455    |
|                                    | 1 : Current value increment with reverse run pulse output | 1                               |               |                                     |        |        |        |



**Pr.5** Pulse output mode

Set the pulse output mode to match the servo amplifier being used.

**IMPORTANT**

The only valid value of the " **Pr.5** Pulse output mode" is the value at the moment when the PLC READY signal [Y0] turns from OFF to ON for the first time after the power is switched ON or the CPU module is reset. Once the PLC READY signal [Y0] has been turned ON, the value will not be reset even if another value is set to the parameter and the PLC READY signal [Y0] is turned from OFF to ON.

Use " **Pr.23** Output signal logic selection" to choose between the positive logic (pulse rising edge detection) and negative logic (pulse falling edge detection). For the output specifications of each pulse output mode, refer to Section 3.4.1, (2) "Output specifications".

An example of the pulse output mode for positive and negative logic is shown below.

**(1) PULSE/SIGN mode**

| Positive logic                                                                                                                                                                                                                                                                | Negative logic                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Forward run and reverse run are controlled with the ON/OFF of the direction sign (SIGN).<br><ul style="list-style-type: none"> <li>The motor will forward run when the direction sign is HIGH.</li> <li>The motor will reverse run when the direction sign is LOW.</li> </ul> | Forward run and reverse run are controlled with the ON/OFF of the direction sign (SIGN).<br><ul style="list-style-type: none"> <li>The motor will forward run when the direction sign is LOW.</li> <li>The motor will reverse run when the direction sign is HIGH.</li> </ul> |
|                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                               |

**(2) CW/CCW mode**

During forward run, the forward run feed pulse (PULSE F) will be output.

During reverse run, the reverse run feed pulse (PULSE R) will be output.

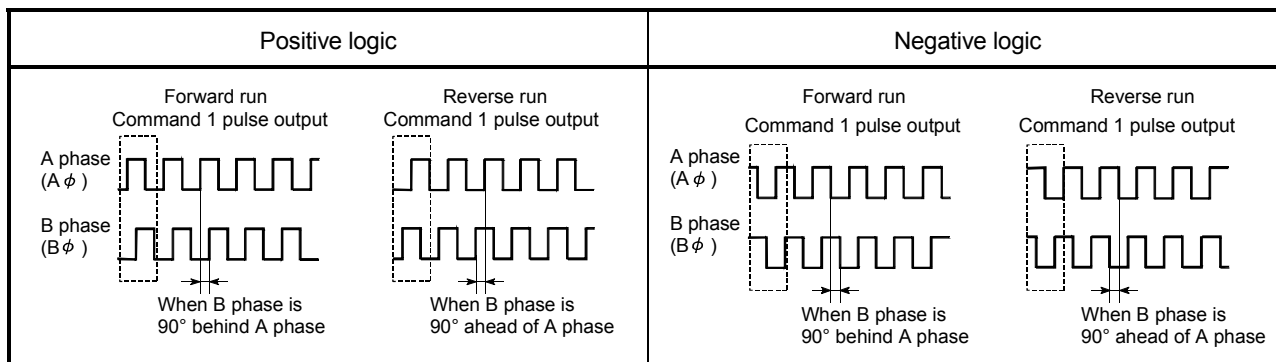
| Positive logic | Negative logic |
|----------------|----------------|
|                |                |

## (3) A phase/B phase mode

Forward run and reverse run are controlled with the phase difference of the A phase (A $\phi$ ) and B phase (B $\phi$ ).

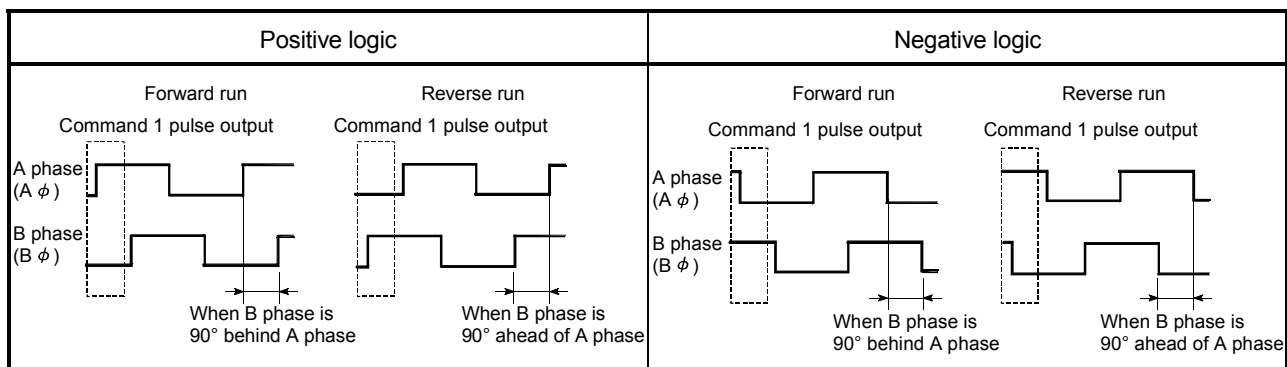
- When the B phase is 90° behind the A phase, the motor will forward run.
- When the B phase is 90° ahead of the A phase, the motor will reverse run.

## 1) For multiple of 1 setting



Example) When the command 1 pulse output is 1 pulse/s, the pulse rises and falls by four times per second.

## 2) For multiple of 4 setting



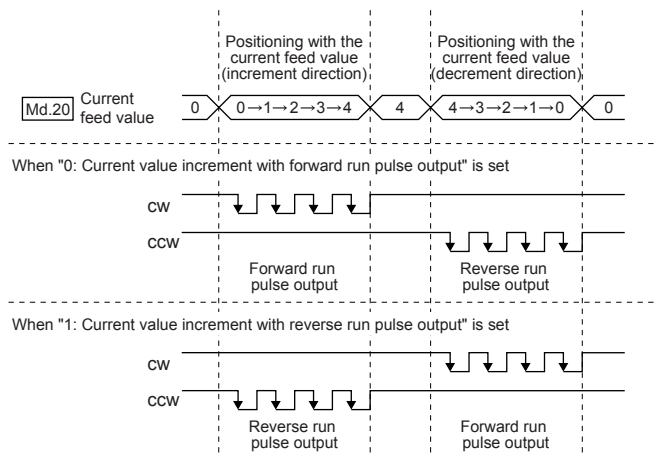
Example) When the command 1 pulse output is 1 pulse/s, the pulse rises and falls by one times per second.

**Pr.6** Rotation direction setting

Set the relation of the positioning direction (**Md.20** Current feed value increment direction/decrement direction) and the pulse output. For the relation of "Forward run pulse output, Reverse run pulse output" and "CW/A phase/PULSE signal, CCW/B phase/SIGN signal", refer to "**Pr.5** Pulse output mode".

**[Example]**

The following figure shows the pulse outputs of when "**Pr.5** Pulse output mode" is set to CW/CCW mode and the positioning with current feed value (increment direction)/(decrement direction) is executed.

**POINT**

When **Pr.6** has been changed from "0" to "1", check that the upper and lower limit switches operate properly by JOG operation.  
If any malfunction is identified, check and correct the wiring.

| Item                               | Setting value, setting range                                                                                                                                                                                                           |                                 | Default value | Setting value buffer memory address |            |            |            |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------|-------------------------------------|------------|------------|------------|
|                                    | Value set with peripheral device                                                                                                                                                                                                       | Value set with sequence program |               | Axis 1                              | Axis 2     | Axis 3     | Axis 4     |
| <b>Pr.7</b><br>Bias speed at start | The setting range differs depending on the " <b>Pr.1</b> Unit setting". Here, the value within the [Table 1] range is set.<br> [Table 1] on next page |                                 | 0             | 6<br>7                              | 156<br>157 | 306<br>307 | 456<br>457 |

**Pr.7 Bias speed at start**

Set the bias speed (minimum speed) upon starting. The bias speed has to be defined to allow the motor to start smoothly especially when a stepping motor is used. (A stepping motor will not start smoothly if a low rotation speed is instructed at the beginning.)

The specified "bias speed at start" will be valid during the following operations:

- Positioning operation
- OPR operation
- JOG operation

Note that the bias speed should not exceed " **Pr.8** Speed limit value".

**■ Precautions for using a stepping motor**

- (1) For the system that uses a stepping motor, executing the S-curve acceleration/deceleration may cause step-out.  
Before using the S-curve acceleration/deceleration, confirm that step-out does not occur.
- (2) In the system that uses a stepping motor, the circular interpolation control cannot be carried out.  
Although setting the bias speed at start is required by the characteristics of the stepping motor, the setting of the bias speed at start is disabled for circular interpolation control.  
Ensure to use a servomotor for both 2 axes when the circular interpolation control is carried out.

**5.2.2 Basic parameters 2**

| Item                                | Setting value, setting range                                                                                                                                                                                                             |                                 | Default value | Setting value buffer memory address |            |            |            |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------|-------------------------------------|------------|------------|------------|
|                                     | Value set with peripheral device                                                                                                                                                                                                         | Value set with sequence program |               | Axis 1                              | Axis 2     | Axis 3     | Axis 4     |
| <b>Pr.8</b><br>Speed limit value    | The setting range differs depending on the " <b>Pr.1</b> Unit setting". Here, the value within the [Table 2] range is set.<br> [Table 2] on next page |                                 | 200000        | 10<br>11                            | 160<br>161 | 310<br>311 | 460<br>461 |
| <b>Pr.9</b><br>Acceleration time 0  | 1 to 8388608 (ms)                                                                                                                                                                                                                        | 1 to 8388608 (ms)               | 1000          | 12<br>13                            | 162<br>163 | 312<br>313 | 462<br>463 |
| <b>Pr.10</b><br>Deceleration time 0 | 1 to 8388608 (ms)                                                                                                                                                                                                                        | 1 to 8388608 (ms)               | 1000          | 14<br>15                            | 164<br>165 | 314<br>315 | 464<br>465 |

[Table 1]

| Pr.1 setting value | Value set with peripheral device (unit) | Value set with sequence program (unit)                                                   |
|--------------------|-----------------------------------------|------------------------------------------------------------------------------------------|
| 0 : mm             | 0 to 20000000.00 (mm/min)               | 0 to 2000000000 ( $\times 10^{-2}$ mm/min)                                               |
| 1 : inch           | 0 to 2000000.000 (inch/min)             | 0 to 2000000000 ( $\times 10^{-3}$ inch/min)                                             |
| 2 : degree         | 0 to 2000000.000 (degree/min)           | 0 to 2000000000 ( $\times 10^{-3}$ degree/min)                                           |
| 3 : pulse          | 0 to 1000000 (pulse/s) * 1              | [QD75P□N/QD75D□N]<br>0 to 4000000 (pulse/s)<br>[QD75P□/QD75D□]<br>0 to 1000000 (pulse/s) |

[Table 2]

| Pr.1 setting value | Value set with peripheral device (unit)                                                                               | Value set with sequence program (unit)                                                   |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 0 : mm             | 0.01 to 20000000.00 (mm/min)                                                                                          | 1 to 2000000000 ( $\times 10^{-2}$ mm/min)                                               |
| 1 : inch           | 0.001 to 2000000.000 (inch/min)                                                                                       | 1 to 2000000000 ( $\times 10^{-3}$ inch/min)                                             |
| 2 : degree         | 0.001 to 2000000.000 (degree/min)                                                                                     | 1 to 2000000000 ( $\times 10^{-3}$ degree/min)                                           |
| 3 : pulse          | [Select type * 2 is QD75P□]<br>1 to 200000 (pulse/s) * 1<br>[Select type * 2 is QD75D□]<br>1 to 1000000 (pulse/s) * 1 | [QD75P□N/QD75D□N]<br>1 to 4000000 (pulse/s)<br>[QD75P□/QD75D□]<br>1 to 1000000 (pulse/s) |

\* 1: In GX Configurator-QP, the QD75P□N/QD75D□N are recognized as the QD75P□/QD75D□. Therefore, the setting ranges are the same as those of the QD75P□/QD75D□. To set a value outside a setting range in GX Configurator-QP, set it through a sequence program or GX Works2. The setting ranges in GX Works2 are the same as those in sequence programs. (Refer to Appendix 1.2)

\* 2: For Select type, refer to GX Configurator-QP Operating Manual.

### Pr.8 Speed limit value

Set the maximum speed during positioning and OPR operations. If the specified speed exceeds speed limit value, positioning will be limited at the speed limit value. Set the speed limit value within the range of the following formula. If the speed limit value exceeds the range, the error "Out of speed limit value range" (error code: 910) will occur.

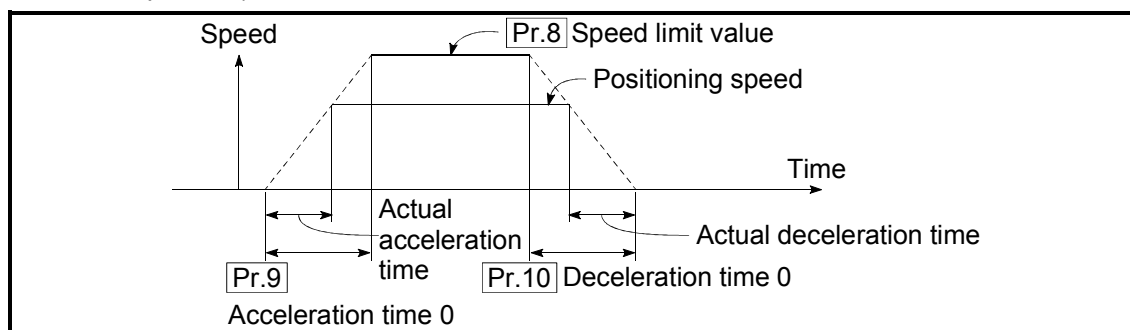
The command pulse frequency  
converted from the speed limit value  $\leq$  Max. output pulse \* 1

\* 1: QD75P□N/QD75P□: 200kpulse/s  
QD75D□N: 4Mpulse/s  
QD75D□: 1Mpulse/s

### Pr.9 Acceleration time 0, Pr.10 Deceleration time 0

" Pr.9 Acceleration time 0" specifies the time for the speed to increase from zero to " Pr.8 Speed limit value" (or " Pr.31 JOG speed limit value" during JOG operation).

" Pr.10 Deceleration time 0" specifies the time for the speed to decrease from " Pr.8 Speed limit value" (or " Pr.31 JOG speed limit value" during JOG operation) to zero.



- 1) If the positioning speed is set lower than the parameter-defined speed limit value, the actual acceleration/deceleration time will be relatively short. Thus, set the maximum positioning speed equal to or only a little lower than the parameter-defined speed limit value.
- 2) These settings are valid for OPR, positioning and JOG operations.
- 3) When the positioning involves interpolation, the acceleration/deceleration time defined for the reference axis is valid.

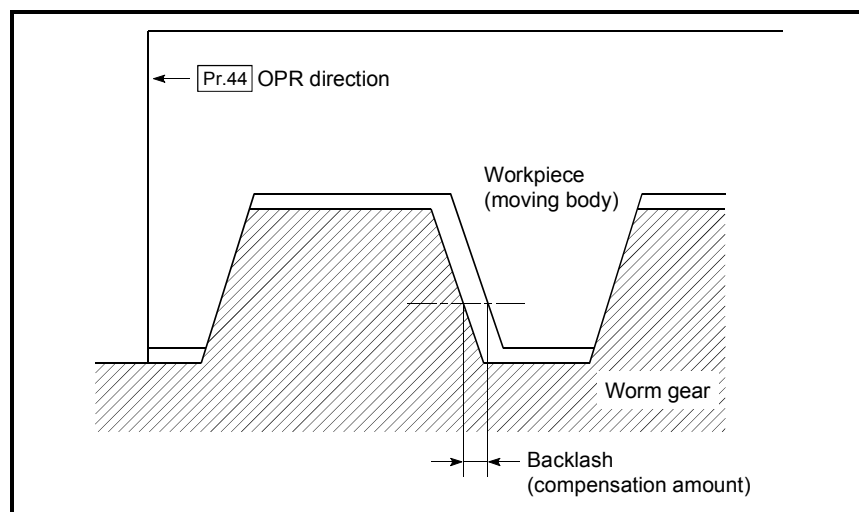
## 5.2.3 Detailed parameters 1

| Item                                                        | Setting value, setting range                                                                                                                                                                                                                    |                                 | Default value | Setting value buffer memory address |            |            |            |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------|-------------------------------------|------------|------------|------------|
|                                                             | Value set with peripheral device                                                                                                                                                                                                                | Value set with sequence program |               | Axis 1                              | Axis 2     | Axis 3     | Axis 4     |
| <b>Pr.11</b><br>Backlash compensation amount                | The setting value range differs according to the " <b>Pr.1</b> Unit setting".<br>Here, the value within the [Table 1] range is set.<br> [Table 1] on next page |                                 | 0             | 17                                  | 167        | 317        | 467        |
| <b>Pr.12</b><br>Software stroke limit upper limit value     | The setting value range differs according to the " <b>Pr.1</b> Unit setting".<br>Here, the value within the [Table 2] range is set.                                                                                                             |                                 | 2147483647    | 18<br>19                            | 168<br>169 | 318<br>319 | 468<br>469 |
| <b>Pr.13</b><br>Software stroke limit lower limit value     |  [Table 2] on next page                                                                                                                                        |                                 | -2147483648   | 20<br>21                            | 170<br>171 | 320<br>321 | 470<br>471 |
| <b>Pr.14</b><br>Software stroke limit selection             | 0 : Apply software stroke limit on current feed value                                                                                                                                                                                           | 0                               | 0             | 22                                  | 172        | 322        | 472        |
|                                                             | 1 : Apply software stroke limit on machine feed value                                                                                                                                                                                           | 1                               |               |                                     |            |            |            |
| <b>Pr.15</b><br>Software stroke limit valid/invalid setting | 0 : Software stroke limit valid during JOG operation, inching operation and manual pulse generator operation                                                                                                                                    | 0                               | 0             | 23                                  | 173        | 323        | 473        |
|                                                             | 1 : Software stroke limit invalid during JOG operation, inching operation and manual pulse generator operation                                                                                                                                  | 1                               |               |                                     |            |            |            |

**Pr.11** Backlash compensation amount

The error that occurs due to backlash when moving the machine via gears can be compensated.

When the backlash compensation amount is set, pulses equivalent to the compensation amount will be output each time the direction changes during positioning.



- 1) The backlash compensation is valid after machine OPR. Thus, if the backlash compensation amount is set or changed, always carry out machine OPR once.
- 2) The backlash compensation amount setting range is 0 to 65535, but it should be set to 255 or less by using the following expression.

$$0 \leq \frac{\text{Backlash compensation amount}}{\text{Movement amount per pulse}} \leq 255$$

[Table 1]

| Pr.1<br>setting value | Value set with peripheral device<br>(unit) | Value set with sequence program<br>(unit) * |
|-----------------------|--------------------------------------------|---------------------------------------------|
| 0 : mm                | 0 to 6553.5 (μm)                           | 0 to 65535 (×10 <sup>-1</sup> μm)           |
| 1 : inch              | 0 to 0.65535 (inch)                        | 0 to 65535 (×10 <sup>-5</sup> inch)         |
| 2 : degree            | 0 to 0.65535 (degree)                      | 0 to 65535 (×10 <sup>-5</sup> degree)       |
| 3 : pulse             | 0 to 65535 (pulse)                         | 0 to 65535 (pulse)                          |

\* 1 to 32767 : Set as a decimal

32768 to 65535 : Convert into hexadecimal and set

[Table 2]

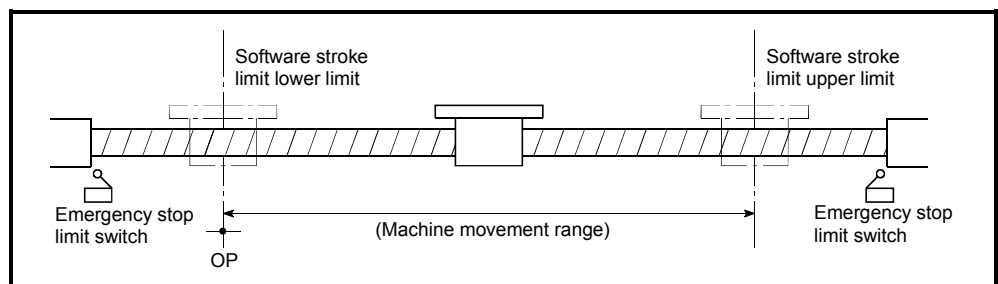
| Pr.1<br>setting value | Value set with peripheral device<br>(unit) | Value set with sequence program<br>(unit)          |
|-----------------------|--------------------------------------------|----------------------------------------------------|
| 0 : mm                | -214748364.8 to 214748364.7 (μm)           | -2147483648 to 2147483647 (×10 <sup>-1</sup> μm)   |
| 1 : inch              | -21474.83648 to 21474.83647 (inch)         | -2147483648 to 2147483647 (×10 <sup>-5</sup> inch) |
| 2 : degree            | 0 to 359.99999 (degree)                    | 0 to 35999999 (×10 <sup>-5</sup> degree)           |
| 3 : pulse             | -2147483648 to 2147483647 (pulse)          | -2147483648 to 2147483647 (pulse)                  |

#### Pr.12 Software stroke limit upper limit value

Set the upper limit for the machine's movement range during positioning control.

#### Pr.13 Software stroke limit lower limit value

Set the lower limit for the machine's movement range during positioning control.



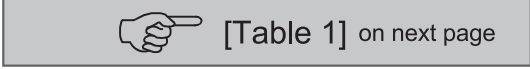
- 1) Generally, the OP is set at the lower limit or upper limit of the stroke limit.
- 2) By setting the upper limit value or lower limit value of the software stroke limit, overrun can be prevented in the software. However, an emergency stop limit switch must be installed nearby outside the range.
- 3) To invalidate the software stroke limit, set the setting value to "upper limit value = lower limit value". (The setting value can be anything in the setting range.)
- 4) When the unit is "degree", the software stroke limit check is invalid during speed control (including speed-position switching control, position-speed switching control) or during manual control.

#### Pr.14 Software stroke limit selection

Set whether to apply the software stroke limit on the "current feed value" or the "machine feed value". The software stroke limit will be validated according to the set value.

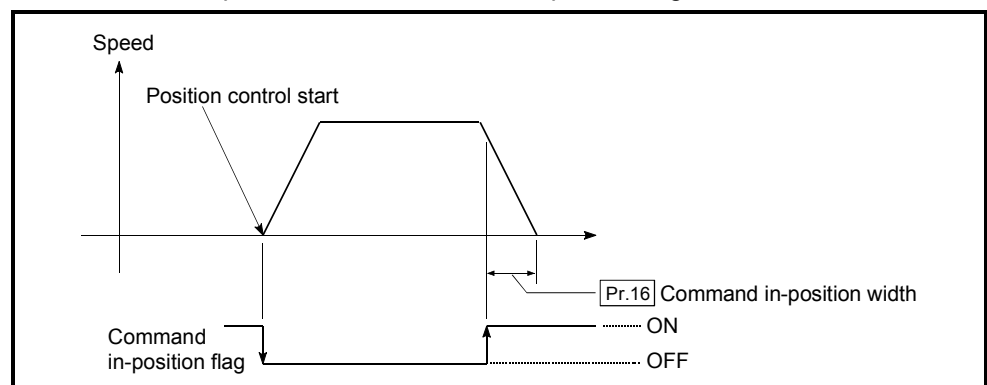
#### Pr.15 Software stroke limit valid/invalid setting

Set whether to validate the software stroke limit during JOG/Inching operation and manual pulse generator operation.

| Item                                    | Setting value, setting range                                                                                                                                                                                         |                                 | Default value | Setting value buffer memory address |            |            |            |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------|-------------------------------------|------------|------------|------------|
|                                         | Value set with peripheral device                                                                                                                                                                                     | Value set with sequence program |               | Axis 1                              | Axis 2     | Axis 3     | Axis 4     |
| Pr.16<br>Command in-position width      | The setting value range differs depending on the " Pr.1 Unit setting".<br>Here, the value within the [Table 1] range is set.<br> |                                 | 100           | 24<br>25                            | 174<br>175 | 324<br>325 | 474<br>475 |
| Pr.17<br>Torque limit setting value     | 1 to 500 (%)                                                                                                                                                                                                         | 1 to 500 (%)                    | 300           | 26                                  | 176        | 326        | 476        |
| Pr.18<br>M code ON signal output timing | 0 : WITH mode                                                                                                                                                                                                        | 0                               | 0             | 27                                  | 177        | 327        | 477        |
|                                         | 1 : AFTER mode                                                                                                                                                                                                       | 1                               |               |                                     |            |            |            |

#### Pr.16 Command in-position width

Set the remaining distance that turns the command in-position ON. The command in-position signal is used as a front-loading signal of the positioning complete signal. When positioning control is started, the "command in-position flag" (3rd flag from right) in " Md.31 Status" turns OFF, and the "command in-position flag" turns ON at the set position of the command in-position signal.





**Pr.17** Torque limit setting value

Set the maximum value of the torque generated by the servomotor as a percentage between 1 and 500%.

\* The torque limit function limits the torque generated by the servomotor within the set range.

If the torque required for control exceeds the torque limit value, it is controlled with the set torque limit value.

**Usage conditions**

Limits for pulse train output type

- (a) A drive unit that can issue a torque limit command with the analog voltage is required.
- (b) The D/A conversion module and the D/A conversion module and drive unit must be wired.
- (c) The set " **Pr.17** Torque limit setting value" is set in the buffer memory " **Md.35** Torque limit stored value", so transmit that " **Md.35** Torque limit stored value" to the D/A conversion module with the sequence program.

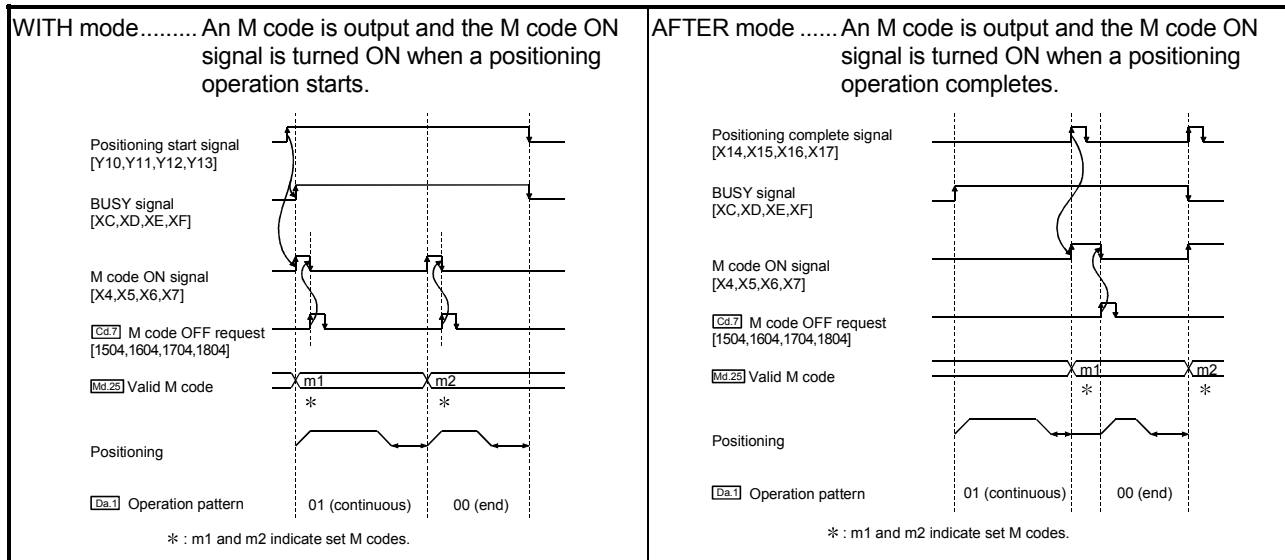
[Table 1]

| <b>Pr.1</b> setting value | Value set with peripheral device<br>(unit) | Value set with sequence program<br>(unit)         |
|---------------------------|--------------------------------------------|---------------------------------------------------|
| 0 : mm                    | 0.1 to 214748364.7 ( $\mu\text{m}$ )       | 1 to 2147483647 ( $\times 10^{-1}\mu\text{m}$ )   |
| 1 : inch                  | 0.00001 to 21474.83647 (inch)              | 1 to 2147483647 ( $\times 10^{-5}\text{inch}$ )   |
| 2 : degree                | 0.00001 to 21474.83647 (degree)            | 1 to 2147483647 ( $\times 10^{-5}\text{degree}$ ) |
| 3 : pulse                 | 1 to 2147483647 (pulse)                    | 1 to 2147483647 (pulse)                           |

**Pr.18** M code ON signal output timing

This parameter sets the M code ON signal output timing.

Choose either WITH mode or AFTER mode as the M code ON signal output timing.



Note: If AFTER mode is used with speed control, an M code will not be output and the M code ON signal will not be turned ON.

An M code ([Da.10]) is a number between 0 and 65535 that can be assigned to each positioning data.

The sequence program can be coded to read an M code from the buffer memory address specified by "[Md.25] Valid M code" whenever the M code ON signal [X4, X5, X6, X7] turns ON so that a command for the sub work (e.g. clamping, drilling, tool change) associated with the M code can be issued.

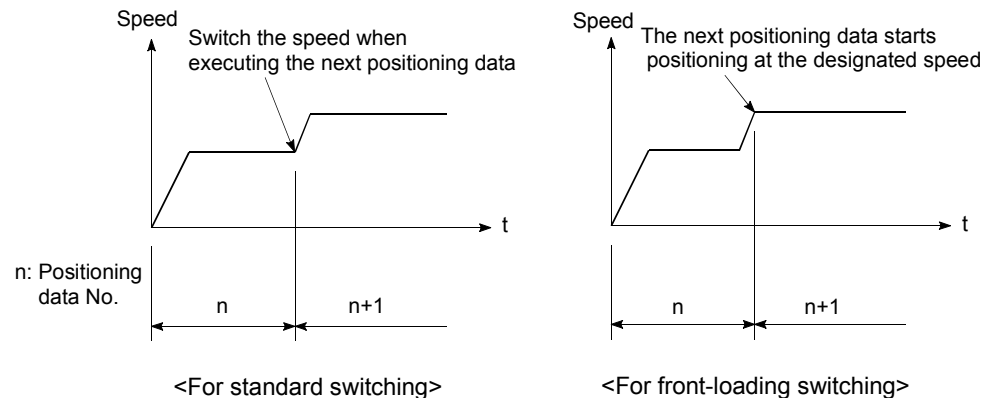
The M code ON signal output timing can be set for each positioning data with the positioning option ("[Da.27] M code ON signal output timing") of the positioning data.

| Item                                                     | Setting value, setting range                                                                                                                                                                                           |                                        | Default value | Setting value buffer memory address |        |        |        |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|---------------|-------------------------------------|--------|--------|--------|
|                                                          | Value set with peripheral device                                                                                                                                                                                       | Value set with sequence program        |               | Axis 1                              | Axis 2 | Axis 3 | Axis 4 |
| <b>Pr.19</b><br>Speed switching mode                     | 0 : Standard speed switching mode                                                                                                                                                                                      | 0                                      | 0             | 28                                  | 178    | 328    | 478    |
|                                                          | 1 : Front-loading speed switching mode                                                                                                                                                                                 | 1                                      |               |                                     |        |        |        |
| <b>Pr.20</b><br>Interpolation speed designation method   | 0 : Composite speed                                                                                                                                                                                                    | 0                                      | 0             | 29                                  | 179    | 329    | 479    |
|                                                          | 1 : Reference axis speed                                                                                                                                                                                               | 1                                      |               |                                     |        |        |        |
| <b>Pr.21</b><br>Current feed value during speed control  | 0 : Do not update current feed value                                                                                                                                                                                   | 0                                      | 0             | 30                                  | 180    | 330    | 480    |
|                                                          | 1 : Update current feed value                                                                                                                                                                                          | 1                                      |               |                                     |        |        |        |
|                                                          | 2 : Clear current feed value to zero                                                                                                                                                                                   | 2                                      |               |                                     |        |        |        |
| <b>Pr.22</b><br>Input signal logic selection             | b0 Lower limit<br>b1 Upper limit<br>b2 Drive unit READY<br>b3 Stop signal<br>b4 External command<br>b5 Zero signal<br>b6 Near-point dog signal<br>b7 Not used<br>b8 Manual pulse generator input<br>b9 to b15 Not used | 0: Negative logic<br>1: Positive logic | 0             | 31                                  | 181    | 331    | 481    |
| <b>Pr.23</b><br>Output signal logic selection            | b0 Command pulse signal<br>b1 Not used<br>b2 Not used<br>b3 Not used<br>b4 Deviation counter clear<br>b5 to b15 Not used                                                                                               | 0: Negative logic<br>1: Positive logic | 0             | 32                                  | 182    | 332    | 482    |
| <b>Pr.24</b><br>Manual pulse generator input selection   | 0: A-phase/B-phase multiplied by 4<br>1: A-phase/B-phase multiplied by 2<br>2: A-phase/B-phase multiplied by 1<br>3: PULSE/SIGN                                                                                        | 0<br>1<br>2<br>3                       | 0             | 33                                  | —      | —      | —      |
| <b>Pr.150</b><br>Speed-position function selection       | 0: Speed-position switching control (INC mode)<br>2: Speed-position switching control (ABS mode)                                                                                                                       | 0<br>2                                 | 0             | 34                                  | 184    | 334    | 484    |
| <b>Pr.70</b><br>Positioning option valid/invalid setting | 0: Invalid<br>1: Valid                                                                                                                                                                                                 | 0<br>1                                 | 0             | 140                                 | —      | —      | —      |

**Pr.19** Speed switching mode

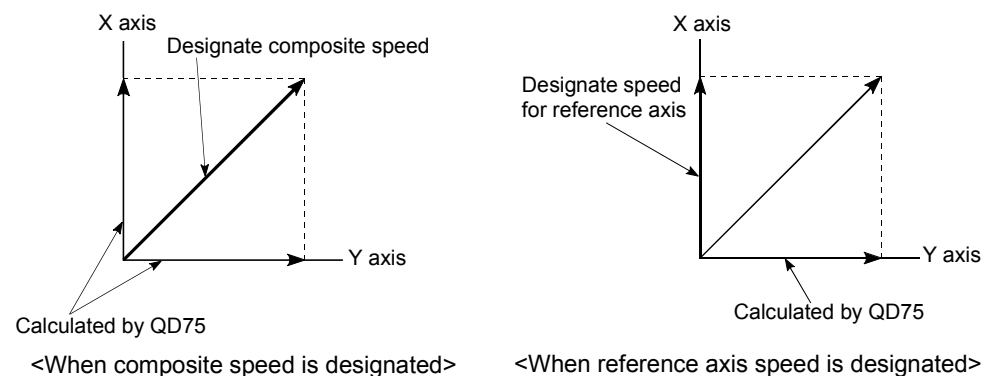
Set whether to switch the speed switching mode with the standard switching or front-loading switching mode.

- 0 : Standard switching..... Switch the speed when executing the next positioning data.  
 1 : Front-loading switching..... The speed switches at the end of the positioning data currently being executed.

**Pr.20** Interpolation speed designation method

When carrying out linear interpolation/circular interpolation, set whether to designate the composite speed or reference axis speed.

- 0: Composite speed ..... The movement speed for the control target is designated, and the speed for each axis is calculated by the QD75.  
 1: Reference axis speed ..... The axis speed set for the reference axis is designated, and the speed for the other axis carrying out interpolation is calculated by the QD75.



**Note:** Always specify the reference axis speed if the 4-axis linear interpolation or 2 to 4 axis speed control has to be performed.

If you specify the composite speed for a positioning operation that involves the 4-axis linear interpolation or 2 to 4 axis speed control, the error "Interpolation mode error" (error code 523) will be output when the positioning operation is attempted.

For a positioning operation that involves the circular interpolation or 3-axis helical interpolation control, specify the composite speed.

The interpolation speed designation method can be set for each positioning data with the positioning option ("Da.29 Interpolation speed designation method") of the positioning data.

**Pr.21** Current feed value during speed control

Specify whether you wish to enable or disable the update of " **Md.20** Current feed value" while operations are performed under the speed control (including the speed-position and position-speed switching control).

0: The update of the current feed value is disabled

The current feed value will not change.

(The value at the beginning of the speed control will be kept.)

1: The update of the current feed value is enabled

The current feed value will be updated.

(The current feed value will change from the initial.)

2: The current feed value is cleared to zero

The current feed will be set initially to zero and change from zero while the speed control is in effect.

Note1: When the speed control is performed over two to four axes, the choice between enabling and disabling the update of " **Md.20** Current feed value" depends on how the reference axis is set.

Note2: Set "1" to exercise speed-position switching control (ABS mode).

**Pr.22** Input signal logic selection, **Pr.23** Output signal logic selection

Set the I/O signal logic that matches the signaling specification of the connected external device.

Note1: A mismatch in the signal logic will disable normal operation. Be careful of this when you change from the default value.

Note2: Set the manual pulse generator input logic selection (b8) to axis 1. (Setting of any of axes 2 to 4 is invalid.)

**Pr.24** Manual pulse generator input selection

Set the manual pulse generator input pulse mode. (Only the value specified against the axis 1 is valid.)

0: A-phase/B-phase; multiplied by 4

1: A-phase/B-phase; multiplied by 2

2: A-phase/B-phase; multiplied by 1

3: PULSE/SIGN

**Pr.150** Speed-position function selection

Select the mode of speed-position switching control.

0: INC mode

2: ABS mode

Note1: If the setting is other than 0 and 2, operation is performed in the INC mode with the setting regarded as 0.

**Pr.70** Positioning option valid/invalid setting

Enable or disable the data set in the positioning option ("Da.27 M code ON signal output timing", "Da.28 ABS direction in degrees", and

"Da.29 Interpolation speed designation method") of the positioning data.

Setting this parameter to "1: Valid" carries out a positioning control according to the data set in the positioning option.

0: Invalid

1: Valid

Note 1: If a value other than 0 and 1 is set, the set value is regarded as 0 and the data is disabled.

Note 2: If "0: Invalid" is set, the setting is ignored even though the positioning option of the positioning data is set.

Note 3: Pr.70 Positioning option valid/invalid setting can be set only with GX Works2 (Version 1.540N or later) or sequence programs.

# MEMO

[illegible]

## 5.2.4 Detailed parameters 2

| Item                                                          | Setting value, setting range                                                                                                   |                                 | Default value | Setting value buffer memory address |        |        |        |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------|-------------------------------------|--------|--------|--------|
|                                                               | Value set with peripheral device                                                                                               | Value set with sequence program |               | Axis 1                              | Axis 2 | Axis 3 | Axis 4 |
| <div>Pr.25</div> Acceleration time 1                          | 1 to 8388608 (ms)                                                                                                              | 1 to 8388608 (ms)               | 1000          | 36                                  | 186    | 336    | 486    |
| <div>Pr.26</div> Acceleration time 2                          |                                                                                                                                |                                 |               | 37                                  | 187    | 337    | 487    |
| <div>Pr.27</div> Acceleration time 3                          |                                                                                                                                |                                 |               | 38                                  | 188    | 338    | 488    |
| <div>Pr.28</div> Deceleration time 1                          |                                                                                                                                |                                 |               | 39                                  | 189    | 339    | 489    |
| <div>Pr.29</div> Deceleration time 2                          |                                                                                                                                |                                 |               | 40                                  | 190    | 340    | 490    |
| <div>Pr.30</div> Deceleration time 3                          |                                                                                                                                |                                 |               | 41                                  | 191    | 341    | 491    |
|                                                               |                                                                                                                                |                                 |               | 42                                  | 192    | 342    | 492    |
|                                                               |                                                                                                                                |                                 |               | 43                                  | 193    | 343    | 493    |
|                                                               |                                                                                                                                |                                 |               | 44                                  | 194    | 344    | 494    |
|                                                               |                                                                                                                                |                                 |               | 45                                  | 195    | 345    | 495    |
|                                                               |                                                                                                                                |                                 |               | 46                                  | 196    | 346    | 496    |
|                                                               |                                                                                                                                |                                 |               | 47                                  | 197    | 347    | 497    |
| <div>Pr.31</div><br>JOG speed limit value                     | The setting range differs depending on the " <div>Pr.1</div> Unit setting". Here, the value within the [Table 1] range is set. |                                 | 20000         | 48                                  | 198    | 348    | 498    |
|                                                               | <div></div>                                  |                                 |               | 49                                  | 199    | 349    | 499    |
| <div>Pr.32</div><br>JOG operation acceleration time selection | 0: <div>Pr.9</div> Acceleration time 0                                                                                         | 0                               | 0             | 50                                  | 200    | 350    | 500    |
|                                                               | 1: <div>Pr.25</div> Acceleration time 1                                                                                        | 1                               |               |                                     |        |        |        |
|                                                               | 2: <div>Pr.26</div> Acceleration time 2                                                                                        | 2                               |               |                                     |        |        |        |
|                                                               | 3: <div>Pr.27</div> Acceleration time 3                                                                                        | 3                               |               |                                     |        |        |        |
| <div>Pr.33</div><br>JOG operation deceleration time selection | 0: <div>Pr.10</div> Deceleration time 0                                                                                        | 0                               | 0             | 51                                  | 201    | 351    | 501    |
|                                                               | 1: <div>Pr.28</div> Deceleration time 1                                                                                        | 1                               |               |                                     |        |        |        |
|                                                               | 2: <div>Pr.29</div> Deceleration time 2                                                                                        | 2                               |               |                                     |        |        |        |
|                                                               | 3: <div>Pr.30</div> Deceleration time 3                                                                                        | 3                               |               |                                     |        |        |        |

**Pr.25** Acceleration time 1 to **Pr.27** Acceleration time 3

These parameters set the time for the speed to increase from zero to " **Pr.8** Speed limit value" (or " **Pr.31** JOG speed limit value" during JOG operation) during a positioning operation.

**Pr.28** Deceleration time 1 to **Pr.30** Deceleration time 3

These parameters set the time for the speed to decrease from " **Pr.8** Speed limit value" (or " **Pr.31** JOG speed limit value" during JOG operation) to zero during a positioning operation.



[Table 1]

| Pr.1 setting value | Value set with peripheral device (unit)                                                                               | Value set with sequence program (unit)                                                   |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 0 : mm             | 0.01 to 20000000.00 (mm/min)                                                                                          | 1 to 2000000000 ( $\times 10^{-2}$ mm/min)                                               |
| 1 : inch           | 0.001 to 2000000.000 (inch/min)                                                                                       | 1 to 2000000000 ( $\times 10^{-3}$ inch/min)                                             |
| 2 : degree         | 0.001 to 2000000.000 (degree/min)                                                                                     | 1 to 2000000000 ( $\times 10^{-3}$ degree/min)                                           |
| 3 : pulse          | [Select type * 1 is QD75P□]<br>1 to 200000 (pulse/s) * 2<br>[Select type * 1 is QD75D□]<br>1 to 1000000 (pulse/s) * 2 | [QD75P□N/QD75D□N]<br>1 to 4000000 (pulse/s)<br>[QD75P□/QD75D□]<br>1 to 1000000 (pulse/s) |

\* 1: For Select type, refer to GX Configurator-QP Operating Manual.

\* 2: In GX Configurator-QP, the QD75P□N/QD75D□N are recognized as the QD75P□/QD75D□. Therefore, the setting ranges are the same as those of the QD75P□/QD75D□. To set a value outside a setting range in GX Configurator-QP, set it through a sequence program or GX Works2. The setting ranges in GX Works2 are the same as those in sequence programs. (Refer to Appendix 1.2)

### Pr.31 JOG speed limit value

Set the maximum speed for JOG operation.

Note) • Set the "JOG speed limit value" to less than " Pr.8 Speed limit value".  
If the "speed limit value" is exceeded, the error "JOG speed limit value error" (error code: 956) will occur.

### Pr.32 JOG operation acceleration time selection

Set which of "acceleration time 0 to 3" to use for the acceleration time during JOG operation.

- 0 : Use value set in " Pr.9 Acceleration time 0".
- 1 : Use value set in " Pr.25 Acceleration time 1".
- 2 : Use value set in " Pr.26 Acceleration time 2".
- 3 : Use value set in " Pr.27 Acceleration time 3".

### Pr.33 JOG operation deceleration time selection

Set which of "deceleration time 0 to 3" to use for the deceleration time during JOG operation.

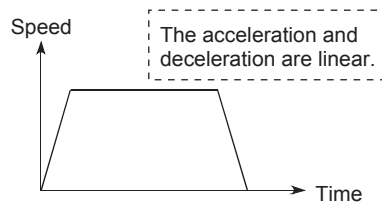
- 0 : Use value set in " Pr.10 Deceleration time 0".
- 1 : Use value set in " Pr.28 Deceleration time 1".
- 2 : Use value set in " Pr.29 Deceleration time 2".
- 3 : Use value set in " Pr.30 Deceleration time 3".

| Item                                                        | Setting value, setting range                                                                     |                                 | Default value | Setting value buffer memory address |            |            |            |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------|---------------|-------------------------------------|------------|------------|------------|
|                                                             | Value set with peripheral device                                                                 | Value set with sequence program |               | Axis 1                              | Axis 2     | Axis 3     | Axis 4     |
| <b>Pr.34</b><br>Acceleration/deceleration process selection | 0 : Trapezoid acceleration/deceleration process<br>1 : S-curve acceleration/deceleration process | 0<br>1                          | 0             | 52                                  | 202        | 352        | 502        |
| <b>Pr.35</b><br>S-curve ratio                               | 1 to 100 (%)                                                                                     | 1 to 100 (%)                    | 100           | 53                                  | 203        | 353        | 503        |
| <b>Pr.36</b><br>Sudden stop deceleration time               | 1 to 8388608 (ms)                                                                                | 1 to 8388608 (ms)               | 1000          | 54<br>55                            | 204<br>205 | 354<br>355 | 504<br>505 |
| <b>Pr.37</b><br>Stop group 1 sudden stop selection          | 0 : Normal deceleration stop                                                                     | 0                               | 0             | 56                                  | 206        | 356        | 506        |
| <b>Pr.38</b><br>Stop group 2 sudden stop selection          |                                                                                                  |                                 |               | 57                                  | 207        | 357        | 507        |
| <b>Pr.39</b><br>Stop group 3 sudden stop selection          | 1 : Sudden stop                                                                                  | 1                               |               | 58                                  | 208        | 358        | 508        |

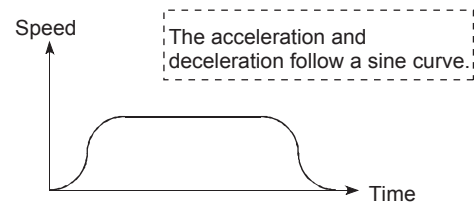
**Pr.34** Acceleration/deceleration process selection

Set whether to use trapezoid acceleration/deceleration or S-curve acceleration/deceleration for the acceleration/deceleration process.

Note) Refer to Section 12.7.6 "Acceleration/deceleration process function" for details.



&lt;Trapezoid acceleration/deceleration&gt;



&lt;S-curve acceleration/deceleration&gt;

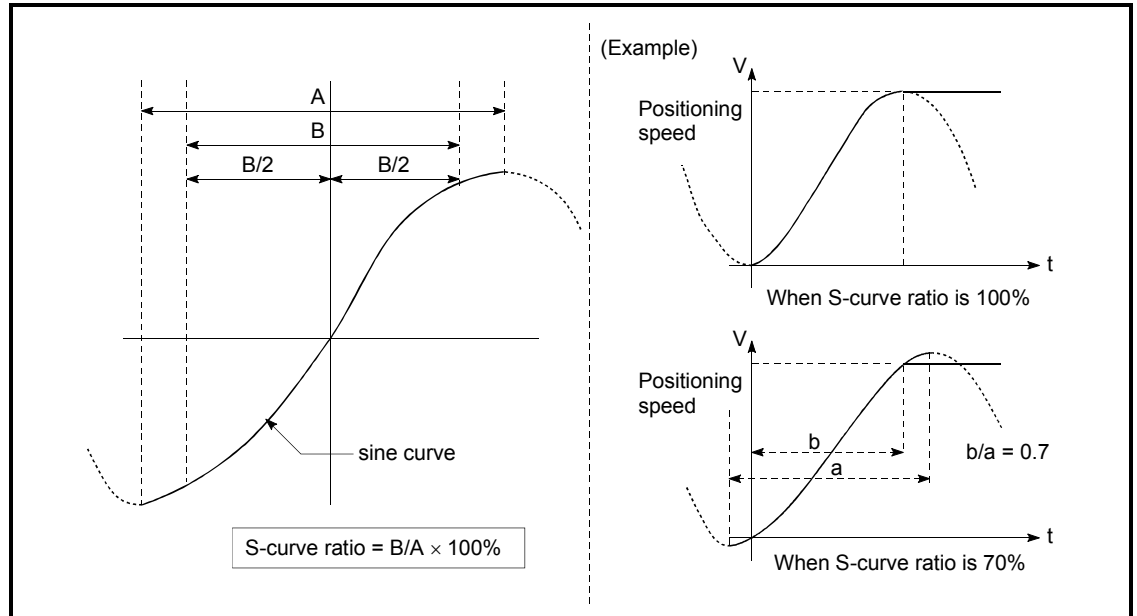
For the system that uses a stepping motor, executing the S-curve acceleration/deceleration may cause step-out.

Before using the S-curve acceleration/deceleration, confirm that step-out does not occur.

**Pr.35** S-curve ratio

Set the S-curve ratio (1 to 100%) for carrying out the S-curve acceleration/deceleration process.

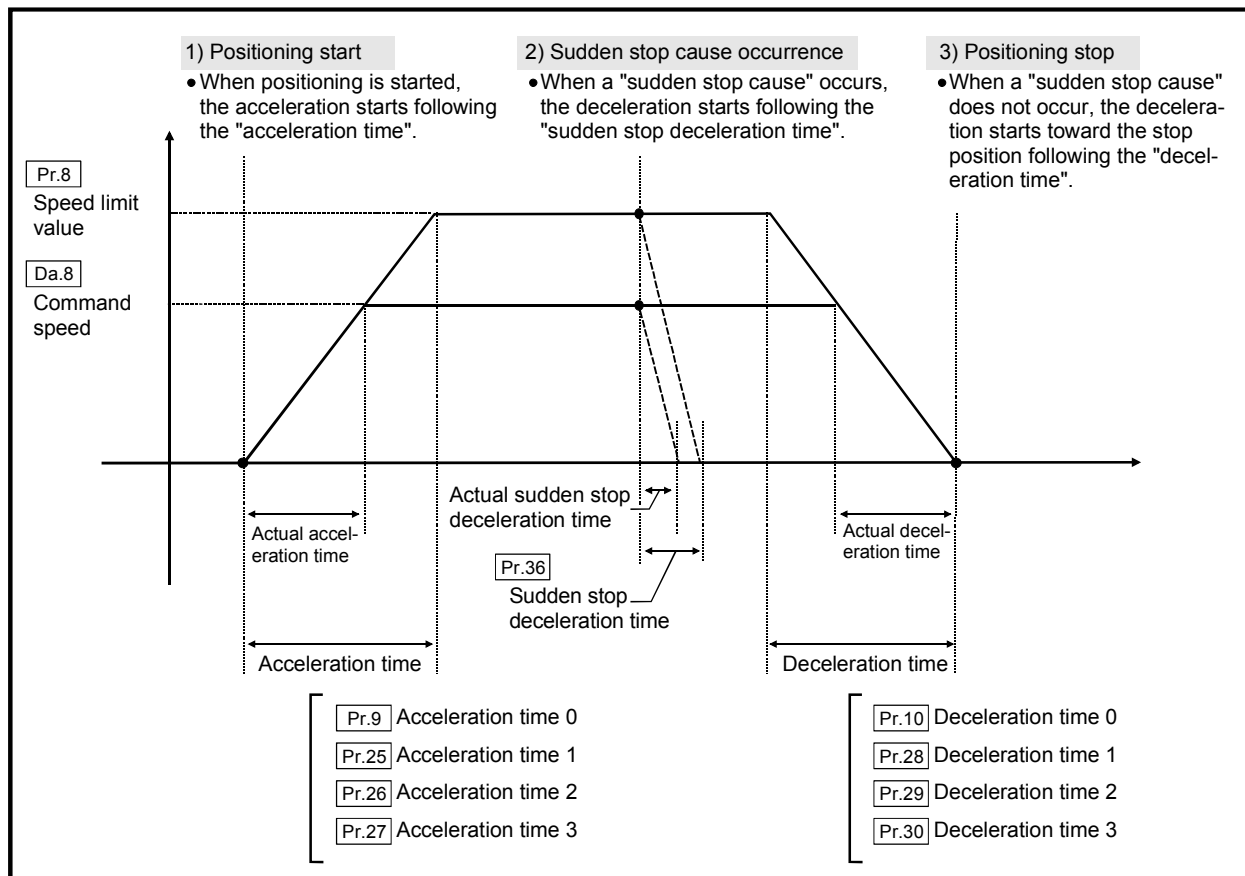
The S-curve ratio indicates where to draw the acceleration/deceleration curve using the sine curve as shown below.



**Pr.36** Sudden stop deceleration time

Set the time to reach speed 0 from " **Pr.8** Speed limit value" (or " **Pr.31** JOG speed limit value" during JOG operation) during the sudden stop.

The illustration below shows the relationships with other parameters.



**Pr.37** Stop group 1 sudden stop selection to **Pr.39** Stop group 3 sudden stop selection

Set the method to stop when the stop causes in the following stop groups occur.

- Stop group 1 ..... Stop with hardware stroke limit
- Stop group 2 ..... CPU module error occurrence, PLC READY signal [Y0] OFF, Fault in test mode
- Stop group 3 ..... External stop signal  
Stop signal from CPU module  
Stop signal from peripheral device  
Error occurrence (excludes errors in stop groups 1 and 2: includes only the software stroke limit errors during JOG operation, speed control, speed-position switching control, and position-speed switching control)

The methods of stopping include "0: Normal deceleration stop" and "1: Sudden stop".

If "1: Sudden stop" is selected, the axis will suddenly decelerate to a stop when the stop cause occurs.

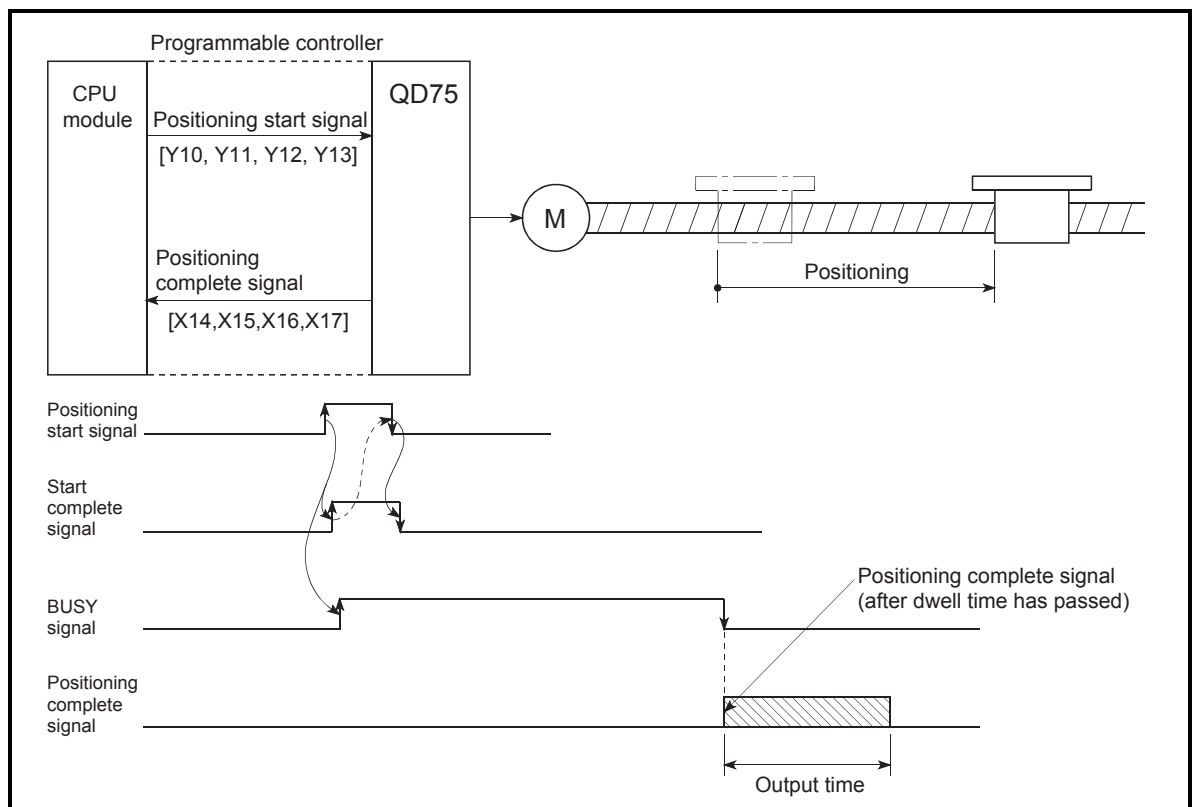
| Item                                                         | Setting value, setting range                                                                                                                                                                                                                    |                                                                                                            | Default value | Setting value buffer memory address |            |            |            |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------|------------|------------|------------|
|                                                              | Value set with peripheral device                                                                                                                                                                                                                | Value set with sequence program                                                                            |               | Axis 1                              | Axis 2     | Axis 3     | Axis 4     |
| <b>Pr.40</b><br>Positioning complete signal output time      | 0 to 65535 (ms)                                                                                                                                                                                                                                 | 0 to 65535 (ms)<br>0 to 32767 :<br>Set as a decimal<br>32768 to 65535:<br>Convert into hexadecimal and set | 300           | 59                                  | 209        | 359        | 509        |
| <b>Pr.41</b><br>Allowable circular interpolation error width | The setting value range differs depending on the " <b>Pr.1</b> Unit setting".<br>Here, the value within the [Table 1] range is set.<br> [Table 1] on next page |                                                                                                            | 100           | 60<br>61                            | 210<br>211 | 360<br>361 | 510<br>511 |
| <b>Pr.42</b><br>External command function selection          | 0: External positioning start                                                                                                                                                                                                                   | 0                                                                                                          | 0             | 62                                  | 212        | 362        | 512        |
|                                                              | 1: External speed change request                                                                                                                                                                                                                | 1                                                                                                          |               |                                     |            |            |            |
|                                                              | 2: Speed-position, position-speed switching request                                                                                                                                                                                             | 2                                                                                                          |               |                                     |            |            |            |
|                                                              | 3: Skip request                                                                                                                                                                                                                                 | 3                                                                                                          |               |                                     |            |            |            |

**Pr.40** Positioning complete signal output time

Set the output time of the positioning complete signal [X14, X15, X16, X17] output from the QD75.

A positioning completes when the specified dwell time has passed after the QD75 had terminated the output.

For the interpolation control, the positioning completed signal of interpolation axis is output only during the time set to the reference axis.



Positioning complete signal output time

[Table 1]

| Pr.1 setting value | Value set with peripheral device<br>(unit) | Value set with sequence program<br>(unit)     |
|--------------------|--------------------------------------------|-----------------------------------------------|
| 0 : mm             | 0 to 10000.0 ( $\mu\text{m}$ )             | 0 to 100000 ( $\times 10^{-1}\mu\text{m}$ )   |
| 1 : inch           | 0 to 1.00000 (inch)                        | 0 to 100000 ( $\times 10^{-5}\text{inch}$ )   |
| 2 : degree         | 0 to 1.00000 (degree)                      | 0 to 100000 ( $\times 10^{-5}\text{degree}$ ) |
| 3 : pulse          | 0 to 100000 (pulse)                        | 0 to 100000 (pulse)                           |

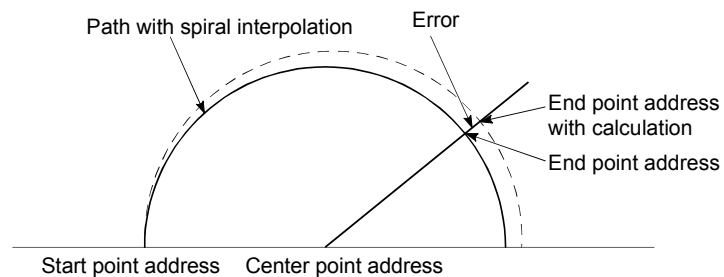
**Pr.41** Allowable circular interpolation error width

With the "allowable circular interpolation error width", the allowable error range of the calculated arc path and end point address is set. \* 1

If the error of the calculated arc path and end point address is within the set range, circular interpolation will be carried out to the set end point address while compensating the error with spiral interpolation.

The allowable circular interpolation error width is set in the following axis buffer memory addresses.

- If axis 1 is the reference axis, set in the axis 1 buffer memory address [60, 61].
- If axis 2 is the reference axis, set in the axis 2 buffer memory address [210, 211].
- If axis 3 is the reference axis, set in the axis 3 buffer memory address [360, 361].
- If axis 4 is the reference axis, set in the axis 4 buffer memory address [510, 511].



\* 1: With circular interpolation control using the center point designation, the arc path calculated with the start point address and center point address and the end point address may deviate.

**Pr.42** External command function selection

Select a command with which the external command signal should be associated.

0: External positioning start

The external command signal input is used to start a positioning operation.

1: External speed change request

The external command signal input is used to change the speed in the current positioning operation. The new speed should be set in the " **Cd.14** New speed value"

2: Speed-position, position-speed switching request

The external command signal input is used to switch from the speed control to the position control while in the speed-position switching control mode, or from the position control to the speed control while in the position-speed switching control mode.

To enable the speed-position switching control, set the " **Cd.24** Speed-position switching enable flag" to "1".

To enable the position-speed switching control, set the " **Cd.26** Position-speed switching enable flag" to "1".

3: Skip request

The external command signal input is used skip the current positioning operation.

**POINT**

To enable the external command signal, set the " **Cd.8** External command enable" to "1".



## 5.2.5 OPR basic parameters

| Item                | Setting value, setting range     |                                 | Default value | Setting value buffer memory address |        |        |        |
|---------------------|----------------------------------|---------------------------------|---------------|-------------------------------------|--------|--------|--------|
|                     | Value set with peripheral device | Value set with sequence program |               | Axis 1                              | Axis 2 | Axis 3 | Axis 4 |
| Pr.43<br>OPR method | 0 : Near-point dog method        | 0                               | 0             | 70                                  | 220    | 370    | 520    |
|                     | 1 : Stopper method 1)            | 1                               |               |                                     |        |        |        |
|                     | 2 : Stopper method 2)            | 2                               |               |                                     |        |        |        |
|                     | 3 : Stopper method 3)            | 3                               |               |                                     |        |        |        |
|                     | 4 : Count method 1)              | 4                               |               |                                     |        |        |        |
|                     | 5 : Count method 2)              | 5                               |               |                                     |        |        |        |

## Pr.43 OPR method

Set the "OPR method" for carrying out machine OPR.

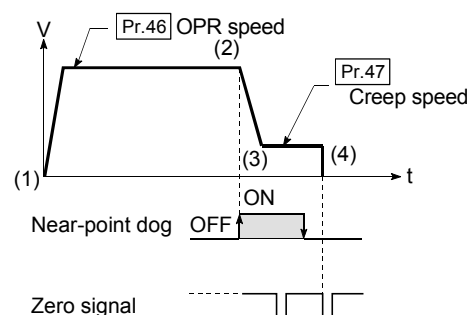
- 0 : Near-point dog method ..... After decelerating at the near-point dog ON, stop at the zero signal and complete the machine OPR.
- 1 : Stopper method 1) ..... After decelerating at the near-point dog ON, stop with the stopper, and complete the machine OPR after the dwell time has passed.
- 2 : Stopper method 2) ..... After decelerating at the near-point dog ON, stop with the stopper, and complete the machine OPR with the zero signal.
- 3 : Stopper method 3) ..... After starting with the creep speed, stop with the stopper, and complete the machine OPR with the zero signal.
- 4 : Count method 1) ..... After decelerating at the near-point dog ON, move the designated distance, and complete the machine OPR with the zero signal.
- 5 : Count method 2) ..... After decelerating at the near-point dog ON, move the designated distance, and complete the machine OPR.

Note) Refer to Section 8.2.2 "Machine OPR method" for details on the OPR methods.

## OPR method

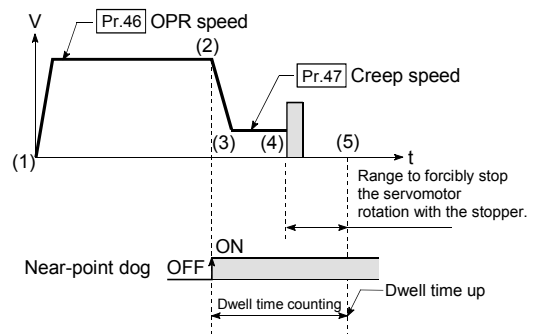
## 0 : Near-point dog method

- (1) Start machine OPR.  
(Start movement at the " Pr.46 OPR speed" in the " Pr.44 OPR direction".)
- (2) Detect the near-point dog ON, and start deceleration.
- (3) Decelerate to " Pr.47 Creep speed", and move with the creep speed.  
(At this time, the near-point dog must be ON. If the near-point dog is OFF, the axis will decelerate to a stop.)
- (4) At the first zero signal (one pulse output at one motor revolution) after the near-point dog OFF, the pulse output from the QD75 stops, and the machine OPR is completed.



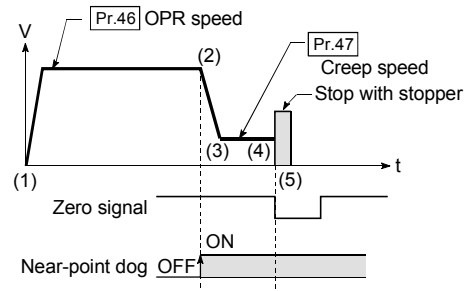
## 1 : Stopper method 1)

- (1) Start machine OPR.  
(Start movement at the " [Pr.46] OPR speed" in the " [Pr.44] OPR direction".)
- (2) Detect the near-point dog ON, and start deceleration.
- (3) Decelerate to " [Pr.47] Creep speed", and move with the creep speed.  
(At this time " [Pr.54] OPR torque limit value" is required. If the torque is not limited, the servomotor could be damaged in step (4).)
- (4) The axis contacts against the stopper at " [Pr.47] Creep speed", and then stops.
- (5) When the near-point dog turns ON and the " [Pr.49] OPR dwell time" is passed, the pulse output from the QD75 stops, and the machine OPR is completed.



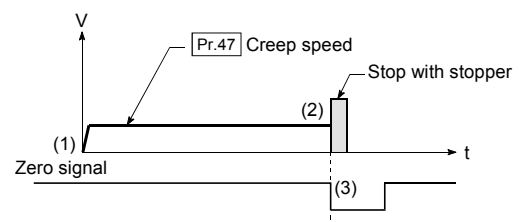
## 2 : Stopper method 2)

- (1) Start machine OPR.  
(Start movement at the " [Pr.46] OPR speed" in the " [Pr.44] OPR direction".)
- (2) Detect the near-point dog ON, and start deceleration.
- (3) Decelerate to " [Pr.47] Creep speed", and move with the creep speed.  
(At this time " [Pr.54] OPR torque limit value" is required. If the torque is not limited, the servomotor could be damaged in step (4).)
- (4) The axis contacts against the stopper at " [Pr.47] Creep speed", and then stops.
- (5) After stopping, the pulse output from the QD75 stops with the zero signal (signal that is output on detection of contact with the stopper. Input externally), and the machine OPR is completed.



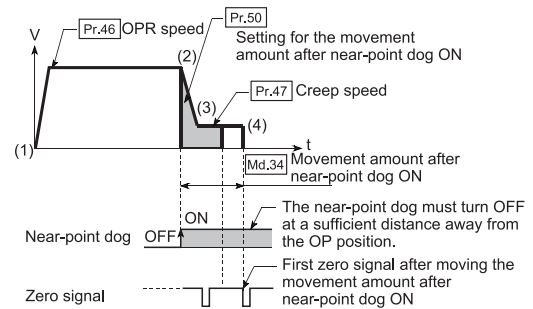
## 3 : Stopper method 3)

- (1) Start machine OPR.  
(Start movement at the " [Pr.47] Creep speed" in the " [Pr.44] OPR direction". At this time " [Pr.54] OPR torque limit value" is required. If the torque is not limited, the servomotor could be damaged in step (2).)
- (2) The axis contacts against the stopper at " [Pr.47] Creep speed", and then stops.
- (3) After stopping, the pulse output from the QD75 stops with the zero signal (signal that is output on detection of contact with the stopper. Input externally), and the machine OPR is completed.



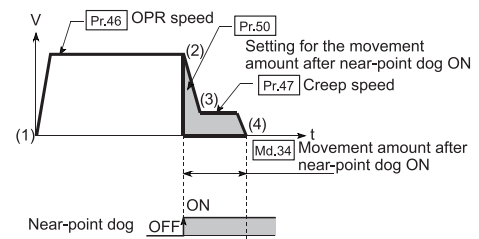
## 4 : Count method 1)

- (1) Start machine OPR.  
(Start movement at the " [Pr.46] OPR speed" in the " [Pr.44] OPR direction".)
- (2) Detect the near-point dog ON, and start deceleration.
- (3) Decelerate to " [Pr.47] Creep speed", and move with the creep speed.
- (4) After the near-point dog turns ON and the movement amount set in " [Pr.50] Setting for the movement amount after near-point dog ON" has passed, the pulse output from the QD75 stops with the first zero signal (one pulse output at one motor revolution), and the machine OPR is completed.



## 5 : Count method 2)

- (1) Start machine OPR.  
(Start movement at the " [Pr.46] OPR speed" in the " [Pr.44] OPR direction".)
- (2) Detect the near-point dog ON, and start deceleration.
- (3) Decelerate to " [Pr.47] Creep speed", and move with the creep speed.
- (4) After the near-point dog turns ON and the movement amount set in " [Pr.50] Setting for the movement amount after near-point dog ON" has passed, the pulse output from the QD75 stops with the first zero signal, and the machine OPR is completed.



| Item                     | Setting value, setting range                                                                                                                                                                                                                 |                                 | Default value | Setting value buffer memory address |            |            |            |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------|-------------------------------------|------------|------------|------------|
|                          | Value set with peripheral device                                                                                                                                                                                                             | Value set with sequence program |               | Axis 1                              | Axis 2     | Axis 3     | Axis 4     |
| [Pr.44]<br>OPR direction | 0 : Positive direction (address increment direction)<br>1 : Negative direction (address decrement direction)                                                                                                                                 | 0<br>1                          | 0             | 71                                  | 221        | 371        | 521        |
| [Pr.45]<br>OP address    | The setting value range differs depending on the " [Pr.1] Unit setting".<br>Here, the value within the [Table 1] range is set.<br> [Table 1] on next page |                                 | 0             | 72<br>73                            | 222<br>223 | 372<br>373 | 522<br>523 |
| [Pr.46]<br>OPR speed     | The setting value range differs depending on the " [Pr.1] Unit setting".<br>Here, the value within the [Table 2] range is set.<br> [Table 2] on next page |                                 | 1             | 74<br>75                            | 224<br>225 | 374<br>375 | 524<br>525 |

**Pr.44** OPR direction

Set the direction to start movement when starting machine OPR.

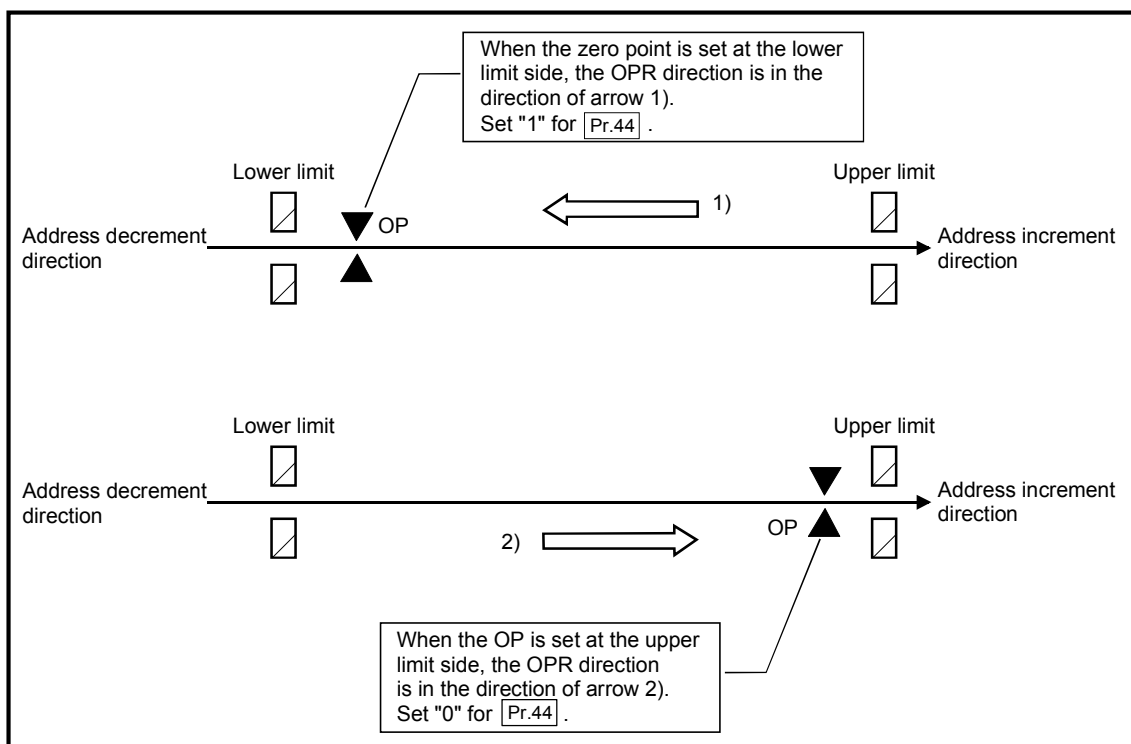
0: Positive direction (address increment direction)

Moves in the direction that the address increments. (Arrow 2))

1: Negative direction (address decrement direction)

Moves in the direction that the address decrements. (Arrow 1))

Normally, the OP is set near the lower limit or the upper limit, so " **Pr.44** OPR direction" is set as shown below.



[Table 1]

| <b>Pr.1</b> setting value | Value set with peripheral device (unit) | Value set with sequence program(unit)                     |
|---------------------------|-----------------------------------------|-----------------------------------------------------------|
| 0 : mm                    | -214748364.8 to 214748364.7 (μm)        | -2147483648 to 2147483647 ( $\times 10^{-1}\mu\text{m}$ ) |
| 1 : inch                  | -21474.83648 to 21474.83647 (inch)      | -2147483648 to 2147483647 ( $\times 10^{-5}\text{inch}$ ) |
| 2 : degree                | 0 to 359.99999 (degree)                 | 0 to 35999999 ( $\times 10^{-5}\text{degree}$ )           |
| 3 : pulse                 | -2147483648 to 2147483647 (pulse)       | -2147483648 to 2147483647 (pulse)                         |

[Table 2]

| <b>Pr.1</b> setting value | Value set with peripheral device (unit) | Value set with sequence program (unit)                                                   |
|---------------------------|-----------------------------------------|------------------------------------------------------------------------------------------|
| 0 : mm                    | 0.01 to 20000000.00 (mm/min)            | 1 to 2000000000 ( $\times 10^{-2}\text{mm/min}$ )                                        |
| 1 : inch                  | 0.001 to 2000000.000 (inch/min)         | 1 to 2000000000 ( $\times 10^{-3}\text{inch/min}$ )                                      |
| 2 : degree                | 0.001 to 2000000.000 (degree/min)       | 1 to 2000000000 ( $\times 10^{-3}\text{degree/min}$ )                                    |
| 3 : pulse                 | 1 to 1000000 (pulse/s) * 1              | [QD75P□N/QD75D□N]<br>1 to 4000000 (pulse/s)<br>[QD75P□/QD75D□]<br>1 to 1000000 (pulse/s) |

\* 1: In GX Configurator-QP, the QD75P□N/QD75D□N are recognized as the QD75P□/QD75D□. Therefore, the setting ranges are the same as those of the QD75P□/QD75D□. To set a value outside a setting range in GX Configurator-QP, set it through a sequence program or GX Works2. The setting ranges in GX Works2 are the same as those in sequence programs. (Refer to Appendix 1.2)

**Pr.45** OP address

Set the address used as the reference point for positioning control (ABS system).  
 (When the machine OPR is completed, the stop position address is changed to the address set in " **Pr.45** OP address". At the same time, the " **Pr.45** OP address" is stored in " **Md.20** Current feed value" and " **Md.21** Machine feed value".)

**Pr.46** OPR speed

Set the speed for OPR.

Note) Set the "OPR speed" to less than " **Pr.8** Speed limit value". If the "speed limit value" is exceeded, the error "Out of speed limit value range" (error code: 910) occurs and OPR is not performed.

The "OPR speed" should be equal to or faster than the " **Pr.7** Bias speed at start" and " **Pr.47** Creep speed".

| Item                        | Setting value, setting range                                                                                                                                                                                                                      |                                 | Default value | Setting value buffer memory address |            |            |            |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------|-------------------------------------|------------|------------|------------|
|                             | Value set with peripheral device                                                                                                                                                                                                                  | Value set with sequence program |               | Axis 1                              | Axis 2     | Axis 3     | Axis 4     |
| <b>Pr.47</b><br>Creep speed | The setting value range differs depending on the " <b>Pr.1</b> Unit setting".<br>Here, the value within the [Table 1] range is set.<br> [Table 1] on next page |                                 | 1             | 76<br>77                            | 226<br>227 | 376<br>377 | 526<br>527 |
| <b>Pr.48</b><br>OPR retry   | 0 : Do not retry OPR with limit switch                                                                                                                                                                                                            | 0                               | 0             | 78                                  | 228        | 378        | 528        |
|                             | 1 : Retry OPR with limit switch                                                                                                                                                                                                                   | 1                               |               |                                     |            |            |            |

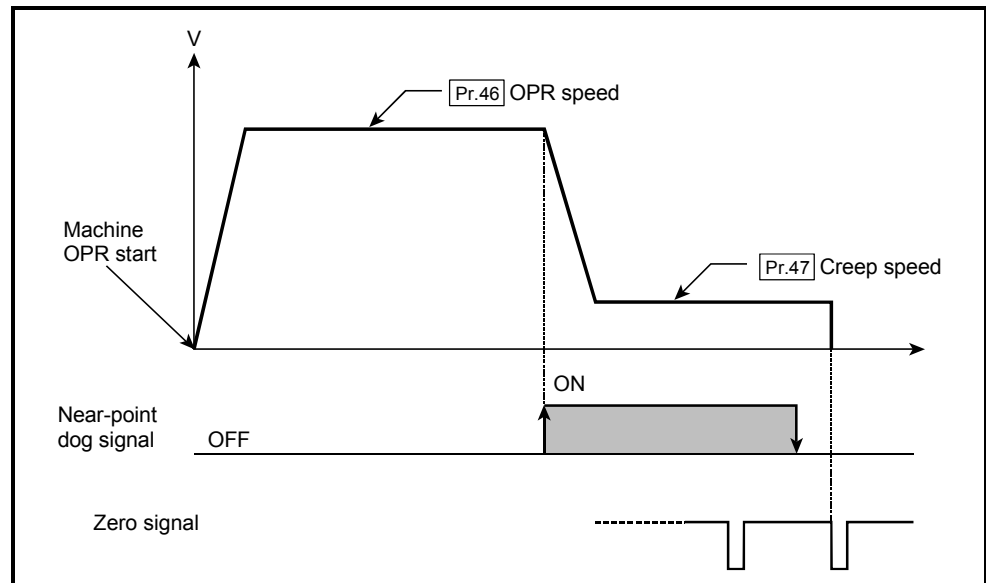
**Pr.47** Creep speed

Set the creep speed after near-point dog ON (the low speed just before stopping after decelerating from the OPR speed).

The creep speed is set within the following range.

$$(\text{Pr.46 OPR speed}) \geq (\text{Pr.47 Creep speed}) \geq (\text{Pr.7 Bias speed at start})$$

Note) The creep speed is related to the detection error when using the OPR method with zero signal, and the size of the collision if a collision occurs during OPR method using the stopper method.



[Table 1]

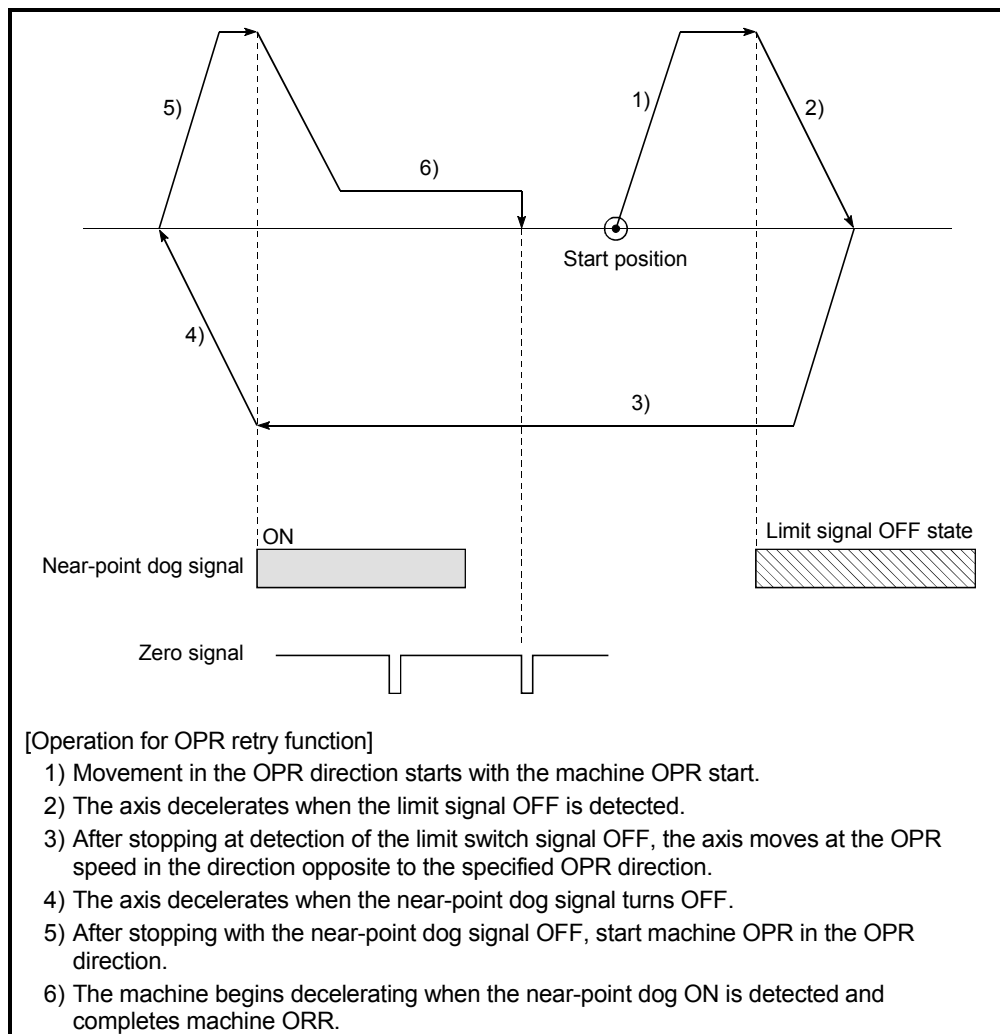
| Pr.1 setting value | Value set with peripheral device (unit) | Value set with sequence program (unit)                                                   |
|--------------------|-----------------------------------------|------------------------------------------------------------------------------------------|
| 0 : mm             | 0.01 to 2000000.00 (mm/min)             | 1 to 200000000 ( $\times 10^{-2}$ mm/min)                                                |
| 1 : inch           | 0.001 to 2000000.000 (inch/min)         | 1 to 200000000 ( $\times 10^{-3}$ inch/min)                                              |
| 2 : degree         | 0.001 to 2000000.000 (degree/min)       | 1 to 200000000 ( $\times 10^{-3}$ degree/min)                                            |
| 3 : pulse          | 1 to 1000000 (pulse/s) * 1              | [QD75P□N/QD75D□N]<br>1 to 4000000 (pulse/s)<br>[QD75P□/QD75D□]<br>1 to 1000000 (pulse/s) |

\* 1: In GX Configurator-QP, the QD75P□N/QD75D□N are recognized as the QD75P□/QD75D□. Therefore, the setting ranges are the same as those of the QD75P□/QD75D□. To set a value outside a setting range in GX Configurator-QP, set it through a sequence program or GX Works2. The setting ranges in GX Works2 are the same as those in sequence programs. (Refer to Appendix 1.2)

**Pr.48** OPR retry

Set whether to carry out OPR retry.

When the OPR retry function is validated and the machine OPR is started, first the axis will move in the OPR direction (1)). If the upper/lower limit signal turns OFF before the near-point dog signal ON is detected (2)), the axis will decelerate to a stop, and then will move in the direction opposite to the specified OPR direction (3)). If the falling edge of the near-point dog signal is detected during movement in the opposite direction, the axis will decelerate to a stop (4)), and will carry out machine OPR again (5)), (6)).



## 5.2.6 OPR detailed parameters

| Item                                                                    | Setting value, setting range                                                                                                                                                                                                                           |                                                                                                             | Default value | Setting value buffer memory address |            |            |            |
|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------|------------|------------|------------|
|                                                                         | Value set with peripheral device                                                                                                                                                                                                                       | Value set with sequence program                                                                             |               | Axis 1                              | Axis 2     | Axis 3     | Axis 4     |
| <b>Pr.49</b><br>OPR dwell time                                          | 0 to 65535 (ms)                                                                                                                                                                                                                                        | 0 to 65535 (ms)<br>0 to 32767 :<br>Set as a decimal<br>32768 to 65535 :<br>Convert into hexadecimal and set | 0             | 79                                  | 229        | 379        | 529        |
| <b>Pr.50</b><br>Setting for the movement amount after near-point dog ON | The setting value range differs depending on the " <b>Pr.1</b> Unit setting".<br>Here, the value within the [Table 1] range is set.<br> <b>[Table 1]</b> on next page |                                                                                                             | 0             | 80<br>81                            | 230<br>231 | 380<br>381 | 530<br>531 |
| <b>Pr.51</b><br>OPR acceleration time selection                         | 0 : <b>Pr.9</b> Acceleration time 0                                                                                                                                                                                                                    | 0                                                                                                           | 0             | 82                                  | 232        | 382        | 532        |
|                                                                         | 1 : <b>Pr.25</b> Acceleration time 1                                                                                                                                                                                                                   | 1                                                                                                           |               |                                     |            |            |            |
|                                                                         | 2 : <b>Pr.26</b> Acceleration time 2                                                                                                                                                                                                                   | 2                                                                                                           |               |                                     |            |            |            |
|                                                                         | 3 : <b>Pr.27</b> Acceleration time 3                                                                                                                                                                                                                   | 3                                                                                                           |               |                                     |            |            |            |
| <b>Pr.52</b><br>OPR deceleration time selection                         | 0 : <b>Pr.10</b> Deceleration time 0                                                                                                                                                                                                                   | 0                                                                                                           | 0             | 83                                  | 233        | 383        | 533        |
|                                                                         | 1 : <b>Pr.28</b> Deceleration time 1                                                                                                                                                                                                                   | 1                                                                                                           |               |                                     |            |            |            |
|                                                                         | 2 : <b>Pr.29</b> Deceleration time 2                                                                                                                                                                                                                   | 2                                                                                                           |               |                                     |            |            |            |
|                                                                         | 3 : <b>Pr.30</b> Deceleration time 3                                                                                                                                                                                                                   | 3                                                                                                           |               |                                     |            |            |            |

**Pr.49** OPR dwell time

When stopper method 1) is set for " **Pr.43** OPR method", set the time for the machine OPR to complete after the near-point dog signal turns ON.  
The setting value must be longer than the movement time from the near-point dog signal ON to stopping with the stopper.  
(If the OPR method is not "stopper method 1)", the " **Pr.49** OPR dwell time" value is irrelevant.)

**Pr.50** Setting for the movement amount after near-point dog ON

When using the count method 1) or 2), set the movement amount to the OP after the near-point dog signal turns ON.  
(The movement amount after near-point dog ON should be equal to or greater than the sum of the "distance covered by the deceleration from the OPR speed to the creep speed" and "distance of movement in 10ms at the OPR speed".)

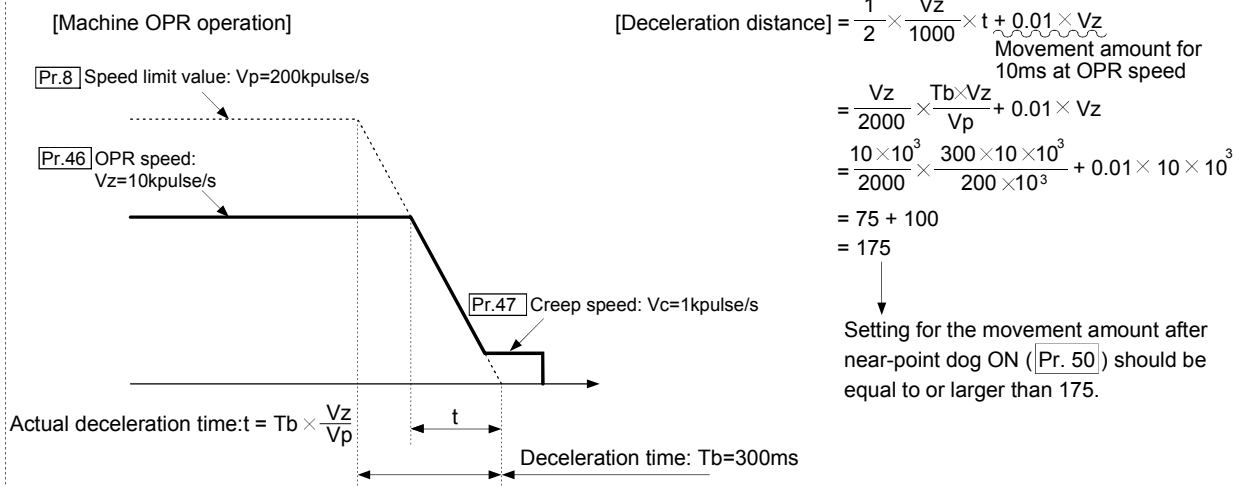


[Table 1]

| [Pr.1] setting value | Value set with peripheral device<br>(unit) | Value set with sequence program<br>(unit)  |
|----------------------|--------------------------------------------|--------------------------------------------|
| 0 : mm               | 0 to 214748364.7 (μm)                      | 0 to 2147483647 (×10 <sup>-1</sup> μm)     |
| 1 : inch             | 0 to 21474.83647 (inch)                    | 0 to 2147483647 (×10 <sup>-5</sup> inch)   |
| 2 : degree           | 0 to 21474.83647 (degree)                  | 0 to 2147483647 (×10 <sup>-5</sup> degree) |
| 3 : pulse            | 0 to 2147483647 (pulse)                    | 0 to 2147483647 (pulse)                    |

Example of setting for "[Pr. 50] Setting for the movement amount after near-point dog ON"

Assuming that the "[Pr. 8] Speed limit value" is set to 200 kpulse/s, "[Pr. 46] OPR speed" to 10 kpulse/s, "[Pr. 47] Creep speed" to 1 kpulse/s, and deceleration time to 300 ms, the minimum value of "[Pr. 50] Setting for the movement amount after near-point dog ON" is calculated as follows:



### [Pr.51] OPR acceleration time selection

Set which of "acceleration time 0 to 3" to use for the acceleration time during OPR.

- 0 : Use the value set in "[Pr.9] Acceleration time 0".
- 1 : Use the value set in "[Pr.25] Acceleration time 1".
- 2 : Use the value set in "[Pr.26] Acceleration time 2".
- 3 : Use the value set in "[Pr.27] Acceleration time 3".

### [Pr.52] OPR deceleration time selection

Set which of "deceleration time 0 to 3" to use for the deceleration time during OPR.

- 0 : Use the value set in "[Pr.10] Deceleration time 0".
- 1 : Use the value set in "[Pr.28] Deceleration time 1".
- 2 : Use the value set in "[Pr.29] Deceleration time 2".
- 3 : Use the value set in "[Pr.30] Deceleration time 3".

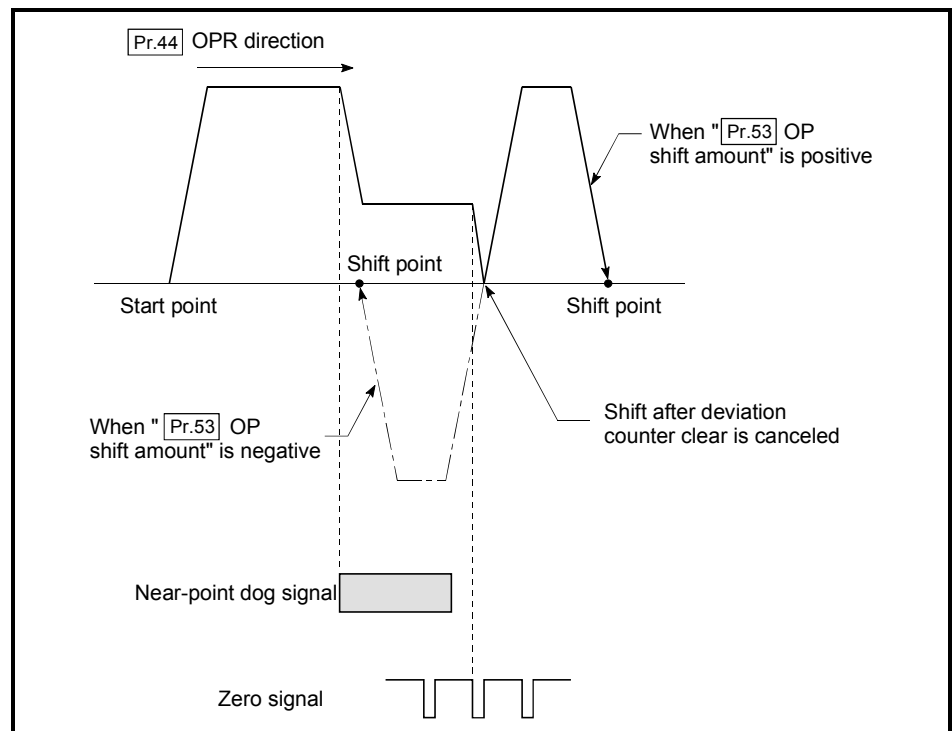
| Item                                                       | Setting value, setting range                                                                                                                                                                                                                                                                      |                                                                                                                     | Default value | Setting value buffer memory address |            |            |            |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------|------------|------------|------------|
|                                                            | Value set with peripheral device                                                                                                                                                                                                                                                                  | Value set with sequence program                                                                                     |               | Axis 1                              | Axis 2     | Axis 3     | Axis 4     |
| <b>Pr.53</b><br>OP shift amount                            | The setting value range differs depending on the " <b>Pr.1</b> Unit setting". Here, the value within the [Table 1] range is set. <div style="text-align: center;">  <b>[Table 1]</b> on next page         </div> |                                                                                                                     | 0             | 84<br>85                            | 234<br>235 | 384<br>385 | 534<br>535 |
| <b>Pr.54</b><br>OPR torque limit value                     | 1 to 300 (%)                                                                                                                                                                                                                                                                                      | 1 to 300 (%)                                                                                                        | 300           | 86                                  | 236        | 386        | 536        |
| <b>Pr.55</b><br>Deviation counter clear signal output time | 1 to 65535 (ms)                                                                                                                                                                                                                                                                                   | 1 to 65535 (ms)<br>1 to 32767: Set the decimal value as it is.<br>32768 to 65535: Convert into hexadecimal and set. | 11            | 87                                  | 237        | 387        | 537        |
| <b>Pr.56</b><br>Speed designation during OP shift          | 0 : OPR speed                                                                                                                                                                                                                                                                                     | 0                                                                                                                   | 0             | 88                                  | 238        | 388        | 538        |
|                                                            | 1 : Creep speed                                                                                                                                                                                                                                                                                   | 1                                                                                                                   |               |                                     |            |            |            |
| <b>Pr.57</b><br>Dwell time during OPR retry                | 0 to 65535 (ms)                                                                                                                                                                                                                                                                                   | 0 to 65535 (ms)<br>0 to 32767 : Set as a decimal<br>32768 to 65535 : Convert into hexadecimal and set               | 0             | 89                                  | 239        | 389        | 539        |

**Pr.53** OP shift amount

Set the amount to shift (move) from the position stopped at with machine OPR.

\* The OP shift function is used to compensate the OP position stopped at with machine OPR.

If there is a physical limit to the OP position, due to the relation of the near-point dog installation position, use this function to compensate the OP to an optimum position.



[Table 1]

| Pr.1 setting value | Value set with peripheral device<br>(unit)    | Value set with sequence program<br>(unit)                   |
|--------------------|-----------------------------------------------|-------------------------------------------------------------|
| 0 : mm             | -214748364.8 to 214748364.7 ( $\mu\text{m}$ ) | -2147483648 to 2147483647 ( $\times 10^{-1}\mu\text{m}$ )   |
| 1 : inch           | -21474.83648 to 21474.83647 (inch)            | -2147483648 to 2147483647 ( $\times 10^{-5}\text{inch}$ )   |
| 2 : degree         | -21474.83648 to 21474.83647 (degree)          | -2147483648 to 2147483647 ( $\times 10^{-5}\text{degree}$ ) |
| 3 : pulse          | -2147483648 to 2147483647 (pulse)             | -2147483648 to 2147483647 (pulse)                           |

**Pr.54** OPR torque limit value

Set the value to limit the servomotor torque after reaching the creep speed during machine OPR.

Refer to Section 12.4.2 "Torque limit function" for details on the torque limits.

**Pr.55** Deviation counter clear signal output time

Set the duration of the deviation counter clear signal output during a machine OPR operation using any of the following methods: the near-point dog method, stopper methods 1) to 3), and count method 1). (For details, refer to your drive unit manual.)

**Pr.56** Speed designation during OP shift

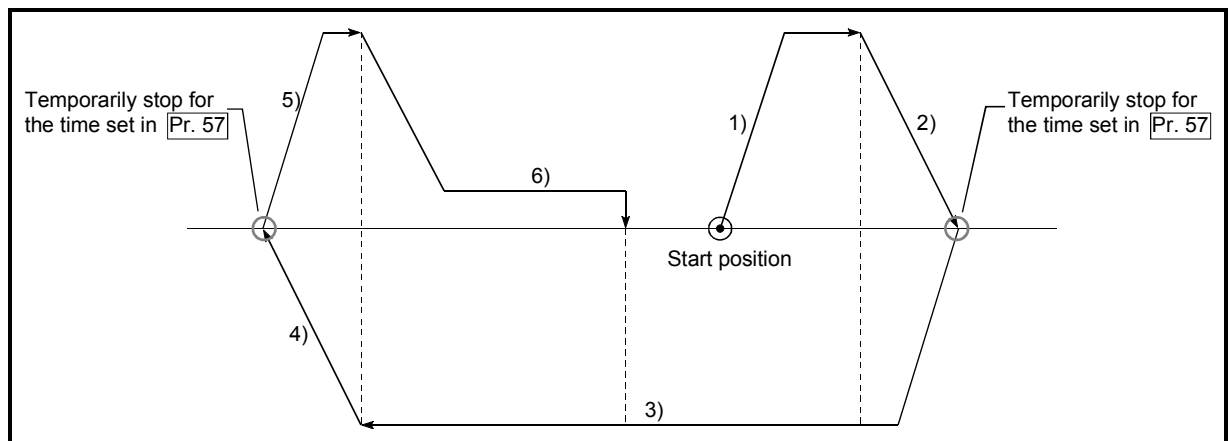
Set the operation speed for when a value other than "0" is set for " **Pr.53** OP shift amount". Select the setting from " **Pr.46** OPR speed" or " **Pr.47** Creep speed".

0 : Designate " **Pr.46** OPR speed" as the setting value.

1 : Designate " **Pr.47** Creep speed" as the setting value.

**Pr.57** Dwell time during OPR retry

When OPR retry is validated (when "1" is set for **Pr.48**), set the stop time after decelerating in 2) and 4) in the following drawing.



## 5.3 List of positioning data

Before explaining the positioning data setting items **Da.1** to **Da.10**, the configuration of the positioning data will be shown below.

Before the explanation on the positioning data setting items **Da.1** to **Da.10** and **Da.27** to **Da.29**, the configuration of the positioning data will be shown below.

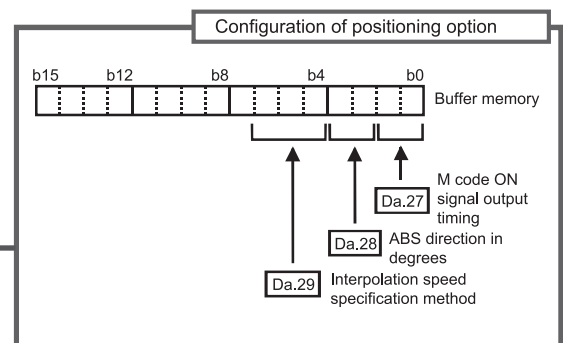
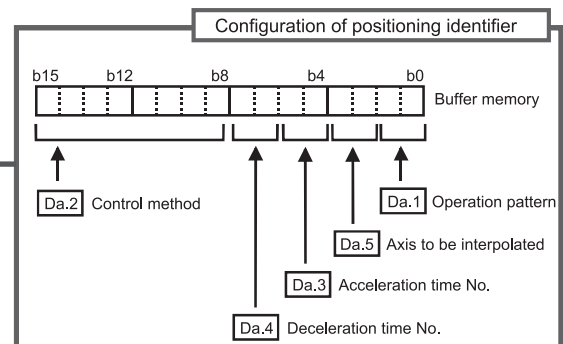
The positioning data stored in the QD75 buffer memory has the following type of configuration.

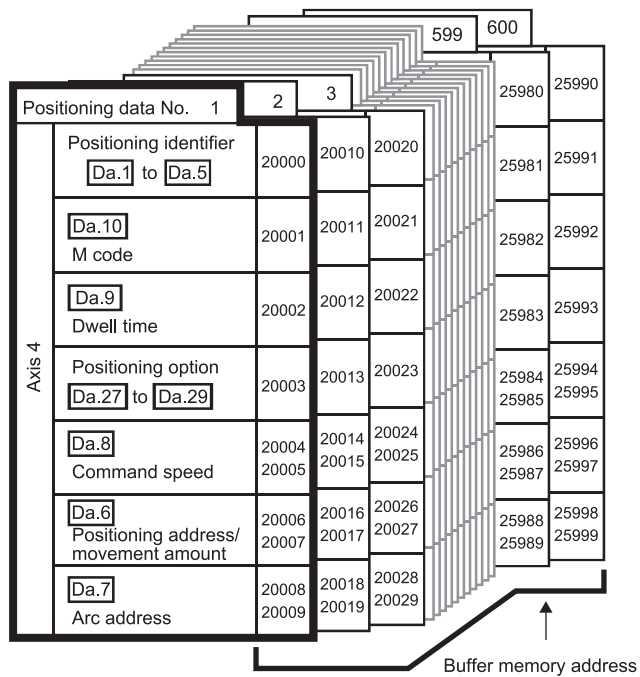
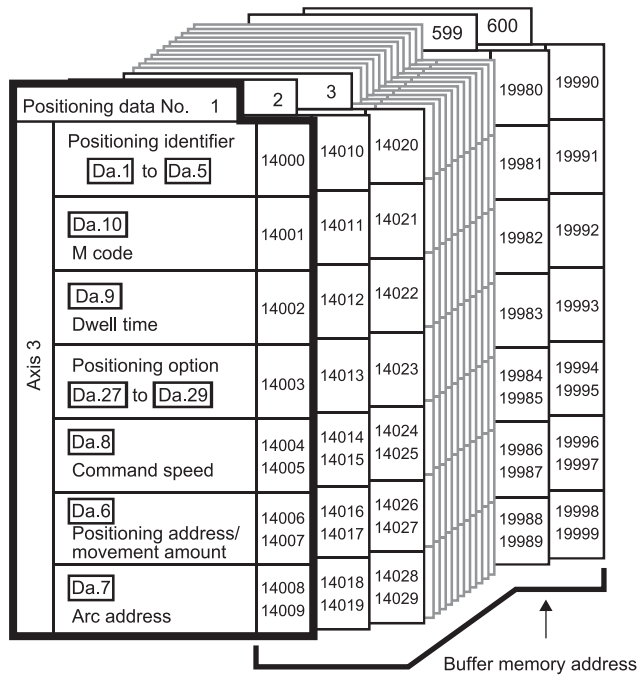
| Positioning data No. |                                                        | 1            | 2            | 3            | ... | 599          | 600          |
|----------------------|--------------------------------------------------------|--------------|--------------|--------------|-----|--------------|--------------|
| Axis 1               | Positioning identifier<br><b>Da.1</b> to <b>Da.5</b>   | 2000         | 2010         | 2020         | ... | 7980         | 7990         |
|                      | <b>Da.10</b><br>M code                                 | 2001         | 2011         | 2021         | ... | 7981         | 7991         |
|                      | <b>Da.9</b><br>Dwell time                              | 2002         | 2012         | 2022         | ... | 7982         | 7992         |
|                      | Positioning option<br><b>Da.27</b> to <b>Da.29</b>     | 2003         | 2013         | 2023         | ... | 7983         | 7993         |
|                      | <b>Da.8</b><br>Command speed                           | 2004<br>2005 | 2014<br>2015 | 2024<br>2025 | ... | 7984<br>7985 | 7994<br>7995 |
|                      | <b>Da.6</b><br>Positioning address/<br>movement amount | 2006<br>2007 | 2016<br>2017 | 2026<br>2027 | ... | 7986<br>7987 | 7996<br>7997 |
|                      | <b>Da.7</b><br>Arc address                             | 2008<br>2009 | 2018<br>2019 | 2028<br>2029 | ... | 7988<br>7989 | 7998<br>7999 |

• Up to 600 positioning data for each axis can be set (stored) in the buffer memory addresses shown on the left. These data are controlled as positioning data No.1 to No.600 for each axis.

• One positioning data is configured of the items shown in the bold box.

| Positioning data No. |                                                        | 1            | 2            | 3            | ... | 599            | 600            |
|----------------------|--------------------------------------------------------|--------------|--------------|--------------|-----|----------------|----------------|
| Axis 2               | Positioning identifier<br><b>Da.1</b> to <b>Da.5</b>   | 8000         | 8010         | 8020         | ... | 13980          | 13990          |
|                      | <b>Da.10</b><br>M code                                 | 8001         | 8011         | 8021         | ... | 13981          | 13991          |
|                      | <b>Da.9</b><br>Dwell time                              | 8002         | 8012         | 8022         | ... | 13982          | 13992          |
|                      | Positioning option<br><b>Da.27</b> to <b>Da.29</b>     | 8003         | 8013         | 8023         | ... | 13983          | 13993          |
|                      | <b>Da.8</b><br>Command speed                           | 8004<br>8005 | 8014<br>8015 | 8024<br>8025 | ... | 13984<br>13985 | 13994<br>13995 |
|                      | <b>Da.6</b><br>Positioning address/<br>movement amount | 8006<br>8007 | 8016<br>8017 | 8026<br>8027 | ... | 13986<br>13987 | 13996<br>13997 |
|                      | <b>Da.7</b><br>Arc address                             | 8008<br>8009 | 8018<br>8019 | 8028<br>8029 | ... | 13988<br>13989 | 13998<br>13999 |





The descriptions that follow relate to the positioning data set items ([Da.1] to [Da.10], [Da.27] to [Da.29]).

(The buffer memory addresses shown are those of the "positioning data No. 1" for the axes 1 to 4.)

## Positioning identifier configuration

The positioning identifier consists of "[Da.1] Operation pattern" to "[Da.5] Axis to be interpolated". The set values of those data are stored in one buffer memory address.

Consider the positioning identifier configuration in the figure below when setting "[Da.1] Operation pattern" to "[Da.5] Axis to be interpolated".

| Positioning identifier configuration | Assignment |                                |
|--------------------------------------|------------|--------------------------------|
|                                      | (1)        | [Da.1] Operation pattern       |
|                                      | (2)        | [Da.5] Axis to be interpolated |
|                                      | (3)        | [Da.3] Acceleration time No.   |
|                                      | (4)        | [Da.4] Deceleration time No.   |
|                                      | (5)        | [Da.2] Control system          |

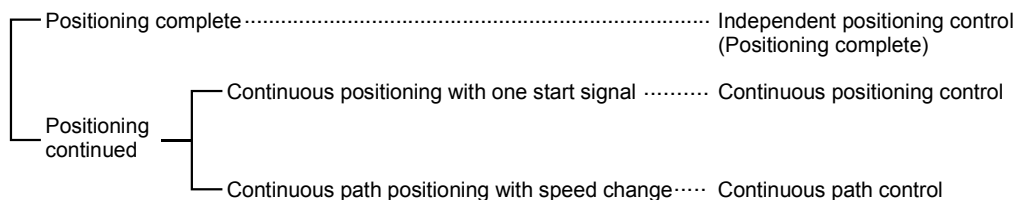
The following table lists the setting range and default value of each buffer memory area of the positioning identifier.

| Item                   | Setting value                          |                                                                               | Default value | Setting value buffer memory address |        |        |        |
|------------------------|----------------------------------------|-------------------------------------------------------------------------------|---------------|-------------------------------------|--------|--------|--------|
|                        | Value set with peripheral device       | Value set with sequence program                                               |               | Axis 1                              | Axis 2 | Axis 3 | Axis 4 |
| Positioning identifier | <b>Da.1</b><br>Operation pattern       | 00: Positioning complete                                                      | 00            |                                     |        |        |        |
|                        |                                        | 01: Continuous positioning control                                            | 01            |                                     |        |        |        |
|                        |                                        | 11: Continuous path control                                                   | 11            |                                     |        |        |        |
|                        | <b>Da.2</b><br>Control system          | ABS1 : 1-axis linear control (ABS)                                            | 01H           |                                     |        |        |        |
|                        |                                        | INC1 : 1-axis linear control (INC)                                            | 02H           |                                     |        |        |        |
|                        |                                        | FEED1: 1-axis fixed-feed control                                              | 03H           |                                     |        |        |        |
|                        |                                        | VF1 : 1-axis speed control (forward run)                                      | 04H           |                                     |        |        |        |
|                        |                                        | VR1 : 1-axis speed control (reverse run)                                      | 05H           |                                     |        |        |        |
|                        |                                        | VPF : Speed-position switching control (forward run)                          | 06H           |                                     |        |        |        |
|                        |                                        | VPR : Speed-position switching control (reverse run)                          | 07H           |                                     |        |        |        |
|                        |                                        | PVF : Position-speed switching control (forward run)                          | 08H           |                                     |        |        |        |
|                        |                                        | PVR : Position-speed switching control (reverse run)                          | 09H           |                                     |        |        |        |
|                        |                                        | ABS2 : 2-axis linear interpolation control (ABS)                              | 0AH           |                                     |        |        |        |
|                        |                                        | INC2 : 2-axis linear interpolation control (INC)                              | 0BH           |                                     |        |        |        |
|                        |                                        | FEED2: Fixed-feed control by 2-axis linear interpolation                      | 0CH           |                                     |        |        |        |
|                        |                                        | ABS : Circular interpolation control with sub point specified (ABS)           | 0DH           |                                     |        |        |        |
|                        |                                        | INC : Circular interpolation control with sub point specified (INC)           | 0EH           |                                     |        |        |        |
|                        |                                        | ABS : Circular interpolation control with center point specified (ABS, CW)    | 0FH           |                                     |        |        |        |
|                        |                                        | ABS : Circular interpolation control with center point specified (ABS, CCW)   | 10H           |                                     |        |        |        |
|                        |                                        | INC : Circular interpolation control with center point specified (INC, CW)    | 11H           |                                     |        |        |        |
|                        |                                        | INC : Circular interpolation control with center point specified (INC, CCW)   | 12H           |                                     |        |        |        |
|                        |                                        | VF2 : 2-axis speed control (forward run)                                      | 13H           |                                     |        |        |        |
|                        |                                        | VR2 : 2-axis speed control (reverse run)                                      | 14H           |                                     |        |        |        |
|                        |                                        | ABS3 : 3-axis linear interpolation control (ABS)                              | 15H           |                                     |        |        |        |
|                        |                                        | INC3 : 3-axis linear interpolation control (INC)                              | 16H           |                                     |        |        |        |
|                        |                                        | FEED3: Fixed-feed control by 3-axis linear interpolation control              | 17H           |                                     |        |        |        |
|                        |                                        | VF3 : 3-axis speed control (forward run)                                      | 18H           |                                     |        |        |        |
|                        |                                        | VR3 : 3-axis speed control (reverse run)                                      | 19H           |                                     |        |        |        |
|                        |                                        | ABSH : Helical interpolation control with sub point designation (ABS)         | 20H           |                                     |        |        |        |
|                        |                                        | INCH : Helical interpolation control with sub point designation (INC)         | 21H           |                                     |        |        |        |
|                        |                                        | ABSH : Helical interpolation control with center point designation (ABS, CW)  | 22H           |                                     |        |        |        |
|                        |                                        | ABSH : Helical interpolation control with center point designation (ABS, CCW) | 23H           |                                     |        |        |        |
|                        |                                        | INCH : Helical interpolation control with center point designation (INC, CW)  | 24H           |                                     |        |        |        |
|                        |                                        | INCH : Helical interpolation control with center point designation (INC, CCW) | 25H           |                                     |        |        |        |
|                        |                                        | ABS4 : 4-axis linear interpolation control (ABS)                              | 1AH           |                                     |        |        |        |
|                        |                                        | INC4 : 4-axis linear interpolation control (INC)                              | 1BH           |                                     |        |        |        |
|                        |                                        | FEED4: Fixed-feed control by 4-axis linear interpolation control              | 1CH           |                                     |        |        |        |
|                        |                                        | VF4 : 4-axis speed control (forward run)                                      | 1DH           |                                     |        |        |        |
|                        |                                        | VR4 : 4-axis speed control (reverse run)                                      | 1EH           |                                     |        |        |        |
|                        |                                        | NOP : NOP instruction                                                         | 80H           |                                     |        |        |        |
|                        |                                        | POS : Current value changing                                                  | 81H           |                                     |        |        |        |
|                        |                                        | JUMP : JUMP instruction                                                       | 82H           |                                     |        |        |        |
|                        |                                        | LOOP : Declares the beginning of LOOP to LEND section                         | 83H           |                                     |        |        |        |
|                        |                                        | LEND : Declares the end of LOOP to LEND section                               | 84H           |                                     |        |        |        |
|                        | <b>Da.3</b><br>Acceleration time No.   | 0: [Pr.9] Acceleration time 0                                                 | 00            |                                     |        |        |        |
|                        |                                        | 1: [Pr.25] Acceleration time 1                                                | 01            |                                     |        |        |        |
|                        |                                        | 2: [Pr.26] Acceleration time 2                                                | 10            |                                     |        |        |        |
|                        |                                        | 3: [Pr.27] Acceleration time 3                                                | 11            |                                     |        |        |        |
|                        | <b>Da.4</b><br>Deceleration time No.   | 0: [Pr.10] Deceleration time 0                                                | 00            |                                     |        |        |        |
|                        |                                        | 1: [Pr.28] Deceleration time 1                                                | 01            |                                     |        |        |        |
|                        |                                        | 2: [Pr.29] Deceleration time 2                                                | 10            |                                     |        |        |        |
|                        |                                        | 3: [Pr.30] Deceleration time 3                                                | 11            |                                     |        |        |        |
|                        | <b>Da.5</b><br>Axis to be interpolated | 0: Axis 1                                                                     | 00            |                                     |        |        |        |
|                        |                                        | 1: Axis 2                                                                     | 01            |                                     |        |        |        |
|                        |                                        | 2: Axis 3                                                                     | 10            |                                     |        |        |        |
|                        |                                        | 3: Axis 4                                                                     | 11            |                                     |        |        |        |
|                        |                                        |                                                                               | 0000H         | 2000                                | 8000   | 14000  | 20000  |

**Da.1** Operation pattern

The operation pattern designates whether positioning of a certain data No. is to be ended with just that data, or whether the positioning for the next data No. is to be carried out in succession.

[Operation pattern]



- 1) Positioning complete..... Set to execute positioning to the designated address, and then complete positioning.
- 2) Continuous positioning control ..... Positioning is carried out successively in order of data Nos. with one start signal. The operation halts at each position indicated by a positioning data.
- 3) Continuous path control..... Positioning is carried out successively in order of data Nos. with one start signal. The operation does not stop at each positioning data.

**Da.2** Control system

Set the "control system" for carrying out positioning control.

- Note)
- When "JUMP instruction" is set for the control system, the " **Da.9** Dwell time" and " **Da.10** M code" setting details will differ.
  - In case you selected "LOOP" as the control system, the " **Da.10** M code" should be set differently from other cases.
  - Refer to CHAPTER 9 "MAJOR POSITIONING CONTROL" for details on the control systems.
  - If "degree" is set for " **Pr.1** Unit setting", circular interpolation control or 3-axis helical interpolation control cannot be carried out. The error "Circular interpolation not possible" (error code: 535) will occur when executed.
  - The 3-axis helical interpolation control can be set only for the QD75N with a serial number (first five digits) of "17102" or later. If the 3-axis helical interpolation control is set for the QD75N with a serial number (first five digits) earlier than "17102", the error "Control system setting error" (error code: 524) will occur.

**Da.3** Acceleration time No.

Set which of "acceleration time 0 to 3" to use for the acceleration time during positioning.

- 0 : Use the value set in " **Pr.9** Acceleration time 0".
- 1 : Use the value set in " **Pr.25** Acceleration time 1".
- 2 : Use the value set in " **Pr.26** Acceleration time 2".
- 3 : Use the value set in " **Pr.27** Acceleration time 3".

**Da.4** Deceleration time No.

Set which of "deceleration time 0 to 3" to use for the deceleration time during positioning.

- 0 : Use the value set in " **Pr.10** Deceleration time 0".
- 1 : Use the value set in " **Pr.28** Deceleration time 1".
- 2 : Use the value set in " **Pr.29** Deceleration time 2".
- 3 : Use the value set in " **Pr.30** Deceleration time 3".



**Da.5** Axis to be interpolated

Set the target axis (partner axis) for operations under the 2-axis interpolation control, and the circular interpolation axis for operations under the 3-axis helical interpolation control.

0 : Selects the axis 1 as the target axis (partner axis).

1 : Selects the axis 2 as the target axis (partner axis).

2 : Selects the axis 3 as the target axis (partner axis).

3 : Selects the axis 4 as the target axis (partner axis).

- Note)
- Do not specify the own axis number or any number except the above. (If you do, the error "Illegal interpolation description command" (error code: 521) will occur during the program execution.)
  - For 3-axis interpolation other than 3-axis helical interpolation control or 4-axis interpolation, this item does not need to be set.
  - The following table lists the target axes for the 3-axis helical interpolation control.

| Reference axis | Circular interpolation axis | Linear interpolation axis |
|----------------|-----------------------------|---------------------------|
| Axis 1         | Axis 2                      | Axis 3                    |
|                | Axis 3                      | Axis 2                    |
| Axis 2         | Axis 3                      | Axis 4                    |
|                | Axis 4                      | Axis 3                    |
| Axis 3         | Axis 4                      | Axis 1                    |
|                | Axis 1                      | Axis 4                    |
| Axis 4         | Axis 1                      | Axis 2                    |
|                | Axis 2                      | Axis 1                    |

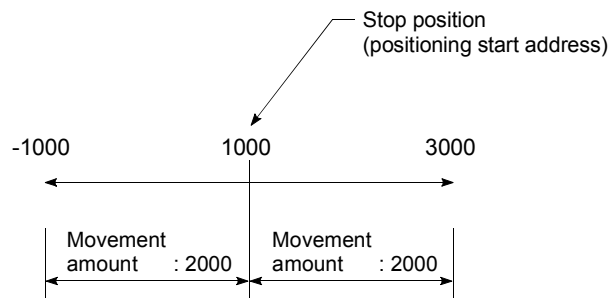
| Item                                                   | Setting value, setting range                                                                                                                             |                                                                                    | Default value | Setting value buffer memory address |              |                |                |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------|-------------------------------------|--------------|----------------|----------------|
|                                                        | Value set with peripheral device                                                                                                                         | Value set with sequence program                                                    |               | Axis 1                              | Axis 2       | Axis 3         | Axis 4         |
| <b>Da.6</b><br>Positioning address/<br>movement amount | The setting value range differs according to the " <b>Da.2</b> Control system".<br>Here, the value within the following range of [Table 1] range is set. |                                                                                    | 0             | 2006<br>2007                        | 8006<br>8007 | 14006<br>14007 | 20006<br>20007 |
|                                                        |                                                                                                                                                          |  |               |                                     |              |                |                |

**Da.6** Positioning address/movement amount

Set the address to be used as the target value for positioning control.  
The setting value range differs according to the " **Da.2** Control system".  
((1) to (4))

**(1) Absolute (ABS) system, current value changing**

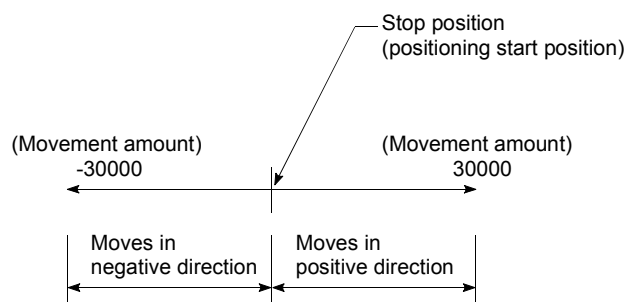
- The setting value (positioning address) for the ABS system and current value changing is set with an absolute address (address from OP).

**(2) Incremental (INC) system, fixed-feed 1, fixed-feed 2, fixed-feed 3, fixed-feed 4**

- The setting value (movement amount) for the INC system is set as a movement amount with sign.

When movement amount is positive: Moves in the positive direction (address increment direction)

When movement amount is negative: Moves in the negative direction (address decrement direction)



[Table 1]

■ When " Pr.1 Unit Setting" is "mm"

The table below lists the control systems that require the setting of the positioning address or movement amount and the associated setting ranges.

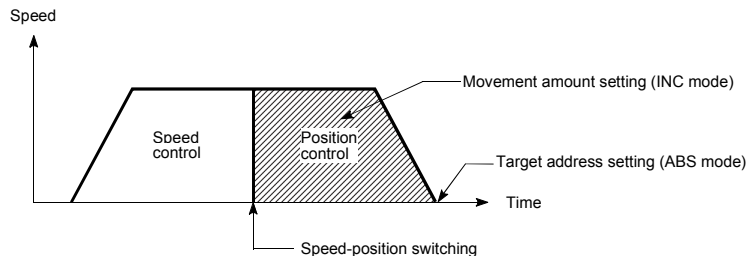
(With any control system excluded from the table below, neither the positioning address nor the movement amount needs to be set.)

| Da.2 setting value                                                                                                                                                           | Value set with peripheral device<br>( $\mu\text{m}$ )    | Value set with sequence program * 1<br>( $\times 10^{-1}\mu\text{m}$ ) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------|
| ABS Linear 1 : 01H<br>ABS Linear 2 : 0AH<br>ABS Linear 3 : 15H<br>ABS Linear 4 : 1AH<br>Current value changing : 81H                                                         | ◇ Set the address<br>-214748364.8 to 214748364.7         | ◇ Set the address<br>-2147483648 to 2147483647                         |
| INC Linear 1 : 02H<br>INC Linear 2 : 0BH<br>INC Linear 3 : 16H<br>INC Linear 4 : 1BH<br>Fixed-feed 1 : 03H<br>Fixed-feed 2 : 0CH<br>Fixed-feed 3 : 17H<br>Fixed-feed 4 : 1CH | ◇ Set the movement amount<br>-214748364.8 to 214748364.7 | ◇ Set the movement amount<br>-2147483648 to 2147483647                 |
| Forward run speed/position: 06H<br>Reverse run speed/position: 07H<br>Forward run position/speed: 08H<br>Reverse run position/speed: 09H                                     | ◇ Set the movement amount<br>0 to 214748364.7            | ◇ Set the movement amount<br>0 to 2147483647                           |
| ABS circular sub : 0DH<br>ABS circular right : 0FH<br>ABS circular left : 10H                                                                                                | ◇ Set the address<br>-214748364.8 to 214748364.7         | ◇ Set the address<br>-2147483648 to 2147483647                         |
| INC circular sub : 0EH<br>INC circular right : 11H<br>INC circular left : 12H                                                                                                | ◇ Set the movement amount<br>-214748364.8 to 214748364.7 | ◇ Set the movement amount<br>-2147483648 to 2147483647                 |
| ABS helical sub : 20H<br>ABS helical right : 22H<br>ABS helical left : 23H                                                                                                   | ◇ Set the address<br>-214748364.8 to 214748364.7         | ◇ Set the address<br>-2147483648 to 2147483647                         |
| INC helical sub : 21H<br>INC helical right : 24H<br>INC helical left : 25H                                                                                                   | ◇ Set the movement amount<br>-214748364.8 to 214748364.7 | ◇ Set the movement amount<br>-2147483648 to 2147483647                 |

\* 1: Set an integer because the sequence program cannot handle fractions.  
(The value will be converted properly within the system.)

## (3) Speed-position switching control

- INC mode:  
Set the amount of movement after the switching from speed control to position control.
- ABS mode:  
Set the absolute address which will be the target value after speed control is switched to position control. (The unit is "degree" only)



## (4) Position-speed switching control

- Set the amount of movement before the switching from position control to speed control.

■ When " Pr.1 Unit Setting" is "degree"

The table below lists the control systems that require the setting of the positioning address or movement amount and the associated setting ranges.

(With any control system excluded from the table below, neither the positioning address nor the movement amount needs to be set.)

| <span style="border: 1px solid black; padding: 0 2px;">Da.2</span> setting value                                                                                             | Value set with peripheral device<br>(degree)                                                                       | Value set with sequence program * 1<br>( $\times 10^{-5}$ degree)                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| ABS Linear 1 : 01H<br>ABS Linear 2 : 0AH<br>ABS Linear 3 : 15H<br>ABS Linear 4 : 1AH<br>Current value changing : 81H                                                         | ◇ Set the address<br>0 to 359.99999                                                                                | ◇ Set the address<br>0 to 35999999                                                                               |
| INC Linear 1 : 02H<br>INC Linear 2 : 0BH<br>INC Linear 3 : 16H<br>INC Linear 4 : 1BH<br>Fixed-feed 1 : 03H<br>Fixed-feed 2 : 0CH<br>Fixed-feed 3 : 17H<br>Fixed-feed 4 : 1CH | ◇ Set the movement amount<br>-21474.83648 to 21474.83647                                                           | ◇ Set the movement amount<br>-2147483648 to 2147483647                                                           |
| ABS helical sub : 20H*2<br>ABS helical right : 22H*2<br>ABS helical left : 23H*2                                                                                             | ◇ Set the address<br>0 to 359.99999                                                                                | ◇ Set the address<br>0 to 35999999                                                                               |
| INC helical sub : 21H*2<br>INC helical right : 24H*2<br>INC helical left : 25H*2                                                                                             | ◇ Set the movement amount<br>-21474.83648 to 21474.83647                                                           | ◇ Set the movement amount<br>-2147483648 to 2147483647                                                           |
| Forward run speed/position : 06H<br>Reverse run speed/position : 07H                                                                                                         | In INC mode<br>◇ Set the movement amount<br>0 to 21474.83647<br>In ABS mode<br>◇ Set the address<br>0 to 359.99999 | In INC mode<br>◇ Set the movement amount<br>0 to 2147483647<br>In ABS mode<br>◇ Set the address<br>0 to 35999999 |
| Forward run position/speed : 08H<br>Reverse run position/speed : 09H                                                                                                         | ◇ Set the movement amount<br>0 to 21474.83647                                                                      | ◇ Set the movement amount<br>0 to 2147483647                                                                     |

\* 1: Set an integer because the sequence program cannot handle fractions.  
(The value will be converted properly within the system.)

\* 2: In the 3-axis helical interpolation control, "degree" can be set only for the linear interpolation axis.

■ When " **Pr.1** Unit Setting" is "pulse"

The table below lists the control systems that require the setting of the positioning address or movement amount and the associated setting ranges.

(With any control system excluded from the table below, neither the positioning address nor the movement amount needs to be set.)

| <b>Da.2</b> setting value                                                                                                                                                    | Value set with peripheral device<br>(pulse)            | Value set with sequence program * 1<br>(pulse)         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|
| ABS Linear 1 : 01H<br>ABS Linear 2 : 0AH<br>ABS Linear 3 : 15H<br>ABS Linear 4 : 1AH<br>Current value changing : 81H                                                         | ◇ Set the address<br>-2147483648 to 2147483647         | ◇ Set the address<br>-2147483648 to 2147483647         |
| INC Linear 1 : 02H<br>INC Linear 2 : 0BH<br>INC Linear 3 : 16H<br>INC Linear 4 : 1BH<br>Fixed-feed 1 : 03H<br>Fixed-feed 2 : 0CH<br>Fixed-feed 3 : 17H<br>Fixed-feed 4 : 1CH | ◇ Set the movement amount<br>-2147483648 to 2147483647 | ◇ Set the movement amount<br>-2147483648 to 2147483647 |
| Forward run speed/position: 06H<br>Reverse run speed/position: 07H<br>Forward run position/speed: 08H<br>Reverse run position/speed: 09H                                     | ◇ Set the movement amount<br>0 to 2147483647           | ◇ Set the movement amount<br>0 to 2147483647           |
| ABS circular sub : 0DH<br>ABS circular right : 0FH<br>ABS circular left : 10H                                                                                                | ◇ Set the address<br>-2147483648 to 2147483647         | ◇ Set the address<br>-2147483648 to 2147483647         |
| INC circular sub : 0EH<br>INC circular right : 11H<br>INC circular left : 12H                                                                                                | ◇ Set the movement amount<br>-2147483648 to 2147483647 | ◇ Set the movement amount<br>-2147483648 to 2147483647 |
| ABS helical sub : 20H<br>ABS helical right : 22H<br>ABS helical left : 23H                                                                                                   | ◇ Set the address<br>-2147483648 to 2147483647         | ◇ Set the address<br>-2147483648 to 2147483647         |
| INC helical sub : 21H<br>INC helical right : 24H<br>INC helical left : 25H                                                                                                   | ◇ Set the movement amount<br>-2147483648 to 2147483647 | ◇ Set the movement amount<br>-2147483648 to 2147483647 |

■ When " Pr.1 Unit Setting" is "inch"

The table below lists the control systems that require the setting of the positioning address or movement amount and the associated setting ranges.

(With any control system excluded from the table below, neither the positioning address nor the movement amount needs to be set.)

| <span style="border: 1px solid black; padding: 0 2px;">Da.2</span> setting value                                                                                                                                                                                             | Value set with peripheral device<br>(inch)               | Value set with sequence program * 1<br>( $\times 10^{-5}$ inch) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------|
| ABS Linear 1 : 01 <sub>H</sub><br>ABS Linear 2 : 0A <sub>H</sub><br>ABS Linear 3 : 15 <sub>H</sub><br>ABS Linear 4 : 1A <sub>H</sub><br>Current value changing : 81 <sub>H</sub>                                                                                             | ◇ Set the address<br>-21474.83648 to 21474.83647         | ◇ Set the address<br>-2147483648 to 2147483647                  |
| INC Linear 1 : 02 <sub>H</sub><br>INC Linear 2 : 0B <sub>H</sub><br>INC Linear 3 : 16 <sub>H</sub><br>INC Linear 4 : 1B <sub>H</sub><br>Fixed-feed 1 : 03 <sub>H</sub><br>Fixed-feed 2 : 0C <sub>H</sub><br>Fixed-feed 3 : 17 <sub>H</sub><br>Fixed-feed 4 : 1C <sub>H</sub> | ◇ Set the movement amount<br>-21474.83648 to 21474.83647 | ◇ Set the movement amount<br>-2147483648 to 2147483647          |
| Forward run speed/position: 06 <sub>H</sub><br>Reverse run speed/position: 07 <sub>H</sub><br>Forward run position/speed: 08 <sub>H</sub><br>Reverse run position/speed: 09 <sub>H</sub>                                                                                     | ◇ Set the movement amount<br>0 to 21474.83647            | ◇ Set the movement amount<br>0 to 2147483647                    |
| ABS circular sub : 0D <sub>H</sub><br>ABS circular right : 0F <sub>H</sub><br>ABS circular left : 10 <sub>H</sub>                                                                                                                                                            | ◇ Set the address<br>-21474.83648 to 21474.83647         | ◇ Set the address<br>-2147483648 to 2147483647                  |
| INC circular sub : 0E <sub>H</sub><br>INC circular right : 11 <sub>H</sub><br>INC circular left : 12 <sub>H</sub>                                                                                                                                                            | ◇ Set the movement amount<br>-21474.83648 to 21474.83647 | ◇ Set the movement amount<br>-2147483648 to 2147483647          |
| ABS helical sub : 20 <sub>H</sub><br>ABS helical right : 22 <sub>H</sub><br>ABS helical left : 23 <sub>H</sub>                                                                                                                                                               | ◇ Set the address<br>-21474.83648 to 21474.83647         | ◇ Set the address<br>-2147483648 to 2147483647                  |
| INC helical sub : 21 <sub>H</sub><br>INC helical right : 24 <sub>H</sub><br>INC helical left : 25 <sub>H</sub>                                                                                                                                                               | ◇ Set the movement amount<br>-21474.83648 to 21474.83647 | ◇ Set the movement amount<br>-2147483648 to 2147483647          |

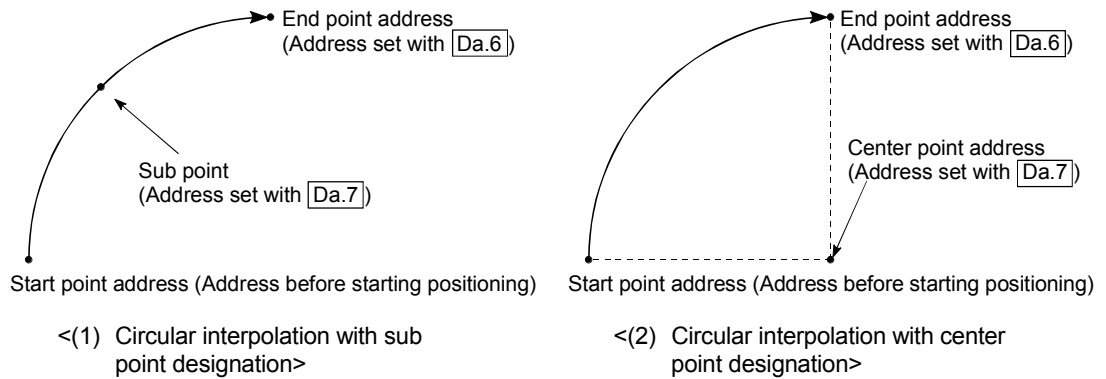
\* 1: Set an integer because the sequence program cannot handle fractions.  
(The value will be converted properly within the system.)

| Item                       | Setting value, setting range                                                                                                                                                                                                                                                                           |                                 | Default value | Setting value buffer memory address |              |                |                |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------|-------------------------------------|--------------|----------------|----------------|
|                            | Value set with peripheral device                                                                                                                                                                                                                                                                       | Value set with sequence program |               | Axis 1                              | Axis 2       | Axis 3         | Axis 4         |
| <b>Da.7</b><br>Arc address | The setting value range differs according to the " <b>Da.2</b> Control system".<br>Here, the value within the [Table 1] range is set. <div style="text-align: center;">  <b>[Table 1]</b> on next page         </div> |                                 | 0             | 2008<br>2009                        | 8008<br>8009 | 14008<br>14009 | 20008<br>20009 |

### **Da.7** Arc address

The arc address is data required only when the circular interpolation control or 3-axis helical interpolation control is carried out.

- (1) When carrying out circular interpolation with sub point designation, set the sub point (passing point) address as the arc address.
- (2) When carrying out circular interpolation with center point designation, set the center point address of the arc as the arc address.



When the circular interpolation control or 3-axis helical interpolation control is not carried out, the value set in "**Da.7** Arc address" will be invalid.

[Table 1]

## ■ When " [Pr.1] Unit Setting" is "mm"

The table below lists the control systems that require the setting of the arc address and shows the setting range.

(With any control system excluded from the table below, the arc address does not need to be set.)

| [Da.2] setting value                                                                                              | Value set with peripheral device (μm)                        | Value set with sequence program * 1 (×10 <sup>-1</sup> μm) |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------|
| ABS circular sub : 0D <sub>H</sub><br>ABS circular right : 0F <sub>H</sub><br>ABS circular left : 10 <sub>H</sub> | ◇ Set the address<br>-214748364.8 to 214748364.7             | ◇ Set the address<br>-2147483648 to 2147483647             |
| INC circular sub : 0E <sub>H</sub><br>INC circular right : 11 <sub>H</sub><br>INC circular left : 12 <sub>H</sub> | ◇ Set the movement amount<br>-214748364.8 to 214748364.7     | ◇ Set the movement amount<br>-2147483648 to 2147483647     |
| ABS helical sub : 20 <sub>H</sub><br>ABS helical right : 22 <sub>H</sub><br>ABS helical left : 23 <sub>H</sub>    | ◇ Set the address<br>-214748364.8 to 214748364.7             | ◇ Set the address<br>-2147483648 to 2147483647             |
| INC helical sub : 21 <sub>H</sub><br>INC helical right : 24 <sub>H</sub><br>INC helical left : 25 <sub>H</sub>    | ◇ Set the movement amount<br>-214748364.8 to 214748364.7 * 2 | ◇ Set the movement amount<br>-2147483648 to 2147483647 * 2 |

\* 1: Set an integer because the sequence program cannot handle fractions.  
(The value will be converted properly within the system.)

\* 2: Note that the maximum radius that circular interpolation control is possible is 536870912, although the setting value can be input within the range shown in the above table, as an arc address.

## ■ When " [Pr.1] Unit Setting" is "degree"

No control system requires the setting of the arc address by "degree".

## ■ When " [Pr.1] Unit Setting" is "pulse"

The table below lists the control systems that require the setting of the arc address and shows the setting range.

(With any control system excluded from the table below, the arc address does not need to be set.)

| [Da.2] setting value                                                                                              | Value set with peripheral device (pulse)                   | Value set with sequence program (pulse)                    |
|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|
| ABS circular sub : 0D <sub>H</sub><br>ABS circular right : 0F <sub>H</sub><br>ABS circular left : 10 <sub>H</sub> | ◇ Set the address<br>-2147483648 to 2147483647             | ◇ Set the address<br>-2147483648 to 2147483647             |
| INC circular sub : 0E <sub>H</sub><br>INC circular right : 11 <sub>H</sub><br>INC circular left : 12 <sub>H</sub> | ◇ Set the movement amount<br>-2147483648 to 2147483647 * 1 | ◇ Set the movement amount<br>-2147483648 to 2147483647 * 1 |
| ABS helical sub : 20 <sub>H</sub><br>ABS helical right : 22 <sub>H</sub><br>ABS helical left : 23 <sub>H</sub>    | ◇ Set the address<br>-2147483648 to 2147483647             | ◇ Set the address<br>-2147483648 to 2147483647             |
| INC helical sub : 21 <sub>H</sub><br>INC helical right : 24 <sub>H</sub><br>INC helical left : 25 <sub>H</sub>    | ◇ Set the movement amount<br>-2147483648 to 2147483647 * 1 | ◇ Set the movement amount<br>-2147483648 to 2147483647 * 1 |

\* 1: Note that the maximum radius that circular interpolation control is possible is 536870912, although the setting value can be input within the range shown in the above table, as an arc address.



■ When " **Pr.1** Unit Setting" is "inch"

The table below lists the control systems that require the setting of the arc address and shows the setting range.

(With any control system excluded from the table below, the arc address does not need to be set.)

| <b>Da.2</b> setting value                                                                                         | Value set with peripheral device<br>(inch)                   | Value set with sequence program *1<br>( $\times 10^{-5}$ inch) |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------|
| ABS circular sub : 0D <sub>H</sub><br>ABS circular right : 0F <sub>H</sub><br>ABS circular left : 10 <sub>H</sub> | ◇ Set the address<br>-21474.83648 to 21474.83647             | ◇ Set the address<br>-2147483648 to 2147483647                 |
| INC circular sub : 0E <sub>H</sub><br>INC circular right : 11 <sub>H</sub><br>INC circular left : 12 <sub>H</sub> | ◇ Set the movement amount<br>-21474.83648 to 21474.83647     | ◇ Set the movement amount<br>-2147483648 to 2147483647         |
| ABS helical sub : 20 <sub>H</sub><br>ABS helical right : 22 <sub>H</sub><br>ABS helical left : 23 <sub>H</sub>    | ◇ Set the address<br>-21474.83648 to 21474.83647             | ◇ Set the address<br>-2147483648 to 2147483647                 |
| INC helical sub : 21 <sub>H</sub><br>INC helical right : 24 <sub>H</sub><br>INC helical left : 25 <sub>H</sub>    | ◇ Set the movement amount<br>-21474.83648 to 21474.83647 * 2 | ◇ Set the movement amount<br>-2147483648 to 2147483647 * 2     |

\* 1: Set an integer because the sequence program cannot handle fractions.  
(The value will be converted properly within the system.)

\* 2: Note that the maximum radius that circular interpolation control is possible is 536870912, although the setting value can be input within the range shown in the above table, as an arc address.

| Item                          |                                       | Setting value, setting range                                                                                                                                                                                                                                                                                                                                         |                                 | Default value | Setting value buffer memory address |              |                |                |
|-------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------|-------------------------------------|--------------|----------------|----------------|
|                               |                                       | Value set with peripheral device                                                                                                                                                                                                                                                                                                                                     | Value set with sequence program |               | Axis 1                              | Axis 2       | Axis 3         | Axis 4         |
| <div>Da.8</div> Command speed |                                       | <div>The setting value range differs depending on the " <div>Pr.1</div> Unit setting".<br/>Here, the value within the [Table 1] range is set.</div> <div> [Table 1] on next page</div> <div>-1: Current speed<br/>(Speed set for previous positioning data No.)</div> <div>-1</div> |                                 | 0             | 2004<br>2005                        | 8004<br>8005 | 14004<br>14005 | 20004<br>20005 |
| <div>Da.9</div> Dwell time    | Dwell time                            | <div>The setting value range differs according to the " <div>Da.2</div> Control system".<br/>Here, the value within the [Table 2] range is set.</div> <div> [Table 2] on next page</div>                                                                                            |                                 | 0             | 2002                                | 8002         | 14002          | 20002          |
|                               | JUMP destination positioning data No. |                                                                                                                                                                                                                                                                                                                                                                      |                                 |               |                                     |              |                |                |
| <div>Da.10</div> M code       | M code                                | <div>The setting value range differs according to the " <div>Da.2</div> Control system".<br/>Here, the value within the [Table 3] range is set.</div> <div> [Table 3] on next page</div>                                                                                           |                                 | 0             | 2001                                | 8001         | 14001          | 20001          |
|                               | Condition data No.                    |                                                                                                                                                                                                                                                                                                                                                                      |                                 |               |                                     |              |                |                |
|                               | No. of LOOP to LEND repetitions       |                                                                                                                                                                                                                                                                                                                                                                      |                                 |               |                                     |              |                |                |

**Da.8 Command speed**

Set the command speed for positioning.

- (1) If the set command speed exceeds " Pr.8 Speed limit value", positioning will be carried out at the speed limit value.
- (2) If "-1" is set for the command speed, the current speed (speed set for previous positioning data No.) will be used for positioning control. Use the current speed for uniform speed control, etc. If "-1" is set for continuing positioning data, and the speed is changed, the following speed will also change.

(Note that when starting positioning, if the "-1" speed is set for the positioning data that carries out positioning control first, the error "No command speed" (error code: 503) will occur, and the positioning will not start.  
Refer to Section 15.3 "List of errors" for details on the errors.)

**Da.10** M code (or condition data No./No. of LOOP to LEND repetitions)

Set an "M code", a "condition data No. ", or the "number of LOOP to LEND repetitions" depending on how the " **Da.2** Control system" is set.

- If a method other than "JUMP instruction" and "LOOP" is selected as the " **Da.2** Control system"
  - ..... Set an "M code".
    - If no "M code" needs to be output, set "0" (default value).
- If "3-axis helical interpolation control" is selected as the " **Da.2** Control system": Set the number of pitch for the linear interpolation axis. Set the rotation speed of the circular interpolation as the number of pitch.
- If "JUMP instruction" is selected as the " **Da.2** Control system"
  - ..... Set the "condition data No." for JUMP.
    - 0 : Unconditional JUMP to the positioning data specified by **Da.9**.
    - 1 to 10 : JUMP performed according to the condition data No. specified (a number between 1 and 10).
- If "LOOP" is selected as the " **Da.2** Control system"
  - ..... Set the number of LOOP to LEND repetitions by a number other than "0".
    - The error "Control system LOOP setting error" (error code: 545) will occur if you specify "0".

\* The condition data specifies the condition for the JUMP instruction to be executed.

(A JUMP will take place when the condition is satisfied.)

[Table 1]

| <b>Pr.1</b> setting value | Value set with peripheral device<br>(unit) | Value set with sequence program<br>(unit)                                               |
|---------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------|
| 0 : mm                    | 0.01 to 20000000.00 (mm/min)               | 1 to 2000000000 ( $\times 10^{-2}$ mm/min)                                              |
| 1 : inch                  | 0.001 to 2000000.000 (inch/min)            | 1 to 2000000000 ( $\times 10^{-3}$ inch/min)                                            |
| 2 : degree                | 0.001 to 2000000.000 (degree/min)          | 1 to 2000000000 ( $\times 10^{-3}$ degree/min)                                          |
| 3 : pulse                 | 1 to 1000000 (pulse/s) * 1                 | [QD75P□/QD75D□N]<br>1 to 4000000 (pulse/s)<br>[QD75P□/QD75D□]<br>1 to 1000000 (pulse/s) |

\* 1: In GX Configurator-QP, the QD75P□/QD75D□N are recognized as the QD75P□/QD75D□. Therefore, the setting ranges are the same as those of the QD75P□/QD75D□. To set a value outside a setting range in GX Configurator-QP, set it through a sequence program or GX Works2. The setting ranges in GX Works2 are the same as those in sequence programs. (Refer to Appendix 1.2)

[Table 2]

| <b>Da.2</b> setting value   | Setting item         | Value set with peripheral device | Value set with sequence program |
|-----------------------------|----------------------|----------------------------------|---------------------------------|
| JUMP instruction: 82H       | Positioning data No. | 1 to 600                         | 1 to 600                        |
| Other than JUMP instruction | Dwell time           | 0 to 65535 (ms)                  | 0 to 65535 (ms)                 |

[Table 3]

| <b>Da.2</b> setting value         | Setting item       | Value set with peripheral device | Value set with sequence program |
|-----------------------------------|--------------------|----------------------------------|---------------------------------|
| JUMP instruction: 82H             | Condition data No. | 0 to 10                          | 0 to 10                         |
| LOOP: 83H                         | Repetition count   | 1 to 65535                       | 1 to 65535                      |
| Helical interpolation: 20H to 25H | Number of pitch    | 0 to 999                         | 0 to 999                        |
| Other than the above              | M code             | 0 to 65535                       | 0 to 65535                      |

**Da.9** Dwell time (JUMP designation positioning data No.)

Set the "dwell time" or "positioning data No." corresponding to the "**Da.2** Control system".

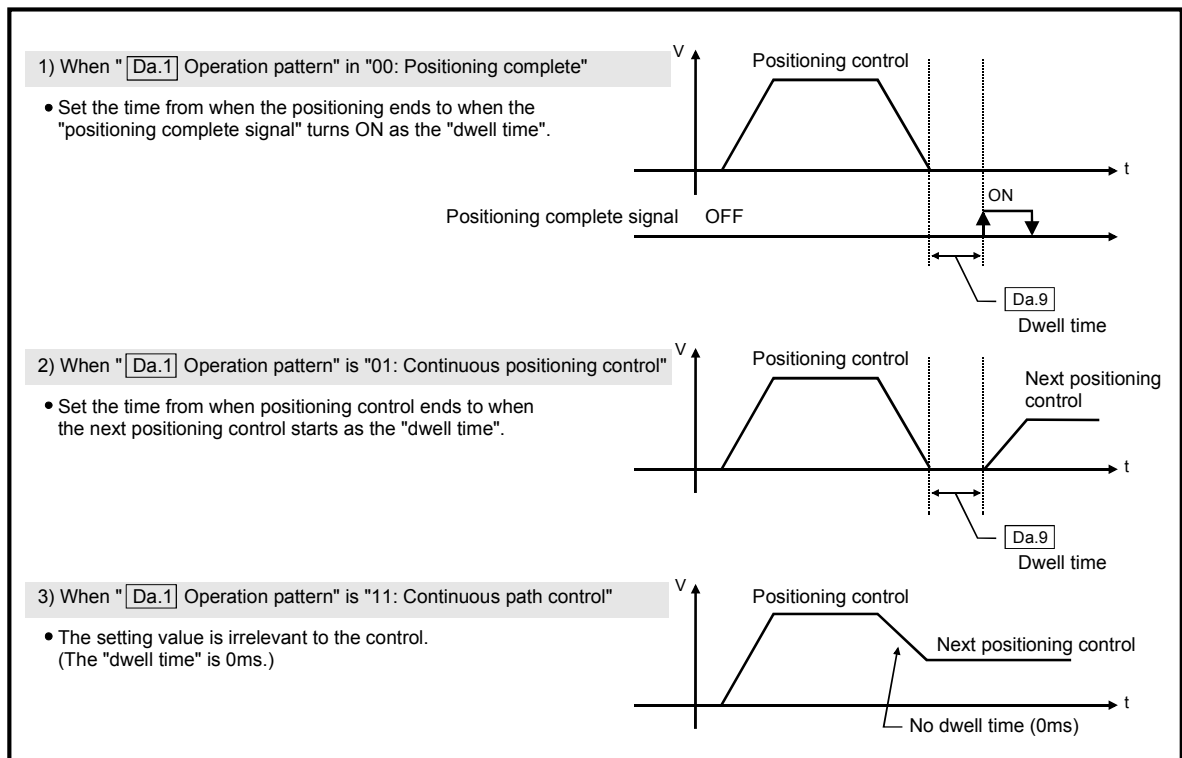
- When a method other than "JUMP instruction " is set for "**Da.2** Control system"

..... Set the "dwell time".

- When "JUMP instruction " is set for "**Da.2** Control system"

..... Set the "positioning data No." for the JUMP destination.

When the "dwell time" is set, the setting details of the "dwell time" will be as follows according to "**Da.1** Operation pattern".



| Item               |                                                 | Setting value, setting range                                              |                                 | Default value | Setting value buffer memory address |        |        |        |
|--------------------|-------------------------------------------------|---------------------------------------------------------------------------|---------------------------------|---------------|-------------------------------------|--------|--------|--------|
|                    |                                                 | Value set with peripheral                                                 | Value set with sequence program |               | Axis 1                              | Axis 2 | Axis 3 | Axis 4 |
| Positioning option | Da.27<br>M code ON signal output timing         | 0: Use the set value in "[Pr.18] M code ON signal output timing".         | 0                               | 0             | 2003                                | 8003   | 14003  | 20003  |
|                    |                                                 | 1: WITH mode                                                              | 1                               |               |                                     |        |        |        |
|                    |                                                 | 2: AFTER mode                                                             | 2                               |               |                                     |        |        |        |
|                    | Da.28<br>ABS direction in degrees               | 0: Use the set value in "[Cd.40] ABS direction in degrees".               | 0                               | 0             |                                     |        |        |        |
|                    |                                                 | 1: ABS clockwise                                                          | 1                               |               |                                     |        |        |        |
|                    |                                                 | 2: ABS counterclockwise                                                   | 2                               |               |                                     |        |        |        |
|                    |                                                 | 3: Takes a shortcut. (direction setting ignored)                          | 3                               |               |                                     |        |        |        |
|                    | Da.29<br>Interpolation speed designation method | 0: Use the set value in "[Pr.20] Interpolation speed designation method". | 0                               | 0             |                                     |        |        |        |
|                    |                                                 | 1: Composite speed                                                        | 1                               |               |                                     |        |        |        |
|                    |                                                 | 2: Reference axis speed                                                   | 2                               |               |                                     |        |        |        |

### Positioning option configuration

The positioning option consists of "Da.27 M code ON signal output timing" to "Da.29 Interpolation speed designation method". The set values of those data are stored in one buffer memory address. Consider the positioning option configuration in the figure below when setting "Da.27 M code ON signal output timing" to "Da.29 Interpolation speed designation method".

| Positioning option configuration | Assignment                                       |
|----------------------------------|--------------------------------------------------|
|                                  | (1) Da.27 M code ON signal output timing         |
|                                  | (2) Da.28 ABS direction in degrees               |
|                                  | (3) Da.29 Interpolation speed designation method |
|                                  | (4) Use prohibited (fixed to 0)                  |

#### Da.27 M code ON signal output timing

Set the M code ON signal output timing for each positioning data.

#### Da.28 ABS direction in degrees

Set the ABS moving direction when "degree" is selected as the unit for each positioning data.

#### Da.29 Interpolation speed designation method

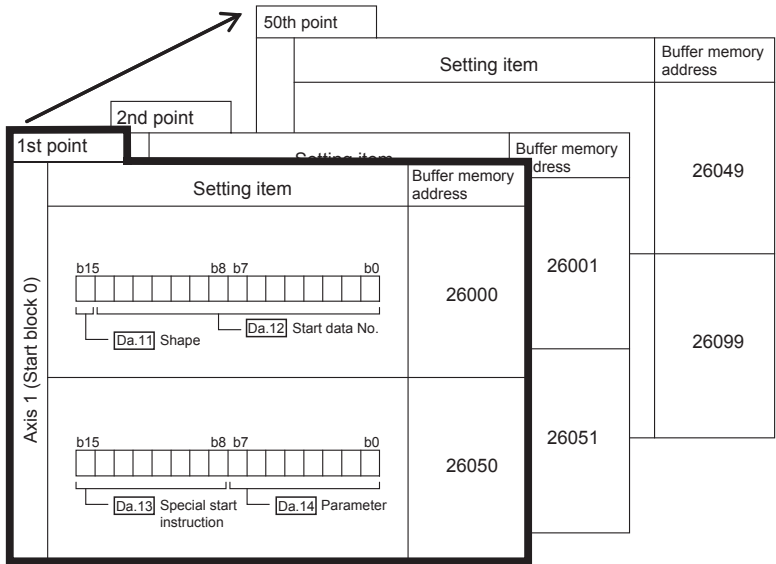
Set the interpolation speed designation method for each positioning data.

### POINTS

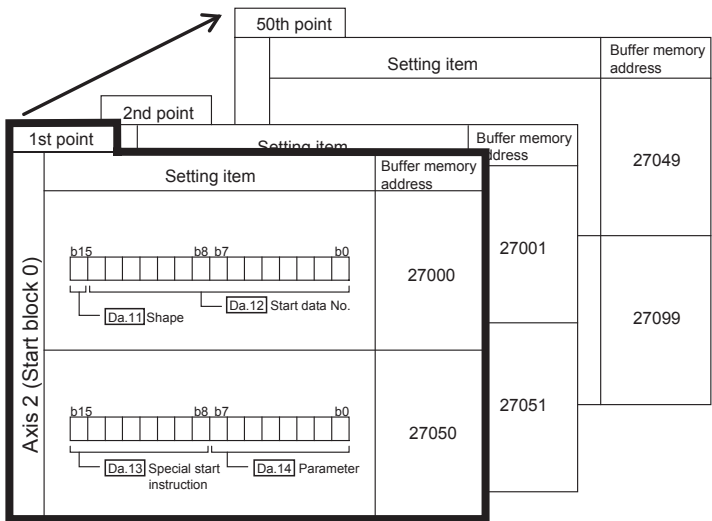
- (1) To use the positioning option, set "1: Valid" in Pr.70 Positioning option valid/invalid setting.
- (2) The positioning option can be set only with GX Works2 (Version 1.540N or later) or sequence programs.
- (3) The positioning option can be set only for the QD75N with a serial number (first five digits) of "17102" or later. If a value is set for the positioning option of the QD75N with a serial number (first five digits) earlier than "17102" and a positioning control is started, the value set in the positioning option is ignored.

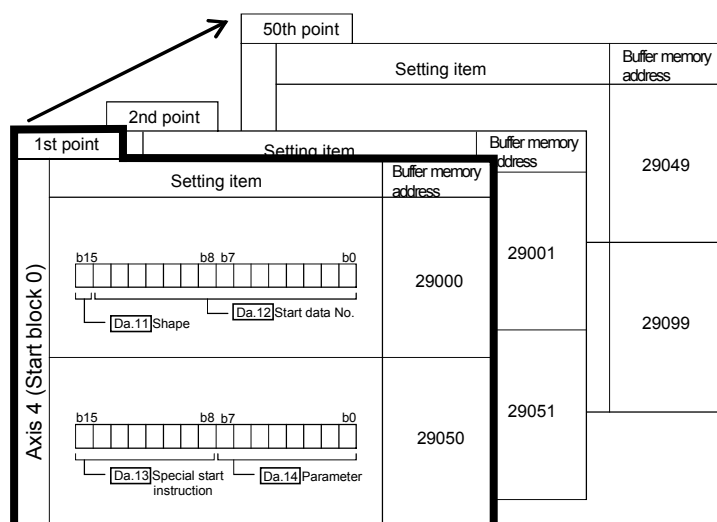
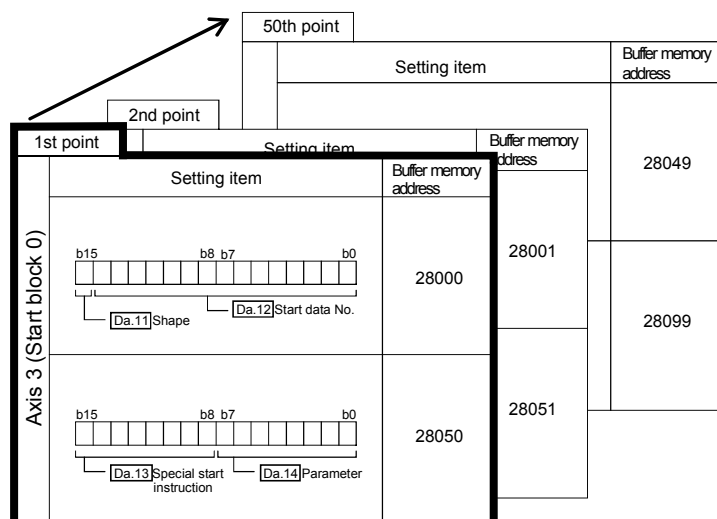
5.4 List of block start data

The illustrations below show the organization of the block start data stored in the QD75 buffer memory. The block start data setting items **Da.11** to **Da.14** are explained in the pages that follow.



- Up to 50 block start data points can be set (stored) for each axis in the buffer memory addresses shown on the left.
  - Items in a single unit of block start data are shown included in a bold frame. **□**
  - Each axis has five start blocks (block Nos. 0 to 4).
- \* For information on the organization of the buffer memory addresses assigned to the start blocks 1 to 4, refer to Appendix 13 "List of buffer memory addresses".





The pages that follow explain the block start data setting items (Da.11 to Da.14).  
 (The buffer memory addresses shown are those of the "1st point block start data (block No. 7000)" for the axes 1 to 4.)

**REMARK**

To perform an high-level positioning control using block start data, set a number between 7000 and 7004 to the " [Cd.3] Positioning start No." and use the " [Cd.4] Positioning starting point No." to specify a point number between 1 and 50, a position counted from the beginning of the block.

The number between 7000 and 7004 specified here is called the "block No.".

With the QD75, up to 50 "block start data" points and up to 10 "condition data" items can be assigned to each "block No.".

| Block No. * | Axis   | Block start data | Condition                | Buffer memory         | GX Configurator-QP    |
|-------------|--------|------------------|--------------------------|-----------------------|-----------------------|
| 7000        | Axis 1 | Start block 0    | Condition data (1 to 10) | Supports the settings | Supports the settings |
|             | Axis 2 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 3 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 4 |                  | Condition data (1 to 10) |                       |                       |
| 7001        | Axis 1 | Start block 1    | Condition data (1 to 10) |                       |                       |
|             | Axis 2 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 3 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 4 |                  | Condition data (1 to 10) |                       |                       |
| 7002        | Axis 1 | Start block 2    | Condition data (1 to 10) |                       |                       |
|             | Axis 2 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 3 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 4 |                  | Condition data (1 to 10) |                       |                       |
| 7003        | Axis 1 | Start block 3    | Condition data (1 to 10) |                       |                       |
|             | Axis 2 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 3 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 4 |                  | Condition data (1 to 10) |                       |                       |
| 7004        | Axis 1 | Start block 4    | Condition data (1 to 10) |                       |                       |
|             | Axis 2 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 3 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 4 |                  | Condition data (1 to 10) |                       |                       |

\* : Setting cannot be made when the "Pre-reading start function" is used. If you set any of Nos. 7000 to 7004 and perform the Pre-reading start function, the error "Outside start No. range" (error code: 543) will occur.  
(For details, refer to Section 12.7.7 "Pre-reading start function".)



| Item                               | Setting value                                                                                                                                 |                                           | Default value     | Setting value buffer memory address |        |        |        |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------|-------------------------------------|--------|--------|--------|
|                                    | Value set with peripheral device                                                                                                              | Value set with sequence program           |                   | Axis 1                              | Axis 2 | Axis 3 | Axis 4 |
| Da.11<br>Shape                     | 0 : End                                                                                                                                       | 0                                         | 0000 <sub>H</sub> | 26000                               | 27000  | 28000  | 29000  |
|                                    | 1 : Continue                                                                                                                                  | 1                                         |                   |                                     |        |        |        |
| Da.12<br>Start data No.            | Positioning data No.:<br>1 to 600<br>(01 <sub>H</sub> to 258 <sub>H</sub> )                                                                   | 01 <sub>H</sub><br>to<br>258 <sub>H</sub> |                   |                                     |        |        |        |
| Da.13<br>Special start instruction | 0 : Block start (normal start)                                                                                                                | 00 <sub>H</sub>                           | 0000 <sub>H</sub> | 26050                               | 27050  | 28050  | 29050  |
|                                    | 1 : Condition start                                                                                                                           | 01 <sub>H</sub>                           |                   |                                     |        |        |        |
|                                    | 2 : Wait start                                                                                                                                | 02 <sub>H</sub>                           |                   |                                     |        |        |        |
|                                    | 3 : Simultaneous start                                                                                                                        | 03 <sub>H</sub>                           |                   |                                     |        |        |        |
|                                    | 4 : FOR loop                                                                                                                                  | 04 <sub>H</sub>                           |                   |                                     |        |        |        |
|                                    | 5 : FOR condition                                                                                                                             | 05 <sub>H</sub>                           |                   |                                     |        |        |        |
|                                    | 6 : NEXT start                                                                                                                                | 06 <sub>H</sub>                           |                   |                                     |        |        |        |
| Da.14<br>Parameter                 | Condition data No.:<br>1 to 10 (01 <sub>H</sub> to 0A <sub>H</sub> )<br>No. of repetitions:<br>0 to 255 (00 <sub>H</sub> to FF <sub>H</sub> ) | 00 <sub>H</sub><br>to<br>FF <sub>H</sub>  |                   |                                     |        |        |        |

**Da.11** Shape

Set whether to carry out only the local "block start data" and then end control, or to execute the "block start data" set in the next point.

| Setting value | Setting details                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| 0 : End       | Execute the designated point's "block start data", and then complete the control.                                             |
| 1 : Continue  | Execute the designated point's "block start data", and after completing control, execute the next point's "block start data". |

**Da.12** Start data No.

Set the "positioning data No." designated with the "block start data".

**Da.13** Special start instruction

Set the "special start instruction " for using "high-level positioning control". (Set how to start the positioning data set in " **Da.12** Start data No.".)

| Setting value                                    | Setting details                                                                                                                                                                                                                                                                  |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00 <sub>H</sub> : Block start (Normal start)     | Execute the random block positioning data in the set order with one start.                                                                                                                                                                                                       |
| 01 <sub>H</sub> : Condition start                | Carry out the condition judgment set in "condition data" for the designated positioning data, and when the conditions are established, execute the "block start data". If not established, ignore that "block start data", and then execute the next point's "block start data". |
| 02 <sub>H</sub> : Wait start                     | Carry out the condition judgment set in "condition data" for the designated positioning data, and when the conditions are established, execute the "block start data". If not established, stop the control (wait) until the conditions are established.                         |
| 03 <sub>H</sub> : Simultaneous start             | Simultaneous execute (output pulses at same timing) the positioning data with the No. designated for the axis designated in the "condition data".<br>Up to four axes can start simultaneously.                                                                                   |
| 04 <sub>H</sub> : Repeated start (FOR loop)      | Repeat the program from the block start data with the "FOR loop" to the block start data with "NEXT" for the designated No. of times.                                                                                                                                            |
| 05 <sub>H</sub> : Repeated start (FOR condition) | Repeat the program from the block start data with the "FOR condition" to the block start data with "NEXT" until the conditions set in the "condition data" are established.                                                                                                      |
| 06 <sub>H</sub> : NEXT start                     | Set the end of the repetition when "05 <sub>H</sub> : Repetition start (FOR loop)" or "06 <sub>H</sub> : Repetition start (FOR condition)" is set.                                                                                                                               |

Refer to CHAPTER 10 "HIGH-LEVEL POSITIONING CONTROL" for details on the control.

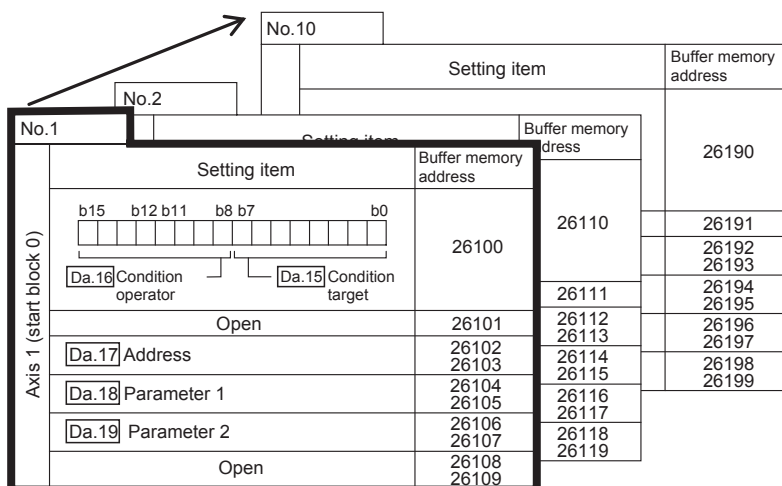
**Da.14** Parameter

Set the value as required for " **Da.13** Special start instruction ".

| <b>Da.13</b> Special start instruction | Setting value | Setting details                                                                                                                                      |
|----------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Block start (Normal start)             | –             | Not used. (There is no need to set.)                                                                                                                 |
| Condition start                        | 1 to 10       | Set the condition data No. (No. of "condition data" set to perform condition judgment)<br>(For details of the condition data, refer to Section 5.5.) |
| Wait start                             |               |                                                                                                                                                      |
| Simultaneous start                     |               |                                                                                                                                                      |
| Repeated start (FOR loop)              | 0 to 255      | Set the No. of repetitions.                                                                                                                          |
| Repeated start (FOR condition)         | 1 to 10       | Set the condition data No. (No. of "condition data" set to perform condition judgment)<br>(For details of the condition data, refer to Section 5.5.) |

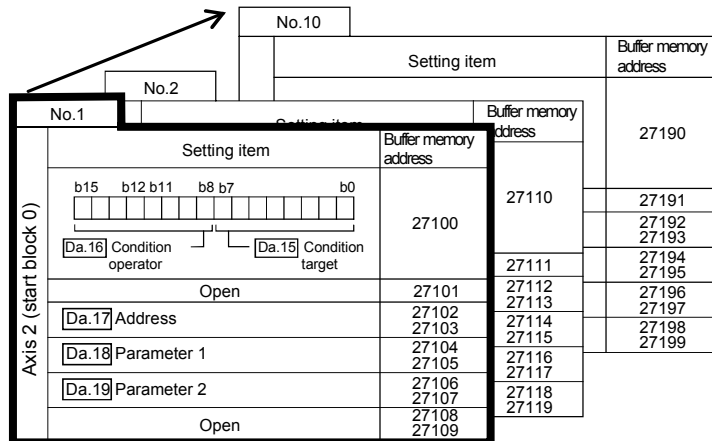
## 5.5 List of condition data

The illustrations below show the organization of the condition data stored in the QD75 buffer memory. The condition data setting items **Da.15** to **Da.19** are explained in the pages that follow.



- Up to 10 block start data points can be set (stored) for each block No. in the buffer memory addresses shown on the left.
- Items in a single unit of condition data are shown included in a bold frame. **Da.15**
- Each axis has five start blocks (block Nos. 0 to 4).

\* For information on the organization of the buffer memory addresses assigned to the start blocks 1 to 4, refer to Appendix 13 "List of buffer memory addresses".



|                        |                                                 |              |              |                       |
|------------------------|-------------------------------------------------|--------------|--------------|-----------------------|
|                        |                                                 | No.10        |              |                       |
|                        |                                                 | No.2         | Setting item | Buffer memory address |
| Axis 3 (start block 0) | No.1                                            | Setting item |              | Buffer memory address |
|                        | b15 b12 b11 b8 b7 b0                            |              | 28100        | 28110                 |
|                        | Da.16 Condition operator Da.15 Condition target |              |              |                       |
|                        | Open                                            |              | 28101        | 28111                 |
|                        | Da.17 Address                                   |              | 28102        | 28112                 |
|                        |                                                 |              | 28103        | 28113                 |
|                        | Da.18 Parameter 1                               |              | 28104        | 28114                 |
|                        |                                                 |              | 28105        | 28115                 |
|                        | Da.19 Parameter 2                               |              | 28106        | 28116                 |
|                        |                                                 |              | 28107        | 28117                 |
|                        | Open                                            |              | 28108        | 28118                 |
|                        |                                                 |              | 28109        | 28119                 |
|                        |                                                 |              |              | 28190                 |
|                        |                                                 |              | 28191        |                       |
|                        |                                                 |              | 28192        |                       |
|                        |                                                 |              | 28193        |                       |
|                        |                                                 |              | 28194        |                       |
|                        |                                                 |              | 28195        |                       |
|                        |                                                 |              | 28196        |                       |
|                        |                                                 |              | 28197        |                       |
|                        |                                                 |              | 28198        |                       |
|                        |                                                 |              | 28199        |                       |

|                        |                                                 |              |                       |                       |
|------------------------|-------------------------------------------------|--------------|-----------------------|-----------------------|
|                        |                                                 | No.10        |                       |                       |
|                        |                                                 | No.2         | Setting item          | Buffer memory address |
| Axis 4 (start block 0) | No.1                                            | Setting item | Buffer memory address | 29190                 |
|                        | Setting item                                    |              | Buffer memory address |                       |
|                        | b15 b12 b11 b8 b7 b0                            |              | 29100                 |                       |
|                        | Da.16 Condition operator Da.15 Condition target |              |                       |                       |
|                        | Open                                            |              | 29101                 |                       |
|                        | Da.17 Address                                   |              | 29102                 |                       |
|                        |                                                 |              | 29103                 |                       |
|                        | Da.18 Parameter 1                               |              | 29104                 |                       |
|                        |                                                 |              | 29105                 |                       |
|                        | Da.19 Parameter 2                               |              | 29106                 |                       |
|                        |                                                 |              | 29107                 |                       |
|                        | Open                                            |              | 29108                 |                       |
|                        |                                                 |              | 29109                 |                       |
|                        |                                                 |              | 29110                 |                       |
|                        |                                                 |              | 29111                 |                       |
|                        |                                                 |              | 29112                 |                       |
|                        |                                                 |              | 29113                 |                       |
|                        |                                                 |              | 29114                 |                       |
|                        |                                                 |              | 29115                 |                       |
|                        |                                                 |              | 29116                 |                       |
|                        |                                                 |              | 29117                 |                       |
|                        |                                                 |              | 29118                 |                       |
|                        |                                                 |              | 29119                 |                       |

The pages that follow explain the condition data setting items (Da.15 to Da.19).  
 (The buffer memory addresses shown are those of the "condition data No. 1 (block No. 7000)" for the axes 1 to 4.)

**REMARK**

To perform an high-level positioning control using block start data, set a number between 7000 and 7004 to the " [Cd.3] Positioning start No." and use the " [Cd.4] Positioning starting point No." to specify a point number between 1 and 50, a position counted from the beginning of the block.

The number between 7000 and 7004 specified here is called the "block No.".

With the QD75, up to 50 "block start data" points and up to 10 "condition data" items can be assigned to each "block No.".

| Block No. * | Axis   | Block start data | Condition                | Buffer memory         | GX Configurator-QP    |
|-------------|--------|------------------|--------------------------|-----------------------|-----------------------|
| 7000        | Axis 1 | Start block 0    | Condition data (1 to 10) | Supports the settings | Supports the settings |
|             | Axis 2 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 3 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 4 |                  | Condition data (1 to 10) |                       |                       |
| 7001        | Axis 1 | Start block 1    | Condition data (1 to 10) |                       |                       |
|             | Axis 2 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 3 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 4 |                  | Condition data (1 to 10) |                       |                       |
| 7002        | Axis 1 | Start block 2    | Condition data (1 to 10) |                       |                       |
|             | Axis 2 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 3 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 4 |                  | Condition data (1 to 10) |                       |                       |
| 7003        | Axis 1 | Start block 3    | Condition data (1 to 10) |                       |                       |
|             | Axis 2 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 3 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 4 |                  | Condition data (1 to 10) |                       |                       |
| 7004        | Axis 1 | Start block 4    | Condition data (1 to 10) |                       |                       |
|             | Axis 2 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 3 |                  | Condition data (1 to 10) |                       |                       |
|             | Axis 4 |                  | Condition data (1 to 10) |                       |                       |

\*: Setting cannot be made when the "Pre-reading start function" is used. If you set any of Nos. 7000 to 7004 and perform the Pre-reading start function, the error "Outside start No. range" (error code: 543) will occur.

(For details, refer to Section 12.7.7 "Pre-reading start function".)

| Item                 |                             | Setting value                                                                                                |                                 | Default value                                                             | Setting value buffer memory address |                |                |                |       |
|----------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------|-------------------------------------|----------------|----------------|----------------|-------|
|                      |                             | Value set with peripheral device                                                                             | Value set with sequence program |                                                                           | Axis 1                              | Axis 2         | Axis 3         | Axis 4         |       |
| Condition identifier | Da.15<br>Condition target   | 01 : Device X                                                                                                | 01H                             | <p>● Condition target</p> <p>● Condition operator</p> <p>b15 b8 b7 b0</p> | 0000H                               | 26100          | 27100          | 28100          | 29100 |
|                      |                             | 02 : Device Y                                                                                                | 02H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 03 : Buffer memory (1-word)                                                                                  | 03H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 04 : Buffer memory (2-word)                                                                                  | 04H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 05 : Positioning data No.                                                                                    | 05H                             |                                                                           |                                     |                |                |                |       |
|                      | Da.16<br>Condition operator | 01 : **=P1                                                                                                   | 01H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 02 : **≠P1                                                                                                   | 02H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 03 : **≤P1                                                                                                   | 03H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 04 : **≥P1                                                                                                   | 04H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 05 : P1≤**≤ P2                                                                                               | 05H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 06 : **≤P1, P2≤**                                                                                            | 06H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 07 : DEV=ON                                                                                                  | 07H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 08 : DEV=OFF                                                                                                 | 08H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 10: Axis 1 selected                                                                                          | 10H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 20: Axis 2 selected                                                                                          | 20H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 30: Axes 1 and 2 selected                                                                                    | 30H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 40: Axis 3 selected                                                                                          | 40H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 50: Axes 1 and 3 selected                                                                                    | 50H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 60: Axes 2 and 3 selected                                                                                    | 60H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 70: Axes 1, 2, and 3 selected                                                                                | 70H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 80: Axis 4 selected                                                                                          | 80H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | 90: Axes 1 and 4 selected                                                                                    | 90H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | A0: Axes 2 and 4 selected                                                                                    | A0H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | B0: Axes 1, 2, and 4 selected                                                                                | B0H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | C0: Axes 3 and 4 selected                                                                                    | C0H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | D0: Axes 1, 3, and 4 selected                                                                                | D0H                             |                                                                           |                                     |                |                |                |       |
|                      |                             | E0: Axes 2, 3, and 4 selected                                                                                | E0H                             |                                                                           |                                     |                |                |                |       |
| Da.17<br>Address     | Buffer memory address       | Example)<br><p>26103 26102</p> <p>b31 (High-order) b16 b15 (Low-order) b0</p> <p>Buffer memory address →</p> |                                 | 0000H                                                                     | 26102<br>26103                      | 27102<br>27103 | 28102<br>28103 | 29102<br>29103 |       |
| Da.18<br>Parameter 1 | Value                       | Example)<br><p>26105 26104</p> <p>b31 (High-order) b16 b15 (Low-order) b0</p> <p>Value →</p>                 |                                 | 0000H                                                                     | 26104<br>26105                      | 27104<br>27105 | 28104<br>28105 | 29104<br>29105 |       |
| Da.19<br>Parameter 2 | Value                       | Example)<br><p>26107 26106</p> <p>b31 (High-order) b16 b15 (Low-order) b0</p> <p>Value →</p>                 |                                 | 0000H                                                                     | 26106<br>26107                      | 27106<br>27107 | 28106<br>28107 | 29106<br>29107 |       |

**Da.15** Condition target

Set the condition target as required for each control.

| Setting value                | Setting details                                                                                                                                                           |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01H : Device X               | Set the input/output signal ON/OFF of the QD75 as the conditions.                                                                                                         |
| 02H : Device Y               |                                                                                                                                                                           |
| 03H : Buffer memory (1-word) | Set the value stored in the buffer memory as the condition.<br>03H: The target buffer memory is "1-word (16 bits)"<br>04H: The target buffer memory is "2-word (32 bits)" |
| 04H : Buffer memory (2-word) |                                                                                                                                                                           |
| 05H : Positioning data No.   | Select only for "simultaneous start".                                                                                                                                     |

**Da.16** Condition operatorSet the condition operator as required for the " **Da.15** Condition target".

| <b>Da.15</b> Condition target                              | Setting value                   | Setting details                                                                                   |
|------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------|
| 01H: Device X                                              | 07H : DEV=ON                    | The state (ON/OFF) of an I/O signal is defined as the condition. Select ON or OFF as the trigger. |
| 02H: Device Y                                              | 08H : DEV=OFF                   |                                                                                                   |
| 03H: Buffer memory (1-word)<br>04H: Buffer memory (2-word) | 01H : **=P1                     | Select how to use the value (**) in the buffer memory as a part of the condition.                 |
|                                                            | 02H : **≠P1                     |                                                                                                   |
|                                                            | 03H : **≤P1                     |                                                                                                   |
|                                                            | 04H : **≥P1                     |                                                                                                   |
|                                                            | 05H : P1≤**≤P2                  |                                                                                                   |
|                                                            | 06H : **≤P1, P2≤**              |                                                                                                   |
| 05H: Positioning data No.                                  | 10H : Axis 1 selected           | If "simultaneous start" is specified, select the axis (or axes) that should start simultaneously. |
|                                                            | 20H : Axis 2 selected           |                                                                                                   |
|                                                            | 30H : Axes 1 and 2 selected     |                                                                                                   |
|                                                            | 40H : Axis 3 selected           |                                                                                                   |
|                                                            | 50H : Axes 1 and 3 selected     |                                                                                                   |
|                                                            | 60H : Axes 2 and 3 selected     |                                                                                                   |
|                                                            | 70H : Axes 1, 2, and 3 selected |                                                                                                   |
|                                                            | 80H : Axis 4 selected           |                                                                                                   |
|                                                            | 90H : Axes 1 and 4 selected     |                                                                                                   |
|                                                            | A0H : Axes 2 and 4 selected     |                                                                                                   |
|                                                            | B0H : Axes 1, 2, and 4 selected |                                                                                                   |
|                                                            | C0H : Axes 3 and 4 selected     |                                                                                                   |
|                                                            | D0H : Axes 1, 3, and 4 selected |                                                                                                   |
|                                                            | E0H : Axes 2, 3, and 4 selected |                                                                                                   |

**Da.17** AddressSet the address as required for the " **Da.15** Condition target".

| <b>Da.15</b> Condition target | Setting value                    | Setting details                                                                                   |
|-------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------|
| 01H : Device X                | —                                | Not used. (There is no need to set.)                                                              |
| 02H : Device Y                |                                  |                                                                                                   |
| 03H : Buffer memory (1-word)  | Value<br>(Buffer memory address) | Set the target "buffer memory address".<br>(For 2 word, set the low-order buffer memory address.) |
| 04H : Buffer memory (2-word)  |                                  |                                                                                                   |
| 05H : Positioning data No.    | —                                | Not used. (There is no need to set.)                                                              |



**Da.18** Parameter 1Set the parameters as required for the " **Da.16** Condition operator".

| <b>Da.16</b> Condition operator | Setting value                | Setting details                                                                                                                                                                                                          |
|---------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01H : $**=P1$                   | Value                        | The value of P1 should be equal to or smaller than the value of P2. ( $P1 \leq P2$ )<br>If P1 is greater than P2 ( $P1 > P2$ ), the error "Condition data error" (error code 533) will occur.                            |
| 02H : $**\neq P1$               |                              |                                                                                                                                                                                                                          |
| 03H : $**\leq P1$               |                              |                                                                                                                                                                                                                          |
| 04H : $**\geq P1$               |                              |                                                                                                                                                                                                                          |
| 05H : $P1 \leq ** \leq P2$      |                              |                                                                                                                                                                                                                          |
| 06H : $** \leq P1, P2 \leq **$  |                              |                                                                                                                                                                                                                          |
| 07H : DEV=ON                    | Value (bit No.)              | Set the device bit No.<br>X: 0H to 1H, 4H to 17H Y: 0, 4H to 17H                                                                                                                                                         |
| 08H : DEV=OFF                   |                              |                                                                                                                                                                                                                          |
| 10H : Axis 1 selected           | Value (positioning data No.) | Set the positioning data No. for starting axis 1 and/or axis 2.<br>Low-order 16-bit<br>: Axis 1 positioning data No. 1 to 600 (01H to 258H)<br>High-order 16-bit<br>: Axis 2 positioning data No. 1 to 600 (01H to 258H) |
| ↓                               |                              |                                                                                                                                                                                                                          |
| E0H : Axes 2, 3, and 4 selected |                              |                                                                                                                                                                                                                          |

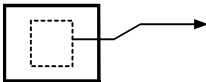
**Da.19** Parameter 2Set the parameters as required for the " **Da.16** Condition operator".

| <b>Da.16</b> Condition operator | Setting value                | Setting details                                                                                                                                                                                                          |
|---------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01H : $**=P1$                   | —                            | Not used. (No need to be set.)                                                                                                                                                                                           |
| 02H : $**\neq P1$               |                              |                                                                                                                                                                                                                          |
| 03H : $**\leq P1$               |                              |                                                                                                                                                                                                                          |
| 04H : $**\geq P1$               |                              |                                                                                                                                                                                                                          |
| 05H : $P1 \leq ** \leq P2$      | Value                        | The value of P2 should be equal to or greater than the value of P1. ( $P1 \leq P2$ )<br>If P1 is greater than P2 ( $P1 > P2$ ), the error "Condition data error" (error code 533) will occur.                            |
| 06H : $** \leq P1, P2 \leq **$  |                              |                                                                                                                                                                                                                          |
| 07H : DEV=ON                    | —                            | Not used. (No need to be set.)                                                                                                                                                                                           |
| 08H : DEV=OFF                   |                              |                                                                                                                                                                                                                          |
| 10H : Axis 1 selected           |                              |                                                                                                                                                                                                                          |
| 20H : Axis 2 selected           |                              |                                                                                                                                                                                                                          |
| 30H : Axes 1 and 2 selected     |                              |                                                                                                                                                                                                                          |
| 40H : Axis 3 selected           | Value (positioning data No.) | Set the positioning data No. for starting axis 3 and/or axis 4.<br>Low-order 16-bit<br>: Axis 3 positioning data No. 1 to 600 (01H to 258H)<br>High-order 16-bit<br>: Axis 4 positioning data No. 1 to 600 (01H to 258H) |
| 50H : Axes 1 and 3 selected     |                              |                                                                                                                                                                                                                          |
| 60H : Axes 2 and 3 selected     |                              |                                                                                                                                                                                                                          |
| 70H : Axes 1, 2, and 3 selected |                              |                                                                                                                                                                                                                          |
| 80H : Axis 4 selected           |                              |                                                                                                                                                                                                                          |
| 90H : Axes 1 and 4 selected     |                              |                                                                                                                                                                                                                          |
| A0H : Axes 2 and 4 selected     |                              |                                                                                                                                                                                                                          |
| B0H : Axes 1, 2, and 4 selected |                              |                                                                                                                                                                                                                          |
| C0H : Axes 3 and 4 selected     |                              |                                                                                                                                                                                                                          |
| D0H : Axes 1, 3, and 4 selected |                              |                                                                                                                                                                                                                          |
| E0H : Axes 2, 3, and 4 selected |                              |                                                                                                                                                                                                                          |

## 5.6 List of monitor data

## 5.6.1 System monitor data

| Storage item                                                                         | Storage details                                                                                                                                                                               |  |
|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <span style="border: 1px solid black; padding: 0 2px;">Md.1</span> In test mode flag | Whether the mode is the test mode from the peripheral device or not is stored. <ul style="list-style-type: none"><li>• When not in test mode : OFF</li><li>• When in test mode : ON</li></ul> |  |

|  | Reading the monitor value                                                                                                                                                                                                    | Default value | Storage buffer<br>memory address<br>(common for axis 1 to<br>axis 4) |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------------------------------|
|  | <p>■ Monitoring is carried out with a decimal.</p> <p>Monitor value</p>  <p>● Storage value<br/>0: Not in test mode<br/>1: In test mode</p> | 0             | 1200                                                                 |

(Unless noted in particular, the monitor value is saved as binary data.)

Starting history (Up to 16 records can be stored)

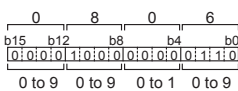
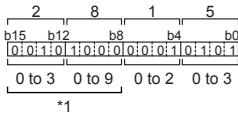
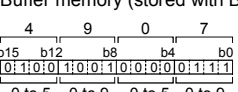
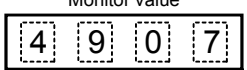
| Storage item                              | Storage details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Reading the monitor value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------|---------------------|-----------------------|------------------------|-----------------|---------------|--------|------------------------|--------|------------------------|--------|------------------------|--------|------------------------|------|------------------------|------|------------------------|------|---------------|------------------------|------|------------------------|------------------------|------|-------------|------------------------|------|----------|------------------------|------|------------------------|------------------------|------|--------------------|------------------------|------|--|
| <div>Md.3</div> Start information         | <div>[Storage details]</div> <div>This area stores the start information (restart flag, start origin, and start axis):</div> <ul style="list-style-type: none"><li>Restart flag: Indicates whether the operation has or has not been halted and restarted.</li><li>Start origin: Indicates the source of the start signal.</li><li>Start axis : Indicates the started axis.</li></ul> <div>[Reading the monitor value] ■ Monitoring is carried out with a hexadecimal display.</div> <div><table><thead><tr><th>Stored contents</th><th>Storage value</th></tr></thead><tbody><tr><td>Restart flag OFF</td><td>0</td></tr><tr><td>Restart flag ON</td><td>1</td></tr></tbody></table><table><thead><tr><th>Stored contents</th><th>Storage value</th></tr></thead><tbody><tr><td>Axis 1</td><td>1</td></tr><tr><td>Axis 2</td><td>2</td></tr><tr><td>Axis 3</td><td>3</td></tr><tr><td>Axis 4</td><td>4</td></tr></tbody></table></div> | Stored contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Storage value   | Restart flag OFF | 0                   | Restart flag ON       | 1                      | Stored contents | Storage value | Axis 1 | 1                      | Axis 2 | 2                      | Axis 3 | 3                      | Axis 4 | 4                      |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Stored contents                           | Storage value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Restart flag OFF                          | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Restart flag ON                           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Stored contents                           | Storage value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Axis 1                                    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Axis 2                                    | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Axis 3                                    | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Axis 4                                    | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| <div>Md.4</div> Start No.                 | <div>The starting No. is stored.</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <div>■ Monitoring is carried out with a hexadecimal display.</div> <div><table><thead><tr><th>Stored contents</th><th>Storage value</th><th>Reference (Decimal)</th></tr></thead><tbody><tr><td rowspan="10">Positioning operation</td><td>A: 0, B: 0, C: 0, D: 1</td><td>001</td></tr><tr><td colspan="2">to</td></tr><tr><td>A: 0, B: 2, C: 5, D: 8</td><td>600</td></tr><tr><td>A: 1, B: 5, C: 8, D: 8</td><td>7000</td></tr><tr><td>A: 1, B: 5, C: 9, D: 8</td><td>7001</td></tr><tr><td>A: 1, B: 5, C: A, D: 8</td><td>7002</td></tr><tr><td>A: 1, B: 5, C: B, D: 8</td><td>7003</td></tr><tr><td>A: 1, B: 5, C: C, D: 8</td><td>7004</td></tr><tr><td>JOG operation</td><td>A: 2, B: 3, C: 2, D: 2</td><td>9010</td></tr><tr><td>Manual pulse generator</td><td>A: 2, B: 3, C: 3, D: 3</td><td>9011</td></tr><tr><td>Machine OPR</td><td>A: 2, B: 3, C: 2, D: 9</td><td>9001</td></tr><tr><td>Fast OPR</td><td>A: 2, B: 3, C: 2, D: A</td><td>9002</td></tr><tr><td>Current value changing</td><td>A: 2, B: 3, C: 2, D: B</td><td>9003</td></tr><tr><td>Simultaneous start</td><td>A: 2, B: 3, C: 2, D: C</td><td>9004</td></tr></tbody></table></div> | Stored contents | Storage value    | Reference (Decimal) | Positioning operation | A: 0, B: 0, C: 0, D: 1 | 001             | to            |        | A: 0, B: 2, C: 5, D: 8 | 600    | A: 1, B: 5, C: 8, D: 8 | 7000   | A: 1, B: 5, C: 9, D: 8 | 7001   | A: 1, B: 5, C: A, D: 8 | 7002 | A: 1, B: 5, C: B, D: 8 | 7003 | A: 1, B: 5, C: C, D: 8 | 7004 | JOG operation | A: 2, B: 3, C: 2, D: 2 | 9010 | Manual pulse generator | A: 2, B: 3, C: 3, D: 3 | 9011 | Machine OPR | A: 2, B: 3, C: 2, D: 9 | 9001 | Fast OPR | A: 2, B: 3, C: 2, D: A | 9002 | Current value changing | A: 2, B: 3, C: 2, D: B | 9003 | Simultaneous start | A: 2, B: 3, C: 2, D: C | 9004 |  |
| Stored contents                           | Storage value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Reference (Decimal)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Positioning operation                     | A: 0, B: 0, C: 0, D: 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
|                                           | to                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
|                                           | A: 0, B: 2, C: 5, D: 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 600                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
|                                           | A: 1, B: 5, C: 8, D: 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
|                                           | A: 1, B: 5, C: 9, D: 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
|                                           | A: 1, B: 5, C: A, D: 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
|                                           | A: 1, B: 5, C: B, D: 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
|                                           | A: 1, B: 5, C: C, D: 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7004                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
|                                           | JOG operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | A: 2, B: 3, C: 2, D: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 9010            |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
|                                           | Manual pulse generator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | A: 2, B: 3, C: 3, D: 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 9011            |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Machine OPR                               | A: 2, B: 3, C: 2, D: 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 9001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Fast OPR                                  | A: 2, B: 3, C: 2, D: A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 9002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Current value changing                    | A: 2, B: 3, C: 2, D: B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 9003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Simultaneous start                        | A: 2, B: 3, C: 2, D: C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 9004                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| <div>Md.50</div> Start<br>(Year:month)    | <div>The starting time (year:month) is stored.<br/>(for the QD75P□N/QD75D□N only)</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <div>■ Monitoring is carried out with a hexadecimal display.</div> <div><div>Buffer memory (stored with BCD code)</div><div><table><thead><tr><th>Monitor value</th></tr></thead><tbody><tr><td>0 8 0 6</td></tr></tbody></table><p>00 to 99 (year) 00 to 12 (month)</p></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Monitor value   | 0 8 0 6          |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Monitor value                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| 0 8 0 6                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| <div>Md.5</div> Start<br>(Day:hour)       | <div>The following are stored.</div> <ul style="list-style-type: none"><li>QD75P□N/QD75D□N: the starting time (day:hour)</li><li>QD75P□/QD75D□: the starting time (hour)</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <div>■ Monitoring is carried out with a hexadecimal display.</div> <div><div>Buffer memory (stored with BCD code)</div><div><table><thead><tr><th>Monitor value</th></tr></thead><tbody><tr><td>2 8 1 5</td></tr></tbody></table><p>01 to 31 (day) 00 to 23 (hour)</p><p>*1: 00(not used) is stored for QD75P□/QD75D□.</p></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Monitor value   | 2 8 1 5          |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Monitor value                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| 2 8 1 5                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| <div>Md.6</div> Start<br>(Minute: second) | <div>The starting time (minute: second) is stored.</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <div>■ Monitoring is carried out with a hexadecimal display.</div> <div><div>Buffer memory (stored with BCD code)</div><div><table><thead><tr><th>Monitor value</th></tr></thead><tbody><tr><td>4 9 0 7</td></tr></tbody></table><p>00 to 59 (minute) 00 to 59 (second)</p></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Monitor value   | 4 9 0 7          |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| Monitor value                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |
| 4 9 0 7                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                  |                     |                       |                        |                 |               |        |                        |        |                        |        |                        |        |                        |      |                        |      |                        |      |               |                        |      |                        |                        |      |             |                        |      |          |                        |      |                        |                        |      |                    |                        |      |  |

Note: If a start signal is issued against an operating axis, a record relating to this event may be output before a record relating to an earlier start signal is output.

| Default value    |                   | Storage buffer memory address (common to axes 1 to 4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           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---|-----------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|--|----------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|--|-----------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
|                  | 0000 <sub>H</sub> | <div style="text-align: center;"><div style="border: 1px solid black; padding: 5px; display: inline-block;"><div style="border: 1px solid black; padding: 2px; margin-bottom: 2px;">Md.8</div>Start history pointer</div>1292</div> <div style="text-align: center; margin-top: 10px;">↑<br/>Indicates a pointer No. that is next to the Pointer No. assigned to the latest of the existing starting history records.</div> <div style="margin-top: 20px;"><div style="display: flex; align-items: center;"><div style="text-align: center; margin-right: 10px;">↓<br/>Pointer No.</div><div style="display: flex; flex-direction: column; align-items: center;"><div style="margin-bottom: 5px;">0</div><div style="margin-bottom: 5px;">1</div><div style="margin-bottom: 5px;">2</div><div style="margin-bottom: 5px;">3</div><div style="margin-bottom: 5px;">4</div><div style="margin-bottom: 5px;">5</div><div style="margin-bottom: 5px;">6</div><div style="margin-bottom: 5px;">7</div><div style="margin-bottom: 5px;">8</div><div style="margin-bottom: 5px;">9</div><div style="margin-bottom: 5px;">10</div><div style="margin-bottom: 5px;">11</div><div style="margin-bottom: 5px;">12</div><div style="margin-bottom: 5px;">13</div><div style="margin-bottom: 5px;">14</div><div style="margin-bottom: 5px;">15</div></div></div><div style="margin-top: 10px;"><table><tr><td rowspan="7">Starting history</td><td>Pointer No.</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td></tr><tr><td>Md.3</td><td>1212</td><td>1217</td><td>1222</td><td>1227</td><td>1232</td><td>1237</td><td>1242</td><td>1247</td><td>1252</td><td>1257</td><td>1262</td><td>1267</td><td>1272</td><td>1277</td><td>1282</td><td>1287</td></tr><tr><td>Start 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|  |  |  |  | Md.4 | 1213 | 1218 | 1223 | 1228 | 1233 | 1238 | 1243 | 1248 | 1253 | 1258 | 1263 | 1268 | 1273 | 1278 | 1283 | 1288 | Start No. |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Md.50 | 1440 | 1441 | 1442 | 1443 | 1444 | 1445 | 1446 | 1447 | 1448 | 1449 | 1450 | 1451 | 1452 | 1453 | 1454 | 1455 | Start year: month |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Md.5 | 1214 | 1219 | 1224 | 1229 | 1234 | 1239 | 1244 | 1249 | 1254 | 1259 | 1264 | 1269 | 1274 | 1279 | 1284 | 1289 | Start day: hour |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Md.6 | 1215 | 1220 | 1225 | 1230 | 1235 | 1240 | 1245 | 1250 | 1255 | 1260 | 1265 | 1270 | 1275 | 1280 | 1285 | 1290 |  | Start min: sec |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Md.7 | 1216 | 1221 | 1226 | 1231 | 1236 | 1241 | 1246 | 1251 | 1256 | 1261 | 1266 | 1271 | 1276 | 1281 | 1286 | 1291 |  | Error judgement |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Starting history | Pointer No.       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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|                  | Md.3              | 1212                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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|                  | Start information |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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|                  | Md.4              | 1213                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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|                  | Start No.         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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|                  | Md.50             | 1440                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1441 | 1442 | 1443 | 1444 | 1445 | 1446 | 1447 | 1448 | 1449 | 1450 | 1451 | 1452 | 1453 | 1454 | 1455 |  |  |                  |             |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |                   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |                   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | 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|                  | Start year: month |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| Md.5             | 1214              | 1219                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1224 | 1229 | 1234 | 1239 | 1244 | 1249 | 1254 | 1259 | 1264 | 1269 | 1274 | 1279 | 1284 | 1289 |      |  |  |                  |             |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |                   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |                   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | 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| Start day: hour  |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| Md.6             | 1215              | 1220                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1225 | 1230 | 1235 | 1240 | 1245 | 1250 | 1255 | 1260 | 1265 | 1270 | 1275 | 1280 | 1285 | 1290 |      |  |  |                  |             |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |                   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |                   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | 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| Start min: sec   |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| Md.7             | 1216              | 1221                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1226 | 1231 | 1236 | 1241 | 1246 | 1251 | 1256 | 1261 | 1266 | 1271 | 1276 | 1281 | 1286 | 1291 |      |  |  |                  |             |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |                   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |                   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | 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| Error judgement  |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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|                  | 0000 <sub>H</sub> | <div style="display: flex; justify-content: space-around; align-items: center;"><div style="text-align: center;">↑<br/>Item</div><div style="text-align: center;">↑<br/>Buffer memory address</div></div> <div style="margin-top: 10px;"><ul style="list-style-type: none"><li>Each group of buffer memory addresses storing a complete starting history record is assigned a pointer No.<br/>Example: Pointer No. 0 = Buffer memory addresses 1212 to 1216, 1440<br/>Pointer No. 1 = Buffer memory addresses 1217 to 1221, 1441<br/>Pointer No. 2 = Buffer memory addresses 1222 to 1226, 1442<br/>⋮<br/>Pointer No. 15 = Buffer memory addresses 1287 to 1291, 1455</li><li>Each history record is assigned a pointer No. in the range between 0 and 15. If the pointer No. 15 has been assigned to a new record, the next record will be assigned the pointer number 0.<br/>(A new record replaces an older record when a pointer No. is reassigned.)</li></ul></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |  |  |                  |             |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |                   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |                   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |                 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |  |                 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                  | 0000 <sub>H</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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|                  | 0000 <sub>H</sub> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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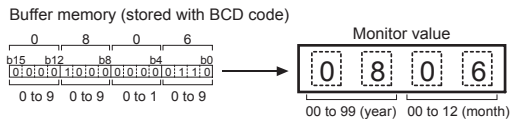
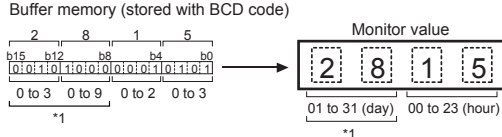
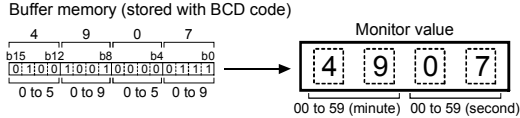
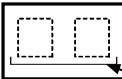
| Storage item                                                                                                                                                                                                                    | Storage details                                                                                                                                                                                                                                                                                                                         | Reading the monitor value                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <div data-bbox="159 705 183 1209" style="writing-mode: vertical-rl; transform: rotate(180deg);">Starting history (Up to 16 records can be stored)</div> <div data-bbox="199 840 311 929"> <b>Md.7</b><br/>Error judgment </div> | <p>[Storage details]</p> <p>This area stores the following results of the error judgment performed upon starting:</p> <ul style="list-style-type: none"> <li>• BUSY start warning flag</li> <li>• Error flag</li> <li>• Error No.</li> </ul> <p>[Reading the monitor value] ■ Monitoring is carried out with a hexadecimal display.</p> |                                                                                  |
| <div data-bbox="199 1489 335 1590"> <b>Md.8</b><br/>Start history pointer </div>                                                                                                                                                | <p>Indicates a pointer No. that is next to the Pointer No. assigned to the latest of the existing starting history records.</p>                                                                                                                                                                                                         | <p>■ Monitoring is carried out with a decimal display.</p> <p>Monitor value </p> |

| Default value |       | Storage buffer memory address (common to axes 1 to 4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|---------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
|               | 0000H | <div><div><div><div><div>Pointer No.</div><div>0</div></div><div><div>1</div><div>2</div><div>3</div><div>4</div><div>5</div><div>6</div><div>7</div><div>8</div><div>9</div><div>10</div><div>11</div><div>12</div><div>13</div><div>14</div><div>15</div></div></div><div><div>Starting history</div><div><div><div>Md.3</div><div>Start infor mation</div></div><div><div>Md.4</div><div>Start No.</div></div><div><div>Md.50</div><div>Start year: month</div></div><div><div>Md.5</div><div>Start day: hour</div></div><div><div>Md.6</div><div>Start min: sec</div></div><div><div>Md.7</div><div>Error judgement</div></div></div><div><div>1212</div><div>1217</div><div>1222</div><div>1227</div><div>1232</div><div>1237</div><div>1242</div><div>1247</div><div>1252</div><div>1257</div><div>1262</div><div>1267</div><div>1272</div><div>1277</div><div>1282</div><div>1287</div></div><div><div>1213</div><div>1218</div><div>1223</div><div>1228</div><div>1233</div><div>1238</div><div>1243</div><div>1248</div><div>1253</div><div>1258</div><div>1263</div><div>1268</div><div>1273</div><div>1278</div><div>1283</div><div>1288</div></div><div><div>1440</div><div>1441</div><div>1442</div><div>1443</div><div>1444</div><div>1445</div><div>1446</div><div>1447</div><div>1448</div><div>1449</div><div>1450</div><div>1451</div><div>1452</div><div>1453</div><div>1454</div><div>1455</div></div><div><div>1214</div><div>1219</div><div>1224</div><div>1229</div><div>1234</div><div>1239</div><div>1244</div><div>1249</div><div>1254</div><div>1259</div><div>1264</div><div>1269</div><div>1274</div><div>1279</div><div>1284</div><div>1289</div></div><div><div>1215</div><div>1220</div><div>1225</div><div>1230</div><div>1235</div><div>1240</div><div>1245</div><div>1250</div><div>1255</div><div>1260</div><div>1265</div><div>1270</div><div>1275</div><div>1280</div><div>1285</div><div>1290</div></div><div><div>1216</div><div>1221</div><div>1226</div><div>1231</div><div>1236</div><div>1241</div><div>1246</div><div>1251</div><div>1256</div><div>1261</div><div>1266</div><div>1271</div><div>1276</div><div>1281</div><div>1286</div><div>1291</div></div></div></div><div><div>↑</div><div>Item</div></div><div><div>↑</div><div>Buffer memory address</div></div><div><div>(<div><div>Md.8</div><div>Start history pointer</div></div><div>1292</div>)</div><div><div>↑</div><div>Indicates a pointer No. that is next to the Pointer No. assigned to the latest of the existing starting history records.</div></div></div></div> |      |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|               |       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1292 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

| Storage item                                   | Storage details                                       | Reading the monitor value                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Error history (Up to 16 records can be stored) | <b>Md.9</b><br>Axis in which the error occurred       | Stores a number (Axis No.) that indicates the axis that encountered an error.                                                                                                         | ■ Monitoring is carried out with a decimal display.<br>Monitor value  → ● Storage value<br>1: Axis 1<br>2: Axis 2<br>3: Axis 3<br>4: Axis 4                                                                                                                                                    |
|                                                | <b>Md.10</b><br>Axis error No.                        | Stores an axis error No.                                                                                                                                                              | ■ Monitoring is carried out with a decimal display.<br>Monitor value  → ● Error No.<br>For details on the error Nos. (error codes), refer to Section 15.3 "List of errors".                                                                                                                    |
|                                                | <b>Md.51</b><br>Axis error occurrence (Year:month)    | Stores the time (year:month) at which an axis error was detected. (for the QD75P□N/QD75D□N only)                                                                                      | ■ Monitoring is carried out with a hexadecimal display.<br>Buffer memory (stored with BCD code)<br> → Monitor value <br>00 to 99 (year) 00 to 12 (month)                                                  |
|                                                | <b>Md.11</b><br>Axis error occurrence (Day:hour)      | The following are stored.<br>• QD75P□N/QD75D□N:<br>the time (day:hour) at which an axis error was detected<br>• QD75P□/QD75D□:<br>the time (hour) at which an axis error was detected | ■ Monitoring is carried out with a hexadecimal display.<br>Buffer memory (stored with BCD code)<br> → Monitor value <br>01 to 31 (day) 00 to 23 (hour)<br>*1: 00(not used) is stored for QD75P□/QD75D□. |
|                                                | <b>Md.12</b><br>Axis error occurrence (Minute:second) | Stores the time (minute:second) at which an axis error was detected.                                                                                                                  | ■ Monitoring is carried out with a hexadecimal display.<br>Buffer memory (stored with BCD code)<br> → Monitor value <br>00 to 59 (minute) 00 to 59 (second)                                             |
|                                                | <b>Md.13</b><br>Error history pointer                 | Indicates a pointer No. that is next to the Pointer No. assigned to the latest of the existing records.                                                                               | ■ Monitoring is carried out with a decimal display.<br>Monitor value  → ● Storage value (Pointer number) 0 to 15                                                                                                                                                                             |



| Default value |  | Storage buffer memory address (common to axes 1 to 4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0             |  | <div><div>(Md.13Error history pointer1357)</div><div>Indicates a pointer No. that is next to the Pointer No. assigned to the latest of the existing error history records.</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 0             |  | <div><div>Pointer No. 0123456789101112131415</div><div>Pointer No. 0123456789101112131415</div><div><div>Error history</div><div><div>Md.9</div><div>Axis in which the error occurred</div><div>1293129713011305130913131317132113251329133313371341134513491353</div></div><div><div>Md.10</div><div>Axis error No.</div><div>1294129813021306131013141318132213261330133413381342134613501354</div></div><div><div>Md.51</div><div>Axis error occurrence year: month</div><div>1456145714581459146014611462146314641465146614671468146914701471</div></div><div><div>Md.11</div><div>Axis error occurrence day: hour</div><div>1295129913031307131113151319132313271331133513391343134713511355</div></div><div><div>Md.12</div><div>Axis error occurrence min: sec</div><div>1296130013041308131213161320132413281332133613401344134813521356</div></div></div></div> |
| 0000H         |  | <div><div>Item</div><div>Buffer memory address</div><div><div>• Each group of buffer memory addresses storing a complete error history record is assigned a pointer</div><div>Example: Pointer No. 0 = Buffer memory addresses 1293 to 1296, 1456</div><div>Pointer No. 1 = Buffer memory addresses 1297 to 1300, 1457</div><div>Pointer No. 2 = Buffer memory addresses 1301 to 1304, 1458</div><div>⋮</div><div>Pointer No. 15 = Buffer memory addresses 1353 to 1356, 1471</div><div>• Each history record is assigned a pointer No. in the range between 0 and 15.</div><div>If the pointer No. 15 has been assigned to a new record, the next record will be assigned the pointer number 0.</div><div>(A new record replaces an older record when a pointer No. is reassigned.)</div></div></div>                                                                   |
| 0000H         |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 0             |  | 1357                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Storage item                                       | Storage details                                                                                                                                                                                           | Reading the monitor value                                                                                                                                                           |                                                                                                                                                                                                                                                   |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Warning history (Up to 16 records can be stored)   | <b>Md.14</b><br>Axis in which the warning occurred                                                                                                                                                        | Stores a number (Axis No.) that indicates the axis that encountered a warning.                                                                                                      | ■ Monitoring is carried out with a decimal display.<br> ● Storage value<br>1: Axis 1<br>2: Axis 2<br>3: Axis 3<br>4: Axis 4                                     |
|                                                    | <b>Md.15</b><br>Axis warning No.                                                                                                                                                                          | Stores an axis warning No.                                                                                                                                                          | ■ Monitoring is carried out with a decimal display.<br> ● Warning No.<br>For details of warning Nos. (warning codes), refer to Section 15.4 "List of warnings". |
|                                                    | <b>Md.52</b><br>Axis warning occurrence (Year:month)                                                                                                                                                      | Stores the time (year:month) at which an axis warning was detected. (for the QD75P□N/QD75D□N only)                                                                                  | ■ Monitoring is carried out with a hexadecimal display.<br>                                                                                                    |
|                                                    | <b>Md.16</b><br>Axis warning occurrence (Day:hour)                                                                                                                                                        | The following are stored.<br>• QD75P□N/QD75D□N: the time (day:hour) at which an axis warning was detected<br>• QD75P□/QD75D□: the time (hour) at which an axis warning was detected | ■ Monitoring is carried out with a hexadecimal display.<br> *1: 00(not used) is stored for QD75P□/QD75D□.                                                     |
|                                                    | <b>Md.17</b><br>Axis warning occurrence (Minute:second)                                                                                                                                                   | Stores the time (minute:second) at which an axis warning was detected.                                                                                                              | ■ Monitoring is carried out with a hexadecimal display.<br>                                                                                                   |
|                                                    | <b>Md.18</b><br>Warning history pointer                                                                                                                                                                   | Indicates a pointer No. that is next to the Pointer No. assigned to the latest of the existing records.                                                                             | ■ Monitoring is carried out with a decimal display.<br> ● Storage value (Pointer number)<br>0 to 15                                                           |
| <b>Md.19</b><br>No. of write accesses to flash ROM | Stores the number of write accesses to the flash ROM after the power is switched ON.<br>The count is cleared to "0" when the number of write accesses reach 26 and an error reset operation is performed. | ■ Monitoring is carried out with a decimal display.<br> ● Storage value<br>0 to 26              |                                                                                                                                                                                                                                                   |

| Default value | Storage buffer memory address (common to axes 1 to 4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 0             | <div style="text-align: center;"> <div style="border: 1px solid black; padding: 5px; display: inline-block;"> Md.18<br/>Warning history<br/>pointer </div> <div style="border: 1px solid black; padding: 5px; display: inline-block; margin-left: 10px;">1422</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |
| 0             | <div style="text-align: center;"> <p>Indicates a pointer No. that is next to the Pointer No. assigned to the latest of the existing warning history records.</p> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
| 0000H         | <div style="text-align: center;"> <p>Item</p> <p>Buffer memory address</p> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
| 0000H         | <ul style="list-style-type: none"> <li>Each group of buffer memory addresses storing a complete warning history record is assigned a pointer No.</li> <li>Example: Pointer No. 0 = Buffer memory addresses 1358 to 1361, 1472</li> <li>Pointer No. 1 = Buffer memory addresses 1362 to 1365, 1473</li> <li>Pointer No. 2 = Buffer memory addresses 1366 to 1369, 1474</li> <li>⋮</li> <li>Pointer No. 15 = Buffer memory addresses 1418 to 1421, 1487</li> <li>Each history record is assigned a pointer No. in the range between 0 and 15. If the pointer No. 15 has been assigned to a new record, the next record will be assigned the pointer number 0. (A new record replaces an older record when a pointer No. is reassigned.)</li> </ul> |  |
| 0             | 1422                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| 0             | 1424<br>1425                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |

## 5.6.2 Axis monitor data

| Storage item                      | Storage details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>[Md.20]</b> Current feed value | <p>The currently commanded address is stored.<br/>(Different from the actual motor position during operation)</p> <p>The current position address is stored.</p> <p>If "degree" is selected as the unit, the addresses will have a ring structure for values between 0 and 359.99999 degrees.</p> <ul style="list-style-type: none"> <li>• Update timing : 0.9ms (QD75P□N/QD75D□N), 1.8ms (QD75P□/QD75D□)</li> <li>• The OP address is stored when the machine OPR is completed.</li> <li>• When the current value is changed with the current value changing function, the changed value is stored.</li> </ul>                                                                                                                                                                                                               |  |
| <b>[Md.21]</b> Machine feed value | <p>The address of the current position according to the machine coordinates will be stored. (Different from the actual motor position during operation)</p> <p>Note that the current value changing function will not change the machine feed value.</p> <p>Under the speed control mode, the machine feed value is constantly updated always, irrespective of the parameter setting.</p> <p>The value will not be cleared to "0" at the beginning of fixed-feed control.</p> <p>Even if "degree" is selected as the unit, the addresses will not have a ring structure for values between 0 and 359.99999 degrees.</p> <ul style="list-style-type: none"> <li>• Machine coordinates: Characteristic coordinates determined with machine</li> <li>• Update timing: 0.9ms (QD75P□N/QD75D□N), 56.8ms (QD75P□/QD75D□)</li> </ul> |  |
| <b>[Md.22]</b> Feedrate           | <p>The command output speed of the operating workpiece is stored. (May be different from the actual motor speed during operation)</p> <ul style="list-style-type: none"> <li>• During interpolation operation, the speed is stored in the following manner.<br/>Reference axis : Composite speed or reference axis speed<br/>(Set with <b>[Pr.20]</b>)<br/>Interpolation axis : 0</li> <li>• Update timing: 0.9ms (QD75P□N/QD75D□N), 56.8ms (QD75P□/QD75D□)</li> </ul>                                                                                                                                                                                                                                                                                                                                                        |  |
| <b>[Md.23]</b> Axis error No.     | <p>When an axis error is detected, the error code corresponding to the error details is stored.</p> <ul style="list-style-type: none"> <li>• The latest error code is always stored.<br/>(When a new axis error occurs, the error code is overwritten.)</li> <li>• When " <b>[Cd.5]</b> Axis error reset" (axis control data) turns ON, the axis error No. is cleared (set to 0).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Reading the monitor value | Default value | Storage buffer memory address |        |        |        |    |        |   |       |   |      |    |        |    |          |    |            |   |         |                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------|-------------------------------|--------|--------|--------|----|--------|---|-------|---|------|----|--------|----|----------|----|------------|---|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                           |               | Axis 1                        | Axis 2 | Axis 3 | Axis 4 |    |        |   |       |   |      |    |        |    |          |    |            |   |         |                                                                                                                                                                                                                                                                                                                                                                                           |
| <div>■ Monitoring is carried out with a hexadecimal.</div> <div><div><div>Monitor value</div><div><div>E</div><div>F</div><div>G</div><div>H</div></div><div><div>A</div><div>B</div><div>C</div><div>D</div></div></div><div><div>Low-order buffer memory Example) 800</div><div><div>b15</div><div>b12</div><div>b8</div><div>b4</div><div>b0</div></div><div><div>E</div><div>F</div><div>G</div><div>H</div></div><div>High-order buffer memory Example) 801</div><div><div>b31</div><div>b28</div><div>b24</div><div>b20</div><div>b16</div></div><div><div>A</div><div>B</div><div>C</div><div>D</div></div></div><div>◇ Sorting</div><div><div>(High-order buffer memory) (Low-order buffer memory)</div><div><div>A</div><div>B</div><div>C</div><div>D</div><div>E</div><div>F</div><div>G</div><div>H</div></div></div><div>◇ Converted from hexadecimal to decimal</div><div><div>Decimal integer value</div><div>R</div></div><div>◇ Unit conversion</div><div><div><math>R \times 10^n</math></div><div>Actual value</div><div><div>Md.20</div>Current feed value</div><div><div>Md.21</div>Machine feed value</div><div><div>Md.22</div>Feedrate</div></div><div><div>● Unit conversion table</div><div>(Md.20 Md.21)</div><table><tr><th>n</th><th>Unit</th></tr><tr><td>-1</td><td>μm</td></tr><tr><td>-5</td><td>inch</td></tr><tr><td>-5</td><td>degree</td></tr><tr><td>0</td><td>pulse</td></tr></table><div>● Unit conversion table</div><div>(Md.22)</div><table><tr><th>n</th><th>Unit</th></tr><tr><td>-2</td><td>mm/min</td></tr><tr><td>-3</td><td>inch/min</td></tr><tr><td>-3</td><td>degree/min</td></tr><tr><td>0</td><td>pulse/s</td></tr></table></div></div> <div><div>0000H</div><div>800<br/>801</div><div>900<br/>901</div><div>1000<br/>1001</div><div>1100<br/>1101</div></div> <div><div>0000H</div><div>802<br/>803</div><div>902<br/>903</div><div>1002<br/>1003</div><div>1102<br/>1103</div></div> <div><div>0000H</div><div>804<br/>805</div><div>904<br/>905</div><div>1004<br/>1005</div><div>1104<br/>1105</div></div> | n                         | Unit          | -1                            | μm     | -5     | inch   | -5 | degree | 0 | pulse | n | Unit | -2 | mm/min | -3 | inch/min | -3 | degree/min | 0 | pulse/s | <div>■ Monitoring is carried out with a decimal.</div> <div><div>Monitor value</div><div><div></div><div></div><div></div><div></div><div></div><div></div></div><div><div>● Error No.</div><div>For details on the error Nos. (error codes), refer to Section 15.3 "List of errors".</div></div></div> <div><div>0</div><div>806</div><div>906</div><div>1006</div><div>1106</div></div> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | n                         | Unit          |                               |        |        |        |    |        |   |       |   |      |    |        |    |          |    |            |   |         |                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -1                        | μm            |                               |        |        |        |    |        |   |       |   |      |    |        |    |          |    |            |   |         |                                                                                                                                                                                                                                                                                                                                                                                           |
| -5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | inch                      |               |                               |        |        |        |    |        |   |       |   |      |    |        |    |          |    |            |   |         |                                                                                                                                                                                                                                                                                                                                                                                           |
| -5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | degree                    |               |                               |        |        |        |    |        |   |       |   |      |    |        |    |          |    |            |   |         |                                                                                                                                                                                                                                                                                                                                                                                           |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | pulse                     |               |                               |        |        |        |    |        |   |       |   |      |    |        |    |          |    |            |   |         |                                                                                                                                                                                                                                                                                                                                                                                           |
| n                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Unit                      |               |                               |        |        |        |    |        |   |       |   |      |    |        |    |          |    |            |   |         |                                                                                                                                                                                                                                                                                                                                                                                           |
| -2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | mm/min                    |               |                               |        |        |        |    |        |   |       |   |      |    |        |    |          |    |            |   |         |                                                                                                                                                                                                                                                                                                                                                                                           |
| -3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | inch/min                  |               |                               |        |        |        |    |        |   |       |   |      |    |        |    |          |    |            |   |         |                                                                                                                                                                                                                                                                                                                                                                                           |
| -3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | degree/min                |               |                               |        |        |        |    |        |   |       |   |      |    |        |    |          |    |            |   |         |                                                                                                                                                                                                                                                                                                                                                                                           |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | pulse/s                   |               |                               |        |        |        |    |        |   |       |   |      |    |        |    |          |    |            |   |         |                                                                                                                                                                                                                                                                                                                                                                                           |

| Storage item                         | Storage details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>[Md.24]</b> Axis warning No.      | <p>Whenever an axis warning is reported, a related warning code is stored.</p> <ul style="list-style-type: none"> <li>• This area stores the latest warning code always. (Whenever an axis warning is reported, a new warning code replaces the stored warning code.)</li> <li>• When the " <b>[Cd.5]</b> Axis error reset" (axis control data) is set to ON, the axis warning No. is cleared to "0".</li> </ul>                                                                                                                                                                                                           |  |
| <b>[Md.25]</b> Valid M code          | <p>This area stores an M code that is currently active (i.e. set to the positioning data relating to the current operation).</p> <ul style="list-style-type: none"> <li>• Update timing : turning ON of the M code ON signal</li> </ul> <p>When the PLC READY signal (Y0) goes OFF, the value is set to "0".</p>                                                                                                                                                                                                                                                                                                           |  |
| <b>[Md.26]</b> Axis operation status | <p>This area stores the axis operation status.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |
| <b>[Md.27]</b> Current speed         | <p>" <b>[Da.8]</b> Command speed" of the positioning data currently in execution is stored.</p> <ul style="list-style-type: none"> <li>• If " <b>[Da.8]</b> Command speed" is set to "-1", this area stores the command speed set by the positioning data used one step earlier.</li> <li>• If " <b>[Da.8]</b> Command speed" is set to a value other than "-1", this area stores the command speed set by the current positioning data.</li> <li>• If the speed change function is executed, " <b>[Cd.14]</b> New speed value" is stored. (For details of the speed change function, refer to Section 12.5.1.)</li> </ul> |  |

|                                                     | Reading the monitor value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Default value | Storage buffer memory address |        |        |        |          |    |            |   |         |   |            |            |              |              |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------|--------|--------|--------|----------|----|------------|---|---------|---|------------|------------|--------------|--------------|
|                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               | Axis 1                        | Axis 2 | Axis 3 | Axis 4 |          |    |            |   |         |   |            |            |              |              |
| ■ Monitoring is carried out with a decimal display. | <div>Monitor value</div> <div><div>□□□□□</div><div>● Warning No.</div><div>For details of warning Nos. (warning codes), refer to Section 15.4 "List of warnings".</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0             | 807                           | 907    | 1007   | 1107   |          |    |            |   |         |   |            |            |              |              |
| ■ Monitoring is carried out with a decimal display. | <div>Monitor value</div> <div><div>□□□□□</div><div>● M code No.</div><div>(0 to 65535)</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0             | 808                           | 908    | 1008   | 1108   |          |    |            |   |         |   |            |            |              |              |
| ■ Monitoring is carried out with a decimal display. | <div>Monitor value</div> <div><div>□□</div><div>● Axis operation status</div><div>-2: Step standby</div><div>-1: Error</div><div>0: Standby</div><div>1: Stopped</div><div>2: Interpolation</div><div>3: JOG operation</div><div>4: Manual pulse generator operation</div><div>5: Analyzing</div><div>6: Special start standby</div><div>7: OPR</div><div>8: Position control</div><div>9: Speed control</div><div>10: Speed control in speed-position switching control</div><div>11: Position control in speed-position switching control</div><div>12: Position control in position-speed switching control</div><div>13: Speed control in position-speed switching control</div></div> | 0             | 809                           | 909    | 1009   | 1109   |          |    |            |   |         |   |            |            |              |              |
| ■ Monitoring is carried out with a decimal display. | <div>Monitor value</div> <div><div>R</div><div>(Decimal integer value)</div><div>◇ Unit conversion</div><div><math>R \times 10^n</math></div><div>● Unit conversion table (Md.27)</div><div><table><tr><th>n</th><th>Unit</th></tr><tr><td>-2</td><td>mm/min</td></tr><tr><td>-3</td><td>inch/min</td></tr><tr><td>-3</td><td>degree/min</td></tr><tr><td>0</td><td>pulse/s</td></tr></table></div><div>Actual value</div><div><div>Md.27</div>Current speed</div></div>                                                                                                                                                                                                                   | n             | Unit                          | -2     | mm/min | -3     | inch/min | -3 | degree/min | 0 | pulse/s | 0 | 810<br>811 | 910<br>911 | 1010<br>1011 | 1110<br>1111 |
| n                                                   | Unit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |                               |        |        |        |          |    |            |   |         |   |            |            |              |              |
| -2                                                  | mm/min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |                               |        |        |        |          |    |            |   |         |   |            |            |              |              |
| -3                                                  | inch/min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |                               |        |        |        |          |    |            |   |         |   |            |            |              |              |
| -3                                                  | degree/min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |                               |        |        |        |          |    |            |   |         |   |            |            |              |              |
| 0                                                   | pulse/s                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |                               |        |        |        |          |    |            |   |         |   |            |            |              |              |

| Storage item                                                     | Storage details                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Md.28</b> Axis feedrate                                       | <ul style="list-style-type: none"> <li>The speed which is actually output as a command at that time in each axis is stored. (May be different from the actual motor speed)</li> <li>"0" is stored when the axis is at a stop.</li> </ul> Update timing: 0.9ms (QD75P□N/QD75D□N), 56.8ms (QD75P□/QD75D□)                                                                                                                                               |  |
| <b>Md.29</b> Speed-position switching control positioning amount | <ul style="list-style-type: none"> <li>The movement amount for the position control to end after changing to position control with the speed-position switching control (INC mode) is stored. When the control method is "Reverse run: position/speed", the negative value is stored.</li> </ul>                                                                                                                                                      |  |
| <b>Md.30</b> External input/output signal                        | The ON/OFF state of the external input/output signal is stored.<br>The following items are stored. <ul style="list-style-type: none"> <li>Upper limit signal</li> <li>Lower limit signal</li> <li>Drive unit READY signal</li> <li>Stop signal</li> <li>External command signal</li> <li>Zero signal</li> <li>Near-point dog signal</li> <li>Deviation counter clear signal</li> </ul> Update timing: 0.9ms (QD75P□N/QD75D□N), 56.8ms (QD75P□/QD75D□) |  |



|                                                 | Reading the monitor value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Default value | Storage buffer memory address |            |              |              |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |       |            |            |              |              |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------|------------|--------------|--------------|----------|----|------------|---|---------|---|------|----|----|----|------|----|--------|---|-------|-------|------------|------------|--------------|--------------|
|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               | Axis 1                        | Axis 2     | Axis 3       | Axis 4       |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |       |            |            |              |              |
| ■ Monitoring is carried out with a hexadecimal. | <div><div>Monitor value</div><div><div>E F G H</div><div>A B C D</div></div><div>Low-order buffer memory Example) 812</div><div><div>b15 b12 b8 b4 b0</div><div>E F G H</div></div><div>High-order buffer memory Example) 813</div><div><div>b31 b28 b24 b20 b16</div><div>A B C D</div></div><div>◇ Sorting</div><div>(High-order buffer memory) (Low-order buffer memory)</div><div><div>A B C D E F G H</div></div><div>◇ Converted from hexadecimal to decimal</div><div>Decimal integer value</div><div>R</div><div>◇ Unit conversion</div><div><math>R \times 10^n</math></div><div>Actual value</div><div><div>Md.28 Axis feedrate</div><div>Md.29 Speed-position switching control positioning amount</div></div><div>Unit conversion table (Md.28)</div><table><tr><th>n</th><th>Unit</th></tr><tr><td>-2</td><td>mm/min</td></tr><tr><td>-3</td><td>inch/min</td></tr><tr><td>-3</td><td>degree/min</td></tr><tr><td>0</td><td>pulse/s</td></tr></table><div>Unit conversion table (Md.29)</div><table><tr><th>n</th><th>Unit</th></tr><tr><td>-1</td><td>μm</td></tr><tr><td>-5</td><td>inch</td></tr><tr><td>-5</td><td>degree</td></tr><tr><td>0</td><td>pulse</td></tr></table></div> | n             | Unit                          | -2         | mm/min       | -3           | inch/min | -3 | degree/min | 0 | pulse/s | n | Unit | -1 | μm | -5 | inch | -5 | degree | 0 | pulse | 0000H | 812<br>813 | 912<br>913 | 1012<br>1013 | 1112<br>1113 |
|                                                 | n                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Unit          |                               |            |              |              |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |       |            |            |              |              |
| -2                                              | mm/min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |                               |            |              |              |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |       |            |            |              |              |
| -3                                              | inch/min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |                               |            |              |              |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |       |            |            |              |              |
| -3                                              | degree/min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |                               |            |              |              |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |       |            |            |              |              |
| 0                                               | pulse/s                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |                               |            |              |              |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |       |            |            |              |              |
| n                                               | Unit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |                               |            |              |              |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |       |            |            |              |              |
| -1                                              | μm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |                               |            |              |              |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |       |            |            |              |              |
| -5                                              | inch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |                               |            |              |              |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |       |            |            |              |              |
| -5                                              | degree                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |                               |            |              |              |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |       |            |            |              |              |
| 0                                               | pulse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |                               |            |              |              |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |       |            |            |              |              |
|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0000H         | 814<br>815                    | 914<br>915 | 1014<br>1015 | 1114<br>1115 |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |       |            |            |              |              |
| ■ Monitoring is carried out with a hexadecimal. | <div><div>Monitor value</div><div><div>0 0 0 0</div></div><div>Buffer memory</div><div><div>b15 b12 b8 b4 b0</div><div>0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0</div></div><div>Not used</div><div><div>b0</div><div>Lower limit signal</div><div>0</div><div rowspan="9">0: OFF<br/>1: ON</div></div><div><div>b1</div><div>Upper limit signal</div><div>0</div></div><div><div>b2</div><div>Drive unit READY signal</div><div>0</div></div><div><div>b3</div><div>Stop signal</div><div>0</div></div><div><div>b4</div><div>External command signal</div><div>0</div></div><div><div>b5</div><div>Zero signal</div><div>0</div></div><div><div>b6</div><div>Near-point dog signal</div><div>0</div></div><div><div>b7</div><div>Not used</div><div>0</div></div><div><div>b8</div><div>Deviation counter clear signal</div><div>0</div></div></div>                                                                                                                                                                                                                                                                                                                                                        | 0000H         | 816                           | 916        | 1016         | 1116         |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |       |            |            |              |              |
|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |                               |            |              |              |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |       |            |            |              |              |

| Storage item              | Storage details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Md.31</b> Status       | <p>This area stores the states (ON/OFF) of various flags.<br/>Information on the following flags is stored.</p> <ul style="list-style-type: none"> <li>● In speed control flag:<br/>This signal that comes ON under the speed control can be used to judge whether the operation is performed under the speed control or position control. The signal goes OFF when the power is switched ON, under the position control, and during JOG operation or manual pulse generator operation. During the speed-position or position-speed switching control, this signal comes ON only when the speed control is in effect. During the speed-position switching control, this signal goes OFF when the speed-position switching signal executes a switching over from speed control to position control. During the position-speed switching control, this signal comes ON when the position-speed switching signal executes a switching over from position control to speed control.</li> <li>● Speed-position switching latch flag:<br/>This signal is used during the speed-position switching control for interlocking the movement amount change function. During the speed-position switching control, this signal comes ON when position control takes over. This signal goes OFF when the next positioning data is processed, and during JOG operation or manual pulse generator operation.</li> <li>● Command in-position flag:<br/>This signal is ON when the remaining distance is equal to or less than the command in-position range (set by a detailed parameter). This signal remains OFF with data that specify the continuous path control (P11) as the operation pattern. The state of this signal is monitored every 0.9ms (QD75P□N/QD75D□N) or 1.8ms (QD75P□/QD75D□) except when the monitoring is canceled under the speed control or while the speed control is in effect during the speed-position or position-speed switching control. While operations are performed with interpolation, this signal comes ON only in respect of the starting axis. (This signal goes OFF in respect of all axes upon starting.)</li> <li>● OPR request flag:<br/>This signal comes ON when the power is switched ON, when the drive unit READY signal goes OFF, when the PLC READY signal goes ON, when a machine OPR operation starts. This signal goes OFF when a machine OPR operation completes.</li> <li>● OPR complete flag:<br/>This signal comes ON when a machine OPR operation completes normally. This signal goes OFF when the operation starts and when the drive unit READY signal goes OFF or PLC READY signal goes ON.</li> <li>● Position-speed switching latch flag:<br/>This signal is used during the position-speed switching control for interlocking the command speed change function. During the position-speed switching control, this signal comes ON when speed control takes over. This signal goes OFF when the next positioning data is processed, and during JOG operation or manual pulse generator operation.</li> <li>● Axis warning detection flag:<br/>This signal comes ON when an axis warning is reported and goes OFF when the axis error reset signal comes ON.</li> <li>● Speed change 0 flag:<br/>This signal comes ON when a speed change request that specifies 0 as the new speed value is issued. This signal comes OFF when a speed change request that specifies a new speed value other than 0 is issued.</li> </ul> |  |
| <b>Md.32</b> Target value | <p>This area stores the target value (<b>Da.6</b> Positioning address/movement amount) for a positioning operation.</p> <ul style="list-style-type: none"> <li>• At the beginning of positioning control and current value changing<br/>: Stores the value of " <b>Da.6</b> Positioning address/movement amount".</li> <li>• At the OP shift operation of OPR control<br/>: Stores the value of OP shift amount.</li> <li>• At other times<br/>: Stores "0".</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |

|              | Reading the monitor value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Default value | Storage buffer memory address |         |        |                       |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------|---------|--------|-----------------------|------|----|-------------------------------------|---|-------|--------------------------|------------|------------|------------------|--------------|----|-------------------|---|----|-------------------------------------|---|----|------------------------|---|-----|---------------------|---|-------|-----|-----|------|------|
|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               | Axis 1                        | Axis 2  | Axis 3 | Axis 4                |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
|              | <p>■ Monitoring is carried out with a hexadecimal display.</p> <div><div>Monitor value</div><div>0008</div><div>Buffer memory</div><div>00000000000000010000</div><div>Not used</div><div>Not used</div><div>b15 b12 b8 b4 b0</div><table><tr><th>Stored items</th><th>Default value</th><th>Meaning</th></tr><tr><td>b0</td><td>In speed control flag</td><td>0</td></tr><tr><td>b1</td><td>Speed-position switching latch flag</td><td>0</td></tr><tr><td>b2</td><td>Command in-position flag</td><td>0</td></tr><tr><td>b3</td><td>OPR request flag</td><td>1</td></tr><tr><td>b4</td><td>OPR complete flag</td><td>0</td></tr><tr><td>b5</td><td>Position-speed switching latch flag</td><td>0</td></tr><tr><td>b9</td><td>Axis warning detection</td><td>0</td></tr><tr><td>b10</td><td>Speed change 0 flag</td><td>0</td></tr></table></div> | Stored items  | Default value                 | Meaning | b0     | In speed control flag | 0    | b1 | Speed-position switching latch flag | 0 | b2    | Command in-position flag | 0          | b3         | OPR request flag | 1            | b4 | OPR complete flag | 0 | b5 | Position-speed switching latch flag | 0 | b9 | Axis warning detection | 0 | b10 | Speed change 0 flag | 0 | 0008H | 817 | 917 | 1017 | 1117 |
| Stored items | Default value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Meaning       |                               |         |        |                       |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
| b0           | In speed control flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0             |                               |         |        |                       |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
| b1           | Speed-position switching latch flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0             |                               |         |        |                       |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
| b2           | Command in-position flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0             |                               |         |        |                       |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
| b3           | OPR request flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1             |                               |         |        |                       |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
| b4           | OPR complete flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0             |                               |         |        |                       |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
| b5           | Position-speed switching latch flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0             |                               |         |        |                       |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
| b9           | Axis warning detection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0             |                               |         |        |                       |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
| b10          | Speed change 0 flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0             |                               |         |        |                       |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
|              | <p>■ Monitoring is carried out with a decimal display.</p> <div><div>Monitor value</div><div>R (Decimal integer value)</div><div>◇ Unit conversion</div><div><math>R \times 10^n</math></div><div>Actual value</div><div>Md.32 Target value</div><div>• Unit conversion table (Md.32)</div><table><tr><th>n</th><th>Unit</th></tr><tr><td>-1</td><td>μm</td></tr><tr><td>-5</td><td>inch</td></tr><tr><td>-5</td><td>degree</td></tr><tr><td>0</td><td>pulse</td></tr></table></div>                                                                                                                                                                                                                                                                                                                                                               | n             | Unit                          | -1      | μm     | -5                    | inch | -5 | degree                              | 0 | pulse | 0                        | 818<br>819 | 918<br>919 | 1018<br>1019     | 1118<br>1119 |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
| n            | Unit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |                               |         |        |                       |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
| -1           | μm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |                               |         |        |                       |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
| -5           | inch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |                               |         |        |                       |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
| -5           | degree                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |                               |         |        |                       |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |
| 0            | pulse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |                               |         |        |                       |      |    |                                     |   |       |                          |            |            |                  |              |    |                   |   |    |                                     |   |    |                        |   |     |                     |   |       |     |     |      |      |

| Storage item                                         | Storage details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Md.33</b> Target speed                            | <ul style="list-style-type: none"> <li>• During operation with positioning data : The actual target speed, considering the override and speed limit value, etc., is stored. "0" is stored when positioning is completed.</li> <li>• During interpolation of position control : The composite speed or reference axis speed is stored in the reference axis address, and "0" is stored in the interpolation axis address.</li> <li>• During interpolation of speed control : The target speeds of each axis are stored in the monitor of the reference axis and interpolation axis.</li> <li>• During JOG operation : The actual target speed, considering the JOG speed limit value for the JOG speed, is stored.</li> <li>• During manual pulse generator operation : "0" is stored.</li> </ul> |  |
| <b>Md.34</b> Movement amount after near-point dog ON | <ul style="list-style-type: none"> <li>• "0" is stored when machine OPR starts.</li> <li>• After machine OPR starts, the movement amount from the near-point dog ON to the machine OPR completion is stored.<br/>(Movement amount: Movement amount to machine OPR completion using near-point dog ON as "0".)</li> <li>• "0" is always stored when using the stopper method 1), 2), or 3).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                            |  |
| <b>Md.35</b> Torque limit stored value               | <p>The " <b>Pr.17</b> Torque limit setting value" or " <b>Cd.22</b> New torque value" is stored.</p> <ul style="list-style-type: none"> <li>• During positioning start, JOG operation start, manual pulse generator operation ...The " <b>Pr.17</b> Torque limit setting value" is stored.</li> <li>• When value other than 0 is set to " <b>Cd.22</b> New torque value" ...The " <b>Cd.22</b> New torque value" is stored.</li> </ul>                                                                                                                                                                                                                                                                                                                                                           |  |

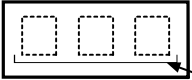
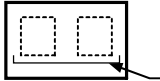
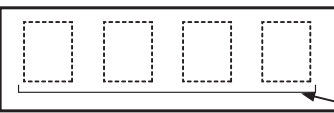
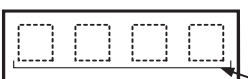
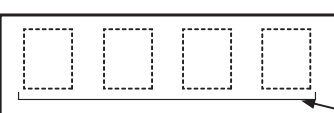
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                                                   | Reading the monitor value | Default value | Storage buffer memory address |        |        |          |    |            |   |         |   |      |    |    |    |      |    |        |   |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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| <div>■ Monitoring is carried out with a hexadecimal display.</div> <div><div><div>Monitor value</div><div><div>E</div><div>F</div><div>G</div><div>H</div></div><div><div>A</div><div>B</div><div>C</div><div>D</div></div></div><div>Low-order buffer memory Example) 820</div><div><div>b15</div><div>b12</div><div>b8</div><div>b4</div><div>b0</div></div><div><div>E</div><div>F</div><div>G</div><div>H</div></div><div>High-order buffer memory Example) 821</div><div><div>b31</div><div>b28</div><div>b24</div><div>b20</div><div>b16</div></div><div><div>A</div><div>B</div><div>C</div><div>D</div></div></div> <div>◇ Sorting</div> <div><div>(High-order buffer memory)</div><div>(Low-order buffer memory)</div></div> <div><div>A</div><div>B</div><div>C</div><div>D</div><div>E</div><div>F</div><div>G</div><div>H</div></div> <div>◇ Converted from hexadecimal to decimal</div> <div><div>Decimal integer value</div><div>R</div></div> <div>◇ Unit conversion <math>R \times 10^n</math></div> <div><div>Actual value</div><div><div>Md.33</div><div>Target speed</div></div><div><div>Md.34</div><div>Movement amount after near-point dog ON</div></div></div> <div><div>Unit conversion table (Md.33)</div><table><tr><th>n</th><th>Unit</th></tr><tr><td>-2</td><td>mm/min</td></tr><tr><td>-3</td><td>inch/min</td></tr><tr><td>-3</td><td>degree/min</td></tr><tr><td>0</td><td>pulse/s</td></tr></table><div>Unit conversion table (Md.34)</div><table><tr><th>n</th><th>Unit</th></tr><tr><td>-1</td><td>μm</td></tr><tr><td>-5</td><td>inch</td></tr><tr><td>-5</td><td>degree</td></tr><tr><td>0</td><td>pulse</td></tr></table></div> <tr><td><div>■ Monitoring is carried out with a decimal display.</div><div><div>Monitor value</div><div><div></div><div></div><div></div></div></div><div>● Storage value 1 to 500 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| Storage item                                                          | Storage details                                                                                                                                                                                                                                                                                                                                           |  |
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| <b>[Md.36]</b> Special start data instruction code setting value      | <ul style="list-style-type: none"> <li>The "instruction code" used with special start and indicated by the start data pointer currently being executed is stored.</li> </ul>                                                                                                                                                                              |  |
| <b>[Md.37]</b> Special start data instruction parameter setting value | <p>The "instruction parameter" used with special start and indicated by the start data pointer currently being executed is stored.</p> <p>The stored value differs according to the value set for <b>[Md.36]</b>.</p>                                                                                                                                     |  |
| <b>[Md.38]</b> Start positioning data No. setting value               | <ul style="list-style-type: none"> <li>The "positioning data No." indicated by the start data pointer currently being executed is stored.</li> </ul>                                                                                                                                                                                                      |  |
| <b>[Md.39]</b> In speed limit flag                                    | <ul style="list-style-type: none"> <li>If the speed exceeds the "<b>[Pr.8]</b> Speed limit value" due to a speed change or override, the speed limit functions, and the in speed limit flag turns ON.</li> <li>When the speed drops to less than "<b>[Pr.8]</b> Speed limit value", or when the axis stops, the in speed limit flag turns OFF.</li> </ul> |  |
| <b>[Md.40]</b> In speed change processing flag                        | <ul style="list-style-type: none"> <li>The speed change process flag turns ON when the speed is changed during positioning control.</li> <li>After the speed change process is completed or when deceleration starts with the stop signal during the speed change process, the in speed change process flag turns OFF.</li> </ul>                         |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Reading the monitor value | Default value   | Storage buffer memory address |          |        |        |                      |                    |         |    |                    |          |   |     |     |      |      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------|-------------------------------|----------|--------|--------|----------------------|--------------------|---------|----|--------------------|----------|---|-----|-----|------|------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                           |                 | Axis 1                        | Axis 2   | Axis 3 | Axis 4 |                      |                    |         |    |                    |          |   |     |     |      |      |
| <div>■ Monitoring is carried out with a decimal display.</div> <div><div>Monitor value<div><div></div><div></div></div></div><div>● Storage value<br/>00: Block start (Normal start)<br/>01: Condition start<br/>02: Wait start<br/>03: Simultaneous start<br/>04: FOR loop<br/>05: FOR condition<br/>06: NEXT</div></div>                                                                                                                                                          | 0                         | 827             | 927                           | 1027     | 1127   |        |                      |                    |         |    |                    |          |   |     |     |      |      |
| <div>■ Monitoring is carried out with a decimal display.</div> <div><div>Monitor value<div><div></div><div></div><div></div></div></div><div>● Storage value<table><tr><th>Md.36 setting value</th><th>Stored contents</th><th>Storage value</th></tr><tr><td>00<br/>06</td><td>None</td><td>None</td></tr><tr><td>01<br/>02<br/>03<br/>05</td><td>Condition data No.</td><td>1 to 10</td></tr><tr><td>04</td><td>No. of repetitions</td><td>0 to 255</td></tr></table></div></div> | Md.36 setting value       | Stored contents | Storage value                 | 00<br>06 | None   | None   | 01<br>02<br>03<br>05 | Condition data No. | 1 to 10 | 04 | No. of repetitions | 0 to 255 | 0 | 828 | 928 | 1028 | 1128 |
| Md.36 setting value                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Stored contents           | Storage value   |                               |          |        |        |                      |                    |         |    |                    |          |   |     |     |      |      |
| 00<br>06                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | None                      | None            |                               |          |        |        |                      |                    |         |    |                    |          |   |     |     |      |      |
| 01<br>02<br>03<br>05                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Condition data No.        | 1 to 10         |                               |          |        |        |                      |                    |         |    |                    |          |   |     |     |      |      |
| 04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | No. of repetitions        | 0 to 255        |                               |          |        |        |                      |                    |         |    |                    |          |   |     |     |      |      |
| <div>■ Monitoring is carried out with a decimal display.</div> <div><div>Monitor value<div><div></div><div></div><div></div><div></div></div></div><div>● Storage value<br/>1 to 600, 9001 to 9003</div></div>                                                                                                                                                                                                                                                                      | 0                         | 829             | 929                           | 1029     | 1129   |        |                      |                    |         |    |                    |          |   |     |     |      |      |
| <div>■ Monitoring is carried out with a decimal display.</div> <div><div>Monitor value<div><div></div><div></div></div></div><div>● Storage value<br/>0: Not in speed limit (OFF)<br/>1: In speed limit (ON)</div></div>                                                                                                                                                                                                                                                            | 0                         | 830             | 930                           | 1030     | 1130   |        |                      |                    |         |    |                    |          |   |     |     |      |      |
| <div>■ Monitoring is carried out with a decimal display.</div> <div><div>Monitor value<div><div></div><div></div></div></div><div>● Storage value<br/>0: Not in speed change (OFF)<br/>1: In speed change (ON)</div></div>                                                                                                                                                                                                                                                          | 0                         | 831             | 931                           | 1031     | 1131   |        |                      |                    |         |    |                    |          |   |     |     |      |      |

| Storage item                                     | Storage details                                                                                                                                                                                                                                                                                                                                        |  |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Md.41</b> Special start repetition counter    | <ul style="list-style-type: none"> <li>This area stores the remaining number of repetitions during "repetitions" specific to special starting.</li> <li>The count is decremented by one (-1) at the loop end.</li> <li>The control comes out of the loop when the count reaches "0".</li> <li>This area stores "0" within an infinite loop.</li> </ul> |  |
| <b>Md.42</b> Control system repetition counter   | <ul style="list-style-type: none"> <li>This area stores the remaining number of repetitions during "repetitions" specific to control system.</li> <li>The count is decremented by one (-1) at the loop start.</li> <li>The loop is terminated with the positioning data of control method "LEND" after the counter becomes "0".</li> </ul>             |  |
| <b>Md.43</b> Start data pointer being executed   | <ul style="list-style-type: none"> <li>This area stores a point No. (1 to 50) attached to the start data currently being executed.</li> <li>This area stores "0" after completion of a positioning operation.</li> </ul>                                                                                                                               |  |
| <b>Md.44</b> Positioning data No. being executed | <ul style="list-style-type: none"> <li>This area stores a positioning data No. attached to the positioning data currently being executed.</li> <li>This area stores "0" after completion of a positioning operation.</li> <li>This area stores "0" when the JOG/inching operation is executed.</li> </ul>                                              |  |
| <b>Md.45</b> Block No. being executed            | <ul style="list-style-type: none"> <li>When the operation is controlled by "block start data", this area stores a block number (7000 to 7004) attached to the block currently being executed.</li> <li>At other times, this area stores "0".</li> <li>This area stores "0" after completion of a positioning operation.</li> </ul>                     |  |
| <b>Md.46</b> Last executed positioning data No.  | <ul style="list-style-type: none"> <li>This area stores the positioning data No. attached to the positioning data that was executed last time.</li> <li>The value is retained until a new positioning operation is executed.</li> <li>This area stores "0" when the JOG/inching operation is executed.</li> </ul>                                      |  |
| <b>Md.47</b> Positioning data being executed     | <ul style="list-style-type: none"> <li>The addresses shown to the right store details of the positioning data currently being executed (positioning data No. given by <b>Md.44</b>).</li> </ul>                                                                                                                                                        |  |
| <b>Md.48</b> Deceleration start flag             | <ul style="list-style-type: none"> <li>"1" is stored when the constant speed status or acceleration status switches to the deceleration status during position control whose operation pattern is "Positioning complete".</li> <li>"0" is stored at the next operation start or manual pulse generator operation enable.</li> </ul>                    |  |

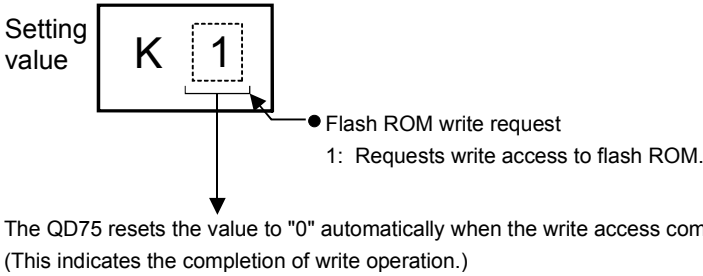
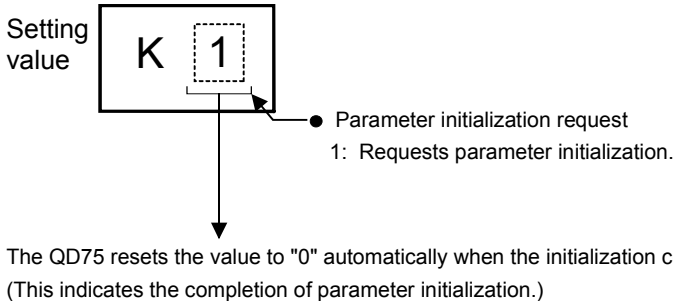
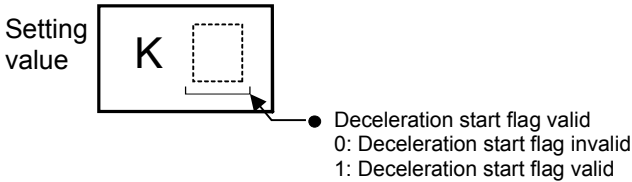
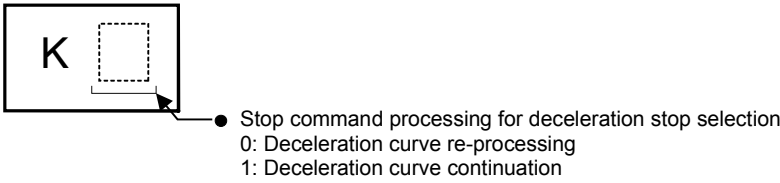


|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Reading the monitor value | Default value | Storage buffer memory address |                     |              |              |                |           |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                           |               | Axis 1                        | Axis 2              | Axis 3       | Axis 4       |                |           |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| <div>■ Monitoring is carried out with a decimal display.</div> <div>Monitor value  ● Storage value 0 to 255</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           | 0             | 832                           | 932                 | 1032         | 1132         |                |           |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| <div>■ Monitoring is carried out with a hexadecimal display.</div> <div>Monitor value  ● Storage value 0 to FFFF</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                           | 0000H         | 833                           | 933                 | 1033         | 1133         |                |           |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| <div>■ Monitoring is carried out with a decimal display.</div> <div>Monitor value  ● Storage value 1 to 50</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                           | 0             | 834                           | 934                 | 1034         | 1134         |                |           |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| <div>■ Monitoring is carried out with a decimal display.</div> <div>Monitor value  ● Storage value 1 to 600, 9001 to 9003</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                           | 0             | 835                           | 935                 | 1035         | 1135         |                |           |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| <div>■ Monitoring is carried out with a decimal display.</div> <div>Monitor value  ● Storage value 7000 to 7004</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                           | 0             | 836                           | 936                 | 1036         | 1136         |                |           |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| <div>■ Monitoring is carried out with a decimal display.</div> <div>Monitor value  ● Storage value 1 to 600, 9001 to 9003</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           | 0             | 837                           | 937                 | 1037         | 1137         |                |           |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| Information is stored in the following addresses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                           | 0             | 838 to 847                    | 938 to 947          | 1038 to 1047 | 1138 to 1147 |                |           |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| <table><tr><th colspan="4">Stored address</th><th rowspan="2">Stored item</th><th rowspan="2">Reference</th></tr><tr><th>Axis1</th><th>Axis2</th><th>Axis3</th><th>Axis4</th></tr><tr><td>838</td><td>938</td><td>1038</td><td>1138</td><td>Positioning identifier</td><td>Da.1 to Da.5</td></tr><tr><td>839</td><td>939</td><td>1039</td><td>1139</td><td>M code</td><td>Da.10</td></tr><tr><td>840</td><td>940</td><td>1040</td><td>1140</td><td>Dwell time</td><td>Da.9</td></tr><tr><td>841</td><td>941</td><td>1041</td><td>1141</td><td>Positioning option * 1</td><td>Da.27 to Da.29</td></tr><tr><td>842</td><td>942</td><td>1042</td><td>1142</td><td rowspan="2">Command speed</td><td rowspan="2">Da.8</td></tr><tr><td>843</td><td>943</td><td>1043</td><td>1143</td></tr><tr><td>844</td><td>944</td><td>1044</td><td>1144</td><td rowspan="2">Positioning address</td><td rowspan="2">Da.6</td></tr><tr><td>845</td><td>945</td><td>1045</td><td>1145</td></tr><tr><td>846</td><td>946</td><td>1046</td><td>1146</td><td rowspan="2">Arc address</td><td rowspan="2">Da.7</td></tr><tr><td>847</td><td>947</td><td>1047</td><td>1147</td></tr></table> |                           |               |                               |                     |              |              | Stored address |           |                        |                | Stored item | Reference | Axis1 | Axis2 | Axis3 | Axis4 | 838 | 938 | 1038 | 1138 | Positioning identifier | Da.1 to Da.5 | 839 | 939 | 1039 | 1139 | M code | Da.10 | 840 | 940 | 1040 | 1140 | Dwell time | Da.9 | 841 | 941 | 1041 | 1141 | Positioning option * 1 | Da.27 to Da.29 | 842 | 942 | 1042 | 1142 | Command speed | Da.8 | 843 | 943 | 1043 | 1143 | 844 | 944 | 1044 | 1144 | Positioning address | Da.6 | 845 | 945 | 1045 | 1145 | 846 | 946 | 1046 | 1146 | Arc address | Da.7 | 847 | 947 | 1047 | 1147 |
| Stored address                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                           |               |                               |                     |              |              | Stored item    | Reference |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| Axis1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Axis2                     |               |                               |                     |              |              |                |           | Axis3                  | Axis4          |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| 838                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 938                       |               |                               |                     |              |              | 1038           | 1138      | Positioning identifier | Da.1 to Da.5   |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| 839                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 939                       |               |                               |                     |              |              | 1039           | 1139      | M code                 | Da.10          |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| 840                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 940                       |               |                               |                     |              |              | 1040           | 1140      | Dwell time             | Da.9           |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| 841                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 941                       |               |                               |                     |              |              | 1041           | 1141      | Positioning option * 1 | Da.27 to Da.29 |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| 842                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 942                       |               |                               |                     |              |              | 1042           | 1142      | Command speed          | Da.8           |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| 843                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 943                       |               |                               |                     |              |              | 1043           | 1143      |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| 844                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 944                       | 1044          | 1144                          | Positioning address | Da.6         |              |                |           |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| 845                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 945                       | 1045          | 1145                          |                     |              |              |                |           |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| 846                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 946                       | 1046          | 1146                          | Arc address         | Da.7         |              |                |           |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| 847                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 947                       | 1047          | 1147                          |                     |              |              |                |           |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| * 1: If "0: Invalid" is set in "[Pr.70] Positioning option valid/invalid setting", 0 is always stored.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                           |               |                               |                     |              |              |                |           |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |
| <div>■ Monitoring is carried out with a decimal display.</div> <div>Monitor value  ● Storage value<br/>0: Status other than below<br/>1: Status from deceleration start to next operation start or manual pulse generator operation enable</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                           | 0             | 899                           | 999                 | 1099         | 1199         |                |           |                        |                |             |           |       |       |       |       |     |     |      |      |                        |              |     |     |      |      |        |       |     |     |      |      |            |      |     |     |      |      |                        |                |     |     |      |      |               |      |     |     |      |      |     |     |      |      |                     |      |     |     |      |      |     |     |      |      |             |      |     |     |      |      |

## 5.7 List of control data

## 5.7.1 System control data

| Setting item                                                             | Setting details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                        |                                     |                                     |  |
|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------|-------------------------------------|--|
| <div>Cd.1</div> Flash ROM write request                                  | <ul style="list-style-type: none"><li>Requests writing of data (parameters, positioning data, and block start data) from the buffer memory to the flash ROM.</li></ul>                                                                                                                                                                                                                                                                                                                                                                       |                                                                        |                                     |                                     |  |
| <div>Cd.2</div> Parameter initialization request                         | <ul style="list-style-type: none"><li>Requests initialization of setting data.<br/>Initialization: Resetting of setting data to default values<br/>Note: After completing the initialization of setting data, reset the CPU module or reboot the programmable controller power.</li></ul> <div>Initialized setting data</div> <table><tr><td>Parameters (<div>Pr.1</div> to <div>Pr.57</div>, <div>Pr.150</div>)</td></tr><tr><td>Positioning data (No. 1 to No. 600)</td></tr><tr><td>Block start data (No. 7000 to 7004)</td></tr></table> | Parameters ( <div>Pr.1</div> to <div>Pr.57</div> , <div>Pr.150</div> ) | Positioning data (No. 1 to No. 600) | Block start data (No. 7000 to 7004) |  |
| Parameters ( <div>Pr.1</div> to <div>Pr.57</div> , <div>Pr.150</div> )   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                        |                                     |                                     |  |
| Positioning data (No. 1 to No. 600)                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                        |                                     |                                     |  |
| Block start data (No. 7000 to 7004)                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                        |                                     |                                     |  |
| <div>Cd.41</div> Deceleration start flag valid                           | Set whether " <div>Md.48</div> Deceleration start flag" is made valid or invalid.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                        |                                     |                                     |  |
| <div>Cd.42</div> Stop command processing for deceleration stop selection | <ul style="list-style-type: none"><li>Set the stop command processing for deceleration stop function (deceleration curve re-processing/deceleration curve continuation).</li></ul>                                                                                                                                                                                                                                                                                                                                                           |                                                                        |                                     |                                     |  |

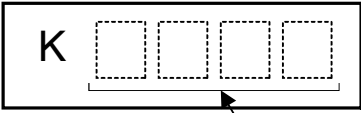
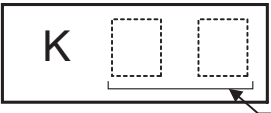
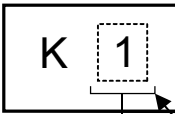
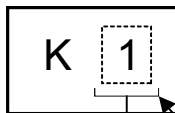
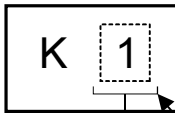
|                       | Setting value                                                                                                                                                                                                                                                                                                                                       | Default value | Storage buffer memory address (common to axes 1 to 4) |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------------|
| ■ Set with a decimal. | <p>Setting value</p>  <p>Flash ROM write request<br/>1: Requests write access to flash ROM.</p> <p>The QD75 resets the value to "0" automatically when the write access completes.<br/>(This indicates the completion of write operation.)</p>                     | 0             | 1900                                                  |
| ■ Set with a decimal. | <p>Setting value</p>  <p>Parameter initialization request<br/>1: Requests parameter initialization.</p> <p>The QD75 resets the value to "0" automatically when the initialization completes.<br/>(This indicates the completion of parameter initialization.)</p> | 0             | 1901                                                  |
| ■ Set with a decimal. | <p>Setting value</p>  <p>Deceleration start flag valid<br/>0: Deceleration start flag invalid<br/>1: Deceleration start flag valid</p>                                                                                                                           | 0             | 1905                                                  |
| ■ Set with a decimal. | <p>Setting value</p>  <p>Stop command processing for deceleration stop selection<br/>0: Deceleration curve re-processing<br/>1: Deceleration curve continuation</p>                                                                                             | 0             | 1907                                                  |

| Setting item                                              | Setting details                                                                                                                                                                                                                                                                                                              |  |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Cd.43</b> Output timing selection of near pass control | <ul style="list-style-type: none"><li>• Select the timing to output the difference (<math>\Delta \ell</math>) between the actual and the set positioning end addresses in continuous path control, in which the difference (<math>\Delta \ell</math>) is output during the execution of the next positioning data.</li></ul> |  |

|  | Setting value                                                                                                                                                                                                                                 | Default value | Storage buffer<br>memory address<br>(common to axes 1<br>to 4) |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------------------------|
|  | <p>■ Set with a decimal.</p> <p>Setting<br/>value</p>  <p>● Output timing selection of near pass control<br/>0: At constant speed<br/>1: At deceleration</p> | 0             | 1934                                                           |

## 5.7.2 Axis control data

| Setting item                               | Setting details                                                                                                                                                                                                                                                                  |  |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Cd.3</b> Positioning start No.          | <ul style="list-style-type: none"> <li>Set the positioning start No.<br/>(Only 1 to 600 for the Pre-reading start function. For details, refer to Section 12.7.7 "Pre-reading start function".)</li> </ul>                                                                       |  |
| <b>Cd.4</b> Positioning starting point No. | <ul style="list-style-type: none"> <li>Set a "starting point No." (1 to 50) if block start data is used for positioning.<br/>(Handled as "1" if the value of other than 1 to 50 is set.)</li> </ul>                                                                              |  |
| <b>Cd.5</b> Axis error reset               | <ul style="list-style-type: none"> <li>Clears the axis error detection, axis error No., axis warning detection and axis warning No.</li> <li>When the QD75 axis operation state is "Error", the error is cleared and the QD75 is returned to the "Standing" state.</li> </ul>    |  |
| <b>Cd.6</b> Restart command                | <ul style="list-style-type: none"> <li>When positioning is stopped for any reason (when axis operation state is "Stopped"), set "1" in <b>Cd.6</b>. Positioning will be carried out again from the stopped position to the end point of the stopped positioning data.</li> </ul> |  |
| <b>Cd.7</b> M code OFF request             | <ul style="list-style-type: none"> <li>The M code ON signal turns OFF.</li> </ul>                                                                                                                                                                                                |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Setting value | Default value | Storage buffer memory address |        |        |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|-------------------------------|--------|--------|--------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |               | Axis 1                        | Axis 2 | Axis 3 | Axis 4 |
| <p>■ Set with a decimal.</p> <p>Setting value</p>  <ul style="list-style-type: none"> <li>● Positioning data No. <ul style="list-style-type: none"> <li>• 1 to 600 : Positioning data No.</li> <li>• 7000 to 7004 : Block start designation</li> <li>• 9001 : Machine OPR</li> <li>• 9002 : Fast-OPR</li> <li>• 9003 : Current value changing</li> <li>• 9004 : Simultaneous starting of multiple axes</li> </ul> </li> </ul> |               | 0             | 1500                          | 1600   | 1700   | 1800   |
| <p>■ Set with a decimal.</p> <p>Setting value</p>  <ul style="list-style-type: none"> <li>● Positioning starting point No.<br/>1 to 50</li> </ul>                                                                                                                                                                                                                                                                             |               | 0             | 1501                          | 1601   | 1701   | 1801   |
| <p>■ Set with a decimal.</p> <p>Setting value</p>  <ul style="list-style-type: none"> <li>● Error reset request<br/>1: Axis error is reset.</li> </ul> <p>After the axis error reset is completed, "0" is stored by the QD75 automatically.<br/>(Indicates that the axis error reset is completed.)</p>                                                                                                                     |               | 0             | 1502                          | 1602   | 1702   | 1802   |
| <p>■ Set with a decimal.</p> <p>Setting value</p>  <ul style="list-style-type: none"> <li>● Restart command<br/>1: Restarts</li> </ul> <p>After restart acceptance is completed, "0" is stored by the QD75 automatically.<br/>(Indicates that the restart acceptance is completed.)</p>                                                                                                                                     |               | 0             | 1503                          | 1603   | 1703   | 1803   |
| <p>■ Set with a decimal.</p> <p>Setting value</p>  <ul style="list-style-type: none"> <li>● M code OFF request<br/>1: M code ON signal turns OFF</li> </ul> <p>After the M code ON signal turns OFF, "0" is stored by the QD75 automatically.<br/>(Indicates that the OFF request is completed.)</p>                                                                                                                        |               | 0             | 1504                          | 1604   | 1704   | 1804   |

| Setting item                           | Setting details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                          |                                              |                                          |                                              |                  |               |                                  |                                  |               |                                  |  |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|----------------------------------------------|------------------------------------------|----------------------------------------------|------------------|---------------|----------------------------------|----------------------------------|---------------|----------------------------------|--|
| <div>Cd.8</div> External command valid | <ul style="list-style-type: none"><li>Validates or in validates external command signals.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |                                              |                                          |                                              |                  |               |                                  |                                  |               |                                  |  |
| <div>Cd.9</div> New current value      | <ul style="list-style-type: none"><li>When changing the "current feed value" using the start No. "9003", use this data item to specify a new feed value.</li><li>Set a value within the following range:</li></ul> <table><tr><td><div>Pr.1</div>Unit setting</td><td>mm<br/>(<math>\times 10^{-1} \mu\text{m}</math>)</td><td>inch<br/>(<math>\times 10^{-5} \text{inch}</math>)</td><td>degree<br/>(<math>\times 10^{-5} \text{degree}</math>)</td><td>pulse<br/>(pulse)</td></tr><tr><td>Setting range</td><td>-2147483648<br/>to<br/>+2147483647</td><td>-2147483648<br/>to<br/>+2147483647</td><td>0 to 35999999</td><td>-2147483648<br/>to<br/>+2147483647</td></tr></table> | <div>Pr.1</div> Unit setting             | mm<br>( $\times 10^{-1} \mu\text{m}$ )       | inch<br>( $\times 10^{-5} \text{inch}$ ) | degree<br>( $\times 10^{-5} \text{degree}$ ) | pulse<br>(pulse) | Setting range | -2147483648<br>to<br>+2147483647 | -2147483648<br>to<br>+2147483647 | 0 to 35999999 | -2147483648<br>to<br>+2147483647 |  |
| <div>Pr.1</div> Unit setting           | mm<br>( $\times 10^{-1} \mu\text{m}$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | inch<br>( $\times 10^{-5} \text{inch}$ ) | degree<br>( $\times 10^{-5} \text{degree}$ ) | pulse<br>(pulse)                         |                                              |                  |               |                                  |                                  |               |                                  |  |
| Setting range                          | -2147483648<br>to<br>+2147483647                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | -2147483648<br>to<br>+2147483647         | 0 to 35999999                                | -2147483648<br>to<br>+2147483647         |                                              |                  |               |                                  |                                  |               |                                  |  |



|                       | Setting value                                                                                                                                                                                                                                                                                                                                                                                                                                            | Default value | Storage buffer memory address |        |        |        |      |    |        |   |       |   |              |              |              |              |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------|--------|--------|--------|------|----|--------|---|-------|---|--------------|--------------|--------------|--------------|
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               | Axis 1                        | Axis 2 | Axis 3 | Axis 4 |      |    |        |   |       |   |              |              |              |              |
| ■ Set with a decimal. | <div>Setting value</div> <div><div>K</div><div><div></div><div></div></div></div> <div>● External command valid<br/>0: Invalidates an external command.<br/>1: Validates an external command.</div>                                                                                                                                                                                                                                                      | 0             | 1505                          | 1605   | 1705   | 1805   |      |    |        |   |       |   |              |              |              |              |
| ■ Set with a decimal. | <div>Actual value</div> <div><div>Cd.9</div><div>New current value</div></div> <div>◇ Conversion into an integer value</div> <div><div>× 10<sup>n</sup></div><div>● Unit conversion table (Cd.9)</div><div><table><tr><th>n</th><th>Unit</th></tr><tr><td>-1</td><td>μm</td></tr><tr><td>-5</td><td>inch</td></tr><tr><td>-5</td><td>degree</td></tr><tr><td>0</td><td>pulse</td></tr></table></div><div>Setting value (Decimal)</div><div>R</div></div> | n             | Unit                          | -1     | μm     | -5     | inch | -5 | degree | 0 | pulse | 0 | 1506<br>1507 | 1606<br>1607 | 1706<br>1707 | 1806<br>1807 |
| n                     | Unit                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |                               |        |        |        |      |    |        |   |       |   |              |              |              |              |
| -1                    | μm                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |                               |        |        |        |      |    |        |   |       |   |              |              |              |              |
| -5                    | inch                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |                               |        |        |        |      |    |        |   |       |   |              |              |              |              |
| -5                    | degree                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |                               |        |        |        |      |    |        |   |       |   |              |              |              |              |
| 0                     | pulse                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |                               |        |        |        |      |    |        |   |       |   |              |              |              |              |

| Setting item                                                                                         | Setting details                                                                                                                                                                                                                                     |  |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div>Cd.10</div> New acceleration time value                                                         | <ul style="list-style-type: none"> <li>When changing the acceleration time during a speed change, use this data item to specify a new acceleration time.</li> </ul> <div> <div>Cd.10</div> setting range (unit) </div> <div>0 to 8388608 (ms)</div> |  |
| <div>Cd.11</div> New deceleration time value                                                         | <ul style="list-style-type: none"> <li>When changing the deceleration time during a speed change, use this data item to specify a new deceleration time.</li> </ul> <div> <div>Cd.11</div> setting range (unit) </div> <div>0 to 8388608 (ms)</div> |  |
| <div>Cd.12</div> Acceleration/deceleration time change during speed change, enable/disable selection | <ul style="list-style-type: none"> <li>Enables or disables modifications to the acceleration/deceleration time during a speed change.</li> </ul>                                                                                                    |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Setting value | Default value | Storage buffer memory address |              |              |              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|-------------------------------|--------------|--------------|--------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |               | Axis 1                        | Axis 2       | Axis 3       | Axis 4       |
| <p>■ Set with a decimal.</p> <p>Setting value</p> <div style="border: 1px solid black; padding: 2px; display: inline-block;"> Cd.10 New acceleration time value<br/> Cd.11 New deceleration time value </div> <p>Example: When the "Cd. 10" New acceleration time value" is set as "60000 ms", the buffer memory stores "60000".</p>                                                                                                                                                                                                                        |               | 0             | 1508<br>1509                  | 1608<br>1609 | 1708<br>1709 | 1808<br>1809 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               | 0             | 1510<br>1511                  | 1610<br>1611 | 1710<br>1711 | 1810<br>1811 |
| <p>■ Set with a decimal.</p> <p>Setting value</p> <div style="border: 1px solid black; padding: 5px; display: inline-block;"> K <div style="border: 1px dashed black; width: 40px; height: 20px; display: inline-block; vertical-align: middle;"></div> </div> <ul style="list-style-type: none"> <li>● Acceleration/deceleration time change during speed change, enable/disable selection</li> <li>1 : Enables modifications to acceleration/deceleration time</li> <li>Other than 1: Disables modifications to acceleration/deceleration time</li> </ul> |               | 0             | 1512                          | 1612         | 1712         | 1812         |

| Setting item                                          | Setting details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                      |                                          |                                                                      |                                          |                    |               |                 |                 |                 |                                                                      |  |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|------------------------------------------|----------------------------------------------------------------------|------------------------------------------|--------------------|---------------|-----------------|-----------------|-----------------|----------------------------------------------------------------------|--|
| <div>Cd.13</div> Positioning operation speed override | <ul style="list-style-type: none"><li>To use the positioning operation speed override function, use this data item to specify an "override" value.<ul style="list-style-type: none"><li>* For details of the override function, refer to Section 12.5.2" Override function".</li></ul></li></ul> <p>If the speed becomes lower than the minimum unit due to override 1% or others, it is raised to the minimum unit. At this time, the warning "Less than minimum speed" (warning code: 110) will occur.</p>                                                                                                                |                                      |                                          |                                                                      |                                          |                    |               |                 |                 |                 |                                                                      |  |
| <div>Cd.14</div> New speed value                      | <ul style="list-style-type: none"><li>When changing the speed, use this data item to specify a new speed.</li><li>The operation halts if you specify "0".</li></ul> <table><tr><td><div>Pr.1</div>Unit setting</td><td>mm<br/>(<math>\times 10^{-2}</math> mm/min)</td><td>inch<br/>(<math>\times 10^{-3}</math> inch/min)</td><td>degree<br/>(<math>\times 10^{-3}</math> degree/min)</td><td>pulse<br/>(pulse/s)</td></tr><tr><td>Setting range</td><td>0 to 2000000000</td><td>0 to 2000000000</td><td>0 to 2000000000</td><td>[QD75P□N/QD75D□N]<br/>0 to 4000000<br/>[QD75P□/QD75D□]<br/>0 to 1000000</td></tr></table> | <div>Pr.1</div> Unit setting         | mm<br>( $\times 10^{-2}$ mm/min)         | inch<br>( $\times 10^{-3}$ inch/min)                                 | degree<br>( $\times 10^{-3}$ degree/min) | pulse<br>(pulse/s) | Setting range | 0 to 2000000000 | 0 to 2000000000 | 0 to 2000000000 | [QD75P□N/QD75D□N]<br>0 to 4000000<br>[QD75P□/QD75D□]<br>0 to 1000000 |  |
| <div>Pr.1</div> Unit setting                          | mm<br>( $\times 10^{-2}$ mm/min)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | inch<br>( $\times 10^{-3}$ inch/min) | degree<br>( $\times 10^{-3}$ degree/min) | pulse<br>(pulse/s)                                                   |                                          |                    |               |                 |                 |                 |                                                                      |  |
| Setting range                                         | 0 to 2000000000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0 to 2000000000                      | 0 to 2000000000                          | [QD75P□N/QD75D□N]<br>0 to 4000000<br>[QD75P□/QD75D□]<br>0 to 1000000 |                                          |                    |               |                 |                 |                 |                                                                      |  |
| <div>Cd.15</div> Speed change request                 | <ul style="list-style-type: none"><li>After setting the "<div>Cd.14</div> New speed value", set this data item to "1" to execute the speed change (through validating the new speed value).</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                       |                                      |                                          |                                                                      |                                          |                    |               |                 |                 |                 |                                                                      |  |

|                       | Setting value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Default value | Storage buffer memory address |        |        |        |          |    |            |   |         |   |              |              |              |              |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------|--------|--------|--------|----------|----|------------|---|---------|---|--------------|--------------|--------------|--------------|
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               | Axis 1                        | Axis 2 | Axis 3 | Axis 4 |          |    |            |   |         |   |              |              |              |              |
| ■ Set with a decimal. | <div>Setting value</div> <div><div>K</div><div><div></div><div></div><div></div></div></div> <div>● Override value (%)<br/>1 to 300</div>                                                                                                                                                                                                                                                                                                                                                                                                                         | 100           | 1513                          | 1613   | 1713   | 1813   |          |    |            |   |         |   |              |              |              |              |
| ■ Set with a decimal. | <div>Actual value</div> <div><div>Cd.14</div>New speed value</div> <div>◇ Conversion into an integer value</div> <div>× 10<sup>n</sup></div> <div>● Unit conversion table (Cd.14)</div> <table><tr><th>n</th><th>Unit</th></tr><tr><td>-2</td><td>mm/min</td></tr><tr><td>-3</td><td>inch/min</td></tr><tr><td>-3</td><td>degree/min</td></tr><tr><td>0</td><td>pulse/s</td></tr></table> <div>Setting value (Decimal)</div> <div>R</div> <div>Example: When the "Cd. 14" New speed value" is set as "20000.00mm /min", the buffer memory stores "2000000".</div> | n             | Unit                          | -2     | mm/min | -3     | inch/min | -3 | degree/min | 0 | pulse/s | 0 | 1514<br>1515 | 1614<br>1615 | 1714<br>1715 | 1814<br>1815 |
| n                     | Unit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |                               |        |        |        |          |    |            |   |         |   |              |              |              |              |
| -2                    | mm/min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |                               |        |        |        |          |    |            |   |         |   |              |              |              |              |
| -3                    | inch/min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |                               |        |        |        |          |    |            |   |         |   |              |              |              |              |
| -3                    | degree/min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |                               |        |        |        |          |    |            |   |         |   |              |              |              |              |
| 0                     | pulse/s                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |                               |        |        |        |          |    |            |   |         |   |              |              |              |              |
| ■ Set with a decimal. | <div>Setting value</div> <div><div>K</div><div><div>1</div></div></div> <div>● Speed change request<br/>1: Executes speed change.</div> <div>The QD75 resets the value to "0" automatically when the speed change request has been processed.<br/>(This indicates the completion of speed change request.)</div>                                                                                                                                                                                                                                                  | 0             | 1516                          | 1616   | 1716   | 1816   |          |    |            |   |         |   |              |              |              |              |

| Setting item                                            | Setting details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div>Cd.16</div> Inching movement amount                | <div><div><div><div></div><div>Pr.1</div></div><div>Unit setting</div></div><div><div>mm</div><div>(<math>\times 10^{-1} \mu\text{m}</math>)</div></div><div><div>inch</div><div>(<math>\times 10^{-5} \text{inch}</math>)</div></div><div><div>degree</div><div>(<math>\times 10^{-5} \text{degree}</math>)</div></div><div><div>pulse</div><div>(pulse)</div></div></div> <div><div>Setting range</div><div>0 to 65535</div><div>0 to 65535</div><div>0 to 65535</div><div>0 to 65535</div></div> <div><div><div></div><div></div></div><div><div></div><div></div></div></div> <div><ul style="list-style-type: none"><li>• Use this data item to set the amount of movement by inching.</li><li>• The machine performs a JOG operation if "0" is set.</li><li>• Set a value within the following range:</li></ul></div>                                    |  |
| <div>Cd.17</div> JOG speed                              | <div><div><div><div></div><div>Pr.1</div></div><div>Unit setting</div></div><div><div>mm</div><div>(<math>\times 10^{-2} \text{mm/min}</math>)</div></div><div><div>inch</div><div>(<math>\times 10^{-3} \text{inch/min}</math>)</div></div><div><div>degree</div><div>(<math>\times 10^{-3} \text{degree/min}</math>)</div></div><div><div>pulse</div><div>(pulse/s)</div></div></div> <div><div>Setting range</div><div>0 to 2000000000</div><div>0 to 2000000000</div><div>0 to 2000000000</div><div><div>[QD75P□N/QD75D□N]</div><div>0 to 4000000</div><div>[QD75P□/QD75D□]</div><div>0 to 1000000</div></div></div> <div><div><div></div><div></div></div><div><div></div><div></div></div></div> <div><ul style="list-style-type: none"><li>• Use this data item to set the JOG speed.</li><li>• Set a value within the following range:</li></ul></div> |  |
| <div>Cd.18</div> Continuous operation interrupt request | <div><div><div></div><div></div></div><div><div></div><div></div></div></div> <div><ul style="list-style-type: none"><li>• To interrupt a continuous operation, set "1" to this data item.</li><li>• After processing the interruption request ("1"), the QD75 automatically resets the value to "0".</li></ul></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |

|                       | Setting value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Default value | Storage buffer memory address |        |        |        |          |    |            |   |         |   |              |              |              |              |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------|--------|--------|--------|----------|----|------------|---|---------|---|--------------|--------------|--------------|--------------|
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               | Axis 1                        | Axis 2 | Axis 3 | Axis 4 |          |    |            |   |         |   |              |              |              |              |
| ■ Set with a decimal. | <div>Actual value<div>Cd.16</div>Inching movement amount</div> <div>◇ Conversion into an integer value</div> <div>Setting value (Decimal)<div>R</div></div> <div>● Unit conversion table (Cd.16)</div> <table><thead><tr><th>n</th><th>Unit</th></tr></thead><tbody><tr><td>-1</td><td>μm</td></tr><tr><td>-5</td><td>inch</td></tr><tr><td>-5</td><td>degree</td></tr><tr><td>0</td><td>pulse</td></tr></tbody></table> <div>Example: When the "Cd. 16" Inching movement amount" is set as "1.0 μm", the buffer memory stores "10".</div> | n             | Unit                          | -1     | μm     | -5     | inch     | -5 | degree     | 0 | pulse   | 0 | 1517         | 1617         | 1717         | 1817         |
| n                     | Unit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |                               |        |        |        |          |    |            |   |         |   |              |              |              |              |
| -1                    | μm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |                               |        |        |        |          |    |            |   |         |   |              |              |              |              |
| -5                    | inch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |                               |        |        |        |          |    |            |   |         |   |              |              |              |              |
| -5                    | degree                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |                               |        |        |        |          |    |            |   |         |   |              |              |              |              |
| 0                     | pulse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |                               |        |        |        |          |    |            |   |         |   |              |              |              |              |
| ■ Set with a decimal. | <div>Actual value<div>Cd.17</div>JOG speed</div> <div>◇ Conversion into an integer value</div> <div>Setting value (Decimal)<div>R</div></div> <div>● Unit conversion table (Cd.17)</div> <table><thead><tr><th>n</th><th>Unit</th></tr></thead><tbody><tr><td>-2</td><td>mm/min</td></tr><tr><td>-3</td><td>inch/min</td></tr><tr><td>-3</td><td>degree/min</td></tr><tr><td>0</td><td>pulse/s</td></tr></tbody></table> <div>Example: When the "Cd. 17" JOG speed" is set as "20000.00mm/min", the buffer memory stores "2000000".</div>  | n             | Unit                          | -2     | mm/min | -3     | inch/min | -3 | degree/min | 0 | pulse/s | 0 | 1518<br>1519 | 1618<br>1619 | 1718<br>1719 | 1818<br>1819 |
| n                     | Unit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |                               |        |        |        |          |    |            |   |         |   |              |              |              |              |
| -2                    | mm/min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |                               |        |        |        |          |    |            |   |         |   |              |              |              |              |
| -3                    | inch/min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |                               |        |        |        |          |    |            |   |         |   |              |              |              |              |
| -3                    | degree/min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |                               |        |        |        |          |    |            |   |         |   |              |              |              |              |
| 0                     | pulse/s                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |                               |        |        |        |          |    |            |   |         |   |              |              |              |              |
| ■ Set with a decimal. | <div>Setting value<div>K<div>1</div></div></div> <div>● Interruption request continuous operation 1: Interrupts continuous operation control or continuous path control.</div> <div>The QD75 resets the value to "0" automatically when the continuous operation interrupt request is processed. (This indicates the completion of continuous operation interruption request.)</div>                                                                                                                                                       | 0             | 1520                          | 1620   | 1720   | 1820   |          |    |            |   |         |   |              |              |              |              |

| Setting item                                                    | Setting details                                                                                                                                                                                                                                                                                      |  |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Cd.19</b> OPR request flag OFF request                       | <ul style="list-style-type: none"> <li>The sequence program can use this data item to forcibly turn the OPR request flag from ON to OFF.</li> </ul>                                                                                                                                                  |  |
| <b>Cd.20</b> Manual pulse generator 1 pulse input magnification | <ul style="list-style-type: none"> <li>This data item determines the factor by which the number of pulses from the manual pulse generator is magnified.</li> <li>Value "1001" or higher: read as "1000". (QD75P□N/QD75D□N)</li> <li>Value "101" or higher: read as "100". (QD75P□/QD75D□)</li> </ul> |  |
| <b>Cd.21</b> Manual pulse generator enable flag                 | <ul style="list-style-type: none"> <li>This data item enables or disables operations using a manual pulse generator.</li> </ul>                                                                                                                                                                      |  |
| <b>Cd.22</b> New torque value                                   | <ul style="list-style-type: none"> <li>When changing the " <b>Md.35</b> Torque limit stored value", use this data item to specify a new torque limit stored value.</li> <li>Set a value within the allowable range of the " <b>Pr.17</b> Torque limit setting value".</li> </ul>                     |  |



|                       | Setting value                                                                                                                                                                                                                                                                                                                    | Default value | Storage buffer memory address |              |              |              |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------|--------------|--------------|--------------|
|                       |                                                                                                                                                                                                                                                                                                                                  |               | Axis 1                        | Axis 2       | Axis 3       | Axis 4       |
| ■ Set with a decimal. | <div>Setting value</div> <div><div>K<div>1</div></div><div>● OPR request flag OFF request<br/>1: Turns the "OPR request flag" from ON to OFF.</div><div>The QD75 resets the value to "0" automatically when the OPR request flag is turned OFF.<br/>(This indicates the completion of OPR request flag OFF request.)</div></div> | 0             | 1521                          | 1621         | 1721         | 1821         |
| ■ Set with a decimal. | <div>Setting value</div> <div><div>K<div><div></div><div></div><div></div><div></div></div></div><div>● Manual pulse generator<br/>1 pulse input magnification<br/>1 to 1000 (QD75P□N/QD75D□N)<br/>1 to 100 (QD75P□/QD75D)</div></div>                                                                                           | 1             | 1522<br>1523                  | 1622<br>1623 | 1722<br>1723 | 1822<br>1823 |
| ■ Set with a decimal. | <div>Setting value</div> <div><div>K<div></div></div><div>● Manual pulse generator enable flag<br/>0: Disable manual pulse generator operation.<br/>1: Enable manual pulse generator operation.</div></div>                                                                                                                      | 0             | 1524                          | 1624         | 1724         | 1824         |
| ■ Set with a decimal. | <div>Setting value</div> <div><div>K<div><div></div><div></div><div></div></div></div><div>● New torque value<br/>1 to <div>Pr.17</div>: Torque limit setting value<br/>0: Torque value is not changed</div></div>                                                                                                               | 0             | 1525                          | 1625         | 1725         | 1825         |

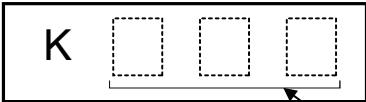
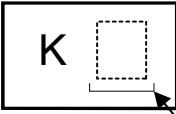
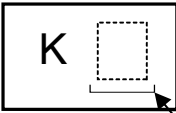
| Setting item                                                                      | Setting details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                       |                                                                                                                                                                                                                       |  |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  |  |
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| <div>Cd.23</div> Speed-position switching control movement amount change register | <div><div><div><div>Pr.1</div><div>Unit setting</div></div><div>mm<br/>(×10<sup>-1</sup> μm)</div></div><div><div><div>inch<br/>(×10<sup>-5</sup> inch)</div></div><div>degree<br/>(×10<sup>-5</sup> degree)</div></div><div><div><div>pulse<br/>(pulse)</div></div><div>Setting range</div></div></div> <div><div>0 to<br/>2147483647</div><div>0 to<br/>2147483647</div><div>0 to<br/>2147483647</div><div>0 to<br/>2147483647</div></div> <div><ul style="list-style-type: none"><li>During the speed control stage of the speed-position switching control (INC mode), it is possible to change the specification of the movement amount during the position control stage. For that, use this data item to specify a new movement amount.</li><li>The new movement amount has to be set during the speed control stage of the speed-position switching control (INC mode).</li><li>The value is reset to "0" when the next operation starts.</li><li>Set a value within the following range:</li></ul></div> <tr><td><div>Cd.24</div>Speed-position switching enable flag</td><td><div><ul style="list-style-type: none"><li>Set whether the external control signal (external command signal [CHG]: "speed-position, position-speed switching request" is selected) is enabled or not.</li></ul></div></td><td></td></tr> <tr><td><div>Cd.25</div>Position-speed switching control speed change register</td><td><div><div><div><div>Pr.1</div><div>Unit setting</div></div><div>mm<br/>(×10<sup>-2</sup> mm/min)</div></div><div><div><div>inch<br/>(×10<sup>-3</sup> inch/min)</div></div><div>degree<br/>(×10<sup>-3</sup> degree/min)</div></div><div><div><div>pulse<br/>(pulse/s)</div></div><div>Setting range</div></div></div><div><div>0 to<br/>2000000000</div><div>0 to<br/>2000000000</div><div>0 to<br/>2000000000</div><div>[QD75P□N/QD75D□N]<br/>0 to 4000000<br/>[QD75P□/QD75D□]<br/>0 to 1000000</div></div><div><ul style="list-style-type: none"><li>During the position control stage of the position-speed switching control, it is possible to change the specification of the speed during the speed control stage. For that, use this data item to specify a new speed.</li><li>The new speed has to be set during the position control stage of the position-speed switching control.</li><li>The value is reset to "0" when the next operation starts.</li><li>Set a value within the following range:</li></ul></div><tr><td></td><td></td><td></td></tr></td></tr> | <div>Cd.24</div> Speed-position switching enable flag | <div><ul style="list-style-type: none"><li>Set whether the external control signal (external command signal [CHG]: "speed-position, position-speed switching request" is selected) is enabled or not.</li></ul></div> |  | <div>Cd.25</div> Position-speed switching control speed change register | <div><div><div><div>Pr.1</div><div>Unit setting</div></div><div>mm<br/>(×10<sup>-2</sup> mm/min)</div></div><div><div><div>inch<br/>(×10<sup>-3</sup> inch/min)</div></div><div>degree<br/>(×10<sup>-3</sup> degree/min)</div></div><div><div><div>pulse<br/>(pulse/s)</div></div><div>Setting range</div></div></div> <div><div>0 to<br/>2000000000</div><div>0 to<br/>2000000000</div><div>0 to<br/>2000000000</div><div>[QD75P□N/QD75D□N]<br/>0 to 4000000<br/>[QD75P□/QD75D□]<br/>0 to 1000000</div></div> <div><ul style="list-style-type: none"><li>During the position control stage of the position-speed switching control, it is possible to change the specification of the speed during the speed control stage. For that, use this data item to specify a new speed.</li><li>The new speed has to be set during the position control stage of the position-speed switching control.</li><li>The value is reset to "0" when the next operation starts.</li><li>Set a value within the following range:</li></ul></div> <tr><td></td><td></td><td></td></tr> |  |  |  |
| <div>Cd.24</div> Speed-position switching enable flag                             | <div><ul style="list-style-type: none"><li>Set whether the external control signal (external command signal [CHG]: "speed-position, position-speed switching request" is selected) is enabled or not.</li></ul></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                       |                                                                                                                                                                                                                       |  |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  |  |
| <div>Cd.25</div> Position-speed switching control speed change register           | <div><div><div><div>Pr.1</div><div>Unit setting</div></div><div>mm<br/>(×10<sup>-2</sup> mm/min)</div></div><div><div><div>inch<br/>(×10<sup>-3</sup> inch/min)</div></div><div>degree<br/>(×10<sup>-3</sup> degree/min)</div></div><div><div><div>pulse<br/>(pulse/s)</div></div><div>Setting range</div></div></div> <div><div>0 to<br/>2000000000</div><div>0 to<br/>2000000000</div><div>0 to<br/>2000000000</div><div>[QD75P□N/QD75D□N]<br/>0 to 4000000<br/>[QD75P□/QD75D□]<br/>0 to 1000000</div></div> <div><ul style="list-style-type: none"><li>During the position control stage of the position-speed switching control, it is possible to change the specification of the speed during the speed control stage. For that, use this data item to specify a new speed.</li><li>The new speed has to be set during the position control stage of the position-speed switching control.</li><li>The value is reset to "0" when the next operation starts.</li><li>Set a value within the following range:</li></ul></div> <tr><td></td><td></td><td></td></tr>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                       |                                                                                                                                                                                                                       |  |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  |  |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Setting value | Default value | Storage buffer memory address |        |        |          |    |            |   |         |   |              |              |              |              |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |               | Axis 1                        | Axis 2 | Axis 3 | Axis 4   |    |            |   |         |   |              |              |              |              |
| <div>■ Set with a decimal.</div> <div><div>Actual value</div><div><div>Cd.23</div><div>Speed-position switching control movement amount change register</div></div><div>◇ Conversion into an integer value</div><div>Setting value (Decimal)</div><div>R</div><div>◇ Conversion into an integer value</div><div><div>× 10<sup>n</sup></div><div>● Unit conversion table (Cd.23)</div><table><tr><th>n</th><th>Unit</th></tr><tr><td>-1</td><td>μm</td></tr><tr><td>-5</td><td>inch</td></tr><tr><td>-5</td><td>degree</td></tr><tr><td>0</td><td>pulse</td></tr></table></div><div><div>Example: If "Cd. 23 Speed-position switching control movement amount change register" is set as "20000.0 μm", the buffer memory stores "200000".</div></div></div> | n             | Unit          | -1                            | μm     | -5     | inch     | -5 | degree     | 0 | pulse   | 0 | 1526<br>1527 | 1626<br>1627 | 1726<br>1727 | 1826<br>1827 |
| n                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Unit          |               |                               |        |        |          |    |            |   |         |   |              |              |              |              |
| -1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | μm            |               |                               |        |        |          |    |            |   |         |   |              |              |              |              |
| -5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | inch          |               |                               |        |        |          |    |            |   |         |   |              |              |              |              |
| -5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | degree        |               |                               |        |        |          |    |            |   |         |   |              |              |              |              |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | pulse         |               |                               |        |        |          |    |            |   |         |   |              |              |              |              |
| <div>■ Set with a decimal.</div> <div><div>Setting value</div><div>K</div><div>● Speed-position switching enable flag</div><div>0: Speed control will not be taken over by position control even when the external command signal comes ON.</div><div>1: Speed control will be taken over by position control when the external command signal comes ON.</div></div>                                                                                                                                                                                                                                                                                                                                                                                       | 0             | 1528          | 1628                          | 1728   | 1828   |          |    |            |   |         |   |              |              |              |              |
| <div>■ Set with a decimal.</div> <div><div>Actual value</div><div><div>Cd.25</div><div>Position-speed switching control speed change register</div></div><div>◇ Conversion into an integer value</div><div>Setting value (Decimal)</div><div>R</div><div>◇ Conversion into an integer value</div><div><div>× 10<sup>n</sup></div><div>● Unit conversion table (Cd.25)</div><table><tr><th>n</th><th>Unit</th></tr><tr><td>-2</td><td>mm/min</td></tr><tr><td>-3</td><td>inch/min</td></tr><tr><td>-3</td><td>degree/min</td></tr><tr><td>0</td><td>pulse/s</td></tr></table></div><div><div>Example: If "Cd. 25 Position-speed switching control speed change register" is set as "2000.00 mm/min", the buffer memory stores "200000".</div></div></div>   | n             | Unit          | -2                            | mm/min | -3     | inch/min | -3 | degree/min | 0 | pulse/s | 0 | 1530<br>1531 | 1630<br>1631 | 1730<br>1731 | 1830<br>1831 |
| n                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Unit          |               |                               |        |        |          |    |            |   |         |   |              |              |              |              |
| -2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | mm/min        |               |                               |        |        |          |    |            |   |         |   |              |              |              |              |
| -3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | inch/min      |               |                               |        |        |          |    |            |   |         |   |              |              |              |              |
| -3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | degree/min    |               |                               |        |        |          |    |            |   |         |   |              |              |              |              |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | pulse/s       |               |                               |        |        |          |    |            |   |         |   |              |              |              |              |

| Setting item                                                | Setting details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                              |                                                  |                                                                      |                                                  |                    |               |                                  |                                  |                    |                                                                      |     |                                  |                                  |                                  |                                  |  |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------|--------------------|---------------|----------------------------------|----------------------------------|--------------------|----------------------------------------------------------------------|-----|----------------------------------|----------------------------------|----------------------------------|----------------------------------|--|
| <div>Cd.26</div> Position-speed switching enable flag       | <ul style="list-style-type: none"><li>Set whether the external control signal (external command signal [CHG]: "speed-position, position-speed switching request" is selected) is enabled or not.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                              |                                                  |                                                                      |                                                  |                    |               |                                  |                                  |                    |                                                                      |     |                                  |                                  |                                  |                                  |  |
| <div>Cd.27</div> Target position change value (New address) | <ul style="list-style-type: none"><li>When changing the target position during a positioning operation, use this data item to specify a new positioning address.</li><li>Set a value within the following range:</li></ul> <table><tr><th><div>Pr.1</div></th><th>mm<br/>(<math>\times 10^{-1} \mu\text{m}</math>)</th><th>inch<br/>(<math>\times 10^{-5} \text{inch}</math>)</th><th>degree<br/>(<math>\times 10^{-5} \text{degree}</math>)</th><th>pulse<br/>(pulse)</th></tr><tr><td>ABS</td><td>-2147483648<br/>to<br/>+2147483647</td><td>-2147483648<br/>to<br/>+2147483647</td><td>0 to 35999999</td><td>-2147483648<br/>to<br/>+2147483647</td></tr><tr><td>INC</td><td>-2147483648<br/>to<br/>+2147483647</td><td>-2147483648<br/>to<br/>+2147483647</td><td>-2147483648<br/>to<br/>+2147483647</td><td>-2147483648<br/>to<br/>+2147483647</td></tr></table> | <div>Pr.1</div>                              | mm<br>( $\times 10^{-1} \mu\text{m}$ )           | inch<br>( $\times 10^{-5} \text{inch}$ )                             | degree<br>( $\times 10^{-5} \text{degree}$ )     | pulse<br>(pulse)   | ABS           | -2147483648<br>to<br>+2147483647 | -2147483648<br>to<br>+2147483647 | 0 to 35999999      | -2147483648<br>to<br>+2147483647                                     | INC | -2147483648<br>to<br>+2147483647 | -2147483648<br>to<br>+2147483647 | -2147483648<br>to<br>+2147483647 | -2147483648<br>to<br>+2147483647 |  |
| <div>Pr.1</div>                                             | mm<br>( $\times 10^{-1} \mu\text{m}$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | inch<br>( $\times 10^{-5} \text{inch}$ )     | degree<br>( $\times 10^{-5} \text{degree}$ )     | pulse<br>(pulse)                                                     |                                                  |                    |               |                                  |                                  |                    |                                                                      |     |                                  |                                  |                                  |                                  |  |
| ABS                                                         | -2147483648<br>to<br>+2147483647                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -2147483648<br>to<br>+2147483647             | 0 to 35999999                                    | -2147483648<br>to<br>+2147483647                                     |                                                  |                    |               |                                  |                                  |                    |                                                                      |     |                                  |                                  |                                  |                                  |  |
| INC                                                         | -2147483648<br>to<br>+2147483647                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -2147483648<br>to<br>+2147483647             | -2147483648<br>to<br>+2147483647                 | -2147483648<br>to<br>+2147483647                                     |                                                  |                    |               |                                  |                                  |                    |                                                                      |     |                                  |                                  |                                  |                                  |  |
| <div>Cd.28</div> Target position change value (New speed)   | <ul style="list-style-type: none"><li>When changing the target position during a positioning operation, use this data item to specify a new speed.</li><li>The speed will not change if "0" is set.</li><li>Set a value within the following range:</li></ul> <table><tr><th><div>Pr.1</div><div>Unit setting</div></th><th>mm<br/>(<math>\times 10^{-2} \text{mm/min}</math>)</th><th>inch<br/>(<math>\times 10^{-3} \text{inch/min}</math>)</th><th>degree<br/>(<math>\times 10^{-3} \text{degree/min}</math>)</th><th>pulse<br/>(pulse/s)</th></tr><tr><td>Setting range</td><td>0 to<br/>2000000000</td><td>0 to<br/>2000000000</td><td>0 to<br/>2000000000</td><td>[QD75P□N/QD75D□N]<br/>0 to 4000000<br/>[QD75P□/QD75D□]<br/>0 to 1000000</td></tr></table>                                                                                                     | <div>Pr.1</div> <div>Unit setting</div>      | mm<br>( $\times 10^{-2} \text{mm/min}$ )         | inch<br>( $\times 10^{-3} \text{inch/min}$ )                         | degree<br>( $\times 10^{-3} \text{degree/min}$ ) | pulse<br>(pulse/s) | Setting range | 0 to<br>2000000000               | 0 to<br>2000000000               | 0 to<br>2000000000 | [QD75P□N/QD75D□N]<br>0 to 4000000<br>[QD75P□/QD75D□]<br>0 to 1000000 |     |                                  |                                  |                                  |                                  |  |
| <div>Pr.1</div> <div>Unit setting</div>                     | mm<br>( $\times 10^{-2} \text{mm/min}$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | inch<br>( $\times 10^{-3} \text{inch/min}$ ) | degree<br>( $\times 10^{-3} \text{degree/min}$ ) | pulse<br>(pulse/s)                                                   |                                                  |                    |               |                                  |                                  |                    |                                                                      |     |                                  |                                  |                                  |                                  |  |
| Setting range                                               | 0 to<br>2000000000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0 to<br>2000000000                           | 0 to<br>2000000000                               | [QD75P□N/QD75D□N]<br>0 to 4000000<br>[QD75P□/QD75D□]<br>0 to 1000000 |                                                  |                    |               |                                  |                                  |                    |                                                                      |     |                                  |                                  |                                  |                                  |  |
| <div>Cd.29</div> Target position change request flag        | <ul style="list-style-type: none"><li>Requests the target position change during a positioning operation.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                              |                                                  |                                                                      |                                                  |                    |               |                                  |                                  |                    |                                                                      |     |                                  |                                  |                                  |                                  |  |

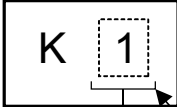
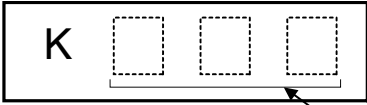
|                       | Setting value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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                                                                                                                     |               | Axis 1                        | Axis 2 | Axis 3 | Axis 4 |
| ■ Set with a decimal. | <div>Setting value<div>K<div></div></div><div>● Position-speed switching enable flag<br/>0: Position control will not be taken over by speed control even when the external command signal [CHG] comes ON.<br/>1: Position control will be taken over by speed control when the external command signal [CHG] comes ON.</div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> <div></div> 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| Setting item                                                                   | Setting details                                                                                                                                                                                                           |  |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Cd.30</b> Simultaneous starting axis start data No. (axis 1 start data No.) | <ul style="list-style-type: none"> <li>• Use these data items to specify a start data No. for each axis that has to start simultaneously.</li> <li>• Set "0" to any axis that should not start simultaneously.</li> </ul> |  |
| <b>Cd.31</b> Simultaneous starting axis start data No. (axis 2 start data No.) |                                                                                                                                                                                                                           |  |
| <b>Cd.32</b> Simultaneous starting axis start data No. (axis 3 start data No.) |                                                                                                                                                                                                                           |  |
| <b>Cd.33</b> Simultaneous starting axis start data No. (axis 4 start data No.) |                                                                                                                                                                                                                           |  |
| <b>Cd.34</b> Step mode                                                         | <ul style="list-style-type: none"> <li>• To perform a step operation, use this data item to specify the units by which the stepping should be performed.</li> </ul>                                                       |  |
| <b>Cd.35</b> Step valid flag                                                   | <ul style="list-style-type: none"> <li>• This data item validates or invalidates step operations.</li> </ul>                                                                                                              |  |

|                                                                                                                                | Setting value | Default value | Storage buffer memory address |        |        |        |
|--------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|-------------------------------|--------|--------|--------|
|                                                                                                                                |               |               | Axis 1                        | Axis 2 | Axis 3 | Axis 4 |
| ■ Set with a decimal.<br><br>Setting value    |               | 0             | 1540                          | 1640   | 1740   | 1840   |
|                                                                                                                                |               | 0             | 1541                          | 1641   | 1741   | 1841   |
|                                                                                                                                |               | 0             | 1542                          | 1642   | 1742   | 1842   |
|                                                                                                                                |               | 0             | 1543                          | 1643   | 1743   | 1843   |
| ■ Set with a decimal.<br><br>Setting value  |               | 0             | 1544                          | 1644   | 1744   | 1844   |
| ■ Set with a decimal.<br><br>Setting value  |               | 0             | 1545                          | 1645   | 1745   | 1845   |

| Setting item                               | Setting details                                                                                                                                                                                                                                                                                                                                       |  |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Cd.36</b> Step start information        | <ul style="list-style-type: none"> <li>To continue the step operation when the step function is used, set "1" in this data item.</li> </ul>                                                                                                                                                                                                           |  |
| <b>Cd.37</b> Skip command                  | <ul style="list-style-type: none"> <li>To skip the current positioning operation, set "1" in this data item.</li> </ul>                                                                                                                                                                                                                               |  |
| <b>Cd.38</b> Teaching data selection       | <ul style="list-style-type: none"> <li>This data item specifies the teaching result write destination.</li> <li>Data are cleared to "0" when the teaching ends.</li> </ul>                                                                                                                                                                            |  |
| <b>Cd.39</b> Teaching positioning data No. | <ul style="list-style-type: none"> <li>This data item specifies data to be produced by teaching.</li> <li>If a value between 1 and 600 is set, a teaching operation is done.</li> <li>The value is cleared to "0" when the QD75 is initialized, when a teaching operation completes, and when an illegal value (601 or higher) is entered.</li> </ul> |  |
| <b>Cd.40</b> ABS direction in degrees      | <ul style="list-style-type: none"> <li>This data item specifies the ABS moving direction carrying out the position control when "degree" is selected as the unit.</li> </ul>                                                                                                                                                                          |  |



|                                                                                                                                                                                                                                                                                                                                                                                  | Setting value | Default value | Storage buffer memory address |        |        |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|-------------------------------|--------|--------|--------|
|                                                                                                                                                                                                                                                                                                                                                                                  |               |               | Axis 1                        | Axis 2 | Axis 3 | Axis 4 |
| <p>■ Set with a decimal.</p> <p>Setting value</p>  <p>● Step start information<br/>1: Continues step operation</p> <p>The QD75 resets the value to "0" automatically when processing of the step start request completes.</p>                                                                   |               | 0             | 1546                          | 1646   | 1746   | 1846   |
| <p>■ Set with a decimal.</p> <p>Setting value</p>  <p>● Skip request<br/>1: Issues a skip request to have the machine decelerate, stop, and then start the next positioning operation.</p> <p>The QD75 resets the value to "0" automatically when processing of the skip request completes.</p> |               | 0             | 1547                          | 1647   | 1747   | 1847   |
| <p>■ Set with a decimal.</p> <p>Setting value</p>  <p>● Teaching data selection<br/>0: Takes the current feed value as a positioning address.<br/>1: Takes the current feed value as an arc data.</p>                                                                                         |               | 0             | 1548                          | 1648   | 1748   | 1848   |
| <p>■ Set with a decimal.</p> <p>Setting value</p>  <p>● Teaching positioning data No.<br/>1 to 600</p>                                                                                                                                                                                        |               | 0             | 1549                          | 1649   | 1749   | 1849   |
| <p>■ Set with a decimal.</p> <p>Setting value</p>  <p>● ABS direction in degrees<br/>0: Shortcut (direction setting ignored)<br/>1: ABS clockwise<br/>2: ABS counterclockwise</p>                                                                                                             |               | 0             | 1550                          | 1650   | 1750   | 1850   |

# MEMO

[illegible]

# CHAPTER 6 SEQUENCE PROGRAM USED FOR POSITIONING CONTROL

The programs required to carry out positioning control with the QD75 are explained in this chapter.

The program required for control is created allowing for the "start conditions", "start time chart", "device settings" and general control configuration.

(The parameters, positioning data, block start data and condition data, etc., must be set in the QD75 according to the control to be executed, and program for setting the control data or a program for starting the various control must be created.)

The first half of this chapter explains the program configuration of general control, and the latter half explains the program details. Create the required program while referring to the various control details explained in PART 2, and to CHAPTER 5 "DATA USED FOR POSITIONING CONTROL".

|       |                                              |       |
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| 6.2   | List of devices used .....                   | 6- 5  |
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## 6.1 Precautions for creating program

The common precautions to be taken when writing data from the CPU module to the QD75 buffer memory are described below.

When diverting any of the program examples introduced in this manual to the actual system, fully verify that there are no problems in the controllability of the target system.

### (1) Reading/writing the data

Setting the data explained in this chapter (various parameters, positioning data, block start data) should be set using GX Configurator-QP.

When set with the sequence program, many sequence programs and devices must be used. This will not only complicate the program, but will also increase the scan time.

When rewriting the positioning data during continuous path control or continuous positioning control, rewrite the data four positioning data items before the actual execution. If the positioning data is not rewritten before the positioning data four items earlier is executed, the process will be carried out as if the data was not rewritten.

### (2) Restrictions to speed change execution interval

Provide an interval of 100ms or more when changing the speed or performing override function with the QD75.

### (3) Process during overrun

Overrun is prevented by the setting of the upper and lower stroke limits with the detail parameter 1.

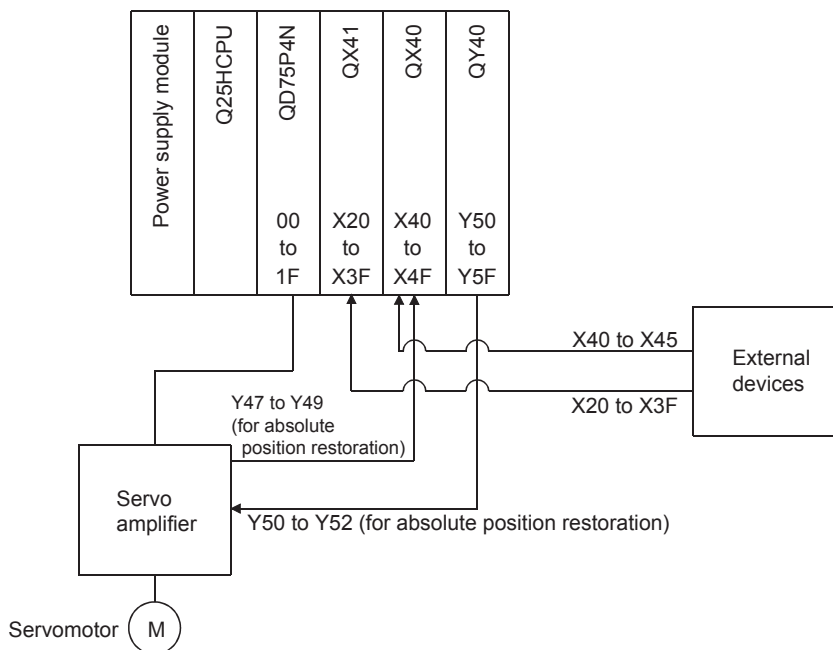
However, this applies only when the QD75 is operating correctly.

It is recommended to create an external circuit including a boundary limit switch to ensure the whole system safety as follows: the external circuit powers OFF the motor when the boundary limit switch operates.

## (4) System configuration

Unless particularly designated, the sequence program for the following system is shown in this chapter and subsequent. In the program, the unit of "0 (mm)" is set for the basic parameter 1.

Refer to Section 6.2 for the application of the devices to be used.



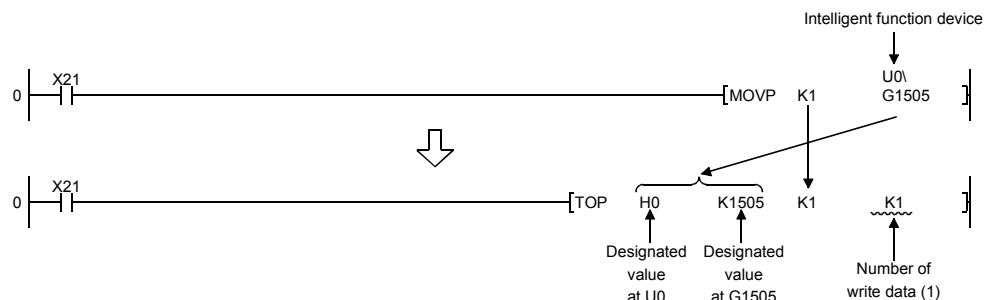
## (5) Communication with QD75

There are two methods for communication with QD75 using the sequence program: a method using an "intelligent function device" and a method using a FROM/TO instruction.

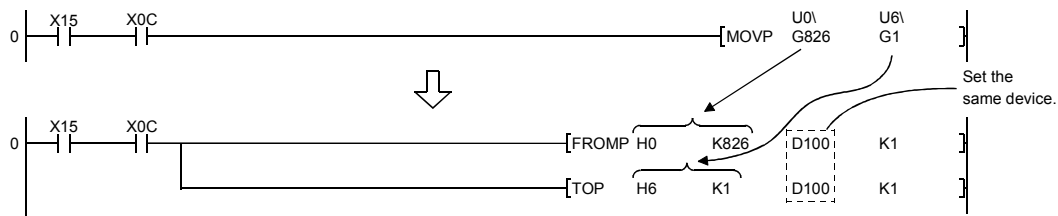
In the sequence program in this chapter and subsequent, the program example using the "intelligent function device" is shown without using a FROM/TO instruction for communication with QD75.

When using the FROM/TO instruction for communication with QD75, change the circuit incorporating the "intelligent function device" as follows.

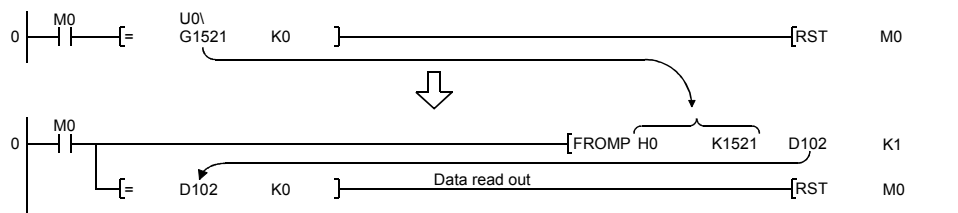
- (a) When the circuit uses the "intelligent function device" on the destination (D) side of a MOV instruction, change the instruction to a TO instruction.



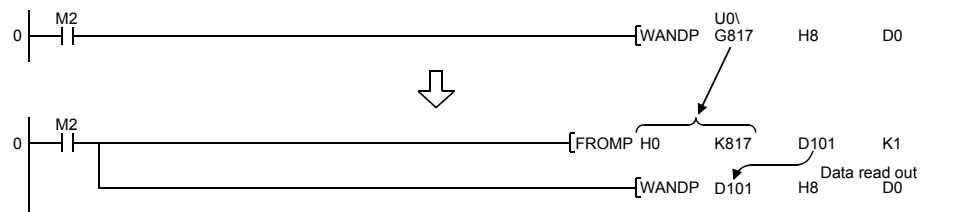
- (b) When the circuit uses the "intelligent function device" on the source(s) side and the destination (D) side of a MOV instruction, change the instruction to a FROM instruction and a TO instruction.



- (c) When the circuit uses the "intelligent function device" for a COMPARISON instruction, change the instruction to a FROM instruction and a COMPARISON instruction.



- (d) When the circuit uses the "intelligent function device" for a WAND instruction, change the instruction to a FROM instruction and a WAND instruction.



### REMARK

Refer to QCPU (Q mode) User's Manual (Functions and Programs Basic Part) for the intelligent function devices.

Refer to QCPU (Q mode) Programming Manual (Common Commands Part) for detail commands used in those programs shown in this chapter and subsequent.

## 6.2 List of devices used

In the sequence programs shown in this chapter and subsequent, the application of the devices used are as follows.

The I/O numbers for QD75 indicate those when QD75 is mounted in the 0-slot of the main base.

If it is mounted in the slot other than the 0-slot of the main base, change the I/O number to that for the position where QD75 was installed.

In addition, change the external inputs, external outputs, internal relays, data registers, and timers according to the system used.

(1) Inputs/outputs, external inputs/external outputs, and internal relays of QD75

| Device name                      |             | Device |        |        |        | Application                                       | Details when ON                                    |
|----------------------------------|-------------|--------|--------|--------|--------|---------------------------------------------------|----------------------------------------------------|
|                                  |             | Axis 1 | Axis 2 | Axis 3 | Axis 4 |                                                   |                                                    |
| Inputs/<br>outputs<br>of<br>QD75 | Input       | X0     |        |        |        | QD75 READY signal                                 | Preparation completed                              |
|                                  |             | X1     |        |        |        | Synchronization flag                              | QD75 buffer memory accessible                      |
|                                  |             | X4     | X5     | X6     | X7     | M code ON signal                                  | M code outputting                                  |
|                                  |             | X8     | X9     | XA     | XB     | Error detection signal                            | Error detected                                     |
|                                  |             | XC     | XD     | XE     | XF     | BUSY signal                                       | BUSY (operating)                                   |
|                                  |             | X10    | X11    | X12    | X13    | Start complete signal                             | Start completed                                    |
|                                  |             | X14    | X15    | X16    | X17    | Positioning complete signal                       | Positioning completed                              |
|                                  | Out-<br>put | Y0     |        |        |        | PLC READY signal                                  | CPU module preparation completed                   |
|                                  |             | Y4     | Y5     | Y6     | Y7     | Axis stop signal                                  | Requesting stop                                    |
|                                  |             | Y8     | YA     | YC     | YE     | Forward run JOG start signal                      | Starting forward run JOG                           |
|                                  |             | Y9     | YB     | YD     | YF     | Reverse run JOG start signal                      | Starting reverse run JOG                           |
|                                  |             | Y10    | Y11    | Y12    | Y13    | Positioning start signal                          | Requesting start                                   |
|                                  |             |        |        |        |        |                                                   |                                                    |
| External<br>input<br>(command)   | X20         | —      |        |        |        | OPR request OFF command                           | Commanding OPR request OFF                         |
|                                  | X21         |        |        |        |        | External command valid command                    | Commanding external command valid setting          |
|                                  | X22         |        |        |        |        | External command invalid command                  | Commanding external command invalid                |
|                                  | X23         |        |        |        |        | Machine OPR command                               | Commanding machine OPR                             |
|                                  | X24         |        |        |        |        | Fast OPR command                                  | Commanding fast OPR                                |
|                                  | X25         |        |        |        |        | Positioning start command                         | Commanding positioning start                       |
|                                  | X26         |        |        |        |        | Speed-position switching operation command        | Commanding speed-position switching operation      |
|                                  | X27         |        |        |        |        | Speed-position switching enable command           | Commanding speed-position switching enable command |
|                                  | X28         |        |        |        |        | Speed-position switching prohibit command         | Commanding speed-position switching prohibit       |
|                                  | X29         |        |        |        |        | Movement amount change command                    | Commanding movement amount change                  |
|                                  | X2A         |        |        |        |        | High-level positioning control start command      | Commanding high-level positioning control start    |
|                                  | X2B         |        |        |        |        | Positioning start command (dedicated instruction) | Commanding positioning start                       |

| Device name                                     | Device |        |        |        | Application                                                 | Details when ON                                                |  |  |
|-------------------------------------------------|--------|--------|--------|--------|-------------------------------------------------------------|----------------------------------------------------------------|--|--|
|                                                 | Axis 1 | Axis 2 | Axis 3 | Axis 4 |                                                             |                                                                |  |  |
| External input (command)                        | X2C    | —      |        |        | M code OFF command                                          | Commanding M code OFF                                          |  |  |
|                                                 | X2D    |        |        |        | JOG operation speed setting command                         | Commanding JOG operation speed setting                         |  |  |
|                                                 | X2E    |        |        |        | Forward run JOG/inching command                             | Commanding forward run JOG/inching operation                   |  |  |
|                                                 | X2F    |        |        |        | Reverse run JOG/inching command                             | Commanding reverse run JOG/inching operation                   |  |  |
|                                                 | X30    |        |        |        | Manual pulse generator operation enable command             | Commanding manual pulse generator operation enable             |  |  |
|                                                 | X31    |        |        |        | Manual pulse generator operation disable command            | Commanding manual pulse generator operation disable            |  |  |
|                                                 | X32    |        |        |        | Speed change command                                        | Commanding speed change                                        |  |  |
|                                                 | X33    |        |        |        | Override command                                            | Commanding override                                            |  |  |
|                                                 | X34    |        |        |        | Acceleration/deceleration time change command               | Commanding acceleration/deceleration time change               |  |  |
|                                                 | X35    |        |        |        | Acceleration/deceleration time change disable command       | Commanding acceleration/deceleration time change disable       |  |  |
|                                                 | X37    |        |        |        | Step operation command                                      | Commanding step operation                                      |  |  |
|                                                 | X38    |        |        |        | Skip operation command                                      | Commanding skip operation                                      |  |  |
|                                                 | X39    |        |        |        | Teaching command                                            | Commanding teaching                                            |  |  |
|                                                 | X3A    |        |        |        | Continuous operation interrupt command                      | Commanding continuous operation interrupt command              |  |  |
|                                                 | X3B    |        |        |        | Restart command                                             | Commanding restart                                             |  |  |
|                                                 | X3C    |        |        |        | Parameter initialization command                            | Commanding parameter initialization                            |  |  |
|                                                 | X3D    |        |        |        | Flash ROM write command                                     | Commanding flash ROM write                                     |  |  |
|                                                 | X3E    | —      |        |        | Error reset command                                         | Commanding error reset                                         |  |  |
|                                                 | X3F    |        |        |        | Stop command                                                | Commanding stop                                                |  |  |
|                                                 | X40    |        |        |        | Position-speed switching operation command                  | Commanding position-speed switching operation                  |  |  |
|                                                 | X41    |        |        |        | Position-speed switching enable command                     | Commanding position-speed switching enable                     |  |  |
|                                                 | X42    |        |        |        | Position-speed switching prohibit command                   | Commanding position-speed switching prohibit                   |  |  |
|                                                 | X43    |        |        |        | Speed change command                                        | Commanding speed change                                        |  |  |
|                                                 | X44    |        |        |        | Inching movement amount setting command                     | Commanding inching movement amount setting                     |  |  |
|                                                 | X45    |        |        |        | Target position change command                              | Commanding target position change                              |  |  |
|                                                 | X4D    |        |        |        | Speed-position switching control (ABS mode) setting command | Commanding speed-position switching control (ABS mode) setting |  |  |
|                                                 | X4E    |        |        |        | Positioning start command (Y start)                         | Positioning start command being given                          |  |  |
| External input (absolute position restoration)  | X47    | —      |        |        | ABS data bit 0                                              | —                                                              |  |  |
|                                                 | X48    |        |        |        | ABS data bit 1                                              | —                                                              |  |  |
|                                                 | X49    |        |        |        | Transmission data READY flag                                | —                                                              |  |  |
| External output (absolute position restoration) | Y50    | —      |        |        | Servo ON signal                                             | —                                                              |  |  |
|                                                 | Y51    |        |        |        | ABS transmission mode                                       | —                                                              |  |  |
|                                                 | Y52    |        |        |        | ABS request mode                                            | —                                                              |  |  |



| Device name    | Device |        |        |        | Application                                      | Details when ON                                     |  |
|----------------|--------|--------|--------|--------|--------------------------------------------------|-----------------------------------------------------|--|
|                | Axis 1 | Axis 2 | Axis 3 | Axis 4 |                                                  |                                                     |  |
| Internal relay | M0     | —      |        |        | OPR request OFF command                          | Commanding OPR request OFF                          |  |
|                | M1     |        |        |        | OPR request OFF command pulse                    | OPR request OFF commanded                           |  |
|                | M2     |        |        |        | OPR request OFF command storage                  | OPR request OFF command held                        |  |
|                | M3     |        |        |        | Fast OPR command                                 | Commanding fast OPR                                 |  |
|                | M4     |        |        |        | Fast OPR command storage                         | Fast OPR command held                               |  |
|                | M5     |        |        |        | Positioning start command pulse                  | Positioning start commanded                         |  |
|                | M6     |        |        |        | Positioning start command storage                | Positioning start command held                      |  |
|                | M7     |        |        |        | In-JOG/Inching operation flag                    | Operating JOG/Inching                               |  |
|                | M8     |        |        |        | Manual pulse generator operation enable command  | Commanding manual pulse generator operation enable  |  |
|                | M9     |        |        |        | Manual pulse generator operating flag            | Operating manual pulse generator                    |  |
|                | M10    |        |        |        | Manual pulse generator operation disable command | Commanding manual pulse generator operation disable |  |
|                | M11    |        |        |        | Speed change command pulse                       | Speed change commanded                              |  |
|                | M12    |        |        |        | Speed change command storage                     | Speed change command held                           |  |
|                | M13    |        |        |        | Override command                                 | Requesting override                                 |  |
|                | M14    |        |        |        | Acceleration/deceleration time change command    | Requesting acceleration/deceleration time change    |  |
|                | M16    |        |        |        | Step operation command pulse                     | Step operation commanded                            |  |
|                | M17    |        |        |        | Skip command pulse                               | Skip commanded                                      |  |
|                | M18    |        |        |        | Skip command storage                             | Skip command held                                   |  |
|                | M19    |        |        |        | Teaching command pulse                           | Teaching commanded                                  |  |
|                | M20    |        |        |        | Teaching command storage                         | Teaching command held                               |  |
|                | M21    |        |        |        | Continuous operation interrupt command           | Requesting continuous operation interrupt           |  |
|                | M22    |        |        |        | Restart command                                  | Requesting restart                                  |  |
|                | M23    |        |        |        | Restart command storage                          | Restart command held                                |  |
|                | M24    |        |        |        | Parameter initialization command pulse           | Parameter initialization commanded                  |  |
|                | M25    |        |        |        | Parameter initialization command storage         | Parameter initialization command held               |  |
|                | M26    |        |        |        | Flash ROM write command pulse                    | Flash ROM write commanded                           |  |
|                | M27    |        |        |        | Flash ROM write command storage                  | Flash ROM write command held                        |  |
|                | M28    | —      |        |        | Error reset                                      | Error reset completed                               |  |
|                | M29    |        |        |        | Stop command pulse                               | Stop commanded                                      |  |
|                | M30    |        |        |        | Target position change command pulse             | Target position change commanded                    |  |
|                | M31    |        |        |        | Target position change command storage           | Target position change command held                 |  |
|                | M32    |        |        |        | ZP.PSTR1 instruction complete device             | ZP.PSTR1 instruction completed                      |  |
|                | M33    |        |        |        | ZP.PSTR1 instruction error failure device        | ZP.PSTR1 instruction failed                         |  |

| Device name    | Device |        |        |        | Application                                       | Details when ON                                |
|----------------|--------|--------|--------|--------|---------------------------------------------------|------------------------------------------------|
|                | Axis 1 | Axis 2 | Axis 3 | Axis 4 |                                                   |                                                |
| Internal relay | M34    | —      | —      | —      | ZP.TEACH1 instruction complete device             | ZP.TEACH1 instruction completed                |
|                | M35    |        |        |        | ZP.TEACH1 instruction failure device              | ZP.TEACH1 instruction failed                   |
|                | M36    |        |        |        | ZP.PINIT instruction complete device              | ZP.PINIT instruction completed                 |
|                | M37    |        |        |        | ZP.PINIT instruction failure device               | ZP.PINIT instruction failed                    |
|                | M38    |        |        |        | ZP.PFWRT instruction complete device              | ZP.PFWRT instruction completed                 |
|                | M39    |        |        |        | ZP.PFWRT instruction failure device               | ZP.PFWRT instruction failed                    |
|                | M40    |        |        |        | Absolute position restoration instruction pulse   | Absolute position restoration commanded        |
|                | M41    |        |        |        | Absolute position restoration instruction storage | Absolute position restoration instruction held |
|                | M42    |        |        |        | Z.ABRST instruction complete device               | Z.ABRST instruction completed                  |
|                | M43    |        |        |        | Z.ABRST instruction failure device                | Z.ABRST instruction failed                     |
|                | M50    |        |        |        | Basic parameter 1 setting complete device         | Basic parameter 1 setting completed            |
|                | M51    |        |        |        | OPR basic parameter setting complete device       | OPR basic parameter setting completed          |

## (2) Data registers and timers

| Device name   | Device |        |        |        | Application                                                     | Details of storage                                                        |
|---------------|--------|--------|--------|--------|-----------------------------------------------------------------|---------------------------------------------------------------------------|
|               | Axis 1 | Axis 2 | Axis 3 | Axis 4 |                                                                 |                                                                           |
| Data register | D0     | —      | —      | —      | OPR request flag                                                | (Md.31) Status (bit 3))                                                   |
|               | D1     |        |        |        | Speed (low-order 16 bits)                                       | (Cd.25) Position-speed switching control speed change resister)           |
|               | D2     |        |        |        | Speed (high-order 16 bits)                                      |                                                                           |
|               | D3     |        |        |        | Movement amount (low-order 16 bits)                             | (Cd.23) Speed-position switching control movement amount change resister) |
|               | D4     |        |        |        | Movement amount (high-order 16 bits)                            |                                                                           |
|               | D5     |        |        |        | Inching movement amount                                         | (Cd.16) Inching movement amount)                                          |
|               | D6     |        |        |        | JOG operation speed (low-order 16 bits)                         | (Cd.17) JOG operation speed)                                              |
|               | D7     |        |        |        | JOG operation speed (high-order 16 bits)                        |                                                                           |
|               | D8     |        |        |        | Manual pulse generator 1 pulse input magnification (low-order)  | (Cd.20) Manual pulse generator 1 pulse input magnification)               |
|               | D9     |        |        |        | Manual pulse generator 1 pulse input magnification (high-order) |                                                                           |
|               | D10    |        |        |        | Manual pulse generator operation enable                         | (Cd.21) Manual pulse generator enable flag)                               |
|               | D11    |        |        |        | Speed change value (low-order 16 bits)                          | (Cd.14) New speed value)                                                  |
|               | D12    |        |        |        | Speed change value (high-order 16 bits)                         |                                                                           |
|               | D13    |        |        |        | Speed change request                                            | (Cd.15) Speed change request)                                             |
|               | D14    |        |        |        | Override value                                                  | (Cd.13) Positioning operation speed override)                             |

| Device name   | Device |        |        |        | Application                                    | Details of storage                                                                                             |
|---------------|--------|--------|--------|--------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
|               | Axis 1 | Axis 2 | Axis 3 | Axis 4 |                                                |                                                                                                                |
| Data register | D15    | —      | —      | —      | Acceleration time setting (low-order 16 bits)  | ( <a href="#">Cd.10</a> ) New acceleration time value)                                                         |
|               | D16    |        |        |        | Acceleration time setting (high-order 16 bits) |                                                                                                                |
|               | D17    |        |        |        | Deceleration time setting (low-order 16 bits)  | ( <a href="#">Cd.11</a> ) New deceleration time value)                                                         |
|               | D18    |        |        |        | Deceleration time setting (high-order 16 bits) |                                                                                                                |
|               | D19    |        |        |        | Acceleration/deceleration time change enable   | ( <a href="#">Cd.12</a> ) Acceleration/deceleration time change during speed change, enable/disable selection) |
|               | D20    |        |        |        | Step mode                                      | ( <a href="#">Cd.34</a> ) Step mode)                                                                           |
|               | D21    |        |        |        | Step valid flag                                | ( <a href="#">Cd.35</a> ) Step valid flag)                                                                     |
|               | D22    |        |        |        | Unused                                         | —                                                                                                              |
|               | D23    |        |        |        | Target position (low-order 16 bits)            | ( <a href="#">Cd.27</a> ) Target position change value (new address))                                          |
|               | D24    |        |        |        | Target position (high-order 16 bits)           |                                                                                                                |
|               | D25    |        |        |        | Target speed (low-order 16 bits)               | ( <a href="#">Cd.28</a> ) Target position change value (new speed))                                            |
|               | D26    |        |        |        | Target speed (high-order 16 bits)              |                                                                                                                |
|               | D27    |        |        |        | Target position change request                 | ( <a href="#">Cd.29</a> ) Target position change request flag)                                                 |
|               | D28    |        |        |        | Unused                                         | —                                                                                                              |
|               | D29    |        |        |        | Unused                                         | —                                                                                                              |
|               | D30    |        |        |        | ZP.PSTRT1 instruction control data             | —                                                                                                              |
|               | D31    |        |        |        | Completion status                              | —                                                                                                              |
|               | D32    |        |        |        | Start number                                   | —                                                                                                              |
|               | D33    |        |        |        | ZP.TEACH1 instruction control data             | —                                                                                                              |
|               | D34    |        |        |        | Completion status                              | —                                                                                                              |
|               | D35    |        |        |        | Teaching data                                  | —                                                                                                              |
|               | D36    |        |        |        | Positioning data No.                           | —                                                                                                              |
|               | D37    |        |        |        | ZP.PINIT instruction control data              | —                                                                                                              |
|               | D38    |        |        |        | Completion status                              | —                                                                                                              |
|               | D39    |        |        |        | ZP.PFWRT instruction control data              | —                                                                                                              |
|               | D40    |        |        |        | Completion status                              | —                                                                                                              |
|               | D41    |        |        |        | Z.ABRST1 control data                          | —                                                                                                              |
|               | D42    |        |        |        | Completion status                              | —                                                                                                              |
|               | D43    |        |        |        | Signals received from servo                    | —                                                                                                              |
|               | D44    |        |        |        | Signals transmitted to servo                   | —                                                                                                              |
|               | D45    |        |        |        | Status                                         | —                                                                                                              |
|               | D46    |        |        |        | System area                                    | —                                                                                                              |
|               | D47    |        |        |        | System area                                    | —                                                                                                              |
|               | D48    |        |        |        | System area                                    | —                                                                                                              |
|               | D49    |        |        |        | Error code                                     | Error code at absolute position restoration                                                                    |
|               | D79    |        |        |        | Error code                                     | ( <a href="#">Md.23</a> ) Axis error No.)                                                                      |
|               | D50    |        |        |        | Unit setting                                   | ( <a href="#">Pr.1</a> ) Unit setting)                                                                         |
|               | D51    |        |        |        | No. of pulses per rotation                     | ( <a href="#">Pr.2</a> ) No. of pulses per rotation)                                                           |

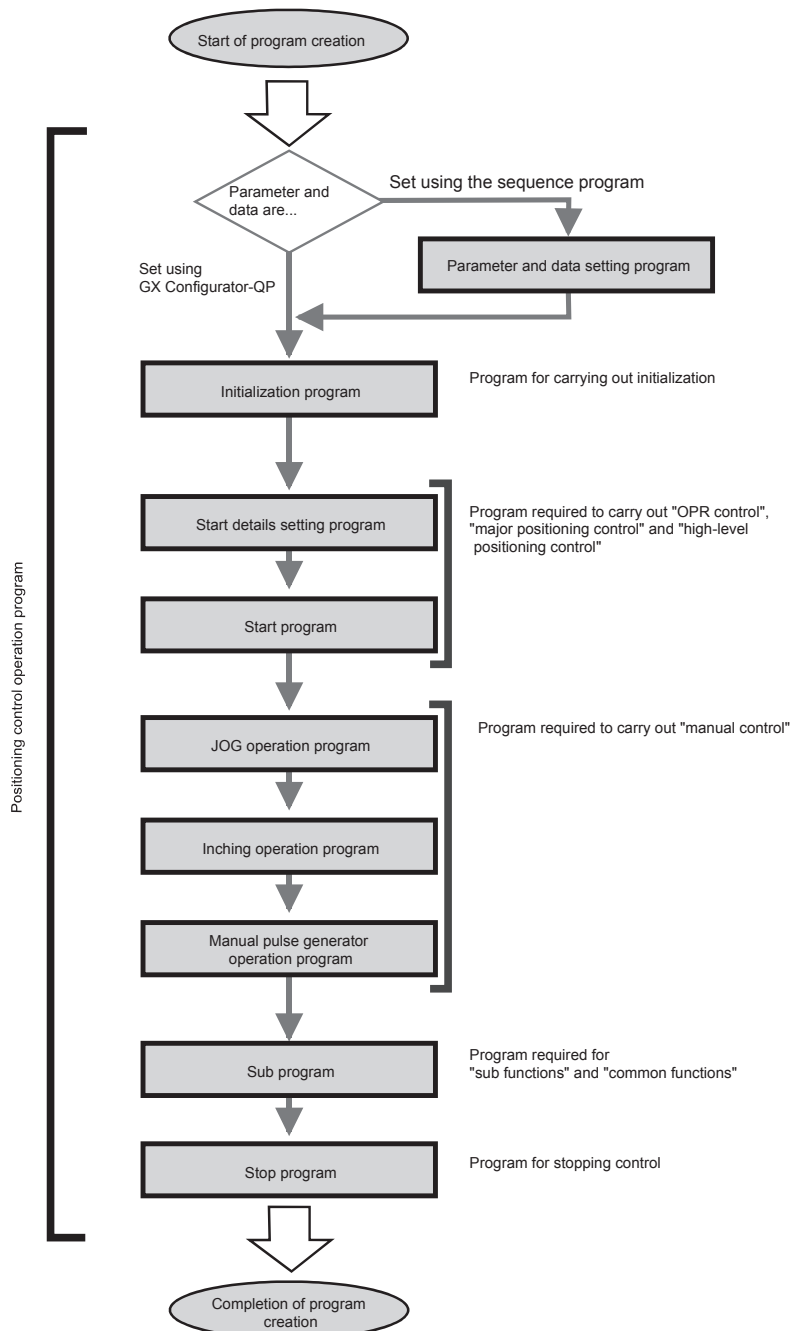
| Device name   | Device |        |        |        | Application                              |                                         | Details of storage                   |                                                                                                                                                          |                                   |                                    |                      |
|---------------|--------|--------|--------|--------|------------------------------------------|-----------------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------------------------------------|----------------------|
|               | Axis 1 | Axis 2 | Axis 3 | Axis 4 |                                          |                                         |                                      |                                                                                                                                                          |                                   |                                    |                      |
| Data register | D52    | —      |        |        | Movement amount per rotation             |                                         | (Pr.3) Movement amount per rotation) |                                                                                                                                                          |                                   |                                    |                      |
|               | D53    |        |        |        | Unit magnification                       |                                         | (Pr.4) Unit magnification)           |                                                                                                                                                          |                                   |                                    |                      |
|               | D54    |        |        |        | Pulse output mode                        |                                         | (Pr.5) Pulse output mode)            |                                                                                                                                                          |                                   |                                    |                      |
|               | D55    |        |        |        | Rotation direction setting               |                                         | (Pr.6) Rotation direction setting)   |                                                                                                                                                          |                                   |                                    |                      |
|               | D56    |        |        |        | Bias speed at start (low-order 16 bits)  |                                         | (Pr.7) Bias speed at start)          |                                                                                                                                                          |                                   |                                    |                      |
|               | D57    |        |        |        | Bias speed at start (high-order 16 bits) |                                         |                                      |                                                                                                                                                          |                                   |                                    |                      |
|               | D200   |        |        |        | OPR method                               |                                         | (Pr.43) OPR method)                  |                                                                                                                                                          |                                   |                                    |                      |
|               | D201   |        |        |        | OPR direction                            |                                         | (Pr.44) OPR direction)               |                                                                                                                                                          |                                   |                                    |                      |
|               | D202   |        |        |        | OP address (low-order 16 bits)           |                                         | (Pr.45) OP address)                  |                                                                                                                                                          |                                   |                                    |                      |
|               | D203   |        |        |        | OP address (high-order 16 bits)          |                                         |                                      |                                                                                                                                                          |                                   |                                    |                      |
|               | D204   |        |        |        | OPR speed (low-order 16 bits)            |                                         | (Pr.46) OPR speed)                   |                                                                                                                                                          |                                   |                                    |                      |
|               | D205   |        |        |        | OPR speed (high-order 16 bits)           |                                         |                                      |                                                                                                                                                          |                                   |                                    |                      |
|               | D206   |        |        |        | Creep speed (low-order 16 bits)          |                                         | (Pr.47) Creep speed)                 |                                                                                                                                                          |                                   |                                    |                      |
|               | D207   |        |        |        | Creep speed (high-order 16 bits)         |                                         |                                      |                                                                                                                                                          |                                   |                                    |                      |
|               | D208   |        |        |        | OPR retry                                |                                         | (Pr.48) OPR retry)                   |                                                                                                                                                          |                                   |                                    |                      |
|               | D58    |        |        |        | Positioning data No.1                    | Positioning identifier                  |                                      | (Da.1) Operation pattern)<br>(Da.2) Control system)<br>(Da.3) Acceleration time No.)<br>(Da.4) Deceleration time No.)<br>(Da.5) Axis to be interpolated) |                                   |                                    |                      |
|               | D59    |        |        |        |                                          | M code                                  |                                      | (Da.10) M code)                                                                                                                                          |                                   |                                    |                      |
|               | D60    |        |        |        |                                          | Dwell time                              |                                      | (Da.9) Dwell time)                                                                                                                                       |                                   |                                    |                      |
|               | D61    |        |        |        |                                          | Positioning option                      |                                      | (Da.27) M code ON signal output timing)<br>(Da.28) ABS direction in degrees)<br>(Da.29) Interpolation speed designation method)                          |                                   |                                    |                      |
|               | D62    |        |        |        |                                          | Command speed (low-order 16 bits)       |                                      | (Da.8) Command speed)                                                                                                                                    |                                   |                                    |                      |
|               | D63    |        |        |        |                                          | Command speed (high-order 16 bits)      |                                      |                                                                                                                                                          |                                   |                                    |                      |
|               | D64    |        |        |        |                                          | Positioning address/<br>movement amount | (low-order 16 bits)                  | (Da.6) Positioning address/movement amount)                                                                                                              |                                   |                                    |                      |
|               | D65    |        |        |        |                                          |                                         | (high-order 16 bits)                 |                                                                                                                                                          |                                   |                                    |                      |
|               | D66    |        |        |        |                                          | Arc address                             | (low-order 16 bits)                  | (Da.7) Arc address)                                                                                                                                      |                                   |                                    |                      |
|               | D67    |        |        |        |                                          |                                         | (high-order 16 bits)                 |                                                                                                                                                          |                                   |                                    |                      |
|               | D68    |        |        |        |                                          | Block start data (Block 0)              | Point 1 (shape, start No.)           |                                                                                                                                                          |                                   |                                    |                      |
|               | D69    |        |        |        |                                          |                                         | Point 2 (shape, start No.)           |                                                                                                                                                          |                                   |                                    |                      |
|               | D70    |        |        |        |                                          |                                         | Point 3 (shape, start No.)           |                                                                                                                                                          |                                   |                                    |                      |
|               | D71    |        |        |        |                                          |                                         | Point 4 (shape, start No.)           |                                                                                                                                                          |                                   | (Da.11) Shape)                     |                      |
|               | D72    |        |        |        |                                          |                                         | Point 5 (shape, start No.)           |                                                                                                                                                          |                                   | (Da.12) Start data No.)            |                      |
|               | D73    |        |        |        |                                          |                                         | Point 1 (special start instruction)  |                                                                                                                                                          |                                   | (Da.13) Special start instruction) |                      |
|               | D74    |        |        |        | Point 2 (special start instruction)      |                                         | (Da.14) Parameter)                   |                                                                                                                                                          |                                   |                                    |                      |
|               | D75    |        |        |        | Point 3 (special start instruction)      |                                         |                                      |                                                                                                                                                          |                                   |                                    |                      |
|               | D76    |        |        |        | Point 4 (special start instruction)      |                                         |                                      |                                                                                                                                                          |                                   |                                    |                      |
|               | D77    |        |        |        | Point 5 (special start instruction)      |                                         |                                      |                                                                                                                                                          |                                   |                                    |                      |
|               | Timer  |        |        |        | T0                                       | —                                       |                                      |                                                                                                                                                          | PLC READY signal OFF confirmation |                                    | PLC READY signal OFF |
|               |        |        |        |        | T1                                       |                                         |                                      |                                                                                                                                                          | PLC READY signal OFF confirmation |                                    |                      |

### 6.3 Creating a program

The "positioning control operation program" actually used is explained in this chapter. The functions and programs explained in PART 2 are assembled into the "positioning control operation program" explained here. (To monitor the control, add the required monitor program that matches the system. Refer to Section 5.6 "List of monitor data" for details on the monitor items.)

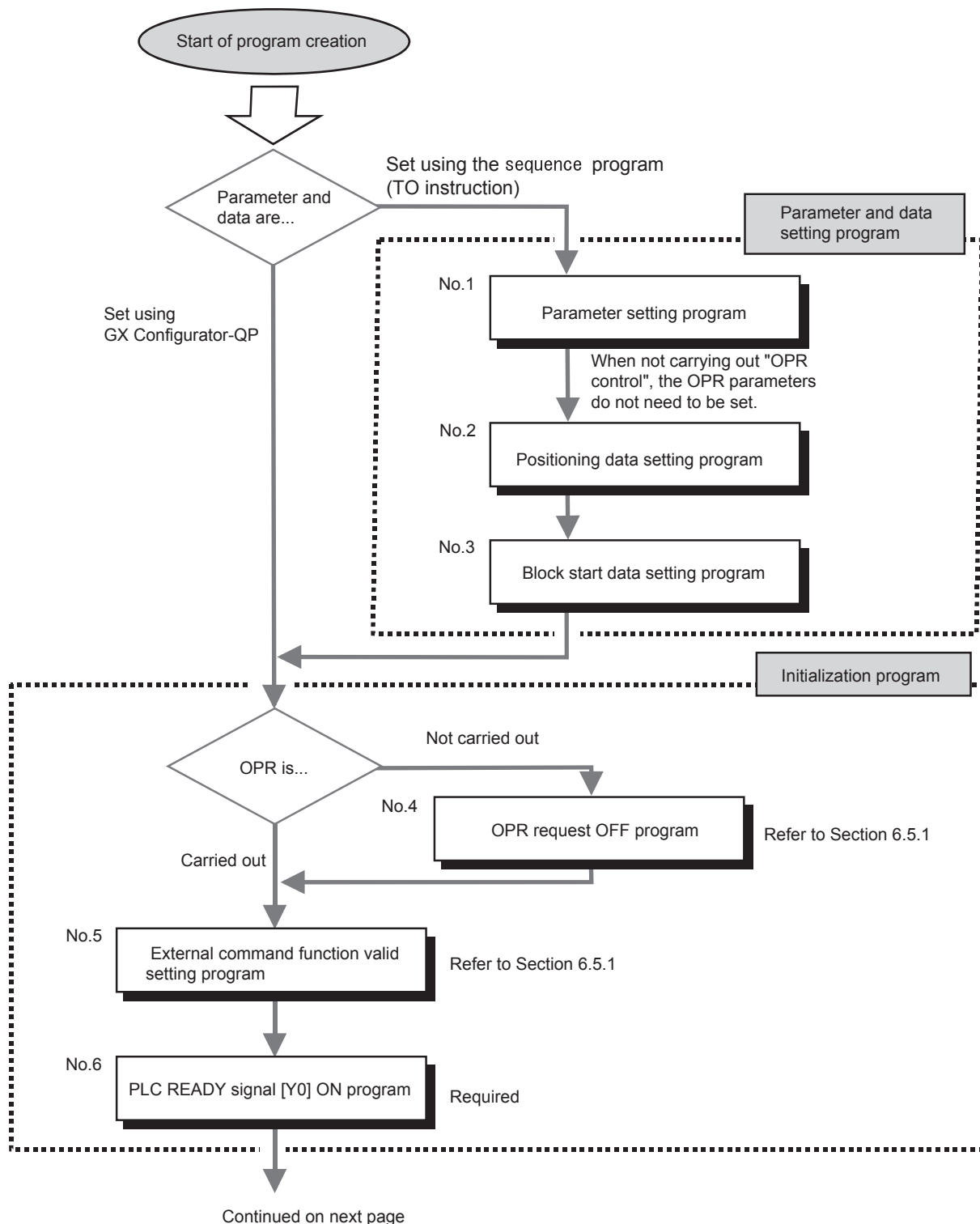
#### 6.3.1 General configuration of program

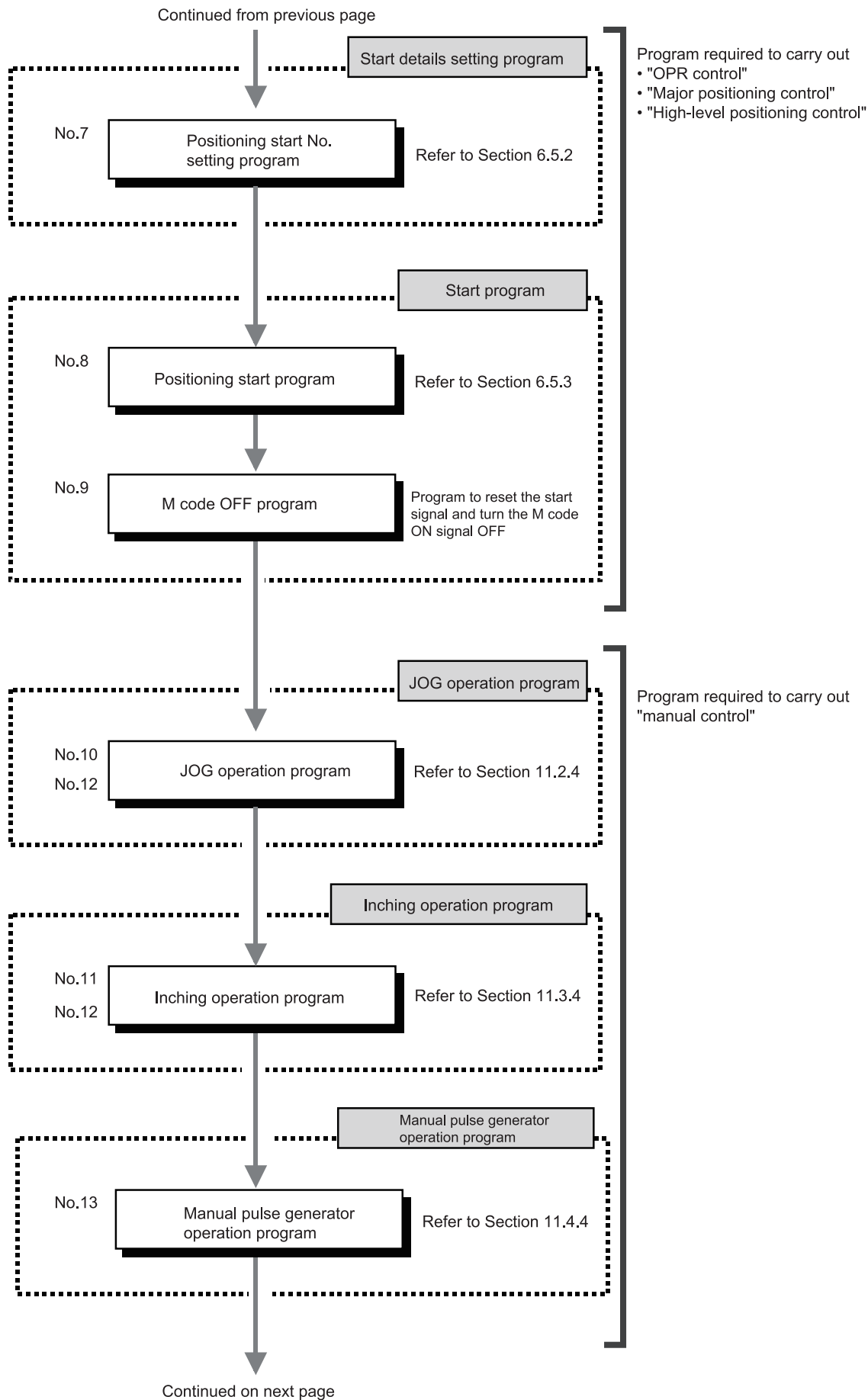
The general configuration of the "positioning control operation program" is shown below.

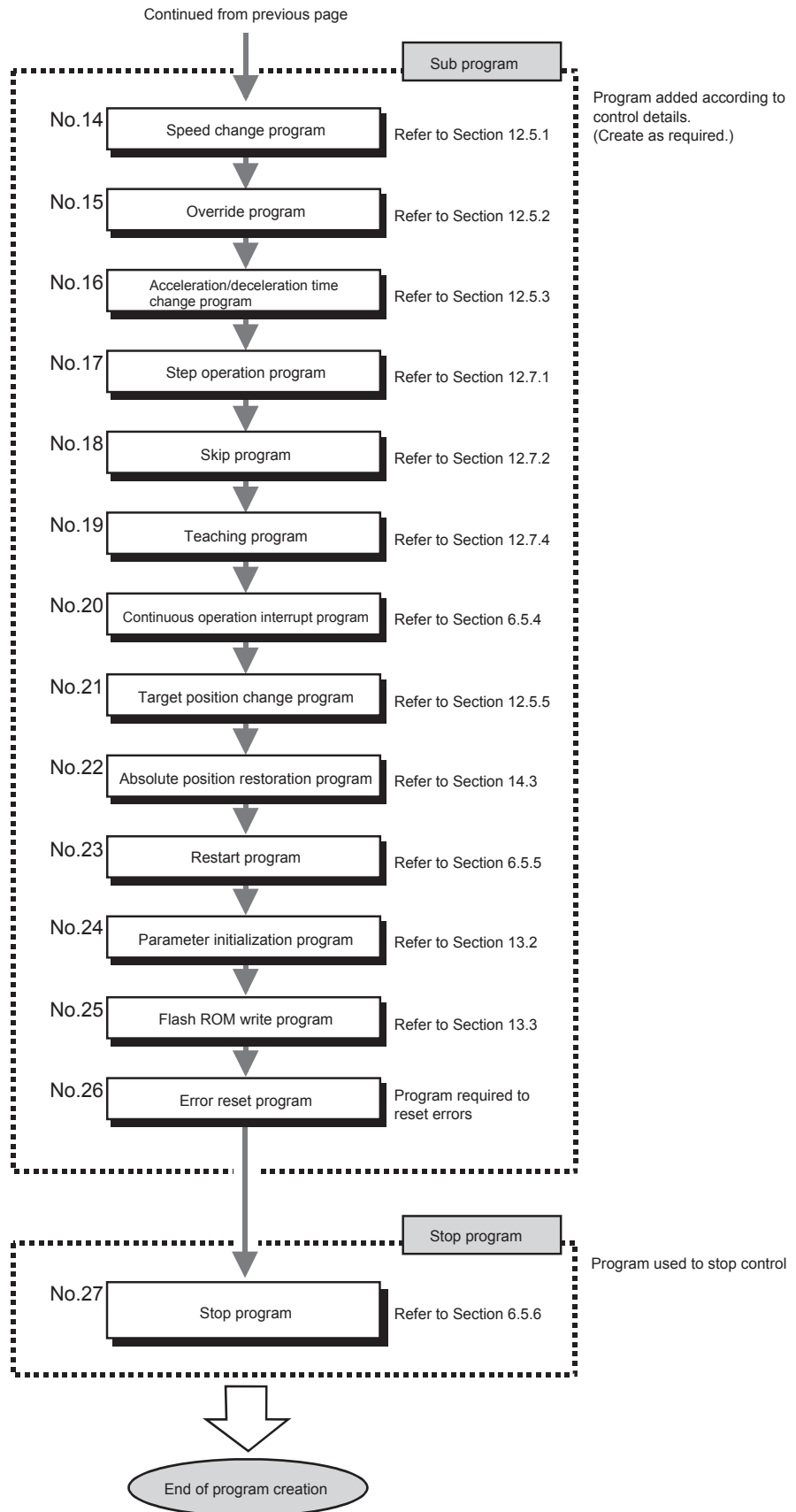


## 6.3.2 Positioning control operation program

The various programs that configure the "positioning control operation program" are shown below. When creating the program, refer to the explanation of each program and Section 6.4 "Positioning program examples", and create an operation program that matches the positioning system. (Numbers are assigned to the following programs. Configuring the program in the order of these numbers is recommended.)







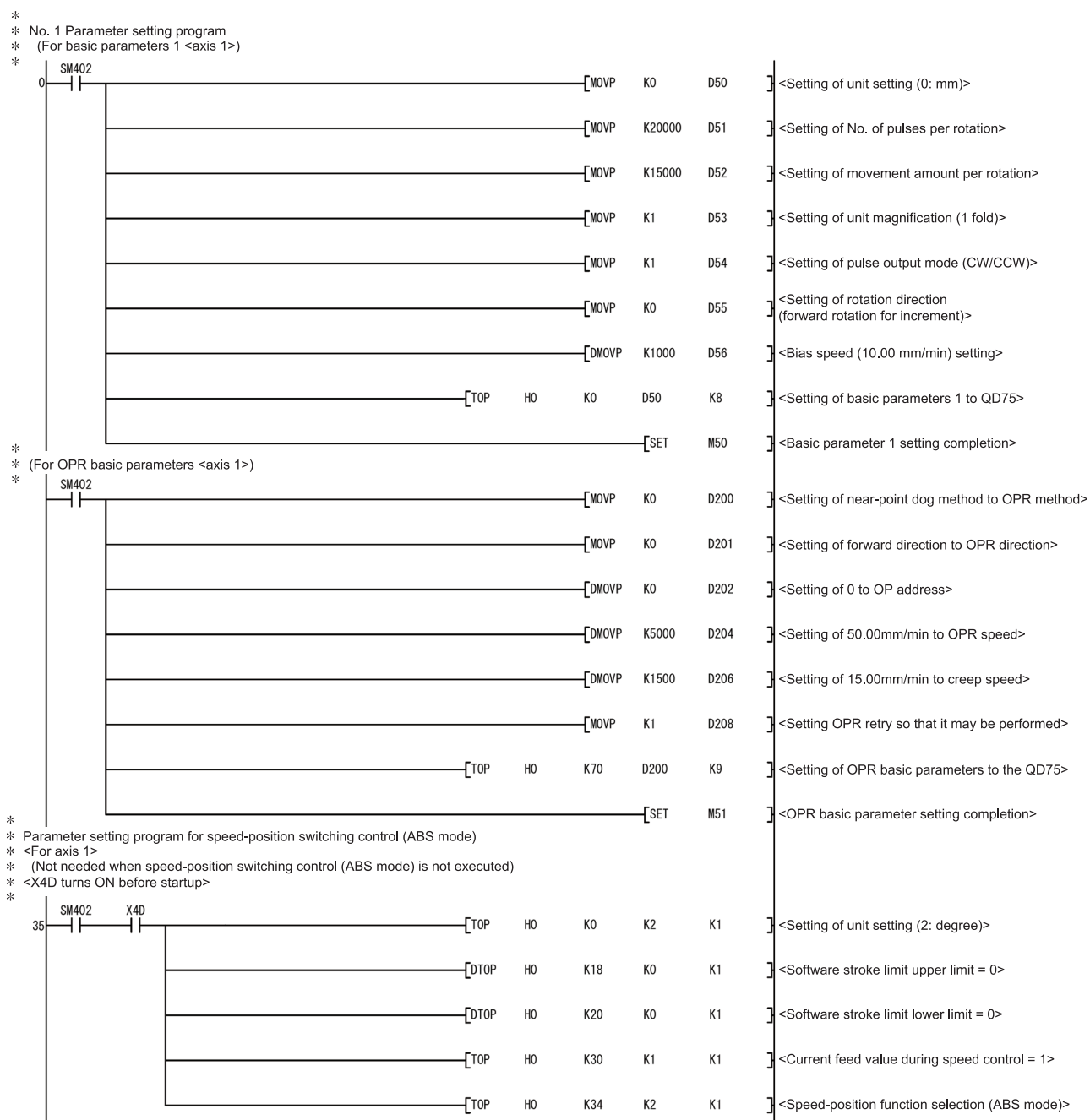


## 6.4 Positioning program examples

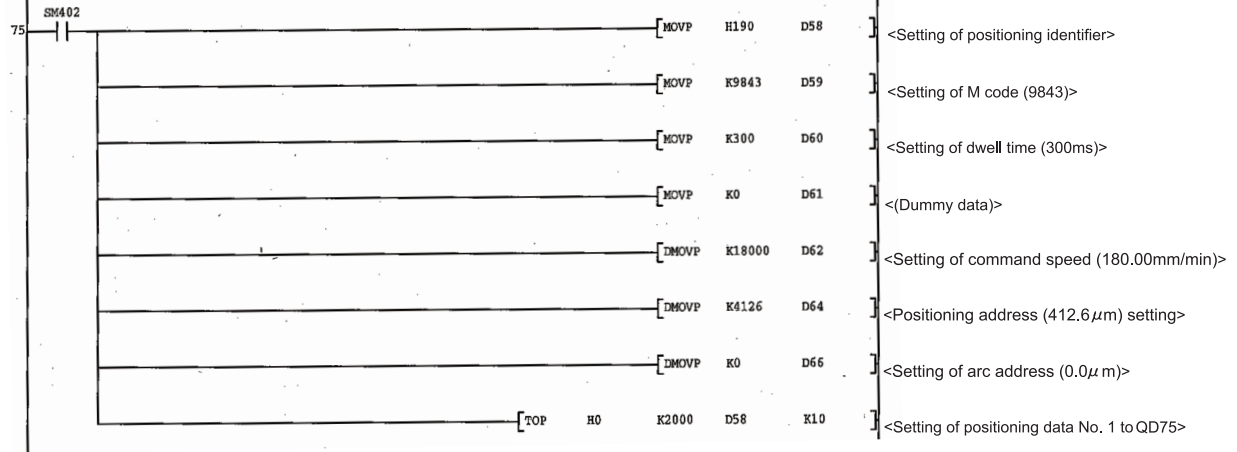
An example of the "Axis 1" positioning program is given in this section.

----- [No. 1] to [No. 3] parameter and data setting program -----

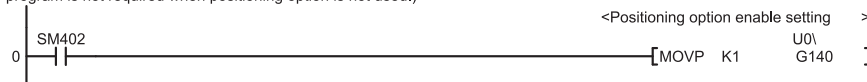
- \* When setting the parameters or data with the sequence program, set them in the QD75 using the TO instruction from the CPU module. (Carry out the settings while the PLC READY signal [Y0] is OFF.)
- \* When setting the parameters or data with GX Configurator-QP, the [No. 1] to [No. 3] program is not necessary.



\*  
 \* No. 2 Positioning data setting program  
 \* (For positioning data No. 1 <axis 1>)  
 \* <Positioning identifier>  
 \* Operation pattern: Positioning complete  
 \* Control system: 1-axis linear control (ABS)  
 \* Acceleration time No.: 1, deceleration time No.: 2  
 \*



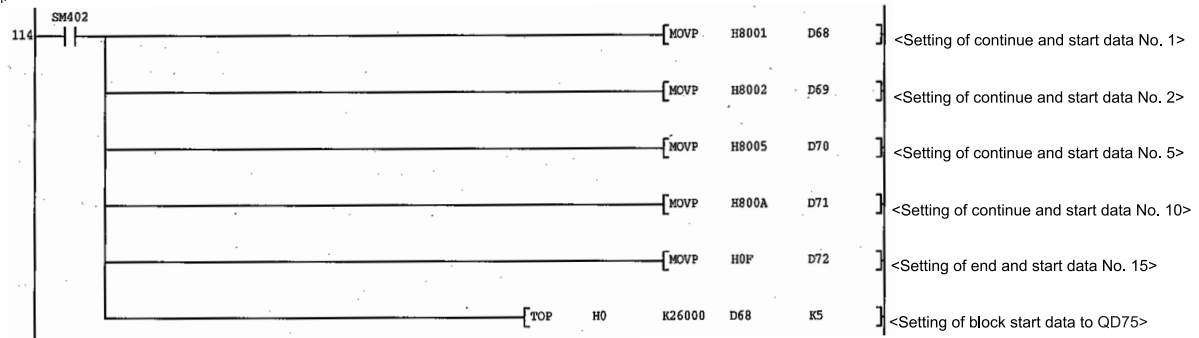
\* Positioning option enable/disable parameter setting program  
 \* (The program is not required when positioning option is not used.)



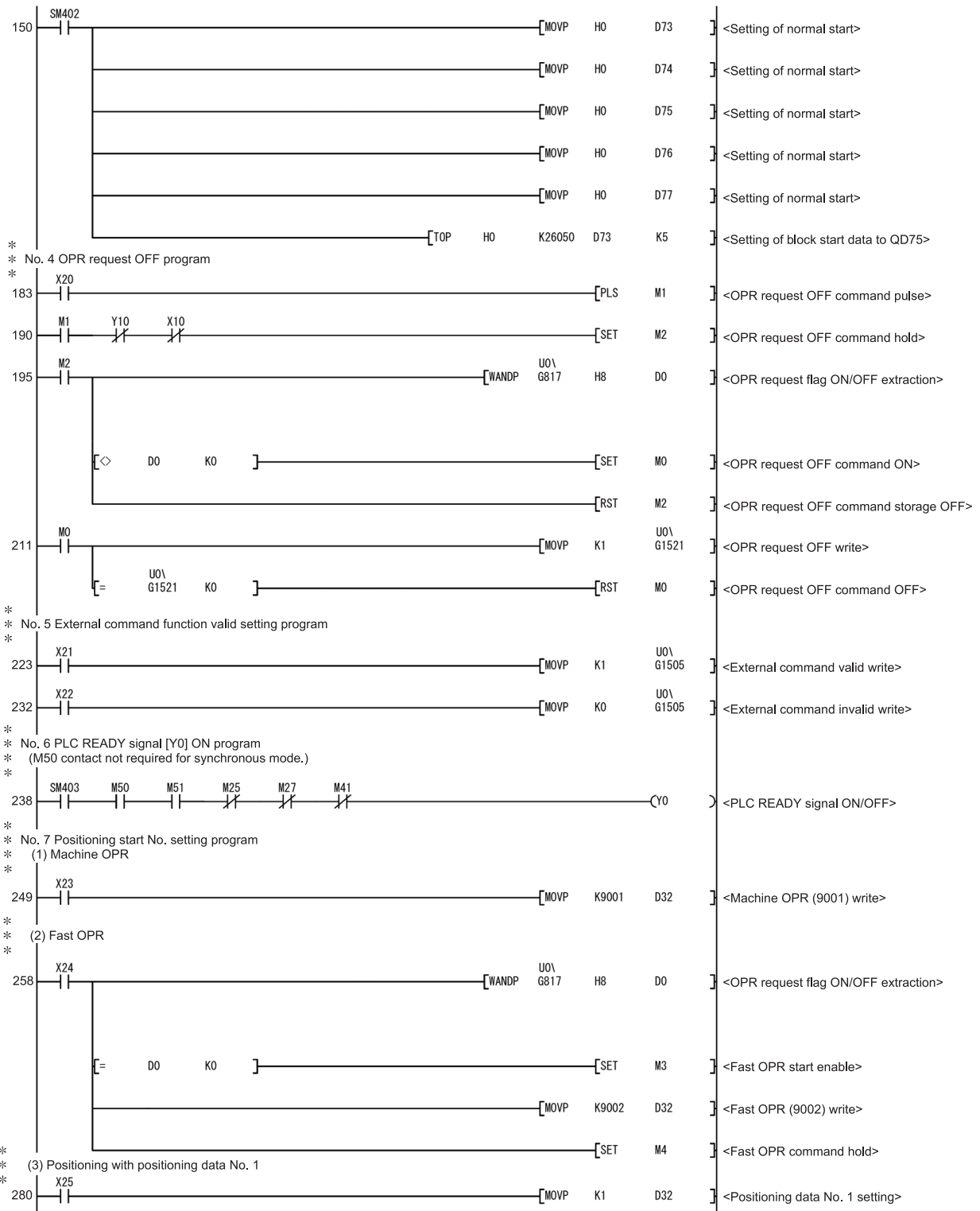
\*  
 \* No. 2 Positioning data setting program  
 \* (For using positioning option)  
 \* (For positioning data No. 1 <axis 1>)  
 \* <Positioning identifier>  
 \* Operation pattern: Positioning complete  
 \* Control system: 1-axis linear control (ABS)  
 \* Acceleration time No.: 1, deceleration time No.: 2  
 \* <Positioning option>  
 \* M code ON signal output timing: WITH mode (1)  
 \* ABS direction in degrees: 0  
 \* Interpolation speed designation method: 0  
 \*

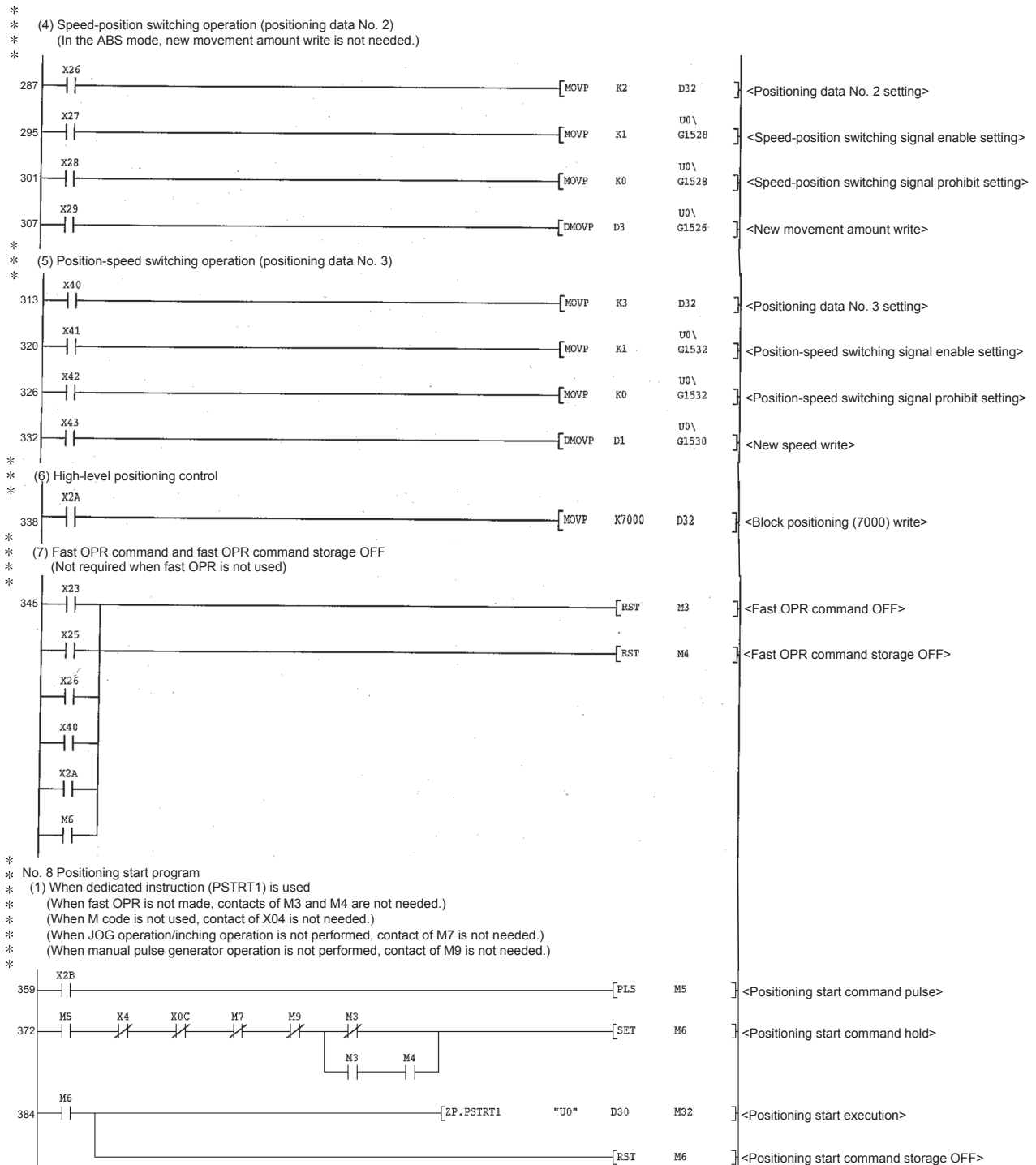


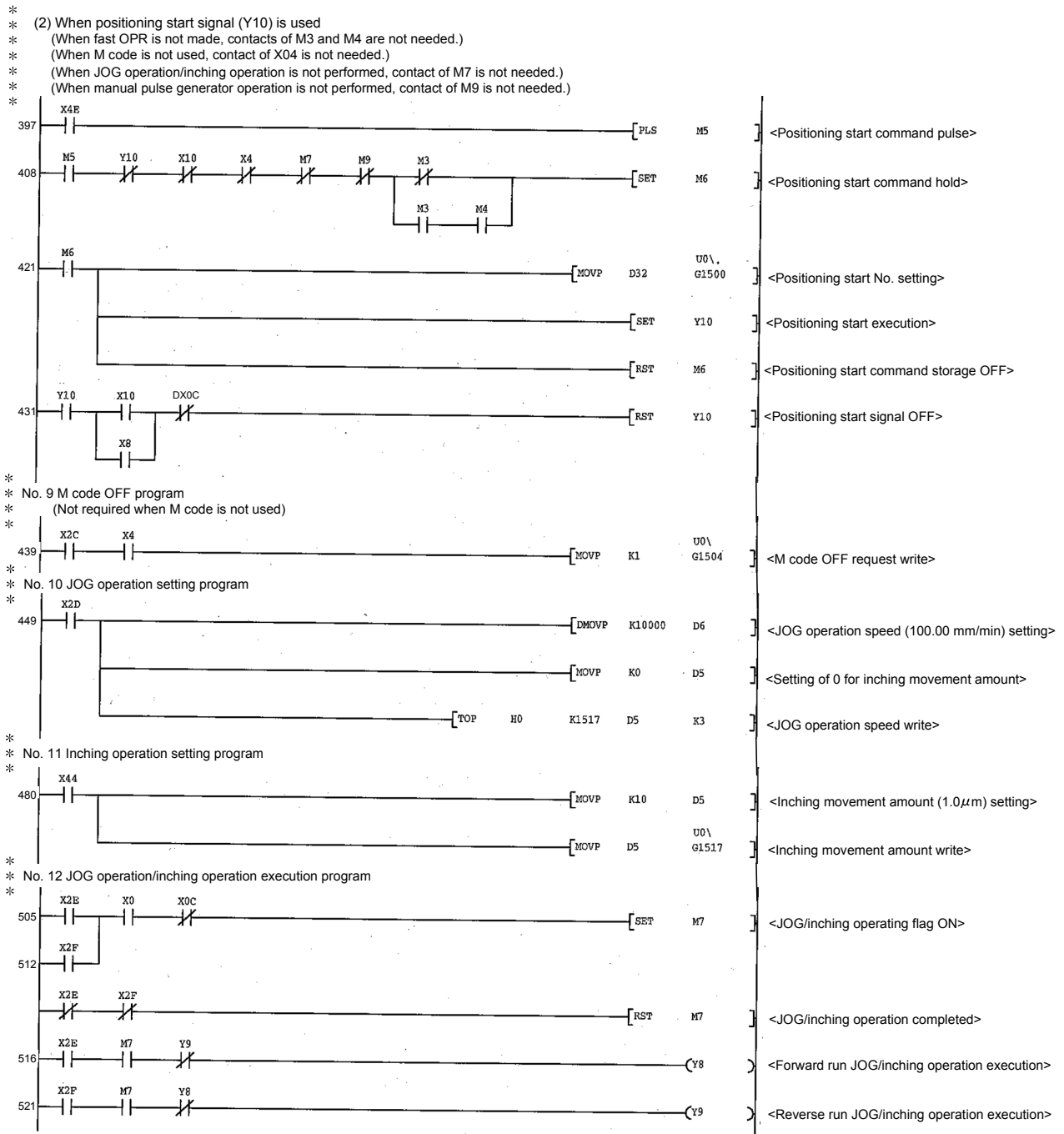
\*  
 \* No. 3 Block start data setting program  
 \* Block start data of start block 0 (axis 1)  
 \* For setting of points 1 to 5  
 \* (Conditions)  
 \* Shape: Continued at points 1 to 4, ended at point 5  
 \* Special start instruction: Normal start at all of points 1 to 5  
 \* <Positioning data are already preset>  
 \*  
 \*  
 \* [Setting of shape and start data No.]  
 \*



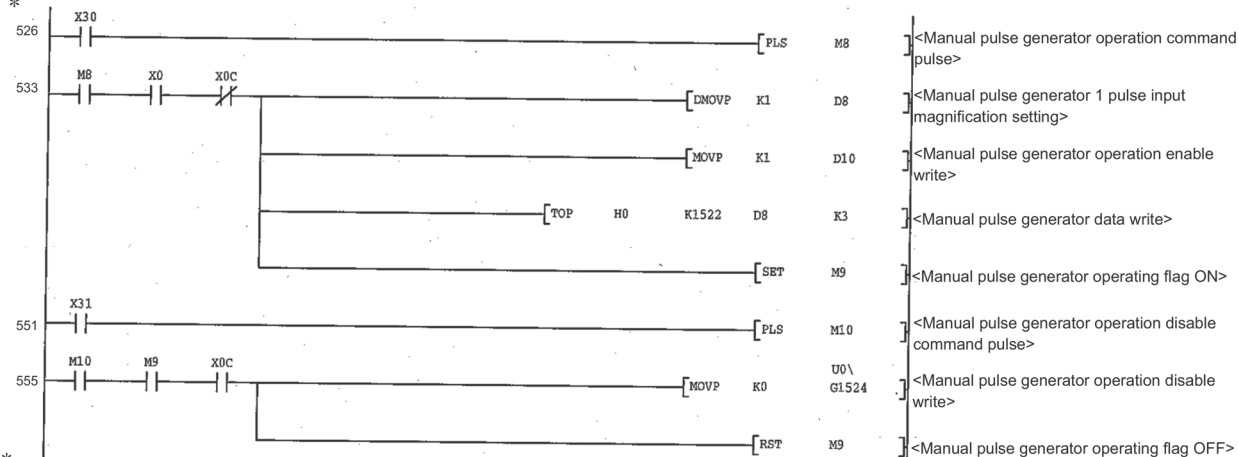
\*  
 \* [Setting of special start instruction to normal start]  
 \*



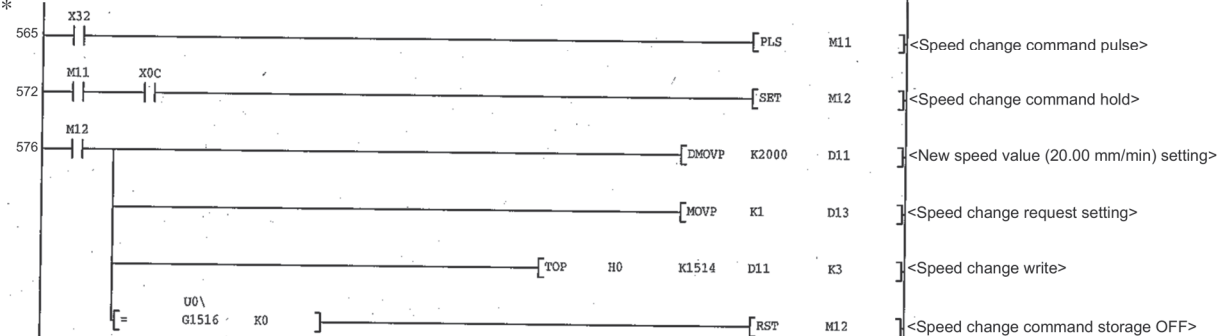




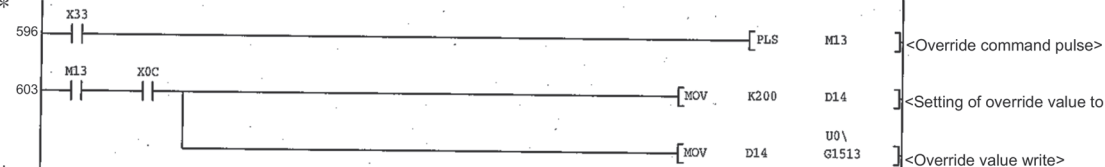
\*  
\* No. 13 Manual pulse generator operation program  
\*



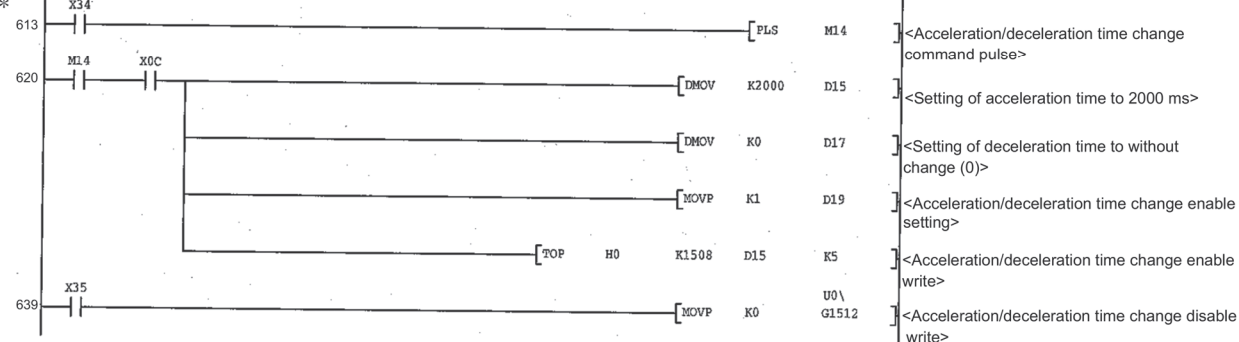
\*  
\* No. 14 Speed change program  
\*

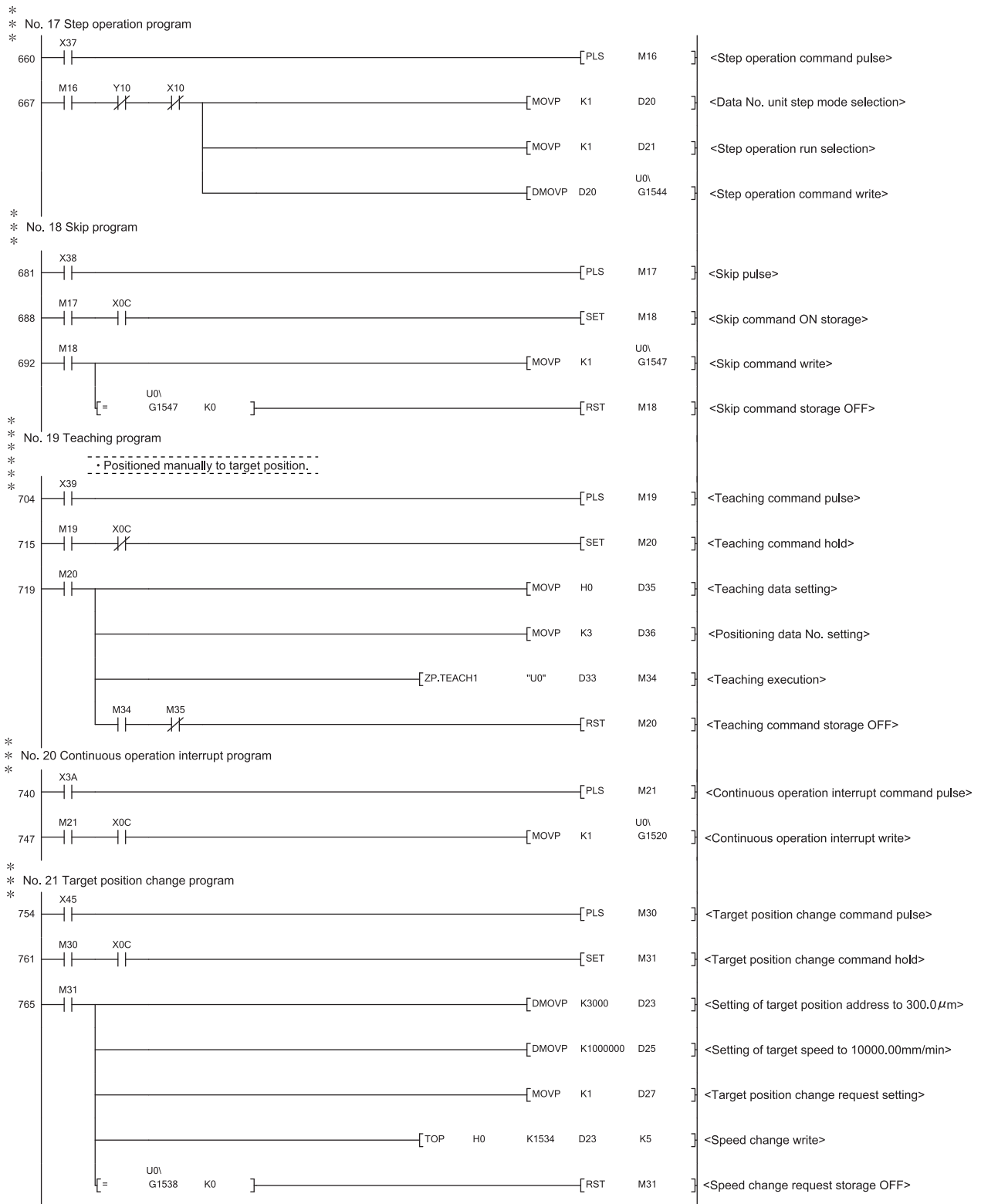


\*  
\* No. 15 Override program  
\*



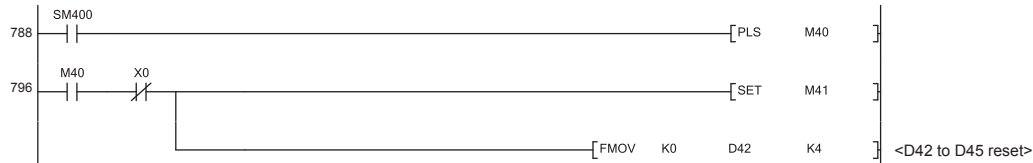
\*  
\* No. 16 Acceleration/deceleration time change program  
\*



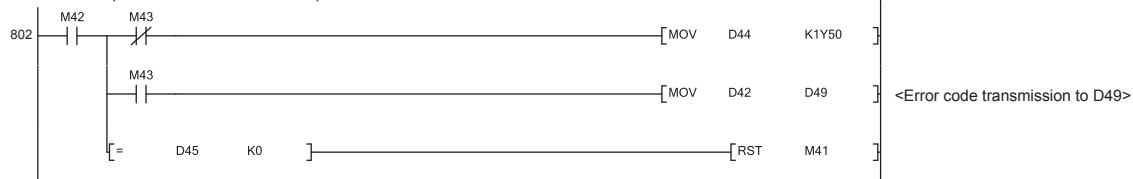




\*  
\* No. 22 Absolute position restoration program  
\* (1) Absolute position restoration command acceptance  
\*



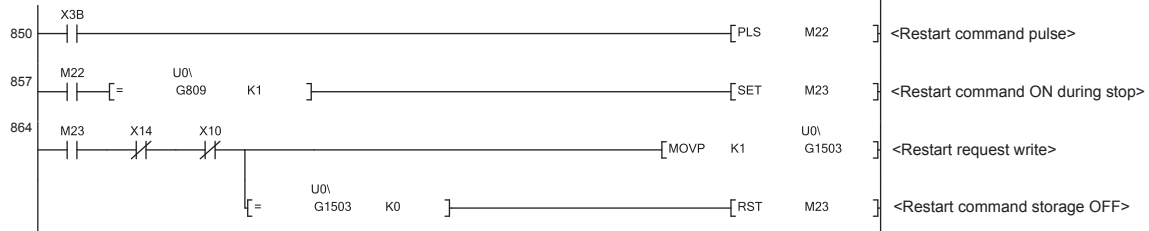
\*  
\* (2) Setting of transmit data to servo-amplifier and confirmation of absolute position restoration completion  
\* ABRST1 instruction completed when M42 is ON and M43 is OFF.  
\* Absolute position data restoration completed when status = 0.  
\*



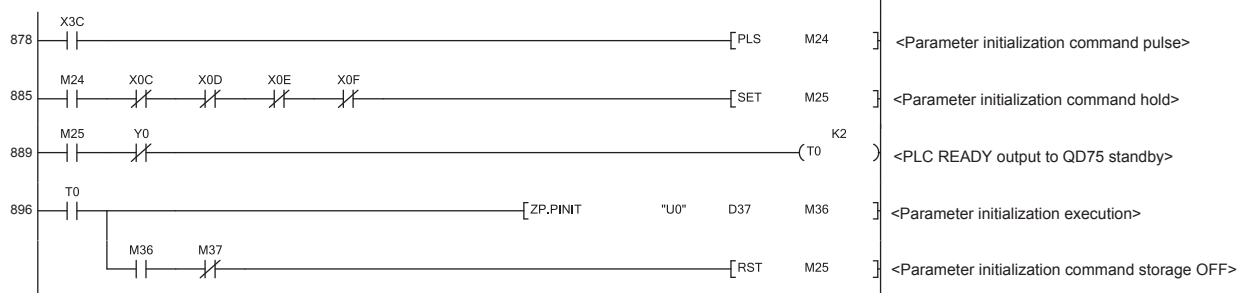
\*  
\* (3) ABS data setting and ABRST1 instruction execution  
\*



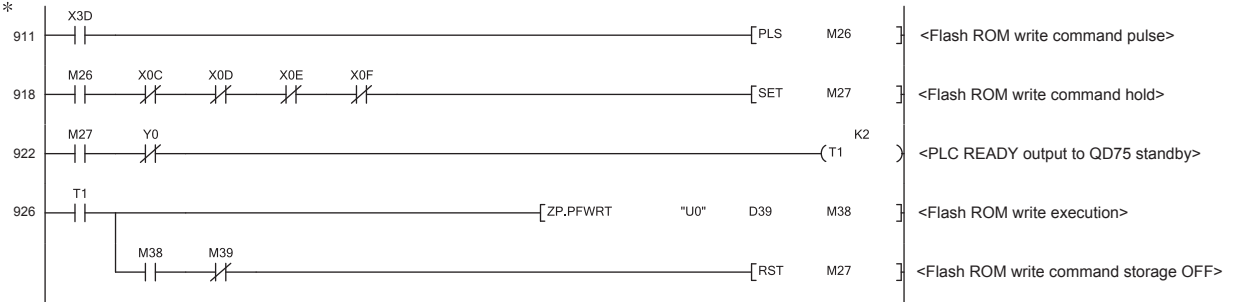
\*  
\* No. 23 Restart program  
\*



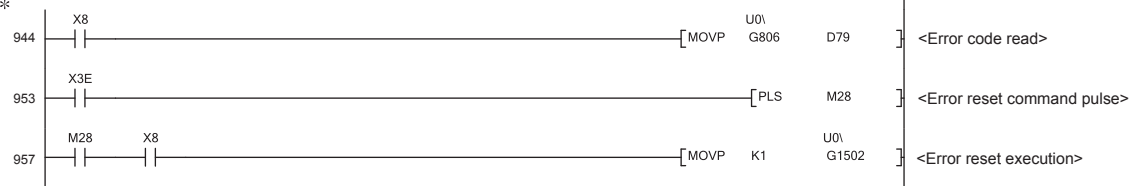
\*  
\* No. 24 Parameter initialization program  
\*



\*  
\* No. 25 Flash ROM write program  
\*



\*  
\* No. 26 Error reset program  
\*



\*  
\* No. 27 Stop program  
\*



## 6.5 Program details

### 6.5.1 Initialization program

#### [1] OPR request OFF program

This program forcibly turns OFF the "OPR request flag" ([Md.31] Status : b3) which is ON.

When using a system that does not require OPR, assemble the program to cancel the "OPR request" made by the QD75 when the power is turned ON, etc.

#### ■ Data requiring setting

Set the following data to use the OPR request flag OFF request.

| Setting item                         | Setting value | Setting details                        | Buffer memory address |        |        |        |
|--------------------------------------|---------------|----------------------------------------|-----------------------|--------|--------|--------|
|                                      |               |                                        | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| [Cd.19] OPR request flag OFF request | 1             | Set to "1: Turn OPR request flag OFF". | 1521                  | 1621   | 1721   | 1821   |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

#### ■ Time chart for OPR request OFF

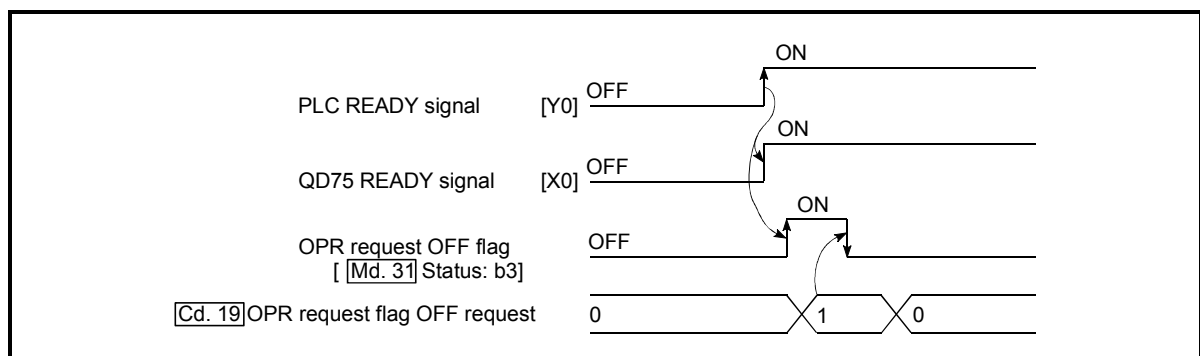


Fig. 6.1 Time chart for OPR request OFF

#### [2] External command function valid setting program

This program is used to validate the "external command signal" beforehand when using the external command functions (external start, speed change, speed-position switching, position-speed switching, skip). (Set which function to use beforehand in " [Pr.42] External command function selection".)

Set the following data to validate the "external command signal".

| Setting item                  | Setting value | Setting details                        | Buffer memory address |        |        |        |
|-------------------------------|---------------|----------------------------------------|-----------------------|--------|--------|--------|
|                               |               |                                        | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| [Cd.8] External command valid | 1             | Set to "1: Validate external command". | 1505                  | 1605   | 1705   | 1805   |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

## 6.5.2 Start details setting program

This program sets which control, out of "OPR", "major positioning control" or "high-level positioning control" to execute. For "high-level positioning control", "fast OPR", "speed-position switching control" and "position-speed switching control", add the respectively required sequence program.

(Refer to CHAPTER 10 for details on starting the "high-level positioning control".)

### ■ Procedures for setting the starting details

- (1) Set the "positioning start No." corresponding to the control to be started in " **Cd.3** Positioning start No.".

| Setting item                      | Setting value | Setting details                                                                                                                                                                                              | Buffer memory address |        |        |        |
|-----------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------|--------|--------|
|                                   |               |                                                                                                                                                                                                              | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <b>Cd.3</b> Positioning start No. | →             | 1 to 600 : Positioning data No.<br>9001 : Machine OPR<br>9002 : Fast OPR<br>9003 : Current value changing<br>9004 : Simultaneous start<br>7000 to 7004 : Block No.<br>(For "high-level positioning control") | 1500                  | 1600   | 1700   | 1800   |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

- (2) For "high-level positioning control", set the "positioning start point No." of the block to be started in " **Cd.4** Positioning starting point No.".

| Setting item                               | Setting value | Setting details                         | Buffer memory address |        |        |        |
|--------------------------------------------|---------------|-----------------------------------------|-----------------------|--------|--------|--------|
|                                            |               |                                         | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <b>Cd.4</b> Positioning starting point No. | →             | 1 to 50 : Point No. of block start data | 1501                  | 1601   | 1701   | 1801   |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

- (3) Set the following control data for "speed-position switching control (INC mode)".  
(Set " **Cd.23** Speed-position switching control movement amount change register as required". Setting is not required in the ABS mode.)

| Setting item                                                                  | Setting value | Setting details                                                                                      | Buffer memory address |              |              |              |
|-------------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------------------------|-----------------------|--------------|--------------|--------------|
|                                                                               |               |                                                                                                      | Axis 1                | Axis 2       | Axis 3       | Axis 4       |
| <b>Cd.23</b> Speed-position switching control movement amount change register | →             | Set the new value when the position control's movement amount is to be changed during speed control. | 1526<br>1527          | 1626<br>1627 | 1726<br>1727 | 1826<br>1827 |
| <b>Cd.24</b> Speed-position switching enable flag                             | 1             | When "1" is set, the speed-position switching signal will be validated.                              | 1528                  | 1628         | 1728         | 1828         |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

- (4) For "position-speed switching control", set the control data shown below.  
 (As required, set the " Cd.25 Position-speed switching control speed change resister".)

| Setting item                                                        |                                                        | Setting value | Setting details                                                           | Buffer memory address |              |              |              |
|---------------------------------------------------------------------|--------------------------------------------------------|---------------|---------------------------------------------------------------------------|-----------------------|--------------|--------------|--------------|
|                                                                     |                                                        |               |                                                                           | Axis 1                | Axis 2       | Axis 3       | Axis 4       |
| <span style="border: 1px solid black; padding: 0 2px;">Cd.25</span> | Position-speed switching control speed change resister | →             | Used to set a new value when speed is changed during positioning control. | 1530<br>1531          | 1630<br>1631 | 1730<br>1731 | 1830<br>1831 |
| <span style="border: 1px solid black; padding: 0 2px;">Cd.26</span> | Position-speed switching enable flag                   | 1             | To validate position-speed switching signal, this is set to 1.            | 1532                  | 1632         | 1732         | 1832         |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

## 6.5.3 Start program

This program is used to start the control with start commands.

The control can be started with the following two methods.

- [1] Starting by inputting positioning start signal [Y10, Y11, Y12, Y13]
- [2] Starting by inputting external command signal

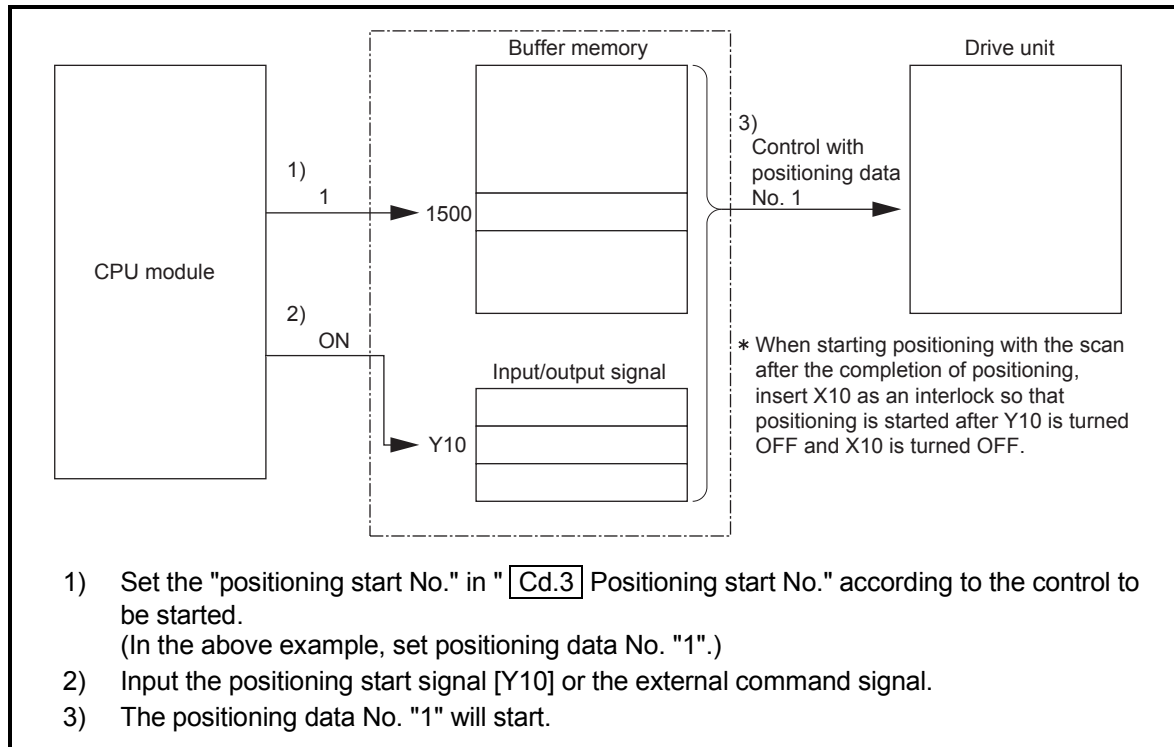


Fig. 6.2 Procedures for starting control (for axis 1)

### ■ Starting conditions

To start the control, the following conditions must be satisfied.

The necessary start conditions must be incorporated in the sequence program so that the control is not started when the conditions are not satisfied.

| Signal name      |                         | Signal state |                                  | Device |        |        |        |
|------------------|-------------------------|--------------|----------------------------------|--------|--------|--------|--------|
|                  |                         |              |                                  | Axis 1 | Axis 2 | Axis 3 | Axis 4 |
| Interface signal | PLC READY signal        | ON           | CPU module preparation completed | Y0     |        |        |        |
|                  | QD75 READY signal       | ON           | QD75 preparation completed       | X0     |        |        |        |
|                  | Synchronization flag *  | ON           | QD75 buffer memory Accessible    | X1     |        |        |        |
|                  | Axis stop signal        | OFF          | Axis stop signal is OFF.         | Y4     | Y5     | Y6     | Y7     |
|                  | M code ON signal        | OFF          | M code ON signal is OFF.         | X4     | X5     | X6     | X7     |
|                  | Error detection signal  | OFF          | No error is present.             | X8     | X9     | XA     | XB     |
|                  | BUSY signal             | OFF          | BUSY signal is OFF.              | XC     | XD     | XE     | XF     |
| External signal  | Start complete signal   | OFF          | Start complete signal is OFF.    | X10    | X11    | X12    | X13    |
|                  | Drive unit READY signal | ON           | Drive unit preparation completed | —      |        |        |        |
|                  | Stop signal             | OFF          | Stop signal is OFF.              | —      |        |        |        |
|                  | Upper limit (FLS)       | ON           | Within limit range               | —      |        |        |        |
|                  | Lower limit (RLS)       | ON           | Within limit range               | —      |        |        |        |

※: When the synchronous setting of the CPU module is made in the nonsynchronous mode, this must be provided as an interlock.

When it is made in the synchronous mode, no interlock must be provided in the program because the flag is turned ON when calculation is run on the CPU module.

## [1] Starting by inputting positioning start signal

## ■ Operation when starting

- (1) When the positioning start signal turns ON, the start complete signal and BUSY signal turn ON, and the positioning operation starts.  
It can be seen that the axis is operating when the BUSY signal is ON.
- (2) When the positioning start signal turns OFF, the start complete signal also turns OFF.  
If the positioning start signal is ON even after positioning is completed, the start complete signal will remain ON.
- (3) If the positioning start signal turns ON again while the BUSY signal is ON, the warning "Start during operation" (warning code: 100) will occur.
- (4) The process taken when positioning is completed will differ according to case (a) and (b) below.
  - (a) When next positioning is not to be carried out
    - If a dwell time is set, the system will wait for the set time to pass, and then positioning will be completed.
    - When positioning is completed, the BUSY signal will turn OFF and the positioning complete signal will turn ON. However, when using speed control or when the positioning complete signal ON time is "0", the signal will not turn ON.
    - When the positioning complete signal ON time is passed, the positioning complete signal will turn OFF.
  - (b) When next positioning is to be carried out
    - If a dwell time is set, the system will wait for the set time to pass.
    - When the set dwell time is passed, the next positioning will start.

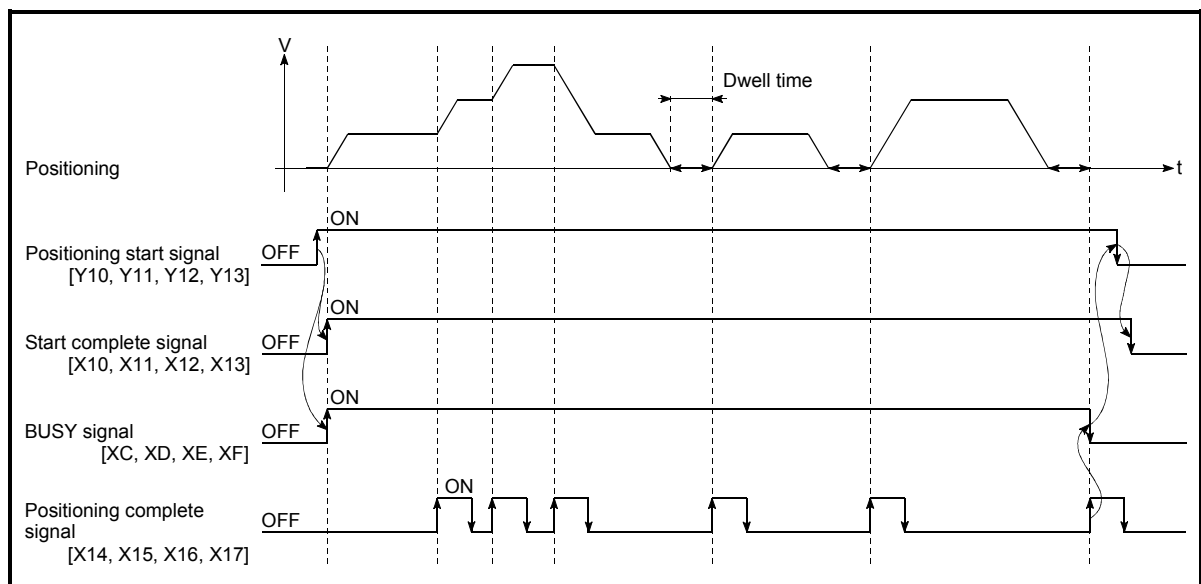


Fig. 6.3 ON/OFF timing of each signal at start of positioning

**POINTS**

The BUSY signal [XC, XD, XE, XF] turns ON even when position control of movement amount 0 is executed. However, since the ON time is short, the ON status may not be detected in the sequence program.

(The ON status of the start complete signal [X10, X11, X12, X13], positioning complete signal [X14, X15, X16, X17] and M code ON signal [X4, X5, X6, X7] can be detected in the sequence program.)



### ■ Starting time chart

The time chart for starting each control is shown below.

#### (1) Time chart for starting "machine OPR"

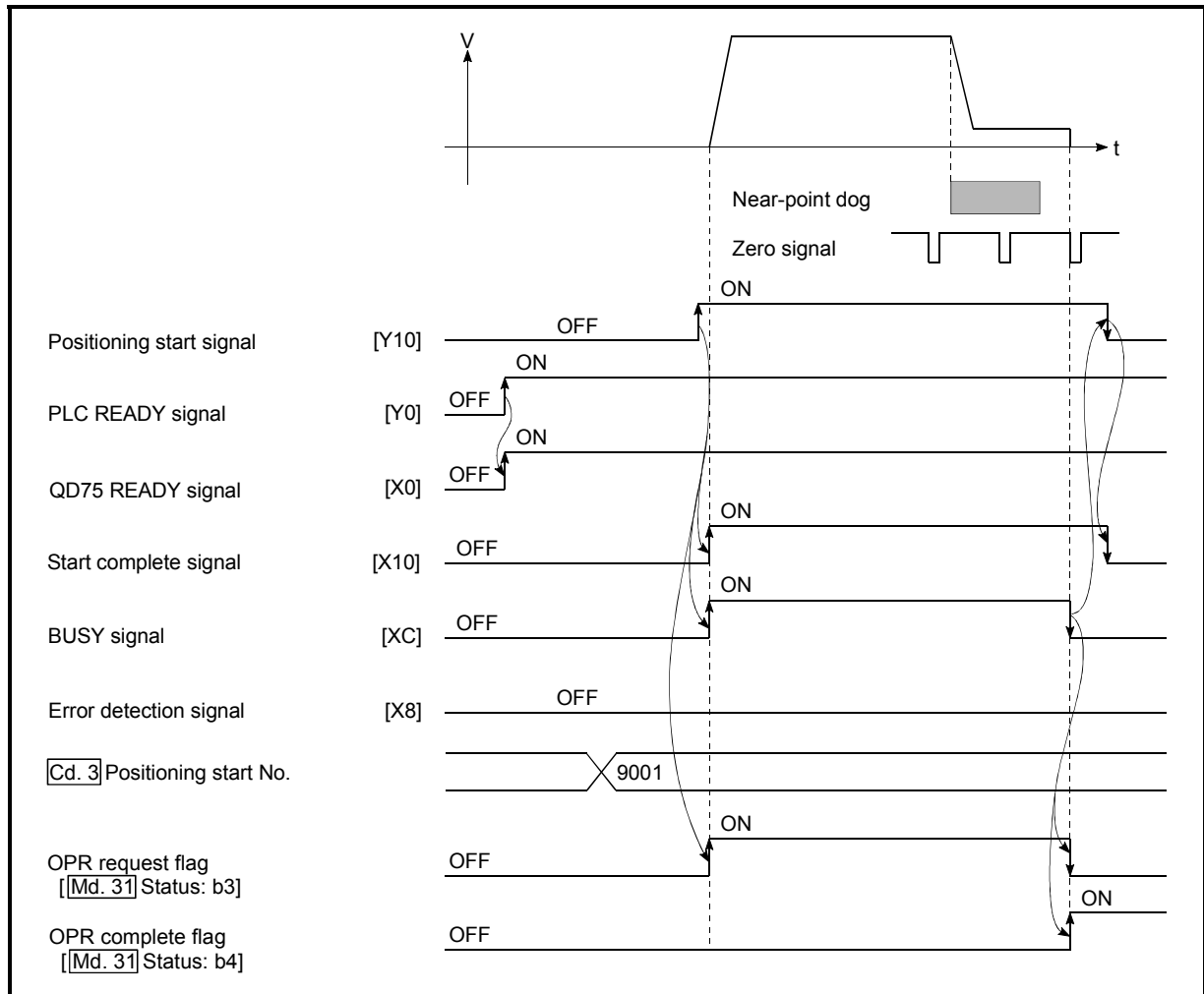


Fig. 6.4 Time chart for starting "machine OPR"

(2) Time chart for starting "fast OPR"

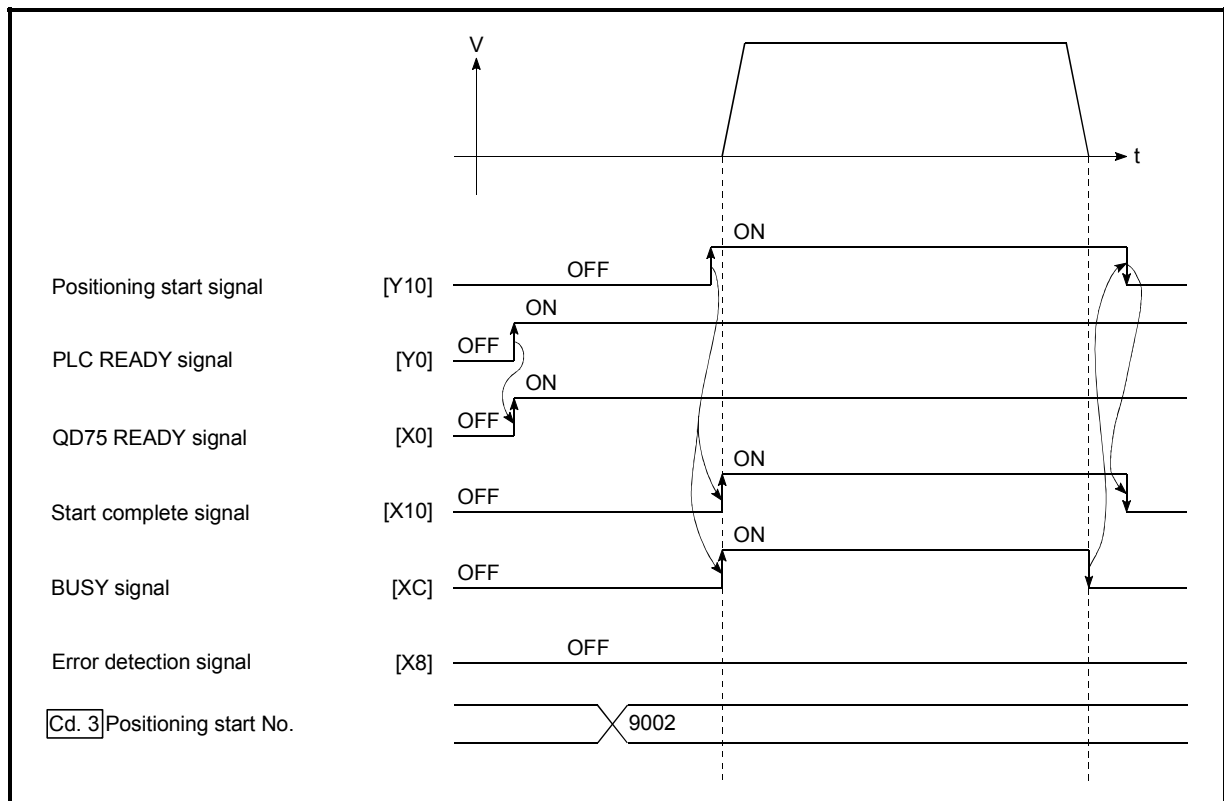


Fig. 6.5 Time chart for starting "fast OPR"

## (3) Time chart for starting "major positioning control"

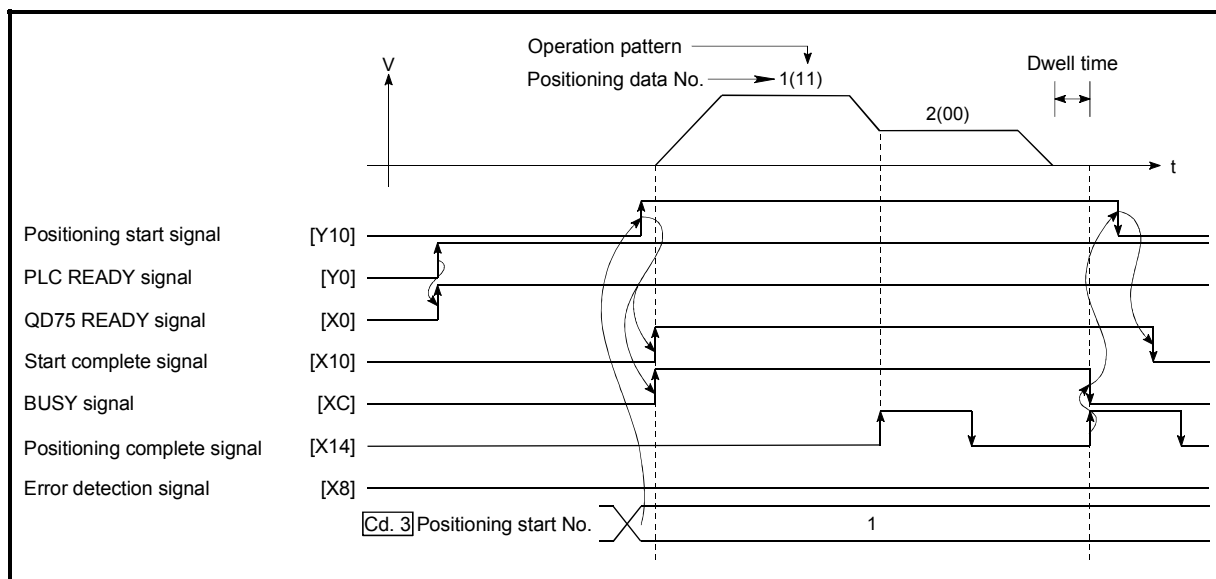


Fig. 6.6 Time chart for starting "major positioning control"

## (4) Time chart for starting "speed-position switching control"

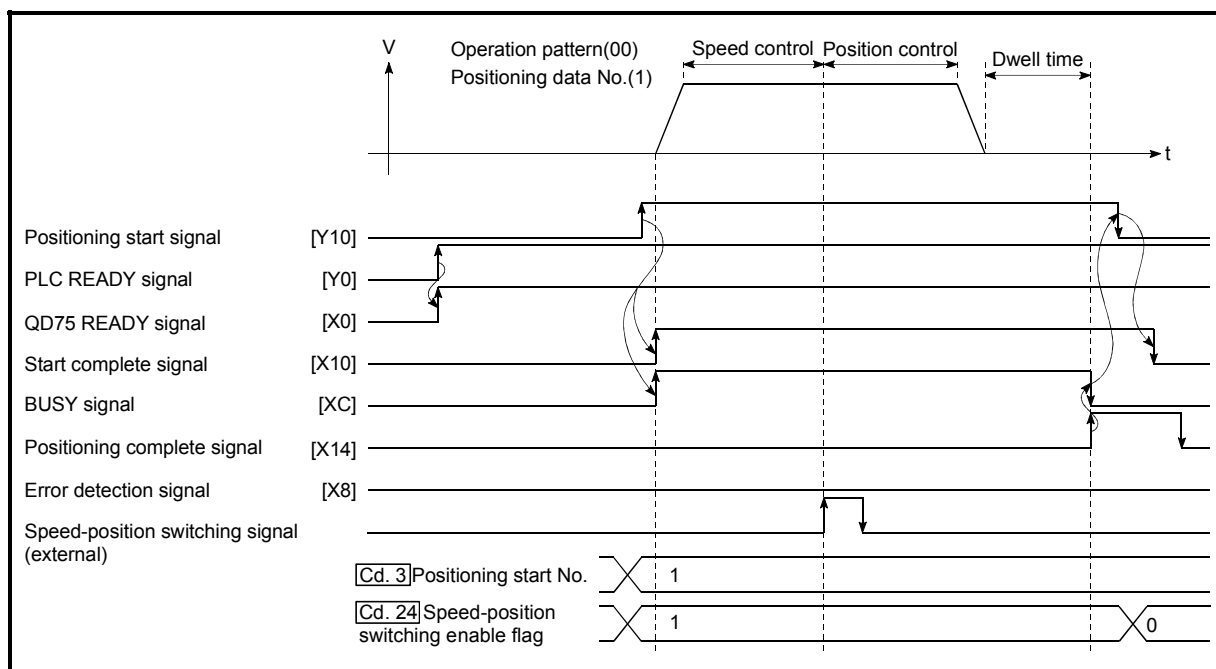


Fig. 6.7 Time chart for starting "speed-position switching control"

(5) Time chart for starting "position-speed switching control"

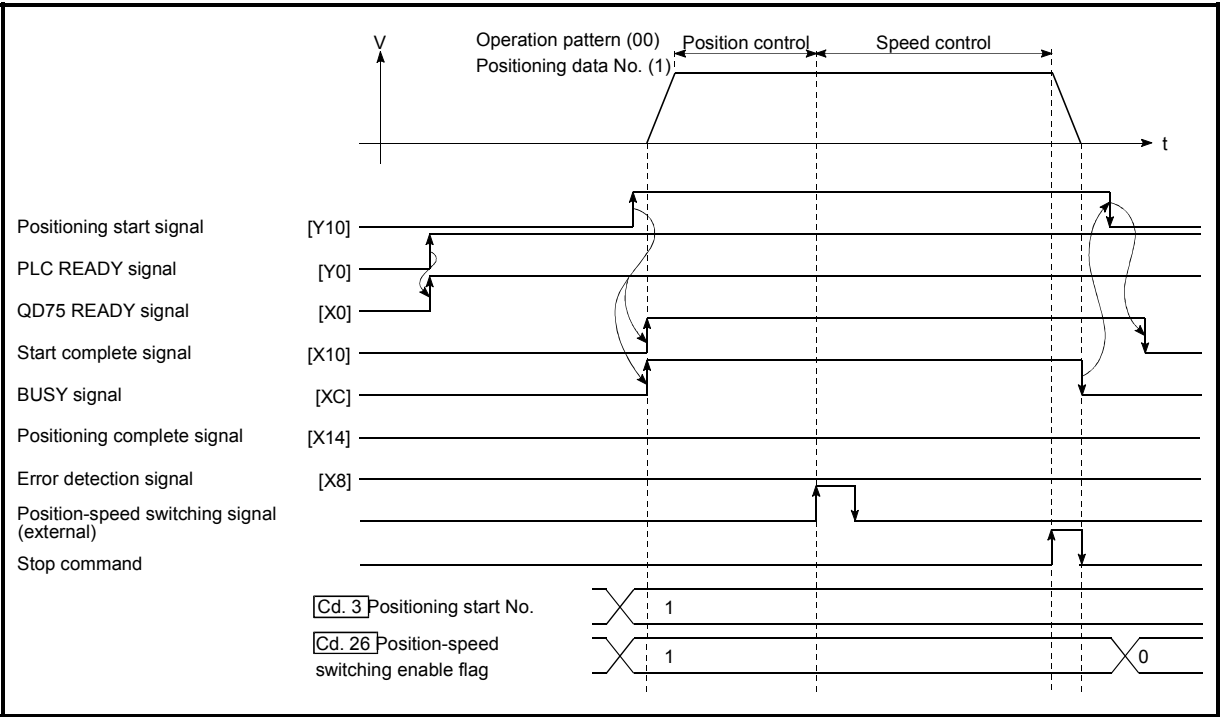


Fig. 6.8 Time chart for starting "position-speed switching control"

### Machine OPR operation timing and process time

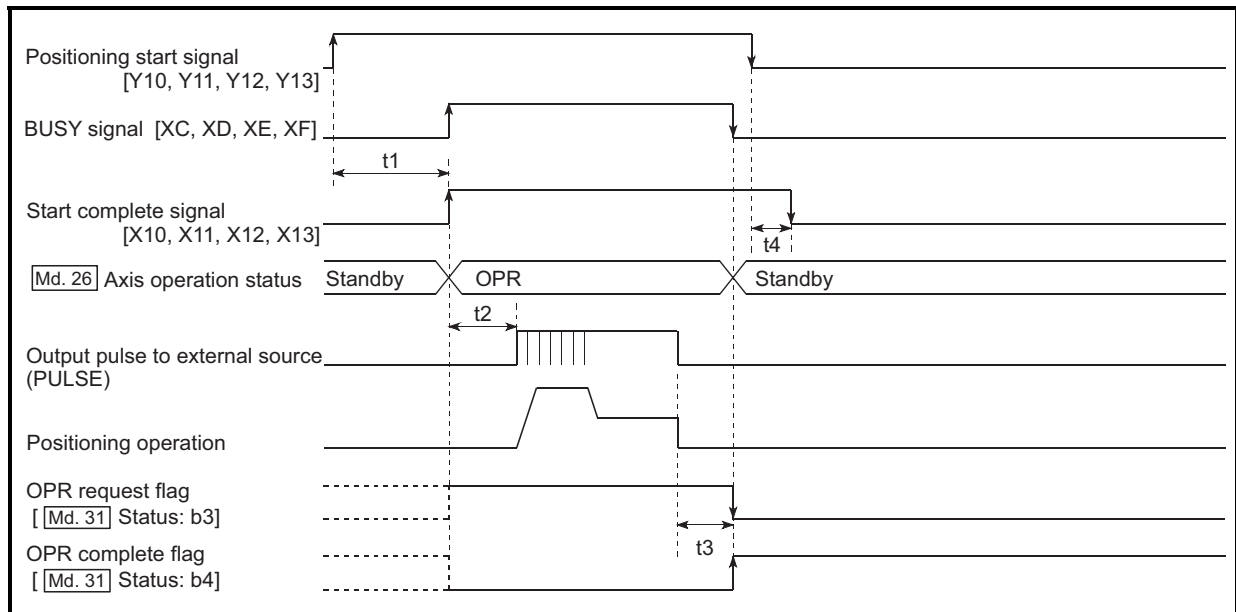


Fig. 6.9 Machine OPR operation timing and process time

#### Normal timing time

| Model           | t1           | t2           | t3         | t4         |
|-----------------|--------------|--------------|------------|------------|
| QD75P□N/QD75D□N | 0.2 to 1.1ms | 0.4 to 1.3ms | 0 to 0.9ms | 0 to 0.9ms |
| QD75P□/QD75D□   | 1.0 to 1.4ms | 2.7 to 4.4ms | 0 to 1.8ms | 0 to 1.8ms |

- The t1 timing time could be delayed depending on the operating conditions of the other axis.

### Position control operation timing and process time

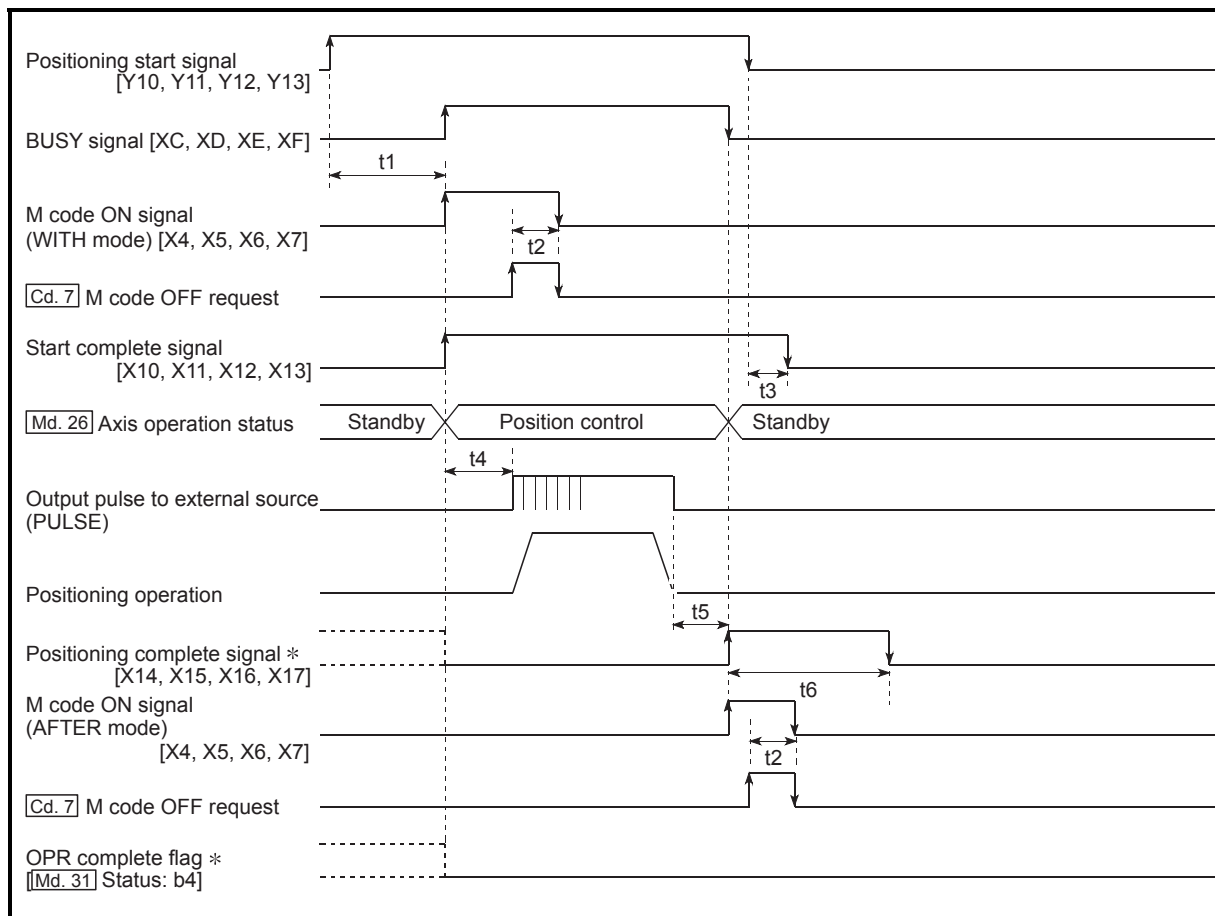


Fig. 6.10 Position control operation timing and process time

- When the positioning start signal turns ON, if all signals marked with an asterisk (\*) are already ON, the signals marked with an asterisk (\*) will turn OFF when the positioning start signal turns ON.

#### Normal timing time

| Model           | t1           | t2         | t3         | t4           | t5         | t6                 |
|-----------------|--------------|------------|------------|--------------|------------|--------------------|
| QD75P□N/QD75D□N | 0.2 to 1.1ms | 0 to 0.9ms | 0 to 0.9ms | 0.4 to 1.3ms | 0 to 0.9ms | Follows parameters |
| QD75P□/QD75D□   | 1.2 to 2.3ms | 0 to 1.8ms | 0 to 1.8ms | 2.7 to 4.4ms | 0 to 1.8ms |                    |

- The t1 timing time could be delayed depending on the operating conditions of the other axis.

## [2] Starting by inputting external command signal

When starting positioning control by inputting the external command signal, the start command can be directly input into the QD75. This allows the variation time equivalent to one scan time of the CPU module to be eliminated. This is an effective procedure when operation is to be started as quickly as possible with the start command or when the starting variation time is to be suppressed. To start positioning control by inputting the external command signal, set the "data required to be set" and then turn ON the external command signal.

### ■ Restrictions

When starting by inputting the external command signal, the start complete signal [X10, X11, X12, X13] will not turn ON.

### ■ Data required to be set

To execute positioning start with the external command signal, set parameter (Pr.42) beforehand, and validate the "external command signal" with the "External command function valid setting program (program No. 5)".

| Setting item                              | Setting value | Setting details                         | Buffer memory address |        |        |        |
|-------------------------------------------|---------------|-----------------------------------------|-----------------------|--------|--------|--------|
|                                           |               |                                         | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| Pr.42 External command function selection | 0             | Set to "0: External positioning start". | 62                    | 212    | 362    | 512    |
| Cd.8 External command valid               | 1             | Set to "1: Validate external command".  | 1505                  | 1605   | 1705   | 1805   |

\* Refer to CHAPTER 5 "DATA USED FOR POSITIONING CONTROL" for details on the setting details.

### ■ Starting time chart

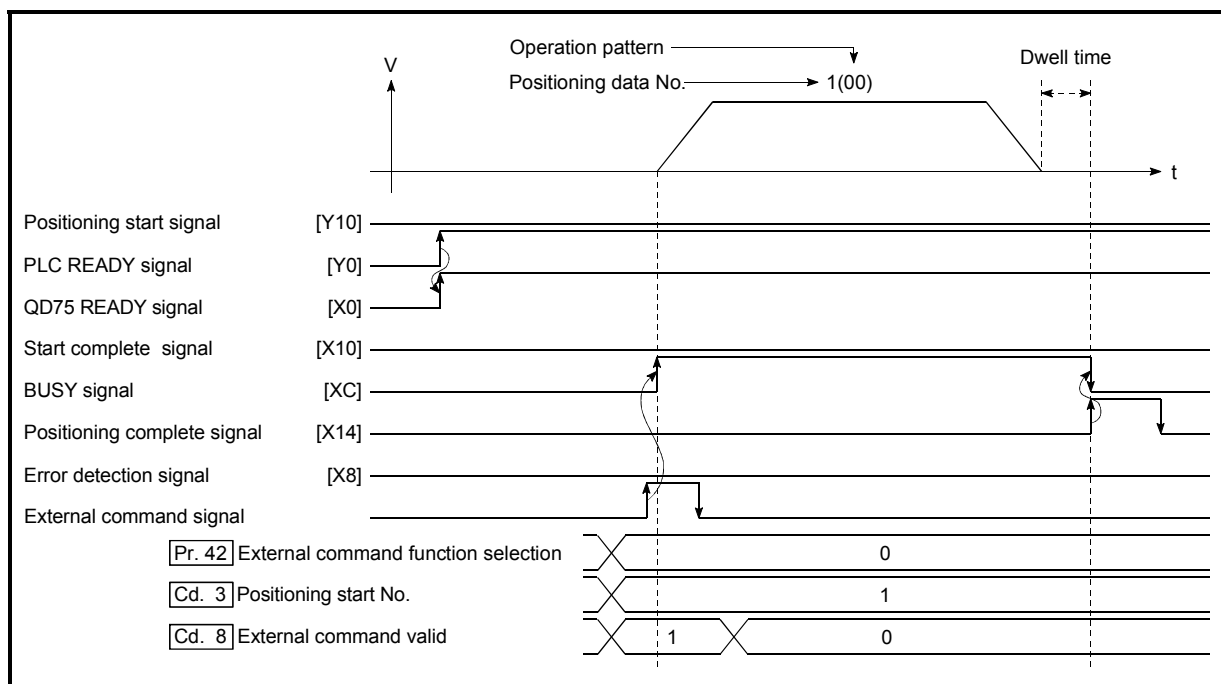


Fig. 6.11 Time chart for starting with external start signal

### 6.5.4 Continuous operation interrupt program

During positioning control, the control can be interrupted during continuous positioning control and continuous path control (continuous operation interrupt function). When "continuous operation interruption" is execution, the control will stop when the operation of the positioning data being executed ends. To execute continuous operation interruption, set "1: Continuous operation interrupt request" for " [Cd.18](#) Continuous operation interrupt request".

#### [1] Operation during continuous operation interruption

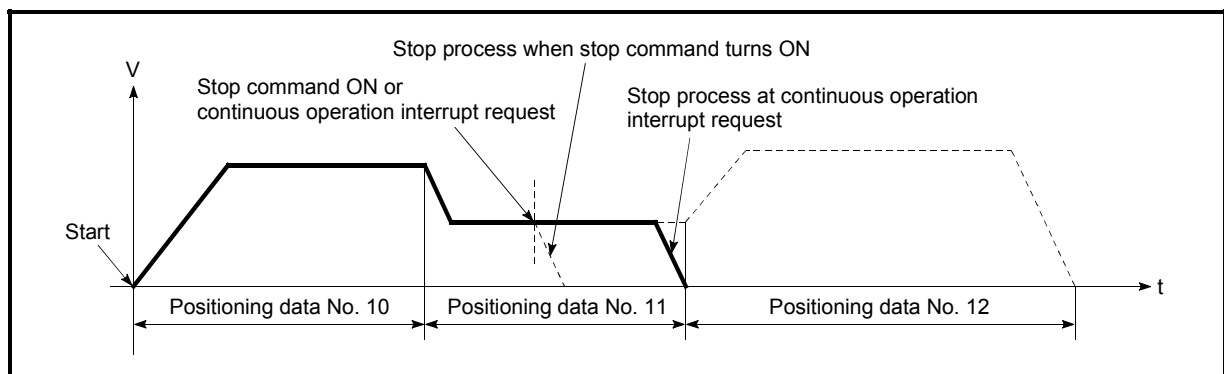


Fig. 6.12 Operation during continuous operation interruption

#### [2] Restrictions

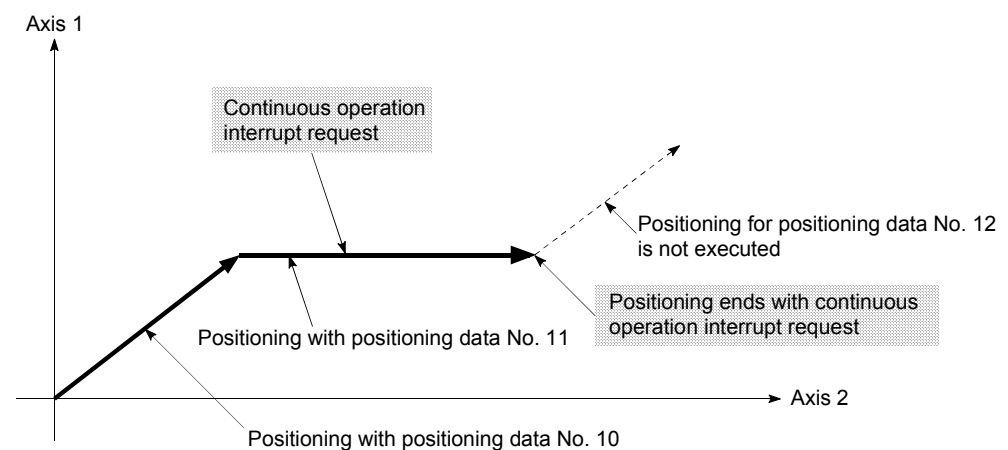
- (1) When the "continuous operation interrupt request" is executed, the positioning will end.

Thus, after stopping, the operation cannot be "restarted".

When " [Cd.6](#) Restart command" is issued, a warning "Restart not possible" (warning code: 104) will occur.

- (2) Even if the stop command is turned ON after executing the "continuous operation interrupt request", the "continuous operation interrupt request" cannot be canceled.

Thus, if "restart" is executed after stopping by turning the stop command ON, the operation will stop when the positioning data No. where "continuous operation interrupt request" was executed is completed.

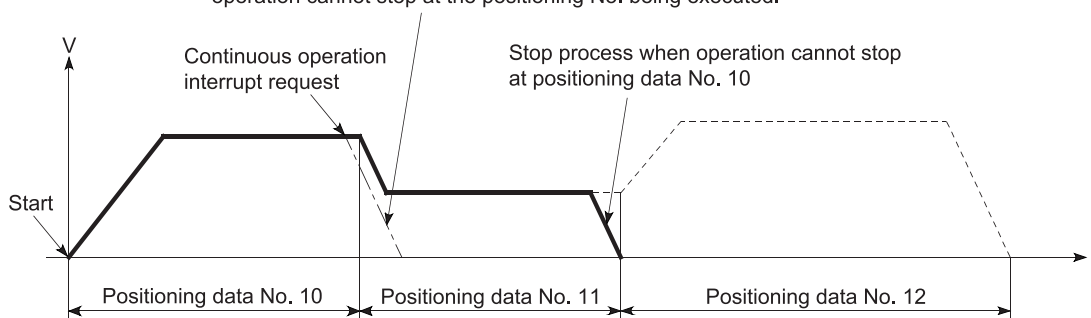




- (3) If the operation cannot be decelerated to a stop because the remaining distance is insufficient when "continuous operation interrupt request" is executed with continuous path control, the interruption of the continuous operation will be postponed until the positioning data shown below.

- Positioning data No. have sufficient remaining distance
- Positioning data No. for positioning complete (pattern: 00)
- Positioning data No. for continuous positioning control (pattern: 01)

Even when the continuous operation interrupt is requested, the remaining distance is insufficient, and thus, the operation cannot stop at the positioning No. being executed.



- (4) When operation is not performed (BUSY signal [XC, XD, XE, XF] is OFF), the interrupt request during continuous operation is not accepted. It is cleared to 0 at a start or restart.

### [3] Control data requiring settings

Set the following data to interrupt continuous operation.

| Setting item                                                                                             | Setting value | Setting details                                  | Buffer memory address |        |        |        |
|----------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------|-----------------------|--------|--------|--------|
|                                                                                                          |               |                                                  | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <span style="border: 1px solid black; padding: 2px;">Cd.18</span> Continuous operation interrupt request | 1             | Set "1: Continuous operation interrupt request". | 1520                  | 1620   | 1720   | 1820   |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

### 6.5.5 Restart program

When a stop factor occurs during position control and the operation stops, the positioning can be restarted from the stopped position to the position control end point by using the "restart command" (**Cd.6** Restart command).

("Restarting" is not possible when "continuous operation is interrupted.")

This instruction is efficient when performing the remaining positioning from the stopped position during position control of incremental system such as the INC Linear 1.

(Calculation of remaining distance is not required.)

#### [1] Restart operation

After a deceleration stop by the stop command is completed, write "1" to the " **Cd.6** Restart command" with " **Md.26** Axis operation status" is "Stopped" and the positioning restarts.

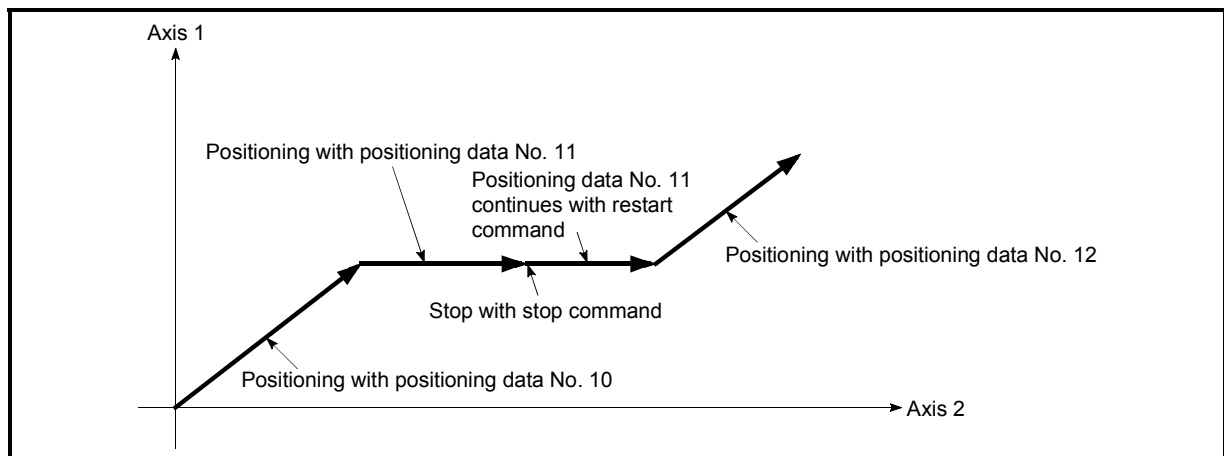


Fig. 6.13 Restart operation

#### [2] Restrictions

- (1) Restarting can be executed only when the " **Md.26** Axis operation status" is "Stopped (the deceleration stop by stop command is completed)".  
If the axis operation is not "stopped", restarting is not possible.  
In this case, a warning "Restart not possible" (warning code: 104) will occur, and the process at that time will be continued.
- (2) Restarting can be executed even while the positioning start signal is ON.  
However, make sure that the positioning start signal does not change from OFF to ON while " **Md.26** Axis operation status" is "Stopped".  
If the signal is changed from OFF to ON while " **Md.26** Axis operation status" is "Stopped", the normal positioning (the positioning data set in " **Cd.3** Positioning start signal") is started.
- (3) If the PLC READY signal is turned ON from OFF while " **Md.26** Axis operation status" is "Stopped", the positioning cannot be restarted. If restart is requested, a warning "Restart not possible" (warning code: 104) will occur.

- (4) Do not execute restart while the stop command is ON.  
If restart is executed while stopped, an error "Stop signal ON at start" (error code: 106) will occur, and the " Md.26 Axis operation status" will change to "Error".  
Thus, even if the error is reset, the operation cannot be restarted.
- (5) If positioning is ended with the continuous operation interrupt request, the operation cannot be restarted.  
If restart is requested, a warning "Restart not possible" (warning code: 104) will occur.
- (6) When stopped with interpolation operation, write "1: Restarts" into " Cd.6 Restart command" for the reference axis, and then restart.
- (7) If any of reference partner axes executes the positioning operation once, a warning "Restart not possible" (warning code: 104) will occur, and the positioning cannot restarts.
- (8) When the machine OPR and fast OPR is stopped, an error "OPR restart not possible" (error code: 209) will occur and the positioning cannot restarts.

## [3] Control data requiring setting

Set the following data to execute restart.

| Setting item                | Setting value | Setting details    | Buffer memory address |        |        |        |
|-----------------------------|---------------|--------------------|-----------------------|--------|--------|--------|
|                             |               |                    | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <b>Cd.6</b> Restart command | 1             | Set "1: Restarts". | 1503                  | 1603   | 1703   | 1803   |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

## [4] Starting conditions

The following conditions must be satisfied when restarting. (Assemble the required conditions into the sequence program as an interlock.)

## (1) Operation state

" **Md.26** Axis operation status" is "1: Stopped"

## (2) Signal state

| Signal name      |                         | Signal state |                                  | Device |        |        |        |
|------------------|-------------------------|--------------|----------------------------------|--------|--------|--------|--------|
|                  |                         |              |                                  | Axis 1 | Axis 2 | Axis 3 | Axis 4 |
| Interface signal | PLC READY signal        | ON           | CPU module preparation completed | Y0     |        |        |        |
|                  | QD75 READY signal       | ON           | QD75 preparation completed       | X0     |        |        |        |
|                  | Synchronization flag *  | ON           | QD75 buffer memory Accessible    | X1     |        |        |        |
|                  | Axis stop signal        | OFF          | Axis stop signal is OFF          | Y4     | Y5     | Y6     | Y7     |
|                  | M code ON signal        | OFF          | M code ON signal is OFF          | X4     | X5     | X6     | X7     |
|                  | Error detection signal  | OFF          | No error is present              | X8     | X9     | XA     | XB     |
|                  | BUSY signal             | OFF          | BUSY signal is OFF               | XC     | XD     | XE     | XF     |
|                  | Start complete signal   | OFF          | Start complete signal is OFF     | X10    | X11    | X12    | X13    |
| External signal  | Drive unit READY signal | ON           | Drive unit preparation completed | —      |        |        |        |
|                  | Stop signal             | OFF          | Stop signal is OFF               | —      |        |        |        |
|                  | Upper limit (FLS)       | ON           | Within limit range               | —      |        |        |        |
|                  | Lower limit (RLS)       | ON           | Within limit range               | —      |        |        |        |

※: When the synchronous setting of the CPU module is made in the nonsynchronous mode, this must be provided as an interlock.  
When it is made in the synchronous mode, no interlock must be provided in the program because the flag is turned ON when calculation is run on the CPU module.

## (5) Time chart for restarting

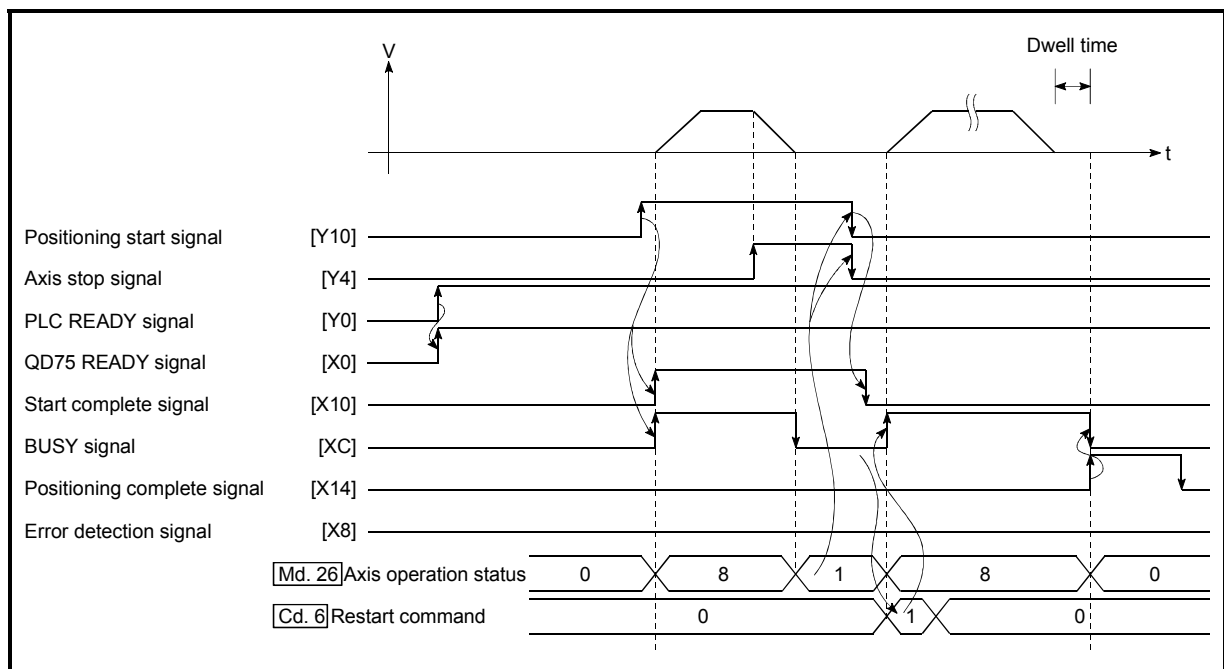


Fig. 6.14 Time chart for restarting

## 6.5.6 Stop program

The axis stop signal [Y4, Y5, Y6, Y7] or a stop signal from an external source is used to stop the control.

Create a program to turn ON the axis stop signal [Y4, Y5, Y6, Y7] as the stop program.

The process for stopping control is explained below.

Each control is stopped in the following cases.

- (1) When each control is completed normally.
- (2) When the drive unit READY signal is turned OFF.
- (3) When a CPU module error occurs
- (4) When the PLC READY signal is turned OFF.
- (5) When an error occurs in QD75.
- (6) When control is intentionally stopped  
(Stop signal from CPU module turned ON, stop signal from peripheral devices)

The stop process for the above cases is shown below.

(Excluding item (1) above "When each control is completed normally".)

## [1] Stop process

| Stop cause                          |                                                                                                  | Stop axis | M code ON signal after stop | Axis operation status (Md.26) after stopping | Stop process                                                                            |                  |                           |                                |                        |                                  |
|-------------------------------------|--------------------------------------------------------------------------------------------------|-----------|-----------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------|------------------|---------------------------|--------------------------------|------------------------|----------------------------------|
|                                     |                                                                                                  |           |                             |                                              | OPR control                                                                             |                  | Major positioning control | High-level positioning control | Manual control         |                                  |
|                                     |                                                                                                  |           |                             |                                              | Machine OPR control                                                                     | Fast OPR control |                           |                                | JOG/ Inching operation | Manual pulse generator operation |
| Forced stop                         | Drive unit READY signal OFF                                                                      | Each axis | No change                   | Error                                        | Immediate stop                                                                          |                  |                           |                                | Deceleration stop      |                                  |
| Fatal stop (Stop group 1)           | Hardware stroke limit upper/lower limit error occurrence                                         | Each axis | No change                   | Error                                        | Deceleration stop/sudden stop (Select with " Pr.37 Stop group 1 sudden stop selection". |                  |                           |                                | Deceleration stop      |                                  |
| Emergency stop (Stop group 2)       | CPU module error occurrence                                                                      | All axes  | No change                   | Error                                        | Delegation stop/sudden stop (Select with " Pr.38 Stop group 2 sudden stop selection".   |                  |                           |                                | Deceleration stop      |                                  |
|                                     | PLC READY signal OFF                                                                             |           | Turns OFF                   |                                              |                                                                                         |                  |                           |                                |                        |                                  |
|                                     | Error in test mode                                                                               |           | No change                   |                                              |                                                                                         |                  |                           |                                |                        |                                  |
| Relatively safe stop (Stop group 3) | Axis error detection (Error other than stop group 1 or 2)<br>"Stop" input from peripheral device | Each axis | No change                   | Error                                        | Deceleration stop/sudden stop (Select with " Pr.39 Stop group 3 sudden stop selection". |                  |                           |                                | Deceleration stop      |                                  |
| Intentional stop (Stop group 3)     | "Stop signal" ON from external source<br>"Axis stop signal" ON from CPU module                   | Each axis | No change                   | Stopped (Standby)                            |                                                                                         |                  |                           |                                |                        |                                  |

## [2] Types of stop processes

The operation can be stopped with deceleration stop, sudden stop or immediate stop.

### (1) Deceleration stop \*1

The operation stops with "deceleration time 0 to 3"

(**Pr.10**, **Pr.28**, **Pr.29**, **Pr.30**).

Which time from "deceleration time 0 to 3" to use for control is set in positioning data (**Da.4**).

### (2) Sudden stop

The operation stops with " **Pr.36** Sudden stop deceleration time".

### (3) Immediate stop

The operation does not decelerate.

The QD75 immediately stops the pulse output, but the operation will coast for the droop pulses accumulated in the drive unit's deviation counter.

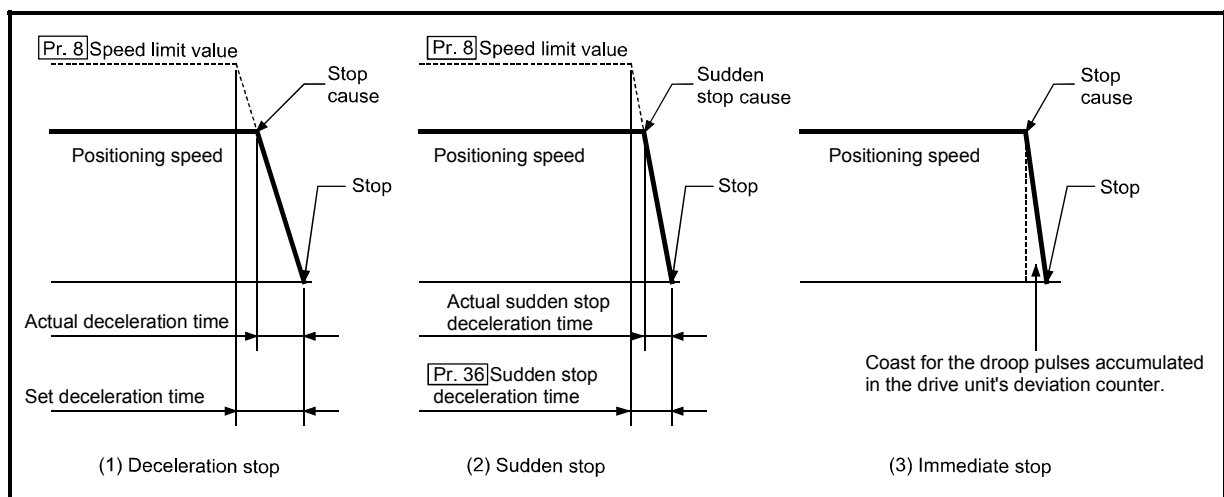


Fig. 6.15 Types of stop processes

## REMARK

\*1 "Deceleration stop" and "sudden stop" are selected with the details parameter 2 "stop group 1 to 3 sudden stop selection". (The default setting is "deceleration stop".)

## [3] Order of priority for stop process

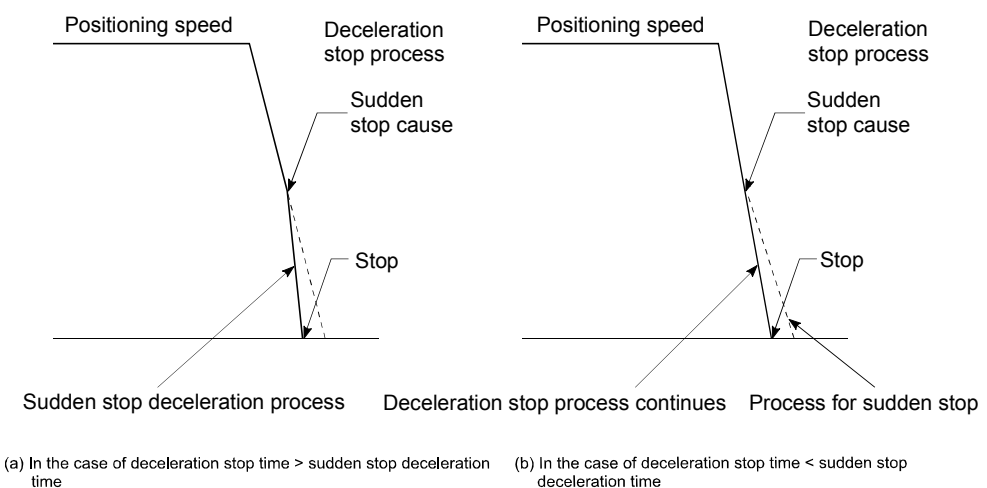
The order of priority for the QD75 stop process is as follows.

Deceleration stop < Sudden stop < Immediate stop

- (1) If the deceleration stop command ON (stop signal ON) or deceleration stop cause occurs during deceleration to speed 0 (including automatic deceleration), operation changes depending on the setting of " Cd.42 Stop command processing for deceleration stop selection".
  - (a) Manual control  
Independently of the Cd.42 setting, a deceleration curve is re-processed from the speed at stop cause occurrence.
  - (b) OPR control, positioning control
    - When Cd.42 = 0 (deceleration curve re-processing):  
A deceleration curve is re-processed from the speed at stop cause occurrence.
    - When Cd.42 = 1 (deceleration curve continuation):  
The current deceleration curve is continued after stop cause occurrence.  
(For details, refer to "Section 12.7.9 Stop command processing for deceleration stop function".)
- (2) If the stop signal ON or stop cause specified for a sudden stop occurs during deceleration, sudden stop process will start at that point.  
However, if the sudden stop deceleration time is longer than the deceleration time, the deceleration stop process will be continued even if a sudden stop cause occurs during the deceleration stop process.

## Example

The process when a sudden stop cause occurs during deceleration stop is shown below.



- (3) Operation will stop immediately if the target reaches the positioning address specified in the currently executed positioning data during deceleration of position control.



# CHAPTER 7 MEMORY CONFIGURATION AND DATA PROCESS

The QD75 memory configuration and data transmission are explained in this chapter.

The QD75 is configured of two memories. By understanding the configuration and roles of two memories, the QD75 internal data transmission process, such as "when the power is turned ON" or "when the PLC READY signal changes from OFF to ON" can be easily understood. This also allows the transmission process to be carried out correctly when saving or changing the data.

|       |                                              |    |   |
|-------|----------------------------------------------|----|---|
| 7.1   | Configuration and roles of QD75 memory.....  | 7- | 2 |
| 7.1.1 | Configuration and roles of QD75 memory ..... | 7- | 2 |
| 7.1.2 | Buffer memory area configuration.....        | 7- | 5 |
| 7.2   | Data transmission process.....               | 7- | 6 |

## 7.1 Configuration and roles of QD75 memory

## 7.1.1 Configuration and roles of QD75 memory

The QD75 is configured of the following two memories.

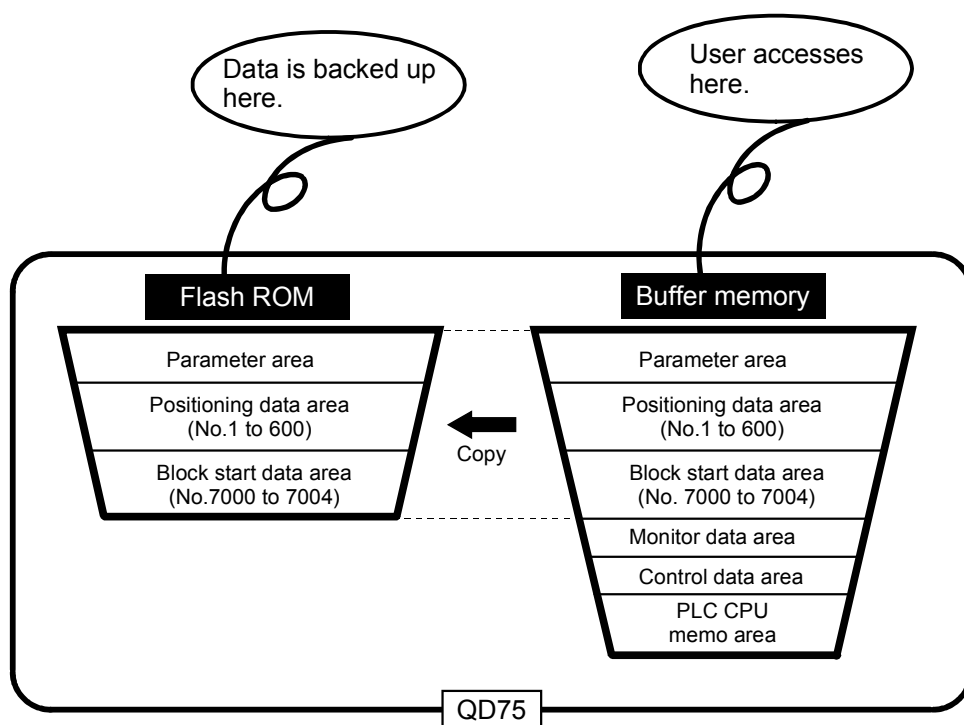
| Memory configuration | Role                                                                      | Area configuration |                   |                   |                                         |                                            |                   | Backup       |
|----------------------|---------------------------------------------------------------------------|--------------------|-------------------|-------------------|-----------------------------------------|--------------------------------------------|-------------------|--------------|
|                      |                                                                           | Parameter area     | Monitor data area | Control data area | Positioning data area<br>(No. 1 to 600) | Block start data area<br>(No.7000 to 7004) | PLC CPU memo area |              |
| • Buffer memory      | Area that can be directly accessed with sequence program from CPU module. | ○                  | ○                 | ○                 | ○                                       | ○                                          | ○                 | Not possible |
| • Flash ROM          | Area for backing up data required for positioning.                        | ○                  | —                 | —                 | ○                                       | ○                                          | —                 | Possible     |

○ : Setting and storage area provided, Not possible: Data is lost when power is turned OFF

— : Setting and storage area not provided, Possible: Data is held even when power is turned OFF

**■ Details of areas**

- Parameter area  
Area where parameters, such as positioning parameters and OPR parameters, required for positioning control are set and stored.  
(Set the items indicated with Pr.1 to Pr.57, Pr.70, and Pr.150 for each axis.)
- Monitor data area  
Area where positioning system or QD75 operation state is stored.  
(Set the items indicated with Md.1 to Md.48, Md.50 to Md.52 \*1.)  
\*1: Md.50 to Md.52 are for the QD75P□N/QD75D□N only.
- Control data area  
Area where data for operating and controlling positioning system is set and stored. (Set the items indicated with Cd.1 to Cd.42.)
- Positioning data area (No.1 to 600)  
Area where positioning data No.1 to 600 is set and stored.  
(Set the items indicated with Da.1 to Da.10 and Da.27 to Da.29 for each positioning data.)
- Block start data area (No.7000 to 7004)  
Area where information required only when carrying out block No. 7000 to 7004 high-level positioning is set and stored. (Set the items indicated with Da.11 to Da.19.)
- PLC CPU memo area  
Area where condition judgment values required for special positioning, etc., are set and stored.



## 7.1.2 Buffer memory area configuration

The QD75 buffer memory is configured of the following types of areas.

| Buffer memory area configuration    |                             | Buffer memory address*1      |                 |                    |                    | Writing possibility |
|-------------------------------------|-----------------------------|------------------------------|-----------------|--------------------|--------------------|---------------------|
|                                     |                             | Axis 1                       | Axis 2          | Axis 3             | Axis 4             |                     |
| Parameter area                      | Basic parameter area        | 0 to 15                      | 150 to 165      | 300 to 315         | 450 to 465         | Possible            |
|                                     | Detailed parameter area     | 17 to 62                     | 167 to 212      | 317 to 362         | 467 to 512         |                     |
|                                     |                             | 140*2                        |                 |                    |                    |                     |
|                                     | OPR basic parameter area    | 70 to 78                     | 220 to 228      | 370 to 378         | 520 to 528         |                     |
|                                     | OPR detailed parameter area | 79 to 89                     | 229 to 239      | 379 to 389         | 529 to 539         |                     |
| Monitor data area                   | System monitor area         | 1200 to 1425, 1440 to 1487*2 |                 |                    |                    | Not possible        |
|                                     | Axis monitor area           | 800 to 847, 899              | 900 to 947, 999 | 1000 to 1047, 1099 | 1100 to 1147, 1199 |                     |
| Control data area                   | System control data area    | 1900, 1901, 1905, 1907       |                 |                    |                    | Possible            |
|                                     | Axis control data area      | 1500 to 1550                 | 1600 to 1650    | 1700 to 1750       | 1800 to 1850       |                     |
| Positioning data area (No.1 to 600) | Positioning data area       | 2000 to 7999                 | 8000 to 13999   | 14000 to 19999     | 20000 to 25999     | Possible            |
| Block start data area (No.7000)     | Block start data area       | 26000 to 26049               | 27000 to 27049  | 28000 to 28049     | 29000 to 29049     |                     |
|                                     |                             | 26050 to 26099               | 27050 to 27099  | 28050 to 28099     | 29050 to 29099     |                     |
|                                     | Condition data area         | 26100 to 26199               | 27100 to 27199  | 28100 to 28199     | 29100 to 29199     |                     |
| Block start data area (No.7001)     | Block start data area       | 26200 to 26249               | 27200 to 27249  | 28200 to 28249     | 29200 to 29249     |                     |
|                                     |                             | 26250 to 26299               | 27250 to 27299  | 28250 to 28299     | 29250 to 29299     |                     |
|                                     | Condition data area         | 26300 to 26399               | 27300 to 27399  | 28300 to 28399     | 29300 to 29399     |                     |
| Block start data area (No.7002)     | Block start data area       | 26400 to 26449               | 27400 to 27449  | 28400 to 28449     | 29400 to 29449     |                     |
|                                     |                             | 26450 to 26499               | 27450 to 27499  | 28450 to 28499     | 29450 to 29499     |                     |
|                                     | Condition data area         | 26500 to 26599               | 27500 to 27599  | 28500 to 28599     | 29500 to 29599     |                     |
| Block start data area (No.7003)     | Block start data area       | 26600 to 26649               | 27600 to 27649  | 28600 to 28649     | 29600 to 29649     |                     |
|                                     |                             | 26650 to 26699               | 27650 to 27699  | 28650 to 28699     | 29650 to 29699     |                     |
|                                     | Condition data area         | 26700 to 26799               | 27700 to 27799  | 28700 to 28799     | 29700 to 29799     |                     |
| Block start data area (No.7004)     | Block start data area       | 26800 to 26849               | 27800 to 27849  | 28800 to 28849     | 29800 to 29849     |                     |
|                                     |                             | 26850 to 26899               | 27850 to 27899  | 28850 to 28899     | 29850 to 29899     |                     |
|                                     | Condition data area         | 26900 to 26999               | 27900 to 27999  | 28900 to 28999     | 29900 to 29999     |                     |
| PLC CPU memo area                   | PLC CPU memo area           | 30000 to 30099               |                 |                    |                    | Possible            |

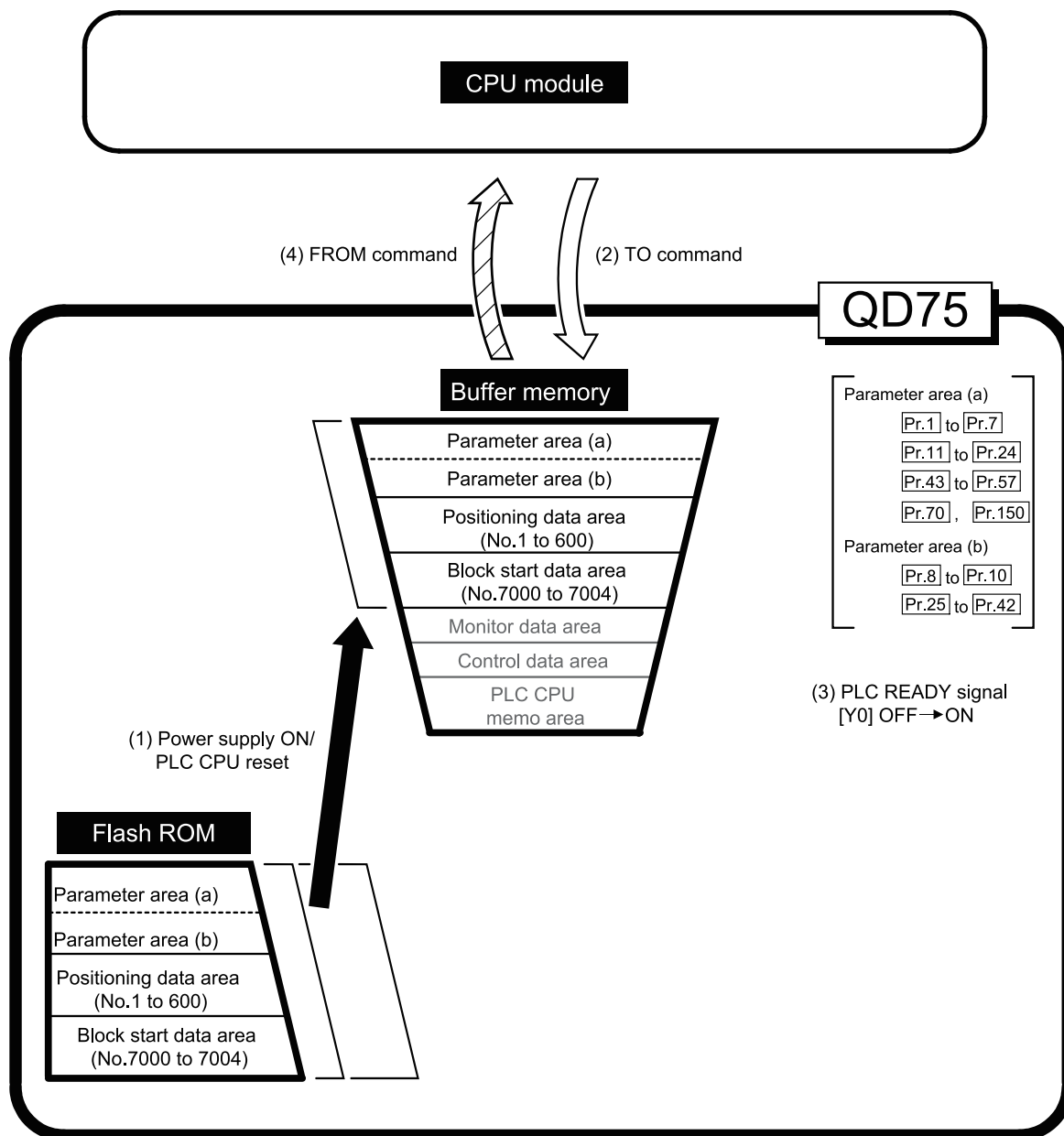
\*1: Use of address Nos. skipped above is prohibited. If used, the system may not operate correctly.

\*2: These addresses are not used for the QD75P□/QD75D□.

## 7.2 Data transmission process

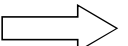
The data is transmitted between the QD75 memories with steps (1) to (8) shown below.

\*The data transmission patterns numbered (1) to (8) on the right page correspond to the numbers (1) to (8) on the left page.



- (1) Transmitting data when power is turned ON or CPU module is reset (  )

When the power is turned ON or the CPU module is reset, the "parameters", "positioning data" and "block start data" stored (backed up) in the flash ROM is transmitted to the buffer memory.

- (2) Transmitting data with TO instruction from CPU module (  )

The parameters or data is written from the CPU module to the buffer memory using the TO instruction. At this time, when the "parameter area (b) \*1", "positioning data (No. 1 to 600)", "block start data (No. 7000 to 7004)", "control data" and "PLC CPU memo area" are written into the buffer memory with the TO instruction, it is simultaneously valid.

\*1 Parameter area (b) ..... Parameters validated with the next each positioning control is started.

( Pr.8 to Pr.10, Pr.25 to Pr.42 )

- (3) Validate parameters when PLC READY signal [Y0] changes from OFF to ON

When the PLC READY signal [Y0] changes from OFF to ON, the data stored in the buffer memory's "parameter area (a) \*2" is validated.

For Pr.5, however, only the data obtained first after the PLC READY signal [Y0] changes from OFF to ON when the power is turned ON or CPU module is reset becomes validate.

(Refer to Section 5.2 "List of parameters" for details.)

\*2: Parameter area (a) .... Parameters validated when PLC READY signal [Y0] changes from OFF to ON.

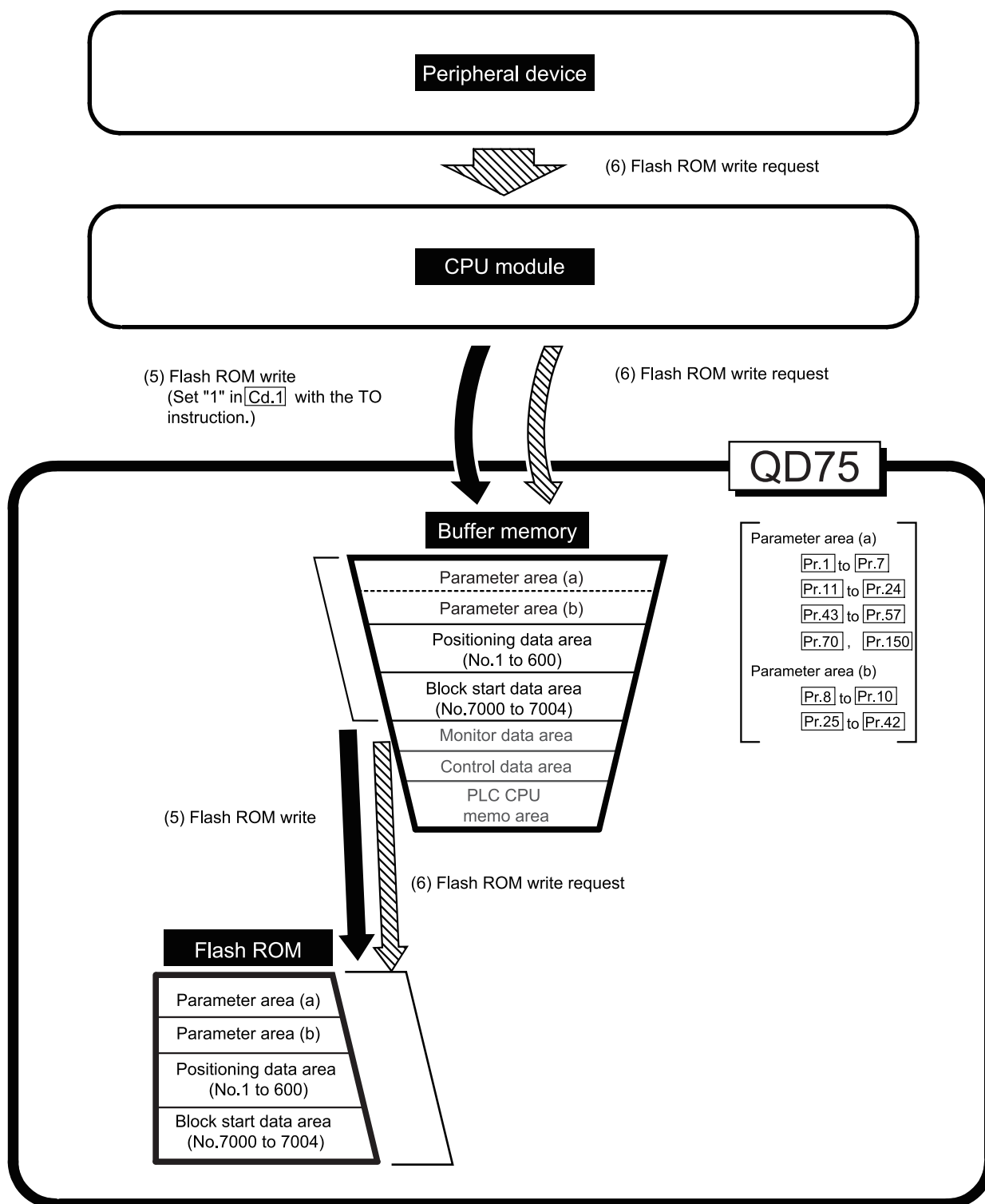
( Pr.1 to Pr.7, Pr.11 to Pr.24, Pr.43 to Pr.57, Pr.70, Pr.150 )

#### POINT

The setting values of the parameters that correspond to parameter area (b) are valid when written into the buffer memory with the TO instruction.  
However, the setting values of the parameters that correspond to parameter area (a) are not validated until the PLC READY signal [Y0] changes from OFF to ON.

- (4) Accessing with FROM instruction from CPU module (  )

The data is read from the buffer memory to the CPU module using the FROM instruction.





(5) Flash ROM write (  )

The following transmission process is carried out by setting "1" in " Cd.1 Flash ROM write request" (buffer memory [1900]).

- 1) The "parameters", "positioning data (No. 1 to 600)" and "block start data (No. 7000 to 7004)" in the buffer memory area are transmitted to the flash ROM. The writing to the flash ROM may also be carried out using a dedicated instruction "ZP.PFWRT". (Refer to CHAPTER 14 "DEDICATED INSTRUCTIONS" for details.)

(6) Flash ROM request (writing) (  )

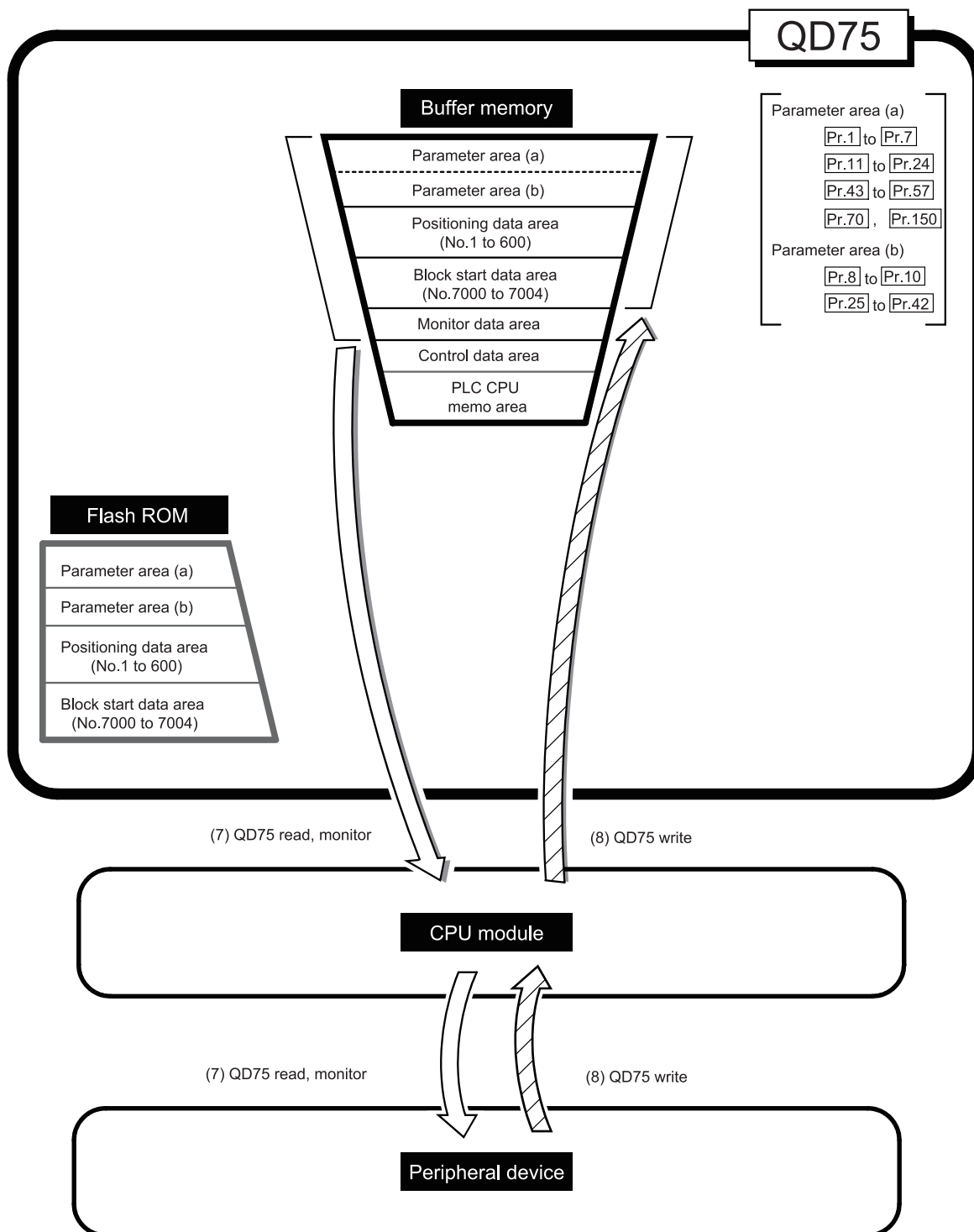
The following transmission processes are carried out with the [flash ROM request] (write) from the peripheral device.

- 1) The "parameters", "positioning data (No. 1 to 600)" and "block start data (No. 7000 to 7004)" in the buffer memory area are transmitted to the flash ROM.

Note) This transmission process is the same as (5) above.

**IMPORTANT**

- (1) Do not turn the power OFF or reset the CPU module while writing to the flash ROM. If the power is turned OFF or the CPU module is reset to forcibly end the process, the data backed up in the flash ROM will be lost.
- (2) Do not write the data to the buffer memory before writing to the flash ROM is completed.
- (3) The number of writes to the flash ROM with the programmable controller program is 25 max. while the power is turned ON. Writing to the flash ROM beyond 25 times will cause the error "Flash ROM write number error" (error code: 805). Refer to Section 15.3 "List of errors" for details.



## (7) Reading data from buffer memory to peripheral device (⇨)

The following transmission processes are carried out with the [Read from module] from the peripheral device.

- 1) The "parameters", "positioning data (No. 1 to 600)" and "block start data (No. 7000 to 7004)" in the buffer memory area are transmitted to the peripheral device via the CPU module.

The following transmission processes are carried out with the [monitor] from the peripheral device.

- 2) The "monitor data" in the buffer memory area is transmitted to the peripheral device via the CPU module.

## (8) Writing data from peripheral device to buffer memory (⇨)

The following transmission processes are carried out with the [Write to module] from the peripheral device.

- 1) The "parameters", "positioning data (No. 1 to 600)" and "block start data (No. 7000 to 7004)" in the peripheral device area transmitted to the buffer memory via the CPU module.

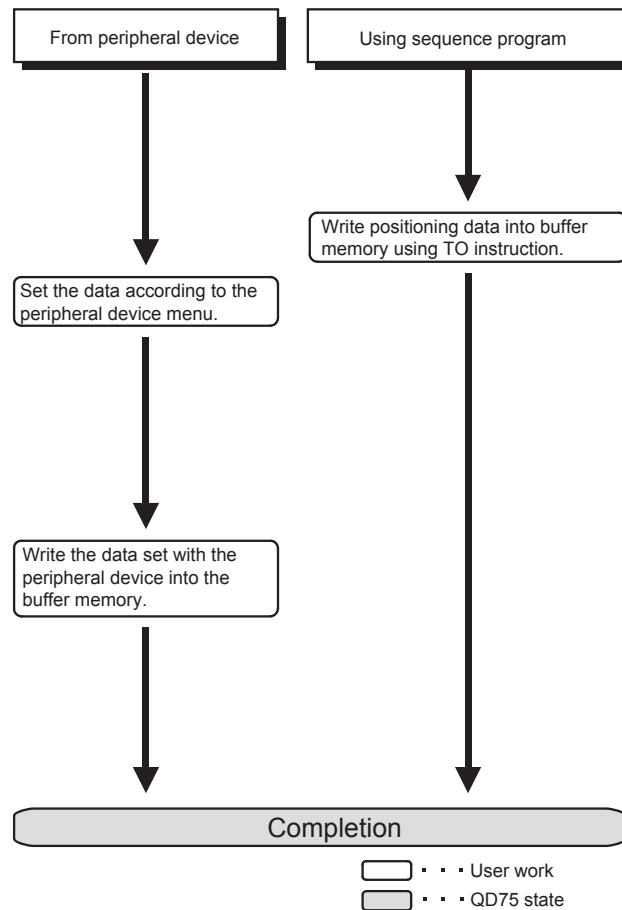
At this time, when [Flash ROM write] is set with the peripheral device, the transmission processes indicated with the following are carried out.

- (5) Flash ROM write

The data transmission is carried out as shown in the previous pages, but the main method of using this data process is shown below.

(Ex.) Setting the positioning data

The following methods can be used to set the positioning data.



## PART 2 CONTROL DETAILS AND SETTING

PART 2 is configured for the following purposes (1) to (3).

- (1) To understand the operation and restrictions of each control.
- (2) To carry out the required settings in each control.
- (3) To deal with errors.

The required settings in each control include parameter setting, positioning data setting, control data setting by a sequence program, etc.

Carry out these settings while referring to CHAPTER 5 "DATA USED FOR POSITIONING CONTROL".

Also refer to CHAPTER 6 "SEQUENCE PROGRAMS USED FOR POSITIONING CONTROL" when creating the programs required in each control, and consider the entire control program configuration when creating each program.

|            |                                      |     |             |
|------------|--------------------------------------|-----|-------------|
| CHAPTER 8  | OPR CONTROL .....                    | 8-  | 1 to 8- 24  |
| CHAPTER 9  | MAJOR POSITIONING CONTROL .....      | 9-  | 1 to 9-140  |
| CHAPTER 10 | HIGH-LEVEL POSITIONING CONTROL ..... | 10- | 1 to 10- 28 |
| CHAPTER 11 | MANUAL CONTROL .....                 | 11- | 1 to 11- 34 |
| CHAPTER 12 | CONTROL SUB FUNCTIONS .....          | 12- | 1 to 12-110 |
| CHAPTER 13 | COMMON FUNCTIONS .....               | 13- | 1 to 13- 8  |
| CHAPTER 14 | DEDICATED INSTRUCTIONS .....         | 14- | 1 to 14- 24 |
| CHAPTER 15 | TROUBLESHOOTING .....                | 15- | 1 to 15- 50 |

# MEMO

[illegible]

# CHAPTER 8 OPR CONTROL

The details and usage of "OPR control" are explained in this chapter.

OPR control includes "machine OPR" that establish a machine OP without using address data, and "fast OPR" that store the coordinates established by the machine OPR, and carry out positioning to that position.

OPR carried out by sequence programs from the CPU module are explained in this chapter.

Refer to GX Configurator-QP Operating Manual for details on OPR using the peripheral device.

|       |                                             |       |
|-------|---------------------------------------------|-------|
| 8.1   | Outline of OPR control .....                | 8- 2  |
| 8.1.1 | Two types of OPR control.....               | 8- 2  |
| 8.2   | Machine OPR .....                           | 8- 4  |
| 8.2.1 | Outline of the machine OPR operation .....  | 8- 4  |
| 8.2.2 | Machine OPR method .....                    | 8- 5  |
| 8.2.3 | OPR method (1): Near-point dog method ..... | 8- 7  |
| 8.2.4 | OPR method (2): Stopper method 1) .....     | 8- 9  |
| 8.2.5 | OPR method (3): Stopper method 2) .....     | 8- 12 |
| 8.2.6 | OPR method (4): Stopper method 3) .....     | 8- 15 |
| 8.2.7 | OPR method (5): Count method 1) .....       | 8- 17 |
| 8.2.8 | OPR method (6): Count method 2) .....       | 8- 20 |
| 8.3   | Fast OPR .....                              | 8- 23 |
| 8.3.1 | Outline of the fast OPR operation .....     | 8- 23 |

## 8.1 Outline of OPR control

### 8.1.1 Two types of OPR control

In "OPR control" a position is established as the starting point (or "OP") when carrying out positioning control, and positioning is carried out toward that starting point.

It is used to return a machine system at any position other than the OP to the OP when the QD75 issues a "OPR request"\* with the power turned ON or others, or after a positioning stop.

In the QD75, the two types of controls shown below are defined as "OPR control", following the flow of the OPR work.

These two types of OPR control can be executed by setting the "OPR parameters", setting "Positioning start No. 9001" and "Positioning start No. 9002" prepared beforehand in the QD75 to "[Cd.3] Positioning start No.", and turning ON the positioning start signal.

The ZP.PSTRT ☐ start numbers of the dedicated instruction can also be set to 9001 or 9002 to execute the OPR control. (For details, refer to CHAPTER 14 "DEDICATED INSTRUCTIONS".)

- (1) Establish a positioning control OP
  - "Machine OPR" (positioning start No. 9001)
- (2) Carry out positioning toward the OP
  - "Fast OPR" (positioning start No. 9002).

\* The "machine OPR" in (1) above must always be carried out before executing the "fast OPR" in (2).

|        |
|--------|
| REMARK |
|--------|

#### OPR request \*

The "OPR request flag" ([Md.31] Status: b3) must be turned ON in the QD75, and a machine OPR must be executed in the following cases.

- When the power is turned ON
- At the ON → OFF of the drive unit READY signal ([Md.30] External input/output signal: b2).
- At the OFF → ON of the PLC READY signal [Y0]

The address information stored in the QD75 cannot be guaranteed while the "OPR request flag" is ON.

The "OPR request flag" turns OFF and the "OPR complete flag" ([Md.31] Status: b4) turns ON if the machine OPR is executed and is completed normally.



### ■ OPR sub functions

Refer to Section 3.2.4 "Combination of QD75 main functions and sub functions" for details on "sub functions" that can be combined with OPR control. Also refer to CHAPTER 12 "CONTROL SUB FUNCTIONS" for details on each sub function.

#### [Remarks]

The following two sub functions are only related to machine OPR.

| Sub function name  | Machine OPR | Fast OPR | Reference      |
|--------------------|-------------|----------|----------------|
| OPR retry function | △           | ×        | Section 12.2.1 |
| OP shift function  | ○           | ×        | Section 12.2.2 |

○ : Combination possible, △: Restricted, ×: Combination not possible

### ■ When an OPR is not required

Control can be carried out ignoring the "OPR request flag" (**[Md.31]** Status: b3) in systems that do not require an OPR.

In this case, the "OPR parameters (**[Pr.43]** to **[Pr.57]**)" must all be set to their initial values or a value at which an error does not occur.

### ■ OPR from peripheral devices

"Machine OPR" and "fast OPR" can be executed from the test mode of the peripheral device.

Refer to GX Configurator-QP Operating Manual for details on OPR from the peripheral device.

## 8.2 Machine OPR

### 8.2.1 Outline of the machine OPR operation

#### Important

Use the OPR retry function when the OP position is not always in the same direction from the workpiece operation area (when the OP is not set near the upper or lower limit of the machine).

\*The machine OPR may not complete unless the OPR retry function is used.

#### Machine OPR operation

In a machine OPR, a machine OP is established.

None of the address information stored in the QD75, CPU module, or servo is used at this time. The position mechanically established after the machine OPR is regarded as the "OP" to be the starting point for positioning control.

The method for establishing an "OP" by a machine OPR differs according to the method set in "Pr.43 OPR method".

The following shows the operation when starting a machine OPR.

|    |                                                                                                                                                                                     |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) | The machine OPR is started.                                                                                                                                                         |
| 2) | The operation starts according to the speed and direction set in the OPR parameters (Pr.43 to Pr.57).                                                                               |
| 3) | The "OP" is established by the method set in "Pr.43 OPR method", and the machine stops. (Refer to Sections 8.2.2 to 8.2.8)                                                          |
| 4) | If "a" is set as "Pr.45 OP address", "a" will be stored as the current position in the "Md.20 Current feed value" and "Md.21 Machine feed value" which are monitoring the position. |
| 5) | The machine OPR is completed.                                                                                                                                                       |

\* The "Pr.45 OP address" is a fixed value set by the user.

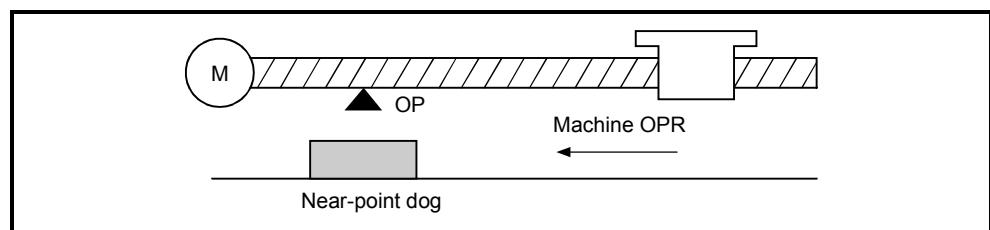


Fig. 8.1 Example of a machine OPR

## 8.2.2 Machine OPR method

The method by which the machine OP is established (method for judging the OP position and machine OPR completion) is designated in the machine OPR according to the configuration and application of the positioning method.

The following table shows the six methods that can be used for this OPR method.

(The OPR method is one of the items set in the OPR parameters. It is set in "Pr.43 OPR method" of the basic parameters for OPR.)

| Pr.43 OPR method      | Operation details                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Near-point dog method | Deceleration starts by the OFF → ON of the near-point dog. (Speed is reduced to "Pr.47 Creep speed".)<br>The operation stops at the first zero signal* after the near-point dog turns from ON → OFF. When a "deviation counter clear output" is completed, the machine OPR is completed.                                                                                                                         |
| Stopper method 1)     | The stopper position is regarded as the OP.<br>After the deceleration starts by the OFF → ON of the near-point dog, the machine presses against the stopper at the "Pr.47 Creep speed" and stops.<br>The machine OPR is regarded as completed on completion of the deviation counter clear output provided after "Pr.49 OPR dwell time" passed after stoppage.                                                   |
| Stopper method 2)     | The stopper position is regarded as the OP.<br>After the deceleration starts by the OFF → ON of the near-point dog, the machine presses against the stopper at the "Pr.47 Creep speed" and stops.<br>The machine OPR is regarded as completed on completion of the deviation counter clear output provided after the zero signal* is detected after stoppage.                                                    |
| Stopper method 3)     | The stopper position is regarded as the OP.<br>The machine starts at the "Pr.47 Creep speed" from the beginning, then presses against the stopper at the "Pr.47 Creep speed" and stops.<br>The machine OPR is regarded as completed on completion of the deviation counter clear output provided after the zero signal* is detected after stoppage.                                                              |
| Count method 1)       | The deceleration starts by the OFF → ON of the near-point dog, and the machine moves at the "Pr.47 Creep speed".<br>The machine stops at the zero signal* after moving the distance set in the "Pr.50 Setting for the movement amount after near-point dog ON" from the near point dog OFF → ON position. When a "deviation counter clear signal output" is completed, the machine OPR is regarded as completed. |
| Count method 2)       | The deceleration starts by the OFF → ON of the near-point dog, and the machine moves at the "Pr.47 Creep speed".<br>The machine moves the distance set in the "Pr.50 Setting for the movement amount after near-point dog ON" from the near point dog OFF → ON position, and stops at that position. The machine OPR is then regarded as completed.                                                              |

\*: The following are the signals input as the zero signals of the QD75 in the corresponding OPR methods.

Near-point dog method, count method 1): Signal that is output as a single pulse at one motor revolution (e.g. Z-phase signal output from the drive unit)

Stopper method 2), 3) : Signal that is output on detection of contact with the stopper. (Input externally)

The following shows the external I/O signals used for machine OPR.

| Pr.43 OPR method      | Signal required for control |             |                            |                                | Torque limit |
|-----------------------|-----------------------------|-------------|----------------------------|--------------------------------|--------------|
|                       | Near-point dog              | Zero signal | Upper/lower limit switches | Deviation counter clear output |              |
| Near-point dog method | ◎                           | ◎           | ○                          | ◎                              | —            |
| Stopper method 1)     | ◎                           | —           | ○                          | ◎                              | ◎            |
| Stopper method 2)     | ◎                           | ◎           | ○                          | ◎                              | ◎            |
| Stopper method 3)     | —                           | ◎           | ○                          | ◎                              | ◎            |
| Count method 1)       | ◎                           | ◎           | ○                          | ◎                              | —            |
| Count method 2)       | ◎                           | —           | ○                          | —                              | —            |

◎: Necessary ○: Necessary as required -: Unnecessary

### REMARK

#### Creep speed

The stopping accuracy is poor when the machine suddenly stops from fast speeds. To improve the machine's stopping accuracy, its must change over to a slow speed before stopping. This speed is set in the " Pr.47 Creep speed".

## 8.2.3 OPR method (1): Near-point dog method

The following shows an operation outline of the "near-point dog method" OPR method.

■ Operation chart

|    |                                                                                                                                                                                                                                                                       |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) | The machine OPR is started.<br>(The machine begins the acceleration designated in " [Pr.51] OPR acceleration time selection", in the direction designated in " [Pr.44] OPR direction". It then moves at the " [Pr.46] OPR speed" when the acceleration is completed.) |
| 2) | The machine begins decelerating when the near-point dog ON is detected.                                                                                                                                                                                               |
| 3) | The machine decelerates to the " [Pr.47] Creep speed", and subsequently moves at that speed.<br>(At this time, the near-point dog must be ON. The workpiece will continue decelerating and stop if the near-point dog is OFF.)                                        |
| 4) | After the near-point dog turns OFF, the pulse output from the QD75 will stop at the first zero signal, outputting a "deviation counter clear signal" to the drive unit.<br>(The "deviation counter clear signal output time" is set in [Pr.55].)                      |
| 5) | After a "deviation counter clear signal" is output to the drive unit, the OPR complete flag ( [Md.31] Status: b4) turns from OFF to ON and the OPR request flag ( [Md.31] Status: b3) turns from ON to OFF.                                                           |

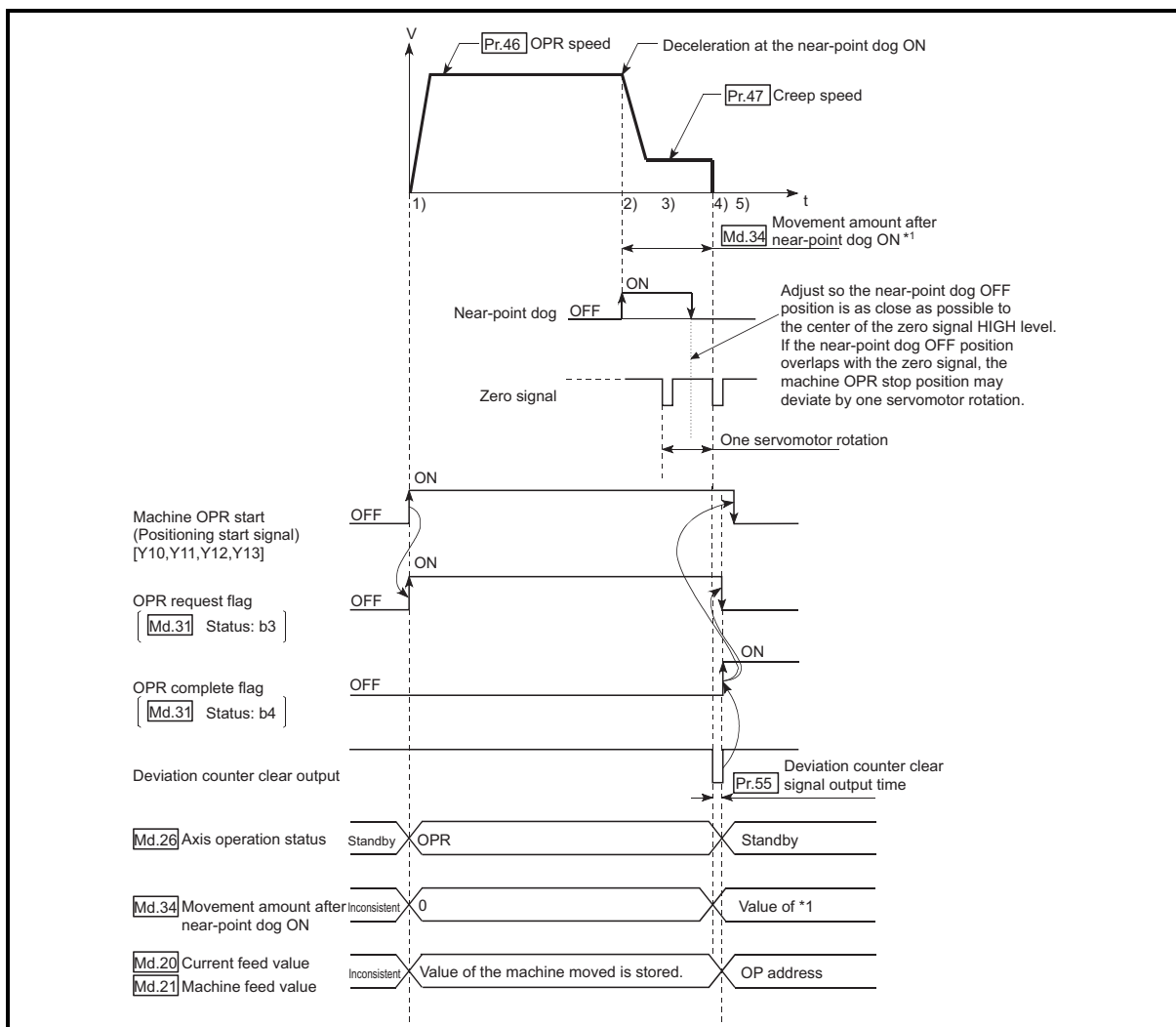


Fig. 8.2 Near-point dog method machine OPR

### ■ Restrictions

A pulse generator with a zero signal is required.

When using a pulse generator without a zero signal, generate a zero signal using an external signal.

### ■ Precautions during operation

- (1) An error "Start at OP (error code: 201)" will occur if another machine OPR is attempted after a machine OPR completion when the OPR retry function is not set ("0" is set in "Pr.48 OPR retry").
  - (2) Machine OPR carried out from the near-point dog ON position will start at the "Pr.47 Creep speed".
  - (3) The near-point dog must be ON during deceleration from the OPR speed "Pr.47 Creep speed".
- \* The workpiece will continue decelerating and stop if the near-point dog is turned OFF before it has decelerated to the creep speed, thus causing an error "Dog detection timing fault" (error code: 203).

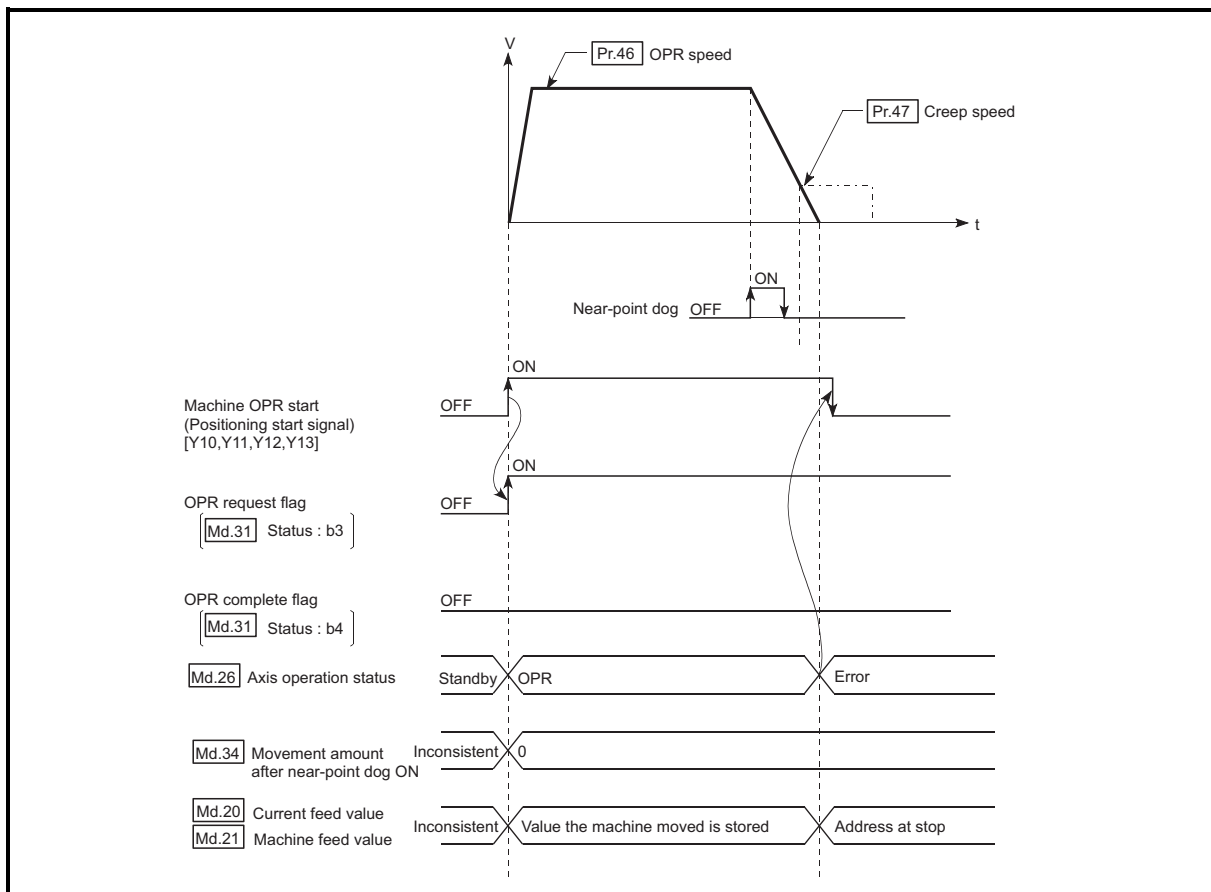


Fig. 8.3 Operation when the near-point dog is turned OFF before the creep speed is reached

- (4) When a machine OPR is stopped with the stop signal, perform a machine OPR again. If the restart command is turned ON after a stop with the stop signal, an error "OPR restart not possible" (error code: 209) occurs.

## 8.2.4 OPR method (2): Stopper method 1)

The following shows an operation outline of the "stopper method 1)" OPR method.

■ Operation chart

|    |                                                                                                                                                                                                                                                                         |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) | The machine OPR is started.<br>(The machine begins the acceleration designated in "[Pr.51] OPR acceleration time selection", in the direction designated in "[Pr.44] OPR direction". It then moves at the "[Pr.46] OPR speed" when the acceleration is completed.)      |
| 2) | The machine begins decelerating when the near-point dog ON is detected.                                                                                                                                                                                                 |
| 3) | The machine decelerates to the "[Pr.47] Creep speed", and subsequently moves at that speed.<br>(Torque limiting is required at this time. If the torque is not limited, the servomotor may fail in step 4).)                                                            |
| 4) | The machine presses against the stopper at the creep speed and stops.                                                                                                                                                                                                   |
| 5) | The pulse output from the QD75 will stop when the "[Pr.49] OPR dwell time" has elapsed after the near-point dog turns ON, outputting the "deviation counter clear output" to the drive unit.<br>(A "deviation counter clear signal output time" is set in the [Pr.55].) |
| 6) | After a "deviation counter clear output" is output to the drive unit, the OPR complete flag ([Md.31] Status: b4) turns from OFF to ON, and the OPR request flag ([Md.31] Status: b3) turns from ON to OFF.                                                              |

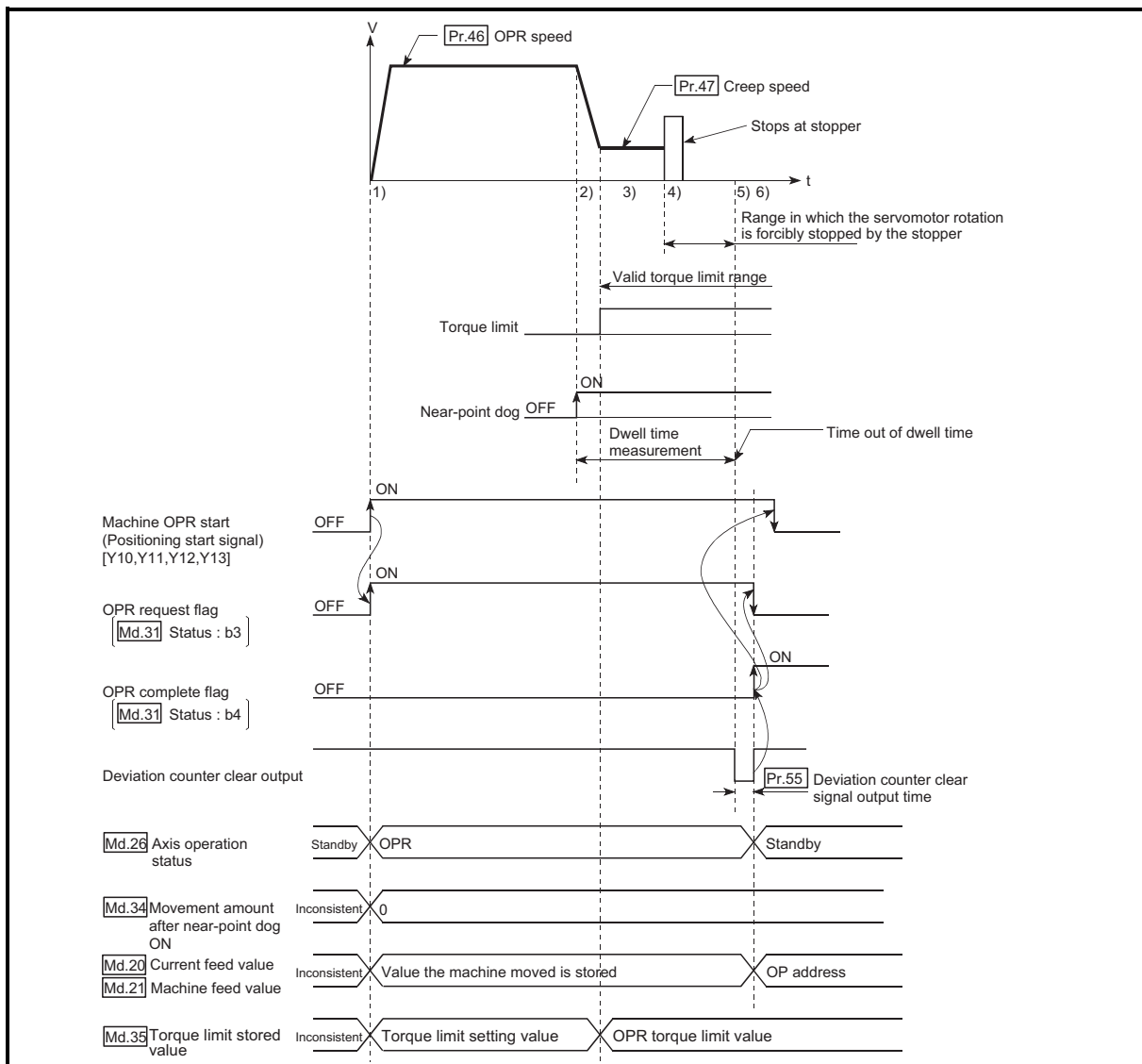


Fig. 8.4 Stopper method 1) machine OPR

### ■ Restrictions

Always limit the servomotor torque after the "[Pr.47] Creep speed" is reached. If the torque is not limited, the servomotor may fail when the machine presses against the stopper. (Refer to Section 12.4.2 "Torque limit function".)

### ■ Precautions during operation

- (1) Set a value in the "[Pr.49] OPR dwell time" that is equal to or higher than the movement time from the near-point dog ON to the time the machine presses against the stopper.
- (2) The workpiece will continue decelerating and stop if the "[Pr.49] OPR dwell time" elapses during deceleration from the "[Pr.46] OPR speed", thus causing an error "Dwell time fault" (error code: 205).

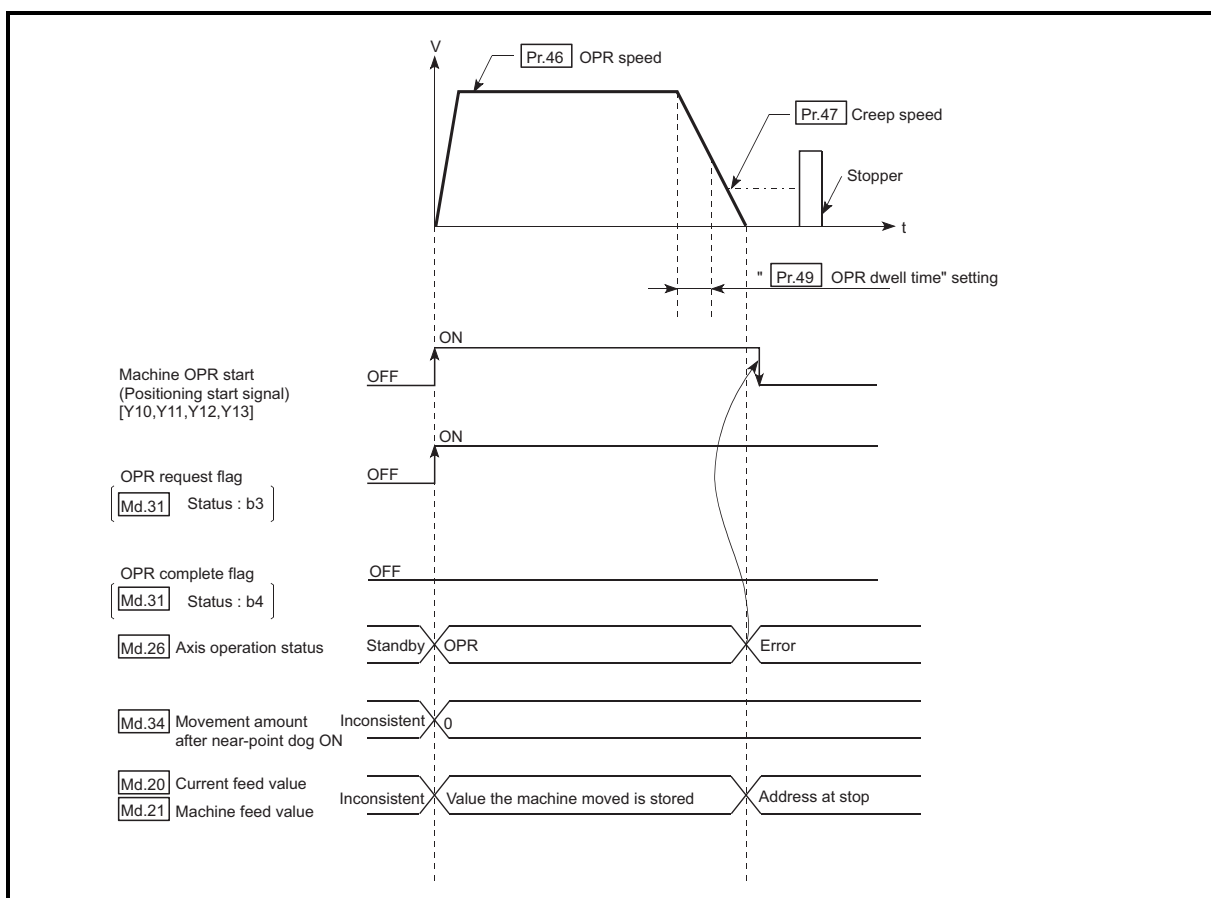


Fig. 8.5 Operation when the dwell time elapses during deceleration from the OPR speed



- (3) If the "Pr.49 OPR dwell time" elapses before the stop at the stopper, the workpiece will stop at that position, and that position will be regarded as the OP. At this time, an error will not occur.

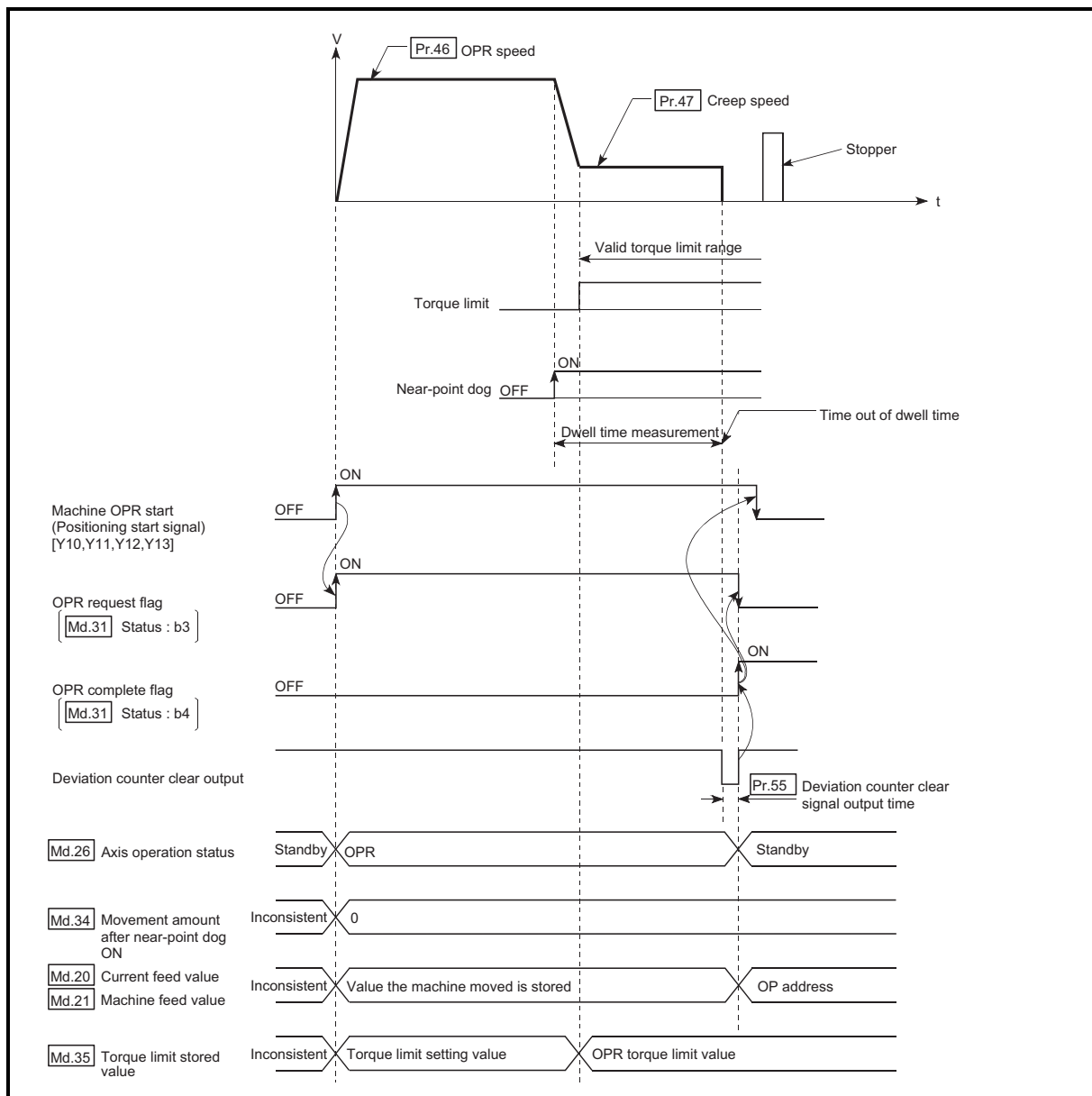


Fig. 8.6 Operation when the dwell time elapses before the stop at the stopper

- (4) The near-point dog must be turned ON until it presses against the stopper. If any range that the near-point is turned OFF exists until it presses against the stopper, executing the machine OPR from the area leads it to be pressed against the stopper at OPR speed.
- (5) Machine OPR started while the near-point dog is ON will start at the "Pr.47 Creep speed".
- (6) When a machine OPR is stopped with the stop signal, perform a machine OPR again. If the restart command is turned ON after a stop with the stop signal, an error "OPR restart not possible" (error code: 209) occurs.

## 8.2.5 OPR method (3): Stopper method 2)

The following shows an operation outline of the "stopper method 2)" OPR method.

■ Operation chart

|    |                                                                                                                                                                                                                                                                       |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) | The machine OPR is started.<br>(The machine begins the acceleration designated in " [Pr.51] OPR acceleration time selection", in the direction designated in " [Pr.44] OPR direction". It then moves at the " [Pr.46] OPR speed" when the acceleration is completed.) |
| 2) | The machine begins decelerating when the near-point dog ON is detected.                                                                                                                                                                                               |
| 3) | The machine decelerates to the " [Pr.47] Creep speed", and subsequently moves at that speed.<br>(Torque limiting is required at this time. If the torque is not limited, the servomotor may fail in step 4).)                                                         |
| 4) | The machine presses against the stopper at the creep speed and stops.                                                                                                                                                                                                 |
| 5) | The pulse output from the QD75 will stop at the zero signal after the machine stops, outputting the "deviation counter clear output" to the drive unit.<br>(A "deviation counter clear signal output time" is set in the [Pr.55].)                                    |
| 6) | After a "deviation counter clear output" is output to the drive unit, the OPR complete flag ( [Md.31] Status: b4) turns from OFF to ON, and the OPR request flag ( [Md.31] Status: b3) turns from ON to OFF.                                                          |

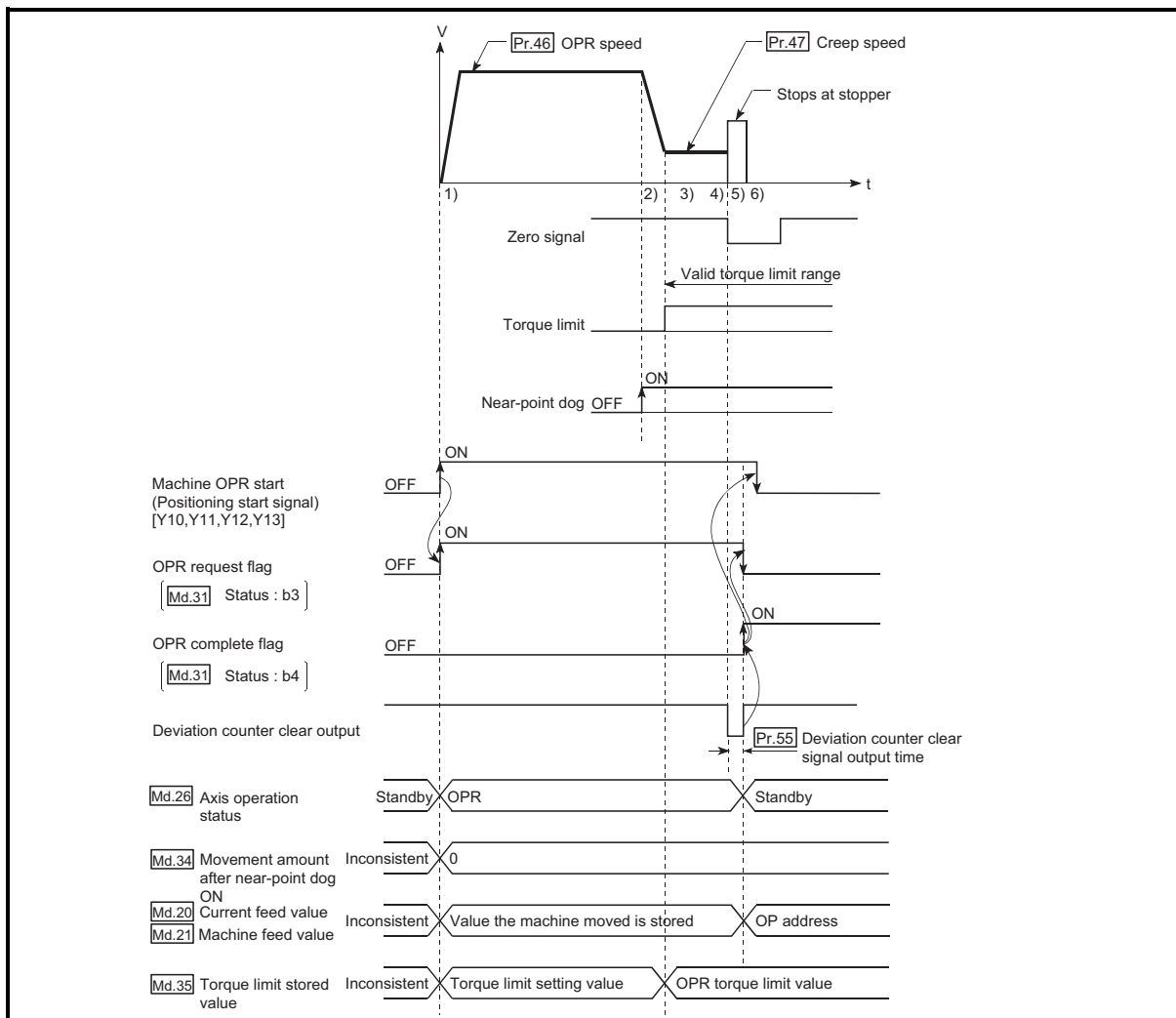


Fig. 8.7 Stopper method 2) machine OPR

### ■ Restrictions

- (1) Always limit the servomotor torque after the " **Pr.47** Creep speed" is reached. If the torque is not limited, the servomotor may fail when the machine presses against the stopper. (Refer to Section 12.4.2 "Torque limit function".)
- (2) Use an external input signal as the zero signal.

### ■ Precautions during operation

- (1) Input a zero signal from an external source after the machine presses against the stopper.  
The workpiece will continue decelerating and stop if a zero signal is input before deceleration to the " **Pr.47** Creep speed". An error "OP detection timing fault" (error code: 204) will occur after the machine stops.

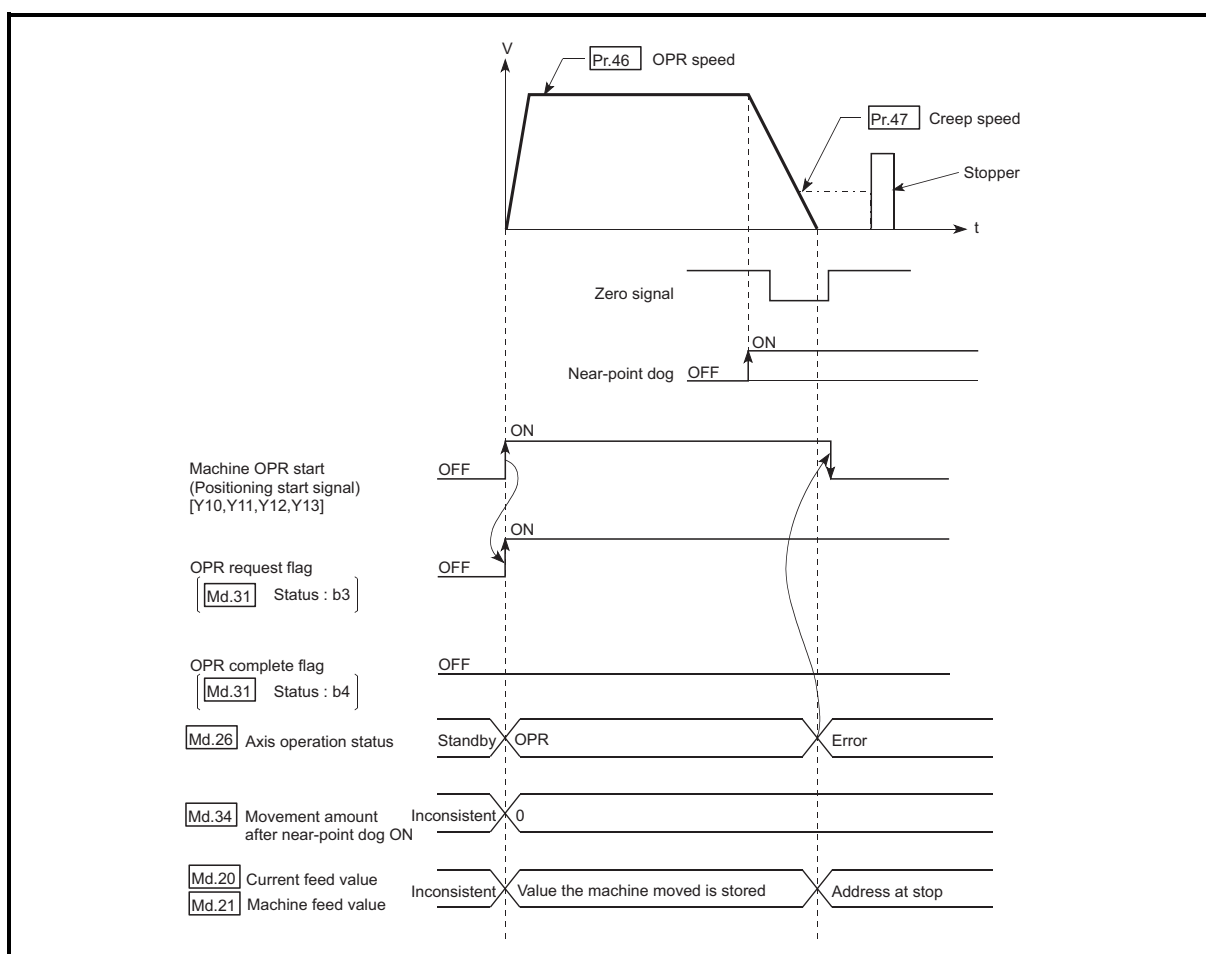


Fig. 8.8 Operation when a zero signal is input before the creep speed is reached

- (2) If the zero signal is input before the workpiece stops at the stopper, the workpiece will stop at that position, and that position will be regarded as the OP.

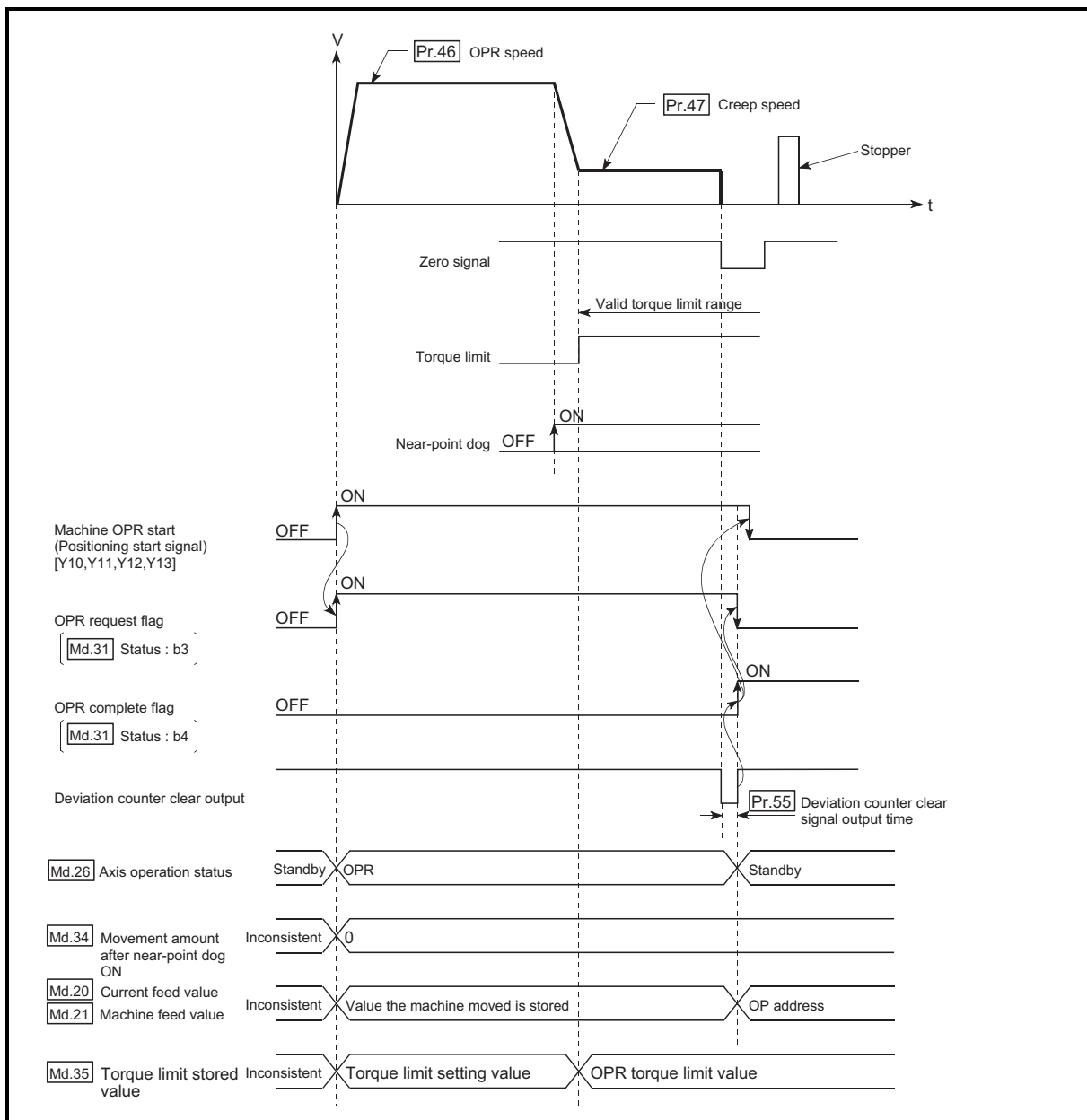


Fig. 8.9 Operation when the zero signal is input before the stop at the stopper

- (3) The near-point dog must be turned ON until it presses against the stopper. If any range that the near-point is turned OFF exists until it presses against the stopper, executing the machine OPR from the area leads it to be pressed against the stopper at ORR speed.
- (4) Machine OPR started while the near-point dog is ON will start at the " [Pr.47] Creep speed".
- (5) When a machine OPR is stopped with the stop signal, perform a machine OPR again. If the restart command is turned ON after a stop with the stop signal, an error "OPR restart not possible" (error code: 209) occurs.

## 8.2.6 OPR method (4): Stopper method 3)

The following shows an operation outline of the "stopper method 3)" OPR method.  
 The "stopper method 3)" method is effective when a near-point dog has not been installed.  
 (Note that the operation is carried out from the start at the "Pr.47 Creep speed", so it will take some time until the machine OPR completion.)

## ■ Operation chart

|    |                                                                                                                                                                                                                                                                                              |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) | The machine OPR is started.<br>(The machine moves at the "Pr.47 Creep speed", in the direction designated in "Pr.44 OPR direction". Torque limiting is required at this time. If the torque is not limited, the servomotor may fail when the machine presses against the stopper in step 2.) |
| 2) | The machine presses against the stopper at the "Pr.47 Creep speed" and stops.                                                                                                                                                                                                                |
| 3) | The pulse output from the QD75 will stop at the zero signal after the machine stops, outputting the "deviation counter clear output" to the drive unit.<br>(A "deviation counter clear signal output time" is set in the Pr.55.)                                                             |
| 4) | After a "deviation counter clear output" is output to the drive unit, the OPR complete flag (Md.31 Status: b4) turns from OFF to ON, and the OPR request flag (Md.31 Status: b3) turns from ON to OFF.                                                                                       |

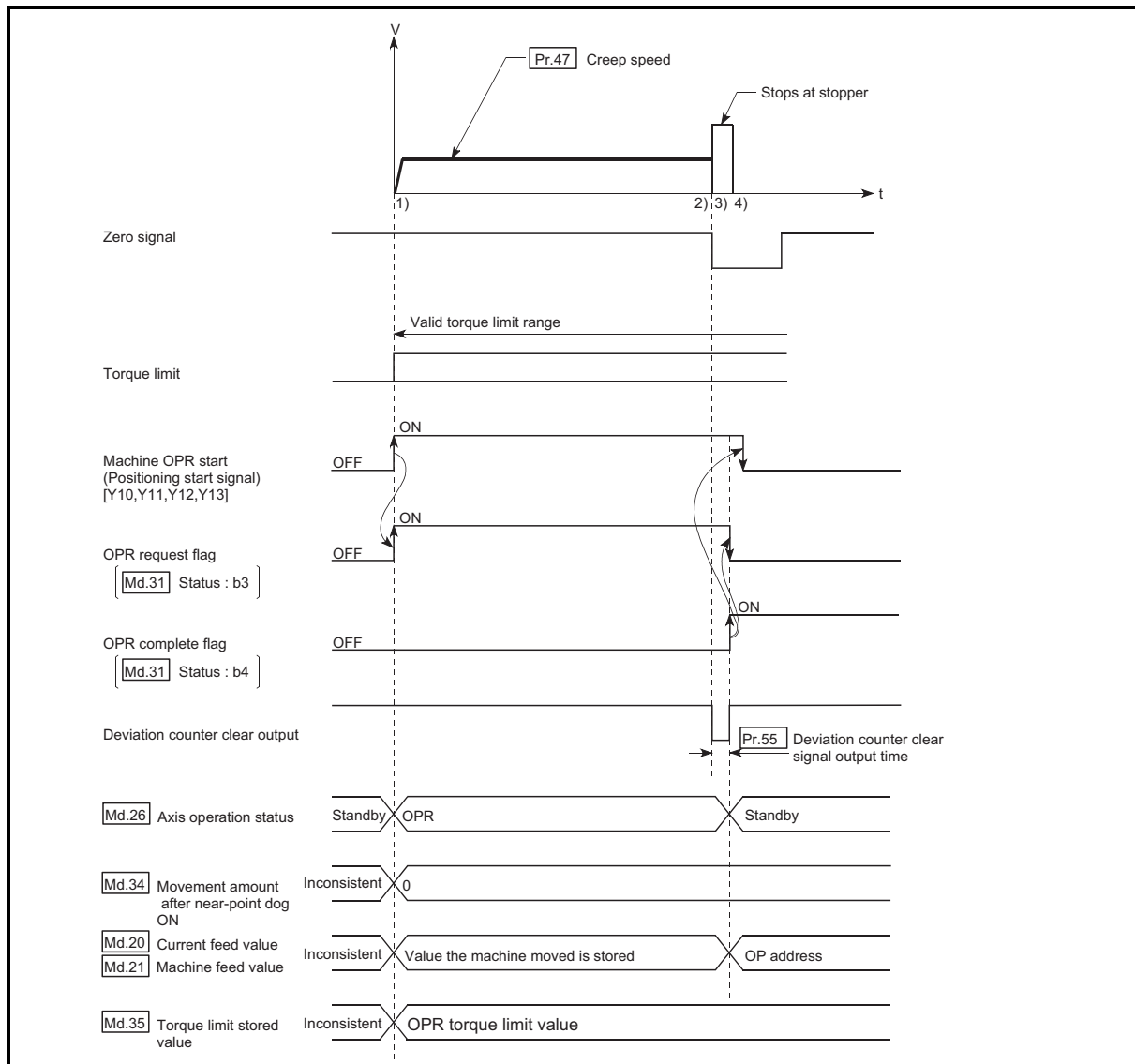


Fig. 8.10 Stopper method 3) machine OPR

### ■ Restrictions

- (1) Always limit the servomotor torque after the "[Pr.47] Creep speed" is reached. If the torque is not limited, the servomotor may fail when the machine presses against the stopper. (Refer to Section 12.4.2 "Torque limit function".)
- (2) Use an external input signal as the zero signal.
- (3) The OPR retry function cannot be used in "stopper stop method 3)".

### ■ Precautions during operation

- (1) If the zero signal is input before the workpiece stops at the stopper, the workpiece will stop at that position, and that position will become the OP. At this time an error will not occur.

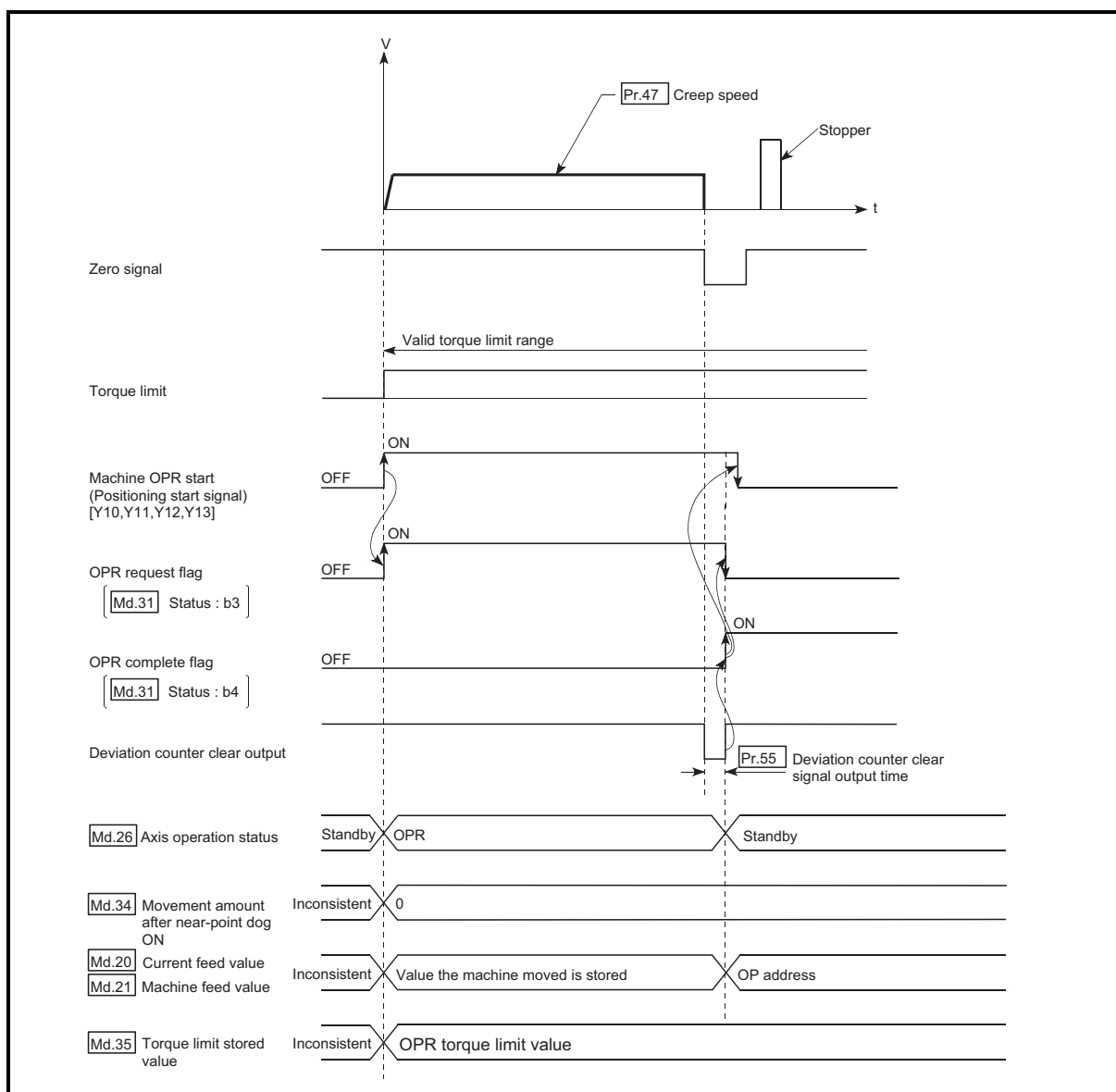


Fig. 8.11 When the zero signal is input before the stop at the stopper

- (2) When a machine OPR is stopped with the stop signal, perform a machine OPR again. If the restart command is turned ON after a stop with the stop signal, an error "OPR restart not possible" (error code: 209) occurs.

## 8.2.7 OPR method (5): Count method1)

The following shows an operation outline of the "count method 1)" OPR method. In the "count method 1)", machine OPR can be performed even in the following situations:

- when near-point dog is ON
- after completion of a machine OPR

■ Operation chart

|    |                                                                                                                                                                                                                                                                                                                                                                            |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) | The machine OPR is started.<br>(The machine begins the acceleration designated in " [Pr.51] OPR acceleration time selection", in the direction designated in " [Pr.44] OPR direction". It then moves at the " [Pr.46] OPR speed" when the acceleration is completed.)                                                                                                      |
| 2) | The machine begins decelerating when the near-point dog ON is detected.                                                                                                                                                                                                                                                                                                    |
| 3) | The machine decelerates to the " [Pr.47] Creep speed", and subsequently moves at that speed.                                                                                                                                                                                                                                                                               |
| 4) | On detection of the first zero signal after the axis has traveled the movement amount set in " [Pr.50] Setting for the movement amount after near-point dog ON" after near-point dog ON, the pulse output from the QD75 stops and the "deviation counter clear output" is output to the drive unit.<br>(A "deviation counter clear signal output time" is set in [Pr.55].) |
| 5) | After a "deviation counter clear output" is output to the drive unit, the OPR complete flag [Md.31] Status: b4) turns from OFF to ON, and the OPR request flag ([Md.31] Status: b3) turns from ON to OFF.                                                                                                                                                                  |

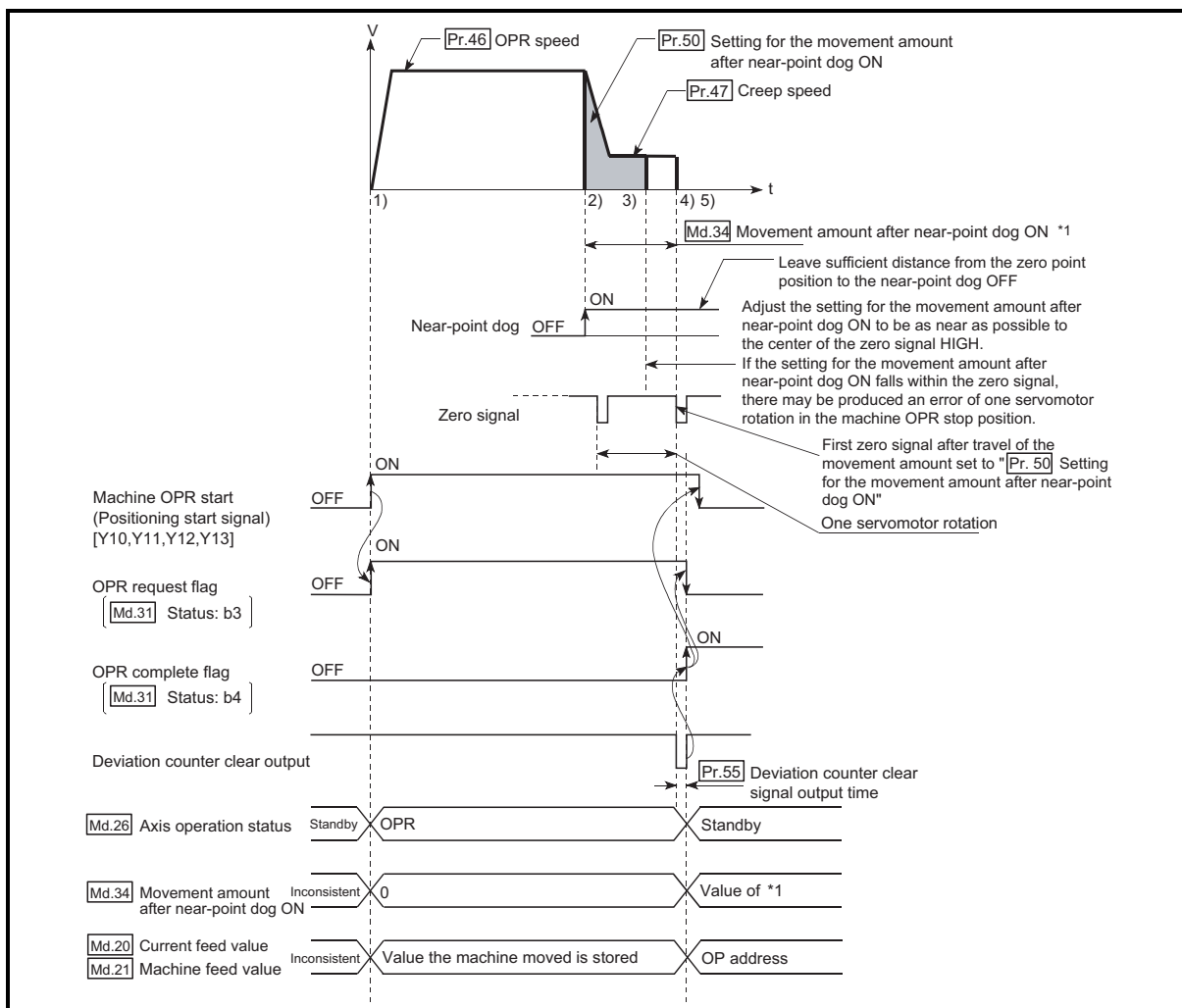


Fig. 8.12 Count method1) machine OPR

### ■ Restrictions

A pulse generator with a zero signal is required.

When using a pulse generator without a zero signal, generate a zero signal using an external signal.

### ■ Precautions during operation

- (1) An error "Count method movement amount fault" (error code: 206) will occur and the operation will not start if the " [Pr.50] Setting for the movement amount after near-point dog ON" is smaller than the deceleration distance from the " [Pr.46] OPR speed" to " [Pr.47] Creep speed".
- (2) If the speed is changed to a speed faster than " [Pr.46] OPR speed" by the speed change function (refer to "12.5.1 Speed change function".) during a machine OPR, the distance to decelerate to " [Pr.47] Creep speed" may not be ensured, depending on the setting value of " [Pr.50] Setting for the movement amount after near-point dog ON". In this case, the error "Count method movement amount fault" (error code: 206) occurs and the Machine OPR is stopped.
- (3) The following shows the operation when a machine OPR is started while the near-point dog is ON.

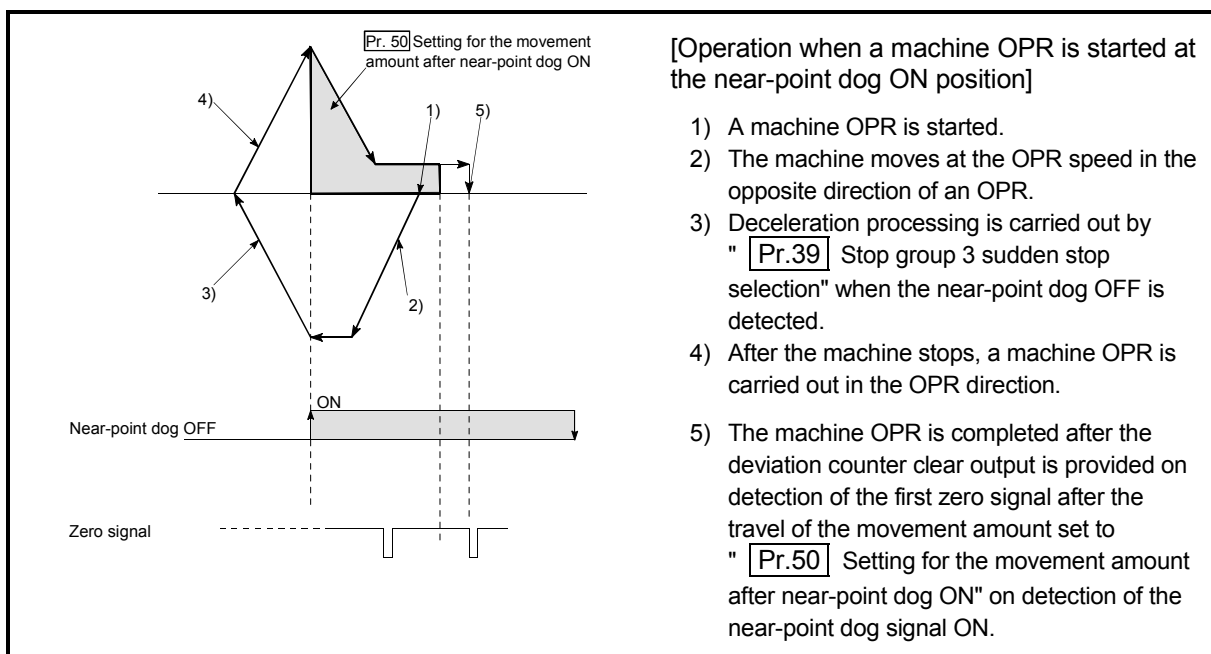


Fig. 8.13 Count method 1) machine OPR on the near-point dog ON position



- (4) Turn OFF the near-point dog at a sufficient distance from the OP.  
Although there is no harm in operation if the near-point dog is turned OFF during a machine OPR, it is recommended to leave a sufficient distance from the OP when the near-point dog is turned OFF for the following reason.  
If machine OPRs are performed consecutively after the near-point dog is turned OFF at the time of machine OPR completion, operation will be performed at the OPR speed until the hardware stroke limit (upper/lower limit) is reached.  
If a sufficient distance cannot be kept, consider the use of the OPR retry function.
- (5) When a machine OPR is stopped with the stop signal, perform a machine OPR again.  
If the restart command is turned ON after a stop with the stop signal, an error "OPR restart not possible" (error code: 209) occurs.

## 8.2.8 OPR method (6): Count method 2)

The following shows an operation outline of the "method 2)" OPR method.

The "count method 2)" method is effective when a "zero signal" cannot be received. (Note that compared to the "count method 1)" method, using this method will result in more deviation in the stop position during machine OPR.)

In the "count method 2)", as well as in the "count method 1)", machine OPR can be performed even in the following situations:

- when near-point dog is ON
- after completion of a machine OPR

#### ■ Operation chart

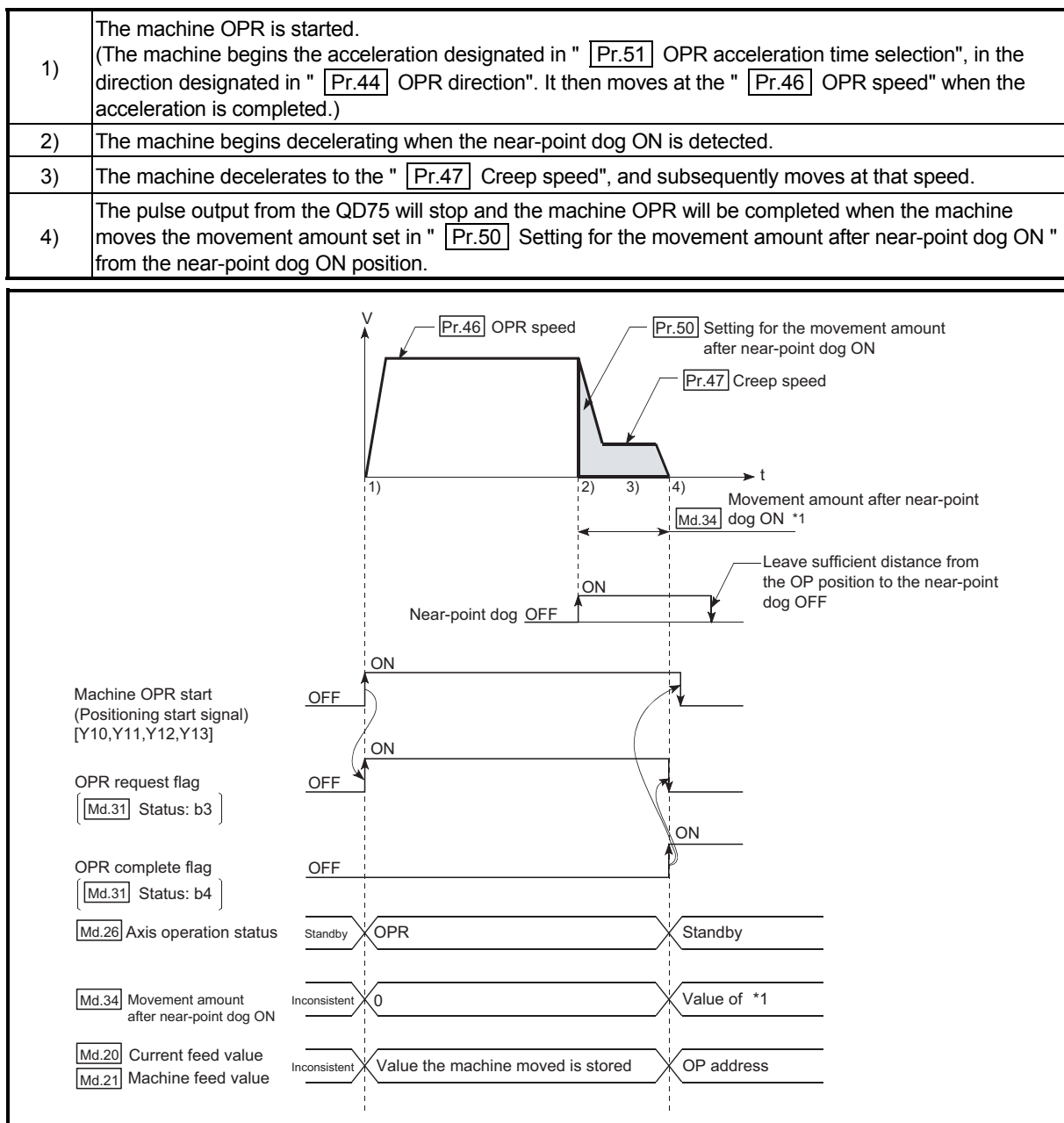


Fig. 8.14 Count method 2) machine OPR

### ■ Restrictions

When this method is used, a deviation will occur in the stop position (OP) compared to other OPR methods because an error of about 1ms occurs in taking in the near-point dog ON.

### ■ Precautions during operation

- (1) An error "Count method movement amount fault" (error code: 206) will occur and the operation will not start if the " [Pr.50] Setting for the movement amount after near-point dog ON" is smaller than the deceleration distance from the " [Pr.46] OPR speed" to " [Pr.47] Creep speed".
- (2) If the speed is changed to a speed faster than " [Pr.46] OPR speed" by the speed change function (refer to Section 12.5.1 "Speed change function".) during a machine OPR, the distance to decelerate to " [Pr.47] Creep speed" may not be ensured, depending on the setting value of " [Pr.50] Setting for the movement amount after near-point dog ON". In this case, the error "Count method movement amount fault" (error code: 206) occurs and the Machine OPR is stopped.
- (3) The following shows the operation when a machine OPR is started while the near-point dog is ON.

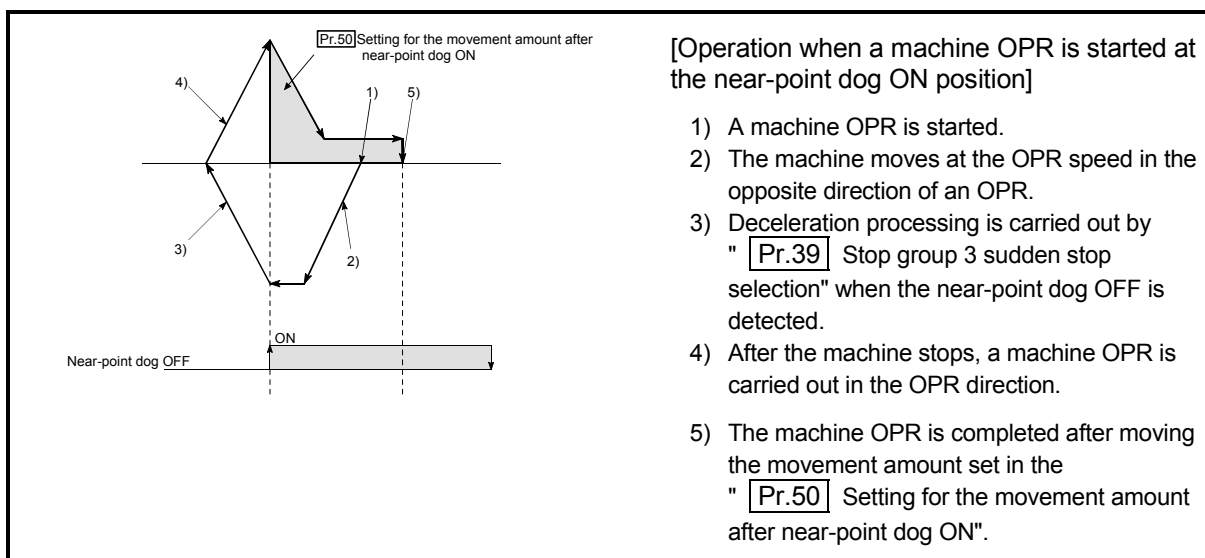


Fig. 8.15 Count method 2) machine OPR on the near-point dog ON position

- (4) Turn OFF the near-point dog at a sufficient distance from the OP.  
Although there is no harm in operation if the near-point dog is turned OFF during a machine OPR, it is recommended to leave a sufficient distance from the OP when the near-point dog is turned OFF for the following reason.  
If machine OPRs are performed consecutively after the near-point dog is turned OFF at the time of machine OPR completion, operation will be performed at the OPR speed until the hardware stroke limit (upper/lower limit) is reached.  
If a sufficient distance cannot be kept, consider the use of the OPR retry function.

- (5) When a machine OPR is stopped with the stop signal, perform a machine OPR again.

If the restart command is turned ON after a stop with the stop signal, an error "OPR restart not possible" (error code: 209) occurs.

### 8.3 Fast OPR

#### 8.3.1 Outline of the fast OPR operation

##### ■ Fast OPR operation

After establishing OP position by a machine OPR, positioning control to the OP position is executed without using a near-point dog or a zero signal. The following shows the operation during a fast OPR start.

- 1) The fast OPR is started.
- 2) Positioning control to the OP position established by a machine OPR begins at the speed set in the OPR parameters (Pr.43 to Pr.57).
- 3) The fast OPR is completed.

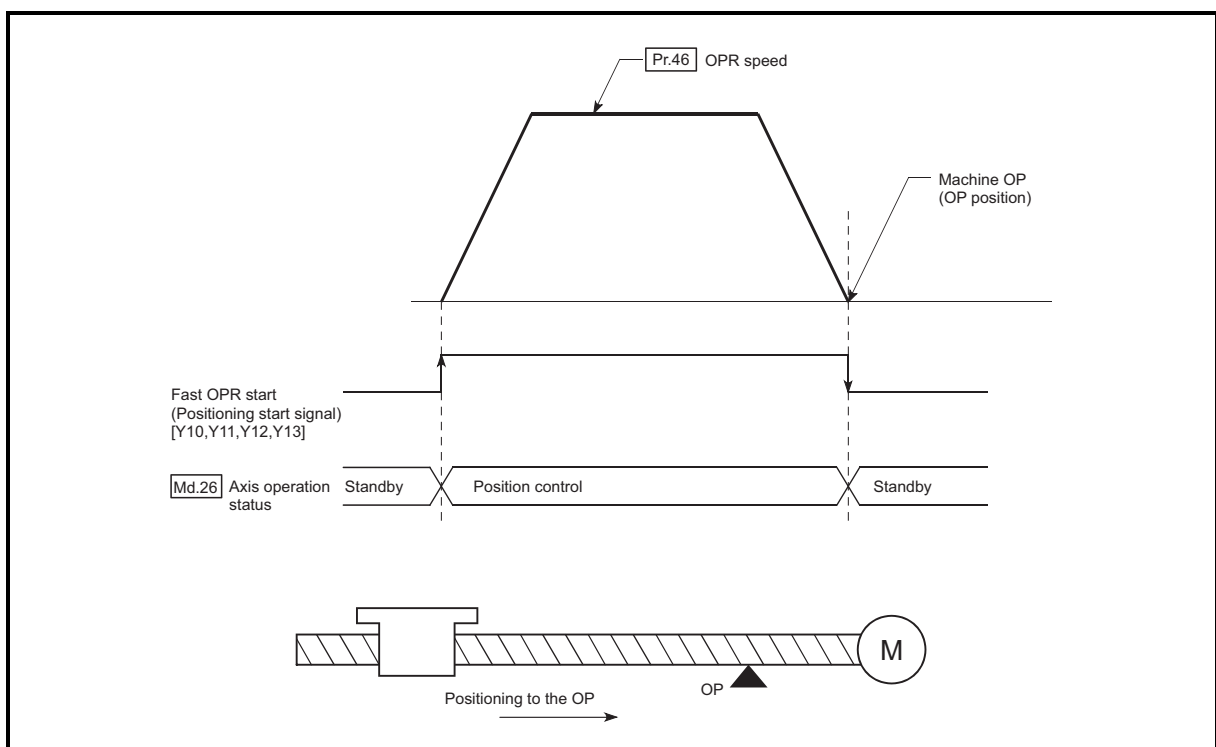


Fig. 8.16 Fast OPR

### ■ Operation timing and processing time of fast OPR

The following shows details about the operation timing and time during fast OPR.

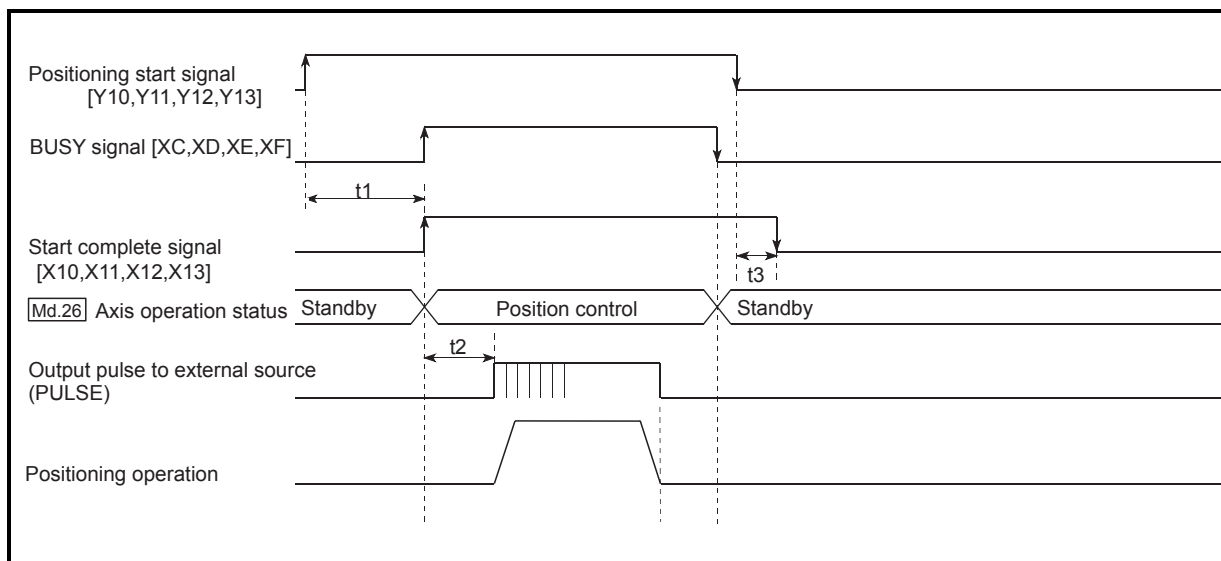


Fig. 8.17 Operation timing and processing time of fast OPR

#### Normal timing time

| Model           | t1           | t2           | t3         |
|-----------------|--------------|--------------|------------|
| QD75P□N/QD75D□N | 0.2 to 1.1ms | 0.4 to 1.3ms | 0 to 0.9ms |
| QD75P□/QD75D□   | 1.0 to 1.3ms | 2.7 to 4.4ms | 0 to 1.8ms |

•The t1 timing time could be delayed by the operation state of other axes.

### ■ Operating restrictions

- (1) The fast OPR can only be executed after the OP position is established by executing the Machine OPR.  
If not, the error "OPR request ON" (error code: 207) will occur.  
(OPR request flag (Md.31 Status: b3) must be turned OFF.)
- (2) If the fraction pulse is cleared to zero using current value changing or fixed-feed control, execute the fast OPR and an error will occur by a cleared amount.
- (3) When unlimited length feed is executed by speed control and the current machine feed value overflows or underflows once, the fast OPR cannot be executed normally.
- (4) The OPR complete flag (Md.31 Status: b4) is not turned ON.
- (5) The axis operation status during fast OPR is "Position control".

# CHAPTER 9 MAJOR POSITIONING CONTROL

The details and usage of the major positioning controls (control functions using the "positioning data") are explained in this chapter.

The major positioning controls include such controls as "positioning control" in which positioning is carried out to a designated position using the address information, "speed control" in which a rotating object is controlled at a constant speed, "speed-position switching control" in which the operation is shifted from "speed control" to "position control" and "position-speed switching control" in which the operation is shifted from "position control" to "speed control".

Carry out the required settings to match each control.

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## 9.1 Outline of major positioning controls

"Major positioning controls" are carried out using the "positioning data" stored in the QD75.

The basic controls such as position control and speed control are executed by setting the required items in this "positioning data", and then starting that positioning data.

The control system for the "major positioning controls" is set in setting item

" [Da.2] Control system" of the positioning data.

Control defined as a "major positioning control" carries out the following types of control according to the " [Da.2] Control system" setting.

| Major positioning control |                                         |                                       | [Da.2] Control system                                                              | Details                                                                                                                                                                                                                  |
|---------------------------|-----------------------------------------|---------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Position control *        | Linear control                          | 1-axis linear control                 | ABS Linear 1<br>INC Linear 1                                                       | Positioning of a designated 1 axis is carried out from the start address (current stop position) to the designated position.                                                                                             |
|                           |                                         | 2-axis linear interpolation control * | ABS Linear 2<br>INC Linear 2                                                       | Using a designated 2 axes, linear interpolation control is carried out from the start address (current stop position) to the designated position.                                                                        |
|                           |                                         | 3-axis linear interpolation control * | ABS Linear 3<br>INC Linear 3                                                       | Using a designated 3 axes, linear interpolation control is carried out from the start address (current stop position) to the designated position.                                                                        |
|                           |                                         | 4-axis linear interpolation control * | ABS Linear 4<br>INC Linear 4                                                       | Using a designated 4 axes, linear interpolation control is carried out from the start address (current stop position) to the designated position.                                                                        |
|                           | Fixed-feed control                      | 1-axis fixed-feed control             | Fixed-feed 1                                                                       | Positioning of a designated 1 axis is carried out from the start address (current stop position).<br>(The " [Md.20] Current feed value" is set to "0" at the start.)                                                     |
|                           |                                         | 2-axis fixed-feed control *           | Fixed-feed 2                                                                       | Using a designated 2 axes, linear interpolation control is carried out from the start address (current stop position).<br>(The " [Md.20] Current feed value" is set to "0" at the start.)                                |
|                           |                                         | 3-axis fixed-feed control *           | Fixed-feed 3                                                                       | Using a designated 3 axes, linear interpolation control is carried out from the start address (current stop position).<br>(The " [Md.20] Current feed value" is set to "0" at the start.)                                |
|                           |                                         | 4-axis fixed-feed control *           | Fixed-feed 4                                                                       | Using a designated 4 axes, linear interpolation control is carried out from the start address (current stop position).<br>(The " [Md.20] Current feed value" is set to "0" at the start.)                                |
|                           | 2-axis circular interpolation control * | Sub point designation                 | ABS Circular sub<br>INC Circular sub                                               | Positioning is carried out in an arc path to a position designated from the start point address (current stop position), using the designated 2 axes.                                                                    |
|                           |                                         | Center point designation              | ABS Circular right<br>ABS Circular left<br>INC Circular right<br>INC Circular left |                                                                                                                                                                                                                          |
|                           | 3-axis helical interpolation control *  | Sub point designation                 | ABS helical sub<br>INC helical sub                                                 | Circular interpolation control is carried out with two of the three axes. The other axis tracks the movement of the circular interpolation control to carry out the positioning in helical or tangential/normal control. |
|                           |                                         | Center point designation              | ABS helical right<br>ABS helical left<br>INC helical right<br>INC helical left     |                                                                                                                                                                                                                          |



| Major positioning control        |                        | Da.2 Control system                                      | Details                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------|------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Speed control *                  | 1-axis speed control   | Forward run speed 1<br>Reverse run speed 1               | The speed control of the designated 1 axis is carried out.                                                                                                                                                                                                                                                                                                                                                 |
|                                  | 2-axis speed control * | Forward run speed 2<br>Reverse run speed 2               | The speed control of the designated 2 axes is carried out.                                                                                                                                                                                                                                                                                                                                                 |
|                                  | 3-axis speed control * | Forward run speed 3<br>Reverse run speed 3               | The speed control of the designated 3 axes is carried out.                                                                                                                                                                                                                                                                                                                                                 |
|                                  | 4-axis speed control * | Forward run speed 4<br>Reverse run speed 4               | The speed control of the 4 axes is carried out.                                                                                                                                                                                                                                                                                                                                                            |
| Speed-position switching control |                        | Forward run speed/position<br>Reverse run speed/position | The control is continued as position control (positioning for the designated address or movement amount) by turning ON the "speed-position switching signal" after first carrying out speed control.                                                                                                                                                                                                       |
| Position-speed switching control |                        | Forward run position/speed<br>Reverse run position/speed | The control is continued as speed control by turning ON the "position-speed switching signal" after first carrying out position control.                                                                                                                                                                                                                                                                   |
| Other control                    | NOP instruction        | NOP instruction                                          | A nonexecutable control system. When this instruction is set, the operation is transferred to the next data operation, and the instruction is not executed.                                                                                                                                                                                                                                                |
|                                  | Current value changing | Current value changing                                   | The current feed value ( <u>Md.20</u> ) is changed to an address set in the positioning data.<br>This can be carried out by either of the following 2 methods.<br>(The machine feed value cannot be changed.) <ul style="list-style-type: none"> <li>• Current value changing using the control system</li> <li>• Current value changing using the current value changing start No. (No. 9003).</li> </ul> |
|                                  | JUMP instruction       | JUMP instruction                                         | An unconditional or conditional JUMP is carried out to a designated positioning data No.                                                                                                                                                                                                                                                                                                                   |
|                                  | LOOP                   | LOOP                                                     | A repeat control is carried out by repeat LOOP to LEND.                                                                                                                                                                                                                                                                                                                                                    |
|                                  | LEND                   | LEND                                                     | Control is returned to the top of the repeat control by repeat LOOP to LEND. After the repeat operation is completed specified times, the next positioning data is run.                                                                                                                                                                                                                                    |

\* In "2-axis linear interpolation control", "3-axis linear interpolation control", "4-axis linear interpolation control", "2-axis fixed-feed control", "3-axis fixed-feed control", "4-axis fixed-feed control", "2-axis circular interpolation control", "3-axis helical interpolation control", "2-axis speed control", "3-axis speed control", and "4-axis speed control", control is carried out so that linear and arc paths are drawn using a motor set in two or more axis directions. This kind of control is called "interpolation control". (Refer to Section 9.1.6 "Interpolation control" for details.)

## 9.1.1 Data required for major positioning control

The following table shows an outline of the "positioning data" configuration and setting details required to carry out the "major positioning controls".

| Setting item           |                                                     | Setting details                                                                                                                                                                                                                         |
|------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positioning data No. 1 | <b>Da.1</b> Operation pattern                       | Set the method by which the continuous positioning data (Ex: positioning data No. 1, No. 2, No. 3) will be controlled. (Refer to Section 9.1.2.)                                                                                        |
|                        | <b>Da.2</b> Control system                          | Set the control system defined as a "major positioning control". (Refer to Section 9.1.)                                                                                                                                                |
|                        | <b>Da.3</b> Acceleration time No.                   | Select and set the acceleration time at control start. (Select one of the four values set in <b>Pr.9</b> , <b>Pr.25</b> , <b>Pr.26</b> , and <b>Pr.27</b> for the acceleration time.)                                                   |
|                        | <b>Da.4</b> Deceleration time No.                   | Select and set the deceleration time at control stop. (Select one of the four values set in <b>Pr.10</b> , <b>Pr.28</b> , <b>Pr.29</b> , and <b>Pr.30</b> for the deceleration time.)                                                   |
|                        | <b>Da.5</b> Axis to be interpolated                 | Set an axis to be interpolated (partner axis) during the 2-axis interpolation operation and a circular interpolation axis during the 3-axis helical interpolation control. (Refer to Section 9.1.6.)                                    |
|                        | <b>Da.6</b> Positioning address/movement amount     | Set the target value during position control. (Refer to Section 9.1.3.)                                                                                                                                                                 |
|                        | <b>Da.7</b> Arc address                             | Set the sub point or center point address during circular interpolation control.                                                                                                                                                        |
|                        | <b>Da.8</b> Command speed                           | Set the speed during the control execution.                                                                                                                                                                                             |
|                        | <b>Da.9</b> Dwell time                              | The time between the command pulse output is completed to the positioning completed signal is turned ON. Set it for absorbing the delay of the mechanical system to the instruction, such as the delay of the servo system (deviation). |
|                        | <b>Da.10</b> M code                                 | Set this item when carrying out sub work (clamp and drill stops, tool replacement, etc.) corresponding to the code No. related to the positioning data execution.                                                                       |
|                        | <b>Da.27</b> M code ON signal output timing         | Set the M code ON signal output timing for each positioning data.                                                                                                                                                                       |
|                        | <b>Da.28</b> ABS direction in degrees               | Set the ABS direction in degrees for each positioning data.                                                                                                                                                                             |
|                        | <b>Da.29</b> Interpolation speed designation method | Set the interpolation speed designation method for each positioning data.                                                                                                                                                               |

\* The settings and setting requirement for the setting details of **Da.1** to **Da.10** and **Da.27** to **Da.29** differ according to the "**Da.2** Control system". (Refer to Section 9.2 "Setting the positioning data".)

#### ■ Major positioning control sub functions

Refer to Section 3.2.4 "Combination of QD75 major functions and sub functions" for details on "sub functions" that can be combined with the major positioning control.

Also refer to CHAPTER 12 "CONTROL SUB FUNCTIONS" for details on each sub function.

#### ■ Major positioning control from peripheral devices

"Major positioning control" can be executed from the peripheral device test mode. Refer to GX Configurator-QP Operating Manual for details on carrying out major positioning control from the peripheral device.

#### REMARK

- 600 positioning data (positioning data No. 1 to 600) items can be set per axis.

## 9.1.2 Operation patterns of major positioning controls

In "major positioning control" (high-level positioning control), " Da.1 Operation pattern" can be set to designate whether to continue executing positioning data after the started positioning data. The "operation pattern" includes the following 3 types.

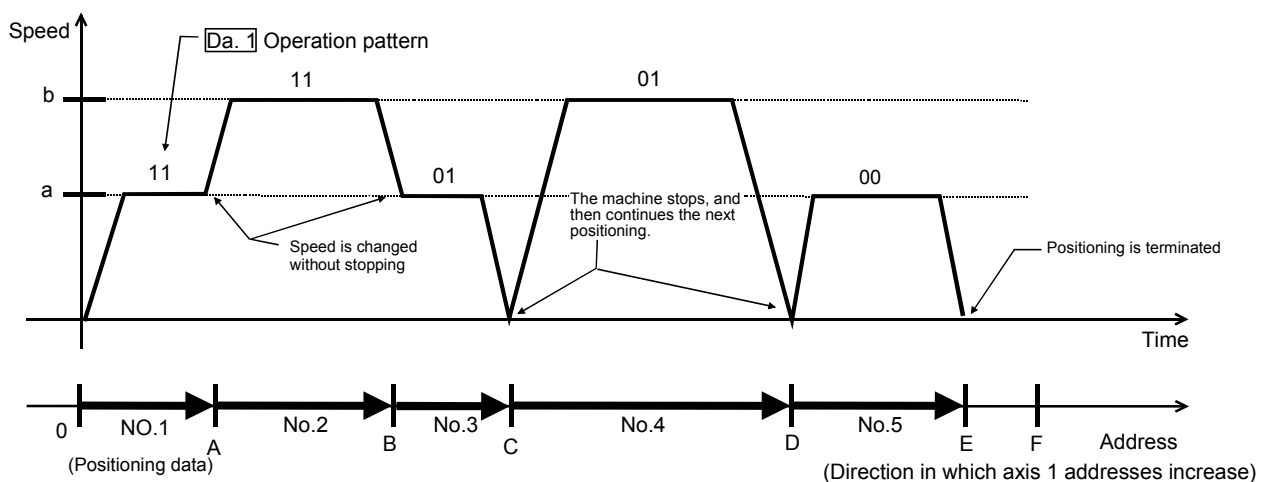
- Positioning complete — (1) Independent positioning control (operation pattern: 00)
- Positioning continue — (2) Continuous positioning control (operation pattern: 01)
- (3) Continuous path control (operation pattern: 11)

The following shows examples of operation patterns when "1-axis linear control (ABS linear 1)" is set in positioning data No. 1 to No. 6 of axis 1. Details of each operation pattern are shown on the following pages.

< Operation example when "1-axis linear positioning" is set in the positioning data of axis 1 >

(Setting details)

|                       |                                                 |                                                                                |              |
|-----------------------|-------------------------------------------------|--------------------------------------------------------------------------------|--------------|
| Positioning data No.1 | Positioning to address [A] at command speed [a] | Operation pattern = 11: Continuous path control                                | No.1 Start   |
| No.2                  | Positioning to address [B] at command speed [b] | Operation pattern = 11: Continuous path control                                |              |
| No.3                  | Positioning to address [C] at command speed [a] | Operation pattern = 01: Continuous positioning control                         |              |
| No.4                  | Positioning to address [D] at command speed [b] | Operation pattern = 01: Continuous positioning control                         |              |
| No.5                  | Positioning to address [E] at command speed [a] | Operation pattern = 00: Independent positioning control (Positioning complete) |              |
| No.6                  | Positioning to address [F] at command speed [a] | Operation pattern = 11: Continuous path control                                | Control stop |



For 1-axis linear control

(One motor is driven, and positioning is carried out to an addresses designated in one direction.)

## POINT

The BUSY signal [XC, XD, XE, XF] turns ON even when position control of movement amount 0 is executed. However, since the ON time is short, the ON status may not be detected in the sequence program.

[1] Independent positioning control (Positioning complete)

This control is set when executing only one designated data item of positioning. If a dwell time is designated, the positioning will complete after the designated time elapses.

This data (operation pattern [00] data) becomes the end of block data when carrying out block positioning. (The positioning stops after this data is executed.)

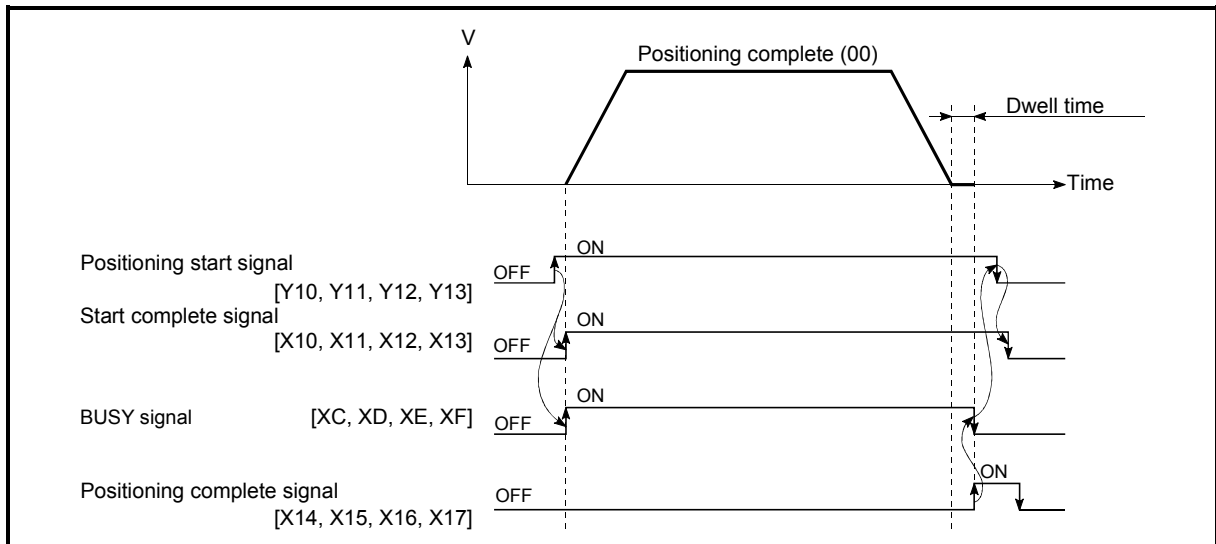


Fig. 9.1 Operation during independent positioning control

## [2] Continuous positioning control

- (1) The machine always automatically decelerates each time the positioning is completed. Acceleration is then carried out after the QD75 command speed reaches 0 to carry out the next positioning data operation. If a dwell time is designated, the acceleration is carried out after the designated time elapses.
- (2) In operation by continuous positioning control (operation pattern "01"), the next positioning No. is automatically executed. Always set operation pattern "00" in the last positioning data to complete the positioning.  
 If the operation pattern is set to positioning continue ("01" or "11"), the operation will continue until operation pattern "00" is found.  
 If the operation pattern "00" cannot be found, the operation may be carried out until the positioning data No. 600. If the operation pattern of the positioning data No. 600 is not completed, the operation will be started again from the positioning data No. 1.

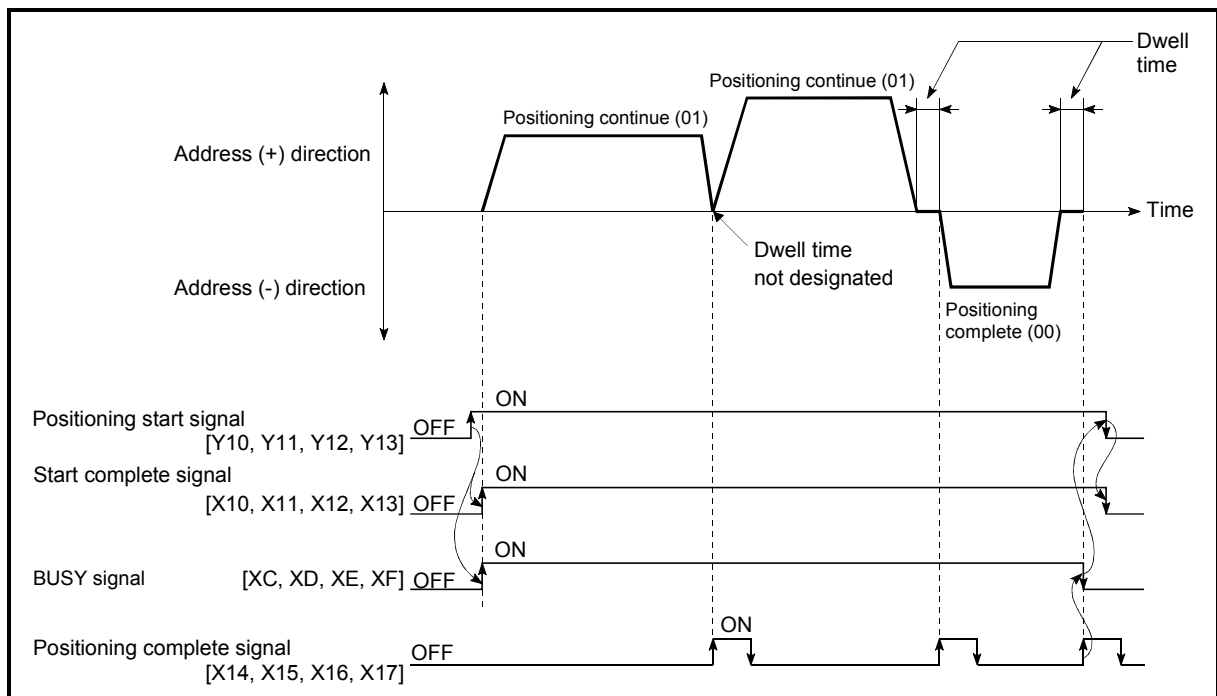


Fig. 9.2 Operation during continuous positioning control

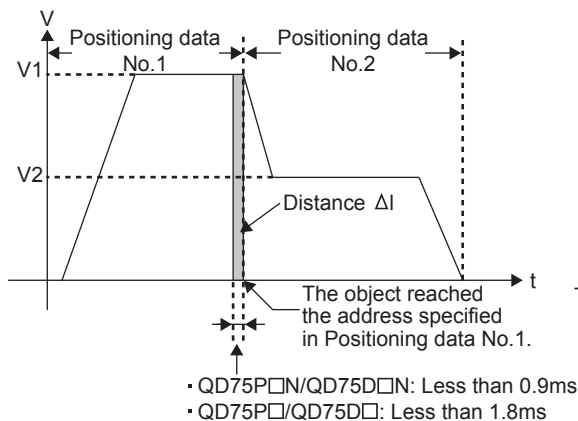


- (f) In continuous path control, the positioning may be completed before the set address/movement amount by the distance  $\Delta \ell$ , and the data may switch to the "positioning data that will run next".

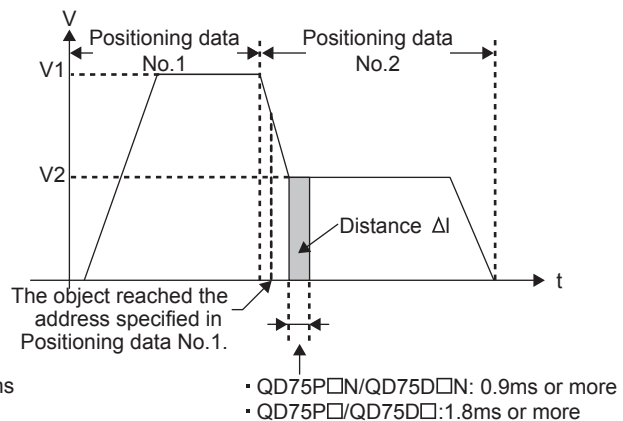
The value of the distance  $\Delta \ell$  is as follows:

- QD75P□N/QD75D□N:  
 $0 \leq \Delta \ell < (\text{Moving distance in 0.9ms at the command speed})$
- QD75P□/QD75D□:  
 $0 \leq \Delta \ell < (\text{Moving distance in 1.8ms at the command speed})$

【Acceleration/deceleration (setting)】



【Acceleration/deceleration (actual one)】



The distance  $\Delta \ell$  is output when the next positioning data is running at the specified speed.

Therefore, the execution time of the next positioning data may be longer than the set execution time.\*1

If this causes a control problem, take one of the following measures.

- 1) Use the function "output timing selection of near pass control".  
 By selecting "At deceleration" as an output timing, the positioning execution time of the next positioning data remains the same as it was set. (Refer to Section 12.3.4 "Output timing selection of near pass control".)
- 2) Use the "speed change function".  
 Change speed using the speed change function instead of continuous path control. (Refer to Section 12.5.1 "Speed change function".)

\*1: When the command speed V1 of a positioning data and the command speed V2 of the next positioning data is different significantly, and  $V1 > V2$

For positioning data in which automatic deceleration is set, the positioning is completed at the set address. Therefore, in continuous path control, the address where a positioning is completed may be different from the set address. However, the address will be the specified one on completion of automatic deceleration for continuous positioning control (01) or positioning complete (00).

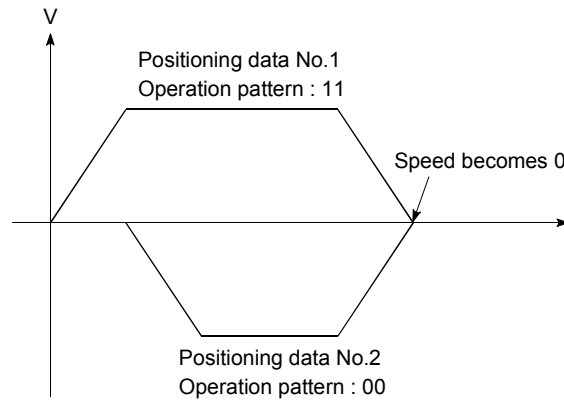
#### POINT

In the continuous path control, a speed variation will not occur using the near-pass function when the positioning data No. is switched (Refer to Section 12.3.3 "Near-pass function").

## (2) Deceleration stop conditions during continuous path control

Deceleration stops are basically not carried out in continuous path control, but the machine will carry out a deceleration stop to speed "0" in the following cases (a) to (c).

- (a) When the operation pattern of the positioning data currently being executed is "continuous path control: 11", and the movement direction of the positioning data currently being executed differs from that of the next positioning data. (Only for 1-axis positioning control (Refer to the "Point" in the next page.))

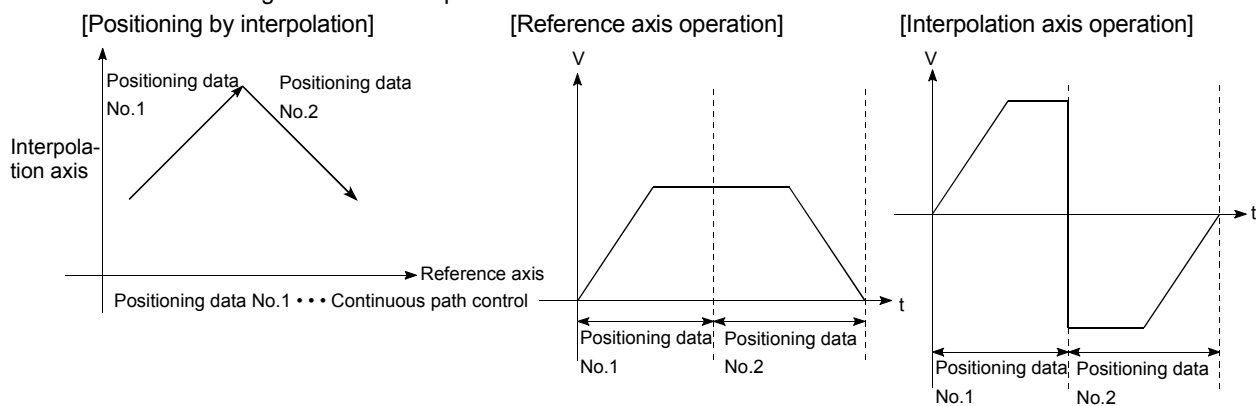




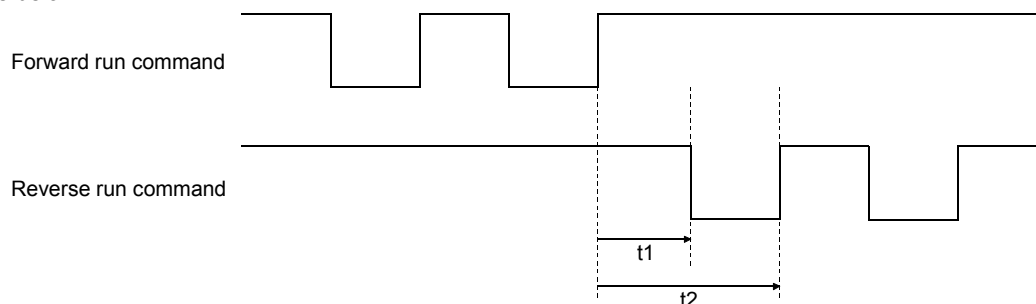
- (b) During operation by step operation.  
(Refer to Section 12.7.1 "Step function".)
- (c) When there is an error in the positioning data to carry out the next operation.

**POINTS**

- (1) The movement direction is not checked during interpolation operations. Thus, automatic deceleration to a stop will not be carried out even if the movement direction is changed (See the figures below). Because of this, the interpolation axis may suddenly reverse direction. To avoid this sudden direction reversal in the interpolation axis, set the pass point to continuous positioning control "01" instead of setting it to continuous path control "11".



- (2) When the interpolation axis reverses direction suddenly, the command pulses from QD75 are output as shown in the figure below.



The  $t_1$  and  $t_2$  are calculated using the following expressions, where a command frequency is  $f$  (pps).

$$t_1 = 1/2 f \text{ (s)} \quad t_2 = 1/f \text{ (s)}$$

A time of  $t_1$  must be maintained by the drive unit for a specified period  $T$  (s).

( $T$  depends on the drive unit specifications.)

If  $t_1$  cannot be maintained for  $T$  or longer, lower the " [Da.8] Command speed" of the positioning data.

- (3) When a "0" is set in the " [Da.6] Positioning address/movement amount" of the continuous path control positioning data, the command speed of about 0.9ms (QD75P□N/QD75D□N) or 2ms (QD75P□/QD75D□) is reduced to 0. When a "0" is set in the " [Da.6] Positioning address/movement amount" to increase the number of speed change points in the future, change the " [Da.2] Control system" to the "NOP instruction" to make the control nonexecutable.

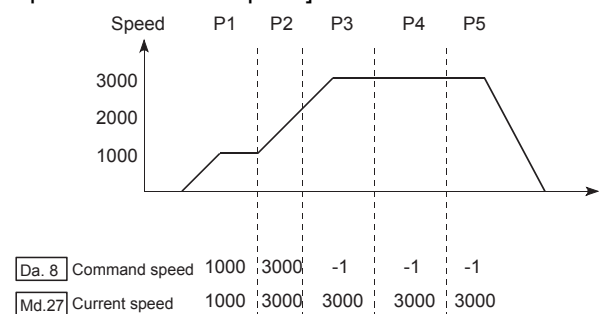
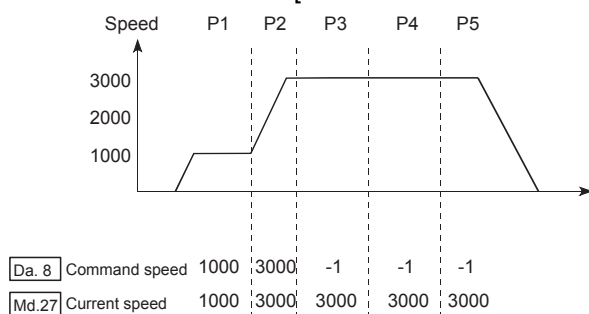
(Refer to Section 9.2.20 "NOP instruction".)

- (4) In the continuous path control positioning data, assure a movement distance so that the execution time with that data is 100ms or longer, or lower the command speed.

## (3) Speed handling

- (a) Continuous path control command speeds are set with each positioning data.  
The QD75 then carries out the positioning at the speed designated with each positioning data.
- (b) The command speed can be set to "-1" in continuous path control.  
The control will be carried out at the speed used in the previous positioning data No. if the command speed is set to "-1".  
(The "current speed" will be displayed in the command speed when the positioning data is set with a peripheral device. The current speed is the speed of the positioning control being executed currently.)
- 1) The speed does not need to be set in each positioning data when carrying out uniform speed control if "-1" is set beforehand in the command speed.
  - 2) If the speed is changed or the override function is executed in the previous positioning data when "-1" is set in the command speed, the operation can be continued at the new speed.
  - 3) An error "No command speed" (error code: 503) occurs and positioning cannot be started if "-1" is set in the command speed of the first positioning data at start.

[Relation between the command speed and current speed]



The current speed is changed even if the command speed is not reached in P2.

## POINTS

- (1) In the continuous path control, a speed variation will not occur using the near-pass function when the positioning data is switched (Refer to Section 12.3.3 "Near-pass function").
- (2) The QD75 holds the command speed set with the positioning data, and the latest value of the speed set with the speed change request as the "Md.27 Current speed". It controls the operation at the "current speed" when "-1" is set in the command speed.  
(Depending on the relation between the movement amount and the speed, the feedrate may not reach the command speed value, but even then the current speed will be updated.)
- (3) When the address for speed change is identified beforehand, generate and execute the positioning data for speed change by the continuous path control to carry out the speed change without requesting the speed change with a sequence program.

## (4) Speed switching

(Refer to " **Pr.19** Speed switching mode".)

The two modes for changing the speed are shown below.

- Standard switching.....Switch the speed when executing the next positioning data.
- Front-loading switching.....The speed switches at the end of the positioning data currently being executed.

## (a) Standard speed switching mode

- 1) If the respective command speeds differ in the "positioning data currently being executed" and the "positioning data to carry out the next operation", the machine will accelerate or decelerate after reaching the positioning point set in the "positioning data currently being executed" and the speed will change over to the speed set in the "positioning data to carry out the next operation".
- 2) The parameters used in acceleration/deceleration to the command speed set in the "positioning data to carry out the next operation" are those of the positioning data to carry out acceleration/deceleration.  
Speed switching will not be carried out if the command speeds are the same.

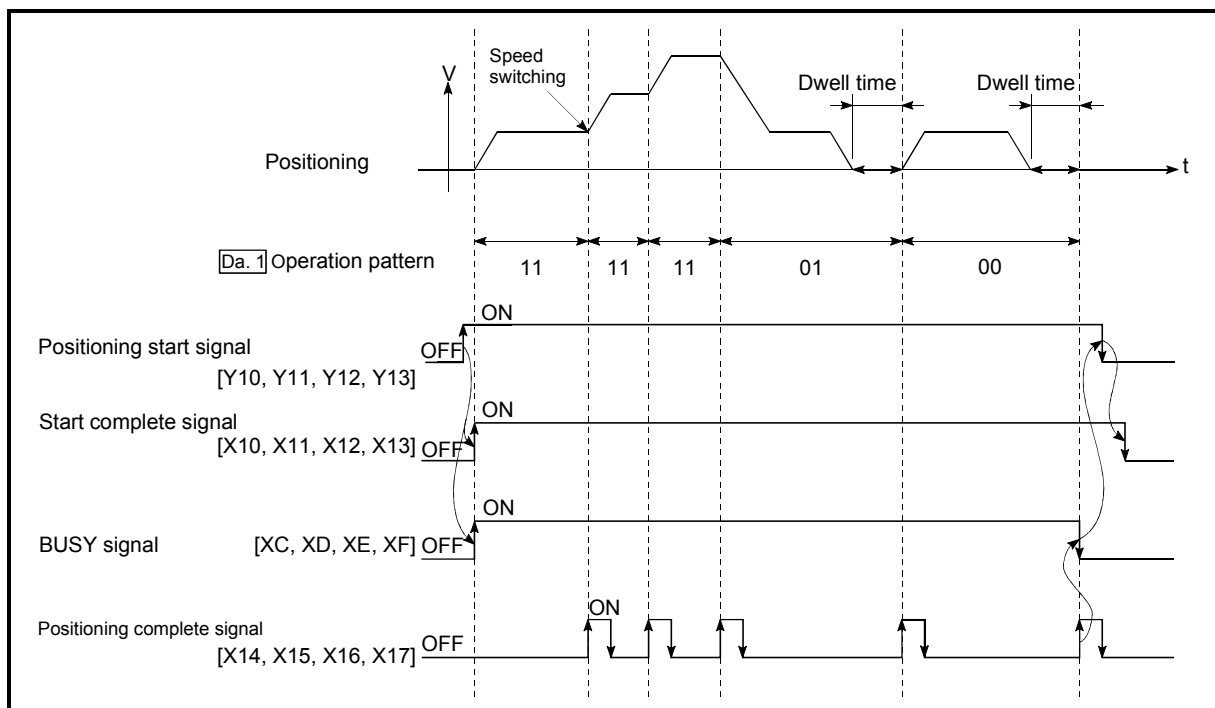


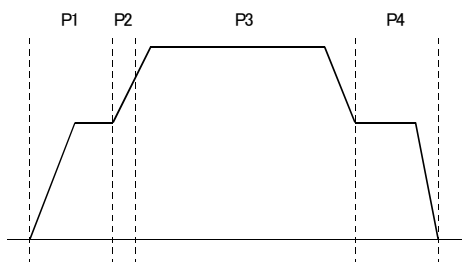
Fig. 9.4 Operation for the standard speed switching mode

## 3) Speed switching condition

If the movement amount is small in regard to the target speed, the current speed may not reach the target speed even if acceleration/deceleration is carried out. In this case, the machine is accelerated/decelerated so that it nears the target speed. If the movement amount will be exceeded when automatic deceleration is required (Ex. Operation patterns "00", "01"), the machine will immediately stop at the designated positioning address, and a warning "Insufficient movement distance" (warning code: 513) will occur.

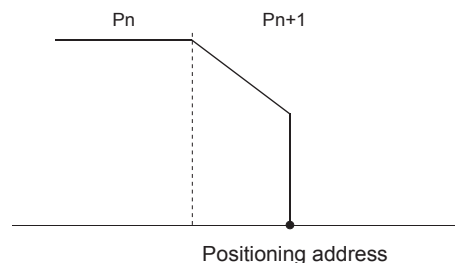
[When the speed cannot change over in P2]

When the relation of the speeds is  $P1 = P4$ ,  $P2 = P3$ ,  $P1 < P2$ .



[When the movement amount is small during automatic deceleration]

The movement amount required to carry out the automatic deceleration cannot be secured, so the machine immediately stops in a speed  $\neq 0$  status.



(b) Front-loading speed switching mode

- 1) If the respective command speeds differ in the "positioning data currently being executed" and the "positioning data to carry out the next operation", the speed will change over to the speed set in the "positioning data to carry out the next operation" at the end of the "positioning data currently being executed".
- 2) The parameters used in acceleration/deceleration to the command speed set in the "positioning data to carry out the next operation" are those of the positioning data to carry out acceleration/deceleration. Speed switching will not be carried out if the command speeds are the same.

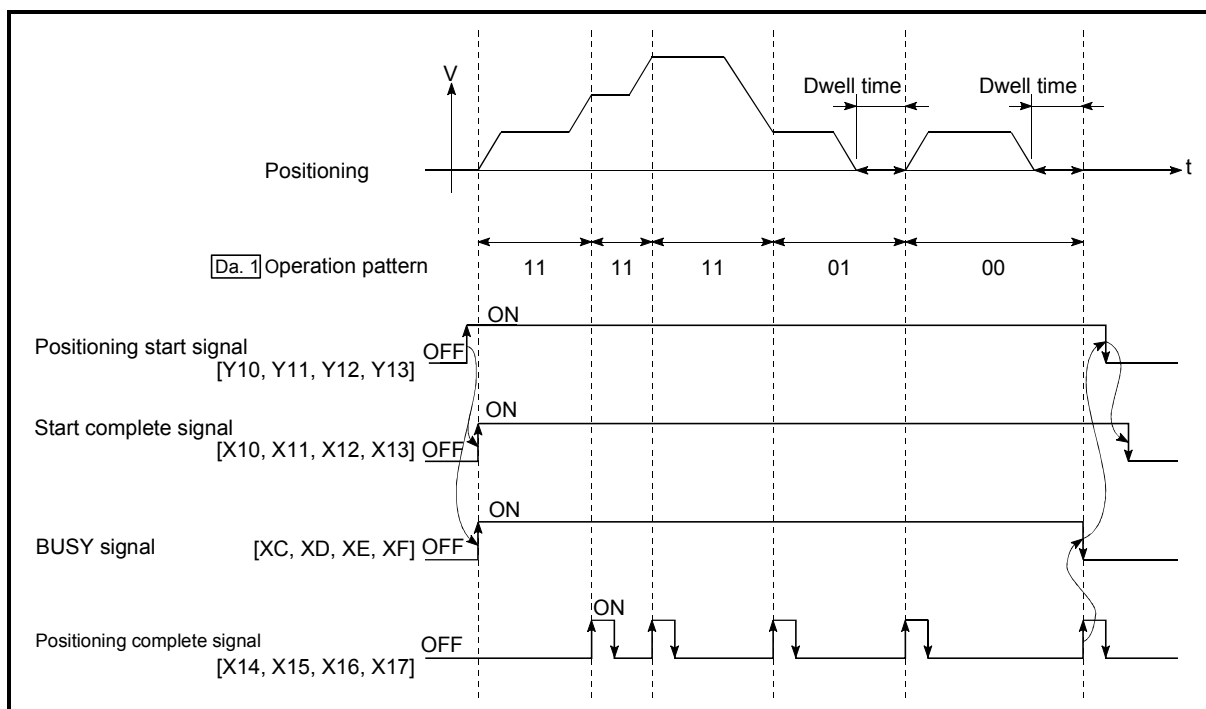


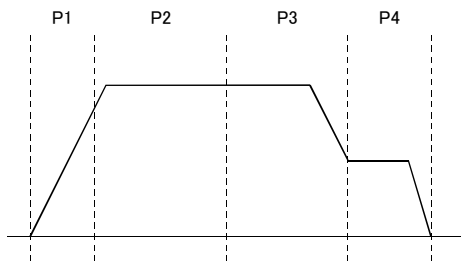
Fig. 9.5 Operation for the front-loading speed switching mode

## 3) Speed switching condition

If the movement amount is small in regard to the target speed, the current speed may not reach the target speed even if acceleration/deceleration is carried out. In this case, the machine is accelerated/decelerated so that it nears the target speed. If the movement amount will be exceeded when automatic deceleration is required (Ex. Operation patterns "00", "01"), the machine will immediately stop at the designated positioning address, and a warning "Insufficient movement distance" (warning code: 513) will occur.

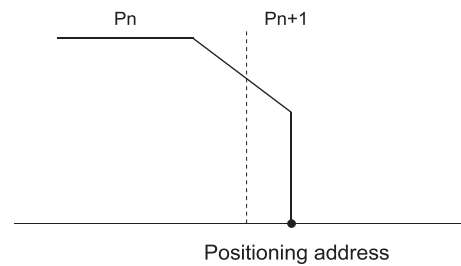
[When the speed cannot change over to the P2 speed in P1]

When the relation of the speeds is  $P1 = P4$ ,  $P2 = P3$ ,  $P1 < P2$ .



[When the movement amount is small during automatic deceleration]

The movement amount required to carry out the automatic deceleration cannot be secured, so the machine immediately stops in a speed  $\neq 0$  status.



### 9.1.3 Designating the positioning address

The following shows the two methods for commanding the position in control using positioning data.

#### ■ Absolute system

Positioning is carried out to a designated position (absolute address) having the OP as a reference. This address is regarded as the positioning address. (The start point can be anywhere.)

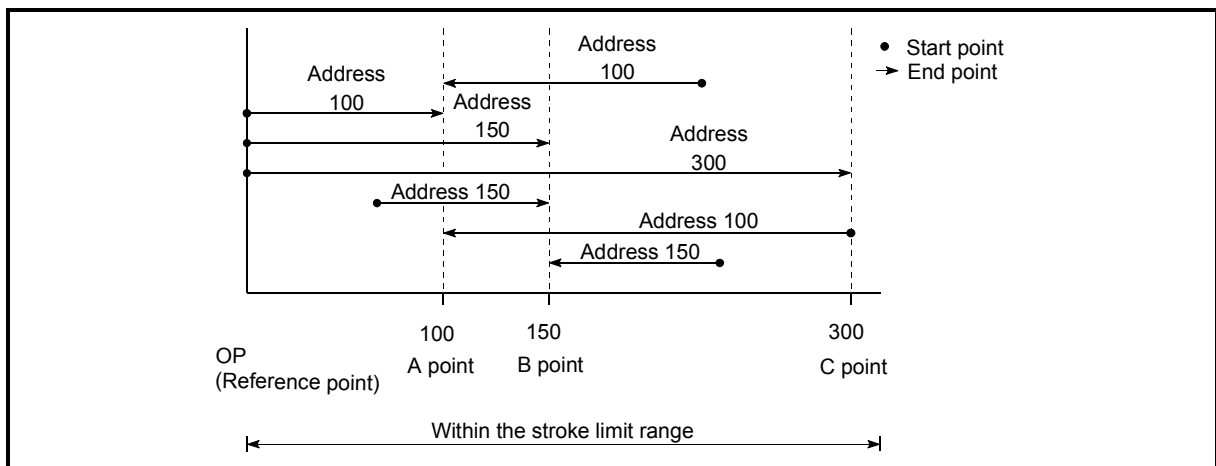


Fig. 9.6 Absolute system positioning

#### ■ Incremental system

The position where the machine is currently stopped is regarded as the start point, and positioning is carried out for a designated movement amount in a designated movement direction.

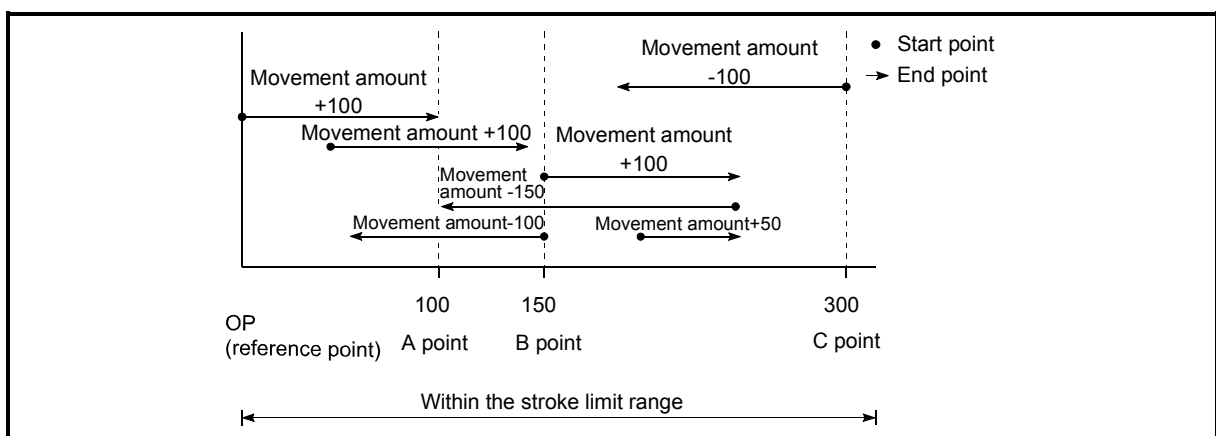


Fig. 9.7 Incremental system positioning

## 9.1.4 Confirming the current value

## ■ Values showing the current value

The following two types of addresses are used as values to show the position in the QD75.

These addresses ("current feed value" and "machine feed value") are stored in the monitor data area, and used in monitoring the current value display, etc.

|                    |                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Current feed value | <ul style="list-style-type: none"> <li>• This is the value stored in " <b>[Md.20]</b> Current feed value".</li> <li>• This value has an address established with a "machine OPR" as a reference, but the address can be changed by changing the current value to a new value.</li> <li>• This value is updated every 0.9ms (QD75P□N/QD75D□N) or 1.8ms (QD75P□/QD75D□).</li> </ul>                |
| Machine feed value | <ul style="list-style-type: none"> <li>• This is the value stored in " <b>[Md.21]</b> Machine feed value".</li> <li>• This value always has an address established with a "machine OPR" as a reference. The address cannot be changed, even if the current value is changed to a new value.</li> <li>• This value is updated every 0.9ms (QD75P□N/QD75D□N) or 56.8ms (QD75P□/QD75D□).</li> </ul> |

The "current feed value" and "machine feed value" are used in monitoring the current value display, etc.

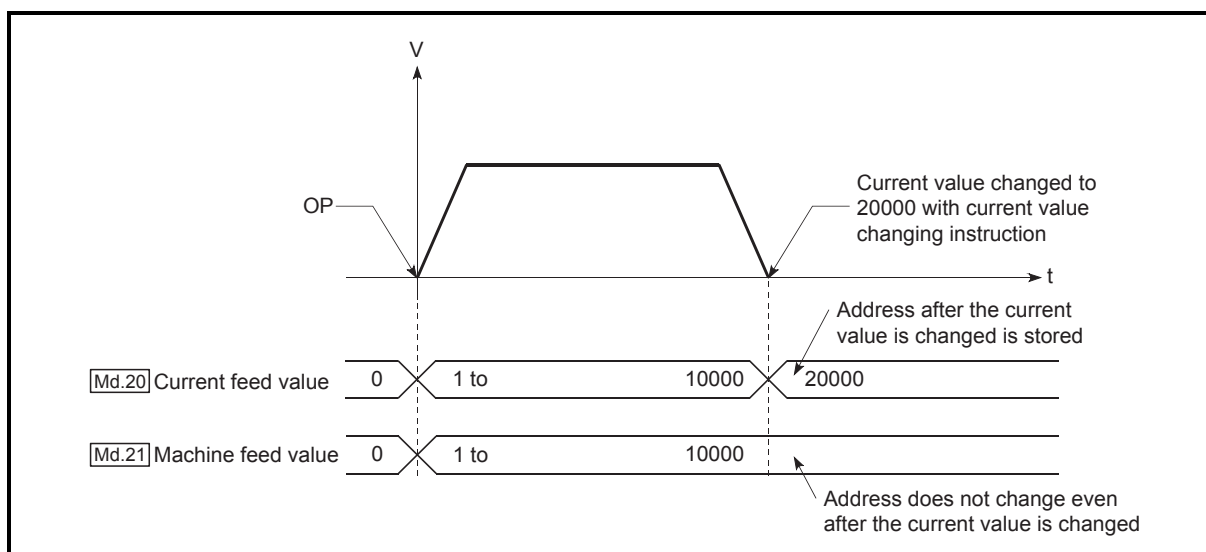


Fig. 9.8 Current feed value and machine feed value

### ■ Restrictions

A 0.9ms (QD75P□N /QD75D□N) or 1.8ms (QD75P□/QD75D□) error will occur in the current value update timing when the stored "current feed value" is used in the control.

A 0.9ms (QD75P□N /QD75D□N) or 56.8ms (QD75P□/QD75D□) error will occur in the current value update timing when the stored "machine feed value" is used in the control.

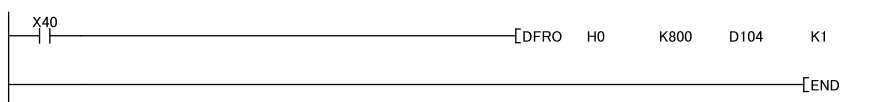
### ■ Monitoring the current value

The "current feed value" and "machine feed value" are stored in the following buffer memory addresses, and can be read using a "DFRO (P) instruction" or "DMOV (P) instruction" from the CPU module.

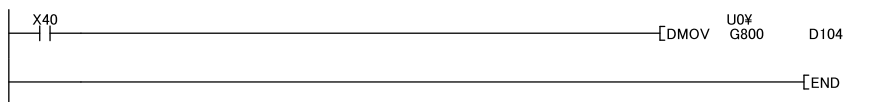
|                                                                   |                    | Buffer memory addresses |          |            |            |
|-------------------------------------------------------------------|--------------------|-------------------------|----------|------------|------------|
|                                                                   |                    | Axis 1                  | Axis 2   | Axis 3     | Axis 4     |
| <span style="border: 1px solid black; padding: 2px;">Md.20</span> | Current feed value | 800, 801                | 900, 901 | 1000, 1001 | 1100, 1101 |
| <span style="border: 1px solid black; padding: 2px;">Md.21</span> | Machine feed value | 802, 803                | 902, 903 | 1002, 1003 | 1102, 1103 |

- (1) The following shows the examples of programs to read out the current feed value of the axis 1 to D104 and D105 when X40 is turned ON.

(a) For the DFRO(P) instruction



(b) For the DMOV(P) instruction





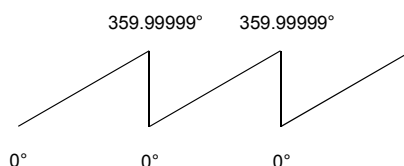
## 9.1.5 Control unit "degree" handling

When the control unit is set to "degree", the following items differ from when other control units are set.

## [1] Current feed value and machine feed value addresses

The address of "Md.20" Current feed value" becomes a ring address from 0 to 359.99999°.

But the address of "Md.21" Machine feed value" doesn't become a ring address.

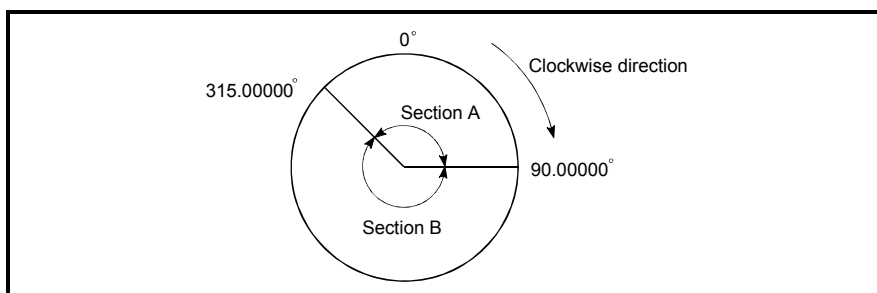


## [2] Software stroke limit valid/invalid setting

With the control unit set to "degree", the software stroke limit upper and lower limit values are 0 to 359.99999.

## (a) Setting to validate software stroke limit

To validate the software stroke limit, set the software stroke limit lower limit value and the upper limit value in a clockwise direction.



## 1) To set the movement range A, set as follows.

- Software stroke limit lower limit value .....315.00000°
- Software stroke limit upper limit value .....90.00000°

## 2) To set the movement range B, set as follows.

- Software stroke limit lower limit value .....90.00000°
- Software stroke limit upper limit value .....315.00000°

## (b) Setting to invalidate software stroke limit

To invalidate the software stroke limit, set the software stroke limit lower limit value equal to the software stroke limit upper limit value.

The control can be carried out irrespective of the setting of the software stroke limit.

## [3] Positioning control method when the control unit is set to "degree"

## 1) Absolute system

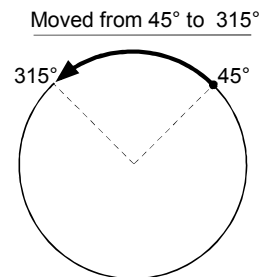
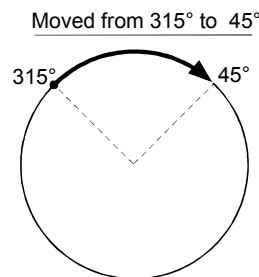
## (a) When the software stroke limit is invalid

Positioning is carried out in the nearest direction to the designated address, using the current value as a reference.

(This is called "shortcut control".)

## ---Example---

- 1) Positioning is carried out in a clockwise direction when the current value is moved from 315° to 45°.
- 2) Positioning is carried out in a counterclockwise direction when the current value is moved from 45° to 315°.



[To specify the positioning direction (not carrying out the shortcut control)]

Use "[Cd.40] ABS direction in degrees" or "[Da.28] ABS direction in degrees" to specify the positioning direction.

This function can be performed when the software stroke limit is invalid. When the software stroke limit is valid, an error "Illegal setting of ABS direction in unit of degree" (error code: 546) occurs and positioning is not started.

In "[Cd.40] ABS direction in degrees" or "[Da.28] ABS direction in degrees", the set value for the reference axis is applied to both the reference axis and interpolation axis. Even if a unit other than degree is set for the reference axis, the setting of the reference axis is applied to the interpolation axis as shown below (for 3-axis linear interpolation control (ABS3)).

| Axis           | Unit setting | "[Cd.40] ABS direction in degrees" | Rotation direction of the axis with degree setting |
|----------------|--------------|------------------------------------|----------------------------------------------------|
| Reference axis | pulse        | 1: ABS clockwise                   | 0                                                  |
| Sub axis 1     | degree       | -                                  | 1: ABS clockwise                                   |
| Sub axis 2     | degree       | -                                  | 1: ABS clockwise                                   |

[To specify the rotation direction in degrees for each positioning data]

To specify the rotation direction for each positioning data in a control executing multiple positioning data sets within one positioning operation continuously, such as continuous positioning control and continuous path control, use "[Da.28] ABS direction in degrees".

[To set the same rotation direction for all positioning data]

Setting "[Cd.40] ABS direction in degrees" enables batch specification of the rotation direction for each positioning data. To use "[Cd.40] ABS direction in degrees", set "0" in "[Da.28] ABS direction in degrees". When a value other than "0" is set, the setting of "[Da.28] ABS direction in degrees" becomes valid. The set value in "[Cd.40] ABS direction in degrees" becomes valid only when the positioning control is started. In the continuous positioning control or continuous path control, the operation is continued with the setting made at a start if the setting is changed during the operation.

| Name                                | Function                                                                                                                                                                                                                               | Buffer memory address     |                           |                            |                            | Initial value |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------|----------------------------|----------------------------|---------------|
|                                     |                                                                                                                                                                                                                                        | Axis 1                    | Axis 2                    | Axis 3                     | Axis 4                     |               |
| [Cd.40]<br>ABS direction in degrees | This data item specifies the ABS moving direction in which the position control carries out when "degree" is selected as the unit.<br>0: Takes a shortcut. (direction setting ignored)<br>1: ABS clockwise<br>2: ABS counterclockwise  | 1550                      | 1650                      | 1750                       | 1850                       | 0             |
| [Da.28]<br>ABS direction in degrees | "[Cd.40] ABS direction in degrees" is set for each positioning data.<br>0: Use the set value in "[Cd.40] ABS direction in degrees".<br>1: ABS clockwise<br>2: ABS counterclockwise<br>3: Takes a shortcut. (direction setting ignored) | 2003<br>+N*<br>(b2 to b3) | 8003<br>+N*<br>(b2 to b3) | 14003<br>+N*<br>(b2 to b3) | 20003<br>+N*<br>(b2 to b3) | 0             |

※: N indicates the offset address of each positioning data.

$N = ((\text{Positioning data No.}) - 1) \times 10$

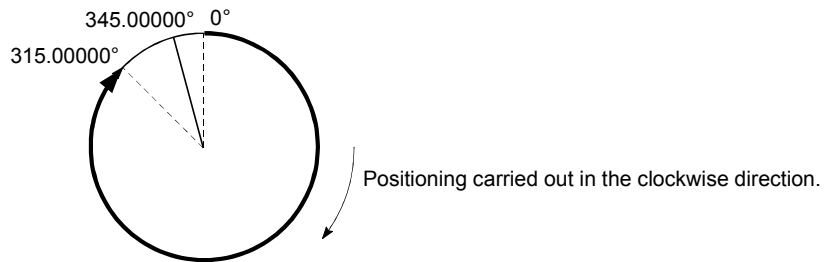
## (b) When the software stroke limit is valid

The positioning is carried out in a clockwise/counterclockwise direction depending on the software stroke limit range setting method.

Because of this, positioning with "shortcut control" may not be possible.

## Example

When the current value is moved from 0° to 315°, positioning is carried out in the clockwise direction if the software stroke limit lower limit value is 0° and the upper limit value is 345°.

**POINT**

Positioning addresses are within a range of 0° to 359.99999°.

Use the incremental system to carry out positioning of one rotation or more.

## 2) Incremental system

Positioning is carried out for a designated movement amount in a designated movement direction when in the incremental system of positioning.

The movement direction is determined by the sign (+, -) of the movement amount.

- For a positive (+) movement direction ....Clockwise
- For a negative (-) movement direction ...Counterclockwise

**POINT**

Positioning of 360° or more can be carried out with the incremental system.

At this time, set as shown below to invalidate the software stroke limit.

(Set the value with in the setting range (0° to 359.99999°).)

[Software stroke limit upper limit value = Software stroke limit lower limit value]

## 9.1.6 Interpolation control

### ■ Meaning of interpolation control

In "2-axis linear interpolation control", "3-axis linear interpolation control", "4-axis linear interpolation control", "2-axis fixed-feed control", "3-axis fixed-feed control", "4-axis fixed-feed control", "2-axis speed control", "3-axis speed control", "4-axis speed control", "2-axis circular interpolation control", and "3-axis helical interpolation control", control is carried out so that a specified path is drawn using a motor set in two to four axis directions.

This kind of control is called "interpolation control".

In interpolation control, the axis in which the control system is set is defined as the "reference axis", and the other axis is defined as the "interpolation axis".

The QD75 controls the "reference axis" following the positioning data set in the "reference axis", and controls the "interpolation axis" corresponding to the reference axis control so that a linear or arc path is drawn.

The following table shows the reference axis and interpolation axis combinations.

| Axis definition<br>Axis set to interpolation control in " Da.2 Control system"                                                | Reference axis             | Interpolation axis                              |
|-------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------|
| 2-axis linear interpolation control, "2-axis fixed-feed control, 2-axis circular interpolation control, 2-axis speed control" | Any of axes 1, 2, 3, and 4 | "Axes to be interpolated" set in reference axis |
| 3-axis linear interpolation control, "3-axis fixed-feed control, 3-axis speed control"                                        | Axis 1                     | Axis 2, Axis 3                                  |
|                                                                                                                               | Axis 2                     | Axis 3, Axis 4                                  |
|                                                                                                                               | Axis 3                     | Axis 4, Axis 1                                  |
|                                                                                                                               | Axis 4                     | Axis 1, Axis 2                                  |
| 4-axis linear interpolation control, "4-axis fixed-feed control, 4-axis speed control"                                        | Axis 1                     | Axis 2, Axis 3, Axis 4                          |
|                                                                                                                               | Axis 2                     | Axis 3, Axis 4, Axis 1                          |
|                                                                                                                               | Axis 3                     | Axis 4, Axis 1, Axis 2                          |
|                                                                                                                               | Axis 4                     | Axis 1, Axis 2, Axis 3                          |

The available axis combinations in the 3-axis helical interpolation control are the same as the ones in the "3-axis linear interpolation control", "3-axis fixed-feed control", and "3-axis speed control". However, the circular interpolation axis can be specified with " Da.5 Axis to be interpolated" of the reference axis.

The following table lists the combinations of the reference axis, circular interpolation axis, and linear interpolation axis for the 3-axis helical interpolation control.

| Axis definition<br>Axis set to interpolation control in " Da.2 Control system" | Reference axis | Circular interpolation axis * 1 | Linear interpolation axis * 2 |
|--------------------------------------------------------------------------------|----------------|---------------------------------|-------------------------------|
| 3-axis helical interpolation control                                           | Axis 1         | Axis 2                          | Axis 3                        |
|                                                                                |                | Axis 3                          | Axis 2                        |
|                                                                                | Axis 2         | Axis 3                          | Axis 4                        |
|                                                                                |                | Axis 4                          | Axis 3                        |
|                                                                                | Axis 3         | Axis 4                          | Axis 1                        |
|                                                                                |                | Axis 1                          | Axis 4                        |
|                                                                                | Axis 4         | Axis 1                          | Axis 2                        |
|                                                                                |                | Axis 2                          | Axis 1                        |

\* 1: Specify with " Da.5 Axis to be interpolated" of the reference axis.

\* 2: An axis which is not specified with " Da.5 Axis to be interpolated" of the reference axis is automatically assigned.

### ■ Setting the positioning data during interpolation control

When carrying out interpolation control, the same positioning data Nos. are set for the "reference axis" and the "interpolation axis".

The following table shows the "positioning data" setting items for the reference axis and interpolation axis.

| Setting item              |       | Axis                                      | Reference axis setting item                                                                                                                                                                    | Interpolation axis setting item                                                                                                 |
|---------------------------|-------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Same positioning data Nos | Da.1  | Operation pattern                         | ◎                                                                                                                                                                                              | —                                                                                                                               |
|                           | Da.2  | Control system                            | Linear 2, 3, 4, Fixed-feed 2, 3, 4,<br>Circular sub, Circular right,<br>Circular left<br>Helical sub, Helical right, Helical<br>left<br>Forward run speed 2, 3, 4<br>Reverse run speed 2, 3, 4 | —                                                                                                                               |
|                           | Da.3  | Acceleration time<br>No.                  | ◎                                                                                                                                                                                              | —                                                                                                                               |
|                           | Da.4  | Deceleration time<br>No.                  | ◎                                                                                                                                                                                              | —                                                                                                                               |
|                           | Da.5  | Axis to be<br>interpolated.               | ○ *                                                                                                                                                                                            | —                                                                                                                               |
|                           | Da.6  | Positioning address/<br>movement amount   | △<br>Forward run speed 2, 3, and 4.<br>Reverse run speed 2, 3, and 4 not<br>required.                                                                                                          | △<br>Forward run speed 2, 3, and 4.<br>Reverse run speed 2, 3, and 4 not<br>required.                                           |
|                           | Da.7  | Arc address                               | △<br>(Only for circular sub, circular<br>right, circular left, helical sub,<br>helical right, and helical left)                                                                                | △<br>(Only for circular sub, circular<br>right, circular left, helical sub,<br>helical right, and helical left)                 |
|                           | Da.8  | Command speed                             | ◎                                                                                                                                                                                              | △<br>Only during forward run speed 2,<br>3, 4 and reverse run speed 2, 3, 4.                                                    |
|                           | Da.9  | Dwell time                                | ○                                                                                                                                                                                              | —                                                                                                                               |
|                           | Da.10 | M code                                    | ○                                                                                                                                                                                              | △<br>Only for helical sub, helical right,<br>and helical left, set the number of<br>pitch for the linear interpolation<br>axis. |
|                           | Da.27 | M code ON signal<br>output timing         | ○                                                                                                                                                                                              | —                                                                                                                               |
|                           | Da.28 | ABS direction in<br>degrees               | ○                                                                                                                                                                                              | —                                                                                                                               |
|                           | Da.29 | Interpolation speed<br>designation method | ○                                                                                                                                                                                              | —                                                                                                                               |

◎ : Setting always required

○ : Set according to requirements

△ : Setting restrictions exist

— : Setting not required (Unrelated setting item, so any setting value will be ignored. Use the initial value or a value within the setting range.)

\* : For 2-axis interpolation, set the partner axis. If the self-axis is set, the error "Illegal interpolation description command" (error code: 521) will occur. For 3-axis interpolation other than 3-axis helical interpolation or 4-axis interpolation, this item does not need to be set. For 3-axis helical interpolation, refer to the previous page for the reference axis and interpolation axis combinations. If axes are set in a combination other than the ones in the previous page, the error "Illegal interpolation description command" (error code: 521) will occur.

Refer to Section 5.3 "List of positioning data" for information on the setting details.

### ■ Starting the interpolation control

The positioning data Nos. of the reference axis (axis in which interpolation control was set in " [Da.2] Control system") are started when starting the interpolation control. (Starting of the interpolation axis is not required.)

The following errors or warnings will occur and the positioning will not start if both reference axis and the interpolation axis are started.

- Reference axis : The error "Interpolation while interpolation axis BUSY" (error code: 519)
- Interpolation axis : The error "Control system setting error" (error code: 524),  
The warning "Start during operation" (warning code: 100)

### ■ Interpolation control continuous positioning

When carrying out interpolation control in which "continuous positioning control" and "continuous path control" are designated in the operation pattern, the positioning method for all positioning data from the started positioning data to the positioning data in which "positioning complete" is set must be set to interpolation control.

The number of the interpolation axes and axes to be interpolated cannot be changed from the intermediate positioning data. An error "Control system setting error" (error code: 524) will occur and the positioning will stop if setting, which changes the number of the interpolation axes and axes to be interpolated, is carried out.

### ■ Speed during interpolation control

Either the "composite speed" or "reference axis speed" can be designated as the speed during interpolation control.

([Pr.20] Interpolation speed designation method)

Only the "Reference axis speed" can be designated in the following interpolation control.

When a "composite speed" is set and positioning is started, the error "Interpolation mode error" (error code: 523) occurs, and the system will not start.

- 4-axis linear interpolation
- 2-axis speed control
- 3-axis speed control
- 4-axis speed control

### ■ Cautions in interpolation control

- (1) If a stepping motor is used, the circular interpolation control or 3-axis helical interpolation control cannot be carried out.  
Make sure to use a servomotor when the circular interpolation control or 3-axis helical interpolation control is carried out.
  - (2) If either of the axes exceeds the " **Pr.8** Speed limit value" in the 2 to 4 axes speed control, the axis which exceeded the speed limit value is controlled by the speed limit value.  
For the other axes which perform interpolation, the speed can be suppressed by the ratio of a command speed.
  - (3) If any axis exceeds " **Pr.8** Speed limit value" in any of the following control, the axis in excess of the speed limit value is controlled at the speed limit value.  
For the other axes performing interpolation, the speed can be suppressed by the ratio of a movement distance.
    - 2- to 4-axis linear interpolation control<sup>\*1</sup>
    - 2- to 4-axis fixed-feed control<sup>\*1</sup>
    - 2-axis circular interpolation control
    - 3-axis helical interpolation control<sup>\*2</sup>
- \*1: The speed limit value for the interpolation axis may not function under the following condition: "Reference axis speed" is set in in " **Pr.20** Interpolation speed designation method" or " **Da.29** Interpolation speed designation method", the reference axis is the minor axis, and the interpolation axis is the major axis.
- \*2: In 3-axis helical interpolation control, control the composite speed of the circular interpolation or the speed of the linear interpolation axis so that the speeds do not exceed " **Pr.8** Speed limit value".
- (4) In 2-axis interpolation, you cannot change the combination of interpolated axes midway through operation.

#### POINT

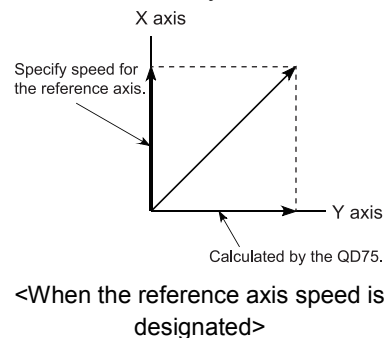
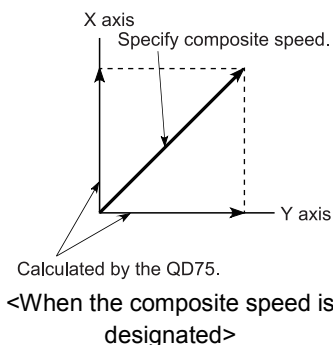
- When the "reference axis speed" is set during interpolation control, set so the major axis side becomes the reference axis. If the minor axis side is set as the reference axis, the major axis side speed may exceed the " **Pr.8** Speed limit value".



### ■ Interpolation speed designation method

To perform interpolation control, set the composite speed or reference axis speed in "[Pr.20] Interpolation speed designation method" or "[Da.29] Interpolation speed designation method" of the reference axis.

- 0: Composite speed ..... The movement speed for the control target is specified, and the speed for each axis is calculated by the QD75.
- 1: Reference axis speed ..... The axis speed set for the reference axis is specified, and the speed for the other axis carrying out interpolation is calculated by the QD75.



[To specify the interpolation speed for each positioning data]

To specify interpolation speed for each positioning data, use "[Da.29] Interpolation speed designation method".

[To set the same interpolation speed for all positioning data]

Setting "[Pr.20] Interpolation speed designation method" enables batch specification of the interpolation speed designation method for each positioning data.

To use "[Pr.20] Interpolation speed designation method", set "0" in "[Da.29] Interpolation speed designation method". When a value other than 0 is set, "[Da.29] Interpolation speed designation method" will be valid.

| Name                                              | Function                                                                                                                                                                                                | Buffer memory address     |                           |                            |                            | Initial value |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------|----------------------------|----------------------------|---------------|
|                                                   |                                                                                                                                                                                                         | Axis 1                    | Axis 2                    | Axis 3                     | Axis 4                     |               |
| [Pr.20]<br>Interpolation speed designation method | When carrying out linear interpolation/circular interpolation, set whether to specify the composite speed or reference axis speed.<br>0: Composite speed<br>1: Reference axis speed                     | 29                        | 179                       | 329                        | 479                        | 0             |
| [Da.29]<br>Interpolation speed designation method | Set the interpolation speed designation method for each positioning data.<br>0: Use the set value in "[Pr.20] Interpolation speed designation method".<br>1: Composite speed<br>2: Reference axis speed | 2003<br>+N*<br>(b4 to b6) | 8003<br>+N*<br>(b4 to b6) | 14003<br>+N*<br>(b4 to b6) | 20003<br>+N*<br>(b4 to b6) | 0             |

\*: N indicates the offset address of each positioning data.

$N = ((\text{Positioning data No.}) - 1) \times 10$

### ■ Limits to interpolation control

There are limits to the interpolation control that can be executed and speed ([Pr.20] Interpolation speed designation method) that can be set, depending on the "[Pr.1] Unit setting" of the reference axis and interpolation axis. (For example, circular interpolation control cannot be executed if the reference axis and interpolation axis units differ.)

The following table shows the interpolation control and speed designation limits.

| " [Da.2] Control system" interpolation control                                   | [Pr.20] Interpolation speed designation method or<br>[Da.29] Interpolation speed designation method *5 | [Pr.1] Unit setting *1                                                                            |                                                       |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------|
|                                                                                  |                                                                                                        | Reference axis and interpolation axis units are the same, or a combination of "mm" and "inch". *3 | Reference axis and interpolation axis units differ *3 |
| Linear 2 (ABS, INC)<br>Fixed-feed 2                                              | Composite speed                                                                                        | ○                                                                                                 | ×                                                     |
|                                                                                  | Reference axis speed                                                                                   | ○                                                                                                 | ○                                                     |
| Circular sub (ABS, INC)<br>Circular right (ABS, INC)<br>Circular left (ABS, INC) | Composite speed                                                                                        | ○ *2                                                                                              | ×                                                     |
|                                                                                  | Reference axis speed                                                                                   | ×                                                                                                 | ×                                                     |
| Helical sub (ABS, INC)<br>Helical right (ABS, INC)<br>Helical left (ABS, INC)    | Composite speed                                                                                        | ○ *2                                                                                              | ○ *4                                                  |
|                                                                                  | Reference axis speed                                                                                   | ×                                                                                                 | ×                                                     |
| Linear 3 (ABS, INC)<br>Fixed-feed 3                                              | Composite speed                                                                                        | ○                                                                                                 | ×                                                     |
|                                                                                  | Reference axis speed                                                                                   | ○                                                                                                 | ○                                                     |
| Linear 4 (ABS, INC)<br>Fixed-feed 4                                              | Composite speed                                                                                        | ×                                                                                                 | ×                                                     |
|                                                                                  | Reference axis speed                                                                                   | ○                                                                                                 | ○                                                     |

○ : Setting possible, × : Setting not possible.

\*1 "mm" and "inch" unit mix possible.

\*2 The error "Circular interpolation not possible" (error code: 535) will occur and the positioning cannot be started if circular interpolation control or 3-axis helical interpolation control is set when the unit is "degree". The machine will immediately stop if "degree" is set during positioning control.

\*3 The unit set in the reference axis will be used for the speed unit during control if the units differ or if "mm" and "inch" are combined.

\*4 Only for the linear interpolation axis, a unit different from the one set for the reference axis can be used.

\*5 When positioning is started with the following setting, the error "Interpolation mode error" (error code: 523) will occur.

- A "composite speed" is set for the 2-axis speed control, 3-axis speed control, or 4-axis linear interpolation control.
- A "reference axis speed" is set for the circular interpolation control or 3-axis helical interpolation control.

### ■ Axis operation status during interpolation control

"Interpolation" will be stored in the "[Md.26] Axis operation status" during interpolation control. "Standby" will be stored when the interpolation operation is terminated. Both the reference axis and interpolation axis will carry out a deceleration stop if an error occurs during control, and "error occurring" will be stored in the operation status.

## 9.2 Setting the positioning data

## 9.2.1 Relation between each control and positioning data

The setting requirements and details for the setting items of the positioning data to be set differ according to the " **Da.2** Control system".

The following table shows the positioning data setting items corresponding to the different types of control. Details and settings for the operation of each control are shown in Section 9.2.2 and subsequent sections.

(In this section, GX Configurator-QP or GX Works2 is used for setting positioning data.)

| Major positioning control      |                                        |                                                        | Position control                                                                                                                           |                                                                                                                  |                                                    |                                                                                                                                                                                      | Speed control                                                  | Speed-position switching control |   |
|--------------------------------|----------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------|---|
|                                |                                        |                                                        | 1-axis linear control<br>2-axis linear interpolation control<br>3-axis linear interpolation control<br>4-axis linear interpolation control | 1-axis fixed-feed control<br>2-axis fixed-feed control<br>3-axis fixed-feed control<br>4-axis fixed-feed control | 2-axis circular interpolation control              | 3-axis helical interpolation control                                                                                                                                                 | 1-axis, 2-axis, 3-axis, 4-axis Speed control                   |                                  |   |
| Positioning data setting items |                                        |                                                        |                                                                                                                                            |                                                                                                                  |                                                    |                                                                                                                                                                                      |                                                                |                                  |   |
| Da.1                           | Operation pattern                      | Independent positioning control (Positioning complete) | ◎                                                                                                                                          | ◎                                                                                                                | ◎                                                  | ◎                                                                                                                                                                                    | ◎                                                              | ◎                                |   |
|                                |                                        | Continuous positioning control                         | ◎                                                                                                                                          | ◎                                                                                                                | ◎                                                  | ◎                                                                                                                                                                                    | ×                                                              | ◎                                |   |
|                                |                                        | Continuous path control                                | ◎                                                                                                                                          | ×                                                                                                                | ◎                                                  | ◎                                                                                                                                                                                    | ×                                                              | ×                                |   |
| Da.2                           | Control system                         | Linear 1<br>Linear 2<br>Linear 3<br>Linear 4<br>*1     | Fixed-feed 1<br>Fixed-feed 2<br>Fixed-feed 3<br>Fixed-feed 4                                                                               | Circular sub<br>Circular right<br>Circular left<br>*1                                                            | Helical sub<br>Helical right<br>Helical left<br>*1 | Forward run speed 1<br>Reverse run speed 1<br>Forward run speed 2<br>Reverse run speed 2<br>Forward run speed 3<br>Reverse run speed 3<br>Forward run speed 4<br>Reverse run speed 4 | Forward run speed/position<br>Reverse run speed/position<br>*1 |                                  |   |
| Da.3                           | Acceleration time No.                  |                                                        | ◎                                                                                                                                          | ◎                                                                                                                | ◎                                                  | ◎                                                                                                                                                                                    | ◎                                                              | ◎                                |   |
| Da.4                           | Deceleration time No.                  |                                                        | ◎                                                                                                                                          | ◎                                                                                                                | ◎                                                  | ◎                                                                                                                                                                                    | ◎                                                              | ◎                                |   |
| Da.5                           | Axis to be interpolated                |                                                        | ◎: 2-axis –: 1, 3, 4-axis                                                                                                                  |                                                                                                                  |                                                    |                                                                                                                                                                                      |                                                                |                                  | – |
| Da.6                           | Positioning address/movement amount    |                                                        | ◎                                                                                                                                          | ◎                                                                                                                | ◎                                                  | ◎                                                                                                                                                                                    | –                                                              | ◎                                |   |
| Da.7                           | Arc address                            |                                                        | –                                                                                                                                          | –                                                                                                                | ◎                                                  | ◎                                                                                                                                                                                    | –                                                              | –                                |   |
| Da.8                           | Command speed                          |                                                        | ◎                                                                                                                                          | ◎                                                                                                                | ◎                                                  | ◎                                                                                                                                                                                    | ◎                                                              | ◎                                |   |
| Da.9                           | Dwell time                             |                                                        | ○                                                                                                                                          | ○                                                                                                                | ○                                                  | ○                                                                                                                                                                                    | –                                                              | ○                                |   |
| Da.10                          | M code                                 |                                                        | ○                                                                                                                                          | ○                                                                                                                | ○                                                  | ○*2                                                                                                                                                                                  | ○                                                              | ○                                |   |
| Da.27                          | M code ON signal output timing         |                                                        | ○                                                                                                                                          | ○                                                                                                                | ○                                                  | ○                                                                                                                                                                                    | ○                                                              | ○                                |   |
| Da.28                          | ABS direction in degrees               |                                                        | ○                                                                                                                                          | ○                                                                                                                | ○                                                  | ○                                                                                                                                                                                    | ○                                                              | ○                                |   |
| Da.29                          | Interpolation speed designation method |                                                        | –: 1-axis control, ○: 2-axis interpolation control, 3-axis interpolation control, 4-axis interpolation control                             |                                                                                                                  |                                                    |                                                                                                                                                                                      |                                                                |                                  |   |

| Major positioning control<br>Positioning data setting items |                                        |                                                        | Position-speed switching control                         | Other control   |                            |                                       |                   |                  |
|-------------------------------------------------------------|----------------------------------------|--------------------------------------------------------|----------------------------------------------------------|-----------------|----------------------------|---------------------------------------|-------------------|------------------|
|                                                             |                                        |                                                        |                                                          | NOP instruction | Current value changing     | JUMP instruction                      | LOOP instruction  | LEND instruction |
| Da.1                                                        | Operation pattern                      | Independent positioning control (Positioning complete) | ◎                                                        | —               | ◎                          | —                                     | —                 | —                |
|                                                             |                                        | Continuous positioning control                         | ×                                                        | —               | ◎                          | —                                     | —                 | —                |
|                                                             |                                        | Continuous path control                                | ×                                                        | —               | ×                          | —                                     | —                 | —                |
| Da.2                                                        | Control system                         |                                                        | Forward run position/speed<br>Reverse run position/speed | NOP instruction | Current value changing     | JUMP instruction                      | LOOP instruction  | LEND instruction |
| Da.3                                                        | Acceleration time No.                  |                                                        | ◎                                                        | —               | —                          | —                                     | —                 | —                |
| Da.4                                                        | Deceleration time No.                  |                                                        | ◎                                                        | —               | —                          | —                                     | —                 | —                |
| Da.5                                                        | Axis to be interpolated                |                                                        | —                                                        | —               | —                          | —                                     | —                 | —                |
| Da.6                                                        | Positioning address/movement amount    |                                                        | ◎                                                        | —               | Change destination address | —                                     | —                 | —                |
| Da.7                                                        | Arc address                            |                                                        | —                                                        | —               | —                          | —                                     | —                 | —                |
| Da.8                                                        | Command speed                          |                                                        | ◎                                                        | —               | —                          | —                                     | —                 | —                |
| Da.9                                                        | Dwell time                             |                                                        | ○                                                        | —               | —                          | JUMP destination-positioning data No. | —                 | —                |
| Da.10                                                       | M code                                 |                                                        | ○                                                        | —               | ○                          | Condition data No. at JUMP            | No. of repetition | —                |
| Da.27                                                       | M code ON signal output timing         |                                                        | ○                                                        | —               | ○                          | —                                     | —                 | —                |
| Da.28                                                       | ABS direction in degrees               |                                                        | ○                                                        | —               | —                          | —                                     | —                 | —                |
| Da.29                                                       | Interpolation speed designation method |                                                        | —                                                        | —               | —                          | —                                     | —                 | —                |

◎ : Always set      ○ : Set as required

× : Setting not possible (If the setting is made, the error "Continuous path control not possible" (error code: 516) or "New current value not possible" (error code: 515) will occur at a start.)

— : Setting not required (Setting value is invalid. Use the initial values or setting values within a range where no error occurs.)

\*1: The "ABS (absolute) system" or "INC (incremental) system" can be used for the control system.

\*2: Set the M code for the reference axis and the number of pitch for the linear interpolation axis.

## REMARK

It is recommended that the "positioning data" be set whenever possible with GX Configurator-QP or GX Works2. Execution by sequence program uses many sequence programs and devices. The execution becomes complicated, and the scan times will increase.

## 9.2.2 1-axis linear control

In "1-axis linear control" (" [Da.2] Control system" = ABS linear 1, INC linear 1), one motor is used to carry out position control in a set axis direction.

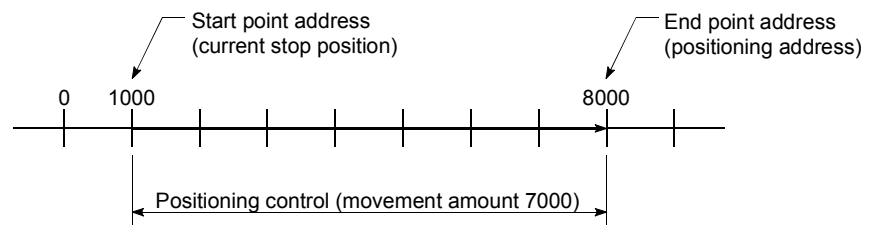
## [1] 1-axis linear control (ABS linear 1)

## ■ Operation chart

In absolute system 1-axis linear control, positioning is carried out from the current stop position (start point address) to the address (end point address) set in " [Da.6] Positioning address/movement amount".

## Example

When the start point address (current stop position) is 1000, and the end point address (positioning address) is 8000, positioning is carried out in the positive direction for a movement amount of 7000 (8000-1000)



## ■ Positioning data setting example

The following table shows setting examples when "1-axis linear control (ABS linear 1)" is set in positioning data No. 1 of axis 1.

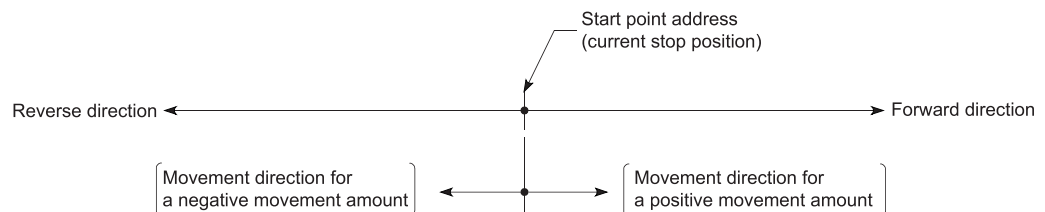
| Setting item |                                            | Setting example      | Setting details                                                                                                                  |
|--------------|--------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Axis 1       | Positioning data No. 1                     |                      |                                                                                                                                  |
|              | [Da.1] Operation pattern                   | Positioning complete | Set "Positioning complete" assuming the next positioning data will not be executed.                                              |
|              | [Da.2] Control system                      | ABS linear 1         | Set absolute system 1-axis linear control.                                                                                       |
|              | [Da.3] Acceleration time No.               | 1                    | Designate the value set in " [Pr.25] Acceleration time 1" as the acceleration time at start.                                     |
|              | [Da.4] Deceleration time No.               | 0                    | Designate the value set in " [Pr.10] Deceleration time 0" as the deceleration time at deceleration.                              |
|              | [Da.5] Axis to be interpolated             | —                    | Setting not required (setting value will be ignored).                                                                            |
|              | [Da.6] Positioning address/movement amount | 8000.0μm             | Set the positioning address. (Assuming "mm" is set in " [Pr.1] Unit setting".)                                                   |
|              | [Da.7] Arc address                         | —                    | Setting not required (setting value will be ignored).                                                                            |
|              | [Da.8] Command speed                       | 6000.00mm/min        | Set the speed during movement to the positioning address.                                                                        |
|              | [Da.9] Dwell time                          | 500ms                | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal. |
|              | [Da.10] M code                             | 10                   | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                            |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

## [2] 1-axis linear control (INC linear 1)

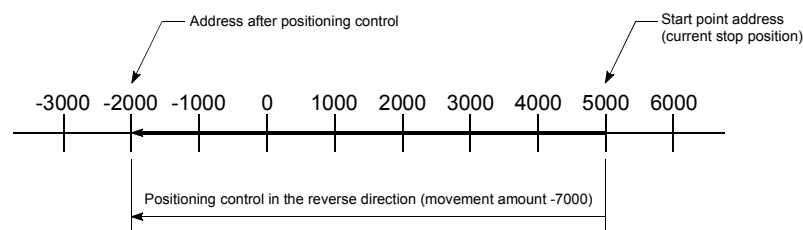
## ■ Operation chart

In incremental system 1-axis linear control, positioning of movement amount set in "Da.6 Positioning address/movement amount" is carried out from the current stop position (start point address). The movement direction is determined by the sign of the movement amount.



## ---Example---

When the start point address is 5000, and the movement amount is -7000, positioning is carried out to the -2000 position.



## ■ Positioning data setting example

The following table shows setting examples when "1-axis linear control (INC linear 1)" is set in positioning data No. 1 of axis 1.

| Setting item                  |                                          | Setting example      | Setting details                                                                                                                  |
|-------------------------------|------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Axis 1 Positioning data No. 1 | Da.1 Operation pattern                   | Positioning complete | Set "Positioning complete" assuming the next positioning data will not be executed.                                              |
|                               | Da.2 Control system                      | INC linear 1         | Set incremental system 1-axis linear control.                                                                                    |
|                               | Da.3 Acceleration time No.               | 1                    | Designate the value set in "Pr.25 Acceleration time 1" as the acceleration time at start.                                        |
|                               | Da.4 Deceleration time No.               | 0                    | Designate the value set in "Pr.10 Deceleration time 0" as the deceleration time at deceleration.                                 |
|                               | Da.5 Axis to be interpolated             | —                    | Setting not required (setting value will be ignored).                                                                            |
|                               | Da.6 Positioning address/movement amount | -7000.0 μm           | Set the movement amount. (Assuming "mm" is set in "Pr.1 Unit setting".)                                                          |
|                               | Da.7 Arc address                         | —                    | Setting not required (setting value will be ignored).                                                                            |
|                               | Da.8 Command speed                       | 6000.00mm/min        | Set the speed during movement.                                                                                                   |
|                               | Da.9 Dwell time                          | 500ms                | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal. |
|                               | Da.10 M code                             | 10                   | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                            |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

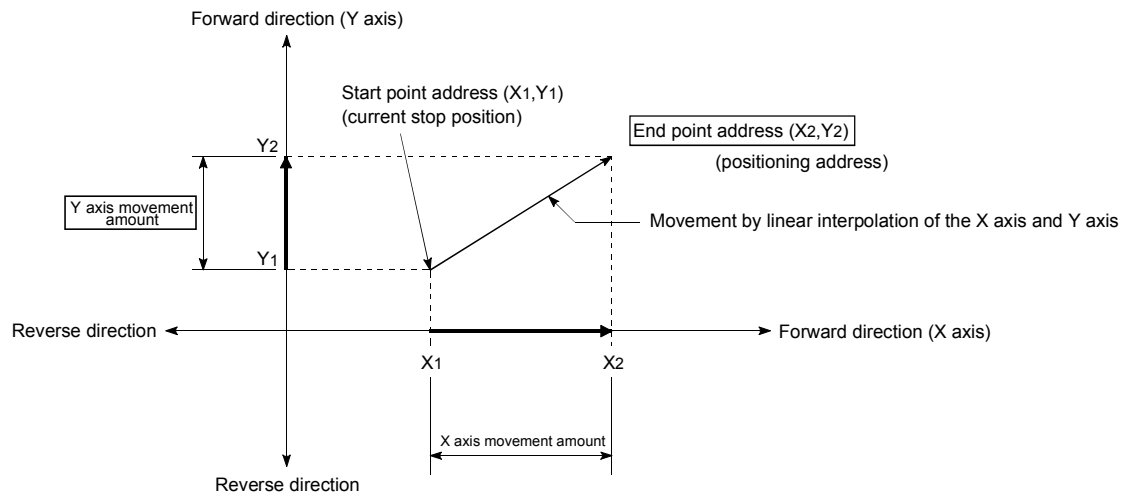
## 9.2.3 2-axis linear interpolation control

In "2-axis linear interpolation control" (" **Da.2** Control system" = ABS linear 2, INC linear 2), two motors are used to carry out position control in a linear path while carrying out interpolation for the axis directions set in each axis.  
(Refer to Section 9.1.6 "Interpolation control" for details on interpolation control.)

## [1] 2-axis linear interpolation control (ABS linear 2)

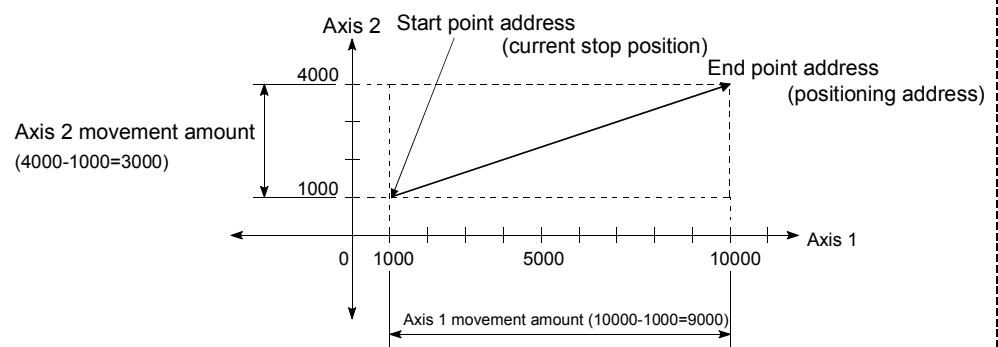
## ■ Operation chart

In absolute system 2-axis linear interpolation control, specified two axes are used to perform linear interpolation positioning from the current stop position (start point address) to the address set in the " **Da.6** Positioning address/movement amount" (end point address).



## --Example--

When the start point address (current stop position) is (1000, 1000) and the end point address (positioning address) is (10000, 4000), positioning is carried out as follows.



### ■ Restrictions

An error will occur and the positioning will not start in the following cases. The machine will immediately stop if the error is detected during a positioning control.

- If the movement amount of each axis exceeds "1073741824 ( $=2^{30}$ )" when "0: Composite speed" is set in " **Pr.20** Interpolation speed designation method" ... The error "Outside linear movement amount range" (error code: 504) occurs at a positioning start.  
(The maximum movement amount that can be set in " **Da.6** Positioning address/movement amount" is "1073741824 ( $=2^{30}$ )".)

### ■ Positioning data setting example

[Reference axis and interpolation axis are designated as axis 1 and axis 2, respectively.]

The following table shows setting examples when "2-axis linear interpolation control (ABS linear 2)" is set in positioning data No. 1 of axis 1. (The required values are also set in positioning data No. 1 of axis 2.)

| Setting item |                        | Axis                                | Axis 1<br>(reference<br>axis) setting<br>example | Axis 2<br>(interpolation<br>axis) setting<br>example | Setting details                                                                                                                  |
|--------------|------------------------|-------------------------------------|--------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Axis 1       | Positioning data No. 1 |                                     |                                                  |                                                      |                                                                                                                                  |
|              | <b>Da.1</b>            | Operation pattern                   | Positioning complete                             | —                                                    | Set "Positioning complete" assuming the next positioning data will not be executed.                                              |
|              | <b>Da.2</b>            | Control system                      | ABS linear 2                                     | —                                                    | Set absolute system 2-axis linear interpolation control.                                                                         |
|              | <b>Da.3</b>            | Acceleration time No.               | 1                                                | —                                                    | Designate the value set in " <b>Pr.25</b> Acceleration time 1" as the acceleration time at start.                                |
|              | <b>Da.4</b>            | Deceleration time No.               | 0                                                | —                                                    | Designate the value set in " <b>Pr.10</b> Deceleration time 0" as the deceleration time at deceleration.                         |
|              | <b>Da.5</b>            | Axis to be interpolated             | Axis 2                                           | —                                                    | Set the axis to be interpolated (partner axis).<br>If the self-axis is set, an error will occur.                                 |
|              | <b>Da.6</b>            | Positioning address/movement amount | 10000.0μm                                        | 4000.0μm                                             | Set the end point address. (Assuming "mm" is set in " <b>Pr.1</b> Unit setting".)                                                |
|              | <b>Da.7</b>            | Arc address                         | —                                                | —                                                    | Setting not required (setting value will be ignored).                                                                            |
|              | <b>Da.8</b>            | Command speed                       | 6000.00 mm/min                                   | —                                                    | Set the speed during movement to the end point address.                                                                          |
|              | <b>Da.9</b>            | Dwell time                          | 500ms                                            | —                                                    | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal. |
|              | <b>Da.10</b>           | M code                              | 10                                               | —                                                    | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                            |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

### POINT

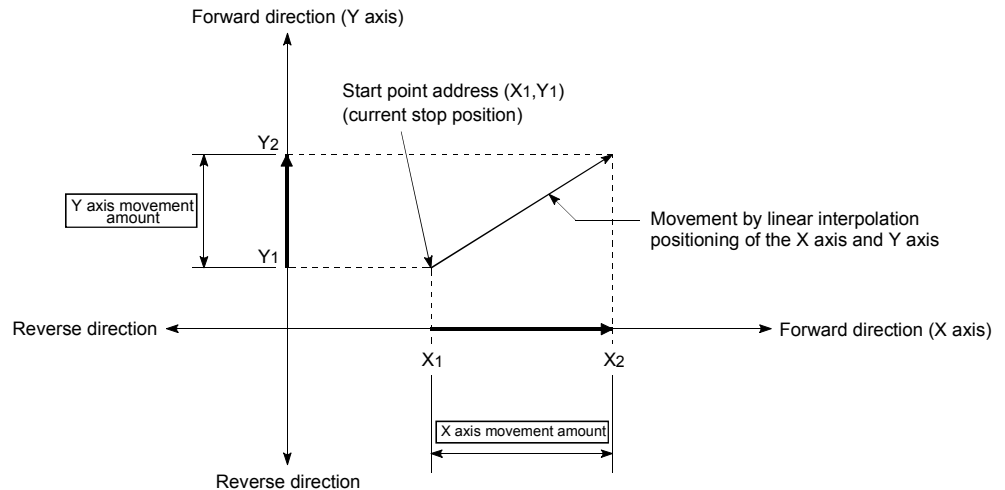
- When the "reference axis speed" is set during 2-axis linear interpolation control, set so the major axis side becomes the reference axis. If the minor axis side is set as the reference axis, the major axis side speed may exceed the " **Pr.8** Speed limit value".



## [2] 2-axis linear interpolation control (INC linear 2)

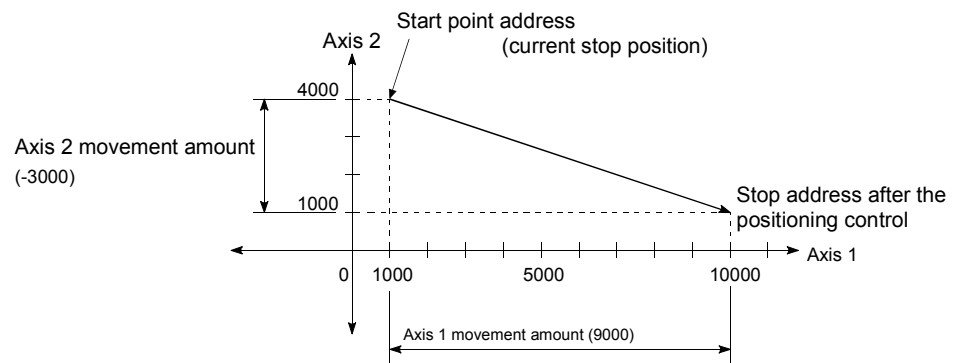
## ■ Operation chart

In incremental system 2-axis linear interpolation control, designated 2 axes are used. Linear interpolation positioning of movement amount set in "Da.6 Positioning address/movement amount" is carried out from the current stop position (start point address). The movement direction is determined by the sign of the movement amount.



## Example

When the axis 1 movement amount is 9000 and the axis 2 movement amount is -3000, positioning is carried out as follows.



### ■ Restrictions

An error will occur and the positioning will not start in the following cases. The machine will immediately stop if the error is detected during a positioning operation.

- If the movement amount of each axis exceeds "1073741824 ( $=2^{30}$ )" when "0: Composite speed" is set in " [Pr.20] Interpolation speed designation method" ... The error "Outside linear movement amount range" (error code: 504) occurs at a positioning start.  
(The maximum movement amount that can be set in " [Da.6] Positioning address/movement amount" is "1073741824 ( $=2^{30}$ )".)

### ■ Positioning data setting example

[Reference axis and interpolation axis are designated as axis 1 and axis 2, respectively.]

The following table shows setting examples when "2-axis linear interpolation control (INC linear 2)" is set in positioning data No. 1 of axis 1. (The required values are also set in positioning data No. 1 of axis 2.)

| Setting item |                        | Axis                                | Axis 1<br>(reference<br>axis) setting<br>example | Axis 2<br>(interpolation<br>axis) setting<br>example | Setting details                                                                                                                  |
|--------------|------------------------|-------------------------------------|--------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Axis 1       | Positioning data No. 1 |                                     |                                                  |                                                      |                                                                                                                                  |
|              | [Da.1]                 | Operation pattern                   | Positioning complete                             | —                                                    | Set "Positioning complete" assuming the next positioning data will not be executed.                                              |
|              | [Da.2]                 | Control system                      | INC linear 2                                     | —                                                    | Set incremental system 2-axis linear interpolation control.                                                                      |
|              | [Da.3]                 | Acceleration time No.               | 1                                                | —                                                    | Designate the value set in " [Pr.25] Acceleration time 1" as the acceleration time at start.                                     |
|              | [Da.4]                 | Deceleration time No.               | 0                                                | —                                                    | Designate the value set in " [Pr.10] Deceleration time 0" as the deceleration time at deceleration.                              |
|              | [Da.5]                 | Axis to be interpolated             | Axis 2                                           | —                                                    | Set the axis to be interpolated (partner axis).<br>If the self-axis is set, an error will occur.                                 |
|              | [Da.6]                 | Positioning address/movement amount | 9000.0μm                                         | -3000.0μm                                            | Set the movement amount. (Assuming "mm" is set in " [Pr.1] Unit setting".)                                                       |
|              | [Da.7]                 | Arc address                         | —                                                | —                                                    | Setting not required (setting value will be ignored).                                                                            |
|              | [Da.8]                 | Command speed                       | 6000.00 mm/min                                   | —                                                    | Set the speed during movement.                                                                                                   |
|              | [Da.9]                 | Dwell time                          | 500ms                                            | —                                                    | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal. |
|              | [Da.10]                | M code                              | 10                                               | —                                                    | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                            |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

### POINT

- When the "reference axis speed" is set during 2-axis linear interpolation control, set so the major axis side becomes the reference axis. If the minor axis side is set as the reference axis, the major axis side speed may exceed the " [Pr.8] Speed limit value".

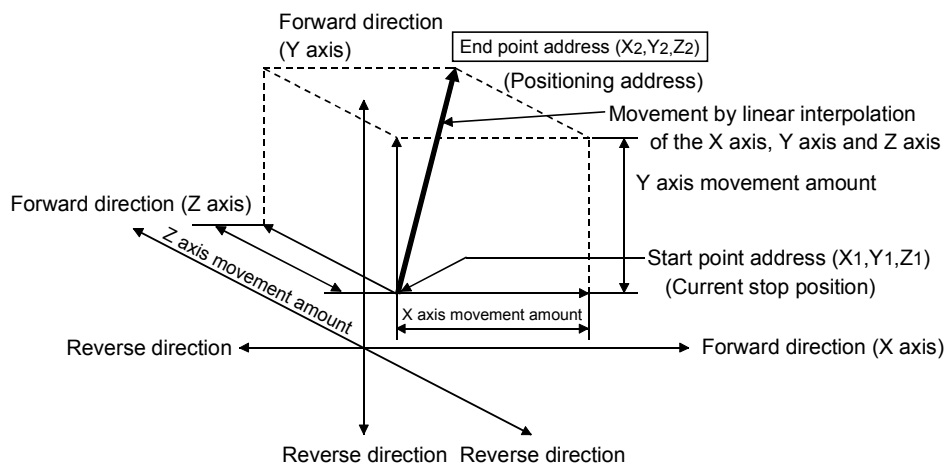
## 9.2.4 3-axis linear interpolation control

In "3-axis linear interpolation control" (" **Da.2** Control system" = ABS linear 3, INC linear 3), three motors are used to carry out position control in a linear path while carrying out interpolation for the axis directions set in each axis.  
(Refer to Section 9.1.6 "Interpolation control" for details on interpolation control.)

## [1] 3-axis linear interpolation control (ABS linear 3)

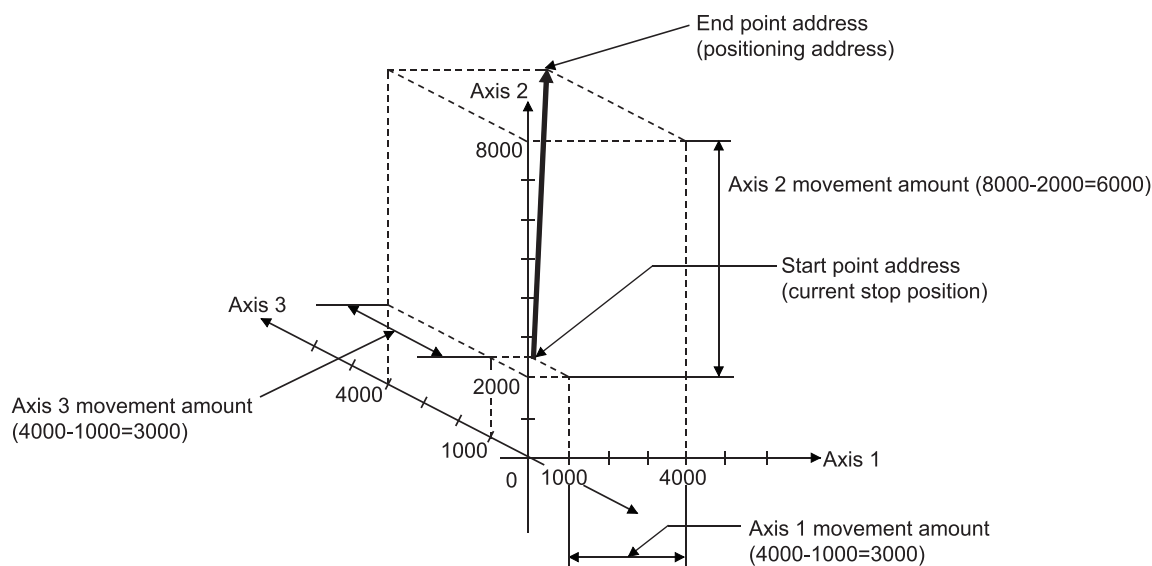
## ■ Operation chart

In absolute system 3-axis linear interpolation control, 3 axes are used. Linear interpolation positioning is carried out from the current stop position (start point address) to the address (end point address) set in " **Da.6** Positioning address/movement amount".



## Example

When the start point address (current stop position) is (1000, 2000, 1000) and the end point address (positioning address) is (4000, 8000, 4000), positioning is carried out as follows.



### ■ Restrictions

An error will occur and the positioning will not start in the following cases. The machine will immediately stop if the error is detected during a positioning control.

- If the movement amount of each axis exceeds "1073741824 ( $=2^{30}$ )" when "0: Composite speed" is set in " **Pr.20** Interpolation speed designation method" ... The error "Outside linear movement amount range" (error code: 504) occurs at a positioning start.  
(The maximum movement amount that can be set in " **Da.6** Positioning address/movement amount" is "1073741824 ( $=2^{30}$ )".)

### ■ Positioning data setting example

[Reference axis is designated as axis 1.]

The following table shows setting examples when "3-axis linear interpolation control (ABS linear 3)" is set in positioning data No. 1 of axis 1. (The required values are also set in positioning data No. 1 of axis 2 and axis 3.)

| Setting item \ Axis           |                                                 | Axis 1<br>(reference axis) setting example | Axis 2<br>(interpolation axis) setting example | Axis 3<br>(interpolation axis) setting example | Setting details                                                                                                                            |
|-------------------------------|-------------------------------------------------|--------------------------------------------|------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1 Positioning data No. 1 | <b>Da.1</b> Operation pattern                   | Positioning complete                       | —                                              | —                                              | Set "Positioning complete" assuming the next positioning data will not be executed.                                                        |
|                               | <b>Da.2</b> Control system                      | ABS linear 3                               | —                                              | —                                              | Set absolute system 3-axis linear interpolation control.                                                                                   |
|                               | <b>Da.3</b> Acceleration time No.               | 1                                          | —                                              | —                                              | Designate the value set in " <b>Pr.25</b> Acceleration time 1" as the acceleration time at start.                                          |
|                               | <b>Da.4</b> Deceleration time No.               | 0                                          | —                                              | —                                              | Designate the value set in " <b>Pr.10</b> Deceleration time 0" as the deceleration time at deceleration.                                   |
|                               | <b>Da.5</b> Axis to be interpolated             | —                                          | —                                              | —                                              | Setting not required (setting value will be ignored).<br>When axis 1 is used as a reference axis, the interpolation axes are axes 2 and 3. |
|                               | <b>Da.6</b> Positioning address/movement amount | 4000.0μm                                   | 8000.0μm                                       | 4000.0μm                                       | Set the end point address. (Assuming "mm" is set in " <b>Pr.1</b> Unit setting".)                                                          |
|                               | <b>Da.7</b> Arc address                         | —                                          | —                                              | —                                              | Setting not required (setting value will be ignored).                                                                                      |
|                               | <b>Da.8</b> Command speed                       | 6000.00 mm/min                             | —                                              | —                                              | Set the speed during movement to the end point address.                                                                                    |
|                               | <b>Da.9</b> Dwell time                          | 500ms                                      | —                                              | —                                              | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal.           |
|                               | <b>Da.10</b> M code                             | 10                                         | —                                              | —                                              | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                                      |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

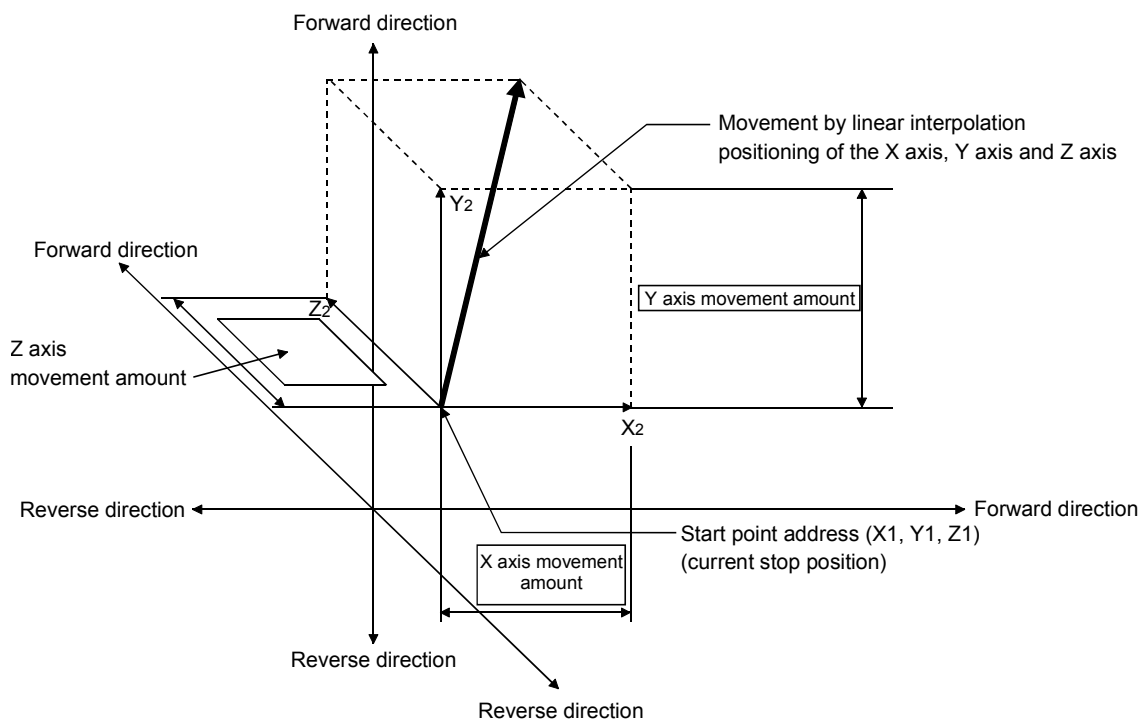
**POINTS**

- When the "reference axis speed" is set during 3-axis linear interpolation control, set so the major axis side becomes the reference axis. If the minor axis side is set as the reference axis, the major axis side speed may exceed the " **Pr.8** Speed limit value".
- Refer to Section 9.1.6 "Interpolation control" for the reference axis and interpolation axis combinations.

## [2] 3-axis linear interpolation control (INC linear 3)

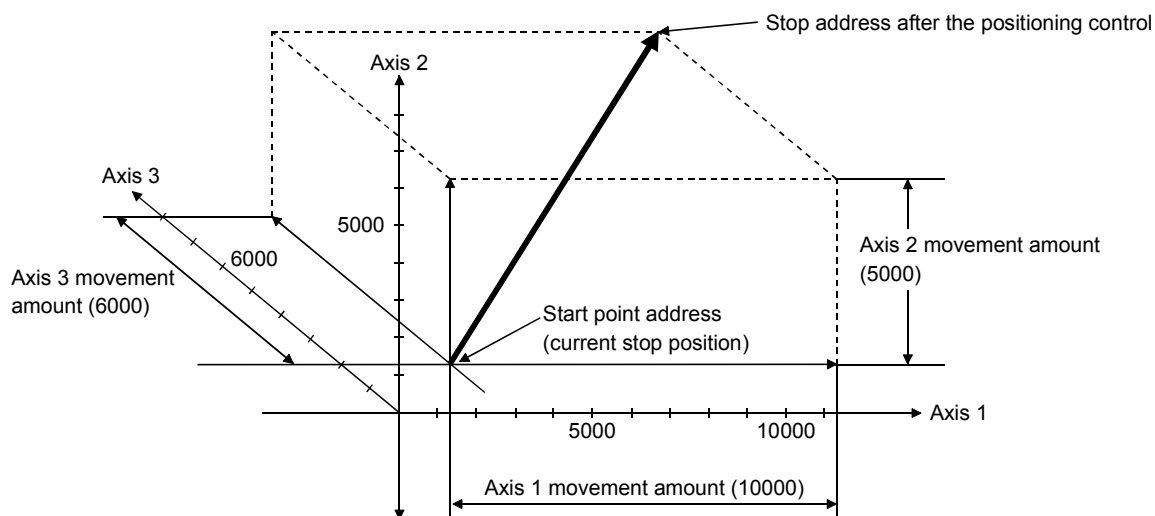
## ■ Operation chart

In incremental system 3-axis linear interpolation control, designated 3 axes are used. Linear interpolation positioning of movement amount set in "Da.6 Positioning address/movement amount" is carried out from the current stop position (start point address). The movement direction is determined by the sign of the movement amount.



## -Example-

When the axis 1 movement amount is 10000, the axis 2 movement amount is 5000 and the axis 3 movement amount is 6000, positioning is carried out as follows.



### ■ Restrictions

An error will occur and the positioning will not start in the following cases. The machine will immediately stop if the error is detected during a positioning operation.

- If the movement amount of each axis exceeds "1073741824 ( $=2^{30}$ )" when "0: Composite speed" is set in " [Pr.20] Interpolation speed designation method" ... The error "Outside linear movement amount range" (error code: 504) occurs at a positioning start.  
(The maximum movement amount that can be set in " [Da.6] Positioning address/movement amount" is "1073741824 ( $=2^{30}$ )".)

### ■ Positioning data setting example

[Reference axis is designated as axis 1.]

The following table shows setting examples when "3-axis linear interpolation control (INC linear 3)" is set in positioning data No. 1 of axis 1. (The required values are also set in positioning data No. 1 of axis 2 and axis 3.)

| Setting item                  |         | Axis                                | Axis 1<br>(reference<br>axis) setting<br>example | Axis 2<br>(interpolation<br>axis) setting<br>example | Axis 3<br>(interpolation<br>axis) setting<br>example | Setting details                                                                                                                            |
|-------------------------------|---------|-------------------------------------|--------------------------------------------------|------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1 Positioning data No. 1 | [Da.1]  | Operation pattern                   | Positioning complete                             | —                                                    | —                                                    | Set "Positioning complete" assuming the next positioning data will not be executed.                                                        |
|                               | [Da.2]  | Control system                      | INC linear 3                                     | —                                                    | —                                                    | Set incremental system 3-axis linear interpolation control.                                                                                |
|                               | [Da.3]  | Acceleration time No.               | 1                                                | —                                                    | —                                                    | Designate the value set in " [Pr.25] Acceleration time 1" as the acceleration time at start.                                               |
|                               | [Da.4]  | Deceleration time No.               | 0                                                | —                                                    | —                                                    | Designate the value set in " [Pr.10] Deceleration time 0" as the deceleration time at deceleration.                                        |
|                               | [Da.5]  | Axis to be interpolated             | —                                                | —                                                    | —                                                    | Setting not required (setting value will be ignored).<br>When axis 1 is used as a reference axis, the interpolation axes are axes 2 and 3. |
|                               | [Da.6]  | Positioning address/movement amount | 10000.0μm                                        | 5000.0μm                                             | 6000.0μm                                             | Set the movement amount. (Assuming "mm" is set in " [Pr.1] Unit setting".)                                                                 |
|                               | [Da.7]  | Arc address                         | —                                                | —                                                    | —                                                    | Setting not required (setting value will be ignored).                                                                                      |
|                               | [Da.8]  | Command speed                       | 6000.00 mm/min                                   | —                                                    | —                                                    | Set the speed during movement.                                                                                                             |
|                               | [Da.9]  | Dwell time                          | 500ms                                            | —                                                    | —                                                    | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal.           |
|                               | [Da.10] | M code                              | 10                                               | —                                                    | —                                                    | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                                      |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

**POINTS**

- (1) When the "reference axis speed" is set during 3-axis linear interpolation control, set so the major axis side becomes the reference axis. If the minor axis side is set as the reference axis, the major axis side speed may exceed the " **Pr.8** Speed limit value".
- (2) Refer to Section 9.1.6 "Interpolation control" for the reference axis and interpolation axis combinations.



### 9.2.5 4-axis linear interpolation control

In "4-axis linear interpolation control" (" [Da.2](#) Control system" = ABS linear 4, INC linear 4), four motors are used to carry out position control in a linear path while carrying out interpolation for the axis directions set in each axis.  
(Refer to Section 9.1.6 "Interpolation control" for details on interpolation control.)

#### [1] 4-axis linear interpolation control (ABS linear 4)

In absolute system 4-axis linear interpolation control, 4 axes are used. Linear interpolation positioning is carried out from the current stop position (start point address) to the address (end point address) set in " [Da.6](#) Positioning address/movement amount".

### ■ Positioning data setting example

[Reference axis is designated as axis 1.]

The following table shows setting examples when "4-axis linear interpolation control (ABS linear 4)" is set in positioning data No. 1 of axis 1. (The required values are also set in positioning data No. 1 of axis 2, axis 3 and axis 4.)

| Setting item                  |       | Axis                                | Axis 1<br>(reference<br>axis) setting<br>example | Axis 2<br>(interpolation<br>axis) setting<br>example | Axis 3<br>(interpolation<br>axis) setting<br>example | Axis 4<br>(interpolation<br>axis) setting<br>example | Setting details                                                                                                                               |
|-------------------------------|-------|-------------------------------------|--------------------------------------------------|------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1 Positioning data No. 1 | Da.1  | Operation pattern                   | Positioning complete                             | —                                                    | —                                                    | —                                                    | Set "Positioning complete" assuming the next positioning data will not be executed.                                                           |
|                               | Da.2  | Control system                      | ABS linear 4                                     | —                                                    | —                                                    | —                                                    | Set absolute system 4-axis linear interpolation control.                                                                                      |
|                               | Da.3  | Acceleration time No.               | 1                                                | —                                                    | —                                                    | —                                                    | Designate the value set in " Pr.25 Acceleration time 1" as the acceleration time at start.                                                    |
|                               | Da.4  | Deceleration time No.               | 0                                                | —                                                    | —                                                    | —                                                    | Designate the value set in " Pr.10 Deceleration time 0" as the deceleration time at deceleration.                                             |
|                               | Da.5  | Axis to be interpolated             | —                                                | —                                                    | —                                                    | —                                                    | Setting not required (setting value will be ignored).<br>When axis 1 is used as a reference axis, the interpolation axes are axes 2, 3 and 4. |
|                               | Da.6  | Positioning address/movement amount | 4000.0μm                                         | 8000.0μm                                             | 4000.0μm                                             | 3000.0μm                                             | Set the end point address. (Assuming "mm" is set in " Pr.1 Unit setting".)                                                                    |
|                               | Da.7  | Arc address                         | —                                                | —                                                    | —                                                    | —                                                    | Setting not required (setting value will be ignored).                                                                                         |
|                               | Da.8  | Command speed                       | 6000.00 mm/min                                   | —                                                    | —                                                    | —                                                    | Set the speed during movement to the end point address.                                                                                       |
|                               | Da.9  | Dwell time                          | 500ms                                            | —                                                    | —                                                    | —                                                    | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal.              |
|                               | Da.10 | M code                              | 10                                               | —                                                    | —                                                    | —                                                    | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                                         |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

### POINTS

- (1) When the "reference axis speed" is set during 4-axis linear interpolation control, set so the major axis side becomes the reference axis. If the minor axis side is set as the reference axis, the major axis side speed may exceed the " Pr.8 Speed limit value".
- (2) Refer to Section 9.1.6 "Interpolation control" for the reference axis and interpolation axis combinations.

**[2] 4-axis linear interpolation control (INC linear 4)****■ Operation chart**

In incremental system 4-axis linear interpolation control, designated 4 axes are used. Linear interpolation positioning of movement amount set in " Da.6 Positioning address/movement amount" is carried out from the current stop position (start point address). The movement direction is determined by the sign of the movement amount.

### ■ Positioning data setting example

[Reference axis is designated as axis 1.]

The following table shows setting examples when "4-axis linear interpolation control (INC linear 4)" is set in positioning data No. 1 of axis 1. (The required values are also set in positioning data No. 1 of axis 2, axis 3 and axis 4.)

| Setting item |                        | Axis                                | Axis 1<br>(reference<br>axis) setting<br>example | Axis 2<br>(interpolation<br>axis) setting<br>example | Axis 3<br>(interpolation<br>axis) setting<br>example | Axis 4<br>(interpolation<br>axis) setting<br>example | Setting details                                                                                                                               |
|--------------|------------------------|-------------------------------------|--------------------------------------------------|------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1       | Positioning data No. 1 |                                     |                                                  |                                                      |                                                      |                                                      |                                                                                                                                               |
|              | Da.1                   | Operation pattern                   | Positioning complete                             | —                                                    | —                                                    | —                                                    | Set "Positioning complete" assuming the next positioning data will not be executed.                                                           |
|              | Da.2                   | Control system                      | INC linear 4                                     | —                                                    | —                                                    | —                                                    | Set incremental system 4-axis linear interpolation control.                                                                                   |
|              | Da.3                   | Acceleration time No.               | 1                                                | —                                                    | —                                                    | —                                                    | Designate the value set in " Pr.25 Acceleration time 1" as the acceleration time at start.                                                    |
|              | Da.4                   | Deceleration time No.               | 0                                                | —                                                    | —                                                    | —                                                    | Designate the value set in " Pr.10 Deceleration time 0" as the deceleration time at deceleration.                                             |
|              | Da.5                   | Axis to be interpolated             | —                                                | —                                                    | —                                                    | —                                                    | Setting not required (setting value will be ignored).<br>When axis 1 is used as a reference axis, the interpolation axes are axes 2, 3 and 4. |
|              | Da.6                   | Positioning address/movement amount | 4000.0μm                                         | 8000.0μm                                             | 4000.0μm                                             | 3000.0μm                                             | Set the movement amount. (Assuming "mm" is set in " Pr.1 Unit setting".)                                                                      |
|              | Da.7                   | Arc address                         | —                                                | —                                                    | —                                                    | —                                                    | Setting not required (setting value will be ignored).                                                                                         |
|              | Da.8                   | Command speed                       | 6000.00 mm/min                                   | —                                                    | —                                                    | —                                                    | Set the speed during movement.                                                                                                                |
|              | Da.9                   | Dwell time                          | 500ms                                            | —                                                    | —                                                    | —                                                    | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal.              |
|              | Da.10                  | M code                              | 10                                               | —                                                    | —                                                    | —                                                    | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                                         |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

### POINTS

- (1) When the "reference axis speed" is set during 4-axis linear interpolation control, set so the major axis side becomes the reference axis. If the minor axis side is set as the reference axis, the major axis side speed may exceed the " Pr.8 Speed limit value".
- (2) Refer to Section 9.1.6 "Interpolation control" for the reference axis and interpolation axis combinations.

## 9.2.6 1-axis fixed-feed control

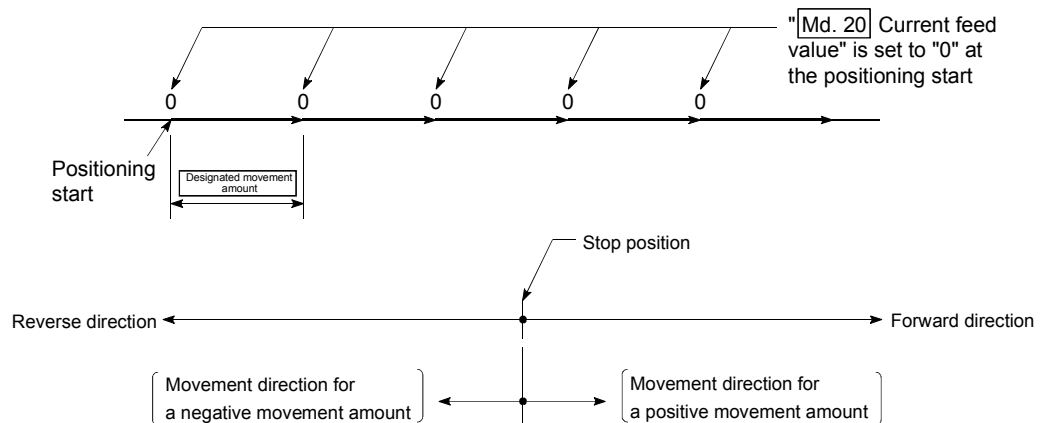
In "1-axis fixed-feed control" (" [Da.2] Control system" = fixed-feed 1), one motor is used to carry out fixed-feed control in a set axis direction.

In fixed-feed control, any remainder of the movement amount designated in the positioning data is rounded down if less than that required for control accuracy to output the same amount of pulses.

### ■ Operation chart

In 1-axis fixed-feed control, the address ([Md.20] Current feed value) of the current stop position (start point address) is set to "0". Positioning is then carried out to a position at the end of the movement amount set in " [Da.6] Positioning address/movement amount".

The movement direction is determined by the movement amount sign.



### ■ Restrictions

- (1) An axis error "Continuous path control not possible" (error code: 516) will occur and the operation cannot start if "continuous path control" is set in " [Da.1] Operation pattern". ("Continuous path control" cannot be set in fixed-feed control.)
- (2) "Fixed-feed" cannot be set in " [Da.2] Control system" in the positioning data when "continuous path control" has been set in " [Da.1] Operation pattern" of the immediately prior positioning data. (For example, if the operation pattern of positioning data No. 1 is "continuous path control", fixed-feed control cannot be set in positioning data No. 2.) An axis error "Continuous path control not possible" (error code: 516) will occur and the machine will carry out a deceleration stop if this type of setting is carried out.

**POINT**

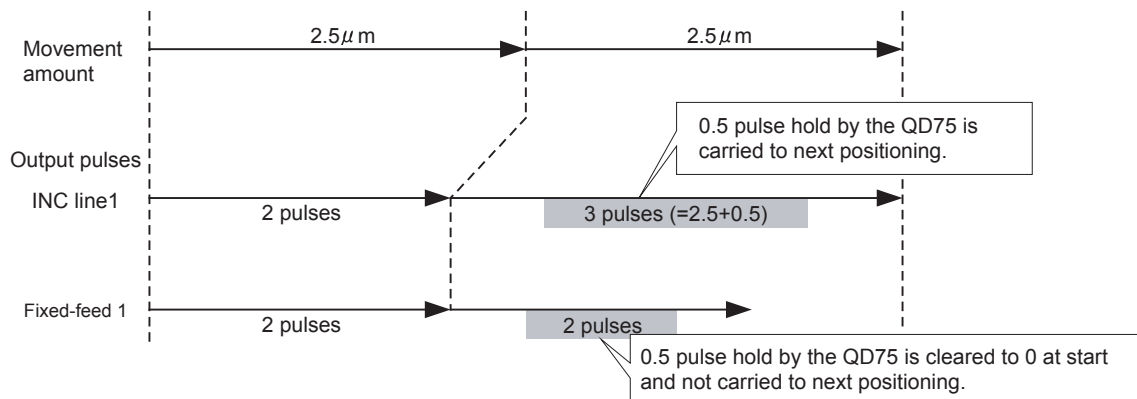
- When the movement amount is converted to the actual number of output pulses, a fraction appears after the decimal point, according to the movement amount per pulse. This fraction is normally retained in the QD75 and reflected at the next positioning.

For the fixed-feed control, since the movement distance is maintained constant (= the output number of pulses is maintained constant), the control is carried out after the fractional pulse is cleared to zero at start.

【Accumulation/cutoff for fractional pulses】

When movement amount per pulse is  $1.0\mu\text{m}$  and movement for  $2.5\mu\text{m}$  is executed two times;

⇒ Conversion to output pulses:  $2.5[\mu\text{m}] \div 1.0 = 2.5$  pulses



### ■ Positioning data setting example

The following table shows setting examples when "1-axis fixed-feed control (fixed-feed 1)" is set in positioning data No. 1 of axis 1.

| Setting item                  |                                          | Setting example      | Setting details                                                                                                                  |
|-------------------------------|------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Axis 1 Positioning data No. 1 | Da.1 Operation pattern                   | Positioning complete | Set "Positioning complete" assuming the next positioning data will not be executed.                                              |
|                               | Da.2 Control system                      | Fixed-feed 1         | Set 1-axis fixed-feed control.                                                                                                   |
|                               | Da.3 Acceleration time No.               | 1                    | Designate the value set in " Pr.25 Acceleration time 1" as the acceleration time at start.                                       |
|                               | Da.4 Deceleration time No.               | 0                    | Designate the value set in " Pr.10 Deceleration time 0" as the deceleration time at deceleration.                                |
|                               | Da.5 Axis to be interpolated             | —                    | Setting not required (setting value will be ignored).                                                                            |
|                               | Da.6 Positioning address/movement amount | 8000.0μm             | Set the positioning address. (Assuming "mm" is set in " Pr.1 Unit setting".)                                                     |
|                               | Da.7 Arc address                         | —                    | Setting not required (setting value will be ignored).                                                                            |
|                               | Da.8 Command speed                       | 6000.00mm/min        | Set the speed during movement to the positioning address.                                                                        |
|                               | Da.9 Dwell time                          | 500ms                | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal. |
|                               | Da.10 M code                             | 10                   | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                            |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

## 9.2.7 2-axis fixed-feed control (interpolation)

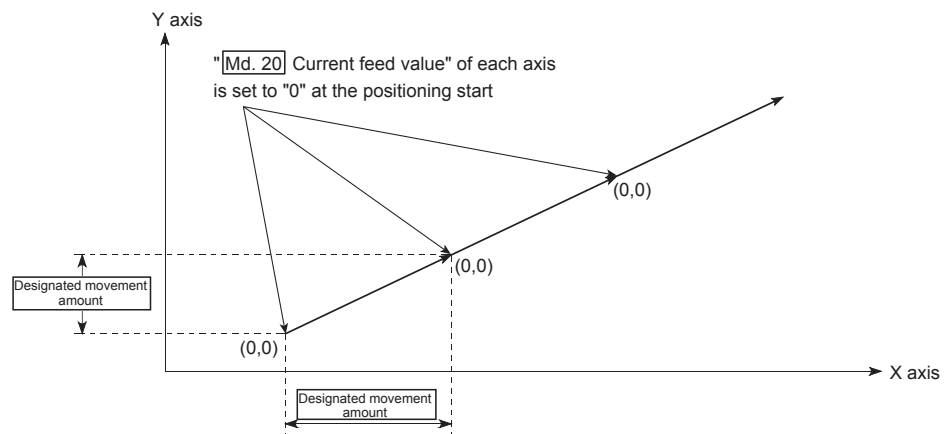
In "2-axis fixed-feed control" (" [Da.2] Control system" = fixed-feed 2), two motors are used to carry out fixed-feed control in a linear path while carrying out interpolation for the axis directions set in each axis.

In fixed-feed control, any remainder of the movement amount designated in the positioning data is rounded down if less than that required for control accuracy to output the same amount of pulses.

(Refer to Section 9.1.6 "Interpolation control" for details on interpolation control.)

### ■ Operation chart

In incremental system 2-axis fixed-feed control, the addresses ([Md.20] Current feed value) of the current stop position (start addresses) of both axes are set to "0". Linear interpolation positioning is then carried out from that position to a position at the end of the movement amount set in " [Da.6] Positioning address/movement amount". The movement direction is determined by the sign of the movement amount.



### ■ Restrictions

- (1) An axis error "Continuous path control not possible" (error code: 516) will occur and the operation cannot start if "continuous path control" is set in " [Da.1] Operation pattern". ("Continuous path control" cannot be set in fixed-feed control.)
- (2) If the movement amount of each axis exceeds "1073741824 ( $=2^{30}$ )" when "0: Composite speed" is set in " [Pr.20] Interpolation speed designation method", the error "Outside linear movement amount range" (error code: 504) occurs at a positioning start and positioning cannot be started. (The maximum movement amount that can be set in " [Da.6] Positioning address/movement amount" is "1073741824 ( $=2^{30}$ )".)
- (3) "Fixed-feed" cannot be set in " [Da.2] Control system" in the positioning data when "continuous path control" has been set in " [Da.1] Operation pattern" of the immediately prior positioning data. (For example, if the operation pattern of positioning data No. 1 is "continuous path control", fixed-feed control cannot be set in positioning data No. 2.) An axis error "Continuous path control not possible" (error code: 516) will occur and the machine will carry out a deceleration stop if this type of setting is carried out.

### ■ Positioning data setting example

[Reference axis and interpolation axis are designated as axis 1 and axis 2, respectively.]

The following table shows setting examples when "2-axis fixed-dimension feed control (fixed-feed 2)" is set in positioning data No. 1 of axis 1. (The required values are also set in positioning data No. 1 of axis 2.)

| Setting item |                        | Axis                                 | Axis 1<br>(reference<br>axis) setting<br>example | Axis 2<br>(interpolation<br>axis) setting<br>example | Setting details                                                                                                                              |
|--------------|------------------------|--------------------------------------|--------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1       | Positioning data No. 1 |                                      |                                                  |                                                      |                                                                                                                                              |
|              | [Da.1]                 | Operation pattern                    | Positioning complete                             | —                                                    | Set "Positioning complete" assuming the next positioning data will not be executed.                                                          |
|              | [Da.2]                 | Control system                       | Fixed-feed 2                                     | —                                                    | Set 2-axis fixed-feed control.                                                                                                               |
|              | [Da.3]                 | Acceleration time No.                | 1                                                | —                                                    | Designate the value set in " [Pr.25] Acceleration time 1" as the acceleration time at start.                                                 |
|              | [Da.4]                 | Deceleration time No.                | 0                                                | —                                                    | Designate the value set in " [Pr.10] Deceleration time 0" as the deceleration time at deceleration.                                          |
|              | [Da.5]                 | Axis to be interpolated              | Axis 2                                           | —                                                    | Set the axis to be interpolated (partner axis).<br>If the self-axis is set, an error will occur.                                             |
|              | [Da.6]                 | Positioning address/ movement amount | 8000.0μm                                         | 6000.0μm                                             | Set the positioning address. (Assuming "mm" is set in " [Pr.1] Unit setting".)                                                               |
|              | [Da.7]                 | Arc address                          | —                                                | —                                                    | Setting not required (setting value will be ignored).                                                                                        |
|              | [Da.8]                 | Command speed                        | 6000.00 mm/min                                   | —                                                    | Set the speed during movement. (Designate the composite speed of reference axis speed in " [Pr.20] Interpolation speed designation method".) |
|              | [Da.9]                 | Dwell time                           | 500ms                                            | —                                                    | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal.             |
|              | [Da.10]                | M code                               | 10                                               | —                                                    | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                                        |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.



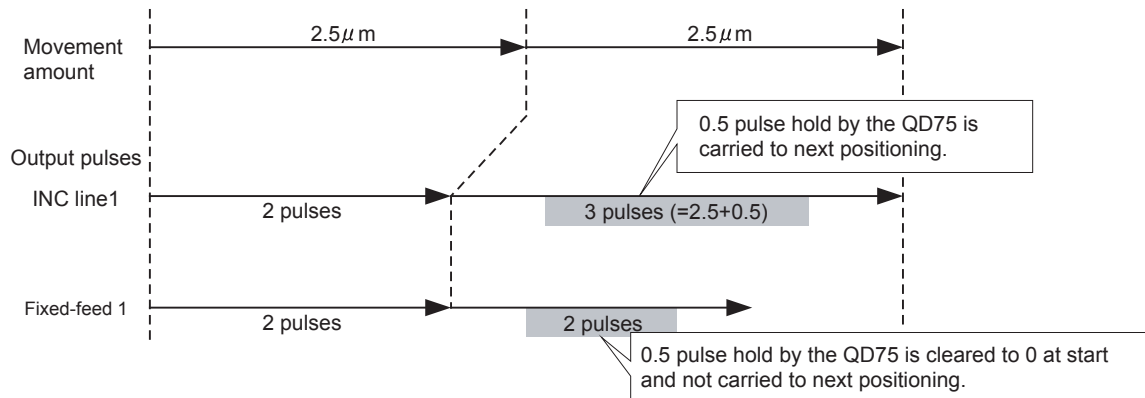
**POINTS**

- (1) When the movement amount is converted to the actual number of output pulses, a fraction appears after the decimal point, according to the movement amount per pulse. This fraction is normally retained in the QD75 and reflected at the next positioning.
- For the fixed-feed control, since the movement distance is maintained constant (= the output number of pulses is maintained constant), the control is carried out after the fractional pulse is cleared to zero at start.

**【Accumulation/cutoff for fractional pulses】**

When movement amount per pulse is  $1.0\mu\text{m}$  and movement for  $2.5\mu\text{m}$  is executed two times;

⇒ Conversion to output pulses:  $2.5[\mu\text{m}] \div 1.0 = 2.5$  pulses



- (2) When the "reference axis speed" is set during 2-axis fixed-feed control, set so the major axis side becomes the reference axis. If the minor axis side is set as the reference axis, the major axis side speed may exceed the " **Pr.8** Speed limit value".

### 9.2.8 3-axis fixed-feed control (interpolation)

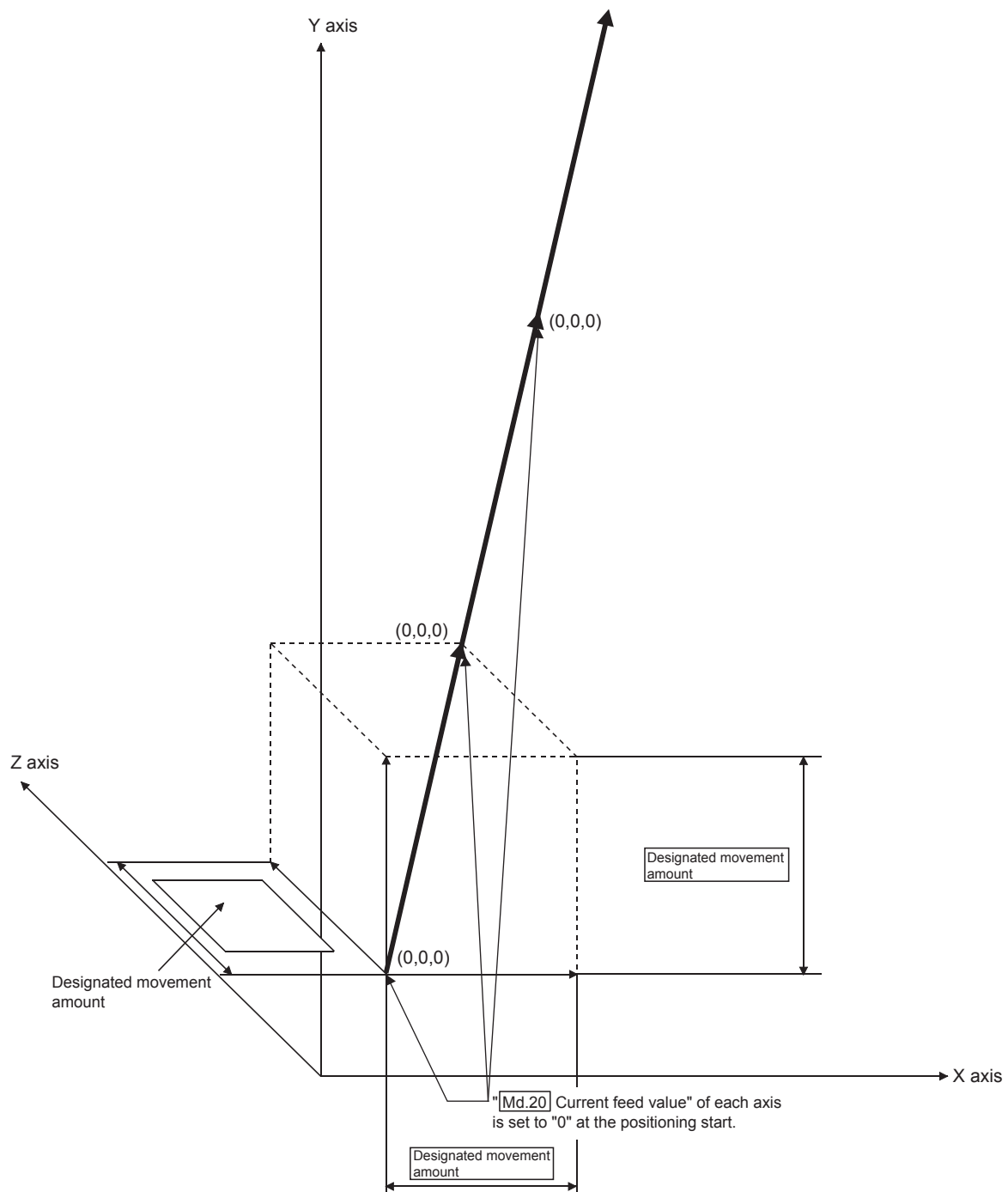
In "3-axis fixed-feed control" (" Da.2 Control system" = fixed-feed 3), three motors are used to carry out fixed-feed control in a linear path while carrying out interpolation for the axis directions set in each axis.

In fixed-feed control, any remainder of the movement amount designated in the positioning data is rounded down if less than that required for control accuracy to output the same amount of pulses.

(Refer to Section 9.1.6 "Interpolation control" for details on interpolation control.)

### ■ Operation chart

In incremental system 3-axis fixed-feed control, the addresses (**[Md.20]** Current feed value) of the current stop position (start addresses) of every axes are set to "0". Linear interpolation positioning is then carried out from that position to a position at the end of the movement amount set in "**[Da.6]** Positioning address/movement amount". The movement direction is determined by the sign of the movement amount.



**■ Restrictions**

- (1) An axis error "Continuous path control not possible" (error code: 516) will occur and the operation cannot start if "continuous path control" is set in "Da.1 Operation pattern". ("Continuous path control" cannot be set in fixed-feed control.)
- (2) If the movement amount of each axis exceeds "1073741824 ( $=2^{30}$ )" when "0: Composite speed" is set in "Pr.20 Interpolation speed designation method", the error "Outside linear movement amount range" (error code: 504) occurs at a positioning start and positioning cannot be started. (The maximum movement amount that can be set in "Da.6 Positioning address/movement amount" is "1073741824 ( $=2^{30}$ )".)
- (3) "Fixed-feed" cannot be set in "Da.2 Control system" in the positioning data when "continuous path control" has been set in "Da.1 Operation pattern" of the immediately prior positioning data. (For example, if the operation pattern of positioning data No. 1 is "continuous path control", fixed-feed control cannot be set in positioning data No. 2.) An axis error "Continuous path control not possible" (error code: 516) will occur and the machine will carry out a deceleration stop if this type of setting is carried out.

**■ Positioning data setting example**

[Reference axis is designated as axis 1.]

The following table shows setting examples when "3-axis fixed-feed control (fixed-feed 3)" is set in positioning data No. 1 of axis 1. (The required values are also set in positioning data No. 1 of axis 2 and axis 3.)

| Setting item                     |       | Axis                                |                      | Axis 1<br>(reference<br>axis) setting<br>example | Axis 2<br>(interpolation<br>axis) setting<br>example | Axis 3<br>(interpolation<br>axis) setting<br>example | Setting details                                                                                                                            |
|----------------------------------|-------|-------------------------------------|----------------------|--------------------------------------------------|------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
|                                  |       |                                     |                      |                                                  |                                                      |                                                      |                                                                                                                                            |
| Axis 1<br>Positioning data No. 1 | Da.1  | Operation pattern                   | Positioning complete | —                                                | —                                                    | —                                                    | Set "Positioning complete" assuming the next positioning data will not be executed.                                                        |
|                                  | Da.2  | Control system                      | Fixed-feed 3         | —                                                | —                                                    | —                                                    | Set 3-axis fixed-feed control.                                                                                                             |
|                                  | Da.3  | Acceleration time No.               | 1                    | —                                                | —                                                    | —                                                    | Designate the value set in " Pr.25 Acceleration time 1" as the acceleration time at start.                                                 |
|                                  | Da.4  | Deceleration time No.               | 0                    | —                                                | —                                                    | —                                                    | Designate the value set in " Pr.10 Deceleration time 0" as the deceleration time at deceleration.                                          |
|                                  | Da.5  | Axis to be interpolated             | —                    | —                                                | —                                                    | —                                                    | Setting not required (setting value will be ignored).<br>When axis 1 is used as a reference axis, the interpolation axes are axes 2 and 3. |
|                                  | Da.6  | Positioning address/movement amount | 10000.0μm            | 5000.0μm                                         | 6000.0μm                                             | —                                                    | Set the positioning address. (Assuming "mm" is set in " Pr.1 Unit setting".)                                                               |
|                                  | Da.7  | Arc address                         | —                    | —                                                | —                                                    | —                                                    | Setting not required (setting value will be ignored).                                                                                      |
|                                  | Da.8  | Command speed                       | 6000.00 mm/min       | —                                                | —                                                    | —                                                    | Set the speed during movement.                                                                                                             |
|                                  | Da.9  | Dwell time                          | 500ms                | —                                                | —                                                    | —                                                    | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal.           |
|                                  | Da.10 | M code                              | 10                   | —                                                | —                                                    | —                                                    | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                                      |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

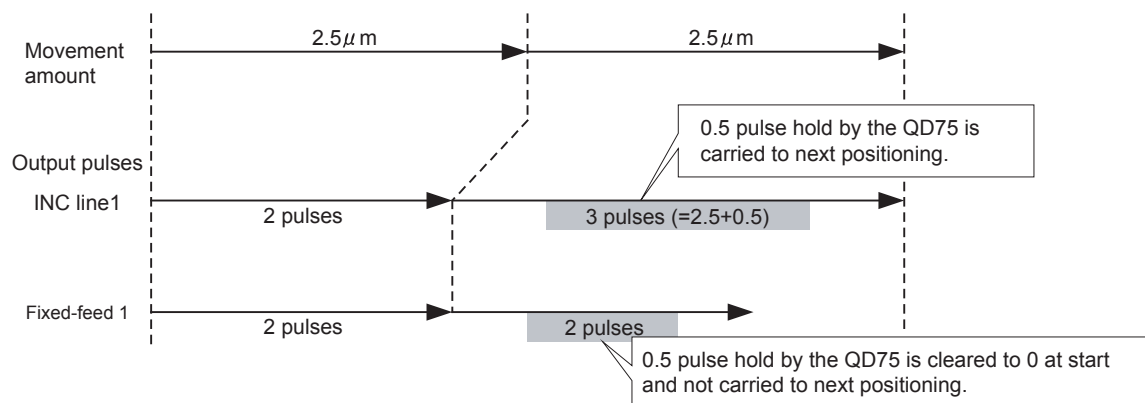
**POINTS**

- (1) When the movement amount is converted to the actual number of output pulses, a fraction appears after the decimal point, according to the movement amount per pulse. This fraction is normally retained in the QD75 and reflected at the next positioning.  
For the fixed-feed control, since the movement distance is maintained constant (= the output number of pulses is maintained constant), the control is carried out after the fractional pulse is cleared to zero at start.

**【Accumulation/cutoff for fractional pulses】**

When movement amount per pulse is  $1.0\mu\text{m}$  and movement for  $2.5\mu\text{m}$  is executed two times;

⇒ Conversion to output pulses:  $2.5[\mu\text{m}] \div 1.0 = 2.5$  pulses



- (2) When the "reference axis speed" is set during 3-axis fixed-feed control, set so the major axis side becomes the reference axis. If the minor axis side is set as the reference axis, the major axis side speed may exceed the " **Pr.8** Speed limit value".
- (3) Refer to Section 9.1.6 "Interpolation control" for the reference axis and interpolation axis combinations.

### 9.2.9 4-axis fixed-feed control (interpolation)

In "4-axis fixed-feed control" (" [Da.2](#) Control system" = fixed-feed 4), four motors are used to carry out fixed-feed control in a linear path while carrying out interpolation for the axis directions set in each axis.

In fixed-feed control, any remainder of the movement amount designated in the positioning data is rounded down if less than that required for control accuracy to output the same amount of pulses.

(Refer to Section 9.1.6 "Interpolation control" for details on interpolation control.)

#### ■ Operation chart

In incremental system 4-axis fixed-feed control, the addresses ([Md.20](#) Current feed value) of the current stop position (start addresses) of every axes are set to "0". Linear interpolation positioning is then carried out from that position to a position at the end of the movement amount set in " [Da.6](#) Positioning address/movement amount". The movement direction is determined by the sign of the movement amount.

#### ■ Restrictions

- (1) An axis error "Continuous path control not possible" (error code: 516) will occur and the operation cannot start if "continuous path control" is set in " [Da.1](#) Operation pattern". ("Continuous path control" cannot be set in fixed-feed control.)
- (2) For 4-axis fixed-feed control, set "1: Reference axis speed" in " [Pr.20](#) Interpolation speed designation method". An error "Interpolation mode error" (error code: 523) will occur and the operation cannot start if "0: Composite speed" is set.
- (3) "Fixed-feed" cannot be set in " [Da.2](#) Control system" in the positioning data when "continuous path control" has been set in " [Da.1](#) Operation pattern" of the immediately prior positioning data. (For example, if the operation pattern of positioning data No. 1 is "continuous path control", fixed-feed control cannot be set in positioning data No. 2.) An axis error "Continuous path control not possible" (error code: 516) will occur and the machine will carry out a deceleration stop if this type of setting is carried out.

### ■ Positioning data setting example

[Reference axis is designated as axis 1.]

The following table shows setting examples when "4-axis fixed-feed control (fixed-feed 4)" is set in positioning data No. 1 of axis 1. (The required values are also set in positioning data No. 1 of axis 2, axis 3 and axis 4.)

| Setting item                  |       | Axis                                | Axis 1<br>(reference<br>axis) setting<br>example | Axis 2<br>(interpolation<br>axis) setting<br>example | Axis 3<br>(interpolation<br>axis) setting<br>example | Axis 4<br>(interpolation<br>axis) setting<br>example | Setting details                                                                                                                               |
|-------------------------------|-------|-------------------------------------|--------------------------------------------------|------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1 Positioning data No. 1 | Da.1  | Operation pattern                   | Positioning complete                             | —                                                    | —                                                    | —                                                    | Set "Positioning complete" assuming the next positioning data will not be executed.                                                           |
|                               | Da.2  | Control system                      | Fixed-feed 4                                     | —                                                    | —                                                    | —                                                    | Set 4-axis fixed-feed control.                                                                                                                |
|                               | Da.3  | Acceleration time No.               | 1                                                | —                                                    | —                                                    | —                                                    | Designate the value set in " Pr.25 Acceleration time 1" as the acceleration time at start.                                                    |
|                               | Da.4  | Deceleration time No.               | 0                                                | —                                                    | —                                                    | —                                                    | Designate the value set in " Pr.10 Deceleration time 0" as the deceleration time at deceleration.                                             |
|                               | Da.5  | Axis to be interpolated             | —                                                | —                                                    | —                                                    | —                                                    | Setting not required (setting value will be ignored).<br>When axis 1 is used as a reference axis, the interpolation axes are axes 2, 3 and 4. |
|                               | Da.6  | Positioning address/movement amount | 4000.0μm                                         | 8000.0μm                                             | 4000.0μm                                             | 3000.0μm                                             | Set the positioning address. (Assuming "mm" is set in " Pr.1 Unit setting".)                                                                  |
|                               | Da.7  | Arc address                         | —                                                | —                                                    | —                                                    | —                                                    | Setting not required (setting value will be ignored).                                                                                         |
|                               | Da.8  | Command speed                       | 6000.00 mm/min                                   | —                                                    | —                                                    | —                                                    | Set the speed during movement.                                                                                                                |
|                               | Da.9  | Dwell time                          | 500ms                                            | —                                                    | —                                                    | —                                                    | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal.              |
|                               | Da.10 | M code                              | 10                                               | —                                                    | —                                                    | —                                                    | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                                         |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.



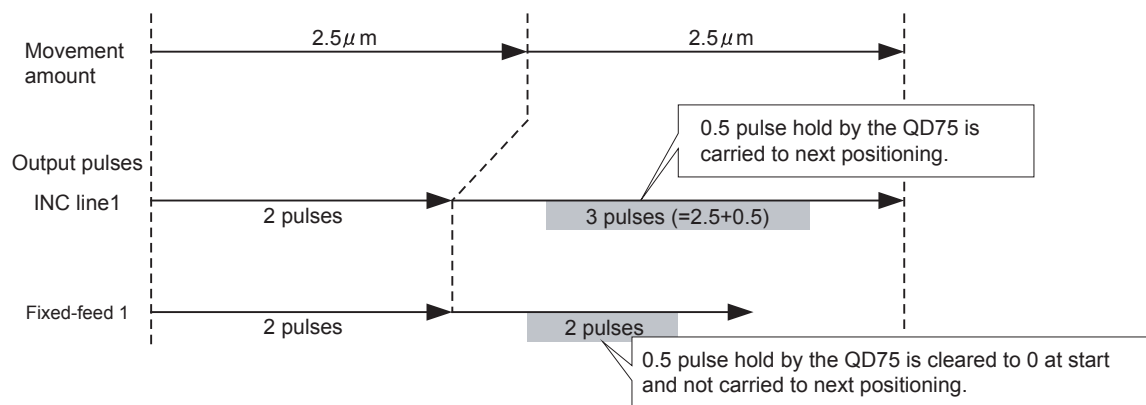
**POINTS**

- (1) When the movement amount is converted to the actual number of output pulses, a fraction appears after the decimal point, according to the movement amount per pulse. This fraction is normally retained in the QD75 and reflected at the next positioning. For the fixed-feed control, since the movement distance is maintained constant (= the output number of pulses is maintained constant), the control is carried out after the fractional pulse is cleared to zero at start.

**【Accumulation/cutoff for fractional pulses】**

When movement amount per pulse is  $1.0\mu\text{m}$  and movement for  $2.5\mu\text{m}$  is executed two times;

⇒ Conversion to output pulses:  $2.5[\mu\text{m}] \div 1.0 = 2.5$  pulses



- (2) For 4-axis fixed-feed control, set the "reference axis speed" and make setting so that the major axis side becomes the reference axis. If the minor axis side is set as the reference axis, the major axis side speed may exceed the " **Pr.8** Speed limit value".
- (3) Refer to Section 9.1.6 "Interpolation control" for the reference axis and interpolation axis combinations.

## 9.2.10 2-axis circular interpolation control with sub point designation

In "2-axis circular interpolation control" (" **Da.2** Control system" = ABS circular sub, INC circular sub), two motors are used to carry out position control in an arc path passing through designated sub points, while carrying out interpolation for the axis directions set in each axis.

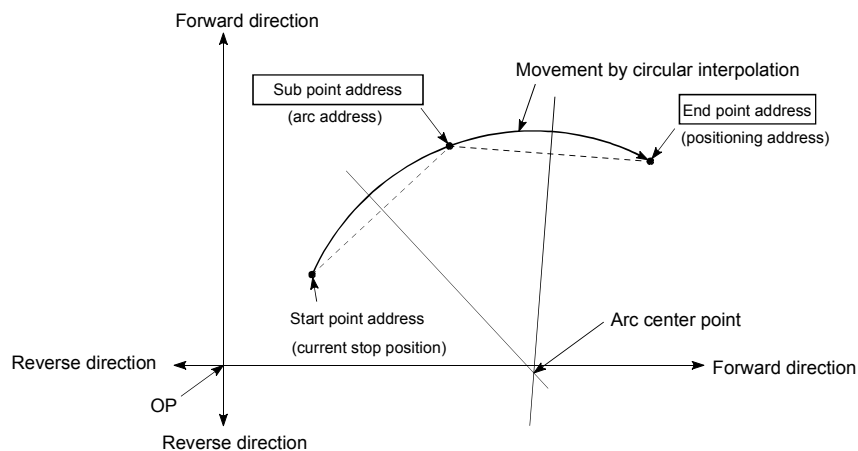
(Refer to Section 9.1.6 "Interpolation control" for details on interpolation control.)

[1] 2-axis circular interpolation control with sub point designation  
(ABS circular sub)

■ Operation chart

In the absolute system, 2-axis circular interpolation control with sub point designation, positioning is carried out from the current stop position (start point address) to the address (end point address) set in " **Da.6** Positioning address/movement amount", in an arc path that passes through the sub point address set in " **Da.7** Arc address".

The resulting control path is an arc having as its center the intersection point of perpendicular bisectors of a straight line between the start point address (current stop position) and sub point address (arc address), and a straight line between the sub point address (arc address) and end point address (positioning address).



**■ Restrictions**

- (1) 2-axis circular interpolation control cannot be set in the following cases.
  - When "degree" is set in " **Pr.1** Unit setting"
  - When the units set in " **Pr.1** Unit setting" are different for the reference axis and interpolation axis. ("mm" and "inch" combinations are possible.)
  - When "reference axis speed" is set in " **Pr.20** Interpolation speed designation method"
- (2) An error will occur and the positioning start will not be possible in the following cases. The machine will immediately stop if the error is detected during positioning control.
  - When the radius exceeds "536870912 ( $=2^{29}$ )". (The maximum radius for which circular interpolation control is possible is "536870912 ( $=2^{29}$ )"
    - ... An error "Outside radius range" (error code: 544) will occur at positioning start.
  - When the center point address is outside the range of "-2147483648 ( $-2^{31}$ ) to 2147483647 ( $2^{31}-1$ )"
    - ... An error "Sub point setting error" (error code: 525) will occur at positioning start.
  - When the start point address is the same as the end point address
    - ... An error "End point setting error" (error code: 526) will occur.
  - When the start point address is the same as the sub point address
    - ... An error "Sub point setting error" (error code: 525) will occur.
  - When the end point address is the same as the sub point address
    - ... An error "Sub point setting error" (error code: 525) will occur.
  - When the start point address, sub point address, and end point address are in a straight line
    - ... An error "Sub point setting error" (error code: 525) will occur.

### ■ Positioning data setting example

[Reference axis and interpolation axis are designated as axis 1 and axis 2, respectively.]

The following table shows setting examples when "2-axis circular interpolation control with sub point designation (ABS circular sub)" is set in positioning data No. 1 of axis 1. (The required values are also set in positioning data No. 1 of axis 2.)

| Setting item                  |       | Axis                                | Axis 1<br>(reference<br>axis) setting<br>example | Axis 2<br>(interpolation<br>axis) setting<br>example | Setting details                                                                                                                         |
|-------------------------------|-------|-------------------------------------|--------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1 Positioning data No. 1 | Da.1  | Operation pattern                   | Positioning complete                             | —                                                    | Set "Positioning complete" assuming the next positioning data will not be executed.                                                     |
|                               | Da.2  | Control system                      | ABS circular sub                                 | —                                                    | Set absolute system, 2-axis circular interpolation control with sub point designation.                                                  |
|                               | Da.3  | Acceleration time No.               | 1                                                | —                                                    | Designate the value set in " Pr.25 Acceleration time 1" as the acceleration time at start.                                              |
|                               | Da.4  | Deceleration time No.               | 0                                                | —                                                    | Designate the value set in " Pr.10 Deceleration time 0" as the deceleration time at deceleration.                                       |
|                               | Da.5  | Axis to be interpolated             | Axis 2                                           | —                                                    | Set the axis to be interpolated (partner axis). If the self-axis is set, an error will occur.                                           |
|                               | Da.6  | Positioning address/movement amount | 8000.0μm                                         | 6000.0μm                                             | Set the positioning address. (Assuming "mm" is set in " Pr.1 Unit setting".)                                                            |
|                               | Da.7  | Arc address                         | 4000.0μm                                         | 3000.0μm                                             | Set the sub point address. (Assuming that the " Pr.1 Unit setting" is set to "mm".)                                                     |
|                               | Da.8  | Command speed                       | 6000.00 mm/min                                   | —                                                    | Set the speed when moving to the end point address. (Designate the composite speed in " Pr.20 Interpolation speed designation method".) |
|                               | Da.9  | Dwell time                          | 500ms                                            | —                                                    | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal.        |
|                               | Da.10 | M code                              | 10                                               | —                                                    | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                                   |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

#### POINT

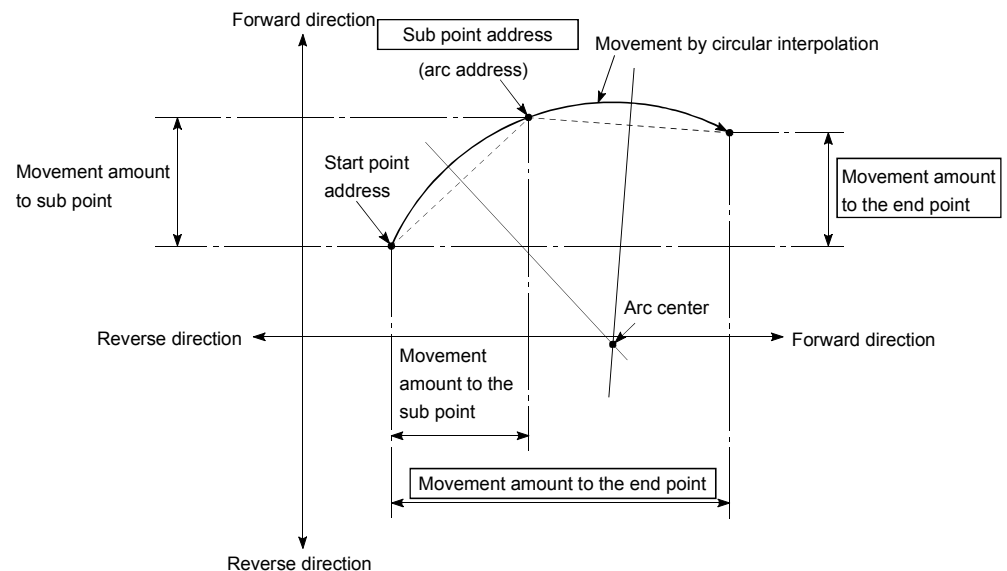
Set a value in " Da.8 Command speed" so that the speed of each axis does not exceed the " Pr.8 Speed limit value". (The speed limit does not function for the speed calculated by the QD75 during interpolation control.)

## [2] 2-axis circular interpolation control with sub point designation (INC circular sub)

### ■ Operation chart

In the incremental system, 2-axis circular interpolation control with sub point designation, positioning is carried out from the current stop position (start point address) to a position at the end of the movement amount set in " **Da.6** Positioning address/movement amount" in an arc path that passes through the sub point address set in " **Da.7** Arc address". The movement direction depends on the sign (+ or -) of the movement amount.

The resulting control path is an arc having as its center the intersection point of perpendicular bisectors of the straight line between the start point address (current stop position) and sub point address (arc address) calculated from the movement amount to the sub point, and a straight line between the sub point address (arc address) and end point address (positioning address) calculated from the movement amount to the end point.



### ■ Restrictions

- (1) 2-axis circular interpolation control cannot be set in the following cases.
  - When "degree" is set in " Pr.1 Unit setting"
  - When the units set in " Pr.1 Unit setting" are different for the reference axis and interpolation axis. ("mm" and "inch" combinations are possible.)
  - When "reference axis speed" is set in " Pr.20 Interpolation speed designation method"
- (2) An error will occur and the positioning start will not be possible in the following cases. The machine will immediately stop if the error is detected during positioning control.
  - When the radius exceeds "536870912 ( $=2^{29}$ )". (The maximum radius for which circular interpolation control is possible is "536870912 ( $=2^{29}$ )"  
 ... An error "Outside radius range" (error code: 544) will occur at positioning start.
  - When the auxiliary point address is outside the range of -2147483648 ( $-2^{31}$ ) to 2147483647 ( $2^{31}-1$ ).  
 ... An error "Sub point setting error" (error code: 525) will occur.
  - When the end point address is outside the range of -2147483648 ( $-2^{31}$ ) to 2147483647 ( $2^{31}-1$ ).  
 ... An error "End point setting error" (error code: 526) will occur.
  - When the auxiliary point address, center point address is outside the range of "-2147483648 ( $-2^{31}$ ) to 2147483647 ( $2^{31}-1$ )"  
 ... An error "Sub point setting error" (error code: 525) will occur at positioning start.
  - When the start point address is the same as the end point address  
 ... An error "End point setting error" (error code: 526) will occur.
  - When the start point address is the same as the sub point address  
 ... An error "Sub point setting error" (error code: 525) will occur.
  - When the end point address is the same as the sub point address  
 ... An error "Sub point setting error" (error code: 525) will occur.
  - When the start point address, sub point address, and end point address are in a straight line  
 ... An error "Sub point setting error" (error code: 525) will occur.

### ■ Positioning data setting example

[Reference axis and interpolation axis are designated as axis 1 and axis 2, respectively.]

The following table shows setting examples when "2-axis circular interpolation control with sub point designation (INC circular sub)" is set in positioning data No. 1 of axis 1. (The required values are also set in positioning data No. 1 of axis 2.)

| Setting item                  |       | Axis                                | Axis 1<br>(reference<br>axis) setting<br>example | Axis 2<br>(interpolation<br>axis) setting<br>example | Setting details                                                                                                                  |
|-------------------------------|-------|-------------------------------------|--------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Axis 1 Positioning data No. 1 | Da.1  | Operation pattern                   | Positioning complete                             | —                                                    | Set "Positioning complete" assuming the next positioning data will not be executed.                                              |
|                               | Da.2  | Control system                      | INC circular sub                                 | —                                                    | Set incremental system, 2-axis circular interpolation control with sub point designation.                                        |
|                               | Da.3  | Acceleration time No.               | 1                                                | —                                                    | Designate the value set in " Pr.25 Acceleration time 1" as the acceleration time at start.                                       |
|                               | Da.4  | Deceleration time No.               | 0                                                | —                                                    | Designate the value set in " Pr.10 Deceleration time 0" as the deceleration time at deceleration.                                |
|                               | Da.5  | Axis to be interpolated             | Axis 2                                           | —                                                    | Set the axis to be interpolated (partner axis).<br>If the self-axis is set, an error will occur.                                 |
|                               | Da.6  | Positioning address/movement amount | 8000.0μm                                         | 6000.0μm                                             | Set the movement amount. (Assuming that the " Pr.1 Unit setting" is set to "mm".)                                                |
|                               | Da.7  | Arc address                         | 4000.0μm                                         | -3000.0μm                                            | Set the sub point address. (Assuming that the " Pr.1 Unit setting" is set to "mm".)                                              |
|                               | Da.8  | Command speed                       | 6000.00 mm/min                                   | —                                                    | Set the speed during movement. (Designate the composite speed in " Pr.20 Interpolation speed designation method".)               |
|                               | Da.9  | Dwell time                          | 500ms                                            | —                                                    | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal. |
|                               | Da.10 | M code                              | 10                                               | —                                                    | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                            |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

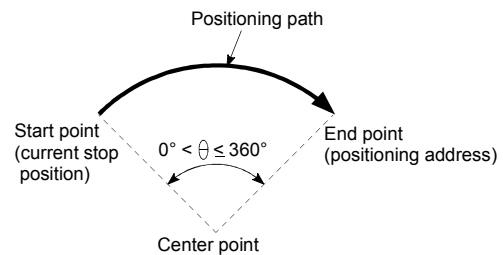
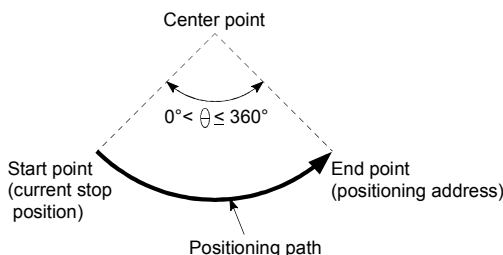
#### POINT

Set a value in " Da.8 Command speed" so that the speed of each axis does not exceed the " Pr.8 Speed limit value". (The speed limit does not function for the speed calculated by the QD75 during interpolation control.)

## 9.2.11 2-axis circular interpolation control with center point designation

In "2-axis circular interpolation control" (" **Da.2** Control system" = ABS circular right, INC circular right, ABS circular left, INC circular left), two motors are used to carry out position control in an arc path having an arc address as a center point, while carrying out interpolation for the axis directions set in each axis.  
(Refer to Section 9.1.6 "Interpolation control" for details on interpolation control.)

The following table shows the rotation directions, arc center angles that can be controlled, and positioning paths for the different control systems.

| Control system     | Rotation direction | Arc center angle that can be controlled | Positioning path                                                                     |
|--------------------|--------------------|-----------------------------------------|--------------------------------------------------------------------------------------|
| ABS circular right | Clockwise          | $0^{\circ} < \theta \leq 360^{\circ}$   |    |
| INC circular right |                    |                                         |                                                                                      |
| ABS circular left  | Counterclockwise   |                                         |  |
| INC circular left  |                    |                                         |                                                                                      |



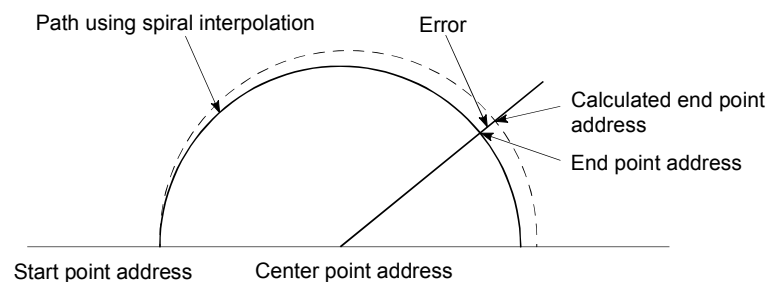
### ■ Circular interpolation error compensation

In circular interpolation control with center point designation, the arc path calculated from the start point address and center point address may deviate from the position of the end point address set in " [Da.6] Positioning address/movement amount".

(Refer to " [Pr.41] Allowable circular interpolation error width".)

- (1) Calculated error  $\leq$  " [Pr.41] Allowable circular interpolation error width"

Circular interpolation control to the set end point address is carried out while the error compensation is carried out. (This is called "spiral interpolation".)



In circular interpolation control with center point designation, an angular velocity is calculated on the assumption that operation is carried out at a command speed on the arc using the radius calculated from the start point address and center point address, and the radius is compensated in proportion to the angular velocity deviated from that at the start point.

Thus, when there is a difference (error) between a radius calculated from the start point address and center point address (start point radius) and a radius calculated from the end point address and center point address (end point radius), the composite speed differs from the command speed as follows.

- \* Start point radius > End point radius: As compared with the speed without error, the speed becomes slower as end point address is reached.
- \* Start point radius < End point radius: As compared with the speed without error, the speed becomes faster as end point address is reached.

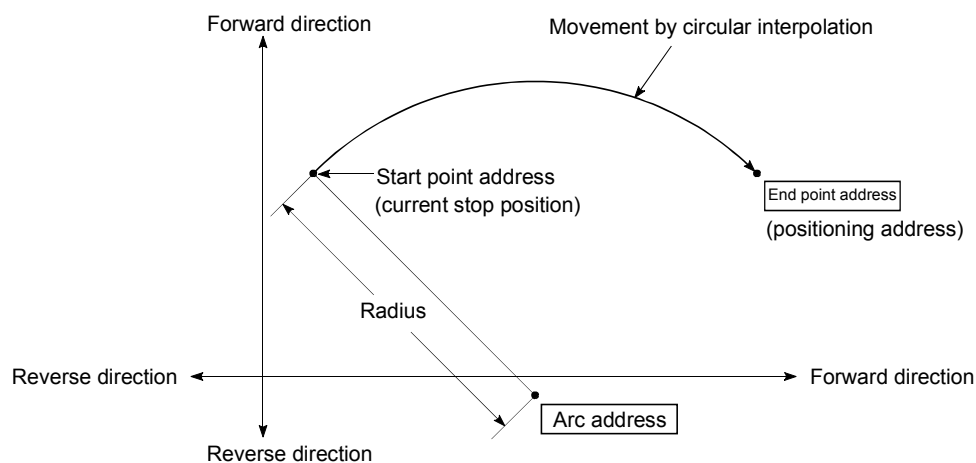
- (2) Calculated error > " [Pr.41] Allowable circular interpolation error width"

At the positioning start, an error "Large arc error deviation" (error code: 506) will occur and the control will not start. The machine will immediately stop if the error is detected during positioning control.

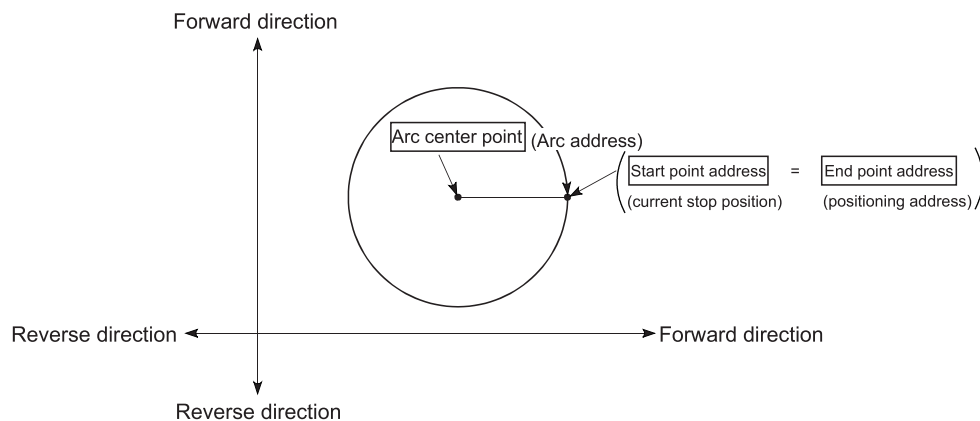
# [1] 2-axis circular interpolation control with center point designation (ABS circular right, ABS circular left)

## ■ Operation chart

In the absolute system, 2-axis circular interpolation control with center point designation, positioning is carried out from the current stop position (start point address) to the address (end point address) set in " **Da.6** Positioning address/movement amount" in an arc path having as its center the address (arc address) of the center point set in " **Da.7** Arc address".



Positioning of a complete round with a radius from the start point address to the arc center point can be carried out by setting the end point address (positioning address) to the same address as the start point address.



In circular interpolation control with center point designation, an angular velocity is calculated on the assumption that operation is carried out at a command speed on the arc using the radius calculated from the start point address and center point address, and the radius is compensated in proportion to the angular velocity deviated from that at the start point.

Thus, when there is a difference (error) between a radius calculated from the start point address and center point address (start point radius) and a radius calculated from the end point address and center point address (end point radius), the composite speed differs from the command speed as follows.

- \* Start point radius > End point radius: As compared with the speed without error, the speed becomes slower as end point address is reached.
- \* Start point radius < End point radius: As compared with the speed without error, the speed becomes faster as end point address is reached.

#### ■ Restrictions

- (1) 2-axis circular interpolation control cannot be set in the following cases.
  - When "degree" is set in " Pr.1 Unit setting"
  - When the units set in " Pr.1 Unit setting" are different for the reference axis and interpolation axis. ("mm" and "inch" combinations are possible.)
  - When "reference axis speed" is set in " Pr.20 Interpolation speed designation method"
- (2) An error will occur and the positioning start will not be possible in the following cases. The machine will immediately stop if the error is detected during positioning control.
  - When the radius exceeds "536870912 ( $=2^{29}$ )". (The maximum radius for which circular interpolation control is possible is "536870912 ( $=2^{29}$ )"  
... An error "Outside radius range" (error code: 544) will occur at positioning start.
  - When the start point address is the same as the center point address  
... An error "Center point setting error" (error code: 527) will occur.
  - When the end point address is the same as the center point address  
... An error "Center point setting error" (error code: 527) will occur.
  - When the center point address is outside the range of  $-2147483648(-2^{31})$  to  $2147483647(2^{31}-1)$   
... An error "Center point setting error" (error code: 527) will occur.

### ■ Positioning data setting examples

[Reference axis and interpolation axis are designated as axis 1 and axis 2, respectively.]

The following table shows setting examples when "2-axis circular interpolation control with center point designation (ABS right arc, ABS left arc)" is set in positioning data No. 1 of axis 1. (The required values are also set in positioning data No. 1 of axis 2.)

| Setting item |                        | Axis                                    | Axis 1<br>(reference<br>axis) setting<br>example | Axis 2<br>(interpolation<br>axis) setting<br>example | Setting details                                                                                                                                            |
|--------------|------------------------|-----------------------------------------|--------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1       | Positioning data No. 1 |                                         |                                                  |                                                      |                                                                                                                                                            |
|              | <a href="#">Da.1</a>   | Operation pattern                       | Positioning complete                             | —                                                    | Set "Positioning complete" assuming the next positioning data will not be executed.                                                                        |
|              | <a href="#">Da.2</a>   | Control system                          | ABS circular right<br>ABS circular left          | —                                                    | Set absolute system, 2-axis circular interpolation control with center point designation. (Select clockwise or counterclockwise according to the control.) |
|              | <a href="#">Da.3</a>   | Acceleration time No.                   | 1                                                | —                                                    | Designate the value set in " <a href="#">Pr.25</a> Acceleration time 1" as the acceleration time at start.                                                 |
|              | <a href="#">Da.4</a>   | Deceleration time No.                   | 0                                                | —                                                    | Designate the value set in " <a href="#">Pr.10</a> Deceleration time 0" as the deceleration time at deceleration.                                          |
|              | <a href="#">Da.5</a>   | Axis to be interpolated                 | Axis 2                                           | —                                                    | Set the axis to be interpolated (partner axis).<br>If the self-axis is set, an error will occur.                                                           |
|              | <a href="#">Da.6</a>   | Positioning address/<br>movement amount | 8000.0μm                                         | 6000.0μm                                             | Set the positioning address. (Assuming "mm" is set in " <a href="#">Pr.1</a> Unit setting".)                                                               |
|              | <a href="#">Da.7</a>   | Arc address                             | 4000.0μm                                         | 3000.0μm                                             | Set the arc address. (Assuming that the " <a href="#">Pr.1</a> Unit setting" is set to "mm".)                                                              |
|              | <a href="#">Da.8</a>   | Command speed                           | 6000.00<br>mm/min                                | —                                                    | Set the speed when moving to the end point address. (Designate the composite speed in " <a href="#">Pr.20</a> Interpolation speed designation method".)    |
|              | <a href="#">Da.9</a>   | Dwell time                              | 500ms                                            | —                                                    | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal.                           |
|              | <a href="#">Da.10</a>  | M code                                  | 10                                               | —                                                    | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                                                      |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

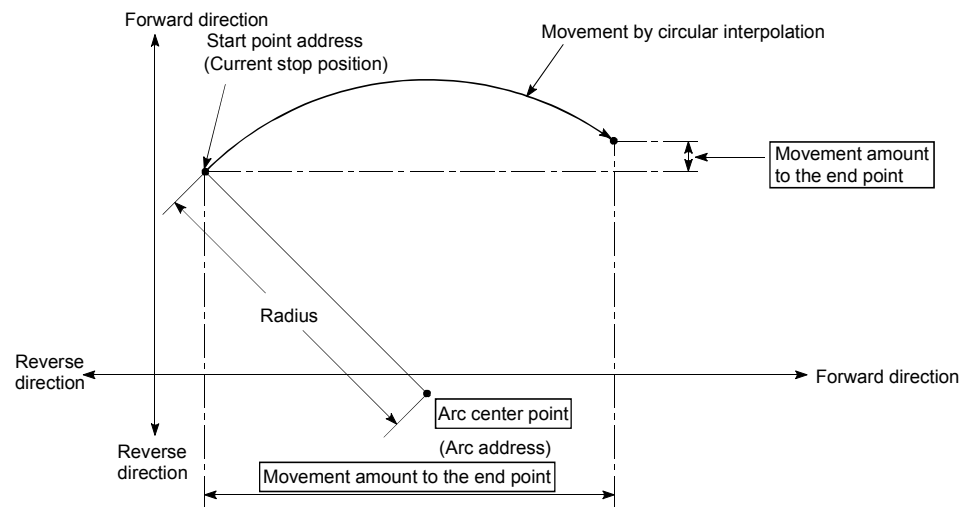
#### POINT

Set a value in " [Da.8](#) Command speed" so that the speed of each axis does not exceed the " [Pr.8](#) Speed limit value". (The speed limit does not function for the speed calculated by the QD75 during interpolation control.)

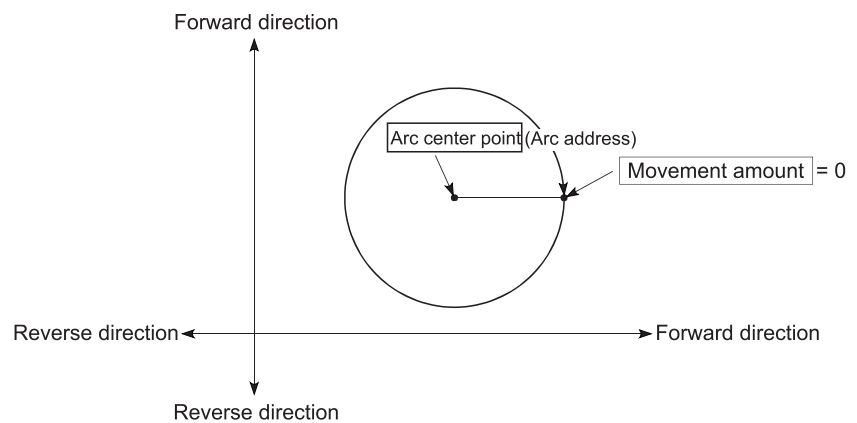
## [2] 2-axis circular interpolation control with center point designation (INC circular right, INC circular left)

### ■ Operation chart

In the incremental system, 2-axis circular interpolation control with center point designation, positioning is carried out from the current stop position (start point address) to a position at the end of the movement amount set in " **Da.6** Positioning address/movement amount" in an arc path having as its center the address (arc address) of the center point set in " **Da.7** Arc address".



Positioning of a complete round with a radius of the distance from the start point address to the arc center point can be carried out by setting the movement amount to "0".



In circular interpolation control with center point designation, an angular velocity is calculated on the assumption that operation is carried out at a command speed on the arc using the radius calculated from the start point address and center point address, and the radius is compensated in proportion to the angular velocity deviated from that at the start point.

Thus, when there is a difference (error) between a radius calculated from the start point address and center point address (start point radius) and a radius calculated from the end point address and center point address (end point radius), the composite speed differs from the command speed as follows.

- \* Start point radius > End point radius: As compared with the speed without error, the speed becomes slower as end point address is reached.
- \* Start point radius < End point radius: As compared with the speed without error, the speed becomes faster as end point address is reached.

### ■ Restrictions

(1) 2-axis circular interpolation control cannot be set in the following cases.

- When "degree" is set in " Pr.1 Unit setting"
- When the units set in " Pr.1 Unit setting" are different for the reference axis and interpolation axis. ("mm" and "inch" combinations are possible.)
- When "reference axis speed" is set in " Pr.20 Interpolation speed designation method"

(2) An error will occur and the positioning start will not be possible in the following cases. The machine will immediately stop if the error is detected during positioning control.

- When the radius exceeds "536870912 ( $=2^{29}$ )". (The maximum radius for which circular interpolation control is possible is "536870912 ( $=2^{29}$ )"  
... An error "Outside radius range" (error code: 544) will occur at positioning start.
- When the end point address is outside the range of  $-2147483648(-2^{31})$  to  $2147483647(2^{31}-1)$   
... An error "End point setting error" (error code: 526) will occur.
- When the start point address is the same as the center point address  
... An error "Center point setting error" (error code: 527) will occur.
- When the end point address is the same as the center point address  
... An error "Center point setting error" (error code: 527) will occur.
- When the center point address is outside the range of  $-2147483648(-2^{31})$  to  $2147483647(2^{31}-1)$   
... An error "Center point setting error" (error code: 527) will occur.

### ■ Positioning data setting examples

[Reference axis and interpolation axis are designated as axis 1 and axis 2, respectively.]

The following table shows setting examples when "2-axis circular interpolation control with center point designation (INC circular right, INC circular left)" is set in positioning data No. 1 of axis 1. (The required values are also set in positioning data No. 1 of axis 2.)

| Setting item |                        | Axis                                    | Axis 1<br>(reference<br>axis) setting<br>example | Axis 2<br>(interpolation<br>axis) setting<br>example | Setting details                                                                                                                                               |
|--------------|------------------------|-----------------------------------------|--------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1       | Positioning data No. 1 |                                         |                                                  |                                                      |                                                                                                                                                               |
|              | <a href="#">Da.1</a>   | Operation pattern                       | Positioning complete                             | —                                                    | Set "Positioning complete" assuming the next positioning data will not be executed.                                                                           |
|              | <a href="#">Da.2</a>   | Control system                          | INC circular right<br>INC circular left          | —                                                    | Set incremental system, 2-axis circular interpolation control with center point designation. (Select clockwise or counterclockwise according to the control.) |
|              | <a href="#">Da.3</a>   | Acceleration time No.                   | 1                                                | —                                                    | Designate the value set in " <a href="#">Pr.25</a> Acceleration time 1" as the acceleration time at start.                                                    |
|              | <a href="#">Da.4</a>   | Deceleration time No.                   | 0                                                | —                                                    | Designate the value set in " <a href="#">Pr.10</a> Deceleration time 0" as the deceleration time at deceleration.                                             |
|              | <a href="#">Da.5</a>   | Axis to be interpolated                 | Axis 2                                           | —                                                    | Set the axis to be interpolated (partner axis).<br>If the self-axis is set, an error will occur.                                                              |
|              | <a href="#">Da.6</a>   | Positioning address/<br>movement amount | 8000.0μm                                         | 6000.0μm                                             | Set the movement amount. (Assuming that the " <a href="#">Pr.1</a> Unit setting" is set to "mm".)                                                             |
|              | <a href="#">Da.7</a>   | Arc address                             | 4000.0μm                                         | 3000.0μm                                             | Set the center point address. (Assuming that the " <a href="#">Pr.1</a> Unit setting" is set to "mm".)                                                        |
|              | <a href="#">Da.8</a>   | Command speed                           | 6000.00<br>mm/min                                | —                                                    | Set the speed when moving to the end point address. (Designate the composite speed in " <a href="#">Pr.20</a> Interpolation speed designation method".)       |
|              | <a href="#">Da.9</a>   | Dwell time                              | 500ms                                            | —                                                    | Set the time the machine dwells after the positioning stop (pulse output stop) to the output of the positioning complete signal.                              |
|              | <a href="#">Da.10</a>  | M code                                  | 10                                               | —                                                    | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                                                         |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

| POINT                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Set a value in " <a href="#">Da.8</a> Command speed" so that the speed of each axis does not exceed the " <a href="#">Pr.8</a> Speed limit value". (The speed limit does not function for the speed calculated by the QD75 during interpolation control.) |

## 9.2.12 3-axis helical interpolation control with sub point designation

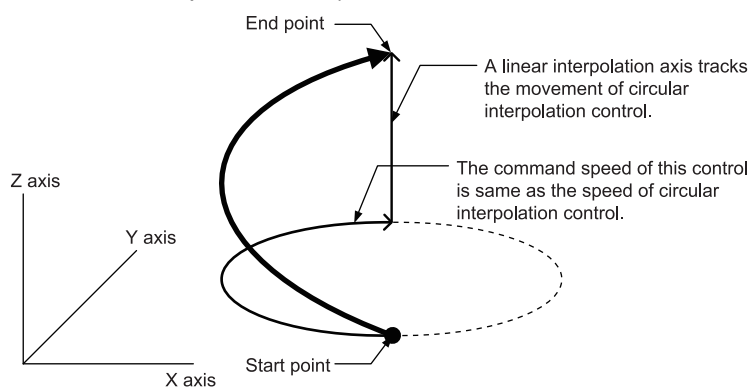
In the 3-axis helical interpolation control, circular interpolation control is carried out with two of the three axes. The other axis tracks the movement of the circular interpolation control to carry out "helical control" or "tangential control". (\* Refer to Section 9.1.6 "Interpolation control" for interpolation control.)

| Control details    | Outline of control | Positioning path                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Helical control    |                    | <p>In this control, the linear interpolation axis (linear control) tracks the movement of circular interpolation control, and a helical path is drawn.</p> <p>■ Unit setting</p> <ul style="list-style-type: none"> <li>• Reference axis: mm/inch, pulse</li> <li>• Circular interpolation axis: Same setting as the one for the reference axis</li> <li>• Linear interpolation axis: mm/inch, pulse</li> </ul>                                                                                                                                                                                                            |
| Tangential control |                    | <p>In this control, the value of the linear interpolation axis (degree) changes according to the movement of circular interpolation control to keep the direction of a workpiece to the center point of the circular interpolation control. The figure on the left shows an example of the control that keeps the direction of the workpiece in the tangential direction.</p> <p>■ Unit setting</p> <ul style="list-style-type: none"> <li>• Reference axis: mm/inch, pulse</li> <li>• Circular interpolation axis: Same setting as the one for the reference axis</li> <li>• Linear interpolation axis: degree</li> </ul> |

X-axis: Reference axis, Y-axis: Circular interpolation axis, Z-axis: Linear interpolation axis

■ Helical interpolation control speed

The command speed of the 3-axis helical interpolation control is the same as the speed of circular interpolation control (composite speed of reference axis and circular interpolation axis).



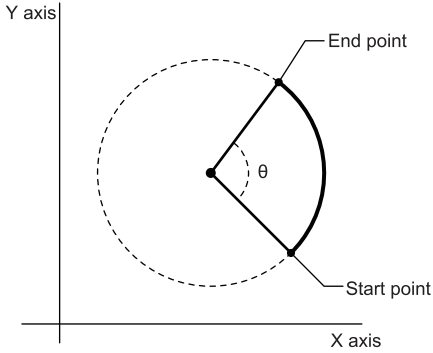


**POINT**

When "0: Composite speed" is set in "[Pr.20] Interpolation speed designation method", the command speed of ABS3/INC3 is the same as the composite speed of the 3 axes (x axis - y axis - z axis). The command speed of 3-axis helical interpolation control is the same as the composite speed of the circular interpolation axis (x axis - y axis). When continuous path control is carried out in ABS3/INC3 and 3-axis helical interpolation control, adjust the command speed to prevent the workpiece from moving unsteadily because the workpiece movement speed may change when positioning data is switched.

■ Rotation angle of the circular interpolation axis (X axis - Y axis)

The following shows the rotation angle of the circular interpolation axis in 3-axis helical interpolation control.

| Number of pitch | Circular interpolation axis control                                                 |  |
|-----------------|-------------------------------------------------------------------------------------|--|
|                 |  |  |
| 0               | $\theta^\circ$                                                                      |  |
| 1               | $360^\circ + \theta^\circ$                                                          |  |
| 2               | $720^\circ + \theta^\circ$                                                          |  |
| to              | to                                                                                  |  |
| n               | $360^\circ \times n + \theta^\circ$                                                 |  |
| to              | to                                                                                  |  |
| 999             | $360^\circ \times 999 + \theta^\circ$                                               |  |

**POINT**

When "degree" is set for the unit setting in the absolute system, the positioning range is 0 to 359.99999°. When the axis rotates 360° or more in circular interpolation control (x axis - y axis), the tangential/normal control cannot be carried out because a degree of 360° or more cannot be set in linear control (z axis: degree). To carry out tangential/normal control in which the axis rotates 360° or more, use 3-axis helical interpolation control (INC).

■ Rotation direction when the unit for the linear axis (z axis) is degree

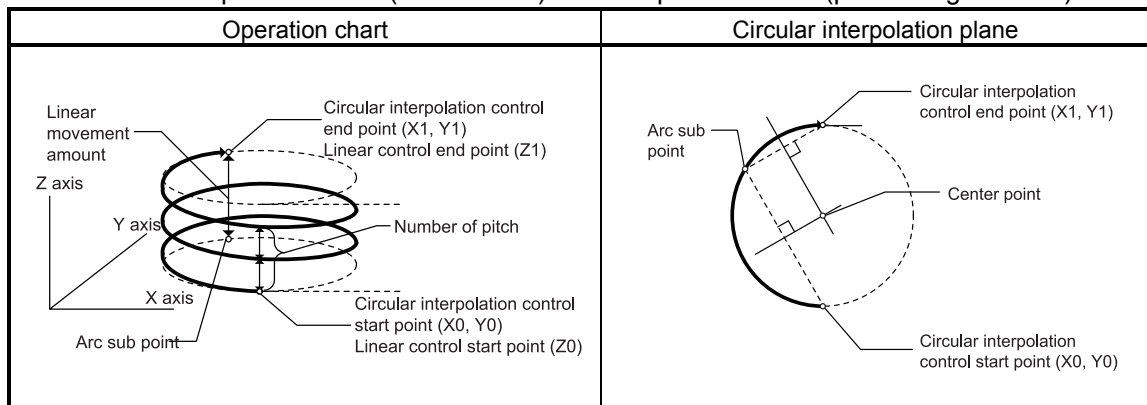
When "[Pr.1] Unit setting" for the linear axis is "degree", the rotation direction is determined according to the axis control data "[Cd.40] ABS direction in degrees" of the reference axis. To set the rotation direction for each positioning data, set the position data "[Da.28] ABS direction in degrees".

# [1] 3-axis helical interpolation control with sub point designation (ABS helical sub)

## ■ Operation chart

In this control, positioning is carried out from the current stop position (X0, Y0, Z0) to the circular end address (X1, Y1) and linear axis end address (Z1) set in "[Da.6] Positioning address/movement amount". Circular interpolation where the arc path passes through the address of the sub point (sub point address) set in "[Da.7] Arc address" is carried out. Simultaneously, linear interpolation is carried out with the other linear axis. The control path is helically rotated for the number of pitch set in "[Da.10] M code" of the linear axis to complete the positioning to the specified position.

The resulting control path is an arc having its center as the intersection point of perpendicular bisectors of a straight line between the start point address (current stop position) and sub point address (arc address), and a straight line between the sub point address (arc address) and end point address (positioning address).



## ■ Restrictions

(1) In the following cases, 3-axis helical interpolation control cannot be set.

- When "degree" is set in "[Pr.1] Unit setting" for the reference axis and circular interpolation axis
- When the units set in "[Pr.1] Unit setting" are different between the reference axis and circular interpolation axis. ("mm" and "inch" combinations are possible.)
- When a method other than "composite speed" is set in "[Pr.20] Interpolation speed designation method" or "[Da.29] Interpolation speed designation method"

- (2) An error will occur and the positioning start will not be possible in the following cases. The machine will immediately stop if the error is detected during positioning control.
- When the radius exceeds " $536870912 (=2^{29})$ "  
(The maximum radius for which circular interpolation control is possible is " $536870912 (=2^{29})$ ".)  
... An error "Outside radius range" (error code: 544) will occur at positioning start.
  - When the center point address is outside the range of " $-2147483648 (-2^{31})$  to  $2147483647 (2^{31}-1)$ "  
... An error "Sub point setting error" (error code: 525) will occur at positioning start.
  - Start point address = End point address  
... Error "End point setting error" (error code: 526)
  - Start point address = Sub point address  
... Error "Sub point setting error" (error code: 525)
  - End point address = Sub point address  
... Error "Sub point setting error" (error code: 525)
  - When the start point address, end point address, and sub point address are in a straight line  
... Error "Sub point setting error" (error code: 525)

### ■ Positioning data setting example

[Reference axis, circular interpolation axis, and linear interpolation axis are specified as axis 1, axis 2, and axis 3, respectively.]

The following table shows setting examples when "helical interpolation control with sub point designation (ABS)" is set in positioning data No. 1 of axis 1.

(The required values are also set in positioning data No. 1 of axis 2 and axis 3.)

| Item \ Axis            |                                          | Axis 1<br>(reference axis) setting example | Axis 2 (circular interpolation axis) setting example | Axis 3 (linear interpolation axis) setting example | Setting details                                                                                                                                                          |
|------------------------|------------------------------------------|--------------------------------------------|------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positioning data No. 1 | Da.1 Operation pattern                   | Positioning complete                       | -                                                    | -                                                  | Set "Positioning complete" assuming that the next positioning data will not be executed.                                                                                 |
|                        | Da.2 Control system                      | ABS helical sub                            | -                                                    | -                                                  | Set absolute system, 3-axis helical interpolation control with sub point designation.                                                                                    |
|                        | Da.3 Acceleration time No.               | 1                                          | -                                                    | -                                                  | Specify the value set in "[Pr.25] Acceleration time 1" as the acceleration time at start.                                                                                |
|                        | Da.4 Deceleration time No.               | 0                                          | -                                                    | -                                                  | Specify the value set in "[Pr.10] Deceleration time 0" as the deceleration time at deceleration.                                                                         |
|                        | Da.5 Axis to be interpolated             | Axis 2                                     | -                                                    | -                                                  | Set the axis to be interpolated (partner axis). If the self-axis is set, an error will occur.                                                                            |
|                        | Da.6 Positioning address/movement amount | 8000.0μm                                   | 6000.0μm                                             | 3000.0μm                                           | Set the end point address. (Assuming "mm" is set in "[Pr.1] Unit setting".)                                                                                              |
|                        | Da.7 Arc address                         | 4000.0μm                                   | 3000.0μm                                             | -                                                  | Set the sub point address. (Assuming "mm" is set in "[Pr.1] Unit setting".)                                                                                              |
|                        | Da.8 Command speed                       | 6000.00mm/min                              | -                                                    | -                                                  | Set the speed during movement to the end point address. (Set the composite speed in "[Pr.20] Interpolation speed designation method".)                                   |
|                        | Da.9 Dwell time                          | 500ms                                      | -                                                    | -                                                  | Set the time the machine dwells from the positioning stop (pulse output stop) to the output of the positioning complete signal.                                          |
|                        | Da.10 M code                             | 10                                         | -                                                    | 50                                                 | Set this item when other sub operation commands are issued in combination with the No. 1 positioning data.<br>Set the number of pitch for the linear interpolation axis. |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

#### POINT

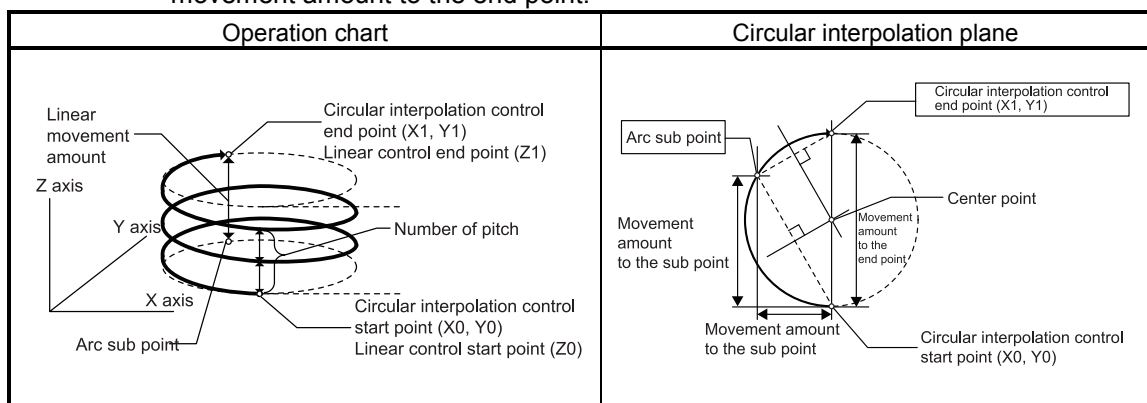
Set a value in "[Da.8] Command speed" so that the speed of each axis does not exceed the "[Pr.8] Speed limit value". (The speed limit does not function for the speed calculated by the QD75N during interpolation control.)

## [2] 3-axis helical interpolation control with sub point designation (INC helical sub)

### ■ Operation chart

In this control, positioning is carried out from the current stop position (X0, Y0, Z0) to the movement amount position (X1, Y1, Z1) set in "[Da.6] Positioning address/movement amount". Circular interpolation where the arc path passes through the address of the sub point (sub point address) set in "[Da.7] Arc address" is carried out. Simultaneously, linear interpolation is carried out with the other linear axis. The control path is helically rotated for the number of pitch set in "[Da.10] M code" of the linear axis to complete the positioning to the specified position. The movement direction is determined by the sign of the movement amount.

The resulting control path is an arc having its center as the intersection point of perpendicular bisectors of the straight line between the start point address (current stop position) and sub point address (arc address) calculated from the movement amount to the sub point, and a straight line between the sub point address (arc address) and end point address (positioning address) calculated from the movement amount to the end point.



### ■ Restrictions

(1) In the following cases, 3-axis helical interpolation control cannot be set.

- When "degree" is set in "[Pr.1] Unit setting" for the reference axis and circular interpolation axis
- When the units set in "[Pr.1] Unit setting" are different between the reference axis and circular interpolation axis. ("mm" and "inch" combinations are possible.)
- When a method other than "composite speed" is set in "[Pr.20] Interpolation speed designation method" or "[Da.29] Interpolation speed designation method"

- (2) An error will occur and the positioning start will not be possible in the following cases. The machine will immediately stop if the error is detected during positioning control.
- When the radius exceeds "536870912 ( $=2^{29}$ )"  
(The maximum radius for which circular interpolation control is possible is "536870912 ( $=2^{29}$ )".)  
... An error "Outside radius range" (error code: 544) will occur at positioning start.
  - When the sub point address is outside the range of -2147483648 ( $-2^{31}$ ) to 2147483647 ( $2^{31}-1$ )  
... An error "Sub point setting error" (error code: 525) will occur at positioning start.
  - When the end point address is outside the range of -2147483648 ( $-2^{31}$ ) to 2147483647 ( $2^{31}-1$ )  
... An error "End point setting error" (error code: 526) will occur at positioning start.
  - When the center point address is outside the range of "-2147483648 ( $-2^{31}$ ) to 2147483647 ( $2^{31}-1$ )"  
... An error "Sub point setting error" (error code: 525) will occur at positioning start.
  - Start point address = End point address  
... Error "End point setting error" (error code: 526)
  - Start point address = Sub point address  
... Error "Sub point setting error" (error code: 525)
  - End point address = Sub point address  
... Error "Sub point setting error" (error code: 525)
  - When the start point address, end point address, and sub point address are in a straight line  
... Error "Sub point setting error" (error code: 525)

### ■ Positioning data setting example

[Reference axis, circular interpolation axis, and linear interpolation axis are specified as axis 1, axis 2, and axis 3, respectively.]

The following table shows setting examples when "helical interpolation control with sub point designation (INC)" is set in positioning data No. 1 of axis 1.

(The required values are also set in positioning data No. 1 of axis 2 and axis 3.)

| Item                   |       | Axis                                 | Axis 1<br>(reference axis)<br>setting example | Axis 2 (circular<br>interpolation<br>axis) setting<br>example | Axis 3 (linear<br>interpolation<br>axis) setting<br>example | Setting details                                                                                                                                                          |
|------------------------|-------|--------------------------------------|-----------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positioning data No. 1 | Da.1  | Operation pattern                    | Positioning complete                          | -                                                             | -                                                           | Set "Positioning complete" assuming that the next positioning data will not be executed.                                                                                 |
|                        | Da.2  | Control system                       | INC helical sub                               | -                                                             | -                                                           | Set incremental system, 3-axis helical interpolation control with sub point designation.                                                                                 |
|                        | Da.3  | Acceleration time No.                | 1                                             | -                                                             | -                                                           | Specify the value set in "[Pr.25] Acceleration time 1" as the acceleration time at start.                                                                                |
|                        | Da.4  | Deceleration time No.                | 0                                             | -                                                             | -                                                           | Specify the value set in "[Pr.10] Deceleration time 0" as the deceleration time at deceleration.                                                                         |
|                        | Da.5  | Axis to be interpolated              | Axis 2                                        | -                                                             | -                                                           | Set the axis to be interpolated (partner axis). If the self-axis is set, an error will occur.                                                                            |
|                        | Da.6  | Positioning address/move-ment amount | 8000.0μm                                      | 6000.0μm                                                      | 3000.0μm                                                    | Set the movement amount. (Assuming "mm" is set in "[Pr.1] Unit setting".)                                                                                                |
|                        | Da.7  | Arc address                          | 4000.0μm                                      | -3000.0μm                                                     | -                                                           | Set the sub point address. (Assuming "mm" is set in "[Pr.1] Unit setting".)                                                                                              |
|                        | Da.8  | Command speed                        | 6000.00mm/min                                 | -                                                             | -                                                           | Set the speed during movement. (Set the composite speed in "[Pr.20] Interpolation speed designation method".)                                                            |
|                        | Da.9  | Dwell time                           | 500ms                                         | -                                                             | -                                                           | Set the time the machine dwells from the positioning stop (pulse output stop) to the output of the positioning complete signal.                                          |
|                        | Da.10 | M code                               | 10                                            | -                                                             | 50                                                          | Set this item when other sub operation commands are issued in combination with the No. 1 positioning data.<br>Set the number of pitch for the linear interpolation axis. |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

#### POINT

Set a value in "[Da.8] Command speed" so that the speed of each axis does not exceed the "[Pr.8] Speed limit value". (The speed limit does not function for the speed calculated by the QD75N during interpolation control.)

## 9.2.13 3-axis helical interpolation control with center point designation

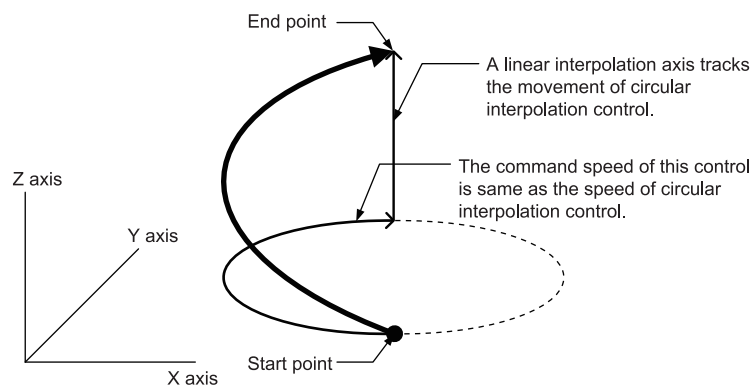
In the 3-axis helical interpolation control, circular interpolation control is carried out with two of the three axes. The other axis tracks the movement of the circular interpolation control to carry out "helical control" or "tangential control". (Refer to Section 9.1.6 "Interpolation control" for interpolation control.)

| Control details    | Outline of control | Positioning path                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Helical control    |                    | <p>In this control, the linear interpolation axis (linear control) tracks the movement of circular interpolation control, and a helical path is drawn.</p> <p>■ Unit setting</p> <ul style="list-style-type: none"> <li>Reference axis: mm/inch, pulse</li> <li>Circular interpolation axis: Same setting as the one for the reference axis</li> <li>Linear interpolation axis: mm/inch, pulse</li> </ul>                                                                                                                                                                                                            |
| Tangential control |                    | <p>In this control, the value of the linear interpolation axis (degree) changes according to the movement of circular interpolation control to keep the direction of a workpiece to the center point of the circular interpolation control. The figure on the left shows an example of the control that keeps the direction of the workpiece in the tangential direction.</p> <p>■ Unit setting</p> <ul style="list-style-type: none"> <li>Reference axis: mm/inch, pulse</li> <li>Circular interpolation axis: Same setting as the one for the reference axis</li> <li>Linear interpolation axis: degree</li> </ul> |

X-axis: Reference axis, Y-axis: Circular interpolation axis, Z-axis: Linear interpolation axis

■ Helical interpolation control speed

The command speed of the 3-axis helical interpolation control is the same as the speed of circular interpolation control (composite speed of reference axis and circular interpolation axis).



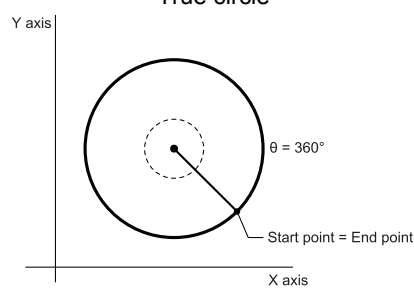
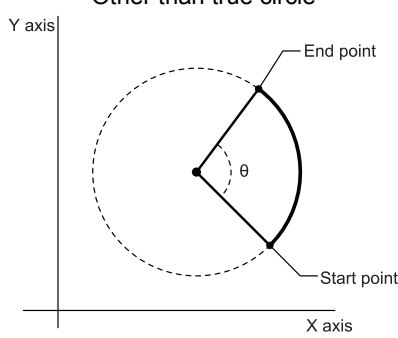


**POINT**

When "0: Composite speed" is set in "[Pr.20] Interpolation speed designation method", the command speed of ABS3/INC3 is the same as the composite speed of the 3 axes (x axis - y axis - z axis). The command speed of 3-axis helical interpolation control is the same as the composite speed of the circular interpolation axis (x axis - y axis). When continuous path control is carried out in ABS3/INC3 and 3-axis helical interpolation control, adjust the command speed to prevent the workpiece from moving unsteadily because the workpiece movement speed may change when positioning data is switched.

■ Rotation angle of the circular interpolation axis (X axis - Y axis)

The following shows the rotation angle of the circular interpolation axis in 3-axis helical interpolation control.

| Number of pitch | Circular interpolation axis control                                                |                                                                                     |
|-----------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                 | True circle                                                                        | Other than true circle                                                              |
|                 |  |  |
| 0               | 360°                                                                               | θ°                                                                                  |
| 1               |                                                                                    | 360° + θ°                                                                           |
| 2               | 720°                                                                               | 720° + θ°                                                                           |
| to              | to                                                                                 | to                                                                                  |
| n               | 360° × n                                                                           | 360° × n + θ°                                                                       |
| to              | to                                                                                 | to                                                                                  |
| 999             | 360° × 999                                                                         | 360° × 999 + θ°                                                                     |

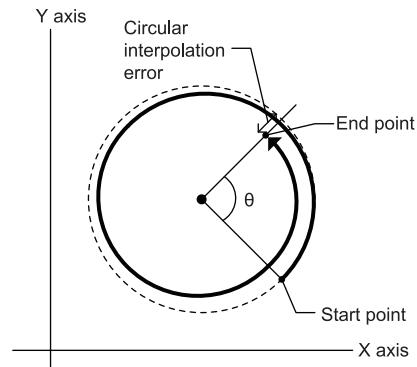
A true circle can be set only when the start point equals to the end point in 3-axis helical interpolation (ABS/INC, center point).

**POINT**

When "degree" is set for the unit setting in the absolute system, the positioning range is 0 to 359.99999°. When the axis rotates 360° or more in circular interpolation control (x axis - y axis), the tangential/normal control cannot be carried out because a degree of 360° or more cannot be set in linear control (z axis: degree). To carry out tangential/normal control in which the axis rotates 360° or more, use 3-axis helical interpolation control (INC).

### ■ Rotation angle of the circular interpolation axis (X axis - Y axis)

In 3-axis helical interpolation control, "[Pr.41] Allowable circular interpolation error width" is made valid as well as in circular interpolation control (2 axes). When a circular interpolation error occurs, the path of the circular interpolation control (x axis - y axis) forms a spiral as shown below.



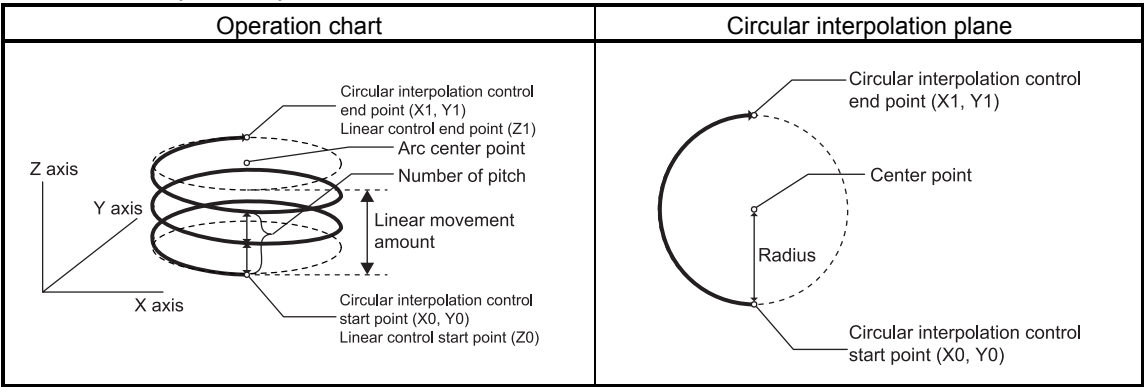
### ■ Rotation direction when the unit for the linear axis (z axis) is degree

When "[Pr.1] Unit setting" for the linear axis is "degree", the rotation direction is determined according to the axis control data "[Cd.40] ABS direction in degrees" of the reference axis. To set the rotation direction for each positioning data, set the position data "[Da.28] ABS direction in degrees".

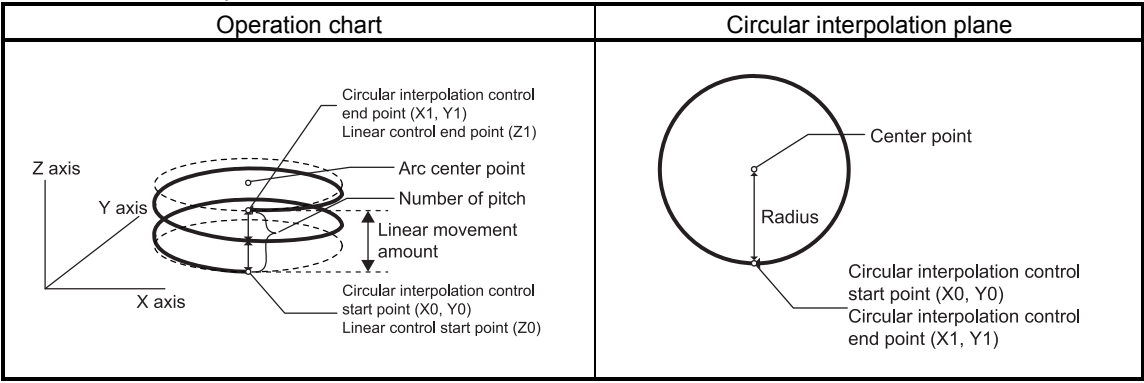
[1] 3-axis helical interpolation control with center point designation  
(ABS helical right, ABS helical left)

■ Operation chart

In this control, positioning is carried out from the current stop position (X0, Y0, Z0) to the circular end address (X1, Y1) and linear axis end address (Z1) set in "Da.6 Positioning address/movement amount". Circular interpolation with an arc path having its center as the address of the center point (arc address) set in "Da.7 Arc address" is carried out. Simultaneously, linear interpolation is carried out with the other linear axis. The control path is helically rotated for the number of pitch set in "Da.10 M code" of the linear axis to complete the positioning to the specified position.



Positioning of a true circle with a radius of the distance from the start point address to the arc center point can be carried out by setting the end point address (positioning address) of the circular interpolation axis to the same address as the start point address.



**■ Restrictions**

- (1) In the following cases, 3-axis helical interpolation control cannot be set.
- When "degree" is set in "[Pr.1] Unit setting" for the reference axis and circular interpolation axis
  - When the units set in "[Pr.1] Unit setting" are different between the reference axis and circular interpolation axis. ("mm" and "inch" combinations are possible.)
  - When the "reference axis speed" is set in "[Pr.20] Interpolation speed designation method" or "[Da.29] Interpolation speed designation method"
- (2) An error will occur and the positioning start will not be possible in the following cases. The machine will immediately stop if the error is detected during positioning control.
- When the radius exceeds " $536870912 (=2^{29})$ "  
(The maximum radius for which circular interpolation control is possible is " $536870912 (=2^{29})$ ".)
    - ... An error "Outside radius range" (error code: 544) will occur at positioning start.
  - Start point address = Center point address
    - ... Error "Center point setting error" (error code: 527)
  - End point address = Center point address
    - ... Error "Center point setting error" (error code: 527)
  - When the center point address is outside the range of " $-2147483648 (-2^{31})$  to  $2147483647 (2^{31}-1)$ "
    - ... Error "Center point setting error" (error code: 527)

### ■ Positioning data setting example

[Reference axis, circular interpolation axis, and linear interpolation axis are specified as axis 1, axis 2, and axis 3, respectively.]

The following table shows setting examples when "helical interpolation control with center point designation (ABS helical right, ABS helical left)" is set in positioning data No. 1 of axis 1.

(The required values are also set in positioning data No. 1 of axis 2 and axis 3.)

| Item \ Axis            |       |                                      | Axis 1<br>(reference axis) setting example | Axis 2<br>(circular interpolation axis) setting example | Axis 3 (linear interpolation axis) setting example | Setting details                                                                                                                                                          |
|------------------------|-------|--------------------------------------|--------------------------------------------|---------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positioning data No. 1 | Da.1  | Operation pattern                    | Positioning complete                       | -                                                       | -                                                  | Set "Positioning complete" assuming that the next positioning data will not be executed.                                                                                 |
|                        | Da.2  | Control system                       | ABS helical right<br>ABS helical left      | -                                                       | -                                                  | Set absolute system, 3-axis helical interpolation control with center point designation. (Select clockwise or counterclockwise according to the control.)                |
|                        | Da.3  | Acceleration time No.                | 1                                          | -                                                       | -                                                  | Specify the value set in "[Pr.25] Acceleration time 1" as the acceleration time at start.                                                                                |
|                        | Da.4  | Deceleration time No.                | 0                                          | -                                                       | -                                                  | Specify the value set in "[Pr.10] Deceleration time 0" as the deceleration time at deceleration.                                                                         |
|                        | Da.5  | Axis to be interpolated              | Axis 2                                     | -                                                       | -                                                  | Set the axis to be interpolated (partner axis). If the self-axis is set, an error will occur.                                                                            |
|                        | Da.6  | Positioning address/move-ment amount | 8000.0μm                                   | 6000.0μm                                                | 3000.0μm                                           | Set the end point address. (Assuming "mm" is set in "[Pr.1] Unit setting".)                                                                                              |
|                        | Da.7  | Arc address                          | 4000.0μm                                   | 3000.0μm                                                | -                                                  | Set the arc address. (Assuming "mm" is set in "[Pr.1] Unit setting".)                                                                                                    |
|                        | Da.8  | Command speed                        | 6000.00mm/min                              | -                                                       | -                                                  | Set the speed during movement to the end point address. (Set the composite speed in "[Pr.20] Interpolation speed designation method".)                                   |
|                        | Da.9  | Dwell time                           | 500ms                                      | -                                                       | -                                                  | Set the time the machine dwells from the positioning stop (pulse output stop) to the output of the positioning complete signal.                                          |
|                        | Da.10 | M code                               | 10                                         | -                                                       | 50                                                 | Set this item when other sub operation commands are issued in combination with the No. 1 positioning data.<br>Set the number of pitch for the linear interpolation axis. |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

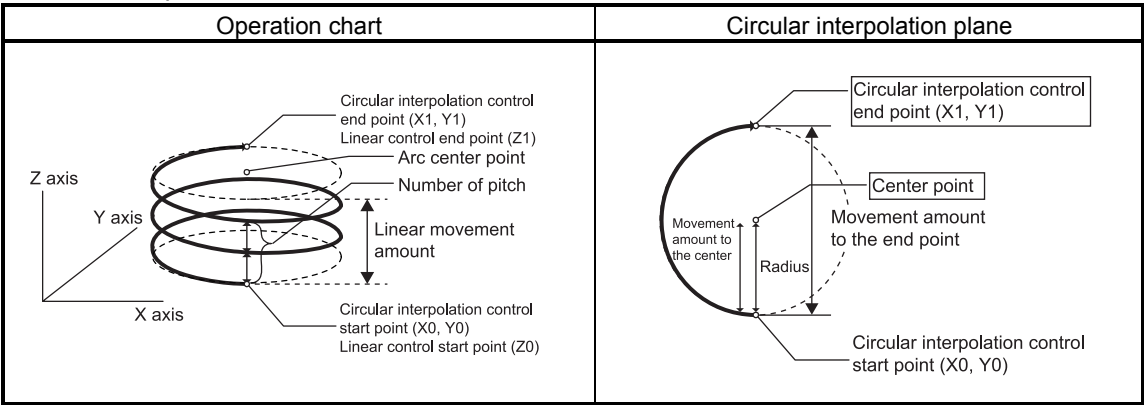
#### POINT

Set a value in "[Da.8] Command speed" so that the speed of each axis does not exceed the "[Pr.8] Speed limit value". (The speed limit does not function for the speed calculated by the QD75N during interpolation control.)

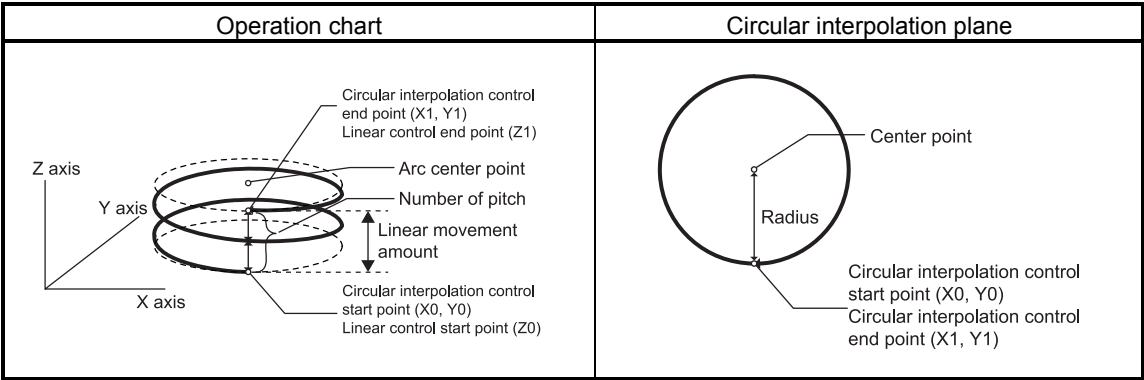
[1] 3-axis helical interpolation control with center point designation  
(INC helical right, INC helical left)

■ Operation chart

In this control, positioning is carried out from the current stop position (X0, Y0, Z0) to the movement amount position (X1, Y1, Z1) set in "[Da.6] Positioning address/movement amount". Circular interpolation with an arc path having its center as the address of the center point (arc address) set in "[Da.7] Arc address" is carried out. Simultaneously, linear interpolation is carried out with the other linear axis. The control path is helically rotated for the number of pitch set in "[Da.10] M code" of the linear axis to complete the positioning to the specified position.



Positioning of a true circle with a radius of the distance from the start point address to the arc center point can be carried out by setting the movement amount of the circular interpolation axis to "0".



### ■ Restrictions

- (1) In the following cases, 3-axis helical interpolation control cannot be set.
  - When "degree" is set in "[Pr.1] Unit setting" for the reference axis and circular interpolation axis
  - When the units set in "[Pr.1] Unit setting" are different between the reference axis and circular interpolation axis. ("mm" and "inch" combinations are possible.)
  - When the "reference axis speed" is set in "[Pr.20] Interpolation speed designation method" or "[Da.29] Interpolation speed designation method"
- (2) An error will occur and the positioning start will not be possible in the following cases. The machine will immediately stop if the error is detected during positioning control.
  - When the radius exceeds "536870912 ( $=2^{29}$ )"  
 (The maximum radius for which circular interpolation control is possible is "536870912 ( $=2^{29}$ )".)  
 ... An error "Outside radius range" (error code: 544) will occur at positioning start.
  - When the end point address is outside the range of  $-2147483648 (-2^{31})$  to  $2147483647 (2^{31}-1)$   
 ... An error "End point setting error" (error code: 526) will occur at positioning start.
  - Start point address = Center point address  
 ... Error "Center point setting error" (error code: 527)
  - End point address = Center point address  
 ... Error "Center point setting error" (error code: 527)
  - When the center point address is outside the range of  $-2147483648 (-2^{31})$  to  $2147483647 (2^{31}-1)$   
 ... Error "Center point setting error" (error code: 527)

### ■ Positioning data setting example

[Reference axis, circular interpolation axis, and linear interpolation axis are specified as axis 1, axis 2, and axis 3, respectively.]

The following table shows setting examples when "helical interpolation control with center point designation (INC helical right, INC helical left)" is set in positioning data No. 1 of axis 1.

(The required values are also set in positioning data No. 1 of axis 2 and axis 3.)

| Item \ Axis            |       |                                      | Axis 1<br>(reference axis) setting example | Axis 2 (circular interpolation axis) setting example | Axis 3 (linear interpolation axis) setting example | Setting details                                                                                                                                                          |
|------------------------|-------|--------------------------------------|--------------------------------------------|------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positioning data No. 1 | Da.1  | Operation pattern                    | Positioning complete                       | -                                                    | -                                                  | Set "Positioning complete" assuming that the next positioning data will not be executed.                                                                                 |
|                        | Da.2  | Control system                       | ABS helical right<br>ABS helical left      | -                                                    | -                                                  | Set absolute system, 3-axis helical interpolation control with center point designation. (Select clockwise or counterclockwise according to the control.)                |
|                        | Da.3  | Acceleration time No.                | 1                                          | -                                                    | -                                                  | Specify the value set in "Pr.25 Acceleration time 1" as the acceleration time at start.                                                                                  |
|                        | Da.4  | Deceleration time No.                | 0                                          | -                                                    | -                                                  | Specify the value set in "Pr.10 Deceleration time 0" as the deceleration time at deceleration.                                                                           |
|                        | Da.5  | Axis to be interpolated              | Axis 2                                     | -                                                    | -                                                  | Set the axis to be interpolated (partner axis). If the self-axis is set, an error will occur.                                                                            |
|                        | Da.6  | Positioning address/move-ment amount | 8000.0μm                                   | 6000.0μm                                             | 3000.0μm                                           | Set the end point address. (Assuming "mm" is set in "Pr.1 Unit setting".)                                                                                                |
|                        | Da.7  | Arc address                          | 4000.0μm                                   | 3000.0μm                                             | -                                                  | Set the arc address. (Assuming "mm" is set in "Pr.1 Unit setting".)                                                                                                      |
|                        | Da.8  | Command speed                        | 6000.00mm/min                              | -                                                    | -                                                  | Set the speed during movement to the end point address. (Set the composite speed in "Pr.20 Interpolation speed designation method".)                                     |
|                        | Da.9  | Dwell time                           | 500ms                                      | -                                                    | -                                                  | Set the time the machine dwells from the positioning stop (pulse output stop) to the output of the positioning complete signal.                                          |
|                        | Da.10 | M code                               | 10                                         | -                                                    | 50                                                 | Set this item when other sub operation commands are issued in combination with the No. 1 positioning data.<br>Set the number of pitch for the linear interpolation axis. |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

#### POINT

Set a value in "[Da.8] Command speed" so that the speed of each axis does not exceed the "[Pr.8] Speed limit value". (The speed limit does not function for the speed calculated by the QD75N during interpolation control.)



## 9.2.14 1-axis speed control

In "1-axis speed control" (" [Da.2] Control system" = Forward run: speed 1, Reverse run: speed 1), control is carried out in the axis direction in which the positioning data has been set by continuously outputting pulses for the speed set in

" [Da.8] Command speed" until the input of a stop command.

The two types of 1-axis speed control are "Forward run: speed 1" in which the control starts in the forward run direction, and "Reverse run: speed 1" in which control starts in the reverse run direction.

#### ■ Operation chart

The following chart shows the operation timing for 1-axis speed control with axis 1 as the reference axis.

The "in speed control" flag ([Md.31] Status: b0) is turned ON during speed control.

The "Positioning complete signal" is not turned ON.

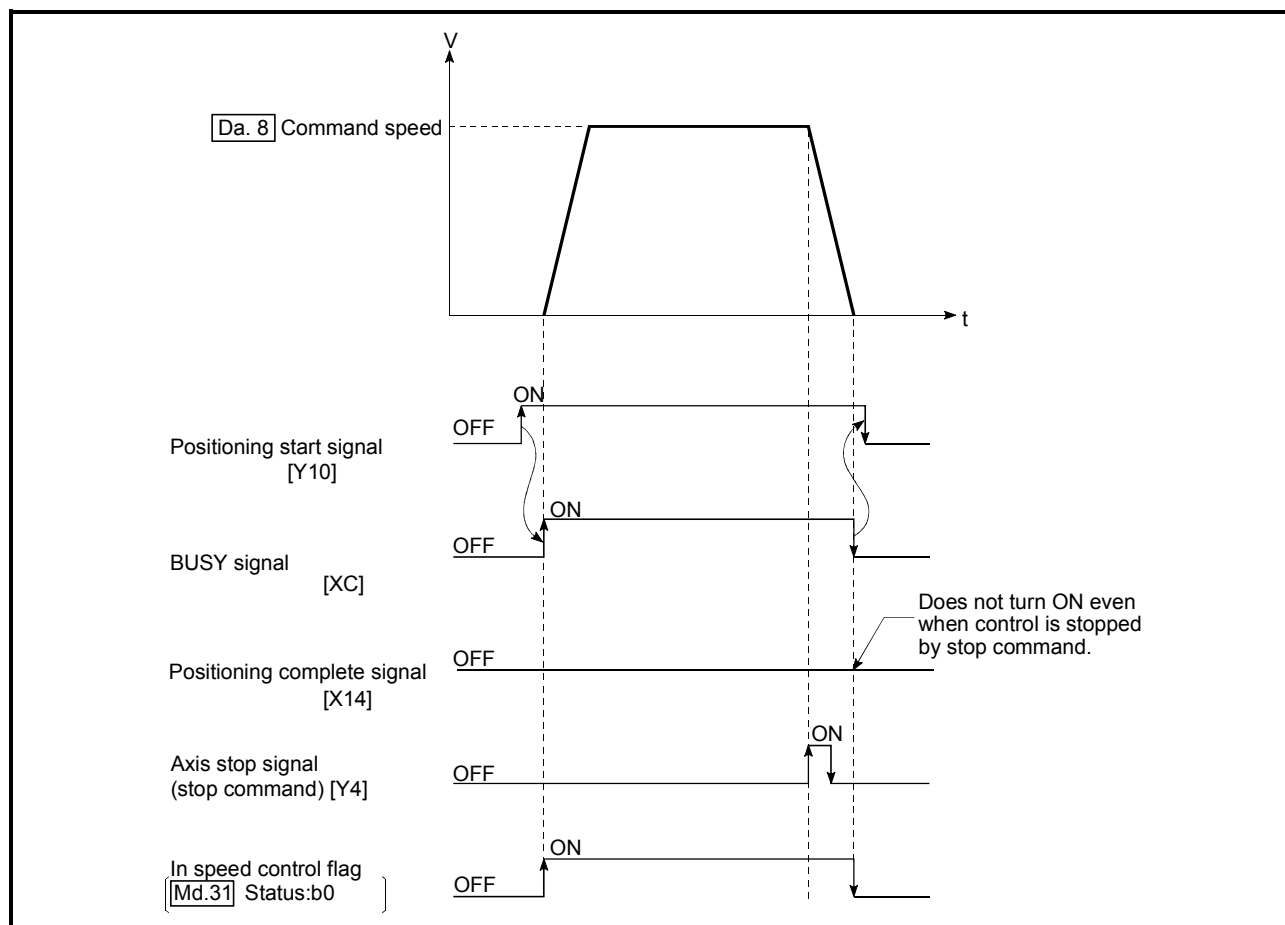
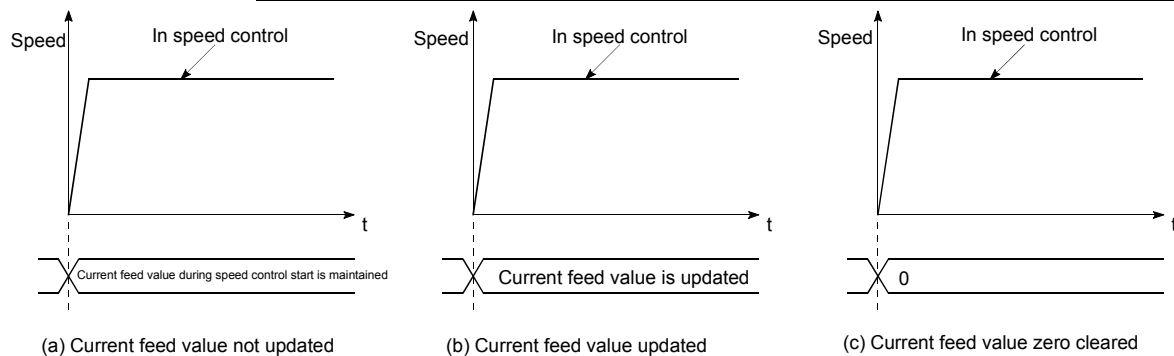


Fig.9.9 1-axis speed control operation timing

### ■ Current feed value during 1-axis speed control

The following table shows the " **Md.20** Current feed value" during 1-axis speed control corresponding to the " **Pr.21** Current feed value during speed control" settings.

| " <b>Pr.21</b> Current feed value during speed control" setting | <b>Md.20</b> Current feed value                              |
|-----------------------------------------------------------------|--------------------------------------------------------------|
| 0: Do not update current feed value                             | The current feed value at speed control start is maintained. |
| 1: Update current feed value                                    | The current feed value is updated.                           |
| 2: Zero clear current feed value                                | The current feed value is fixed at 0.                        |



### ■ Restrictions

- (1) Set "Positioning complete" in " **Da.1** Operation pattern". An axis error "Continuous path control not possible" (error code: 516) will occur and the operation cannot start if "continuous positioning control" or "continuous path control" is set in " **Da.1** Operation pattern". ("Continuous positioning control" and "continuous path control" cannot be set in speed control.)
- (2) Set the WITH mode in " **Pr.18** M code ON signal output timing" when using an M code. The M code will not be output, and the M code ON signal will not turn ON if the AFTER mode is set.
- (3) An error "No command speed" (error code: 503) will occur if the current speed (-1) is set in " **Da.8** Command speed".
- (4) The software stroke limit check will not be carried out if the control unit is set to "degree".

### ■ Positioning data setting examples

The following table shows the setting examples when "1-axis speed control (forward run: speed 1)" is set in the positioning data No. 1 of axis 1.

| Setting item |                        | Setting example                     | Setting details                                                                                                                                                                         |
|--------------|------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1       | Positioning data No. 1 |                                     |                                                                                                                                                                                         |
|              | Da.1                   | Operation pattern                   | Positioning complete                                                                                                                                                                    |
|              |                        |                                     | Setting other than "Positioning complete" is not possible in speed control.                                                                                                             |
|              | Da.2                   | Control system                      | Forward run speed 1                                                                                                                                                                     |
|              |                        |                                     | Set 1-axis speed control.                                                                                                                                                               |
|              | Da.3                   | Acceleration time No.               | 1                                                                                                                                                                                       |
|              |                        |                                     | Designate the value set in " Pr.25 Acceleration time 1" as the acceleration time at start.                                                                                              |
|              | Da.4                   | Deceleration time No.               | 0                                                                                                                                                                                       |
|              |                        |                                     | Designate the value set in " Pr.10 Deceleration time 0" as the deceleration time at deceleration.                                                                                       |
|              | Da.5                   | Axis to be interpolated             | –                                                                                                                                                                                       |
|              |                        |                                     | Setting not required (setting value will be ignored).                                                                                                                                   |
|              | Da.6                   | Positioning address/movement amount | –                                                                                                                                                                                       |
|              |                        |                                     | Setting not required (setting value will be ignored).                                                                                                                                   |
|              | Da.7                   | Arc address                         | –                                                                                                                                                                                       |
|              |                        |                                     | Setting not required (setting value will be ignored).                                                                                                                                   |
|              | Da.8                   | Command speed                       | 6000.00mm/min                                                                                                                                                                           |
|              |                        |                                     | Set the speed to be commanded.                                                                                                                                                          |
|              | Da.9                   | Dwell time                          | –                                                                                                                                                                                       |
|              |                        |                                     | Setting not required (setting value will be ignored).                                                                                                                                   |
|              | Da.10                  | M code                              | 10                                                                                                                                                                                      |
|              |                        |                                     | Set this when other sub operation commands are issued in combination with the No. 1 positioning data. (" Pr.18 M code ON signal output timing" setting only possible in the WITH mode.) |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

## 9.2.15 2-axis speed control

In "2-axis speed control" (" [Da.2] Control system" = Forward run: speed 2, Reverse run: speed 2), control is carried out in the 2-axis direction in which the positioning data has been set by continuously outputting pulses for the speed set in " [Da.8] Command speed" until the input of a stop command.

The two types of 2-axis speed control are "Forward run: speed 2" in which the control starts in the forward run direction, and "Reverse run: speed 2" in which control starts in the reverse run direction.

(Refer to Section 9.1.6 "Interpolation control" for the combination of the reference axis with the interpolation axis.)

#### ■ Operation chart

The following chart shows the operation timing for 2-axis (axes 1 and 2) speed control with axis 1 as the reference axis. The "in speed control" flag ([Md.31] Status: b0) is turned ON during speed control.

The "positioning complete signal" is not turned ON.

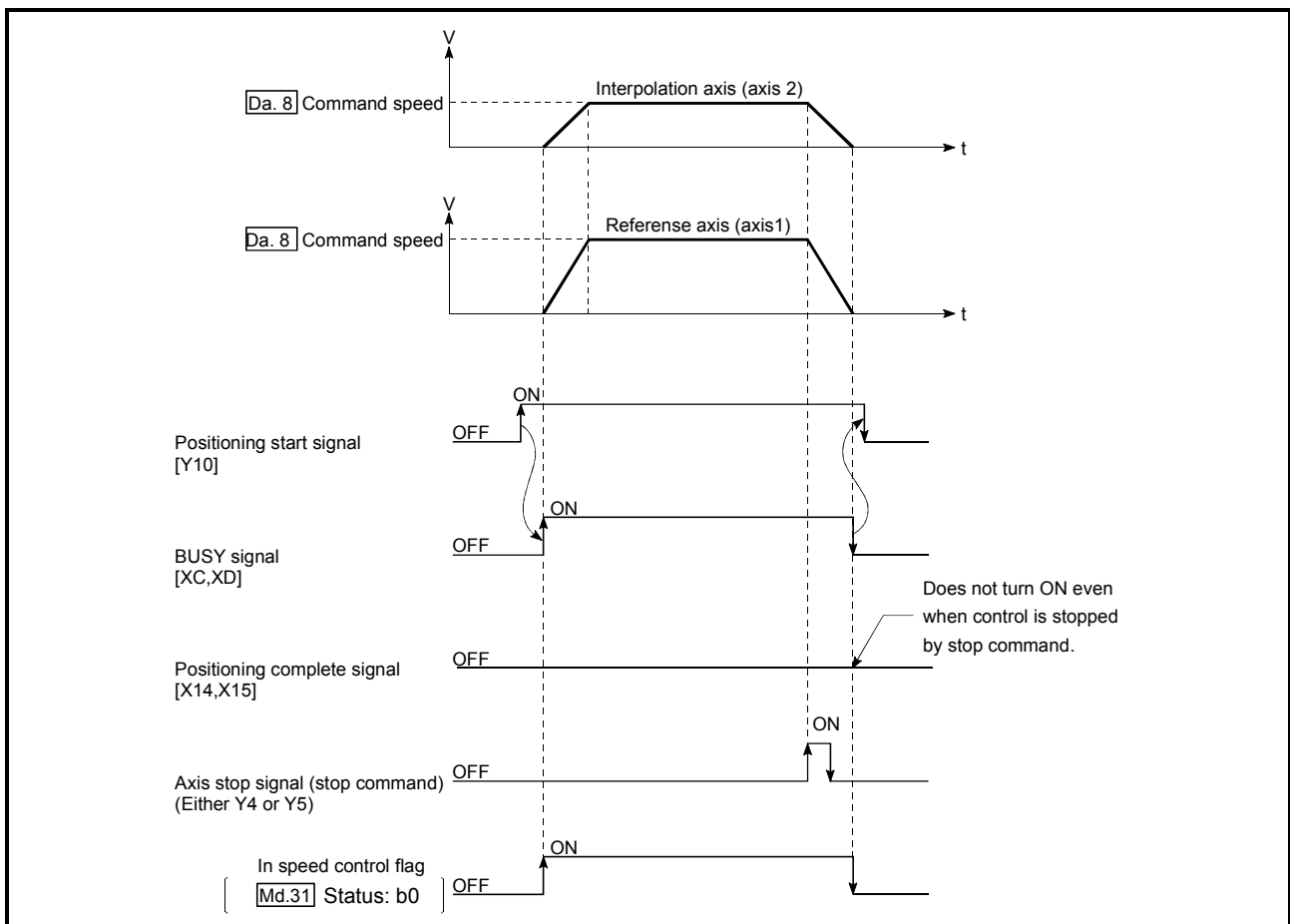
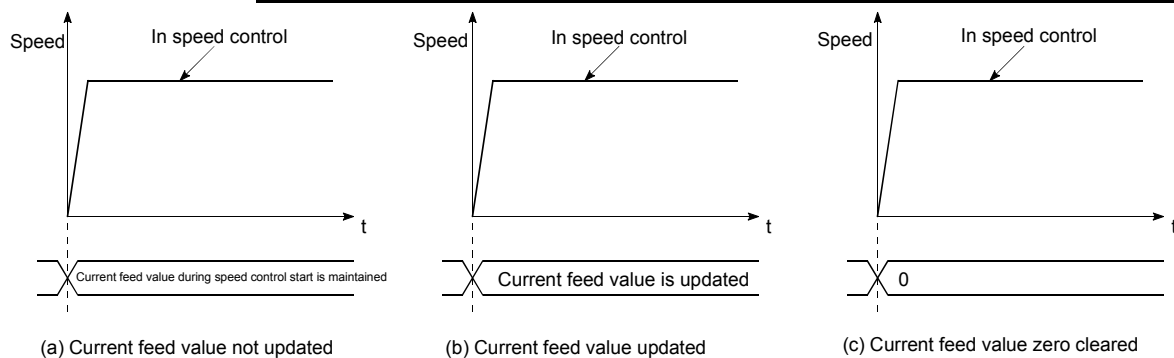


Fig. 9.10 2-axis speed control operation timing

### ■ Current feed value during 2-axis speed control

The following table shows the " [Md.20] Current feed value" during 2-axis speed control corresponding to the " [Pr.21] Current feed value during speed control" settings. (Note that the reference axis setting values are used for parameters.)

| " [Pr.21] Current feed value during speed control" setting | [Md.20] Current feed value                                   |
|------------------------------------------------------------|--------------------------------------------------------------|
| 0: Do not update current feed value                        | The current feed value at speed control start is maintained. |
| 1: Update current feed value                               | The current feed value is updated.                           |
| 2: Zero clear current feed value                           | The current feed value is fixed at 0.                        |



### ■ Restrictions

- (1) Set "Positioning complete" in " [Da.1] Operation pattern". An axis error "Continuous path control not possible" (error code: 516) will occur and the operation cannot start if "continuous positioning control" or "continuous path control" is set.  
("Continuous positioning control" and "continuous path control" cannot be set in speed control.)
- (2) Set the WITH mode in " [Pr.18] M code ON signal output timing" when using an M code. The M code will not be output, and the M code ON signal will not turn ON if the AFTER mode is set.
- (3) Set the "reference axis speed" in " [Pr.20] Interpolation speed designation method". An error "Interpolation mode error" (error code: 523) will occur and the operation cannot start if a composite speed is set.
- (4) When either of two axes exceeds the speed limit, that axis is controlled with the speed limit value. The speeds of the other axes are limited at the ratios of " [Da.8] Command speed".  
(Examples)

| Axis         |                   | Axis 1 setting | Axis 2 setting |
|--------------|-------------------|----------------|----------------|
| Setting item |                   |                |                |
| [Pr.8]       | Speed limit value | 4000.00mm/min  | 5000.00mm/min  |
| [Da.8]       | Command speed     | 8000.00mm/min  | 6000.00mm/min  |

With the settings shown above, the operation speed in speed control is as follows.

Axis 1: 4000.00 mm/min (Speed is limited by [Pr.8]).

Axis 2: 3000.00 mm/min (Speed is limited at a ratio of an axis 1 command speed to an axis 2 command speed).

Note: Operation runs at speed 1 when a reference axis speed is less than 1 as a result of speed limit.

In addition, when a bias speed is set, it will be the minimum speed.

- (5) An error "No command speed" (error code: 503) occurs if a current speed (-1) is set in " **Da.8** Command speed".
- (6) The software stroke limit check is not carried out when the control unit is set to "degree".

#### ■ Positioning data setting examples

[Setting examples when the reference axis and interpolation axis are designated as axes 1 and 2, respectively.]

The following table shows the setting examples when "2-axis speed control (forward run: speed 2)" is set in the positioning data No. 1 of axis 1 (reference axis).

| Setting item |                        | Axis                                | Axis 1<br>(reference<br>axis) setting<br>example | Axis 2<br>(interpolation<br>axis) setting<br>example | Setting details                                                                                                                                                                                |
|--------------|------------------------|-------------------------------------|--------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1       | Positioning data No. 1 |                                     |                                                  |                                                      |                                                                                                                                                                                                |
|              | <b>Da.1</b>            | Operation pattern                   | Positioning complete                             | —                                                    | Setting other than "Positioning complete" is not possible in speed control.                                                                                                                    |
|              | <b>Da.2</b>            | Control system                      | Forward run speed 2                              | —                                                    | Set 2-axis speed control.                                                                                                                                                                      |
|              | <b>Da.3</b>            | Acceleration time No.               | 1                                                | —                                                    | Designate the value set in " <b>Pr.25</b> Acceleration time 1" as the acceleration time at start.                                                                                              |
|              | <b>Da.4</b>            | Deceleration time No.               | 0                                                | —                                                    | Designate the value set in " <b>Pr.10</b> Deceleration time 0" as the deceleration time at deceleration.                                                                                       |
|              | <b>Da.5</b>            | Axis to be interpolated             | Axis 2                                           | —                                                    | Set the axis to be interpolated (partner axis).<br>If the self-axis is set, an error will occur.                                                                                               |
|              | <b>Da.6</b>            | Positioning address/movement amount | —                                                | —                                                    | Setting not required (setting value will be ignored).                                                                                                                                          |
|              | <b>Da.7</b>            | Arc address                         | —                                                | —                                                    | Setting not required (setting value will be ignored).                                                                                                                                          |
|              | <b>Da.8</b>            | Command speed                       | 6000.00 mm/min                                   | 3000.00 mm/min                                       | Set the speed to be commanded.                                                                                                                                                                 |
|              | <b>Da.9</b>            | Dwell time                          | —                                                | —                                                    | Setting not required (setting value will be ignored).                                                                                                                                          |
|              | <b>Da.10</b>           | M code                              | 10                                               | —                                                    | Set this when other sub operation commands are issued in combination with the No. 1 positioning data. (" <b>Pr.18</b> M code ON signal output timing" setting only possible in the WITH mode.) |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

## 9.2.16 3-axis speed control

In "3-axis speed control" (" [Da.2] Control system" = Forward run: speed 3, Reverse run: speed 3), control is carried out in the 3-axis direction in which the positioning data has been set by continuously outputting pulses for the speed set in

" [Da.8] Command speed" until the input of a stop command.

The two types of 3-axis speed control are "Forward run: speed 3" in which the control starts in the forward run direction, and "Reverse run: speed 3" in which control starts in the reverse run direction.

(Refer to Section 9.1.6 "Interpolation control" for the combination of the reference axis with the interpolation axes.)

#### ■ Operation chart

The following chart shows the operation timing for 3-axis (axes 1, 2, and 3) speed control with axis 1 as the reference axis.

The "in speed control" flag ([Md.31] Status: b0) is turned ON during speed control.

The "positioning complete signal" is not turned ON.

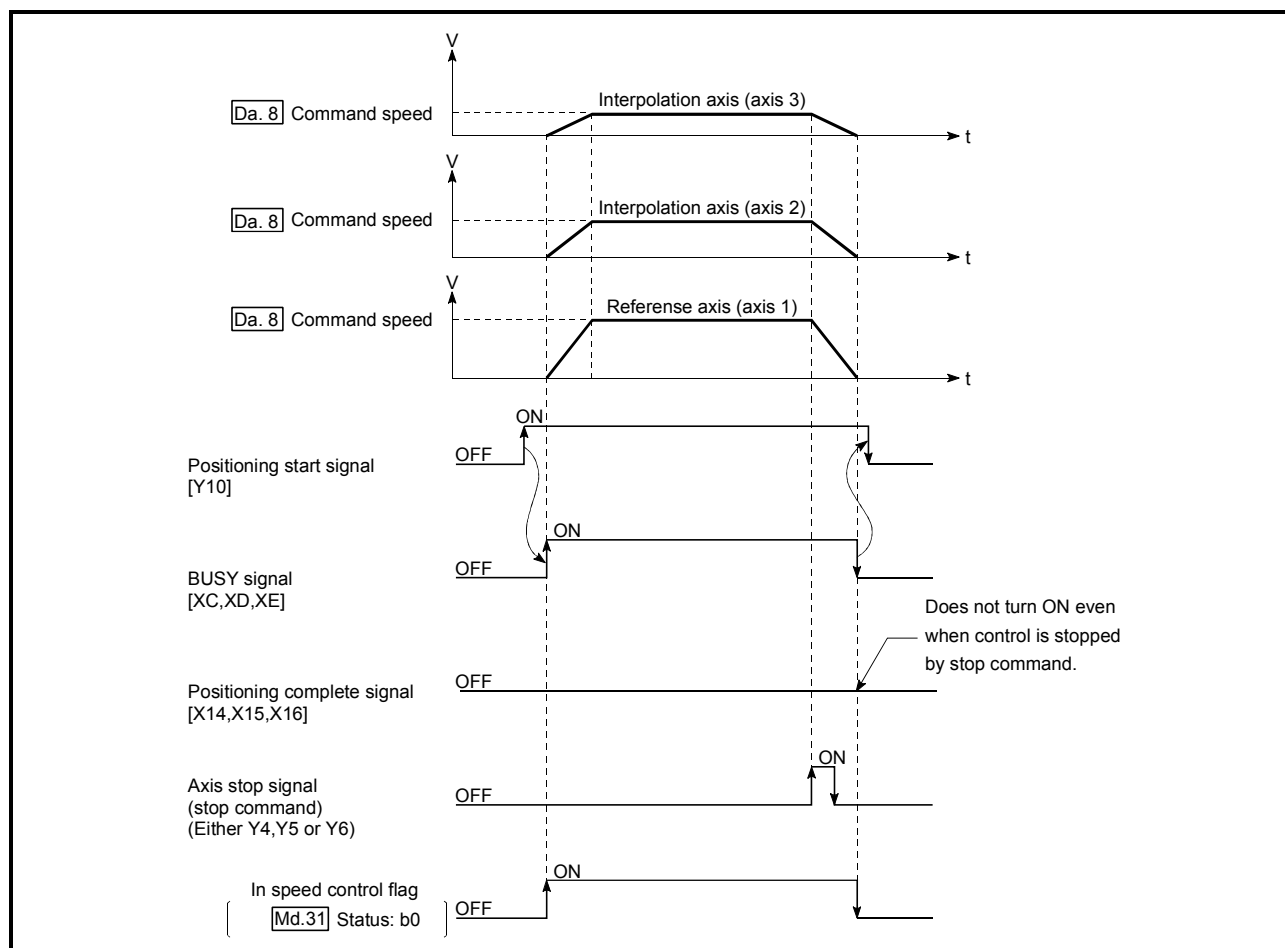
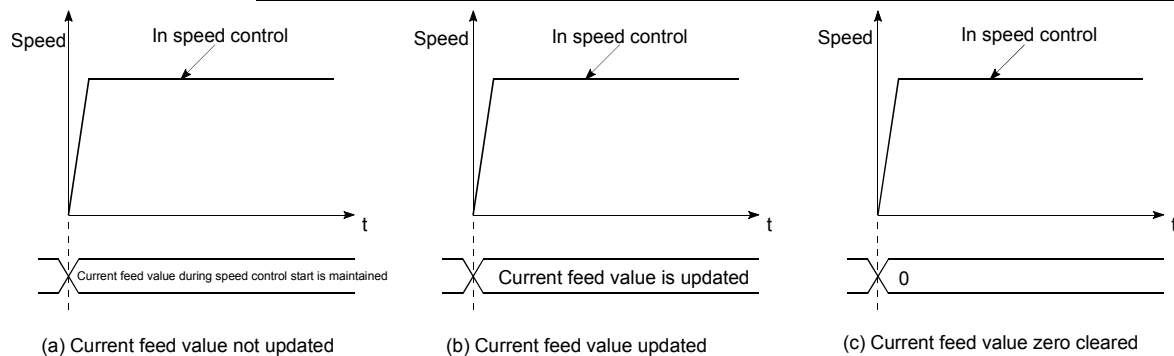


Fig. 9.11 3-axis speed control operation timing

### ■ Current feed value during 3-axis speed control

The following table shows the " [Md.20](#) Current feed value" during 3-axis speed control corresponding to the " [Pr.21](#) Current feed value during speed control" settings. (Note that the reference axis setting values are used for parameters.)

| " <a href="#">Pr.21</a> Current feed value during speed control" setting | <a href="#">Md.20</a> Current feed value                     |
|--------------------------------------------------------------------------|--------------------------------------------------------------|
| 0: Do not update current feed value                                      | The current feed value at speed control start is maintained. |
| 1: Update current feed value                                             | The current feed value is updated.                           |
| 2: Zero clear current feed value                                         | The current feed value is fixed at 0.                        |



### ■ Restrictions

- (1) Set "Positioning complete" in " [Da.1](#) Operation pattern". An axis error "Continuous path control not possible" (error code: 516) will occur and the operation cannot start if "continuous positioning control" or "continuous path control" is set. ("Continuous positioning control" and "continuous path control" cannot be set in speed control.)
- (2) Set the WITH mode in " [Pr.18](#) M code ON signal output timing" when using an M code. The M code will not be output, and the M code ON signal will not turn ON if the AFTER mode is set.
- (3) Set the "reference axis speed" in " [Pr.20](#) Interpolation speed designation method". An error "Interpolation mode error" (error code: 523) will occur and the operation cannot start if a composite speed is set.



- (4) When either of three axes exceeds the speed limit, that axis is controlled with the speed limit value. The speeds of the other axes are limited at the ratios of " [Da.8] Command speed".

(Examples)

| Setting item \ Axis |                   | Axis 1 setting | Axis 2 setting | Axis 3 setting |
|---------------------|-------------------|----------------|----------------|----------------|
| [Pr.8]              | Speed limit value | 4000.00mm/min  | 5000.00mm/min  | 6000.00mm/min  |
| [Da.8]              | Command speed     | 8000.00mm/min  | 6000.00mm/min  | 4000.00mm/min  |

With the settings shown above, the operation speed in speed control is as follows.

Axis 1: 4000.00 mm/min (Speed is limited by [Pr.8]).

Axis 2: 3000.00 mm/min (Speed is limited at ratios in axes 1, 2, and 3 command speeds).

Axis 3: 2000.00 mm/min (Speed is limited at ratios in axes 1, 2, and 3 command speeds).

Note: Operation runs at speed 1 when a reference axis speed is less than 1 as a result of speed limit.

In addition, when a bias speed is set, it will be the minimum speed.

- (5) An error "No command speed" (error code: 503) will occur if a current speed (-1) is set in " [Da.8] Command speed".
- (6) The software stroke limit check is not carried out when the control unit is set to "degree".

### ■ Positioning data setting examples

The following table shows the setting examples when "3-axis speed control (forward run: speed 3)" is set in the positioning data No. 1 of axis 1 (reference axis).

| Setting item \ Axis              |                                          | Axis 1<br>(reference axis) setting example | Axis 2<br>(interpolation axis) setting example | Axis 3<br>(interpolation axis) setting example | Setting details                                                                                                                                                                         |
|----------------------------------|------------------------------------------|--------------------------------------------|------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1<br>Positioning data No. 1 | Da.1 Operation pattern                   | Positioning complete                       | —                                              | —                                              | Setting other than "Positioning complete" is not possible in speed control.                                                                                                             |
|                                  | Da.2 Control system                      | Forward run speed 3                        | —                                              | —                                              | Set 3-axis speed control.                                                                                                                                                               |
|                                  | Da.3 Acceleration time No.               | 1                                          | —                                              | —                                              | Designate the value set in " Pr.25 Acceleration time 1" as the acceleration time at start.                                                                                              |
|                                  | Da.4 Deceleration time No.               | 0                                          | —                                              | —                                              | Designate the value set in " Pr.10 Deceleration time 0" as the deceleration time at deceleration.                                                                                       |
|                                  | Da.5 Axis to be interpolated             | —                                          | —                                              | —                                              | Setting not required (setting value will be ignored).<br>When axis 1 is used as a reference axis, the interpolation axes are axes 2 and 3.                                              |
|                                  | Da.6 Positioning address/movement amount | —                                          | —                                              | —                                              | Setting not required (setting value will be ignored).                                                                                                                                   |
|                                  | Da.7 Arc address                         | —                                          | —                                              | —                                              | Setting not required (setting value will be ignored).                                                                                                                                   |
|                                  | Da.8 Command speed                       | 6000.00 mm/min                             | 3000.00 mm/min                                 | 2000.00 mm/min                                 | Set the speed to be commanded.                                                                                                                                                          |
|                                  | Da.9 Dwell time                          | —                                          | —                                              | —                                              | Setting not required (setting value will be ignored).                                                                                                                                   |
|                                  | Da.10 M code                             | 10                                         | —                                              | —                                              | Set this when other sub operation commands are issued in combination with the No. 1 positioning data. (" Pr.18 M code ON signal output timing" setting only possible in the WITH mode.) |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

## 9.2.17 4-axis speed control

In "4-axis speed control" (" Da.2 Control system" = Forward run: speed 4, Reverse run: speed 4), control is carried out in the 4-axis direction in which the positioning data has been set by continuously outputting pulses for the speed set in

" Da.8 Command speed" until the input of a stop command.

The two types of 4-axis speed control are "Forward run: speed 4" in which the control starts in the forward run direction, and "Reverse run: speed 4" in which control starts in the reverse run direction.

(Refer to Section 9.1.6 "Interpolation control" for the combination of the reference axis with the interpolation axes.)

### ■ Operation chart

The following chart shows the operation timing for 4-axis speed control with axis 1 as the reference axis.

The "in speed control" flag ([Md.31] Status: b0) is turned ON during speed control.  
The "positioning complete signal" is not turned ON.

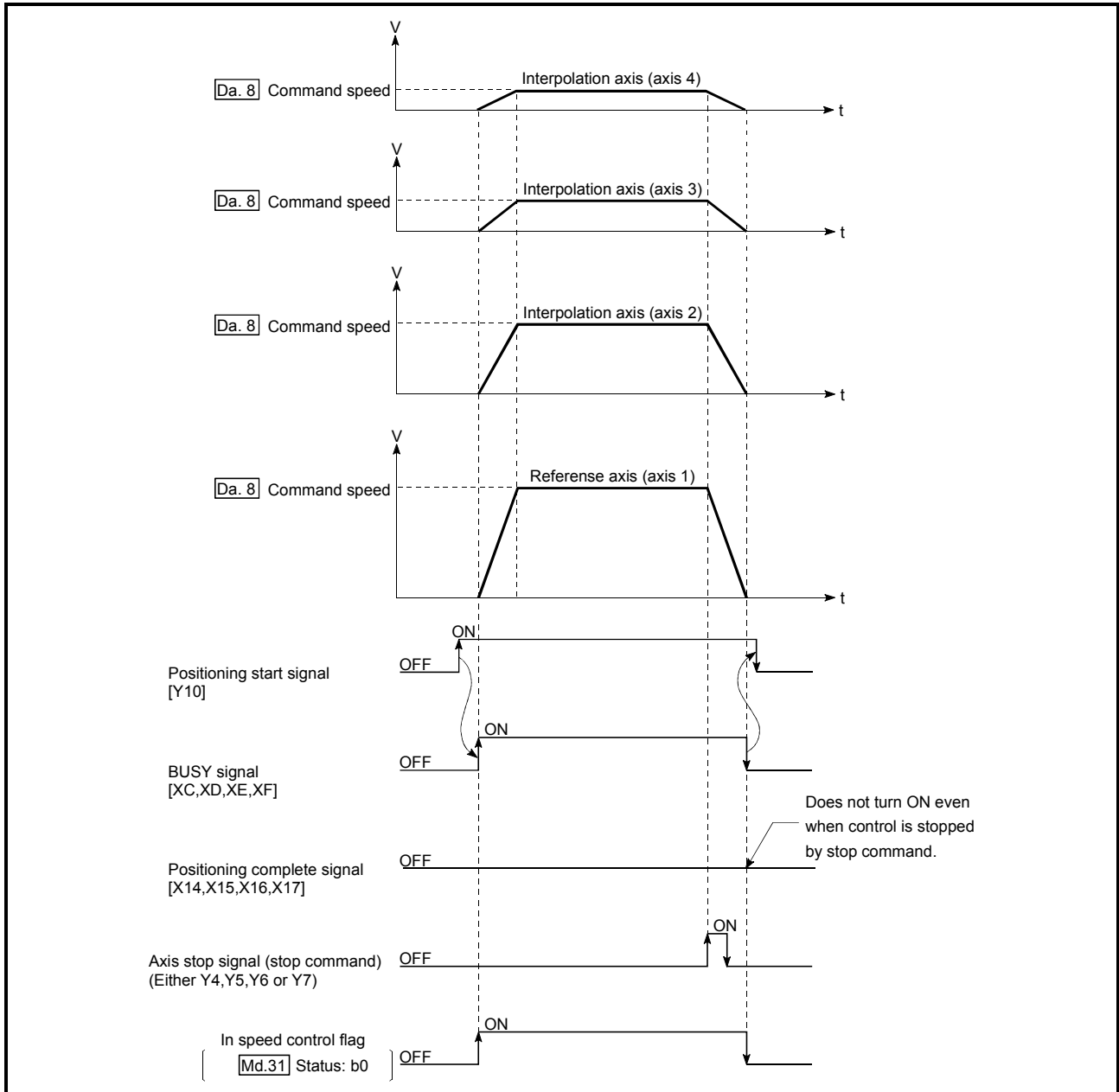
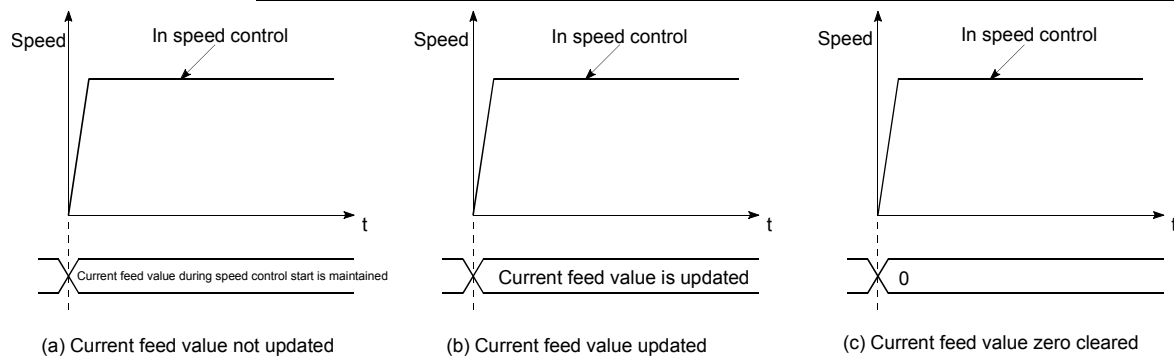


Fig. 9.12 4-axis speed control operation timing

### ■ Current feed value during 4-axis speed control

The following table shows the " [Md.20](#) Current feed value" during 4-axis speed control corresponding to the " [Pr.21](#) Current feed value during speed control" settings. (Note that the reference axis setting values are used for parameters.)

| " <a href="#">Pr.21</a> Current feed value during speed control" setting | <a href="#">Md.20</a> Current feed value                     |
|--------------------------------------------------------------------------|--------------------------------------------------------------|
| 0: Do not update current feed value                                      | The current feed value at speed control start is maintained. |
| 1: Update current feed value                                             | The current feed value is updated.                           |
| 2: Zero clear current feed value                                         | The current feed value is fixed at 0.                        |



### ■ Restrictions

- (1) Set "Positioning complete" in " [Da.1](#) Operation pattern". An axis error "Continuous path control not possible" (error code: 516) will occur and the operation cannot start if "continuous positioning control" or "continuous path control" is set. ("Continuous positioning control" and "continuous path control" cannot be set in speed control.)
- (2) Set the WITH mode in " [Pr.18](#) M code ON signal output timing" when using an M code. The M code will not be output, and the M code ON signal will not turn ON if the AFTER mode is set.
- (3) Set the "reference axis speed" in " [Pr.20](#) Interpolation speed designation method". An error "Interpolation mode error" (error code: 523) will occur and the operation cannot start if a composite speed is set.

- (4) When either of four axes exceeds the speed limit, that axis is controlled with the speed limit value. The speeds of the other axes are limited at the ratios of "Da.8 Command speed".

(Examples)

| Setting item \ Axis |                   | Axis 1 setting | Axis 2 setting | Axis 3 setting | Axis 4 setting |
|---------------------|-------------------|----------------|----------------|----------------|----------------|
| Pr.8                | Speed limit value | 4000.00mm/min  | 5000.00mm/min  | 6000.00mm/min  | 8000.00mm/min  |
| Da.8                | Command speed     | 8000.00mm/min  | 6000.00mm/min  | 4000.00mm/min  | 1500.00mm/min  |

With the settings shown above, the operation speed in speed control is as follows.

Axis 1: 4000.00 mm/min (Speed is limited by Pr.8).

Axis 2: 3000.00 mm/min (Speed is limited at ratios in axes 1, 2, 3 and 4 command speeds).

Axis 3: 2000.00 mm/min (Speed is limited at ratios in axes 1, 2, 3 and 4 command speeds).

Axis 4: 750.00 mm/min (Speed is limited at ratios in axes 1, 2, 3 and 4 command speeds).

Note: Operation runs at speed 1 when a reference axis speed is less than 1 as a result of speed limit.

In addition, when a bias speed is set, it will be the minimum speed.

- (5) An error "No command speed" (error code: 503) will occur if a current speed (-1) is set in "Da.8 Command speed".
- (6) The software stroke limit check is not carried out when the control unit is set to "degree".

### ■ Positioning data setting examples

The following table shows the setting examples when "4-axis speed control (forward run: speed 4)" is set in the positioning data No. 1 of axis 1 (reference axis).

| Setting item \ Axis           |                                          | Axis 1<br>(reference axis) setting example | Axis 2<br>(interpolation axis) setting example | Axis 3<br>(interpolation axis) setting example | Axis 4<br>(interpolation axis) setting example | Setting details                                                                                                                                                                            |
|-------------------------------|------------------------------------------|--------------------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1 Positioning data No. 1 | Da.1 Operation pattern                   | Positioning complete                       | —                                              | —                                              | —                                              | Setting other than "Positioning complete" is not possible in speed control.                                                                                                                |
|                               | Da.2 Control system                      | Forward run speed 4                        | —                                              | —                                              | —                                              | Set 4-axis speed control.                                                                                                                                                                  |
|                               | Da.3 Acceleration time No.               | 1                                          | —                                              | —                                              | —                                              | Designate the value set in " Pr.25 Acceleration time 1" as the acceleration time at start.                                                                                                 |
|                               | Da.4 Deceleration time No.               | 0                                          | —                                              | —                                              | —                                              | Designate the value set in " Pr.10 Deceleration time 0" as the deceleration time at deceleration.                                                                                          |
|                               | Da.5 Axis to be interpolated             | —                                          | —                                              | —                                              | —                                              | Setting not required (setting value will be ignored).<br>When axis 1 is used as a reference axis, the interpolation axes are axes 2, 3 and 4.                                              |
|                               | Da.6 Positioning address/movement amount | —                                          | —                                              | —                                              | —                                              | Setting not required (setting value will be ignored).                                                                                                                                      |
|                               | Da.7 Arc address                         | —                                          | —                                              | —                                              | —                                              | Setting not required (setting value will be ignored).                                                                                                                                      |
|                               | Da.8 Command speed                       | 6000.00 mm/min                             | 3000.00 mm/min                                 | 2000.00 mm/min                                 | 1000.00 mm/min                                 | Set the speed to be commanded.                                                                                                                                                             |
|                               | Da.9 Dwell time                          | —                                          | —                                              | —                                              | —                                              | Setting not required (setting value will be ignored).                                                                                                                                      |
|                               | Da.10 M code                             | 10                                         | —                                              | —                                              | —                                              | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.<br>(" Pr.18 M code ON signal output timing" setting only possible in the WITH mode.) |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

## 9.2.18 Speed-position switching control (INC mode)

In "speed-position switching control (INC mode)" (" **Da.2** Control system" = Forward run: speed/position, Reverse run: speed/position), the pulses of the speed set in " **Da.8** Command speed" are kept output on the axial direction set to the positioning data. When the "speed-position switching signal" is input, position control of the movement amount set in " **Da.6** Positioning address/movement amount" is exercised.

"Speed-position switching control (INC mode)" is available in two different types: "forward run: speed/position" which starts the axis in the forward run direction and "reverse run: speed/position" which starts the axis in the reverse run direction.

Use the detailed parameter 1 " **Pr.150** Speed-position function selection" with regard to the choice for "speed-position switching control (INC mode)".

| Setting item  |                                   | Setting value | Setting details                             | Buffer memory address |        |        |        |
|---------------|-----------------------------------|---------------|---------------------------------------------|-----------------------|--------|--------|--------|
|               |                                   |               |                                             | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <b>Pr.150</b> | Speed-position function selection | 0             | Speed-position switching control (INC mode) | 34                    | 184    | 334    | 484    |

\* If the set value is other than 0 and 2, it is regarded as 0 and operation is performed in the INC mode.  
For details of the setting, refer to Section 5.2 "List of parameters".

#### ■ Switching over from speed control to position control

- (1) The control is switched over from speed control to position control by executing the external command signal set in "speed-position switching signal".
- (2) Besides setting the positioning data, the " **Cd.24** Speed-position switching enable flag" must also be turned ON to switch over from speed control to position control. (If the " **Cd.24** Speed-position switching enable flag" turns ON after the speed-position switching signal turns ON, the control will continue as speed control without switching over to position control. The control will be switched over from speed control to position control when the speed-position switching signal turns from OFF to ON again. Only position control will be carried out when the " **Cd.24** Speed-position switching enable flag" and speed-position switching signal are ON at the operation start.)

| Setting item |                                      | Setting value | Setting details                                                                                         | Buffer memory address |        |        |        |
|--------------|--------------------------------------|---------------|---------------------------------------------------------------------------------------------------------|-----------------------|--------|--------|--------|
|              |                                      |               |                                                                                                         | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <b>Cd.24</b> | Speed-position switching enable flag | 1             | Set "1: Switch from speed control to position control when the external command signal [CHG] turns ON." | 1528                  | 1628   | 1728   | 1828   |



### ■ Speed-position switching signal setting

The following table shows the items that must be set to use the external command signals (CHG) as speed-position switching signals.

| Setting item       |                                     | Setting value | Setting details                                                    | Buffer memory address |        |        |        |
|--------------------|-------------------------------------|---------------|--------------------------------------------------------------------|-----------------------|--------|--------|--------|
|                    |                                     |               |                                                                    | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <span>Pr.42</span> | External command function selection | 2             | Set the "2: speed-position and position-speed switching requests". | 62                    | 212    | 362    | 512    |
| <span>Cd.8</span>  | External command valid              | 1             | Set "1: Validate external command".                                | 1505                  | 1605   | 1705   | 1805   |

\* Refer to Section 5.2 "List of parameters" and Section 5.7 "List of control data" for information on the setting details.

### ■ Operation chart

The following chart (Fig.9.13) shows the operation timing for speed-position switching control (INC mode). The "in speed control flag" ([Md.31] Status: b0) is turned ON during speed control of speed-position switching control (INC mode).

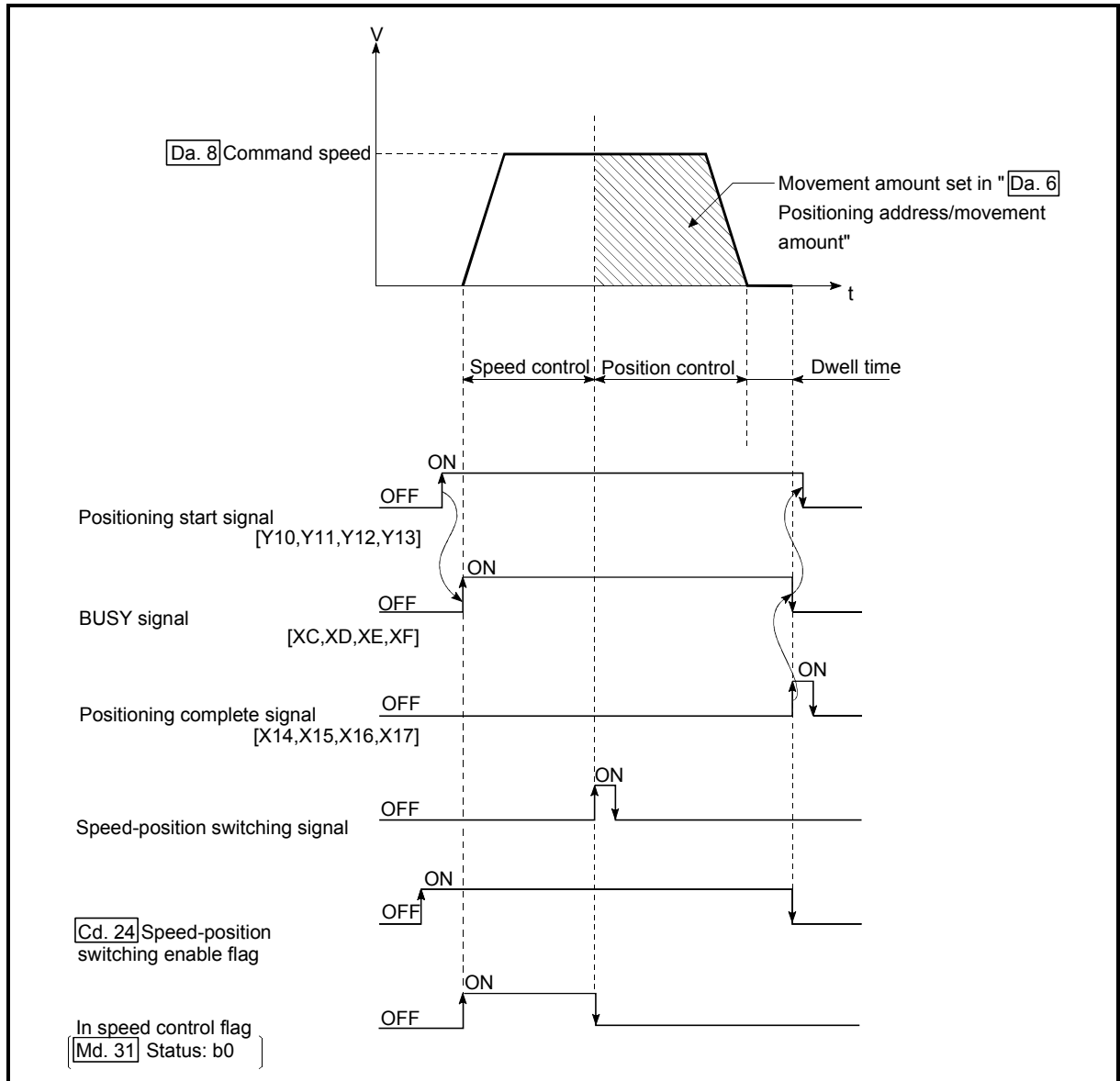
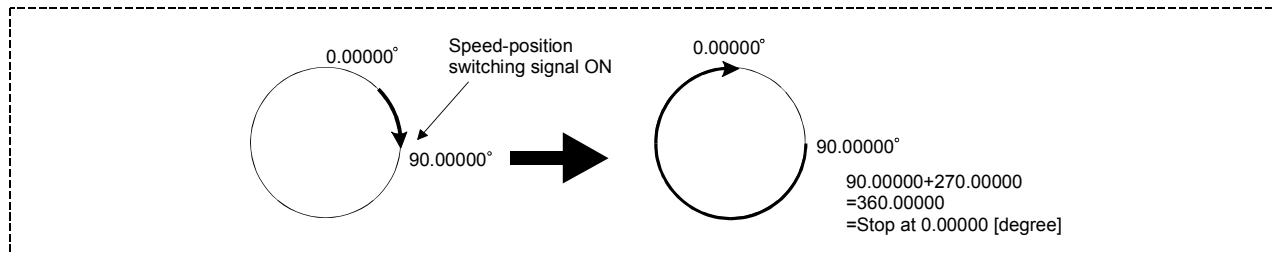


Fig. 9.13 Speed-position switching control (INC mode) operation timing

## [Operation example]

The following operation assumes that the speed-position switching signal is input at the position of the current feed value of 90.00000 [degree] during execution of " [Da.2] Control system "Forward run: speed/position" at " [Pr.1] Unit setting " of "2: degree" and " [Pr.21] Current feed value during speed control" setting of "1: Update current feed value".

(The value set in " [Da.6] Positioning address/movement amount" is 270.00000 [degree])



### ■ Operation timing and processing time during speed-position switching control (INC mode)

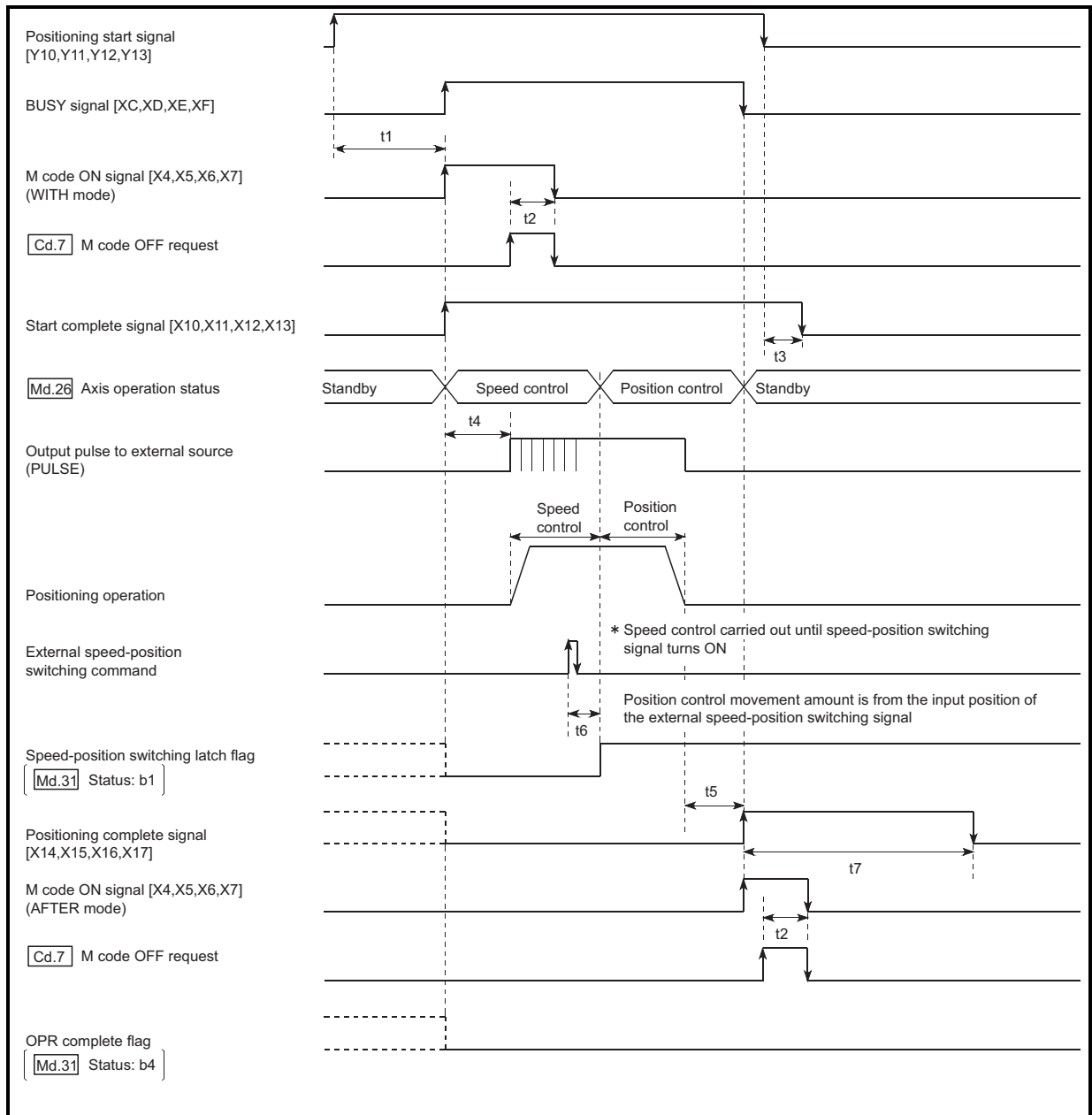


Fig. 9.14 Operation timing and processing time during speed-position switching control (INC mode)

#### Normal timing time

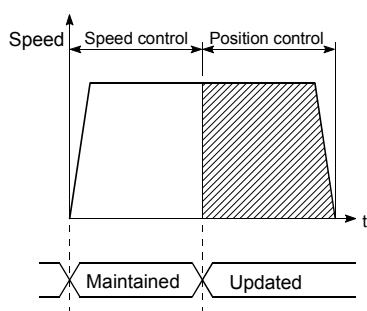
| Model           | t1           | t2         | t3         | t4           | t5         | t6    | t7                 |
|-----------------|--------------|------------|------------|--------------|------------|-------|--------------------|
| QD75P□N/QD75D□N | 0.2 to 1.1ms | 0 to 0.9ms | 0 to 0.9ms | 0.4 to 1.3ms | 0 to 0.9ms | 1.0ms | Follows parameters |
| QD75P□/QD75D□   | 1.1ms        | 0 to 1.8ms | 0 to 1.8ms | 2.7 to 4.4ms | 0 to 1.8ms | 1.0ms |                    |

- The t1 timing time could be delayed by the operation state of other axes.

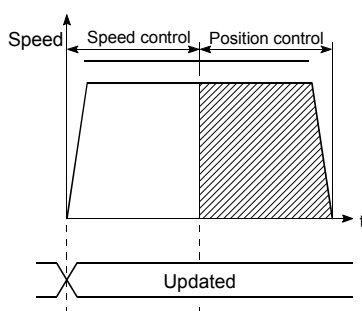
### ■ Current feed value during speed-position switching control (INC mode)

The following table shows the "Md.20 Current feed value" during speed-position switching control (INC mode) corresponding to the "Pr.21 Current feed value during speed control" settings.

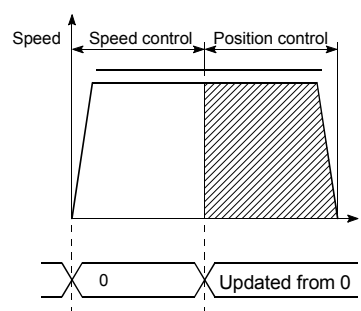
| "Pr.21 Current feed value during speed control" setting | Md.20 Current feed value                                                                                                        |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 0: Do not update current feed value                     | The current feed value at control start is maintained during speed control, and updated from the switching to position control. |
| 1: Update current feed value                            | The current feed value is updated during speed control and position control.                                                    |
| 2: Zero clear current feed value                        | The current feed value is cleared (set to "0") at control start, and updated from the switching to position control.            |



(a) Current feed value not updated



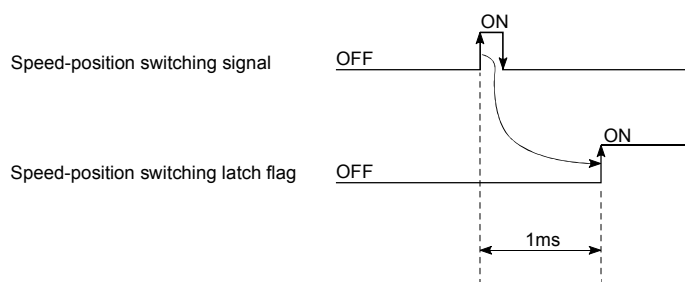
(b) Current feed value updated



(c) Current feed value zero cleared

### ■ Switching time from speed control to position control

There is 1ms from the time the speed-position switching signal is turned ON to the time the speed-position switching latch flag (Md.31 Status: b1) turns ON.



### ■ Changing the position control movement amount

In "speed-position switching control (INC mode)", the position control movement amount can be changed during the speed control section.

- (1) The "new movement amount" is stored in " **Cd.23** Speed-position switching control movement amount change register" by the sequence program during speed control.

When the speed-position switching signal is turned ON, the movement amount for position control is stored in " **Cd.23** Speed-position switching control movement amount change register".

- (2) The movement amount is stored in the " **Md.29** Speed-position switching control positioning amount" of the axis monitor area from the point where the control changes to position control by the input of a speed-position switching signal from an external source.

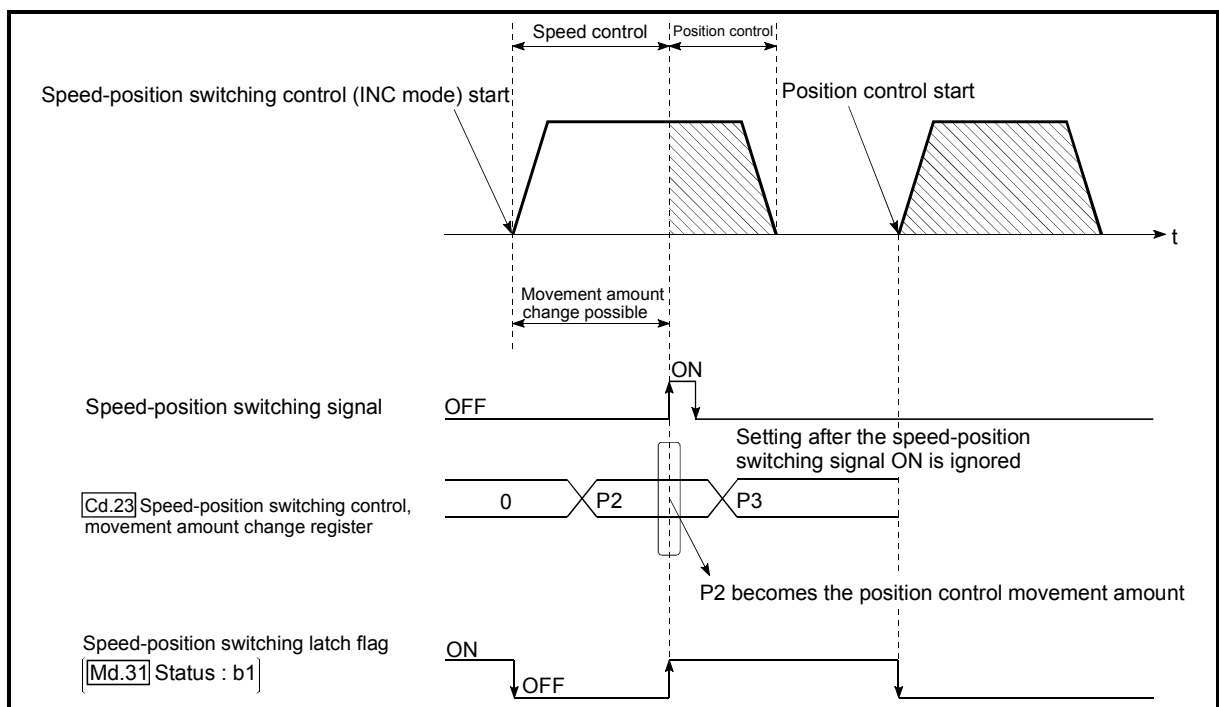


Fig. 9.15 Position control movement amount change timing

## POINT

- The machine recognizes the presence of a movement amount change request when the data is written to " [Cd.23] Speed-position switching control movement amount change register" with the sequence program.
- The new movement amount is validated after execution of the speed-position switching control (INC mode), before the input of the speed-position switching signal.
- The movement amount change can be enable/disable with the interlock function in position control using the "speed-position switching latch flag" ([Md.31] Status : b1) of the axis monitor area.

### ■ Restrictions

- (1) The axis error "Continuous path control not possible" (error code: 516) will occur, and the operation cannot start if "continuous path control" is set in " [Da.1] Operation pattern".
- (2) "Speed-position switching control" cannot be set in " [Da.2] Control system" of the positioning data when "continuous path control" has been set in " [Da.1] Operation pattern" of the immediately prior positioning data. (For example, if the operation pattern of positioning data No. 1 is "continuous path control", "speed-position switching control" cannot be set in positioning data No. 2.) The axis error "Continuous path control not possible" (error code: 516) will occur and the machine will carry out a deceleration stop if this type of setting is carried out.
- (3) The error "No command speed" (error code: 503) will occur if "current speed (-1)" is set in " [Da.8] Command speed".
- (4) The software stroke limit range check during speed control is made only when the following (a) and (b) are satisfied:
  - (a) When " [Pr.21] Current feed value during speed control" is "1: Update current feed value"
 

If the movement amount exceeds the software stroke limit range during speed control in cases of settings other than the one above, the error "Software stroke limit+" (error code: 507) or "Software stroke limit-" (error code: 508) will occur as soon as speed control is changed to position control, and the axis will decelerate to a stop.
  - (b) When " [Pr.1] Unit setting" is other than "2: degree"
 

If the unit is "degree", the software stroke limit range check is not performed.
- (5) If the value set in " [Da.6] Positioning address/movement amount" is negative, the error "Outside address range" (error code: 530) will occur.
- (6) Deceleration processing is carried out from the point where the speed-position switching signal is input if the position control movement amount set in " [Da.6] Positioning address/movement amount" is smaller than the deceleration distance from the " [Da.8] Command speed".
- (7) Turn ON the speed-position switching signal in the speed stabilization region (constant speed status). The warning "Speed-position switching (during acceleration) signal ON" (warning code: 508) will occur because of large deviation in the droop pulse amount if the signal is turned ON during acceleration.
 

During use of the servomotor, the actual movement amount after switching of speed control to position control is the "preset movement amount + droop pulse amount". If the signal is turned ON during acceleration/deceleration, the stop position will vary due to large variation of the droop pulse amount. Even though " [Md.29] Speed-position switching control positioning amount" is the same, the stop position will change due to a change in droop pulse amount when " [Da.8] Command speed" is different.

### ■ Positioning data setting examples

The following table shows setting examples when "speed-position switching control (INC mode) by forward run" is set in positioning data No. 1 of axis 1.

| Setting item                     |                                                 | Setting example             | Setting details                                                                                                                                                                                      |
|----------------------------------|-------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1<br>Positioning data No. 1 | <b>Da.1</b> Operation pattern                   | Positioning complete        | Set "Positioning complete" assuming the next positioning data will not be executed. ("Continuous path control" cannot be set in "speed-position switching control (INC mode)".)                      |
|                                  | <b>Da.2</b> Control system                      | Forward run: speed/position | Set speed-position switching control by forward run.                                                                                                                                                 |
|                                  | <b>Da.3</b> Acceleration time No.               | 1                           | Designate the value set in " <b>Pr.25</b> Acceleration time 1" as the acceleration time at start.                                                                                                    |
|                                  | <b>Da.4</b> Deceleration time No.               | 0                           | Designate the value set in " <b>Pr.10</b> Deceleration time 0" as the deceleration time at deceleration.                                                                                             |
|                                  | <b>Da.5</b> Axis to be interpolated             | –                           | Setting not required. (Setting value is ignored.)                                                                                                                                                    |
|                                  | <b>Da.6</b> Positioning address/movement amount | 10000.0μm                   | INC mode ( <b>Pr.150</b> = 0)<br>Set the movement amount after the switching to position control. (Assuming that the " <b>Pr.1</b> Unit setting" is set to "mm".)                                    |
|                                  | <b>Da.7</b> Arc address                         | –                           | Setting not required. (Setting value is ignored.)                                                                                                                                                    |
|                                  | <b>Da.8</b> Command speed                       | 6000.00mm/min               | Set the speed to be controlled.                                                                                                                                                                      |
|                                  | <b>Da.9</b> Dwell time                          | 500ms                       | Set a time from the positioning stop (pulse output stop) by position control until the positioning complete signal is output. When the system is stopped by speed control, ignore the setting value. |
|                                  | <b>Da.10</b> M code                             | 10                          | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                                                                                                |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.



## 9.2.19 Speed-position switching control (ABS mode)

In case of "speed-position switching control (ABS mode)" (" **Da.2** Control system" = Forward run: speed/position, Reverse run: speed/position), the pulses of the speed set in " **Da.8** Command speed" are kept output in the axial direction set to the positioning data. When the "speed-position switching signal" is input, position control to the address set in " **Da.6** Positioning address/movement amount" is exercised.

"Speed-position switching control (ABS mode)" is available in two different types: "forward run: speed/position" which starts the axis in the forward run direction and "reverse run: speed/position" which starts the axis in the reverse run direction.

"Speed-position switching control (ABS mode)" is valid only when " **Pr.1** Unit setting" is "2: degree".

| <b>Pr.1</b> Unit setting          | mm | inch | degree | pulse |
|-----------------------------------|----|------|--------|-------|
| Speed-position function selection |    |      |        |       |
| INC mode                          | ○  | ○    | ○      | ○     |
| ABS mode                          | ×  | ×    | ○      | ×     |

○: Setting allowed,

×: Setting disallowed (If setting is made, the error "Speed-position function selection error" (error code: 935) will occur when the PLC READY (Y0) turns ON.)

Use the detailed parameter 1 " **Pr.150** Speed-position function selection" to choose "speed-position switching control (ABS mode)".

| Setting item  |                                   | Setting value | Setting details                             | Buffer memory address |        |        |        |
|---------------|-----------------------------------|---------------|---------------------------------------------|-----------------------|--------|--------|--------|
|               |                                   |               |                                             | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <b>Pr.150</b> | Speed-position function selection | 2             | Speed-position switching control (ABS mode) | 34                    | 184    | 334    | 484    |

\* If the set value is other than 0 and 2, it is regarded as 0 and operation is performed in the INC mode.

For details of the setting, refer to Section 5.2 "List of parameters".

### ■ Switching over from speed control to position control

- (1) The control is switched over from speed control to position control by executing the external command signal set in "speed-position switching signal".
- (2) Besides setting the positioning data, the " Cd.24 Speed-position switching enable flag" must also be turned ON to switch over from speed control to position control. (If the " Cd.24 Speed-position switching enable flag" turns ON after the speed-position switching signal turns ON, the control will continue as speed control without switching over to position control. The control will be switched over from speed control to position control when the speed-position switching signal turns from OFF to ON again. Only position control will be carried out when the " Cd.24 Speed-position switching enable flag" and speed-position switching signal are ON at the operation start.)

| Setting item                                                        |                                      | Setting value | Setting details                                                                                         | Buffer memory address |        |        |        |
|---------------------------------------------------------------------|--------------------------------------|---------------|---------------------------------------------------------------------------------------------------------|-----------------------|--------|--------|--------|
|                                                                     |                                      |               |                                                                                                         | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <span style="border: 1px solid black; padding: 0 2px;">Cd.24</span> | Speed-position switching enable flag | 1             | Set "1: Switch from speed control to position control when the external command signal [CHG] turns ON." | 1528                  | 1628   | 1728   | 1828   |

### ■ Speed-position switching signal setting

The following table shows the items that must be set to use the external command signals (CHG) as speed-position switching signals.

| Setting item                                                        |                                     | Setting value | Setting details                                                    | Buffer memory address |        |        |        |
|---------------------------------------------------------------------|-------------------------------------|---------------|--------------------------------------------------------------------|-----------------------|--------|--------|--------|
|                                                                     |                                     |               |                                                                    | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <span style="border: 1px solid black; padding: 0 2px;">Pr.42</span> | External command function selection | 2             | Set the "2: speed-position and position-speed switching requests". | 62                    | 212    | 362    | 512    |
| <span style="border: 1px solid black; padding: 0 2px;">Cd.8</span>  | External command valid              | 1             | Set "1: Validate external command".                                | 1505                  | 1605   | 1705   | 1805   |

\* Refer to Section 5.2 "List of parameters" and Section 5.7 "List of control data" for information on the setting details.

### ■ Operation chart

The following chart (Fig.9.16) shows the operation timing for speed-position switching control (ABS mode). The "in speed control flag" (Md.31 Status: b0) is turned ON during speed control of speed-position switching control (ABS mode).

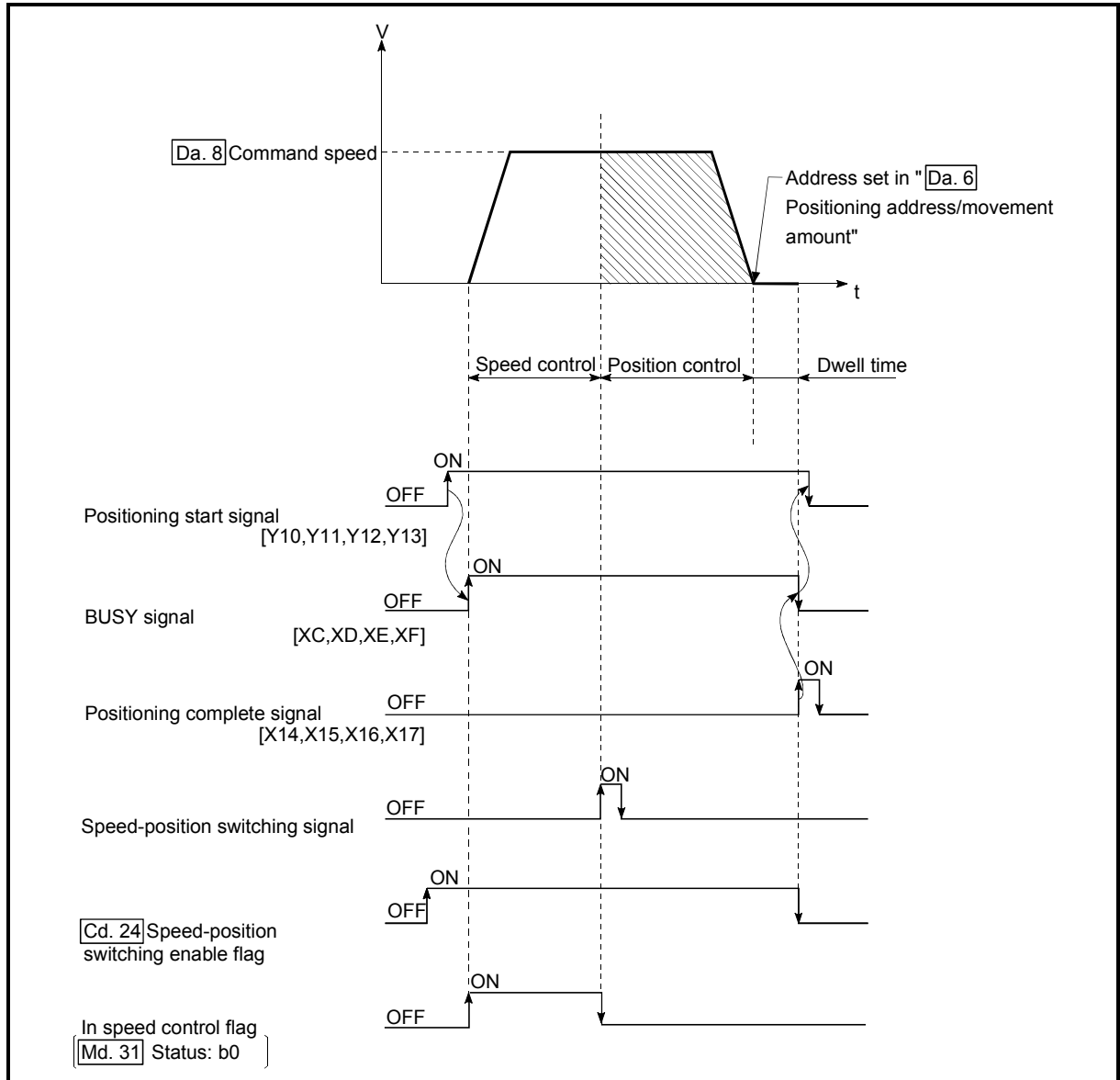
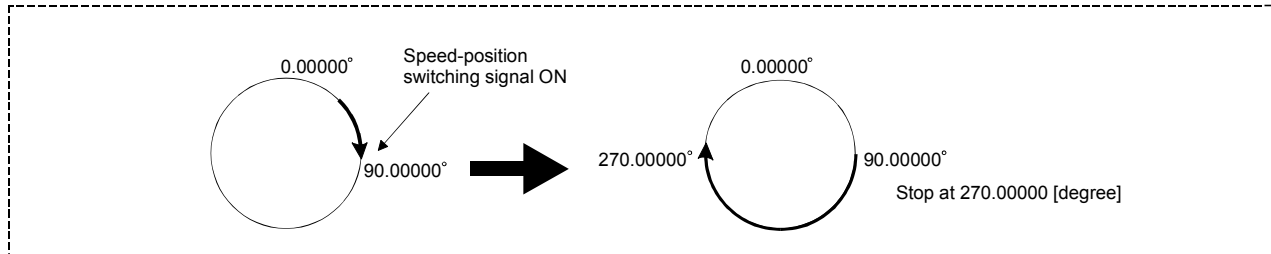


Fig. 9.16 Speed-position switching control (ABS mode) operation timing

## [Operation example]

The following operation assumes that the speed-position switching signal is input at the position of the current feed value of 90.00000 [degree] during execution of " [Da.2] Control system" "Forward run: speed/position" at " [Pr.1] Unit setting" of "2: degree" and " [Pr.21] Current feed value during speed control" setting of "1: Update current feed value".

(The value set in " [Da.6] Positioning address/movement amount" is 270.00000 [degree])



■ Operation timing and processing time during speed-position switching control (ABS mode)

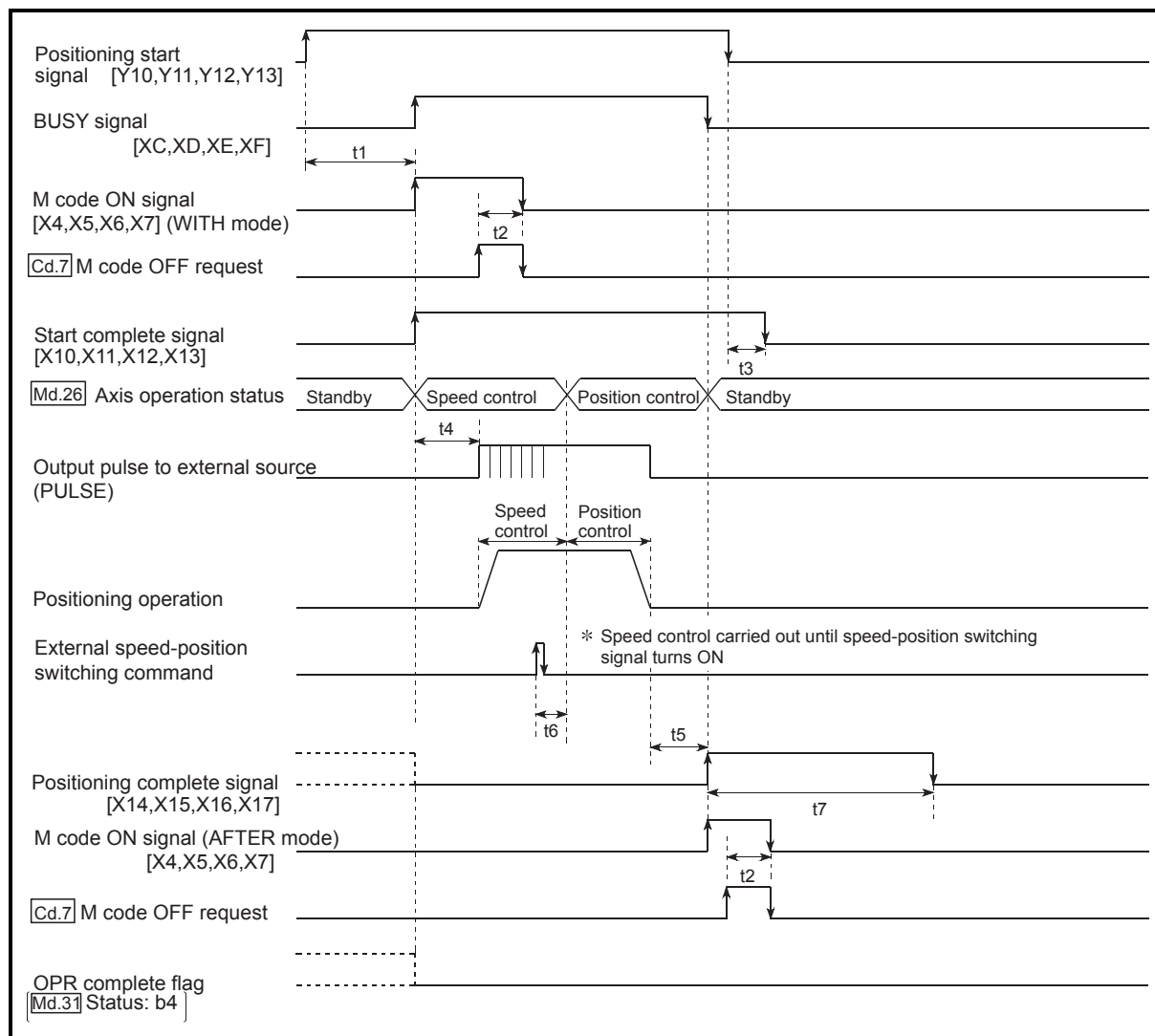


Fig. 9.17 Operation timing and processing time during speed-position switching control (ABS mode)

Normal timing time

| Model           | t1           | t2         | t3         | t4           | t5         | t6    | t7                 |
|-----------------|--------------|------------|------------|--------------|------------|-------|--------------------|
| QD75P□N/QD75D□N | 0.2 to 1.1ms | 0 to 0.9ms | 0 to 0.9ms | 0.4 to 1.3ms | 0 to 0.9ms | 1.0ms | Follows parameters |
| QD75P□/QD75D□   | 1.1ms        | 0 to 1.8ms | 0 to 1.8ms | 2.7 to 4.4ms | 0 to 1.8ms | 1.0ms |                    |

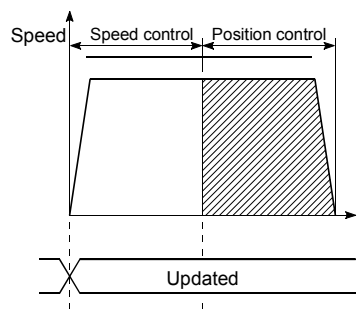
- The t1 timing time could be delayed by the operation state of other axes.

### ■ Current feed value during speed-position switching control (ABS mode)

The following table shows the " [Md.20] Current feed value" during speed-position switching control (ABS mode) corresponding to the " [Pr.21] Current feed value during speed control" settings.

| " [Pr.21] Current feed value during speed control" setting | [Md.20] Current feed value                                                   |
|------------------------------------------------------------|------------------------------------------------------------------------------|
| 1: Update current feed value                               | The current feed value is updated during speed control and position control. |

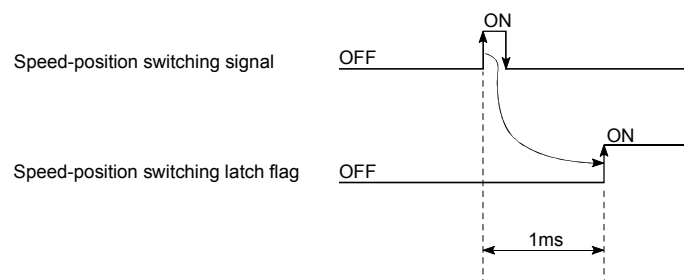
Only "1: Update current value" is valid for the setting of " [Pr.21] Current feed value during speed control" in speed-position switching control (ABS mode). The error "Speed-position function selection error" (error code: 935) will occur if the " [Pr.21] Current feed value during speed control" setting is other than 1.



Current feed value updated

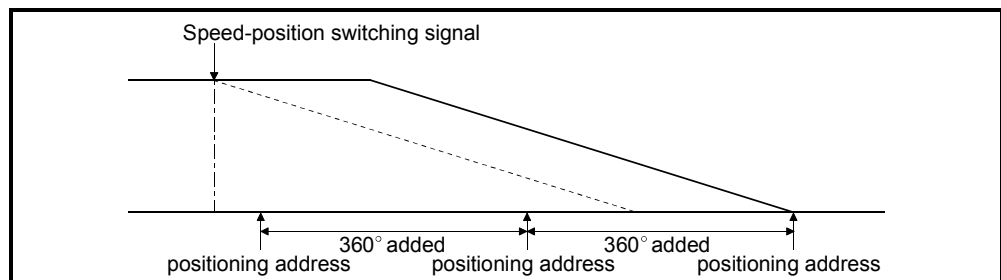
### ■ Switching time from speed control to position control

There is 1ms from the time the speed-position switching signal is turned ON to the time the speed-position switching latch flag ([Md.31] Status: b1) turns ON.



### ■ Restrictions

- (1) The axis error "Continuous path control not possible" (error code: 516) will occur, and the operation cannot start if "continuous path control" is set in " [Da.1] Operation pattern".
- (2) "Speed-position switching control" cannot be set in " [Da.2] Control system" of the positioning data when "continuous path control" has been set in " [Da.1] Operation pattern" of the immediately prior positioning data. (For example, if the operation pattern of positioning data No. 1 is "continuous path control", "speed-position switching control" cannot be set in positioning data No. 2.) The axis error "Continuous path control not possible" (error code: 516) will occur and the machine will carry out a deceleration stop if this type of setting is carried out.
- (3) The error "No command speed" (error code: 503) will occur if "current speed (-1)" is set in " [Da.8] Command speed".
- (4) If the value set in " [Da.6] Positioning address/movement amount" is negative, the error "Outside address range" (error code: 530) will occur.
- (5) Even though the axis control data " [Cd.23] Speed-position switching control movement amount change register" was set in speed-position switching control (ABS mode), it would not function. The set value is ignored.
- (6) To exercise speed-position switching control (ABS mode), the following conditions must be satisfied:
  - (a) " [Pr.1] Unit setting" is "2: degree"
  - (b) The software stroke limit function is invalid (upper limit value = lower limit value)
  - (c) " [Pr.21] Current feed value during speed control" is "1: Update current feed value"
  - (d) The setting range of " [Da.6] Positioning address/movement amount" is within 0 to 359.99999 (degree)  
(If the value is outside of the range, the error "Outside address range" (error code: 530) will occur at the start.)
  - (e) The " [Pr.150] Speed-position function selection" setting is "2: Speed-position switching control (ABS mode)".
- (7) If any of the conditions in (6)(a) to (6)(c) is not satisfied in the case of (6)(e), the error "Speed-position function selection error" (error code: 935) will occur when the PLC READY [Y0] turns from OFF to ON.
- (8) If the axis reaches the positioning address midway through deceleration after automatic deceleration started at the input of the speed-position switching signal, the axis will not stop immediately at the positioning address. The axis will stop at the positioning address after N revolutions so that automatic deceleration can always be made. (N: Natural number)  
In the following example, since making deceleration in the path of dotted line will cause the axis to exceed the positioning addresses twice, the axis will decelerate to a stop at the third positioning address.



### ■ Positioning data setting examples

The following table shows setting examples when "speed-position switching control (ABS mode) by forward run" is set in positioning data No. 1 of axis 1.

| Setting item                     |                                                 | Setting example             | Setting details                                                                                                                                                                                      |
|----------------------------------|-------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1<br>Positioning data No. 1 | <b>Da.1</b> Operation pattern                   | Positioning complete        | Set "Positioning complete" assuming the next positioning data will not be executed. ("Continuous path control" cannot be set in "speed-position switching control (ABS mode)".)                      |
|                                  | <b>Da.2</b> Control system                      | Forward run: speed/position | Set speed-position switching control by forward run.                                                                                                                                                 |
|                                  | <b>Da.3</b> Acceleration time No.               | 1                           | Designate the value set in " <b>Pr.25</b> Acceleration time 1" as the acceleration time at start.                                                                                                    |
|                                  | <b>Da.4</b> Deceleration time No.               | 0                           | Designate the value set in " <b>Pr.10</b> Deceleration time 0" as the deceleration time at deceleration.                                                                                             |
|                                  | <b>Da.5</b> Axis to be interpolated             | –                           | Setting not required. (Setting value is ignored.)                                                                                                                                                    |
|                                  | <b>Da.6</b> Positioning address/movement amount | 270.00000 degree            | ABS mode ( <b>Pr.150</b> = 2)<br>Set the address after the switching to position control. (Assuming that the " <b>Pr.1</b> Unit setting" is set to "degree".)                                        |
|                                  | <b>Da.7</b> Arc address                         | –                           | Setting not required. (Setting value is ignored.)                                                                                                                                                    |
|                                  | <b>Da.8</b> Command speed                       | 6000.00mm/min               | Set the speed to be controlled.                                                                                                                                                                      |
|                                  | <b>Da.9</b> Dwell time                          | 500ms                       | Set a time from the positioning stop (pulse output stop) by position control until the positioning complete signal is output. When the system is stopped by speed control, ignore the setting value. |
|                                  | <b>Da.10</b> M code                             | 10                          | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                                                                                                |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.



## 9.2.20 Position-speed switching control

In "position-speed switching control" (" [Da.2] Control system" = Forward run: position/speed, Reverse run: position/speed), before the position-speed switching signal is input, position control is carried out for the movement amount set in " [Da.6] Positioning address/movement amount" in the axis direction in which the positioning data has been set. When the position-speed switching signal is input, the position control is carried out by continuously outputting the pulses for the speed set in " [Da.8] Command speed" until the input of a stop command.

The two types of position-speed switching control are "Forward run: position/speed" in which the control starts in the forward run direction, and "Reverse run: position/speed" in which control starts in the reverse run direction.

#### ■ Switching over from position control to speed control

- (1) The control is switched over from position control to speed control by executing the external command signal set in "position-speed switching signal".
- (2) Besides setting the positioning data, the " [Cd.26] Position-speed switching enable flag" must also be turned ON to switch over from position control to speed control. (If the " [Cd.26] Position-speed switching enable flag" turns ON after the position-speed switching signal turns ON, the control will continue as position control without switching over to speed control. The control will be switched over from position control to speed control when the position-speed switching signal turns from OFF to ON again. Only speed control will be carried out when the " [Cd.26] Position-speed switching enable flag" and position-speed switching signal are ON at the operation start.)

| Setting item |                                      | Setting value | Setting details                                                                                         | Buffer memory address |        |        |        |
|--------------|--------------------------------------|---------------|---------------------------------------------------------------------------------------------------------|-----------------------|--------|--------|--------|
|              |                                      |               |                                                                                                         | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| [Cd.26]      | Position-speed switching enable flag | 1             | Set "1: Switch from position control to speed control when the external command signal [CHG] turns ON". | 1532                  | 1632   | 1732   | 1832   |

#### ■ Position-speed switching signal setting

The following table shows the items that must be set to use the external command signals (CHG) as position-speed switching signals.

| Setting item |                                     | Setting value | Setting details                                                    | Buffer memory address |        |        |        |
|--------------|-------------------------------------|---------------|--------------------------------------------------------------------|-----------------------|--------|--------|--------|
|              |                                     |               |                                                                    | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| [Pr.42]      | External command function selection | 2             | Set the "2: speed-position and position-speed switching requests". | 62                    | 212    | 362    | 512    |
| [Cd.8]       | External command valid              | 1             | Set "1: Validate external command".                                | 1505                  | 1605   | 1705   | 1805   |

\* Refer to Section 5.2 "List of parameters" and Section 5.7 "List of control data" for information on the setting details.

### ■ Operation chart

The following chart shows the operation timing for position-speed switching control.

The "in speed control" flag (**[Md.31]** Status: b0) is turned ON during speed control of position-speed switching control.

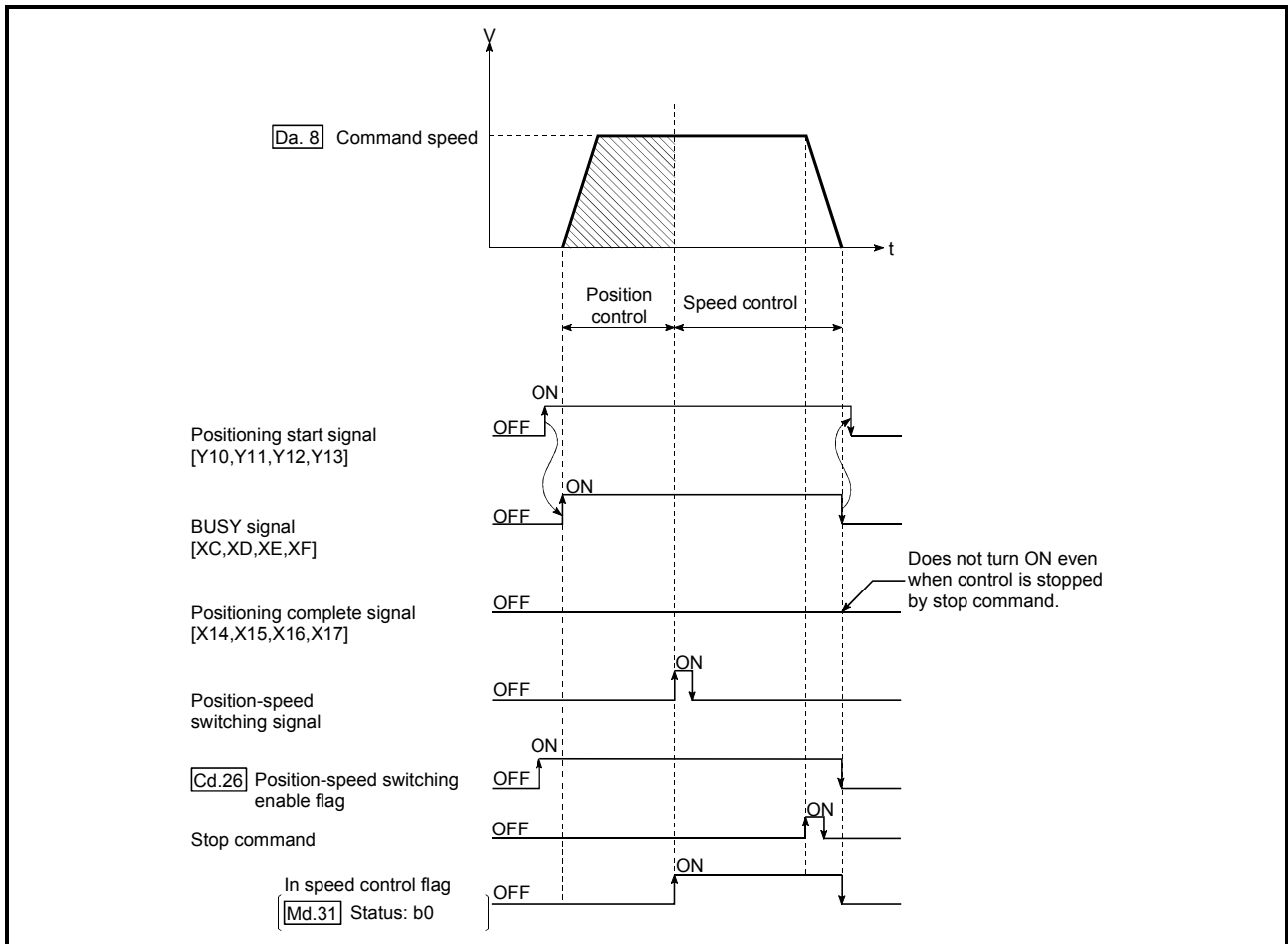


Fig. 9.18 Position-speed switching control operation timing

### ■ Operation timing and processing time during position-speed switching control

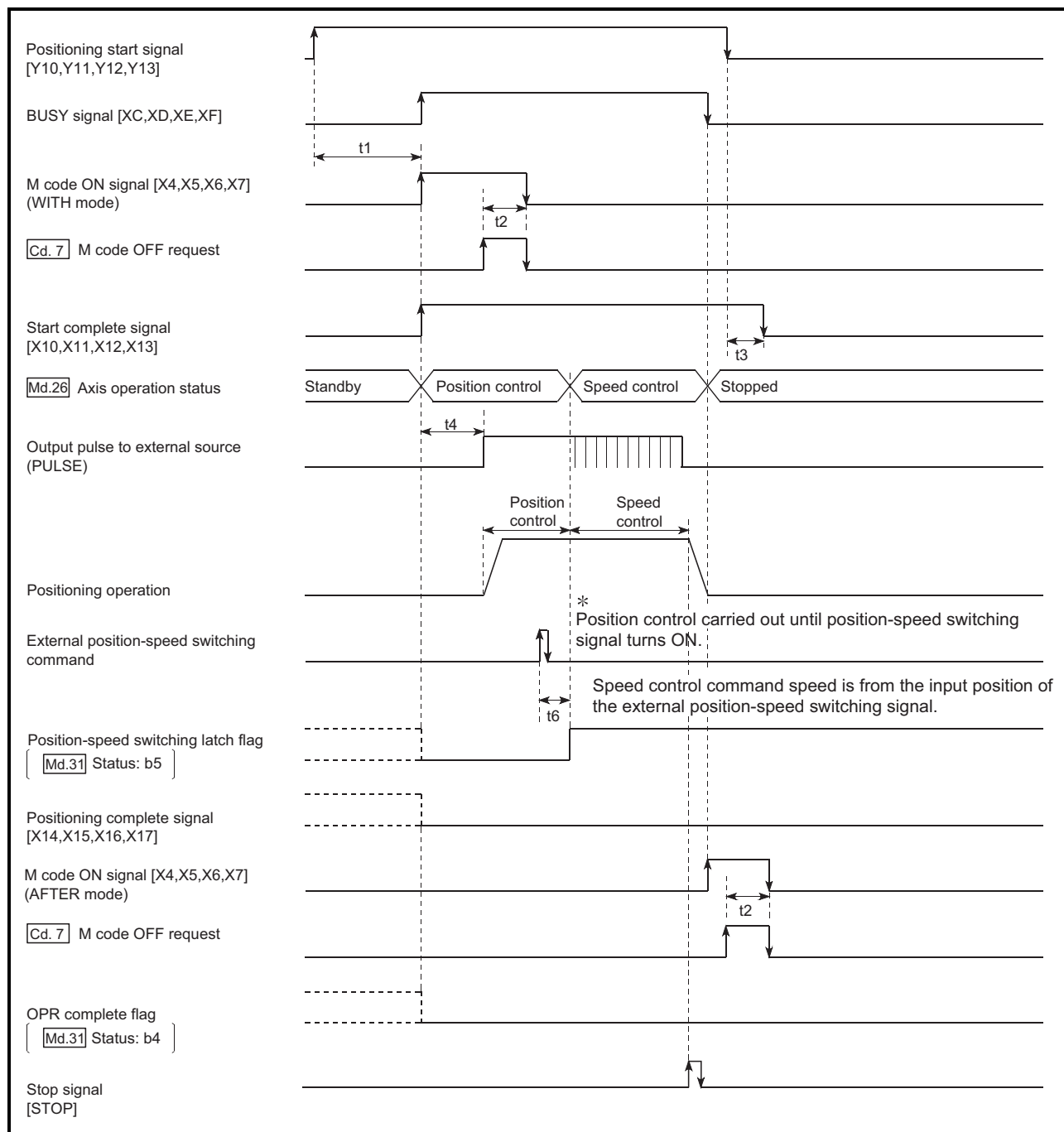


Fig. 9.19 Operation timing and processing time during position-speed switching control

#### Normal timing time

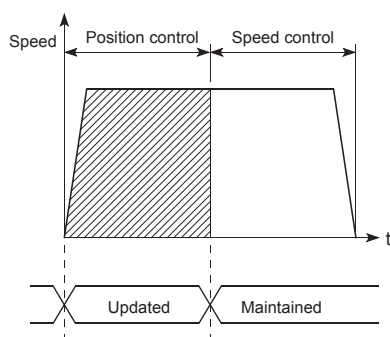
| Model           | t1           | t2         | t3         | t4           | t5 | t6    |
|-----------------|--------------|------------|------------|--------------|----|-------|
| QD75P□N/QD75D□N | 0.2 to 1.1ms | 0 to 0.9ms | 0 to 0.9ms | 0.4 to 1.3ms | -  | 1.0ms |
| QD75P□/QD75D□   | 1.0 to 1.4ms | 0 to 1.8ms | 0 to 1.8ms | 2.7 to 4.4ms | -  | 1.0ms |

- The t1 timing time could be delayed by the operation state of other axes.

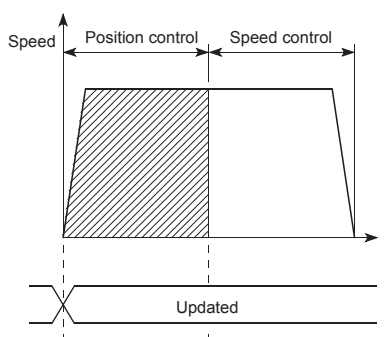
### ■ Current feed value during position-speed switching control

The following table shows the " [Md.20] Current feed value" during position-speed switching control corresponding to the " [Pr.21] Current feed value during speed control" settings.

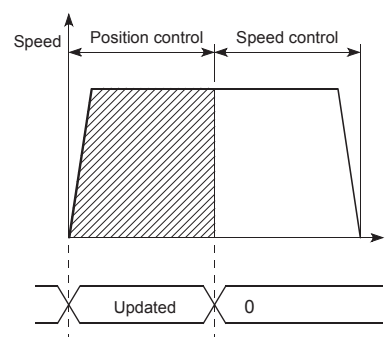
| " [Pr.21] Current feed value during speed control" setting | [Md.20] Current feed value                                                                                                                                                             |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0: Do not update current feed value                        | The current feed value is updated during position control, and the current feed value at the time of switching is maintained as soon as position control is switched to speed control. |
| 1: Update current feed value                               | The current feed value is updated during position control and speed control.                                                                                                           |
| 2: Zero clear current feed value                           | The current feed value is updated during positioning control, and the current feed value is cleared (to "0") as soon as position control is switched to speed control.                 |



(a) Current feed value not updated



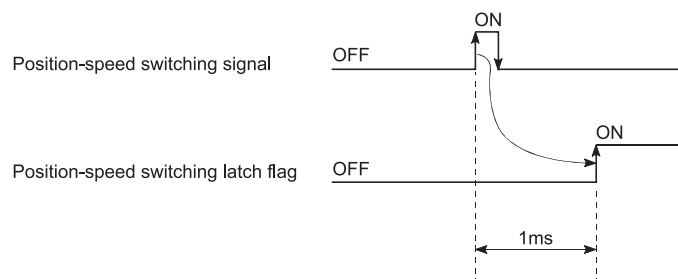
(b) Current feed value updated



(c) Current feed value zero cleared

### ■ Switching time from position control to speed control

There is 1ms from the time the position-speed switching signal is turned ON to the time the position-speed switching latch flag ([Md.31] Status: b5) turns ON.



### ■ Changing the speed control command speed

In "position-speed switching control", the speed control command speed can be changed during the position control.

- (1) The speed control command speed can be changed during the position control of position-speed switching control.

A command speed change request will be ignored unless issued during the position control of the position-speed switching control.

- (2) The "new command speed" is stored in " Cd.25 Position-speed switching control speed change register" by the sequence program during position control.

This value then becomes the speed control command speed when the position-speed switching signal turns ON.

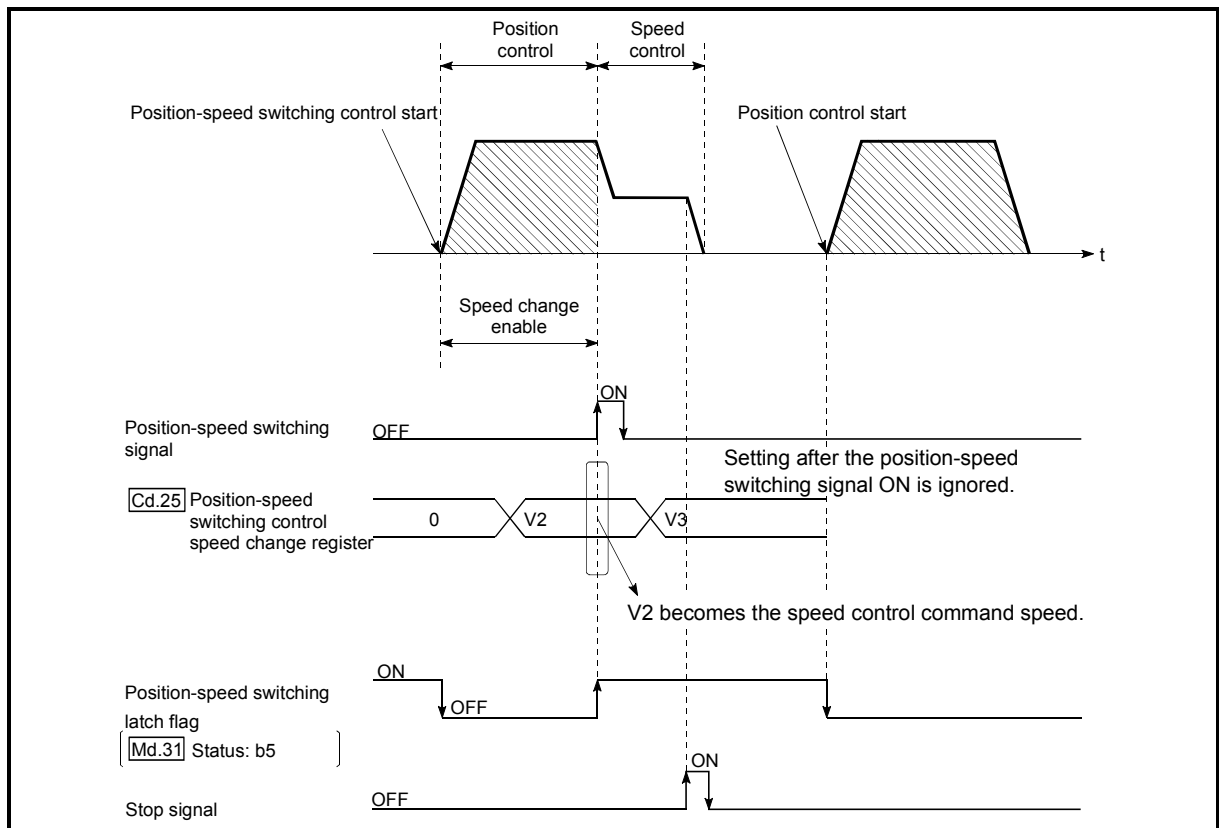


Fig. 9.20 Speed control speed change timing

**POINTS**

- The machine recognizes the presence of a command speed change request when the data is written to " [Cd.25] Position-speed switching control speed change register" with the sequence program.
- The new command speed is validated after execution of the position-speed switching control before the input of the position-speed switching signal.
- The command speed change can be enabled/disabled with the interlock function in speed control using the "position-speed switching latch flag" ([Md.31] Status: b5) of the axis monitor area.

**■ Restrictions**

- (1) The axis error "Continuous path control not possible" (error code: 516) will occur and the operation cannot start if "continuous positioning control" or "continuous path control" is set in " [Da.1] Operation pattern".
- (2) "Position-speed switching control" cannot be set in " [Da.2] Control system" of the positioning data when "continuous path control" has been set in " [Da.1] Operation pattern" of the immediately prior positioning data. (For example, if the operation pattern of positioning data No. 1 is "continuous path control", "position-speed switching control" cannot be set in positioning data No. 2.) The axis error "Continuous path control not possible" (error code: 516) will occur and the machine will carry out a deceleration stop if this type of setting is carried out.
- (3) The software stroke limit range is only checked during speed control if the "1: Update current feed value" is set in " [Pr.21] Current feed value during speed control".  
The software stroke limit range is not checked when the control unit is set to "degree".
- (4) The error "Software stroke limit+" or "Software stroke limit-" (error code: 507 or 508) will occur and the operation cannot start if the start point address or end point address for position control exceeds the software stroke limit range.
- (5) Deceleration stop will be carried out if the position-speed switching signal is not input before the machine is moved by a specified movement amount.  
When the position-speed switching signal is input during automatic deceleration by positioning control, acceleration is carried out again to the command speed to continue speed control.  
When the position-speed switching signal is input during deceleration to a stop with the stop signal, the control is switched to the speed control to stop the machine.  
Restart is carried out by speed control using the restart command.
- (6) The warning "Speed limit value over" (warning code: 501) will occur and control is continued by " [Pr.8] Speed limit value" if a new speed exceeds " [Pr.8] Speed limit value" at the time of change of the command speed.
- (7) If the value set in " [Da.6] Positioning address/movement amount" is negative, the error "Outside address range" (error code: 530) will occur.

### ■ Positioning data setting examples

The following table shows setting examples when "position-speed switching control (forward run: position/speed)" is set in positioning data No. 1 of axis 1.

| Setting item                     |              | Setting example                     | Setting details             |
|----------------------------------|--------------|-------------------------------------|-----------------------------|
| Axis 1<br>Positioning data No. 1 | <b>Da.1</b>  | Operation pattern                   | Positioning complete        |
|                                  | <b>Da.2</b>  | Control system                      | Forward run: position/speed |
|                                  | <b>Da.3</b>  | Acceleration time No.               | 1                           |
|                                  | <b>Da.4</b>  | Deceleration time No.               | 0                           |
|                                  | <b>Da.5</b>  | Axis to be interpolated             | –                           |
|                                  | <b>Da.6</b>  | Positioning address/movement amount | 10000.0μm                   |
|                                  | <b>Da.7</b>  | Arc address                         | –                           |
|                                  | <b>Da.8</b>  | Command speed                       | 6000.00mm/min               |
|                                  | <b>Da.9</b>  | Dwell time                          | 500ms                       |
|                                  | <b>Da.10</b> | M code                              | 10                          |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

## 9.2.21 Current value changing

When the current value is changed to a new value, control is carried out in which the " [Md.20] Current feed value" of the stopped axis is changed to a random address set by the user. (The " [Md.21] Machine feed value" is not changed when the current value is changed.)

The two methods for changing the current value are shown below.

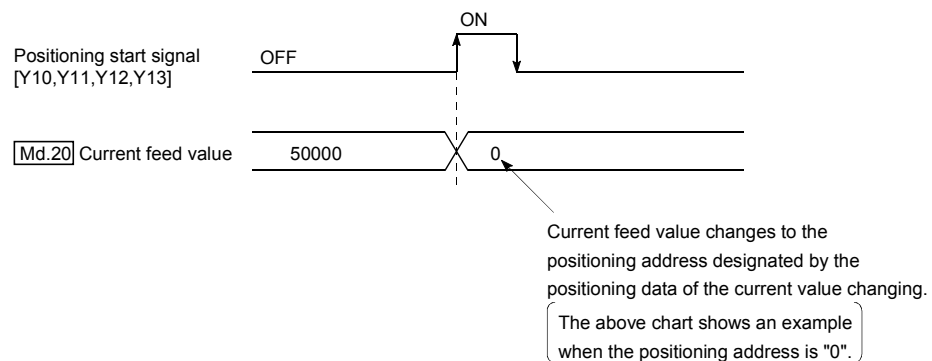
- [1] Changing to a new current value using the positioning data
- [2] Changing to a new current value using the start No. (No. 9003) for a current value changing

The current value changing using method [1] is used during continuous positioning of multiple blocks, etc.

## [1] Changing to a new current value using the positioning data

## ■ Operation chart

The following chart shows the operation timing for a current value changing. The " [Md.20] Current feed value" is changed to the value set in " [Da.6] Positioning address/movement amount" when the positioning start signal turns ON.



## ■ Restrictions

- (1) An axis error "New current value not possible" (error code: 515) will occur and the operation cannot start if "continuous path control" is set in " [Da.1] Operation pattern". ("Continuous path control" cannot be set in current value changing.)
- (2) "Current value changing" cannot be set in " [Da.2] Control system" of the positioning data when "continuous path control" has been set in " [Da.1] Operation pattern" of the immediately prior positioning data. (For example, if the operation pattern of positioning data No. 1 is "continuous path control", "current value changing" cannot be set in positioning data No. 2.) An axis error "New current value not possible" (error code: 515) will occur and the machine will carry out a deceleration stop if this type of setting is carried out.
- (3) An axis error "Outside new current value range" (error code: 514) will occur and the operation cannot start if "degree" is set in " [Pr.1] Unit setting" and the value set in " [Da.6] Positioning address/movement amount" is outside the setting range (0 to 359.99999 [degree]).



- (4) If the value set in " **Da.6** Positioning address/movement amount" is outside the software stroke limit ( **Pr.12**, **Pr.13** ) setting range, an error "Software stroke limit +" or "Software stroke limit –" (error code: 507 or 508) will occur at the positioning start, and the operation will not start.

#### ■ Positioning data setting examples

The following table shows the setting examples when " current value changing" is set in the positioning data No. 1 of axis 1.

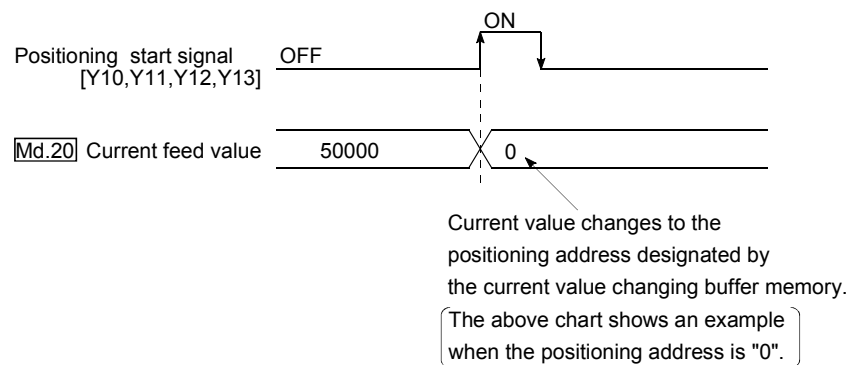
| Setting item                        |              |                                               | Setting example        | Setting details                                                                                                                                         |
|-------------------------------------|--------------|-----------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1<br>positioning<br>data No. 1 | <b>Da.1</b>  | Operation<br>pattern                          | Positioning complete   | Set "Positioning complete" assuming that the next positioning data will be executed. ("Continuous path control" cannot be set by current value change.) |
|                                     | <b>Da.2</b>  | Control<br>system                             | Current value changing | Set the current value changing.                                                                                                                         |
|                                     | <b>Da.3</b>  | Acceleration<br>time No.                      | –                      | Setting not required (Setting value is ignored.)                                                                                                        |
|                                     | <b>Da.4</b>  | Deceleration<br>time No.                      | –                      | Setting not required (Setting value is ignored.)                                                                                                        |
|                                     | <b>Da.5</b>  | Axis to be<br>interpolated                    | –                      | Setting not required (Setting value is ignored.)                                                                                                        |
|                                     | <b>Da.6</b>  | Positioning<br>address/<br>movement<br>amount | 10000.0 $\mu$ m        | Set the address to which address change is desired. (Assuming that the " <b>Pr.1</b> Unit setting" is set to "mm".)                                     |
|                                     | <b>Da.7</b>  | Arc address                                   | –                      | Setting not required (Setting value is ignored.)                                                                                                        |
|                                     | <b>Da.8</b>  | Command<br>speed                              | –                      | Setting not required (Setting value is ignored.)                                                                                                        |
|                                     | <b>Da.9</b>  | Dwell time                                    | –                      | Setting not required (Setting value is ignored.)                                                                                                        |
|                                     | <b>Da.10</b> | M code                                        | 10                     | Set this when other sub operation commands are issued in combination with the No. 1 positioning data.                                                   |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

## [2] Changing to a new current value using the start No. (No. 9003) for a current value changing

### ■ Operation chart

The current value is changed by setting the new current value in the current value changing buffer memory " [Cd.9] Current value changing", setting "9003" in the " [Cd.3] Positioning start No.", and turning ON the positioning start signal.

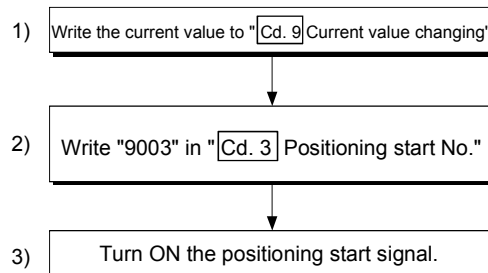


### ■ Restrictions

- (1) An axis error "Outside new current value range" (error code: 514) will occur if the designated value is outside the setting range when "degree" is set in "Unit setting".
- (2) An error "Software stroke limit +" or "Software stroke limit -" (error code: 507 or 508) will occur if the designated value is outside the software stroke limit range.
- (3) The current value cannot be changed during stop commands and while the M code ON signal is ON.
- (4) The M code output function is made invalid.

### ■ Current value changing procedure

The following shows the procedure for changing the current value to a new value.



### ■ Setting method for the current value changing function

The following shows an example of a sequence program and data setting to change the current value to a new value with the positioning start signal. (The " **Md.20** " Current feed value is changed to "5000.0μm" in the example shown.)

(1) Set the following data.

(Set with the sequence program shown in (3), while referring to the start time chart shown in (2).)

| Setting item                       | Setting value | Setting details                                     | Buffer memory address |              |              |              |
|------------------------------------|---------------|-----------------------------------------------------|-----------------------|--------------|--------------|--------------|
|                                    |               |                                                     | Axis 1                | Axis 2       | Axis 3       | Axis 4       |
| <b>Cd.3</b> Positioning start No.  | 9003          | Set the start No. "9003" for the new current value. | 1500                  | 1600         | 1700         | 1800         |
| <b>Cd.9</b> Current value changing | 50000         | Set the new " <b>Md.20</b> " Current feed value".   | 1506<br>1507          | 1606<br>1607 | 1706<br>1707 | 1806<br>1807 |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

(2) The following shows a start time chart.

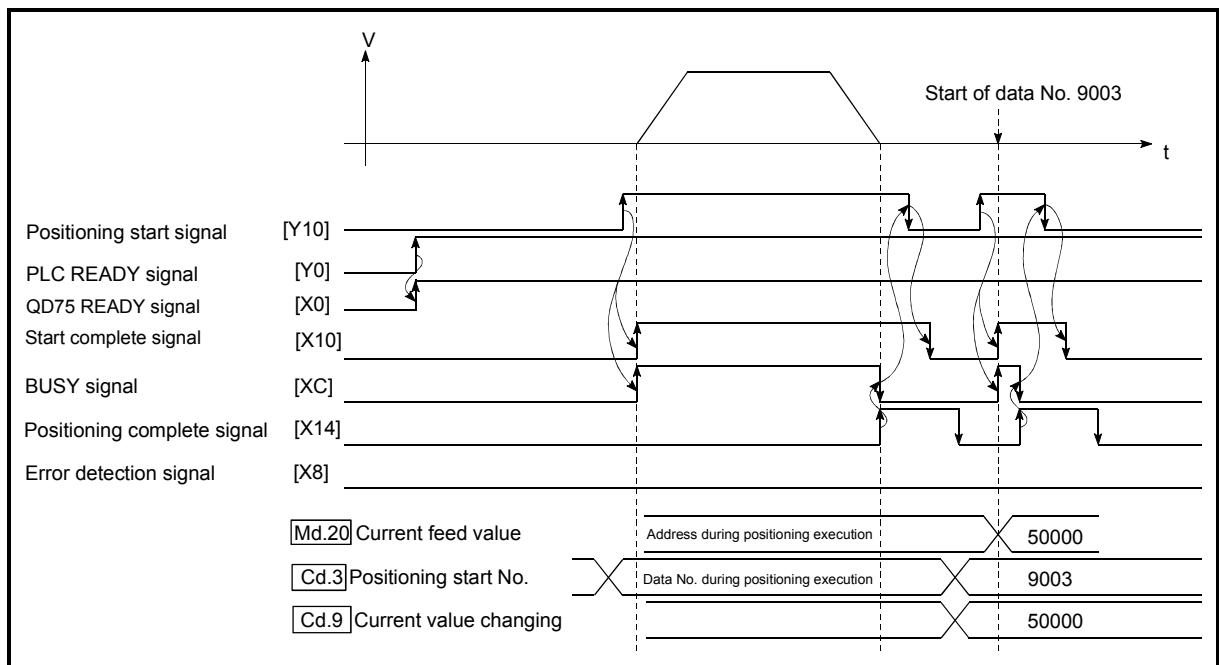
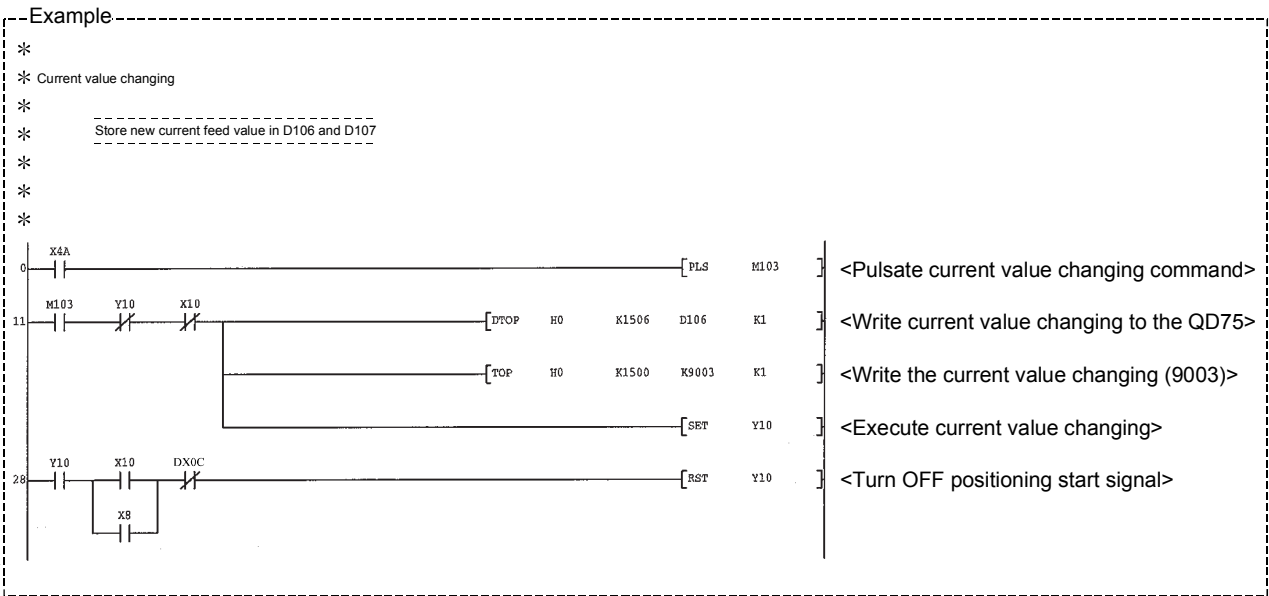


Fig. 9.21 Changing to a new current value using the start No. (No. 9003) for a current value changing

(3) Add the following sequence program to the control program, and write it to the CPU module.



## 9.2.22 NOP instruction

The NOP instruction is used for the nonexecutable control system.

### ■ Operation

The positioning data No. to which the NOP instruction is set transfers, without any processing, to the operation for the next positioning data No.

### ■ Positioning data setting examples

The following table shows the setting examples when "NOP instruction" is set in positioning data No. 1 of axis 1.

| Setting item                        |                                                       | Setting example | Setting details                                  |
|-------------------------------------|-------------------------------------------------------|-----------------|--------------------------------------------------|
| Axis 1<br>positioning<br>data No. 1 | Da.1<br>Operation<br>pattern                          | —               | Setting not required (Setting value is ignored.) |
|                                     | Da.2<br>Control<br>system                             | NOP instruction | Set the NOP instruction                          |
|                                     | Da.3<br>Acceleration<br>time No.                      | —               | Setting not required (Setting value is ignored.) |
|                                     | Da.4<br>Deceleration<br>time No.                      | —               | Setting not required (Setting value is ignored.) |
|                                     | Da.5<br>Axis to be<br>interpolated                    | —               | Setting not required (Setting value is ignored.) |
|                                     | Da.6<br>Positioning<br>address/<br>movement<br>amount | —               | Setting not required (Setting value is ignored.) |
|                                     | Da.7<br>Arc address                                   | —               | Setting not required (Setting value is ignored.) |
|                                     | Da.8<br>Command<br>speed                              | —               | Setting not required (Setting value is ignored.) |
|                                     | Da.9<br>Dwell time                                    | —               | Setting not required (Setting value is ignored.) |
|                                     | Da.10<br>M code                                       | —               | Setting not required (Setting value is ignored.) |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

### ■ Restrictions

An error "Control system setting error" (error code: 524) will occur if the "NOP instruction" is set for the control system of the positioning data No. 600.

| POINT                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>&lt;Use example of NOP instruction&gt;</p> <p>If there is a possibility of speed switching or temporary stop (automatic deceleration) at a point between two points during positioning, that data can be reserved with the NOP instruction to change the data merely by the replacement of the identifier.</p> |

### 9.2.23 JUMP instruction

The JUMP instruction is used to control the operation so it jumps to a positioning data No. set in the positioning data during "continuous positioning control" or "continuous path control".

JUMP instruction include the following two types of JUMP.

- (1) Unconditional JUMP  
When no execution conditions are set for the JUMP instruction  
(When "0" is set as the condition data No.)
- (2) Conditional JUMP  
When execution conditions are set for the JUMP instruction  
(The conditions are set in the "condition data" used with "high-level positioning control".)

Using the JUMP instruction enables repeating of the same positioning control, or selection of positioning data by the execution conditions during "continuous positioning control" or "continuous path control".

#### ■ Operation

- (1) Unconditional JUMP  
The JUMP instruction is unconditionally executed. The operation jumps to the positioning data No. set in " **Da.9** Dwell time".
- (2) Conditional JUMP  
The block start condition data is used as the JUMP instruction execution conditions.
  - When block positioning data No. 7000 to 7004 is started:  
Each block condition data is used.
  - When positioning data No. 1 to 600 is started:  
Start block 0 condition data is used.
  - When the execution conditions set in " **Da.10** M code" of the JUMP instruction have been established:  
the JUMP instruction is executed to jump the operation to the positioning data No. set in " **Da.9** Dwell time".
  - When the execution conditions set in " **Da.10** M code" of the JUMP instruction have not been established:  
the JUMP instruction is ignored, and the next positioning data No. is executed.

#### ■ Restrictions

- (1) When using a conditional JUMP instruction, establish the JUMP instruction execution conditions by the 4th positioning data No. before the JUMP instruction positioning data No.  
If the JUMP instruction execution conditions are not established by the time the 4th positioning control is carried out before the JUMP instruction positioning data No., the operation will be processed as an operation without established JUMP instruction execution conditions.  
(During execution of continuous path control/continuous positioning control, the QD75 calculates the positioning data of the positioning data No. four items ahead of the current positioning data.)

- (2) Positioning control such as loops cannot be executed by conditional JUMP instructions alone until the conditions have been established.  
As the target of the JUMP instruction, specify a positioning data that is controlled by other than JUMP and NOP instructions.

#### ■ Positioning data setting example

The following table shows setting examples when "JUMP instruction" is set in positioning data No. 1 of axis 1.

| Setting item                     |                  | Setting example                     | Setting details  |                                                                                                                                                                                                                                      |
|----------------------------------|------------------|-------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1<br>Positioning data No. 1 | <div>Da.1</div>  | Operation pattern                   | —                | Setting not required. (Setting value is ignored.)                                                                                                                                                                                    |
|                                  | <div>Da.2</div>  | Control system                      | JUMP instruction | Set the JUMP instruction.                                                                                                                                                                                                            |
|                                  | <div>Da.3</div>  | Acceleration time No.               | —                | Setting not required. (Setting value is ignored.)                                                                                                                                                                                    |
|                                  | <div>Da.4</div>  | Deceleration time No.               | —                | Setting not required. (Setting value is ignored.)                                                                                                                                                                                    |
|                                  | <div>Da.5</div>  | Axis to be interpolated             | —                | Setting not required. (Setting value is ignored.)                                                                                                                                                                                    |
|                                  | <div>Da.6</div>  | Positioning address/movement amount | —                | Setting not required. (Setting value is ignored.)                                                                                                                                                                                    |
|                                  | <div>Da.7</div>  | Arc address                         | —                | Setting not required. (Setting value is ignored.)                                                                                                                                                                                    |
|                                  | <div>Da.8</div>  | Command speed                       | —                | Setting not required. (Setting value is ignored.)                                                                                                                                                                                    |
|                                  | <div>Da.9</div>  | Dwell time                          | 500              | Set the positioning data No. 1 to 600 for the JUMP destination. (The positioning data No. of the JUMP instruction cannot be set. Setting its own positioning data No. will result in an error "Illegal data No." (error code: 502).) |
|                                  | <div>Da.10</div> | M code                              | 1                | Set the JUMP instruction execution conditions with the condition data No.<br>0 : Unconditional JUMP<br>1 to 10 : Condition data No.<br>("Simultaneous start" condition data cannot be set.)                                          |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

## 9.2.24 LOOP

The LOOP is used for loop control by the repetition of LOOP to LEND.

### ■ Operation

The LOOP to LEND loop is repeated by set repeat cycles.

### ■ Positioning data setting examples

The following table shows the setting examples when "LOOP" is set in positioning data No. 1 of axis 1.

| Setting item                     |       | Setting example                     | Setting details |                                                   |
|----------------------------------|-------|-------------------------------------|-----------------|---------------------------------------------------|
| Axis 1<br>Positioning data No. 1 | Da.1  | Operation pattern                   | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.2  | Control system                      | LOOP            | Set the LOOP.                                     |
|                                  | Da.3  | Acceleration time No.               | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.4  | Deceleration time No.               | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.5  | Axis to be interpolated             | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.6  | Positioning address/movement amount | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.7  | Arc address                         | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.8  | Command speed                       | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.9  | Dwell time                          | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.10 | M code                              | 5               | Set the LOOP to LEND repeat cycles.               |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

### ■ Restrictions

- (1) An error "Control system LOOP setting error" (error code: 545) will occur if a "0" is set for the repeat cycles.
- (2) Even if LEND is absent after LOOP, no error will occur, but repeat processing will not be carried out.
- (3) Nesting is not allowed between LOOP-LEND's. If such setting is made, only the inner LOOP-LEND is processed repeatedly.

| POINT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The setting by this control system is easier than that by the special start "FOR loop" of "High-level Positioning Control" (refer to CHAPTER 10).</p> <p>&lt;Setting data&gt;</p> <ul style="list-style-type: none"> <li>• For special start: Positioning start data, special start data, condition data, and positioning data</li> <li>• For control system: Positioning data</li> </ul> <p>For the special start FOR to NEXT, the positioning data is required for each of FOR and NEXT points. For the control system, loop can be executed even only by one data.</p> <p>Also, nesting is enabled by using the control system LOOP to LEND in combination with the special start FOR to NEXT.</p> <p>However LOOP to LEND cannot be set across block. Always set LOOP to LEND so that the processing ends within one block.</p> <p>(For details of the "block", refer to Section 10.1 "Outline of high-level positioning control".)</p> |



## 9.2.25 LEND

The LEND is used to return the operation to the top of the repeat (LOOP to LEND) loop.

### ■ Operation

When the repeat cycle designated by the LOOP becomes 0, the loop is terminated, and the next positioning data No. processing is started. (The operation pattern, if set to "Positioning complete", will be ignored.)

When the operation is stopped after the repeat operation is executed by designated cycles, the dummy positioning data (for example, incremental positioning without movement amount) is set next to LEND.

| Positioning data No. | Operation pattern       | Control system | Conditions               | Operation                                                                                                                                                              |
|----------------------|-------------------------|----------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | Continuous control      | ABS2           |                          | Executed in the order of the positioning data No. 1 → 2 → 3 → 4 → 5 → 2 → 3 → 4 → 5 → 6.<br>(The operation patterns of the positioning data Nos. 2 and 5 are ignored.) |
| 2                    | Positioning complete    | LOOP           | Number of loop cycles: 2 |                                                                                                                                                                        |
| 3                    | Continuous path control | ABS2           |                          |                                                                                                                                                                        |
| 4                    | Continuous control      | ABS2           |                          |                                                                                                                                                                        |
| 5                    | Positioning complete    | LEND           |                          |                                                                                                                                                                        |
| 6                    | Positioning complete    | ABS2           |                          |                                                                                                                                                                        |

### ■ Positioning data setting examples

The following table shows the setting examples when "LEND" is set in positioning data No. 8 of axis 1.

| Setting item                     |                                          | Setting example | Setting details                                   |
|----------------------------------|------------------------------------------|-----------------|---------------------------------------------------|
| Axis 1<br>Positioning data No. 8 | Da.1 Operation pattern                   | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.2 Control system                      | LEND            | Set the LEND.                                     |
|                                  | Da.3 Acceleration time No.               | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.4 Deceleration time No.               | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.5 Axis to be interpolated             | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.6 Positioning address/movement amount | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.7 Arc address                         | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.8 Command speed                       | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.9 Dwell time                          | –               | Setting not required. (Setting value is ignored.) |
|                                  | Da.10 M code                             | –               | Setting not required. (Setting value is ignored.) |

\* Refer to Section 5.3 "List of positioning data" for information on the setting details.

### ■ Restrictions

(1) Ignore the "LEND" before the "LOOP" is executed.

# MEMO

[illegible]

# CHAPTER 10 HIGH-LEVEL POSITIONING CONTROL

The details and usage of high-level positioning control (control functions using the "block start data") are explained in this chapter.

High-level positioning control is used to carry out applied control using the "positioning data". Examples of applied control are using conditional judgment to control "positioning data" set with the major positioning control, or simultaneously starting "positioning data" for several different axes.

Read the execution procedures and settings for each control, and set as required.

10

|        |                                                                     |        |
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## 10.1 Outline of high-level positioning control

In "high-level positioning control" the execution order and execution conditions of the "positioning data" are set to carry out more applied positioning. (The execution order and execution conditions are set in the "block start data" and "condition data".)  
The following applied positioning controls can be carried out with "high-level positioning control".

| High-level positioning control   | Details                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Block *1 start<br>(Normal start) | With one start, executes the positioning data in a random block with the set order.                                                                                                                                                                                                                                                       |
| Condition start                  | Carries out condition judgment set in the "condition data" for the designated positioning data, and then executes the "block start data".<br>• When the condition is established, the "block start data" is executed.<br>• When not established, that "block start data" is ignored, and the next point's "block start data" is executed. |
| Wait start                       | Carries out condition judgment set in the "condition data" for the designated positioning data, and then executes the "block start data".<br>• When the condition is established, the "block start data" is executed.<br>• When not established, stops the control until the condition is established. (Waits.)                           |
| Simultaneous start<br>*2         | Simultaneously executes the positioning data having the No. for the axis designated with the "condition data". (Outputs pulses at the same timing.)                                                                                                                                                                                       |
| Repeated start (FOR loop)        | Repeats the program from the "block start data" set with the "FOR loop" to the "block start data" set in "NEXT" for the designated No. of times.                                                                                                                                                                                          |
| Repeated start (FOR condition)   | Repeats the program from the "block start data" set with the "FOR condition" to the "block start data" set in "NEXT" until the conditions set in the "condition data" are established.                                                                                                                                                    |

### ■ High-level positioning control sub functions

"High-level positioning control" uses the "positioning data" set with the "major positioning control". Refer to Section 3.2.4 "Combination of QD75 main functions and sub functions" for details on sub functions that can be combined with the major positioning control.

Note that the sub function Section 12.7.7 "Pre-reading start function" cannot be used together with "high-level positioning control".

### ■ High-level positioning control from peripheral devices

"High-level positioning control" (start of the "block start data") can be executed from GX Configurator-QP test mode.

Refer to GX Configurator-QP Operating Manual for details on starting of the "block start data" from GX Configurator-QP.

### REMARK

Block \*1:

"1 block" is defined as all the data continuing from the positioning data in which "continuous positioning control" or "continuous path control" is set in the operation pattern (Da.1) to the positioning data in which "independent positioning control (Positioning complete)" is set.

Simultaneous start \*2:

Besides the simultaneous start of "block start data" system, the "simultaneous starts" include the "multiple axes simultaneous start control" of control system.  
Refer to Section 10.5 "Multiple axes simultaneous start control" for details.

## 10.1.1 Data required for high-level positioning control

"High-level positioning control" is executed by setting the required items in the "block start data" and "condition data", then starting that "block start data". Judgment about whether execution is possible, etc., is carried out at execution using the "condition data" designated in the "block start data".

"Block start data" can be set for each No. from 7000 to 7004 (called "block Nos."), and up to 50 points can be set for each axis. (This data is controlled with Nos. called "points" to distinguish it from the positioning data. For example, the 1st block start data item is called the "1st point block start data" or "point No. 1 block start data".)

"Condition data" can be set for each No. from 7000 to 7004 (called "block Nos."), and up to 10 data items can be set for each block No.

The "block start data" and "condition data" are set as 1 set for each block No.

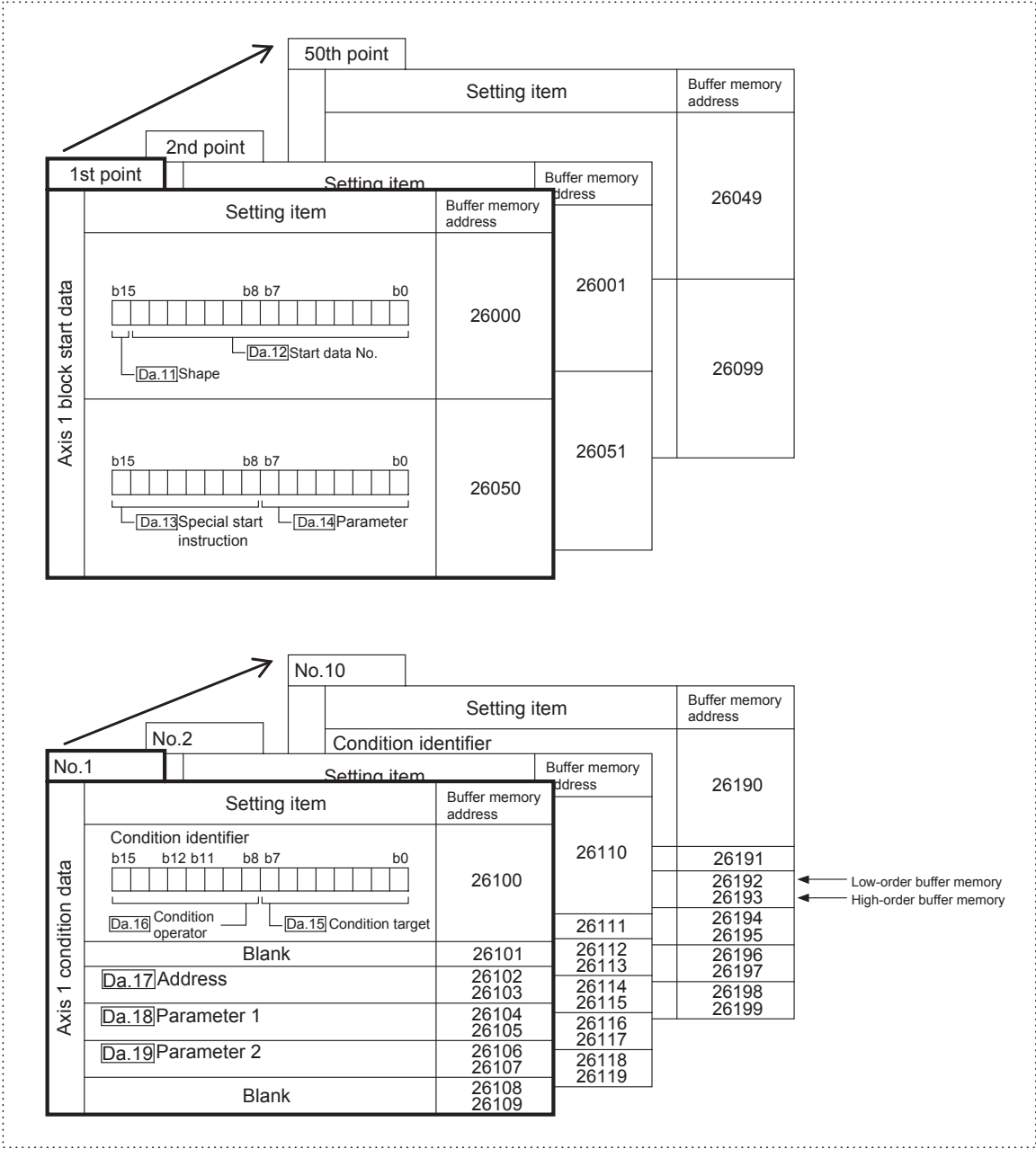
The following table shows an outline of the "block start data" and "condition data" stored in the QD75.

| Setting item     |       |                           | Setting details                                                                                                                                                     |
|------------------|-------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Block start data | Da.11 | Shape                     | Set whether to end the control after executing only the "block start data" of the shape itself, or continue executing the "block start data" set in the next point. |
|                  | Da.12 | Start data No.            | Set the "positioning data No." to be executed.                                                                                                                      |
|                  | Da.13 | Special start instruction | Set the method by which the positioning data set in Da.12 will be started.                                                                                          |
|                  | Da.14 | Parameter                 | Set the conditions by which the start will be executed according to the commands set in Da.13. (Designate the "condition data No." and "No. of repetitions".)       |

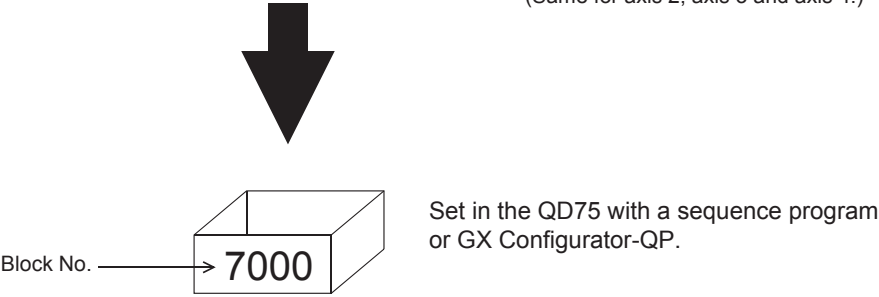
| Setting item   |       |                    | Setting details                                                                                                                                    |
|----------------|-------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Condition data | Da.15 | Condition target   | Designate the "device", "buffer memory storage details", and "positioning data No." elements for which the conditions are set.                     |
|                | Da.16 | Condition operator | Set the judgment method carried out for the target set in Da.15.                                                                                   |
|                | Da.17 | Address            | Set the buffer memory address in which condition judgment is carried out (only when the details set in Da.15 are "buffer memory storage details"). |
|                | Da.18 | Parameter 1        | Set the required conditions according to the details set in Da.15 and Da.16.                                                                       |
|                | Da.19 | Parameter 2        | Set the required conditions according to the details set in Da.15 and Da.16.                                                                       |

10.1.2 "Block start data" and "condition data" configuration

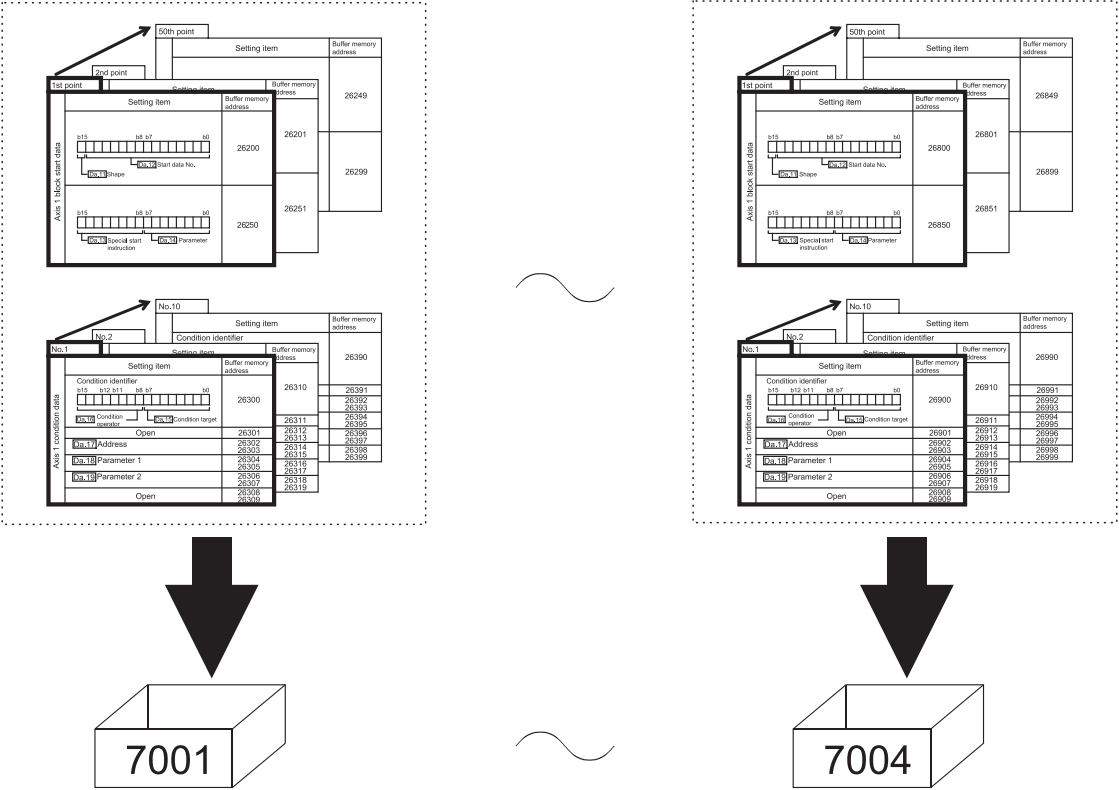
The "block start data" and "condition data" corresponding to "block No. 7000" can be stored in the buffer memory. (The following drawing shows an example for axis 1.)



(Same for axis 2, axis 3 and axis 4.)

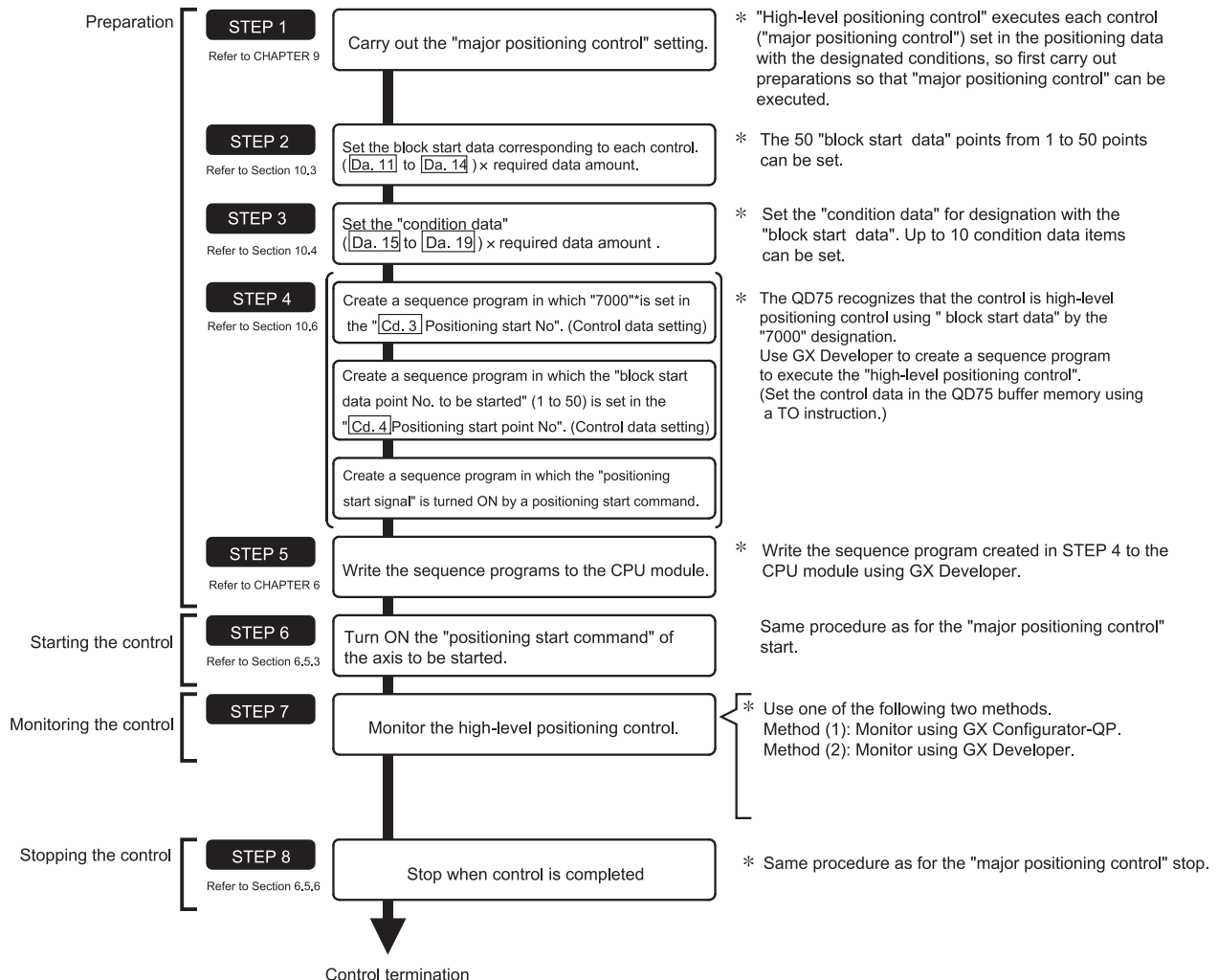


Set in QD75 the " block start data" and "condition data" corresponding to the following "block Nos. 7001 to 7004" using GX Configurator-QP or the sequence program. (The following drawing shows an example for axis 1.)



## 10.2 High-level positioning control execution procedure

High-level positioning control is carried out using the following procedure.

**REMARK**

- \* (1) Five sets of "block start data (50 points)" and "condition data (10 items)" corresponding to "7000" to "7004" are set with a sequence program.
- (2) Five sets of data from "7000" to "7004" can be set when GX Configurator-QP is used. If GX Configurator-QP is used to set the "block start data" and "condition data" corresponding to "7000" to "7004" and write the data to the QD75, "7000" to "7004" can be set in "[Cd. 3] Positioning start No." in STEP 4.



### 10.3 Setting the block start data

#### 10.3.1 Relation between various controls and block start data

The "block start data" must be set to carry out "high-level positioning control". The setting requirements and details of each "block start data" item to be set differ according to the "Da.13 Special start instruction" setting.

The following shows the "block start data" setting items corresponding to various control systems. The operation details of each control type are explained starting in Section 10.3.2. Also refer to Section 10.4 "Setting the condition data for details on "condition data" with which control execution is judged.

(The "block start data" settings in this chapter are assumed to be carried out using GX Configurator-QP.)

| High-level positioning control<br>Block start data setting items |                           |              | Block start<br>(Normal start) | Condition start    | Wait start | Simultaneous start | Repeated start<br>(FOR loop) | Repeated start<br>(FOR condition) | NEXT start * |
|------------------------------------------------------------------|---------------------------|--------------|-------------------------------|--------------------|------------|--------------------|------------------------------|-----------------------------------|--------------|
| Da.11                                                            | Shape                     | 0 : End      | ◎                             | ◎                  | ◎          | ◎                  | ×                            | ×                                 | ◎            |
|                                                                  |                           | 1 : Continue | ◎                             | ◎                  | ◎          | ◎                  | ◎                            | ◎                                 | ◎            |
| Da.12                                                            | Start data No.            |              | 1 to 600                      |                    |            |                    |                              |                                   |              |
| Da.13                                                            | Special start instruction |              | 0                             | 1                  | 2          | 3                  | 4                            | 5                                 | 6            |
| Da.14                                                            | Parameter                 |              | —                             | Condition data No. |            |                    | No. of repetitions           | Condition data No.                | —            |

◎ : One of the two setting items must be set.

○ : Set when required

×

— : Setting not required (Setting value will be ignored. Use the initial value or a value within the setting range.)

\* The "NEXT start" instruction is used in combination with "repeated start (FOR loop)" and "repeated start (FOR condition)". Control using only the "NEXT start" will not be carried out.

#### REMARK

It is recommended that the "block start data" be set whenever possible with GX Configurator-QP. Execution by sequence program uses many sequence programs and devices. The execution becomes complicated, and the scan times will increase.

## 10.3.2 Block start (normal start)

In a "block start (normal start)", the positioning data groups of a block are continuously executed in a set sequence starting from the positioning data set in " [Da.12] Start data No." by one start.

Section [2] shows a control example where the " block start data" and "positioning data" are set as shown in section [1].

## [1] Setting examples

## (1) Block start data setting example

| Axis 1 block start data | [Da.11]<br>Shape | [Da.12]<br>Start data No. | [Da.13]<br>Special start instruction | [Da.14]<br>Parameter |
|-------------------------|------------------|---------------------------|--------------------------------------|----------------------|
| 1st point               | 1: Continue      | 1                         | 0: Block start                       | —                    |
| 2nd point               | 1: Continue      | 2                         | 0: Block start                       | —                    |
| 3rd point               | 1: Continue      | 5                         | 0: Block start                       | —                    |
| 4th point               | 1: Continue      | 10                        | 0: Block start                       | —                    |
| 5th point               | 0: End           | 15                        | 0: Block start                       | —                    |
| •                       |                  |                           |                                      |                      |
| •                       |                  |                           |                                      |                      |

## (2) Positioning data setting example

| Axis 1 positioning data No. | [Da.1]<br>Operation pattern        |             |
|-----------------------------|------------------------------------|-------------|
| 1                           | 00: Positioning complete           |             |
| 2                           | 11: Continuous path control        | } 1 block * |
| 3                           | 01: Continuous positioning control |             |
| 4                           | 00: Positioning complete           | } 1 block   |
| 5                           | 11: Continuous path control        |             |
| 6                           | 00: Positioning complete           |             |
| •                           |                                    |             |
| 10                          | 00: Positioning complete           |             |
| •                           |                                    |             |
| 15                          | 00: Positioning complete           |             |
| •                           |                                    |             |

**REMARK**

Block \* :

"1 block" is defined as all the data continuing from the positioning data in which "continuous positioning control" or "continuous path control" is set in the operation pattern ([Da.1]) to the positioning data in which "independent positioning control (Positioning complete)" is set.

## [2] Control examples

The following shows the control executed when the "block start data" of the 1st point of axis 1 is set as shown in section [1] and started.

<1> The positioning data is executed in the following order before stopping.  
Axis 1 positioning data No. 1 → 2 → 3 → 4 → 5 → 6 → 10 → 15.

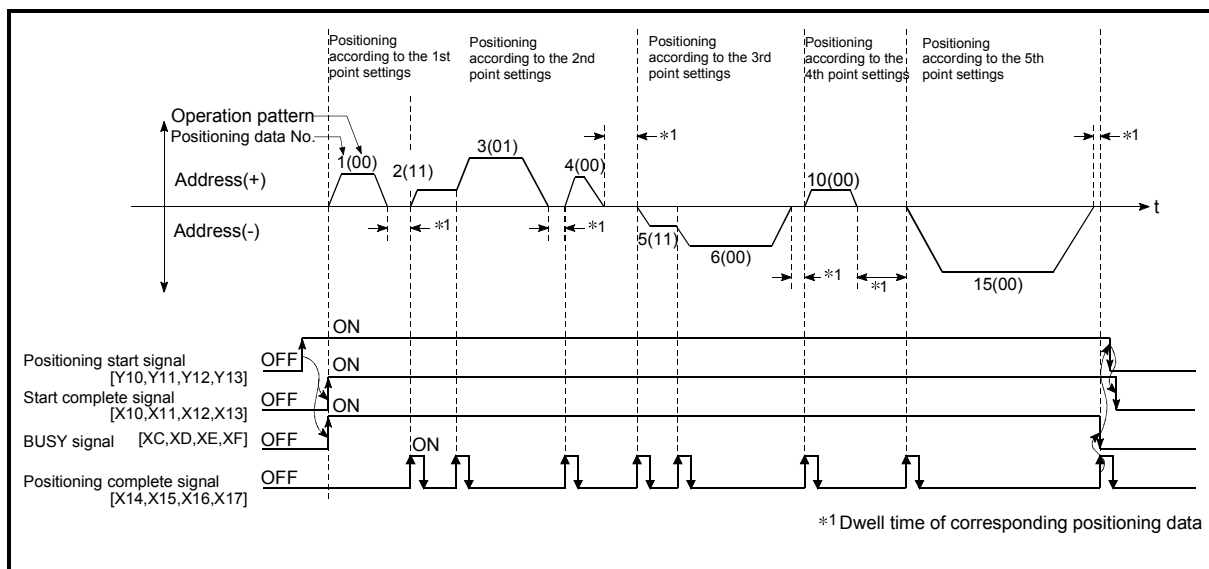


Fig. 10.1 Block start control example

## 10.3.3 Condition start

In a "condition start", the "condition data" conditional judgment designated in " [Da.14] Parameter" is carried out for the positioning data set in " [Da.12] Start data No.". If the conditions have been established, the " block start data" set in "1: condition start" is executed. If the conditions have not been established, that " block start data" will be ignored, and the "block start data" of the next point will be executed.

Section [2] shows a control example where the " block start data" and "positioning data" are set as shown in section [1].

## [1] Setting examples

## (1) Block start data setting example

| Axis 1 block start data | [Da.11]<br>Shape | [Da.12]<br>Start data No. | [Da.13]<br>Special start instruction | [Da.14]<br>Parameter |
|-------------------------|------------------|---------------------------|--------------------------------------|----------------------|
| 1st point               | 1: Continue      | 1                         | 1: Condition start                   | 1                    |
| 2nd point               | 1: Continue      | 10                        | 1: Condition start                   | 2                    |
| 3rd point               | 0: End           | 50                        | 0: Block start                       | —                    |
| •                       |                  |                           |                                      |                      |
| •                       |                  |                           |                                      |                      |

\* The "condition data Nos." have been set in " [Da.14] Parameter".

## (2) Positioning data setting example

| Axis 1 position-<br>ing data No. | [Da.1]<br>Operation pattern        |
|----------------------------------|------------------------------------|
| 1                                | 01: Continuous positioning control |
| 2                                | 01: Continuous positioning control |
| 3                                | 00: Positioning complete           |
| •                                |                                    |
| 10                               | 11: Continuous path control        |
| 11                               | 11: Continuous path control        |
| 12                               | 00: Positioning complete           |
| •                                |                                    |
| 50                               | 00: Positioning complete           |
| •                                |                                    |

## [2] Control examples

The following shows the control executed when the " block start data" of the 1st point of axis 1 is set as shown in section [1] and started.

<1> The conditional judgment set in "condition data No. 1" is carried out before execution of the axis 1 "positioning data No. 1".

→ Conditions established → Execute positioning data No. 1, 2, and 3 → Go to <2>.

→ Conditions not established → Go to <2>.

<2> The conditional judgment set in "condition data No. 2" is carried out before execution of the axis 1 "positioning data No. 10".

→ Conditions established → Execute positioning data No. 10, 11, and 12 → Go to <3>.

→ Conditions not established → Go to <3>.

<3> Execute axis 1 "positioning data No. 50" and stop the control.

## 10.3.4 Wait start

In a "wait start", the "condition data" conditional judgment designated in " [Da.14] Parameter" is carried out for the positioning data set in " [Da.12] Start data No.". If the conditions have been established, the " block start data" is executed. If the conditions have not been established, the control stops (waits) until the conditions are established.

Section [2] shows a control example where the " block start data" and "positioning data" are set as shown in section [1].

## [1] Setting examples

## (1) Block start data setting example

| Axis 1 block start data | [Da.11]<br>Shape | [Da.12]<br>Start data No. | [Da.13]<br>Special start instruction | [Da.14]<br>Parameter |
|-------------------------|------------------|---------------------------|--------------------------------------|----------------------|
| 1st point               | 1: Continue      | 1                         | 2: Wait start                        | 3                    |
| 2nd point               | 1: Continue      | 10                        | 0: Block start                       | —                    |
| 3rd point               | 0: End           | 50                        | 0: Block start                       | —                    |
| •                       |                  |                           |                                      |                      |
| •                       |                  |                           |                                      |                      |

\* The "condition data Nos." have been set in " [Da.14] Parameter".

## (2) Positioning data setting example

| Axis 1 position-<br>ing data No. | [Da.1]<br>Operation pattern        |
|----------------------------------|------------------------------------|
| 1                                | 01: Continuous positioning control |
| 2                                | 01: Continuous positioning control |
| 3                                | 00: Positioning complete           |
| •                                |                                    |
| 10                               | 11: Continuous path control        |
| 11                               | 11: Continuous path control        |
| 12                               | 00: Positioning complete           |
| •                                |                                    |
| 50                               | 00: Positioning complete           |
| •                                |                                    |

## [2] Control examples

The following shows the control executed when the " block start data" of the 1st point of axis 1 is set as shown in section [1] and started.

<1> The condition judgment set in "condition data No. 3" is performed to "positioning data No. 1" of the axis 1.

→ Conditions established → Execute positioning data No. 1, 2, and 3 → Go to <2>.

→ Conditions not established → Control stops (waits) until conditions are established → Go to <1>.

<2> Execute the axis 1 "positioning data No. 10, 11, 12, and 50" and stop the control.

## 10.3.5 Simultaneous start

In a "simultaneous start", the positioning data set in the "Da.12 Start data No." and positioning data of other axes set in the "condition data" are simultaneously executed (pulses are output with the same timing).

(The "condition data" is designated with "Da.14 Parameter".)

Section [2] shows a control example where the "block start data" and "positioning data" are set as shown in section [1].

## [1] Setting examples

## (1) Block start data setting example

| Axis 1 block start data | Da.11<br>Shape | Da.12<br>Start data No. | Da.13<br>Special start instruction | Da.14<br>Parameter |
|-------------------------|----------------|-------------------------|------------------------------------|--------------------|
| 1st point               | 0: End         | 1                       | 3: Simultaneous start              | 4                  |
| •                       |                |                         |                                    |                    |
| •                       |                |                         |                                    |                    |
| •                       |                |                         |                                    |                    |
| •                       |                |                         |                                    |                    |

\* It is assumed that the "axis 2 positioning data" for simultaneous starting is set in the "condition data" designated with "Da.14 Parameter".

## (2) Positioning data setting example

| Axis 1 positioning data No. | Da.1<br>Operation pattern          |
|-----------------------------|------------------------------------|
| 1                           | 01: Continuous positioning control |
| 2                           | 01: Continuous positioning control |
| 3                           | 00: Positioning complete           |
| •                           |                                    |
| •                           |                                    |
| •                           |                                    |
| •                           |                                    |
| •                           |                                    |
| •                           |                                    |
| •                           |                                    |

## [2] Control examples

The following shows the control executed when the "block start data" of the 1st point of axis 1 is set as shown in section [1] and started.

<1> Check the axis operation status of axis 2 which is regarded as the simultaneously started axis.

→ Axis 2 is standing by → Go to <2>.

→ Axis 2 is carrying out positioning. → An error occurs and simultaneous start will not be carried out.

<2> Simultaneously start the axis 1 "positioning data No. 1" and axis 2 positioning data set in "condition data No. 4".

### [3] Precautions

Positioning data No. executed by simultaneously started axes is set to condition data (" [Da.18](#) Parameter 1", " [Da.19](#) Parameter 2", but the setting value of start axis (the axis which carries out positioning start) should be "0". If the setting value is set to other than "0", the positioning data set in " [Da.18](#) Parameter 1", " [Da.19](#) Parameter 2" is given priority to be executed rather than " [Da.12](#) Start data No.".

(For details, refer to section 5.5 "List of condition data".)

## 10.3.6 Repeated start (FOR loop)

In a "repeated start (FOR loop)", the data between the " block start data" in which "4: FOR loop" is set in " Da.13 Special start instruction" and the "block start data" in which "6: NEXT start" is set in " Da.13 Special start instruction " is repeatedly executed for the No. of times set in " Da.14 Parameter". An endless loop will result if the No. of repetitions is set to "0".

(The No. of repetitions is set in " Da.14 Parameter" of the " block start data" in which "4: FOR loop" is set in " Da.13 Special start instruction".)

Section [2] shows a control example where the " block start data" and "positioning data" are set as shown in section [1].

## [1] Setting examples

## (1) Block start data setting example

| Axis 1 block start data | Da.11<br>Shape | Da.12<br>Start data No. | Da.13<br>Special start instruction | Da.14<br>Parameter |
|-------------------------|----------------|-------------------------|------------------------------------|--------------------|
| 1st point               | 1: Continue    | 1                       | 4: FOR loop                        | 2                  |
| 2nd point               | 1: Continue    | 10                      | 0: Block start                     | —                  |
| 3rd point               | 0: End         | 50                      | 6: NEXT start                      | —                  |
| •                       |                |                         |                                    |                    |
| •                       |                |                         |                                    |                    |

\* The "condition data Nos." have been set in " Da.14 Parameter".

## (2) Positioning data setting example

| Axis 1 positioning data No. | Da.1<br>Operation pattern          |
|-----------------------------|------------------------------------|
| 1                           | 01: Continuous positioning control |
| 2                           | 01: Continuous positioning control |
| 3                           | 00: Positioning complete           |
| •                           |                                    |
| 10                          | 11: Continuous path control        |
| 11                          | 00: Positioning complete           |
| •                           |                                    |
| 50                          | 01: Continuous positioning control |
| 51                          | 00: Positioning complete           |
| •                           |                                    |

## [2] Control examples

The following shows the control executed when the " block start data" of the 1st point of axis 1 is set as shown in section [1] and started.

<1> Execute the axis 1 "positioning data No. 1, 2, 3, 10, 11, 50, and 51".

<2> Return to the axis 1 "1st point block start data". Again execute the axis 1 "positioning data No. 1, 2, 3, 10, 11, 50 and 51", and then stop the control. (Repeat for the No. of times (2 times) set in Da.14.)



## 10.3.7 Repeated start (FOR condition)

In a "repeated start (FOR condition)", the data between the " block start data" in which "5: FOR condition" is set in " Da.13 Special start instruction" and the " block start data" in which "6: NEXT start" is set in " Da.13 Special start instruction" is repeatedly executed until the establishment of the conditions set in the "condition data". Conditional judgment is carried out as soon as switching to the point of "6: NEXT start" (before positioning of NEXT start point). (The "condition data" designation is set in " Da.14 Parameter" of the " block start data" in which "5: FOR condition" is set in " Da.13 Special start instruction".)

Section [2] shows a control example where the " block start data" and "positioning data" are set as shown in section [1].

## [1] Setting examples

## (1) Block start data setting example

| Axis 1 block start data | Da.11<br>Shape | Da.12<br>Start data No. | Da.13<br>Special start instruction | Da.14<br>Parameter |
|-------------------------|----------------|-------------------------|------------------------------------|--------------------|
| 1st point               | 1: Continue    | 1                       | 5: FOR condition                   | 5                  |
| 2nd point               | 1: Continue    | 10                      | 0: Block start                     | —                  |
| 3rd point               | 0: End         | 50                      | 6: NEXT start                      | —                  |
| •                       |                |                         |                                    |                    |
| •                       |                |                         |                                    |                    |

\* The "condition data Nos." have been set in " Da.14 Parameter".

## (2) Positioning data setting example

| Axis 1 position-<br>ing data No. | Da.1<br>Operation pattern          |
|----------------------------------|------------------------------------|
| 1                                | 01: Continuous positioning control |
| 2                                | 01: Continuous positioning control |
| 3                                | 00: Positioning complete           |
| •                                |                                    |
| 10                               | 11: Continuous path control        |
| 11                               | 00: Positioning complete           |
| •                                |                                    |
| 50                               | 01: Continuous positioning control |
| 51                               | 00: Positioning complete           |
| •                                |                                    |

## [2] Control examples

The following shows the control executed when the " block start data" of the 1st point of axis 1 is set as shown in section [1] and started.

<1> Execute axis 1 "Positioning data No. 1, 2, 3, 10, 11".

<2> Execute the conditional judgment set in axis 1 "Condition data No.5". \*1  
 → Conditions not established → Execute "Positioning data No. 50, 51".  
 Go to <1>  
 → Conditions established → Execute "Positioning data No.50, 51" and complete the positioning.

\*1: Conditional judgment is carried out as soon as switching to NEXT start point (before positioning of NEXT start point).

## 10.3.8 Restrictions when using the NEXT start

The "NEXT start" is a instruction indicating the end of the repetitions when executing Section 10.3.6 "Repeated start (FOR loop)" and Section 10.3.7 "Repeated start (FOR condition)".

The following shows the restrictions when setting "6: NEXT start" in the " block start data".

- (1) The processing when "6: NEXT start" is set before execution of "4: FOR loop" or "5: FOR condition" is the same as that for a "0: block start".
- (2) Repeated processing will not be carried out if there is no "6: NEXT start" instruction after the "4: FOR loop" or "5: FOR condition" instruction. (Note that an "error" will not occur.)
- (3) Nesting is not possible between "4: FOR loop" and "6: NEXT start", or between "5: FOR condition" and "6: NEXT start". A warning "FOR to NEXT nest construction" (warning code: 506) will occur if nesting is attempted.

[Operating examples without nesting structure]

| Start block data | Da.13 Special start instruction |
|------------------|---------------------------------|
| 1st point        | Normal start                    |
| 2nd point        | FOR ←                           |
| 3rd point        | Normal start                    |
| 4th point        | NEXT —                          |
| 5th point        | Normal start                    |
| 6th point        | Normal start                    |
| 7th point        | FOR ←                           |
| 8th point        | Normal start                    |
| 9th point        | NEXT —                          |
| .                |                                 |
| .                |                                 |

[Operating examples with nesting structure]

| Start block data | Da.13 Special start instruction |
|------------------|---------------------------------|
| 1st point        | Normal start                    |
| 2nd point        | FOR                             |
| 3rd point        | Normal start                    |
| 4th point        | FOR ←                           |
| 5th point        | Normal start                    |
| 6th point        | Normal start                    |
| 7th point        | NEXT —                          |
| 8th point        | Normal start                    |
| 9th point        | NEXT                            |
| .                |                                 |
| .                |                                 |

A warning will occur when starting the 4th point "FOR".

The JUMP destination of the 7th point "NEXT" is the 4th point. The 9th point "NEXT" is processed as normal start.

## 10.4 Setting the condition data

### 10.4.1 Relation between various controls and the condition data

"Condition data" is set in the following cases.

- (1) When setting conditions during execution of Section 9.2.23 "JUMP instruction" (major positioning control)
- (2) When setting conditions during execution of "high-level positioning control"

The "condition data" to be set includes the 5 setting items from [Da.15] to [Da.19], but the setting requirements and details differ according to the control system and setting conditions.

The following shows the "condition data" " [Da.15] Condition target" corresponding to the different types of control.

(The "condition data" settings in this chapter are assumed to be carried out using GX Configurator-QP.)

| Control type<br>[Da.15]<br>Setting item | High-level positioning control |            |                    |                                | Major positioning control |
|-----------------------------------------|--------------------------------|------------|--------------------|--------------------------------|---------------------------|
|                                         | Block start                    | Wait start | Simultaneous start | Repeated start (For condition) | JUMP instruction          |
| 01: Device X*1                          | ◎                              | ◎          | ×                  | ◎                              | ◎                         |
| 02: Device Y*1                          | ◎                              | ◎          | ×                  | ◎                              | ◎                         |
| 03: Buffer memory (1 word)              | ◎                              | ◎          | ×                  | ◎                              | ◎                         |
| 04: Buffer memory (2 words)             | ◎                              | ◎          | ×                  | ◎                              | ◎                         |
| 05: Positioning data No.                | ×                              | ×          | ◎                  | ×                              | ×                         |

◎ : One of the setting items must be set.

× : Setting not possible

\*1: Refers to buffer memories and devices X/Y which belongs to QD75.

#### REMARK

It is recommended that the "condition data" be set whenever possible with GX Configurator-QP. Execution by sequence program uses many sequence programs and devices. The execution becomes complicated, and the scan times will increase.

The setting requirements and details of the following "condition data" [Da.16] to [Da.19] setting items differ according to the "[Da.15] Condition target" setting. The following shows the [Da.16] to [Da.19] setting items corresponding to the "[Da.15] Condition target".

| <div>Other setting item</div> <div><div>Da.15</div>Setting item</div>                                                                                                                                                                                                                      | <div>Da.16</div> Condition operator  | <div>Da.17</div> Address                                                                                                                                                                                                                             | <div>Da.18</div> Parameter 1 |                   | <div>Da.19</div> Parameter 2                                              |                   |                                |                                |                    |                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------|---------------------------------------------------------------------------|-------------------|--------------------------------|--------------------------------|--------------------|--------------------------------|
| 01H: Device X                                                                                                                                                                                                                                                                              | 07H : DEV=ON                         | —                                                                                                                                                                                                                                                    | 0 to 1FH (bit No.)           |                   | —                                                                         |                   |                                |                                |                    |                                |
| 02H: Device Y                                                                                                                                                                                                                                                                              | 08H : DEV=OFF                        |                                                                                                                                                                                                                                                      | 0 to 1FH (bit No.)           |                   |                                                                           |                   |                                |                                |                    |                                |
| 03H: Buffer memory (1 word) *1                                                                                                                                                                                                                                                             | 01H : **=P1<br>02H : **≠P1           | Buffer memory address                                                                                                                                                                                                                                | P1 (numeric value)           |                   | P2 (numeric value)<br>(Set only when <div>Da.16</div> is [05H] or [06H].) |                   |                                |                                |                    |                                |
| 04H: Buffer memory (2 words) *1                                                                                                                                                                                                                                                            | 03H : **≤P1<br>04H : **≥P1           |                                                                                                                                                                                                                                                      |                              |                   |                                                                           |                   |                                |                                |                    |                                |
|                                                                                                                                                                                                                                                                                            | 05H : P1≤**≤P2<br>06H : **≤P1, P2≤** |                                                                                                                                                                                                                                                      |                              |                   |                                                                           |                   |                                |                                |                    |                                |
|                                                                                                                                                                                                                                                                                            | 05H: Positioning data No.            | 10H : Axis 1 designation<br>20H : Axis 2 designation<br>30H : Axis 1 and axis 2 designation<br>40H : Axis 3 designation<br>50H : Axis 1 and axis 3 designation<br>60H : Axis 2 and axis 3 designation<br>70H : Axis 1, axis 2 and axis 3 designation | —                            | Low-order 16 bits | Axis 1 positioning data No. *2                                            | Low-order 16 bits | Axis 3 positioning data No. *2 |                                |                    |                                |
| 80H : Axis 4 designation<br>90H : Axis 1 and axis 4 designation<br>A0H : Axis 2 and axis 4 designation<br>B0H : Axis 1, axis 2 and axis 4 designation<br>C0H : Axis 3 and axis 4 designation<br>D0H : Axis 1, axis 3 and axis 4 designation<br>E0H : Axis 2, axis 3 and axis 4 designation |                                      |                                                                                                                                                                                                                                                      |                              |                   |                                                                           |                   |                                |                                |                    |                                |
|                                                                                                                                                                                                                                                                                            |                                      | High-order 16 bits                                                                                                                                                                                                                                   |                              |                   |                                                                           |                   |                                | Axis 2 positioning data No. *2 | High-order 16 bits | Axis 4 positioning data No. *2 |
|                                                                                                                                                                                                                                                                                            |                                      |                                                                                                                                                                                                                                                      |                              |                   |                                                                           |                   |                                |                                |                    |                                |
|                                                                                                                                                                                                                                                                                            |                                      |                                                                                                                                                                                                                                                      |                              |                   |                                                                           |                   |                                |                                |                    |                                |
|                                                                                                                                                                                                                                                                                            |                                      |                                                                                                                                                                                                                                                      |                              |                   |                                                                           |                   |                                |                                |                    |                                |
|                                                                                                                                                                                                                                                                                            |                                      |                                                                                                                                                                                                                                                      |                              |                   |                                                                           |                   |                                |                                |                    |                                |
|                                                                                                                                                                                                                                                                                            |                                      |                                                                                                                                                                                                                                                      |                              |                   |                                                                           |                   |                                |                                |                    |                                |
|                                                                                                                                                                                                                                                                                            |                                      |                                                                                                                                                                                                                                                      |                              |                   |                                                                           |                   |                                |                                |                    |                                |

— : Setting not required (Setting value will be ignored. Use the initial value or a value within the setting range.)

\*\* : Value stored in buffer memory designated in [Da.17].

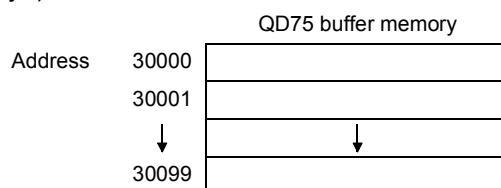
\*1 : Comparison of ≤ and ≥ is judged as signed values. Refer to Section 5.5 "List of condition data" for the setting contents.

\*2 : The setting value of start axis (the axis which executes positioning start) should be "0". If the setting value is set to other than "0", the positioning data set in [Da.18] Parameter 1 and [Da.19] Parameter 2 is given priority to be executed rather than "[Da.12] Start data No.".

Judgment whether the condition operator is "=" or "≠" at the wait start.  
Judgment on data is carried out for each control cycle of the QD75. Thus, in the judgment on the data such as current feed value which varies continuously, the operator "=" may not be detected. If this occurs, use a range operator.

**REMARK**

The "PLC CPU memory area" can be designated as the buffer memory address to be designated in [Da.17](#). (Refer to Section 7.1.1 "Configuration and roles of QD75 memory".)



## 10.4.2 Condition data setting examples

The following shows setting examples for "condition data".

## (1) Setting the device ON/OFF as a condition

[Condition] Device "XC" (= Axis 1 Busy signal) is OFF

| Da.15<br>Condition target | Da.16<br>Condition operator | Da.17<br>Address | Da.18<br>Parameter 1 | Da.19<br>Parameter 2 |
|---------------------------|-----------------------------|------------------|----------------------|----------------------|
| 01H: Device X             | 08H: DEV=OFF                | —                | 0CH                  | —                    |

## (2) Setting the numeric value stored in the "buffer memory" as a condition

[Condition]

The value stored in buffer memory addresses "800, 801" (= "Md.20 Current feed value") is "1000" or larger.

| Da.15<br>Condition target    | Da.16<br>Condition operator | Da.17<br>Address | Da.18<br>Parameter 1 | Da.19<br>Parameter 2 |
|------------------------------|-----------------------------|------------------|----------------------|----------------------|
| 04H: Buffer memory (2 words) | 04H: * * ≥ P1               | 800              | 1000                 | —                    |

## (3) Designating the axis and positioning data No. to be simultaneously started in "simultaneous start"

[Condition]

Simultaneously starting "axis 2 positioning data No.3".

| Da.15<br>Condition target | Da.16<br>Condition operator | Da.17<br>Address | Da.18<br>Parameter 1          | Da.19<br>Parameter 2 |
|---------------------------|-----------------------------|------------------|-------------------------------|----------------------|
| 05H: Positioning data No. | 20H: Axis 2 designation     | —                | High-order 16 bits "0003H" *1 | — *1                 |

\*1: The setting value of start axis (the axis which executes positioning start) should be "0000H".

## 10.5 Multiple axes simultaneous start control

The "multiple axes simultaneous start control" starts and controls the multiple axes simultaneously by outputting pulses to the axis to be started at the same timing as the start axis.

The maximum of four axes can be started simultaneously.

### [1] Control details

Multiple axes simultaneous start control is executed as follows.

Set the start data No. to the buffer memory areas " [Cd.30] to [Cd.33]

Simultaneous starting axis start data No. (axis 1 to 4 start data No.).

Set "9004" to " [Cd.3] Positioning start No." of the start axis.

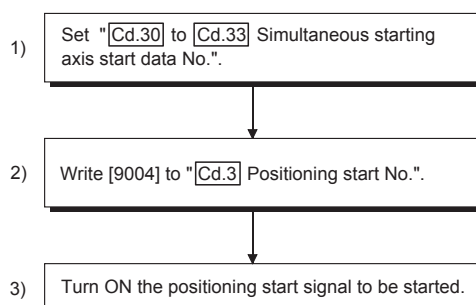
Turn ON the positioning start signal.

### [2] Restrictions

- (1) The error "Error before simultaneous start" (error code: 501) will occur and all simultaneously started axes will not start if the simultaneously started axis start data No. is not set to the axis control data on the start axis or set outside the setting range.
- (2) The error "Error before simultaneous start" (error code: 501) will occur and all simultaneously started axes will not start if either of the simultaneously started axes is BUSY.
- (3) The error "Error before simultaneous start" (error code: 501) will occur and all simultaneously started axes will not start if an error occurs during the analysis of the positioning data on the simultaneously started axes.
- (4) No error or warning will occur if only the start axis is the simultaneously started axis.
- (5) This function cannot be used with the sub function Section 12.7.7 "Pre-reading start function".

### [3] Multiple axes simultaneous start control procedure

The procedure for multiple axes simultaneous start control is as follows.



## [4] Multiple axes simultaneous start control function setting method

The following shows the setting of the data used to execute the multiple axes simultaneous start control with positioning start signals (The axis control data on the start axis is set).

| Setting item                                                                   | Setting value                                                                                                            | Setting details                                                    | Buffer memory address |        |        |        |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------|--------|--------|--------|
|                                                                                |                                                                                                                          |                                                                    | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <b>Cd.3</b> Positioning start No.                                              | 9004                                                                                                                     | Set the multiple axes simultaneous start control start No. "9004". | 1500                  | 1600   | 1700   | 1800   |
| <b>Cd.30</b> Simultaneous starting axis start data No. (Axis 1 start data No.) | Set the simultaneously started axis start data No.<br>Set a "0" for the axis other than the simultaneously started axes. |                                                                    | 1540                  | 1640   | 1740   | 1840   |
| <b>Cd.31</b> Simultaneous starting axis start data No. (Axis 2 start data No.) |                                                                                                                          |                                                                    | 1541                  | 1641   | 1741   | 1841   |
| <b>Cd.32</b> Simultaneous starting axis start data No. (Axis 3 start data No.) |                                                                                                                          |                                                                    | 1542                  | 1642   | 1742   | 1842   |
| <b>Cd.33</b> Simultaneous starting axis start data No. (Axis 4 start data No.) |                                                                                                                          |                                                                    | 1543                  | 1643   | 1743   | 1843   |

\* Refer to Section 5.7 "List of control data" for information on setting details.

## [5] Setting examples

The following shows the setting examples in which the axis 1 is used as the start axis and the simultaneously started axes are used as the axes 2 and 4.

| Setting item                                                                   | Setting value | Setting details                                                                                   | Buffer memory address<br>(Axis 1) |
|--------------------------------------------------------------------------------|---------------|---------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>Cd.3</b> Positioning start No.                                              | 9004          | Set the multiple axes simultaneous start control start No. "9004".                                | 1500                              |
| <b>Cd.30</b> Simultaneous starting axis start data No. (Axis 1 start data No.) | 100           | The axis 1 starts the positioning data No. 100.                                                   | 1540                              |
| <b>Cd.31</b> Simultaneous starting axis start data No. (Axis 2 start data No.) | 200           | Immediately after the start of the axis 1, the axis 2 starts the axis 2 positioning data No. 200. | 1541                              |
| <b>Cd.32</b> Simultaneous starting axis start data No. (Axis 3 start data No.) | 0             | Will not start simultaneously.                                                                    | 1542                              |
| <b>Cd.33</b> Simultaneous starting axis start data No. (Axis 4 start data No.) | 300           | Immediately after the start of the axis 1, the axis 4 starts the axis 4 positioning data No. 300. | 1543                              |



|        |
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| POINTS |
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- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(1) The "multiple axes simultaneous start control" carries out an operation equivalent to the "simultaneous start" using the "block start data".</p> <p>(2) The setting of the "multiple axes simultaneous start control" is easier than that of the "simultaneous start" using the "block start data".</p> <ul style="list-style-type: none"><li>• Setting items for "simultaneous start" using "block start data"<br/>Positioning start data, block start data, condition data, and positioning data</li><li>• Setting items for "multiple axes simultaneous start control"<br/>Positioning data and axis control data</li></ul> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 10.6 Start program for high-level positioning control

### 10.6.1 Starting high-level positioning control

To execute high-level positioning control, a sequence program must be created to start the control in the same method as for major positioning control.

The following shows the procedure for starting the "1st point block start data" (regarded as block No. 7000) set in axis 1.

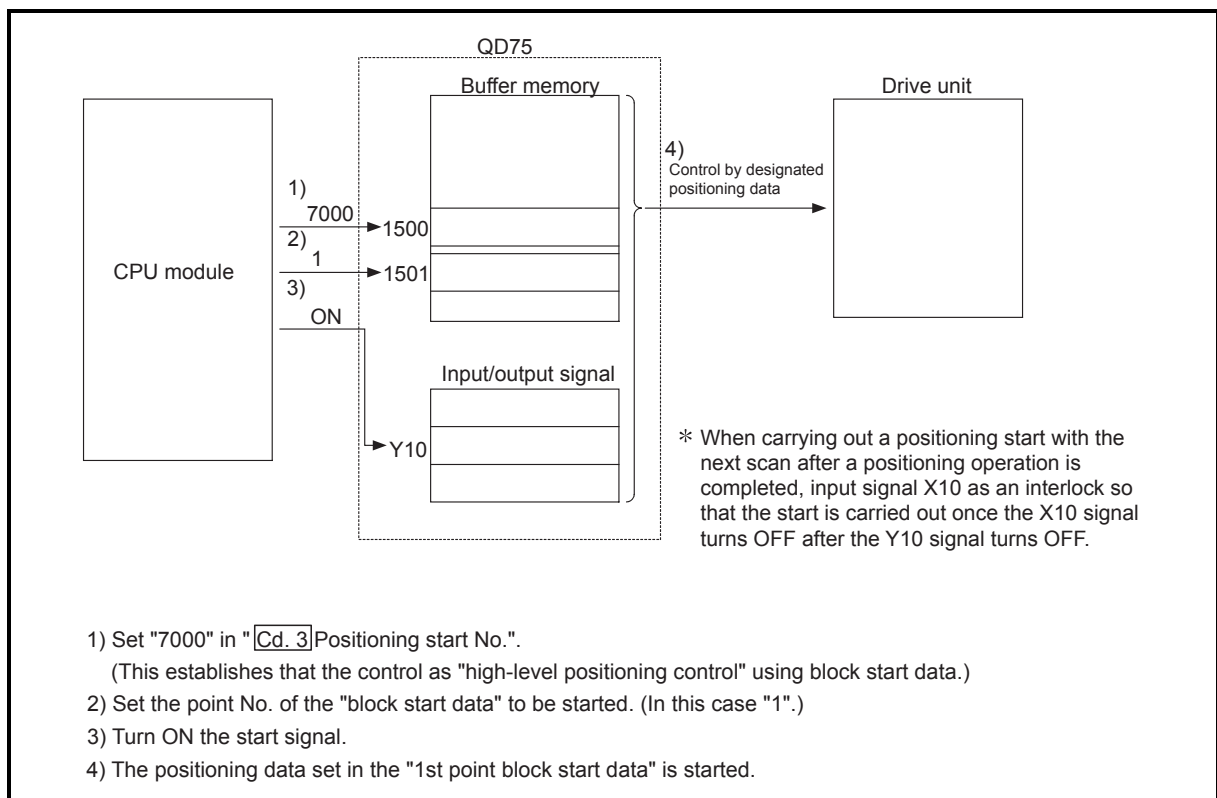


Fig. 10.2 High-level positioning control start procedure

## 10.6.2 Example of a start program for high-level positioning control

The following shows an example of a start program for high-level positioning control in which the 1st point "block start data" of axis 1 is started. (The block No. is regarded as "7000".)

### ■ Control data that require setting

The following control data must be set to execute high-level positioning control. The setting is carried out using a sequence program.

| Setting item                               | Setting value | Setting details                                            | Buffer memory address |        |        |        |
|--------------------------------------------|---------------|------------------------------------------------------------|-----------------------|--------|--------|--------|
|                                            |               |                                                            | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <b>Cd.3</b> Positioning start No.          | 7000          | Set "7000" to indicate control using "block start data".   | 1500                  | 1600   | 1700   | 1800   |
| <b>Cd.4</b> Positioning starting point No. | 1             | Set the point No. of the "block start data" to be started. | 1501                  | 1601   | 1701   | 1801   |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

### ■ Start conditions

The following conditions must be fulfilled when starting the control. The required conditions must also be integrated into the sequence program, and configured so the control does not start unless the conditions are fulfilled.

| Signal name      |                         | Signal state |                                               | Device |        |        |        |
|------------------|-------------------------|--------------|-----------------------------------------------|--------|--------|--------|--------|
|                  |                         |              |                                               | Axis 1 | Axis 2 | Axis 3 | Axis 4 |
| Interface signal | PLC READY signal        | ON           | CPU module preparation completed              | Y0     |        |        |        |
|                  | QD75 READY signal       | ON           | QD75 preparation completed                    | X0     |        |        |        |
|                  | Synchronization flag    | ON           | QD75 buffer memory<br>The access is possible. | X1     |        |        |        |
|                  | Axis stop signal        | OFF          | Axis stop signal is OFF                       | Y4     | Y5     | Y6     | Y7     |
|                  | Start complete signal   | OFF          | Start complete signal is OFF                  | X10    | X11    | X12    | X13    |
|                  | BUSY signal             | OFF          | BUSY signal is OFF                            | XC     | XD     | XE     | XF     |
|                  | Error detection signal  | OFF          | There is no error                             | X8     | X9     | XA     | XB     |
| External signal  | M code ON signal        | OFF          | M code ON signal is OFF                       | X4     | X5     | X6     | X7     |
|                  | Drive unit READY signal | ON           | Drive unit preparation completed              | —      |        |        |        |
|                  | Stop signal             | OFF          | Stop signal is OFF                            | —      |        |        |        |
|                  | Upper limit (FLS)       | ON           | Within limit range                            | —      |        |        |        |
|                  | Lower limit (RLS)       | ON           | Within limit range                            | —      |        |        |        |

### Start time chart

The following chart shows a time chart in which the positioning data No. 1, 2, 10, 11, and 12 of axis 1 are continuously executed as an example.

#### (1) Block start data setting example

| Axis 1 block start data | Da.11<br>Shape | Da.12<br>Start data No. | Da.13<br>Special start instruction | Da.14<br>Parameter |
|-------------------------|----------------|-------------------------|------------------------------------|--------------------|
| 1st point               | 1: Continue    | 1                       | 0: Block start                     | —                  |
| 2nd point               | 0: End         | 10                      | 0: Block start                     | —                  |
| •                       |                |                         |                                    |                    |
| •                       |                |                         |                                    |                    |

#### (2) Positioning data setting example

| Axis 1 position-<br>ing data No. | Da.1<br>Operation pattern   |
|----------------------------------|-----------------------------|
| 1                                | 11: Continuous path control |
| 2                                | 00: Positioning complete    |
| •                                |                             |
| 10                               | 11: Continuous path control |
| 11                               | 11: Continuous path control |
| 12                               | 00: Positioning complete    |
| •                                |                             |

#### (3) Start time chart

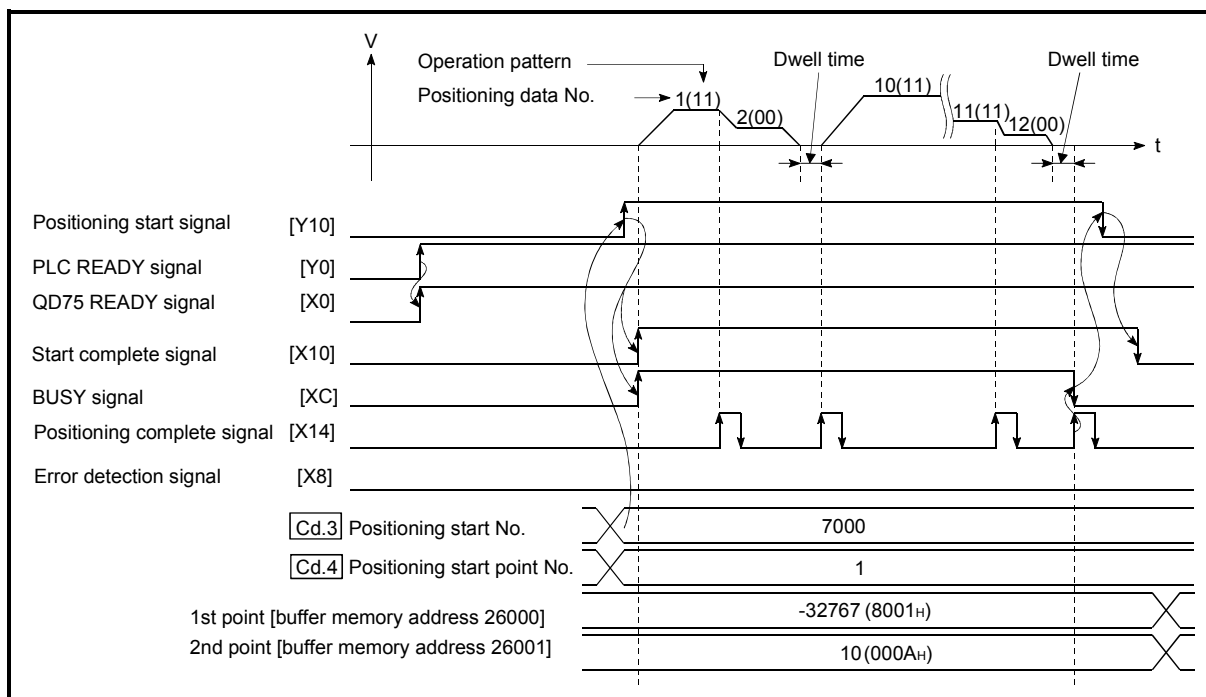
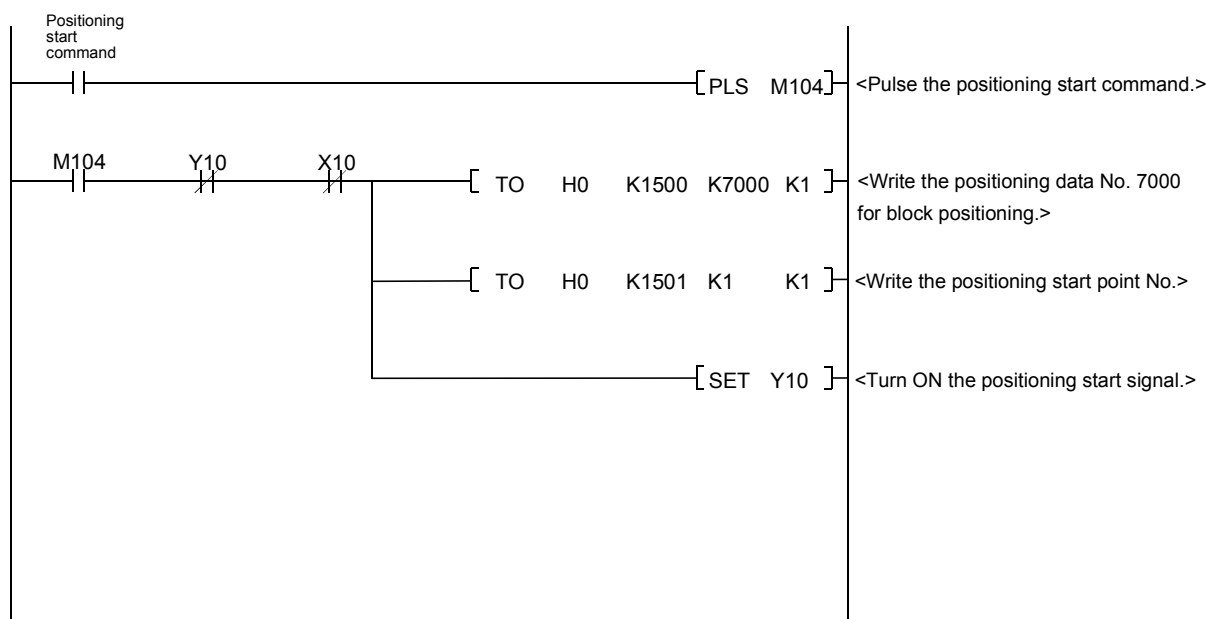


Fig. 10.3 Start time chart for high-level positioning control (block start)

### ■ Creating the program

#### Example

Set the block start data beforehand.



Y10: Positioning start signal  
X10: Start complete signal  
M104: Positioning start command pulse

# MEMO

[illegible]

# CHAPTER 11 MANUAL CONTROL

The details and usage of manual control are explained in this chapter.  
In manual control, pulse output commands are issued during a JOG operation and an inching operation executed by the turning ON of the JOG START signal, or from a manual pulse generator connected to the QD75.  
Manual control using a sequence program from the CPU module is explained in this chapter. Refer to GX Configurator-QP Operating Manual for an explanation of manual control (JOG operation, inching operation and manual pulse generator operation) using the peripheral devices.

|        |                                                                                 |        |
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| 11.4.4 | Creating a program to enable/disable the manual pulse generator operation ..... | 11- 32 |

## 11.1 Outline of manual control

### 11.1.1 Three manual control methods

"Manual control" refers to control in which positioning data is not used, and a positioning operation is carried out in response to signal input from an external source. The three types of this "manual control" are explained below.

#### [1] JOG operation

"JOG operation" is a control method in which the machine is moved by only a movement amount (pulses are continuously outputted while the JOG START signal is ON). This operation is used to move the workpiece in the direction in which the limit signal is ON, when the operation is stopped by turning the limit signal OFF to confirm the positioning system connection and obtain the positioning data address (Refer to Section 12.7.4 "Teaching function").

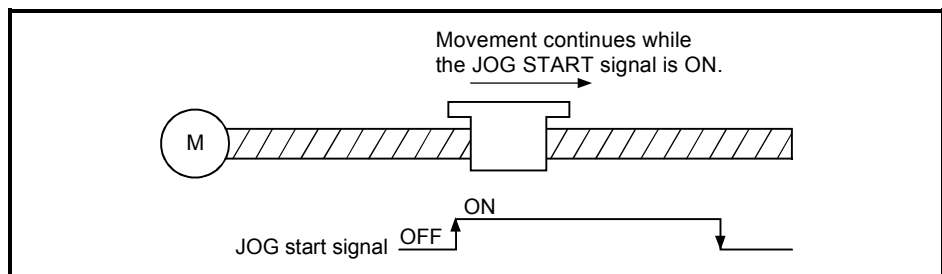


Fig. 11.1 JOG operation

#### [2] Inching operation

"Inching operation" is a control method in which a minute movement amount of pulses is output manually at 1.8ms.

When the "inching movement amount" of the axis control data is set by JOG operation, the workpiece is moved by a set movement amount. (When the "inching movement amount" is set to "0", the operation is performed as the JOG operation.)

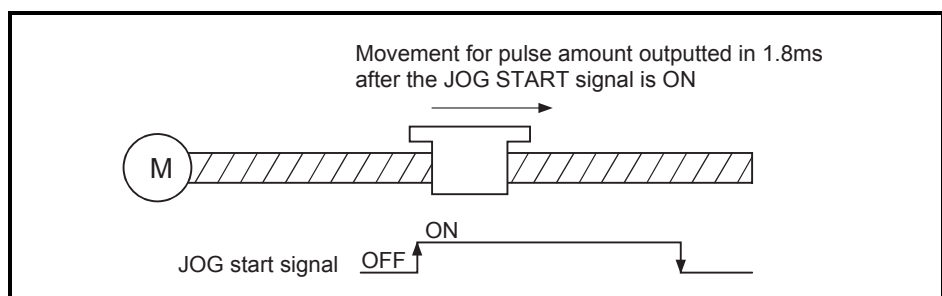


Fig. 11.2 Inching operation



### [3] Manual pulse generator operation

"Manual pulse generator operation" is a control method in which positioning is carried out in response to the No. of pulses input from a manual pulse generator (the No. of input pulses is output). This operation is used for manual fine adjustment, etc., when carrying out accurate positioning to obtain the positioning address.

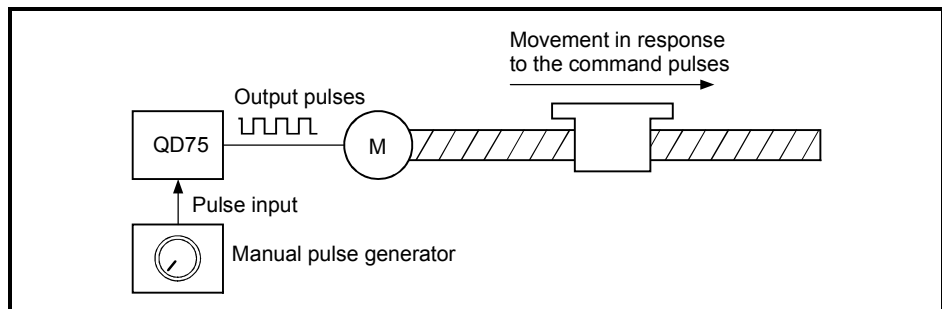


Fig. 11.3 Manual pulse generator control

#### Manual control sub functions

Refer to Section 3.2.4 "Combination of QD75 major functions and sub functions" for details on "sub functions" that can be combined with manual control. Also refer to CHAPTER 12 "CONTROL SUB FUNCTIONS" for details on each sub function.

#### Carrying out manual control from peripheral devices

"JOG operation", "Inching operation" and enabling/disabling of the "manual pulse generator operation" can be executed from GX Configurator-QP test mode. Refer to GX Configurator-QP Operating Manual for details on manual control from GX Configurator-QP.

#### Monitoring manual control

Refer to Section 5.6 "List of monitor data" when directly monitoring the buffer memory using GX Developer. Also refer to GX Configurator-QP Operating Manual when monitoring with the monitor functions of GX Configurator-QP.

## 11.2 JOG operation

### 11.2.1 Outline of JOG operation

#### Important

Use the hardware stroke limit function when carrying out JOG operation near the upper or lower limits. (Refer to Section 12.4.4).

\* If the hardware stroke limit function is not used, the workpiece may exceed the moving range, causing an accident.

#### ■ JOG operation

In JOG operation, the FORWARD run JOG start signal (Y8, YA, YC, YE) or REVERSE run JOG start signal (Y9, YB, YD, YF) turns ON, causing pulses to be output to the drive unit from the QD75 while the signal is ON. The workpiece is then moved in the designated direction.

The following shows examples of JOG operation.

|    |                                                                                                                                                                                                                                                                        |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) | When the START signal turns ON, acceleration begins in the direction designated by the START signal, and continues for the acceleration time designated in " [Pr.32] JOG operation acceleration time selection". At this time, the BUSY signal changes from OFF to ON. |
| 2) | When the workpiece being accelerated reaches the speed set in " [Cd.17] JOG speed", the movement continues at this speed.<br>The constant speed movement takes place at 2) and 3).                                                                                     |
| 3) | When the START signal is turned OFF, deceleration begins from the speed set in " [Cd.17] JOG speed", and continues for the deceleration time designated in " [Pr.33] JOG operation deceleration time selection".                                                       |
| 4) | The operation stops when the speed becomes "0". At this time, the BUSY signal changes from ON to OFF.                                                                                                                                                                  |

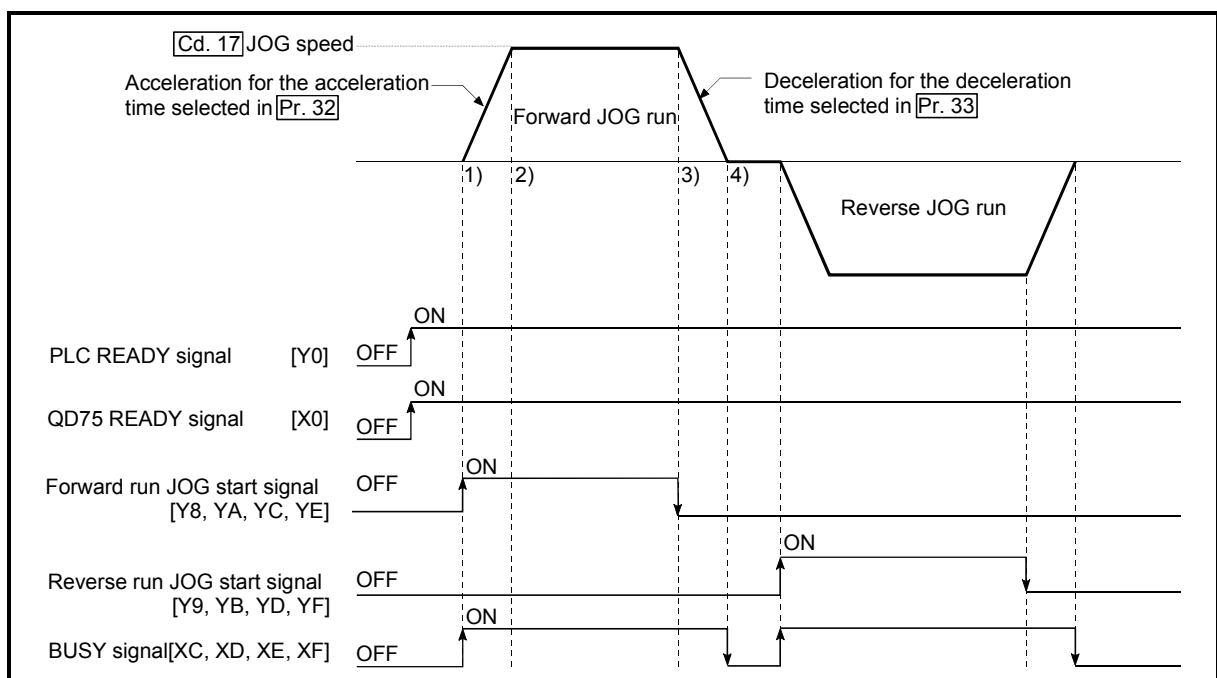


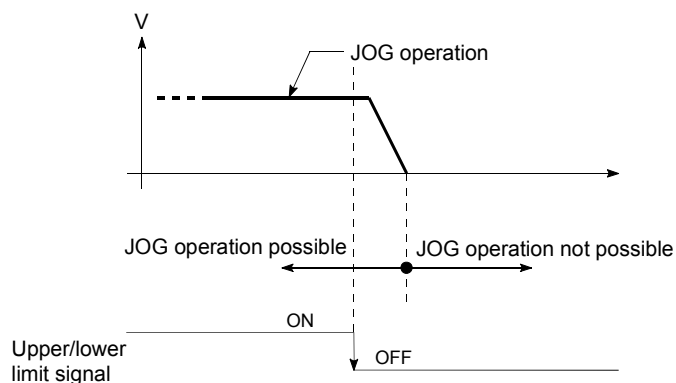
Fig. 11.4 JOG operation

### ■ Precautions during operation

- (1) For safety, first set " **Cd.17** JOG speed" to a smaller value and check the movement. Then gradually increase the value.
- (2) The axis error "Outside JOG speed range" (error code: 300) will occur and the operation will not start if the "JOG speed" is outside the setting range at the JOG start.
- (3) The axis error "JOG speed limit value error" (error code: 956) will occur and the operation will not start if " **Pr.31** JOG speed limit value" is set to a value larger than " **Pr.8** Speed limit value".
- (4) If " **Cd.17** JOG speed" exceeds the speed set in " **Pr.31** JOG speed limit value", the workpiece will move at the " **Pr.31** JOG speed limit value" and the warning "JOG speed limit value" (warning code: 301) will occur in the QD75.
- (5) The JOG operation can be continued even if an "Axis warning" has occurred.
- (6) Set a "0" in " **Cd.16** Inching movement amount". If a value other than "0" is set, the operation will become an inching operation (Refer to Section 11.3 "Inching operation").

### ■ Operations when stroke limit error occurs

When the operation is stopped by hardware stroke limit error or software stroke limit error, the JOG operation can execute in an opposite way (direction within normal limits) after an error reset. (An error will occur again if JOG start signal is turned ON in a direction to outside the stroke limit.)



### ■ JOG operation timing and processing time

The following drawing shows details of the JOG operation timing and processing time.

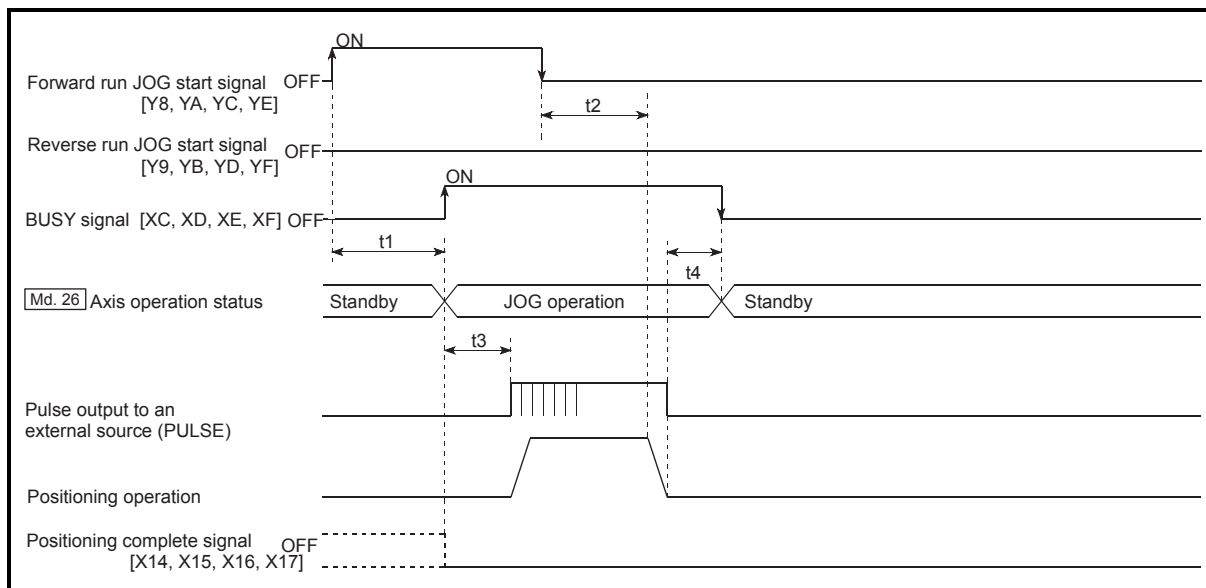


Fig. 11.5 JOG operation timing and processing times

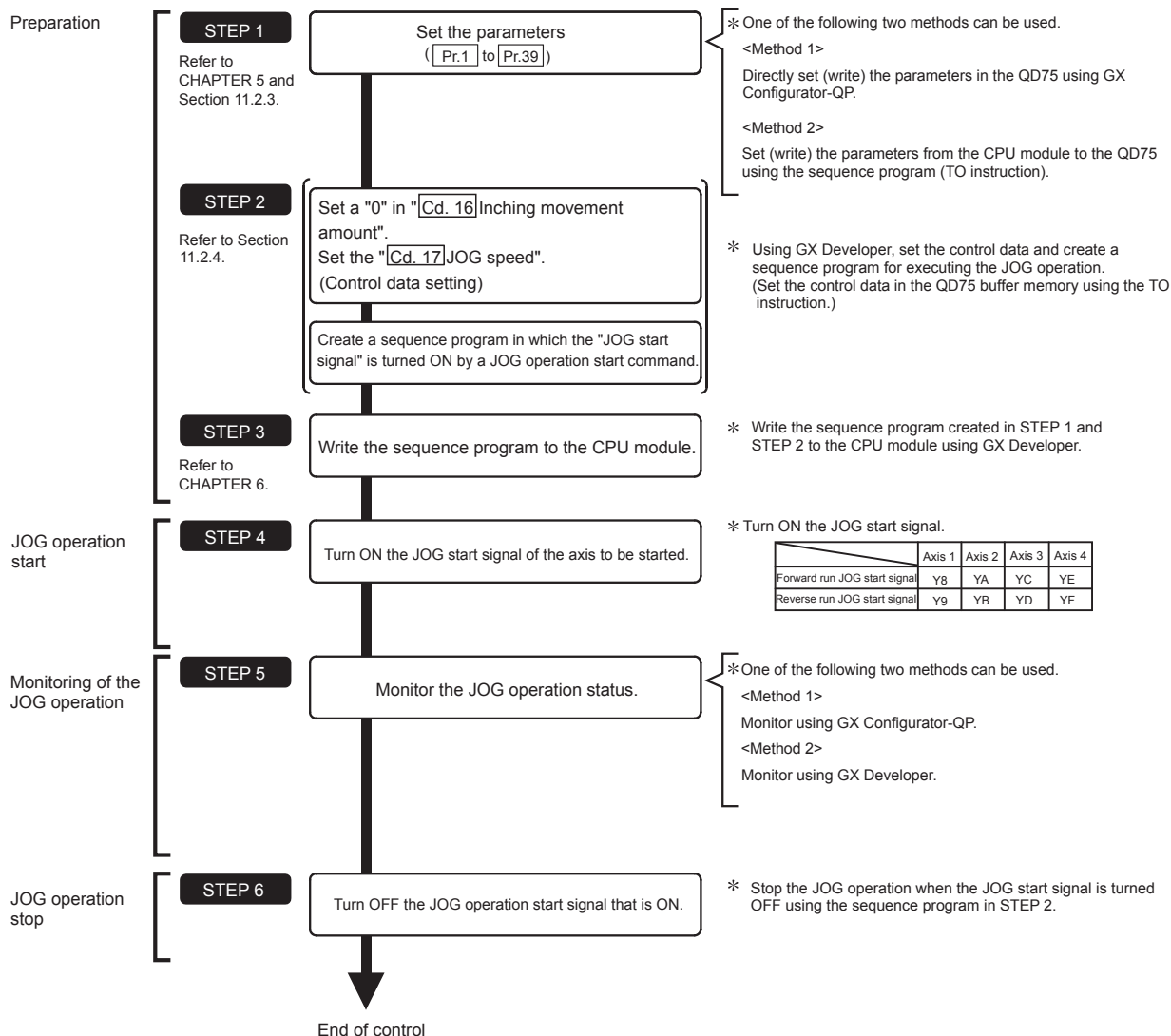
#### Normal timing times

| Model           | t1           | t2         | t3           | t4         |
|-----------------|--------------|------------|--------------|------------|
| QD75P□N/QD75D□N | 1.0 to 3.0ms | 0 to 0.9ms | 1.3 to 2.2ms | 0 to 0.9ms |
| QD75P□/QD75D□   | 1.0 to 3.0ms | 0 to 1.8ms | 2.7 to 4.4ms | 0 to 1.8ms |

- Delays may occur in the t1 timing time due to the operation status of other axes.

## 11.2.2 JOG operation execution procedure

The JOG operation is carried out by the following procedure.

**REMARK**

- Mechanical elements such as limit switches are considered as already installed.
- Parameter settings work in common for all control using the QD75.

## 11.2.3 Setting the required parameters for JOG operation

The "Parameters" must be set to carry out JOG operation.

The following table shows the setting items of the required parameters for carrying out JOG operation. When only JOG operation will be carried out, no parameters other than those shown below need to be set. (Use the initial values or setting values within a range where no error occurs for trouble-free operation.)

| Setting item |                                                             | Setting requirement | Factory-set initial value<br>(setting details)          |
|--------------|-------------------------------------------------------------|---------------------|---------------------------------------------------------|
| Parameters   | Pr.1 Unit setting                                           | ◎                   | 3 (pulse)                                               |
|              | Pr.2 No. of pulses per rotation (Ap) (Unit: pulse)          | ◎                   | 20000                                                   |
|              | Pr.3 Movement amount per rotation (Al) (Unit: pulse)        | ◎                   | 20000                                                   |
|              | Pr.4 Unit magnification (Am)                                | ◎                   | 1 (1-fold)                                              |
|              | Pr.5 Pulse output mode                                      | ◎                   | 1 (CW/CCW mode)                                         |
|              | Pr.6 Rotation direction setting                             | ◎                   | 0 (current value increases by forward run pulse output) |
|              | Pr.7 Bias speed at start (Unit: pulse/s)                    | ○                   | 0                                                       |
|              | Pr.8 Speed limit value (Unit: pulse/s)                      | ◎                   | 200000                                                  |
|              | Pr.9 Acceleration time 0 (Unit: ms)                         | ◎                   | 1000                                                    |
|              | Pr.10 Deceleration time 0 (Unit: ms)                        | ◎                   | 1000                                                    |
|              | Pr.11 Backlash compensation amount (Unit: pulse)            | ○                   | 0                                                       |
|              | Pr.12 Software stroke limit upper limit value (Unit: pulse) | ○                   | 2147483647                                              |
|              | Pr.13 Software stroke limit lower limit value (Unit: pulse) | ○                   | -2147483648                                             |
|              | Pr.14 Software stroke limit selection                       | ○                   | 0 (current feed value)                                  |
|              | Pr.15 Software stroke limit valid/invalid setting           | ○                   | 0 (valid)                                               |
|              | Pr.17 Torque limit setting value (Unit: %)                  | ○                   | 300                                                     |
|              | Pr.23 Output signal logic selection                         | ○                   | 0 (Pulse output to drive unit is negative logic.)       |

◎ : Setting always required.

○ : Set according to requirements (Leave set to the initial value when not used.)

**REMARK**

- Parameter settings work in common for all control using the QD75. When carrying out other control ("major positioning control", "high-level positioning control", "OPR positioning control"), the respective setting items must also be matched and set.
- Parameters are set for each axis.
- Refer to CHAPTER 5 "DATA USED FOR POSITIONING CONTROL" for setting details.

| Setting item |       |                                             | Setting requirement | Factory-set initial value<br>(setting details)           |
|--------------|-------|---------------------------------------------|---------------------|----------------------------------------------------------|
| Parameters   | Pr.25 | Acceleration time 1 (Unit: ms)              | ○                   | 1000                                                     |
|              | Pr.26 | Acceleration time 2 (Unit: ms)              | ○                   | 1000                                                     |
|              | Pr.27 | Acceleration time 3 (Unit: ms)              | ○                   | 1000                                                     |
|              | Pr.28 | Deceleration time 1 (Unit: ms)              | ○                   | 1000                                                     |
|              | Pr.29 | Deceleration time 2 (Unit: ms)              | ○                   | 1000                                                     |
|              | Pr.30 | Deceleration time 3 (Unit: ms)              | ○                   | 1000                                                     |
|              | Pr.31 | JOG speed limit value (Unit: pulse/s)       | ◎                   | 20000                                                    |
|              | Pr.32 | JOG operation acceleration time selection   | ◎                   | 0 (acceleration time 0)                                  |
|              | Pr.33 | JOG operation deceleration time selection   | ◎                   | 0 (deceleration time 0)                                  |
|              | Pr.34 | Acceleration/deceleration process selection | ○                   | 0 (trapezoidal acceleration/<br>deceleration processing) |
|              | Pr.35 | S-curve ratio (Unit: %)                     | ○                   | 100                                                      |
|              | Pr.36 | Sudden stop deceleration time (Unit: ms)    | ○                   | 1000                                                     |
|              | Pr.37 | Stop group 1 sudden stop selection          | ○                   | 0 (deceleration stop)                                    |
|              | Pr.38 | Stop group 2 sudden stop selection          | ○                   | 0 (deceleration stop)                                    |
|              | Pr.39 | Stop group 3 sudden stop selection          | ○                   | 0 (deceleration stop)                                    |

◎ : Setting always required.

○ : Set according to requirements (Leave set to the initial value when not used.)

### 11.2.4 Creating start programs for JOG operation

A sequence program must be created to execute a JOG operation. Consider the "required control data setting", "start conditions" and "start time chart" when creating the program.

The following shows an example when a JOG operation is started for axis 1.  
(" [Cd.17] JOG speed" is set to "100.00mm/min" in the example shown.)

#### ■ Required control data setting

The control data shown below must be set to execute a JOG operation. The setting is carried out with the sequence program.

| Setting item                    | Setting value | Setting details                                                                                                             | Buffer memory address |        |        |        |
|---------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------|--------|--------|--------|
|                                 |               |                                                                                                                             | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| [Cd.16] Inching movement amount | 0             | Set "0".                                                                                                                    | 1517                  | 1617   | 1717   | 1817   |
| [Cd.17] JOG speed               | 10000         | Set a value equal to or above the " [Pr.7] Bias speed at start" and equal to or below the " [Pr.31] JOG speed limit value". | 1518                  | 1618   | 1718   | 1818   |
|                                 |               |                                                                                                                             | 1519                  | 1619   | 1719   | 1819   |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

#### ■ Start conditions

The following conditions must be fulfilled when starting. The required conditions must also be assembled in the sequence program, and the sequence program must be configured so the operation will not start if the conditions are not fulfilled.

| Signal name      |                         | Signal state |                                               | Device |        |        |        |
|------------------|-------------------------|--------------|-----------------------------------------------|--------|--------|--------|--------|
|                  |                         |              |                                               | Axis 1 | Axis 2 | Axis 3 | Axis 4 |
| Interface signal | PLC READY signal        | ON           | CPU module preparation completed              | Y0     |        |        |        |
|                  | QD75 READY signal       | ON           | QD75 preparation completed                    | X0     |        |        |        |
|                  | Synchronization flag *  | ON           | QD75 buffer memory<br>The access is possible. | X1     |        |        |        |
|                  | Axis stop signal        | OFF          | Axis stop signal is OFF                       | Y4     | Y5     | Y6     | Y7     |
|                  | Start complete signal   | OFF          | Start complete signal is OFF                  | X10    | X11    | X12    | X13    |
|                  | BUSY signal             | OFF          | QD75 is not operating                         | XC     | XD     | XE     | XF     |
|                  | Error detection signal  | OFF          | There is no error                             | X8     | X9     | XA     | XB     |
|                  | M code ON signal        | OFF          | M code ON signal is OFF                       | X4     | X5     | X6     | X7     |
| External signal  | Drive unit READY signal | ON           | Drive unit preparation completed              | —      |        |        |        |
|                  | Stop signal             | OFF          | Stop signal is OFF                            | —      |        |        |        |
|                  | Upper limit (FLS)       | ON           | Within limit range                            | —      |        |        |        |
|                  | Lower limit (RLS)       | ON           | Within limit range                            | —      |        |        |        |

\* If the CPU module is set to the asynchronous mode in the synchronization setting, this must be inserted in the program for interlocking. If it is set to the synchronous mode, it must not be inserted in the program for interlocking because it is turned ON when the CPU module executes calculation.



■ Start time chart

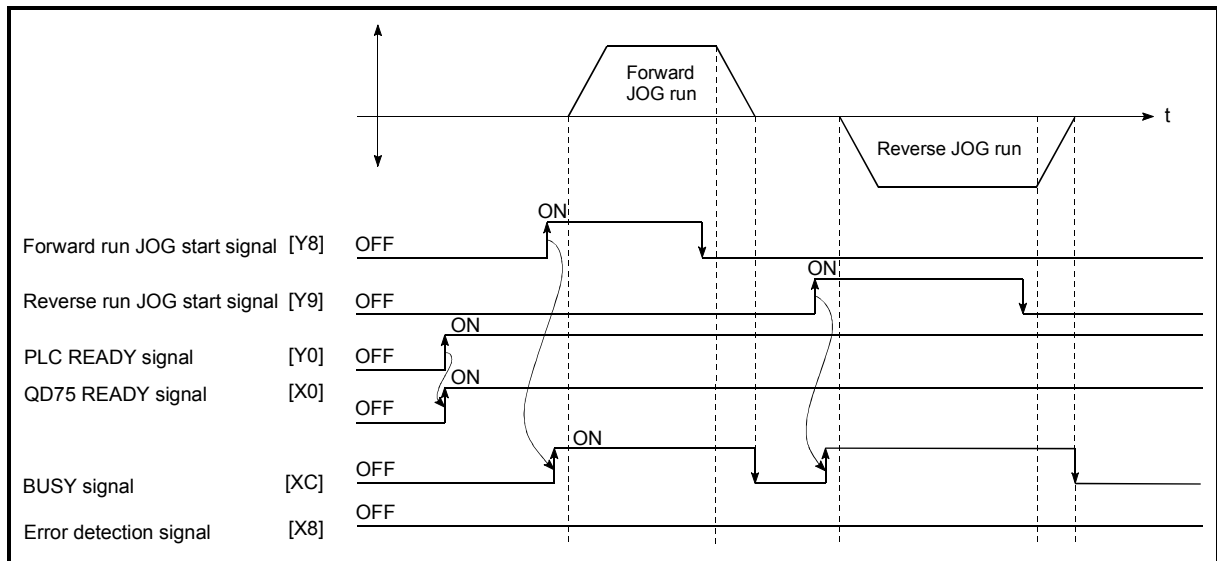


Fig. 11.6 JOG operation start time chart

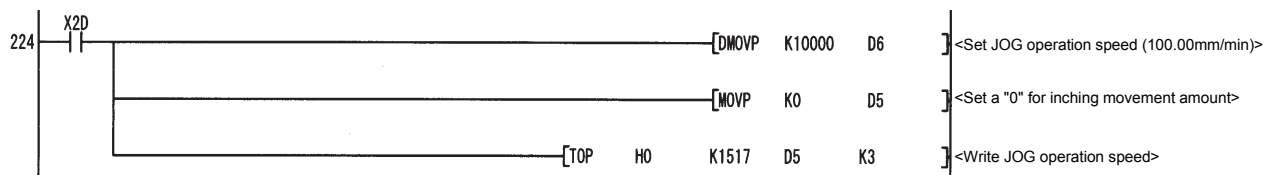
## ■ Creating the program

### Example

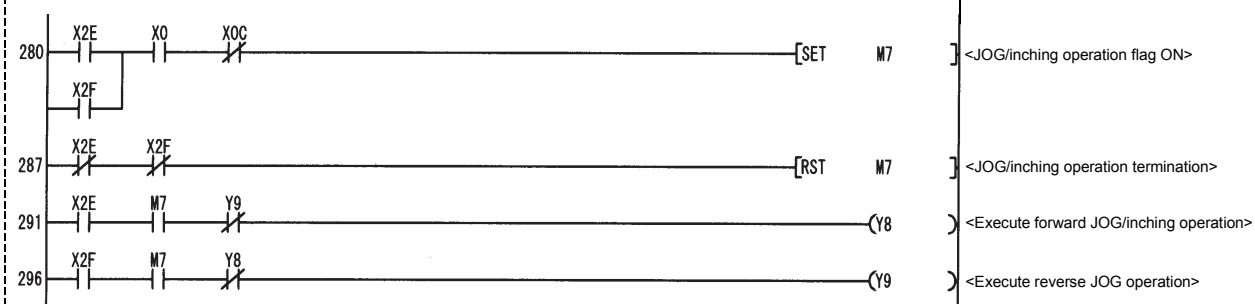
\*

\* No. 10 JOG operation setting program

\*



\* No.12 JOG operation/inching operation execution program



## 11.2.5 JOG operation example

■ When the "stop signal" is turned ON during JOG operation

When the "stop signal" is turned ON during JOG operation, the JOG operation will stop by the "deceleration stop" method.

An error "Stop signal ON at start" (error code 106) will occur if JOG start signal is turned ON while the stop signal is ON.

The operation can be started by turning the stop signal OFF, and turning the JOG start signal from OFF to ON again.

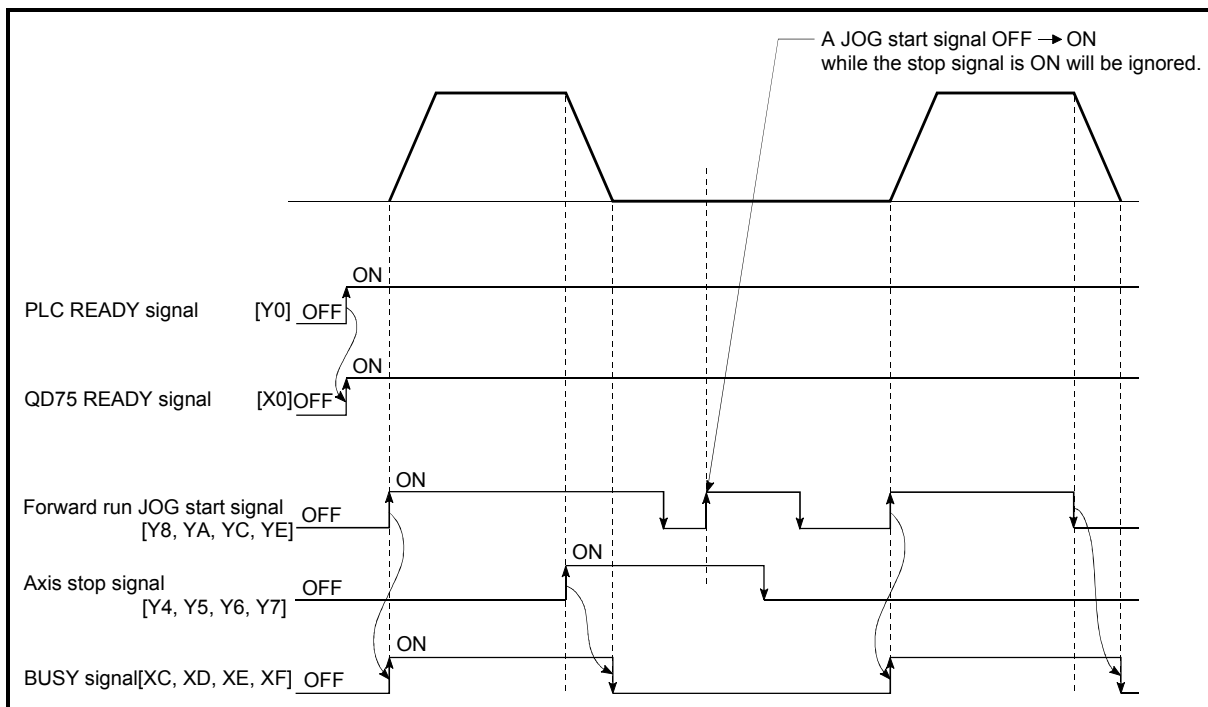


Fig. 11.7 Operation when the stop signal is turned ON during JOG operation

■ When both the "forward run JOG start signal" and "reverse run JOG start signal" are turned ON simultaneously for one axis

When both the "forward run JOG start signal" and "reverse run JOG start signal" are turned ON simultaneously for one axis, the "forward run JOG start signal" is given priority. In this case, the "reverse run JOG start signal" is validated when the QD75 BUSY signal is turned OFF.

If the forward run JOG operation is stopped due to stop or axis error by a stop signal, the reverse run JOG operation will not be executed even if the "reverse run JOG start signal" turns ON.

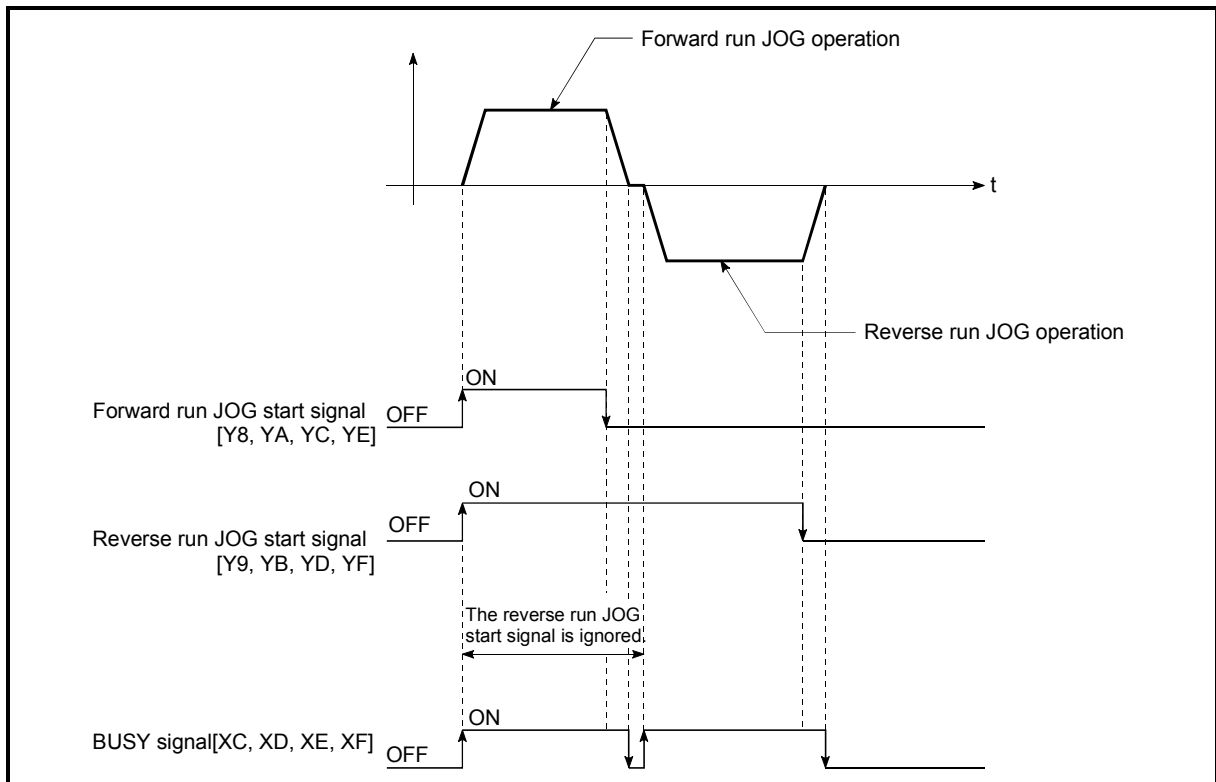


Fig. 11.8 Operation when both the forward run JOG start signal and reverse run JOG start signal are turned ON simultaneously

- When the "JOG start signal" is turned ON again during deceleration caused by the ON → OFF of the "JOG start signal"

When the "JOG start signal" is turned ON again during deceleration caused by the ON → OFF of the "JOG start signal", the JOG operation will be carried out from the time the "JOG start signal" is turned ON.

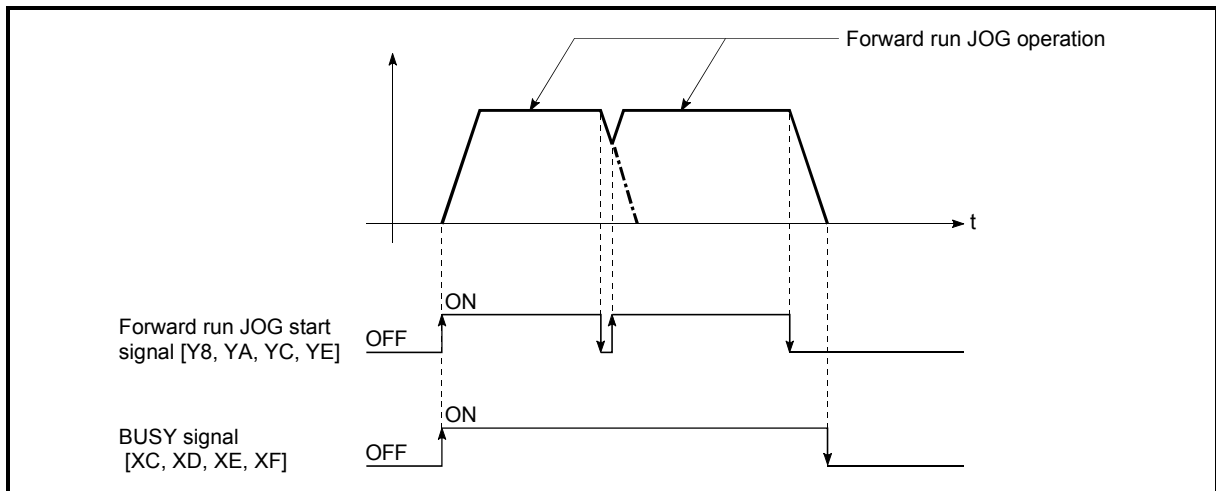


Fig. 11.9 Operation when the JOG start signal is turned ON during deceleration

- When the "JOG start signal" is turned ON during a peripheral device test mode

When the "JOG start signal" is turned ON during a peripheral device test mode, it will be ignored and the JOG operation will not be carried out.

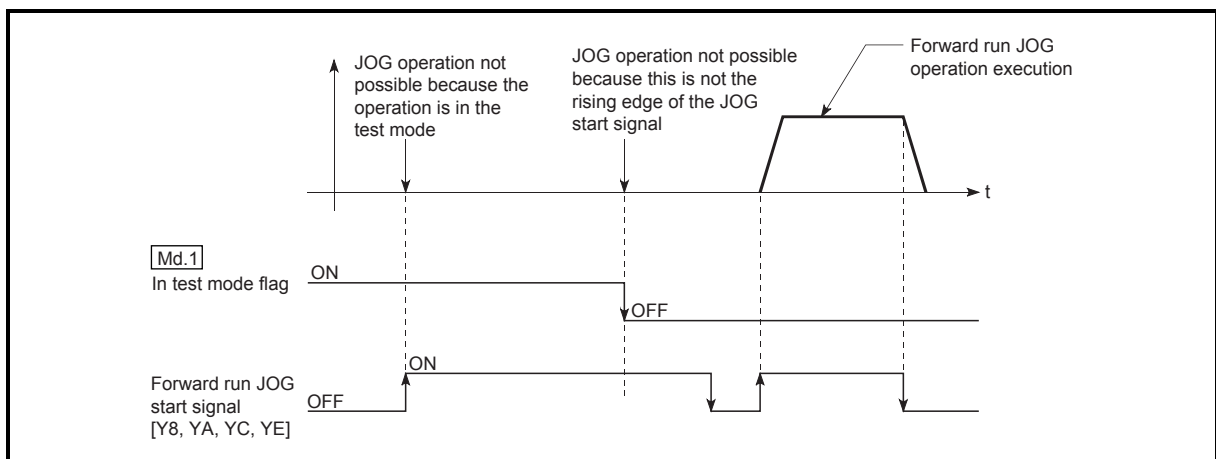


Fig. 11.10 Operation when the JOG start signal is turned ON during a test mode

## 11.3 Inching operation

### 11.3.1 Outline of inching operation

#### Important

When the inching operation is carried out near the upper or lower limit, use the hardware stroke limit function (Refer to Section 12.4.4).

\* If the hardware stroke limit function is not used, the workpiece may exceed the movement range, and an accident may result.

#### ■ Inching operation

In inching operation, pulses are input to the drive unit at 1.8ms to move the workpiece by a designated movement amount after the forward run JOG start signal [Y8, YA, YC, YE] or reverse JOG start signal [Y9, YB, YD, YF] is turned ON. The following shows the example of inching operation.

|    |                                                                                                                                                                           |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) | When the start signal is turned ON, inching operation is carried out in the direction designated by the start signal. In this case, BUSY signal is turned from OFF to ON. |
| 2) | The workpiece is moved by a movement amount set in " <u>Cd.16</u> Inching movement amount".                                                                               |
| 3) | The workpiece movement stops when the speed becomes "0". In this case, BUSY signal is turned from ON to OFF. The positioning complete signal is turned from OFF to ON.    |
| 4) | The positioning complete signal is turned from ON to OFF after a time set in " <u>Pr.40</u> Positioning complete signal output time" has been elapsed.                    |

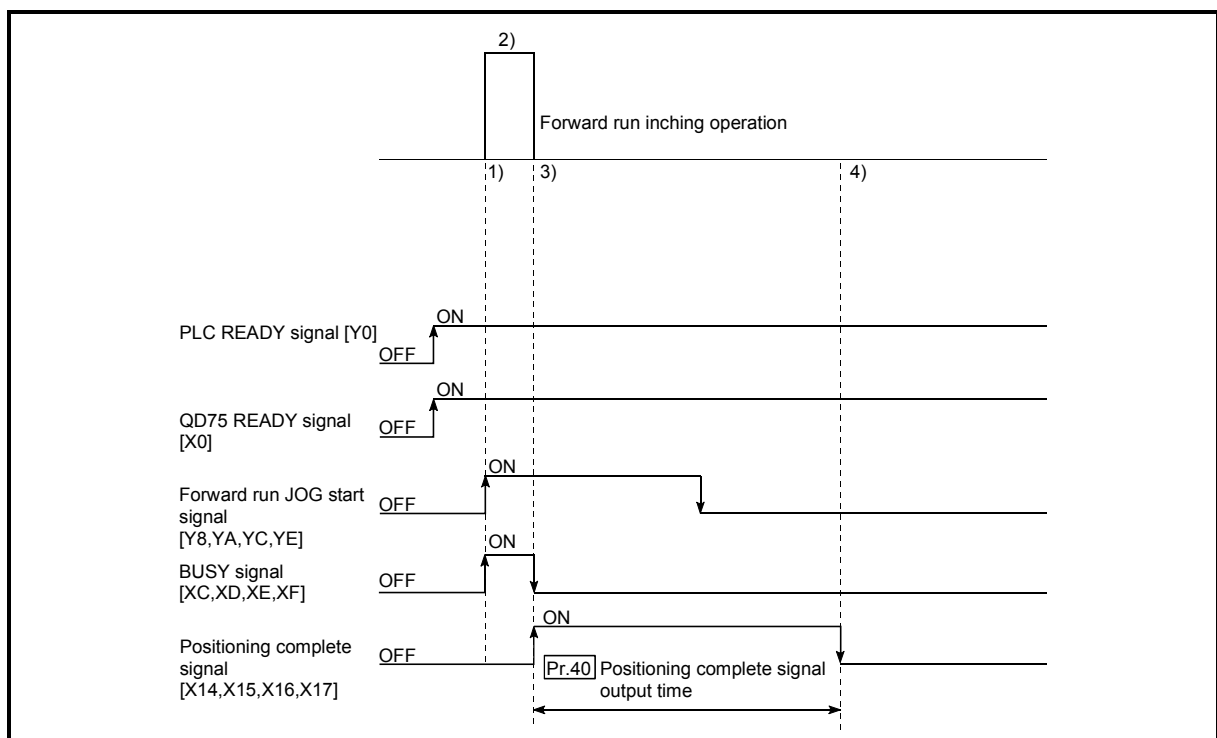


Fig. 11.11 Inching operation

### ■ Precautions during operation

- (1) Acceleration/deceleration processing is not carried out during inching operation. (Pulses corresponding to the designated inching movement amount are output at 1.8ms. The movement direction of inching operation is reversed and, when backlash compensation is carried out, first pulses corresponding to the backlash amount are output at 1.8ms and then pulses corresponding to the designated inching movement amount are output in the next 1.8ms.)  
The " [Cd.17] JOG speed" is ignored even if it is set. However, the error "Inching movement amount error" (error code: 301) will occur in the following cases:

$([Cd.16] \text{ Inching movement amount}) \times (A) > ([Pr.31] \text{ JOG speed limit value})$

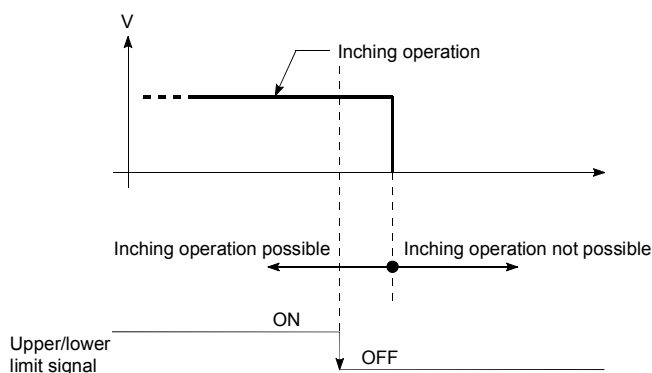
Where (A) is as follows.

- When the unit is pulse: 562.5
- When the unit is other than pulse: 337.5

- (2) Set a value other than a "0" in " [Cd.16] Inching movement amount".  
If a "0" is set, the operation will become JOG operation (Refer to Section 11.2 "JOG operation").

### ■ Operations when stroke limit error occurs

When the operation is stopped by hardware stroke limit error or software stroke limit error, the inching operation can be performed in an opposite way (direction within normal limits) after an error reset. (An error will occur again if JOG start signal is turned ON in a direction to outside the stroke limit.)



### ■ Inching operation timing and processing times

The following drawing shows the details of the inching operation timing and processing time.

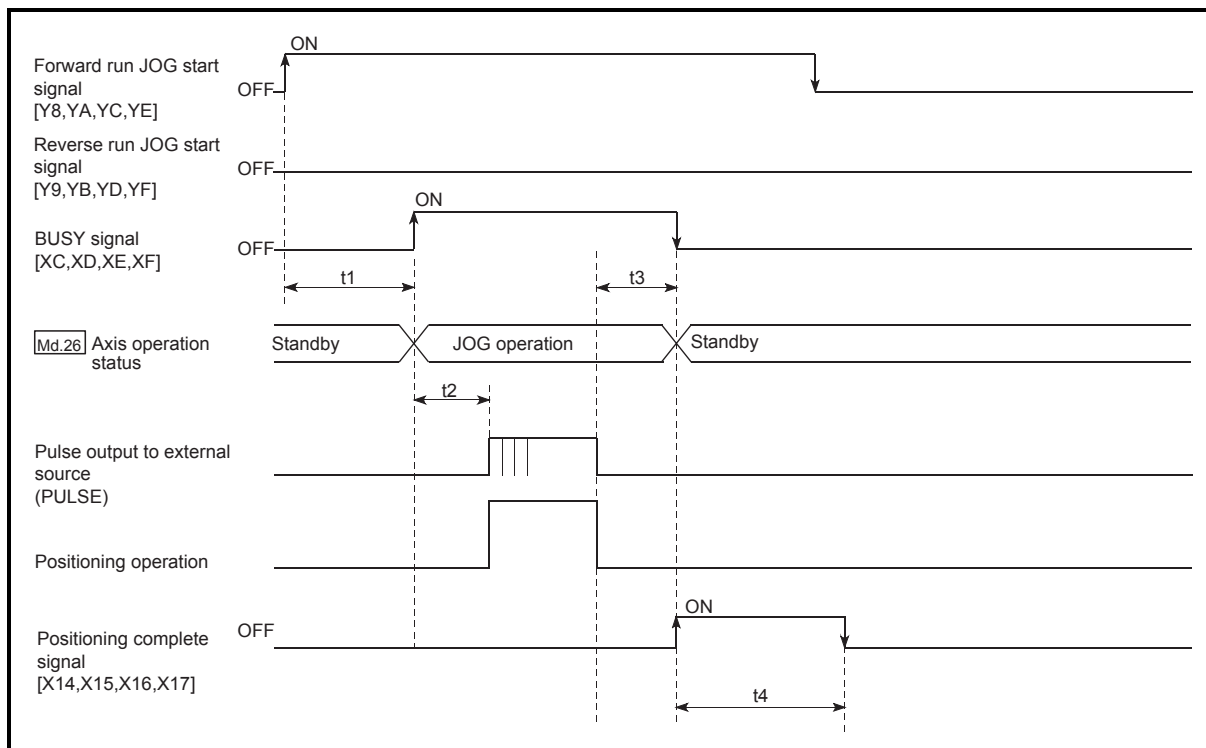


Fig. 11.12 Inching operation timing and processing times

#### Normal timing times

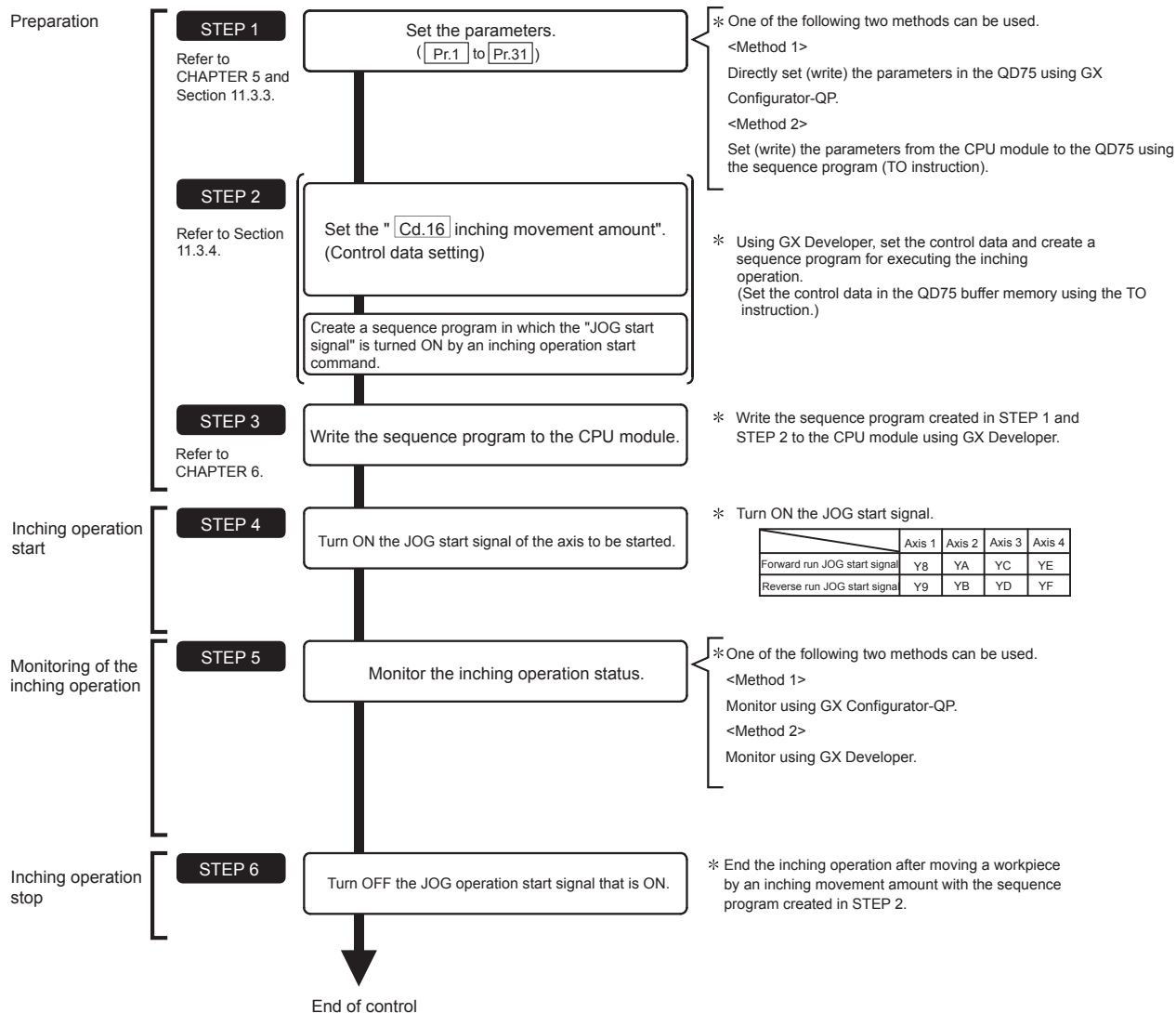
| Model           | t1           | t2           | t3         | t4                      |
|-----------------|--------------|--------------|------------|-------------------------|
| QD75P□N/QD75D□N | 1.0 to 3.0ms | 1.3 to 2.2ms | 0 to 0.9ms | Depending on parameters |
| QD75P□/QD75D□   | 1.0 to 3.0ms | 2.7 to 4.4ms | 0 to 1.8ms |                         |

- Depending on the operating statuses of the other axes, delay may occur in the t1 timing time.



## 11.3.2 Inching operation execution procedure

The inching operation is carried out by the following procedure.

**REMARK**

- Mechanical elements such as limit switches are considered as already installed.
- Parameter settings work in common for all control using the QD75.

## 11.3.3 Setting the required parameters for inching operation

The "Parameters" must be set to carry out inching operation.

The following table shows the setting items of the required parameters for carrying out inching operation. When only inching operation will be carried out, no parameters other than those shown below need to be set. (Use the initial values or setting values within a range where no error occurs for trouble-free operation.)

| Setting item |                                                             | Setting requirement | Factory-set initial value<br>(setting details)          |
|--------------|-------------------------------------------------------------|---------------------|---------------------------------------------------------|
| Parameters   | Pr.1 Unit setting                                           | ◎                   | 3 (pulse)                                               |
|              | Pr.2 No. of pulses per rotation (Ap) (Unit: pulse)          | ◎                   | 20000                                                   |
|              | Pr.3 Movement amount per rotation (Al) (Unit: pulse)        | ◎                   | 20000                                                   |
|              | Pr.4 Unit magnification (Am)                                | ◎                   | 1 (1-fold)                                              |
|              | Pr.5 Pulse output mode                                      | ◎                   | 1 (CW/CCW mode)                                         |
|              | Pr.6 Rotation direction setting                             | ◎                   | 0 (current value increases by forward run pulse output) |
|              | Pr.11 Backlash compensation amount (Unit: pulse)            | ○                   | 0                                                       |
|              | Pr.12 Software stroke limit upper limit value (Unit: pulse) | ○                   | 2147483647                                              |
|              | Pr.13 Software stroke limit lower limit value (Unit: pulse) | ○                   | -2147483648                                             |
|              | Pr.14 Software stroke limit selection                       | ○                   | 0 (current feed value)                                  |
|              | Pr.15 Software stroke limit valid/invalid setting           | ○                   | 0 (valid)                                               |
|              | Pr.17 Torque limit setting value (Unit: %)                  | ○                   | 300                                                     |
|              | Pr.23 Output signal logic selection                         | ○                   | 0 (Pulse output to the drive unit is negative logic)    |
|              | Pr.31 JOG speed limit value (Unit: pulse/s)                 | ◎                   | 20000                                                   |

◎ : Setting always required.

○ : Set according to requirements (Leave set to the initial value when not used.)

**REMARK**

- Parameter settings work in common for all control using the QD75. When carrying out other controls ("major positioning control", "high-level positioning control", and "OPR positioning control"), the respective setting items must also be set.
- Parameters are set for each axis.
- Refer to CHAPTER 5 "DATA USED FOR POSITIONING CONTROL" for setting details.

### 11.3.4 Creating a program to enable/disable the inching operation

A sequence program must be created to execute an inching operation. Consider the "required control data setting", "start conditions", and "start time chart" when creating the program.

The following shows an example when an inching operation is started for axis 1. (The example shows the inching operation when a "10.0  $\mu$ m" is set in "Cd.16 Inching movement amount".)

#### ■ Required control data setting

The control data shown below must be set to execute an inching operation. The setting is carried out with the sequence program.

| Setting item | Setting value           | Setting details                                                                                                                                                                                        | Buffer memory address |        |        |        |
|--------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------|--------|--------|
|              |                         |                                                                                                                                                                                                        | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| Cd.16        | Inching movement amount | 100<br>Set the setting value so that the command pulse is not increased larger than the maximum output pulse.<br>(The max. output pulse<br>QD75D□N: 4Mpps<br>QD75D□: 1Mpps<br>QD75P□N/QD75P□: 200kpps) | 1517                  | 1617   | 1717   | 1817   |

\* Refer to Section 5.7 "List of control data" for information on setting details.

#### ■ Start conditions

The following conditions must be fulfilled when starting. The required conditions must also be assembled in the sequence program, and the sequence program must be configured so the operation will not start if the conditions are not fulfilled.

| Signal name      |                             | Signal state |                                    | Device |        |        |        |
|------------------|-----------------------------|--------------|------------------------------------|--------|--------|--------|--------|
|                  |                             |              |                                    | Axis 1 | Axis 2 | Axis 3 | Axis 4 |
| Interface signal | PLC READY signal            | ON           | CPU module preparation completed   | Y0     |        |        |        |
|                  | QD75 READY signal           | ON           | QD75 preparation completed         | X0     |        |        |        |
|                  | Synchronization flag *      | ON           | Accessible to QD75 buffer memory   | X1     |        |        |        |
|                  | Axis stop signal            | OFF          | Axis stop signal is OFF            | Y4     | Y5     | Y6     | Y7     |
|                  | Start complete signal       | OFF          | Start complete signal is OFF       | X10    | X11    | X12    | X13    |
|                  | BUSY signal                 | OFF          | QD75 is not operating              | XC     | XD     | XE     | XF     |
|                  | Positioning complete signal | OFF          | Positioning complete signal is OFF | X14    | X15    | X16    | X17    |
|                  | Error detection signal      | OFF          | There is no error                  | X8     | X9     | XA     | XB     |
| External signal  | M code ON signal            | OFF          | M code ON signal is OFF            | X4     | X5     | X6     | X7     |
|                  | Drive unit READY signal     | ON           | Drive unit preparation completed   | —      |        |        |        |
|                  | Stop signal                 | OFF          | Stop signal is OFF                 | —      |        |        |        |
|                  | Upper limit (FLS)           | ON           | Within limit range                 | —      |        |        |        |
|                  | Lower limit (RLS)           | ON           | Within limit range                 | —      |        |        |        |

\* If the CPU module is set to the asynchronous mode in the synchronization setting, this must be inserted in the program for interlocking. If it is set to the synchronous mode, it must not be inserted in the program for interlocking because it is turned ON when the CPU module executes calculation.

■ Start time chart

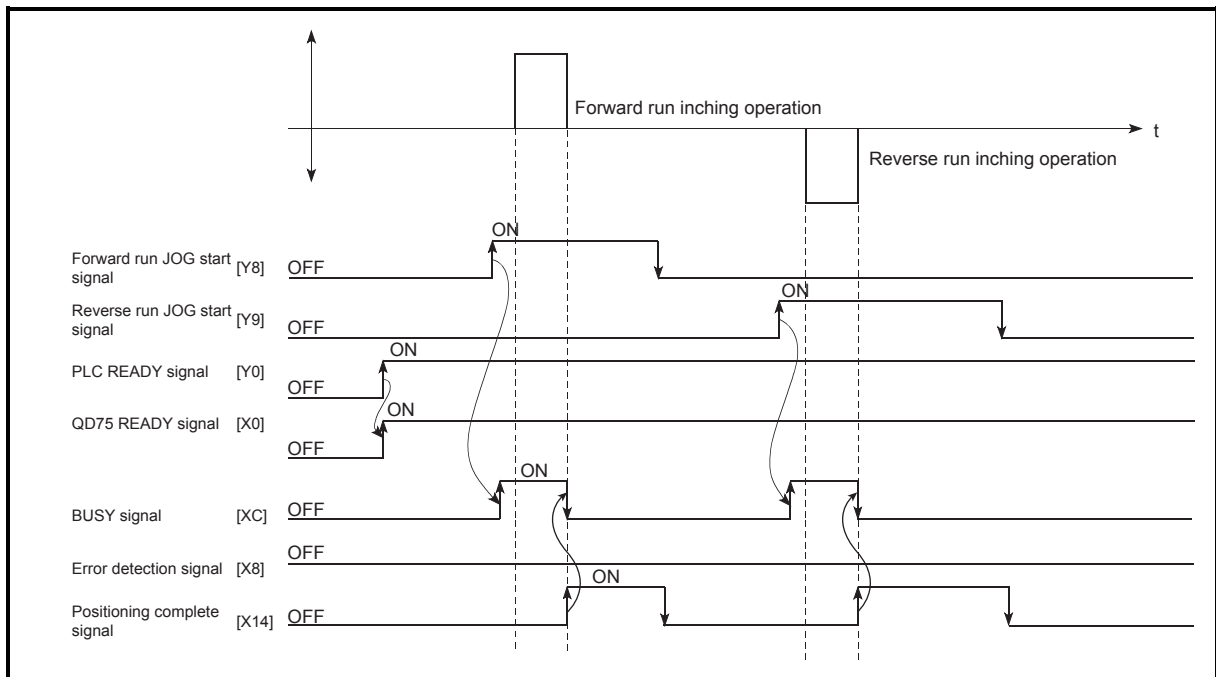
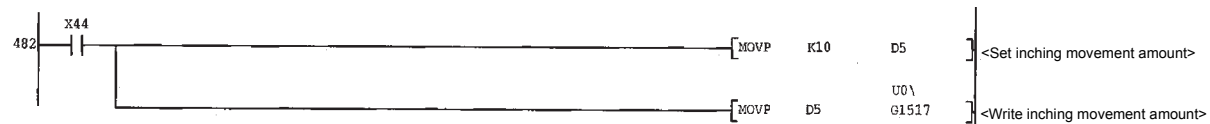


Fig. 11.13 Inching operation start time chart

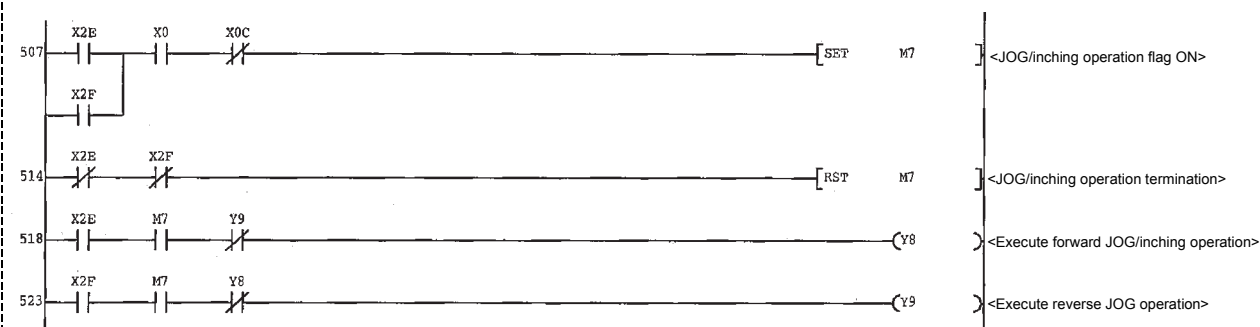
### ■ Creating the program

#### Example

\*  
\* No.11 Inching operation setting program  
\*



\* No.12 JOG operation/inching operation execution program



## 11.3.5 Inching operation example

■ When executing inching operation while stop signal is turned ON:

If the JOG start signal is turned ON while the stop signal is ON, an error "Stop signal ON at start" (error code: 106) will occur.

The inching operation can be re-started when the stop signal is turned OFF and the JOG start signal is turned OFF and then turned ON.

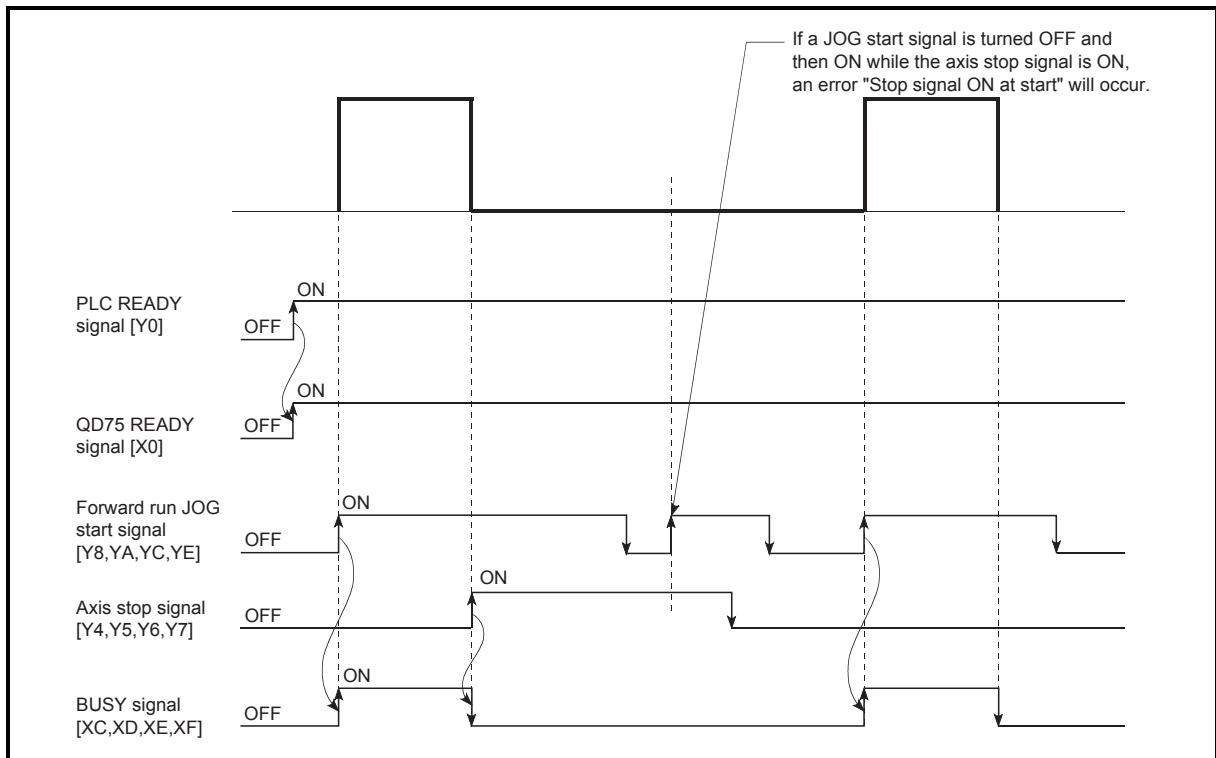


Fig. 11.14 Operation when executing inching operation while stop signal is ON

■ When "JOG start signal" is turned ON when peripheral devices are in the test mode:

If "JOG start signal" is turned ON when peripheral devices are in the test mode, the "JOG start signal" will be ignored and inching operation will not be carried out.

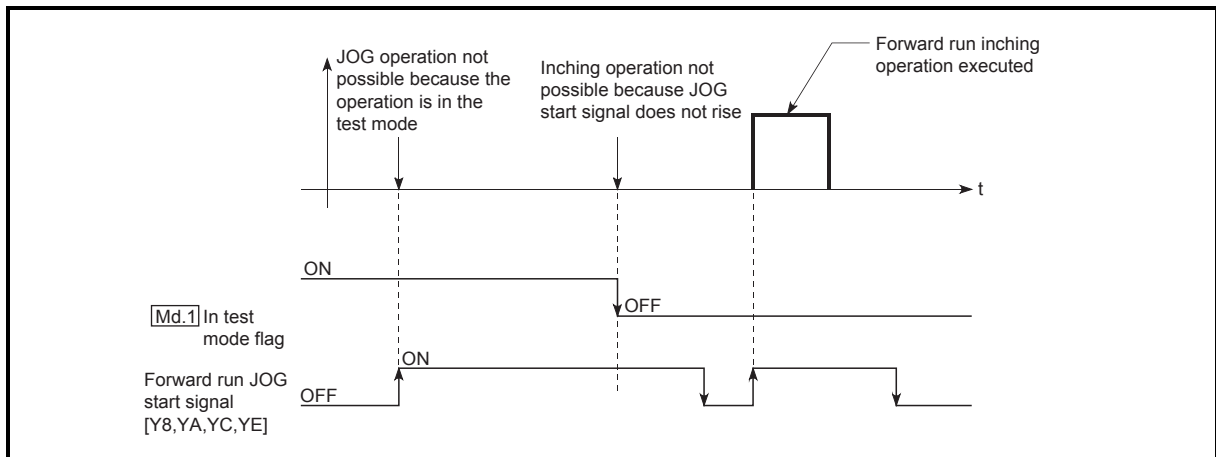


Fig. 11.15 Operation when JOG start signal is turned ON in test mode

## 11.4 Manual pulse generator operation

### 11.4.1 Outline of manual pulse generator operation

#### Important

Create the sequence program so that " **Cd.21** Manual pulse generator enable flag" is always set to "0" (disabled) when a manual pulse generator operation is not carried out.

\* Mistakenly touching the manual pulse generator when the manual pulse generator enable flag is set to "1" (enable) can cause accidents or incorrect positioning.

#### Manual pulse generator operation

In manual pulse generator operations, pulses are input to the QD75 from the manual pulse generator. This causes the same No. of input pulses to be output from the QD75 to the servo amplifier, and the workpiece is moved in the designated direction.

The following shows an example of manual pulse generator operation.

|    |                                                                                                                                                            |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) | When the " <b>Cd.21</b> Manual pulse generator enable flag" is set to "1", the BUSY signal turns ON and the manual pulse generator operation is enabled.   |
| 2) | The workpiece is moved corresponding to the No. of pulses input from the manual pulse generator.                                                           |
| 3) | The workpiece movement stops when no more pulses are input from the manual pulse generator.                                                                |
| 4) | When the " <b>Cd.21</b> Manual pulse generator enable flag" is set to "0", the BUSY signal turns OFF and the manual pulse generator operation is disabled. |

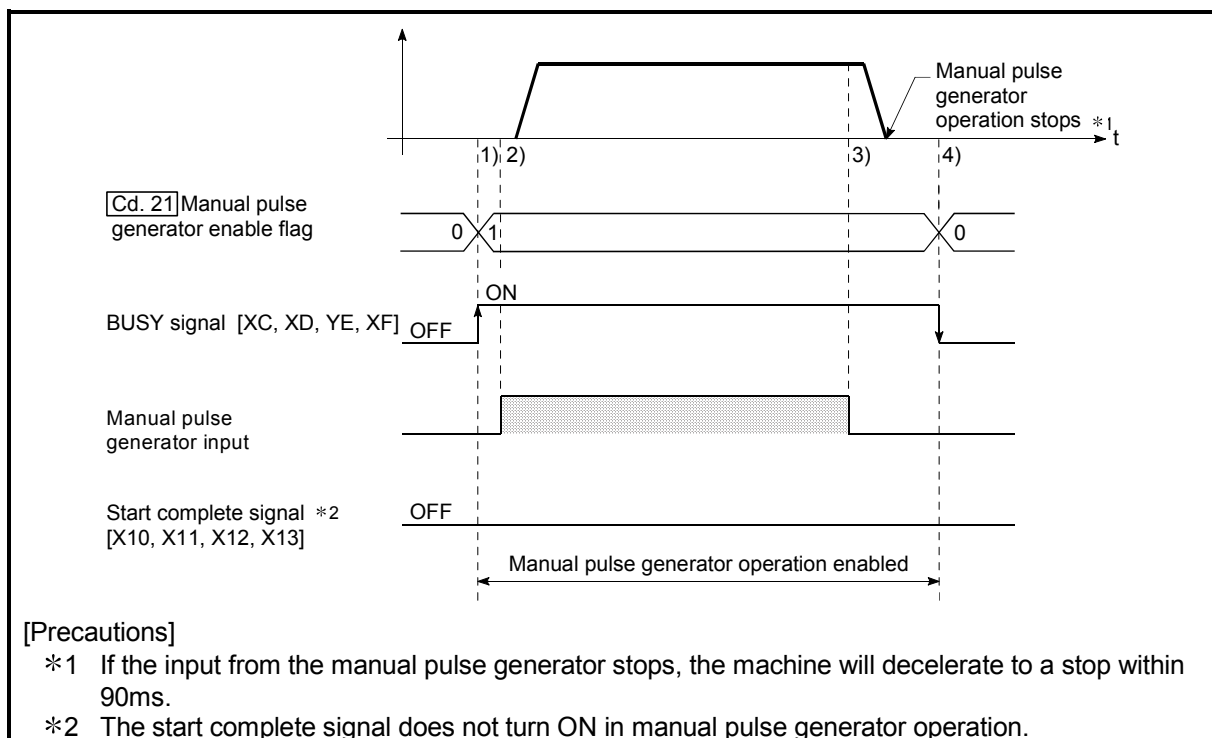


Fig. 11.16 Manual pulse generator operation



**■ Restricted items**

A manual pulse generator is required to carry out manual pulse generator operation.

**■ Precautions during operation**

- (1) The speed during manual pulse generator operation is not limited by the " **Pr.8** Speed limit value".
- (2) If the " **Cd.21** Manual pulse generator enable flag" is turned ON while the QD75 is BUSY (BUSY signal ON), a warning "Start during operation" (warning code: 100) will occur.
- (3) If a stop factor occurs during manual pulse generator operation, the operation will stop, and the BUSY signal will turn OFF.  
At this time, the " **Cd.21** Manual pulse generator enable flag" will be left ON, but manual pulse generator operation will not be possible. To carry out manual pulse generator operation again, measures must be carried out to eliminate the stop factor. Once eliminated, the operation can be carried out again by turning the " **Cd.21** Manual pulse generator enable flag" ON → OFF → ON.  
(Note that this excludes when hardware/software stroke limit errors occurs.)
- (4) Pulses will not be output if an error occurs when the manual pulse generator operation starts.

**REMARK**

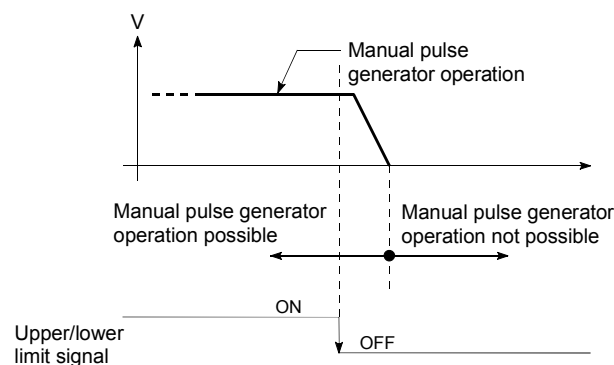
- One QD75 module can be connected to one manual pulse generator.
- The QD75 module can simultaneously output pulses to the axis 1 to axis 4 drive units by one manual pulse generator.  
(1- to 4-axis simultaneous operation is possible.)

### Operations when stroke limit error occurs

When the hardware stroke limit error or the software stroke limit error is detected during operation, the operation will decelerate to a stop. However, "Md.26 Axis operation status" will keep the status "Manual pulse generator operation" in that case. \*1

After stopping, manual pulse generator input pulses to the outside direction of the limit range are not accepted, but operation can be executed within the range.

\*1: Only when the current feed value or the current machine feed value overflows or underflows during deceleration, the manual pulse generator operation will terminate as "Md.26 Axis operation status" is changed to "Error". To carry out manual pulse generator operation again, "Cd.21 Manual pulse generator enable flag" must be turned OFF once and turn ON.



### Manual pulse generator operation timing and processing time

The following drawing shows details of the manual pulse generator operation timing and processing time.

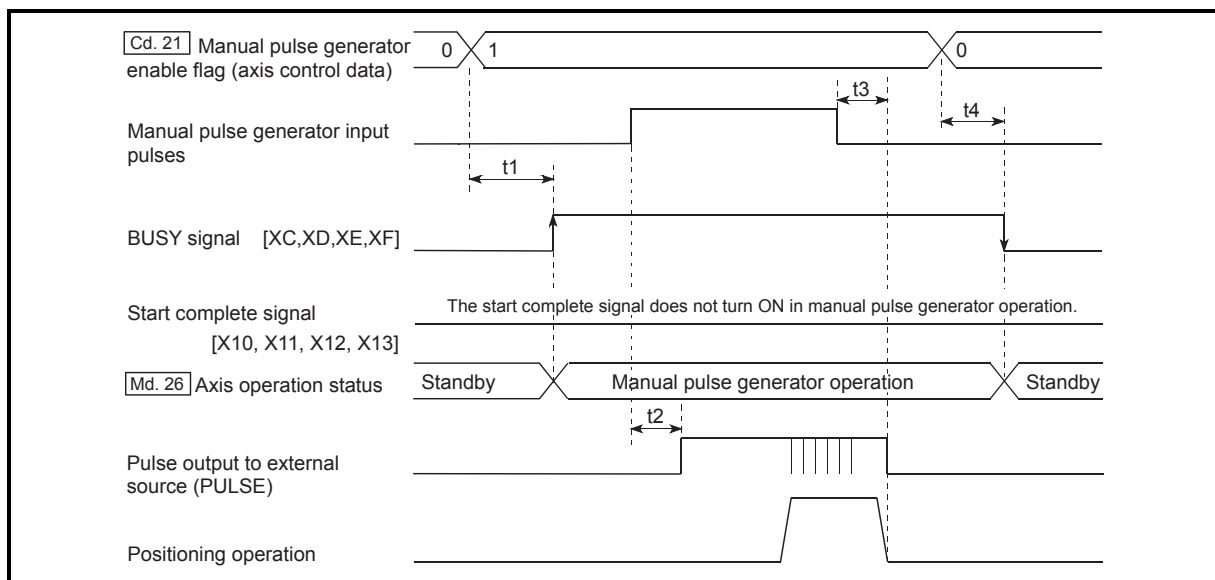


Fig. 11.17 Manual pulse generator operation timing and processing times

#### Normal timing times

| Model           | t1         | t2            | t3             | t4             |
|-----------------|------------|---------------|----------------|----------------|
| QD75P□N/QD75D□N | 0 to 0.9ms | 1.7 to 30.2ms | 58.6 to 87.6ms | 28.4 to 57.2ms |
| QD75P□/QD75D□   | 0 to 2.6ms | 6.2 to 32ms   | 60.4 to 90ms   | 28.4 to 57.6ms |

• Delays may occur in the t1 timing time due to the operation status of other axes.

### ■ Position control by manual pulse generator operation

In manual pulse generator operation, the position is moved by a "manual pulse generator 1 pulse movement amount" per pulse.

The current feed value in the positioning control by manual pulse generator operation can be calculated using the expression shown below.

$$\text{Current feed value} = \text{Number of input pulses} \times \boxed{\text{Cd.20}} \text{ Manual pulse generator 1 pulse input magnification} \times \text{Manual pulse generator 1 pulse movement amount}$$

| $\boxed{\text{Pr.1}}$ Unit setting             | mm                | inch        | degree        | pulse  |
|------------------------------------------------|-------------------|-------------|---------------|--------|
| Manual pulse generator 1 pulse movement amount | 0.1 $\mu\text{m}$ | 0.00001inch | 0.00001degree | 1pulse |

For example, when " $\boxed{\text{Pr.1}}$  Unit setting" is mm and " $\boxed{\text{Cd.20}}$  Manual pulse generator 1 pulse input magnification" is 2, and 100 pulses are input from the manual pulse generator, the current feed value is as follows.

$$\begin{aligned} 100 \times 2 \times 0.1 &= 20 [\mu\text{m}] \\ &= 200 [\text{Current feed value}] \end{aligned}$$

The number of pulses output actually to the drive unit is "Manual pulse generator 1 pulse movement amount/movement amount per pulse \*". For example, when " $\boxed{\text{Pr.1}}$  Unit setting" is mm and the movement amount per pulse is 1  $\mu\text{m}$ ,  $0.1/1 = 1/10$ , i.e., the output to the drive unit per pulse from the manual pulse generator is 1/10 pulse. Thus, the QD75 outputs 1 pulse to the drive unit after receiving 10 pulses from the manual pulse generator.

$$* \text{ Movement amount per pulse} = \frac{\boxed{\text{Pr.3}} \text{ Movement amount per rotation}}{\boxed{\text{Pr.2}} \text{ No. of pulses per rotation}} \times \boxed{\text{Pr.4}} \text{ Unit magnification}$$

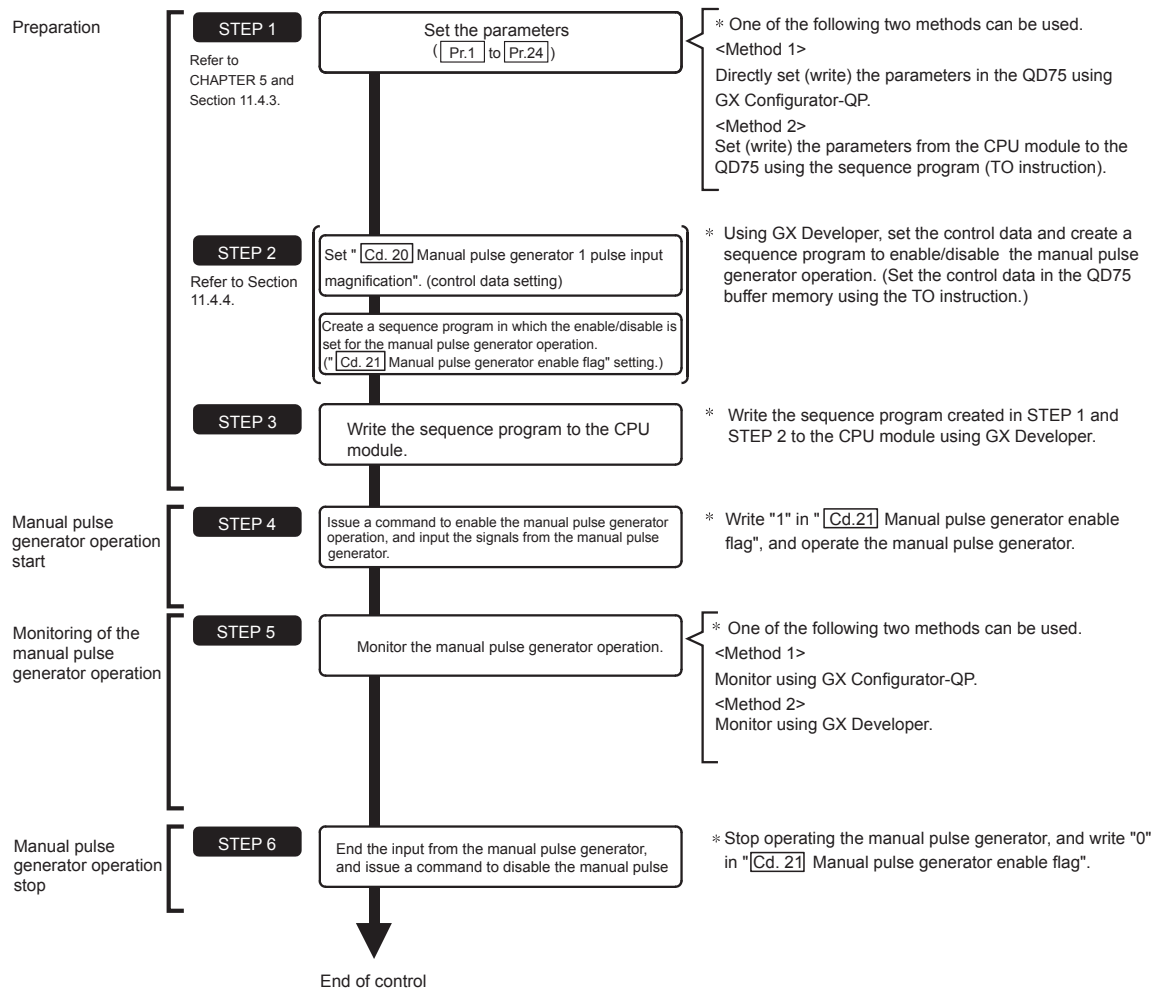
### ■ Speed control by manual pulse generation operation

The speed during positioning control by manual pulse generator operation is a speed corresponding to the No. of input pulses per unit time, and can be obtained using the following equation.

$$\text{Output command frequency} = \text{Input frequency} \times \boxed{\text{Cd.20}} \text{ Manual pulse generator 1 pulse input magnification}$$

## 11.4.2 Manual pulse generator operation execution procedure

The manual pulse generator operation is carried out by the following procedure.

**REMARK**

- Mechanical elements such as limit switches are considered as already installed.
- Parameter settings work in common for all control using the QD75.

## 11.4.3 Setting the required parameters for manual pulse generator operation

The "Parameters" must be set to carry out manual pulse generator operation.

The following table shows the setting items of the required parameters for carrying out manual pulse generator operation. When only manual pulse generator operation will be carried out, no parameters other than those shown below need to be set. (Use the initial values or setting values within a range where no error occurs for trouble-free operation.)

| Setting item |                                                                    | Setting requirement | Factory-set initial value<br>(setting details)          |
|--------------|--------------------------------------------------------------------|---------------------|---------------------------------------------------------|
| Parameters   | <b>Pr.1</b> Unit setting                                           | ◎                   | 3 (pulse)                                               |
|              | <b>Pr.2</b> No. of pulses per rotation (Ap) (Unit: pulse)          | ◎                   | 20000                                                   |
|              | <b>Pr.3</b> Movement amount per rotation (Al) (Unit: pulse)        | ◎                   | 20000                                                   |
|              | <b>Pr.4</b> Unit magnification (Am)                                | ◎                   | 1 (1-fold)                                              |
|              | <b>Pr.5</b> Pulse output mode                                      | ◎                   | 1 (CW/CCW mode)                                         |
|              | <b>Pr.6</b> Rotation direction setting                             | ◎                   | 0 (current value increases by forward run pulse output) |
|              | <b>Pr.11</b> Backlash compensation amount (Unit: pulse)            | ○                   | 0                                                       |
|              | <b>Pr.12</b> Software stroke limit upper limit value (Unit: pulse) | ○                   | 2147483647                                              |
|              | <b>Pr.13</b> Software stroke limit lower limit value (Unit: pulse) | ○                   | -2147483648                                             |
|              | <b>Pr.14</b> Software stroke limit selection                       | ○                   | 0 (current feed value)                                  |
|              | <b>Pr.15</b> Software stroke limit valid/invalid setting           | ○                   | 0 (valid)                                               |
|              | <b>Pr.17</b> Torque limit setting value (Unit: %)                  | ○                   | 300                                                     |
|              | <b>Pr.22</b> Input signal logic selection                          | ○                   | 0 (Manual pulse generator input is negative logic.)     |
|              | <b>Pr.23</b> Output signal logic selection                         | ○                   | 0 (Pulse output to drive unit is negative logic.)       |
|              | <b>Pr.24</b> Manual pulse generator input selection                | ○                   | 0 (4 times multiplication of A phase/B phase)           |

◎ : Setting always required.

○ : Set according to requirements (Leave set to the initial value when not used.)

|        |
|--------|
| REMARK |
|--------|

- Parameter settings work in common for all control using the QD75. When carrying out other control ("major positioning control", "high-level positioning control", "OPR positioning control"), the respective setting items must also be matched and set.
- Parameters are set for each axis. But **Pr.22** Manual pulse generator input logic (b8), **Pr.24** is set only for axis 1. (The setting for axes 2,3, and 4 is ignored.)
- Refer to CHAPTER 5 "DATA USED FOR POSITIONING CONTROL" for setting details.

#### 11.4.4 Creating a program to enable/disable the manual pulse generator operation

A sequence program must be created to execute a manual pulse generator operation. Consider the "required control data setting", "start conditions" and "start time chart" when creating the program.

The following shows an example when a manual pulse generator operation is started for axis 1.

##### ■ Required control data setting

The control data shown below must be set to execute a manual pulse generator operation. The setting is carried out with the sequence program.

| Setting item |                                                    | Setting value | Setting details                                                                                                                                                | Buffer memory address |              |              |              |
|--------------|----------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------|--------------|--------------|
|              |                                                    |               |                                                                                                                                                                | Axis 1                | Axis 2       | Axis 3       | Axis 4       |
| <b>Cd.20</b> | Manual pulse generator 1 pulse input magnification | 1             | Set the manual pulse generator 1 pulse input magnification.<br>(QD75P□N/QD75D□N: 1 to 1000 times<br>QD75P□/QD75D□: 1 to 100 times)                             | 1522<br>1523          | 1622<br>1623 | 1722<br>1723 | 1822<br>1823 |
| <b>Cd.21</b> | Manual pulse generator enable flag                 | 1 (0)         | Set "1: Enable manual pulse generator operation". (Set "0: Disable manual pulse generator operation" when finished with the manual pulse generator operation.) | 1524                  | 1624         | 1724         | 1824         |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

##### ■ Start conditions

The following conditions must be fulfilled when starting. The required conditions must also be assembled in the sequence program, and the sequence program must be configured so the operation will not start if the conditions are not fulfilled.

| Signal name      |                         | Signal state |                                               | Device |        |        |        |
|------------------|-------------------------|--------------|-----------------------------------------------|--------|--------|--------|--------|
|                  |                         |              |                                               | Axis 1 | Axis 2 | Axis 3 | Axis 4 |
| Interface signal | PLC READY signal        | ON           | CPU module preparation completed              | Y0     |        |        |        |
|                  | QD75 READY signal       | ON           | QD75 preparation completed                    | X0     |        |        |        |
|                  | Synchronization flag *  | ON           | QD75 buffer memory<br>The access is possible. | X1     |        |        |        |
|                  | Axis stop signal        | OFF          | Axis stop signal is OFF                       | Y4     | Y5     | Y6     | Y7     |
|                  | Start complete signal   | OFF          | Start complete signal is OFF                  | X10    | X11    | X12    | X13    |
|                  | BUSY signal             | OFF          | QD75 is not operating                         | XC     | XD     | XE     | XF     |
|                  | Error detection signal  | OFF          | There is no error                             | X8     | X9     | XA     | XB     |
| External signal  | M code ON signal        | OFF          | M code ON signal is OFF                       | X4     | X5     | X6     | X7     |
|                  | Drive unit READY signal | ON           | Drive unit preparation completed              | —      |        |        |        |
|                  | Stop signal             | OFF          | Stop signal is OFF                            | —      |        |        |        |
|                  | Upper limit (FLS)       | ON           | Within limit range                            | —      |        |        |        |
|                  | Lower limit (RLS)       | ON           | Within limit range                            | —      |        |        |        |

\* If the CPU module is set to the asynchronous mode in the synchronization setting, this must be inserted in the program for interlocking. If it is set to the synchronous mode, it must not be inserted in the program for interlocking because it is turned ON when the CPU module executes calculation.

■ Start time chart

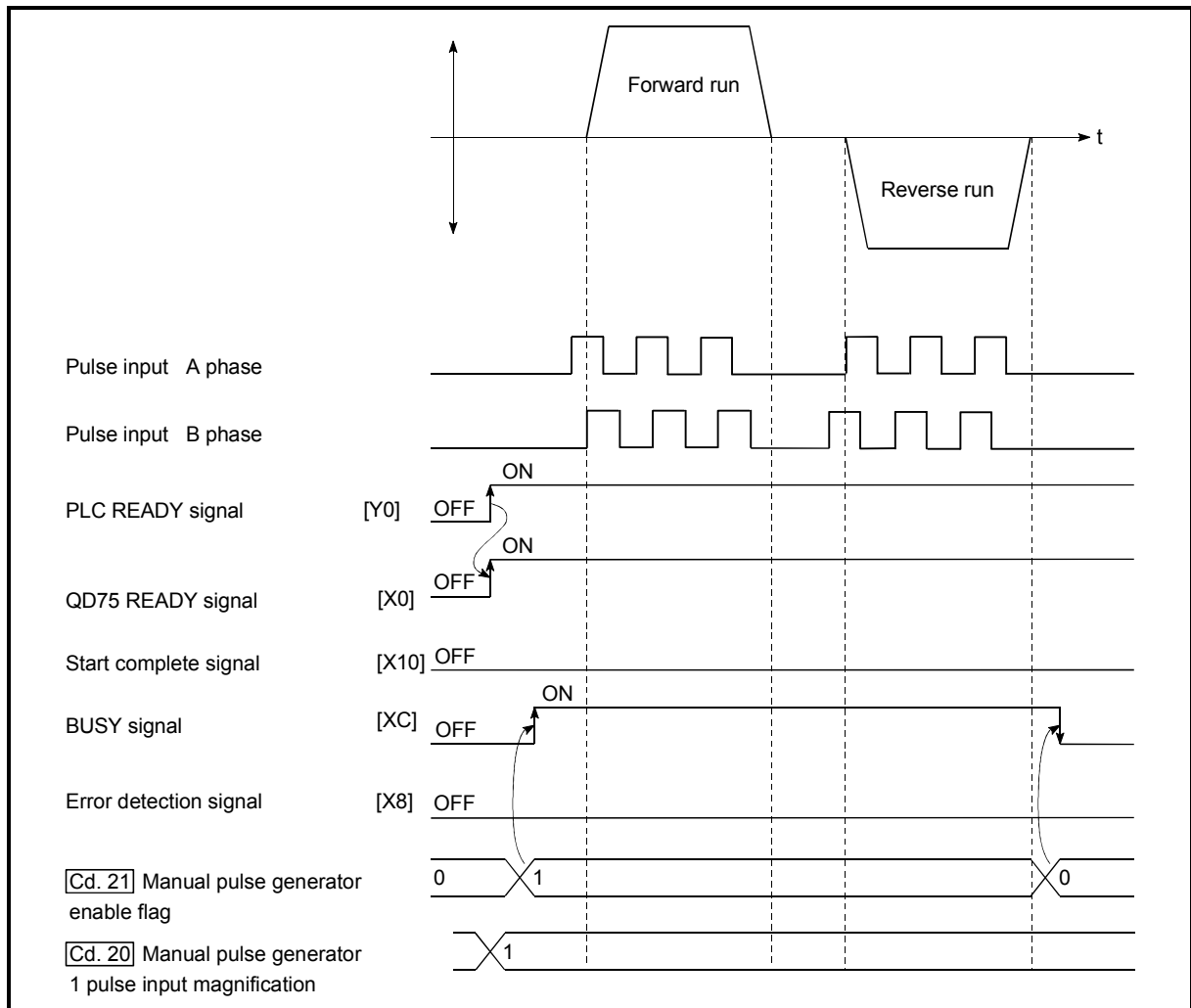
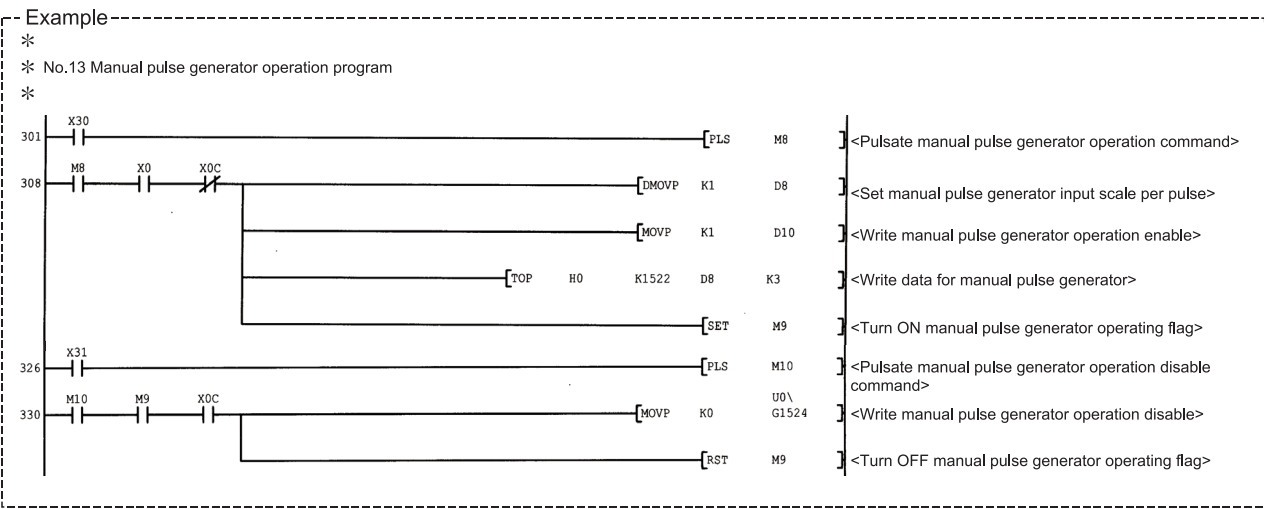


Fig. 11.18 Manual pulse generator operation start time chart

■ Creating the program





# CHAPTER 12 CONTROL SUB FUNCTIONS

The details and usage of the "sub functions" added and used in combination with the main functions are explained in this chapter.

A variety of sub functions are available, including functions specifically for machine OPR and generally related functions such as control compensation, etc. More appropriate, finer control can be carried out by using these sub functions. Each sub function is used together with a main function by creating matching parameter settings and sequence programs. Read the execution procedures and settings for each sub function, and set as required.

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## 12.1 Outline of sub functions

"Sub functions" are functions that compensate, limit, add functions, etc., to the control when the main functions are executed. These sub functions are executed by parameter settings, commands from GX Configurator-QP, sub function sequence programs, etc.

## 12.1.1 Outline of sub functions

The following table shows the types of sub functions available.

| Sub function                               |                                                | Details                                                                                                                                                                                                                                                                                   |
|--------------------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sub functions specifically for machine OPR | OPR retry function                             | This function retries the machine OPR with the upper/lower limit switches during machine OPR. This allows machine OPR to be carried out even if the axis is not returned to before the near-point dog with JOG operation, etc.                                                            |
|                                            | OP shift function                              | After returning to the machine OP, this function offsets the position by the designated distance from the machine OP position and sets that position as the OP address.                                                                                                                   |
| Functions for compensating the control     | Backlash compensation function                 | This function compensates the mechanical backlash. Feed pulses equivalent to the set backlash amount are output each time the movement direction changes.                                                                                                                                 |
|                                            | Electronic gear function                       | By setting the movement amount per pulse, this function can freely change the machine movement amount per commanded pulse.<br>A flexible positioning system that matches the machine system can be structured with this function.                                                         |
|                                            | Near pass function * 1                         | This function suppresses the machine vibration when the positioning data is switched during continuous path control in the interpolation control.                                                                                                                                         |
|                                            | Output timing selection of near pass control   | This function allows the user to select the timing to output the difference ( $\Delta \ell$ ) between the actual and the set positioning end addresses in continuous path control, in which the difference ( $\Delta \ell$ ) is output during the execution of the next positioning data. |
| Functions to limit the control             | Speed limit function                           | If the command speed exceeds " [Pr.8] Speed limit value" during control, this function limits the commanded speed to within the " [Pr.8] Speed limit value" setting range.                                                                                                                |
|                                            | Torque limit function * 2                      | If the torque generated by the servomotor exceeds " [Pr.17] Torque limit setting value" during control, this function limits the generated torque to within the " [Pr.17] Torque limit setting value" setting range.                                                                      |
|                                            | Software stroke limit function                 | If a command outside of the upper/lower limit stroke limit setting range, set in the parameters, is issued, this function will not execute positioning for that command.                                                                                                                  |
|                                            | Hardware stroke limit function                 | This function carries out deceleration stop with the limit switch connected to the QD75 external device connector.                                                                                                                                                                        |
| Functions to change the control details    | Speed change function                          | This function changes the speed during positioning.<br>Set the changed speed in the speed change buffer memory ( [Cd.14] New speed value), and change the speed with the speed change request ( [Cd.15] Speed change request).                                                            |
|                                            | Override function                              | This function changes the speed within a percentage of 1 to 300% during positioning. This is executed using " [Cd.13] Positioning operation speed override".                                                                                                                              |
|                                            | Acceleration/deceleration time change function | This function changes the acceleration/deceleration time during speed change. (function added to the speed change function and override function)                                                                                                                                         |
|                                            | Torque change function                         | This function changes the "torque limit value" during control.                                                                                                                                                                                                                            |
|                                            | Target position change function                | This function changes the target position during the execution of positioning. At the same time, this also can change the speed.                                                                                                                                                          |

\* 1 The near pass function is validated only when the machine of the standard specification carries out the position control with the continuous path control mode.

It cannot be invalidated with parameters.

\* 2 To carry out "torque limit", the "D/A conversion module" and a "drive unit capable of the torque limit command with an analog voltage" must be prepared.

| Sub function                               |                                                        | Details                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute position restoration function * 3 |                                                        | This function restores the absolute position of designated axis. By this function, the OPR after power ON from OFF is not required once the OPR is executed when the system operation is started.                                                                                                                                     |
| Other functions                            | Step function                                          | This function temporarily stops the operation to confirm the positioning operation during debugging, etc.<br>The operation can be stopped at each "automatic deceleration" or "positioning data".                                                                                                                                     |
|                                            | Skip function                                          | This function stops the positioning being executed (decelerates to a stop) when the skip signal is input, and carries out the next positioning.                                                                                                                                                                                       |
|                                            | M code output function                                 | This function issues a sub work (clamp or drill stop, tool change, etc.) according to the code No. (0 to 65535) set for each positioning data.                                                                                                                                                                                        |
|                                            | Teaching function                                      | This function stores the address positioned with manual control into the positioning address ([Da.6] Positioning address/movement amount) having the designated positioning data No.                                                                                                                                                  |
|                                            | Command in-position function                           | At each automatic deceleration, this function calculates the remaining distance for the QD75 to reach the positioning stop position, and when the value is less than the set value, sets the "command in-position flag" to 1. When using another sub work before ending the control, use this function as a trigger for the sub work. |
|                                            | Acceleration/deceleration processing function          | This function adjusts the control acceleration/deceleration. (acceleration/deceleration time and curve).                                                                                                                                                                                                                              |
|                                            | Pre-reading start function                             | This function shortens the virtual start time.                                                                                                                                                                                                                                                                                        |
|                                            | Deceleration start flag function                       | Function that turns ON the flag when the constant speed status or acceleration status switches to the deceleration status during position control, whose operation pattern is "Positioning complete", to make the stop timing known.                                                                                                  |
|                                            | Stop command processing for deceleration stop function | Function that selects a deceleration curve when a stop cause occurs during deceleration stop processing to speed 0.                                                                                                                                                                                                                   |

\* 3 To execute the "absolute position restoration function", the following are required: "16-point input module"; "16-point output module"; and "drive unit capable of configuring an absolute position detection system, which is a Mitsubishi Electric General-Purpose AC Servo having an absolute position detection function (absolute position data transference protocol) equivalent to that of MR-J3-□A".

## 12.2 Sub functions specifically for machine OPR

The sub functions specifically for machine OPR include the "OPR retry function" and "OP shift function". Each function is executed by parameter setting.

### 12.2.1 OPR retry function

When the workpiece goes past the OP without stopping during positioning control, it may not move back in the direction of the OP although a machine OPR is commanded, depending on the workpiece position. This normally means the workpiece has to be moved to a position before the near-point dog by a JOG operation, etc., to start the machine OPR again. However, by using the OPR retry function, a machine OPR can be carried out regardless of the workpiece position.

The details shown below explain about the "OPR retry function".

- [1] Control details
- [2] Control Precautions
- [3] Setting method

#### [1] Control details

The following drawing shows the operation of the OPR retry function.

- (1) OPR retry point return retry operation when the workpiece is within the range between the upper/lower limits.

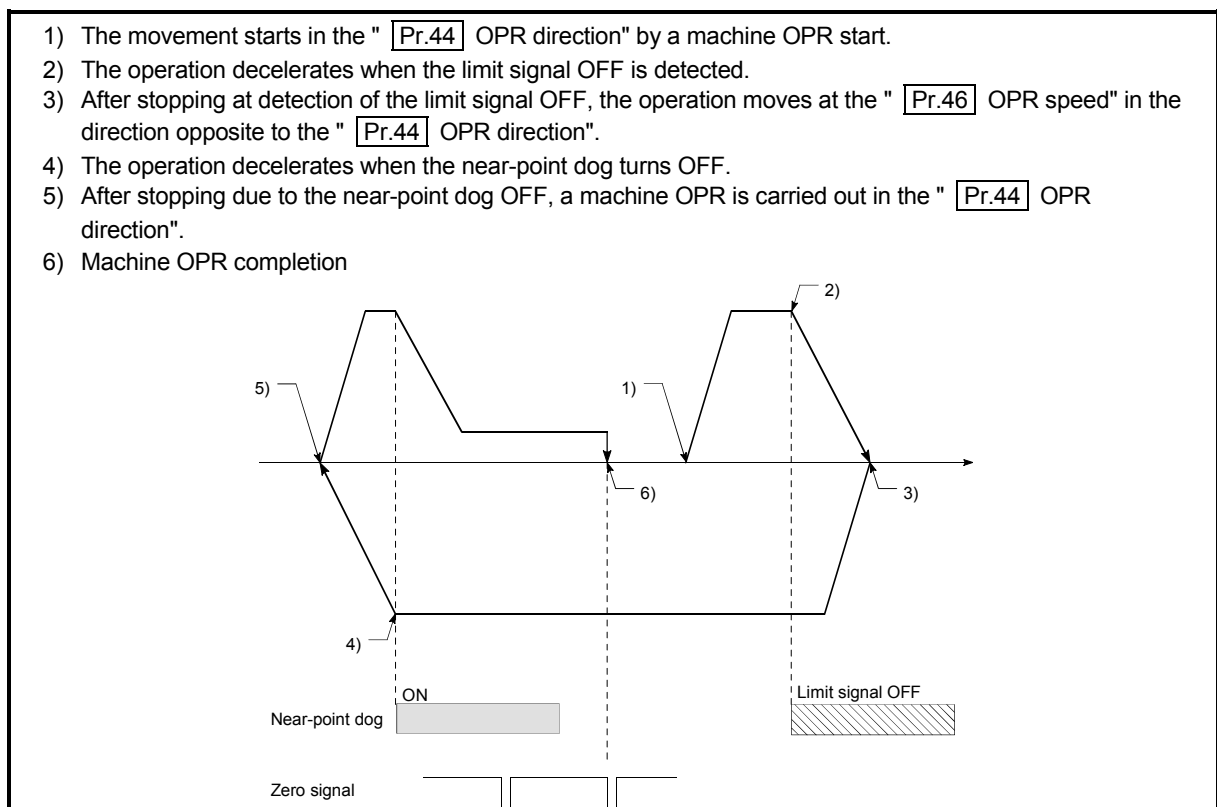
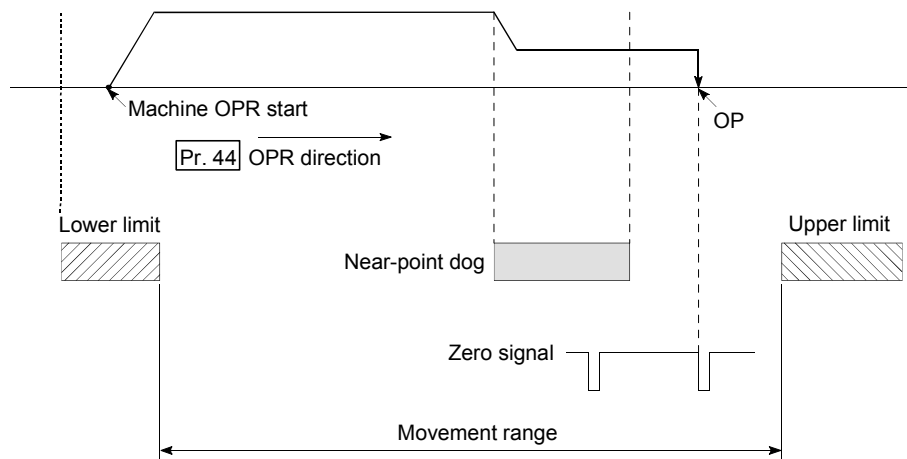


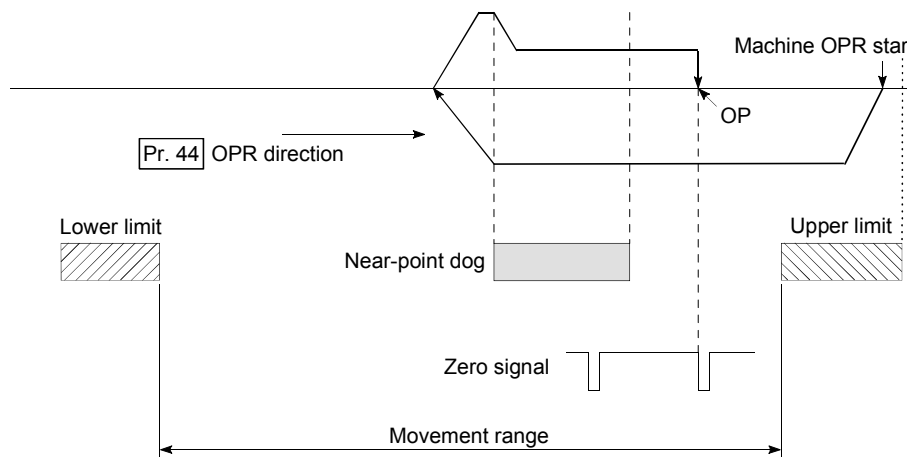
Fig. 12.1 OPR retry operation by limit signal detection

- (2) OPR retry operation when the workpiece is outside the range between the upper/lower limits.

(a) When the direction from the workpiece to the OP is the same as the "Pr.44 OPR direction", a normal machine OPR is carried out.



(b) When the direction from the workpiece to the OP is the opposite direction from the "Pr.44 OPR direction", the operation carries out a deceleration stop when the near-point dog turns OFF, and then carries out a machine OPR in the direction set in "Pr.44 OPR direction".



Note) The above figures in (a) and (b) are examples where "Pr.44 OPR direction" is set to "0: Positive direction". When "0: Positive direction" is set in Pr.44, check that the limit switch placed in the OPR direction acts as the upper limit. When "1: Negative direction" is set in Pr.44, check that the limit switch placed in the OPR direction acts as the lower limit. Incorrect wiring of these limit switches may cause improper OPR retry operation. If any malfunction is identified, check and correct "Pr.6 Rotation direction setting" and the wiring.

Fig. 12.2 OPR retry operation from on limit (limit switch OFF)

## (3) Setting the dwell time during an OPR retry

The OPR retry function can perform such function as the dwell time using "Pr.57 Dwell time at OPR retry" when the reverse run operation is carried out due to detection by the limit signal for upper/lower limits and when the machine OPR is executed after the near point dog is turned OFF to stop the operation.

"Pr.57 Dwell time during OPR" is validated when the operation stops at the "A" and "B" positions in the following drawing. (The dwell time is the same value at both positions "A" and "B".)

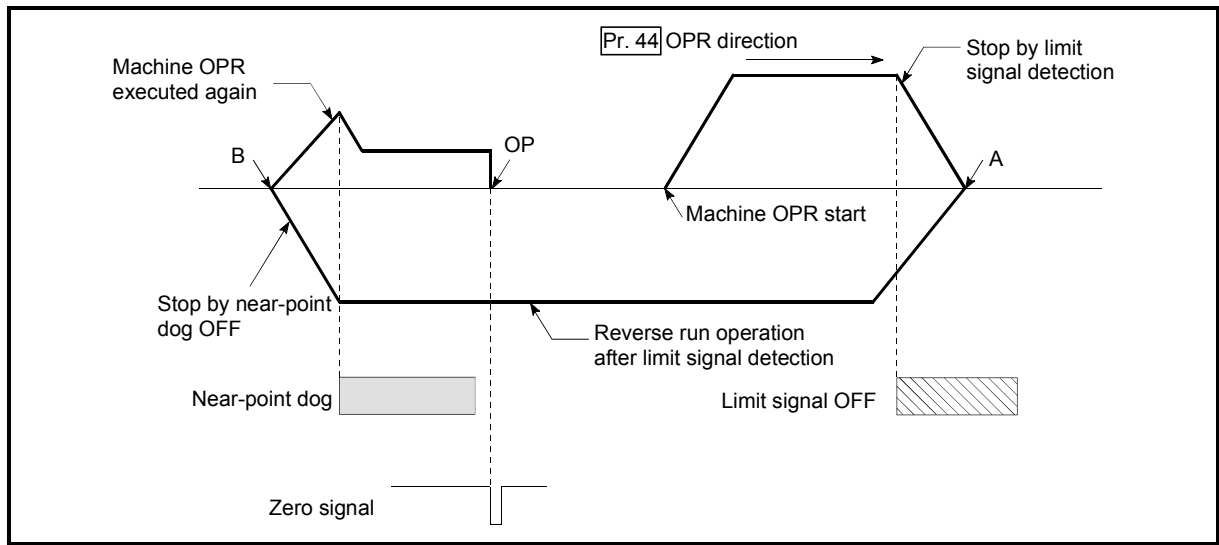


Fig. 12.3 Setting the dwell time during an OPR retry

## [2] Control Precautions

- (1) The following table shows whether the OPR retry function may be executed by the " **Pr.43** OPR method".

| <b>Pr.43</b> OPR method | Execution status of OPR retry function |
|-------------------------|----------------------------------------|
| Near-point dog method   | ○ : Execution possible                 |
| Stopper method 1)       | ○ : Execution possible * 1             |
| Stopper method 2)       | ○ : Execution possible * 1             |
| Stopper method 3)       | × : Execution not possible             |
| Count method 1)         | ○ : Execution possible                 |
| Count method 2)         | ○ : Execution possible                 |

\* 1 Starting can be executed from the position of the limit switch installed in the opposition direction of " **Pr.44** OPR direction". (The limit signal is OFF.) However, the retry operation by the limit switch in the OPR direction cannot be executed since a stopper is installed in the OPR direction.

- (2) Always establish upper/lower limit switches at the upper/lower limit positions of the machine, and connect them to the QD75. If the OPR retry function is used without hardware stroke limit switches, the motor will continue rotation until a hardware stroke limit signal is detected.
- (3) Do not configure a system so that the drive unit power turns OFF by the upper/lower limit switches connected to the QD75. If the drive unit power is turned OFF, the OPR retry cannot be carried out.

## [3] Setting method

To use the "OPR retry function", set the required details in the parameters shown in the following table, and write them to the QD75.

When the parameters are set, the OPR retry function will be added to the machine OPR control. The set details are validated at the rising edge (OFF → ON) of the PLC READY signal (Y0). (Set " **Pr.57** Dwell time during OPR retry" according to the user's requirements.)

| Setting item                             | Setting value | Setting details                                                                          | Factory-set initial value |
|------------------------------------------|---------------|------------------------------------------------------------------------------------------|---------------------------|
| <b>Pr.48</b> OPR retry                   | 1             | Set "1: Carry out OPR retry by limit switch".                                            | 0                         |
| <b>Pr.57</b> Dwell time during OPR retry | →             | Set the deceleration stop time during OPR retry. (Random value between 0 and 65535 (ms)) | 0                         |

\* Refer to Section 5.2 "List of parameters" for setting details.

**REMARK**

- Parameters are set for each axis.
- It is recommended that the parameters be set whenever possible with GX Configurator-QP. Execution by sequence program uses many sequence programs and devices. The execution becomes complicated, and the scan times will increase.

## 12.2.2 OP shift function

When a machine OPR is carried out, the OP is normally established using the near-point dog, stopper, and zero signal. However, by using the OP shift function, the machine can be moved a designated movement amount from the position where the zero signal was detected. A mechanically established OP can then be interpreted at that point.

The OP shift function can be used without relation to " **Pr.43** OPR method".

The details shown below explain about the "OP shift function".

- [1] Control details
- [2] Setting range for the OP shift amount
- [3] Movement speed during OP shift
- [4] Control Precautions
- [5] Setting method

## [1] Control details

The following drawing shows the operation of the OP shift function.

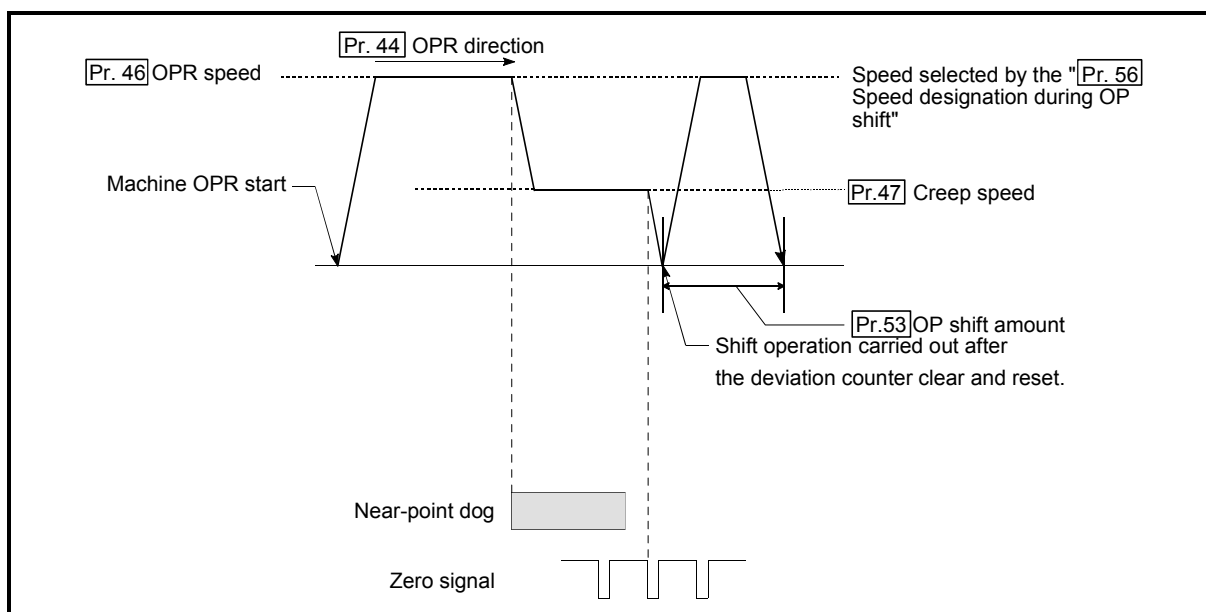


Fig. 12.4 OP shift operation



### [2] Setting range for the OP shift amount

Set the OP shift amount within the range from the detected zero signal to the upper/lower limit switches.

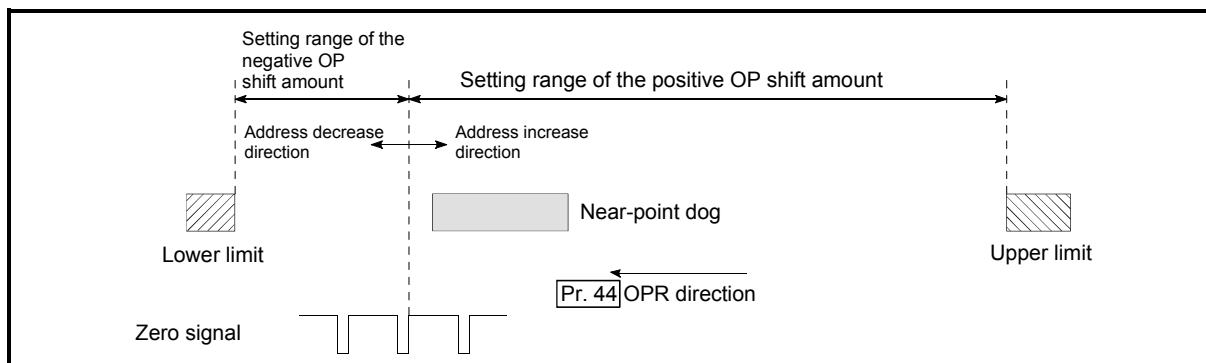


Fig. 12.5 Setting range for the OP shift amount

### [3] Movement speed during OP shift

When using the OP shift function, the movement speed during the OP shift is set in "Pr.56 Speed designation during OP shift". The movement speed during the OP shift is selected from either the "Pr.46 OPR speed" or the "Pr.47 Creep speed".

The following drawings show the movement speed during the OP shift when a mechanical OPR is carried out by the near-point dog method.

(1) OP shift operation at the "Pr.46 OPR speed"

(When "Pr.56 Speed designation during OP shift" is 0)

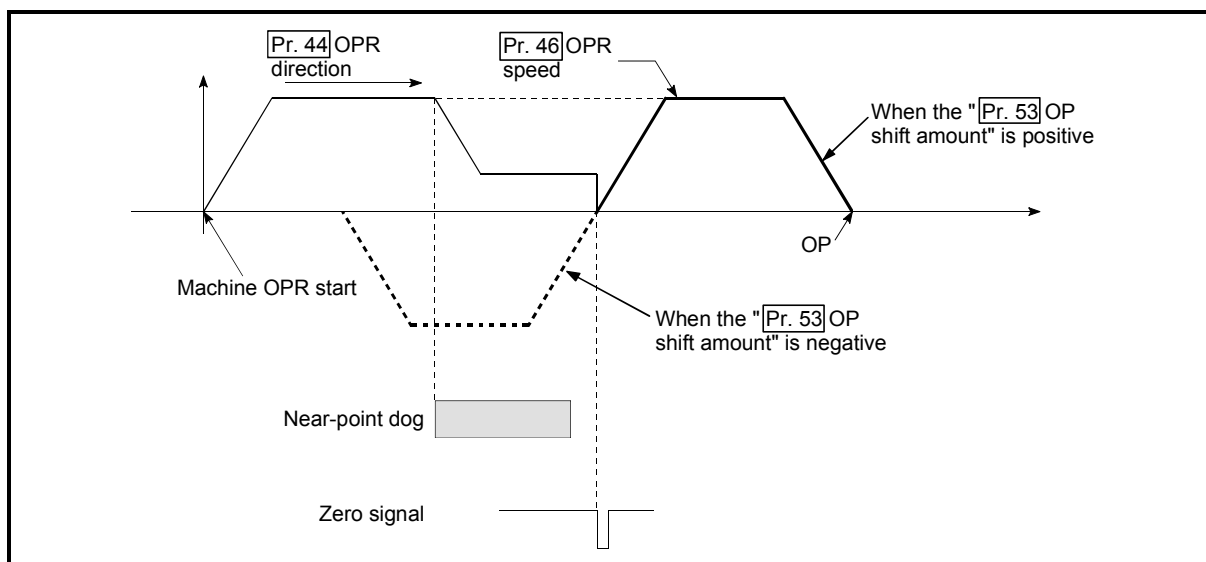


Fig. 12.6 OP shift operation at the OPR speed

- (2) OP shift operation at the " **Pr.47** Creep speed"  
(When " **Pr.56** Speed designation during OP shift" is 1)

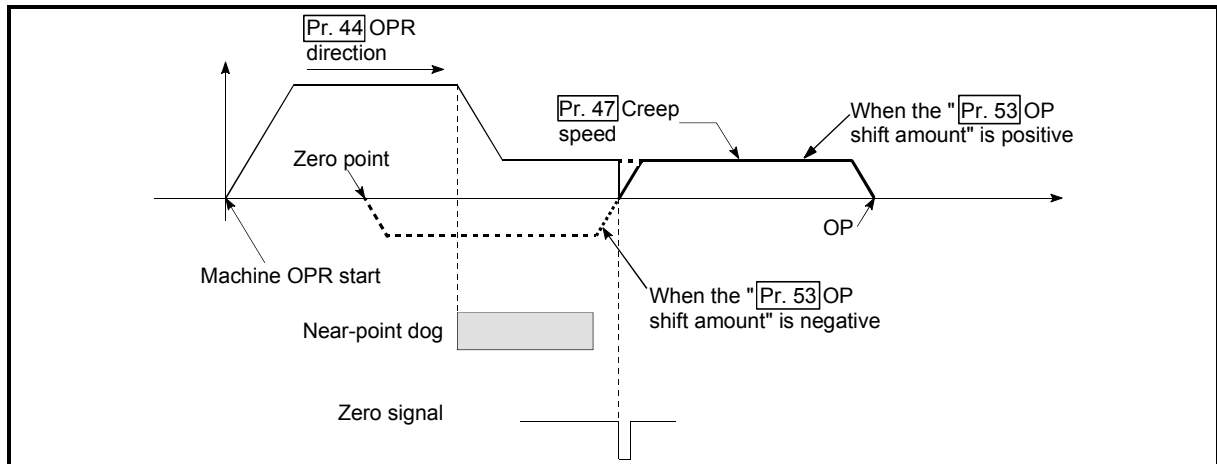


Fig. 12.7 OP shift operation at the creep speed

#### [4] Control Precautions

- (1) The following data are set after the OP shift amount is complete.
  - OPR complete flag (" **Md.31** Status: b4)
  - **Md.20** Current feed value
  - **Md.21** Machine feed value
  - **Md.26** Axis operation status
 OPR request flag (**Md.31** Status: b3) is reset after completion of the OP shift.
- (2) " **Pr.53** OP shift amount" is not added to " **Md.34** Movement amount after near-point dog ON". The movement amount immediately before the OP shift operation, considering near-point dog ON as "0", is stored. For the stopper method (1) 2) 3)), the movement amount is not changed from "0".
- (3) When using the OP shift function with the stopper method (1) 2) 3)) selected for the OPR method, configure the OP shift operation in the opposite direction of the OPR direction.  
It cannot shift to the OPR direction due to a mechanical stopper in the OPR direction.

#### [5] Setting method

To use the "OP shift function", set the required details in the parameters shown in the following table, and write them to the QD75.

When the parameters are set, the OP shift function will be added to the machine OPR control. The set details are validated at the rising edge (OFF → ON) of the PLC READY signal (Y0).

| Setting item                                   | Setting value | Setting details                                                                                  | Factory-set initial value |
|------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------|---------------------------|
| <b>Pr.53</b> OP shift amount                   | →             | Set the shift amount during the OP shift.                                                        | 0                         |
| <b>Pr.56</b> Speed designation during OP shift | →             | Select the speed during the OP shift<br>0: <b>Pr.46</b> OPR speed<br>1: <b>Pr.47</b> Creep speed | 0                         |

\* Refer to Section 5.2 "List of parameters" for setting details.

|        |
|--------|
| REMARK |
|--------|

- Parameters are set for each axis.
- It is recommended that the parameters be set whenever possible with GX Configurator-QP. Execution by sequence program uses many sequence programs and devices. The execution becomes complicated, and the scan times will increase.

### 12.3 Functions for compensating the control

The sub functions for compensating the control include the "backlash compensation function", "electronic gear function", "near pass function", and "output timing selection of near pass control". Each function is executed by parameter setting or sequence program creation and writing.

#### 12.3.1 Backlash compensation function

The "backlash compensation function" compensates the backlash amount in the mechanical system. When the backlash compensation amount is set, an extra amount of pulses equivalent to the set backlash amount is output every time the movement direction changes.

The details shown below explain about the "backlash compensation function".

- [1] Control details
- [2] Control Precautions
- [3] Setting method

##### [1] Control details

The following drawing shows the operation of the backlash compensation function.

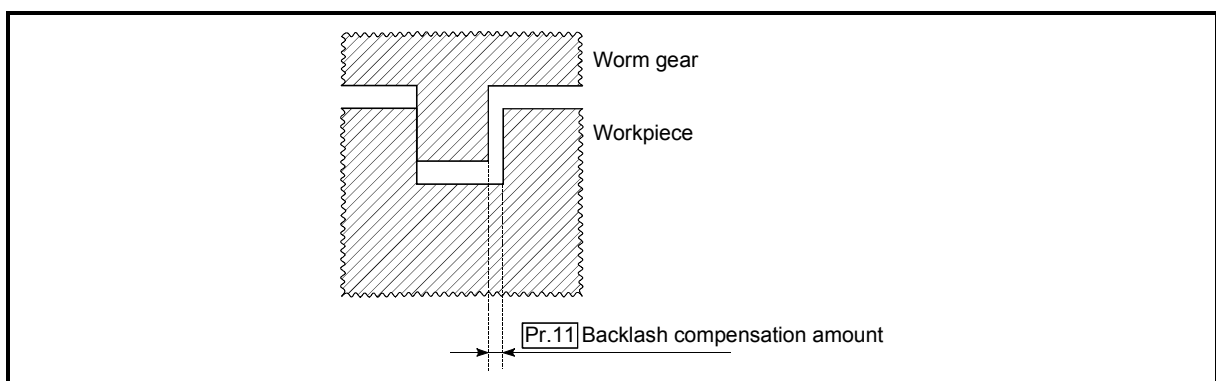


Fig. 12.8 Backlash compensation amount

## [2] Control Precautions

- (1) The feed pulses of the backlash compensation amount are not added to the " [Md.20] Current feed value" or " [Md.21] Machine feed value".
- (2) Always carry out a machine OPR before starting the control when using the backlash compensation function (when " [Pr.11] Backlash compensation amount" is set). The backlash in the mechanical system cannot be correctly compensated if a machine OPR is not carried out.
- (3) Set the No. of pulses output in one backlash compensation (value in which the " [Pr.11] Backlash compensation amount" is divided by the "movement amount per pulse") to a value of 255 or lower. An error "Backlash compensation amount error" (error code: 920) will occur if a value over 255 is set. (Depending on the connected servo, tracking may not be possible if a large amount of pulses is output at once.)

$$0 \leq \frac{\text{Backlash compensation amount}}{\text{Movement amount per pulse}} \leq 255$$

(Omit values after the decimal point.)

- (4) Backlash compensation, which includes the movement amount and " [Pr.11] Backlash compensation amount", is output the moment at the moving direction changes.
- (5) Backlash compensation cannot be made when the stepping motor is used. The backlash compensation function cannot be used for an axis where a stepping motor is connected. Set 0 (initial value) in " [Pr.11] Backlash compensation amount".

## [3] Setting method

To use the "backlash compensation function", set the "backlash compensation amount" in the parameter shown in the following table, and write it to the QD75. The set details are validated at the rising edge (OFF → ON) of the PLC READY signal (Y0).

| Setting item |                              | Setting value | Setting details                       | Factory-set initial value |
|--------------|------------------------------|---------------|---------------------------------------|---------------------------|
| [Pr.11]      | Backlash compensation amount | →             | Set the backlash compensation amount. | 0                         |

\* Refer to Section 5.2 "List of parameters" for setting details.

## REMARK

- Parameters are set for each axis.
- It is recommended that the parameters be set whenever possible with GX Configurator-QP. Execution by sequence program uses many sequence programs and devices. The execution becomes complicated, and the scan times will increase.

### 12.3.2 Electronic gear function

The "electronic gear function" adjusts the pulses calculated and output according to the parameters set in the QD75 with the actual machine movement amount.

The "electronic gear function" has the following four functions.

- [A] The function converts the command value (speed, movement amount from the start point to the end point), which is set in mm units, to pulse units, and determines the pulse frequency and pulse number of the command pulse.

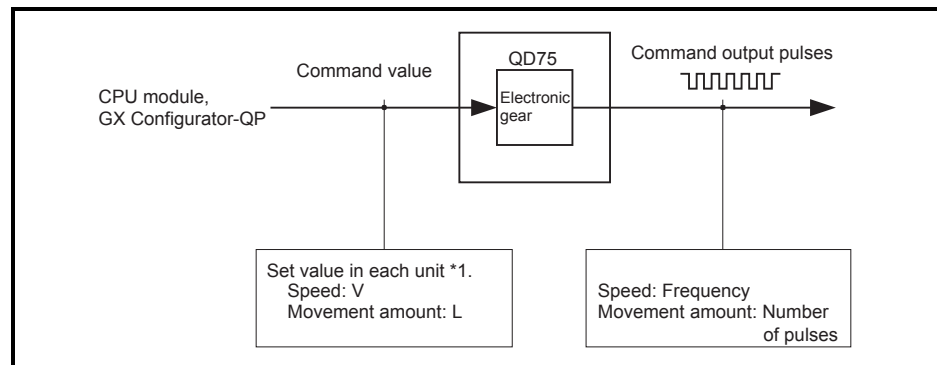


Fig. 12.9 Conversion to pulse units

\* 1: Unit specified with " Pr.1 Unit setting" (mm, inch, degree, pulse)

- [B] If values less than one pulse are generated by converting the movement amount from the start point to the end point to pulse units, the values are not output and the machine stops at the front side of the positioning direction. The function accumulates values less than one pulse that could not be pulse output in the QD75 and outputs one pulse when the total cumulative value reached one pulse or more.
- [C] When machine OPR is completed, current value changing is completed, speed control is started (except when current feed value change is present), or fixed-feed control is started, the function clears to "0" the cumulative values of less than one pulse which could not be output. (If the cumulative value is cleared, an error will occur by a cleared amount in the feed machine value. Control can be constantly carried out at the same machine movement amount, even when the fixed-feed control is continued.)
- [D] The function compensates the mechanical system error of the command movement amount and actual movement amount by adjusting the "movement amount per pulse".  
 (The "movement amount per pulse" value is defined by " Pr.2 No. of pulses per rotation (Ap)", " Pr.3 Movement amount per rotation (Al)", and " Pr.4 Unit magnification (Am)".)

The QD75 automatically carries out the processing for [A] and [C].

The details shown below explain about the "electronic gear function", including the method for compensating the error in [D] above, etc.

- [1] Movement amount per pulse
- [2] Error compensation method
- [3] Control Precautions

### [1] Movement amount per pulse

" [Pr.2] No. of pulses per rotation (Ap)", " [Pr.3] Movement amount per rotation (Al)", and " [Pr.4] Unit magnification (Am)" are the items for determining how many rotations (equivalent to how many pulses) a motor should operate to move a machine for movement amount set in a program.

The drive unit controls the motor with the pulse number.

The following shows the details of control of the QD75.

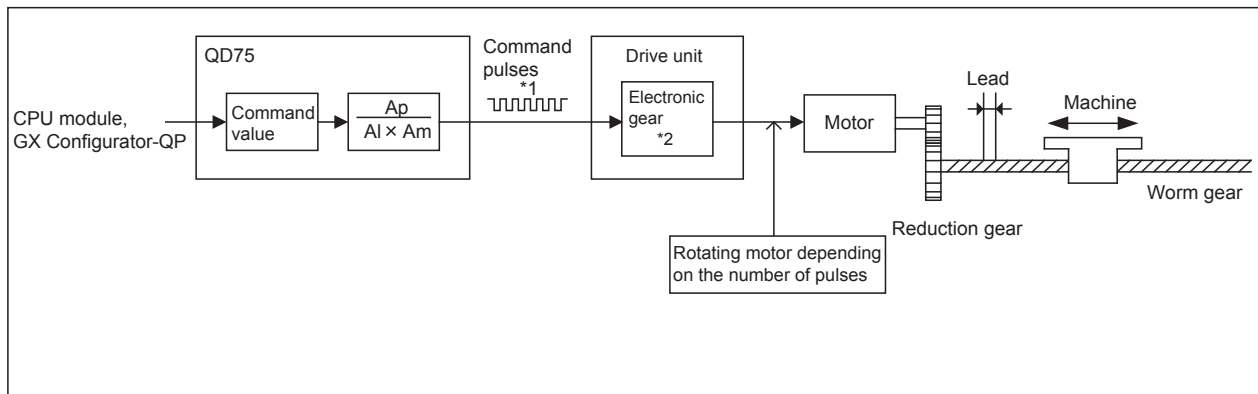


Fig. 12.10 System using a ball screw

\*1: The upper limits of the command pulse frequency are as follows

- QD75D□N : 4Mpulse/s
- QD75D□ : 1Mpulse/s
- QD75P□N/QD75P□ : 200kpulse/s

\*2: For a drive unit without electronic gear function, or when not using electronic gear function, this value is 1 (single).

Consider a system with the motor connected to a ball screw, such as the figure above.

1 is set to the electronic gear of the drive unit.

The movement amount of the machine is in mm or inch unit. Set the command value in mm or inch units to the QD75 for the CPU module program.

The motor is controlled by the drive unit in pulse number units.

Consequently, since the command value in mm or inch units is converted to pulse units, set Ap, Al, and Am so that the following relational expressions are satisfied.

$$\begin{aligned} \text{Number of pulses per rotation of motor} &= A_p \\ \text{Movement amount per rotation of motor} &= A_l \times A_m \end{aligned}$$

In this case, the machine movement amount for the command 1 pulse output from the QD75 is calculated by the following expression.

$$\text{Movement amount per pulse (A)} = \frac{A_l \times A_m}{A_p}$$

| POINT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The command frequency from the QD75 is limited by a ceiling. If the command frequency exceeds the upper limit, increase the "Movement amount per pulse (A)" greater (N times) to decrease the command frequency.</p> <p>In this case, the electronic gear on the drive unit must be increased by N times as well.</p> <p>The command pulse from the QD75 is changes to 1/N times. Therefore, multiply it by N on the drive unit side to keep the number of motor rotations.</p> <p>Since the "movement amount per pulse (A)" is increased, the position accuracy (command resolution) for the command 1 pulse from the QD75 decreases.</p> <p>Consider to decrease the command speed when the position accuracy is required.</p> |

### (1) Setting range of Ap, Al, Am

Determined setting ranges are available for Ap, Al, and Am. The following shows the setting ranges.

| Setting item |                                   | Setting range |                                |
|--------------|-----------------------------------|---------------|--------------------------------|
| Pr.2         | No. of pulses per rotation (Ap)   | 1 to 65535    |                                |
| Pr.3         | Movement amount per rotation (Al) | 1 to 65535    | $\times 10^{-1} \mu\text{m}$   |
|              |                                   |               | $\times 10^{-5} \text{inch}$   |
|              |                                   |               | $\times 10^{-5} \text{degree}$ |
|              |                                   |               | pulse                          |
| Pr.4         | Unit magnification (Am)           | 1/10/100/1000 |                                |

For details of the setting, refer to Section 5.2 "List of parameters". In addition, use the value set in " Pr.3 Movement amount per rotation (Al)" as movement amount per rotation (Al) to calculate movement amount per pulse (A).

Example: Movement amount per pulse (A) when " Pr.1 Unit setting" is "0: mm"

" Pr.2 No. of pulses per rotation": 20000

" Pr.3 Movement amount per rotation": 40000

" Pr.4 Unit magnification": 1

$$\text{Movement amount per pulse (A)} = \frac{40000 \times 10^{-1}}{20000} \times 1 = 0.2 \mu\text{m}$$

If the value is beyond the setting range, setting values of each parameter can be decreased by canceling the numerator and denominator with the movement amount per pulse (A) kept.



## [2] Error compensation method

When position control is carried out by the "movement amount per pulse" set in the QD75 parameters, an error sometimes occurs between the command movement amount (L) and the actual movement amount (L').

That error is compensated in the QD75 by adjusting the values in " [Pr.2] No. of pulses per rotation (Ap)", " [Pr.3] Movement amount per rotation (Al)", and " [Pr.4] Unit magnification (Am)". (When " [Pr.1] Unit setting" is "0: mm")

### (1) Definition

The "error compensation amount" used to carry out the error compensation is defined as follows.

$$\text{Error compensation amount} = \frac{\text{Actual movement amount (L')}}{\text{Designated movement amount (L)}}$$

The QD75 "movement amount per pulse" is calculated with the following equation.

Movement amount per pulse is "A", " [Pr.2] No. of pulses per rotation" is (Ap), " [Pr.3] Movement amount per rotation" is (Al), and " [Pr.4] Unit magnification" is (Am).

$$A = \frac{Al}{Ap} \times Am$$

### (2) Procedure

- (a) Set the "command movement amount (L)", and carry out positioning. (Set the "movement amount per pulse (A)" according to Section 5.2 "List of parameters".)
- (b) After positioning, measure the "actual movement amount (L)'".
- (c) Calculate the "error compensation amount".

$$\text{Error compensation amount} = \frac{L'}{L}$$

- (d) Calculate the post-compensation " [Pr.2] No. of pulses per rotation (Ap')", " [Pr.3] Movement amount per rotation (Al')", and " [Pr.4] Unit magnification (Am')" from the "post-compensation movement amount per pulse (A)'".

$$\begin{aligned} A &= A \times \text{Error compensation amount} \\ &= \frac{Al}{Ap} \times Am \times \frac{L'}{L} \\ &= \frac{Al'}{Ap'} \times Am' \end{aligned}$$

(Adjust with Am' so that Al' and Ap' do not exceed the setting range.)

|                                                                                                                                       |                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Calculation example                                                                                                                   |                                                                                                |
| (Conditions)                                                                                                                          |                                                                                                |
| Movement amount per pulse                                                                                                             | : 500 (μm/rev)                                                                                 |
| No. of pulses per rotation                                                                                                            | : 12000 (pulse/rev)                                                                            |
| Unit magnification                                                                                                                    | : 1                                                                                            |
| (Positioning results)                                                                                                                 |                                                                                                |
| Command movement amount                                                                                                               | : 100mm                                                                                        |
| Actual movement amount                                                                                                                | : 101mm                                                                                        |
| (Compensation amount)                                                                                                                 |                                                                                                |
| $\frac{AL'}{AP'} = \frac{5 \times 10^3}{12000} \times \frac{101 \times 10^3}{101 \times 10^3} = \frac{5050}{12000} = \frac{101}{240}$ |                                                                                                |
| Movement amount per pulse                                                                                                             | : 101 (μm/rev) [Set in <span style="border: 1px solid black; padding: 0 2px;">Pr.3</span> ]    |
| No. of pulses per rotation                                                                                                            | : 240 (pulse/rev) [Set in <span style="border: 1px solid black; padding: 0 2px;">Pr.2</span> ] |
| Unit magnification                                                                                                                    | : 1 [Set in <span style="border: 1px solid black; padding: 0 2px;">Pr.4</span> ]               |

- (e) Set the post-compensation " Pr.2 No. of pulses per rotation (Ap')", " Pr.3 Movement amount per rotation (Al')", and " Pr.4 Unit magnification (Am')" in the parameters, and write them to the QD75. The set details are validated at the rising edge (OFF → ON) of the PLC READY signal (Y0).

| Setting item                                                                                    | Setting value | Setting details                  | Pre-compensation value |
|-------------------------------------------------------------------------------------------------|---------------|----------------------------------|------------------------|
| <span style="border: 1px solid black; padding: 0 2px;">Pr.2</span> No. of pulses per rotation   | Ap'           | Set the post-compensation value. | Ap                     |
| <span style="border: 1px solid black; padding: 0 2px;">Pr.3</span> Movement amount per rotation | Al'           | Set the post-compensation value. | Al                     |
| <span style="border: 1px solid black; padding: 0 2px;">Pr.4</span> Unit magnification           | Am'           | Set the post-compensation value. | Am                     |

\* Refer to Section 5.2 "List of parameters" for setting details.

### [3] Control Precautions

If values less than one pulse are generated by converting the movement amount from the start point to the end point to pulse units with the electronic gear function, the values are not output and the machine stops at the front side of the positioning direction. The function accumulates values less than one pulse that could not be pulse output in the QD75 and outputs one pulse when the total cumulative value reached one pulse or more.

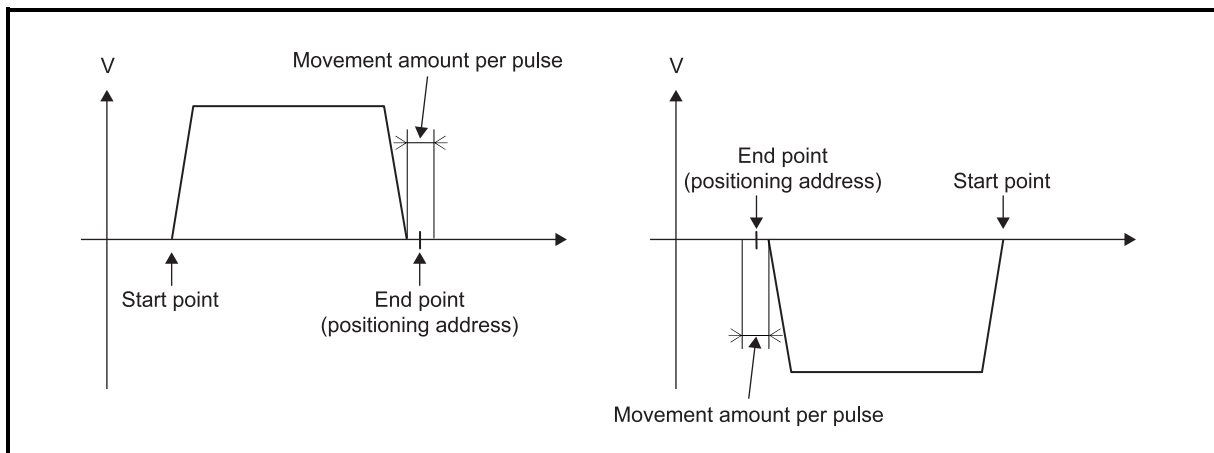


Fig. 12.11 Operation when values less than one pulse are generated

To prevent values less than one pulse from being generated, set the positioning address so that the value, calculated by multiplying the inverse number of movement amount per pulse (A) to the movement amount to the end point, becomes an integer.

It is recommended that the "movement amount per pulse (A)" be set to a value close to "1" for the following reasons. The "movement amount per pulse" of "1" means the minimum value in each " Pr.1 Unit setting". (0.1 [μm] for the unit [mm])

- (1) When the setting of the movement amount per pulse is decreased, the command frequency may increase. Be careful when changing the setting.
- (2) If the setting of the movement amount per pulse is less than 1, the mechanical system may oscillate. Be sure to use the movement amount per pulse among the range indicated below.

$$\text{Movement amount per pulse (A)} \geq \frac{1}{500}$$

If the mechanical system oscillates, also use the electronic gear function of the drive unit and the movement amount per pulse greater.

- (3) Set the movement amount per pulse so that the pulse output frequency for the drive unit becomes a value in the following table.

|                                       | QD75P□N/QD75P□      | QD75D□N           | QD75D□            |
|---------------------------------------|---------------------|-------------------|-------------------|
| Pulse output frequency for drive unit | 200kpulse/s or less | 4Mpulse/s or less | 1Mpulse/s or less |

If the setting of pulse output frequency for the drive unit exceeds a value in the table, the QD75 may not operate correctly.

**REMARK**

In the QD75, the general term for the functions in this section is defined as the "electronic gear function". Refer to the User's Manual for the servomotor for the definition of the "electronic gear" on the servomotor side.

### 12.3.3 Near pass function

When continuous pass control is carried out using interpolation control, the near pass function is carried out.

The "near pass function" is a function to suppress the mechanical vibration occurring at the time of switching the positioning data when continuous pass control is carried out using interpolation control.

#### [Near pass function]

The extra movement amount occurring at the end of each positioning data unit being continuously executed is carried over to the next positioning data unit. Alignment is not carried out, and thus the output speed drops are eliminated, and the mechanical vibration occurring during speed changes can be suppressed. Because alignment is not carried out, the operation is controlled on a path that passes near the position set in " Da.6 Positioning address/movement amount".

The details shown below explain about the "near pass function".

#### [1] Control details

#### [2] Control Precautions

#### [1] Control details

The following drawing shows the path of the continuous path control by the 2-axis linear interpolation control.

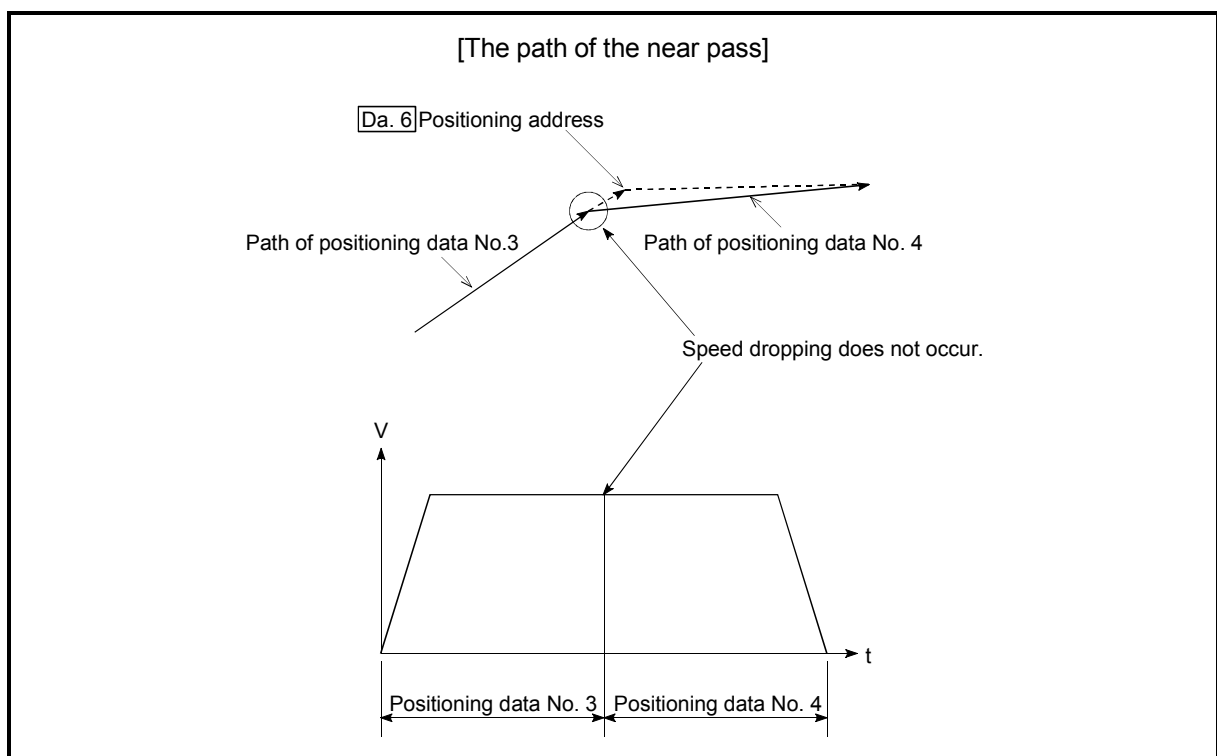


Fig. 12.12 The path of the continuous path control

## [2] Control Precautions

- (1) If the movement amount designated by the positioning data is small when the continuous path control is executed, the output speed may not reach the designated speed.
- (2) The movement direction is not checked during interpolation operation. Therefore, a deceleration stops are not carried out even the movement direction changes. (See below) For this reason, the output will suddenly reverse when the reference axis movement direction changes. To prevent the sudden output reversal, assign not the continuous path control "11", but the continuous positioning control "01" to the positioning data of the passing point.

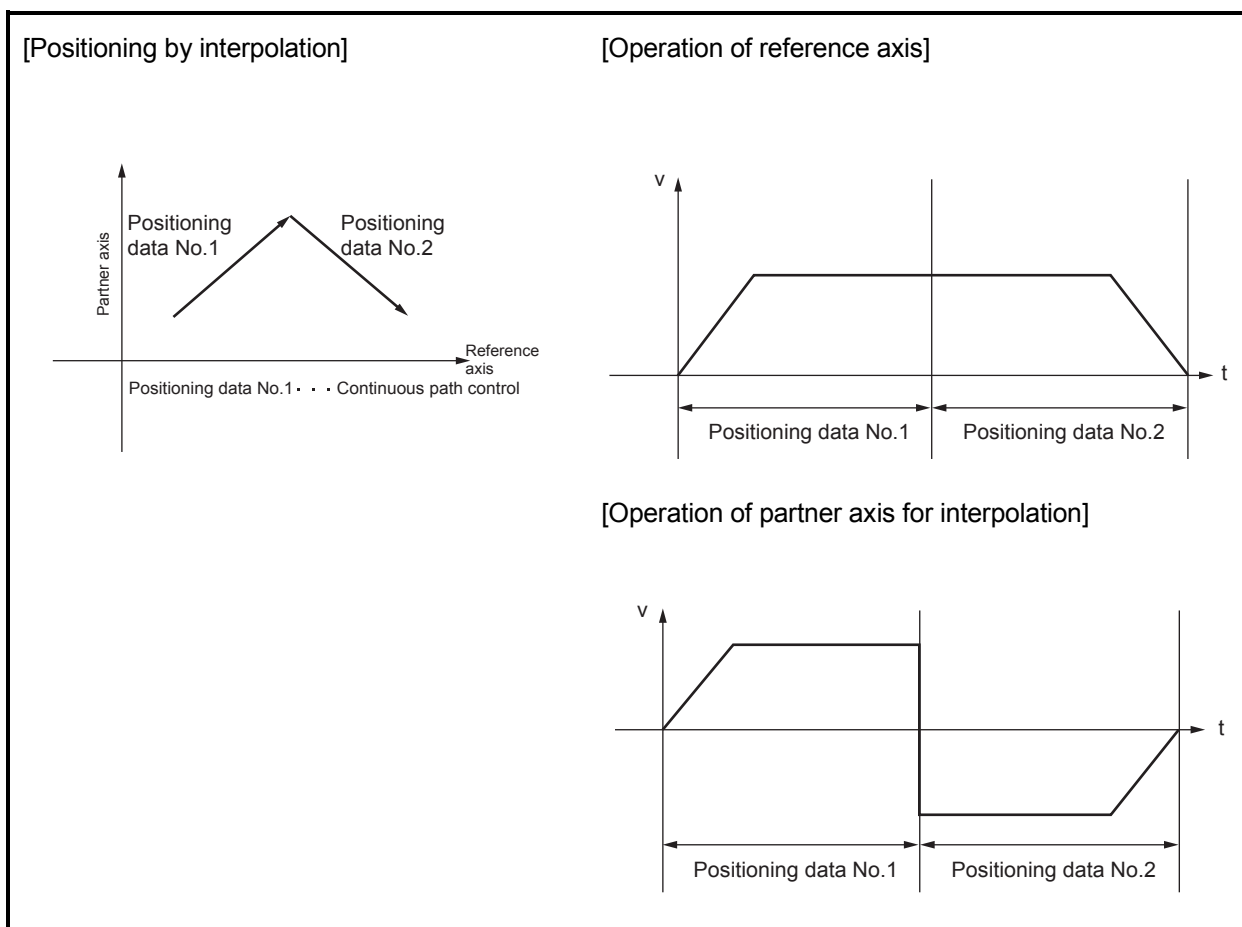


Fig. 12.13 Path and output speed of various axes when movement direction varies during continuous path control

### 12.3.4 Output timing selection of near pass control

The function "output timing selection of near pass control" allows the user to select the timing to output the difference ( $\Delta \ell$ ) between the actual and the set positioning end addresses in continuous path control, in which the difference ( $\Delta \ell$ ) is output during the execution of the next positioning data.

The details shown below explain about the function "output timing selection of near pass control".

- [1] Control details
- [2] Control precautions
- [3] Setting method

#### [1] Control details

There are two options for an output timing of near pass control: "At constant speed" and "At deceleration".

- At constant speed: The distance  $\Delta \ell$  is output when the next positioning data is executed in constant speed.
- At deceleration: The distance  $\Delta \ell$  is output when the next positioning speed decelerates from V1 to V2.

When "At constant speed" is selected in a condition where the specified speed of positioning data No.1 and No.2 in the figure 12.14 are "V1 > V2", the distance  $\Delta \ell$  is output when the next positioning data is executed in constant speed; therefore, the positioning execution time is extended.

When "At deceleration" is selected, the positioning execution time is not extended and remains the same as it was set.

The following drawing shows the operation of the function "output timing selection of near pass control".

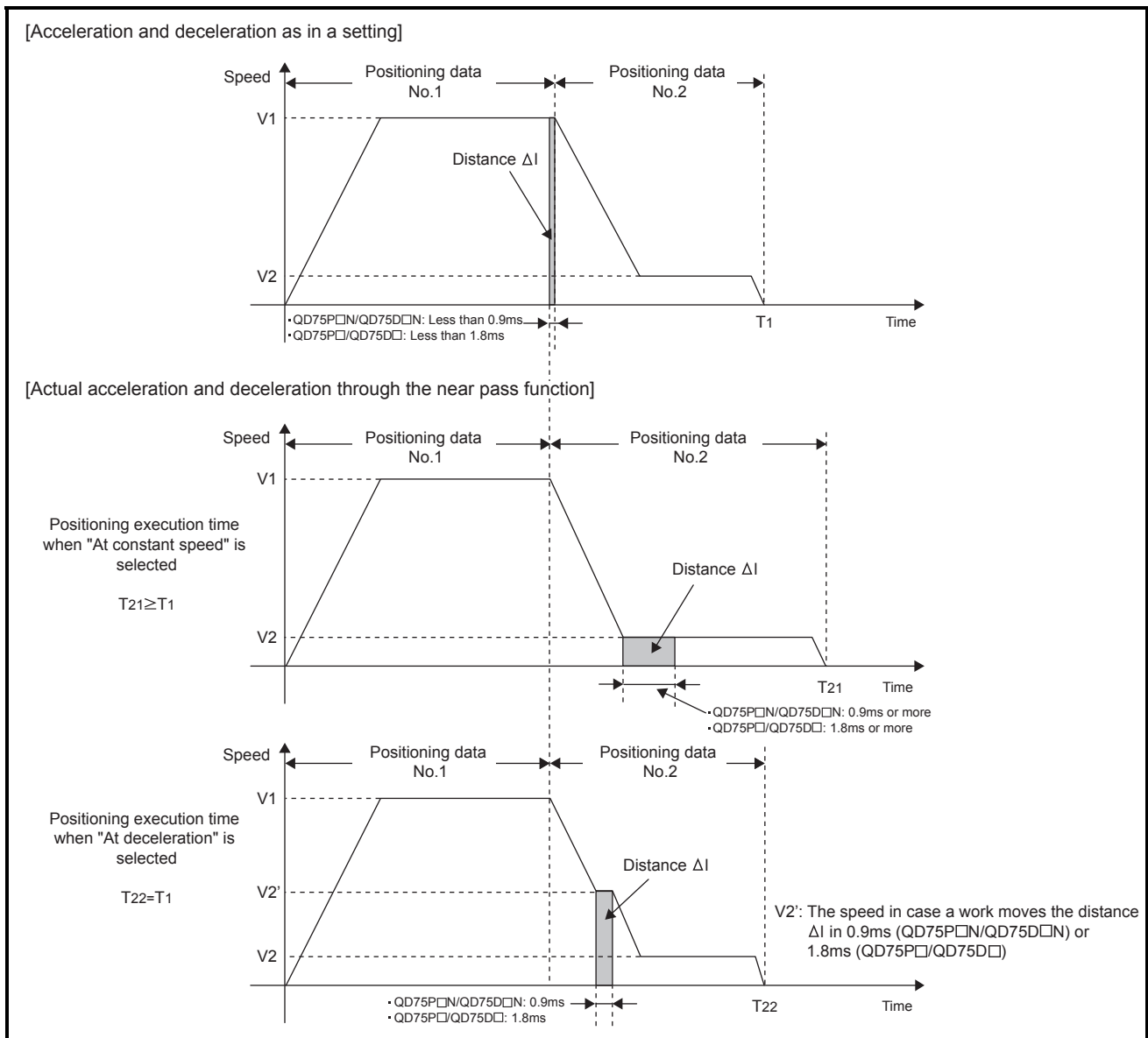


Fig. 12.14 Acceleration and deceleration depending on the output timing selection of near pass control



## [2] Control precautions

When command speed  $V1$  and  $V2$  are in the condition 1) or 2) below, the command output is that of "At constant speed" even though "At deceleration" was selected.

1) When " $V1 \leq V2$ "

2) When " $(\text{Speed value converted from distance } \Delta \ell)^{*1} \leq V2 < V1$ "

\*1: The height of the grey area in the graph on the bottom right

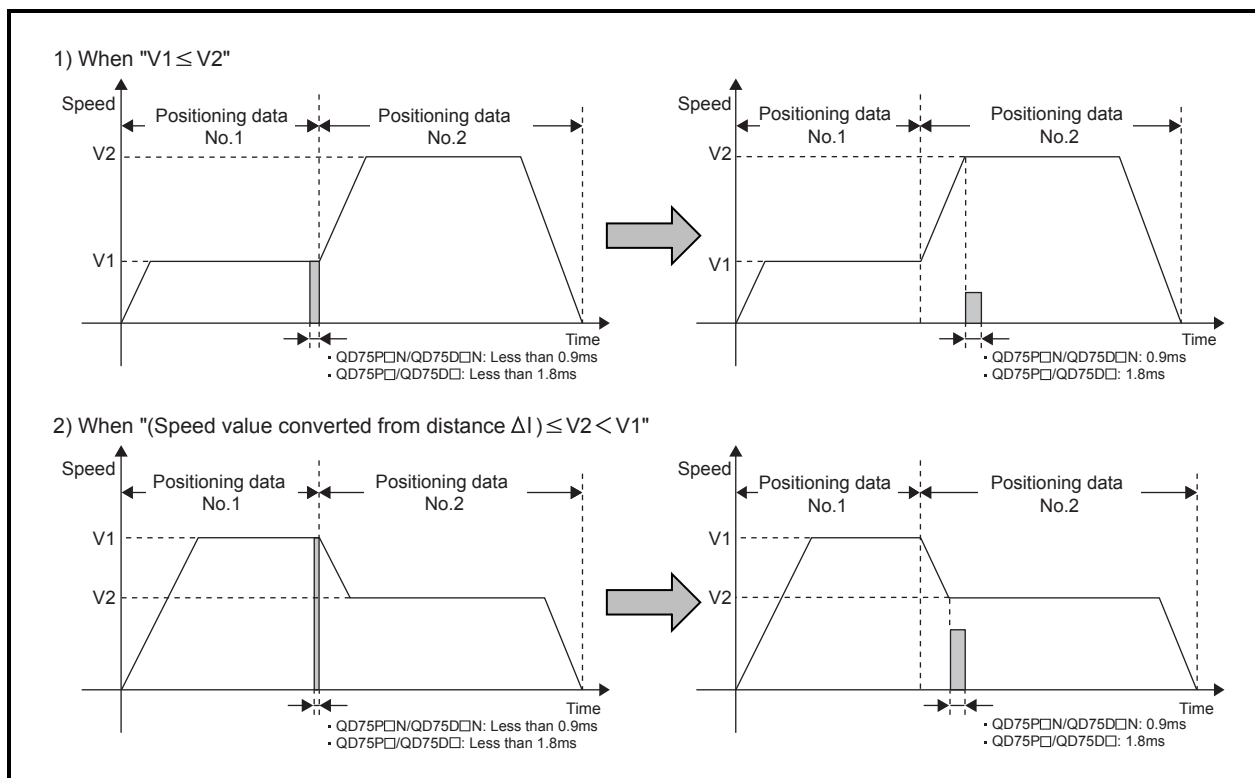


Fig. 12.15 Examples of when the command output is that of "At constant speed" even though "At deceleration" was selected

## [3] Setting method

To use the function "output timing selection of near pass control", set the following control data in a sequence program.

The set detail is validated at the rising edge (OFF → ON) of the PLC READY signal (Y0).

| Setting item |                                              | Setting value | Setting details                                                                                                                                                                                                                                                                                        | Buffer memory address |
|--------------|----------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Cd.43        | Output timing selection of near pass control | →             | Select the timing to output the difference ( $\Delta \ell$ ) between the actual and the set positioning end addresses in continuous path control, in which the difference ( $\Delta \ell$ ) is output during the execution of the next positioning data.<br>0: At constant speed<br>1: At deceleration | 1934                  |

\* Refer to Section 5.7 "List of control data" for setting details.

## 12.4 Functions to limit the control

Functions to limit the control include the "speed limit function", "torque limit function", "software stroke limit", and "hardware stroke limit". Each function is executed by parameter setting or sequence program creation and writing.

### 12.4.1 Speed limit function

The "speed limit function" limits the command speed to a value within the "speed limit value" setting range when the command speed during control exceeds the "speed limit value".

The details shown below explain about the "speed limit function".

[1] Relation between the speed limit function and various controls

[2] Control precautions

[3] Setting method

#### [1] Relation between the speed limit function and various controls

The following table shows the relation of the "speed limit function" and various controls.

| Control type              |                                                                    |                                                 | Speed limit function | Speed limit value             |
|---------------------------|--------------------------------------------------------------------|-------------------------------------------------|----------------------|-------------------------------|
| OPR control               | Machine OPR control                                                |                                                 | ⊙                    | [Pr.8] Speed limit value      |
|                           | Fast OPR control                                                   |                                                 | ⊙                    |                               |
| Major positioning control | Position control                                                   | 1-axis linear control                           | ⊙                    |                               |
|                           |                                                                    | 2- to 4-axis linear interpolation control       | ⊙                    |                               |
|                           |                                                                    | 1-axis fixed-feed control                       | ⊙                    |                               |
|                           |                                                                    | 2- to 4-axis fixed-feed control (interpolation) | ⊙                    |                               |
|                           |                                                                    | 2-axis circular interpolation control           | ⊙                    |                               |
|                           |                                                                    | 3-axis helical interpolation control            | ⊙                    |                               |
|                           | 1- to 4-axis Speed control                                         |                                                 | ⊙                    |                               |
|                           | Speed-position switching control, Position-speed switching control |                                                 | ⊙                    |                               |
|                           | Other control                                                      | Current value changing                          | —                    | Setting value invalid         |
|                           |                                                                    | JUMP instruction, NOP instruction, LOOP to LEND | —                    |                               |
| Manual control            | JOG operation, Inching operation                                   |                                                 | ⊙                    | [Pr.31] JOG speed limit value |
|                           | Manual pulse generator operation                                   |                                                 | —                    | Setting is invalid            |

⊙ : Always set

— : Setting not required (Setting value is invalid. Use the initial values or setting values within a range where no error occurs.)

## [2] Control precautions

- If any axis exceeds "[Pr.8] Speed limit value" during 2- to 4-axis speed control, the axis in excess of the speed limit value is controlled at the speed limit value. For the other axes which perform interpolation, the speed can be suppressed by the ratio of a command speed.
- If any axis exceeds "[Pr.8] Speed limit value" during 2- to 4-axis linear interpolation control, 2- to 4-axis fixed-feed control, 2-axis circular interpolation control, or 3-axis helical interpolation control, the axis exceeding the speed limit value is controlled at the speed limit value. For the other axes performing interpolation, the speed can be suppressed by the ratio of a movement distance.
- In 2- to 4-axis linear interpolation control or 2- to 4-axis fixed-feed control, the speed limit value for the interpolation axis may not function under the following condition: "1: Reference axis speed" is set in "[Pr.20] Interpolation speed designation method" or "[Da.29] Interpolation speed designation method", and the reference axis is the minor axis and the interpolation axis is the major axis.
- In 3-axis helical interpolation control, control the composite speed of the circular interpolation axis or the speed of the linear interpolation axis so that the speeds do not exceed "[Pr.8] Speed limit value".

## [3] Setting method

To use the "speed limit function", set the "speed limit value" in the parameters shown in the following table, and write them to the QD75.

The set details are validated after they are written to the QD75.

| Setting item                  | Setting value | Setting details                                                                                                                                                                    | Factory-set initial value |
|-------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| [Pr.8] Speed limit value      | →             | Set the speed limit value (max. speed during control).                                                                                                                             | 200000                    |
| [Pr.31] JOG speed limit value | →             | Set the speed limit value during JOG operation (max. speed during control). (Note that "[Pr.31] JOG speed limit value" shall be less than or equal to "[Pr.8] Speed limit value".) | 20000                     |

\* Refer to Section 5.2 "List of parameters" for setting details.

**REMARK**

- Parameters are set for each axis.
- It is recommended that the parameters be set whenever possible with GX Configurator-QP. Execution by sequence program uses many sequence programs and devices. The execution becomes complicated, and the scan times will increase.

## 12.4.2 Torque limit function

The "torque limit function" limits the generated torque to a value within the "torque limit value" setting range when the torque generated in the servomotor exceeds the "torque limit value".

The "torque limit function" protects the deceleration function, limits the power of the operation pressing against the stopper, etc. It controls the operation so that unnecessary force is not applied to the load and machine.

The QD75 does not carry out the torque limit of servo motor directly with this function. The torque limit command to servo amplifier is carried out by the D/A converter module.

The details shown below explain about the "torque limit function".

- [1] System configuration for carrying out the torque limit
- [2] Relation between the torque limit function and various controls
- [3] Control details
- [4] Control precautions
- [5] Setting method

### [1] System configuration for carrying out the torque limit

Carry out the torque limit in the configuration in the figure below. (The following modules are required.)

- D/A converter module
- Drive unit capable of torque limit control by analog voltage input

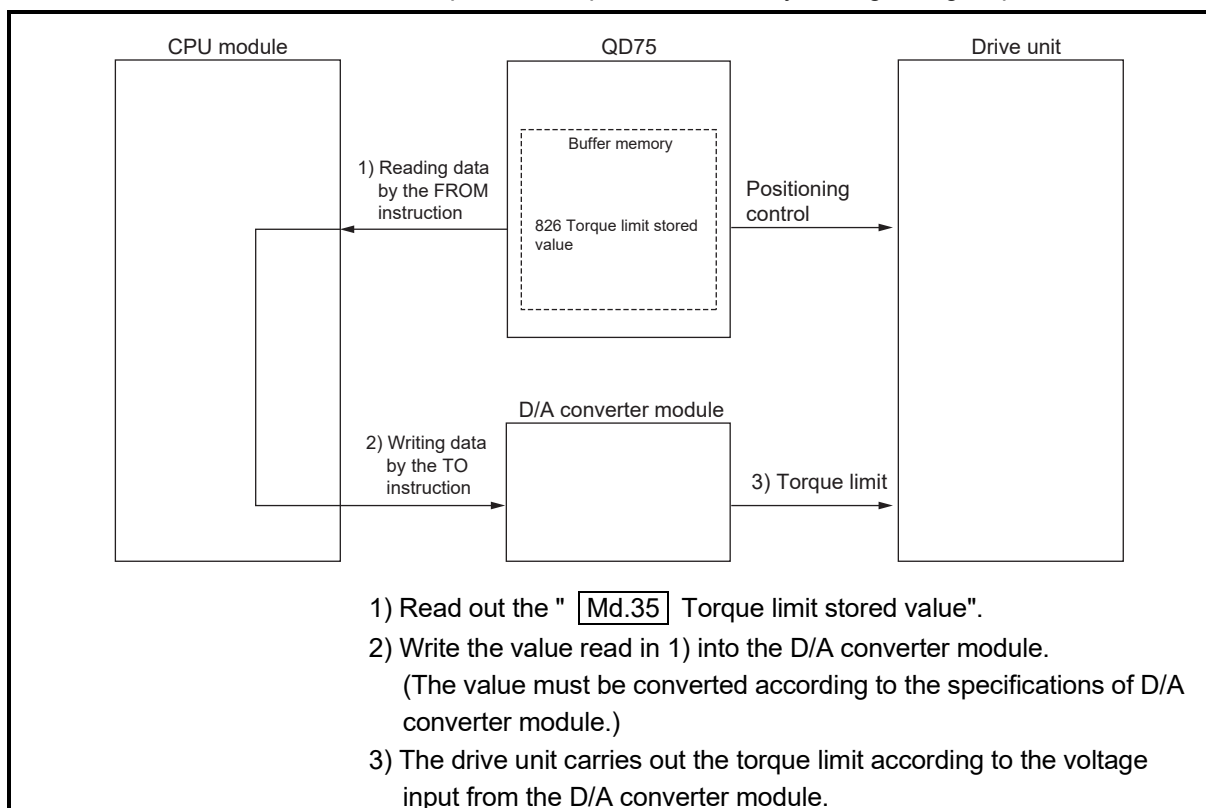


Fig. 12.16 Torque limit to the drive unit (axis 1)

| POINT                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The QD75 monitors the creep speed reach during the OPR control and updates the " <a href="#">Md.35</a> Torque limit stored value" to the " <a href="#">Pr.54</a> OPR torque limit value".</p> <p>Monitoring this value prevents the need to monitor the creep speed reach by the program. If all controls of torque limit value are carried out by the program (1)</p> <p>Reading data by the FROM instruction, this function must not be used.</p> |

## [2] Relation between the torque limit function and various controls

The following table shows the relation of the "torque limit function" and various controls.

| Control type              |                                  |                                                 | Torque limit function | Torque limit value *                                                                                                                                                                     |
|---------------------------|----------------------------------|-------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OPR control               | Machine OPR control              |                                                 | ○                     | <a href="#">Pr.17</a> Torque limit setting value<br>* After the " <a href="#">Pr.47</a> Creep speed" is reached, this value becomes the " <a href="#">Pr.54</a> OPR torque limit value". |
|                           | Fast OPR control                 |                                                 | ○                     |                                                                                                                                                                                          |
| Major positioning control | Position control                 | 1-axis linear control                           | ○                     | <a href="#">Pr.17</a> Torque limit setting value                                                                                                                                         |
|                           |                                  | 2- to 4-axis linear interpolation control       | ○                     |                                                                                                                                                                                          |
|                           |                                  | 1-axis fixed-feed control                       | ○                     |                                                                                                                                                                                          |
|                           |                                  | 2- to 4-axis fixed-feed control (interpolation) | ○                     |                                                                                                                                                                                          |
|                           |                                  | 2-axis circular interpolation control           | ○                     |                                                                                                                                                                                          |
|                           |                                  | 3-axis helical interpolation control            | ○                     |                                                                                                                                                                                          |
|                           | 1- to 4-axis Speed control       |                                                 | ○                     |                                                                                                                                                                                          |
|                           | Speed-position switching control |                                                 | ○                     |                                                                                                                                                                                          |
|                           | Position-speed switching control |                                                 | ○                     |                                                                                                                                                                                          |
|                           | Other control                    | Current value changing                          | —                     | Setting value is invalid.                                                                                                                                                                |
|                           |                                  | JUMP instruction, NOP instruction, LOOP to LEND | —                     |                                                                                                                                                                                          |
| Manual control            | JOG operation, Inching operation |                                                 | ○                     | <a href="#">Pr.17</a> Torque limit setting value                                                                                                                                         |
|                           | Manual pulse generator operation |                                                 | ○                     | <a href="#">Pr.17</a> Torque limit setting value                                                                                                                                         |

○ : Set when required

— : Setting not required (Setting value is invalid. Use the initial values or setting values within a range where no error occurs.)

\* : Shows the torque limit value when " [Cd.22](#) New torque value" is set to "0".

## [3] Control details

The following drawing shows the operation of the torque limit function.

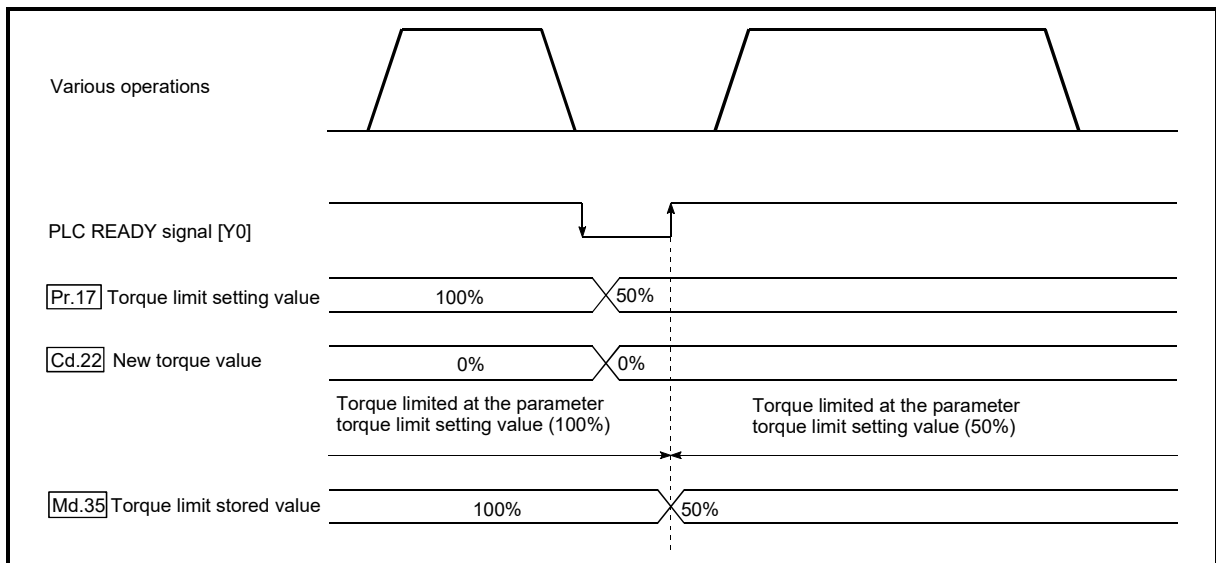


Fig. 12.17 Torque limit function operation

## [4] Control precautions

- (1) When limiting the torque at the "[Pr.17] Torque limit setting value", confirm that "[Cd.22] New torque value" is set to "0". If this parameter is set to a value besides "0", the "[Cd.22] New torque value" will be validated, and the torque will be limited at that value. (Refer to Section 12.5.4 "Torque change function" for details about the "new torque value".)
- (2) When "[Pr.54] OPR shift amount" exceeds "[Pr.17] Torque limit setting value", the error "OPR torque limit value error" (error code: 995) will occur.
- (3) When the operation is stopped by torque limiting, the droop pulse will remain in the deviation counter. If a "deviation counter clear" is carried out by issuing an external signal at this time, positional deviation will occur when the operation is continued. If the load torque is eliminated, operation for the amount of droop pulses will be carried out.

## [5] Setting method

- (1) To use the "torque limit function", set the "torque limit value" in the parameters shown in the following table, and write them to the QD75. The set details are validated at the rising edge (OFF → ON) of the PLC READY signal (Y0).

| Setting item                            | Setting value | Setting details                                                                                   | Factory-set initial value |
|-----------------------------------------|---------------|---------------------------------------------------------------------------------------------------|---------------------------|
| <b>Pr.17</b> Torque limit setting value | →             | Set the torque limit value as a percentage.                                                       | 300                       |
| <b>Pr.54</b> OPR torque limit value     | →             | Set the torque limit value after the " <b>Pr.47</b> Creep speed" is reached. Set as a percentage. | 300                       |

\* Refer to Section 5.2 "List of parameters" for setting details.

The following table shows the " **Md.35** Torque limit stored value" of the buffer memory address.

| Monitor item                           | Monitor value | Storage details                                                                                          | Buffer memory address |        |        |        |
|----------------------------------------|---------------|----------------------------------------------------------------------------------------------------------|-----------------------|--------|--------|--------|
|                                        |               |                                                                                                          | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <b>Md.35</b> Torque limit stored value | →             | The "torque limit value" valid at that time is stored. ( <b>Pr.17</b> , <b>Pr.54</b> , or <b>Cd.22</b> ) | 826                   | 926    | 1026   | 1126   |

\* Refer to Section 5.6 "List of monitor data" for information on the setting details.

**REMARK**

- Parameters are set for each axis.
- It is recommended that the parameters be set whenever possible with GX Configurator-QP. Execution by sequence program uses many sequence programs and devices. The execution becomes complicated, and the scan times will increase.



### 12.4.3 Software stroke limit function

In the "software stroke limit function" the address established by a machine OPR is used to set the upper/lower limits of the moveable range of the workpiece. Movement commands issued to addresses outside that setting range will not be executed. In the QD75, the "current feed value" and "machine feed value" are used as the addresses indicating the current position. However, in the "software stroke limit function", the address used to carry out the limit check is designated in the " [Pr.14] Software stroke limit selection". (Refer to Section 9.1.4 "Confirming the current value" or details on the "current feed value" and "machine feed value".)

The upper/lower limits of the moveable range of the workpiece are set in " [Pr.12] Software stroke limit upper limit value"/ " [Pr.13] Software stroke limit lower limit value".

The details shown below explain about the "software stroke limit function".

- [1] Differences in the moveable range when "current feed value" and "machine feed value" are selected.
- [2] Software stroke limit check details
- [3] Relation between the software stroke limit function and various controls
- [4] Precautions during software stroke limit check
- [5] Setting method
- [6] Invalidating the software stroke limit
- [7] Setting when the control unit is "degree"

- [1] Differences in the moveable range when "current feed value" and "machine feed value" are selected.

The following drawing shows the moveable range of the workpiece when the software stroke limit function is used.

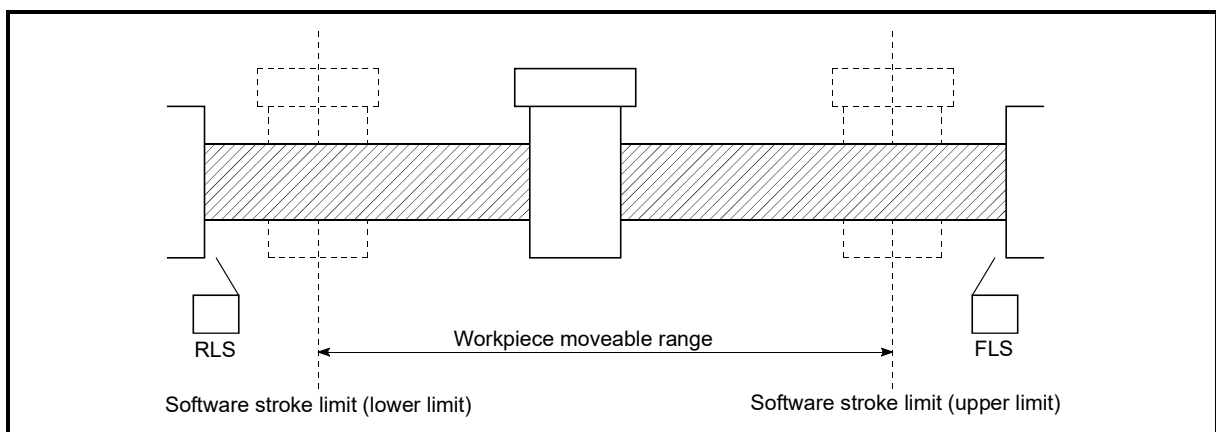
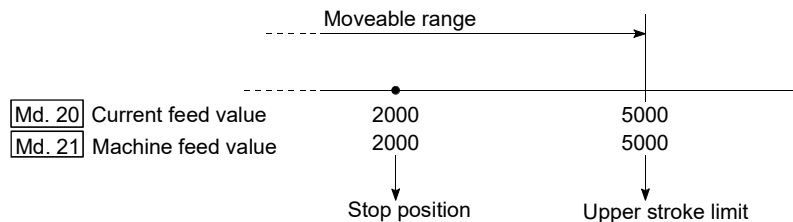


Fig. 12.18 Workpiece moveable range

The following drawing shows the differences in the operation when "Md.20 Current feed value" and "Md.21 Machine feed value" are used in the moveable range limit check.

**[Conditions]**

Assume the current stop position is 2000, and the upper stroke limit is set to 5000.

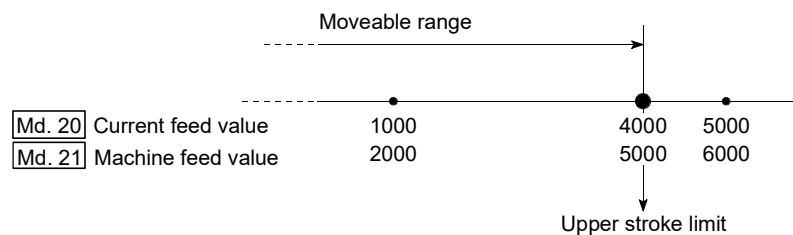


**[Current value changing]**

When the current value is changed by a new current value command from 2000 to 1000, the current value will change to 1000, but the machine feed value will stay the same at 2000.

- 1) When the machine feed value is set at the limit

The machine feed value of 5000 (current feed value: 4000) becomes the upper stroke limit.



- 2) When the current feed value is set at the limit

The current feed value of 5000 (machine feed value: 6000) becomes the upper stroke limit.

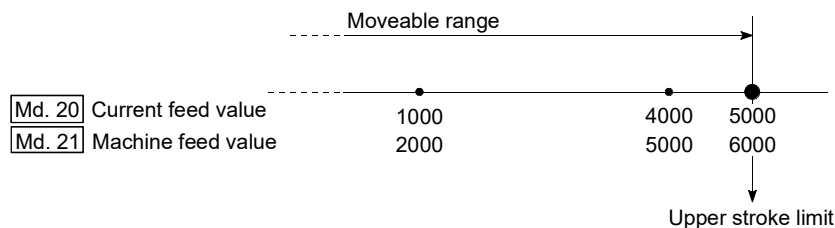


Fig. 12.19 Software stroke limits of the current feed value and machine feed value

**POINT**

When "machine feed value" is set in "Pr.14 Software stroke limit selection", the moveable range becomes an absolute range referenced on the OP. When "current feed value" is set, the moveable range is the relative range from the "current feed value".

## [2] Software stroke limit check details

| Check details |                                                                                                                                                                        | Processing when an error occurs                                                                      |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 1)            | An error shall occur if the current value *1 is outside the software stroke limit range *2.<br>(Check " [Md.20] Current feed value" or " [Md.21] Machine feed value".) | An error "Software stroke limit +" or "Software stroke limit -" (error code: 507 or 508) will occur. |
| 2)            | An error shall occur if the command address is outside the software stroke limit range.<br>(Check " [Da.6] Positioning address/movement amount".)                      |                                                                                                      |

\*1: Check whether the " [Md.20] Current feed value" or " [Md.21] Machine feed value" is set in " [Pr.14] Software stroke limit selection".

\*2: Moveable range from the " [Pr.12] Software stroke limit upper limit value" to the " [Pr.13] Software stroke limit lower limit value".

## [3] Relation between the software stroke limit function and various controls

| Control type                                    |                                  |                                                 | Limit check            | Processing at check                                                                                                                                                                                                                                                               |
|-------------------------------------------------|----------------------------------|-------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OPR control                                     | Machine OPR control              |                                                 | —                      | Check not carried out.                                                                                                                                                                                                                                                            |
|                                                 | Fast OPR control                 |                                                 | —                      |                                                                                                                                                                                                                                                                                   |
| Major positioning control                       | Position control                 | 1-axis linear control                           | ⊙                      | Checks 1) and 2) in the previous section [2] are carried out.<br>For speed control: The axis decelerates to a stop when it exceeds the software stroke limit range.<br>For position control: The axis comes to an immediate stop when it exceeds the software stroke limit range. |
|                                                 |                                  | 2- to 4-axis axis linear interpolation control  | ⊙                      |                                                                                                                                                                                                                                                                                   |
|                                                 |                                  | 1-axis fixed-feed control                       | ⊙                      |                                                                                                                                                                                                                                                                                   |
|                                                 |                                  | 2- to 4-axis fixed-feed control (interpolation) | ⊙                      |                                                                                                                                                                                                                                                                                   |
|                                                 |                                  | 2-axis circular interpolation control           | ⊙                      |                                                                                                                                                                                                                                                                                   |
|                                                 |                                  | 3-axis helical interpolation control            | ⊙                      |                                                                                                                                                                                                                                                                                   |
|                                                 | 1- to 4-axis speed control       |                                                 | ○<br>*3, 4             |                                                                                                                                                                                                                                                                                   |
|                                                 | Speed-position switching control |                                                 | ○                      |                                                                                                                                                                                                                                                                                   |
|                                                 | Position-speed switching control |                                                 | *3, 4                  |                                                                                                                                                                                                                                                                                   |
|                                                 | Other control                    | Current value changing                          | ⊙                      |                                                                                                                                                                                                                                                                                   |
| JUMP instruction, NOP instruction, LOOP to LEND |                                  | —                                               | Check not carried out. |                                                                                                                                                                                                                                                                                   |
| Manual control                                  | JOG operation, Inching operation |                                                 | △ *5                   | Check 1) in the previous section [2] is carried out.<br>The machine will carry out a deceleration stop when the software stroke limit range is exceeded. If the address falls out of the software stroke limit range, the operation can be started only toward the movable range. |
|                                                 | Manual pulse generator operation |                                                 | △ *5                   |                                                                                                                                                                                                                                                                                   |

◎ : Check valid

○ : Check is not made when the current feed value is not updated (Refer to [Pr.21] at the setting of " current feed value" in " [Pr.14] Software stroke limit selection" during speed control.

— : Check not carried out (check invalid).

△ : Valid only when "0:valid" is set in the " [Pr.15] Software stroke limit valid/invalid setting".

\*3: The value in " [Md.20] Current feed value" will differ according to the " [Pr.21] Current feed value during speed control" setting.

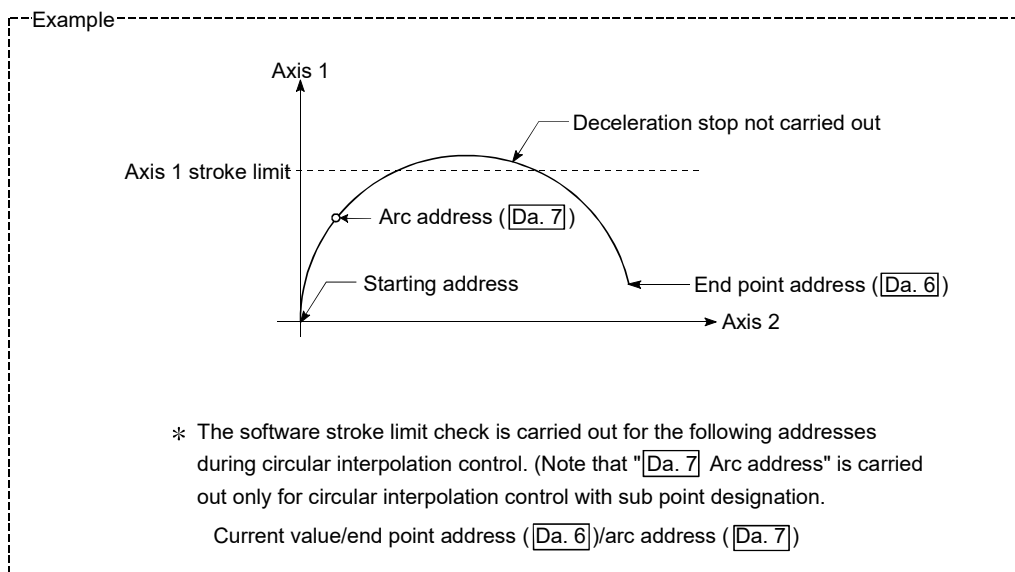
\*4: When the unit is "degree", check is not made during speed control.

\*5: When the unit is "degree", check is not carried out.

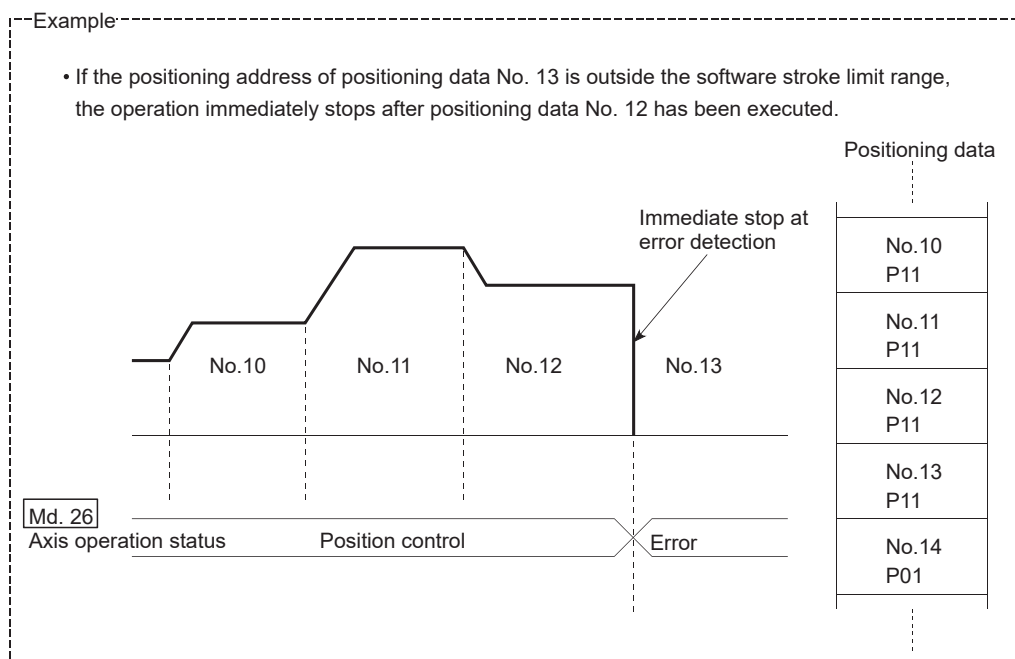
#### [4] Precautions during software stroke limit check

- (1) A machine OPR must be executed beforehand for the "software stroke limit function" to function properly.
- (2) During interpolation control, a stroke limit check is carried out for the every current value of both the reference axis and the interpolation axis. Every axis will not start if an error occurs, even if it only occurs in one axis.
- (3) During circular interpolation control, the " Pr.12 Software stroke limit upper limit value"/" Pr.13 Software stroke limit lower limit value" may be exceeded.

In this case, a deceleration stop will not be carried out even if the stroke limit is exceeded. Always install an external limit switch if there is a possibility the stroke limit will be exceeded.



- (4) If an error is detected during continuous path control, the axis stops immediately on completion of execution of the positioning data located right before the positioning data in error.



- (5) During simultaneous start, a stroke limit check is carried out for the current values of every axis to be started. Every axis will not start if an error occurs, even if it only occurs in one axis.

#### [5] Setting method

To use the "software stroke limit function", set the required values in the parameters shown in the following table, and write them to the QD75.

The set details are validated at the rising edge (OFF → ON) of the PLC READY signal (Y0).

| Setting item                                             | Setting value | Setting details                                                                                                                                               | Factory-set initial value |
|----------------------------------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>Pr.12</b> Software stroke limit upper limit value     | →             | Set the upper limit value of the moveable range.                                                                                                              | 2147483647                |
| <b>Pr.13</b> Software stroke limit lower limit value     | →             | Set the lower limit value of the moveable range.                                                                                                              | -2147483648               |
| <b>Pr.14</b> Software stroke limit selection             | →             | Set whether to use the " <b>Md.20</b> Current feed value" or " <b>Md.21</b> Machine feed value" as the "current value".                                       | 0: Current feed value     |
| <b>Pr.15</b> Software stroke limit valid/invalid setting | 0: Valid      | Set whether the software stroke limit is validated or invalidated during manual control (JOG operation, Inching operation, manual pulse generator operation). | 0: valid                  |

\* Refer to Section 5.2 "List of parameters" for setting details.

#### [6] Invalidating the software stroke limit

To invalidate the software stroke limit, set the following parameters as shown, and write them to the QD75. (Set the value within the setting range.)

$$\boxed{\text{Pr.12}} \text{ Software stroke limit upper limit value} = \boxed{\text{Pr.13}} \text{ Software stroke limit lower limit value}$$

To invalidate only the manual operation, set "1: software stroke limit invalid" in the " **Pr.15** Software stroke limit valid/invalid setting".

The set details are validated at the rising edge (OFF → ON) of the PLC READY signal (Y0).

When the unit is "degree", the software stroke limit check is not performed during speed control (including speed control in speed-position switching control or position-speed switching control) or during manual control, independently of the values set in **Pr.12**, **Pr.13** and **Pr.15**.

#### REMARK

- Parameters are set for each axis.
- It is recommended that the parameters be set whenever possible with GX Configurator-QP. Execution by sequence program uses many sequence programs and devices. The execution becomes complicated, and the scan times will increase.

## [7] Setting when the control unit is "degree"

## ■ Current value address

The " Md.20 Current feed value" address is a ring address between 0 and 359.99999°.

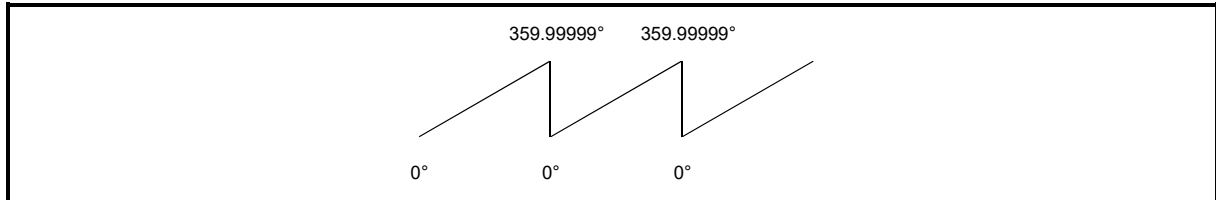


Fig. 12.20 Current value address when the control unit is "degree"

## ■ Setting the software stroke limit

The upper limit value/lower limit value of the software stroke limit is a value between 0 and 359.99999°.

## (1) Setting when the software stroke limit is to be validated.

When the software stroke limit is to be validated, set the upper limit value in a clockwise direction from the lower limit value.

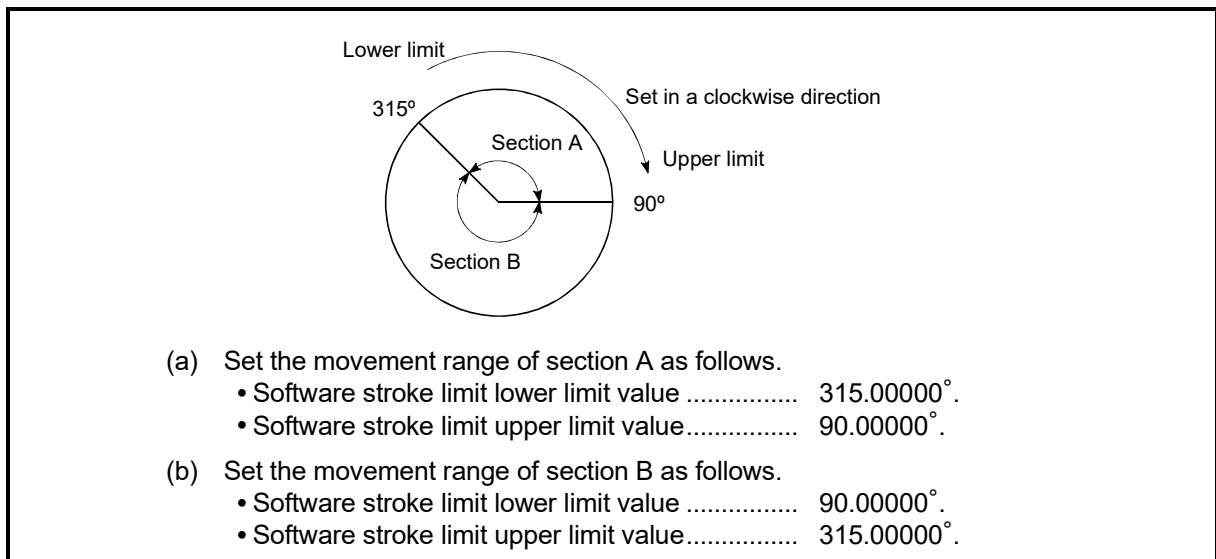


Fig. 12.21 Software stroke limit when the control unit is "degree"

## 12.4.4 Hardware stroke limit function

**⚠ WARNING**

When the hardware stroke limit is required to be wired, ensure to wire it in the negative logic using b-contact. If it is set in positive logic using a-contact, a serious accident may occur.

In the "hardware stroke limit function", limit switches are set at the upper/lower limit of the physical moveable range, and the control is stopped (by deceleration stop) by the input of a signal from the limit switch. Damage to the machine can be prevented by stopping the control before the upper/lower limit of the physical moveable range is reached.

Hardware stroke limit switches are normally "installed inside the stroke limit/stroke end on the drive unit side", and the control is stopped before the stroke limit/stroke end on the drive unit side is reached.

The details shown below explain about the "hardware stroke limit function".

- [1] Control details
- [2] Wiring the hardware stroke limit
- [3] Control Precautions
- [4] When the hardware stroke limit is not used

**[1] Control details**

The following drawing shows the operation of the hardware stroke limit function.

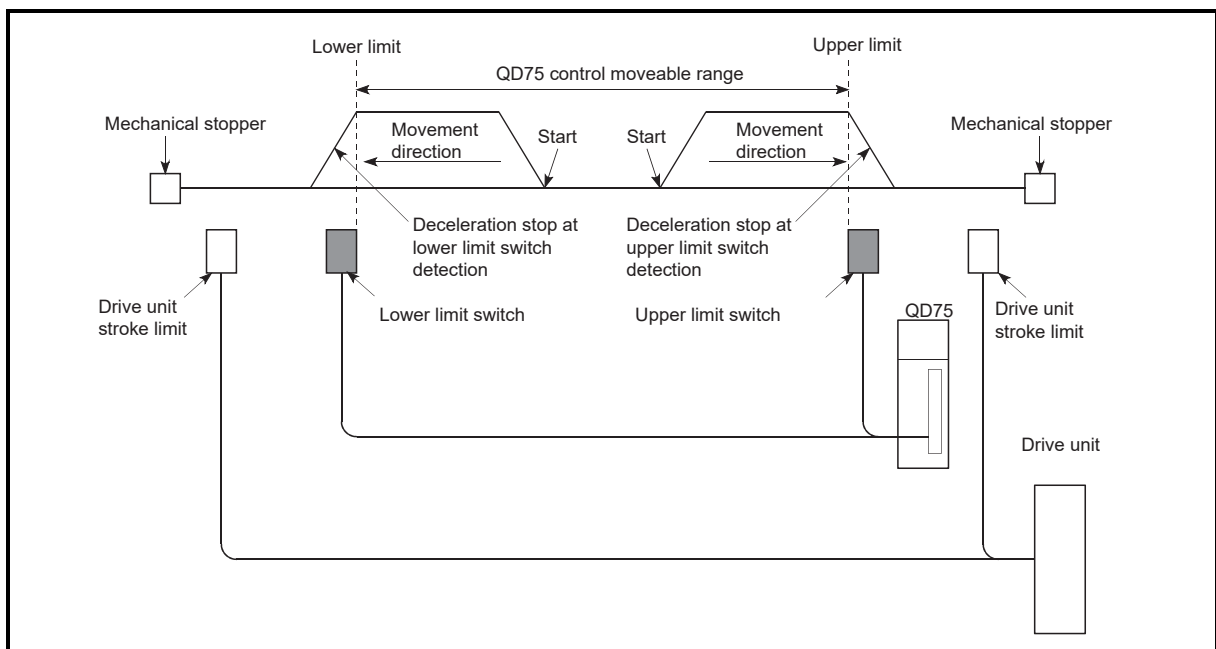


Fig. 12.22 Hardware stroke limit function operation

### [2] Wiring the hardware stroke limit

When using the hardware stroke limit function, wire the terminals of the QD75 upper/lower limit stroke limit as shown in the following drawing.

(When " **Pr.22** Input signal logic selection" is set to the initial value)

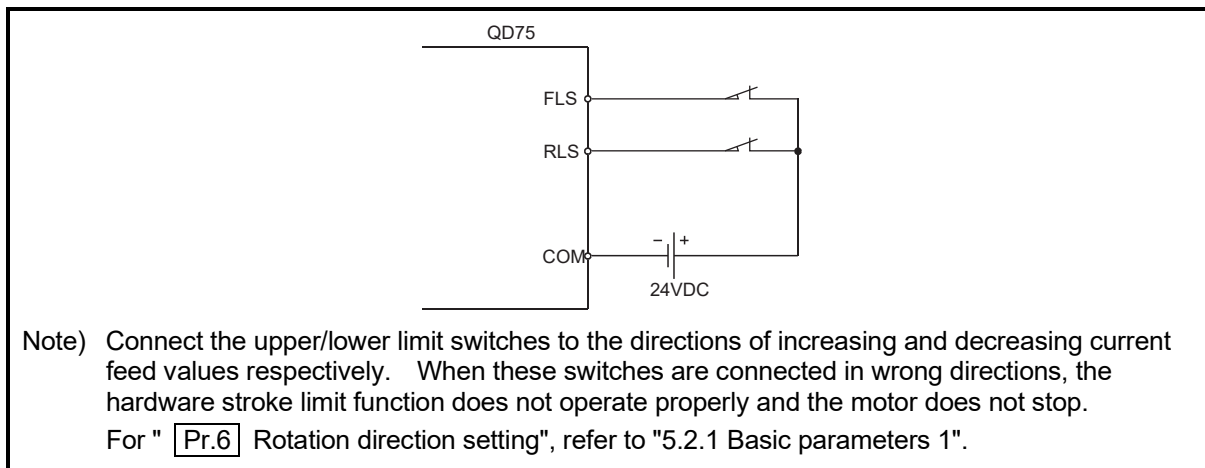


Fig. 12.23 Wiring when using the hardware stroke limit

### [3] Control Precautions

- (1) If the machine is stopped outside the QD75 control range (outside the upper/lower limit switches), or if stopped by hardware stroke limit detection, the "OPR control", "major positioning control", and "high-level positioning control" cannot start. To carry out these types of control again, return the workpiece to the QD75 control range by a "JOG operation", "inching operation" or "manual pulse generator operation".
- (2) When " **Pr.22** Input signal logic selection" is set to the initial value, the QD75 cannot carry out the positioning control if FLS (upper limit signal) is separated from COM or RLS (lower limit signal) is separated from COM (including when wiring is not carried out).

### [4] When the hardware stroke limit function is not used

When not using the hardware stroke limit function, wire the terminals of the QD75 upper/lower limit stroke limit as shown in the following drawing.

When the logic of FLS and RLS is set to "positive logic" using " **Pr.22** Input signal logic selection", positioning control can be carried out even if FLS and RLS are not wired.

(For details, refer to Section 13.4 "External I/O signal logic switching function".)

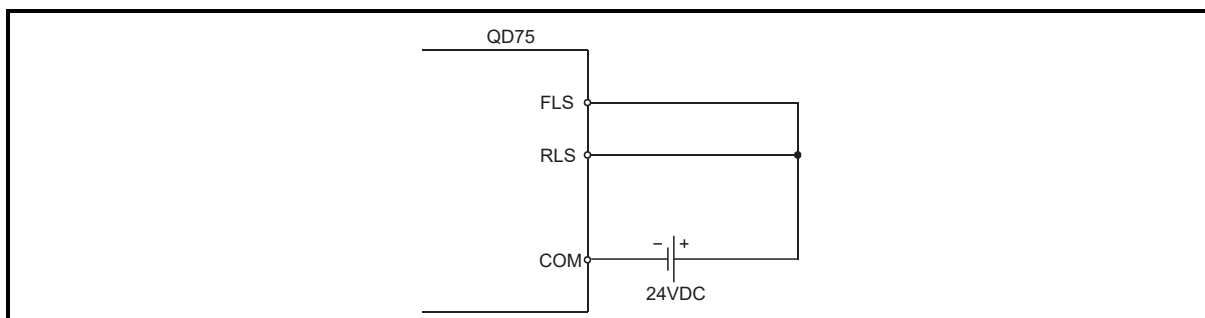


Fig. 12.24 Wiring when not using the hardware stroke limit function  
(When " **Pr.22** Input signal logic selection" is the initial value)



## 12.5 Functions to change the control details

Functions to change the control details include the "speed change function", "override function", "acceleration/deceleration time change function" and "torque change function". Each function is executed by parameter setting or sequence program creation and writing.

Both the "speed change function" or "override function" change the speed, but the differences between the functions are shown below. Use the function that corresponds to the application.

### "Speed change function"

- The speed is changed at any time, only in the control being executed.
- The new speed is directly set.

### "Override function"

- The speed is changed for all control to be executed. (Note that this excludes manual pulse generator operation.)
- The new speed is set as a percent (%) of the command speed.

### 12.5.1 Speed change function

The "Speed change function" is used to change the speed during control to a newly designated speed at any time.

The new speed is directly set in the buffer memory, and the speed is changed by a speed change command ([Cd.15](#) Speed change request) or external command signal.

During the machine OPR, a speed change to the creep speed cannot be carried out after deceleration start because the near point dog ON is detected.

The details shown below explain about the "speed change function".

- [1] Control details
- [2] Control precautions
- [3] Setting the speed change function from the CPU module
- [4] Setting the speed change function using an external command signal

#### [1] Control details

The following drawing shows the operation during a speed change.

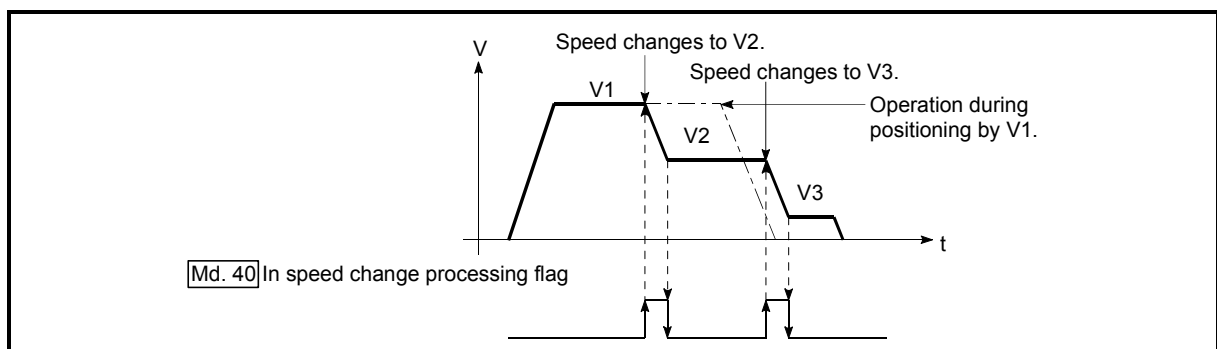


Fig. 12.25 Speed change operation

## [2] Control precautions

- (1) Control is carried out as follows at the speed change during continuous path control.
  - a) When no speed designation (current speed) is provided in the next positioning data:
    - The next positioning data is controlled at the " Cd.14 New speed value".
  - b) When a speed designation is provided in the next positioning data:
    - The next positioning data is controlled at its command speed ( Da.8 ).

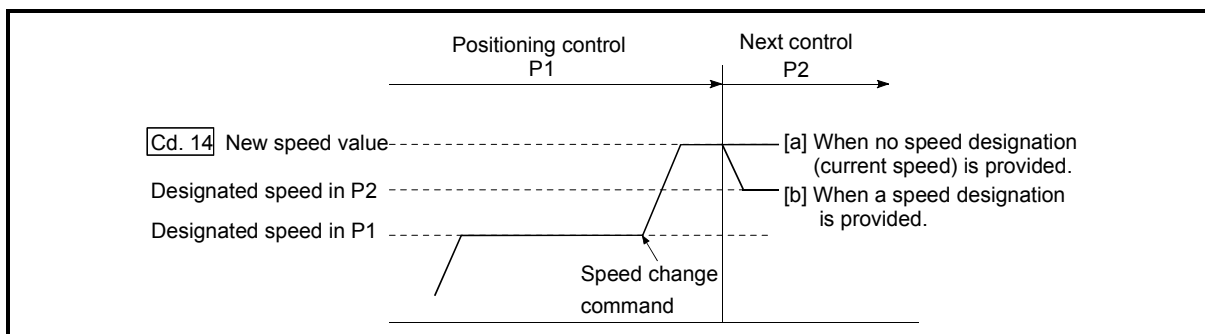


Fig. 12.26 Speed change during continuous path control

- (2) When changing the speed during continuous path control, the speed change will be ignored if there is not enough distance remaining to carry out the change.
- (3) When the stop command was given to make a stop after a speed change that had been made during position control, the restarting speed depends on the function version.

| Function version | Restarting speed                                                                    |
|------------------|-------------------------------------------------------------------------------------|
| A                | <span style="border: 1px solid black; padding: 0 2px;">Da.8</span> Command speed    |
| B or later       | <span style="border: 1px solid black; padding: 0 2px;">Cd.14</span> New speed value |

Refer to Section 2.4 for the way to check the function version.

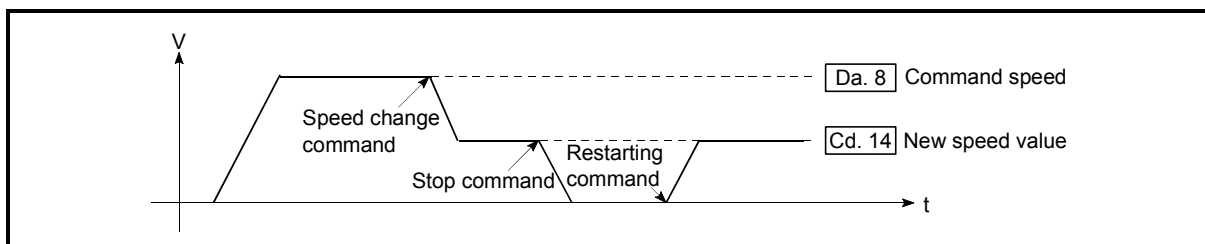


Fig. 12.27 Restarting speed after speed change made during position control (Function version B)

- (4) When the speed is changed by setting " **Cd.14** New speed value" to "0", the operation is carried out as follows.
- When " **Cd.15** Speed change request" is turned ON, the speed change 0 flag ( **Md.31** Status: b10) turns ON.  
(During interpolation control, the speed change 0 flag on the reference axis side turns ON.)
  - The axis stops, but " **Md.26** Axis operation status" does not change, and the BUSY signal remains ON. (If a stop signal is input, the BUSY signal will turn OFF, and " **Md.26** Axis operation status" will change to "Stopped".)
- \*In this case, setting the " **Cd.14** New speed value" to a value besides "0" will turn OFF the speed change 0 flag ( **Md.31** Status: b10), and enable continued operation.

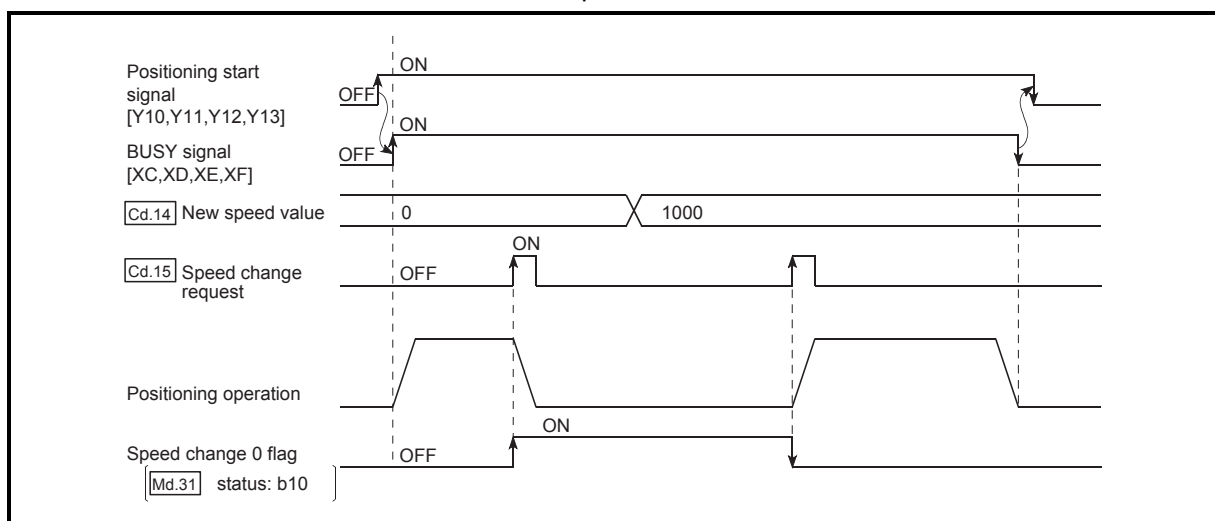


Fig. 12.28 Speed change at new speed value "0"

- (5) A warning "Deceleration/stop speed change" (warning code: 500) occurs and the speed cannot be changed in the following cases.
- During deceleration by a stop command
  - During automatic deceleration during positioning control
- (6) A warning "Speed limit value over" (warning code: 501) occurs and the speed is controlled at the " **Pr.8** Speed limit value" when the value set in " **Cd.14** New speed value" exceeds the " **Pr.8** Speed limit value".
- (7) When the speed is changed during interpolation control, the required speed is set in the reference axis.
- (8) When carrying out consecutive speed changes, be sure there is an interval between the speed changes of 100ms or more.  
(If the interval between speed changes is short, the QD75 will not be able to track, and it may become impossible to carry out commands correctly.)
- (9) When a speed change is requested simultaneously for multiple axes, change the speed one by one. Therefore, the start timing of speed change is different for each axis.
- (10) The speed cannot be changed to 0 during the machine OPR. The speed change request is ignored.
- (11) When deceleration is started by the speed change function, the deceleration start flag does not turn ON.

### [3] Setting the speed change function from the CPU module

The following shows the data settings and sequence program example for changing the control speed of axis 1 from the CPU module. (In this example, the control speed is changed to "20.00mm/min".)

(1) Set the following data.

(Use the start time chart shown in section (2) below as a reference, and set using the sequence program shown in section (3).)

| Setting item                      | Setting value | Setting details            | Buffer memory address |              |              |              |
|-----------------------------------|---------------|----------------------------|-----------------------|--------------|--------------|--------------|
|                                   |               |                            | Axis 1                | Axis 2       | Axis 3       | Axis 4       |
| <b>Cd.14</b> New speed value      | 2000          | Set the new speed.         | 1514<br>1515          | 1614<br>1615 | 1714<br>1715 | 1814<br>1815 |
| <b>Cd.15</b> Speed change request | 1             | Set "1: Change the speed". | 1516                  | 1616         | 1716         | 1816         |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

(2) The following shows the speed change time chart.

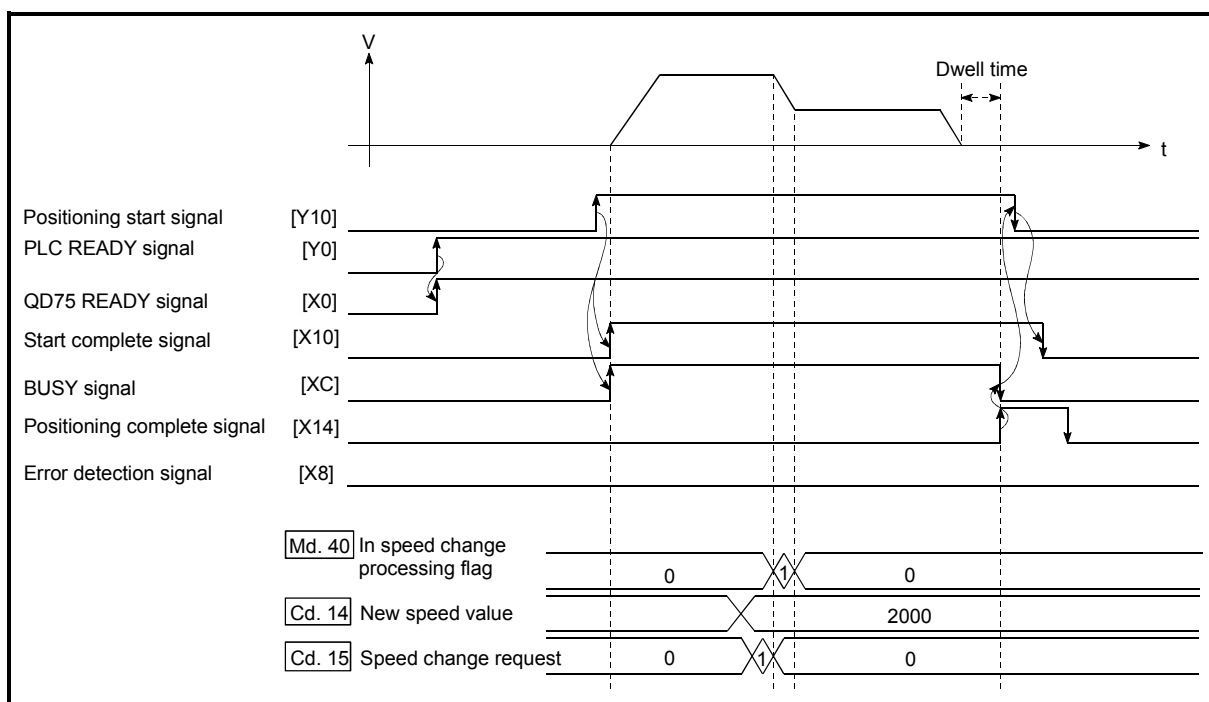
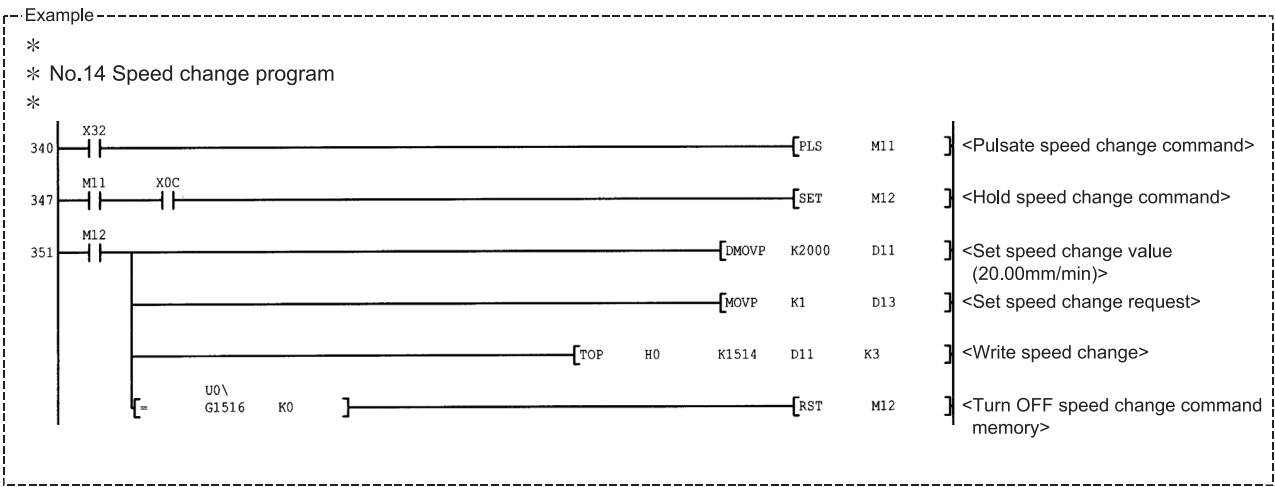


Fig. 12.29 Time chart for changing the speed from the CPU module

(3) Add the following sequence program to the control program, and write it to the CPU module.



#### [4] Setting the speed change function using an external command signal

The speed can also be changed using an "external command signal".

The following shows the data settings and sequence program example for changing the control speed of axis 1 using an "external command signal". (In this example, the control speed is changed to "10000.00mm/min".)

- (1) Set the following data to change the speed using an external command signal.

(Use the start time chart shown in section (2) below as a reference, and set using the sequence program shown in section (3).)

| Setting item                              | Setting value | Setting details                         | Buffer memory address |              |              |              |
|-------------------------------------------|---------------|-----------------------------------------|-----------------------|--------------|--------------|--------------|
|                                           |               |                                         | Axis 1                | Axis 2       | Axis 3       | Axis 4       |
| Pr.42 External command function selection | 1             | Set "1: External speed change request". | 62                    | 212          | 362          | 512          |
| Cd.8 External command valid               | 1             | Set "1: Validate the external command". | 1505                  | 1605         | 1705         | 1805         |
| Cd.14 New speed value                     | 1000000       | Set the new speed.                      | 1514<br>1515          | 1614<br>1615 | 1714<br>1715 | 1814<br>1815 |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

- (2) The following shows the speed change time chart.

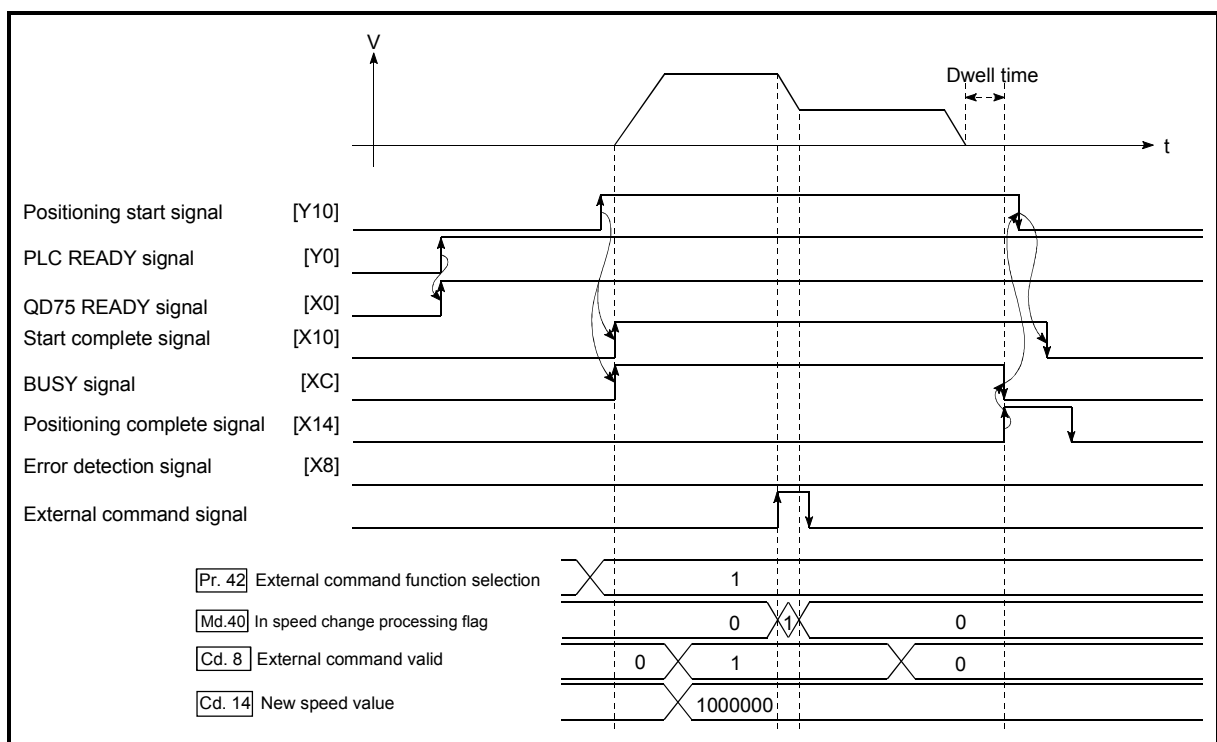
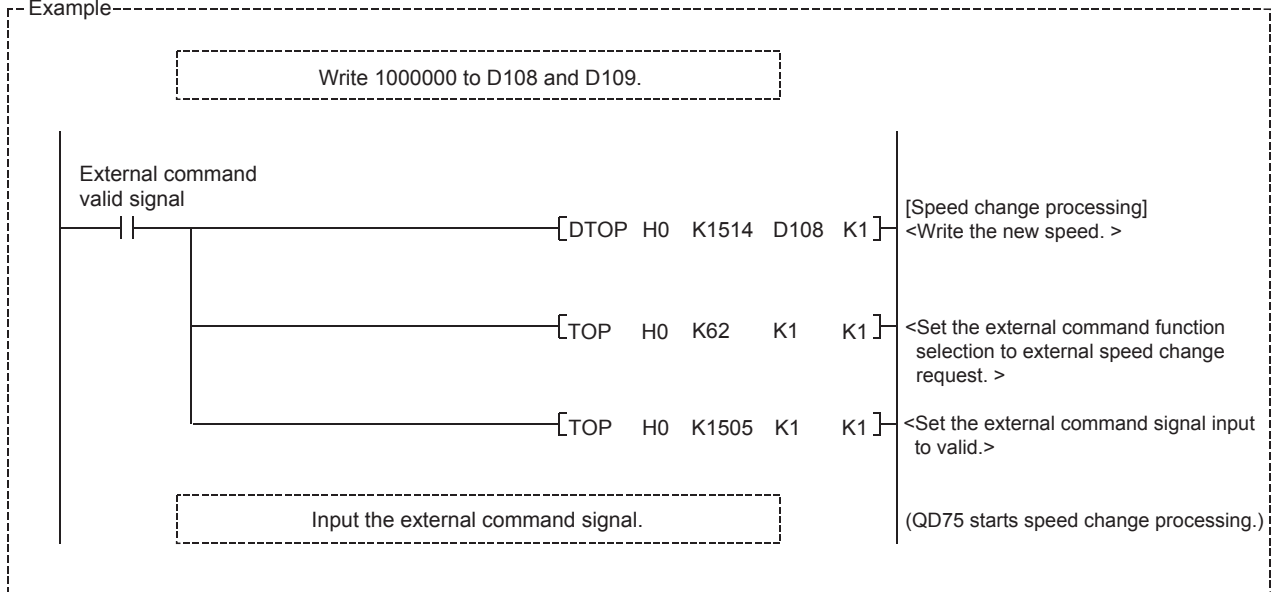


Fig. 12.30 Time chart for changing the speed using an external command signal

- (3) Add the following sequence program to the control program, and write it to the CPU module.

Example



## 12.5.2 Override function

The "override function" changes the command speed by a designated percentage (1 to 300%) for all control to be executed.

The speed can be changed by setting the percentage (%) by which the speed is changed in " [Cd.13] Positioning operation speed override".

- [1] Control details
- [2] Control precautions
- [3] Setting method

## [1] Control details

The following shows that operation of the override function.

- 1) A value changed by the override function is monitored by " [Md.22] Feedrate". (When [Md.22] " becomes "0", the warning "Less than minimum speed" (warning code: 110) is generated and the axis is controlled in the then speed unit of "1".)
- 2) If " [Cd.13] Positioning operation speed override" is set to 100%, the speed will not change.
- 3) If " [Cd.13] Positioning operation speed override" is set a value 100% or less, the warning "Less than minimum speed" (warning code: 110) is generated, and control will be carried out at speed unit "1" at the time "Feedrate" becomes a value of "1" or less.
- 4) If there is not enough remaining distance to change the speed by the "override function" during the "position control" or position control by the "speed-position switching control" or "position-speed switching control", the operation will be carried out at the possible speed for the distance.
- 5) If the speed changed by the "override function" is greater than the " [Pr.8] Speed limit value", a warning "Speed limit value over" (warning code: 501) will occur and the speed will be controlled at the " [Pr.8] Speed limit value". The " [Md.39] In speed limit flag" will turn ON.

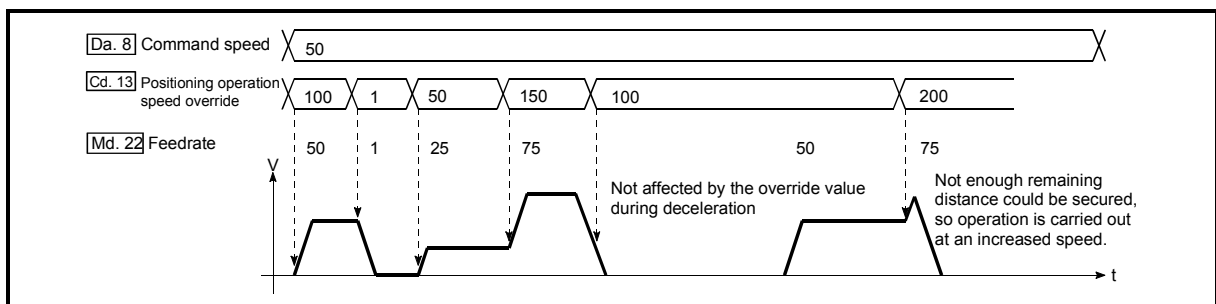


Fig. 12.31 Override function operation



## [2] Control precautions

- (1) When changing the speed by the "override function" during continuous path control, the speed change will be ignored if there is not enough distance remaining to carry out the change.
- (2) A warning "Deceleration/stop speed change" (warning code: 500) occurs and the speed cannot be changed by the "override function" in the following cases.  
(The value set in " Cd.13 Positioning operation speed override" is validated after a deceleration stop.)
  - During deceleration by a stop command
  - During automatic deceleration during positioning control
- (3) When the speed is changed by the "override function" during interpolation control, the required speed is set in the reference axis.
- (4) When carrying out consecutive speed changes by the "override function", be sure there is an interval between the speed changes of 100ms or more.  
(If the interval between speed changes is short, the QD75 will not be able to track, and it may become impossible to carry out commands correctly.)
- (5) When a machine OPR is performed, the speed change by the "override function" cannot be carried out after a deceleration start to the creep speed following the detection of near-point dog ON. In this case, a request for speed change is ignored.
- (6) When deceleration is started by the override function, the deceleration start flag does not turn ON.

## [3] Setting method

The following shows the data settings and sequence program example for setting the override value of axis 1 to "200%".

- (1) Set the following data. (Use the start time chart shown in section (2) below as a reference, and set using the sequence program shown in section (3).)

| Setting item                                                        |                                      | Setting value | Setting details                        | Buffer memory address |        |        |        |
|---------------------------------------------------------------------|--------------------------------------|---------------|----------------------------------------|-----------------------|--------|--------|--------|
|                                                                     |                                      |               |                                        | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <span style="border: 1px solid black; padding: 0 2px;">Cd.13</span> | Positioning operation speed override | 200           | Set the new speed as a percentage (%). | 1513                  | 1613   | 1713   | 1813   |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

The diagram shows the relationship between voltage (V) and time (t) during a positioning operation. The voltage profile starts at zero, ramps up to a peak, and then ramps down to zero. A horizontal dashed line indicates the dwell time at the peak voltage. Below the voltage profile, several digital signals are shown as step functions:

- Positioning start signal [Y10]:** Transitions from low to high at the start of the ramp-up phase and returns to low at the end of the dwell time.
- PLC READY signal [Y0]:** Transitions from low to high at the start of the ramp-up phase and returns to low at the end of the dwell time.
- QD75 READY signal [X0]:** Transitions from low to high at the start of the ramp-up phase and returns to low at the end of the dwell time.
- Start complete signal [X10]:** Transitions from low to high at the start of the ramp-up phase and returns to low at the end of the dwell time.
- BUSY signal [XC]:** Transitions from high to low at the start of the ramp-up phase and returns to high at the end of the dwell time.
- Positioning complete signal [X14]:** Transitions from low to high at the end of the dwell time and returns to low at the end of the dwell time.
- Error detection signal [X8]:** Remains low throughout the operation.

At the bottom, a box labeled "Cd. 13" is shown, with a horizontal line indicating the positioning operation speed override. The value "200" is shown in a box, indicating the speed override factor.

Fig. 12.32 Time chart for changing the speed using the override function

**Example**

```
*
* No.15 Override program
*
```

| Rung | Contacts           | Coils                               | Description                                           |
|------|--------------------|-------------------------------------|-------------------------------------------------------|
| 371  | X33 (NO)           | [PLS M13]                           | <Pulsate override command>                            |
| 378  | M13 (NO), X0C (NC) | [MOV K200 D14], [MOV U0\ G1513 D14] | <Set override value (200%)><br><Write override value> |

### 12.5.3 Acceleration/deceleration time change function

The "acceleration/deceleration time change function" is used to change the acceleration/deceleration time during a speed change to a random value when carrying out the speed change by the "speed change function" and "override function". In a normal speed change (when the acceleration/deceleration time is not changed), the acceleration/deceleration time previously set in the parameters ([Pr.9], [Pr.10], and [Pr.25] to [Pr.30] values) is set in the positioning parameter data items [Da.3] and [Da.4], and control is carried out with that acceleration/deceleration time. However, by setting the new acceleration/deceleration time ([Cd.10], [Cd.11]) in the control data, and issuing an acceleration/deceleration time change enable command ([Cd.12] Acceleration/deceleration time change during speed change, enable/disable selection) to change the speed when the acceleration/deceleration time change is enabled, the speed will be changed with the new acceleration/deceleration time ([Cd.10], [Cd.11]).

The details shown below explain about the "acceleration/deceleration time change function".

- [1] Control details
- [2] Control precautions
- [3] Setting method

## [1] Control details

After setting the following two items, carry out the speed change to change the acceleration/deceleration time during the speed change.

- Set change value of the acceleration/deceleration time (" [Cd.10] New acceleration time value", " [Cd.11] New deceleration time value")
- Setting acceleration/deceleration time change to enable (" [Cd.12] Acceleration/deceleration time change during speed change, enable/disable selection")

The following drawing shows the operation during an acceleration/deceleration time change.

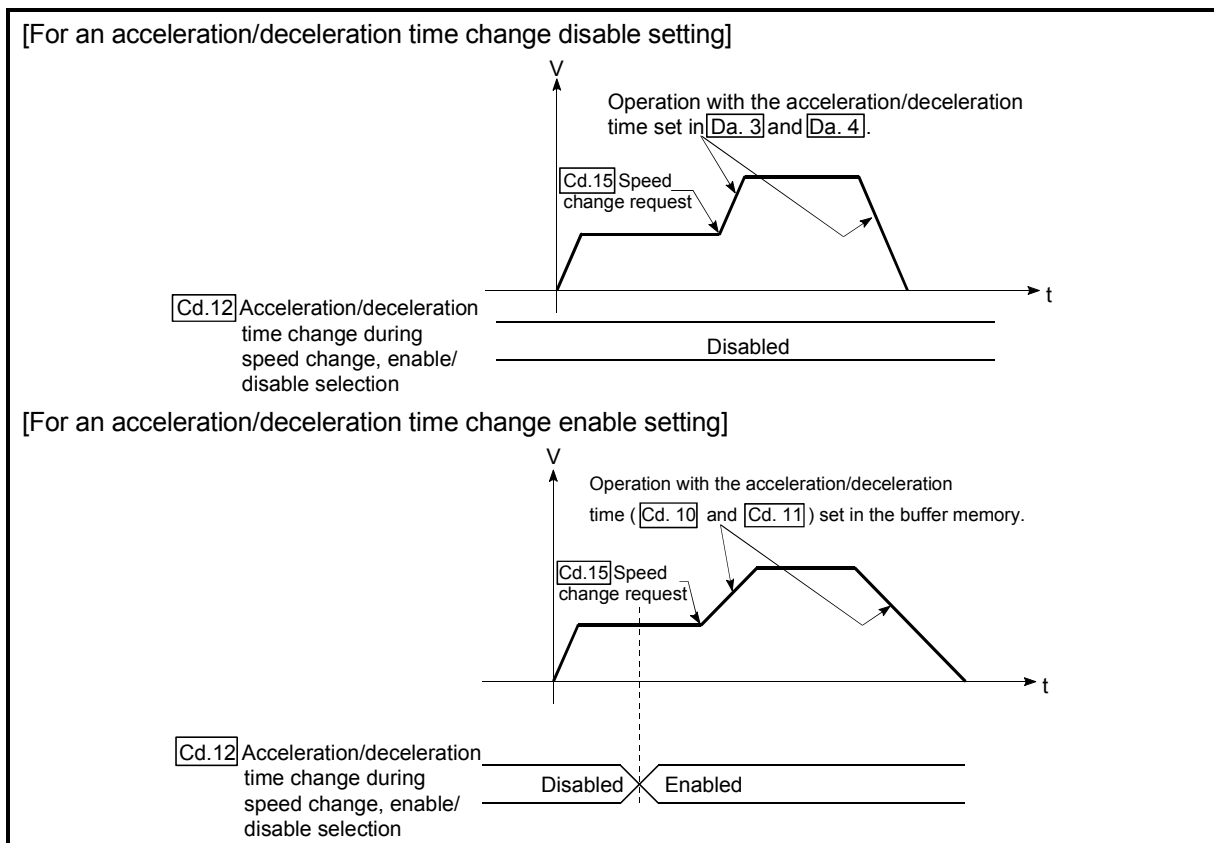
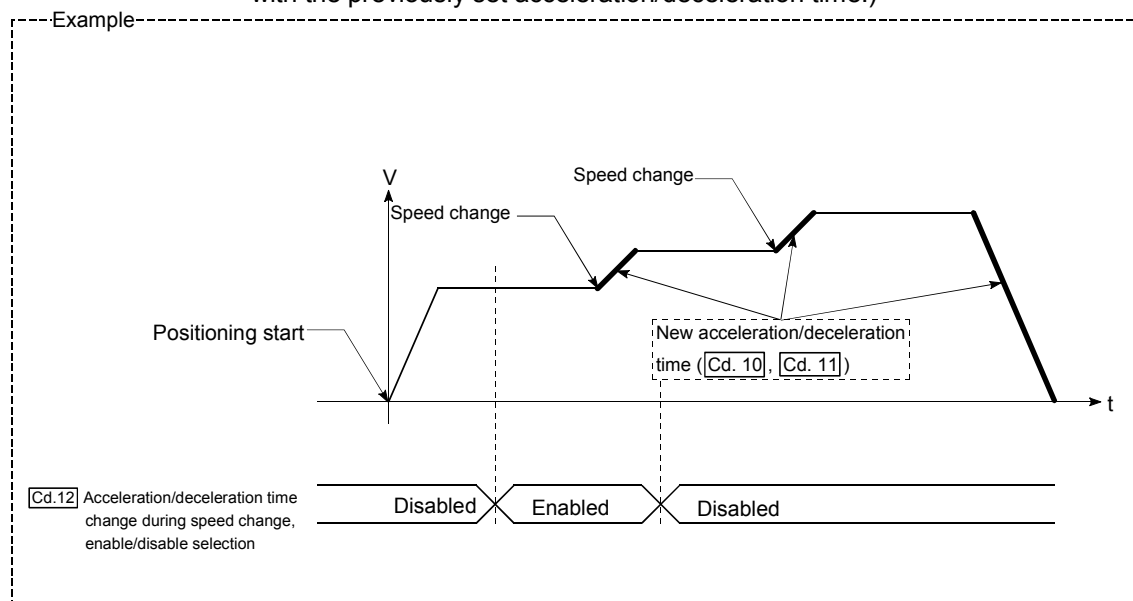


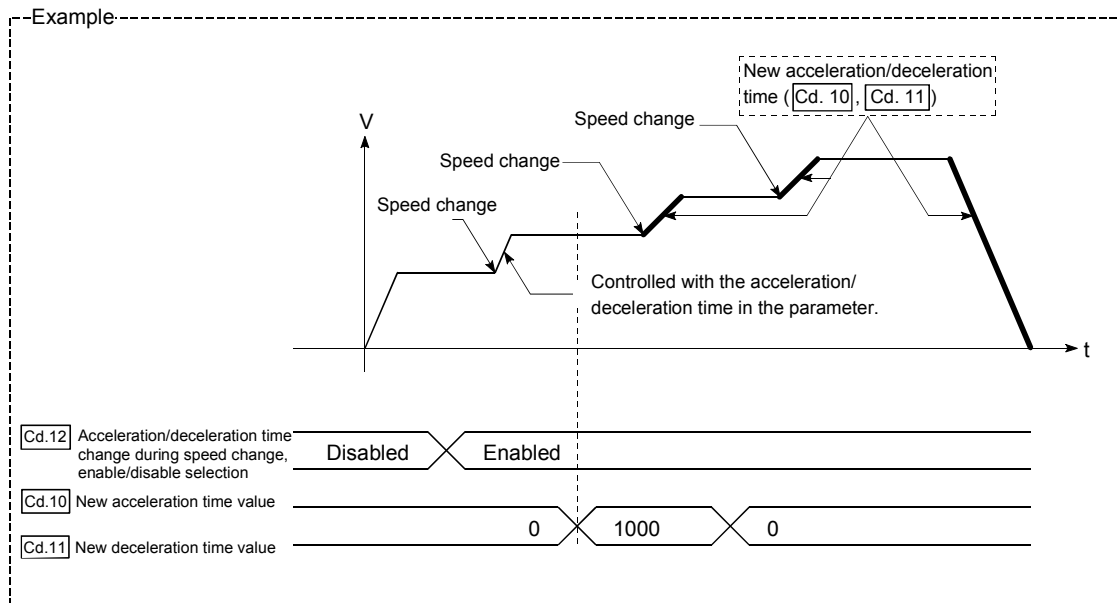
Fig. 12.33 Operation during an acceleration/deceleration time change

## [2] Control precautions

- (1) When "0" is set in " Cd.10 New acceleration time value" and " Cd.11 New deceleration time value", the acceleration/deceleration time will not be changed even if the speed is changed. In this case, the operation will be controlled at the acceleration/deceleration time previously set in the parameters.
- (2) The "new acceleration/deceleration time" is valid during execution of the positioning data for which the speed was changed. In continuous positioning control and continuous path control, the speed is changed and control is carried out with the previously set acceleration/deceleration time at the changeover to the next positioning data, even if the acceleration/deceleration time is changed to the "new acceleration/deceleration time ( Cd.10, Cd.11 )".
- (3) Even if the acceleration/deceleration time change is set to disable after the "new acceleration/deceleration time" is validated, the positioning data for which the "new acceleration/deceleration time" was validated will continue to be controlled with that value. (The next positioning data will be controlled with the previously set acceleration/deceleration time.)



- (4) If the "new acceleration/deceleration time" is set to "0" and the speed is changed after the "new acceleration/deceleration time" is validated, the operation will be controlled with the previous "new acceleration/deceleration time".



### POINT

If the speed is changed when an acceleration/deceleration change is enabled, the "new acceleration/deceleration time" will become the acceleration/deceleration time of the positioning data being executed. The "new acceleration/deceleration time" remains valid until the changeover to the next positioning data. (The automatic deceleration processing at the completion of the positioning will also be controlled by the "new acceleration/deceleration time".)

### [3] Setting method

To use the "acceleration/deceleration time change function", write the data shown in the following table to the QD75 using the sequence program.

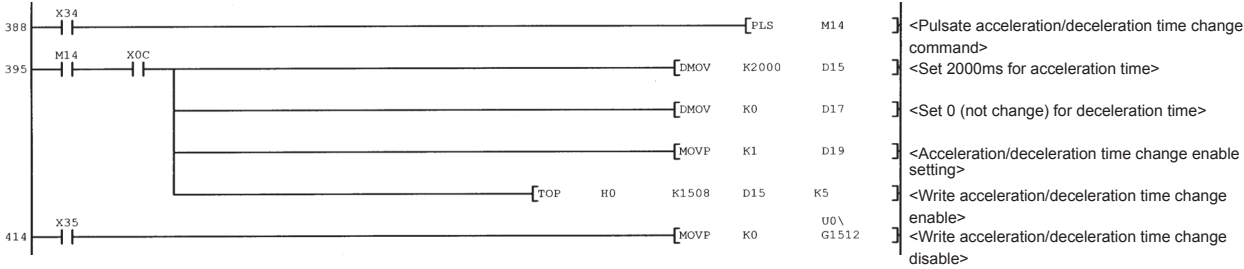
The set details are validated when a speed change is executed after the details are written to the QD75.

| Setting item                                                                               | Setting value | Setting details                                        | Buffer memory address |              |              |              |
|--------------------------------------------------------------------------------------------|---------------|--------------------------------------------------------|-----------------------|--------------|--------------|--------------|
|                                                                                            |               |                                                        | Axis 1                | Axis 2       | Axis 3       | Axis 4       |
| Cd. 10 New acceleration time value                                                         | →             | Set the new acceleration time.                         | 1508<br>1509          | 1608<br>1609 | 1708<br>1709 | 1808<br>1809 |
| Cd. 11 New deceleration time value                                                         | →             | Set the new deceleration time.                         | 1510<br>1511          | 1610<br>1611 | 1710<br>1711 | 1810<br>1811 |
| Cd. 12 Acceleration/deceleration time change during speed change, enable/disable selection | 1             | Set "1: Acceleration/deceleration time change enable". | 1512                  | 1612         | 1712         | 1812         |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

Example

\*  
\* No.16 Acceleration/deceleration time change program  
\*



### 12.5.4 Torque change function

The "torque change function" is used to change the torque limit value during torque limiting.

The torque limit value during torque limiting is normally the value set in the " [Pr.17] Torque limit setting value" that was previously set in the parameters.

However, by setting the new torque limit value in the axis control data " [Cd.22] New torque value", and writing it to the QD75, the torque generated by the servomotor during control can be limited with the new torque value.

(The " [Cd.22] New torque value" is validated when written to the QD75.)

The details shown below explain about the "torque change function".

- [1] Control details
- [2] Control precautions
- [3] Setting method

#### [1] Control details

The torque value of the axis control data can be changed at all times. The torque can be limited with a new torque value from the time the new torque value has been written to the QD75. (a torque change is made only during operation.) The torque limiting is not carried out from the time the power supply is turned ON to the time the PLC READY signal (Y0) is turned ON.

The torque setting range is from 0 to " [Pr.17] Torque limit setting value".

When the new torque value is 0, a torque change is considered not to be carried out.

The torque change range is 1 to " [Pr.17] Torque limit setting value".

The following drawing shows the torque change operation.

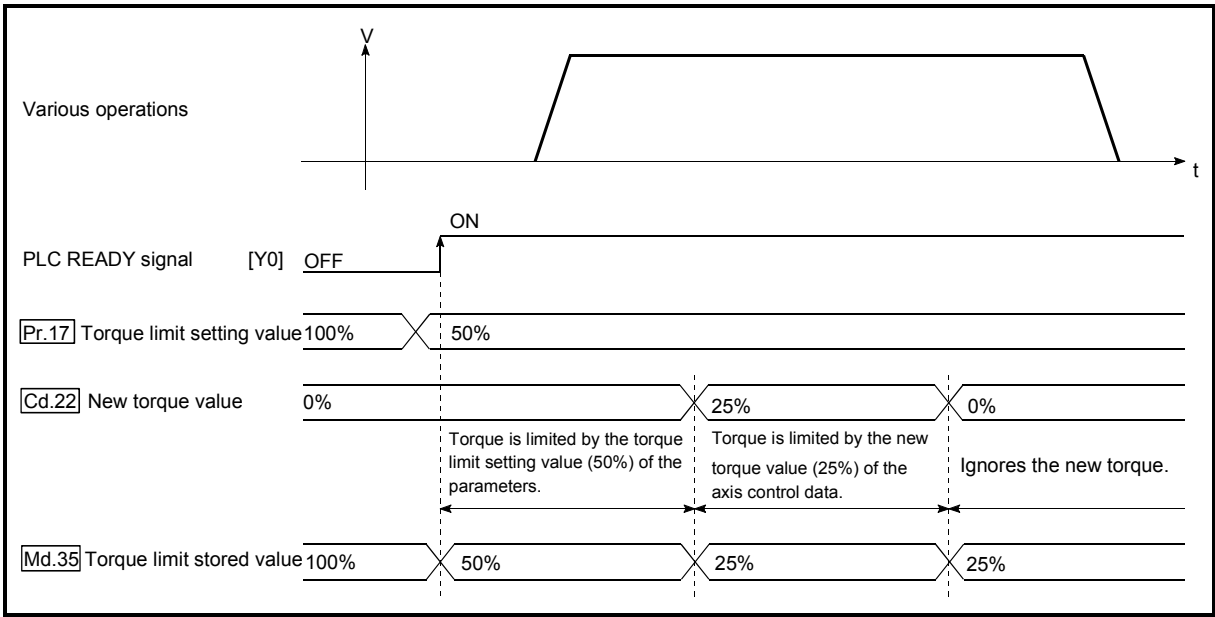


Fig. 12.34 Torque change operation



## [2] Control precautions

- (1) If a value besides "0" is set in the " Cd.22 New torque value", the torque generated by the servomotor will be limited by that value. To limit the torque with the value set in " Pr.17 Torque limit setting value", set the " Cd.22 New torque value" to "0".
- (2) The " Cd.22 New torque value" is validated when written to the QD75. (Note that it is not validated from the time the power supply is turned ON to the time the PLC READY signal (Y0) is turned ON.)
- (3) If the setting value is outside the setting range, an axis warning "Outside new torque value range" (warning code: 113) will occur and the torque will not be changed.
- (4) If the time to hold the new torque value is not more than 100ms, a torque change may not be executed.

## [3] Setting method

To use the "torque change function", write the data shown in the following table to the QD75 using the sequence program.

The set details are validated when written to the QD75.

| Setting item                                                                         | Setting value | Setting details                 | Buffer memory address |        |        |        |
|--------------------------------------------------------------------------------------|---------------|---------------------------------|-----------------------|--------|--------|--------|
|                                                                                      |               |                                 | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <span style="border: 1px solid black; padding: 0 2px;">Cd.22</span> New torque value | →             | Set the new torque limit value. | 1525                  | 1625   | 1725   | 1825   |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

## 12.5.5 Target position change function

The "target position change function" is a function to change a target position to a newly designated target position at any timing during the position control (1-axis linear control). A command speed can also be changed simultaneously.

The target position and command speed changed are set directly in the buffer memory, and the target position change is executed by turning ON " Cd.29 Target position change request flag".

The following shows the details of the "target position change function".

[1] Control details

[2] Control precautions

[3] Setting method

## [1] Control details

The following charts show the details of control of the target position change function.

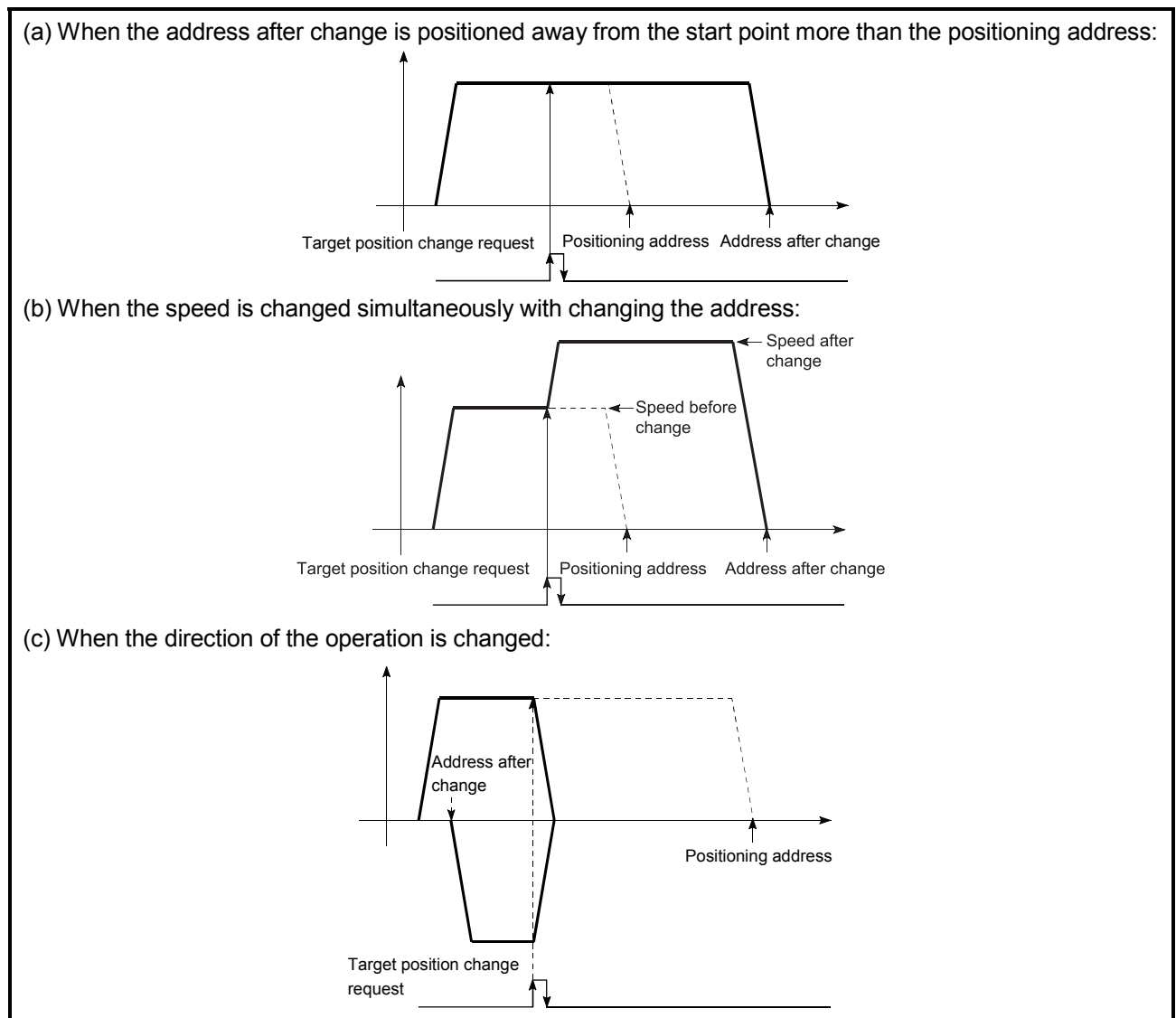


Fig. 12.35 Target position change operation

**[2] Control precautions**

- (1) If the positioning movement direction from the stop position to a new target position is reversed, stop the operation once and then position to the new target position. (Refer to Fig. 12.35 (c).)
- (2) If a command speed exceeding the speed limit value is set to change the command speed, the warning "Speed limit value over" (warning code: 501) will occur, and the new command speed will be the speed limit value. Also, if the command speed change disables the remaining distance to the target value from being assured, the warning "Insufficient remaining distance" (warning code: 509) will occur.
- (3) In the following cases, the target position change request is ignored, and the warning "Target position change not possible" (warning code: 518) will occur.
  - During interpolation control
  - A new target position value (address) is outside the software stroke limit range.
  - The axis is decelerating to a stop by a stop cause.
  - While the positioning data whose operation pattern is continuous path control is executed.
  - When the speed change 0 flag (**Md.31** Status: b10) is ON.
- (4) When a command speed is changed, the current speed is also changed. When the next positioning speed uses the current speed in the continuous positioning, the next positioning operation is carried out at the new speed value. When the speed is set with the next positioning data, that speed becomes the current speed and the operation is carried out at the current speed.
- (5) When a target position change request is given during automatic deceleration in position control, positioning control to a new position is exercised after the axis has stopped once if the moving direction is reversed. If the moving direction is not reversed, the axis is accelerated to the command speed again and positioned to the new position.
- (6) If the constant speed status is regained or the output is reversed by a target position change made while " **Md.48** Deceleration start flag" is ON, the deceleration start flag remains ON. (For details, refer to Section 12.7.8.)
- (7) Carrying out the target position change to the ABS linear 1 in degrees may carry out the positioning to the new target position after the operation decelerates to stop once, even the movement direction is not reversed.

**POINT**

When carrying out the target position change continuously, take an interval of 100ms or longer between the times of the target position changes. Also, take an interval of 100ms or longer when the speed change and override is carried out after changing the target position or the target position change is carried out after the speed change and override.

### [3] Setting method

The following table and chart show the example of a data setting and program used to change the target position of the axis 1 by the command from CPU module, respectively."

(In this example, the target position and command speed is changed to "300.0  $\mu$ m" and "10000.00 mm/min".)

- (1) The following data is set.  
(Referring to the starting time chart shown in item (2) below, carry out the setting with the program shown in item (3).)

| Setting item                                            | Setting value | Setting details                  | Buffer memory address |              |              |              |
|---------------------------------------------------------|---------------|----------------------------------|-----------------------|--------------|--------------|--------------|
|                                                         |               |                                  | Axis 1                | Axis 2       | Axis 3       | Axis 4       |
| <b>Cd.27</b> Target position change value (new address) | 3000          | Set the new address.             | 1534<br>1535          | 1634<br>1635 | 1734<br>1735 | 1834<br>1835 |
| <b>Cd.28</b> Target position change value (new speed)   | 1000000       | Set the new speed.               | 1536<br>1537          | 1636<br>1637 | 1736<br>1737 | 1836<br>1837 |
| <b>Cd.29</b> Target position change request flag        | 1             | Set "1: Carry out speed change". | 1538                  | 1638         | 1738         | 1838         |

Refer to Section 5.7 "List of control data" for details on the setting details.

- (2) The following shows the time chart for target position change.

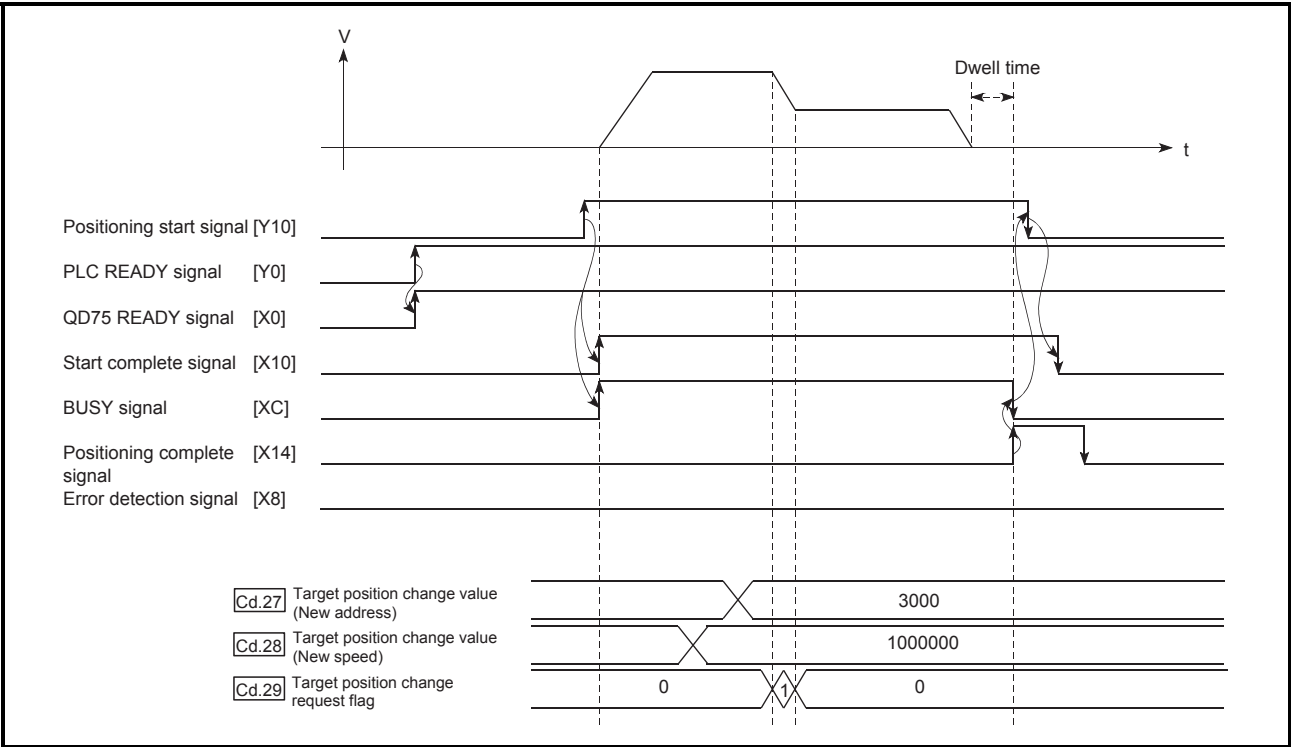
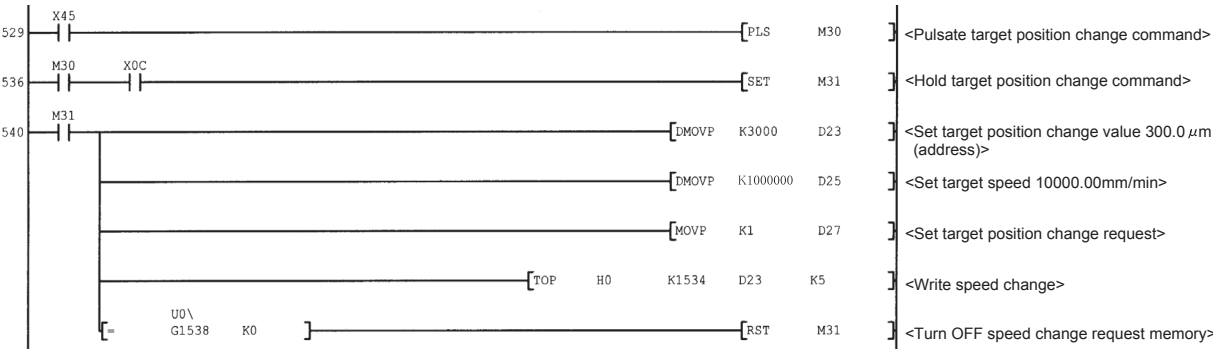


Fig. 12.36 Time chart for target position change from CPU module

(3) The following program is added to the control program, and written to the CPU module.

Example

\*  
\* No.21 Target position change program  
\*



## 12.6 Absolute position restoration function

### ⚠ CAUTION

An absolute position restoration by the positioning function may turn off the servo-on signal (servo off) for approximately 60ms + scan time, and the motor may run unexpectedly. If this causes a problem, provide an electromagnetic brake to lock the motor during absolute position restoration.

The "absolute position restoration function" is a function to restore the absolute position of the designated axis by the absolute position detection system. By this function, the OPR after power OFF such as instantaneous power interruption and emergency stop is not required, and the restoration operation at site can be carried out easily.

The details of the "absolute position restoration function" are described below.

- [1] Configuration and preparation of absolute position detection system
- [2] Outline of absolute position detection data communication
- [3] Absolute position signal transmission procedure
- [4] Control precautions

#### [1] Configuration and preparation of absolute position detection system

##### (1) Configuration

The following drawing shows the configuration of the absolute position detection system.

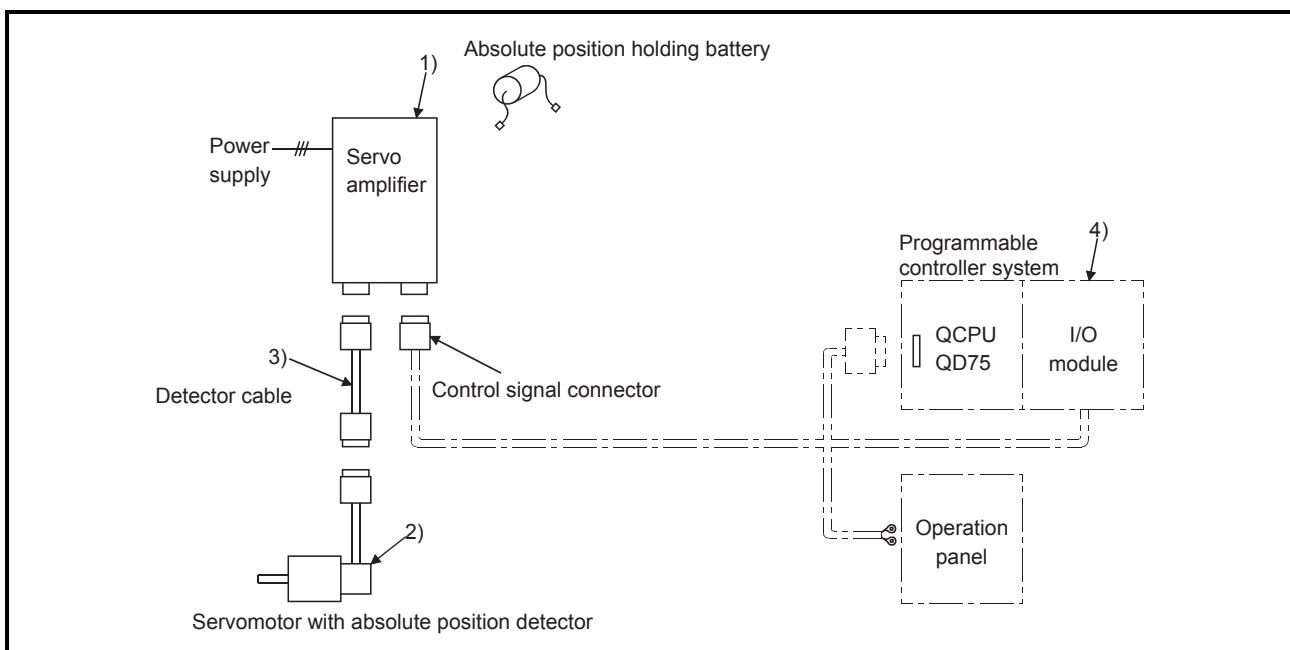


Fig. 12.37 Configuration of absolute position detection system

## (2) Preparation

Prepare the absolute position detection system taking care of the following.

| Component                         | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) Servo amplifier                | <ul style="list-style-type: none"> <li>• Use a Mitsubishi Electric General-Purpose AC Servo which has an absolute position detection function (absolute position data transference protocol) equivalent to that of MR-J3-□A).</li> <li>• Install the battery to the servo amplifier.</li> <li>• Validate the absolute position detection function of the servo amplifier.</li> </ul> <p>Refer to the servo amplifier manual for details.</p>             |
| 2) Servomotor                     | <ul style="list-style-type: none"> <li>• Use a servomotor with absolute position detector.</li> </ul> <p>Refer to the servomotor manual for details.</p>                                                                                                                                                                                                                                                                                                 |
| 3) Detector cable                 | <ul style="list-style-type: none"> <li>• Add a battery power connection cable (BAT/LG signal) for wiring the incremental detector cable.</li> </ul> <p>Refer to the cable operation manual for details.</p>                                                                                                                                                                                                                                              |
| 4) Programmable controller system | <ul style="list-style-type: none"> <li>• Carry out the transmission and receiving of the absolute position detection data by the I/O modules (input 3 points/output 3 points).</li> <li>• As input/output modules, use the "16 point input module" and "16 point output module".</li> <li>• Place the 3 points of input signals so that the □ □ is the same among the input number □ □0 to □ □F. The same for the 3 points of output signals.</li> </ul> |

## [2] Outline of absolute position detection data communication

As shown in Fig. 12.38 System block diagram, the detector comprises an encoder for detecting its position in one rotation in addition to the A, B, Z phase signal for position control in normal operation and a cumulative rotation counter for detecting the number of rotations.

The absolute position detection system detects the absolute position of the machine constantly and stores it with the backup of the battery irrespective of whether the programmable controller system power is turned ON/OFF. Thus, once the OP initial setting is carried out at the time of installation of the machine, the OPR is not required even when the power is later turned ON. The restoration can be carried out easily when an instantaneous power interruption or emergency stop occurs.

In addition, because the absolute position data is backed up by a super condenser built in the detector, the absolute position data will be stored for a specified time even if the cable is disconnected or broken.

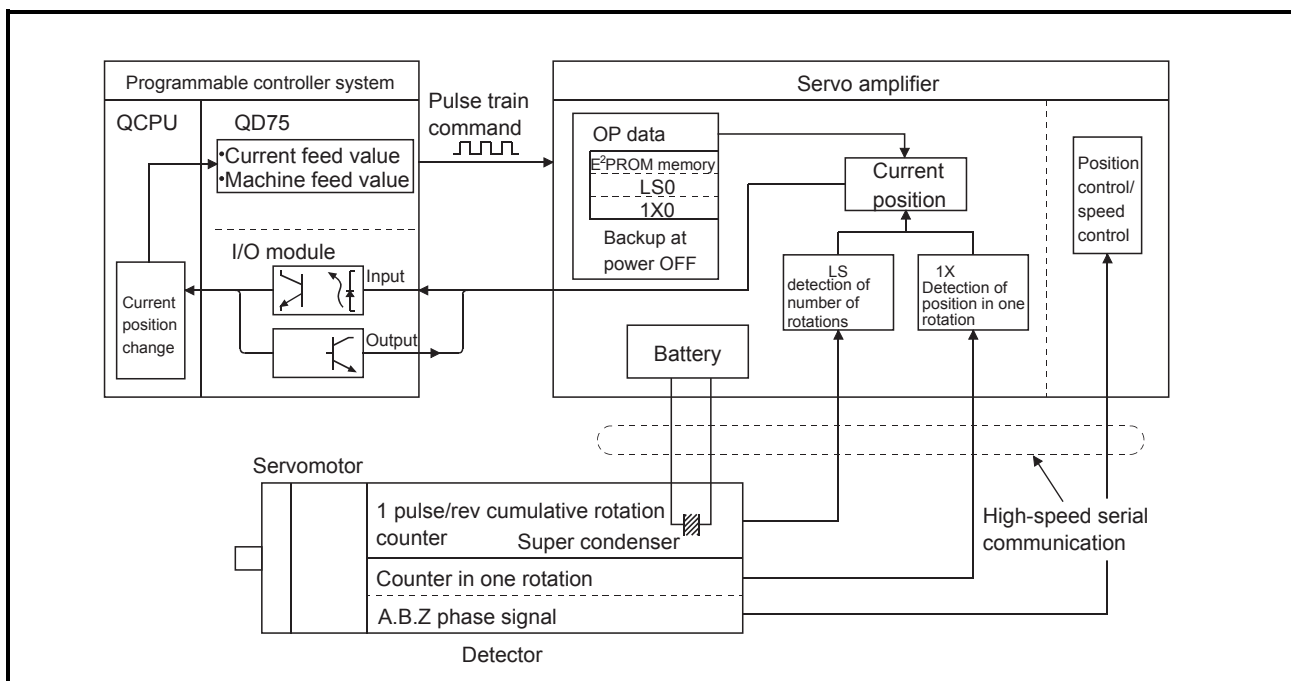


Fig. 12.38 System block diagram



### [3] Absolute position signal transmission procedure

(1) Figure 12.39 shows the outline of the absolute position signal transmission procedure between the servo amplifier and the programmable controller system (CPU module, QD75, I/O module).

Refer to the operation manual of the servo amplifier for details on the communication between the servo amplifier and the programmable controller system.

(2) Errors during communication

(a) If the system expires during communication, an error "ABS transmission time" (error code: 213) will occur.

(b) If the transmission data causes a sum error, an error "ABS transmission SUM" (error code: 214) will occur.

\*Refer to Section 15.3 "List of errors" for the remedial method to be taken when an error occurs.

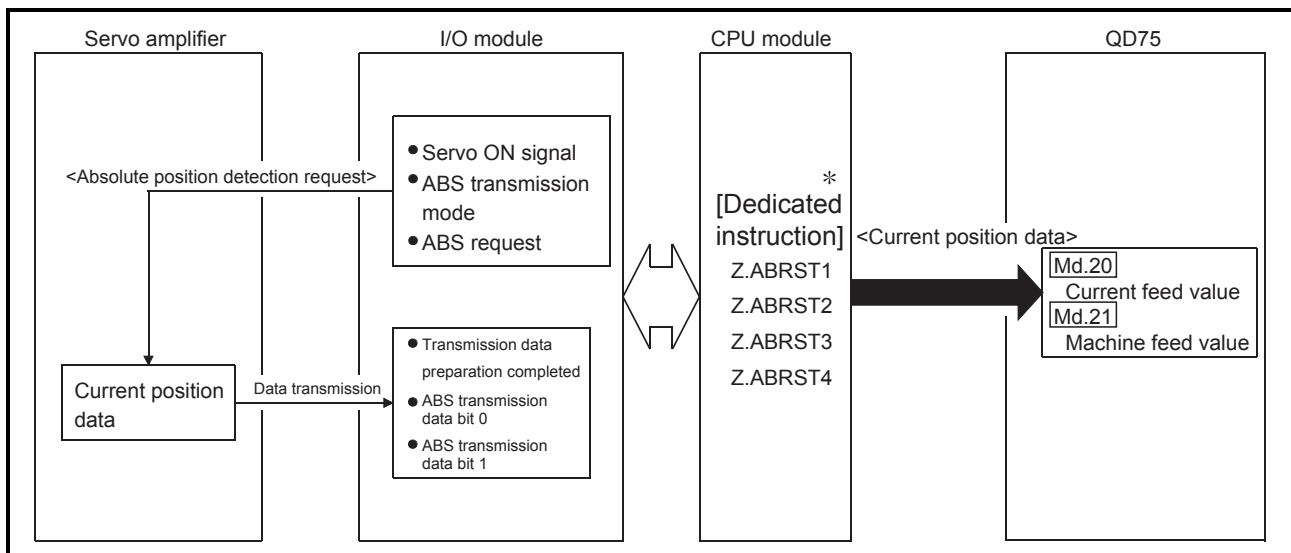
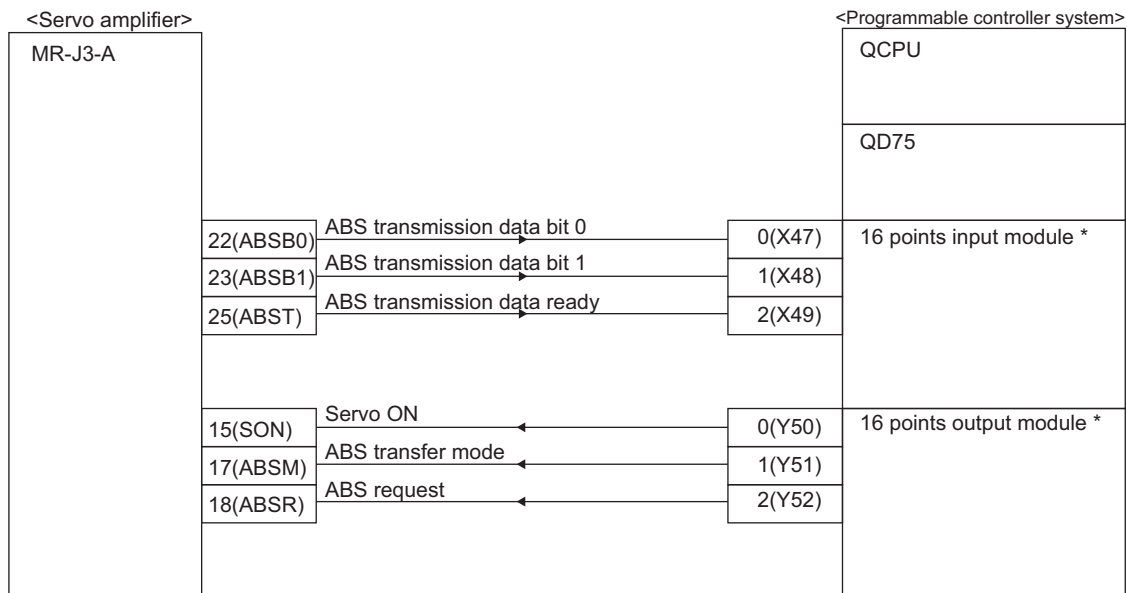


Fig. 12.39 Absolute position signal transmission procedure

\* Refer to CHAPTER 14 "DEDICATED INSTRUCTIONS" for details of the dedicated instructions.

## (3) Connection example

The following diagram shows the example of connection between the programmable controller system and the Mitsubishi Electric servo amplifier (MR-J3-A).



\*: The X and Y devices can be set arbitrarily with the program.

## Details of servo amplifier connector pins

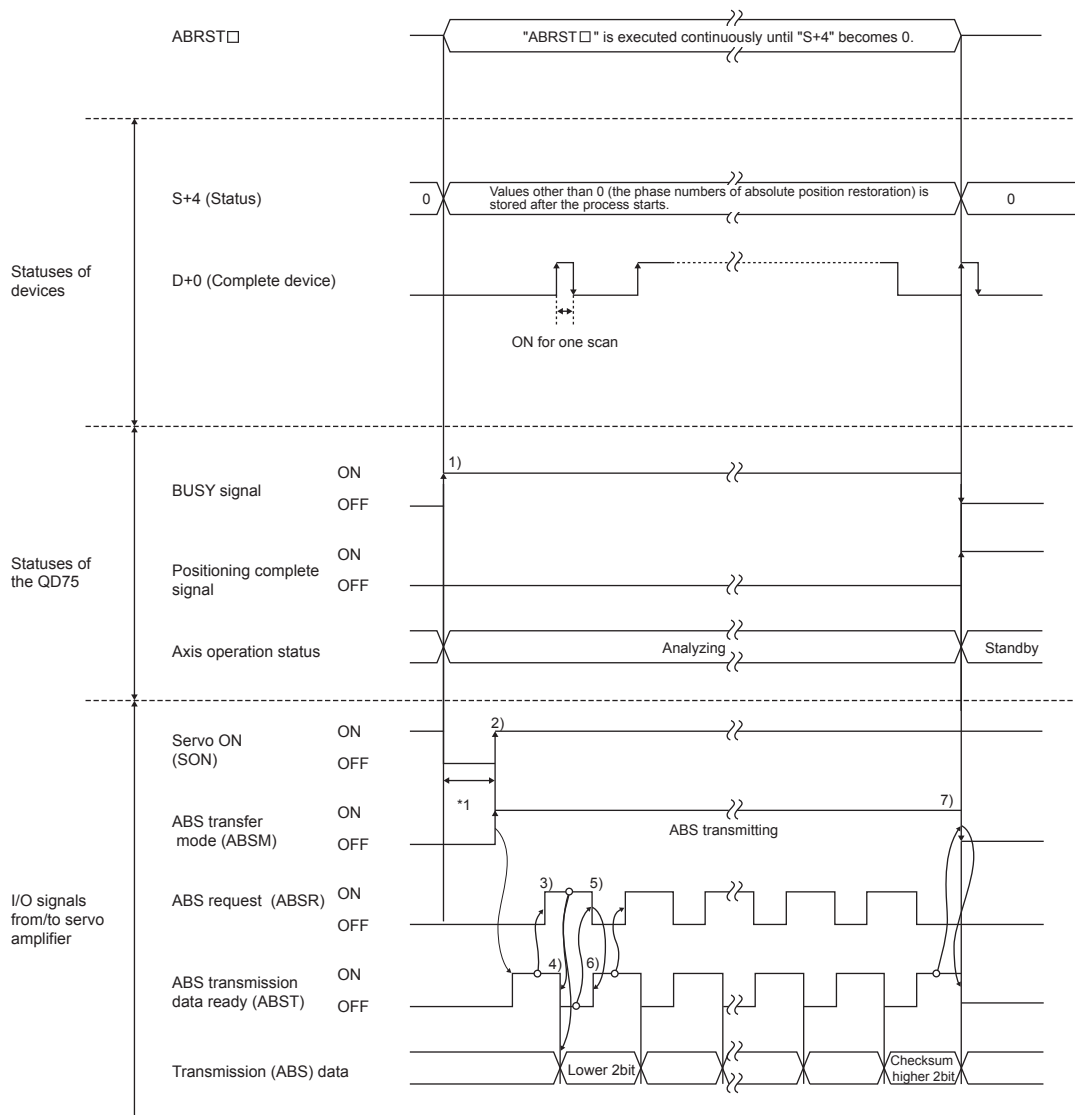
The following table shows the pins for setting the "absolute position detection system".

| Signal name                 | Abbreviation | Pin No. | Function and application                                                                                                                                                                                           |
|-----------------------------|--------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ABS transfer mode           | ABSM         | 17*     | While the ABSM is turned ON, the servo amplifier is in the ABS transfer mode, and the CN1-22 • 23 • 25 functions are as shown in this table.                                                                       |
| ABS request                 | ABSR         | 18*     | The ABSR is turned ON when ABS data is requested in the ABS transfer mode.                                                                                                                                         |
| ABS transmission data bit 0 | ABSB0        | 22      | This indicates the lower order bit among 2 bits of ABS data to be transferred from the servo amplifier to the programmable controller system in the ABS transfer mode. The ABSB0 turns ON when this signal occurs. |
| ABS transmission data bit 1 | ABSB1        | 23      | This indicates the upper order bit among 2 bits of ABS data to be transferred from the servo amplifier to the programmable controller system in the ABS transfer mode. The ABSB1 turns ON when this signal occurs. |
| ABS Transmission data ready | ABST         | 25      | This indicates the ABS transmission data preparation completed in the ABS transfer mode. When the preparation is completed, the ABST is turned ON.                                                                 |

\* When "use in the absolute position detection system" is selected in the parameter No. PA03, the pin 17 is the ABS transfer mode (ABSM), and the pin 18 is the ABS request (ABSR). The signals do not return to the original signal even after the data transfer is completed.

For details on signals of the pin 17 and 18 when the ABS transfer mode is off and I/O interface, refer to the manual of the servo amplifier (MR-J3-A).

The following drawing shows an operation when data is transferred to the servo amplifier.



\*1: Approx. 60ms + scan time

Fig. 12.40 Signal state during the absolute position detection

- 1) The BUSY signal is turned ON and the axis operation status is set to "Analyzing" by the dedicated instruction "ABRST□".  
At this time, the signal is controlled to turn OFF the servo amplifier.  
The servo is powered OFF in 60ms + scan time.
- 2) When the servo is turned ON, the ABS transfer mode is simultaneously turned ON.  
After receiving the ABS transfer mode, detecting the absolute position and calculating the absolute position, the servo amplifier will turn ON the ABS transmission data ready (ABST) and answer back to QD75 notifying that the send data is ready.
- 3) After recognizing that the ABS transmission data ready (ABST) turned ON, QD75 turns ON the ABS request (ABSR).

- 4) The servo amplifier outputs the ABS lower 2 bits and ABS transmission data ready (ABST) OFF by the ABS request (ABSR).
- 5) After recognizing that the ABS transmission data ready (ABST) turned OFF (the ABS2bit data is output), QD75 reads the lower 2bits of ABS and turns OFF the ABS request (ABSR).
- 6) The servo amplifier turns ON the ABS transmission data ready (ABST) and prepares for the next transmission. After that, procedures 3) to 6) are repeated until the data corresponding to 32bits and the checksum corresponding to 6bits are sent.
- 7) After the sum check, the QD75 turns OFF the ABS transfer mode (ABSM). If the ABS transfer mode (ABSM) is turned OFF during the data transmission, the ABS transfer mode will be interrupted.

#### [4] Control precautions

- (1) When an absolute position detection system is constructed, absolute position restoration must be made at least once after power supply on or resetting. Also, the servo amplifier does not switch on unless the absolute position restoration is completed.
- (2) For use of positioning in an absolute position detection system, the following controls cannot be carried out:
  - Limitless-feed control which exercises control only in the fixed direction, e.g. turntable
  - Control where the movement amount from the OP address exceeds the range of the following conditions 1 and 2

When performing positioning in an absolute position detection system, use it in the range which satisfies Conditions 1 and 2 given below.

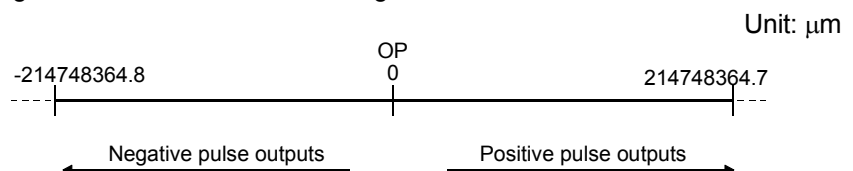
In the range which does not satisfy Conditions 1 and 2, positioning cannot be used in the absolute position detection system since the normal current value cannot be provided during absolute position restoration.

#### Condition 1. Number of output pulses

- a) Indicates the number of pulses that may be output to a servo amplifier when an axis is to be positioned from the OP in an absolute position detection system. In the absolute position detection system, pulses within the range of the following expression can be output to the servo amplifier.

$$[-32768 \times (\text{number of feedback pulses})] \leq [\text{number of output pulses}] \leq [32768 \times (\text{number of feedback pulses}) - 1]$$

- b) Pulses are positive in the address increasing direction from the OP, and negative in the address decreasing direction from the OP.



- c) The number of output pulses varies with the number of feedback pulses as indicated below:

- When the number of feedback pulses is 8192 (pulse):  
-268435456 (pulse) to 268435455 (pulse)
- When the number of feedback pulses is 16384 (pulse):  
-536870912 (pulse) to 536870911 (pulse)

When the electronic gear of the servo amplifier is used, the electronic gear ratio must be considered. The actual range of the number of output pulses is the value of the range of output pulses above multiplied by the inverse number of the electronic gear ratio.

#### Condition 2. Positioning address

- a) The following positioning addresses can be specified on the QD75:

- When the unit is mm: -214748364.8 ( $\mu\text{m}$ ) to 214748364.7 ( $\mu\text{m}$ )
- When the unit is inch: -21474.83648 (inch) to 21474.83647 (inch)
- When the unit is pulse: -2147483648 (pulse) to 2147483647 (pulse)
- When the unit is degree:  $0^\circ$  to  $359.99999^\circ$

### [Calculation of positioning address and concept of absolute position detection system]

Use the following expression to calculate the positioning address.

$$(\text{Positioning address}) = (\text{movement amount per pulse}) \times (\text{number of output pulses}) + (\text{OP address}) \dots\dots\dots \text{Expression 1}$$

#### 1. Concept for the unit of mm, inch or pulse

The range which satisfies Conditions 1 and 2 can be used as the positioning address of the absolute position detection system.

The range which does not satisfy Conditions 1 and 2 cannot be used as the positioning address of the absolute position detection system.

The concepts of the positioning addresses in mm, inch and pulse are identical, the following examples provide those of the mm unit.

#### Example 1.

(1) There are the following conditions to calculate the positioning address:

- Movement amount per pulse: 0.1 (μm)
- OP address: 0.0 (μm)
- Feedback pulses = 8192 (pulse)

(2) Calculate the upper/lower limit values of the positioning address which can be specified from the output pulse count using range in Condition 1 and the positioning address calculation expression (Expression 1).

- Lower limit value of positioning address (the number of negative pulses in Condition 1 is used for calculation)

$$\begin{aligned} (\text{Positioning address}) &= (\text{Movement amount per pulse}) \\ &\quad \times (\text{number of output pulses}) + (\text{OP address}) \\ &= 0.1 \times (-268435456) + 0.0 \\ &= -26843545.6(\mu\text{m}) \end{aligned}$$

- Upper limit value of positioning address (the number of positive pulses in Condition 1 is used for calculation)

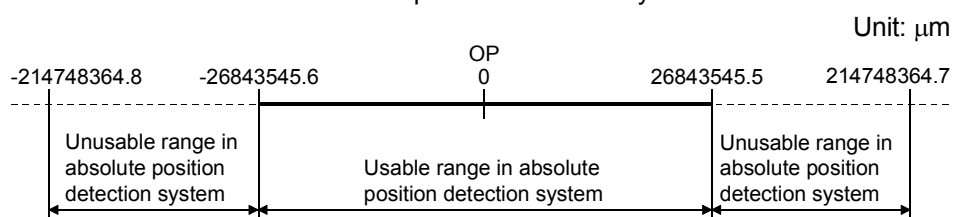
$$\begin{aligned} (\text{Positioning address}) &= (\text{Movement amount per pulse}) \\ &\quad \times (\text{number of output pulses}) + (\text{OP address}) \\ &= 0.1 \times (268435456) + 0.0 \\ &= 26843545.5(\mu\text{m}) \end{aligned}$$

(3) The upper/lower limit values of the positioning address calculated are within the range of Condition 2.

Hence, the positioning range [-26843545.6 (μm) to 26843545.5 (μm)]

calculated in Condition 1 can be used in the absolute position detection system.

Positioning outside of the range -26843545.6 (μm) to 26843545.5 (μm) cannot be used in the absolute position detection system.



**Example 2.**

(1) Using Expression 1, calculate the positioning address which can be specified in the system where the OP address in Example 1 is 214740000.0 (μm).

- Lower limit value of positioning address

$$\begin{aligned} \text{(Positioning address)} &= 0.1 \times (-268435456) + 214740000.0 \\ &= 187896454.4 \text{ (}\mu\text{m)} \end{aligned}$$

- Upper limit value of positioning address

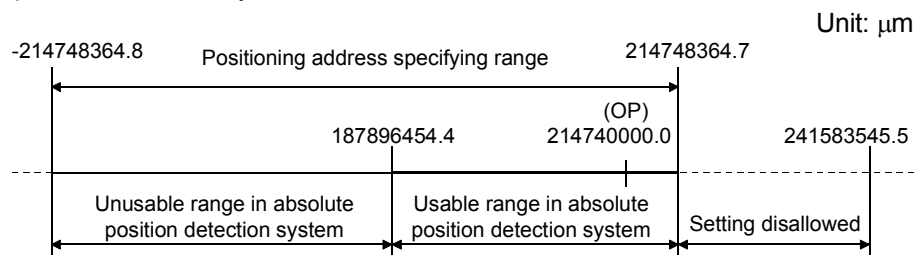
$$\begin{aligned} \text{(Positioning address)} &= 0.1 \times 268435455 + 214740000.0 \\ &= 241583545.5 \text{ (}\mu\text{m)} \end{aligned}$$

(2) Since the lower limit value of the positioning address calculated is within the range of Condition 2, the calculated address 187896454.4 (μm) is the lower limit for positioning in the absolute position detection system.

Since the upper limit of the positioning address calculated is outside of the range of Condition 2, the upper limit value 214748364.7 (μm) is the upper limit for positioning in the absolute position detection system.

In the absolute position detection system, use the positioning address within the range 187896454.4 (μm) to 214748364.7 (μm).

Positioning in excess of 187896454.4 (μm) cannot be used in the absolute position detection system.

**Example 3.**

(1) There are the following conditions to calculate the positioning address:

- Movement amount per pulse: 0.9 (μm)
- OP address: 0.0 (μm)
- Feedback pulses = 8192 (pulse)

(2) Calculate the positioning address from the output pulse count using range in Condition 1 and the positioning address calculation expression (Expression 1).

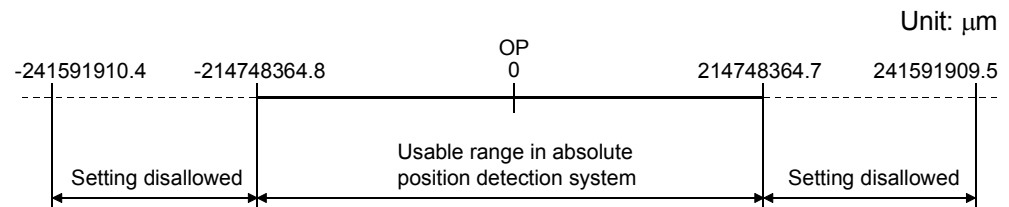
- Lower limit value of positioning address (the number of negative pulses in Condition 1 is used for calculation)

$$\begin{aligned} \text{(Positioning address)} &= (\text{Movement amount per pulse}) \\ &\quad \times (\text{number of output pulses}) + (\text{OP address}) \\ &= 0.9 \times (-268435456) + 0.0 \\ &= -241591910.4 \text{ (}\mu\text{m)} \end{aligned}$$

- Upper limit value of positioning address (the number of positive pulses in Condition 1 is used for calculation)

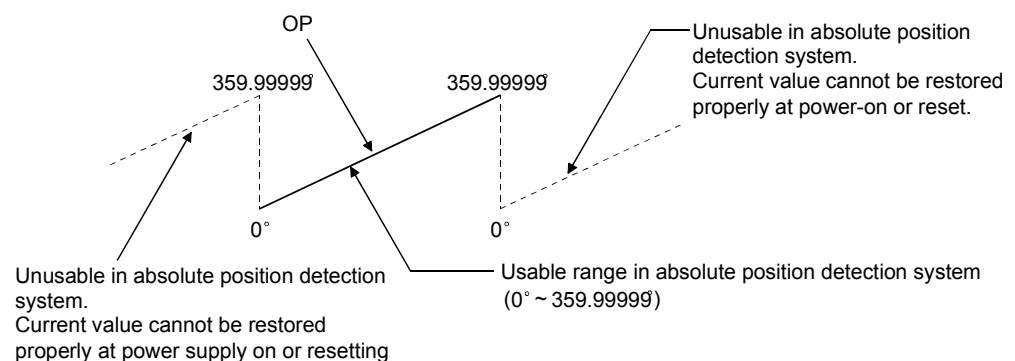
$$\begin{aligned} \text{(Positioning address)} &= (\text{Movement amount per pulse}) \\ &\quad \times (\text{number of output pulses}) + (\text{OP address}) \\ &= 0.9 \times 268435455 + 0.0 \\ &= 241591909.5 \text{ (}\mu\text{m)} \end{aligned}$$

- (3) Since the upper/lower limit values of the positioning address calculated are outside of the range of Condition 2, use the positioning address within the positioning range of Condition 2 (-214748364.8( $\mu\text{m}$ ) to 214748364.7( $\mu\text{m}$ )).



## 2. Concept for the unit of degree

- The positioning address is within the range  $0^\circ$  to  $359.99999^\circ$  at the position to which a machine OPR was made.  
When the OP position is not  $0^\circ$ , the range is also  $0^\circ$  to  $359.99999^\circ$ .
- For positioning in the same direction, control from maximum to minimum (for address increase:  $359.99999^\circ$  to  $0^\circ$ /for address decrease:  $0^\circ$  to  $359.99999^\circ$ ) cannot be exercised. (See below)



- When positioning is to be used in the absolute position detection system, set the upper/lower limit values of the software stroke limit within the range  $0^\circ$  to  $359.99999^\circ$ .



## 12.7 Other functions

Other functions include the "step function", "skip function", "M code output function", "teaching function", "command in-position function", "acceleration/deceleration processing function", "pre-reading start function", "deceleration start flag function" and "stop command processing for deceleration stop function". Each function is executed by parameter setting or sequence program creation and writing.

### 12.7.1 Step function

The "step function" is used to confirm each operation of the positioning control one by one.

It is used in debugging work for major positioning control, etc.

A positioning operation in which a "step function" is used is called a "step operation". In step operations, the timing for stopping the control can be set. (This is called the "step mode".) Control stopped by a step operation can be continued by setting "step continue" (to continue the control)" in the "step start information".

The details shown below explain about the "step function".

- [1] Relation between the step function and various controls
- [2] Step mode
- [3] Step start information
- [4] Using the step operation
- [5] Control details
- [6] Control precautions
- [7] Setting method

#### [1] Relation between the step function and various controls

The following table shows the relation between the "step function" and various controls.

| Control type              |                                                                      |                                                 | Step function | Step applicability          |
|---------------------------|----------------------------------------------------------------------|-------------------------------------------------|---------------|-----------------------------|
| OPR control               | Machine OPR control                                                  |                                                 | ×             | Step operation not possible |
|                           | Fast OPR control                                                     |                                                 | ×             |                             |
| Major positioning control | Position control                                                     | 1-axis linear control                           | ○             | Step operation possible     |
|                           |                                                                      | 2- to 4-axis linear interpolation control       | ○             |                             |
|                           |                                                                      | 1-axis fixed-feed control                       | ○             |                             |
|                           |                                                                      | 2- to 4-axis fixed-feed control (interpolation) | ○             |                             |
|                           |                                                                      | 2-axis circular interpolation control           | ○             |                             |
|                           |                                                                      | 3-axis helical interpolation control            | ○             |                             |
|                           | 1- to 4-axis Speed control                                           |                                                 | ×             | Step operation not possible |
|                           | Speed-position switching control<br>Position-speed switching control |                                                 | ○             | Step operation possible     |
|                           | Other control                                                        | Current value changing                          | ○             |                             |
|                           |                                                                      | JUMP instruction, NOP instruction, LOOP to LEND | ×             | Step operation not possible |
| Manual control            | JOG operation, Inching operation                                     |                                                 | ×             | Step operation not possible |
|                           | Manual pulse generator operation                                     |                                                 | ×             |                             |

○ : Set when required. × : Setting not possible

**[2] Step mode**

In step operations, the timing for stopping the control can be set. This is called the "step mode". (The "step mode" is set in the control data " [Cd.34](#) Step mode".)

The following shows the two types of "step mode" functions.

**(1) Deceleration unit step**

The operation stops at positioning data requiring automatic deceleration. (A normal operation will be carried out until the positioning data requiring automatic deceleration is found. Once found, that positioning data will be executed, and the operation will then automatically decelerate and stop.)

**(2) Data No. unit step**

The operation automatically decelerates and stops for each positioning data. (Even in continuous path control, an automatic deceleration and stop will be forcibly carried out.)

**[3] Step start information**

Control stopped by a step operation can be continued by setting "step continue" (to continue the control) in the "step start information". (The "step start information" is set in the control data " [Cd.36](#) Step start information".)

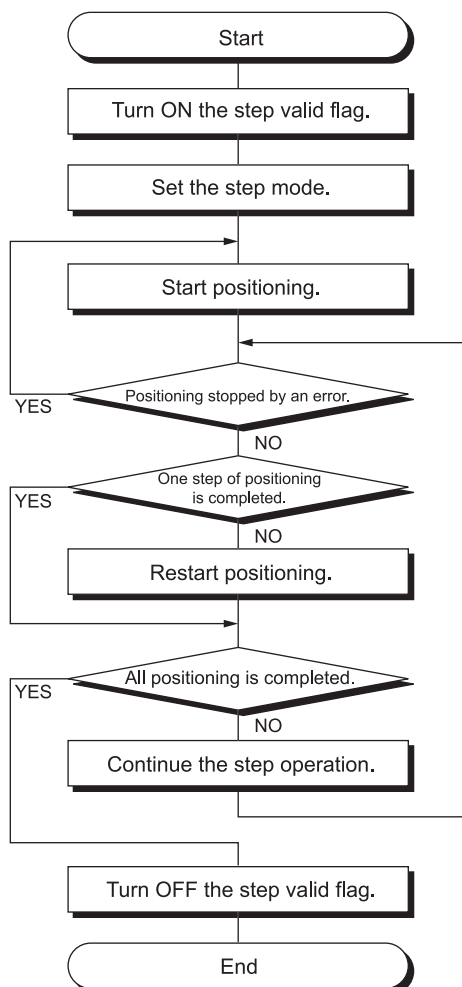
The following table shows the results of starts using the "step start information" during step operation.

| Stop status in the step operation      | <a href="#">Md.26</a> Axis operation status | <a href="#">Cd.36</a> Step start information | Step start results                     |
|----------------------------------------|---------------------------------------------|----------------------------------------------|----------------------------------------|
| 1 step of positioning stopped normally | Step standby                                | 1: Step continue                             | The next positioning data is executed. |

The warnings "Step not possible" (warning code: 511) will occur and the step operation will not be continued if the " [Md.26](#) Axis operation status" is other than "Step standby" or the step valid flag is OFF when step start information is set.

#### [4] Using the step operation

The following shows the procedure for checking positioning data using the step operation.



Write "1" (carry out step operation) in "[Cd.35]Step valid flag".

Set in "[Cd.34]Step mode".

Write "1" (restart) to "[Cd.6]Restart command" and check whether the positioning data operates normally.

Write "1" (step continue) to "[Cd.36]Step start information", and check whether the next positioning data operates normally.

Write "0" (carry out no step operation) in "[Cd.35]Step valid flag".

## [5] Control details

- (1) The following drawing shows a step operation during a "deceleration unit step".

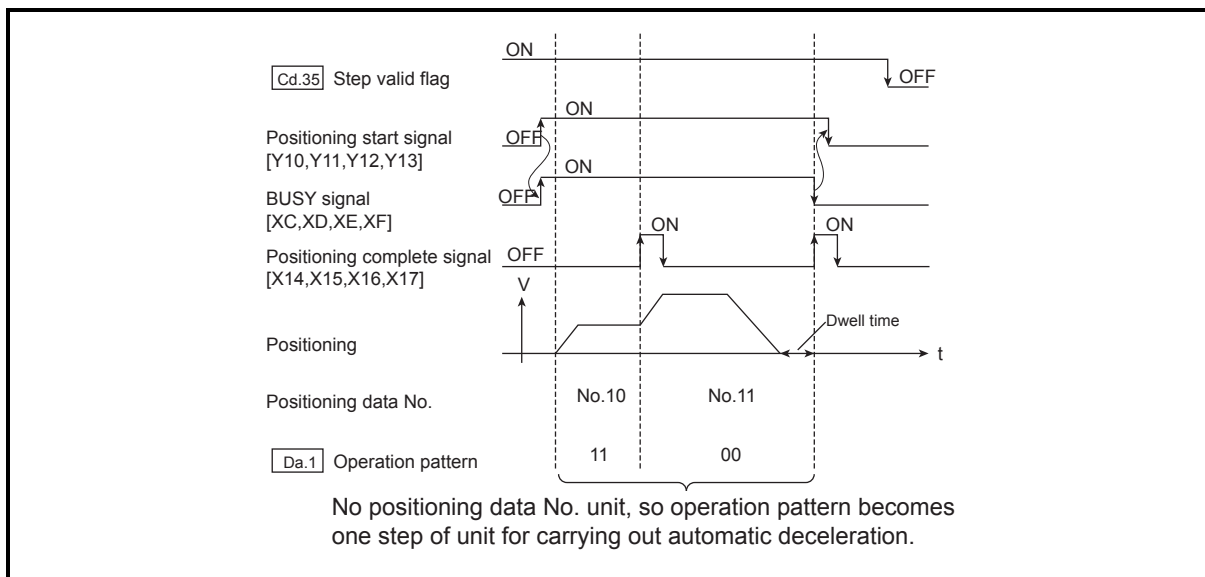


Fig. 12.41 Operation during step execution by deceleration unit step

- (2) The following drawing shows a step operation during a "data No. unit step".

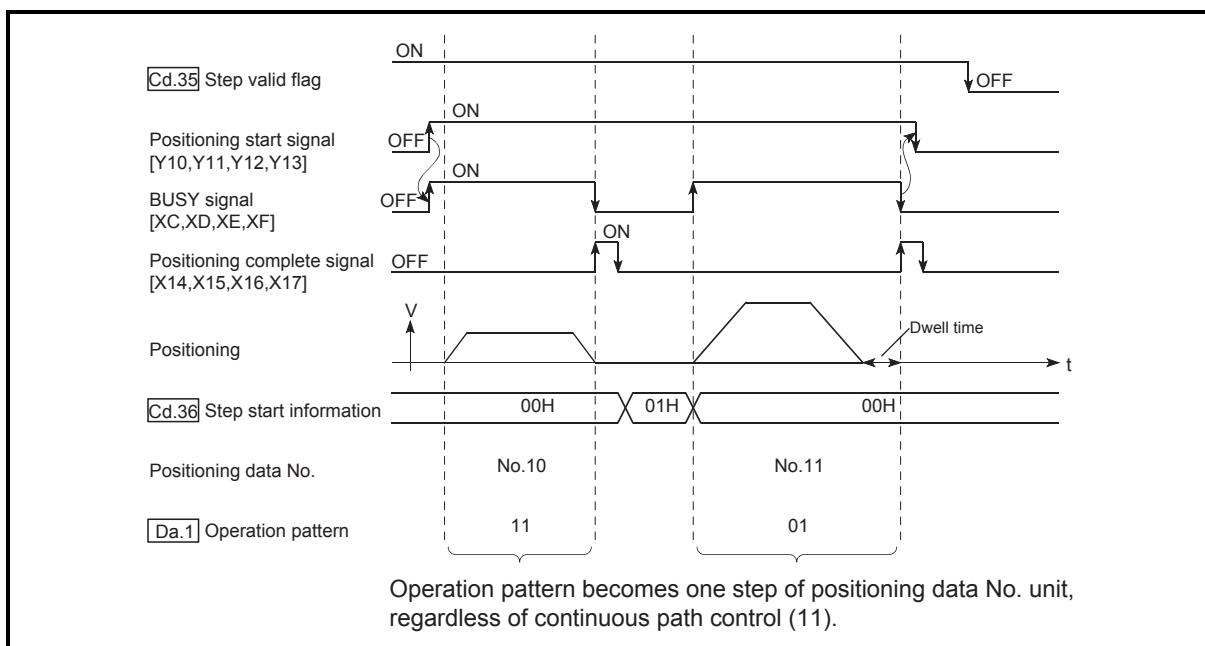


Fig. 12.42 Operation during step execution positioning data No. unit step

**[6] Control precautions**

- (1) When step operation is carried out using interpolation control positioning data, the step function settings are carried out for the reference axis.
- (2) When the step valid flag is ON, the step operation will start from the beginning if the positioning start signal is turned ON while " [Md.26](#) Axis operation status" is "Step standby". (The step operation will be carried out from the positioning data set in " [Cd.3](#) Positioning start No.".)

**[7] Setting method**

To use the "step function", write the data shown in the following table to the QD75 using the sequence program. Refer to section [4] "Using the step operation" for the timing of the settings.

The set details are validated when written to the QD75.

| Setting item          |                        | Setting value | Setting details                                             | Buffer memory address |        |        |        |
|-----------------------|------------------------|---------------|-------------------------------------------------------------|-----------------------|--------|--------|--------|
|                       |                        |               |                                                             | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <a href="#">Cd.34</a> | Step mode              | →             | Set "0: Deceleration unit step" or "1: Data No. unit step". | 1544                  | 1644   | 1744   | 1844   |
| <a href="#">Cd.35</a> | Step valid flag        | 1             | Set "1: Carry out step operation".                          | 1545                  | 1645   | 1745   | 1845   |
| <a href="#">Cd.36</a> | Step start information | →             | Set "1: Step continue", depending on the stop status.       | 1546                  | 1646   | 1746   | 1846   |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

## 12.7.2 Skip function

The "skip function" is used to stop (deceleration stop) the control of the positioning data being executed at the time of the skip signal input, and execute the next positioning data.

A skip is executed by a skip command ([Cd.37] Skip command) or external command signal.

The "skip function" can be used during control in which positioning data is used.

The details shown below explain about the "skip function".

[1] Control details

[2] Control precautions

[3] Setting the skip function from the CPU module

[4] Setting the skip function using an external command signal

### [1] Control details

The following drawing shows the skip function operation.

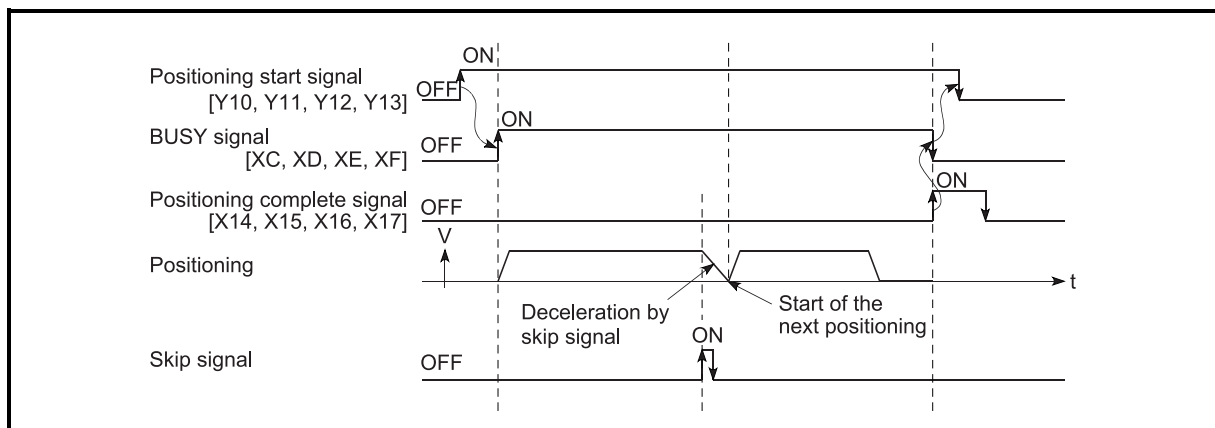


Fig. 12.43 Operation when a skip signal is input during positioning control

### [2] Control precautions

- (1) If the skip signal is turned ON at the last of an operation, a deceleration stop will occur and the operation will be terminated.
- (2) When a control is skipped (when the skip signal is turned ON during a control), the positioning complete signals (X14, X15, X16, X17) will not turn ON.
- (3) When the skip signal is turned ON during the dwell time, the remaining dwell time will be ignored, and the next positioning data will be executed.
- (4) When a control is skipped during interpolation control, the reference axis skip signal is turned ON. When the reference axis skip signal is turned ON, a deceleration stop will be carried out for every axis, and the next reference axis positioning data will be executed.
- (5) The M code ON signals (X4, X5, X6, X7) will not turn ON when the M code output is set to the AFTER mode (when "1: AFTER mode" is set in " [Pr.18] M code ON signal output timing").  
(In this case, the M code will not be stored in " [Md.25] Valid M code".)
- (6) The skip cannot be carried out by the speed control and position-speed switching control.
- (7) If the skip signal is turned ON with the M code signal turned ON, the transition to the next data is not carried out until the M code signal is turned OFF.

### [3] Setting the skip function from the CPU module

The following shows the settings and sequence program example for skipping the control being executed in axis 1 with a command from the CPU module.

(1) Set the following data.

(The setting is carried out using the sequence program shown below in section (2)).

| Setting item              | Setting value | Setting details        | Buffer memory address |        |        |        |
|---------------------------|---------------|------------------------|-----------------------|--------|--------|--------|
|                           |               |                        | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <b>Cd.37</b> Skip command | 1             | Set "1: Skip request". | 1547                  | 1647   | 1747   | 1847   |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

(2) Add the following sequence program to the control program, and write it to the CPU module.

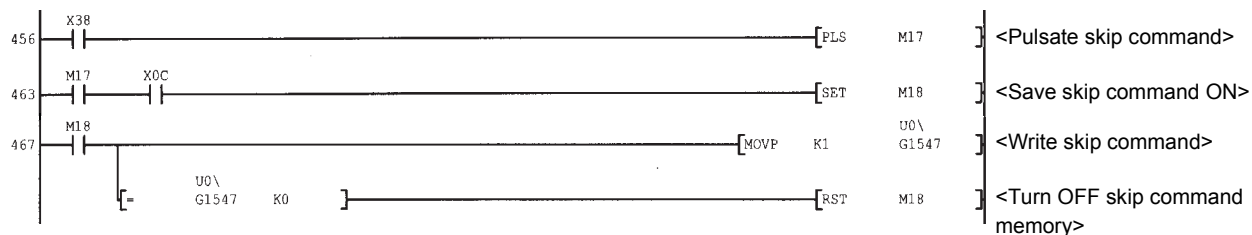
- 1) When the "skip command" is input, the value "1" (skip request) set in "**Cd.37** Skip command" is written to the QD75 buffer memory (1547).

#### Example

\*

\* No.19 Skip program

\*



#### [4] Setting the skip function using an external command signal

The skip function can also be executed using an "external command signal".

The following shows the settings and sequence program example for skipping the control being executed in axis 1 using an "external command signal".

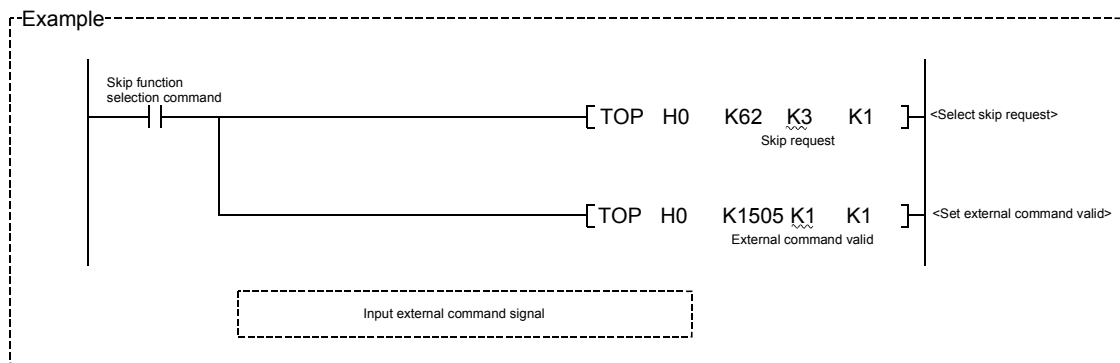
- (1) Set the following data to execute the skip function using an external command signal.

(The setting is carried out using the sequence program shown below in section (2)).

| Setting item                                     | Setting value | Setting details                     | Buffer memory address |        |        |        |
|--------------------------------------------------|---------------|-------------------------------------|-----------------------|--------|--------|--------|
|                                                  |               |                                     | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <b>Pr.42</b> External command function selection | 3             | Set "3: Skip request".              | 62                    | 212    | 362    | 512    |
| <b>Cd.8</b> External command valid               | 1             | Set "1: Validate external command". | 1505                  | 1605   | 1705   | 1805   |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

- (2) Add the following sequence program to the control program, and write it to the CPU module.





## 12.7.3 M code output function

The "M code output function" is used to command sub work (clamping, drill rotation, tool replacement, etc.) related to the positioning data being executed.

When the M code ON signal (X4, X5, X6, X7) is turned ON during positioning execution, a No. called the M code is stored in " [Md.25] Valid M code".

These " [Md.25] Valid M code" are read from the CPU module, and used to command auxiliary work. M codes can be set for each positioning data. (Set in setting item " [Da.10] M code" of the positioning data.)

The timing for outputting (storing) the M codes can also be set in the "M code output function".

The details shown below explain about the "M code output function".

- [1] M code ON signal output timing
- [2] M code OFF request
- [3] Control precautions
- [4] Setting method
- [5] Reading M codes

## [1] M code ON signal output timing

The timing for outputting (storing) the M codes can be set in the "M code output function". (The M code is stored in " [Md.25] Valid M code" when the M code ON signal is turned ON.)

The following shows the two types of timing for outputting M codes: the "WITH" mode and the "AFTER" mode.

## (1) WITH mode

The M code ON signal (X4, X5, X6, X7) is turned ON at the positioning start, and the M code is stored in " [Md.25] Valid M code".

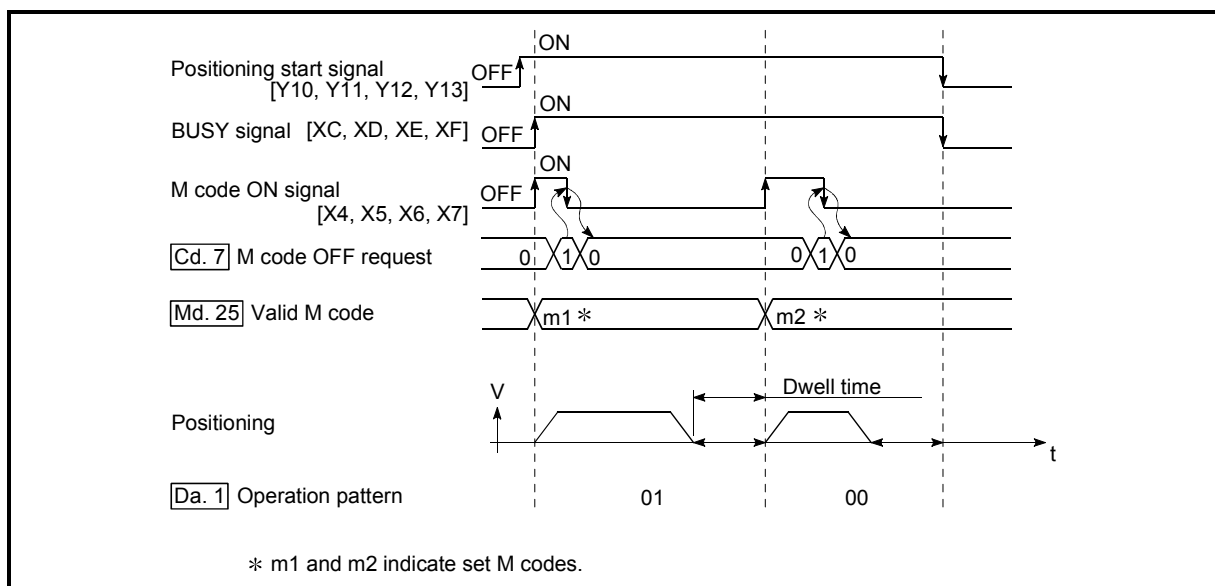


Fig. 12.44 M code ON/OFF timing (WITH mode)

(2) AFTER mode

The M code ON signal (X4, X5, X6, X7) is turned ON at the positioning completion, and the M code is stored in " **[Md.25]** Valid M code".

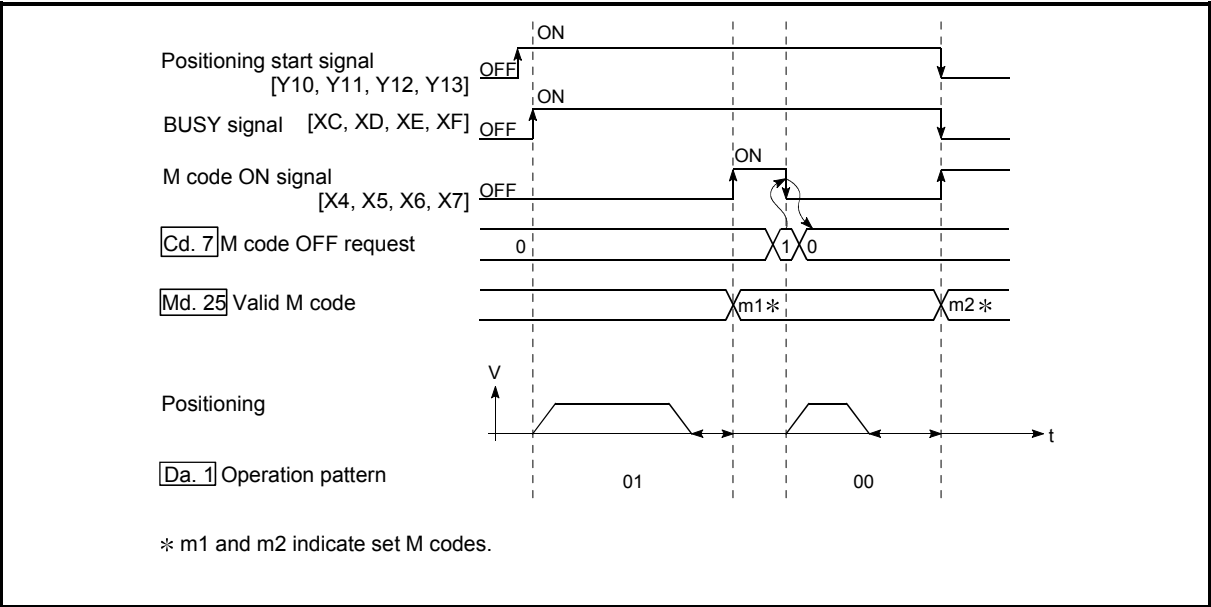


Fig. 12.45 M code ON/OFF timing (AFTER mode)

[2] M code OFF request

When the M code ON signal (X4, X5, X6, X7) is ON, it must be turned OFF by the sequence program.

To turn OFF the M code ON signal, set "1" (turn OFF the M code signal) in " **[Cd.7]** M code OFF request".

| Setting item                     | Setting value | Setting details                         | Buffer memory address |        |        |        |
|----------------------------------|---------------|-----------------------------------------|-----------------------|--------|--------|--------|
|                                  |               |                                         | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <b>[Cd.7]</b> M code OFF request | 1             | Set "1: Turn OFF the M code ON signal". | 1504                  | 1604   | 1704   | 1804   |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

The next positioning data will be processed as follows if the M code ON signal is not turned OFF. (The processing differs according to the " **[Da.1]** Operation pattern".)

| <b>[Da.1]</b> Operation pattern |                                                        | Processing                                                                                                                                                                                  |
|---------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00                              | Independent positioning control (Positioning complete) | The next positioning data will not be executed until the M code ON signal is turned OFF.                                                                                                    |
| 01                              | Continuous positioning control                         |                                                                                                                                                                                             |
| 11                              | Continuous path control                                | The next positioning data will be executed. If the M code is set to the next positioning data, a warning "M code ON signal ON start" (warning code: 503) will occur. (Refer to Fig. 12.46.) |

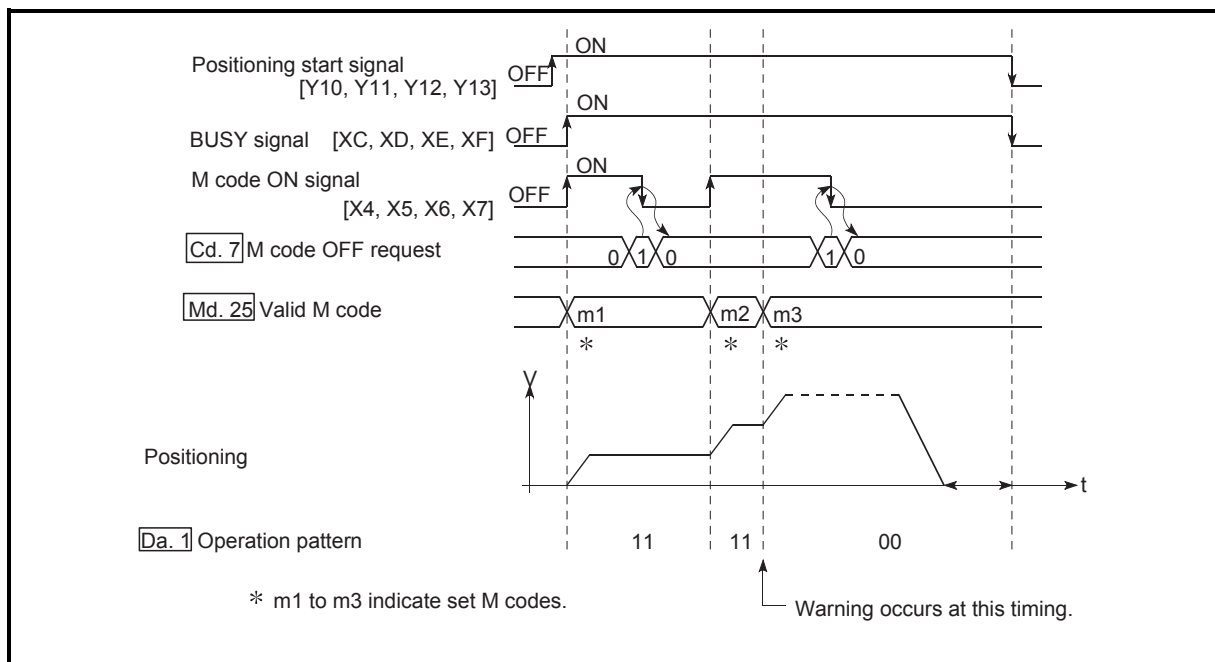


Fig. 12.46 Warning due to an M code ON signal during continuous path control

**POINT**

If the M code output function is not required, set a "0" in setting item " [Da.10] M code" of the positioning data.

**[3] Control precautions**

- (1) During interpolation control, the reference axis M code ON signal is turned ON.
- (2) The M code ON signal will not turn ON if "0" is set in " [Da.10] M code".  
(The M code will not be output, and the previously output value will be held in " [Md.25] Valid M code".)
- (3) If the M code ON signal is ON at the positioning start, an error "M code ON signal start" (error code: 536) will occur, and the positioning will not start.
- (4) If the PLC READY signal (Y0) is turned OFF, the M code ON signal will turn OFF and "0" will be stored in " [Md.25] Valid M code".
- (5) If the positioning operation time is short during continuous path control, there will not be enough time to turn OFF the M code ON signal, and a warning "M code ON signal ON" (warning code: 503) may occur. In this case, set a "0" in the " [Da.10] M code" of that section's positioning data.
- (6) In the AFTER mode during speed control, the M code is not output and the M code ON signal does not turn ON.
- (7) If current value changing where "9003" has been set to " [Cd.3] Positioning start No." is performed, the M code output function is made invalid.

#### [4] Setting method

The following shows the settings to use the "M code output function".

- (1) To specify the M code ON signal output timing for each positioning data  
Use "[Da.27] M code ON signal output timing".

The following shows the settings to use "[Da.27] M code ON signal output timing".

- Set the M code No. in the positioning data "[Da.10] M code".
- Set the timing to output the M code ON signal [X4, X5, X6, X7] in the positioning data "[Da.27] M code ON signal output timing".

- (2) To set the same M code ON signal output timing for all positioning data  
Setting "[Pr.18] M code ON signal output timing" enables batch specification of M code ON signal output timing for each positioning data. Set 0 in "[Da.27] M code ON signal output timing" when using "[Pr.18] M code ON signal output timing". When a value other than 0 is set, "[Da.27] M code ON signal output timing" will be valid. (The setting in "[Pr.18] M code ON signal output timing" becomes valid on the rising edge (OFF to ON) of the PLC READY signal [Y0].)

The following shows the settings to use "[Pr.18] M code ON signal output timing".

- Set the M code No. in the positioning data "[Da.10] M code".
- Set "0: Use the set value in "[Pr.18] M code ON signal output timing". (initial value) in the positioning data "[Da.27] M code ON signal output timing".
- Set the timing to output the M code ON signal [X4, X5, X6, X7] in the detailed parameter "[Pr.18] M code ON signal output timing".

| Setting item |                                | Setting value | Setting details                                                                                                                                                         | Buffer memory address    |                          |                           |                           |
|--------------|--------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|---------------------------|---------------------------|
|              |                                |               |                                                                                                                                                                         | Axis 1                   | Axis 2                   | Axis 3                    | Axis 4                    |
| [Pr.18]      | M code ON signal output timing | →             | Set the timing to output the M code ON signal.<br>0: WITH mode<br>1: AFTER mode                                                                                         | 27                       | 177                      | 327                       | 477                       |
| [Da.27]      | M code ON signal output timing | →             | Set the M code ON signal output timing for each positioning data.<br>0: Use the set value in "[Pr.18] M code ON signal output timing".<br>1: WITH mode<br>2: AFTER mode | 2003 + N*1<br>(b0 to b1) | 8003 + N*1<br>(b0 to b1) | 14003 + N*1<br>(b0 to b1) | 20003 + N*1<br>(b0 to b1) |

\*1 N indicates the offset address of each positioning data.

$N = ((\text{Positioning data No.}) - 1) \times 10$

\* Refer to Section 5.2 "List of parameters" for the information on detail settings.

## [5] Reading M codes

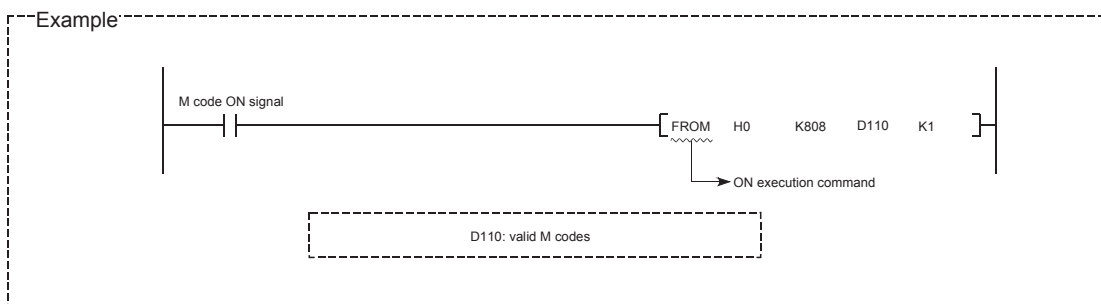
"M codes" are stored in the following buffer memory when the M code ON signal turns ON.

| Monitor item | Monitor value | Storage details                                                                | Buffer memory address |        |        |        |
|--------------|---------------|--------------------------------------------------------------------------------|-----------------------|--------|--------|--------|
|              |               |                                                                                | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <b>Md.25</b> | Valid M code  | → The M code No. ( <b>Da.10</b> M code) set in the positioning data is stored. | 808                   | 908    | 1008   | 1108   |

\* Refer to Section 5.6 "List of monitor data" for information on the storage details.

The following shows a sequence program example for reading the " **Md.25** Valid M code" to the CPU module data register (D110). (The read value is used to command the sub work.)

Read M codes not as "rising edge commands", but as "ON execution commands".



### 12.7.4 Teaching function

The "teaching function" is used to set addresses aligned using the manual control (JOG operation, inching operation, and manual pulse generator operation) in the positioning data addresses ("Da.6" Positioning address/movement amount", "Da.7" Arc address").

The details shown below explain about the "teaching function".

- [1] Control details
- [2] Control precautions
- [3] Data used in teaching
- [4] Teaching procedure
- [5] Teaching program example

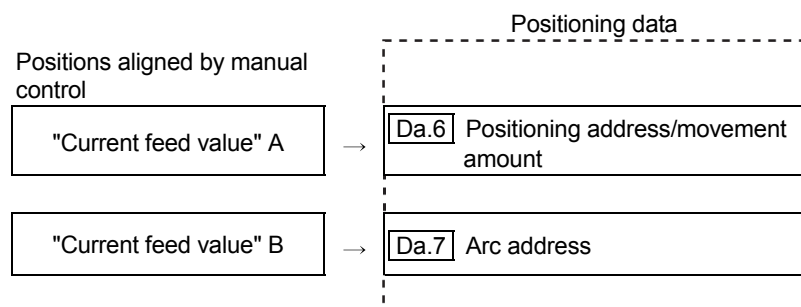
#### [1] Control details

##### (1) Teaching timing

Teaching is executed using the sequence program when the BUSY signal (XC, XD, XE, XF) is OFF. (During manual control, teaching can be carried out as long as the axis is not BUSY, even when an error or warning has occurred.)

##### (2) Addresses for which teaching is possible

The addresses for which teaching is possible are "current feed values" (Md.20 Current feed value) having the OP as a reference. The settings of the "movement amount" used in incremental system positioning cannot be used. In the teaching function, these "current feed values" are set in the "Da.6" Positioning address/movement amount" or "Da.7" Arc address".



##### (3) Dedicated instructions "ZP.TEACH 1, ZP.TEACH 2, ZP.TEACH 3, ZP.TEACH 4, ZP.PFWRT"

When the dedicated instructions "ZP.TEACH 1, ZP.TEACH 2, ZP.TEACH 3, ZP.TEACH 4, ZP.PFWRT" are used to execute the teaching function, the programming becomes easier. Refer to CHAPTER 14 "DEDICATED INSTRUCTIONS" for details.

## [2] Control precautions

- (1) Before teaching, a "machine OPR" must be carried out to establish the OP.  
(When a current value changing, etc., is carried out, " [Md.20] Current feed value" may not show absolute addresses having the OP as a reference.)
- (2) Teaching cannot be carried out for positions to which movement cannot be executed by manual control (positions to which the workpiece cannot physically move). (During center point designation circular interpolation control, etc., teaching of " [Da.7] Arc address" cannot be carried out if the center point of the arc is not within the moveable range of the workpiece.)
- (3) Writing to the flash ROM can be executed up to 100,000 times.  
If writing to the flash ROM exceeds 100,000 times, the writing may become impossible (assured value is up to 100,000 times).  
If the error "Flash ROM write number error" (error code: 805) occurs when writing to the flash ROM has been completed, check whether or not the program is created so as to write continuously to the flash ROM.

## [3] Data used in teaching

The following control data is used in teaching.

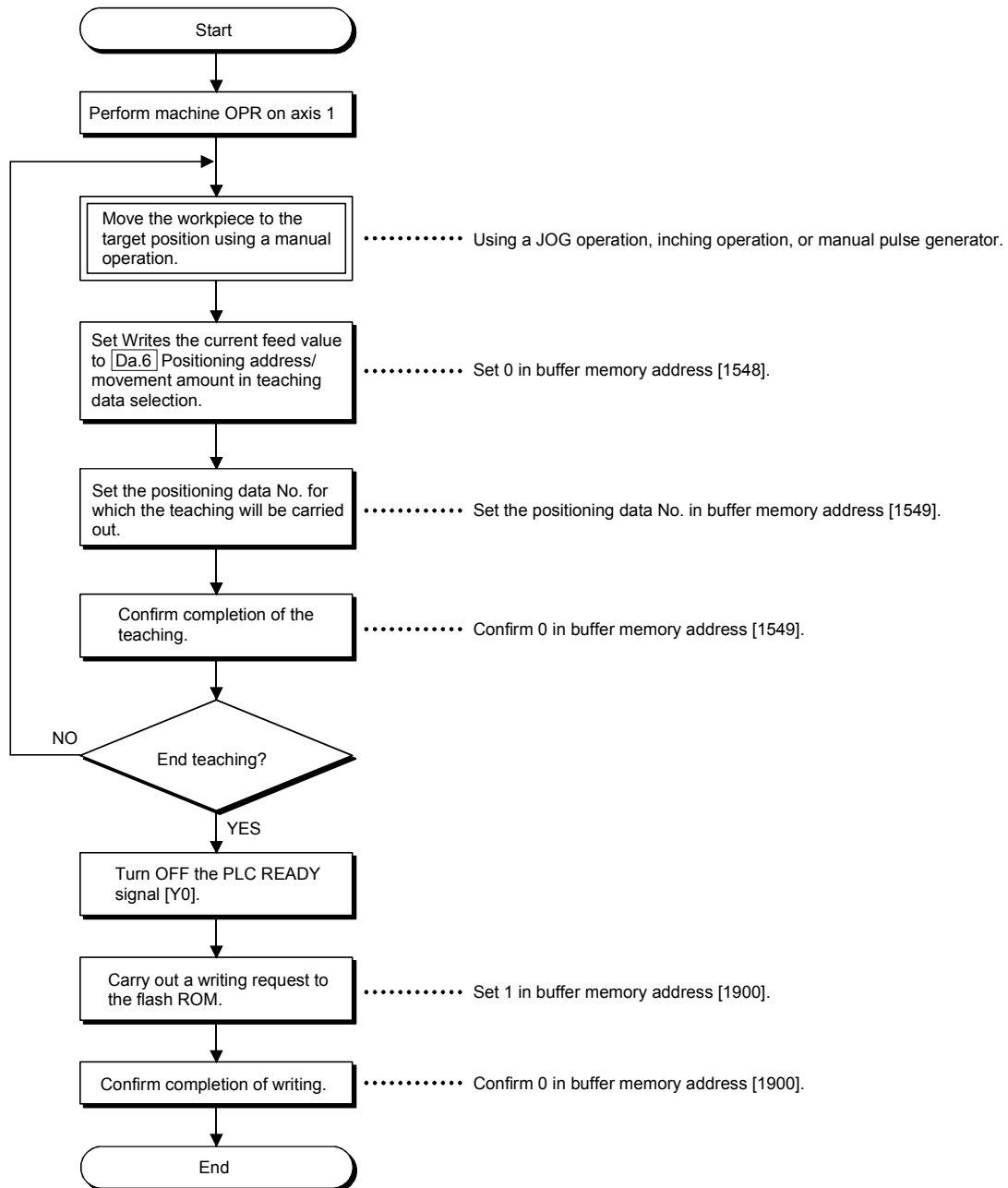
| Setting item |                               | Setting value | Setting details                                                                                                                                                       | Buffer memory address |        |        |        |
|--------------|-------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------|--------|--------|
|              |                               |               |                                                                                                                                                                       | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| [Cd.1]       | Flash ROM write request       | 1             | Write the set details to the flash ROM (backup the changed data).                                                                                                     | 1900                  |        |        |        |
| [Cd.38]      | Teaching data selection       | →             | Sets to which "current feed value" is written.<br>0: Written to " [Da.6] Positioning address/movement amount".<br>1: Written to " [Da.7] Arc address".                | 1548                  | 1648   | 1748   | 1848   |
| [Cd.39]      | Teaching positioning data No. | →             | Designates the data to be taught.<br>(Teaching is carried out when the setting value is 1 to 600.)<br>When teaching has been completed, this data is cleared to zero. | 1549                  | 1649   | 1749   | 1849   |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

#### [4] Teaching procedure

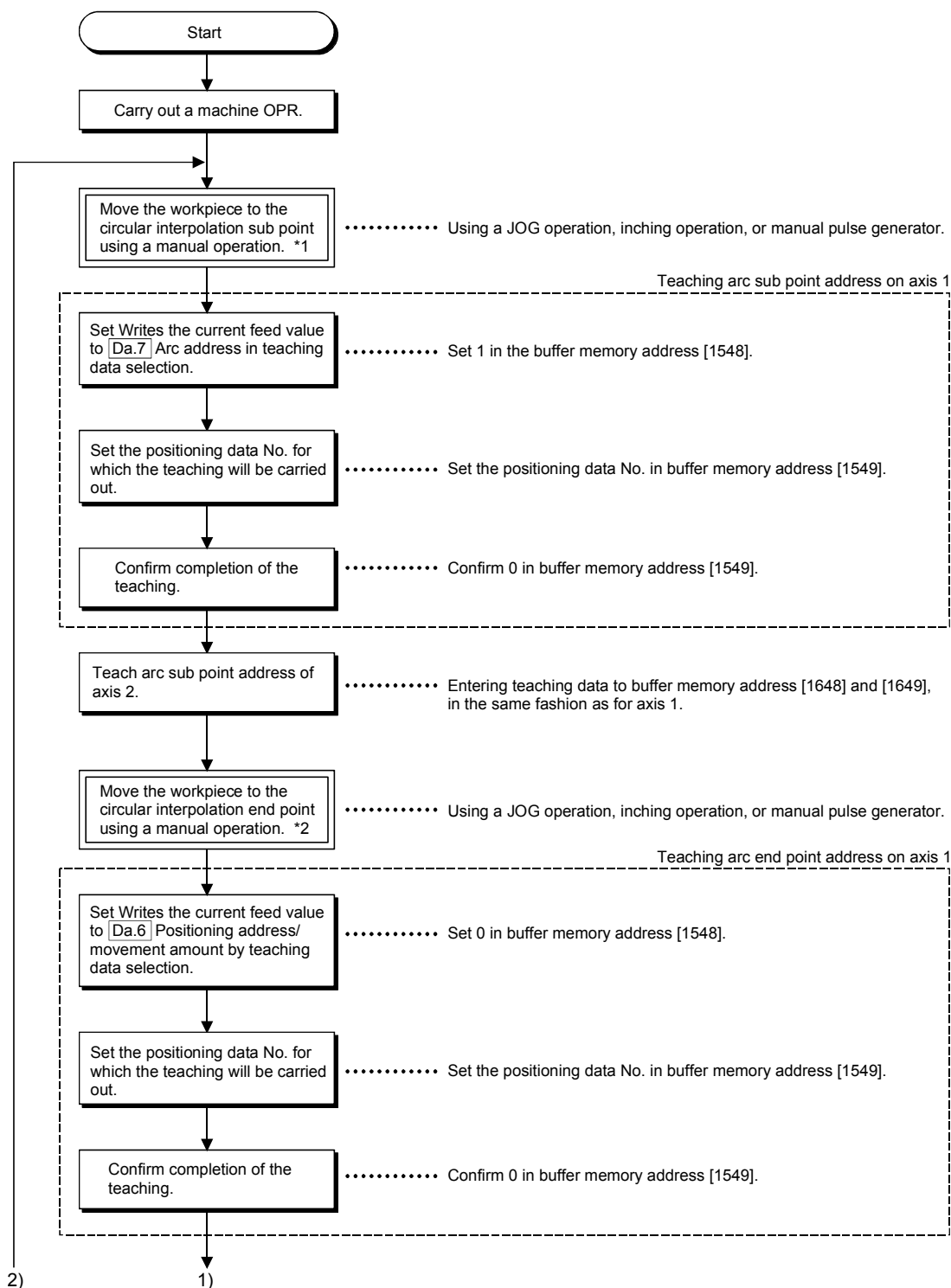
The following shows the procedure for a teaching operation.

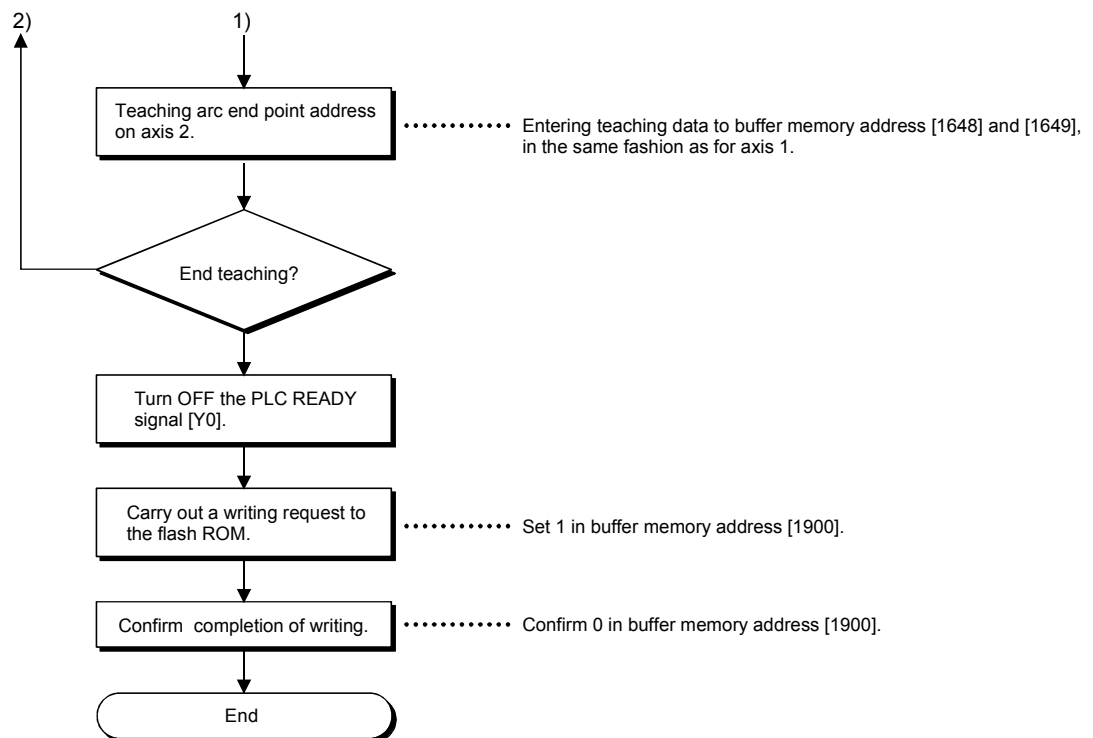
- (1) When teaching to the " [Da.6] Positioning address/movement amount"  
(Teaching example on axis 1)



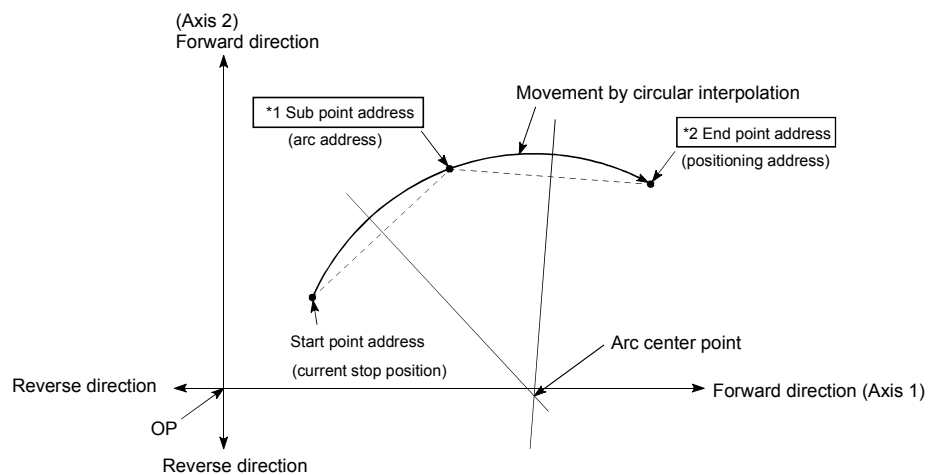


- (2) When entering teaching data into " Da.7 Arc address" and then into " Da.6 Positioning address/movement amount" (Teaching example for 2-axis circular interpolation control with sub point designation on axes 1 and 2)





## [Motion path]



### [5] Teaching program example

The following shows a sequence program example for setting (writing) the positioning data obtained with the teaching function to the QD75.

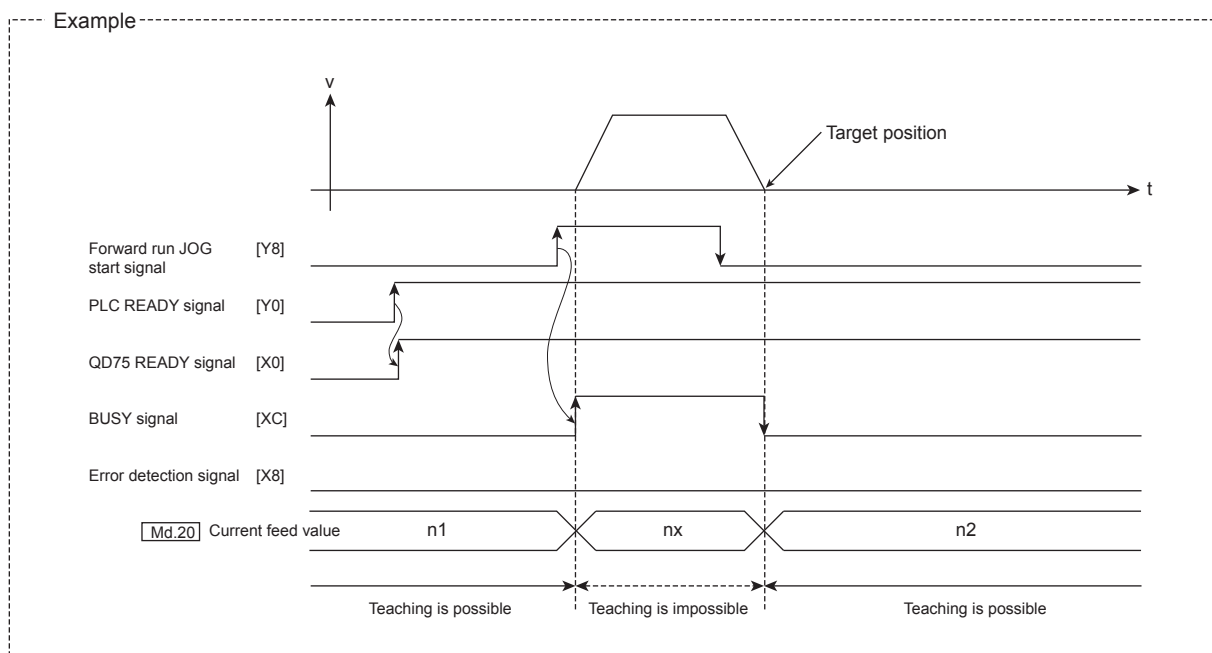
#### (1) Setting conditions

- When setting the current feed value as the positioning address, write it when the BUSY signal is OFF.

#### (2) Program example

- The following example shows a program to carry out the teaching of axis 1 by the dedicated instruction "ZP.TEACH 1".

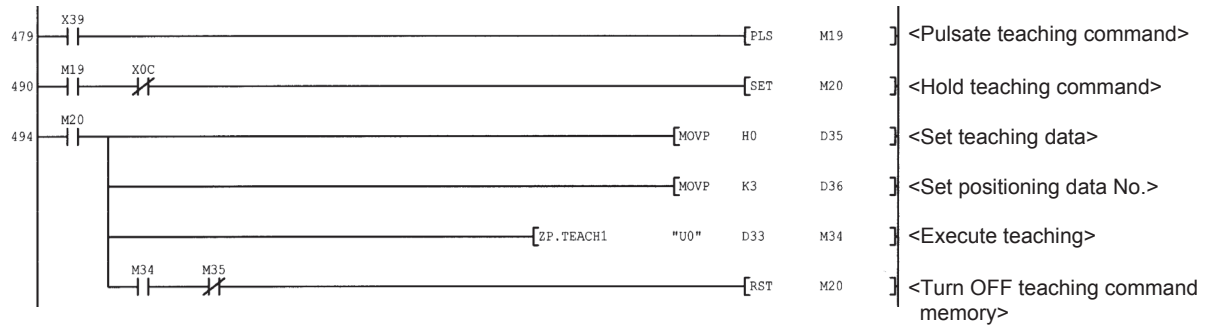
- 1) Move the workpiece to the target position using a JOG operation (or an inching operation, a manual pulse generator operation).



2) Carry out the teaching operation with the following program.

#### Example

\*  
 \* No.19 Teaching program  
 \*  
 \*       •Position to the target position with manual operation.  
 \*  
 \*  
 \*



#### POINT

- (1) Confirm the teaching function and teaching procedure before setting the positioning data.
- (2) The positioning addresses that are written are absolute address (ABS) values.
- (3) If the positioning operation is correctly completed with the written positioning data, it is recommended that the positioning data be registered in the QD75 flash ROM.

### 12.7.5 Command in-position function

The "command in-position function" checks the remaining distance to the stop position during the automatic deceleration of positioning control, and set a flag to 1. This flag is called the "command in-position flag". The command in-position flag is used as a front-loading signal indicating beforehand the completion of the position control.

The details shown below explain about the "command in-position function".

- [1] Control details
- [2] Control precautions
- [3] Setting method
- [4] Confirming the command in-position flag

#### [1] Control details

The following shows control details of the command in-position function.

- (1) When the remaining distance to the stop position during the automatic deceleration of positioning control becomes equal to or less than the value set in " **Pr.16** Command in-position width", "1" is stored in the command in-position flag (**Md.31** Status: b2).

(Command in-position width check)

Remaining distance  $\leq$  " **Pr.16** Command in-position width" setting value

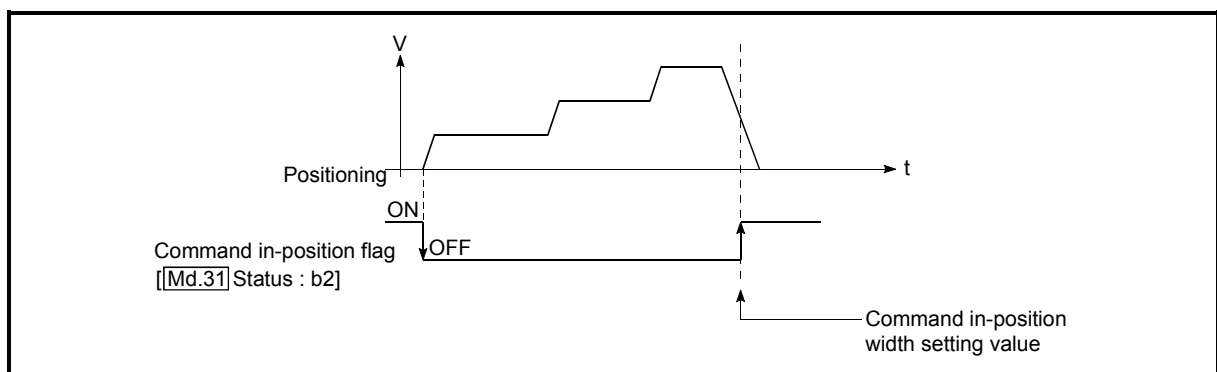


Fig. 12.47 Command in-position operation

- (2) A command in-position width check is carried out every 0.9ms (QD75P□N/QD75D□N) or 1.8ms (QD75P□/QD75D□).

## [2] Control precautions

(1) A command in-position width check will not be carried out in the following cases.

- During speed control
- During speed control in speed-position switching control
- During speed control in position-speed switching control

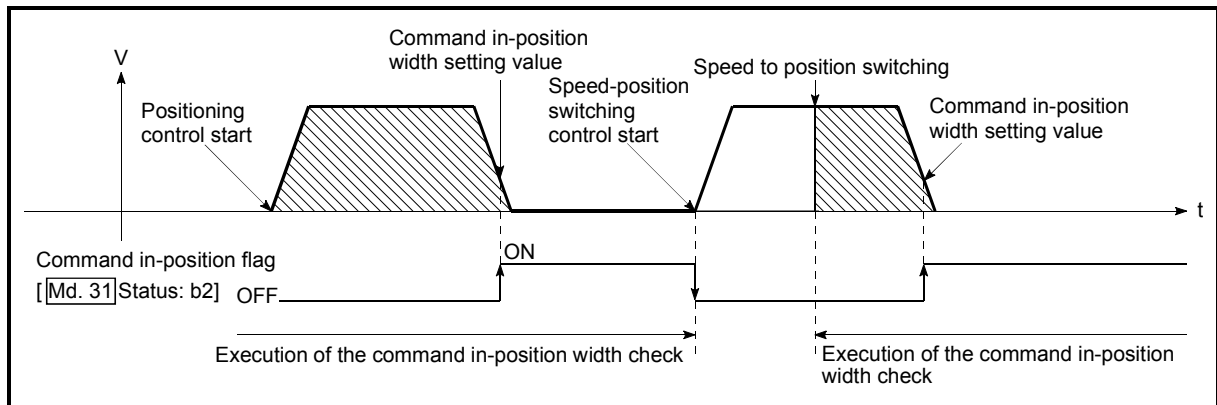


Fig. 12.48 Command in-position width check

(2) The command in-position flag will be turned OFF in the following cases.

("0" will be stored in " [Md.31] Status: b2".)

- At the positioning control start
- At the speed control start
- At the speed-position switching control, position-speed switching control start
- At the OPR control start
- At the JOG operation start
- At the inching operation start
- When the manual pulse generator operation is enabled.

(3) The " [Pr.16] Command in-position width" and command in-position flag ([Md.31] Status: b2) of the reference axis are used during interpolation control. When the " [Pr.20] Interpolation speed designation method" is "Composite speed", the command in-position width check is carried out in the remaining distance on the composite axis (line/arc connecting the start point address and end point address).

## [3] Setting method

To use the "command in-position function", set the required value in the parameter shown in the following table, and write it to the QD75.

The set details are validated at the rising edge (OFF → ON) of the PLC READY signal (Y0).

| Setting item |                           | Setting value | Setting details                                                                                                    | Factory-set initial value |
|--------------|---------------------------|---------------|--------------------------------------------------------------------------------------------------------------------|---------------------------|
| Pr.16        | Command in-position width | →             | Turn ON the command in-position flag, and set the remaining distance to the stop position of the position control. | 100                       |

\* Refer to Section 5.2 "List of parameters" for setting details.

## [4] Confirming the command in-position flag

The "command in-position flag" is stored in the following buffer memory.

| Monitor item | Monitor value | Storage details | Buffer memory address                                        |        |        |        |
|--------------|---------------|-----------------|--------------------------------------------------------------|--------|--------|--------|
|              |               |                 | Axis 1                                                       | Axis 2 | Axis 3 | Axis 4 |
| Md.31        | Status        | →               | The command in-position flag is stored in the "b2" position. |        |        |        |
|              |               |                 | 817                                                          | 917    | 1017   | 1117   |

\* Refer to Section 5.6 "List of monitor data" for information on the storage details.

**REMARK**

- Parameters are set for each axis.
- It is recommended that the parameters be set whenever possible with GX Configurator-QP. Execution by sequence program uses many sequence programs and devices. The execution becomes complicated, and the scan times will increase.

### 12.7.6 Acceleration/deceleration processing function

The "acceleration/deceleration processing function" adjusts the acceleration/deceleration of each control to the acceleration/deceleration curve suitable for devices.

Setting the acceleration/deceleration time changes the slope of the acceleration/deceleration curve.

The following two methods can be selected for the acceleration/deceleration curve:

- Trapezoidal acceleration/deceleration
- S-curve acceleration/deceleration

The details shown below explain about the "acceleration/deceleration processing function".

[1] "Acceleration/deceleration time 0 to 3" control details and setting

[2] "Acceleration/deceleration method setting" control details and setting

#### [1] "Acceleration/deceleration time 0 to 3" control details and setting

In the QD75, four types each of acceleration time and deceleration time can be set. By using separate acceleration/deceleration times, control can be carried out with different acceleration/deceleration times for positioning control, JOG operation, OPR, etc.

Set the required values for the acceleration/deceleration time in the parameters shown in the following table, and write them to the QD75.

The set details are validated when written to the QD75.

| Setting item |                     | Setting value | Setting details                                                          | Factory-set initial value |
|--------------|---------------------|---------------|--------------------------------------------------------------------------|---------------------------|
| Pr.9         | Acceleration time 0 | →             | Set the acceleration time at a value within the range of 1 to 8388608ms. | 1000                      |
| Pr.25        | Acceleration time 1 | →             |                                                                          | 1000                      |
| Pr.26        | Acceleration time 2 | →             |                                                                          | 1000                      |
| Pr.27        | Acceleration time 3 | →             |                                                                          | 1000                      |
| Pr.10        | Deceleration time 0 | →             | Set the deceleration time at a value within the range of 1 to 8388608ms. | 1000                      |
| Pr.28        | Deceleration time 1 | →             |                                                                          | 1000                      |
| Pr.29        | Deceleration time 2 | →             |                                                                          | 1000                      |
| Pr.30        | Deceleration time 3 | →             |                                                                          | 1000                      |

\* Refer to Section 5.2 "List of parameters" for setting details.



## [2] "Acceleration/deceleration method setting" control details and setting

In the "acceleration/deceleration method setting", the acceleration/deceleration processing method is selected and set. The set acceleration/deceleration processing is applied to all acceleration/deceleration (except for inching operation and manual pulse generator operation).

The two types of "acceleration/deceleration method setting" are shown below.

### (1) Trapezoidal acceleration/deceleration processing method

This is a method in which linear acceleration/deceleration is carried out based on the acceleration time, deceleration time, and speed limit value set by the user.

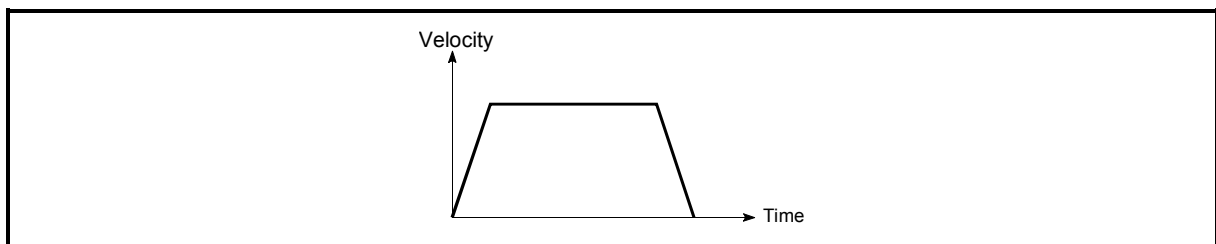


Fig. 12.49 Trapezoidal acceleration/deceleration processing method

### (2) S-curve acceleration/deceleration processing method

In this method, the motor burden is reduced during starting and stopping.

This is a method in which acceleration/deceleration is carried out gradually, based on the acceleration time, deceleration time, speed limit value, and " Pr.35 S-curve ratio" (1 to 100%) set by the user.

When the stepping motor is used, the acceleration around the inflection point on the s shape is faster compared with the trapezoidal acceleration/deceleration. This may cause step-out. \*1

In this case, adjust the acceleration/deceleration time to decrease the acceleration around the inflection point, or use a servo motor.

\*1: When comparing the trapezoidal acceleration/deceleration and S-curve acceleration/deceleration processing method in the same acceleration/deceleration time.

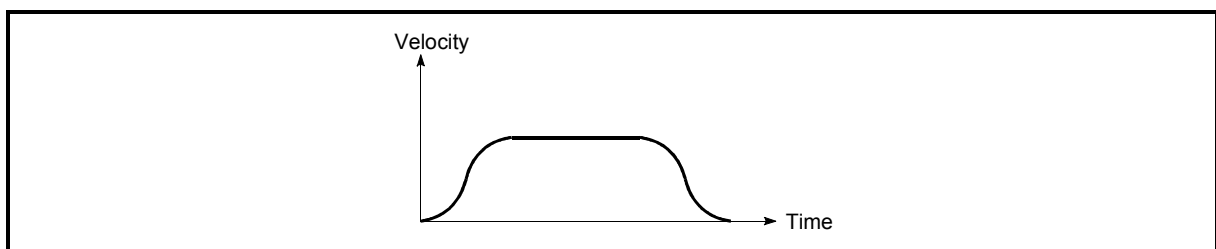


Fig. 12.50 S-curve acceleration/deceleration processing method

When a speed change request or override request is given during S-curve acceleration/deceleration processing, S-curve acceleration/deceleration processing begins at the start of a speed change request or an override request.

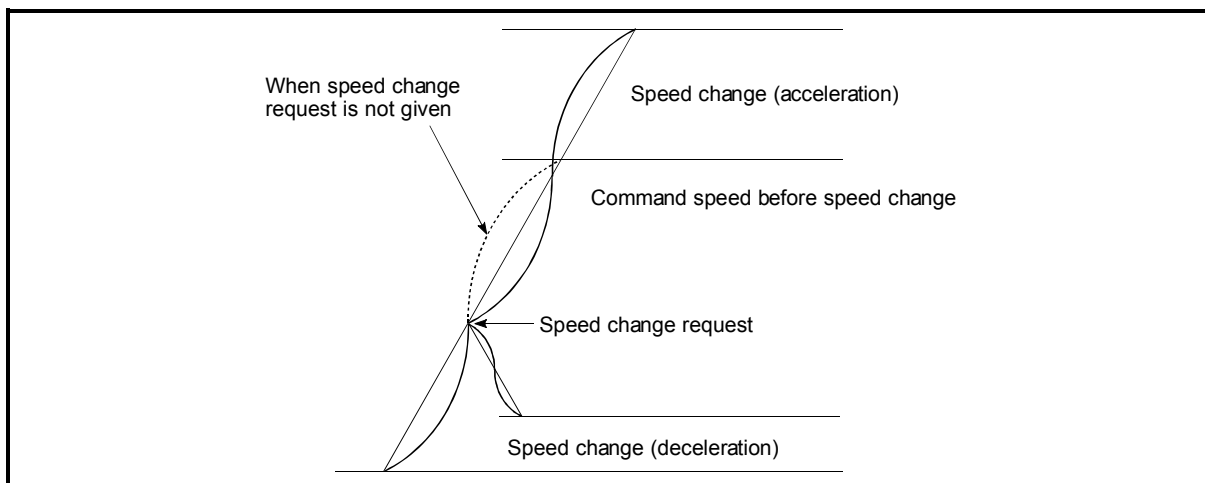


Fig. 12.51 Speed change during S-curve acceleration/deceleration processing

Set the required values for the "acceleration/deceleration method setting" in the parameters shown in the following table, and write them to the QD75. The set details are validated when written to the QD75.

| Setting item                                      | Setting value | Setting details                                                                                                                                     | Factory-set initial value |
|---------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Pr.34 Acceleration/deceleration process selection | →             | Set the acceleration/deceleration method.<br>0: Trapezoidal acceleration/deceleration processing<br>1: S-curve acceleration/deceleration processing | 0                         |
| Pr.35 S-curve ratio                               | →             | Set the acceleration/deceleration curve when "1" is set in " Pr.34 Acceleration/deceleration processing selection".                                 | 100                       |

\* Refer to Section 5.2 "List of parameters" for setting details.

#### REMARK

- Parameters are set for each axis.
- It is recommended that the parameters be set whenever possible with GX Configurator-QP. Execution by sequence program uses many sequence programs and devices. The execution becomes complicated, and the scan times will increase.

## 12.7.7 Pre-reading start function

The "pre-reading start function" does not output pulses while the execution prohibition flag is ON if a positioning start request is given with the execution prohibition flag ON, and starts outputting pulses within 1.3ms (QD75P□N/QD75D□N) or 3ms (QD75P□/QD75D□) after OFF of the execution prohibition flag is detected. The positioning start request is given when the axis is in a standby status, and the execution prohibition flag is turned OFF at the axis operating timing. This shortens the virtual start time. The QD75 normally takes 1.5 to 2.0ms (QD75P□N/QD75D□N) or 6 to 7ms (QD75P□/QD75D□) from when it receives a positioning start request until it starts pulse output (start time). Some systems often need the start time to be shortened. This "pre-reading start function" can improve the tact time of the system.

The "pre-reading start function" will be explained below.

- [1] Control details
- [2] Control precautions
- [3] Program examples

## [1] Control details

The pre-reading start function is performed by turning ON the positioning start signal [Y10, Y11, Y12, Y13] with the execution prohibition flag [Y14, Y15, Y16, Y17] ON, or by executing the dedicated instruction (ZP.PSTRT1, ZP.PSTRT2, ZP.PSTRT3, ZP.PSTRT4). However, if positioning is started with the execution prohibition flag ON, the positioning data is analyzed but pulse output is not provided. While the execution prohibition flag is ON, " [Md.26] Axis operation status" remains unchanged from "5: Analyzing". Pulse output starts within 1.3ms (QD75P□N/QD75D□N) or 3ms (QD75P□/QD75D□) after the execution prohibition flag [Y14, Y15, Y16, Y17] has turned OFF, and " [Md.26] Axis operation status" changes to the status (e.g. "Position control", "Speed control") that matches the control system. (Refer to Fig. 12.52)

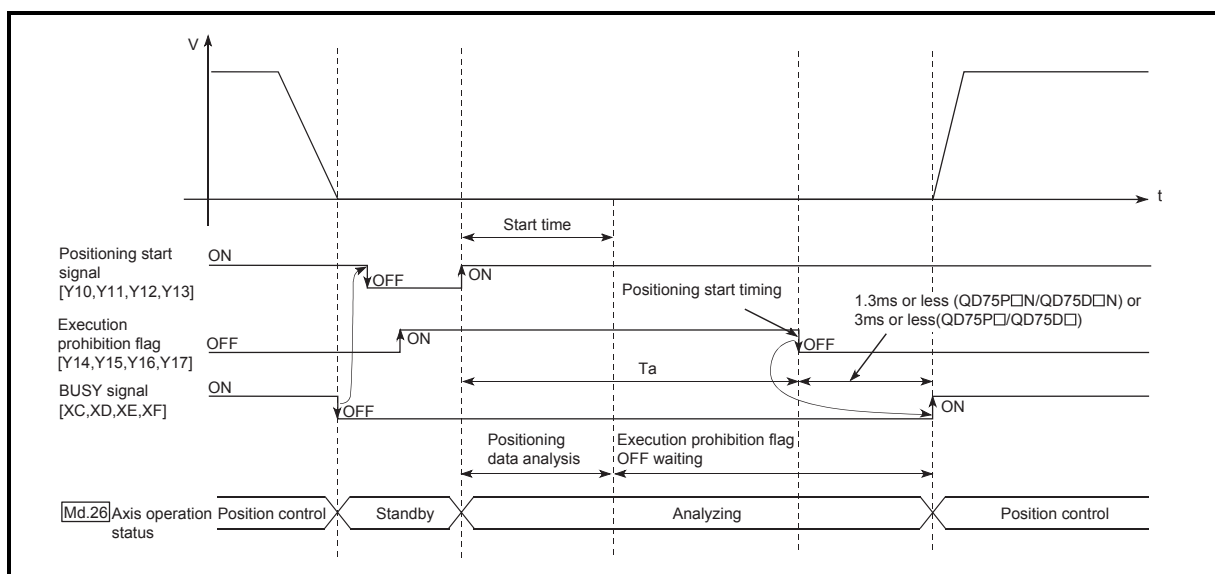


Fig. 12.52 Operations of pre-reading start function

The pre-reading start function is effective for the system as shown below.

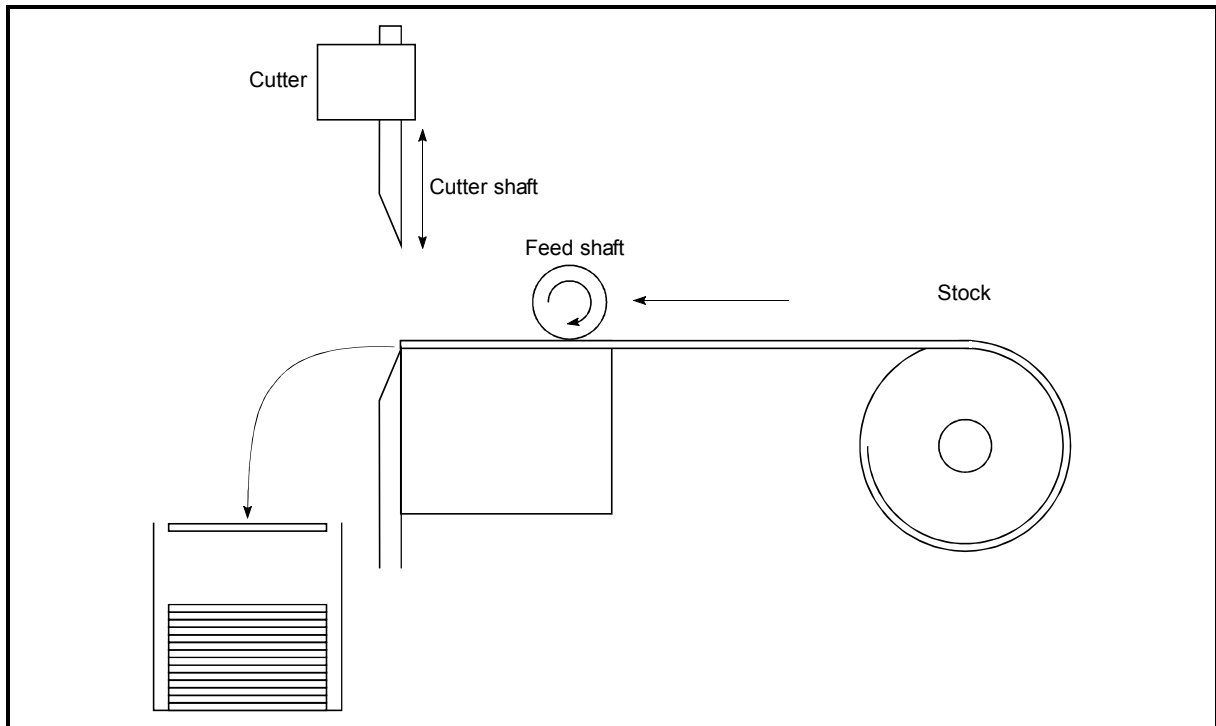


Fig. 12.53 System example using pre-reading start function

Fig. 12.53 shows a system example which repeats:

- 1) Feeding a stock with a feed shaft; and
- 2) Cutting it with a cutter

to cut the stock to fixed size. The operations of the feed shaft and cutter shaft are represented as shown in Fig. 12.54.

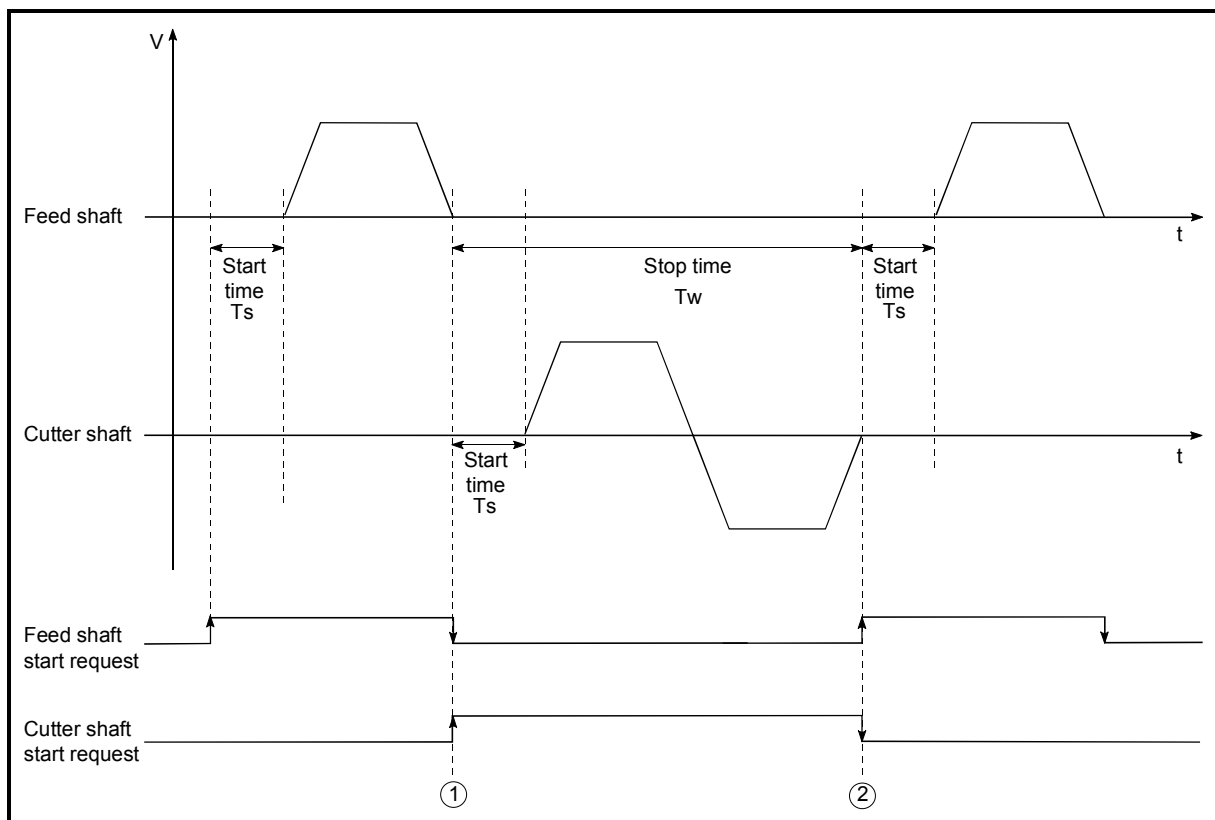


Fig. 12.54 Operation timings of system example

The cutter shaft starts from the moment the feed shaft has completed feeding the stock "①", and the feed shaft starts from the moment the cutter shaft has returned to the standby position "②". Actually, however, there is a delay of start time  $T_s$  (1.5 to 2.0ms (QD75P□N/QD75D□N) or 6 to 7ms (QD75P□/QD75D□)) from when the QD75 receives a start request until it outputs pulses. The system's tact time can be reduced by the shortening of this delay with the Pre-reading start function.

In Fig. 12.54, the feed shaft stands by during the stop time  $T_w$ .

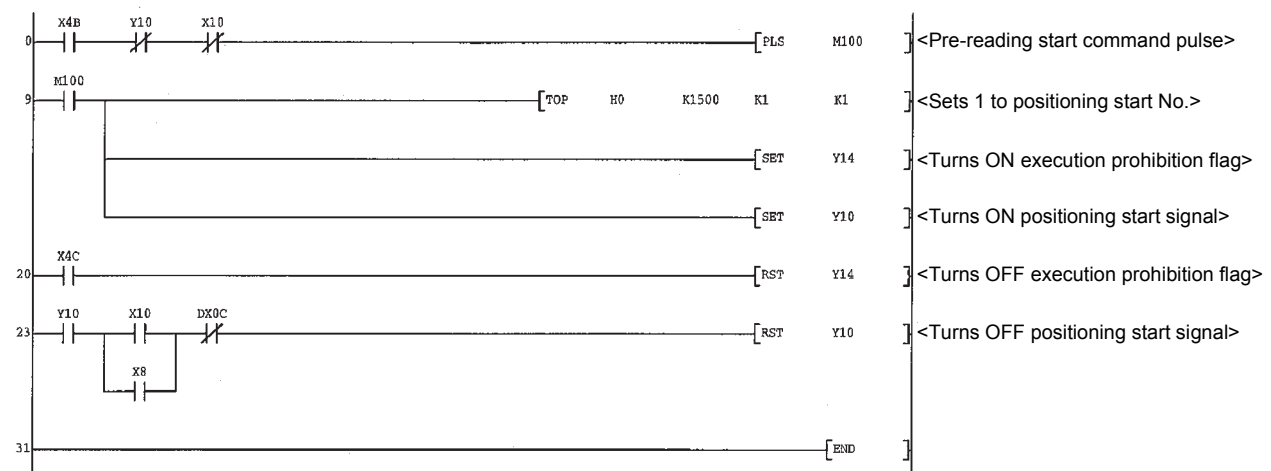
Hence, pre-reading of the next data starts during the stop time  $T_w$ . If  $T_w$  is a certain period of time, the analysis of the next data is completed during that period, and the system is placed in an execution prohibition flag OFF waiting status. Therefore, replacing the positioning start timing at ② with the execution prohibition flag OFF allows the time from when the axis operation request turns ON until pulse output starts to be reduced to within 1.3ms (QD75P□N/QD75D□N) or 3ms (QD75P□/QD75D□) in the sequence program. (Refer to Fig. 12.52.)

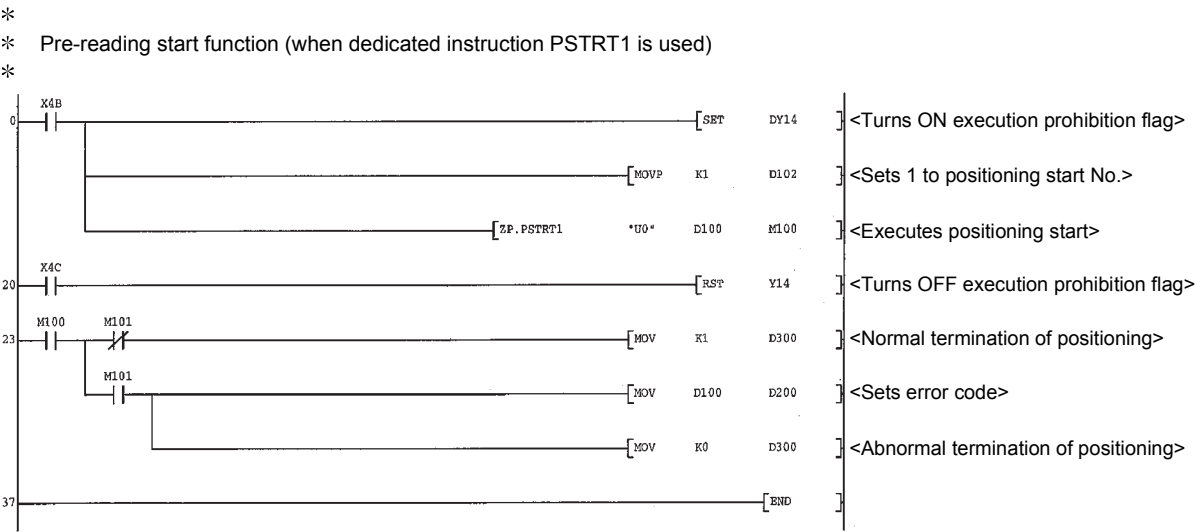
## [2] Control precautions

- (1) The time required to analyze the positioning data is up to 7ms.
- (2) After positioning data analysis, the system is put in an execution prohibition flag OFF waiting status. Any change made to the positioning data in the execution prohibition flag OFF waiting status is not reflected on the positioning data. Change the positioning data before turning ON the positioning start signal.
- (3) The pre-reading start function is invalid if the execution prohibition flag is turned OFF between when the positioning start signal has turned ON and when positioning data analysis is completed ( $T_a < \text{start time}$ ,  $T_a$ : Refer to Fig. 12.52).
- (4) The data No. which can be executed positioning start using "Cd.3 Positioning start No." with the pre-reading start function are No. 1 to 600 only. Performing the pre-reading start function at the setting of No. 7000 to 7004 or 9001 to 9004 will result in an error "Outside start No. range" (error code: 543).
- (5) Always turn ON the execution prohibition flag at the same time or before turning ON the positioning start signal. Pre-reading may not be started if the execution prohibition flag is turned ON during  $T_a$  after the positioning start signal is turned ON. The pre-reading start function is invalid if the execution prohibition flag is turned ON after positioning start (pulse output) with the execution prohibition flag OFF. (It is made valid at the next positioning start.)

## [3] Program examples

\*  
\* Pre-reading start function (when positioning start signal Y10 is used)  
\*





## 12.7.8 Deceleration start flag function

The "deceleration start flag function" turns ON the flag when the constant speed status or acceleration status switches to the deceleration status during position control whose operation pattern is "Positioning complete". This function can be used as a signal to start the operation to be performed by other equipment at each end of position control or to perform preparatory operation, etc. for the next position control.

For the "deceleration start flag function", the following will be explained.

- [1] Control details
- [2] Control precautions
- [3] Setting method
- [4] Checking of deceleration start flag

## [1] Control details

When deceleration for a stop is started in the position control whose operation pattern is "Positioning complete", "1" is stored into " Md.48 Deceleration start flag". When the next operation start is made or the manual pulse generator operation enable status is gained, "0" is stored. (Refer to Fig. 12.55.)

(1) Start made with positioning data No. specified

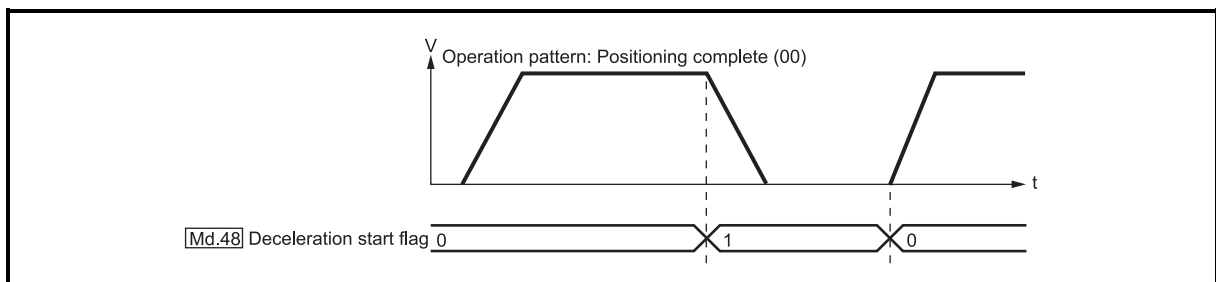


Fig. 12.55 Operation of deceleration start flag



## (2) Block start

At a block start, this function is valid for only the position control whose operation pattern is "Positioning complete" at the point whose shape has been set to "End". (Refer to Fig. 12.56.)

The following table indicates the operation of the deceleration start flag in the case of the following block start data and positioning data.

| Block start data | Da.11<br>Shape | Da.12<br>Start data No. | Da.13<br>Special start instruction |
|------------------|----------------|-------------------------|------------------------------------|
| 1st point        | 1: Continue    | 1                       | 0: Block start                     |
| 2nd point        | 1: Continue    | 3                       | 0: Block start                     |
| 3rd point        | 0: End         | 4                       | 0: Block start                     |
| •                |                |                         |                                    |
| •                |                |                         |                                    |

| Positioning Data No. | Da.1<br>Operation pattern          |
|----------------------|------------------------------------|
| 1                    | 01: Continuous positioning control |
| 2                    | 00: Positioning complete           |
| 3                    | 00: Positioning complete           |
| 4                    | 11: Continuous path control        |
| 5                    | 00: Positioning complete           |
| •                    |                                    |
| •                    |                                    |

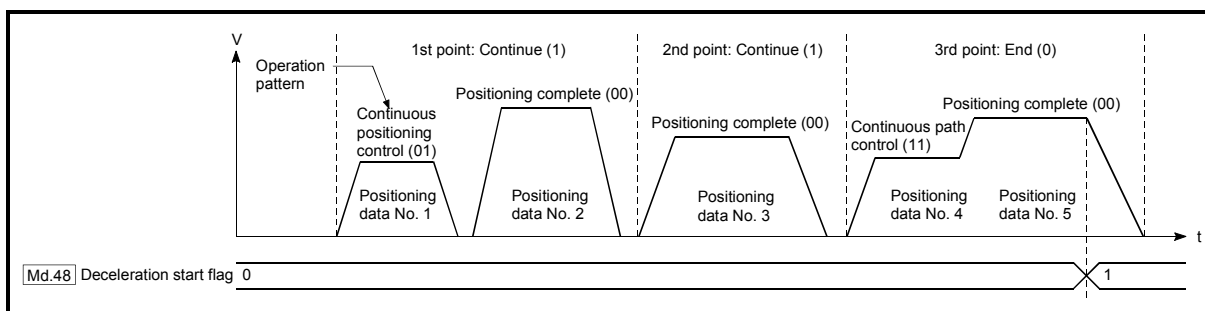
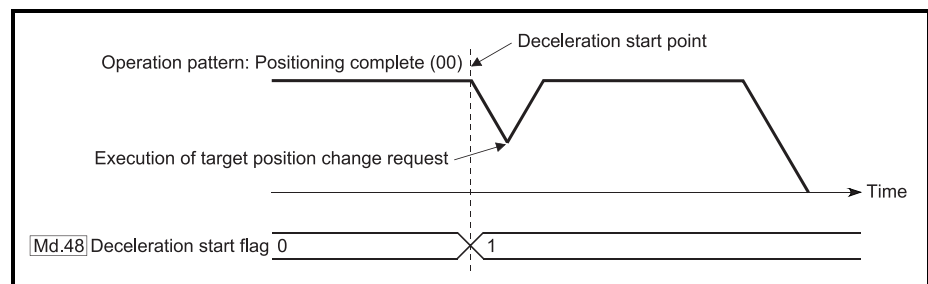


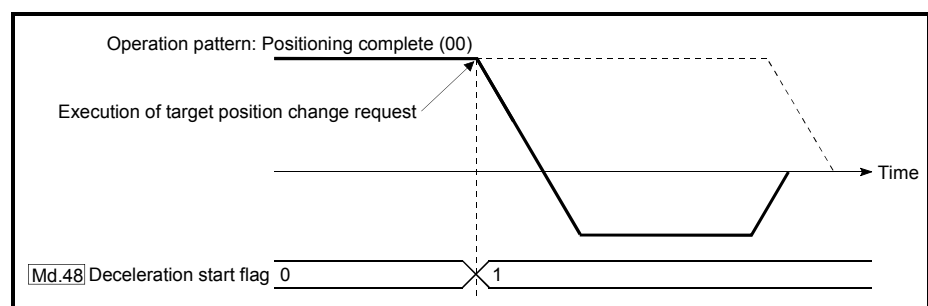
Fig. 12.56 Operation of deceleration start flag at block start

## [2] Control precautions

- (1) The deceleration start flag function is valid for the control system of "1-axis linear control", "2-axis linear interpolation control", "3-axis linear interpolation control", "4-axis linear interpolation control", "speed-position switching control" or "position-speed switching control". (In the case of linear interpolation control, the function is valid for only the reference axis.) Refer to Section 3.2.4 "Combination of QD75 main functions and sub functions".
- (2) The deceleration start flag does not turn ON when the operation pattern is "continuous positioning control" or "continuous path control".
- (3) The deceleration start flag function is invalid for an OPR, JOG operation, inching operation, manual pulse generator operation, and deceleration made with a stop signal.
- (4) The deceleration start flag does not turn ON when a speed change or override is used to make deceleration.
- (5) If a target position change is made while the deceleration start flag is ON, the deceleration start flag remains ON.



- (6) When the movement direction is reversed by a target position change, the deceleration start flag turns ON.



- (7) During position control of position-speed switching control, the deceleration start flag is turned ON by automatic deceleration. The deceleration start flag remains ON if position control is switched to speed control by the position-speed switching signal after the deceleration start flag has turned ON.
- (8) If the condition start of a block start is not made since the condition is not satisfied, the deceleration start flag turns ON when the shape is "End".
- (9) When a continuous operation interrupt request is issued, the deceleration start flag turns ON at a start of deceleration in the positioning data being executed.

## [3] Setting method

To use the "deceleration start flag function", set "1" to the following control data using a sequence program.

The set data is made valid on the rising edge (OFF to ON) of the PLC READY signal [Y0].

| Setting item |                               | Setting value | Setting details                                                                                                                                      | Buffer memory address |
|--------------|-------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Cd.41</b> | Deceleration start flag valid | →             | Set whether the deceleration start flag function is made valid or invalid.<br>0: Deceleration start flag invalid<br>1: Deceleration start flag valid | 1905                  |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

## [4] Checking of deceleration start flag

The "deceleration start flag" is stored into the following buffer memory addresses.

| Monitor item |                         | Monitor value | Storage details                                                                         | Buffer memory address |        |        |        |
|--------------|-------------------------|---------------|-----------------------------------------------------------------------------------------|-----------------------|--------|--------|--------|
|              |                         |               |                                                                                         | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| Md.48        | Deceleration start flag | →             | 0: Status other than below<br>1: Status from deceleration start to next operation start | 899                   | 999    | 1099   | 1199   |

\* Refer to Section 5.6 "List of monitor data" for information on the storage details.

### 12.7.9 Stop command processing for deceleration stop function

The "stop command processing for deceleration stop function" is provided to set the deceleration curve if a stop cause occurs during deceleration stop processing (including automatic deceleration).

This function is valid for both trapezoidal and S-curve acceleration/deceleration processing methods.

(For the stop cause, refer to Section 1.2.3 Outline of stopping.)

The "stop command processing for deceleration stop function" performs the following two operations:

- (1) **Deceleration curve re-processing**  
Re-processes a deceleration curve starting from the speed at stop cause occurrence to stop, according to the preset deceleration time.
- (2) **Deceleration curve continuation**  
Continues the current deceleration curve after a stop cause has occurred.

This section explains the "stop command processing for deceleration stop function" as follows:

- [1] Control details
- [2] Control precautions
- [3] Setting method

#### [1] Control details

The operation of "stop command processing for deceleration stop function" is explained below.

##### (1) Deceleration curve re-processing

A deceleration curve is re-processed starting from the speed at stop cause occurrence to stop, according to the preset deceleration time.

If a stop cause occurs during automatic deceleration of position control, the deceleration stop processing stops as soon as the target has reached the positioning address specified in the positioning data that is currently executed.

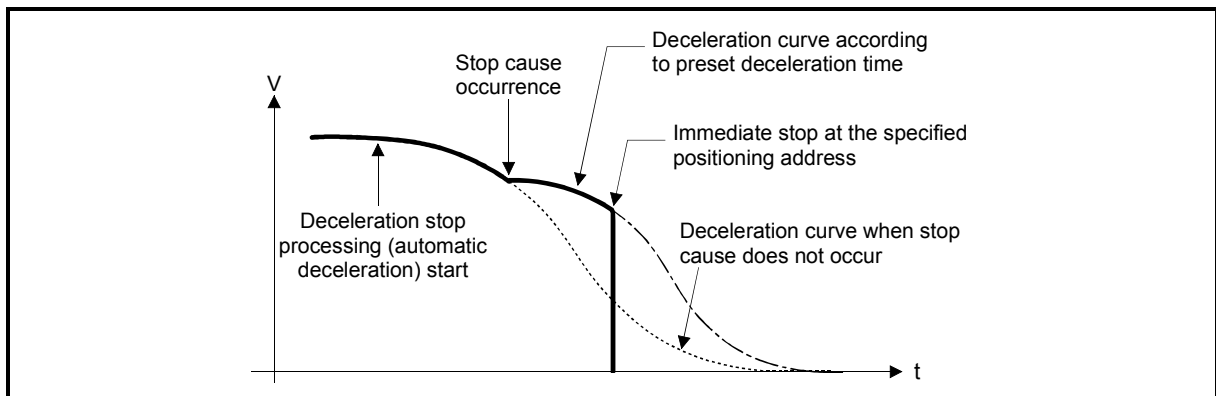


Fig. 12.57 Deceleration curve re-processing operation (for position control or S-curve acceleration/deceleration processing)

**(2) Deceleration curve continuation**

The current deceleration curve is continued after a stop cause has occurred.

If a stop cause occurs during automatic deceleration of position control, the deceleration stop processing may be complete before the target has reached the positioning address specified in the positioning data that is currently executed.

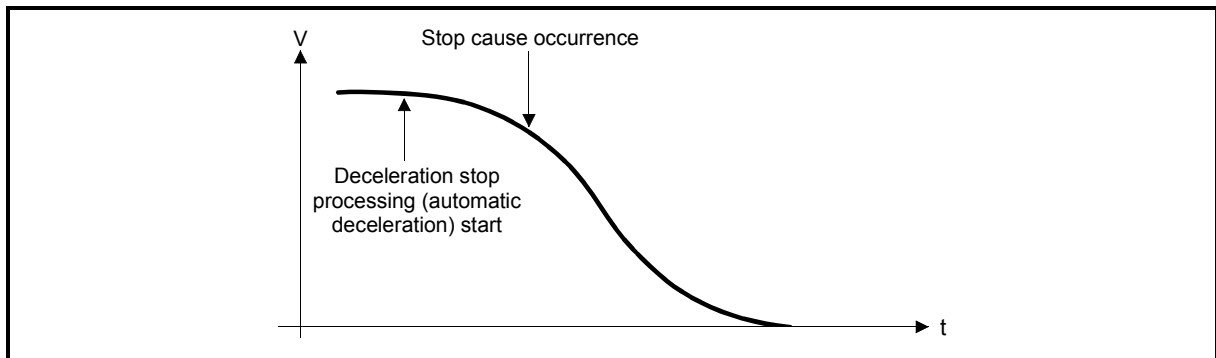


Fig. 12.58 Deceleration curve continuation operation (for position control or S-curve acceleration/deceleration processing)

**[2] Control precautions**

- (1) In manual control (JOG operation, inching operation, manual pulse generator operation), the stop command processing for deceleration stop function is invalid.
- (2) The stop command processing for deceleration stop function is valid when "0: Normal deceleration stop" is set in " [Pr.37](#) Stop group 1 sudden stop selection" to " [Pr.39](#) Stop group 3 sudden stop selection" as the stopping method for stop cause occurrence.
- (3) The stop command processing for deceleration stop function is invalid when "1: Sudden stop" is set in " [Pr.37](#) Stop group 1 sudden stop selection" to " [Pr.39](#) Stop group 3 sudden stop selection". (A deceleration curve is re-processed, according to the " [Pr.36](#) Sudden stop deceleration time" (starting from the speed at stop cause occurrence to a stop))  
In the position control (including position control of speed/position changeover control or position/speed changeover control) mode, positioning may stop immediately depending on the stop cause occurrence timing and " [Pr.36](#) Sudden stop deceleration time" setting.

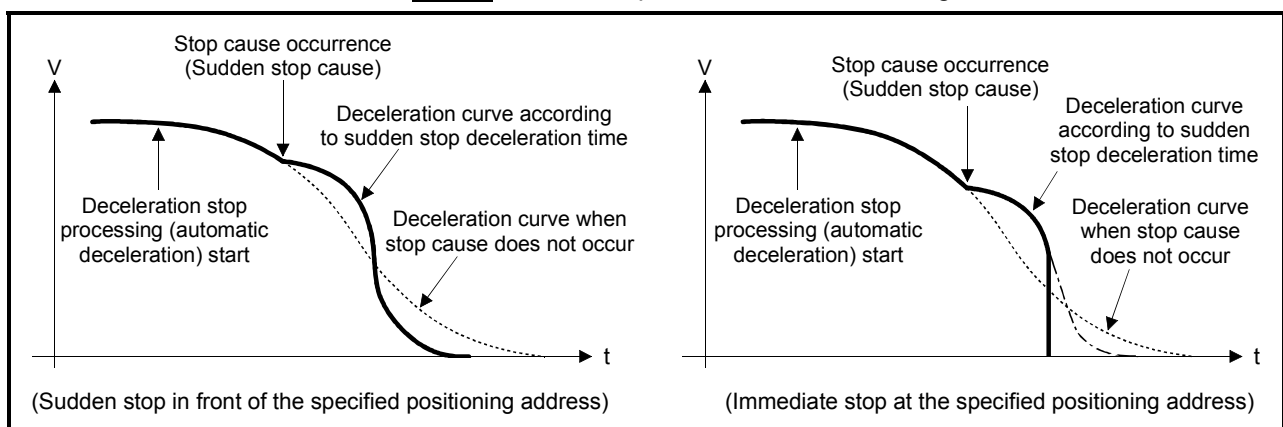


Fig. 12.59 Sudden stop operation (for position control or S-curve acceleration/deceleration processing)

## [3] Setting method

To use the "stop command processing for deceleration stop function", set the following control data in a sequence program.

The set data are made valid as soon as they are written to the buffer memory.

The PLC READY signal [Y0] is irrelevant.

| Setting item |                                                         | Setting value | Setting details                                                                                                                              | Buffer memory address |
|--------------|---------------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Cd.42        | Stop command processing for deceleration stop selection | →             | Set the stop command processing for deceleration stop function.<br>0: Deceleration curve re-processing<br>1: Deceleration curve continuation | 1907                  |

\* For details of the setting details, refer to "Section 5.7 Control data list".

# CHAPTER 13 COMMON FUNCTIONS

The details and usage of the "common functions" executed according to the user's requirements are explained in this chapter.

Common functions include functions required when using the QD75, such as parameter initialization and execution data backup.

Read the setting and execution procedures for each common function indicated in this chapter thoroughly, and execute the appropriate function where required.

|      |                                                   |     |   |
|------|---------------------------------------------------|-----|---|
| 13.1 | Outline of common functions.....                  | 13- | 2 |
| 13.2 | Parameter initialization function .....           | 13- | 3 |
| 13.3 | Execution data backup function .....              | 13- | 5 |
| 13.4 | External I/O signal logic switching function..... | 13- | 7 |
| 13.5 | External I/O signal monitor function .....        | 13- | 8 |

## 13.1 Outline of common functions

"Common functions" are executed according to the user's requirements, regardless of the control system, etc. These common functions are executed by peripheral devices or using sequence programs.

The following table shows the functions included in the "common functions".

| Common function                              | Details                                                                                                                                                                                                                                                                                                             | Means            |                    |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------|
|                                              |                                                                                                                                                                                                                                                                                                                     | Sequence program | Peripheral devices |
| Parameter initialization function            | This function returns the parameter stored in the QD75 buffer memory and flash ROM to the factory-set initial value.                                                                                                                                                                                                | ○                | ○                  |
| Execution data backup function               | This function writes the "execution data", currently being used for control, to the flash ROM.                                                                                                                                                                                                                      | ○                | ○                  |
| External I/O signal logic switching function | This function switches I/O signal logic according to the equipment connected to the QD75.<br>For the system in which drive unit READY with b-contact, upper limit switch, and lower limit switch are not used, the parameter logic setting can be controlled without wiring if it is changed to a "positive logic". | ○                | ○                  |
| External I/O signal monitor function         | This function monitors the external I/O signal monitor information in the module's detailed information which can be displayed on the system monitor of GX Developer*.                                                                                                                                              | —                | —                  |

\* For the QD75P□/QD75D□, this function is available in GX Developer (SW6D5C-GPPW-E or later).

For the QD75P□N/QD75D□N, external I/O signals cannot be monitored on GX Developer. Use the system monitor of GX Works2.



## 13.2 Parameter initialization function

The "parameter initialization function" is used to return the setting data set in the QD75 buffer memory and flash ROM to their factory-set initial values.

The details shown below explain about the "parameter initialization function".

[1] Parameter initialization means

[2] Control details

[3] Control Precautions

[4] Parameter initialization method

### [1] Parameter initialization means

- Initialization is executed with a sequence program.
- Initialization is executed by peripheral device.

Refer to GX Configurator-QP Operating Manual for the execution method by peripheral device.

### [2] Control details

The following table shows the setting data initialized by the "parameter initialization function".

(The data initialized are "buffer memory" and " flash ROM " setting data.)

| Setting data                                            |
|---------------------------------------------------------|
| Basic parameters (Pr.1 to Pr.10)                        |
| Detailed parameters<br>(Pr.11 to Pr.42, Pr.70 , Pr.150) |
| OPR basic parameters (Pr.43 to Pr.48)                   |
| OPR detailed parameters (Pr.49 to Pr.57)                |
| Positioning data (No.1 to No.600)                       |
| Block start data (No.7000 to No.7004)                   |

### [3] Control Precautions

- (1) Parameter initialization is only executed when the positioning control is not carried out (when the PLC READY signal (Y0) is OFF).  
The warning "In PLC READY" (warning code: 111) will occur if executed when the PLC READY [Y0] is ON.
- (2) Writing to the flash ROM can be executed up to 100,000 times.  
If writing to the flash ROM exceeds 100,000 times, the writing may become impossible, and the error "Flash ROM write error" (error code: 801) will occur.
- (3) A "CPU module reset" or "CPU module power restart" must be carried out after the parameters are initialized.
- (4) If an error occurs on the parameter set in the QD75 when the PLC READY signal [Y0] is turned ON, the QD75 READY signal [X0] will not be turned ON and the control cannot be carried out.

**Important**

Parameter initialization takes about 10 seconds. (Up to 30 seconds are sometimes required.)

Do not turn the power ON/OFF; reset the CPU module, etc., during parameter initialization. If the power is turned OFF or the CPU module is reset to forcibly end the process, the data backed up in the flash ROM will be lost.

**[4] Parameter initialization method**

(1) Parameter initialization is carried out using the dedicated instruction "ZP.PINIT".

(Refer to CHAPTER 14 "DEDICATED INSTRUCTIONS" for details.)

(2) Parameter initialization can also be carried out by the writing of the data shown in the table below to the buffer memory using the TO instruction /intelligent function device.

The initialization of the parameter is executed at the time point the data is written to the QD75 buffer memory.

| Setting item                                 | Setting value | Setting details                             | Buffer memory address |        |        |        |
|----------------------------------------------|---------------|---------------------------------------------|-----------------------|--------|--------|--------|
|                                              |               |                                             | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <b>Cd.2</b> Parameter initialization request | 1             | Set "1" (parameter initialization request). | 1901                  |        |        |        |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

When the initialization is complete, "0" will be set in " **Cd.2** Parameter initialization request" by the QD75 automatically.

13.3 Execution data backup function

When the QD75 buffer memory data is rewritten from the CPU module, "the data backed up in the QD75 flash ROM" may differ from "the data (buffer memory data) for which control is being executed".

In cases like these, the data being executed will be lost when the programmable controller power is turned OFF. (Refer to CHAPTER 7 "MEMORY CONFIGURATION AND DATA PROCESS")

In cases like these, the "execution data backup function" backs up the data being executed by writing it to the flash ROM. The data that was backed up is then written to the buffer memory when the power is turned ON next.

The details shown below explain about the "execution data backup function".

- [1] Execution data backup means
- [2] Control details
- [3] Control Precautions
- [4] Execution data backup method

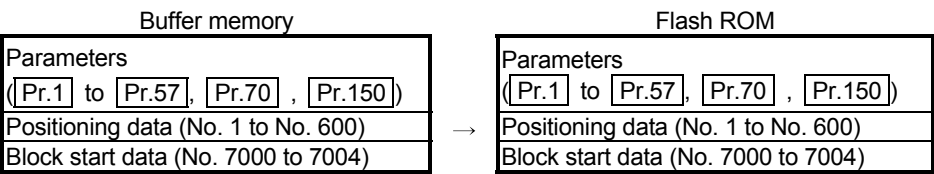
[1] Execution data backup (written to flash ROM) means

- The backup is executed with a sequence program.
- The backup is executed by peripheral device.

Refer to GX Configurator-QP Operating Manual for execution data backup method by peripheral device.

[2] Control details

The following shows the data that can be written to the flash ROM using the "execution data backup function".



[3] Control Precautions

- (1) Parameter initialization is only executed when the positioning control is not carried out (when the PLC READY signal (Y0) is OFF). The warning "In PLC READY" (warning code: 111) will occur if executed when the PLC READY [Y0] is ON.
- (2) Writing to the flash ROM can be executed up to 100,000 times. If writing to the flash ROM exceeds 100,000 times, the writing may become impossible, and the error "Flash ROM write error" (error code: 801) will occur.
- (3) After one power ON/CPU module reset operation, writing to the flash ROM using a sequence program is limited to up to 25 times. If the 26th writing is executed, the error "Flash ROM write number error" (error code: 805) will occur. If this error occurs, carry out the power OFF→ON/CPU module reset operation again. Refer to Md.19 of Section 5.1.7 "Types and roles of monitor data" for details.

|                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Important</b>                                                                                                                     |
| If the power is turned OFF or the CPU module is reset to forcibly end the process, the data backed up in the flash ROM will be lost. |

## [4] Execution data backup method

- (1) Execution data backup (writing to the flash ROM) is carried out using the dedicated instruction "PFWRT". (Refer to CHAPTER 14 "DEDICATED INSTRUCTIONS" for details.)
- (2) Refer to Section 7.2 "Data transmission process" for the data transmission processing at the backup of the execution data.
- (3) Execution data backup can also be carried out by the writing of the data shown in the table below to the QD75 buffer memory using the TO instruction /intelligent function device.

The writing to the flash ROM is executed at the time point the data is written to the QD75 buffer memory.

| Setting item                                                                               | Setting value | Setting details                    | Buffer memory address |        |        |        |
|--------------------------------------------------------------------------------------------|---------------|------------------------------------|-----------------------|--------|--------|--------|
|                                                                                            |               |                                    | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <span style="border: 1px solid black; padding: 0 2px;">Cd.1</span> Flash ROM write request | 1             | Set "1" (flash ROM write request). | 1900                  |        |        |        |

\* Refer to Section 5.7 "List of control data" for details on the setting details.

When the writing to the flash ROM is complete, "0" will be set in " Cd.1 Flash ROM write request" by the QD75 automatically.

### 13.4 External I/O signal logic switching function

This function switches the signal logic according to the external equipment connected to the QD75.

For the system in which drive unit READY with b-contact, upper limit switch, and lower limit switch are not used, the parameter logic setting can be controlled without wiring if it is changed to a "positive logic".

When the drive unit READY, upper limit switch, and lower limit switch are used, ensure to use them with b-contact.

The details shown below explain about the "external I/O signal logic switching function".

[1] Parameter setting details

[2] Precautions on parameter setting

#### [1] Parameter setting details

To use the "external I/O signal logic switching function", set the parameters shown in the following table.

| Setting item |                               | Setting details                                                     |                                | Factory-set initial value | Buffer memory address |        |        |        |                                         |                                         |
|--------------|-------------------------------|---------------------------------------------------------------------|--------------------------------|---------------------------|-----------------------|--------|--------|--------|-----------------------------------------|-----------------------------------------|
|              |                               |                                                                     |                                |                           | Axis 1                | Axis 2 | Axis 3 | Axis 4 |                                         |                                         |
| Pr.22        | Input signal logic selection  | • Selection of logic of signals input from external source to QD75  |                                | 0                         | 31                    | 181    | 331    | 481    |                                         |                                         |
|              |                               | b0                                                                  | Lower limit                    |                           |                       |        |        |        | 0: Negative logic,<br>1: Positive logic |                                         |
|              |                               | b1                                                                  | Upper limit                    |                           |                       |        |        |        |                                         |                                         |
|              |                               | b2                                                                  | Drive unit READY               |                           |                       |        |        |        |                                         |                                         |
|              |                               | b3                                                                  | Stop signal                    |                           |                       |        |        |        |                                         |                                         |
|              |                               | b4                                                                  | External command signal        |                           |                       |        |        |        |                                         |                                         |
|              |                               | b5                                                                  | Zero signal                    |                           |                       |        |        |        |                                         |                                         |
|              |                               | b6                                                                  | Near-point dog signal          |                           |                       |        |        |        |                                         |                                         |
|              |                               | b7                                                                  | Unused                         |                           |                       |        |        |        |                                         | Set "0".                                |
|              |                               | b8                                                                  | Manual pulse generator input   |                           |                       |        |        |        |                                         | 0: Negative logic,<br>1: Positive logic |
| b9 to b15    | Unused                        | Set "0".                                                            |                                |                           |                       |        |        |        |                                         |                                         |
| Pr.23        | Output signal logic selection | • Selection of logic of signals output from QD75 to external source |                                | 0                         | 32                    | 182    | 332    | 482    |                                         |                                         |
|              |                               | b0                                                                  | Command pulse signal           |                           |                       |        |        |        | 0: Negative logic,<br>1: Positive logic |                                         |
|              |                               | b1 to b3                                                            | Unused                         |                           |                       |        |        |        | Set "0".                                |                                         |
|              |                               | b4                                                                  | Deviation counter clear signal |                           |                       |        |        |        | 0: Negative logic,<br>1: Positive logic |                                         |
|              |                               | b5 to b15                                                           | Unused                         |                           |                       |        |        |        | Set "0".                                |                                         |

\* Refer to Section 5.2 "List of parameters" for the information on detail settings.

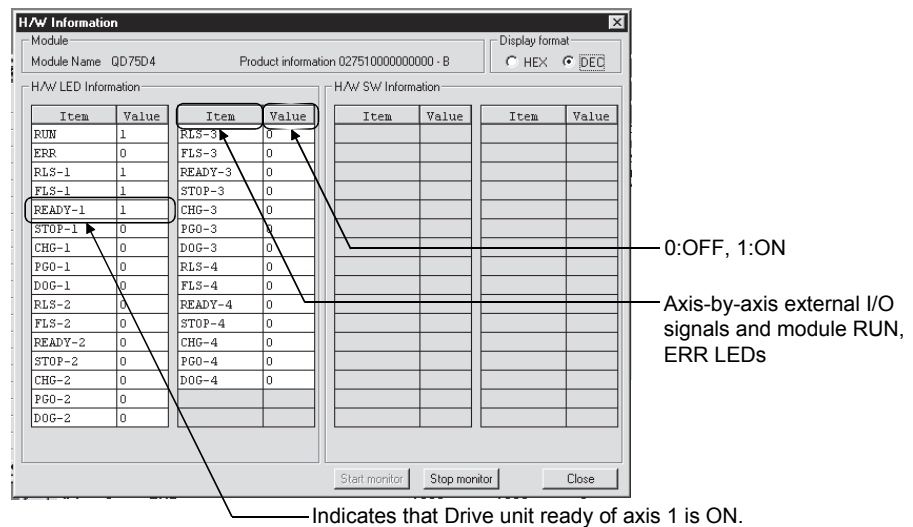
#### [2] Precautions on parameter setting

- (1) The external I/O signal logic switching parameters are validated when the PLC READY signal [Y0] is turned OFF to ON. (The logic is negative right after power-on.)
- (2) If each signal logic is set erroneously, the operation may not be carried out correctly.  
Before setting, check the specifications of the equipment to be used.

## 13.5 External I/O signal monitor function

The "external I/O signal monitor function" monitors the module's information and external I/O signal monitor information in the module's detailed information which can be displayed on the system monitor of GX Developer\*1.

The information that can be monitored are the module's information (same as the QD75 front "RUN", "ERR" LED indicators) and the following external I/O signals. (Set the logic of the external I/O signals in " [Pr.22](#) Input signal logic selection" and " [Pr.23](#) Output signal logic selection".)



\*1: For the QD75P□/QD75D□, this function is available in GX Developer (SW6D5C-GPPW-E or later). For details, refer to GX Developer Operating Manual.

For the QD75P□N/QD75D□N, external I/O signals cannot be monitored on GX Developer. Use the system monitor of GX Works2. For details on the system monitor of GX Works2, refer to GX Works2 Version 1 Operating Manual (Common).

# CHAPTER 14 DEDICATED INSTRUCTIONS

The QD75 dedicated instructions are explained in this chapter.

These instructions are used to facilitate the programming for the use of the functions of the intelligent function module.

Using the dedicated instructions, the programming can be carried out without being aware of the QD75 buffer memory address and interlock signal.

|      |                                                          |        |
|------|----------------------------------------------------------|--------|
| 14.1 | List of dedicated instructions .....                     | 14- 2  |
| 14.2 | Interlock during dedicated instruction is executed ..... | 14- 2  |
| 14.3 | Z.ABRST1, Z.ABRST2, Z.ABRST3, Z.ABRST4 .....             | 14- 3  |
| 14.4 | ZP.PSTRT1, ZP.PSTRT2, ZP.PSTRT3, ZP.PSTRT4 .....         | 14- 8  |
| 14.5 | ZP.TEACH1, ZP.TEACH2, ZP.TEACH3, ZP.TEACH4 .....         | 14- 12 |
| 14.6 | ZP.PFWRT .....                                           | 14- 16 |
| 14.7 | ZP.PINIT .....                                           | 14- 20 |

## 14.1 List of dedicated instructions

The dedicated instructions explained in this Chapter are listed in Table 14.1.

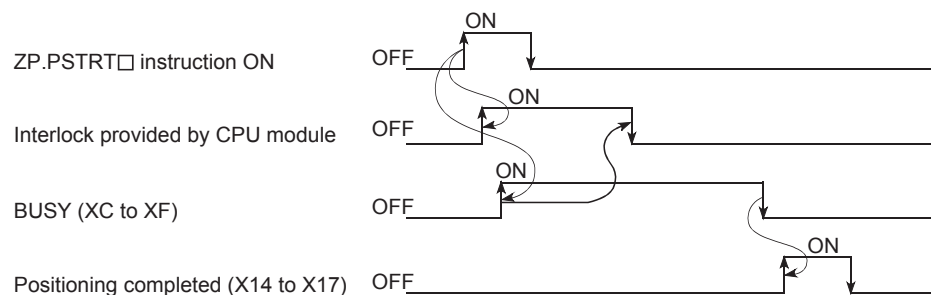
Table 14.1 List of dedicated instructions

| Application                   | Dedicated instruction | Outline of functions                                                                                             | Reference    |
|-------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------|--------------|
| Absolute position restoration | Z.ABRST1              | This function restores the absolute position of the designated axis of the QD75.                                 | Section 14.3 |
|                               | Z.ABRST2              |                                                                                                                  |              |
|                               | Z.ABRST3              |                                                                                                                  |              |
|                               | Z.ABRST4              |                                                                                                                  |              |
| Positioning start             | ZP.PSTR1              | This function starts the positioning control of the designated axis of the QD75.                                 | Section 14.4 |
|                               | ZP.PSTR2              |                                                                                                                  |              |
|                               | ZP.PSTR3              |                                                                                                                  |              |
|                               | ZP.PSTR4              |                                                                                                                  |              |
| Teaching                      | ZP.TEACH1             | This function carries out teaching the designated axis of the QD75.                                              | Section 14.5 |
|                               | ZP.TEACH2             |                                                                                                                  |              |
|                               | ZP.TEACH3             |                                                                                                                  |              |
|                               | ZP.TEACH4             |                                                                                                                  |              |
| Writing to flash ROM          | ZP.PFWRT              | This function writes the buffer memory parameters, positioning data and block start data to the flash ROM.       | Section 14.6 |
| Parameter initialization      | ZP.PINIT              | This function initializes the buffer memory and flash ROM setting data to the factory-set data (initial values). | Section 14.7 |

## 14.2 Interlock during dedicated instruction is executed

The absolute position restoration instruction (Z.ABRST□), positioning start instruction (ZP.PSTR□) and teaching instruction (ZP.TEACH□) cannot be executed simultaneously in each axis. If they are executed at the same time, the second and later instructions are ignored by an internal interlock (no error will occur).

The timing of the positioning start dedicated instruction (ZP.PSTR□) is as shown below.

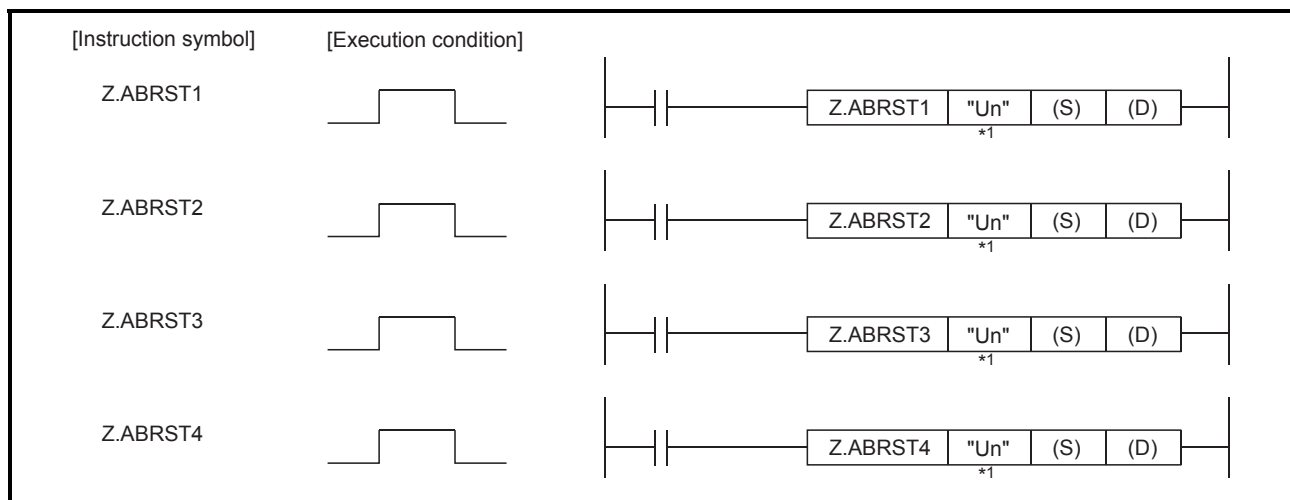




## 14.3 Z.ABRST1, Z.ABRST2, Z.ABRST3, Z.ABRST4

These dedicated instructions restore the absolute position of the designated axis.

| Setting data | Usable device   |      |               |                        |      |                                   |                   |          |        |
|--------------|-----------------|------|---------------|------------------------|------|-----------------------------------|-------------------|----------|--------|
|              | Internal device |      | File register | Link direct device J□□ |      | Intelligent function module U□\G□ | Index register Zn | Constant | Others |
|              | Bit             | Word |               | Bit                    | Word |                                   |                   | K, H     |        |
| (S)          | —               | ○    |               | —                      |      |                                   |                   | —        | —      |
| (D)          | ○               | ○    | —             | —                      |      |                                   |                   | —        | —      |



\*1 If the originating station is a Basic model QCPU (function version B or later), universal model QCPU, or safety CPU, "" (double quotation) of the first argument can be omitted.

When Z.ABRST1, Z.ABRST2, Z.ABRST3, and Z.ABRST4 are common to each other, they are designated as "Z.ABRST□".

## [Setting data]

| Setting data | Setting details                                                                                                                                                                                     | Setting side (*1) | Data type   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------|
| "Un"         | QD75 head I/O number<br>(00 to FE: High-order two digits of I/O number expressed in three digits)                                                                                                   | User              | BIN 16 bits |
| (S)          | Head number of a device in which control data is stored                                                                                                                                             | —                 | Device name |
| (D)          | Head number of a bit device which turns ON the operation by one scan at the time of completion of the instruction.<br>If the instruction is completed abnormally, ((D) + 1) will also be turned ON. | System            | Bit         |

Note) The file register of each of the local device and the program cannot be used as a device for setting data.

\*1: The data on the setting side is as follows.

- User : Data before the execution of dedicated instructions is stored by user.
- System : Data after the execution of dedicated instruction is stored by CPU module.

## [Control data]

| Device         | Item                                  | Setting data                                                                                                                                                                                                                                                                    | Setting range                 | Setting side (*1) |
|----------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------|
| (S)+0          | System area                           | —                                                                                                                                                                                                                                                                               | —                             | —                 |
| (S)+1          | Complete status                       | The state at the time of completion is stored.<br>• 0 : Normal completion<br>• Other than 0 : Abnormal completion (error code) (*2)                                                                                                                                             | —                             | System            |
| (S)+2          | Signal received from servo amplifier  | The following signal states taken in from the servo amplifier to the input module are written.<br>• b0: ABS data bit 0<br>• b1: ABS data bit 1<br>• b2: Transmission data READY flag                                                                                            | b0: 0/1<br>b1: 0/1<br>b2: 0/1 | User              |
| (S)+3          | Signal transmitted to servo amplifier | The ON/OFF states of the following data which are calculated using the dedicated instructions by the "signals received from the servo amplifier" and output to the servo amplifier are stored.<br>• b0: Servo amplifier ON<br>• b1: ABS transfer mode<br>• b2: ABS request flag | —                             | System            |
| (S)+4          | Status                                | Status of communication with servo amplifier<br>• 0 : Communication completed (Set by user at communication start)<br>• Other than 0 : During communication (System stores)                                                                                                     | 0                             | User/system       |
| (S)+5 to (S)+7 | System area                           | —                                                                                                                                                                                                                                                                               | —                             | —                 |

\*1: The data on the setting side is as follows.

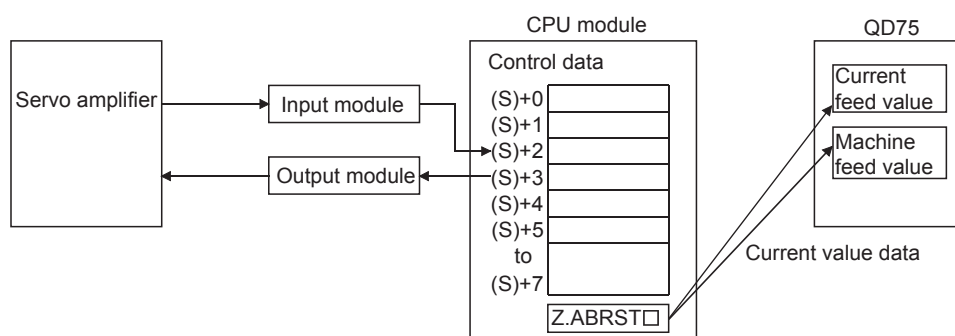
- User : Data before the execution of dedicated instructions is stored by user.
- System : Data after the execution of dedicated instruction is stored by CPU module.

\*2: Refer to Section 15.3 for error codes at abnormal completion.

## [Functions]

- (1) The positioning data is read from the servo amplifier capable of processing the absolute positions of the axes to be set (See below), and the values converted with respect to a unit are stored in " **[Md.20]** Current feed value" and " **[Md.21]** Machine feed value" area of the QD75.
- Z.ABRST1: Axis 1
  - Z.ABRST2: Axis 2
  - Z.ABRST3: Axis 3
  - Z.ABRST4: Axis 4

For absolute position detection system, carry out the absolute position restoration operation each time the power is turned ON or CPU module is reset.



- (2) An I/O module is used for communication (data read/write) with the servo amplifier capable of processing the absolute positions.

When using the Z.ABRST□, prepare the input/output with the following number of points, for each axis, for communication with the servo amplifier.

- Input : 3 points
- Output : 3 points

Refer to Section 12.6 for wiring of I/O signals.

- (3) The Z.ABRST□ instruction completion can be confirmed using the complete devices ((D)+0) and ((D)+1).

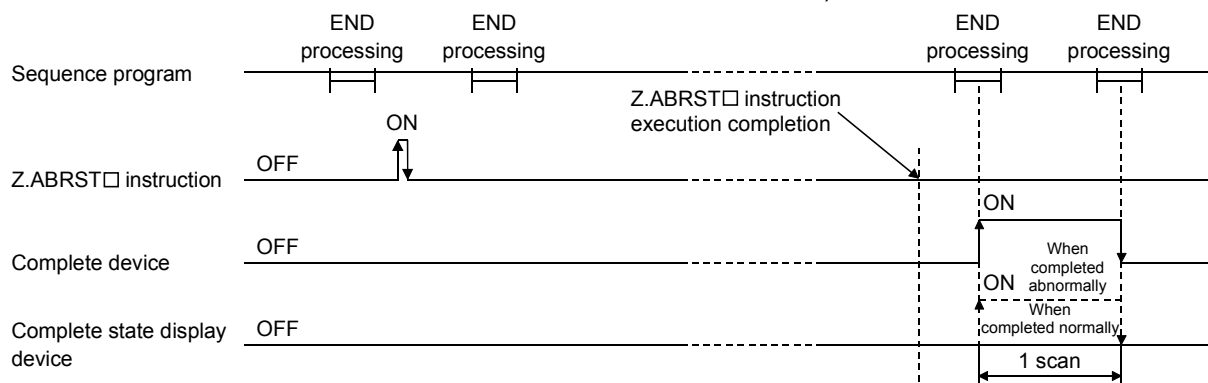
- (a) Complete device ((D)+0)

This device is turned ON by the END processing of the scan for which Z.ABRST□ instruction is completed, and turned OFF by the next END processing.

- (b) Complete state display device ((D)+1)

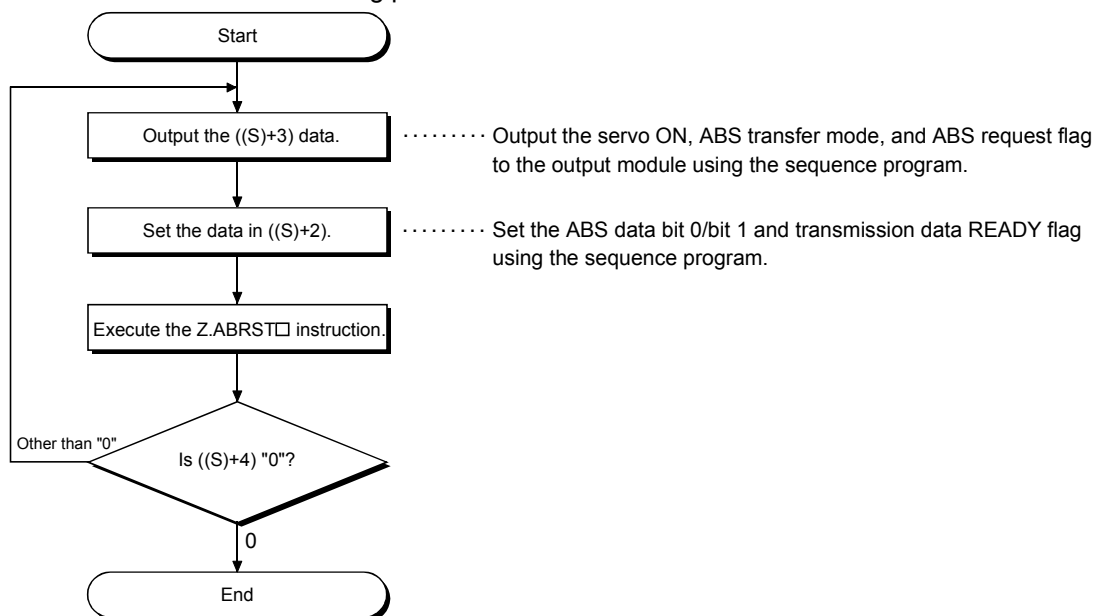
This device is turned ON and OFF according to the state in which Z.ABRST□ instruction is completed.

- When completed normally : Kept unchanged at OFF.
- When completed abnormally : This device is turned ON by the END processing of the scan for which Z.ABRST□ instruction is completed, and turned OFF by the next END processing (same ON/OFF operation as complete device).



The completion of absolute position restoration can be confirmed using the ((S)+4) "status".

- (4) Using the Z.ABRST□ instruction, the absolute position restoration is carried out in the following procedure.



#### [Errors]

- (1) When a dedicated instruction is completed abnormally, the error complete signal ((D)+1) is turned ON, and the error code is stored in the complete status ((S)+1). Check and take a measure against the error referring to Section 15.3 "List of error codes".

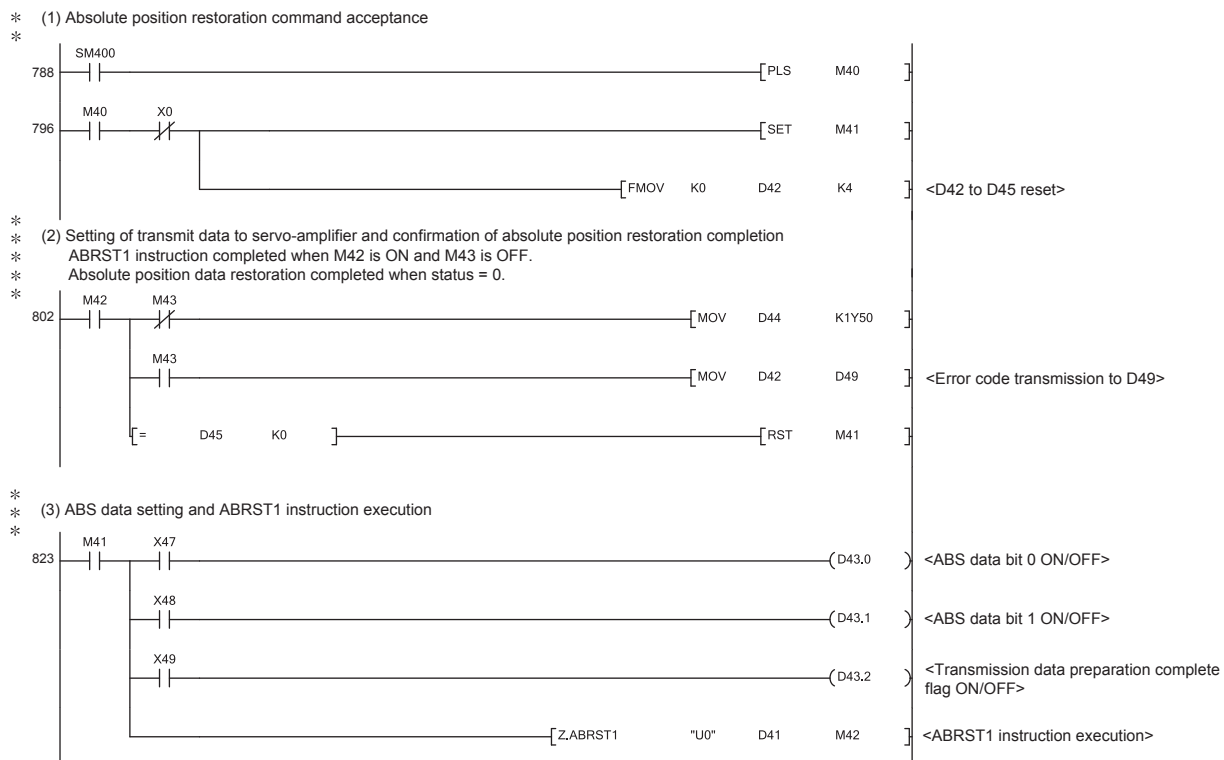
#### [Precautions]

- (1) After the absolute position detection system is configured, the absolute position restoration must be carried out at least once after the power is turned ON or reset. Unless the absolute position restoration of the QD75 is completed, the servo amplifier will not be turned ON.
- (2) Execute absolute position restoration when the PLC READY signal [Y0] is OFF.
- (3) The restoration of the absolute position (Z.ABRST□ instruction execution) can also be carried out when the servo amplifier is turned ON. If this is carried out, however, the servo ON signal is turned OFF for approx. 60ms + scan time (servo OFF) and the motor may move.  
When carrying out the absolute position restoration during servo OFF, install an electromagnetic brake and output the Z.ABRST□ instruction to that brake during the Z.ABRST□ instruction execution.
- (4) The following dedicated instructions cannot be executed simultaneously for the same axis.  
(Can be executed simultaneously for different axes.)
  - Positioning start instructions (ZP.PSTRT1 to ZP.PSTRT4)
  - Absolute position restoration instructions (Z.ABRST1 to Z.ABRST4)
  - Teaching instructions (ZP.TEACH1 to ZP.TEACH4)
- (5) When the remote I/O station\* (Q Corresponding MELSECNET/H network remote I/O module) is used, the dedicated instruction (Z.ABRST□) is unusable.  
(For the sequence program example of when the dedicated instruction is not used, refer to the "Absolute Position Detection System" Installation Guide of the Mitsubishi Electric General-Purpose AC Servo.)  
\*: For details of the remote I/O station, refer to Q Corresponding MELSECNET/H Network System Reference Manual (Remote I/O Network).

- [Program examples]

The X47 to X49 and Y50 to Y52 are used for communication with the servo amplifier.

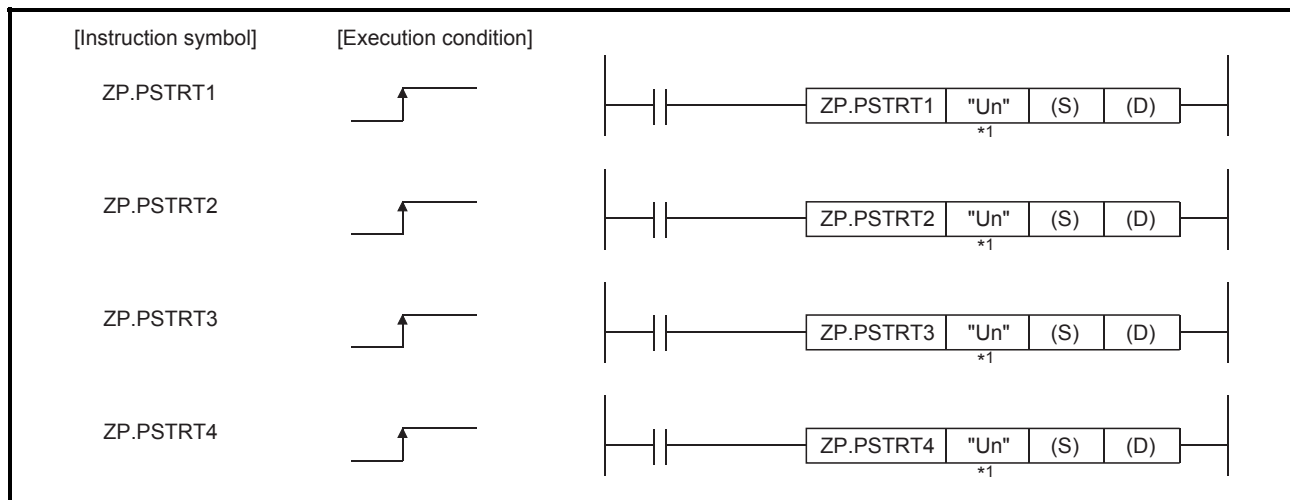
X47: ABS data bit 0  
X48: ABS data bit 1  
X49: Transmission data READY flag  
Y50: Servo ON signal  
Y51: ABS transfer mode  
Y52: ABS request flag



## 14.4 ZP.PSTRT1, ZP.PSTRT2, ZP.PSTRT3, ZP.PSTRT4

These dedicated instructions are used to start the positioning of the designated axis.

| Setting data | Usable device   |      |               |                        |      |                                  |                   |          |        |
|--------------|-----------------|------|---------------|------------------------|------|----------------------------------|-------------------|----------|--------|
|              | Internal device |      | File register | Link direct device J□□ |      | Intelligent function module U□G□ | Index register Zn | Constant | Others |
|              | Bit             | Word |               | Bit                    | Word |                                  |                   | K, H     |        |
| (S)          | —               | ○    |               | —                      |      |                                  | —                 | —        |        |
| (D)          | ○               | ○    | —             | —                      |      |                                  | —                 | —        |        |



\*1 If the originating station is a Basic model QCPU (function version B or later), universal model QCPU, or safety CPU, "" (double quotation) of the first argument can be omitted.

When ZP.PSTRT1, ZP.PSTRT2, ZP.PSTRT3, and ZP.PSTRT4 are common to each other, they are designated as "ZP.PSTRT□".

## [Setting data]

| Setting data | Setting details                                                                                                                                                                                     | Setting side (*1) | Data type   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------|
| "Un"         | QD75 head I/O number<br>(00 to FE: High-order two digits of I/O number expressed in three digits)                                                                                                   | User              | BIN 16 bits |
| (S)          | Head number of a device in which control data is stored                                                                                                                                             | —                 | Device name |
| (D)          | Head number of a bit device which turns ON the operation by one scan at the time of completion of the instruction.<br>If the instruction is completed abnormally, ((D) + 1) will also be turned ON. | System            | Bit         |

Note) The file register of each of the local device and the program cannot be used as a device for setting data.

\*1: The data on the setting side is as follows.

- User : Data before the execution of dedicated instructions is stored by user.
- System : Data after the execution of dedicated instruction is stored by CPU module.

## [Control data]

| Device | Item            | Setting data                                                                                                                                                                                                                                                                        | Setting range                            | Setting side (*1) |
|--------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------------|
| (S)+0  | System area     | —                                                                                                                                                                                                                                                                                   | —                                        | —                 |
| (S)+1  | Complete status | The state at the time of completion is stored.<br>• 0 : Normal completion<br>• Other than 0: Abnormal completion (error code)(*2)                                                                                                                                                   | —                                        | System            |
| (S)+2  | Start No.       | The following data Nos. to be started by the ZP.PSTRT□ instruction are designated.<br>• Positioning data No. : 1 to 600<br>• Block start : 7000 to 7004<br>• Machine OPR : 9001<br>• Fast OPR : 9002<br>• Current value changing : 9003<br>• Multiple axes simultaneous start: 9004 | 1 to 600<br>7000 to 7004<br>9001 to 9004 | User              |

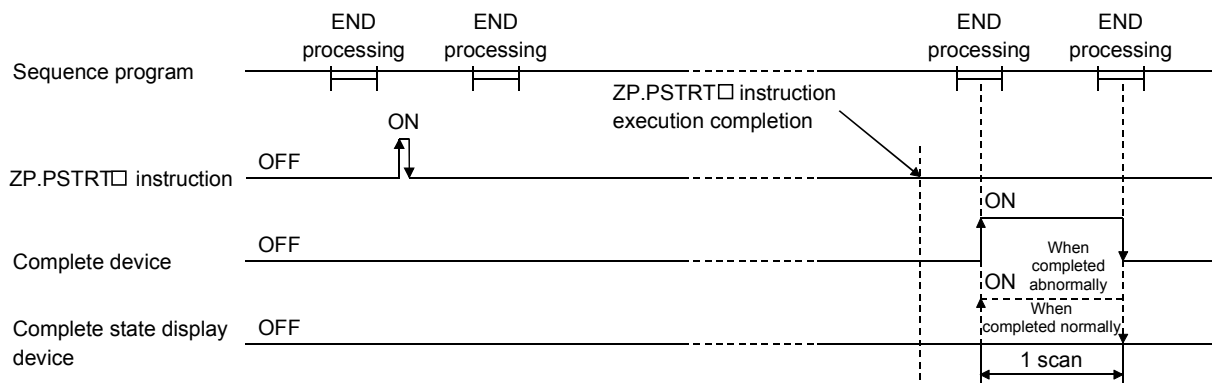
\*1: The data on the setting side is as follows.

- User : Data before the execution of dedicated instructions is stored by user.
- System : Data after the execution of dedicated instruction is stored by CPU module.

\*2: Refer to Section 15.3 for error codes at abnormal completion.

## [Functions]

- (1) The positioning start of the axes to be processed (See below) is carried out.
  - ZP.PSTRT1: Axis 1
  - ZP.PSTRT2: Axis 2
  - ZP.PSTRT3: Axis 3
  - ZP.PSTRT4: Axis 4
- (2) The block start, OPR start, current value changing, and multiple axes simultaneous start can be carried out by the setting of "start number" 7000 to 7004/9001 to 9004 in ((S)+2).
- (3) The ZP.PSTRT□ instruction completion can be confirmed using the complete devices ((D)+0) and ((D)+1).
  - (a) Complete device ((D)+0)  
This device is turned ON by the END processing of the scan for which ZP.PSTRT□ instruction is completed, and turned OFF by the next END processing.
  - (b) Complete state display device ((D)+1)  
This device is turned ON and OFF according to the state in which ZP.PSTRT□ instruction is completed.
    - When completed normally : Kept unchanged at OFF.
    - When completed abnormally : This device is turned ON by the END processing of the scan for which ZP.PSTRT□ instruction is completed, and turned OFF by the next END processing (same ON/OFF operation as complete device).



### [Errors]

- (1) When an ZP.PSTRT□ instruction is completed abnormally, the error complete signal ((D)+1) is turned ON, and the error code is stored in the complete status ((S)+1).  
Check and take a measure against the error referring to Section 15.3 "List of error codes".

### [Precautions]

- (1) When positioning is started by the ZP.PSTRT□ instruction, the positioning start signals (Y10 to Y13) will not turn ON.  
Therefore, the start complete signal (X10 to X13) cannot be used for confirming completion of start of positioning.  
To confirm that positioning control is being executed, use the ZP.PSTRT□ start command or BUSY signals (XC to XF).
- (2) If, after positioning has been started by the ZP.PSTRT□ instruction, a stop command is input before positioning is complete, the complete device (D) turns ON one scan and the execution of the ZP.PSTRT□ instruction is completed.
- (3) The following dedicated instructions cannot be executed simultaneously for the same axis.  
(Can be executed simultaneously for different axes.)
  - Positioning start instructions (ZP.PSTRT1 to ZP.PSTRT4)
  - Absolute position restoration instructions (Z.ABRST1 to Z.ABRST4)
  - Teaching instructions (ZP.TEACH1 to ZP.TEACH4)
- (4) The ZP.PSTRT□ instruction can only be executed when the QD75 READY signal (X0) is turned ON.  
Even if the ZP.PSTRT□ instruction execution request is given when the QD75 READY signal is turned OFF, the ZP.PSTRT□ instruction will not be executed. (not processed.)  
Before executing the ZP.PSTRT□ instruction, turn ON the PLC READY signal (Y0), and turn ON the QD75 READY signal (X0).

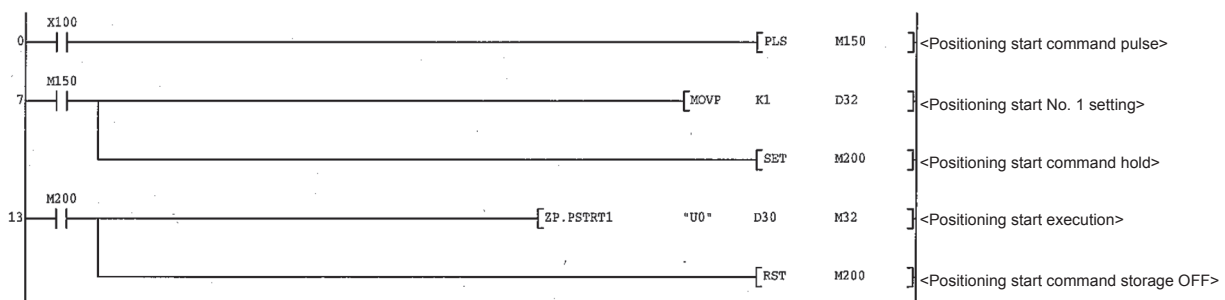


- (5) When the remote I/O station\* (Q Corresponding MELSECNET/H network remote I/O module) is used, the dedicated instruction (ZP.PSTRT□) is unusable.  
 \*: For details of the remote I/O station, refer to Q Corresponding MELSECNET/H Network System Reference Manual (Remote I/O Network).
- (6) If the ZP.PSTRT□ instruction is executed in either of the following cases, an error "Dedicated instruction error" (error code: 804) will occur and positioning cannot be started.
- Any value other than 1 to 600, 7000 to 7004, and 9001 to 9004 is set to "Starting number" (device: (S)+2) of the control data.
  - The instruction for a non-existent axis is specified.
- (Example: The ZP.PSTRT2 instruction is specified when the QD75P1 is used.)
- (7) When the multiple axes simultaneous start is executed by ZP.PSTRT□ instruction, the completion device (D) will turn ON when the positioning of the axes executed by ZP.PSTRT□ instructions is completed. (When the instructions is ZP.PSTRT1, the axis will be axis 1.)

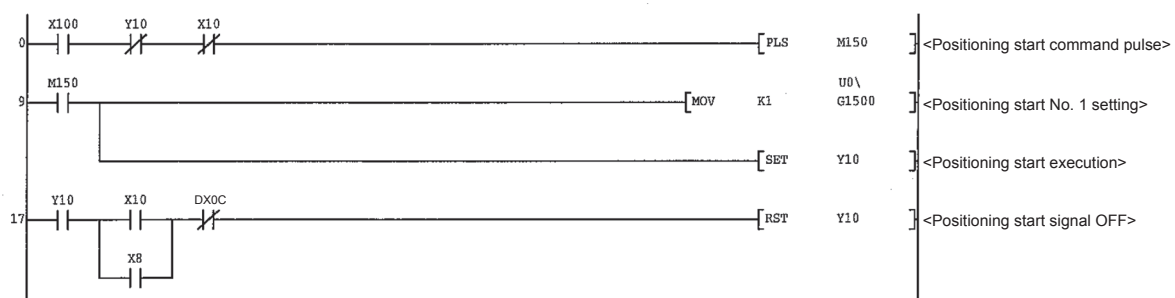
### [Program examples]

- The following program executes the positioning start of positioning data No. 1 when X100 turns ON.  
 Use D30 to D32 as the control data devices of positioning data No. 1, and M32 and M200 as the completion devices.

#### (1) Positioning start program



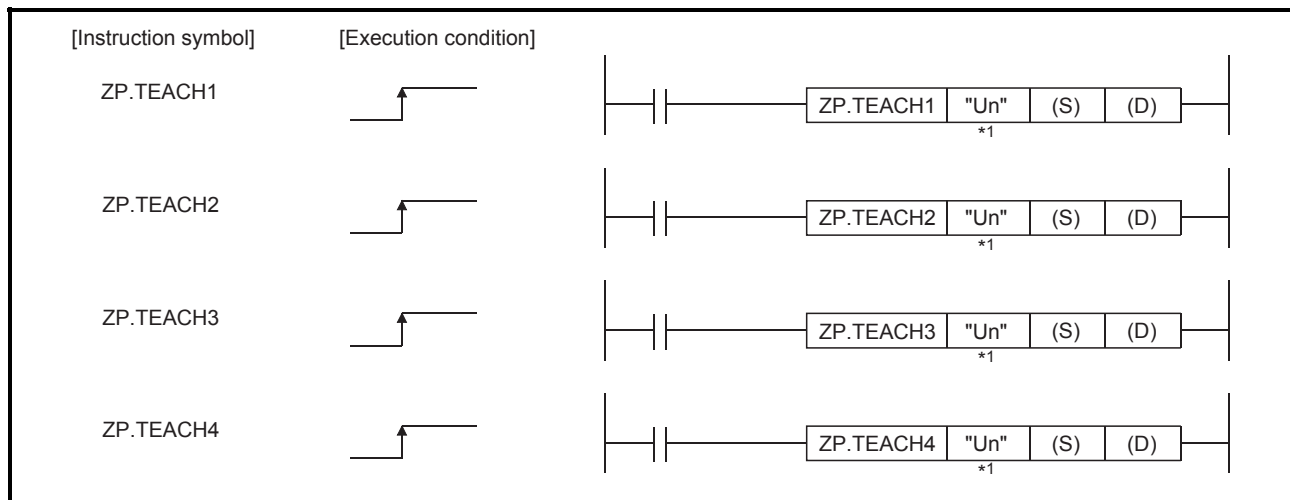
#### (2) Positioning start program (when the dedicated instruction is not used)



## 14.5 ZP.TEACH1, ZP.TEACH2, ZP.TEACH3, ZP.TEACH4

This dedicated instruction is used to teach the designated axis.

| Setting data | Usable device   |      |               |                        |      |                                   |    |          |        |
|--------------|-----------------|------|---------------|------------------------|------|-----------------------------------|----|----------|--------|
|              | Internal device |      | File register | Link direct device J□□ |      | Intelligent function module U□\G□ | Zn | Constant | Others |
|              | Bit             | Word |               | Bit                    | Word |                                   |    | K, H     |        |
| (S)          | —               | ○    |               | —                      |      |                                   |    | —        | —      |
| (D)          | ○               | ○    | —             | —                      |      |                                   |    | —        | —      |



\*1 If the originating station is a Basic model QCPU (function version B or later), universal model QCPU, or safety CPU, "" (double quotation) of the first argument can be omitted.

When ZP.TEACH1, ZP.TEACH2, ZP.TEACH3, and ZP.TEACH4 are common to each other, they are designated as "ZP.TEACH□".

## [Setting data]

| Setting data | Setting details                                                                                                                                                                                     | Setting side (*1) | Data type   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------|
| "Un"         | QD75 head I/O number<br>(00 to FE: High-order two digits of I/O number expressed in three digits)                                                                                                   | User              | BIN 16 bits |
| (S)          | Head number of a device in which control data is stored                                                                                                                                             | —                 | Device name |
| (D)          | Head number of a bit device which turns ON the operation by one scan at the time of completion of the instruction.<br>If the instruction is completed abnormally, ((D) + 1) will also be turned ON. | System            | Bit         |

Note) The file register of each of the local device and the program cannot be used as a device for setting data.

\*1: The data on the setting side is as follows.

- User : Data before the execution of dedicated instructions is stored by user.
- System : Data after the execution of dedicated instruction is stored by CPU module.

## [Control data]

| Device | Item                    | Setting data                                                                                                                                                                                                     | Setting range | Setting side (*1) |
|--------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------|
| (S)+0  | System area             | —                                                                                                                                                                                                                | —             | —                 |
| (S)+1  | Complete status         | The state at the time of completion is stored.<br>0 : Normal completion<br>Other than 0 : Abnormal completion (error code)(*2)                                                                                   | —             | System            |
| (S)+2  | Teaching data selection | The address (positioning address/arc address) to which the current feed value is written is set.<br>0: Current feed value is written to positioning address.<br>1: Current feed value is written to arc address. | 0, 1          | User              |
| (S)+3  | Positioning data No.    | The positioning data No. for which teaching is carried out is set.                                                                                                                                               | 1 to 600      | User              |

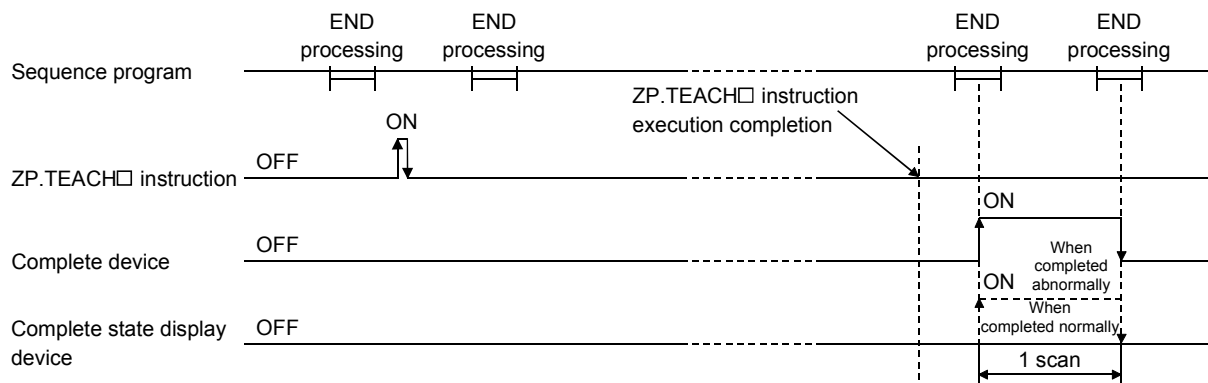
\*1: The data on the setting side is as follows.

- User : Data before the execution of dedicated instructions is stored by user.
- System : Data after the execution of dedicated instruction is stored by CPU module.

\*2: Refer to Section 15.3 for error codes at abnormal completion.

## [Functions]

- (1) The "current feed value" of the axes to be set (See below) is set in the positioning address or arc address.  
The positioning data other than the positioning addresses and arc addresses are set by peripheral device or using a sequence program.
  - ZP.TEACH1: Axis 1
  - ZP.TEACH2: Axis 2
  - ZP.TEACH3: Axis 3
  - ZP.TEACH4: Axis 4
- (2) Teaching can be carried out for the positioning data No. 1 to 600.
- (3) The movement of the machine to the address (position) set in the positioning address/arc address of the positioning data is carried out by the JOG operation, inching operation, or manual pulse generator operation.
- (4) The ZP.TEACH□ instruction completion can be confirmed using the complete devices ((D)+0) and ((D)+1).
  - (a) Complete device ((D)+0)  
This device is turned ON by the END processing of the scan for which ZP.TEACH□ instruction is completed, and turned OFF by the next END processing.
  - (b) Complete state display device ((D)+1)  
This device is turned ON and OFF according to the state in which ZP.TEACH□ instruction is completed.
    - When completed normally : Kept unchanged at OFF.
    - When completed abnormally : This device is turned ON by the END processing of the scan for which ZP.TEACH□ instruction is completed, and turned OFF by the next END processing (same ON/OFF operation as complete device).



### [Errors]

- (1) When a ZP.TEACH instruction is completed abnormally, the error complete signal ((D)+1) is turned ON, and the error code is stored in the complete status (S)+1. Check and take a measure against the error referring to Section 15.3 "List of error codes".

### [Precautions]

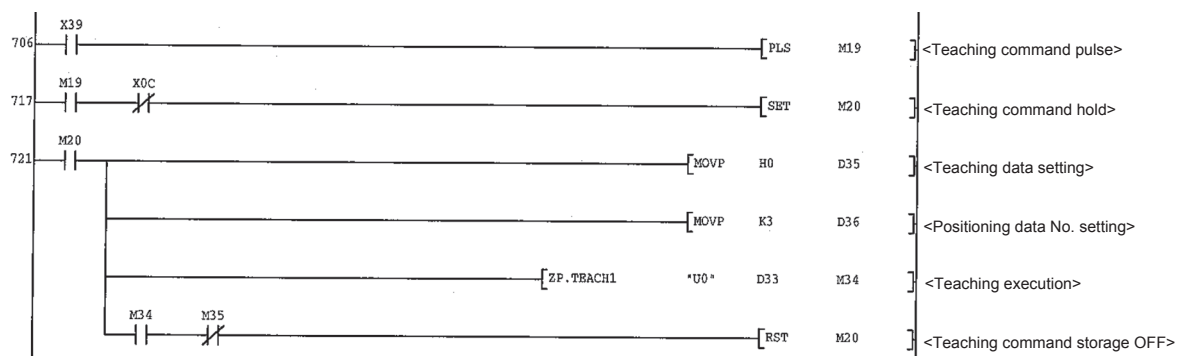
- (1) The following dedicated instructions cannot be executed simultaneously for the same axis.  
(Can be executed simultaneously for different axes.)
  - Positioning start instructions (ZP.PSTRT1 to ZP.PSTRT4)
  - Absolute position restoration instructions (Z.ABRST1 to Z.ABRST4)
  - Teaching instructions (ZP.TEACH1 to ZP.TEACH4)
- (2) The ZP.TEACH instruction can only be executed when the BUSY signal (XC, XD, XE, XF) is turned OFF.  
When the BUSY signal is turned ON, the ZP.TEACH instruction will not be executed. (not processed.)  
Before executing the ZP.TEACH instruction, make sure that the BUSY signal for the axis to be processed is turned OFF.
- (3) When the remote I/O station\* (Q Corresponding MELSECNET/H network remote I/O module) is used, the dedicated instruction (ZP.TEACH) is unusable.  
\*: For details of the remote I/O station, refer to Q Corresponding MELSECNET/H Network System Reference Manual (Remote I/O Network).
- (4) If the ZP.TEACH instruction is executed in any of the following cases, an error "Dedicated instruction error" (error code: 804) will occur and teaching cannot be performed.
  - Any value other than 0 and 1 is set to "Teaching selection" (device: (S)+2) of the control data.
  - Any value other than 1 to 600 is set to "Positioning No." (device: (S)+3) of the control data.
  - The instruction for a non-existent axis is specified.  
(Example: The ZP.TEACH2 instruction is specified when the QD75P1 is used.)

## [Program example]

Program to execute the teaching of the positioning data No. 3 of the axis 1 when X39 is turned ON.

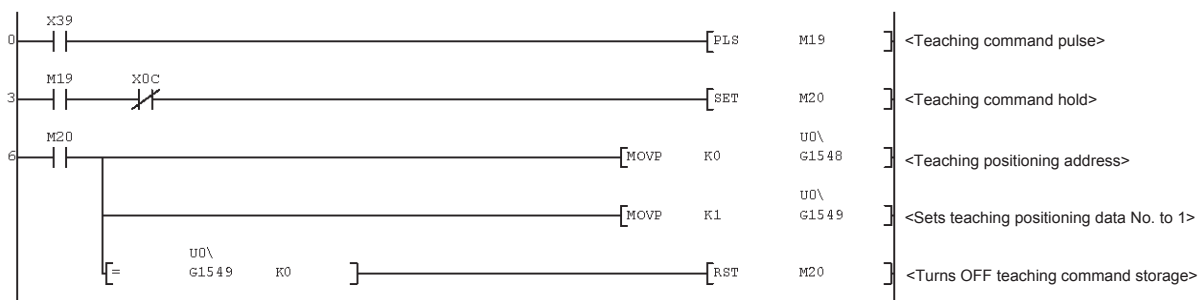
## (1) Teaching program

Positioning is carried out for a target position by manual operation.



## (2) Teaching program (when the dedicated instruction is not used)

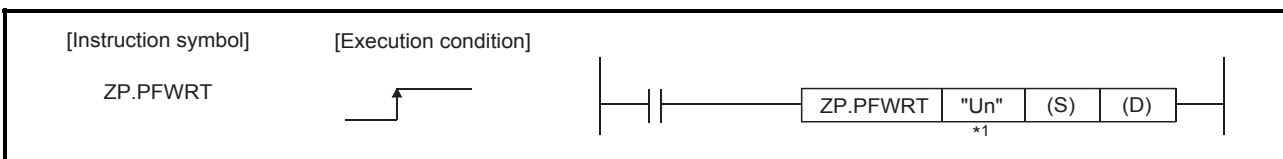
Positioning is carried out for a target position by manual operation.



## 14.6 ZP.PFWRT

These dedicated instructions are used to write the QD75 parameters, positioning data and block start data to the flash ROM.

| Setting data | Usable device   |      |               |                        |      |                                   |                   |          |        |
|--------------|-----------------|------|---------------|------------------------|------|-----------------------------------|-------------------|----------|--------|
|              | Internal device |      | File register | Link direct device J□□ |      | Intelligent function module U□\G□ | Index register Zn | Constant | Others |
|              | Bit             | Word |               | Bit                    | Word |                                   |                   | K, H     |        |
| (S)          | —               | ○    |               | —                      |      |                                   | —                 | —        |        |
| (D)          | ○               | ○    | —             | —                      |      |                                   | —                 | —        |        |



\*1 If the originating station is a Basic model QCPU (function version B or later), universal model QCPU, or safety CPU, "" (double quotation) of the first argument can be omitted.

## [Setting data]

| Setting data | Setting details                                                                                                                                                                                     | Setting side (*1) | Data type   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------|
| "Un"         | QD75 head I/O number<br>(00 to FE: High-order two digits of I/O number expressed in three digits)                                                                                                   | User              | BIN 16 bits |
| (S)          | Head number of a device in which control data is stored                                                                                                                                             | —                 | Device name |
| (D)          | Head number of a bit device which turns ON the operation by one scan at the time of completion of the instruction.<br>If the instruction is completed abnormally, ((D) + 1) will also be turned ON. | System            | Bit         |

Note) The file register of each of the local device and the program cannot be used as a device for setting data.

\*1: The data on the setting side is as follows.

- User : Data before the execution of dedicated instructions is stored by user.
- System : Data after the execution of dedicated instruction is stored by CPU module.

## [Control data]

| Device | Item            | Setting data                                                                                                                   | Setting Range | Setting side (*1) |
|--------|-----------------|--------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------|
| (S)+0  | System area     | —                                                                                                                              | —             | —                 |
| (S)+1  | Complete status | The state at the time of completion is stored.<br>0 : Normal completion<br>Other than 0 : Abnormal completion (error code)(*2) | —             | System            |

\*1: The data on the setting side is as follows.

- User : Data before the execution of dedicated instructions is stored by user.
- System : Data after the execution of dedicated instruction is stored by CPU module.

\*2: Refer to Section 15.3 for error codes at abnormal completion.

**[Functions]**

(1) The ZP.PFWRT instruction completion can be confirmed using the complete devices ((D)+0) and ((D)+1).

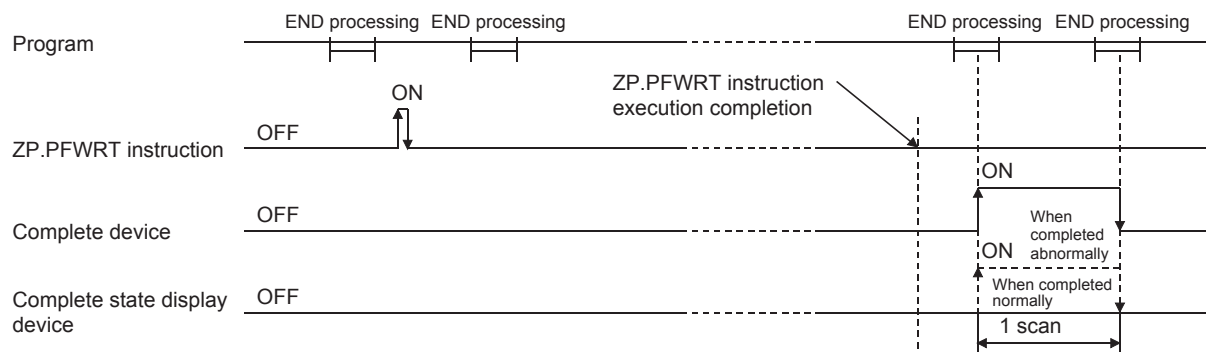
(a) Complete device ((D)+0)

This device is turned ON by the END processing of the scan for which ZP.PFWRT instruction is completed, and turned OFF by the next END processing.

(b) Complete state display device ((D)+1)

This device is turned ON and OFF according to the state in which ZP.PFWRT instruction is completed.

- When completed normally : Kept unchanged at OFF.
- When completed abnormally : This device is turned ON by the END processing of the scan for which ZP.PFWRT instruction is completed, and turned OFF by the next END processing (same ON/OFF operation as complete device).

**[Errors]**

(1) When a dedicated instruction is completed abnormally, the error complete signal ((D)+1) is turned ON, and the error code is stored in the complete status ((S)+1). Check and take measures against the error referring to Section 15.3 "List of error codes".

### [Precautions]

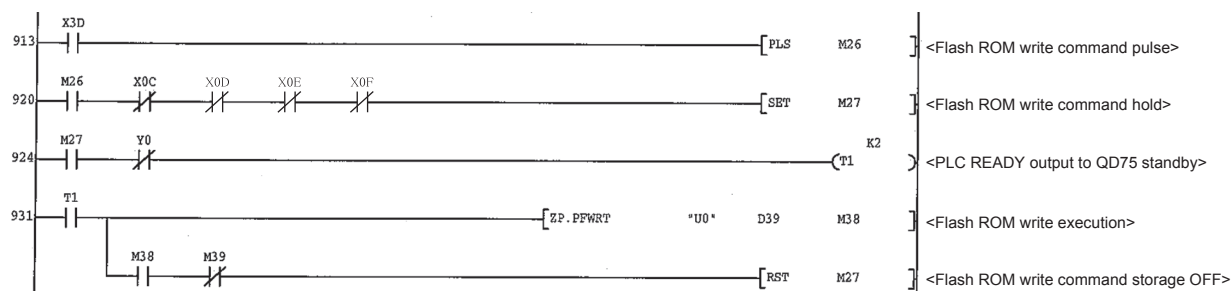
- (1) Do not turn ON the power and reset the CPU module while parameters, positioning data and block start data are written to the flash ROM using the ZP.PFWRT instruction.  
A parameter error will occur or normal positioning start will become impossible because the parameters, positioning data and block start data are not written normally to the flash ROM.  
If this occurs, restart the operation by the method shown below.
  - For GX Configurator-QP or GX Works2, write the parameters, positioning data and block start data again to the flash ROM.
  - For a sequence program, write the parameters, positioning data and block start data to the QD75 after initializing the parameters (ZP.PINIT instruction execution and others).  
Then execute the ZP.PFWRT instruction again.
- (2) Writing to the flash ROM can be executed up to 100,000 times.  
If writing to the flash ROM exceeds 100,000 times, the writing may become impossible.
- (3) After the power ON and CPU module reset operation, writing to the flash ROM using a sequence program is limited to up to 25 times. (Not limited to up to 25 times when writing to the flash ROM is carried out by peripheral device.)  
If the 26th or more writing is requested after the power ON/CPU module reset operation, an error "flash ROM write number error" (error code: 805) will occur, and the writing will be disabled. If a flash ROM write error occurs by one writing to the flash ROM, check and correct the flash ROM writing program. Then reset the error or turn ON the power and reset the CPU module again.
- (4) The ZP.PFWRT instruction can only be executed when the QD75 READY signal (X0) is turned OFF.  
When the QD75 READY signal is turned ON, the ZP.PFWRT instruction cannot be executed.  
Before executing the ZP.PFWRT instruction, turn OFF the PLC READY signal (Y0) and then turn OFF the QD75 READY signal.
- (5) When the remote I/O station\* (Q Corresponding MELSECNET/H network remote I/O module) is used, the dedicated instruction (ZP.PFWRT) is unusable.  
\*: For details of the remote I/O station, refer to Q Corresponding MELSECNET/H Network System Reference Manual (Remote I/O Network).



## [Program example]

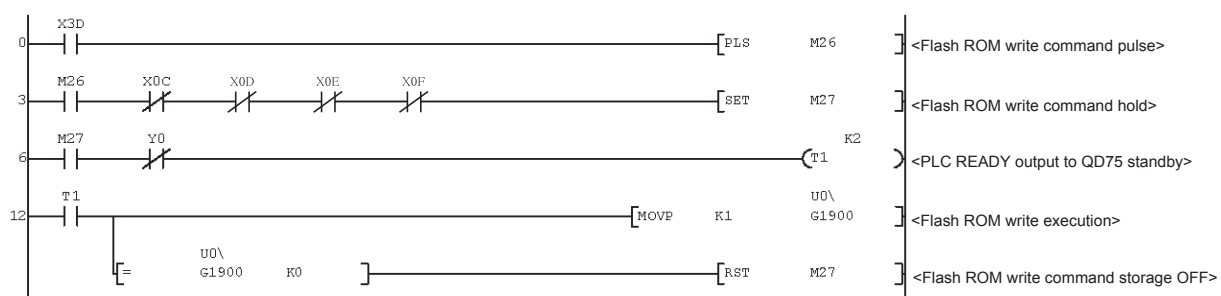
Program used to write the parameters and positioning data stored in the buffer memory to the flash ROM when X3D is turned ON.

## (1) Flash ROM write program



## [Program example for use when dedicated instruction is not used]

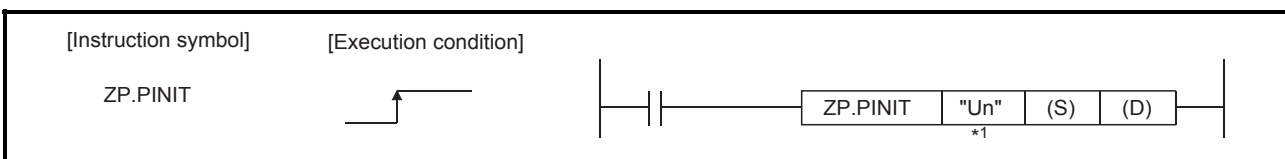
## (2) Flash ROM write program (when the dedicated instruction is not used)



## 14.7 ZP.PINIT

This dedicated instruction is used to initialize the setting data of the QD75.

| Setting data | Usable device   |      |               |                        |      |       |                   |          |        |
|--------------|-----------------|------|---------------|------------------------|------|-------|-------------------|----------|--------|
|              | Internal device |      | File register | Link direct device J□□ |      | U□\G□ | Index register Zn | Constant | Others |
|              | Bit             | Word |               | Bit                    | Word |       |                   | K, H     |        |
| (S)          | —               | ○    |               | —                      |      |       | —                 | —        | —      |
| (D)          | ○               | ○    | —             | —                      |      |       | —                 | —        | —      |



\*1 If the originating station is a Basic model QCPU (function version B or later), Universal model QCPU, or safety CPU, "" (double quotation) of the first argument can be omitted.

## [Setting data]

| Setting data | Setting details                                                                                                                                                                                     | Setting side (*1) | Data type   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------|
| "Un"         | QD75 head I/O number<br>(00 to FE: High-order two digits of I/O number expressed in three digits)                                                                                                   | User              | BIN 16 bits |
| (S)          | Head number of a device in which control data is stored                                                                                                                                             | —                 | Device name |
| (D)          | Head number of a bit device which turns ON the operation by one scan at the time of completion of the instruction.<br>If the instruction is completed abnormally, ((D) + 1) will also be turned ON. | System            | Bit         |

Note) The file register of each of the local device and the program cannot be used as a device for setting data.

\*1: The data on the setting side is as follows.

- User : Data before the execution of dedicated instructions is stored by user.
- System : Data after the execution of dedicated instruction is stored by CPU module.

## [Control data]

| Device | Item            | Setting data                                                                                                                  | Setting range | Setting side (*1) |
|--------|-----------------|-------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------|
| (S)+0  | System area     | —                                                                                                                             | —             | —                 |
| (S)+1  | Complete status | The state at the time of completion is stored.<br>0 : Normal completion<br>Other than 0: Abnormal completion (error code)(*2) | —             | System            |

\*1: The data on the setting side is as follows.

- User : Data before the execution of dedicated instructions is stored by user.
- System : Data after the execution of dedicated instruction is stored by CPU module.

\*2: Refer to Section 15.3 for error codes at abnormal completion.

## [Functions]

- (1) This dedicated instruction is used to return the setting data set in the QD75 buffer memory and flash ROM to their factory-set data (initial values).

|                                     |                                |
|-------------------------------------|--------------------------------|
| Initialized setting data            |                                |
| Parameters                          | (Pr.1 to Pr.57, Pr.70, Pr.150) |
| Positioning data (No. 1 to No. 600) |                                |
| Block start data (No. 7000 to 7004) |                                |

- (2) The ZP.PINIT instruction completion can be confirmed using the complete devices ((D)+0) and ((D)+1).

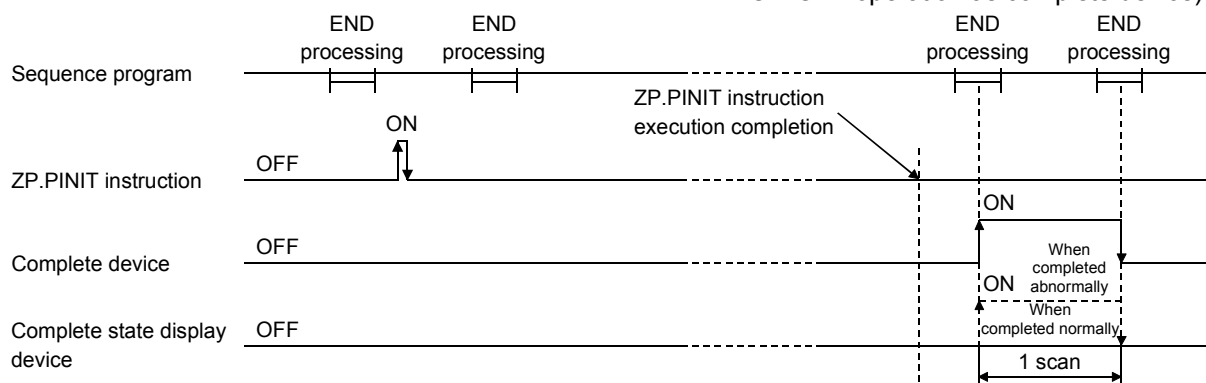
- (a) Complete device ((D)+0)

This device is turned ON by the END processing of the scan for which ZP.PINIT instruction is completed, and turned OFF by the next END processing.

- (b) Complete state display device ((D)+1)

This device is turned ON and OFF according to the state in which ZP.PINIT instruction is completed.

- When completed normally : Kept unchanged at OFF.
- When completed abnormally : This device is turned ON by the END processing of the scan for which ZP.PINIT instruction is completed, and turned OFF by the next END processing (same ON/OFF operation as complete device).



## [Errors]

- (1) When a dedicated instruction is completed abnormally, the error complete signal ((D)+1) is turned ON, and the error code is stored in the complete status ((S)+1). Check and take measures against the error referring to Section 15.3 "List of error codes".

## [Precautions]

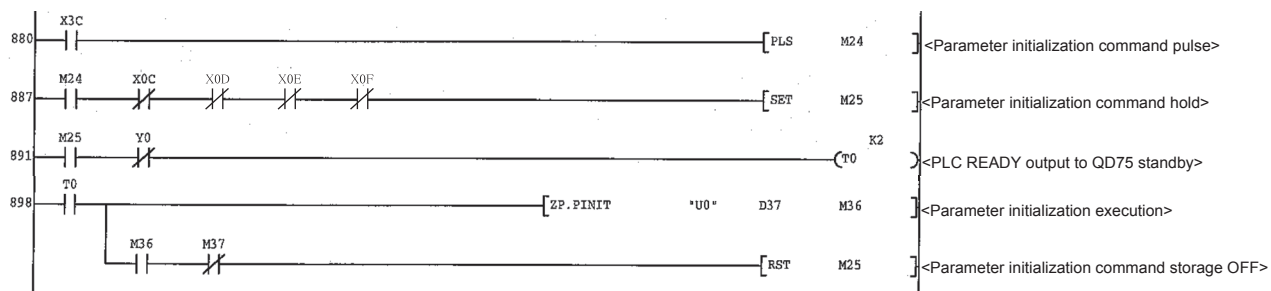
- (1) The ZP.PINIT instruction can only be executed when the QD75 READY signal (X0) is turned OFF.  
When the QD75 READY signal is turned ON, the ZP.PINIT instruction cannot be executed.  
Before executing the ZP.PINIT instruction, turn OFF the PLC READY signal (Y0) and then turn OFF the QD75 READY signal.
- (2) When the remote I/O station\* (Q Corresponding MELSECNET/H network remote I/O module) is used, the dedicated instruction (ZP.PINIT) is unusable.  
\*: For details of the remote I/O station, refer to Q Corresponding MELSECNET/H Network System Reference Manual (Remote I/O Network).

- (3) Writing to the flash ROM can be executed up to 100,000 times.  
If writing to the flash ROM exceeds 100,000 times, the writing may become impossible.
- (4) After the power ON and CPU module reset operation, writing to the flash ROM using a sequence program is limited to up to 25 times. (Not limited to up to 25 times when writing to the flash ROM is carried out by peripheral device.)  
If the 26th or more writing is requested after the power ON/CPU module reset operation, an error "flash ROM write number error" (error code: 805) will occur, and the writing will be disabled. If a flash ROM write error occurs by one writing to the flash ROM, check and correct the flash ROM writing program. Then reset the error or turn ON the power and reset the CPU module again.

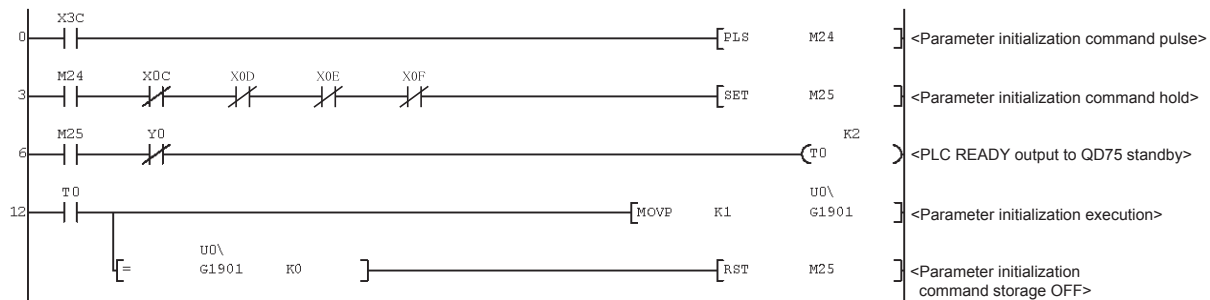
## [Program example]

The following program initializes the parameters in buffer memory and flash ROM when X3C turns ON.

## (1) Parameter initialization program



## (2) Parameter initialization program (when the dedicated instruction is not used)



[illegible]

# CHAPTER 15 TROUBLESHOOTING

The "errors" and "warnings" detected by the QD75 are explained in this chapter.

Errors can be confirmed with the QD75 LED display and peripheral devices.  
When an error or warning is detected, confirm the detection details and carry out the required measures.

|      |                                |        |
|------|--------------------------------|--------|
| 15.1 | Troubleshooting .....          | 15- 2  |
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## 15.1 Troubleshooting

## (1) Troubleshooting using the LEDs

Check items and corrective actions for troubleshooting using the indicator LEDs of the QD75 are described below.

## (a) When the RUN LED turns off

| Check item                               | Action                                                                                                                                                                                     |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Is the power supplied?                   | Check that the voltage supplied to the power supply module is within the rated range.                                                                                                      |
| Is the power supply capacity sufficient? | Calculate the total current consumption of the connected modules (CPU module, I/O modules, and intelligent function modules) and check that the power supply capacity is not insufficient. |
| Is the module connected correctly?       | Check that the module is correctly connected to the base unit.                                                                                                                             |

If there is no problem on the above check items, a watchdog timer error may have occurred. Reset the CPU module and check that the RUN LED turns on.

If not, the possible cause is a hardware failure. Please consult your local Mitsubishi representative.

## (b) When the ERR. LED turns on

| Check item               | Action                                                                                      |
|--------------------------|---------------------------------------------------------------------------------------------|
| Is there a system error? | An error may have occurred in the CPU module. Check the error and take a corrective action. |

## (c) When the ERR. LED and axis LED flash

| Check item              | Action                                                              |
|-------------------------|---------------------------------------------------------------------|
| Is there an axis error? | Check the error code and take the action described in Section 15.3. |

## (d) When all LEDs turn on

Reset the CPU module and check that the module is in the normal status.

If all LEDs still turn on, the possible cause is a hardware failure. Please consult your local Mitsubishi representative.



## (2) Troubleshooting when a motor does not rotate

Check items and corrective actions for troubleshooting when a motor does not rotate are described below.

| POINT                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The following signals must be ON for the QD75 to operate (excluding when the positioning test function of GX Works2 is used).</p> <ul style="list-style-type: none"> <li>• QD75 READY signal (X0)</li> <li>• Drive unit READY signal</li> <li>• Upper limit signal and Lower limit signal</li> </ul> <p>The status of the drive unit READY signal and upper/lower limit signals can be checked in b0 to b2 of " <b>[Md.30]</b> External I/O signal".</p> |

| Check item                                                                                           | Action                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Are the QD75 READY signal (X0), drive unit READY signal, and upper/lower limit signals ON?           | Review and correct the program and wiring so that all the QD75 READY signal (X0), drive unit READY signal, and upper/lower limit signals turn ON.                               |
| Is there an error in the QD75? (ERR. LED is on or flashing)                                          | Check the error code and take a corrective action.                                                                                                                              |
| Is the drive unit powered ON?                                                                        | Power on the drive unit.                                                                                                                                                        |
| Is there an error in the drive unit?                                                                 | Check the error code of the drive unit and take a corrective action.                                                                                                            |
| Is the wiring between the QD75 and drive unit correct?                                               | Check the wiring between the QD75 and drive unit, and correct it.                                                                                                               |
| Is the wiring between the drive unit and motor correct?                                              | Check the wiring between the drive unit and motor, and correct it.                                                                                                              |
| Is the value in " <b>[Md.20]</b> Current feed value" changed after positioning control is performed? | Review the start program.                                                                                                                                                       |
| Is the number of the input pulses for monitoring changed after positioning control is performed? *1  | Refer to the drive unit operating manual and check that the function to suppress the motor rotation is not working.                                                             |
| Isn't the value in " <b>[Md.26]</b> Axis operation status" "1: Stopped"?                             | <ul style="list-style-type: none"> <li>• Review the stop program.</li> <li>• Check that the stop signal (STOP) is not input.</li> </ul>                                         |
| Does the pulse output mode setting meet the specifications of the drive unit?                        | Set the value in " <b>[Pr.5]</b> Pulse output mode" so that it satisfies the specifications of the drive unit.                                                                  |
| Does the output logic setting of the command pulse signal meet the specifications of the drive unit? | Set the value of the logic selection of the command pulse signal ( <b>[Pr.23]</b> Output signal logic selection: b0) so that it satisfies the specifications of the drive unit. |

\*1: This check item applies only for the drive unit having a monitor function for the number of input pulses.

If a motor does not rotate even after the above items are checked, the possible cause is a hardware failure.

Please consult your local Mitsubishi representative.

### (3) Troubleshooting when a motor does not rotate as intended.

Check items and corrective actions for troubleshooting when a motor does not rotate as intended are described below.

#### (a) When a motor rotates only in one direction

| Check item                                                                    | Action                                                                                                                              |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Is the wiring correct?                                                        | Check that the signal line of the pulse output (for axis 1, connector pin No. "1A15 to 18") is correctly wired or not disconnected. |
| Does the pulse output mode setting meet the specifications of the drive unit? | Set the value in " <b>Pr.5</b> Pulse output mode" so that it satisfies the specifications of the drive unit.                        |

#### (b) When a motor rotates in the opposite direction

| Check item                                                                                                                                                                    | Action                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Is the wiring correct?                                                                                                                                                        | Check that the signal line of the pulse output (for axis 1, connector pin No. "1A15 to 18") is wired correctly. (CW and CCW or phase A and phase B should be correctly wired.)        |
| Do the values in " <b>Pr.6</b> Rotation direction setting" and " <b>Pr.23</b> Output signal logic selection (b0: Command pulse signal)" match the settings of the drive unit? | Check that the values in " <b>Pr.6</b> Rotation direction setting" and " <b>Pr.23</b> Output signal logic selection (b0: Command pulse signal)" match the settings of the drive unit. |

## (c) When a motor does not rotate at the set speed

| Check item                                                      | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Does the value in "Md.28 Axis feedrate" indicate the set speed? | [When "Md.28 Axis feedrate" indicates the set speed]<br><ul style="list-style-type: none"> <li>• Check that the values in "Pr.2 No. of pulses per rotation", "Pr.3 Movement amount per rotation", and "Pr.4 Unit magnification" meet the system.</li> <li>• When the drive unit has the electronic gear function, check that the settings meet the system.</li> </ul>                                                                                 |
|                                                                 | [When "Md.28 Axis feedrate" does not indicate the set speed]<br><ul style="list-style-type: none"> <li>• Check that the speed is not limited by the value in "Pr.8 Speed limit value".</li> <li>• In the JOG operation, Check that the speed is not limited by the value in "Pr.31 JOG speed limit value".</li> <li>• In the JOG operation, check that Forward/Reverse run JOG start signals (Y8 to YF) do not repeatedly turn ON and OFF.</li> </ul> |

## (d) When the set position is not reached

| Check item                                                                                        | Action                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Does the value in "Md.20 Current feed value" indicate the intended position when the motor stops? | [When "Md.20 Current feed value" reaches the set position]<br><ul style="list-style-type: none"> <li>• Check that the values in "Pr.2 No. of pulses per rotation", "Pr.3 Movement amount per rotation", and "Pr.4 Unit magnification" meet the system.</li> <li>• When the drive unit has the electronic gear function, check that the settings meet the system.</li> </ul> |
|                                                                                                   | [When "Md.20 Current feed value" does not reaches the set position]<br><ul style="list-style-type: none"> <li>• Check that the motor is not stopped by Axis stop signals (Y4 to Y7) or a stop signal (STOP). If a motor is stopped by them, the value "1: Stopped" is stored in "Md.26 Axis operation status".</li> </ul>                                                   |

## 15.2 Error and warning details

## [1] Errors

## ■ Types of errors

Errors detected by the QD75 include parameter setting range errors and errors at the operation start or during operation.

## (1) Parameter setting range errors

The parameters are checked when the power is turned ON and at the rising edge (OFF → ON) of the PLC READY signal [Y0]. An error will occur if there is a mistake in the parameter setting details at that time.

When this kind of error occurs, the QD75 READY signal does not turn ON. To cancel this kind of error, set the correct value in the parameter for which the error occurred, and then turn ON the PLC READY signal [Y0].

## (2) Errors at the operation start or during operation

These are errors that occur at the operation start or during operation when the positioning control, JOG operation, or inching operation is used. If an axis error occurs during interpolation operation, the error No. will be stored in both the reference axis and the interpolation axis.

Note that, in the following cases (a) and (b), the axis error No. will be stored only in the reference axis during analysis of the positioning data set in each point of the positioning start data table.

(a) When the interpolation axis is BUSY.

(b) When the error occurred in positioning data or parameters unrelated to interpolation control.

If the error occurred at the simultaneous start of a positioning operation, the axis error storage details will differ depending on whether the error occurred before or after the simultaneous start.

- If the error occurred before the simultaneous start (illegal axis No., other axis BUSY, etc.), an error "Error before simultaneous start" (error code: 501) will occur.
- If the error occurred after the simultaneous start (positioning data error, software stroke limit error, etc.), an error code corresponding to the axis in which the error occurred will be stored. Because a simultaneous start cannot be carried out due to this, an error "Simultaneous start not possible" (error code: 532) error code will be stored in all axes in which an error has not occurred.

The axis operation status will be displayed as "error occurring" for axes in which an error occurred.

If an error occurs during operation, any moving axes will deceleration stop, and their operation status will be displayed as "error occurring".

All axes will decelerate to a stop during interpolation operations, even if the error occurs in only one axis.

## (3) Types of error codes

| Error code | Error type                                             |
|------------|--------------------------------------------------------|
| 001 to 009 | Fatal error                                            |
| 100 to 199 | Common error                                           |
| 200 to 299 | Error at OPR or absolute position restoration          |
| 300 to 399 | Error during JOG operation or during inching operation |
| 500 to 599 | Error during positioning operation                     |
| 800 to 899 | I/F (Interface) error                                  |
| 900 to 999 | Error during parameter setting range check             |

### ■ Error storage

When an error occurs, the error detection signal turns ON, and the error code corresponding to the error details is stored in the following buffer memory address ([Md.23] Axis error No.) for axis error No. storage.

| Axis No. | Error detection signal | Buffer memory address |
|----------|------------------------|-----------------------|
| 1        | X8                     | 806                   |
| 2        | X9                     | 906                   |
| 3        | XA                     | 1006                  |
| 4        | XB                     | 1106                  |

A new error code is stored in the buffer memory address ([Md.23] Axis error No.) for axis error storage every time an error occurs.

#### POINT

When any of the following errors is detected, it is stored in the axis error No. of axis 1.  
Error code: 001, 002, 107, 800, 801, 802, 805

## [2] Warnings

### ■ Types of warnings

- (1) Warnings include system warnings and axis warnings.

The types of system warnings are shown below.

- System control data setting warnings  
An axis warning for axis 1 will occur.
- Positioning data setting warnings  
An axis warning for each axis will occur.  
Note that a warning will occur for the reference axis when an interpolation designation or axis setting warning occurs.

- (2) Axis warnings occur due to setting warnings from operations such as positioning operations, JOG operations, manual pulse generator operations, or system errors. They can be canceled by turning ON the "[Cd.5] Axis error reset".

Note that some warnings cannot be canceled unless the cause of the warning is eliminated.

The axis operation status does not change even if an axis warning occurs.

- (3) Types of warning codes

| Warning code | Warning type                                    |
|--------------|-------------------------------------------------|
| 100 to 199   | Common warning                                  |
| 300 to 399   | Warning during JOG operation                    |
| 400 to 499   | Warning during manual pulse generator operation |
| 500 to 599   | Warning during positioning operation            |

### ■ Warning storage

- (1) When an axis warning occurs, the warning code corresponding to the warning details is stored in the following buffer memory ([Md.24] Axis warning No.) for axis warning No. storage.

| Axis No. | Buffer memory address |
|----------|-----------------------|
| 1        | 807                   |
| 2        | 907                   |
| 3        | 1007                  |
| 4        | 1107                  |

- (2) When an axis warning occurs in a positioning operation, etc., "1" is set in bit 9 (b9) of the following buffer memory ([Md.31] Status) for axis status storage.

| Axis No. | Buffer memory address |
|----------|-----------------------|
| 1        | 817                   |
| 2        | 917                   |
| 3        | 1017                  |
| 4        | 1117                  |

### [3] Resetting errors and warnings

Remove the cause of error or warning following the actions described in Section 15.3 and 15.4, before cancel an error or warning state by resetting the error.

### ■ How to clear errors or warnings

An error or warning state is canceled after the following processing has been carried out by setting a "1" in the address [1502 (for axis 1)], [1602 (for axis 2)], [1702 (for axis 3)], and [1802 (for axis 4)] of the buffer memory for axis error resetting ([Cd.5] Axis error reset).

- Axis error detection signal turned OFF
- "[Md.23] Axis error No." cleared
- "[Md.24] Axis warning No." cleared
- Changing of "[Md.26] Axis operation status" from "Error" to "Standby".
- Axis warning detection "[Md.31] Status : b9" turned OFF

### [4] Confirming the error and warning definitions

The error and warning definitions can be confirmed with the error and warning codes. Confirming them requires GX Developer or GX Configurator-QP. For details, refer to GX Developer Operating Manual or GX Configurator-QP Operating Manual. (Refer to Section 15.3 and Section 15.4 for details of the error and warning codes.)

# MEMO

[illegible]

## 15.3 List of errors

The following table shows the error details and remedies to be taken when an error occurs.

| Classification of errors | Error code | Error name                        | Error                                                                                                  | Operation status at error occurrence                                                                                                                                                                                              |  |
|--------------------------|------------|-----------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| —                        | 000        | (Normal status)                   | —                                                                                                      | —                                                                                                                                                                                                                                 |  |
| Fatal errors             | 001        | Faults                            | Hardware is faulty.                                                                                    | The system stops.                                                                                                                                                                                                                 |  |
|                          | 002        | Internal circuit fault            |                                                                                                        |                                                                                                                                                                                                                                   |  |
| Common                   | 101        | PLC READY OFF during operation    | A PLC READY signal (Y0) is turned OFF during operation.                                                | The system stops with the setting (deceleration stop/sudden stop) of the detailed parameter 2 Sudden stop selection (stop group 2).<br>(Note that the deceleration stop only occurs during the manual pulse generator operation.) |  |
|                          | 102        | Drive unit READY OFF              | The drive unit READY signal is turned OFF during operation.                                            | The system stops immediately.                                                                                                                                                                                                     |  |
|                          |            |                                   | The start of an operation is requested when the drive unit READY signal is OFF.                        | The system does not start.                                                                                                                                                                                                        |  |
|                          | 103        | Test mode faults during operation | The personal computer cannot communicate with the CPU unit.                                            | The system stops with the setting (deceleration stop/sudden stop) of the detailed parameter 2 Sudden stop selection (stop group 2).<br>(Note that the deceleration stop only occurs during the manual pulse generator operation.) |  |
|                          | 104        | Hardware stroke limit (+)         | The hardware stroke limit (upper limit signal FLS) is turned OFF during operation.                     | The system stops with the setting (deceleration stop/sudden stop) of the detailed parameter 2 Sudden stop selection (stop group 1).<br>(Note that the deceleration stop only occurs during the manual pulse generator operation.) |  |
|                          |            |                                   | The start of an operation is requested when the hardware stroke limit (upper limit signal FLS) is OFF. | The system does not start.                                                                                                                                                                                                        |  |
|                          |            |                                   |                                                                                                        |                                                                                                                                                                                                                                   |  |
|                          |            |                                   |                                                                                                        |                                                                                                                                                                                                                                   |  |



|  | Related buffer memory address |        |        |        | Set range<br>(Setting with sequence program) | Remedy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--|-------------------------------|--------|--------|--------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Axis 1                        | Axis 2 | Axis 3 | Axis 4 |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|  | —                             | —      | —      | —      | —                                            | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  | —                             | —      | —      | —      | —                                            | Check that there is no influence from noise.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  | —                             | —      | —      | —      | —                                            | Review the sequence program which turns ON/OFF PLC READY signal (Y0).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|  | —                             | —      | —      | —      | —                                            | Check the power, wiring, and connector connection status of the drive unit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|  | —                             | —      | —      | —      | —                                            | <ul style="list-style-type: none"> <li>• Check the power, wiring, and connector connection status of the drive unit.</li> <li>• Check the values in <b>Pr.22</b> Input signal logic selection.</li> <li>• When a drive unit without READY signal output is used, wire the system so that the drive unit READY signal input of the QD75 is always ON.</li> </ul>                                                                                                                                                                                                                                                                 |
|  | —                             | —      | —      | —      | —                                            | <ul style="list-style-type: none"> <li>• Check that there is no error on the personal computer side I/F to which a cable is connected.</li> <li>• Because communication takes a long time between the personal computer and a CPU module, ensure high-speed communication by setting the transmission speed to high at the transfer setup of GX Configurator-QP or GX Works2, or connecting the personal computer directly to the communication I/F of the CPU module.</li> <li>• Reduce the load of communication by stopping the access to a CPU module from GX Developer or GX Works2 (such as monitor function).</li> </ul> |
|  | —                             | —      | —      | —      | —                                            | After making an axis error reset (refer to [3] in Section 15.2), perform manual control operation (refer to CHAPTER 11) to move the axis to the other position in order that the upper limit signal (FLS) will not turn OFF.                                                                                                                                                                                                                                                                                                                                                                                                    |
|  | —                             | —      | —      | —      | —                                            | <ul style="list-style-type: none"> <li>• Check the wiring of the upper limit signal (FLS).</li> <li>• Check if the values in <b>Pr.22</b> Input signal logic selection meet the specifications of the limit switch.</li> <li>• In a system which does not require a hardware stroke limit (limit switch), wire the system so that the upper limit signal (FLS) input of the QD75 is always ON.</li> </ul>                                                                                                                                                                                                                       |

| Classification of errors | Error code | Error name                         | Error                                                                                                                                                                                   | Operation status at error occurrence                                                                                                                                                                                              |  |
|--------------------------|------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Common                   | 105        | Hardware stroke limit (–)          | The hardware stroke limit (lower limit signal RLS) is turned OFF during operation.                                                                                                      | The system stops with the setting (deceleration stop/sudden stop) of the detailed parameter 2 Sudden stop selection (stop group 1).<br>(Note that the deceleration stop only occurs during the manual pulse generator operation.) |  |
|                          |            |                                    | The start of an operation is requested when the hardware stroke limit (lower limit signal RLS) is OFF.                                                                                  | The system does not start.                                                                                                                                                                                                        |  |
|                          | 106        | Stop signal ON at start            | Start is requested when a stop signal is turned ON.                                                                                                                                     | The system does not start.                                                                                                                                                                                                        |  |
|                          | 107        | READY OFF → ON during BUSY         | The PLC READY signal is turned from OFF to ON when BUSY signal is turned ON.                                                                                                            | The QD75 READY signal (X0) is not turned ON.                                                                                                                                                                                      |  |
| OPR                      | 201        | Start at OP                        | When the OPR retry invalid is set, the near-point dog machine OPR is started with the OPR complete flag turned ON.                                                                      | The machine OPR is not started.                                                                                                                                                                                                   |  |
|                          | 203        | Dog detection timing fault         | The near-point dog signal is turned OFF during the deceleration from an OPR speed to a creep speed by the near-point dog machine OPR.                                                   | The system stops with the setting (deceleration stop/sudden stop) of the detailed parameter 2 Sudden stop selection (stop group 3).<br>(Note that the deceleration stop only occurs during the manual pulse generator operation.) |  |
|                          | 204        | OP detection timing fault          | The zero signal is turned OFF during the deceleration from an OPR speed to a creep speed by the stopper method 2)-OPR.                                                                  |                                                                                                                                                                                                                                   |  |
|                          | 205        | Dwell time fault                   | A dwell time is passed during the deceleration from a creep speed to a OPR speed by the stopper method 1)-OPR.                                                                          |                                                                                                                                                                                                                                   |  |
|                          | 206        | Count method movement amount fault | In the count method 1 and 2 machine OPR, a parameter "Setting of movement amount after near-point dog ON" is smaller than a distance necessary for deceleration stop from an OPR speed. | The machine OPR is not started.                                                                                                                                                                                                   |  |

|  | Related buffer memory address |            |            |            | Set range<br>(Setting with sequence program)                                                                                                                                                                                                                          | Remedy                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--|-------------------------------|------------|------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Axis 1                        | Axis 2     | Axis 3     | Axis 4     |                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|  | —                             | —          | —          | —          | —                                                                                                                                                                                                                                                                     | After making an axis error reset (refer to [3] in Section 15.2), perform manual control operation (refer to CHAPTER 11) to move the axis to the other position in order that the lower limit signal (RLS) will not turn OFF.                                                                                                                                                                                                                     |
|  | —                             | —          | —          | —          | —                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>• Check the wiring of the lower limit signal (RLS).</li> <li>• Check if the values in [Pr.22] Input signal logic selection meet the specifications of the limit switch.</li> <li>• In a system which does not require a hardware stroke limit (limit switch), wire the system so that the lower limit signal (RLS) input of the QD75 is always ON.</li> </ul>                                             |
|  | —                             | —          | —          | —          | —                                                                                                                                                                                                                                                                     | <p>The start timing is reviewed so that the operation starts after the stop command is reset.</p> <ul style="list-style-type: none"> <li>• Output signals to QD75<br/>Axis 1: Y4, Axis 2: Y5, Axis 3: Y6, Axis 4: Y7</li> <li>• External input<br/>Connectors for external device connection:<br/>Stop signals (STOP)</li> </ul>                                                                                                                 |
|  | —                             | —          | —          | —          | —                                                                                                                                                                                                                                                                     | Turn ON the PLC READY signal (Y0) with the BUSY signals of all axes OFF.                                                                                                                                                                                                                                                                                                                                                                         |
|  | 78                            | 228        | 378        | 528        | <OPR retry><br>0, 1                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>• Validate the OPR retry function (set value: 1). (Refer to Section 12.1.1)</li> <li>• Move the workpiece from the current position (on OP) using the manual control operation (refer to CHAPTER 11), then carry out a machine OPR again.</li> </ul>                                                                                                                                                      |
|  | 74<br>75                      | 224<br>225 | 374<br>375 | 524<br>525 | <p>&lt;OPR speed&gt;<br/>QD75P□N/QD75D□N:<br/>1 to 4000000 [pulse/s]<br/>1 to 2000000000<br/>[<math>\times 10^{-2}</math>mm/min or others]</p>                                                                                                                        | <ul style="list-style-type: none"> <li>• Lower the OPR speed.</li> <li>• Increase the dog signal input time. (Refer to Section 8.2.3)</li> </ul>                                                                                                                                                                                                                                                                                                 |
|  | 74<br>75                      | 224<br>225 | 374<br>375 | 524<br>525 | <p>QD75P□/QD75D□:<br/>1 to 1000000 [pulse/s]<br/>1 to 2000000000<br/>[<math>\times 10^{-2}</math>mm/min or others]</p>                                                                                                                                                | <ul style="list-style-type: none"> <li>• Lower the OPR speed.</li> <li>• Input external zero signals during the movement at a creep speed. (Refer to Section 8.2.5)</li> </ul>                                                                                                                                                                                                                                                                   |
|  | 74<br>75                      | 224<br>225 | 374<br>375 | 524<br>525 |                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>• Lower the OPR speed.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                         |
|  | 79                            | 229        | 379        | 529        | <OPR dwell time><br>0 to 65535                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• Increase the OPR dwell time. (Refer to Section 8.2.4)</li> </ul>                                                                                                                                                                                                                                                                                                                                        |
|  | 80<br>81                      | 230<br>231 | 380<br>381 | 530<br>531 | <Movement amount setting after near-point dog ON><br>0 to 2147483647                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>• Calculate the movement distance using a speed limit, OPR speed, and deceleration time, and set the movement amount after near-point dog ON so that the distance becomes a deceleration distance or longer.</li> <li>• Lower the OPR speed.</li> <li>• Adjust the near-point dog position so that the movement amount after near-point dog ON becomes longer. (Refer to Section 8.2.7, 8.2.8)</li> </ul> |
|  | 74<br>75                      | 224<br>225 | 374<br>375 | 524<br>525 | <p>&lt;OPR speed&gt;<br/>QD75P□N/QD75D□N:<br/>1 to 4000000 [pulse/s]<br/>1 to 2000000000<br/>[<math>\times 10^{-2}</math>mm/min or others]</p> <p>QD75P□/QD75D□:<br/>1 to 1000000 [pulse/s]<br/>1 to 2000000000<br/>[<math>\times 10^{-2}</math>mm/min or others]</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Classification of errors | Error code | Error name                    | Error                                                                                                                                                                                                                                                                                                                                                | Operation status at error occurrence                                                                           |  |
|--------------------------|------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--|
| OPR                      | 207        | OPR request ON                | The OPR request flag is turned ON when a fast-OPR is started (positioning start No. 9002).                                                                                                                                                                                                                                                           | The fast OPR is not started.                                                                                   |  |
|                          | 209        | OPR restart not possible      | The restart command is turned ON after the machine OPR is stopped using a stop signal.                                                                                                                                                                                                                                                               | The restart is not carried out.                                                                                |  |
|                          | 213        | ABS transmission time         | Communication cannot be carried out with the servo-amplifier using an absolute position restoration instruction.                                                                                                                                                                                                                                     | The absolute position restoration is not carried out.                                                          |  |
|                          | 214        | ABS transmission SUM          |                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                |  |
| JOG inching              | 300        | Outside JOG speed range       | At the time of JOG starting, the JOG speed comes out of a specified range.                                                                                                                                                                                                                                                                           | The JOG operation is not carried out when the JOG speed is outside the setting range at the time of JOG start. |  |
|                          | 301        | Inching movement amount error | <p>The inching movement amount does not meet the setting condition. (The setting value is larger.)</p> <p>Setting condition:</p> <p>"Inching movement amount <math>\times</math> (A) <math>\leq</math> JOG speed limit value"</p> <p>(A)...</p> <p>When the setting unit is pulse: 562.5</p> <p>When the setting unit is other than pulse: 337.5</p> | The inching operation is not carried out.                                                                      |  |

|  | Related buffer memory address |              |              |              | Set range<br>(Setting with sequence program)                                                                                                                                                                                | Remedy                                                                                                                                                                                                                 |
|--|-------------------------------|--------------|--------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Axis 1                        | Axis 2       | Axis 3       | Axis 4       |                                                                                                                                                                                                                             |                                                                                                                                                                                                                        |
|  | 1500                          | 1600         | 1700         | 1800         | <Positioning start No.><br>1 to 600<br>7000 to 7004<br>9001 to 9004                                                                                                                                                         | Execute the machine OPR (positioning No. 9001).<br>(Refer to Section 8.2)                                                                                                                                              |
|  | 1500                          | 1600         | 1700         | 1800         | <Positioning start No.><br>1 to 600<br>7000 to 7004<br>9001 to 9004                                                                                                                                                         | Start the machine OPR (positioning start No. 9001) again. (Refer to Section 8.2)                                                                                                                                       |
|  | —                             | —            | —            | —            | —                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>• Review the wiring. (Refer to Section 12.6)</li> <li>• Review the sequence program.</li> </ul>                                                                                 |
|  | —                             | —            | —            | —            | —                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>• Review the wiring. (Refer to Section 12.6)</li> <li>• Review the sequence program.</li> <li>• Review the dedicated instruction parameters. (Refer to Section 14.3)</li> </ul> |
|  | 1518<br>1519                  | 1618<br>1619 | 1718<br>1719 | 1818<br>1819 | <JOG speed><br>QD75P□N/QD75D□N:<br>1 to 4000000 [pulse/s]<br>1 to 2000000000<br>[ $\times 10^{-2}$ mm/min or others]<br>QD75P□/QD75D□:<br>1 to 1000000 [pulse/s]<br>1 to 2000000000<br>[ $\times 10^{-2}$ mm/min or others] | Bring the JOG speed into the setting range.<br>(Refer to Section 11.2)                                                                                                                                                 |
|  | 1517                          | 1617         | 1717         | 1817         | <Inching movement amount><br>0 to 65535                                                                                                                                                                                     | Reduce the inching movement amount to satisfy setting conditions.<br>(Refer to Section 11.3)                                                                                                                           |

| Classification of errors | Error code | Error name                      | Error                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Operation status at error occurrence                                                         |  |
|--------------------------|------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--|
| Positioning operation    | 500        | Illegal condition data No.      | The condition data No. is outside the setting range when a block using the condition data is started by a special starting (conditional start, wait start, simultaneous start, FOR (condition)).<br>(1 ≤ Condition data No. ≤ 10)                                                                                                                                                                                                                                                                                                                                                         | The operation is terminated.                                                                 |  |
|                          | 501        | Error before simultaneous start | <p>&lt;When blocks are started simultaneously&gt;</p> <ul style="list-style-type: none"> <li>• The partner axis for simultaneous start is BUSY.</li> </ul> <p>&lt;When multiple axes are started and controlled simultaneously&gt;</p> <ul style="list-style-type: none"> <li>• The partner axis for simultaneous start is BUSY.</li> <li>• The "Simultaneous start axis start data No." of the start axis is 0 or is outside the setting range.</li> <li>• The "Simultaneous start axis start data No." of those axes other than the start axis is outside the setting range.</li> </ul> | At start: The system will not operate.<br>During operation:<br>The system stops immediately. |  |
| Positioning operation    | 502        | Illegal data No.                | <ul style="list-style-type: none"> <li>• The positioning data No. tried to be executed is outside the ranges of 1 to 600, 7000 to 7004, and 9001 to 9004.</li> <li>• The designation of a JUMP destination is executed currently.</li> <li>• The designation of a JUMP destination is outside the ranges of 1 to 600.</li> </ul>                                                                                                                                                                                                                                                          | The positioning data is not executed.                                                        |  |
|                          | 503        | No command speed                | <ul style="list-style-type: none"> <li>• At the start of positioning, a current speed (–1) is set for the command speed of the positioning data to be initially executed.</li> <li>• The current speed is set by speed control.</li> <li>• The current speed is set for speed-position switching control or position-speed switching control.</li> </ul>                                                                                                                                                                                                                                  | At the start of positioning, operation does not start.                                       |  |

| Related buffer memory address                      |        |        |        | Set range<br>(Setting with sequence program)                                                                                                                                                                                                                                                                  |                                                              | Remedy                                                                                                                            |
|----------------------------------------------------|--------|--------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Axis 1                                             | Axis 2 | Axis 3 | Axis 4 |                                                                                                                                                                                                                                                                                                               |                                                              |                                                                                                                                   |
| Refer to Section 5.4<br>"List of block start data" |        |        |        | <Condition data No.><br>1 to 10                                                                                                                                                                                                                                                                               |                                                              | Review the condition data No.<br>(Refer to Section 5.4 <span style="border: 1px solid black; padding: 0 2px;">Da.14</span> )      |
| Refer to Section 5.5<br>"List of condition data"   |        |        |        | <Condition operators><br>Axis designation:<br>10 <sub>H</sub> , 20 <sub>H</sub> , 30 <sub>H</sub> , 40 <sub>H</sub> , 50 <sub>H</sub> ,<br>60 <sub>H</sub> , 70 <sub>H</sub> , 80 <sub>H</sub> , 90 <sub>H</sub> , A0 <sub>H</sub> ,<br>B0 <sub>H</sub> , C0 <sub>H</sub> , D0 <sub>H</sub> , E0 <sub>H</sub> |                                                              | Normalize the condition operators.<br>(Refer to Section 5.5 <span style="border: 1px solid black; padding: 0 2px;">Da.16</span> ) |
| 1540                                               | 1640   | 1740   | 1840   | Axis 1 start data No.                                                                                                                                                                                                                                                                                         | Simultaneous<br>start axis start<br>data No.<br><br>0 to 600 | Normalize the simultaneous start axis start data No.<br>(Refer to Section 10.5)                                                   |
| 1541                                               | 1641   | 1741   | 1841   | Axis 2 start data No.                                                                                                                                                                                                                                                                                         |                                                              |                                                                                                                                   |
| 1542                                               | 1642   | 1742   | 1842   | Axis 3 start data No.                                                                                                                                                                                                                                                                                         |                                                              |                                                                                                                                   |
| 1543                                               | 1643   | 1743   | 1843   | Axis 4 start data No.                                                                                                                                                                                                                                                                                         |                                                              |                                                                                                                                   |
| 1500                                               | 1600   | 1700   | 1800   | <Positioning start No.><br>1 to 600,<br>7000 to 7004,<br>9001 to 9004                                                                                                                                                                                                                                         |                                                              | Normalize the positioning start No., positioning start data (in block start), and positioning data (in JUMP instruction).         |
|                                                    |        |        |        | <JUMP destination><br>1 to 600                                                                                                                                                                                                                                                                                |                                                              |                                                                                                                                   |
| Refer to Section 5.3<br>"List of positioning data" |        |        |        | <Command speed><br>QD75P□N/QD75D□N:<br>1 to 4000000 [pulse/s]<br>1 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others]<br>QD75P□/QD75D□:<br>1 to 1000000 [pulse/s]<br>1 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others]                                                                               |                                                              | Normalize the positioning data.                                                                                                   |

| Classification of errors | Error code | Error name                           | Error                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Operation status at error occurrence                                                             |  |
|--------------------------|------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--|
| Positioning operation    | 504        | Outside linear movement amount range | <ul style="list-style-type: none"> <li>When the parameter or positioning data "Interpolation speed designation method" performs a linear interpolation in setting a "composite speed", the axis movement amount for each positioning data exceeds 1073741824(2<sup>30</sup>).</li> <li>The positioning address is -360.00000 or less or 360.00000 or more using INC instruction, where the control unit is set to "degree" and software stroke limit upper limit is not equal to the software stroke limit lower limit.</li> </ul> | At start: The system will not operate.<br>During operation:<br>The system stops immediately.     |  |
|                          | 506        | Large arc error deviation            | When an arc or helical is interpolated with center point designation, a difference between a radius of start point-center point and a radius of end point-center point exceeds the parameter "Allowable circular interpolation error width".                                                                                                                                                                                                                                                                                       | At start: The system will not operate.<br>During operation:<br>The system will stop immediately. |  |



|  | Related buffer memory address                      |        |        |        | Set range<br>(Setting with sequence program)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Remedy                          |
|--|----------------------------------------------------|--------|--------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
|  | Axis 1                                             | Axis 2 | Axis 3 | Axis 4 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |
|  | Refer to Section 5.3<br>"List of positioning data" |        |        |        | <Positioning address/movement amount><br>• ABS<br>unit [mm] [pulse] [inch]<br>–2147483648 to 2147483647<br>Unit [degree] 0 to 35999999<br>• INC<br>(When software stroke limits are valid)<br>Unit [degree]:<br>–35999999 to 35999999<br>Unit [mm], [pulse], [inch]:<br>–2147483648 to 2147483647<br>(When software stroke limits are invalid)<br>–2147483648 to 2147483647<br>• Speed-position switching<br>INC mode: 0 to 2147483647<br>ABS mode: 0 to 35999999<br>• Position-speed switching<br>0 to 2147483647<br><Arc address><br>–2147483648 to 2147483647 | Review the positioning address. |
|  |                                                    |        |        |        | • Center point address (arc address)<br>• End address (positioning address)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | } Correct.                      |

| Classification of errors | Error code | Error name             | Error                                                                                                                                                                                                                                                                                                                                                                                                 | Operation status at error occurrence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|--------------------------|------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Positioning operation    | 507        | Software stroke limit+ | <ul style="list-style-type: none"> <li>Positioning is carried out at a position beyond the software stroke limit upper limit.</li> <li>The positioning address and new current value exceed the software stroke limit upper limit.</li> <li>The sub point exceeds the software stroke limit upper limit in the circular interpolation or helical interpolation with sub point designation.</li> </ul> | <p>At start: The system will not operate.</p> <p>In the analysis of new current value: Current value is not changed.</p> <p>During operation:</p> <ul style="list-style-type: none"> <li>The system stops immediately when the positioning address during position control (including position control in speed-position switching control or position-speed switching control) is switched to the data outside the software stroke limit range.</li> <li>The system makes a stop at the setting (normal deceleration stop only) of sudden stop selection (stop group 3) in the detailed parameter 2 when the current feed value or machine feed value during speed control (including speed control in speed-position switching control or position-speed switching control) or during manual control falls outside the software stroke limit range.</li> </ul> |  |
|                          | 508        | Software stroke limit- | <ul style="list-style-type: none"> <li>Positioning is carried out at a position beyond the software stroke limit lower limit.</li> <li>The positioning address and new current value exceed the software stroke limit lower limit.</li> <li>The sub point exceeds the software stroke limit lower limit in the circular interpolation or helical interpolation with sub point designation.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |

| Related buffer memory address     |              |              |              | Set range<br>(Setting with sequence program)                                                                                                                                                                                | Remedy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------|--------------|--------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1                            | Axis 2       | Axis 3       | Axis 4       |                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| New current value                 |              |              |              | <div>&lt;New current value&gt;</div> <div>&lt;Software stroke upper and lower limits&gt;</div> <div><div>• [mm] [inch] [pulse]</div><div>–2147483648 to 2147483647</div><div>• [degree]</div><div>0 to 35999999</div></div> | <div>At start: Bring the current feed value into the software stroke limit using the manual control operation.<br/>(Refer to CHAPTER 11)<br/>Correct the positioning address (Also check the arc address in the circular interpolation or helical interpolation with sub point designation.).</div> <div>New current value:<br/>Bring the new current value into the software stroke limit. (Refer to Section 9.2.21)</div> <div>During operation:<br/>Correct the positioning address.<br/>(For the positioning and arc addresses, refer to <span>Da.6</span> and <span>Da.7</span> in Section 5.3.)</div> |
| 1506<br>1507                      | 1606<br>1607 | 1706<br>1707 | 1806<br>1807 |                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Software stroke limit upper limit |              |              |              |                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 18<br>19                          | 168<br>169   | 318<br>319   | 468<br>469   |                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Software stroke limit lower limit |              |              |              |                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 20<br>21                          | 170<br>171   | 320<br>321   | 470<br>471   |                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Classification of errors | Error code | Error name                                  | Error                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Operation status at error occurrence                                                                                                                                                                                                                                                             |  |
|--------------------------|------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Positioning operation    | 514        | Outside new current value range             | The new current address is outside the ranges of 0 to 359.99999, where the control unit is set to "degree".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | The current value is not changed.                                                                                                                                                                                                                                                                |  |
|                          | 515        | New current value not possible              | <ul style="list-style-type: none"> <li>The operation pattern "continuous path control" is set in the positioning data in which the control system "current value changing" is set.</li> <li>The control system "current value changing" is set in the positioning data following the positioning data in which the operation pattern "continuous path control" is set.</li> </ul>                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                  |  |
|                          | 516        | Continuous path control not possible        | <ul style="list-style-type: none"> <li>The continuous path control is designated using a control system which is not allowed to use for continuous path control such as speed control, speed-position switching control, position-speed switching control, fixed-feed, and current value change.</li> <li>The previous data such as those on speed control, speed-position switching control, position-speed switching control, fixed-feed, and current value change shows a continuous path control.</li> <li>The continuous positioning control is designated for speed control or position-speed switching control.</li> </ul> | At start, the system will not operate.                                                                                                                                                                                                                                                           |  |
|                          | 518        | Outside operation pattern range             | <ul style="list-style-type: none"> <li>The operation pattern set value is 2.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | At start: The system will not operate.<br>During operation:<br>The system stops with the setting (deceleration stop/sudden stop) of the detailed parameter 2 Sudden stop selection (stop group 3).<br>(Note that the deceleration stop only occurs during the manual pulse generator operation.) |  |
|                          | 519        | Interpolation while interpolation axis BUSY | Interpolation is started during the operation of the interpolation axis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                  |  |
|                          | 520        | Unit group unmatched                        | When the "composite speed" has been set in "Interpolation speed designation method" of the parameter or positioning data, the units of the reference axis and interpolation axis differ.                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                  |  |
|                          | 521        | Illegal interpolation description command   | <ul style="list-style-type: none"> <li>The self-axis or a non-existent axis is set as the axis to be interpolated in the 2-axis interpolation or helical interpolation.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                  |  |

| Related buffer memory address                      |              |              |              |  | Set range<br>(Setting with sequence program)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Remedy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------|--------------|--------------|--------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1                                             | Axis 2       | Axis 3       | Axis 4       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1506<br>1507                                       | 1606<br>1607 | 1706<br>1707 | 1806<br>1807 |  | <New current value><br>[degree] 0 to 35999999                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Bring the new current value into the setting range.<br>(Refer to Section 9.2.21)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Refer to Section 5.3<br>"List of positioning data" |              |              |              |  | <Control system><br>01 <sub>H</sub> to 1E <sub>H</sub> , 80 <sub>H</sub> to 84 <sub>H</sub><br>• 03 <sub>H</sub> , 0C <sub>H</sub> , 17 <sub>H</sub> , 1C <sub>H</sub> :<br>1 to 4 axis fixed-feed control<br>• 04 <sub>H</sub> , 05 <sub>H</sub> , 13 <sub>H</sub> , 14 <sub>H</sub> , 18 <sub>H</sub> , 19 <sub>H</sub> , 1D <sub>H</sub> ,<br>1E <sub>H</sub> :<br>1 to 4 axis speed control<br>• 81 <sub>H</sub> : current value changing<br>• Speed-position switching control:<br>06 <sub>H</sub> , 07 <sub>H</sub><br>• Position-speed switching control:<br>08 <sub>H</sub> , 09 <sub>H</sub><br><Operation pattern> 00, 01, 11<br>• 01: Continuous positioning control<br>• 11: Continuous path control | • Do not designate the "continuous path control" using the "current value changing".<br>• Do not designate the "current value changing" using the positioning data following the positioning data that the "continuous path control" is designated.<br>(Refer to Section 9.2.21)                                                                                                                                                                                                                                                                                                     |
|                                                    |              |              |              |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | • Do not designate a speed control, fixed-feed, speed-position switching control, position-speed switching control, and current value change using the positioning data following the continuous path control data.<br>• Do not carry out the fixed-feed, speed control, speed-position switching control, position-speed switching control, and current value change using the continuous path control operation pattern.<br>• Do not carry out the speed control and position-speed switching control using the continuous path control operation pattern.<br>(Refer to CHAPTER 9) |
| Same as error codes 515 to 516                     |              |              |              |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Correct the operation pattern.<br>(Refer to Section 5.3 <a href="#">Da.1</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                    |              |              |              |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Correct the control system.<br>(Refer to Section 5.3 <a href="#">Da.2</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 0                                                  | 150          | 300          | 450          |  | <Unit setting><br>0, 1, 2, 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Correct the positioning data or change the parameter "Unit setting" of the axis to be interpolated.<br>(Refer to Section 9.1.6)                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Same as error codes 515 to 516                     |              |              |              |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | • Correct the control system.<br>(Refer to Section 5.3 <a href="#">Da.2</a> )<br>• Correct the axis to be interpolated.<br>(Refer to Section 5.3 <a href="#">Da.5</a> )                                                                                                                                                                                                                                                                                                                                                                                                              |

| Classification of errors | Error code | Error name                   | Error                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Operation status at error occurrence                                                                                                                                                                                                                                                                           |  |
|--------------------------|------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Positioning operation    | 522        | Command speed setting error  | <p>The command speed is outside the setting range.</p> <p>Linear interpolation, circular interpolation, helical interpolation:<br/>Reference axis is outside the setting range.</p> <p>Speed control interpolation:<br/>Either of reference axis and interpolation axis is outside the speed range.</p>                                                                                                                                                                                                                                                                                                                                                       | <p>At start: The system will not operate.</p> <p>During operation:<br/>The system stops with the setting (deceleration stop/sudden stop) of the detail parameter 2 Sudden stop selection (stop group 3).</p> <p>(Note that the deceleration stop only occurs during the manual pulse generator operation.)</p> |  |
|                          | 523        | Interpolation mode error     | <ul style="list-style-type: none"> <li>In the interpolation control of the speed control, 4-axis linear interpolation control, or 4-axis fixed-feed control, the system is started while a composite speed is set in "Interpolation speed designation method" of the parameter or positioning data of the reference axis.</li> <li>In the circular interpolation control or helical interpolation control, the system is started while a reference axis speed is set in "Interpolation speed designation method" of the parameter or positioning data of the reference axis.</li> </ul>                                                                       |                                                                                                                                                                                                                                                                                                                |  |
|                          | 524        | Control system setting error | <ul style="list-style-type: none"> <li>The control system setting value is outside the specified limit.</li> <li>The number of control axes or the axis to be interpolated differs from the previous data when the operation is continuously performed by setting the continuous positioning control or continuous path control.</li> <li>Machine OPR, fast OPR, or speed-position or position-speed switching control was performed in the wiring-less mode.</li> <li>The NOP instruction was set to the control system of positioning data No. 600.</li> <li>Any value other than 0 was set at buffer memory address 1906 (use prohibited area).</li> </ul> |                                                                                                                                                                                                                                                                                                                |  |
|                          | 525        | Sub point setting error      | <p>One of the following settings is configured in circular interpolation or helical interpolation with sub point designation.</p> <ul style="list-style-type: none"> <li>Start point = sub point</li> <li>End point = sub point</li> <li>Start point, end point, and sub point are in line with each other.</li> <li>Sub point address and center point address are outside the range of -2147483648 to 2147483647.</li> </ul>                                                                                                                                                                                                                                | <p>At start: The system will not operate.</p> <p>During operation:<br/>The system stops immediately.</p>                                                                                                                                                                                                       |  |

| Related buffer memory address                                    |        |        |        |  | Set range<br>(Setting with sequence program)                                                                                                                                                                                    | Remedy                                                                                                                                                                                                                                         |
|------------------------------------------------------------------|--------|--------|--------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1                                                           | Axis 2 | Axis 3 | Axis 4 |  |                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                |
| Command speed storage addresses of positioning data No. 1 to 600 |        |        |        |  | <Command speed><br>QD75P□N/QD75D□N:<br>1 to 4000000 [pulse/s]<br>1 to 2000000000<br>[ $\times 10^{-2}$ mm/min or others]<br>QD75P□/QD75D□:<br>1 to 1000000 [pulse/s]<br>1 to 2000000000<br>[ $\times 10^{-2}$ mm/min or others] | Correct the command speed.<br>(Refer to Section 5.3 [Da.8])                                                                                                                                                                                    |
| 29                                                               | 179    | 329    | 479    |  | <Interpolation speed designation method><br>0: Composite speed<br>1: Reference axis speed                                                                                                                                       | Set the "Interpolation speed designation method" correctly.<br>(Refer to Section 9.1.6)                                                                                                                                                        |
| Same as error codes 515 to 516                                   |        |        |        |  |                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Correct the control system axis to be interpolated, or parameter.<br/>(Refer to Section 9.1.6, 9.2.22)</li> <li>• Do not make setting at buffer memory address 1906 (use prohibited area).</li> </ul> |
| Refer to Section 5.3 "List of positioning data"                  |        |        |        |  | <Positioning address/movement amount><br>• unit [mm] [pulse] [inch]<br>–2147483648 to 2147483647<br>(Unit [degree] cannot be set.)<br><Arc address><br>–2147483648 to 2147483647                                                | Correct the sub address (arc address).<br>(Refer to Section 9.2.10, 9.2.12)                                                                                                                                                                    |

| Classification of errors | Error code | Error name                          | Error                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Operation status at error occurrence                                                                                                                                                                                                                                                                         |  |
|--------------------------|------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Positioning operation    | 526        | End point setting error             | <ul style="list-style-type: none"> <li>The start point is equal to the end point in circular interpolation or helical interpolation with sub point designation.</li> <li>The end point address is outside the range of -2147483648 to 2147483647 in circular interpolation or helical interpolation with sub point designation or center point designation.</li> </ul>                                                                                                                                                                                                                                                                                                        | At start: The system will not operate.<br>During operation:<br>The system stops immediately.                                                                                                                                                                                                                 |  |
|                          | 527        | Center point setting error          | One of the following settings is configured in circular interpolation or helical interpolation with center point designation. <ul style="list-style-type: none"> <li>Start point = Center point</li> <li>End point = Center point</li> <li>Center point address is outside the range of -2147483648 to 2147483647.</li> </ul>                                                                                                                                                                                                                                                                                                                                                 | At start: The system will not operate.<br>During operation:<br>The system stops immediately.                                                                                                                                                                                                                 |  |
|                          | 530        | Outside address range               | <ul style="list-style-type: none"> <li>In the speed-position switching control and the position-speed switching control, the setting value of a positioning address is negative.</li> <li>In the 1-axis linear control (ABS), 2- to 4-axis linear interpolation control (ABS), or helical interpolation control (ABS), the setting value of a positioning address is outside the range of 0 to 359.99999 [degree].</li> </ul>                                                                                                                                                                                                                                                 | At start: The system will not operate.<br>During operation:<br>The system stops immediately with the setting (deceleration stop/sudden stop) of the detailed parameter 2 sudden stop selection (stop group 3).<br>(Note that the deceleration stop only occurs during the manual pulse generator operation.) |  |
|                          | 532        | Simultaneous start not possible     | Among the axes to be started simultaneously, there is an axis on which an error other than this error occurs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                              |  |
|                          | 533        | Condition data error                | <ul style="list-style-type: none"> <li>The condition setting values are not set or outside the setting range.</li> <li>The condition operator setting values are not set or outside the setting range.</li> <li>The condition operator is a bit operator, and the parameter 1 is 32 or more.</li> <li>An unusable condition operator is set for the set condition.</li> <li>The conditional operator has been [parameter 1 &gt; parameter 2] with 05H (<math>P1 \leq ** \leq P2</math>).</li> <li>The value in 'address' is out of the setting range when the condition target is set to 'Buffer memory (1-word/2-word)'. (1-word: 0 to 32767, 2-word: 0 to 32766)</li> </ul> | The operation is terminated.                                                                                                                                                                                                                                                                                 |  |
|                          | 534        | Special start instruction error     | No applicable special start instruction is present.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                              |  |
|                          | 535        | Circular interpolation not possible | The circular interpolation or helical interpolation is carried out for the axis with the "degree" unit setting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                              |  |



|  | Related buffer memory address                                                               |        |        |        | Set range<br>(Setting with sequence program)                      | Remedy                                                                                                                                                               |
|--|---------------------------------------------------------------------------------------------|--------|--------|--------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Axis 1                                                                                      | Axis 2 | Axis 3 | Axis 4 |                                                                   |                                                                                                                                                                      |
|  | Refer to Section 5.3<br>"List of positioning data"                                          |        |        |        |                                                                   | Correct the end address (positioning address).<br>(Refer to Section 9.2.10, 9.2.12)                                                                                  |
|  |                                                                                             |        |        |        | Same as in error codes 525.                                       |                                                                                                                                                                      |
|  |                                                                                             |        |        |        |                                                                   | Correct the center point address (arc address).<br>(Refer to Section 9.2.11, 9.2.13)                                                                                 |
|  |                                                                                             |        |        |        | Same as in error codes 504, 506.                                  | Correct the positioning address.<br>(Refer to Section 9.2.18, 9.2.19, 9.2.20)                                                                                        |
|  | Refer to Section 5.3<br>"List of positioning data"<br>and Section 5.4<br>"Block start data" |        |        |        | —                                                                 | In the error history, check the axis where the error other than this error occurred, and remove the error factor. Correct the block start data and positioning data. |
|  | Refer to Section 5.4<br>"Block start data"                                                  |        |        |        | —                                                                 | Normalize the block start data.                                                                                                                                      |
|  |                                                                                             |        |        |        | <Special start instruction><br>00 <sub>H</sub> to 06 <sub>H</sub> | Correct the special start instruction code.<br>(Refer to Section 5.4 <a href="#">Da.13</a> )                                                                         |
|  |                                                                                             |        |        |        | —                                                                 | Correct the control system.<br>(Refer to Section 5.3 <a href="#">Da.2</a> )                                                                                          |

| Classification of errors | Error code | Error name                                         | Error                                                                                                                                                                                                                                                                                                                                             | Operation status at error occurrence                                                                                                                                                                                                                                                                              |  |
|--------------------------|------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Positioning operation    | 536        | M code ON signal start                             | The positioning start is carried out when an M code ON signal (X4 to X7) is turned ON.                                                                                                                                                                                                                                                            | At start, the system will not operate.                                                                                                                                                                                                                                                                            |  |
|                          | 537        | PLC READY OFF start                                | The positioning start is carried out when the PLC READY signal (Y0) is turned OFF.                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                   |  |
|                          | 538        | READY OFF start                                    | The positioning start is carried out when the QD75 READY signal (X0) is turned OFF.                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                   |  |
|                          | 543        | Outside start No. range                            | <ul style="list-style-type: none"> <li>At the start of positioning, the setting value of the "positioning start No." of the axis control data is outside the ranges of 1 to 600, 7000 to 7004, and 9001 to 9004.</li> <li>At a Pre-reading start, the "positioning start No." setting of the axis control data is other than 1 to 600.</li> </ul> |                                                                                                                                                                                                                                                                                                                   |  |
|                          | 544        | Outside radius range                               | The arc radius exceeds 536870912.                                                                                                                                                                                                                                                                                                                 | At start: The system will not operate.<br>During operation:<br>The system stops immediately.                                                                                                                                                                                                                      |  |
|                          | 545        | Control system LOOP setting error                  | A "0" is set for number of repetitions of the control system "LOOP".                                                                                                                                                                                                                                                                              | The operation is terminated.                                                                                                                                                                                                                                                                                      |  |
|                          | 546        | Illegal setting of ABS direction in unit of degree | <p>The setting value of ABS direction in the unit of degree is as follows.</p> <ul style="list-style-type: none"> <li>Set outside the setting range.</li> <li>A figure other than "0" is set when the software stroke limit is valid.</li> </ul>                                                                                                  | At start: The system will not operate.<br>During operation:<br>The system decelerates to a stop.<br>(Note that, in the continuous positioning control and continuous path control, the system continues operating with the setting set at the time of start even if the setting is changed during the operation.) |  |
|                          | 553        | M code ON timing error                             | The setting value of the positioning data "M code ON signal output timing" is outside the range.                                                                                                                                                                                                                                                  | At start: The system will not operate.<br>During operation:<br>The system stops with the setting (deceleration stop/sudden stop) of the detailed parameter 2 sudden stop selection (stop group 3).                                                                                                                |  |
|                          | 554        | Interpolation speed designation method error       | The setting value of the positioning data "Interpolation speed designation method" is outside the range.                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                   |  |
|                          | 555        | Outside range of the number of pitches             | In the helical interpolation, the number of pitches set in the positioning data "M code" of the linear interpolation axis is outside the range.                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                   |  |

| Related buffer memory address                      |        |        |        |                                                                       | Set range<br>(Setting with sequence program)                                                                                      | Remedy                                                                                                                                                                                |
|----------------------------------------------------|--------|--------|--------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1                                             | Axis 2 | Axis 3 | Axis 4 |                                                                       |                                                                                                                                   |                                                                                                                                                                                       |
| 1504                                               | 1604   | 1704   | 1804   | <M code OFF request><br>1: M code ON signal is turned OFF             | After turning OFF the M code ON signal, start the system. (Refer to Section 12.7.3)                                               |                                                                                                                                                                                       |
| —                                                  | —      | —      | —      | —                                                                     | Check the sequence program which turns ON/OFF the PLC READY signal (Y0), and turn ON the PLC READY signal. Then start the system. |                                                                                                                                                                                       |
| —                                                  | —      | —      | —      | —                                                                     | Check the QD75 READY ON signal, and then start the system. (Refer to Section 3.3.2)                                               |                                                                                                                                                                                       |
| 1500                                               | 1600   | 1700   | 1800   | <Positioning start No.><br>1 to 600,<br>7000 to 7004,<br>9001 to 9004 | Normalize the positioning start No.<br>(Refer to Section 12.7.7)                                                                  |                                                                                                                                                                                       |
| Refer to Section 5.3<br>"List of positioning data" |        |        |        |                                                                       | <Maximum radius><br>536870912                                                                                                     | Correct the positioning data.<br>(Refer to Section 9.2.10, 9.2.11)                                                                                                                    |
|                                                    |        |        |        |                                                                       | <LOOP to LEND><br>1 to 65535                                                                                                      | Set 1 to 65535 for number of repetitions of the "LOOP".<br>(Refer to Section 9.2.22)                                                                                                  |
| ABS setting direction in the unit of degree        |        |        |        |                                                                       | 0: Shortcut<br>1: ABS clockwise<br>2: ABS counterclockwise                                                                        | • Set the ABS setting direction in the unit of degree within the setting range.<br>• Set "0" when the software stroke limits are valid.<br>(Refer to Section 9.1.5)                   |
| 1550                                               | 1650   | 1750   | 1850   |                                                                       |                                                                                                                                   |                                                                                                                                                                                       |
| Software stroke limit upper limit                  |        |        |        |                                                                       | • [mm] [inch] [pulse]<br>–2147483648 to 2147483647<br><br>• [degree]<br>0 to 35999999                                             | Invalidate the software stroke limit.<br>(To invalidate, set the software stroke limit upper limit value to the software stroke limit lower limit value.)<br>(Refer to Section 9.1.5) |
| 18                                                 | 168    | 318    | 468    |                                                                       |                                                                                                                                   |                                                                                                                                                                                       |
| 19                                                 | 169    | 319    | 469    |                                                                       |                                                                                                                                   |                                                                                                                                                                                       |
| Software stroke limit lower limit                  |        |        |        |                                                                       |                                                                                                                                   |                                                                                                                                                                                       |
| 20                                                 | 170    | 320    | 470    |                                                                       |                                                                                                                                   |                                                                                                                                                                                       |
| 21                                                 | 171    | 321    | 471    |                                                                       |                                                                                                                                   |                                                                                                                                                                                       |
| Refer to Section 5.3<br>"List of positioning data" |        |        |        |                                                                       | 0: Use the set value in "Pr.18" M code ON signal output timing".<br>1: WITH mode<br>2: AFTER mode                                 | Correct the positioning data "M code ON signal output timing" to 0 to 2.                                                                                                              |
|                                                    |        |        |        |                                                                       | 0: Use the set value in "Pr.20" Interpolation speed designation method".<br>1: Composite speed<br>2: Reference axis speed         | Correct the positioning data "Interpolation speed designation method" to 0 to 2.                                                                                                      |
|                                                    |        |        |        |                                                                       | <Number of pitch><br>0 to 999                                                                                                     | Correct the number of pitch set in the positioning data "M code" of the linear interpolation axis to 0 to 999.                                                                        |

| Classification of errors | Error code | Error name                | Error                                                                               | Operation status at error occurrence                                                                                                                                                                                                                                                             |  |
|--------------------------|------------|---------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| I/F                      | 800        | Hold error                | In the CPU module parameter "Output at error stop", the setting for QD75 is "Hold". | At start: The system will not operate.<br>During operation:<br>The system stops with the setting (deceleration stop/sudden stop) of the detailed parameter 2 sudden stop selection (stop group 3).<br>(Note that the deceleration stop only occurs during the manual pulse generator operation.) |  |
|                          | 801        | Flash ROM write error     | Data is not written to the flash ROM.                                               |                                                                                                                                                                                                                                                                                                  |  |
|                          | 802        | Flash ROM sum check error | While data is written to the flash ROM, the power is turned OFF.                    | At start: The system will not operate.                                                                                                                                                                                                                                                           |  |

|  | Related buffer memory address |        |        |        | Set range<br>(Setting with sequence program)                                   | Remedy                                                                                                    |
|--|-------------------------------|--------|--------|--------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
|  | Axis 1                        | Axis 2 | Axis 3 | Axis 4 |                                                                                |                                                                                                           |
|  | —                             | —      | —      | —      | —                                                                              | Clear the setting of the CPU module parameter "Output at error stop".                                     |
|  | —                             | —      | —      | —      | —                                                                              | The flash ROM is expected to be at the end of its writable life.<br>Replace the flash ROM with a new one. |
|  | 1901                          |        |        |        | <Parameter initialization request><br>1: Parameter initialization is requested | Return the parameter to that set at the time of delivery from the plant. (Refer to Section 13.2)          |

| Classification of errors | Error code | Error name                      | Error                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Operation status at error occurrence                                                                                                                                                                                                                                                             |  |
|--------------------------|------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| I/F                      | 803        | PLC CPU error                   | The CPU module resulted in an error.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | At start: The system will not operate.<br>During operation:<br>The system stops with the setting (deceleration stop/sudden stop) of the detailed parameter 2 sudden stop selection (stop group 2).<br>(Note that the deceleration stop only occurs during the manual pulse generator operation.) |  |
|                          | 804        | Dedicated instruction error     | <ul style="list-style-type: none"> <li>The Z.ABRST□ instruction is executed with the status set to other than 0 (at the start of communication with the servo-amplifier).</li> <li>The status of the Z.ABRST□ instruction is changed during absolute position restoration (during communication with the servo-amplifier).</li> <li>The ZP.PSTRT□ instruction is executed with the start No. set to other than 1 to 600, 7000 to 7004 and 9001 to 9004.</li> <li>The ZP.TEACH□ instruction is executed with the teaching data selection set to other than 0 and 1.</li> <li>The ZP.TEACH□ instruction is executed with the positioning data No. set to other than 1 to 600.</li> <li>The instruction of a non-existent axis is specified by the Z.ABRST□, ZP.PSTRT□ or ZP.TEACH□ instruction.</li> </ul> | The function corresponding to the instruction is not executed.                                                                                                                                                                                                                                   |  |
|                          | 805        | Flash ROM write number error    | Data is written to the flash ROM continuously 25 times or more from the sequence program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | At start: The system will not operate.                                                                                                                                                                                                                                                           |  |
|                          | 806        | Dedicated instruction I/F error | Mismatching occurs between the CPU module and QD75.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                  |  |

|  | Related buffer memory address |        |        |        | Set range<br>(Setting with sequence program)                                                                                                                                                                                                                                                                                                                                                                                                   | Remedy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--|-------------------------------|--------|--------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Axis 1                        | Axis 2 | Axis 3 | Axis 4 |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  | —                             | —      | —      | —      | —                                                                                                                                                                                                                                                                                                                                                                                                                                              | Check the error code of CPU module and refer to the QCPU User's Manual.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|  | —                             | —      | —      | —      | <p>&lt; Z.ABRST□ status &gt;<br/> 0: Communication complete<br/> (received from the servo amplifier)<br/> &lt; ZP.PSTRT□ start No. &gt;<br/> 1 to 600<br/> 7000 to 7004<br/> 9001 to 9004<br/> &lt; ZP.TEACH□ teaching data selection &gt;<br/> 0: The current feed value is written to the positioning address.<br/> 1: The current feed value is written to the arc address.<br/> &lt; ZP.TEACH□ positioning data No. &gt;<br/> 1 to 600</p> | <ul style="list-style-type: none"> <li>• When executing the Z.ABRST□ instruction, set the status to 0 (refer to Section 14.3).</li> <li>• Do not change the status during absolute position restoration by the Z.ABRST□ instruction.</li> <li>• When executing the ZP.PSTRT□ instruction, set the start No. to within the setting range (refer to Section 14.4).</li> <li>• When executing the ZP.TEACH□ instruction, set the teaching data selection and positioning data No. to within the setting range (refer to Section 14.5).</li> <li>• Do not specify the instruction of a non-existent axis by the Z.ABRST□, ZP.PSTRT□ and ZP.TEACH□ instructions (refer to Section 14.3 to Section 14.5).</li> </ul> |
|  | —                             | —      | —      | —      | —                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Review the sequence program so that data is not written continuously to the flash ROM.<br/> (Using <a href="#">Md.19</a> in Section 5.6.1, the number of flash ROM write times can be monitored.)<br/> (If this error has occurred in a proper using method, writing is enabled by resetting the error, switching power OFF, then ON, or resetting the CPU module.)</p>                                                                                                                                                                                                                                                                                                                                     |
|  | —                             | —      | —      | —      | —                                                                                                                                                                                                                                                                                                                                                                                                                                              | A trouble occurs. Repair.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Classification of errors | Error code | Error name                                 | Error                                                                                                                                                                                                                                                                                                                                                           | Operation status at error occurrence                                                                                                                                        |  |
|--------------------------|------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Parameter                | 900        | Outside unit setting range                 | The set value of the basic parameter 1 "Unit setting" is outside the setting range.                                                                                                                                                                                                                                                                             | The QD75 READY signal (X0) is not turned ON.                                                                                                                                |  |
|                          | 901        | Outside pulse number per rotation range    | The set value of the basic parameter 1 "No. of pulses per rotation" is outside the setting range.                                                                                                                                                                                                                                                               |                                                                                                                                                                             |  |
|                          | 902        | Outside movement amount per rotation range | The set value of the basic parameter 1 "Movement amount per rotation" is outside the setting range.                                                                                                                                                                                                                                                             |                                                                                                                                                                             |  |
|                          | 903        | Outside unit magnification range           | The set value of the basic parameter 1 "Unit magnification" is outside the setting range.                                                                                                                                                                                                                                                                       |                                                                                                                                                                             |  |
|                          | 904        | Pulse output mode error                    | The set value of the basic parameter 1 "Pulse output mode" is outside the setting range.                                                                                                                                                                                                                                                                        |                                                                                                                                                                             |  |
|                          | 905        | Rotation direction setting error           | The set value of the basic parameter 1 "Rotation direction setting" is outside the setting range.                                                                                                                                                                                                                                                               |                                                                                                                                                                             |  |
|                          | 906        | Outside bias speed range                   | <ul style="list-style-type: none"> <li>The set value of the basic parameter 1 "Bias speed at start" is outside the setting range.</li> <li>The bias speed exceeds the speed limit.</li> </ul>                                                                                                                                                                   | When the power is turned ON or PLC READY signal (Y0) is turned from OFF to ON:<br>QD75 READY signal (X0) is not turned ON.<br><br>At start:<br>The system will not operate. |  |
|                          | 910        | Outside speed limit value range            | <ul style="list-style-type: none"> <li>The set value of the basic parameter 2 "Speed limit value" is outside the setting range.</li> <li>The value obtained by the conversion of the speed limit value with respect to the frequency exceeds the maximum output frequency of the unit.</li> <li>The speed limit value is smaller than the OPR speed.</li> </ul> |                                                                                                                                                                             |  |
|                          | 911        | Outside acceleration time 0 range          | The set value of the basic parameter 2 "Acceleration time 0" is outside the setting range.                                                                                                                                                                                                                                                                      |                                                                                                                                                                             |  |
|                          | 912        | Outside deceleration time 0 range          | The set value of the basic parameter 2 "Deceleration time 0" is outside the setting range.                                                                                                                                                                                                                                                                      |                                                                                                                                                                             |  |
|                          | 920        | Backlash compensation amount error         | The value converted into pulse number using the movement amount per pulse is 256 pulses or more.                                                                                                                                                                                                                                                                |                                                                                                                                                                             |  |
|                          | 921        | Software stroke limit upper limit          | <ul style="list-style-type: none"> <li>In the unit of degree, the set value of the detailed parameter 1 "Software stroke limit upper limit value" is outside the setting range.</li> <li>In a unit other than degree, the software stroke limit upper limit value is smaller than the software stroke limit lower limit value.</li> </ul>                       | The QD75 READY signal (X0) is not turned ON.                                                                                                                                |  |



|  | Related buffer memory address |            |            |            | Set range<br>(Setting with sequence program)                                                                                                                                                                 | Remedy                                                                                                                                                                                                                                                                                                                           |
|--|-------------------------------|------------|------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Axis 1                        | Axis 2     | Axis 3     | Axis 4     |                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                  |
|  | 0                             | 150        | 300        | 450        | 0, 1, 2, 3                                                                                                                                                                                                   | With the setting brought into the setting range, turn the PLC READY signal [Y0] from OFF to ON.                                                                                                                                                                                                                                  |
|  | 1                             | 151        | 301        | 451        | 1 to 65535                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                  |
|  | 2                             | 152        | 302        | 452        | 1 to 65535                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                  |
|  | 3                             | 153        | 303        | 453        | 1, 10, 100, 1000                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                  |
|  | 4                             | 154        | 304        | 454        | 0, 1, 2, 3                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                  |
|  | 5                             | 155        | 305        | 455        | 0, 1                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                  |
|  | 6<br>7                        | 156<br>157 | 306<br>307 | 456<br>457 | QD75P□N/QD75D□N:<br>0 to 4000000 [pulse/s]<br>0 to 2000000000<br>[ $\times 10^{-2}$ mm/min or others]<br>QD75P□/QD75D□:<br>0 to 1000000 [pulse/s]<br>0 to 2000000000<br>[ $\times 10^{-2}$ mm/min or others] | Set the bias speed to not more than the speed limit value.<br>With the setting brought into the setting range, turn the PLC READY signal [Y0] from OFF to ON.                                                                                                                                                                    |
|  | 10<br>11                      | 160<br>161 | 310<br>311 | 460<br>461 | QD75P□N/QD75D□N:<br>1 to 4000000 [pulse/s]<br>1 to 2000000000<br>[ $\times 10^{-2}$ mm/min or others]<br>QD75P□/QD75D□:<br>1 to 1000000 [pulse/s]<br>1 to 2000000000<br>[ $\times 10^{-2}$ mm/min or others] | <ul style="list-style-type: none"> <li>• The value converted into the frequency should not exceed the maximum output frequency of the module.</li> <li>• Set a value which is not less than the OPR speed.</li> <li>• With the setting brought into the setting range, turn the PLC READY signal [Y0] from OFF to ON.</li> </ul> |
|  | 12<br>13                      | 162<br>163 | 312<br>313 | 462<br>463 | 1 to 8388608                                                                                                                                                                                                 | With the setting brought into the setting range, turn the PLC READY signal [Y0] from OFF to ON.                                                                                                                                                                                                                                  |
|  | 14<br>15                      | 164<br>165 | 314<br>315 | 464<br>465 | 1 to 8388608                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                  |
|  | 17                            | 167        | 317        | 467        | 0 to 255                                                                                                                                                                                                     | Set the value converted into the pulse number using "the movement amount per pulse" to less than 256 pulses. (Refer to Section 12.3.1)                                                                                                                                                                                           |
|  | 18<br>19                      | 168<br>169 | 318<br>319 | 468<br>469 | <ul style="list-style-type: none"> <li>• [mm] [inch] [pulse]<br/>-2147483648 to 2147483647</li> <li>• [degree] 0 to 35999999</li> </ul>                                                                      | <ul style="list-style-type: none"> <li>• Bring the setting into the setting range.</li> <li>• In a unit other than degree, set the setting so that the lower limit value is smaller than the upper limit value.</li> </ul>                                                                                                       |

| Classification of errors | Error code | Error name                                   | Error                                                                                                                                                                                                                                                                                                                                     | Operation status at error occurrence                                                                                                                                                                                                                                                                     |  |
|--------------------------|------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Parameter                | 922        | Software stroke limit lower limit            | <ul style="list-style-type: none"> <li>In the unit of degree, the set value of the detailed parameter 1 "Software stroke limit lower limit value" is outside the setting range.</li> <li>In a unit other than degree, the software stroke limit upper limit value is smaller than the software stroke limit lower limit value.</li> </ul> | The QD75 READY signal (X0) is not turned ON.                                                                                                                                                                                                                                                             |  |
|                          | 923        | Software stroke limit selection              | The set value of the detailed parameter 1 "Software stroke limit selection" is outside the setting range.                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                          |  |
|                          | 924        | Software stroke limit valid/invalid setting  | The set value of the detailed parameter 1 "Software stroke limit valid/invalid setting" is outside the setting range.                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                          |  |
|                          | 925        | Command in-position width                    | The set value of the detailed parameter 1 "Command in-position width" is outside the setting range.                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                          |  |
|                          | 926        | Illegal torque limit setting value           | The set value of the detailed parameter 1 "Torque limit setting value" is outside the setting range.                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                          |  |
|                          | 927        | M code ON timing error                       | The set value of the detailed parameter 1 "M code ON signal output timing" is outside the setting range.                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                          |  |
|                          | 928        | Speed changeover mode error                  | The set value of the detailed parameter 1 "Speed changeover mode" is outside the setting range.                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          |  |
|                          | 929        | Interpolation speed designation method error | The set value of the detailed parameter 1 "Interpolation speed designation method" is outside the setting range.                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                          |  |
|                          | 930        | Current value update request error           | The set value of the detailed parameter 1 "Current feed value during speed control" is outside the setting range.                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                          |  |
|                          | 932        | Manual pulse generator input mode error      | The set value of the detailed parameter 1 "Manual pulse generator input selection" is outside the setting range.                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                          |  |
|                          | 935        | Speed-position function selection error      | The detailed parameter 1 "speed-position function selection" is preset to 2 and the following three conditions are not satisfied:<br>1) Unit is "degree".<br>2) Software stroke limits are invalid.<br>3) Update current feed value.                                                                                                      |                                                                                                                                                                                                                                                                                                          |  |
|                          | 950        | Acceleration time 1 setting error            | The set value of the detailed parameter 2 "Acceleration time 1" is outside the setting range.                                                                                                                                                                                                                                             | At start: The system will not operate.<br><br>During operation:<br>The system stops with the setting (deceleration stop/sudden stop) of the detailed parameter 2 sudden stop selection (stop group 3).<br><br>(Note that the deceleration stop only occurs during the manual pulse generator operation.) |  |
|                          | 951        | Acceleration time 2 setting error            | The set value of the detailed parameter 2 "Acceleration time 2" is outside the setting range.                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                          |  |
|                          | 952        | Acceleration time 3 setting error            | The set value of the detailed parameter 2 "Acceleration time 3" is outside the setting range.                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                          |  |
|                          | 953        | Deceleration time 1 setting error            | The set value of the detailed parameter 2 "Deceleration time 1" is outside the setting range.                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                          |  |

| Related buffer memory address |            |            |            |                                                                                                                                      | Set range<br>(Setting with sequence program)                                                                                                                                                                | Remedy |
|-------------------------------|------------|------------|------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Axis 1                        | Axis 2     | Axis 3     | Axis 4     |                                                                                                                                      |                                                                                                                                                                                                             |        |
| 20<br>21                      | 170<br>171 | 320<br>321 | 470<br>471 | <ul style="list-style-type: none"><li>• [mm] [inch] [pulse]<br/>–2147483648 to 2147483647</li><li>• [degree] 0 to 35999999</li></ul> | <ul style="list-style-type: none"><li>• Bring the setting into the setting range.</li><li>• In a unit other than degree, set so that the lower limit value is smaller than the upper limit value.</li></ul> |        |
| 22                            | 172        | 322        | 472        | 0, 1                                                                                                                                 | With the setting brought into the setting range, turn the PLC READY signal [Y0] from OFF to ON.                                                                                                             |        |
| 23                            | 173        | 323        | 473        | 0, 1                                                                                                                                 |                                                                                                                                                                                                             |        |
| 24<br>25                      | 174<br>175 | 324<br>325 | 474<br>475 | 1 to 2147483647                                                                                                                      |                                                                                                                                                                                                             |        |
| 26                            | 176        | 326        | 476        | 1 to 500                                                                                                                             |                                                                                                                                                                                                             |        |
| 27                            | 177        | 327        | 477        | 0, 1                                                                                                                                 |                                                                                                                                                                                                             |        |
| 28                            | 178        | 328        | 478        | 0, 1                                                                                                                                 |                                                                                                                                                                                                             |        |
| 29                            | 179        | 329        | 479        | 0, 1                                                                                                                                 |                                                                                                                                                                                                             |        |
| 30                            | 180        | 330        | 480        | 0, 1, 2                                                                                                                              |                                                                                                                                                                                                             |        |
| 33                            | —          | —          | —          | 0, 1, 2, 3                                                                                                                           |                                                                                                                                                                                                             |        |
| 34                            | 184        | 334        | 484        | 0, 2                                                                                                                                 |                                                                                                                                                                                                             |        |
| 36<br>37                      | 186<br>187 | 336<br>337 | 486<br>487 | 1 to 8388608                                                                                                                         | With the setting brought into the setting range, turn the PLC READY signal [Y0] from OFF to ON.                                                                                                             |        |
| 38<br>39                      | 188<br>189 | 338<br>339 | 488<br>489 | 1 to 8388608                                                                                                                         |                                                                                                                                                                                                             |        |
| 40<br>41                      | 190<br>191 | 340<br>341 | 490<br>491 | 1 to 8388608                                                                                                                         |                                                                                                                                                                                                             |        |
| 42<br>43                      | 192<br>193 | 342<br>343 | 492<br>493 | 1 to 8388608                                                                                                                         |                                                                                                                                                                                                             |        |

| Classification of errors | Error code | Error name                                                 | Error                                                                                                                                                                                                                                                 | Operation status at error occurrence                                                                                                                                                                                                                                                                          |  |
|--------------------------|------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Parameter                | 954        | Deceleration time 2 setting error                          | The set value of the detailed parameter 2 "Deceleration time 2" is outside the setting range.                                                                                                                                                         | <p>At start: The system will not operate.</p> <p>During operation:<br/>The system stops with the setting (deceleration stop/sudden stop) of the detailed parameter 2 Sudden stop selection (stop group 3).<br/>(Note that the deceleration stop only occurs during the manual pulse generator operation.)</p> |  |
|                          | 955        | Deceleration time 3 setting error                          | The set value of the detailed parameter 2 "Deceleration time 3" is outside the setting range.                                                                                                                                                         |                                                                                                                                                                                                                                                                                                               |  |
|                          | 956        | JOG speed limit value error                                | <ul style="list-style-type: none"> <li>The set value of the detailed parameter 2 "JOG speed limit value" is outside the setting range.</li> <li>The set range of the detailed parameter 2 "JOG speed limit value" exceeds the speed limit.</li> </ul> |                                                                                                                                                                                                                                                                                                               |  |
|                          | 957        | JOG acceleration time selection setting error              | The set value of the detailed parameter 2 "JOG acceleration time selection setting" is outside the setting range.                                                                                                                                     |                                                                                                                                                                                                                                                                                                               |  |
|                          | 958        | JOG deceleration time selection setting error              | The set value of the detailed parameter 2 "JOG deceleration time selection setting" is outside the setting range.                                                                                                                                     |                                                                                                                                                                                                                                                                                                               |  |
|                          | 959        | Acceleration/ deceleration process selection setting error | The set value of the detailed parameter 2 "Acceleration/deceleration process selection setting" is outside the setting range.                                                                                                                         |                                                                                                                                                                                                                                                                                                               |  |
|                          | 960        | S-curve ratio setting error                                | The set value of the detailed parameter 2 "S-curve ratio" is outside the setting range.                                                                                                                                                               |                                                                                                                                                                                                                                                                                                               |  |
|                          | 961        | Illegal sudden stop deceleration time                      | The set value of the detailed parameter 2 "Sudden stop deceleration time" is outside the setting range.                                                                                                                                               |                                                                                                                                                                                                                                                                                                               |  |
|                          | 962        | Stop group 1 sudden stop selection error                   | The set value of the detailed parameter 2 "Stop group 1 sudden stop selection" is outside the setting range.                                                                                                                                          |                                                                                                                                                                                                                                                                                                               |  |
|                          | 963        | Stop group 2 sudden stop selection error                   | The set value of the detailed parameter 2 "Stop group 2 sudden stop selection" is outside the setting range.                                                                                                                                          |                                                                                                                                                                                                                                                                                                               |  |
|                          | 964        | Stop group 3 sudden stop selection error                   | The set value of the detailed parameter 2 "Stop group 3 sudden stop selection" is outside the setting range.                                                                                                                                          |                                                                                                                                                                                                                                                                                                               |  |
|                          | 966        | Outside allowance circular interpolation error width       | The set value of the detailed parameter 2 "Allowance circular interpolation error width" is outside the setting range.                                                                                                                                |                                                                                                                                                                                                                                                                                                               |  |
|                          | 967        | External command function selection error                  | The set value of the detailed parameter 2 "External command function selection" is outside the setting range.                                                                                                                                         |                                                                                                                                                                                                                                                                                                               |  |
|                          | 980        | OPR method error                                           | The set value of the OPR basic parameter "OPR method" is outside the setting range.                                                                                                                                                                   | The QD75 READY signal (X0) is not turned ON.                                                                                                                                                                                                                                                                  |  |

| Related buffer memory address |            |            |            |                                                                                                                                                                                                              | Set range<br>(Setting with sequence program) | Remedy                                                                                                                                                                                                                |
|-------------------------------|------------|------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1                        | Axis 2     | Axis 3     | Axis 4     |                                                                                                                                                                                                              |                                              |                                                                                                                                                                                                                       |
| 44<br>45                      | 194<br>195 | 344<br>345 | 494<br>495 |                                                                                                                                                                                                              | 1 to 8388608                                 | With the setting brought into the setting range, turn the PLC READY signal [Y0] from OFF to ON.                                                                                                                       |
| 46<br>47                      | 196<br>197 | 346<br>347 | 496<br>497 |                                                                                                                                                                                                              | 1 to 8388608                                 |                                                                                                                                                                                                                       |
| 48<br>49                      | 198<br>199 | 348<br>349 | 498<br>499 | QD75P□N/QD75D□N:<br>1 to 4000000 [pulse/s]<br>1 to 2000000000<br>[ $\times 10^{-2}$ mm/min or others]<br>QD75P□/QD75D□:<br>1 to 1000000 [pulse/s]<br>1 to 2000000000<br>[ $\times 10^{-2}$ mm/min or others] |                                              | <ul style="list-style-type: none"> <li>• With the setting brought into the setting range, turn the PLC READY signal [Y0] from OFF to ON.</li> <li>• Bring the setting into the speed limit value or below.</li> </ul> |
| 50                            | 200        | 350        | 500        |                                                                                                                                                                                                              | 0, 1, 2, 3                                   | With the setting brought into the setting range, turn the PLC READY signal [Y0] from OFF to ON.                                                                                                                       |
| 51                            | 201        | 351        | 501        |                                                                                                                                                                                                              | 0, 1, 2, 3                                   |                                                                                                                                                                                                                       |
| 52                            | 202        | 352        | 502        |                                                                                                                                                                                                              | 0, 1                                         |                                                                                                                                                                                                                       |
| 53                            | 203        | 353        | 503        |                                                                                                                                                                                                              | 1 to 100                                     |                                                                                                                                                                                                                       |
| 54<br>55                      | 204<br>205 | 354<br>355 | 504<br>505 |                                                                                                                                                                                                              | 1 to 8388608                                 |                                                                                                                                                                                                                       |
| 56                            | 206        | 356        | 506        |                                                                                                                                                                                                              | 0, 1                                         |                                                                                                                                                                                                                       |
| 57                            | 207        | 357        | 507        |                                                                                                                                                                                                              | 0, 1                                         |                                                                                                                                                                                                                       |
| 58                            | 208        | 358        | 508        |                                                                                                                                                                                                              | 0, 1                                         |                                                                                                                                                                                                                       |
| 60<br>61                      | 210<br>211 | 360<br>361 | 510<br>511 |                                                                                                                                                                                                              | 0 to 10000                                   |                                                                                                                                                                                                                       |
| 62                            | 212        | 362        | 512        |                                                                                                                                                                                                              | 0, 1, 2, 3                                   |                                                                                                                                                                                                                       |
| 70                            | 220        | 370        | 520        |                                                                                                                                                                                                              | 0, 1, 2, 3, 4, 5                             |                                                                                                                                                                                                                       |

| Classification of errors | Error code | Error name                                                    | Error                                                                                                                                                                                                                                                                                                                                         | Operation status at error occurrence         |  |
|--------------------------|------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|--|
| Parameter                | 981        | OPR direction error                                           | The set value of the OPR basic parameter "OPR direction" is outside the setting range.                                                                                                                                                                                                                                                        | The QD75 READY signal (X0) is not turned ON. |  |
|                          | 982        | OP address setting error                                      | The set value of the OPR basic parameter "OP address" is outside the setting range.                                                                                                                                                                                                                                                           |                                              |  |
|                          | 983        | OPR speed error                                               | <ul style="list-style-type: none"> <li>The set value of the OPR basic parameter "OPR speed" is outside the setting range.</li> <li>The set value of the OPR basic parameter "OPR speed" is smaller than the bias speed at start.</li> </ul>                                                                                                   |                                              |  |
|                          | 984        | Creep speed error                                             | <ul style="list-style-type: none"> <li>The set value of the OPR basic parameter "Creep speed" is outside the setting range.</li> <li>The set value of the OPR basic parameter "Creep speed" is larger than the OPR speed.</li> <li>The set value of the OPR basic parameter "Creep speed" is smaller than the bias speed at start.</li> </ul> |                                              |  |
|                          | 985        | OPR retry error                                               | The set value of the OPR basic parameter "OPR retry" is outside the setting range.                                                                                                                                                                                                                                                            |                                              |  |
|                          | 991        | Setting for the movement amount after near-point dog ON error | The set value of the OPR detailed parameter "Setting for the movement amount after near-point dog ON" is outside the setting range.                                                                                                                                                                                                           |                                              |  |
|                          | 992        | OPR acceleration time selection error                         | The set value of the OPR detailed parameter "OPR acceleration time selection" is outside the setting range.                                                                                                                                                                                                                                   |                                              |  |
|                          | 993        | OPR deceleration time selection error                         | The set value of the OPR detailed parameter "OPR deceleration time selection" is outside the setting range.                                                                                                                                                                                                                                   |                                              |  |
|                          | 995        | OPR torque limit value error                                  | <ul style="list-style-type: none"> <li>The set value of the OPR detailed parameter "OPR torque limit value" is outside the setting range.</li> <li>The OPR detailed parameter "OPR torque limit value" has exceeded the detailed parameter 1 "Torque limit setting value".</li> </ul>                                                         |                                              |  |
|                          | 996        | Deviation counter clear signal output time setting error      | The set value of the OPR detailed parameter "Deviation counter clear signal output time" is outside the setting range.                                                                                                                                                                                                                        |                                              |  |
|                          | 997        | Speed designation during OP shift error                       | The set value of the OPR detailed parameter "Speed designation during OP shift" is outside the setting range.                                                                                                                                                                                                                                 |                                              |  |

| Related buffer memory address |            |            |            |  | Set range<br>(Setting with sequence program)                                                                                                                                                                 | Remedy                                                                                                                                                                                                                                          |
|-------------------------------|------------|------------|------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1                        | Axis 2     | Axis 3     | Axis 4     |  |                                                                                                                                                                                                              |                                                                                                                                                                                                                                                 |
| 71                            | 221        | 371        | 521        |  | 0, 1                                                                                                                                                                                                         | With the setting brought into the setting range, turn the PLC READY signal [Y0] from OFF to ON.                                                                                                                                                 |
| 72<br>73                      | 222<br>223 | 372<br>373 | 522<br>523 |  | <ul style="list-style-type: none"> <li>• [mm] [inch] [pulse]<br/>-2147483648 to 2147483647</li> <li>• [degree] 0 to 35999999</li> </ul>                                                                      |                                                                                                                                                                                                                                                 |
| 74<br>75                      | 224<br>225 | 374<br>375 | 524<br>525 |  | QD75P□N/QD75D□N:<br>1 to 4000000 [pulse/s]<br>1 to 2000000000<br>[ $\times 10^{-2}$ mm/min or others]<br>QD75P□/QD75D□:<br>1 to 1000000 [pulse/s]<br>1 to 2000000000<br>[ $\times 10^{-2}$ mm/min or others] | <ul style="list-style-type: none"> <li>• Bring the setting into the setting range.</li> <li>• Set the speed to the bias speed at start or higher.<br/>(Refer to Section 5.2.5)</li> </ul>                                                       |
| 76<br>77                      | 226<br>227 | 376<br>377 | 526<br>527 |  |                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Bring the setting into the setting range.</li> <li>• Set the speed to that below the OPR speed.</li> <li>• Set the value to the bias speed at start or higher.<br/>(Refer to Section 5.2.5)</li> </ul> |
| 78                            | 228        | 378        | 528        |  | 0, 1                                                                                                                                                                                                         | With the setting brought into the setting range, turn the PLC READY signal [Y0] from OFF to ON.                                                                                                                                                 |
| 80<br>81                      | 230<br>231 | 380<br>381 | 530<br>531 |  | 0 to 2147483647                                                                                                                                                                                              |                                                                                                                                                                                                                                                 |
| 82                            | 232        | 382        | 532        |  | 0, 1, 2, 3                                                                                                                                                                                                   |                                                                                                                                                                                                                                                 |
| 83                            | 233        | 383        | 533        |  | 0, 1, 2, 3                                                                                                                                                                                                   |                                                                                                                                                                                                                                                 |
| 86                            | 236        | 386        | 536        |  | 1 to 300                                                                                                                                                                                                     |                                                                                                                                                                                                                                                 |
| 87                            | 237        | 387        | 537        |  | 1 to 65535                                                                                                                                                                                                   |                                                                                                                                                                                                                                                 |
| 88                            | 238        | 388        | 538        |  | 0, 1                                                                                                                                                                                                         |                                                                                                                                                                                                                                                 |

## 15.4 List of warnings

The following table shows the warning details and remedies to be taken when a warning occurs.

| Classification of warnings | Warning code | Warning name                    | Warning                                                                             | Operation status at warning occurrence                                                                                                                   |  |
|----------------------------|--------------|---------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| —                          | 000          | (Normal status)                 | —                                                                                   | —                                                                                                                                                        |  |
| Common                     | 100          | Start during operation          | The start request is issued while the axis is BUSY.                                 | Continue the operation.                                                                                                                                  |  |
|                            | 102          | Deviation counter clear request | The deviation counter clear request is issued while the axis is BUSY.               | The deviation counter clear request is ignored.                                                                                                          |  |
|                            | 104          | Restart not possible            | The restart command is issued when the axis operation status is not "Stopped".      | Continue the operation.                                                                                                                                  |  |
|                            | 109          | Teaching in BUSY                | The teaching request is issued while the axis is BUSY.                              | The warning is issued for the axis designated at the time of the teaching request.                                                                       |  |
|                            | 110          | Less than minimum speed         | The overridden speed becomes "0".                                                   | The system is controlled with the currently executing unit of 1.                                                                                         |  |
|                            | 111          | In PLC READY                    | The request for writing to the flash ROM is issued when the PLC READY is turned ON. | The warning for axis 1 is issued.                                                                                                                        |  |
|                            | 112          | Illegal override value          | A value other than 1 to 300 is set for the override value.                          | <ul style="list-style-type: none"> <li>Controlled at a setting value of 0 : 100.</li> <li>Controlled at a setting value of 301 or over : 300.</li> </ul> |  |
|                            | 113          | Outside new torque value range  | The new torque value exceeds the torque limit setting value.                        | Set a value smaller than the torque limit setting value to the new torque value.                                                                         |  |
|                            | 114          | Below bias speed                | The command speed is below the bias speed at start.                                 | Operate by the bias speed at start.                                                                                                                      |  |



| Related buffer memory address                                           |              |              |              |  | Set range<br>(Setting with sequence program)                                                                                                                                                                                          | Remedy                                                                                                                                           |
|-------------------------------------------------------------------------|--------------|--------------|--------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1                                                                  | Axis 2       | Axis 3       | Axis 4       |  |                                                                                                                                                                                                                                       |                                                                                                                                                  |
| —                                                                       | —            | —            | —            |  | —                                                                                                                                                                                                                                     | —                                                                                                                                                |
| —                                                                       | —            | —            | —            |  | —                                                                                                                                                                                                                                     | Normalize the start request ON timing.                                                                                                           |
| —                                                                       | —            | —            | —            |  | —                                                                                                                                                                                                                                     | Do not carry out the deviation counter clear while the axis is running. (Refer to CHAPTER 8)                                                     |
| 1503                                                                    | 1603         | 1703         | 1803         |  | <Restart command><br>1: Restart                                                                                                                                                                                                       | Normalize the start request ON timing.<br>(Refer to Section 6.5.5)<br>(Do not issue the restart command when the axis operation is not stopped.) |
| 1548<br>1549                                                            | 1648<br>1649 | 1748<br>1749 | 1848<br>1849 |  | <Teaching data selection><br>0, 1<br><Teaching positioning data No.><br>1 to 600                                                                                                                                                      | Carry out the teaching request when the axis is not BUSY. (Refer to Section 12.7.4)                                                              |
| 1513                                                                    | 1613         | 1713         | 1813         |  | <Positioning operation speed override><br>1 to 300                                                                                                                                                                                    | Prevent the overridden speed from being reduced to 0. (Refer to Section 12.5.2)                                                                  |
| 1900<br>1901                                                            |              |              |              |  | <Flash ROM write request><br>1: Flash ROM write request<br><Parameter initialization request><br>1: Parameter initialization request                                                                                                  | Issue a write request when PLC READY signal (Y0) is OFF.                                                                                         |
| 1513                                                                    | 1613         | 1713         | 1813         |  | <Positioning operation speed override><br>1 to 300                                                                                                                                                                                    | Set a value within the setting range.                                                                                                            |
| 1525                                                                    | 1625         | 1725         | 1825         |  | <New torque value><br>1 to [Torque limit set value]                                                                                                                                                                                   | Set a value smaller than the torque limit setting value to the new torque value.                                                                 |
| 26                                                                      | 176          | 326          | 476          |  | <Torque limit set value><br>1 to 500                                                                                                                                                                                                  |                                                                                                                                                  |
| Refer to Section 5.3<br>"List of positioning data"<br>for command speed |              |              |              |  | <Command speed><br>QD75P□N/QD75D□N:<br>1 to 4000000 [pulse/s]<br>1 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others]<br>QD75P□/QD75D□:<br>1 to 1000000 [pulse/s]<br>1 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others]       | Re-set the command speed/bias speed at start so that the command speed is equal to or larger than the bias speed at start.                       |
|                                                                         |              |              |              |  | <Bias speed at start><br>QD75P□N/QD75D□N:<br>0 to 4000000 [pulse/s]<br>0 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others]<br>QD75P□/QD75D□:<br>0 to 1000000 [pulse/s]<br>0 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others] |                                                                                                                                                  |
| Bias speed at start                                                     |              |              |              |  |                                                                                                                                                                                                                                       |                                                                                                                                                  |
| 6<br>7                                                                  | 156<br>157   | 306<br>307   | 456<br>457   |  |                                                                                                                                                                                                                                       |                                                                                                                                                  |

| Classification of warnings | Warning code | Warning name                                             | Warning                                                                                                                                                                                                                            | Operation status at warning occurrence                                                                                                                                                                                                                                                             |  |
|----------------------------|--------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| JOG                        | 300          | Speed change during deceleration                         | The speed change request is issued during deceleration stop with JOG start signal OFF.                                                                                                                                             | The speed change is not carried out.                                                                                                                                                                                                                                                               |  |
|                            | 301          | JOG speed limit value                                    | <ul style="list-style-type: none"> <li>• The JOG speed exceeds the JOG speed limit value at the JOG start.</li> <li>• The new speed value exceeds the JOG speed limit value when the speed is changed during operation.</li> </ul> | <ul style="list-style-type: none"> <li>• When the speed exceeds the JOG speed limit, the JOG operation is carried out with the JOG speed limit value.</li> <li>• While the speed is limited by the JOG speed limit value, the "Speed limiting flag" is turned ON.</li> </ul>                       |  |
| Manual pulse generator     | 401          | Outside manual pulse generator input magnification range | <p>The manual pulse generator 1 pulse input magnification is set as follows.</p> <ul style="list-style-type: none"> <li>• 0 or 1001 or higher (QD75P□N/QD75D□N)</li> <li>• 0 or 101 or higher (QD75P□/QD75D□)</li> </ul>           | <ul style="list-style-type: none"> <li>• When input magnification is set at 1001 or higher: Re-set to 1000. (QD75P□N/QD75D□N)</li> <li>• When input magnification is set at 101 or higher: Re-set to 100. (QD75P□/QD75D□)</li> <li>• When input magnification is set at 0: Re-set to 1.</li> </ul> |  |

| Related buffer memory address |        |        |        |                                                                                                                                                                                                              | Set range<br>(Setting with sequence program)                                                                                                                                                                 | Remedy                                                                                   |
|-------------------------------|--------|--------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Axis 1                        | Axis 2 | Axis 3 | Axis 4 |                                                                                                                                                                                                              |                                                                                                                                                                                                              |                                                                                          |
| 1516                          | 1616   | 1716   | 1816   |                                                                                                                                                                                                              | <Speed change request><br>1: Speed change is requested                                                                                                                                                       | Do not carry out the JOG speed change during deceleration with the JOG start signal OFF. |
| New speed value               |        |        |        |                                                                                                                                                                                                              | QD75P□N/QD75D□N:<br>0 to 4000000 [pulse/s]<br>0 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others]<br>QD75P□/QD75D□:<br>0 to 1000000 [pulse/s]<br>0 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others] | Bring the set value into the setting range.                                              |
| 1514                          | 1614   | 1714   | 1814   |                                                                                                                                                                                                              |                                                                                                                                                                                                              |                                                                                          |
| 1515                          | 1615   | 1715   | 1815   |                                                                                                                                                                                                              |                                                                                                                                                                                                              |                                                                                          |
| JOG speed limit value         |        |        |        |                                                                                                                                                                                                              |                                                                                                                                                                                                              |                                                                                          |
| 48                            | 198    | 348    | 498    | QD75P□N/QD75D□N:<br>1 to 4000000 [pulse/s]<br>1 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others]<br>QD75P□/QD75D□:<br>1 to 1000000 [pulse/s]<br>1 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others] |                                                                                                                                                                                                              |                                                                                          |
| 49                            | 199    | 349    | 499    |                                                                                                                                                                                                              |                                                                                                                                                                                                              |                                                                                          |
| 1522                          | 1622   | 1722   | 1822   | <Manual pulse generator 1 pulse input magnification><br>1 to 1000 (QD75P□N/QD75D□N)<br>1 to 100 (QD75P□/QD75D□)                                                                                              | Set the manual pulse generator 1 pulse input magnification to within the setting range.                                                                                                                      |                                                                                          |
| 1523                          | 1623   | 1723   | 1823   |                                                                                                                                                                                                              |                                                                                                                                                                                                              |                                                                                          |

| Classification of warnings | Warning code | Warning name                                             | Warning                                                                                                                                                                                                                                                                           | Operation status at warning occurrence                                                                                                                                                                                                                                                                                                                                                                             |  |
|----------------------------|--------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Positioning operation      | 500          | Deceleration/stop speed change                           | The speed change request is issued during deceleration stop.                                                                                                                                                                                                                      | The speed change is not carried out.                                                                                                                                                                                                                                                                                                                                                                               |  |
|                            | 501          | Speed limit value over                                   | The new value exceeds the speed limit value when the speed is changed during operation.                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>• The speed is controlled with the speed limit value.</li> <li>• The "speed limiting flag" is turned ON.</li> </ul>                                                                                                                                                                                                                                                         |  |
|                            | 503          | M code ON signal ON start                                | The M code ON signal is turned ON when the positioning data is executed.                                                                                                                                                                                                          | Continue executing the positioning data.                                                                                                                                                                                                                                                                                                                                                                           |  |
|                            | 505          | No operation termination setting                         | In the positioning by block starting, the 50th point of the positioning start data is set to CONTINUE.                                                                                                                                                                            | The operation is terminated.                                                                                                                                                                                                                                                                                                                                                                                       |  |
|                            | 506          | FOR to NEXT nest construction                            | FOR to NEXT is nested.                                                                                                                                                                                                                                                            | The operation is continued.                                                                                                                                                                                                                                                                                                                                                                                        |  |
|                            | 508          | Speed-position switching (during acceleration) signal ON | The switching signal for speed-position switching control (INC mode) is turned ON during acceleration.                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
|                            | 509          | Insufficient remaining distance                          | <ul style="list-style-type: none"> <li>• At a continuous operation interrupt request, the distance required deceleration stop is not long enough.</li> <li>• At a speed change request, the remaining distance is shorter than the distance required for speed change.</li> </ul> | <ul style="list-style-type: none"> <li>• When a command speed is changed:<br/>Change to a value as near a new speed value as possible.</li> <li>• When a target position is changed:<br/>Adjust the speed to a value as near the command speed as possible, and then change to a target position.</li> </ul> <p>(When the operation pattern is a continuous path control, ignore the operations stated above.)</p> |  |

| Related buffer memory address                      |              |              |              |                                                                                                                                                                                                              | Set range<br>(Setting with sequence program)                                                                                                                                                                 | Remedy                                                                                                                                              |
|----------------------------------------------------|--------------|--------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis 1                                             | Axis 2       | Axis 3       | Axis 4       |                                                                                                                                                                                                              |                                                                                                                                                                                                              |                                                                                                                                                     |
| 1516                                               | 1616         | 1716         | 1816         |                                                                                                                                                                                                              | <Speed change request><br>1: Speed change is requested                                                                                                                                                       | Do not carry out the speed change during deceleration with a stop command, during stoppage, or during automatic deceleration with position control. |
| New speed value                                    |              |              |              |                                                                                                                                                                                                              | QD75P□N/QD75D□N:<br>0 to 4000000 [pulse/s]<br>0 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others]<br>QD75P□/QD75D□:<br>0 to 1000000 [pulse/s]<br>0 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others] | Set the new speed value to a range of 0 to "speed limit value".                                                                                     |
| 1514<br>1515                                       | 1614<br>1615 | 1714<br>1715 | 1814<br>1815 |                                                                                                                                                                                                              |                                                                                                                                                                                                              |                                                                                                                                                     |
| Speed limit value                                  |              |              |              |                                                                                                                                                                                                              |                                                                                                                                                                                                              |                                                                                                                                                     |
| 10<br>11                                           | 160<br>161   | 310<br>311   | 460<br>461   | QD75P□N/QD75D□N:<br>1 to 4000000 [pulse/s]<br>1 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others]<br>QD75P□/QD75D□:<br>1 to 1000000 [pulse/s]<br>1 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others] |                                                                                                                                                                                                              |                                                                                                                                                     |
| 1504                                               | 1604         | 1704         | 1804         |                                                                                                                                                                                                              | <M code OFF request><br>1: M code ON signal is turned OFF                                                                                                                                                    | Normalize the ON and OFF timings of the "M code OFF request". (Refer to Section 12.7.3)                                                             |
| Refer to Section 5.3<br>"List of positioning data" |              |              |              |                                                                                                                                                                                                              | <Operation pattern><br>00: Positioning complete<br>01: Continuous positioning control<br>11: Continuous path control                                                                                         | Set the operation termination to the 50th point.<br>(Refer to CHAPTER 10)                                                                           |
| —                                                  | —            | —            | —            |                                                                                                                                                                                                              | —                                                                                                                                                                                                            | Make 1 nest construction for FOR to NEXT.<br>(Refer to Section 10.3.8)                                                                              |
| —                                                  | —            | —            | —            |                                                                                                                                                                                                              | —                                                                                                                                                                                                            | Do not turn ON the speed-position switching signal during acceleration.<br>(Refer to Section 9.2.16)                                                |
| —                                                  | —            | —            | —            |                                                                                                                                                                                                              | —                                                                                                                                                                                                            | Give a request at the position where there is an enough remaining distance.                                                                         |

| Classification of warnings | Warning code | Warning name                        | Warning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Operation status at warning occurrence                                                                                                                         |  |
|----------------------------|--------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Positioning operation      | 511          | Step not possible                   | Code 1 is set for the step start information when the step is outside standby.                                                                                                                                                                                                                                                                                                                                                                                                                                                     | The step will not start.                                                                                                                                       |  |
|                            | 512          | Illegal external command function   | The detailed parameter 2 "External command function selection" setting range is exceeded.                                                                                                                                                                                                                                                                                                                                                                                                                                          | Even if the external command signal is turned ON, the system will not perform anything.                                                                        |  |
|                            | 513          | Insufficient movement amount        | The movement amount is not large enough for automatic deceleration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                | The system stops immediately after it reaches the positioning address.                                                                                         |  |
|                            | 514          | Outside command speed range         | The command speed exceeds the speed limit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>• The command speed is controlled at the "speed limit value".</li> <li>• The "speed limiting flag" turns ON.</li> </ul> |  |
|                            | 516          | Illegal teaching data No.           | The positioning data No. is set outside the setting range.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Teaching is not carried out when the set value is 0 or 601 or more. (A "0" is canceled by the QD75 automatically even when a "0" or "601" or more is set.)     |  |
|                            | 517          | Illegal teaching data selection     | The teaching data selection set value is outside the setting range.                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Teaching is not carried out.                                                                                                                                   |  |
|                            | 518          | Target position change not possible | <ul style="list-style-type: none"> <li>• A target position change request was given for the control system other than ABS1 and INC1.</li> <li>• The target position change request is turned ON during continuous path control.</li> <li>• A new target position address is outside the software stroke limit range.</li> <li>• A target position change request was given during deceleration to a stop.</li> <li>• A target position change request is given while the speed change 0 flag (Md.31 Status: b10) is ON.</li> </ul> | The target position change is not carried out.                                                                                                                 |  |

| Related buffer memory address                                            |            |            |            |                                                                                | Set range<br>(Setting with sequence program)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Remedy                                                                   |
|--------------------------------------------------------------------------|------------|------------|------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Axis 1                                                                   | Axis 2     | Axis 3     | Axis 4     |                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                          |
| 1546                                                                     | 1646       | 1746       | 1846       | <Step start information><br>1: Step is continued<br>2: Re-start is carried out | Do not set a "1" to the step start information when the step is not in standby state. (Refer to Section 12.7.1)                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                          |
| 62                                                                       | 212        | 362        | 512        | <External command function selection><br>0, 1, 2, 3                            | Set the detailed parameter 2 "External command function selection" to within the setting range.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                          |
| Refer to Section 5.3<br>"List of positioning data"                       |            |            |            |                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Set a decelerating address or a movement amount to the positioning data. |
| For command speed,<br>refer to Section 5.3<br>"List of positioning data" |            |            |            |                                                                                | QD75P□N/QD75D□N:<br>1 to 4000000 [pulse/s]<br>1 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others]<br>QD75P□/QD75D□:<br>1 to 1000000 [pulse/s]<br>1 to 2000000000<br>[×10 <sup>-2</sup> mm/min or others]                                                                                                                                                                                                                                                                                                                                                         | Set the command speed to within the setting range.                       |
| Speed limit value                                                        |            |            |            |                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                          |
| 10<br>11                                                                 | 160<br>161 | 310<br>311 | 460<br>461 |                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                          |
| 1549                                                                     | 1649       | 1749       | 1849       | <Teaching positioning data No.><br>1 to 600                                    | Set the positioning data No. to within the setting range.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                          |
| 1548                                                                     | 1648       | 1748       | 1848       | <Teaching data selection><br>0, 1                                              | Set the teaching data selection set value to within the setting range.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |
| 1538                                                                     | 1638       | 1738       | 1838       | <Target position change request flag><br>1: Target position change request     | <ul style="list-style-type: none"><li>• Do not turn ON the target position change request in the following cases.</li><li>• An operating pattern "continuous path control" is used.</li><li>• A control system other than ABS1, and INC1 is used.</li><li>• During deceleration stop.</li><li>• When the speed change 0 flag (<span style="border: 1px solid black;">Md.31</span> Status: b10) is ON.</li><li>• When the target position change address is outside the stroke limit, correct the target position change address. (Refer to Section 12.5.5)</li></ul> |                                                                          |

[illegible]



# APPENDICES

|                                                                                                               |               |
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## Appendix 1 Version up of the functions

## Appendix 1.1 Comparison of functions according to function versions

The following tables list the QD75 functions compared according to function versions and the buffer memory for their additional functions.

## (1) Function comparison

| Item                                                     | Function versions   |               |       | Reference                        |
|----------------------------------------------------------|---------------------|---------------|-------|----------------------------------|
|                                                          | QD75P□N/<br>QD75D□N | QD75P□/QD75D□ |       |                                  |
|                                                          | A                   | A             | B     |                                  |
| Speed-position switching control (ABS mode)              | ○                   | ×             | ○     | Section 9.2.19                   |
| Pre-reading start function                               | ○                   | ×             | ○     | Section 12.7.7                   |
| External I/O signal monitor function                     | ○ * 1               | ×             | ○ * 2 | Section 13.5                     |
| Multiple programmable controller compatible function * 3 | ○                   | ×             | ○     | —                                |
| Deceleration start flag function                         | ○                   | ×             | ○ * 4 | Section 12.7.8                   |
| Stop command processing for deceleration stop function   | ○                   | ×             | ○ * 5 | Section 12.7.9                   |
| Output timing selection of near pass control             | ○ * 6               | ×             | ○ * 7 | Section 12.3.4                   |
| 3-axis helical interpolation control                     | ○ * 8               | ×             | ×     | Section 9.2.12<br>Section 9.2.13 |
| Positioning option                                       | ○ * 8               | ×             | ×     | Section 5.3                      |

○:Available × :Not available

- \*1: For the QD75P□N/QD75D□N, external I/O signals can not be monitored on GX Developer. Use the system monitor of GX Works2.
- \*2: Usable on GX Developer of SW6D5C-GPPW-E or later.
- \*3: Compatible with QCPU of function version B.
- \*4: Supported by the module whose first five digits of SERIAL No. are "03042" or later.
- \*5: Supported by the module whose first five digits of SERIAL No. are "05072" or later.
- \*6: Supported by the module whose first five digits of SERIAL No. are "14032" or later.
- \*7: Supported by the module whose first five digits of SERIAL No. are "14012" or later.
- \*8: Supported by the module whose first five digits of the serial number are "17102" or later. GX Configurator-QP does not support these functions. Only GX Works2 (Version 1.540N or later) supports the functions.

## (2) Additional buffer memory

| Buffer memory addresses |             |              |              | Name                                                    | Used function                                          |
|-------------------------|-------------|--------------|--------------|---------------------------------------------------------|--------------------------------------------------------|
| Axis 1                  | Axis 2      | Axis 3       | Axis 4       |                                                         |                                                        |
| 34                      | 184         | 334          | 484          | Speed-position function selection                       | Speed-position switching control                       |
| 140                     |             |              |              | Positioning option valid/invalid setting                | Positioning option                                     |
| 899                     | 999         | 1099         | 1199         | Deceleration start flag                                 | Deceleration start flag function                       |
| 1905                    |             |              |              | Deceleration start flag valid                           |                                                        |
| 1907                    |             |              |              | Stop command processing for deceleration stop selection | Stop command processing for deceleration stop function |
| 2003<br>* 1             | 8003<br>* 1 | 14003<br>* 1 | 20003<br>* 1 | M code ON signal output timing                          | Positioning option                                     |
|                         |             |              |              | ABS direction in degrees                                |                                                        |
|                         |             |              |              | Interpolation speed designation method                  |                                                        |

\*1: The listed buffer memory addresses are those of the positioning data No. 1 of each axis.  
For the buffer memory address of other positioning data No., refer to Appendix 3.

## (3) Additional devices

| Device No. | Name   |                            | Used function              |
|------------|--------|----------------------------|----------------------------|
| Y14        | Axis 1 | Execution prohibition flag | Pre-reading start function |
| Y15        | Axis 2 |                            |                            |
| Y16        | Axis 3 |                            |                            |
| Y17        | Axis 4 |                            |                            |

## Appendix 1.2 Precautions for the replacement of the QD75P□/QD75D□ with the QD75P□N/QD75D□N

### (1) Specification differences between the QD75P□N/QD75D□N and the QD75P□/QD75D□

The following table shows the differences. The specifications not listed below are the same for the both models.

| Item                                               |                                                      | QD75P□N/QD75D□N                                                                                                | QD75P□/QD75D□                                                                                            |
|----------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Max. output pulse                                  |                                                      | 4Mpulse/s<br>(QD75D□N)                                                                                         | 1Mpulse/s<br>(Q75D□)                                                                                     |
| Speed command (pulse unit)                         |                                                      | 1 to 4000000pulse/s                                                                                            | 1 to 1000000pulse/s                                                                                      |
| Starting time (1-axis linear control)              |                                                      | Trapezoidal acceleration/<br>deceleration: 1.5ms<br>S-curve acceleration/<br>deceleration: 1.6ms               | Trapezoidal acceleration/<br>deceleration: 6ms<br>S-curve acceleration/<br>deceleration: 6.5ms           |
| Monitor data<br>refreshing cycle                   | Current feed value                                   | 0.9ms                                                                                                          | 1.8ms                                                                                                    |
|                                                    | Other axis monitors<br>(except external I/O signals) | 0.9ms                                                                                                          | 56.8ms                                                                                                   |
| Manual pulse generator 1 pulse input magnification |                                                      | 1 to 1000                                                                                                      | 1 to 100                                                                                                 |
| ON<br>voltage/current of<br>external input         | External command signal                              | 19VDC or more/2.7mA or more                                                                                    | 17.5VDC or more/<br>3.5mA or more                                                                        |
| OFF<br>voltage/current of<br>external input        | External command signal                              | 7VDC or less/<br>0.8mA or less                                                                                 | 7VDC or less/<br>1.7mA or less                                                                           |
| Input resistance of<br>external input              | Zero signal (5VDC)                                   | Approx. 620Ω                                                                                                   | Approx. 300Ω                                                                                             |
|                                                    | Manual pulse generator A/B<br>phase                  | Approx. 1.1kΩ                                                                                                  | Approx. 1.5kΩ                                                                                            |
|                                                    | External command signal                              | Approx. 7.7kΩ                                                                                                  | Approx. 4.3kΩ                                                                                            |
| Internal current consumption (5VDC)                |                                                      | QD75P1N: 0.29A<br>QD75P2N: 0.30A<br>QD75P4N: 0.36A<br>QD75D1N: 0.43A<br>QD75D2N: 0.45A<br>QD75D4N: 0.66A       | QD75P1: 0.40A<br>QD75P2: 0.46A<br>QD75P4: 0.58A<br>QD75D1: 0.52A<br>QD75D2: 0.56A<br>QD75D4: 0.82A       |
| Weight                                             |                                                      | QD75P1N: 0.14kg<br>QD75P2N: 0.14kg<br>QD75P4N: 0.16kg<br>QD75D1N: 0.15kg<br>QD75D2N: 0.15kg<br>QD75D4N: 0.16kg | QD75P1: 0.15kg<br>QD75P2: 0.15kg<br>QD75P4: 0.16kg<br>QD75D1: 0.15kg<br>QD75D2: 0.15kg<br>QD75D4: 0.16kg |

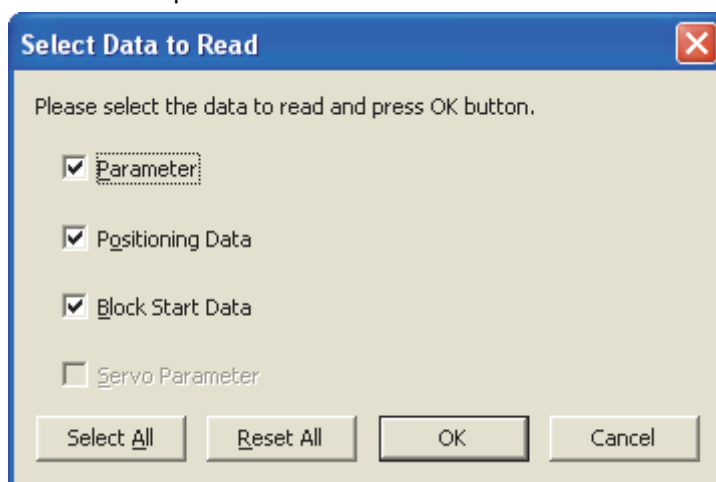
## (2) Precaution on the use of sequence programs

The QD75P□N/QD75D□N is upgraded from the QD75P□/QD75D□. Therefore, the recognized sequence programs for the QD75P□/QD75D□ can be applied to the QD75P□N/QD75D□N. Note that specifications such as time takes for start-up and data update cycle are improved. When applying a sequence program to the QD75P□N/QD75D□N, modify the sequence program if necessary, checking the processing timing.

## (3) Transferring the set data of the QD75P□/QD75D□ using GX Works2

When GX Works2 is used, the set data of the QD75P□/QD75D□ can be transferred to the QD75P□N/QD75D□N in the following procedure.

- (a) Saving the set data of the QD75P□/QD75D□ from "Save the Positioning Module Data..."
  - 1) In the project view, select the QD75P□/QD75D□ from where the set data is transferred.
  - 2) Go to [Project] → [Intelligent Function Module] → [Save the Positioning Module Data...].
  - 3) Input the file name, and save the set data.
- (b) Reading the set data to the QD75P□N/QD75D□N from "Read from the Positioning Module Data..."
  - 1) In the project view, select the QD75P□N/QD75D□N to where the saved data is transferred.
  - 2) Go to [Project] → [Intelligent Function Module] → [Read from the Positioning Module Data...].
  - 3) Select the name of the file saved in step (a), and open it. The following window opens.



- 4) Check the data to read and click . The set data is read to the QD75P□N/QD75D□N.

## (4) Precaution on the use of GX Configurator-QP

To use the QD75P□N/QD75D□N with GX Configurator-QP, select the QD75P□/QD75D□ in "Select module type". The QD75P□N/QD75D□N can be used in the same manner as the QD75P□/QD75D□. Note that a speed exceeding 1000000pulse/s cannot be set in the following items when "Pulse" is set in "[Pr.1] Unit setting"). To set a value outside a setting range in GX Configurator-QP, set it through a sequence program or GX Works2 of the version 1.64S or later.

| Setting item                           | Setting range in<br>GX Configurator-QP | Setting range in<br>GX Works2 and sequence programs |
|----------------------------------------|----------------------------------------|-----------------------------------------------------|
| <div>Pr.7</div> Bias speed at start    | 0 to 1000000(pulse/s)                  | 0 to 4000000(pulse/s)                               |
| <div>Pr.8</div> Speed limit value      | [QD75P□N]<br>1 to 200000(pulse/s)      | 1 to 4000000(pulse/s)                               |
| <div>Pr.31</div> JOG speed limit value | [QD75D□N]<br>1 to 1000000(pulse/s)     |                                                     |
| <div>Pr.46</div> OPR speed             | 1 to 1000000(pulse/s)                  |                                                     |
| <div>Pr.47</div> Creep speed           |                                        |                                                     |
| <div>Da.8</div> Command speed          |                                        |                                                     |

## Appendix 2 Format sheets

## Appendix 2.1 Positioning Module operation chart

[illegible]

Axis address  
mm, inch, degree, pulse

Axis address  
mm, inch, degree, pulse

Axis address  
mm. inch. degree. pulse

Axis address  
mm, inch, degree, pulse



# MEMO

[illegible]

## Appendix 2.2 Parameter setting value entry table

| Item                  |                                                   | Setting range                                                                                                                                                                                                                  |                                                    |                                                      |                                                                                    |
|-----------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------|
|                       |                                                   | mm                                                                                                                                                                                                                             | inch                                               | degree                                               | pulse                                                                              |
| Basic parameters 1    | Pr.1 Unit setting                                 | 0                                                                                                                                                                                                                              | 1                                                  | 2                                                    | 3                                                                                  |
|                       | Pr.2 No. of pulses per rotation (Ap)              | 1 to 65535 pulse                                                                                                                                                                                                               |                                                    |                                                      |                                                                                    |
|                       | Pr.3 Movement amount per rotation (Al)            | 1 to 65535<br>$\times 10^{-1}\mu\text{m}$                                                                                                                                                                                      | 1 to 65535<br>$\times 10^{-5}\text{inch}$          | 1 to 65535<br>$\times 10^{-5}\text{degree}$          | 1 to 65535<br>pulse                                                                |
|                       | Pr.4 Unit magnification (Am)                      | 1: 1-fold, 10: 10-fold, 100: 100-fold, 1000: 1000-fold                                                                                                                                                                         |                                                    |                                                      |                                                                                    |
|                       | Pr.5 Pulse output mode                            | 0: PULSE/SIGN mode, 1: CW/CCW mode,<br>2: A phase/B phase (multiple of 4), 3: A phase/B phase (multiple of 1)                                                                                                                  |                                                    |                                                      |                                                                                    |
|                       | Pr.6 Rotation direction setting                   | 0: Current value increment with forward run pulse output<br>1: Current value increment with reverse run pulse output                                                                                                           |                                                    |                                                      |                                                                                    |
|                       | Pr.7 Bias speed at start                          | 0 to 2000000000<br>$\times 10^{-2}\text{mm/min}$                                                                                                                                                                               | 0 to 2000000000<br>$\times 10^{-3}\text{inch/min}$ | 0 to 2000000000<br>$\times 10^{-3}\text{degree/min}$ | QD75P□N/QD75D□N:<br>0 to 4000000 pulse/s<br>QD75P□/QD75D□:<br>0 to 1000000 pulse/s |
| Basic parameters 2    | Pr.8 Speed limit value                            | 1 to 2000000000<br>$\times 10^{-2}\text{mm/min}$                                                                                                                                                                               | 1 to 2000000000<br>$\times 10^{-3}\text{mm/min}$   | 1 to 2000000000<br>$\times 10^{-3}\text{degree/min}$ | QD75P□N/QD75D□N:<br>1 to 4000000 pulse/s<br>QD75P□/QD75D□:<br>1 to 1000000 pulse/s |
|                       | Pr.9 Acceleration time 0                          | 1 to 8388608ms                                                                                                                                                                                                                 |                                                    |                                                      |                                                                                    |
|                       | Pr.10 Deceleration time 0                         |                                                                                                                                                                                                                                |                                                    |                                                      |                                                                                    |
| Detailed parameters 1 | Pr.11 Backlash compensation amount                | 0 to 65535<br>$\times 10^{-1}\mu\text{m}$                                                                                                                                                                                      | 0 to 65535<br>$\times 10^{-5}\text{inch}$          | 0 to 65535<br>$\times 10^{-5}\text{degree}$          | 0 to 65535<br>pulse                                                                |
|                       | Pr.12 Software stroke limit upper limit value     | -2147483648 to 2147483647                                                                                                                                                                                                      | -2147483648 to 2147483647                          | 0 to 35999999                                        | -2147483648 to 2147483647<br>pulse                                                 |
|                       | Pr.13 Software stroke limit lower limit value     | $\times 10^{-1}\mu\text{m}$                                                                                                                                                                                                    | $\times 10^{-5}\text{inch}$                        | $\times 10^{-5}\text{degree}$                        |                                                                                    |
|                       | Pr.14 Software stroke limit selection             | 0: Apply software stroke limit on current feed value<br>1: Apply software stroke limit on machine feed value                                                                                                                   |                                                    |                                                      |                                                                                    |
|                       | Pr.15 Software stroke limit valid/invalid setting | 0: Software stroke limit valid during JOG operation, inching operation, and manual pulse generator operation<br>1: Software stroke limit invalid during JOG operation, inching operation, and manual pulse generator operation |                                                    |                                                      |                                                                                    |
|                       | Pr.16 Command in-position width                   | 1 to 2147483647<br>$\times 10^{-1}\mu\text{m}$                                                                                                                                                                                 | 1 to 2147483647<br>$\times 10^{-5}\text{inch}$     | 1 to 2147483647<br>$\times 10^{-5}\text{degree}$     | 1 to 2147483647<br>pulse                                                           |
|                       | Pr.17 Torque limit setting value                  | 1 to 500%                                                                                                                                                                                                                      |                                                    |                                                      |                                                                                    |
|                       | Pr.18 M code ON signal output timing              | 0: WITH mode, 1: AFTER mode                                                                                                                                                                                                    |                                                    |                                                      |                                                                                    |
|                       | Pr.19 Speed switching mode                        | 0: Standard speed switching mode<br>1: Front-loading speed switching mode                                                                                                                                                      |                                                    |                                                      |                                                                                    |
|                       | Pr.20 Interpolation speed designation method      | 0: Composite speed, 1: Reference axis speed                                                                                                                                                                                    |                                                    |                                                      |                                                                                    |
|                       | Pr.21 Current feed value during speed control     | 0: Do not update current feed value, 1: Update current feed value<br>2: Clear current feed value                                                                                                                               |                                                    |                                                      |                                                                                    |

|  | Initial value | Axis 1 | Axis 2 | Axis 3 | Axis 4 | Remarks |
|--|---------------|--------|--------|--------|--------|---------|
|  | 3             |        |        |        |        |         |
|  | 20000         |        |        |        |        |         |
|  | 20000         |        |        |        |        |         |
|  | 1             |        |        |        |        |         |
|  | 1             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 20000         |        |        |        |        |         |
|  | 1000          |        |        |        |        |         |
|  | 1000          |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 2147483647    |        |        |        |        |         |
|  | -2147483648   |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 100           |        |        |        |        |         |
|  | 300           |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |

| Item                  |                                              | Setting range                               |                                                                                                                                |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|-----------------------|----------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------|---------------|------------------------------|-----------------------------------------------------------------------------------------------------|
|                       |                                              | mm                                          |                                                                                                                                | inch                                        |                                               | degree                                                                             |               | pulse                        |                                                                                                     |
| Detailed parameters 1 | Pr.22                                        | Input signal logic selection                | b0                                                                                                                             | Lower limit                                 | b3                                            | Stop signal                                                                        | b6            | Near-point dog signal        | Setting of each bit value<br>0: Negative logic<br>1: Positive logic<br>("0" is set for unused bit.) |
|                       |                                              |                                             | b1                                                                                                                             | Upper limit                                 | b4                                            | External command                                                                   | b7, b9 to b15 | Unused                       |                                                                                                     |
|                       |                                              |                                             | b2                                                                                                                             | Drive unit READY                            | b5                                            | Zero signal                                                                        | b8            | Manual pulse generator input |                                                                                                     |
|                       | Pr.23                                        | Output signal logic selection               | b0                                                                                                                             | Command pulse signal                        | b2                                            | Unused                                                                             | b4            | Deviation counter clear      |                                                                                                     |
|                       |                                              |                                             | b1                                                                                                                             | Unused                                      | b3                                            | Unused                                                                             | b5 to b15     | Unused                       |                                                                                                     |
|                       | Pr.24                                        | Manual pulse generator input selection      | 0: A phase/B phase multiple of 4<br>1: A phase/B phase multiple of 2<br>2: A phase/B phase multiple of 1<br>3: PULSE/SIGN mode |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
| Pr.70                 | Positioning option valid/invalid setting     | 0: Invalid<br>1: Valid                      |                                                                                                                                |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
| Detailed parameters 2 | Pr.150                                       | Speed -position function selection          | 0: Speed-position switching control (INC mode)<br>2: Speed-position switching control (ABS mode)                               |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|                       | Pr.25                                        | Acceleration time 1                         | 1 to 8388608ms                                                                                                                 |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|                       | Pr.26                                        | Acceleration time 2                         |                                                                                                                                |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|                       | Pr.27                                        | Acceleration time 3                         |                                                                                                                                |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|                       | Pr.28                                        | Deceleration time 1                         |                                                                                                                                |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|                       | Pr.29                                        | Deceleration time 2                         |                                                                                                                                |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|                       | Pr.30                                        | Deceleration time 3                         |                                                                                                                                |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|                       | Pr.31                                        | JOG speed limit value                       | 1 to 2000000000 × 10 <sup>-2</sup> mm/min                                                                                      | 1 to 2000000000 × 10 <sup>-3</sup> inch/min | 1 to 2000000000 × 10 <sup>-3</sup> degree/min | QD75P□N/QD75D□N:<br>1 to 4000000 pulse/s<br>QD75P□/QD75D□:<br>1 to 1000000 pulse/s |               |                              |                                                                                                     |
|                       | Pr.32                                        | JOG operation acceleration time selection   | 0 to 3                                                                                                                         |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|                       | Pr.33                                        | JOG operation deceleration time selection   |                                                                                                                                |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|                       | Pr.34                                        | Acceleration/deceleration process selection | 0: Trapezoidal acceleration/deceleration process<br>1: S-curve acceleration/deceleration process                               |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|                       | Pr.35                                        | S-curve ratio                               | 1 to 100%                                                                                                                      |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|                       | Pr.36                                        | Sudden stop deceleration time               | 1 to 8388608ms                                                                                                                 |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|                       | Pr.37                                        | Stop group 1 sudden stop selection          | 0: Normal deceleration stop<br>1: Sudden stop                                                                                  |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|                       | Pr.38                                        | Stop group 2 sudden stop selection          |                                                                                                                                |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|                       | Pr.39                                        | Stop group 3 sudden stop selection          |                                                                                                                                |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
|                       | Pr.40                                        | Positioning complete signal output time     | 0 to 65535ms                                                                                                                   |                                             |                                               |                                                                                    |               |                              |                                                                                                     |
| Pr.41                 | Allowable circular interpolation error width | 1 to 100000 × 10 <sup>-1</sup> μm           |                                                                                                                                | 1 to 100000 × 10 <sup>-5</sup> inch         |                                               | 1 to 100000 × 10 <sup>-5</sup> degree                                              |               | 1 to 100000 pulse            |                                                                                                     |

|  | Initial value | Axis 1 | Axis 2 | Axis 3 | Axis 4 | Remarks |
|--|---------------|--------|--------|--------|--------|---------|
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 1000          |        |        |        |        |         |
|  | 1000          |        |        |        |        |         |
|  | 1000          |        |        |        |        |         |
|  | 1000          |        |        |        |        |         |
|  | 1000          |        |        |        |        |         |
|  | 1000          |        |        |        |        |         |
|  | 20000         |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 100           |        |        |        |        |         |
|  | 1000          |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 300           |        |        |        |        |         |
|  | 100           |        |        |        |        |         |

| Item                    |                                                               | Setting range                                                                                                                                                                                                                                                                     |                                                          |                                                            |                                          |
|-------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------|------------------------------------------|
|                         |                                                               | mm                                                                                                                                                                                                                                                                                | inch                                                     | degree                                                     | pulse                                    |
| Detailed parameters 2   | Pr.42 External command function selection                     | 0: External positioning start 1: External speed change request<br>2: Speed-position or position-speed switching request<br>3: Skip request                                                                                                                                        |                                                          |                                                            |                                          |
|                         |                                                               |                                                                                                                                                                                                                                                                                   |                                                          |                                                            |                                          |
| OPR basic parameters    | Pr.43 OPR method                                              | 0: Near-point dog method<br>1: Stopper method 1) (By dwell time elapse)<br>2: Stopper method 2) (By OP signal when stopper is hit)<br>3: Stopper method 3) (Without near-point dog method)<br>4: Count method 1) (Use zero signal)<br>5: Count method 2) (Do not use zero signal) |                                                          |                                                            |                                          |
|                         | Pr.44 OPR direction                                           | 0: Positive direction (address increment direction)<br>1: Negative direction (address decrement direction)                                                                                                                                                                        |                                                          |                                                            |                                          |
|                         | Pr.45 OP address                                              | -2147483648 to 2147483647<br>$\times 10^{-1}\mu\text{m}$                                                                                                                                                                                                                          | -2147483648 to 2147483647<br>$\times 10^{-5}\text{inch}$ | 0 to 35999999<br>$\times 10^{-5}\text{degree}$             | -2147483648 to 2147483647<br>pulse       |
|                         | Pr.46 OPR speed                                               | 1 to 2000000000<br>$\times 10^{-2}\text{mm/min}$                                                                                                                                                                                                                                  | 1 to 2000000000<br>$\times 10^{-3}\text{inch/min}$       | 1 to 2000000000<br>$\times 10^{-3}\text{degree/min}$       | QD75P□N/QD75D□N:<br>1 to 4000000 pulse/s |
|                         | Pr.47 Creep speed                                             |                                                                                                                                                                                                                                                                                   |                                                          |                                                            | QD75P□/QD75D□:<br>1 to 1000000 pulse/s   |
|                         | Pr.48 OPR retry                                               | 0: Do not retry OPR with upper/lower limit switch<br>1: Retry OPR with upper/lower limit switch                                                                                                                                                                                   |                                                          |                                                            |                                          |
| OPR detailed parameters | Pr.49 OPR dwell time                                          | 0 to 65535ms                                                                                                                                                                                                                                                                      |                                                          |                                                            |                                          |
|                         | Pr.50 Setting for the movement amount after near-point dog ON | 0 to 2147483647<br>$\times 10^{-1}\mu\text{m}$                                                                                                                                                                                                                                    | 0 to 2147483647<br>$\times 10^{-5}\text{inch}$           | 0 to 2147483647<br>$\times 10^{-5}\text{degree}$           | 0 to 2147483647<br>pulse                 |
|                         | Pr.51 OPR acceleration time selection                         | 0 to 3                                                                                                                                                                                                                                                                            |                                                          |                                                            |                                          |
|                         | Pr.52 OPR deceleration time selection                         |                                                                                                                                                                                                                                                                                   |                                                          |                                                            |                                          |
|                         | Pr.53 OP shift amount                                         | -2147483648 to 2147483647<br>$\times 10^{-1}\mu\text{m}$                                                                                                                                                                                                                          | -2147483648 to 2147483647<br>$\times 10^{-5}\text{inch}$ | -2147483648 to 2147483647<br>$\times 10^{-5}\text{degree}$ | -2147483648 to 2147483647<br>pulse       |
|                         | Pr.54 OPR torque limit value                                  | 1 to 300%                                                                                                                                                                                                                                                                         |                                                          |                                                            |                                          |
|                         | Pr.55 Deviation counter clear signal output time              | 1 to 65535ms                                                                                                                                                                                                                                                                      |                                                          |                                                            |                                          |
|                         | Pr.56 Speed designation during OP shift                       | 0: OPR speed 1: Creep speed                                                                                                                                                                                                                                                       |                                                          |                                                            |                                          |
|                         | Pr.57 Dwell time during OPR retry                             | 0 to 65535ms                                                                                                                                                                                                                                                                      |                                                          |                                                            |                                          |

|  | Initial value | Axis 1 | Axis 2 | Axis 3 | Axis 4 | Remarks |
|--|---------------|--------|--------|--------|--------|---------|
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 1             |        |        |        |        |         |
|  | 1             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 300           |        |        |        |        |         |
|  | 11            |        |        |        |        |         |
|  | 0             |        |        |        |        |         |
|  | 0             |        |        |        |        |         |















## (1) For axis 1 (positioning option)

| Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option |
|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|
| 1        | 2003               | 51       | 2503               | 101      | 3003               | 151      | 3503               | 201      | 4003               | 251      | 4503               |
| 2        | 2013               | 52       | 2513               | 102      | 3013               | 152      | 3513               | 202      | 4013               | 252      | 4513               |
| 3        | 2023               | 53       | 2523               | 103      | 3023               | 153      | 3523               | 203      | 4023               | 253      | 4523               |
| 4        | 2033               | 54       | 2533               | 104      | 3033               | 154      | 3533               | 204      | 4033               | 254      | 4533               |
| 5        | 2043               | 55       | 2543               | 105      | 3043               | 155      | 3543               | 205      | 4043               | 255      | 4543               |
| 6        | 2053               | 56       | 2553               | 106      | 3053               | 156      | 3553               | 206      | 4053               | 256      | 4553               |
| 7        | 2063               | 57       | 2563               | 107      | 3063               | 157      | 3563               | 207      | 4063               | 257      | 4563               |
| 8        | 2073               | 58       | 2573               | 108      | 3073               | 158      | 3573               | 208      | 4073               | 258      | 4573               |
| 9        | 2083               | 59       | 2583               | 109      | 3083               | 159      | 3583               | 209      | 4083               | 259      | 4583               |
| 10       | 2093               | 60       | 2593               | 110      | 3093               | 160      | 3593               | 210      | 4093               | 260      | 4593               |
| 11       | 2103               | 61       | 2603               | 111      | 3103               | 161      | 3603               | 211      | 4103               | 261      | 4603               |
| 12       | 2113               | 62       | 2613               | 112      | 3113               | 162      | 3613               | 212      | 4113               | 262      | 4613               |
| 13       | 2123               | 63       | 2623               | 113      | 3123               | 163      | 3623               | 213      | 4123               | 263      | 4623               |
| 14       | 2133               | 64       | 2633               | 114      | 3133               | 164      | 3633               | 214      | 4133               | 264      | 4633               |
| 15       | 2143               | 65       | 2643               | 115      | 3143               | 165      | 3643               | 215      | 4143               | 265      | 4643               |
| 16       | 2153               | 66       | 2653               | 116      | 3153               | 166      | 3653               | 216      | 4153               | 266      | 4653               |
| 17       | 2163               | 67       | 2663               | 117      | 3163               | 167      | 3663               | 217      | 4163               | 267      | 4663               |
| 18       | 2173               | 68       | 2673               | 118      | 3173               | 168      | 3673               | 218      | 4173               | 268      | 4673               |
| 19       | 2183               | 69       | 2683               | 119      | 3183               | 169      | 3683               | 219      | 4183               | 269      | 4683               |
| 20       | 2193               | 70       | 2693               | 120      | 3193               | 170      | 3693               | 220      | 4193               | 270      | 4693               |
| 21       | 2203               | 71       | 2703               | 121      | 3203               | 171      | 3703               | 221      | 4203               | 271      | 4703               |
| 22       | 2213               | 72       | 2713               | 122      | 3213               | 172      | 3713               | 222      | 4213               | 272      | 4713               |
| 23       | 2223               | 73       | 2723               | 123      | 3223               | 173      | 3723               | 223      | 4223               | 273      | 4723               |
| 24       | 2233               | 74       | 2733               | 124      | 3233               | 174      | 3733               | 224      | 4233               | 274      | 4733               |
| 25       | 2243               | 75       | 2743               | 125      | 3243               | 175      | 3743               | 225      | 4243               | 275      | 4743               |
| 26       | 2253               | 76       | 2753               | 126      | 3253               | 176      | 3753               | 226      | 4253               | 276      | 4753               |
| 27       | 2263               | 77       | 2763               | 127      | 3263               | 177      | 3763               | 227      | 4263               | 277      | 4763               |
| 28       | 2273               | 78       | 2773               | 128      | 3273               | 178      | 3773               | 228      | 4273               | 278      | 4773               |
| 29       | 2283               | 79       | 2783               | 129      | 3283               | 179      | 3783               | 229      | 4283               | 279      | 4783               |
| 30       | 2293               | 80       | 2793               | 130      | 3293               | 180      | 3793               | 230      | 4293               | 280      | 4793               |
| 31       | 2303               | 81       | 2803               | 131      | 3303               | 181      | 3803               | 231      | 4303               | 281      | 4803               |
| 32       | 2313               | 82       | 2813               | 132      | 3313               | 182      | 3813               | 232      | 4313               | 282      | 4813               |
| 33       | 2323               | 83       | 2823               | 133      | 3323               | 183      | 3823               | 233      | 4323               | 283      | 4823               |
| 34       | 2333               | 84       | 2833               | 134      | 3333               | 184      | 3833               | 234      | 4333               | 284      | 4833               |
| 35       | 2343               | 85       | 2843               | 135      | 3343               | 185      | 3843               | 235      | 4343               | 285      | 4843               |
| 36       | 2353               | 86       | 2853               | 136      | 3353               | 186      | 3853               | 236      | 4353               | 286      | 4853               |
| 37       | 2363               | 87       | 2863               | 137      | 3363               | 187      | 3863               | 237      | 4363               | 287      | 4863               |
| 38       | 2373               | 88       | 2873               | 138      | 3373               | 188      | 3873               | 238      | 4373               | 288      | 4873               |
| 39       | 2383               | 89       | 2883               | 139      | 3383               | 189      | 3883               | 239      | 4383               | 289      | 4883               |
| 40       | 2393               | 90       | 2893               | 140      | 3393               | 190      | 3893               | 240      | 4393               | 290      | 4893               |
| 41       | 2403               | 91       | 2903               | 141      | 3403               | 191      | 3903               | 241      | 4403               | 291      | 4903               |
| 42       | 2413               | 92       | 2913               | 142      | 3413               | 192      | 3913               | 242      | 4413               | 292      | 4913               |
| 43       | 2423               | 93       | 2923               | 143      | 3423               | 193      | 3923               | 243      | 4423               | 293      | 4923               |
| 44       | 2433               | 94       | 2933               | 144      | 3433               | 194      | 3933               | 244      | 4433               | 294      | 4933               |
| 45       | 2443               | 95       | 2943               | 145      | 3443               | 195      | 3943               | 245      | 4443               | 295      | 4943               |
| 46       | 2453               | 96       | 2953               | 146      | 3453               | 196      | 3953               | 246      | 4453               | 296      | 4953               |
| 47       | 2463               | 97       | 2963               | 147      | 3463               | 197      | 3963               | 247      | 4463               | 297      | 4963               |
| 48       | 2473               | 98       | 2973               | 148      | 3473               | 198      | 3973               | 248      | 4473               | 298      | 4973               |
| 49       | 2483               | 99       | 2983               | 149      | 3483               | 199      | 3983               | 249      | 4483               | 299      | 4983               |
| 50       | 2493               | 100      | 2993               | 150      | 3493               | 200      | 3993               | 250      | 4493               | 300      | 4993               |

## (1) For axis 1 (positioning option)

| Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option |
|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|
| 301      | 5003               | 351      | 5503               | 401      | 6003               | 451      | 6503               | 501      | 7003               | 551      | 7503               |
| 302      | 5013               | 352      | 5513               | 402      | 6013               | 452      | 6513               | 502      | 7013               | 552      | 7513               |
| 303      | 5023               | 353      | 5523               | 403      | 6023               | 453      | 6523               | 503      | 7023               | 553      | 7523               |
| 304      | 5033               | 354      | 5533               | 404      | 6033               | 454      | 6533               | 504      | 7033               | 554      | 7533               |
| 305      | 5043               | 355      | 5543               | 405      | 6043               | 455      | 6543               | 505      | 7043               | 555      | 7543               |
| 306      | 5053               | 356      | 5553               | 406      | 6053               | 456      | 6553               | 506      | 7053               | 556      | 7553               |
| 307      | 5063               | 357      | 5563               | 407      | 6063               | 457      | 6563               | 507      | 7063               | 557      | 7563               |
| 308      | 5073               | 358      | 5573               | 408      | 6073               | 458      | 6573               | 508      | 7073               | 558      | 7573               |
| 309      | 5083               | 359      | 5583               | 409      | 6083               | 459      | 6583               | 509      | 7083               | 559      | 7583               |
| 310      | 5093               | 360      | 5593               | 410      | 6093               | 460      | 6593               | 510      | 7093               | 560      | 7593               |
| 311      | 5103               | 361      | 5603               | 411      | 6103               | 461      | 6603               | 511      | 7103               | 561      | 7603               |
| 312      | 5113               | 362      | 5613               | 412      | 6113               | 462      | 6613               | 512      | 7113               | 562      | 7613               |
| 313      | 5123               | 363      | 5623               | 413      | 6123               | 463      | 6623               | 513      | 7123               | 563      | 7623               |
| 314      | 5133               | 364      | 5633               | 414      | 6133               | 464      | 6633               | 514      | 7133               | 564      | 7633               |
| 315      | 5143               | 365      | 5643               | 415      | 6143               | 465      | 6643               | 515      | 7143               | 565      | 7643               |
| 316      | 5153               | 366      | 5653               | 416      | 6153               | 466      | 6653               | 516      | 7153               | 566      | 7653               |
| 317      | 5163               | 367      | 5663               | 417      | 6163               | 467      | 6663               | 517      | 7163               | 567      | 7663               |
| 318      | 5173               | 368      | 5673               | 418      | 6173               | 468      | 6673               | 518      | 7173               | 568      | 7673               |
| 319      | 5183               | 369      | 5683               | 419      | 6183               | 469      | 6683               | 519      | 7183               | 569      | 7683               |
| 320      | 5193               | 370      | 5693               | 420      | 6193               | 470      | 6693               | 520      | 7193               | 570      | 7693               |
| 321      | 5203               | 371      | 5703               | 421      | 6203               | 471      | 6703               | 521      | 7203               | 571      | 7703               |
| 322      | 5213               | 372      | 5713               | 422      | 6213               | 472      | 6713               | 522      | 7213               | 572      | 7713               |
| 323      | 5223               | 373      | 5723               | 423      | 6223               | 473      | 6723               | 523      | 7223               | 573      | 7723               |
| 324      | 5233               | 374      | 5733               | 424      | 6233               | 474      | 6733               | 524      | 7233               | 574      | 7733               |
| 325      | 5243               | 375      | 5743               | 425      | 6243               | 475      | 6743               | 525      | 7243               | 575      | 7743               |
| 326      | 5253               | 376      | 5753               | 426      | 6253               | 476      | 6753               | 526      | 7253               | 576      | 7753               |
| 327      | 5263               | 377      | 5763               | 427      | 6263               | 477      | 6763               | 527      | 7263               | 577      | 7763               |
| 328      | 5273               | 378      | 5773               | 428      | 6273               | 478      | 6773               | 528      | 7273               | 578      | 7773               |
| 329      | 5283               | 379      | 5783               | 429      | 6283               | 479      | 6783               | 529      | 7283               | 579      | 7783               |
| 330      | 5293               | 380      | 5793               | 430      | 6293               | 480      | 6793               | 530      | 7293               | 580      | 7793               |
| 331      | 5303               | 381      | 5803               | 431      | 6303               | 481      | 6803               | 531      | 7303               | 581      | 7803               |
| 332      | 5313               | 382      | 5813               | 432      | 6313               | 482      | 6813               | 532      | 7313               | 582      | 7813               |
| 333      | 5323               | 383      | 5823               | 433      | 6323               | 483      | 6823               | 533      | 7323               | 583      | 7823               |
| 334      | 5333               | 384      | 5833               | 434      | 6333               | 484      | 6833               | 534      | 7333               | 584      | 7833               |
| 335      | 5343               | 385      | 5843               | 435      | 6343               | 485      | 6843               | 535      | 7343               | 585      | 7843               |
| 336      | 5353               | 386      | 5853               | 436      | 6353               | 486      | 6853               | 536      | 7353               | 586      | 7853               |
| 337      | 5363               | 387      | 5863               | 437      | 6363               | 487      | 6863               | 537      | 7363               | 587      | 7863               |
| 338      | 5373               | 388      | 5873               | 438      | 6373               | 488      | 6873               | 538      | 7373               | 588      | 7873               |
| 339      | 5383               | 389      | 5883               | 439      | 6383               | 489      | 6883               | 539      | 7383               | 589      | 7883               |
| 340      | 5393               | 390      | 5893               | 440      | 6393               | 490      | 6893               | 540      | 7393               | 590      | 7893               |
| 341      | 5403               | 391      | 5903               | 441      | 6403               | 491      | 6903               | 541      | 7403               | 591      | 7903               |
| 342      | 5413               | 392      | 5913               | 442      | 6413               | 492      | 6913               | 542      | 7413               | 592      | 7913               |
| 343      | 5423               | 393      | 5923               | 443      | 6423               | 493      | 6923               | 543      | 7423               | 593      | 7923               |
| 344      | 5433               | 394      | 5933               | 444      | 6433               | 494      | 6933               | 544      | 7433               | 594      | 7933               |
| 345      | 5443               | 395      | 5943               | 445      | 6443               | 495      | 6943               | 545      | 7443               | 595      | 7943               |
| 346      | 5453               | 396      | 5953               | 446      | 6453               | 496      | 6953               | 546      | 7453               | 596      | 7953               |
| 347      | 5463               | 397      | 5963               | 447      | 6463               | 497      | 6963               | 547      | 7463               | 597      | 7963               |
| 348      | 5473               | 398      | 5973               | 448      | 6473               | 498      | 6973               | 548      | 7473               | 598      | 7973               |
| 349      | 5483               | 399      | 5983               | 449      | 6483               | 499      | 6983               | 549      | 7483               | 599      | 7983               |
| 350      | 5493               | 400      | 5993               | 450      | 6493               | 500      | 6993               | 550      | 7493               | 600      | 7993               |















## (2) For axis 2 (positioning option)

| Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option |
|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|
| 1        | 8003               | 51       | 8503               | 101      | 9003               | 151      | 9503               | 201      | 10003              | 251      | 10503              |
| 2        | 8013               | 52       | 8513               | 102      | 9013               | 152      | 9513               | 202      | 10013              | 252      | 10513              |
| 3        | 8023               | 53       | 8523               | 103      | 9023               | 153      | 9523               | 203      | 10023              | 253      | 10523              |
| 4        | 8033               | 54       | 8533               | 104      | 9033               | 154      | 9533               | 204      | 10033              | 254      | 10533              |
| 5        | 8043               | 55       | 8543               | 105      | 9043               | 155      | 9543               | 205      | 10043              | 255      | 10543              |
| 6        | 8053               | 56       | 8553               | 106      | 9053               | 156      | 9553               | 206      | 10053              | 256      | 10553              |
| 7        | 8063               | 57       | 8563               | 107      | 9063               | 157      | 9563               | 207      | 10063              | 257      | 10563              |
| 8        | 8073               | 58       | 8573               | 108      | 9073               | 158      | 9573               | 208      | 10073              | 258      | 10573              |
| 9        | 8083               | 59       | 8583               | 109      | 9083               | 159      | 9583               | 209      | 10083              | 259      | 10583              |
| 10       | 8093               | 60       | 8593               | 110      | 9093               | 160      | 9593               | 210      | 10093              | 260      | 10593              |
| 11       | 8103               | 61       | 8603               | 111      | 9103               | 161      | 9603               | 211      | 10103              | 261      | 10603              |
| 12       | 8113               | 62       | 8613               | 112      | 9113               | 162      | 9613               | 212      | 10113              | 262      | 10613              |
| 13       | 8123               | 63       | 8623               | 113      | 9123               | 163      | 9623               | 213      | 10123              | 263      | 10623              |
| 14       | 8133               | 64       | 8633               | 114      | 9133               | 164      | 9633               | 214      | 10133              | 264      | 10633              |
| 15       | 8143               | 65       | 8643               | 115      | 9143               | 165      | 9643               | 215      | 10143              | 265      | 10643              |
| 16       | 8153               | 66       | 8653               | 116      | 9153               | 166      | 9653               | 216      | 10153              | 266      | 10653              |
| 17       | 8163               | 67       | 8663               | 117      | 9163               | 167      | 9663               | 217      | 10163              | 267      | 10663              |
| 18       | 8173               | 68       | 8673               | 118      | 9173               | 168      | 9673               | 218      | 10173              | 268      | 10673              |
| 19       | 8183               | 69       | 8683               | 119      | 9183               | 169      | 9683               | 219      | 10183              | 269      | 10683              |
| 20       | 8193               | 70       | 8693               | 120      | 9193               | 170      | 9693               | 220      | 10193              | 270      | 10693              |
| 21       | 8203               | 71       | 8703               | 121      | 9203               | 171      | 9703               | 221      | 10203              | 271      | 10703              |
| 22       | 8213               | 72       | 8713               | 122      | 9213               | 172      | 9713               | 222      | 10213              | 272      | 10713              |
| 23       | 8223               | 73       | 8723               | 123      | 9223               | 173      | 9723               | 223      | 10223              | 273      | 10723              |
| 24       | 8233               | 74       | 8733               | 124      | 9233               | 174      | 9733               | 224      | 10233              | 274      | 10733              |
| 25       | 8243               | 75       | 8743               | 125      | 9243               | 175      | 9743               | 225      | 10243              | 275      | 10743              |
| 26       | 8253               | 76       | 8753               | 126      | 9253               | 176      | 9753               | 226      | 10253              | 276      | 10753              |
| 27       | 8263               | 77       | 8763               | 127      | 9263               | 177      | 9763               | 227      | 10263              | 277      | 10763              |
| 28       | 8273               | 78       | 8773               | 128      | 9273               | 178      | 9773               | 228      | 10273              | 278      | 10773              |
| 29       | 8283               | 79       | 8783               | 129      | 9283               | 179      | 9783               | 229      | 10283              | 279      | 10783              |
| 30       | 8293               | 80       | 8793               | 130      | 9293               | 180      | 9793               | 230      | 10293              | 280      | 10793              |
| 31       | 8303               | 81       | 8803               | 131      | 9303               | 181      | 9803               | 231      | 10303              | 281      | 10803              |
| 32       | 8313               | 82       | 8813               | 132      | 9313               | 182      | 9813               | 232      | 10313              | 282      | 10813              |
| 33       | 8323               | 83       | 8823               | 133      | 9323               | 183      | 9823               | 233      | 10323              | 283      | 10823              |
| 34       | 8333               | 84       | 8833               | 134      | 9333               | 184      | 9833               | 234      | 10333              | 284      | 10833              |
| 35       | 8343               | 85       | 8843               | 135      | 9343               | 185      | 9843               | 235      | 10343              | 285      | 10843              |
| 36       | 8353               | 86       | 8853               | 136      | 9353               | 186      | 9853               | 236      | 10353              | 286      | 10853              |
| 37       | 8363               | 87       | 8863               | 137      | 9363               | 187      | 9863               | 237      | 10363              | 287      | 10863              |
| 38       | 8373               | 88       | 8873               | 138      | 9373               | 188      | 9873               | 238      | 10373              | 288      | 10873              |
| 39       | 8383               | 89       | 8883               | 139      | 9383               | 189      | 9883               | 239      | 10383              | 289      | 10883              |
| 40       | 8393               | 90       | 8893               | 140      | 9393               | 190      | 9893               | 240      | 10393              | 290      | 10893              |
| 41       | 8403               | 91       | 8903               | 141      | 9403               | 191      | 9903               | 241      | 10403              | 291      | 10903              |
| 42       | 8413               | 92       | 8913               | 142      | 9413               | 192      | 9913               | 242      | 10413              | 292      | 10913              |
| 43       | 8423               | 93       | 8923               | 143      | 9423               | 193      | 9923               | 243      | 10423              | 293      | 10923              |
| 44       | 8433               | 94       | 8933               | 144      | 9433               | 194      | 9933               | 244      | 10433              | 294      | 10933              |
| 45       | 8443               | 95       | 8943               | 145      | 9443               | 195      | 9943               | 245      | 10443              | 295      | 10943              |
| 46       | 8453               | 96       | 8953               | 146      | 9453               | 196      | 9953               | 246      | 10453              | 296      | 10953              |
| 47       | 8463               | 97       | 8963               | 147      | 9463               | 197      | 9963               | 247      | 10463              | 297      | 10963              |
| 48       | 8473               | 98       | 8973               | 148      | 9473               | 198      | 9973               | 248      | 10473              | 298      | 10973              |
| 49       | 8483               | 99       | 8983               | 149      | 9483               | 199      | 9983               | 249      | 10483              | 299      | 10983              |
| 50       | 8493               | 100      | 8993               | 150      | 9493               | 200      | 9993               | 250      | 10493              | 300      | 10993              |

## (2) For axis 2 (positioning option)

| Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option |
|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|
| 301      | 11003              | 351      | 11503              | 401      | 12003              | 451      | 12503              | 501      | 13003              | 551      | 13503              |
| 302      | 11013              | 352      | 11513              | 402      | 12013              | 452      | 12513              | 502      | 13013              | 552      | 13513              |
| 303      | 11023              | 353      | 11523              | 403      | 12023              | 453      | 12523              | 503      | 13023              | 553      | 13523              |
| 304      | 11033              | 354      | 11533              | 404      | 12033              | 454      | 12533              | 504      | 13033              | 554      | 13533              |
| 305      | 11043              | 355      | 11543              | 405      | 12043              | 455      | 12543              | 505      | 13043              | 555      | 13543              |
| 306      | 11053              | 356      | 11553              | 406      | 12053              | 456      | 12553              | 506      | 13053              | 556      | 13553              |
| 307      | 11063              | 357      | 11563              | 407      | 12063              | 457      | 12563              | 507      | 13063              | 557      | 13563              |
| 308      | 11073              | 358      | 11573              | 408      | 12073              | 458      | 12573              | 508      | 13073              | 558      | 13573              |
| 309      | 11083              | 359      | 11583              | 409      | 12083              | 459      | 12583              | 509      | 13083              | 559      | 13583              |
| 310      | 11093              | 360      | 11593              | 410      | 12093              | 460      | 12593              | 510      | 13093              | 560      | 13593              |
| 311      | 11103              | 361      | 11603              | 411      | 12103              | 461      | 12603              | 511      | 13103              | 561      | 13603              |
| 312      | 11113              | 362      | 11613              | 412      | 12113              | 462      | 12613              | 512      | 13113              | 562      | 13613              |
| 313      | 11123              | 363      | 11623              | 413      | 12123              | 463      | 12623              | 513      | 13123              | 563      | 13623              |
| 314      | 11133              | 364      | 11633              | 414      | 12133              | 464      | 12633              | 514      | 13133              | 564      | 13633              |
| 315      | 11143              | 365      | 11643              | 415      | 12143              | 465      | 12643              | 515      | 13143              | 565      | 13643              |
| 316      | 11153              | 366      | 11653              | 416      | 12153              | 466      | 12653              | 516      | 13153              | 566      | 13653              |
| 317      | 11163              | 367      | 11663              | 417      | 12163              | 467      | 12663              | 517      | 13163              | 567      | 13663              |
| 318      | 11173              | 368      | 11673              | 418      | 12173              | 468      | 12673              | 518      | 13173              | 568      | 13673              |
| 319      | 11183              | 369      | 11683              | 419      | 12183              | 469      | 12683              | 519      | 13183              | 569      | 13683              |
| 320      | 11193              | 370      | 11693              | 420      | 12193              | 470      | 12693              | 520      | 13193              | 570      | 13693              |
| 321      | 11203              | 371      | 11703              | 421      | 12203              | 471      | 12703              | 521      | 13203              | 571      | 13703              |
| 322      | 11213              | 372      | 11713              | 422      | 12213              | 472      | 12713              | 522      | 13213              | 572      | 13713              |
| 323      | 11223              | 373      | 11723              | 423      | 12223              | 473      | 12723              | 523      | 13223              | 573      | 13723              |
| 324      | 11233              | 374      | 11733              | 424      | 12233              | 474      | 12733              | 524      | 13233              | 574      | 13733              |
| 325      | 11243              | 375      | 11743              | 425      | 12243              | 475      | 12743              | 525      | 13243              | 575      | 13743              |
| 326      | 11253              | 376      | 11753              | 426      | 12253              | 476      | 12753              | 526      | 13253              | 576      | 13753              |
| 327      | 11263              | 377      | 11763              | 427      | 12263              | 477      | 12763              | 527      | 13263              | 577      | 13763              |
| 328      | 11273              | 378      | 11773              | 428      | 12273              | 478      | 12773              | 528      | 13273              | 578      | 13773              |
| 329      | 11283              | 379      | 11783              | 429      | 12283              | 479      | 12783              | 529      | 13283              | 579      | 13783              |
| 330      | 11293              | 380      | 11793              | 430      | 12293              | 480      | 12793              | 530      | 13293              | 580      | 13793              |
| 331      | 11303              | 381      | 11803              | 431      | 12303              | 481      | 12803              | 531      | 13303              | 581      | 13803              |
| 332      | 11313              | 382      | 11813              | 432      | 12313              | 482      | 12813              | 532      | 13313              | 582      | 13813              |
| 333      | 11323              | 383      | 11823              | 433      | 12323              | 483      | 12823              | 533      | 13323              | 583      | 13823              |
| 334      | 11333              | 384      | 11833              | 434      | 12333              | 484      | 12833              | 534      | 13333              | 584      | 13833              |
| 335      | 11343              | 385      | 11843              | 435      | 12343              | 485      | 12843              | 535      | 13343              | 585      | 13843              |
| 336      | 11353              | 386      | 11853              | 436      | 12353              | 486      | 12853              | 536      | 13353              | 586      | 13853              |
| 337      | 11363              | 387      | 11863              | 437      | 12363              | 487      | 12863              | 537      | 13363              | 587      | 13863              |
| 338      | 11373              | 388      | 11873              | 438      | 12373              | 488      | 12873              | 538      | 13373              | 588      | 13873              |
| 339      | 11383              | 389      | 11883              | 439      | 12383              | 489      | 12883              | 539      | 13383              | 589      | 13883              |
| 340      | 11393              | 390      | 11893              | 440      | 12393              | 490      | 12893              | 540      | 13393              | 590      | 13893              |
| 341      | 11403              | 391      | 11903              | 441      | 12403              | 491      | 12903              | 541      | 13403              | 591      | 13903              |
| 342      | 11413              | 392      | 11913              | 442      | 12413              | 492      | 12913              | 542      | 13413              | 592      | 13913              |
| 343      | 11423              | 393      | 11923              | 443      | 12423              | 493      | 12923              | 543      | 13423              | 593      | 13923              |
| 344      | 11433              | 394      | 11933              | 444      | 12433              | 494      | 12933              | 544      | 13433              | 594      | 13933              |
| 345      | 11443              | 395      | 11943              | 445      | 12443              | 495      | 12943              | 545      | 13443              | 595      | 13943              |
| 346      | 11453              | 396      | 11953              | 446      | 12453              | 496      | 12953              | 546      | 13453              | 596      | 13953              |
| 347      | 11463              | 397      | 11963              | 447      | 12463              | 497      | 12963              | 547      | 13463              | 597      | 13963              |
| 348      | 11473              | 398      | 11973              | 448      | 12473              | 498      | 12973              | 548      | 13473              | 598      | 13973              |
| 349      | 11483              | 399      | 11983              | 449      | 12483              | 499      | 12983              | 549      | 13483              | 599      | 13983              |
| 350      | 11493              | 400      | 11993              | 450      | 12493              | 500      | 12993              | 550      | 13493              | 600      | 13993              |















## (3) For axis 3 (positioning option)

| Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option |
|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|
| 1        | 14003              | 51       | 14503              | 101      | 15003              | 151      | 15503              | 201      | 16003              | 251      | 16503              |
| 2        | 14013              | 52       | 14513              | 102      | 15013              | 152      | 15513              | 202      | 16013              | 252      | 16513              |
| 3        | 14023              | 53       | 14523              | 103      | 15023              | 153      | 15523              | 203      | 16023              | 253      | 16523              |
| 4        | 14033              | 54       | 14533              | 104      | 15033              | 154      | 15533              | 204      | 16033              | 254      | 16533              |
| 5        | 14043              | 55       | 14543              | 105      | 15043              | 155      | 15543              | 205      | 16043              | 255      | 16543              |
| 6        | 14053              | 56       | 14553              | 106      | 15053              | 156      | 15553              | 206      | 16053              | 256      | 16553              |
| 7        | 14063              | 57       | 14563              | 107      | 15063              | 157      | 15563              | 207      | 16063              | 257      | 16563              |
| 8        | 14073              | 58       | 14573              | 108      | 15073              | 158      | 15573              | 208      | 16073              | 258      | 16573              |
| 9        | 14083              | 59       | 14583              | 109      | 15083              | 159      | 15583              | 209      | 16083              | 259      | 16583              |
| 10       | 14093              | 60       | 14593              | 110      | 15093              | 160      | 15593              | 210      | 16093              | 260      | 16593              |
| 11       | 14103              | 61       | 14603              | 111      | 15103              | 161      | 15603              | 211      | 16103              | 261      | 16603              |
| 12       | 14113              | 62       | 14613              | 112      | 15113              | 162      | 15613              | 212      | 16113              | 262      | 16613              |
| 13       | 14123              | 63       | 14623              | 113      | 15123              | 163      | 15623              | 213      | 16123              | 263      | 16623              |
| 14       | 14133              | 64       | 14633              | 114      | 15133              | 164      | 15633              | 214      | 16133              | 264      | 16633              |
| 15       | 14143              | 65       | 14643              | 115      | 15143              | 165      | 15643              | 215      | 16143              | 265      | 16643              |
| 16       | 14153              | 66       | 14653              | 116      | 15153              | 166      | 15653              | 216      | 16153              | 266      | 16653              |
| 17       | 14163              | 67       | 14663              | 117      | 15163              | 167      | 15663              | 217      | 16163              | 267      | 16663              |
| 18       | 14173              | 68       | 14673              | 118      | 15173              | 168      | 15673              | 218      | 16173              | 268      | 16673              |
| 19       | 14183              | 69       | 14683              | 119      | 15183              | 169      | 15683              | 219      | 16183              | 269      | 16683              |
| 20       | 14193              | 70       | 14693              | 120      | 15193              | 170      | 15693              | 220      | 16193              | 270      | 16693              |
| 21       | 14203              | 71       | 14703              | 121      | 15203              | 171      | 15703              | 221      | 16203              | 271      | 16703              |
| 22       | 14213              | 72       | 14713              | 122      | 15213              | 172      | 15713              | 222      | 16213              | 272      | 16713              |
| 23       | 14223              | 73       | 14723              | 123      | 15223              | 173      | 15723              | 223      | 16223              | 273      | 16723              |
| 24       | 14233              | 74       | 14733              | 124      | 15233              | 174      | 15733              | 224      | 16233              | 274      | 16733              |
| 25       | 14243              | 75       | 14743              | 125      | 15243              | 175      | 15743              | 225      | 16243              | 275      | 16743              |
| 26       | 14253              | 76       | 14753              | 126      | 15253              | 176      | 15753              | 226      | 16253              | 276      | 16753              |
| 27       | 14263              | 77       | 14763              | 127      | 15263              | 177      | 15763              | 227      | 16263              | 277      | 16763              |
| 28       | 14273              | 78       | 14773              | 128      | 15273              | 178      | 15773              | 228      | 16273              | 278      | 16773              |
| 29       | 14283              | 79       | 14783              | 129      | 15283              | 179      | 15783              | 229      | 16283              | 279      | 16783              |
| 30       | 14293              | 80       | 14793              | 130      | 15293              | 180      | 15793              | 230      | 16293              | 280      | 16793              |
| 31       | 14303              | 81       | 14803              | 131      | 15303              | 181      | 15803              | 231      | 16303              | 281      | 16803              |
| 32       | 14313              | 82       | 14813              | 132      | 15313              | 182      | 15813              | 232      | 16313              | 282      | 16813              |
| 33       | 14323              | 83       | 14823              | 133      | 15323              | 183      | 15823              | 233      | 16323              | 283      | 16823              |
| 34       | 14333              | 84       | 14833              | 134      | 15333              | 184      | 15833              | 234      | 16333              | 284      | 16833              |
| 35       | 14343              | 85       | 14843              | 135      | 15343              | 185      | 15843              | 235      | 16343              | 285      | 16843              |
| 36       | 14353              | 86       | 14853              | 136      | 15353              | 186      | 15853              | 236      | 16353              | 286      | 16853              |
| 37       | 14363              | 87       | 14863              | 137      | 15363              | 187      | 15863              | 237      | 16363              | 287      | 16863              |
| 38       | 14373              | 88       | 14873              | 138      | 15373              | 188      | 15873              | 238      | 16373              | 288      | 16873              |
| 39       | 14383              | 89       | 14883              | 139      | 15383              | 189      | 15883              | 239      | 16383              | 289      | 16883              |
| 40       | 14393              | 90       | 14893              | 140      | 15393              | 190      | 15893              | 240      | 16393              | 290      | 16893              |
| 41       | 14403              | 91       | 14903              | 141      | 15403              | 191      | 15903              | 241      | 16403              | 291      | 16903              |
| 42       | 14413              | 92       | 14913              | 142      | 15413              | 192      | 15913              | 242      | 16413              | 292      | 16913              |
| 43       | 14423              | 93       | 14923              | 143      | 15423              | 193      | 15923              | 243      | 16423              | 293      | 16923              |
| 44       | 14433              | 94       | 14933              | 144      | 15433              | 194      | 15933              | 244      | 16433              | 294      | 16933              |
| 45       | 14443              | 95       | 14943              | 145      | 15443              | 195      | 15943              | 245      | 16443              | 295      | 16943              |
| 46       | 14453              | 96       | 14953              | 146      | 15453              | 196      | 15953              | 246      | 16453              | 296      | 16953              |
| 47       | 14463              | 97       | 14963              | 147      | 15463              | 197      | 15963              | 247      | 16463              | 297      | 16963              |
| 48       | 14473              | 98       | 14973              | 148      | 15473              | 198      | 15973              | 248      | 16473              | 298      | 16973              |
| 49       | 14483              | 99       | 14983              | 149      | 15483              | 199      | 15983              | 249      | 16483              | 299      | 16983              |
| 50       | 14493              | 100      | 14993              | 150      | 15493              | 200      | 15993              | 250      | 16493              | 300      | 16993              |

## (3) For axis 3 (positioning option)

| Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option |
|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|
| 301      | 17003              | 351      | 17503              | 401      | 18003              | 451      | 18503              | 501      | 19003              | 551      | 19503              |
| 302      | 17013              | 352      | 17513              | 402      | 18013              | 452      | 18513              | 502      | 19013              | 552      | 19513              |
| 303      | 17023              | 353      | 17523              | 403      | 18023              | 453      | 18523              | 503      | 19023              | 553      | 19523              |
| 304      | 17033              | 354      | 17533              | 404      | 18033              | 454      | 18533              | 504      | 19033              | 554      | 19533              |
| 305      | 17043              | 355      | 17543              | 405      | 18043              | 455      | 18543              | 505      | 19043              | 555      | 19543              |
| 306      | 17053              | 356      | 17553              | 406      | 18053              | 456      | 18553              | 506      | 19053              | 556      | 19553              |
| 307      | 17063              | 357      | 17563              | 407      | 18063              | 457      | 18563              | 507      | 19063              | 557      | 19563              |
| 308      | 17073              | 358      | 17573              | 408      | 18073              | 458      | 18573              | 508      | 19073              | 558      | 19573              |
| 309      | 17083              | 359      | 17583              | 409      | 18083              | 459      | 18583              | 509      | 19083              | 559      | 19583              |
| 310      | 17093              | 360      | 17593              | 410      | 18093              | 460      | 18593              | 510      | 19093              | 560      | 19593              |
| 311      | 17103              | 361      | 17603              | 411      | 18103              | 461      | 18603              | 511      | 19103              | 561      | 19603              |
| 312      | 17113              | 362      | 17613              | 412      | 18113              | 462      | 18613              | 512      | 19113              | 562      | 19613              |
| 313      | 17123              | 363      | 17623              | 413      | 18123              | 463      | 18623              | 513      | 19123              | 563      | 19623              |
| 314      | 17133              | 364      | 17633              | 414      | 18133              | 464      | 18633              | 514      | 19133              | 564      | 19633              |
| 315      | 17143              | 365      | 17643              | 415      | 18143              | 465      | 18643              | 515      | 19143              | 565      | 19643              |
| 316      | 17153              | 366      | 17653              | 416      | 18153              | 466      | 18653              | 516      | 19153              | 566      | 19653              |
| 317      | 17163              | 367      | 17663              | 417      | 18163              | 467      | 18663              | 517      | 19163              | 567      | 19663              |
| 318      | 17173              | 368      | 17673              | 418      | 18173              | 468      | 18673              | 518      | 19173              | 568      | 19673              |
| 319      | 17183              | 369      | 17683              | 419      | 18183              | 469      | 18683              | 519      | 19183              | 569      | 19683              |
| 320      | 17193              | 370      | 17693              | 420      | 18193              | 470      | 18693              | 520      | 19193              | 570      | 19693              |
| 321      | 17203              | 371      | 17703              | 421      | 18203              | 471      | 18703              | 521      | 19203              | 571      | 19703              |
| 322      | 17213              | 372      | 17713              | 422      | 18213              | 472      | 18713              | 522      | 19213              | 572      | 19713              |
| 323      | 17223              | 373      | 17723              | 423      | 18223              | 473      | 18723              | 523      | 19223              | 573      | 19723              |
| 324      | 17233              | 374      | 17733              | 424      | 18233              | 474      | 18733              | 524      | 19233              | 574      | 19733              |
| 325      | 17243              | 375      | 17743              | 425      | 18243              | 475      | 18743              | 525      | 19243              | 575      | 19743              |
| 326      | 17253              | 376      | 17753              | 426      | 18253              | 476      | 18753              | 526      | 19253              | 576      | 19753              |
| 327      | 17263              | 377      | 17763              | 427      | 18263              | 477      | 18763              | 527      | 19263              | 577      | 19763              |
| 328      | 17273              | 378      | 17773              | 428      | 18273              | 478      | 18773              | 528      | 19273              | 578      | 19773              |
| 329      | 17283              | 379      | 17783              | 429      | 18283              | 479      | 18783              | 529      | 19283              | 579      | 19783              |
| 330      | 17293              | 380      | 17793              | 430      | 18293              | 480      | 18793              | 530      | 19293              | 580      | 19793              |
| 331      | 17303              | 381      | 17803              | 431      | 18303              | 481      | 18803              | 531      | 19303              | 581      | 19803              |
| 332      | 17313              | 382      | 17813              | 432      | 18313              | 482      | 18813              | 532      | 19313              | 582      | 19813              |
| 333      | 17323              | 383      | 17823              | 433      | 18323              | 483      | 18823              | 533      | 19323              | 583      | 19823              |
| 334      | 17333              | 384      | 17833              | 434      | 18333              | 484      | 18833              | 534      | 19333              | 584      | 19833              |
| 335      | 17343              | 385      | 17843              | 435      | 18343              | 485      | 18843              | 535      | 19343              | 585      | 19843              |
| 336      | 17353              | 386      | 17853              | 436      | 18353              | 486      | 18853              | 536      | 19353              | 586      | 19853              |
| 337      | 17363              | 387      | 17863              | 437      | 18363              | 487      | 18863              | 537      | 19363              | 587      | 19863              |
| 338      | 17373              | 388      | 17873              | 438      | 18373              | 488      | 18873              | 538      | 19373              | 588      | 19873              |
| 339      | 17383              | 389      | 17883              | 439      | 18383              | 489      | 18883              | 539      | 19383              | 589      | 19883              |
| 340      | 17393              | 390      | 17893              | 440      | 18393              | 490      | 18893              | 540      | 19393              | 590      | 19893              |
| 341      | 17403              | 391      | 17903              | 441      | 18403              | 491      | 18903              | 541      | 19403              | 591      | 19903              |
| 342      | 17413              | 392      | 17913              | 442      | 18413              | 492      | 18913              | 542      | 19413              | 592      | 19913              |
| 343      | 17423              | 393      | 17923              | 443      | 18423              | 493      | 18923              | 543      | 19423              | 593      | 19923              |
| 344      | 17433              | 394      | 17933              | 444      | 18433              | 494      | 18933              | 544      | 19433              | 594      | 19933              |
| 345      | 17443              | 395      | 17943              | 445      | 18443              | 495      | 18943              | 545      | 19443              | 595      | 19943              |
| 346      | 17453              | 396      | 17953              | 446      | 18453              | 496      | 18953              | 546      | 19453              | 596      | 19953              |
| 347      | 17463              | 397      | 17963              | 447      | 18463              | 497      | 18963              | 547      | 19463              | 597      | 19963              |
| 348      | 17473              | 398      | 17973              | 448      | 18473              | 498      | 18973              | 548      | 19473              | 598      | 19973              |
| 349      | 17483              | 399      | 17983              | 449      | 18483              | 499      | 18983              | 549      | 19483              | 599      | 19983              |
| 350      | 17493              | 400      | 17993              | 450      | 18493              | 500      | 18993              | 550      | 19493              | 600      | 19993              |















## (4) For axis 4 (positioning option)

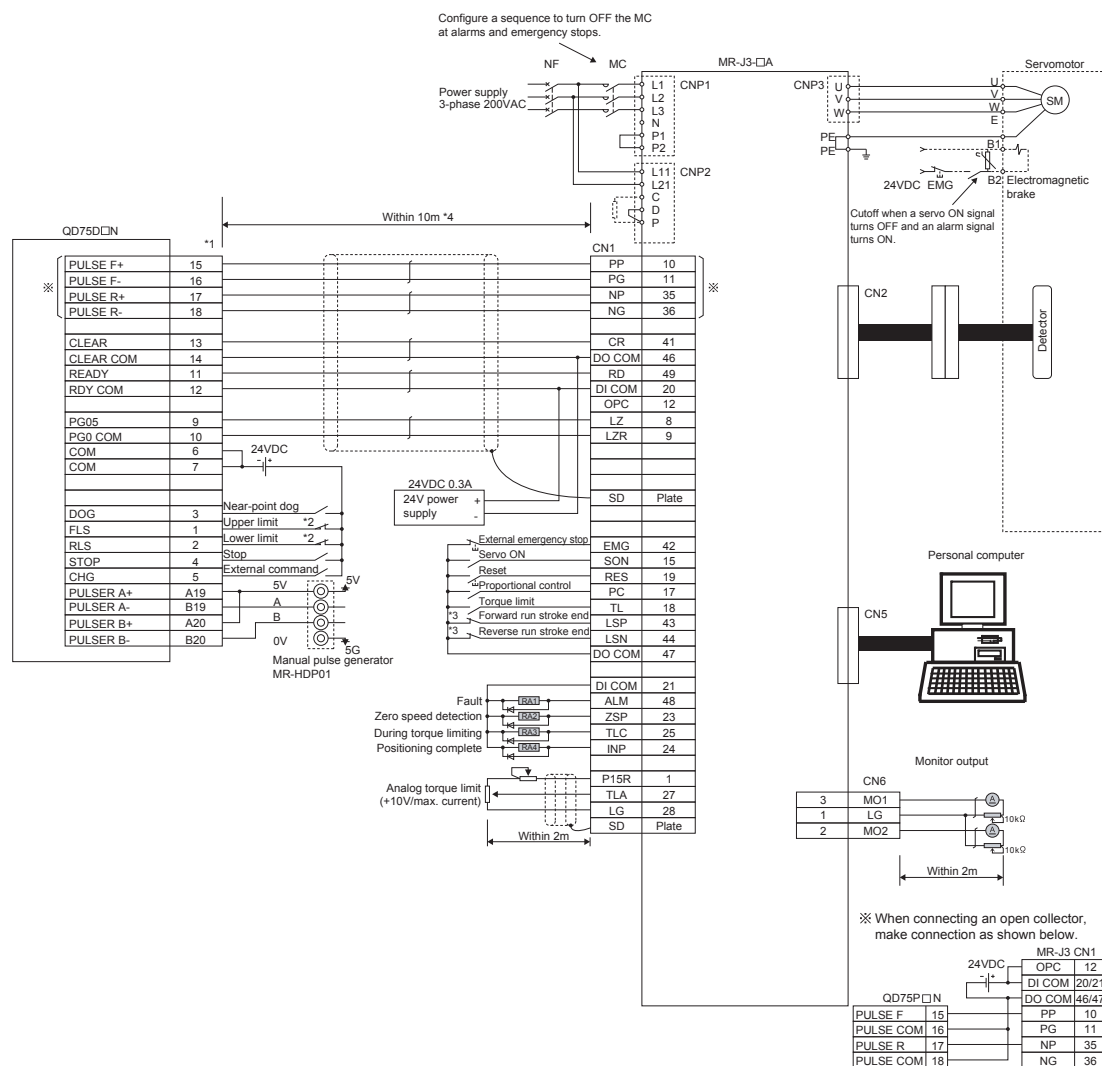
| Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option |
|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|
| 1        | 20003              | 51       | 20503              | 101      | 21003              | 151      | 21503              | 201      | 22003              | 251      | 22503              |
| 2        | 20013              | 52       | 20513              | 102      | 21013              | 152      | 21513              | 202      | 22013              | 252      | 22513              |
| 3        | 20023              | 53       | 20523              | 103      | 21023              | 153      | 21523              | 203      | 22023              | 253      | 22523              |
| 4        | 20033              | 54       | 20533              | 104      | 21033              | 154      | 21533              | 204      | 22033              | 254      | 22533              |
| 5        | 20043              | 55       | 20543              | 105      | 21043              | 155      | 21543              | 205      | 22043              | 255      | 22543              |
| 6        | 20053              | 56       | 20553              | 106      | 21053              | 156      | 21553              | 206      | 22053              | 256      | 22553              |
| 7        | 20063              | 57       | 20563              | 107      | 21063              | 157      | 21563              | 207      | 22063              | 257      | 22563              |
| 8        | 20073              | 58       | 20573              | 108      | 21073              | 158      | 21573              | 208      | 22073              | 258      | 22573              |
| 9        | 20083              | 59       | 20583              | 109      | 21083              | 159      | 21583              | 209      | 22083              | 259      | 22583              |
| 10       | 20093              | 60       | 20593              | 110      | 21093              | 160      | 21593              | 210      | 22093              | 260      | 22593              |
| 11       | 20103              | 61       | 20603              | 111      | 21103              | 161      | 21603              | 211      | 22103              | 261      | 22603              |
| 12       | 20113              | 62       | 20613              | 112      | 21113              | 162      | 21613              | 212      | 22113              | 262      | 22613              |
| 13       | 20123              | 63       | 20623              | 113      | 21123              | 163      | 21623              | 213      | 22123              | 263      | 22623              |
| 14       | 20133              | 64       | 20633              | 114      | 21133              | 164      | 21633              | 214      | 22133              | 264      | 22633              |
| 15       | 20143              | 65       | 20643              | 115      | 21143              | 165      | 21643              | 215      | 22143              | 265      | 22643              |
| 16       | 20153              | 66       | 20653              | 116      | 21153              | 166      | 21653              | 216      | 22153              | 266      | 22653              |
| 17       | 20163              | 67       | 20663              | 117      | 21163              | 167      | 21663              | 217      | 22163              | 267      | 22663              |
| 18       | 20173              | 68       | 20673              | 118      | 21173              | 168      | 21673              | 218      | 22173              | 268      | 22673              |
| 19       | 20183              | 69       | 20683              | 119      | 21183              | 169      | 21683              | 219      | 22183              | 269      | 22683              |
| 20       | 20193              | 70       | 20693              | 120      | 21193              | 170      | 21693              | 220      | 22193              | 270      | 22693              |
| 21       | 20203              | 71       | 20703              | 121      | 21203              | 171      | 21703              | 221      | 22203              | 271      | 22703              |
| 22       | 20213              | 72       | 20713              | 122      | 21213              | 172      | 21713              | 222      | 22213              | 272      | 22713              |
| 23       | 20223              | 73       | 20723              | 123      | 21223              | 173      | 21723              | 223      | 22223              | 273      | 22723              |
| 24       | 20233              | 74       | 20733              | 124      | 21233              | 174      | 21733              | 224      | 22233              | 274      | 22733              |
| 25       | 20243              | 75       | 20743              | 125      | 21243              | 175      | 21743              | 225      | 22243              | 275      | 22743              |
| 26       | 20253              | 76       | 20753              | 126      | 21253              | 176      | 21753              | 226      | 22253              | 276      | 22753              |
| 27       | 20263              | 77       | 20763              | 127      | 21263              | 177      | 21763              | 227      | 22263              | 277      | 22763              |
| 28       | 20273              | 78       | 20773              | 128      | 21273              | 178      | 21773              | 228      | 22273              | 278      | 22773              |
| 29       | 20283              | 79       | 20783              | 129      | 21283              | 179      | 21783              | 229      | 22283              | 279      | 22783              |
| 30       | 20293              | 80       | 20793              | 130      | 21293              | 180      | 21793              | 230      | 22293              | 280      | 22793              |
| 31       | 20303              | 81       | 20803              | 131      | 21303              | 181      | 21803              | 231      | 22303              | 281      | 22803              |
| 32       | 20313              | 82       | 20813              | 132      | 21313              | 182      | 21813              | 232      | 22313              | 282      | 22813              |
| 33       | 20323              | 83       | 20823              | 133      | 21323              | 183      | 21823              | 233      | 22323              | 283      | 22823              |
| 34       | 20333              | 84       | 20833              | 134      | 21333              | 184      | 21833              | 234      | 22333              | 284      | 22833              |
| 35       | 20343              | 85       | 20843              | 135      | 21343              | 185      | 21843              | 235      | 22343              | 285      | 22843              |
| 36       | 20353              | 86       | 20853              | 136      | 21353              | 186      | 21853              | 236      | 22353              | 286      | 22853              |
| 37       | 20363              | 87       | 20863              | 137      | 21363              | 187      | 21863              | 237      | 22363              | 287      | 22863              |
| 38       | 20373              | 88       | 20873              | 138      | 21373              | 188      | 21873              | 238      | 22373              | 288      | 22873              |
| 39       | 20383              | 89       | 20883              | 139      | 21383              | 189      | 21883              | 239      | 22383              | 289      | 22883              |
| 40       | 20393              | 90       | 20893              | 140      | 21393              | 190      | 21893              | 240      | 22393              | 290      | 22893              |
| 41       | 20403              | 91       | 20903              | 141      | 21403              | 191      | 21903              | 241      | 22403              | 291      | 22903              |
| 42       | 20413              | 92       | 20913              | 142      | 21413              | 192      | 21913              | 242      | 22413              | 292      | 22913              |
| 43       | 20423              | 93       | 20923              | 143      | 21423              | 193      | 21923              | 243      | 22423              | 293      | 22923              |
| 44       | 20433              | 94       | 20933              | 144      | 21433              | 194      | 21933              | 244      | 22433              | 294      | 22933              |
| 45       | 20443              | 95       | 20943              | 145      | 21443              | 195      | 21943              | 245      | 22443              | 295      | 22943              |
| 46       | 20453              | 96       | 20953              | 146      | 21453              | 196      | 21953              | 246      | 22453              | 296      | 22953              |
| 47       | 20463              | 97       | 20963              | 147      | 21463              | 197      | 21963              | 247      | 22463              | 297      | 22963              |
| 48       | 20473              | 98       | 20973              | 148      | 21473              | 198      | 21973              | 248      | 22473              | 298      | 22973              |
| 49       | 20483              | 99       | 20983              | 149      | 21483              | 199      | 21983              | 249      | 22483              | 299      | 22983              |
| 50       | 20493              | 100      | 20993              | 150      | 21493              | 200      | 21993              | 250      | 22493              | 300      | 22993              |

## (4) For axis 4 (positioning option)

| Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option | Data No. | Positioning option |
|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|----------|--------------------|
| 301      | 23003              | 351      | 23503              | 401      | 24003              | 451      | 24503              | 501      | 25003              | 551      | 25503              |
| 302      | 23013              | 352      | 23513              | 402      | 24013              | 452      | 24513              | 502      | 25013              | 552      | 25513              |
| 303      | 23023              | 353      | 23523              | 403      | 24023              | 453      | 24523              | 503      | 25023              | 553      | 25523              |
| 304      | 23033              | 354      | 23533              | 404      | 24033              | 454      | 24533              | 504      | 25033              | 554      | 25533              |
| 305      | 23043              | 355      | 23543              | 405      | 24043              | 455      | 24543              | 505      | 25043              | 555      | 25543              |
| 306      | 23053              | 356      | 23553              | 406      | 24053              | 456      | 24553              | 506      | 25053              | 556      | 25553              |
| 307      | 23063              | 357      | 23563              | 407      | 24063              | 457      | 24563              | 507      | 25063              | 557      | 25563              |
| 308      | 23073              | 358      | 23573              | 408      | 24073              | 458      | 24573              | 508      | 25073              | 558      | 25573              |
| 309      | 23083              | 359      | 23583              | 409      | 24083              | 459      | 24583              | 509      | 25083              | 559      | 25583              |
| 310      | 23093              | 360      | 23593              | 410      | 24093              | 460      | 24593              | 510      | 25093              | 560      | 25593              |
| 311      | 23103              | 361      | 23603              | 411      | 24103              | 461      | 24603              | 511      | 25103              | 561      | 25603              |
| 312      | 23113              | 362      | 23613              | 412      | 24113              | 462      | 24613              | 512      | 25113              | 562      | 25613              |
| 313      | 23123              | 363      | 23623              | 413      | 24123              | 463      | 24623              | 513      | 25123              | 563      | 25623              |
| 314      | 23133              | 364      | 23633              | 414      | 24133              | 464      | 24633              | 514      | 25133              | 564      | 25633              |
| 315      | 23143              | 365      | 23643              | 415      | 24143              | 465      | 24643              | 515      | 25143              | 565      | 25643              |
| 316      | 23153              | 366      | 23653              | 416      | 24153              | 466      | 24653              | 516      | 25153              | 566      | 25653              |
| 317      | 23163              | 367      | 23663              | 417      | 24163              | 467      | 24663              | 517      | 25163              | 567      | 25663              |
| 318      | 23173              | 368      | 23673              | 418      | 24173              | 468      | 24673              | 518      | 25173              | 568      | 25673              |
| 319      | 23183              | 369      | 23683              | 419      | 24183              | 469      | 24683              | 519      | 25183              | 569      | 25683              |
| 320      | 23193              | 370      | 23693              | 420      | 24193              | 470      | 24693              | 520      | 25193              | 570      | 25693              |
| 321      | 23203              | 371      | 23703              | 421      | 24203              | 471      | 24703              | 521      | 25203              | 571      | 25703              |
| 322      | 23213              | 372      | 23713              | 422      | 24213              | 472      | 24713              | 522      | 25213              | 572      | 25713              |
| 323      | 23223              | 373      | 23723              | 423      | 24223              | 473      | 24723              | 523      | 25223              | 573      | 25723              |
| 324      | 23233              | 374      | 23733              | 424      | 24233              | 474      | 24733              | 524      | 25233              | 574      | 25733              |
| 325      | 23243              | 375      | 23743              | 425      | 24243              | 475      | 24743              | 525      | 25243              | 575      | 25743              |
| 326      | 23253              | 376      | 23753              | 426      | 24253              | 476      | 24753              | 526      | 25253              | 576      | 25753              |
| 327      | 23263              | 377      | 23763              | 427      | 24263              | 477      | 24763              | 527      | 25263              | 577      | 25763              |
| 328      | 23273              | 378      | 23773              | 428      | 24273              | 478      | 24773              | 528      | 25273              | 578      | 25773              |
| 329      | 23283              | 379      | 23783              | 429      | 24283              | 479      | 24783              | 529      | 25283              | 579      | 25783              |
| 330      | 23293              | 380      | 23793              | 430      | 24293              | 480      | 24793              | 530      | 25293              | 580      | 25793              |
| 331      | 23303              | 381      | 23803              | 431      | 24303              | 481      | 24803              | 531      | 25303              | 581      | 25803              |
| 332      | 23313              | 382      | 23813              | 432      | 24313              | 482      | 24813              | 532      | 25313              | 582      | 25813              |
| 333      | 23323              | 383      | 23823              | 433      | 24323              | 483      | 24823              | 533      | 25323              | 583      | 25823              |
| 334      | 23333              | 384      | 23833              | 434      | 24333              | 484      | 24833              | 534      | 25333              | 584      | 25833              |
| 335      | 23343              | 385      | 23843              | 435      | 24343              | 485      | 24843              | 535      | 25343              | 585      | 25843              |
| 336      | 23353              | 386      | 23853              | 436      | 24353              | 486      | 24853              | 536      | 25353              | 586      | 25853              |
| 337      | 23363              | 387      | 23863              | 437      | 24363              | 487      | 24863              | 537      | 25363              | 587      | 25863              |
| 338      | 23373              | 388      | 23873              | 438      | 24373              | 488      | 24873              | 538      | 25373              | 588      | 25873              |
| 339      | 23383              | 389      | 23883              | 439      | 24383              | 489      | 24883              | 539      | 25383              | 589      | 25883              |
| 340      | 23393              | 390      | 23893              | 440      | 24393              | 490      | 24893              | 540      | 25393              | 590      | 25893              |
| 341      | 23403              | 391      | 23903              | 441      | 24403              | 491      | 24903              | 541      | 25403              | 591      | 25903              |
| 342      | 23413              | 392      | 23913              | 442      | 24413              | 492      | 24913              | 542      | 25413              | 592      | 25913              |
| 343      | 23423              | 393      | 23923              | 443      | 24423              | 493      | 24923              | 543      | 25423              | 593      | 25923              |
| 344      | 23433              | 394      | 23933              | 444      | 24433              | 494      | 24933              | 544      | 25433              | 594      | 25933              |
| 345      | 23443              | 395      | 23943              | 445      | 24443              | 495      | 24943              | 545      | 25443              | 595      | 25943              |
| 346      | 23453              | 396      | 23953              | 446      | 24453              | 496      | 24953              | 546      | 25453              | 596      | 25953              |
| 347      | 23463              | 397      | 23963              | 447      | 24463              | 497      | 24963              | 547      | 25463              | 597      | 25963              |
| 348      | 23473              | 398      | 23973              | 448      | 24473              | 498      | 24973              | 548      | 25473              | 598      | 25973              |
| 349      | 23483              | 399      | 23983              | 449      | 24483              | 499      | 24983              | 549      | 25483              | 599      | 25983              |
| 350      | 23493              | 400      | 23993              | 450      | 24493              | 500      | 24993              | 550      | 25493              | 600      | 25993              |

## Appendix 4 Connection examples with servo amplifiers manufactured by MITSUBISHI Electric Corporation

### Appendix 4.1 Connection example of QD75D□N and MR-J3-□ A (Differential driver) \*5

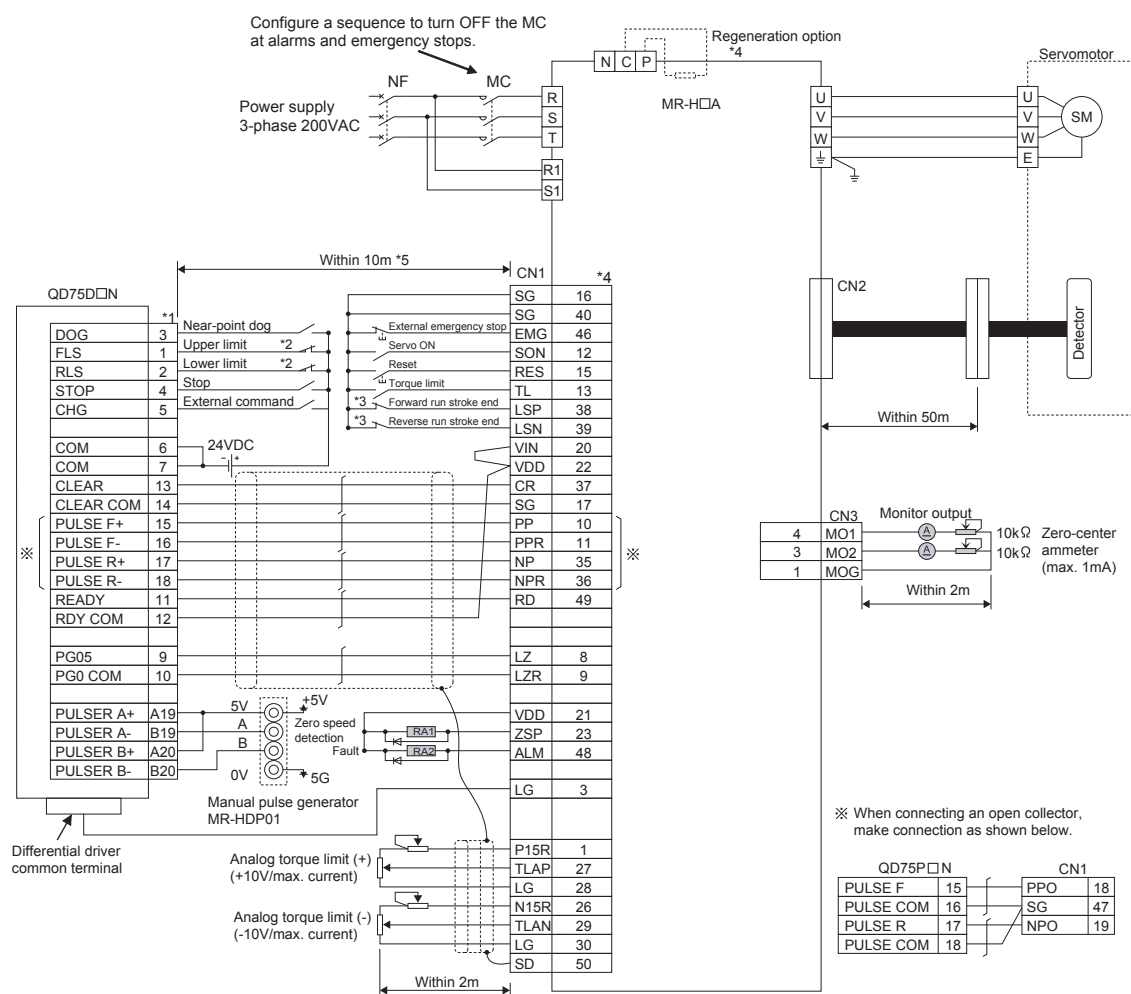


#### REMARK

- (1) It is recommended to make differential driver connection since differential driver connection is more excellent than open collector connection in max. output pulse and max. connection distance between servos. (Refer to Section 3.1 "Performance specifications".)
- (2) \*1: The logic for each I/O terminal can be changed with " **Pr.22** Input signal logic selection" and " **Pr.23** Output signal logic selection" in detailed parameters 1. (Negative logic is used for all terminals in the example above.)
- (3) \*2: The QD75D□N upper limit (FLS) and lower limit (RLS) are used in the OPR retry function. Set these signals inside the servo amplifier limit switches.
- (4) \*3: These are limit switches for the servo (for stop).
- (5) \*4: This indicates the distance between the QD75D□N and servo amplifier.
- (6) \*5: When used, check the logic for the QD75D□N and servo amplifier in Section 3.4.1, (2) "Output specifications". The QD75D□N is initially set to negative logic.
- (7) "FA-CBLQ75M2J3(-P) cable" can be used for connecting the QD75D□N and MR-J3-□A.



## Appendix 4.2 Connection example of QD75D□N and MR-H □ A (Differential driver) \*6

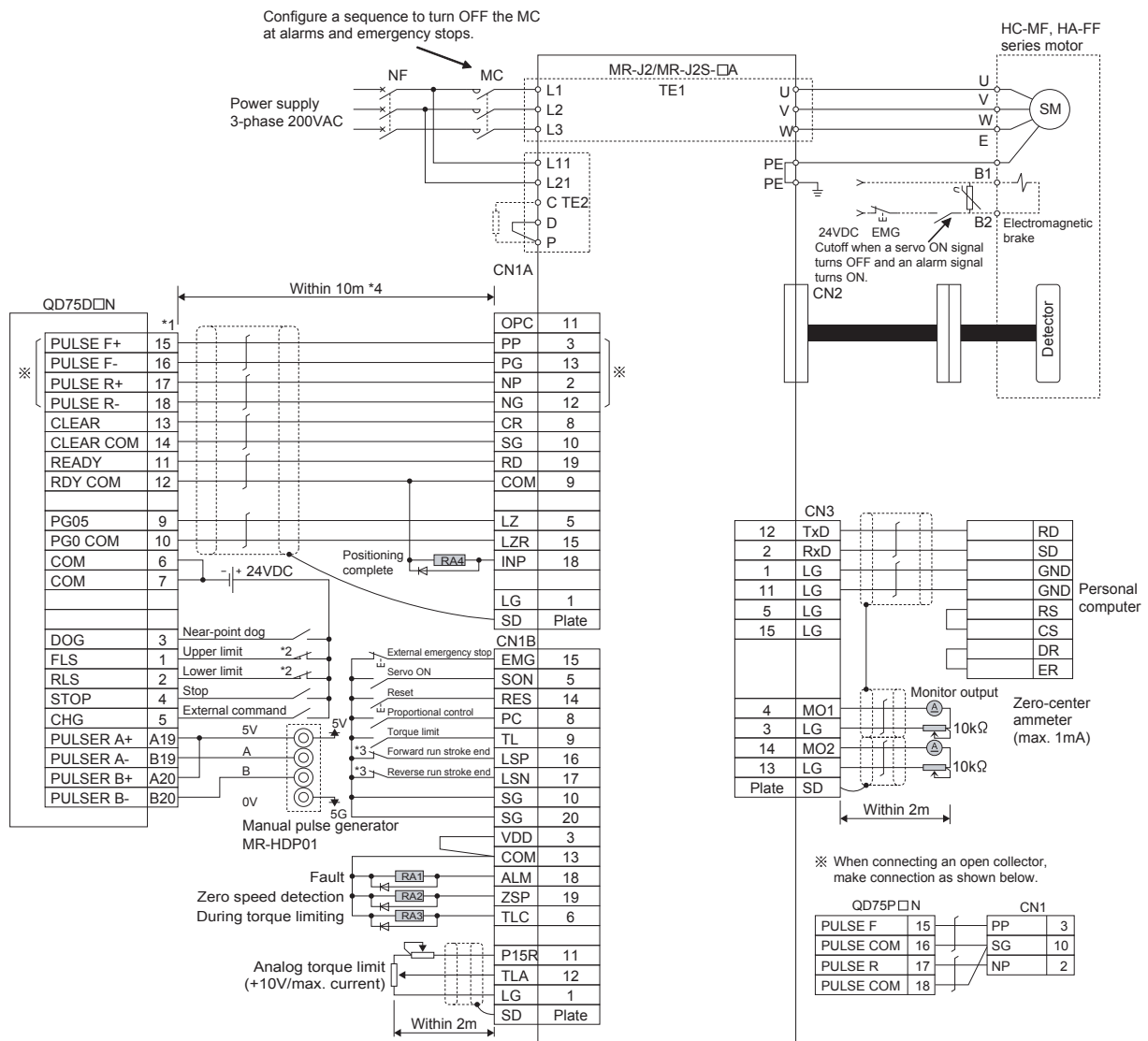


### REMARK

- (1) It is recommended to make differential driver connection since differential driver connection is more excellent than open collector connection in max. output pulse and max. connection distance between servos. (Refer to Section 3.1 "Performance specifications".)
- (2) \*1: The logic for each I/O terminal can be changed with " **Pr.22** Input signal logic selection" and " **Pr.23** Output signal logic selection" in detailed parameters 1. (Negative logic is used for all terminals in the example above.)
- (3) \*2: The QD75D□N upper limit (FLS) and lower limit (RLS) are used in the OPR retry function. Set these signals inside the servo amplifier limit switches.
- (4) \*3: These are limit switches for the servo (for stop).
- (5) \*4: Refer to the specification and handling instruction manual of the servo amplifier MR-H for details on connection.
- (6) \*5: This indicates the distance between the QD75D□N and servo amplifier.
- (7) \*6: Use the same logic (positive logic/negative logic) for the QD75D□N and servo amplifier. The QD75D□N is initially set to negative logic.

## Appendix 4.3 Connection example of QD75D□N and MR-J2/J2S- □ A (Differential driver)

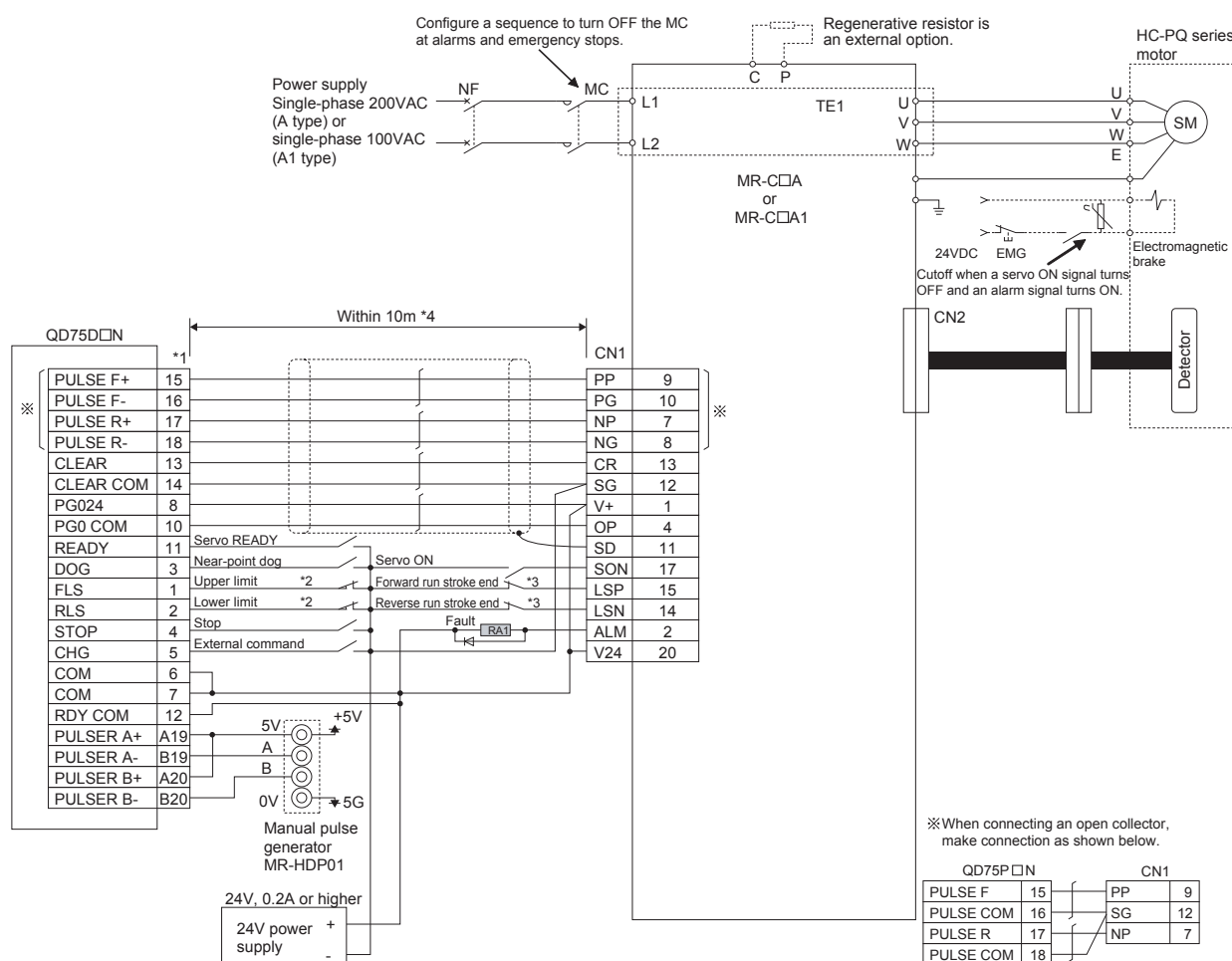
\*5



## REMARK

- (1) It is recommended to make differential driver connection since differential driver connection is more excellent than open collector connection in max. output pulse and max. connection distance between servos. (Refer to Section 3.1 "Performance specifications".)
- (2) \*1: The logic for each I/O terminal can be changed with " Pr.22 Input signal logic selection" and " Pr.23 Output signal logic selection" in detailed parameters 1. (Negative logic is used for all terminals in the example above.)
- (3) \*2: The QD75D□N upper limit (FLS) and lower limit (RLS) are used in the OPR retry function. Set these signals inside the servo amplifier limit switches.
- (4) \*3: These are limit switches for the servo amplifier (for stop).
- (5) \*4: This indicates the distance between the QD75D□N and servo amplifier.
- (6) \*5: Use the same logic (positive logic/negative logic) for the QD75D□N and servo amplifier. The QD75D□N is initially set to negative logic.

# Appendix 4.4 Connection example of QD75D□N and MR-C □ A (Differential driver)\*5

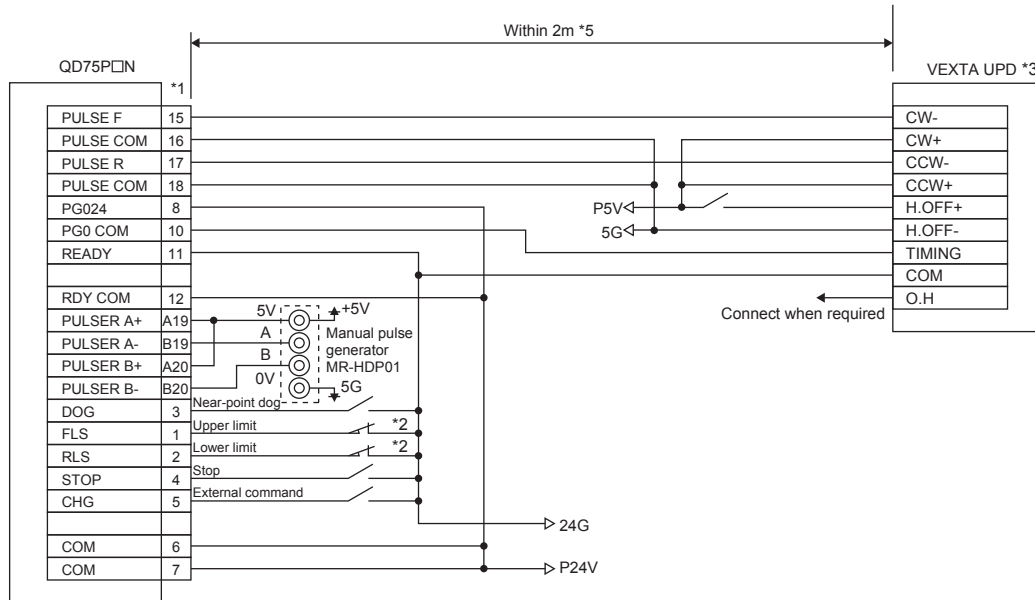


## REMARK

- (1) It is recommended to make differential driver connection since differential driver connection is more excellent than open collector connection in max. output pulse and max. connection distance between servos. (Refer to Section 3.1 "Performance specifications".)
- (2) \*1: The logic for each I/O terminal can be changed with " **Pr.22** Input signal logic selection" and " **Pr.23** Output signal logic selection" in detailed parameters 1. (Negative logic is used for all terminals in the example above.)
- (3) \*2: The QD75D□N upper limit (FLS) and lower limit (RLS) are used in the OPR retry function. Set these signals inside the servo amplifier limit switches.
- (4) \*3: These are limit switches for the servo amplifier (for stop).
- (5) \*4: This indicates the distance between the QD75D□N and servo amplifier.
- (6) \*5: Use the same logic (positive logic/negative logic) for the QD75D□N and servo amplifier. The QD75D□N is initially set to negative logic.

## Appendix 5 Connection examples with stepping motors manufactured by ORIENTAL MOTOR Co., Ltd.

### Appendix 5.1 Connection example of QD75P□N and VEXTA UPD (Open collector) \*4

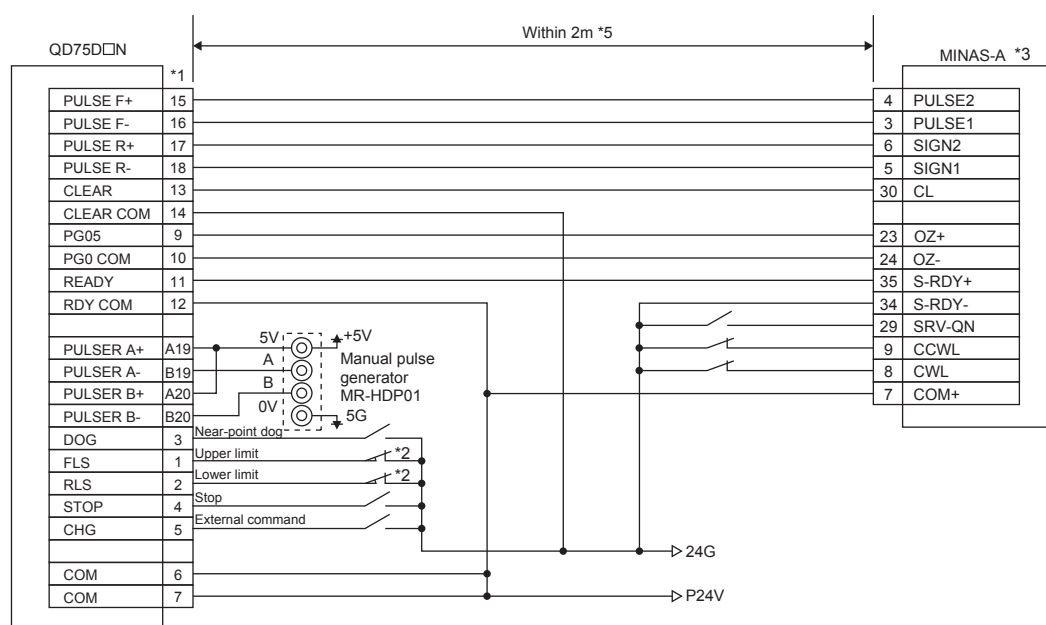


#### REMARK

- (1) \*1: The logic for each I/O terminal can be changed with " **Pr.22** Input signal logic selection" and " **Pr.23** Output signal logic selection" in detailed parameters 1. (Negative logic is used for all terminals in the example above.)
- (2) \*2: The QD75P□N upper limit (FLS) and lower limit (RLS) are used in the OPR retry function.
- (3) \*3: Refer to the manual of the stepping motor drive for information on the stepping motor drive side wiring and various signal wire shields not shown above.
- (4) \*4: Use the same logic (positive logic/negative logic) for the QD75P□N and stepping motor. The QD75P□N is initially set to negative logic.
- (5) \*5: This indicates the distance between the QD75P□N and VEXTA UPD.

## Appendix 6 Connection examples with servo amplifiers manufactured by Panasonic Corporation

### Appendix 6.1 Connection example of QD75D□N and MINAS-A series (Differential driver) \*4

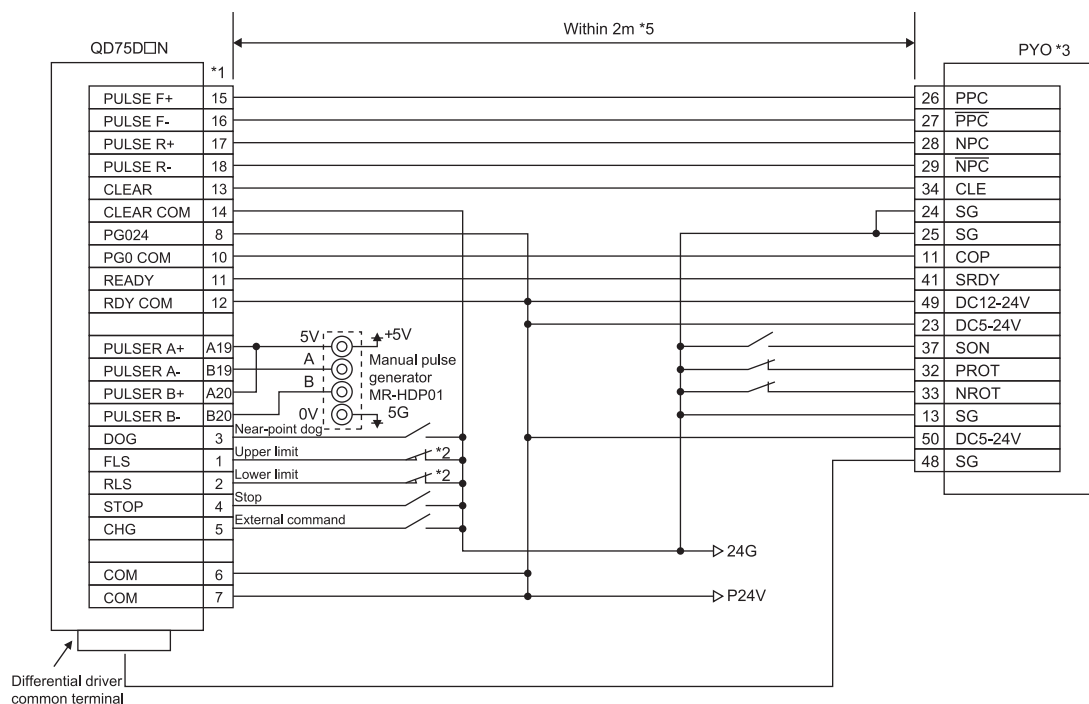


#### REMARK

- (1) \*1: The logic for each I/O terminal can be changed with " **Pr.22** Input signal logic selection" and " **Pr.23** Output signal logic selection" in detailed parameters 1. (Negative logic is used for all terminals in the example above.)
- (2) \*2: The QD75D□N upper limit (FLS) and lower limit (RLS) are used in the OPR retry function. Set these signals inside the servo amplifier limit switches.
- (3) \*3: Refer to the manual of the servo amplifier for information on the servo amplifier side wiring and various signal wire shields not shown above.
- (4) \*4: Use the same logic (positive logic/negative logic) for the QD75D□N and servo amplifier. The QD75D□N is initially set to negative logic.
- (5) \*5: This indicates the distance between the QD75D□N and MINAS-A.

## Appendix 7 Connection examples with servo amplifiers manufactured by SANYO DENKI Co., Ltd.

### Appendix 7.1 Connection example of QD75D□N and PYO series (Differential driver) \*4

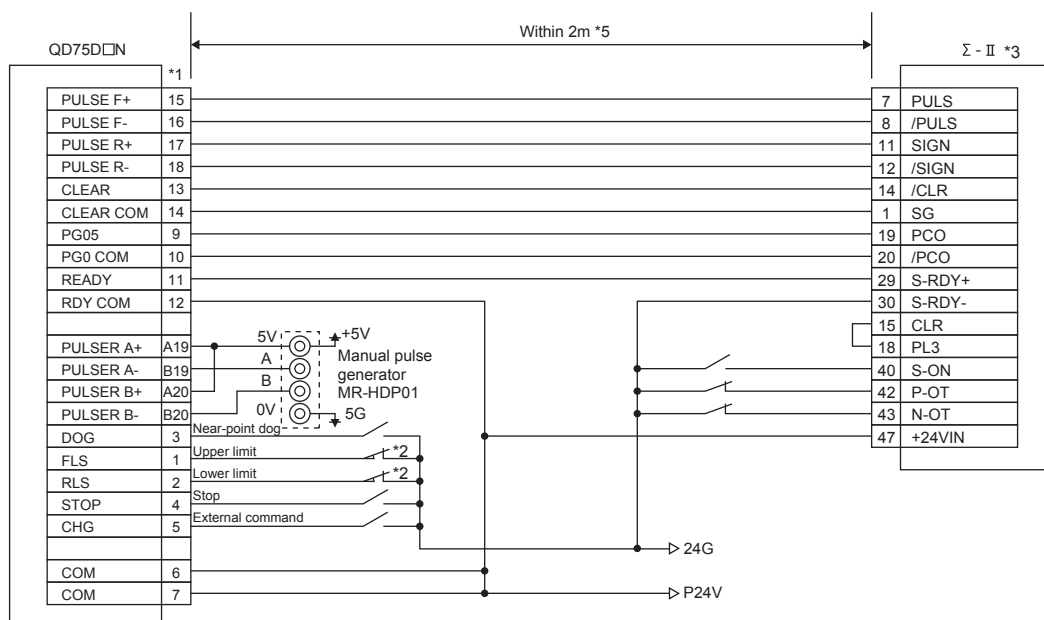


#### REMARK

- (1) \*1: The logic for each I/O terminal can be changed with " Pr.22 Input signal logic selection" and " Pr.23 Output signal logic selection" in detailed parameters 1. (Negative logic is used for all terminals in the example above.)
- (2) \*2: The QD75D□N upper limit (FLS) and lower limit (RLS) are used in the OPR retry function. Set these signals inside the servo amplifier limit switches.
- (3) \*3: Refer to the manual of the servo amplifier for information on the servo amplifier side wiring and various signal wire shields not shown above.
- (4) \*4: Use the same logic (positive logic/negative logic) for the QD75D□N and servo amplifier. The QD75D□N is initially set to negative logic.
- (5) \*5: This indicates the distance between the QD75D□N and PYO.

## Appendix 8 Connection examples with servo amplifiers manufactured by YASKAWA Electric Corporation

### Appendix 8.1 Connection example of QD75D□N and $\Sigma$ -II series (Differential driver)\*<sup>4</sup>



#### REMARK

- (1) \*1: The logic of each I/O terminal can be changed with " **Pr.22** Input signal logic selection" and " **Pr.23** Output signal logic selection" in detailed parameters 1. (Negative logic is used for all terminals in the example above.)
- (2) \*2: The QD75D□N upper limit (FLS) and lower limit (RLS) are used in the OPR retry function. Set these signals inside the servo amplifier limit switches.
- (3) \*3: Refer to the manual of the servo amplifier for information on the servo amplifier side wiring and various signal wire shields not shown above.
- (4) \*4: Use the same logic (positive logic/negative logic) for the QD75D□N and servo amplifier. The QD75D□N is initially set to negative logic.
- (5) \*5: This indicates the distance between the QD75D□N and  $\Sigma$ -II.

## Appendix 9 Comparisons with conventional positioning modules

## Appendix 9.1 Comparisons with A1SD71S2 model

The following shows comparisons with the conventional positioning module A1SD71S2, centered on the QD75 specifications.

| Model                                        |                                       | QD75P1N/QD75P1<br>QD75D1N/QD75D1                                           | QD75P2N/QD75P2<br>QD75D2N/QD75D2 | QD75P4N/QD75P4<br>QD75D4N/QD75D4 | A1SD71S2                            |
|----------------------------------------------|---------------------------------------|----------------------------------------------------------------------------|----------------------------------|----------------------------------|-------------------------------------|
| Item                                         |                                       |                                                                            |                                  |                                  |                                     |
| No. of control axes                          |                                       | 1                                                                          | 2                                | 4                                | 2                                   |
| No. of positioning data items                |                                       | 600/axis                                                                   |                                  |                                  | 400/axis                            |
| Interpolation functions                      | 2-axis linear interpolation           | ×                                                                          | ○                                | ○                                | ○                                   |
|                                              | 3-axis linear interpolation           | ×                                                                          | ×                                | ○                                | ×                                   |
|                                              | 4-axis linear interpolation           | ×                                                                          | ×                                | ○                                | ×                                   |
|                                              | 2-axis circular interpolation         | ×                                                                          | ○                                | ○                                | ×                                   |
| Positioning systems                          | Position control                      | ○                                                                          |                                  |                                  | ○                                   |
|                                              | Speed control                         | ○                                                                          |                                  |                                  | ○                                   |
|                                              | Speed-position switching control      | ○ (INC/ABS mode)                                                           |                                  |                                  | ○ (Only INC mode)                   |
|                                              | Position-speed switching control      | ○                                                                          |                                  |                                  | ×                                   |
| Machine OPR function (OPR method)            |                                       | ○ (6 types)                                                                |                                  |                                  | ○ (3 types)                         |
| JOG operation                                |                                       | ○                                                                          |                                  |                                  | ○                                   |
| Inching operation                            |                                       | ○                                                                          |                                  |                                  | ×                                   |
| Manual pulse generator function              |                                       | ○                                                                          |                                  |                                  | ×                                   |
| Acceleration/deceleration processing         | Trapezoidal acceleration/deceleration | ○                                                                          |                                  |                                  | ○                                   |
|                                              | S-curve acceleration/deceleration     | ○                                                                          |                                  |                                  | ×                                   |
| Acceleration/deceleration time               |                                       | Acceleration time and deceleration time setting possible (4 patterns each) |                                  |                                  | Acceleration/deceleration time same |
| Compensation                                 |                                       | Electronic gears, backlash compensation                                    |                                  |                                  | Backlash compensation               |
| Error display                                |                                       | Error LED                                                                  |                                  |                                  | Error LED                           |
| History data storage (Start, error, warning) |                                       | Provided (3 types, 16 items/axis)                                          |                                  |                                  | None                                |
| Data storage destination                     |                                       | Flash ROM (battery-less backup)                                            |                                  |                                  | Buffer memory (battery backup)      |
| No. of occupied I/O points                   |                                       | 32                                                                         |                                  |                                  | 48                                  |
| No. of module occupied slots                 |                                       | 1                                                                          |                                  |                                  | 2                                   |
| Peripheral devices (data setting, etc.)      | AD71TU                                | ×                                                                          |                                  |                                  | ○                                   |
|                                              | AD75TU                                | ×                                                                          |                                  |                                  | ×                                   |
|                                              | A6GPP, A6PHP                          | ×                                                                          |                                  |                                  | ○                                   |
|                                              | DOS/V personal computer               | ○                                                                          |                                  |                                  | ○                                   |

○ : Possible, × : Not possible



## Appendix 9.2 Comparisons with A1SD75P1-S3/ A1SD75P2-S3/ A1SD75P3-S3 models

The following shows the comparisons between the QD75 and the conventional positioning modules A1SD75P1-S3/A1SD75P2-S3/A1SD75P3-S3.

## (1) Comparisons of performance specifications

| Model                                    |                                  | QD75P1N/QD75P1<br>QD75D1N/QD75D1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | QD75P2N/QD75P2<br>QD75D2N/QD75D2 | QD75P4N/QD75P4<br>QD75D4N/QD75D4 | A1SD75P1<br>-S3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | A1SD75P2<br>-S3 | A1SD75P3<br>-S3 |
|------------------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|
| Item                                     |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                  |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                 |                 |
| No. of control axes                      |                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2                                | 4                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2               | 3               |
| No. of positioning data items            |                                  | 600/axis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                  |                                  | 600/axis * 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 |                 |
| Position control interpolation functions | 2-axis linear interpolation      | ×                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ○                                | ○                                | ×                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ○               | ○               |
|                                          | 3-axis linear interpolation      | ×                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ×                                | ○                                | ×                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ×               | ×               |
|                                          | 4-axis linear interpolation      | ×                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ×                                | ○                                | ×                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ×               | ×               |
|                                          | 2-axis circular interpolation    | ×                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ○                                | ○                                | ×                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ○               | ○               |
| Positioning systems                      | Position control                 | ○                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                  |                                  | ○                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 |                 |
|                                          | Speed control                    | 1-axis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | ○                                | ○                                | ○                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ○               | ○               |
|                                          |                                  | 2-axis linear interpolation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ×                                | ○                                | ×                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ×               | ×               |
|                                          |                                  | 3-axis linear interpolation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ×                                | ×                                | ×                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ×               | ×               |
|                                          |                                  | 4-axis linear interpolation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ×                                | ×                                | ○                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ×               | ×               |
|                                          | Speed-position switching control | ○                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                  |                                  | ○                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 |                 |
|                                          | Position-speed switching control | ○                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                  |                                  | ×                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 |                 |
| Positioning range * 2                    |                                  | <Absolute system><br>-214748364.8 to 214748364.7 (μm)<br>-21474.83648 to 21474.83647 (inch)<br>0 to 359.99999 (degree)<br>-2147483648 to 2147483647 (pulse)<br><Incremental system ><br>-214748364.8 to 214748364.7 (μm)<br>-21474.83648 to 21474.83647 (inch)<br>-21474.83648 to 21474.83647 (degree)<br>-2147483648 to 2147483647 (pulse)<br><Speed-position or position-speed switching controls><br>0 to 214748364.7 (μm)<br>0 to 21474.83647 (inch)<br>0 to 21474.83647 (degree)/<br>0 to 359.99999 (degree) * 3<br>0 to 2147483647 (pulse) |                                  |                                  | <Absolute system ><br>-214748364.8 to 214748364.7 (μm)<br>/-13421772.8 to 13421772.7 (μm)<br>-21474.83648 to 21474.83647 (inch)<br>/-1342.17728 to 1342.17727 (inch)<br>0 to 359.99999 (degree)<br>/0 to 359.99999 (degree)<br>-2147483648 to 2147483647 (pulse)<br>/-134217728 to 134217727 (pulse)<br><Incremental system ><br>-214748364.8 to 214748364.7 (μm)<br>/-13421772.8 to 13421772.7 (μm)<br>-21474.83648 to 21474.83647 (inch)<br>/-1342.17728 to 1342.17727 (inch)<br>-21474.83648 to 21474.83647 (degree)<br>/-1342.17728 to 1342.17727 (degree)<br>-2147483648 to 2147483647 (pulse)<br>/-134217728 to 134217727 (pulse)<br><Speed-position switching control><br>0 to 214748364.7 (μm)<br>/0 to 13421772.7 (μm)<br>0 to 21474.83647 (inch)<br>/0 to 1342.17727 (inch)<br>0 to 21474.83647 (degree)<br>/0 to 1342.17727 (degree)<br>0 to 2147483647 (pulse)<br>/0 to 134217727 (pulse) |                 |                 |

| Model                                        |                                                                                    | QD75P1N/QD75P1<br>QD75D1N/QD75D1                                                                                                                                                                                                                                                                      | QD75P2N/QD75P2<br>QD75D2N/QD75D2 | QD75P4N/QD75P4<br>QD75D4N/QD75D4                                | A1SD75P1<br>-S3                                                                                                                                                                                                                                           | A1SD75P2<br>-S3 | A1SD75P3<br>-S3 |
|----------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|
| Item                                         |                                                                                    |                                                                                                                                                                                                                                                                                                       |                                  |                                                                 |                                                                                                                                                                                                                                                           |                 |                 |
| Speed command range * 2                      |                                                                                    | QD75P□N/QD75D□N:<br>0.01 to 2000000.00 (mm/min)<br>0.001 to 2000000.000 (inch/min)<br>0.001 to 2000000.000 (degree/min)<br>1 to 4000000 (pulse/s)<br>QD75P□/QD75D□ :<br>0.01 to 2000000.00 (mm/min)<br>0.001 to 2000000.000 (inch/min)<br>0.001 to 2000000.000 (degree/min)<br>1 to 1000000 (pulse/s) |                                  |                                                                 | 0.01 to 6000000.00 (mm/min)<br>/0.01 to 375000.00 (mm/min)<br>0.001 to 600000.000 (inch/min)<br>/0.001 to 37500.000 (inch/min)<br>0.001 to 600000.000 (degree/min)<br>/0.001 to 37500.000 (degree/min)<br>1 to 1000000 (pulse/s)<br>/1 to 62500 (pulse/s) |                 |                 |
| Machine OPR function (OPR method)            |                                                                                    | ○ (6 types) * 4                                                                                                                                                                                                                                                                                       |                                  |                                                                 | ○ (6 types)                                                                                                                                                                                                                                               |                 |                 |
| JOG operation                                |                                                                                    | ○                                                                                                                                                                                                                                                                                                     |                                  |                                                                 | ○                                                                                                                                                                                                                                                         |                 |                 |
| Inching operation                            |                                                                                    | ○                                                                                                                                                                                                                                                                                                     |                                  |                                                                 | ×                                                                                                                                                                                                                                                         |                 |                 |
| Manual pulse generator function              |                                                                                    | 1 generator/module * 5                                                                                                                                                                                                                                                                                |                                  |                                                                 | 1 generator/axis * 6                                                                                                                                                                                                                                      |                 |                 |
| Acceleration/deceleration processing         | Trapezoidal acceleration/deceleration                                              | ○                                                                                                                                                                                                                                                                                                     |                                  |                                                                 | ○                                                                                                                                                                                                                                                         |                 |                 |
|                                              | S-curve acceleration/deceleration                                                  | ○                                                                                                                                                                                                                                                                                                     |                                  |                                                                 | ○                                                                                                                                                                                                                                                         |                 |                 |
| Acceleration/deceleration time               | No. of patterns                                                                    | Acceleration time and deceleration time can be set independently. (4 patterns each)                                                                                                                                                                                                                   |                                  |                                                                 | Acceleration time and deceleration time can be set independently. (4 patterns each)                                                                                                                                                                       |                 |                 |
|                                              | Setting range                                                                      | 1 to 8388608ms                                                                                                                                                                                                                                                                                        |                                  |                                                                 | 1 to 65535ms/1 to 8388608ms selectable                                                                                                                                                                                                                    |                 |                 |
| Compensation                                 |                                                                                    | Electronic gears, backlash compensation, near pass * 6                                                                                                                                                                                                                                                |                                  |                                                                 | Electronic gears, backlash compensation, near pass * 6                                                                                                                                                                                                    |                 |                 |
| Error display                                |                                                                                    | Error LED                                                                                                                                                                                                                                                                                             |                                  |                                                                 | 17-segment LED                                                                                                                                                                                                                                            |                 |                 |
| History data storage (Start, error, warning) |                                                                                    | Provided (3 types, 16 items/axis)                                                                                                                                                                                                                                                                     |                                  |                                                                 | Provided (4 types, 16 items/axis)                                                                                                                                                                                                                         |                 |                 |
| Data storage destination                     |                                                                                    | Flash ROM (battery-less backup)                                                                                                                                                                                                                                                                       |                                  |                                                                 | Flash ROM (battery-less backup)                                                                                                                                                                                                                           |                 |                 |
| Connection connector                         | A6CON1 (Soldering type, sold separately)                                           |                                                                                                                                                                                                                                                                                                       |                                  | 10136-3000VE (Soldering type, supplied)                         |                                                                                                                                                                                                                                                           |                 |                 |
|                                              | A6CON2 (Crimp type, sold separately)                                               |                                                                                                                                                                                                                                                                                                       |                                  | 10136-6000EL (Crimp type, sold separately)                      |                                                                                                                                                                                                                                                           |                 |                 |
|                                              | A6CON4 (Soldering type, usable for straight out and diagonal out, sold separately) |                                                                                                                                                                                                                                                                                                       |                                  |                                                                 |                                                                                                                                                                                                                                                           |                 |                 |
| Applicable wire size                         | A6CON1, A6CON4: 0.3mm <sup>2</sup> (22AWG)                                         |                                                                                                                                                                                                                                                                                                       |                                  | 10136-3000VE: Approx. 0.05 to 0.24mm <sup>2</sup> (30 to 24AWG) |                                                                                                                                                                                                                                                           |                 |                 |
|                                              | A6CON2: 0.088 to 0.24mm <sup>2</sup> (28 to 24AWG)                                 |                                                                                                                                                                                                                                                                                                       |                                  | 10136-6000EL: Approx. 0.088mm <sup>2</sup> (28AWG)              |                                                                                                                                                                                                                                                           |                 |                 |
| Command pulse output system                  |                                                                                    | QD75P□N/QD75P□: Open collector<br>QD75D□N/QD75D□: Differential driver                                                                                                                                                                                                                                 |                                  |                                                                 | Both differential driver and open collector available                                                                                                                                                                                                     |                 |                 |
| Max. output pulse                            |                                                                                    | When connected with open collector: 200kpps<br>When connected with differential driver: 4Mpps (QD75D□N), 1Mpps (QD75D□)                                                                                                                                                                               |                                  |                                                                 | When connected with open collector: 200kpps<br>When connected with differential driver: 400kpps                                                                                                                                                           |                 |                 |
| Max. connection distance between servos      |                                                                                    | When connected with open collector: 2m (6.56ft)<br>When connected with differential driver: 10m (32.79ft)                                                                                                                                                                                             |                                  |                                                                 | When connected with open collector: 2m (6.56ft)<br>When connected with differential driver: 10m (32.79ft)                                                                                                                                                 |                 |                 |
| Internal current consumption [5VDC]          | QD75P1N: 0.29A                                                                     |                                                                                                                                                                                                                                                                                                       | QD75P2N: 0.30A                   | QD75P4N: 0.36A                                                  | 0.7A or less                                                                                                                                                                                                                                              |                 |                 |
|                                              | QD75D1N: 0.43A                                                                     |                                                                                                                                                                                                                                                                                                       | QD75D2N: 0.45A                   | QD75D4N: 0.66A                                                  |                                                                                                                                                                                                                                                           |                 |                 |
|                                              | QD75P1: 0.40A                                                                      |                                                                                                                                                                                                                                                                                                       | QD75P2: 0.46A                    | QD75P4: 0.58A                                                   |                                                                                                                                                                                                                                                           |                 |                 |
|                                              | QD75D1: 0.52A                                                                      |                                                                                                                                                                                                                                                                                                       | QD75D2: 0.56A                    | QD75D4: 0.82A                                                   |                                                                                                                                                                                                                                                           |                 |                 |
| Flash ROM write count                        |                                                                                    | Max. 100000 times                                                                                                                                                                                                                                                                                     |                                  |                                                                 | Max. 100000 times                                                                                                                                                                                                                                         |                 |                 |
| No. of occupied I/O points                   |                                                                                    | 32                                                                                                                                                                                                                                                                                                    |                                  |                                                                 | 32                                                                                                                                                                                                                                                        |                 |                 |
| No. of module occupied slots                 |                                                                                    | 1                                                                                                                                                                                                                                                                                                     |                                  |                                                                 | 1                                                                                                                                                                                                                                                         |                 |                 |
| Outline dimensions                           |                                                                                    | 98(3.86)(H)×27.4(1.08)(W)×90(3.55)(D)mm (inch)                                                                                                                                                                                                                                                        |                                  |                                                                 | 130(5.12)(H)×34.5(1.36)(W)×93.6(3.69)(D)mm (inch)                                                                                                                                                                                                         |                 |                 |
| Weight                                       | QD75P1N: 0.14kg (0.31lb)                                                           |                                                                                                                                                                                                                                                                                                       | QD75P2N: 0.14kg (0.31lb)         | QD75P4N: 0.16kg (0.35lb)                                        | 0.35kg (0.77lb)                                                                                                                                                                                                                                           |                 |                 |
|                                              | QD75D1N: 0.15kg (0.33lb)                                                           |                                                                                                                                                                                                                                                                                                       | QD75D2N: 0.15kg (0.33lb)         | QD75D4N: 0.16kg (0.35lb)                                        |                                                                                                                                                                                                                                                           |                 |                 |
|                                              | QD75P1: 0.15kg (0.33lb)                                                            |                                                                                                                                                                                                                                                                                                       | QD75P2: 0.15kg (0.33lb)          | QD75P4: 0.16kg (0.35lb)                                         |                                                                                                                                                                                                                                                           |                 |                 |
|                                              | QD75D1: 0.15kg (0.33lb)                                                            |                                                                                                                                                                                                                                                                                                       | QD75D2: 0.15kg (0.33lb)          | QD75D4: 0.16kg (0.35lb)                                         |                                                                                                                                                                                                                                                           |                 |                 |

| Model                                   |                                    | QD75P1N/QD75P1<br>QD75D1N/QD75D1                                                                                   | QD75P2N/QD75P2<br>QD75D2N/QD75D2 | QD75P4N/QD75P4<br>QD75D4N/QD75D4 | A1SD75P1<br>-S3                  | A1SD75P2<br>-S3 | A1SD75P3<br>-S3 |
|-----------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------------|----------------------------------|-----------------|-----------------|
| Item                                    |                                    |                                                                                                                    |                                  |                                  |                                  |                 |                 |
| I/O signal for external devices         | STRT signal                        | × (integrated into "CHG")                                                                                          |                                  |                                  | ○ (External start signal)        |                 |                 |
|                                         | CHG signal                         | External command signal (External start or speed-position switching selectable with parameters)                    |                                  |                                  | Speed-position switching signal  |                 |                 |
|                                         | In-position (INP)                  | ×                                                                                                                  |                                  |                                  | ○ (for monitor)                  |                 |                 |
|                                         | Signal logic switching             | ○                                                                                                                  |                                  |                                  | Command pulse output signal only |                 |                 |
| Peripheral devices (data setting, etc.) | Connection with peripheral devices | Connection via CPU module, Q Corresponding Serial Communication Module, Q Corresponding MELSEC/H Remote I/O Module |                                  |                                  | Direct connection                |                 |                 |
|                                         | AD71TU                             | ×                                                                                                                  |                                  |                                  | ×                                |                 |                 |
|                                         | AD75TU                             | ×                                                                                                                  |                                  |                                  | ○                                |                 |                 |
|                                         | A6GPP, A6PHP                       | ×                                                                                                                  |                                  |                                  | ○                                |                 |                 |
|                                         | DOS/V personal computer            | ○                                                                                                                  |                                  |                                  | ○                                |                 |                 |

○ : Possible, × : Not possible

- \* 1: Up to 100 data items/axis of positioning data (No. 1 to 100) can be set using the buffer memory. The positioning data in the buffer memory is not backed up.
- \* 2: Indicates the standard mode/stepping motor mode about A1SD75P□-S3.
- \* 3: Indicates the INC mode/ABS mode in speed-position switching control.
- \* 4: The deviation counter clear signal output time can be set with parameters.
- \* 5: The movement amount against the number of pulses input from manual pulsar is as follows:
- QD75: Movement amount = "Number of pulses" × "Unit setting" × "Manual pulse generator 1 pulse input magnification" (refer to Section 11.4.1.)
  - A1SD75: Movement amount = "Number of pulses" × "Manual pulse generator 1 pulse input magnification"
- \* 6: The near pass function is valid only during the continuous path control. (A1SD75: Selected with parameters, QD75: Standard function)

## (2) Function comparisons

Functions added to those of A1SD75P1-S3/A1SD75P2-S3/A1SD75P3-S3

| Added functions                                                  |                                                                                                                                                                                                                                                                                                           | Remarks                                                                   |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| External I/O signal logic switching function                     |                                                                                                                                                                                                                                                                                                           | Refer to Section 13.4.                                                    |
| Inching operation                                                |                                                                                                                                                                                                                                                                                                           | Refer to Section 11.3.                                                    |
| Target position change function                                  |                                                                                                                                                                                                                                                                                                           | Refer to Section 12.5.5.                                                  |
| Multiple axes simultaneous start control                         |                                                                                                                                                                                                                                                                                                           | Refer to Section 10.5.                                                    |
| Control systems                                                  | 3-axis linear interpolation control<br>4-axis linear interpolation control<br>3-axis fixed-feed control<br>4-axis fixed-feed control<br>2-axis speed control<br>3-axis speed control<br>4-axis speed control<br>Position-speed switching control<br>NOP instruction<br>LOOP instruction, LEND instruction | Refer to CHAPTER 9.                                                       |
| Dedicated instructions                                           | Absolute position restoration<br>Positioning start<br>Teaching<br>Flash ROM writing<br>Parameter initialization                                                                                                                                                                                           | Refer to CHAPTER 14.                                                      |
| Automatic refresh of intelligent function modules                |                                                                                                                                                                                                                                                                                                           | Refer to GX Configurator-QP Operating Manual.                             |
| Output hold/clear parameter setting during CPU module error stop |                                                                                                                                                                                                                                                                                                           | Refer to QCPU User's Manual (Function Explanation, Program Fundamentals). |
| Flash ROM write limit                                            |                                                                                                                                                                                                                                                                                                           | Refer to Section 13.3.                                                    |
| Speed-position switching control (ABS mode)                      |                                                                                                                                                                                                                                                                                                           | Refer to Section 9.2.19.                                                  |
| Pre-reading start function                                       |                                                                                                                                                                                                                                                                                                           | Refer to Section 12.7.7.                                                  |
| External I/O signal monitor function                             |                                                                                                                                                                                                                                                                                                           | Refer to Section 13.5.                                                    |
| Multiple PLC compatible function                                 |                                                                                                                                                                                                                                                                                                           | Refer to QCPU User's Manual (Multiple CPU System).                        |
| Deceleration start flag function                                 |                                                                                                                                                                                                                                                                                                           | Refer to Section 12.7.8                                                   |
| Stop command processing for deceleration stop function           |                                                                                                                                                                                                                                                                                                           | Refer to Section 12.7.9                                                   |

## Functions deleted from those of A1SD75P1-S3/A1SD75P2-S3/A1SD75P3-S3

| Deleted functions                                 | Remarks                                                                                                                                                                                                                     |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stepping motor mode                               | —                                                                                                                                                                                                                           |
| OPR operation error (Error code: 208)             | —                                                                                                                                                                                                                           |
| Fast machine OPR                                  | —                                                                                                                                                                                                                           |
| Special start (stop)                              | —                                                                                                                                                                                                                           |
| Indirect designation                              | In the QD75, the start block area on the buffer memory is expanded to five blocks (0 to 4). Each start block can be directly designated with positioning start No. (7000 to 7004).                                          |
| Block transfer                                    | With the A1SD75, this interface is used to set positioning data Nos. 101 to 600 that do not exist on the buffer memory. Since all positioning data can be set in the buffer memory with the QD75, this function is deleted. |
| Positioning data I/F                              |                                                                                                                                                                                                                             |
| Start history storage during error                | The contents are the same as those of the start history. Therefore, the QD75 stores only the start history.                                                                                                                 |
| System monitor data (module type, OS information) | These data were deleted because they can be displayed in system monitor "Module's detailed information" of GX Developer. (Refer to GX Developer Operating Manual.)                                                          |

## Functions changed from those of A1SD75P1-S3/A1SD75P2-S3/A1SD75P3-S3

| Changed functions                         | Descriptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Software stroke limit function            | <ol style="list-style-type: none"> <li>1. The limit check of arc address is carried out only when a sub point is designated. It is not carried out when a center point is designated.</li> <li>2. The software stroke limit check during speed control is carried out in the following cases: <ul style="list-style-type: none"> <li>• When the software stroke limit is applied to the current feed value with [Pr.14] and the current feed value is updated with [Pr.21].</li> <li>• When the software stroke limit is applied to the machine feed value.</li> </ul> </li> <li>3. If an attempt is made to change the current value but the designated address is out of the software stroke limit range, the attempt is considered as an error and the current value is not changed.</li> <li>4. The conventional models feature three types of software stroke limit error codes for upper limit and lower limit respectively. With the QD75, errors for the software stroke upper limit are integrated into error code 507, and errors for the lower limit are integrated into error code 508. Error codes 509 to 512 are deleted.</li> </ol> |
| Current value changing<br>M code function | <ol style="list-style-type: none"> <li>1. An error occurs when the designated new current value is out of the software stroke limit range.</li> <li>2. The M code setting value is valid during the positioning data current value changing instruction.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Acceleration/deceleration control         | <ol style="list-style-type: none"> <li>1. An error occurs when the command frequency value calculated from the speed limit value is out of the maximum command frequency of the positioning module being used.</li> <li>2. Only two-word type (1 to 8388608ms) can be used as the setting value for the acceleration/deceleration time. (The switch between 1-word type and 2-word type is eliminated.)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Changed functions                                                                                                                                                                                   | Descriptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                      |                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------|
| Stop process and restart after stop<br>Positioning operation stop                                                                                                                                   | 1. "Peripheral side (emergency) stop" is deleted from the stop causes of Stop group 2 sudden stop selection. "Test mode fault" in the stop causes of Stop group 3 sudden stop selection is changed to be in the stop causes of Stop group 2 sudden stop selection.<br>2. "Stop (QD75 peripheral)" is added to the stop causes of Stop group 3 sudden stop selection.<br>3. Error code 100 (Peripheral device stop during operation) is deleted.<br>4. "CPU module error occurrence" is added to the stop causes of Stop group 2 sudden stop selection. |                      |                      |
| READY signal (X0)                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | A1SD75               | QD75                 |
|                                                                                                                                                                                                     | OFF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Normal (READY)       | Not READY /WDT error |
|                                                                                                                                                                                                     | ON                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Not READY /WDT error | Normal (READY)       |
| Manual pulse generator operation                                                                                                                                                                    | The No. of connectable manual pulse generator is changed from 1 generator/1 axis to 1 generator/1 module.                                                                                                                                                                                                                                                                                                                                                                                                                                              |                      |                      |
| Axis operation status                                                                                                                                                                               | "Step stopped" is changed to "Stopped" and "Step error occurring" is changed to "Error occurring".<br>"In position control for position-speed switching control", "In speed control for position-speed switching control", and "In absolute position restoration" are added.                                                                                                                                                                                                                                                                           |                      |                      |
| Continuous path control                                                                                                                                                                             | A1SD75: If the reference axis operates in reverse direction, the control is internally changed into the continuous positioning control (restart after deceleration stop).<br>QD75: Even if the reference axis operates in reverse direction with interpolation, the control remains as the continuous path control. (In single-axis operation, the operation is the same as that of the A1SD75.)                                                                                                                                                       |                      |                      |
| Near pass                                                                                                                                                                                           | For the continuous path control, only the near pass function is available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                      |                      |
| 2-axis interpolation <ul style="list-style-type: none"> <li>• 2-axis linear interpolation</li> <li>• 2-axis fixed-feed</li> <li>• Circular interpolation</li> <li>• 2-axis speed control</li> </ul> | The interpolation target axis can be randomly set with a positioning identifier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                      |
| Step function                                                                                                                                                                                       | 1. "Step stopped" is changed to "Stopped" and "Step error occurring" is changed to "Error occurring" in the axis operation status parameters.<br>2. The restart command for step start information (02H) is deleted.<br>3. The step operation is restarted with the restart command.                                                                                                                                                                                                                                                                   |                      |                      |
| Command in-position function                                                                                                                                                                        | The command in-position width is expanded.<br>• A1SD75 : 1 to 32767000<br>• QD75 : 1 to 2147483647                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                      |                      |
| Control unit "degree" handling                                                                                                                                                                      | The operating direction can be designated even for the ABS control in the unit of "degree".                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                      |                      |

| Changed functions                                                                        | Descriptions                                                                                                                                                            |                                                     |                                                     |
|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|
| Positioning start No.                                                                    | No. 9004 (Multiple axes simultaneous start control) is added.<br>Nos. 7004 to 7010 (block start designation) and 8000 to 8049 (indirect designation) are deleted.       |                                                     |                                                     |
| Block start data                                                                         | With the QD75, the number of blocks is changed to 5 (7000 to 7004). (With the A1SD75, this data is called "positioning start information".)                             |                                                     |                                                     |
| Special start data "Simultaneous start"                                                  | The simultaneous start is possible up to 4 axes.                                                                                                                        |                                                     |                                                     |
| List of errors/warnings                                                                  | Refer to the following "Error code comparisons" and "Warning code comparisons".                                                                                         |                                                     |                                                     |
| Start history                                                                            | The configuration of "start information" and "start No." is changed so that the start No. can be directly checked.                                                      |                                                     |                                                     |
| Synchronization flag (X1)                                                                | When the CPU module starting method is set to asynchronous, interlock is established with the synchronization flag (X1) signal.                                         |                                                     |                                                     |
| Basic parameter 1 " <span>Pr.5</span> Pulse output mode"                                 | When the programmable controller is turned ON or the CPU module is reset, the valid value is only the first value after the PLC READY signal (Y0) turns from OFF to ON. |                                                     |                                                     |
| Detailed parameters<br>" <span>Pr.15</span> Software stroke limit valid/invalid setting" |                                                                                                                                                                         | A1SD75                                              | QD75                                                |
|                                                                                          | 0 (Factory setting)                                                                                                                                                     | Software stroke limits invalid for manual operation | Software stroke limits valid for manual operation   |
|                                                                                          | 1                                                                                                                                                                       | Software stroke limits valid for manual operation   | Software stroke limits invalid for manual operation |

## Error code comparisons

| Error name                                                    | Error code |      |
|---------------------------------------------------------------|------------|------|
|                                                               | A1SD75     | QD75 |
| Outside bias speed range                                      | 913        | 906  |
| Illegal sudden stop deceleration time                         | 962        | 961  |
| Stop group 1 sudden stop selection error                      | 963        | 962  |
| Stop group 2 sudden stop selection error                      | 964        | 963  |
| Stop group 3 sudden stop selection error                      | 965        | 964  |
| OPR torque limit value error                                  | 991        | 995  |
| Setting for the movement amount after near-point dog ON error | 992        | 991  |
| OPR acceleration time selection error                         | 993        | 992  |
| OPR deceleration time selection error                         | 994        | 993  |

| Error type                                                                                                                               | Added              | Deleted                 |
|------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------|
| Fatal error                                                                                                                              | 002                | 51, 52                  |
| Common                                                                                                                                   | —                  | 100, 108                |
| OPR, Absolute position restoration                                                                                                       | 213, 214           | 208                     |
| JOG/Inching operation                                                                                                                    | 301                | —                       |
| Manual pulse generator operation                                                                                                         | —                  | —                       |
| Positioning operation                                                                                                                    | 523, 535, 545, 546 | 509 to 512              |
| I/F                                                                                                                                      | 800 to 806         | —                       |
| Parameter                                                                                                                                | 920, 935, 990, 996 | 931, 933, 938, 971, 999 |
| Error whose name is changed: Error code 967<br>A1SD75: External start selection error<br>QD75: External command function selection error |                    |                         |

## Warning code comparisons

| Warning type                                                                                                                            | Added         | Deleted              |
|-----------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------|
| Fatal warning                                                                                                                           | —             | 51, 52               |
| Common                                                                                                                                  | 110           | 101, 105 to 108, 115 |
| OPR, Absolute position restoration                                                                                                      | —             | —                    |
| JOG operation/Inching operation                                                                                                         | —             | —                    |
| Manual pulse generator operation                                                                                                        | —             | 402                  |
| Positioning operation                                                                                                                   | 516, 517, 518 | —                    |
| I/F                                                                                                                                     | —             | —                    |
| Parameter                                                                                                                               | —             | 900                  |
| Warning whose name is changed: Warning code 512<br>A1SD75 : Illegal external start function<br>QD75 : Illegal external command function |               |                      |

\*: Refer to "Section 15.3 List of errors" and "Section 15.4 List of warnings" for details on error codes and warning codes.

## (3) Input/output (X/Y) comparisons

| Input (X)            |                      |                   |                        |
|----------------------|----------------------|-------------------|------------------------|
| Name                 |                      | A1SD75            | QD75                   |
| (QD75/A1SD75) READY  |                      | X00 <sup>*1</sup> | X00 <sup>*1</sup>      |
| Axis 1               | Start complete       | X01               | X10                    |
| Axis 2               |                      | X02               | X11                    |
| Axis 3               |                      | X03               | X12                    |
| Axis 4               |                      | —                 | X13                    |
| Axis 1               | BUSY                 | X04 <sup>*2</sup> | X0C                    |
| Axis 2               |                      | X05 <sup>*2</sup> | X0D                    |
| Axis 3               |                      | X06 <sup>*2</sup> | X0E                    |
| Axis 4               |                      | —                 | X0F                    |
| Axis 1               | Positioning complete | X07               | X14                    |
| Axis 2               |                      | X08               | X15                    |
| Axis 3               |                      | X09               | X16                    |
| Axis 4               |                      | —                 | X17                    |
| Axis 1               | Error detection      | X0A               | X08                    |
| Axis 2               |                      | X0B               | X09                    |
| Axis 3               |                      | X0C               | X0A                    |
| Axis 4               |                      | —                 | X0B                    |
| Axis 1               | M code ON            | X0D               | X04                    |
| Axis 2               |                      | X0E               | X05                    |
| Axis 3               |                      | X0F               | X06                    |
| Axis 4               |                      | —                 | X07                    |
| Synchronization flag |                      | —                 | X01                    |
| Use prohibited       |                      | X10 to X1F        | X02, X03<br>X18 to X1F |

| Output (Y)     |                            |                          |                          |
|----------------|----------------------------|--------------------------|--------------------------|
| Name           |                            | A1SD75                   | QD75                     |
| Axis 1         | Positioning start          | Y10                      | Y10                      |
| Axis 2         |                            | Y11                      | Y11                      |
| Axis 3         |                            | Y12                      | Y12                      |
| Axis 4         |                            | —                        | Y13                      |
| Axis 1         | Axis stop                  | Y13                      | Y04                      |
| Axis 2         |                            | Y14                      | Y05                      |
| Axis 3         |                            | Y1C                      | Y06                      |
| Axis 4         |                            | —                        | Y07                      |
| Axis 1         | Forward run JOG start      | Y16                      | Y08                      |
| Axis 1         | Reverse run JOG start      | Y17                      | Y09                      |
| Axis 2         | Forward run JOG start      | Y18                      | Y0A                      |
| Axis 2         | Reverse run JOG start      | Y19                      | Y0B                      |
| Axis 3         | Forward run JOG start      | Y1A                      | Y0C                      |
| Axis 3         | Reverse run JOG start      | Y1B                      | Y0D                      |
| Axis 4         | Forward run JOG start      | —                        | Y0E                      |
| Axis 4         | Reverse run JOG start      | —                        | Y0F                      |
| PLC READY      |                            | Y1D                      | Y00                      |
| Axis 1         | Execution prohibition flag | —                        | Y14                      |
| Axis 2         |                            | —                        | Y15                      |
| Axis 3         |                            | —                        | Y16                      |
| Axis 4         |                            | —                        | Y17                      |
| Use prohibited |                            | Y00 to Y0F<br>Y1E to Y1F | Y01 to Y03<br>Y18 to Y1F |

\* 1: The ON/OFF statuses for READY are different between the QD75 and A1SD75.

|        | Not READY/WDT error | READY |
|--------|---------------------|-------|
| QD75   | OFF                 | ON    |
| A1SD75 | ON                  | OFF   |

\* 2: When using a program example of No.10 Reset program described in Section 6.4 Positioning program examples of the "A1SD75P1-S3/P2-S3/P3-S3, AD75P1-S3/P2-S3/P3-S3 Positioning Module User's Manual" for the QD75, replace "X4 (BUSY signal for Axis 1)" with "DXC (Direct access input of BUSY signal for Axis 1)". Do the same thing for programs for Axis 2 and Axis 3.

## (4) Buffer memory address comparisons

The following table shows the buffer memory addresses of the QD75 (Axis 1 to Axis 3) corresponding to the items of the A1SD75.

The shaded area shows the differences between the A1SD75 and QD75.

| Items of A1SD75                                          | Buffer memory address |        |        |        |        |        |
|----------------------------------------------------------|-----------------------|--------|--------|--------|--------|--------|
|                                                          | A1SD75                |        |        | QD75   |        |        |
|                                                          | Axis 1                | Axis 2 | Axis 3 | Axis 1 | Axis 2 | Axis 3 |
| Pr.1 Unit setting                                        | 0                     | 150    | 300    | 0      | 150    | 300    |
| Pr.2 No. of pulses per rotation (Ap)                     | 1                     | 151    | 301    | 1      | 151    | 301    |
| Pr.3 Movement amount per rotation (Al)                   | 2                     | 152    | 302    | 2      | 152    | 302    |
| Pr.4 Unit magnification (Am)                             | 3                     | 153    | 303    | 3      | 153    | 303    |
| Pr.5 Pulse output mode                                   | 4                     | 154    | 304    | 4      | 154    | 304    |
| Pr.6 Rotation direction setting                          | 5                     | 155    | 305    | 5      | 155    | 305    |
| Pr.7 Speed limit value                                   | 6                     | 156    | 306    | 10     | 160    | 310    |
|                                                          | 7                     | 157    | 307    | 11     | 161    | 311    |
| Pr.8 Acceleration time 0                                 | 8                     | 158    | 308    | 12     | 162    | 312    |
|                                                          | 9                     | 159    | 309    | 13     | 163    | 313    |
| Pr.9 Deceleration time 0                                 | 10                    | 160    | 310    | 14     | 164    | 314    |
|                                                          | 11                    | 161    | 311    | 15     | 165    | 315    |
| Pr.10 Bias speed at start                                | 12                    | 162    | 312    | 6      | 156    | 306    |
|                                                          | 13                    | 163    | 313    | 7      | 157    | 307    |
| Pr.11 Stepping motor mode selection amount               | 14                    | 164    | 314    | –      | –      | –      |
| Pr.12 Backlash compensation amount                       | 15                    | 165    | 315    | 17     | 167    | 317    |
| Pr.13 Software stroke limit upper limit value            | 16                    | 166    | 316    | 18     | 168    | 318    |
|                                                          | 17                    | 167    | 317    | 19     | 169    | 319    |
| Pr.14 Software stroke limit lower limit value            | 18                    | 168    | 318    | 20     | 170    | 320    |
|                                                          | 19                    | 169    | 319    | 21     | 171    | 321    |
| Pr.15 Software stroke limit selection                    | 20                    | 170    | 320    | 22     | 172    | 322    |
| Pr.16 Software stroke limit valid/invalid setting        | 21                    | 171    | 321    | 23     | 173    | 323    |
| Pr.17 Command in-position width                          | 22                    | 172    | 322    | 24     | 174    | 324    |
|                                                          | 23                    | 173    | 323    | 25     | 175    | 325    |
| Pr.18 Torque limit setting value                         | 24                    | 174    | 324    | 26     | 176    | 326    |
| Pr.19 M code ON signal output timing                     | 25                    | 175    | 325    | 27     | 177    | 327    |
| Pr.20 Speed switching mode                               | 26                    | 176    | 326    | 28     | 178    | 328    |
| Pr.21 Interpolation speed designation method             | 27                    | 177    | 327    | 29     | 179    | 329    |
| Pr.22 Current feed value during speed control            | 28                    | 178    | 328    | 30     | 180    | 330    |
| Pr.23 Manual pulse generator selection                   | 29                    | 179    | 329    | –      | –      | –      |
| Pr.24 Logic selection for pulse output to the drive unit | 30                    | 180    | 330    | –      | –      | –      |
| Pr.25 Size selection for acceleration/deceleration time  | 31                    | 181    | 331    | –      | –      | –      |
| Pr.26 Acceleration time 1                                | 36                    | 186    | 336    | 36     | 186    | 336    |
|                                                          | 37                    | 187    | 337    | 37     | 187    | 337    |
| Pr.27 Acceleration time 2                                | 38                    | 188    | 338    | 38     | 188    | 338    |
|                                                          | 39                    | 189    | 339    | 39     | 189    | 339    |



| Items of A1SD75                                                 | Buffer memory address |        |        |        |        |        |
|-----------------------------------------------------------------|-----------------------|--------|--------|--------|--------|--------|
|                                                                 | A1SD75                |        |        | QD75   |        |        |
|                                                                 | Axis 1                | Axis 2 | Axis 3 | Axis 1 | Axis 2 | Axis 3 |
| [Pr.28] Acceleration time 3                                     | 40                    | 190    | 340    | 40     | 190    | 340    |
|                                                                 | 41                    | 191    | 341    | 41     | 191    | 341    |
| [Pr.29] Deceleration time 1                                     | 42                    | 192    | 342    | 42     | 192    | 342    |
|                                                                 | 43                    | 193    | 343    | 43     | 193    | 343    |
| [Pr.30] Deceleration time 2                                     | 44                    | 194    | 344    | 44     | 194    | 344    |
|                                                                 | 45                    | 195    | 345    | 45     | 195    | 345    |
| [Pr.31] Deceleration time 3                                     | 46                    | 196    | 346    | 46     | 196    | 346    |
|                                                                 | 47                    | 197    | 347    | 47     | 197    | 347    |
| [Pr.32] JOG speed limit value                                   | 48                    | 198    | 348    | 48     | 198    | 348    |
|                                                                 | 49                    | 199    | 349    | 49     | 199    | 349    |
| [Pr.33] JOG operation acceleration time selection               | 50                    | 200    | 350    | 50     | 200    | 350    |
| [Pr.34] JOG operation deceleration time selection               | 51                    | 201    | 351    | 51     | 201    | 351    |
| [Pr.35] Acceleration/deceleration process selection             | 52                    | 202    | 352    | 52     | 202    | 352    |
| [Pr.36] S-curve ratio                                           | 53                    | 203    | 353    | 53     | 203    | 353    |
| [Pr.37] Sudden stop deceleration time                           | 54                    | 204    | 354    | 54     | 204    | 354    |
|                                                                 | 55                    | 205    | 355    | 55     | 205    | 355    |
| [Pr.38] Stop group 1 sudden stop selection                      | 56                    | 206    | 356    | 56     | 206    | 356    |
| [Pr.39] Stop group 2 sudden stop selection                      | 57                    | 207    | 357    | 57     | 207    | 357    |
| [Pr.40] Stop group 3 sudden stop selection                      | 58                    | 208    | 358    | 58     | 208    | 358    |
| [Pr.41] Positioning complete signal output time                 | 59                    | 209    | 359    | 59     | 209    | 359    |
| [Pr.42] Allowable circular interpolation error width            | 60                    | 210    | 360    | 60     | 210    | 360    |
|                                                                 | 61                    | 211    | 361    | 61     | 211    | 361    |
| [Pr.43] External start function selection                       | 62                    | 212    | 362    | 62     | 212    | 362    |
| (QD75: [Pr.42] External command function selection)             |                       |        |        |        |        |        |
| [Pr.44] Near pass mode selection for path control               | 66                    | 216    | 366    | —      | —      | —      |
| [Pr.45] OPR method                                              | 70                    | 220    | 370    | 70     | 220    | 370    |
| [Pr.46] OPR direction                                           | 71                    | 221    | 371    | 71     | 221    | 371    |
| [Pr.47] OP address                                              | 72                    | 222    | 372    | 72     | 222    | 372    |
|                                                                 | 73                    | 223    | 373    | 73     | 223    | 373    |
| [Pr.48] OPR speed                                               | 74                    | 224    | 374    | 74     | 224    | 374    |
|                                                                 | 75                    | 225    | 375    | 75     | 225    | 375    |
| [Pr.49] Creep speed                                             | 76                    | 226    | 376    | 76     | 226    | 376    |
|                                                                 | 77                    | 227    | 377    | 77     | 227    | 377    |
| [Pr.50] OPR retry                                               | 78                    | 228    | 378    | 78     | 228    | 378    |
| [Pr.51] OPR dwell time                                          | 79                    | 229    | 379    | 79     | 229    | 379    |
| [Pr.52] Setting for the movement amount after near-point dog ON | 80                    | 230    | 380    | 80     | 230    | 380    |
|                                                                 | 81                    | 231    | 381    | 81     | 231    | 381    |
| [Pr.53] OPR acceleration time selection                         | 82                    | 232    | 382    | 82     | 232    | 382    |
| [Pr.54] OPR deceleration time selection                         | 83                    | 233    | 383    | 83     | 233    | 383    |
| [Pr.55] OP shift amount                                         | 84                    | 234    | 384    | 84     | 234    | 384    |
|                                                                 | 85                    | 235    | 385    | 85     | 235    | 385    |
| [Pr.56] OPR torque limit value                                  | 86                    | 236    | 386    | 86     | 236    | 386    |
| [Pr.57] Speed designation during OP shift                       | 88                    | 238    | 388    | 88     | 238    | 388    |
| [Pr.58] Dwell time during OPR retry                             | 89                    | 239    | 389    | 89     | 239    | 389    |

| Items of A1SD75                                                                                                 |                             | Buffer memory address   |                            |
|-----------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------|----------------------------|
|                                                                                                                 |                             | A1SD75                  | QD75                       |
|                                                                                                                 |                             | Common for axis 1, 2, 3 | Common for axis 1, 2, 3, 4 |
| <b>Md.1</b> In test mode flag                                                                                   |                             | 450                     | 1200                       |
| <b>Md.2</b> Module name                                                                                         |                             | 451                     | —                          |
| <b>Md.3</b> OS type                                                                                             |                             | 452 453<br>454 455      | —                          |
| <b>Md.4</b> OS version                                                                                          |                             | 456<br>457              | —                          |
| <b>Md.5</b> Clock data (hour: minute)                                                                           |                             | 460                     | —                          |
| <b>Md.6</b> Clock data (second: 100ms)                                                                          |                             | 461                     | —                          |
| (Pointer number)                                                                                                |                             | (0) to (15)             |                            |
| <b>Md.7</b> Start axis<br>(QD75: <b>Md.3</b> Start information)                                                 | Start history               | 462 to 537              | 1212 to 1287               |
| <b>Md.8</b> Operation type<br>(QD75: <b>Md.4</b> Start No.)                                                     |                             | 463 to 538              | 1213 to 1288               |
| <b>Md.9</b> Start Hour: minute<br>(QD75: <b>Md.5</b> Start Day: Hour)                                           |                             | 464 to 539              | 1214 to 1289               |
| <b>Md.10</b> Start Second: 100ms<br>(QD75: <b>Md.3</b> Start Minute: second)                                    |                             | 465 to 540              | 1215 to 1290               |
| <b>Md.11</b> Error judgment                                                                                     |                             | 466 to 541              | 1216 to 1291               |
| <b>Md.12</b> Start history pointer                                                                              |                             | 542                     | 1292                       |
| (Pointer number)                                                                                                |                             | (0) to (15)             | —                          |
| <b>Md.13</b> Start axis                                                                                         | Start history during errors | 543 to 618              | —                          |
| <b>Md.14</b> Operation type                                                                                     |                             | 544 to 619              | —                          |
| <b>Md.15</b> Start Hour: minute                                                                                 |                             | 545 to 620              | —                          |
| <b>Md.16</b> Start Second: 100ms                                                                                |                             | 546 to 621              | —                          |
| <b>Md.17</b> Error judgment                                                                                     |                             | 547 to 622              | —                          |
| <b>Md.18</b> Start history storage during error                                                                 |                             | 623                     | —                          |
| (Pointer number)                                                                                                |                             | (0) to (15)             |                            |
| <b>Md.19</b> Axis in which the error occurred                                                                   | Error history               | 624 to 684              | 1293 to 1353               |
| <b>Md.20</b> Axis error No.                                                                                     |                             | 625 to 685              | 1294 to 1354               |
| <b>Md.21</b> Axis error occurrence Hour: minute<br>(QD75: <b>Md.11</b> Axis error occurrence (Day: Hour))       |                             | 626 to 686              | 1295 to 1355               |
| <b>Md.22</b> Axis error occurrence Second: 100ms<br>(QD75: <b>Md.12</b> Axis error occurrence (Minute: second)) |                             | 627 to 687              | 1296 to 1356               |
| <b>Md.23</b> Error history pointer                                                                              |                             | 688                     | 1357                       |

| Items of A1SD75                                           |                 | Buffer memory address   |                            |
|-----------------------------------------------------------|-----------------|-------------------------|----------------------------|
|                                                           |                 | A1SD75                  | QD75                       |
|                                                           |                 | Common for axis 1, 2, 3 | Common for axis 1, 2, 3, 4 |
| (Pointer number)                                          | Warning history | (0) to (15)             |                            |
| Md.24 Axis in which the warning occurred                  |                 | 689 to 749              | 1358 to 1418               |
| Md.25 Axis warning No.                                    |                 | 690 to 750              | 1359 to 1419               |
| Md.26 Axis warning occurrence Hour:<br>minute             |                 | 691 to 751              | 1360 to 1420               |
| (QD75: Md.16 Axis warning occurrence<br>(Day: Hour))      |                 |                         |                            |
| Md.27 Axis warning occurrence Second:<br>100ms            |                 | 692 to 752              | 1361 to 1421               |
| (QD75: Md.17 Axis warning occurrence<br>(Minute: second)) |                 |                         |                            |
| Md.28 Warning history pointer                             |                 | 753                     | 1422                       |

| Items of A1SD75                                              | Buffer memory address |            |              |            |            |              |
|--------------------------------------------------------------|-----------------------|------------|--------------|------------|------------|--------------|
|                                                              | A1SD75                |            |              | QD75       |            |              |
|                                                              | Axis 1                | Axis 2     | Axis 3       | Axis 1     | Axis 2     | Axis 3       |
| Md.29 Current feed value                                     | 800                   | 900        | 1000         | 800        | 900        | 1000         |
|                                                              | 801                   | 901        | 1001         | 801        | 901        | 1001         |
| Md.30 Machine feed value                                     | 802                   | 902        | 1002         | 802        | 902        | 1002         |
|                                                              | 803                   | 903        | 1003         | 803        | 903        | 1003         |
| Md.31 Feedrate                                               | 804                   | 904        | 1004         | 804        | 904        | 1004         |
|                                                              | 805                   | 905        | 1005         | 805        | 905        | 1005         |
| Md.32 Valid M code                                           | 806                   | 906        | 1006         | 808        | 908        | 1008         |
| Md.33 Axis error No.                                         | 807                   | 907        | 1007         | 806        | 906        | 1006         |
| Md.34 Axis warning No.                                       | 808                   | 908        | 1008         | 807        | 907        | 1007         |
| Md.35 Axis operation status                                  | 809                   | 909        | 1009         | 809        | 909        | 1009         |
| Md.36 Current speed                                          | 810                   | 910        | 1010         | 810        | 910        | 1010         |
|                                                              | 811                   | 911        | 1011         | 811        | 911        | 1011         |
| Md.37 Axis feedrate                                          | 812                   | 912        | 1012         | 812        | 912        | 1012         |
|                                                              | 813                   | 913        | 1013         | 813        | 913        | 1013         |
| Md.38 Speed-position switching control positioning amount    | 814                   | 914        | 1014         | 814        | 914        | 1014         |
|                                                              | 815                   | 915        | 1015         | 815        | 915        | 1015         |
| Md.39 External input/output signal                           | 816                   | 916        | 1016         | 816        | 916        | 1016         |
| Md.40 Status                                                 | 817                   | 917        | 1017         | 817        | 917        | 1017         |
| Md.41 Target value                                           | 818                   | 918        | 1018         | 818        | 918        | 1018         |
|                                                              | 819                   | 919        | 1019         | 819        | 919        | 1019         |
| Md.42 Target speed                                           | 820                   | 920        | 1020         | 820        | 920        | 1020         |
|                                                              | 821                   | 921        | 1021         | 821        | 921        | 1021         |
| Md.43 OP absolute position                                   | 822                   | 922        | 1022         | —          | —          | —            |
|                                                              | 823                   | 923        | 1023         | —          | —          | —            |
| Md.44 Movement amount after near-point dog ON                | 824                   | 924        | 1024         | 824        | 924        | 1024         |
|                                                              | 825                   | 925        | 1025         | 825        | 925        | 1025         |
| Md.45 Torque limit stored value                              | 826                   | 926        | 1026         | 826        | 926        | 1026         |
| Md.46 Special start data instruction code setting value      | 827                   | 927        | 1027         | 827        | 927        | 1027         |
| Md.47 Special start data instruction parameter setting value | 828                   | 928        | 1028         | 828        | 928        | 1028         |
| Md.48 Start positioning data No. setting value               | 829                   | 929        | 1029         | 829        | 929        | 1029         |
| Md.49 In speed limit flag                                    | 830                   | 930        | 1030         | 830        | 930        | 1030         |
| Md.50 In speed change processing flag                        | 831                   | 931        | 1031         | 831        | 931        | 1031         |
| Md.51 Start data pointer being executed                      | 832                   | 932        | 1032         | 834        | 934        | 1034         |
| Md.52 Last executed positioning data No.                     | 833                   | 933        | 1033         | 837        | 937        | 1037         |
| Md.53 Repetition counter                                     | 834                   | 934        | 1034         | 832        | 932        | 1032         |
| (QD75: Md.41 Special start repetition counter)               |                       |            |              |            |            |              |
| Md.54 Positioning data No. being executed                    | 835                   | 935        | 1035         | 835        | 935        | 1035         |
| Md.55 Block No. being executed                               | 836                   | 936        | 1036         | 836        | 936        | 1036         |
| Md.56 Positioning data being executed                        | 838 to 847            | 938 to 947 | 1038 to 1047 | 838 to 847 | 938 to 947 | 1038 to 1047 |

| Items of A1SD75                                                        | Buffer memory address |        |        |        |        |        |
|------------------------------------------------------------------------|-----------------------|--------|--------|--------|--------|--------|
|                                                                        | A1SD75                |        |        | QD75   |        |        |
|                                                                        | Axis 1                | Axis 2 | Axis 3 | Axis 1 | Axis 2 | Axis 3 |
| Cd.1 Clock data setting (hour)                                         | 1100                  |        |        | —      |        |        |
| Cd.2 Clock data setting (minute, second)                               | 1101                  |        |        | —      |        |        |
| Cd.3 Clock data writing                                                | 1102                  |        |        | —      |        |        |
| Cd.4 Target axis                                                       | 1103                  |        |        | —      |        |        |
| Cd.5 Positioning data No.                                              | 1104                  |        |        | —      |        |        |
| Cd.6 Write pattern                                                     | 1105                  |        |        | —      |        |        |
| Cd.7 Read/write request                                                | 1106                  |        |        | —      |        |        |
| Cd.8 Read/write positioning data I/F                                   | 1108 to 1137          |        |        | —      |        |        |
| Cd.9 Flash ROM write request                                           | 1138                  |        |        | —      |        |        |
| Cd.10 Parameter initialization request                                 | 1139                  |        |        | 1901   |        |        |
| Cd.11 Positioning start No.                                            | 1150                  | 1200   | 1250   | 1500   | 1600   | 1700   |
| Cd.12 Axis error reset                                                 | 1151                  | 1201   | 1251   | 1502   | 1602   | 1702   |
| Cd.13 Restart command                                                  | 1152                  | 1202   | 1252   | 1503   | 1603   | 1703   |
| Cd.14 M code OFF request                                               | 1153                  | 1203   | 1253   | 1504   | 1604   | 1704   |
| Cd.15 New current value                                                | 1154                  | 1204   | 1254   | 1506   | 1606   | 1706   |
|                                                                        | 1155                  | 1205   | 1255   | 1507   | 1607   | 1707   |
| Cd.16 New speed value                                                  | 1156                  | 1206   | 1256   | 1514   | 1614   | 1714   |
|                                                                        | 1157                  | 1207   | 1257   | 1515   | 1615   | 1715   |
| Cd.17 Speed change request                                             | 1158                  | 1208   | 1258   | 1516   | 1616   | 1716   |
| Cd.18 Positioning operation speed override                             | 1159                  | 1209   | 1259   | 1513   | 1613   | 1713   |
| Cd.19 JOG speed                                                        | 1160                  | 1210   | 1260   | 1518   | 1618   | 1718   |
|                                                                        | 1161                  | 1211   | 1261   | 1519   | 1619   | 1719   |
| Cd.20 Speed-position switching enable flag                             | 1163                  | 1213   | 1263   | 1528   | 1628   | 1728   |
| Cd.21 Speed-position switching control movement amount change register | 1164                  | 1214   | 1264   | 1526   | 1626   | 1726   |
|                                                                        | 1165                  | 1215   | 1265   | 1527   | 1627   | 1727   |
| Cd.22 Manual pulse generator enable flag                               | 1167                  | 1217   | 1267   | 1524   | 1624   | 1724   |
| Cd.23 Manual pulse generator 1 pulse input magnification               | 1168                  | 1218   | 1268   | 1522   | 1622   | 1722   |
|                                                                        | 1169                  | 1219   | 1269   | 1523   | 1623   | 1723   |
| Cd.24 OPR return request flag OFF request                              | 1170                  | 1220   | 1270   | 1521   | 1621   | 1721   |
| Cd.25 External start valid                                             | 1171                  | 1221   | 1271   | 1505   | 1605   | 1705   |
| (QD75: Cd.8 External command valid)                                    |                       |        |        |        |        |        |
| Cd.26 Step valid flag                                                  | 1172                  | 1222   | 1272   | 1545   | 1645   | 1745   |
| Cd.27 Step mode                                                        | 1173                  | 1223   | 1273   | 1544   | 1644   | 1744   |
| Cd.28 Step start information                                           | 1174                  | 1224   | 1274   | 1546   | 1646   | 1746   |
| Cd.29 Skip command                                                     | 1175                  | 1225   | 1275   | 1547   | 1647   | 1747   |
| Cd.30 New torque value                                                 | 1176                  | 1226   | 1276   | 1525   | 1625   | 1725   |
| Cd.31 Positioning starting point No.                                   | 1178                  | 1228   | 1278   | 1501   | 1601   | 1701   |
| Cd.32 Interrupt request during continuous operation                    | 1181                  | 1231   | 1281   | 1520   | 1620   | 1720   |
| Cd.33 New acceleration time value                                      | 1184                  | 1234   | 1284   | 1508   | 1608   | 1708   |
|                                                                        | 1185                  | 1235   | 1285   | 1509   | 1609   | 1709   |

| Items of A1SD75      |                                                                                     | Buffer memory address |              |              |              |              |                |
|----------------------|-------------------------------------------------------------------------------------|-----------------------|--------------|--------------|--------------|--------------|----------------|
|                      |                                                                                     | A1SD75                |              |              | QD75         |              |                |
|                      |                                                                                     | Axis 1                | Axis 2       | Axis 3       | Axis 1       | Axis 2       | Axis 3         |
| Cd.34                | New deceleration time value                                                         | 1186                  | 1236         | 1286         | 1510         | 1610         | 1710           |
|                      |                                                                                     | 1187                  | 1237         | 1287         | 1511         | 1611         | 1711           |
| Cd.35                | Acceleration/deceleration time change during speed change, enable/disable selection | 1188                  | 1238         | 1288         | 1512         | 1612         | 1712           |
| Positioning data * 1 | Da.1 Operation pattern                                                              | 1300                  | 2300         | 3300         | 2000         | 8000         | 14000          |
|                      | Da.2 Control system                                                                 |                       |              |              |              |              |                |
|                      | Da.3 Acceleration time No.                                                          |                       |              |              |              |              |                |
|                      | Da.4 Deceleration time No.                                                          |                       |              |              |              |              |                |
|                      | Da.9 M code/condition data                                                          | 1301                  | 2301         | 3301         | 2001         | 8001         | 14001          |
|                      | Da.8 Dwell time/JUMP destination positioning data No.                               | 1302                  | 2302         | 3302         | 2002         | 8002         | 14002          |
|                      | Da.7 Command speed                                                                  | 1304                  | 2304         | 3304         | 2004         | 8004         | 14004          |
|                      |                                                                                     | 1305                  | 2305         | 3305         | 2005         | 8005         | 14005          |
|                      | Da.5 Positioning address/movement amount                                            | 1306                  | 2306         | 3306         | 2006         | 8006         | 14006          |
|                      |                                                                                     | 1307                  | 2307         | 3307         | 2007         | 8007         | 14007          |
|                      | Da.6 Arc address                                                                    | 1308                  | 2308         | 3308         | 2008         | 8008         | 14008          |
|                      |                                                                                     | 1309                  | 2309         | 3309         | 2009         | 8009         | 14009          |
|                      | No.2                                                                                | 1310 to 1319          | 2310 to 2319 | 3310 to 3319 | 2010 to 2019 | 8010 to 8019 | 14010 to 14019 |
|                      | No.3                                                                                | 1320 to 1329          | 2320 to 2329 | 3320 to 3329 | 2020 to 2029 | 8020 to 8029 | 14020 to 14029 |
|                      | to                                                                                  | to                    | to           | to           | to           | to           | to             |
|                      | No.100                                                                              | 2290 to 2299          | 3290 to 3299 | 4290 to 4299 | 2990 to 2999 | 8990 to 8999 | 14990 to 14999 |

\* 1: With the QD75, the positioning data buffer memory addresses are Nos. 1 to 600.

Refer to Appendix 13 "List of buffer memory addresses" for details.

| Items of A1SD75                   |                               |                                                      |                           | Buffer memory address |           |              |      |              |      |                |       |                |       |                |       |       |              |
|-----------------------------------|-------------------------------|------------------------------------------------------|---------------------------|-----------------------|-----------|--------------|------|--------------|------|----------------|-------|----------------|-------|----------------|-------|-------|--------------|
|                                   |                               |                                                      |                           | A1SD75                |           |              |      |              |      | QD75           |       |                |       |                |       |       |              |
|                                   |                               |                                                      |                           | Axis 1                |           | Axis 2       |      | Axis 3       |      | Axis 1         |       | Axis 2         |       | Axis 3         |       |       |              |
| Positioning start information * 2 | Start block data * 2          | Da.10                                                | Shape                     | 1st point             | 4300      | 4350         | 4550 | 4600         | 4800 | 4850           | 26000 | 26050          | 27000 | 27050          | 28000 | 28050 |              |
|                                   |                               | Da.11                                                | Start data No.            |                       |           |              |      |              |      |                |       |                |       |                |       |       |              |
|                                   |                               | Da.12                                                | Special start instruction |                       | 2nd point | 4301         | 4351 | 4551         | 4601 | 4801           | 4851  | 26001          | 26051 | 27001          | 27051 | 28001 | 28051        |
|                                   |                               | Da.13                                                | Parameter                 |                       |           |              |      |              |      |                |       |                |       |                |       |       |              |
|                                   |                               | 3rd point                                            |                           |                       | 4302      | 4352         | 4552 | 4602         | 4802 | 4852           | 26002 | 26052          | 27002 | 27052          | 28002 | 28052 |              |
|                                   |                               | to                                                   |                           |                       | to        |              | to   |              | to   |                | to    |                | to    |                | to    |       |              |
|                                   |                               | 50th point                                           |                           |                       | 4349      | 4399         | 4599 | 4649         | 4849 | 4899           | 26049 | 26099          | 27049 | 27099          | 28049 | 28099 |              |
|                                   |                               | Condition data                                       | Da.14                     | Condition target      | No. 1     | 4400         |      | 4650         |      | 4900           |       | 26100          |       | 27100          |       | 28100 |              |
|                                   | Da.15                         |                                                      | Condition operator        |                       |           |              |      |              |      |                |       |                |       |                |       |       |              |
|                                   | Da.16                         |                                                      | Address                   | 4402                  |           | 4652         |      | 4902         |      | 26102          |       | 27102          |       | 28102          |       |       |              |
|                                   | Da.17                         |                                                      | Parameter 1               |                       |           |              |      |              |      |                |       |                |       |                |       |       |              |
|                                   |                               |                                                      |                           | 4403                  |           | 4653         |      | 4903         |      | 26103          |       | 27103          |       | 28103          |       |       |              |
|                                   | 4404                          |                                                      | 4654                      |                       |           |              |      |              |      |                |       |                |       |                |       | 4904  |              |
|                                   |                               |                                                      |                           |                       | 4405      |              | 4655 |              | 4905 |                | 26105 |                | 27105 |                | 28105 |       |              |
|                                   | 4406                          |                                                      | 4656                      |                       |           |              |      |              |      |                |       |                |       |                |       |       | 4906         |
|                                   |                               |                                                      |                           |                       | 4407      |              | 4657 |              | 4907 |                | 26107 |                | 27107 |                | 28107 |       |              |
|                                   | No. 2                         |                                                      |                           | 4410 to 4419          |           |              |      |              |      |                |       |                |       |                |       |       | 4660 to 4669 |
|                                   | No. 3                         |                                                      |                           | 4420 to 4429          |           | 4670 to 4679 |      | 4920 to 4929 |      | 26120 to 26129 |       | 27120 to 27129 |       | 28120 to 28129 |       |       |              |
|                                   | to                            |                                                      |                           | to                    |           | to           |      | to           |      | to             |       | to             |       | to             |       |       |              |
|                                   | No. 10                        |                                                      |                           | 4490 to 4499          |           | 4740 to 4749 |      | 4990 to 4999 |      | 26190 to 26199 |       | 27190 to 27199 |       | 28190 to 28199 |       |       |              |
|                                   | Positioning start information |                                                      | Indirect specification    | Start No. 8001        |           |              | 4500 |              | 4750 |                | 5000  |                | —     |                | —     |       | —            |
|                                   |                               | Start No. 8002                                       |                           |                       | 4501      |              | 4751 |              | 5001 |                | —     |                | —     |                | —     |       |              |
| to                                |                               |                                                      |                           | to                    |           | to           |      | to           |      | to             |       | to             |       | to             |       |       |              |
| Start No. 8050                    |                               |                                                      |                           | 4549                  |           | 4799         |      | 5049         |      | —              |       | —              |       | —              |       |       |              |
| PLC CPU memo area                 |                               | Condition judgment target data of the condition data |                           |                       | 5050      |              |      |              |      |                | 30000 |                |       |                |       |       |              |
|                                   |                               |                                                      |                           |                       | to        |              |      |              |      |                | to    |                |       |                |       |       |              |
|                                   |                               |                                                      |                           |                       | 5099      |              |      |              |      |                | 30099 |                |       |                |       |       |              |
| Target axis                       |                               |                                                      |                           | 5100                  |           |              |      |              |      | —              |       |                |       |                |       |       |              |
| Head positioning block No.        |                               |                                                      |                           | 5101                  |           |              |      |              |      | —              |       |                |       |                |       |       |              |
| No. of read/write data items      |                               |                                                      |                           | 5102                  |           |              |      |              |      | —              |       |                |       |                |       |       |              |
| Read/write request                |                               |                                                      |                           | 5103                  |           |              |      |              |      | —              |       |                |       |                |       |       |              |
| Read/write block                  |                               |                                                      |                           | 5110 to 6109          |           |              |      |              |      | —              |       |                |       |                |       |       |              |

\* 2: With the QD75, it is called "block start data".

\* 3: With the QD75, the "block start data" and "condition data" in the shaded area are called "start block 0". There are five start blocks: 0 to 4.

Refer to Appendix 13 "List of buffer memory addresses" for details.

## (5) Data indication No. comparisons

The following table shows the comparisons of numbers for each symbols (Pr., Md., Cd., and Da.) indicating parameters or positioning data items.

The shaded sections indicate the added or changed items with the QD75.

## Parameters

| Item                                               | A1SD75 | QD75  |
|----------------------------------------------------|--------|-------|
| Unit setting                                       |        | Pr.1  |
| No. of pulses per rotation (Ap)                    |        | Pr.2  |
| Movement amount per rotation (Al)                  |        | Pr.3  |
| Unit magnification (Am)                            |        | Pr.4  |
| Pulse output mode                                  |        | Pr.5  |
| Rotation direction setting                         |        | Pr.6  |
| Speed limit value                                  | Pr.7   | Pr.8  |
| Acceleration time 0                                | Pr.8   | Pr.9  |
| Deceleration time 0                                | Pr.9   | Pr.10 |
| Bias speed at start                                | Pr.10  | Pr.7  |
| Stepping motor mode selection                      | Pr.11  | –     |
| Backlash compensation amount                       | Pr.12  | Pr.11 |
| Software stroke limit upper limit value            | Pr.13  | Pr.12 |
| Software stroke limit lower limit value            | Pr.14  | Pr.13 |
| Software stroke limit selection                    | Pr.15  | Pr.14 |
| Software stroke limit valid/invalid setting        | Pr.16  | Pr.15 |
| Command in-position width                          | Pr.17  | Pr.16 |
| Torque limit setting value                         | Pr.18  | Pr.17 |
| M code ON signal output timing                     | Pr.19  | Pr.18 |
| Speed switching mode                               | Pr.20  | Pr.19 |
| Interpolation speed designation method             | Pr.21  | Pr.20 |
| Current feed value during speed control            | Pr.22  | Pr.21 |
| Manual pulse generator selection                   | Pr.23  | –     |
| Logic selection for pulse output to the drive unit | Pr.24  | –     |
| Input signal logic selection                       | –      | Pr.22 |
| Output signal logic selection                      | –      | Pr.23 |
| Manual pulse generator input selection             | –      | Pr.24 |
| Size selection for acceleration/deceleration time  | Pr.25  | –     |
| Acceleration time 1                                | Pr.26  | Pr.25 |
| Acceleration time 2                                | Pr.27  | Pr.26 |



| Item                                                                                   | A1SD75 | QD75   |
|----------------------------------------------------------------------------------------|--------|--------|
| Acceleration time 3                                                                    | Pr.28  | Pr.27  |
| Deceleration time 1                                                                    | Pr.29  | Pr.28  |
| Deceleration time 2                                                                    | Pr.30  | Pr.29  |
| Deceleration time 3                                                                    | Pr.31  | Pr.30  |
| JOG speed limit value                                                                  | Pr.32  | Pr.31  |
| JOG operation acceleration time selection                                              | Pr.33  | Pr.32  |
| JOG operation deceleration time selection                                              | Pr.34  | Pr.33  |
| Acceleration/deceleration process selection                                            | Pr.35  | Pr.34  |
| S-curve ratio                                                                          | Pr.36  | Pr.35  |
| Sudden stop deceleration time                                                          | Pr.37  | Pr.36  |
| Stop group 1 sudden stop selection                                                     | Pr.38  | Pr.37  |
| Stop group 2 sudden stop selection                                                     | Pr.39  | Pr.38  |
| Stop group 3 sudden stop selection                                                     | Pr.40  | Pr.39  |
| Positioning complete signal output time                                                | Pr.41  | Pr.40  |
| Allowable circular interpolation error width                                           | Pr.42  | Pr.41  |
| External start function selection<br>(QD75: Pr.42 External command function selection) | Pr.43  | Pr.42  |
| Near pass mode selection for path control                                              | Pr.44  | —      |
| OPR method                                                                             | Pr.45  | Pr.43  |
| OPR direction                                                                          | Pr.46  | Pr.44  |
| OP address                                                                             | Pr.47  | Pr.45  |
| OPR speed                                                                              | Pr.48  | Pr.46  |
| Creep speed                                                                            | Pr.49  | Pr.47  |
| OPR retry                                                                              | Pr.50  | Pr.48  |
| OPR dwell time                                                                         | Pr.51  | Pr.49  |
| Setting for the movement amount after near-point dog ON                                | Pr.52  | Pr.50  |
| OPR acceleration time selection                                                        | Pr.53  | Pr.51  |
| OPR deceleration time selection                                                        | Pr.54  | Pr.52  |
| OP shift amount                                                                        | Pr.55  | Pr.53  |
| OPR torque limit value                                                                 | Pr.56  | Pr.54  |
| Deviation counter clear signal output time                                             | —      | Pr.55  |
| Speed designation during OP shift                                                      | Pr.57  | Pr.56  |
| Dwell time during OPR retry                                                            | Pr.58  | Pr.57  |
| Speed-position function selection                                                      | —      | Pr.150 |

## Monitor data

| Item                                                                                            |                             | A1SD75 | QD75  |
|-------------------------------------------------------------------------------------------------|-----------------------------|--------|-------|
| In test mode flag                                                                               |                             | Md.1   |       |
| Module name                                                                                     |                             | Md.2   | —     |
| OS type                                                                                         |                             | Md.3   | —     |
| OS version                                                                                      |                             | Md.4   | —     |
| Clock data (hour: minute)                                                                       |                             | Md.5   | —     |
| Clock data (second: 100ms)                                                                      |                             | Md.6   | —     |
| Start axis (QD75: Md.3 Start information)                                                       | Start history               | Md.7   | Md.3  |
| Operation type (QD75: Md.4 Start No.)                                                           |                             | Md.8   | Md.4  |
| Start Hour: minute (QD75: Md.5 Start Hour)                                                      |                             | Md.9   | Md.5  |
| Start Second: 100ms (QD75: Md.6 Start Minute: second)                                           |                             | Md.10  | Md.6  |
| Error judgment                                                                                  |                             | Md.11  | Md.7  |
| Start history pointer                                                                           |                             | Md.12  | Md.8  |
| Start axis                                                                                      | Start history during errors | Md.13  | —     |
| Operation type                                                                                  |                             | Md.14  | —     |
| Start Hour: minute                                                                              |                             | Md.15  | —     |
| Start Second: 100ms                                                                             |                             | Md.16  | —     |
| Error judgment                                                                                  |                             | Md.17  | —     |
| Start history storage during error                                                              |                             | Md.18  | —     |
| Axis in which the error occurred                                                                | Error history               | Md.19  | Md.9  |
| Axis error No.                                                                                  |                             | Md.20  | Md.10 |
| Axis error occurrence Hour: minute<br>(QD75: Md.11 Axis error occurrence (Hour))                |                             | Md.21  | Md.11 |
| Axis error occurrence Second: 100ms<br>(QD75: Md.12 Axis error occurrence (Minute: second))     |                             | Md.22  | Md.12 |
| Pointer                                                                                         |                             | Md.23  | Md.13 |
| Axis in which the warning occurred                                                              | Warning history             | Md.24  | Md.14 |
| Axis warning No.                                                                                |                             | Md.25  | Md.15 |
| Axis warning occurrence Hour: minute<br>(QD75: Md.16 Axis warning occurrence (Hour))            |                             | Md.26  | Md.16 |
| Axis warning occurrence Second: 100ms<br>(QD75: Md.17 Axis warning occurrence (Minute: second)) |                             | Md.27  | Md.17 |
| Pointer                                                                                         |                             | Md.28  | Md.18 |

| Item                                                                              | A1SD75                | QD75                  |
|-----------------------------------------------------------------------------------|-----------------------|-----------------------|
| No. of write accesses to flash ROM                                                | –                     | <a href="#">Md.19</a> |
| Current feed value                                                                | <a href="#">Md.29</a> | <a href="#">Md.20</a> |
| Machine feed value                                                                | <a href="#">Md.30</a> | <a href="#">Md.21</a> |
| Feedrate                                                                          | <a href="#">Md.31</a> | <a href="#">Md.22</a> |
| Valid M code                                                                      | <a href="#">Md.32</a> | <a href="#">Md.25</a> |
| Axis error No.                                                                    | <a href="#">Md.33</a> | <a href="#">Md.23</a> |
| Axis warning No.                                                                  | <a href="#">Md.34</a> | <a href="#">Md.24</a> |
| Axis operation status                                                             | <a href="#">Md.35</a> | <a href="#">Md.26</a> |
| Current speed                                                                     | <a href="#">Md.36</a> | <a href="#">Md.27</a> |
| Axis feedrate                                                                     | <a href="#">Md.37</a> | <a href="#">Md.28</a> |
| Speed-position switching control positioning amount                               | <a href="#">Md.38</a> | <a href="#">Md.29</a> |
| External input/output signal                                                      | <a href="#">Md.39</a> | <a href="#">Md.30</a> |
| Status                                                                            | <a href="#">Md.40</a> | <a href="#">Md.31</a> |
| Target value                                                                      | <a href="#">Md.41</a> | <a href="#">Md.32</a> |
| Target speed                                                                      | <a href="#">Md.42</a> | <a href="#">Md.33</a> |
| OP absolute position                                                              | <a href="#">Md.43</a> | –                     |
| Movement amount after near-point dog ON                                           | <a href="#">Md.44</a> | <a href="#">Md.34</a> |
| Torque limit stored value                                                         | <a href="#">Md.45</a> | <a href="#">Md.35</a> |
| Special start data instruction code setting value                                 | <a href="#">Md.46</a> | <a href="#">Md.36</a> |
| Special start data instruction parameter setting value                            | <a href="#">Md.47</a> | <a href="#">Md.37</a> |
| Start positioning data No. setting value                                          | <a href="#">Md.48</a> | <a href="#">Md.38</a> |
| In speed control flag                                                             | <a href="#">Md.49</a> | <a href="#">Md.39</a> |
| In speed change processing flag                                                   | <a href="#">Md.50</a> | <a href="#">Md.40</a> |
| Start data pointer being executed                                                 | <a href="#">Md.51</a> | <a href="#">Md.43</a> |
| Last executed positioning data No.                                                | <a href="#">Md.52</a> | <a href="#">Md.46</a> |
| Repetition counter (QD75: <a href="#">Md.41</a> Special start repetition counter) | <a href="#">Md.53</a> | <a href="#">Md.41</a> |
| Control system repetition counter                                                 | –                     | <a href="#">Md.42</a> |
| Positioning data No. being executed                                               | <a href="#">Md.54</a> | <a href="#">Md.44</a> |
| Block No. being executed                                                          | <a href="#">Md.55</a> | <a href="#">Md.45</a> |
| Positioning data being executed                                                   | <a href="#">Md.56</a> | <a href="#">Md.47</a> |
| Deceleration start flag                                                           | –                     | <a href="#">Md.48</a> |

## Control data

| Item                                                                     | A1SD75                | QD75                  |
|--------------------------------------------------------------------------|-----------------------|-----------------------|
| Clock data setting (hour)                                                | <a href="#">Cd.1</a>  | –                     |
| Clock data setting (minute, second)                                      | <a href="#">Cd.2</a>  | –                     |
| Clock data writing                                                       | <a href="#">Cd.3</a>  | –                     |
| Target axis                                                              | <a href="#">Cd.4</a>  | –                     |
| Positioning data No.                                                     | <a href="#">Cd.5</a>  | –                     |
| Write pattern                                                            | <a href="#">Cd.6</a>  | –                     |
| Read/write request                                                       | <a href="#">Cd.7</a>  | –                     |
| Read/write positioning data I/F                                          | <a href="#">Cd.8</a>  | –                     |
| Flash ROM write request                                                  | <a href="#">Cd.9</a>  | <a href="#">Cd.1</a>  |
| Parameter initialization request                                         | <a href="#">Cd.10</a> | <a href="#">Cd.2</a>  |
| Positioning start No.                                                    | <a href="#">Cd.11</a> | <a href="#">Cd.3</a>  |
| Axis error reset                                                         | <a href="#">Cd.12</a> | <a href="#">Cd.5</a>  |
| Restart command                                                          | <a href="#">Cd.13</a> | <a href="#">Cd.6</a>  |
| M code OFF request                                                       | <a href="#">Cd.14</a> | <a href="#">Cd.7</a>  |
| New current value                                                        | <a href="#">Cd.15</a> | <a href="#">Cd.9</a>  |
| New speed value                                                          | <a href="#">Cd.16</a> | <a href="#">Cd.14</a> |
| Speed change request                                                     | <a href="#">Cd.17</a> | <a href="#">Cd.15</a> |
| Positioning operation speed override                                     | <a href="#">Cd.18</a> | <a href="#">Cd.13</a> |
| JOG speed                                                                | <a href="#">Cd.19</a> | <a href="#">Cd.17</a> |
| Inching movement amount                                                  | –                     | <a href="#">Cd.16</a> |
| Speed-position switching enable flag                                     | <a href="#">Cd.20</a> | <a href="#">Cd.24</a> |
| Speed-position switching control movement amount change register         | <a href="#">Cd.21</a> | <a href="#">Cd.23</a> |
| Position-speed switching enable flag                                     | –                     | <a href="#">Cd.26</a> |
| Position-speed switching control speed change register                   | –                     | <a href="#">Cd.25</a> |
| Target position change request flag                                      | –                     | <a href="#">Cd.29</a> |
| Target position change value (new address)                               | –                     | <a href="#">Cd.27</a> |
| Target position change value (new speed)                                 | –                     | <a href="#">Cd.28</a> |
| ABS direction in degrees                                                 | –                     | <a href="#">Cd.40</a> |
| Manual pulse generator enable flag                                       | <a href="#">Cd.22</a> | <a href="#">Cd.21</a> |
| Manual pulse generator 1 pulse input magnification                       | <a href="#">Cd.23</a> | <a href="#">Cd.20</a> |
| OPR request flag OFF request                                             | <a href="#">Cd.24</a> | <a href="#">Cd.19</a> |
| External start valid (QD75: <a href="#">Cd.8</a> External command valid) | <a href="#">Cd.25</a> | <a href="#">Cd.8</a>  |

| Item                                                                                | A1SD75                | QD75                  |
|-------------------------------------------------------------------------------------|-----------------------|-----------------------|
| Step valid flag                                                                     | <a href="#">Cd.26</a> | <a href="#">Cd.35</a> |
| Step mode                                                                           | <a href="#">Cd.27</a> | <a href="#">Cd.34</a> |
| Step start information                                                              | <a href="#">Cd.28</a> | <a href="#">Cd.36</a> |
| Skip command                                                                        | <a href="#">Cd.29</a> | <a href="#">Cd.37</a> |
| New torque value                                                                    | <a href="#">Cd.30</a> | <a href="#">Cd.22</a> |
| Positioning starting point No.                                                      | <a href="#">Cd.31</a> | <a href="#">Cd.4</a>  |
| Interrupt request during continuous operation                                       | <a href="#">Cd.32</a> | <a href="#">Cd.18</a> |
| Simultaneous starting axis start data No. (Axis 1 start data No.)                   | –                     | <a href="#">Cd.30</a> |
| Simultaneous starting axis start data No. (Axis 2 start data No.)                   | –                     | <a href="#">Cd.31</a> |
| Simultaneous starting axis start data No. (Axis 3 start data No.)                   | –                     | <a href="#">Cd.32</a> |
| Simultaneous starting axis start data No. (Axis 4 start data No.)                   | –                     | <a href="#">Cd.33</a> |
| Teaching data selection                                                             | –                     | <a href="#">Cd.38</a> |
| Teaching positioning data No.                                                       | –                     | <a href="#">Cd.39</a> |
| New acceleration time value                                                         | <a href="#">Cd.33</a> | <a href="#">Cd.10</a> |
| New deceleration time value                                                         | <a href="#">Cd.34</a> | <a href="#">Cd.11</a> |
| Acceleration/deceleration time change during speed change, enable/disable selection | <a href="#">Cd.35</a> | <a href="#">Cd.12</a> |
| Stop command processing for deceleration stop selection                             | –                     | <a href="#">Cd.42</a> |
| Deceleration start flag valid                                                       | –                     | <a href="#">Cd.41</a> |

## Positioning data, block start data, condition data

| Item                            |                                                  |                           | A1SD75 | QD75  |
|---------------------------------|--------------------------------------------------|---------------------------|--------|-------|
| Positioning data                | Operation pattern                                |                           | Da.1   |       |
|                                 | Control system                                   |                           | Da.2   |       |
|                                 | Acceleration time No.                            |                           | Da.3   |       |
|                                 | Deceleration time No.                            |                           | Da.4   |       |
|                                 | Axis to be interpolated                          |                           | —      | Da.5  |
|                                 | Positioning address/movement amount              |                           | Da.5   | Da.6  |
|                                 | Arc address                                      |                           | Da.6   | Da.7  |
|                                 | Command speed                                    |                           | Da.7   | Da.8  |
|                                 | Dwell time/JUMP destination positioning data No. |                           | Da.8   | Da.9  |
|                                 | M code/condition data                            |                           | Da.9   | Da.10 |
| Positioning start information * | Start block data *                               | Shape                     | Da.10  | Da.11 |
|                                 |                                                  | Start data No.            | Da.11  | Da.12 |
|                                 |                                                  | Special start instruction | Da.12  | Da.13 |
|                                 |                                                  | Parameter                 | Da.13  | Da.14 |
|                                 | Condition data                                   | Condition target          | Da.14  | Da.15 |
|                                 |                                                  | Condition operator        | Da.15  | Da.16 |
|                                 |                                                  | Address                   | Da.16  | Da.17 |
|                                 |                                                  | Parameter 1               | Da.17  | Da.18 |
|                                 |                                                  | Parameter 2               | Da.18  | Da.19 |
|                                 |                                                  |                           |        |       |

\*: Called "block start data" with the QD75.

## (6) Input/output signal comparisons

## Input signal comparisons

| Name                                                                | A1SD75                            |                              | QD75                              |                              |
|---------------------------------------------------------------------|-----------------------------------|------------------------------|-----------------------------------|------------------------------|
|                                                                     | Logic (initial status)            | Logic switch with parameters | Logic (initial status)            | Logic switch with parameters |
| Drive unit READY                                                    | Negative logic                    | Not possible                 | Negative logic                    | Possible                     |
| In-position signal                                                  | Negative logic                    | Not possible                 | —                                 | —                            |
| Zero signal                                                         | Negative logic                    | Not possible                 | Negative logic                    | Possible                     |
| Manual pulse generator A phase<br>Manual pulse generator B phase *1 | Negative logic<br>(multiple of 4) | Not possible                 | Negative logic<br>(multiple of 4) | Possible                     |
| Near-point dog signal                                               | Negative logic                    | Not possible                 | Negative logic                    | Possible                     |
| Stop signal                                                         | Negative logic                    | Not possible                 | Negative logic                    | Possible                     |
| Upper limit                                                         | Negative logic                    | Not possible                 | Negative logic                    | Possible                     |
| Lower limit                                                         | Negative logic                    | Not possible                 | Negative logic                    | Possible                     |
| External start *2                                                   | Negative logic                    | Not possible                 | Negative logic                    | Possible                     |
| Speed-position switching signal *2                                  | Negative logic                    | Not possible                 | Negative logic                    | Possible                     |

## Output signal comparisons

| Name                    | A1SD75                        |                              | QD75                          |                              |
|-------------------------|-------------------------------|------------------------------|-------------------------------|------------------------------|
|                         | Logic (initial status)        | Logic switch with parameters | Logic (initial status)        | Logic switch with parameters |
| Command pulse *3        | Positive logic<br>CW/CCW mode | Possible                     | Negative logic<br>CW/CCW mode | Possible                     |
| Deviation counter clear | Negative logic                | Not possible                 | Negative logic                | Possible                     |

\*1: Comparisons about manual pulse generator A phase/B phase

|                                            | A1SD75             | QD75                                                                   |
|--------------------------------------------|--------------------|------------------------------------------------------------------------|
| No. of connectable manual pulse generators | 1 generator/1 axis | 1 generator/1 module                                                   |
| Mode selection (with parameter)            | Not possible       | Possible<br>Multiple of 1 mode, multiple of 4 mode,<br>PULSE/SIGN mode |

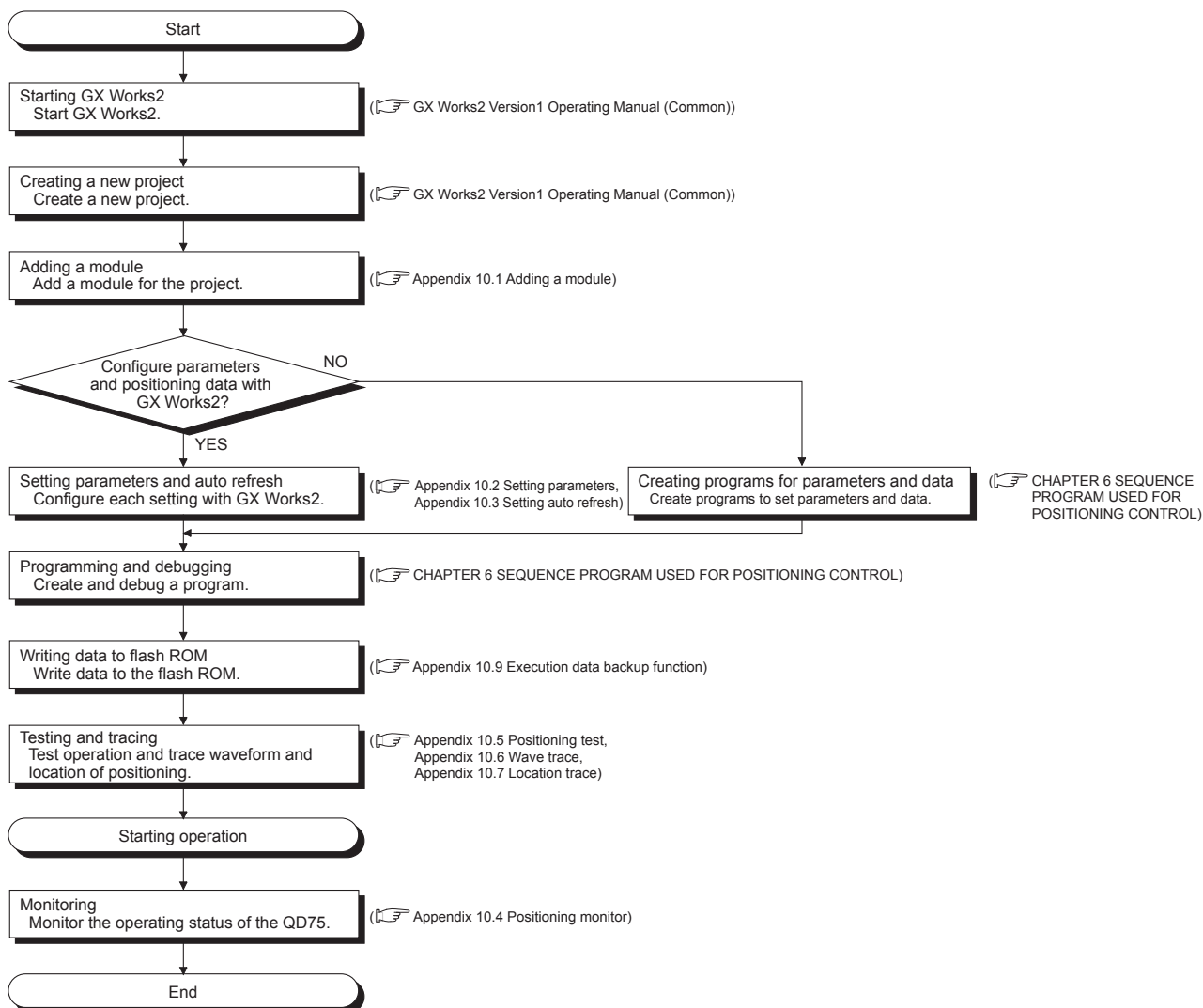
\*2: With the QD75, the "external start signal" and "speed-position switching signal" are combined into the "external command signal".

\*3: Comparisons about command pulse

|                                 | A1SD75                                                                                                                  | QD75                                                                             |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Mode selection (with parameter) | Possible<br>PULSE/SIGN mode, A phase/B phase (multiple of 4) mode,<br>A phase/B phase (multiple of 1) mode, CW/CCW mode |                                                                                  |
| Max. command frequency          | Open collector: 200kpps<br>Differential driver: 400kpps                                                                 | Open collector: 200kpps<br>Differential driver: 4Mpps(QD75D□N)<br>1Mpps (QD75D□) |

## Appendix 10 When using GX Works2

The following shows the procedure for positioning operation when GX Works2 is used.



For details on the operation method of GX Works2, refer to the following.

GX Works2 Version 1 Operating Manual (Common)

For details on operation in GX Works2 for the QD75, refer to the following.

GX Works2 Version 1 Operating Manual (Intelligent Function Module)



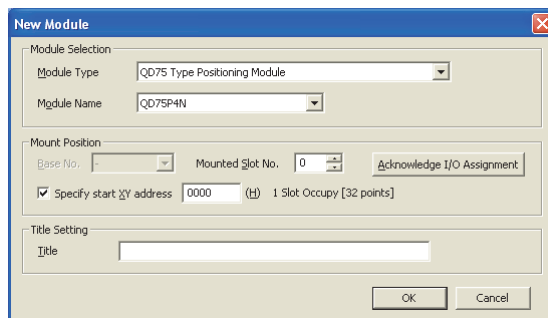
## Appendix 10.1 Adding a module

Add the model name of the positioning module to be used in the project.

## (1) Operating procedure

1. Open the "New Module..." dialog box.

Project window → Right-click [Intelligent Function Module] → "New Module..."



2. Configure settings

Set the following items.

| Item             |                           | Contents                                                                                                                                     |
|------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Module Selection | Module Type               | Select "QD75 Type Positioning Module".                                                                                                       |
|                  | Module Name               | Select the model name of the module to be connected.                                                                                         |
| Mount Position   | Mounted Slot No.          | Select the number of the slot where the module is mounted.                                                                                   |
|                  | Specify start X/Y address | The start I/O number (hexadecimal) of the module to be mounted on the slot set in "Mounted Slot No." is displayed. The value can be changed. |
| Title Setting    | Title                     | Enter a title.                                                                                                                               |

## Appendix 10.2 Setting parameters

Set parameters for the positioning module.

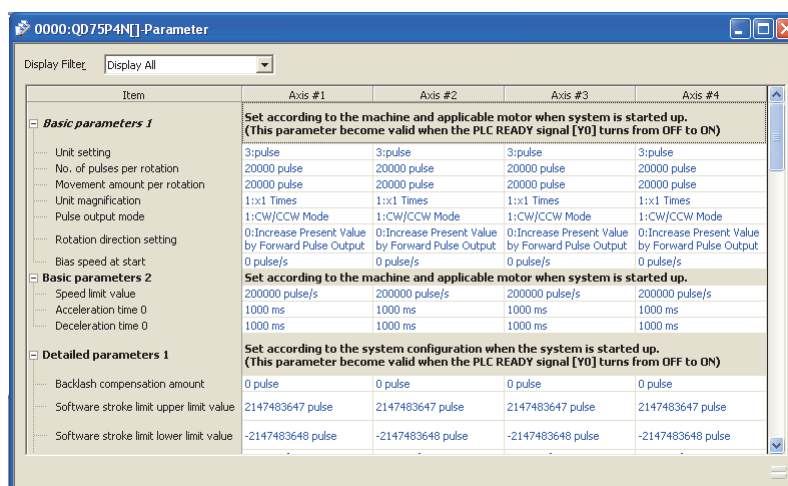
By setting parameters, the parameter setting by sequence program is not needed.

### ■ Parameter setting

#### (1) Operating procedure

1. Open the "Parameter" window.

Project window → [Intelligent Function Module] → Module name → "Parameter"



2. Configure settings

Double-click the setting-target item and select or enter a value.

- Items with a pull-down list  
Double-click the item and select an item in the displayed pull-down list.
- Items without a pull-down list  
Double-click the item and enter a value.

For details on the setting items, refer to the following.

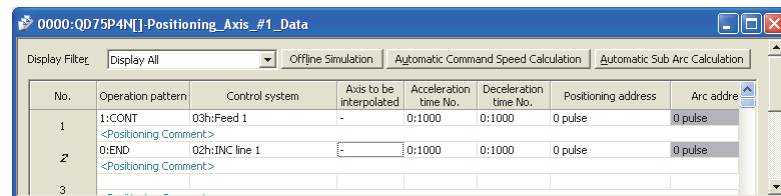
| Parameter               | Reference     |
|-------------------------|---------------|
| Basic parameters 1      | Section 5.2.1 |
| Basic parameters 2      | Section 5.2.2 |
| Detailed parameters 1   | Section 5.2.3 |
| Detailed parameters 2   | Section 5.2.4 |
| OPR basic parameters    | Section 5.2.5 |
| OPR detailed parameters | Section 5.2.6 |

## ■ Setting positioning data

### (1) Operating procedure

1. Open the "Parameter" window.

Project window → [Intelligent Function Module] → Module name →  
"Positioning\_Axis\_#□\_Data"



2. Configure settings.

Double-click the setting-target item and select or enter a value.

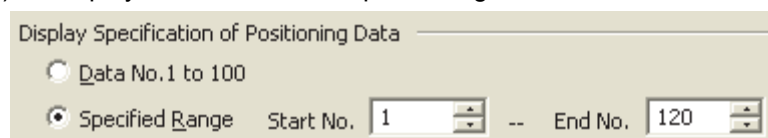
- Items with a pull-down list  
Double-click the item and select an item in the displayed pull-down list.
- Items without a pull-down list  
Double-click the item and enter a value.

### POINT

In the default status, the data displayed on "Positioning\_Axis\_#□\_Data" are only No. 1 to No. 100 of positioning data. To display the data No. 101 and later, change the display range under "Display Specification of Positioning data".

[Tool] → [Options...] → [Intelligent Function Module] → "QD75/LD75 Type Positioning"

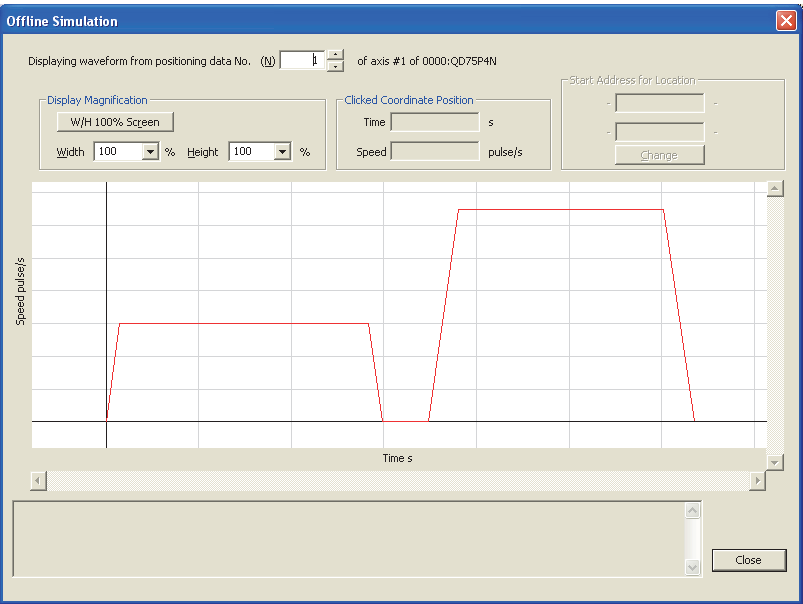
(Example) To display No.1 to No. 120 of positioning data



(2) Sub-function

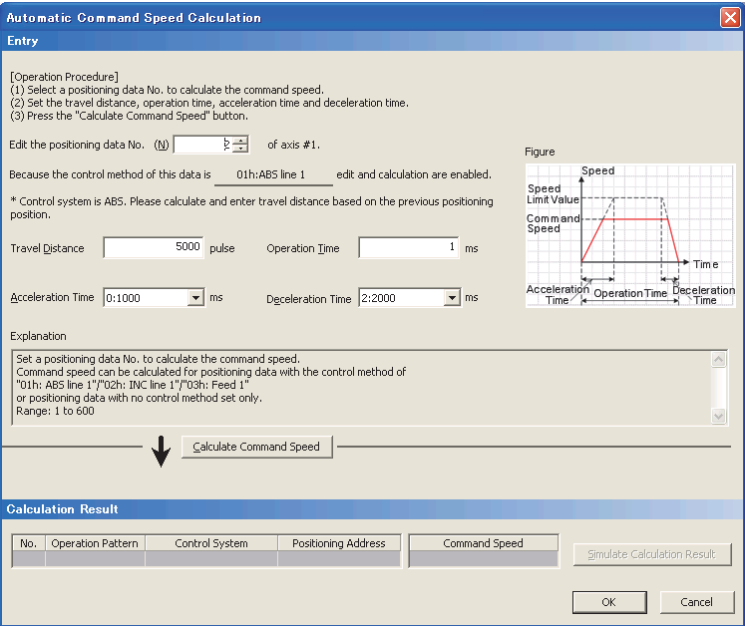
- Offline Simulation

The locus and waveform of configured positioning data can be checked.



- Automatic Command Speed Calculation

Constant speed is automatically calculated by setting the time for positioning from the starting position to the target position.



- Automatic Sub Arc Calculation

The circular interpolation control data for two positioning data interpolation is automatically created, by selecting two positioning data and setting the radius.

**Automatic Sub Arc Calculation**

**Entry**

[Operation Procedure]  
 (1) Select a positioning data No. to calculate the sub arc.  
 (2) Set the P0 (start point) and radius.  
 (3) Press the "Calculate Sub Arc" button.

Execute circular interpolation for positioning data No.  and positioning data No.   
 Edit and calculation are not enabled because circular interpolation cannot be executed with the control system of this data.

P0 (start point) Address  
☒ Use simulation result  
☐ Directly specify start point address

Reference Axis  pulse Interpolation Axis  pulse Radius  pulse

**Illustration**

| No. | Axis | Operation Pattern | Control System | Axis to be Interpolated | Positioning Address |
|-----|------|-------------------|----------------|-------------------------|---------------------|
| P1  | 2    | Axis #1           | 0:END          | 02h:INC line 1          | -                   |
| P2  | 3    | Axis #1           |                |                         | 5000 pulse          |

**Explanation**  
 Set the start positioning data No. for circular interpolation.  
 Sub arc can be calculated only when both of the control systems of specified positioning data and the next positioning data are any of "0Ah:ABS line 2", "0Bh:INC line 2" or "0Ch:Feed 2".  
 Range: 1 to 598

**Calculation Result**

| No. | Axis | Operation Pattern | Control System | Axis to be Interpolated | Positioning Address | Arc Address |
|-----|------|-------------------|----------------|-------------------------|---------------------|-------------|
|     |      |                   |                |                         |                     |             |
|     |      |                   |                |                         |                     |             |
|     |      |                   |                |                         |                     |             |
|     |      |                   |                |                         |                     |             |
|     |      |                   |                |                         |                     |             |
|     |      |                   |                |                         |                     |             |
|     |      |                   |                |                         |                     |             |
|     |      |                   |                |                         |                     |             |
|     |      |                   |                |                         |                     |             |
|     |      |                   |                |                         |                     |             |

- M Code Comment Edit

Set and display M code comments of the positioning module.

**M Code Comment Edit 0000:QD75P4N**

| No. | M Code | M Code Comment |
|-----|--------|----------------|
| 1   |        |                |
| 2   |        |                |
| 3   |        |                |
| 4   |        |                |
| 5   |        |                |
| 6   |        |                |
| 7   |        |                |
| 8   |        |                |
| 9   |        |                |
| 10  |        |                |

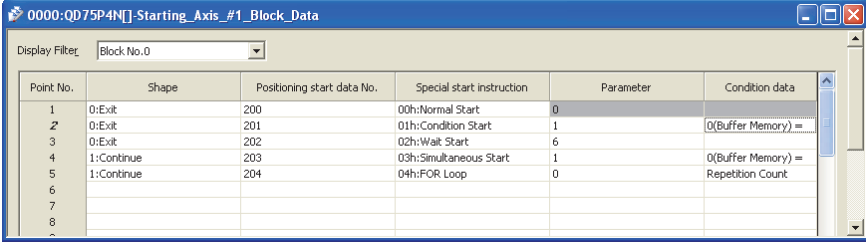
For details on each function, refer to the GX Works2 Version1 Operating Manual (Intelligent Function Module).

## ■ Setting block start data

### (1) Operating procedure

1. Open the "Parameter" window.

Project window → [Intelligent Function Module] → Module name →  
"Starting\_Axis\_#□\_Block\_Data"



| Point No. | Shape      | Positioning start data No. | Special start instruction | Parameter | Condition data     |
|-----------|------------|----------------------------|---------------------------|-----------|--------------------|
| 1         | 0:Exit     | 200                        | 00h:Normal Start          | 0         |                    |
| 2         | 0:Exit     | 201                        | 01h:Condition Start       | 1         | 0(Buffer Memory) = |
| 3         | 0:Exit     | 202                        | 02h:Wait Start            | 6         |                    |
| 4         | 1:Continue | 203                        | 03h:Simultaneous Start    | 1         | 0(Buffer Memory) = |
| 5         | 1:Continue | 204                        | 04h:FOR Loop              | 0         | Repetition Count   |
| 6         |            |                            |                           |           |                    |
| 7         |            |                            |                           |           |                    |
| 8         |            |                            |                           |           |                    |

2. Configure settings.

Double-click the setting-target item and select or enter a value.

- Items with a pull-down list  
Double-click the item and select an item in the displayed pull-down list.
- Items without a pull-down list  
Double-click the item and enter a value.

For details on the setting items, refer to Section 5.4 "List of block start data".

## Appendix 10.3 Setting auto refresh

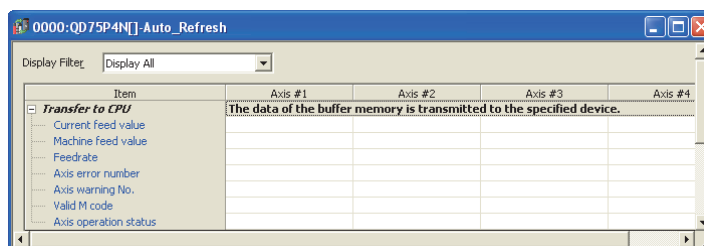
Transfer data in the buffer memory of the positioning module to specified devices in the CPU module.

By setting auto refresh, reading by sequence program is not needed.

## (1) Operating procedure

1. Open the "Auto\_Refresh" window.

Project window → [Intelligent Function Module] → Module name → "Auto\_Refresh"



2. Enter devices.

Click the text box of the item to be set, and enter the auto refresh target device.

**POINT**

To validate the auto refresh data after writing them into the CPU module, power off and then on or reset the CPU module.

## Appendix 10.4 Positioning monitor

With the positioning monitor function, the QD75 operating status can be confirmed, and debugging can be performed.

The following five types of monitors are available in this function:

- Axis Monitor : The actual status of each axis can be monitored.
- Starting History : 16 starting history logs of operations such as positioning operation, JOG operation, and manual pulse generator operation can be monitored.
- Error History : 16 error history logs can be monitored.
- Warning History : 16 warning history logs can be monitored.
- Module Information List : Signals and flag ON/OFF status of each axis can be monitored.

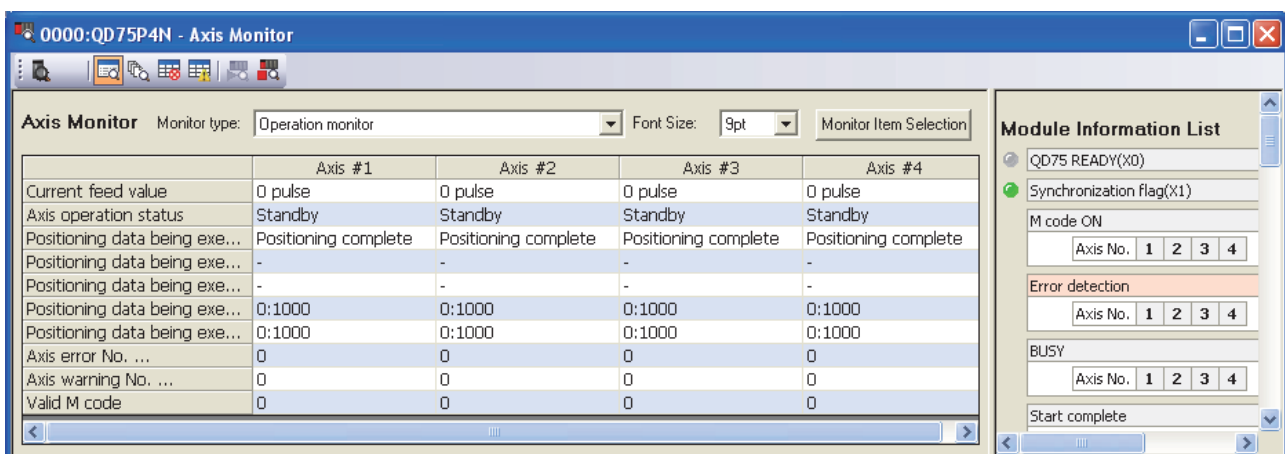
### ■ Axis Monitor

#### (1) Operating procedure

1. Starting the "Positioning Monitor" window

Display the "Positioning Monitor" window.

[Tool] → [Intelligent Function Module Tool] → [QD75/LD75 Positioning Module] → [Positioning Monitor...]



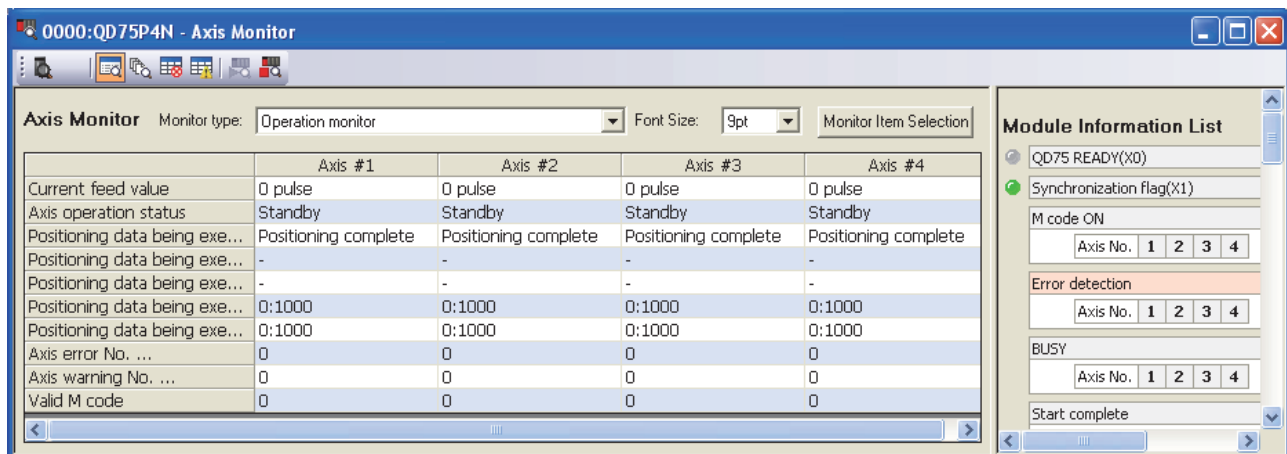
By default, the "Axis Monitor" window is displayed. When displaying the "Axis Monitor" window from the status which another window is selected, proceed according to the 2. below.



## 2. Switching to the "Axis Monitor" window

Click the "Axis Monitor" button (  ) on the toolbar.

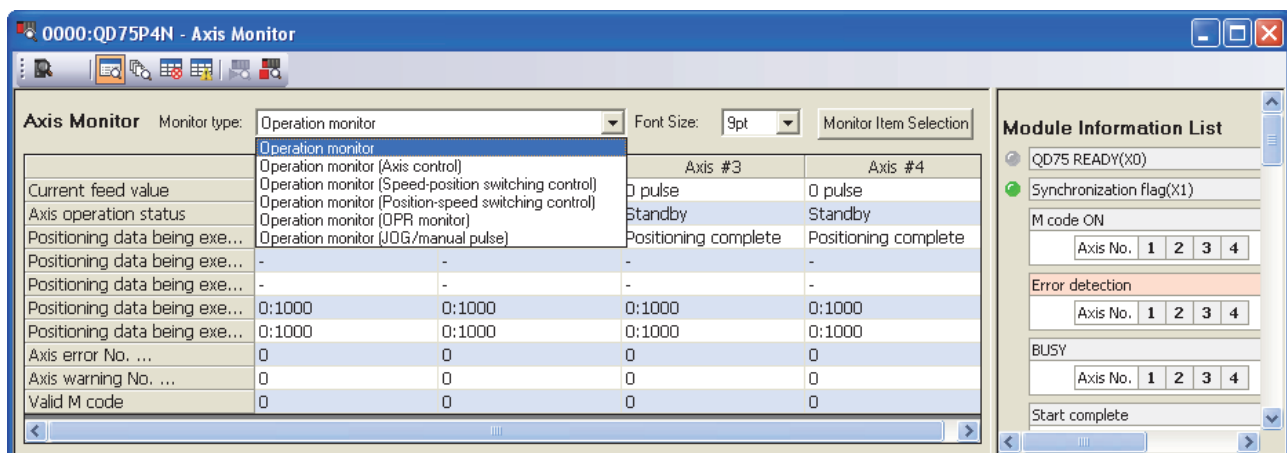
The display switches to the "Axis Monitor" window.



## 3. Selecting the monitor type

Select the monitor type from the pull-down menu of "Monitor type".

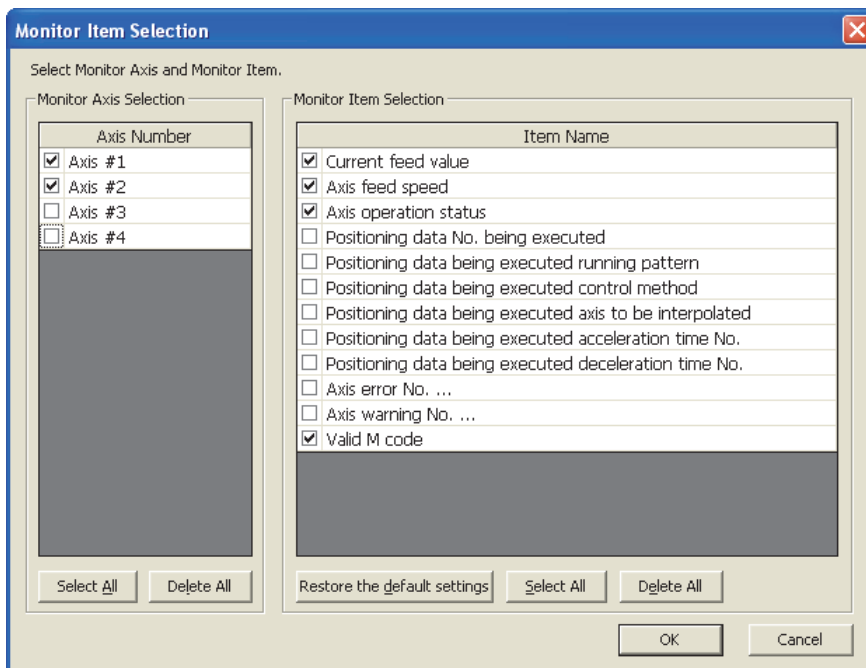
- "Operation monitor"
- "Operation monitor (Axis control)"
- "Operation monitor (Speed-position switching control)"
- "Operation monitor (Position-speed switching control)"
- "Operation monitor (OPR monitor)"
- "Operation monitor (JOG/manual pulse)"



## 4. Selecting the monitor axis and monitor item

1) Click the **Monitor Item Selection** button.

The "Monitor Item Selection" window is displayed.

2) Select the ☐ of the axes and items to be monitored.

The following shows the selectable monitor items for each monitor type.

• For "Operation monitor":

| Monitor item                                            | Symbol of reference buffer memory | Reference     |
|---------------------------------------------------------|-----------------------------------|---------------|
| Current feed value                                      | Md.20                             | Section 5.6.2 |
| Axis feed speed                                         | Md.28                             |               |
| Axis operation status                                   | Md.26                             |               |
| Positioning data No. being executed                     | Md.44                             |               |
| Positioning data being executed running pattern         | Md.47                             |               |
| Positioning data being executed control method          |                                   |               |
| Positioning data being executed axis to be interpolated |                                   |               |
| Positioning data being executed acceleration time No.   |                                   |               |
| Positioning data being executed deceleration time No.   |                                   |               |
| Axis error No. ...                                      | Md.23                             |               |
| Axis warning No. ...                                    | Md.24                             |               |
| Valid M code                                            | Md.25                             |               |

• For "Operation monitor (Axis control)":

| Monitor item                         | Symbol of reference buffer memory | Reference     |
|--------------------------------------|-----------------------------------|---------------|
| Current feed value                   | Md.20                             | Section 5.6.2 |
| Axis feed speed                      | Md.28                             |               |
| Axis operation status                | Md.26                             |               |
| Target value                         | Md.32                             |               |
| Machine feed value                   | Md.21                             |               |
| New current value                    | Cd.9                              | Section 5.7.2 |
| New speed value                      | Cd.14                             |               |
| Positioning operation speed override | Cd.13                             |               |
| Step mode                            | Cd.34                             |               |
| Step valid flag                      | Cd.35                             |               |
| Speed change being processed flag    | Md.40                             | Section 5.6.2 |
| Status speed change 0 flag           | Md.31                             |               |
| External command valid               | Cd.8                              | Section 5.7.2 |
| Skip command                         | Cd.37                             |               |

• For "Operation monitor (Speed-position switching control)":

| Monitor item                                                     | Symbol of reference buffer memory | Reference     |
|------------------------------------------------------------------|-----------------------------------|---------------|
| Current feed value                                               | Md.20                             | Section 5.6.2 |
| Axis feed speed                                                  | Md.28                             |               |
| Axis operation status                                            | Md.26                             |               |
| Target speed                                                     | Md.33                             |               |
| Feedrate                                                         | Md.22                             |               |
| Current speed                                                    | Md.27                             |               |
| Positioning amount of speed and position switch control          | Md.29                             | Section 5.7.2 |
| Speed-position switching control movement amount change register | Cd.23                             |               |
| Status speed and position change latch flag                      | Md.31                             | Section 5.6.2 |
| Speed-position switching enable flag                             | Cd.24                             | Section 5.7.2 |
| Status speed controlling flag                                    | Md.31                             | Section 5.6.2 |

• For "Operation monitor (Position-speed switching control)":

| Monitor item                                           | Symbol of reference buffer memory | Reference     |
|--------------------------------------------------------|-----------------------------------|---------------|
| Current feed value                                     | Md.20                             | Section 5.6.2 |
| Axis feed speed                                        | Md.28                             |               |
| Axis operation status                                  | Md.26                             |               |
| Target speed                                           | Md.33                             |               |
| Feedrate                                               | Md.22                             |               |
| Current speed                                          | Md.27                             |               |
| Position-speed switching control speed change register | Cd.25                             | Section 5.7.2 |
| Status position and speed change latch flag            | Md.31                             | Section 5.6.2 |
| Position-speed switching enable flag                   | Cd.26                             | Section 5.7.2 |
| Status speed controlling flag                          | Md.31                             | Section 5.6.2 |

• For "Operation monitor (OPR monitor)":

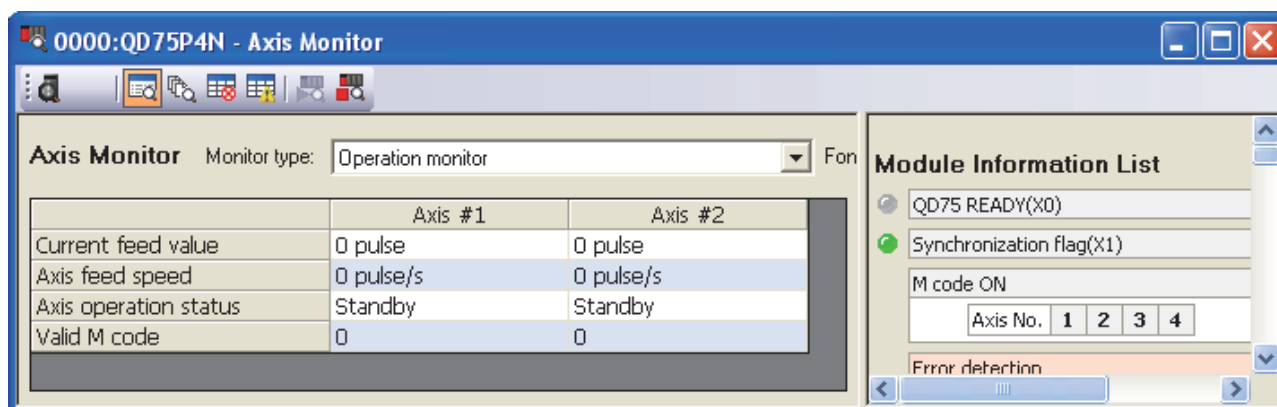
| Monitor item                                | Symbol of reference buffer memory | Reference     |
|---------------------------------------------|-----------------------------------|---------------|
| Current feed value                          | Md.20                             | Section 5.6.2 |
| Axis feed speed                             | Md.28                             |               |
| Axis operation status                       | Md.26                             |               |
| Movement amount after near-point dog ON     | Md.34                             |               |
| Torque limitation stored value              | Md.35                             |               |
| Status command in-position flag             | Md.31                             |               |
| Status OPR request flag                     |                                   |               |
| Status OPR complete flag                    |                                   |               |
| External I/O signal lower limit             | Md.30                             |               |
| External I/O signal upper limit             |                                   |               |
| External I/O signal zero signal             |                                   |               |
| External I/O signal near-point dog signal   |                                   |               |
| External I/O signal deviation counter clear |                                   |               |

• For "Operation monitor (JOG/manual pulse)":

| Monitor item                                          | Symbol of reference buffer memory<br>or device No. of the output signal |        |        |        | Reference     |
|-------------------------------------------------------|-------------------------------------------------------------------------|--------|--------|--------|---------------|
|                                                       | Axis 1                                                                  | Axis 2 | Axis 3 | Axis 4 |               |
| Current feed value                                    | Md.20                                                                   |        |        |        | Section 5.6.2 |
| Axis feed speed                                       | Md.28                                                                   |        |        |        |               |
| Axis operation status                                 | Md.26                                                                   |        |        |        |               |
| Forward JOG start                                     | Y8                                                                      | YA     | YC     | YE     | Section 3.3.3 |
| Reverse JOG start                                     | Y9                                                                      | YB     | YD     | YF     |               |
| JOG speed                                             | Cd.17                                                                   |        |        |        | Section 5.7.2 |
| JOG speed limit value                                 | Pr.31                                                                   |        |        |        | Section 5.2.4 |
| JOG acceleration time                                 | Pr.32                                                                   |        |        |        |               |
| JOG deceleration time                                 | Pr.33                                                                   |        |        |        |               |
| Manual pulse generator enable flag                    | Cd.21                                                                   |        |        |        | Section 5.7.2 |
| Manual pulse generator input selection                | Pr.24                                                                   |        |        |        | Section 5.2.3 |
| Manual pulse generator 1 pulse input<br>magnification | Cd.20                                                                   |        |        |        | Section 5.7.2 |

3) Click the  button to close the "Monitor Item Selection" window.

The selected items are reflected to the "Axis Monitor" window.



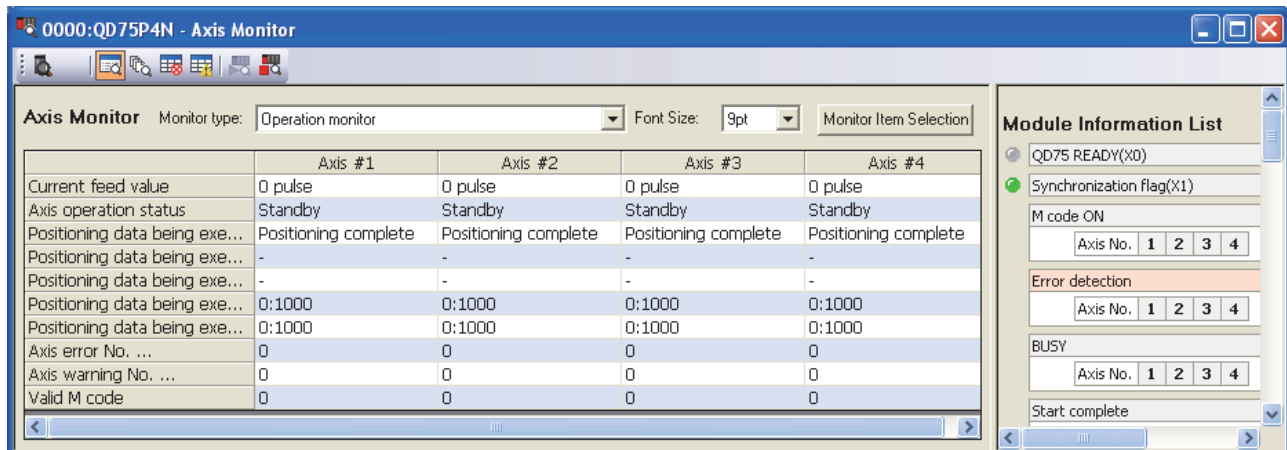
## ■ Starting History

### (1) Operating procedure

#### 1. Starting the "Positioning Monitor" window

Display the "Positioning Monitor" window.

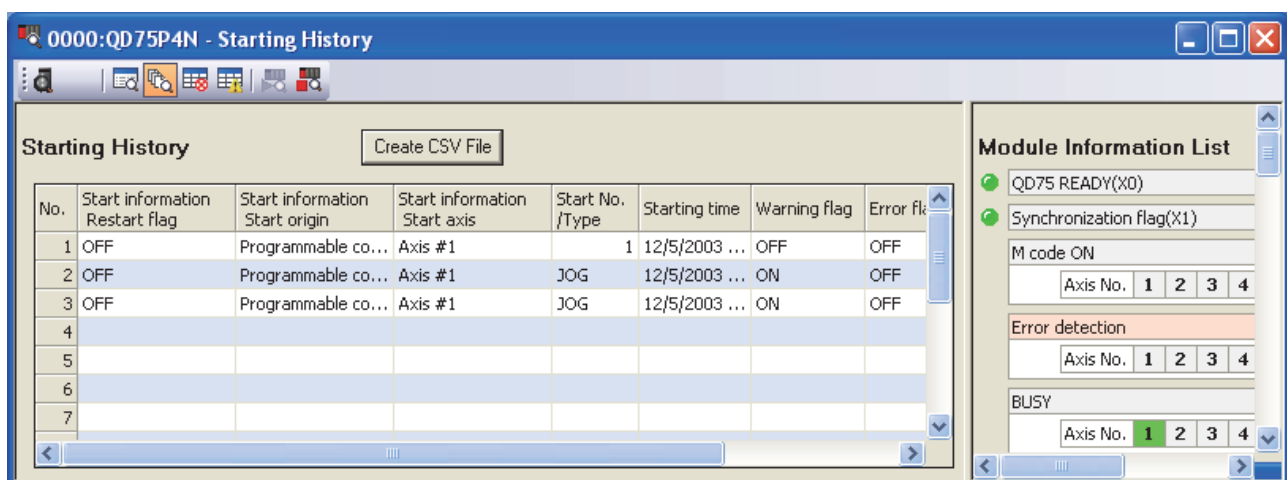
[Tool] → [Intelligent Function Module Tool] → [QD75/LD75 Positioning Module] → [Positioning Monitor...]



#### 2. Switching to the "Starting History" window

Click the "Starting History" button (  ) on the toolbar.

The display switches to the "Starting History" window.



When the number of logs exceeds 16, the latest log overwrites the oldest log so that the latest 16 starting histories can be monitored all the time.

Also, the error details can be confirmed by clicking the error number.

For details on the starting history, refer to Section 5.6.1.

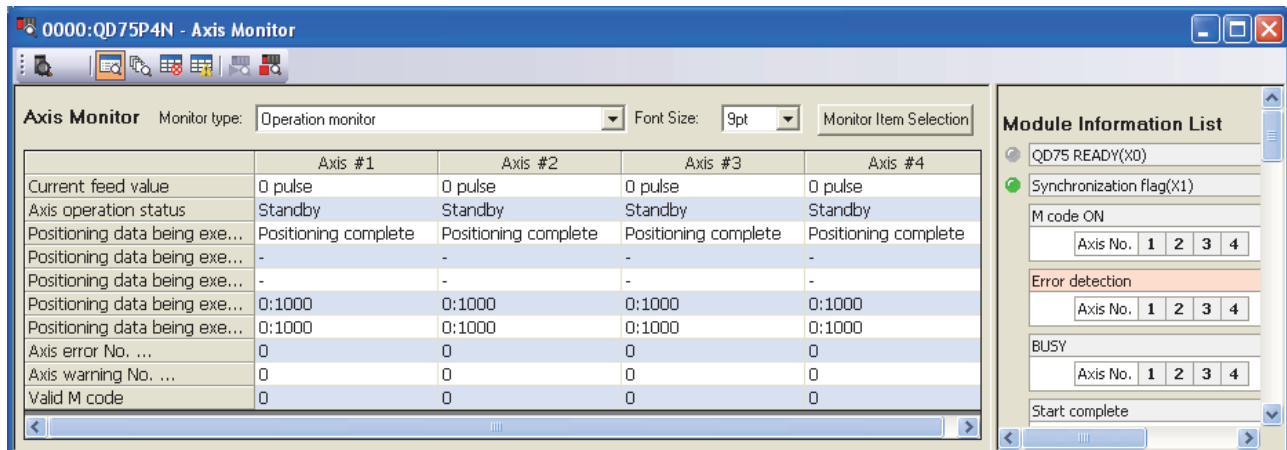
## ■ Error History

### (1) Operating procedure

#### 1. Starting the "Positioning Monitor" window

Display the "Positioning Monitor" window.

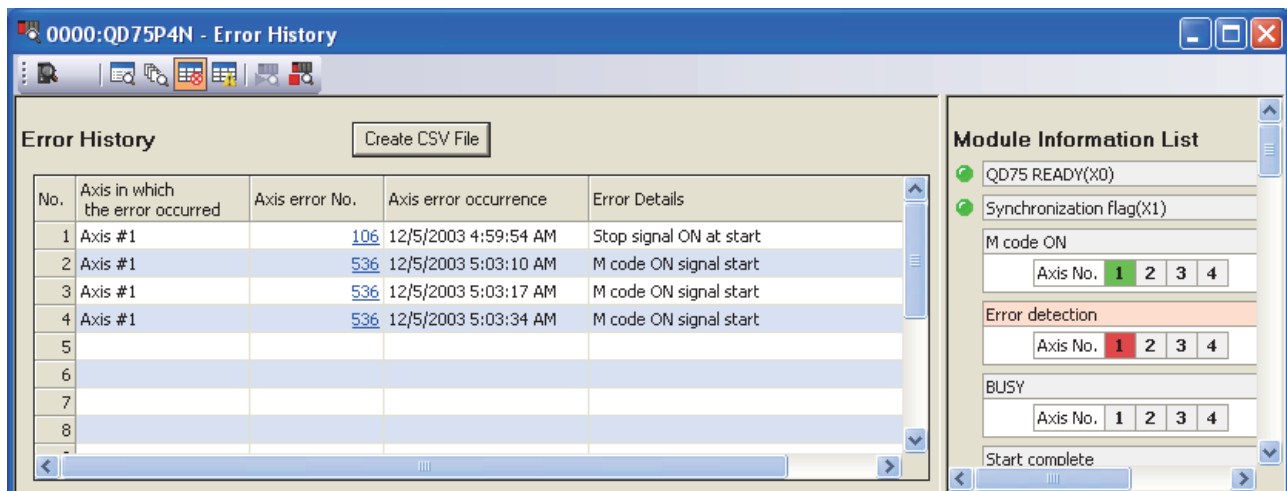
[Tool] → [Intelligent Function Module Tool] → [QD75/LD75 Positioning Module] → [Positioning Monitor...]



#### 2. Switching to the "Error History" window

Click the "Error History" button (  ) on the toolbar.

The display switches to the "Error History" window.



When the number of logs exceeds 16, the latest log overwrites the oldest log so that the latest 16 error histories can be monitored all the time.

Also, the error details can be confirmed by clicking the axis error No.

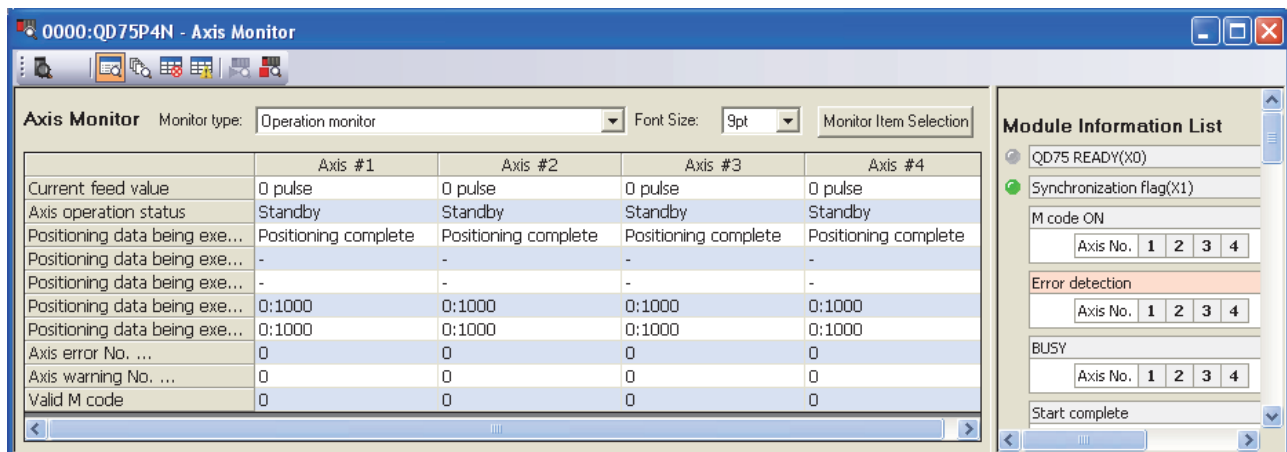
## ■ Warning History

### (1) Operating procedure

#### 1. Starting the "Positioning Monitor" window

Display the "Positioning Monitor" window.

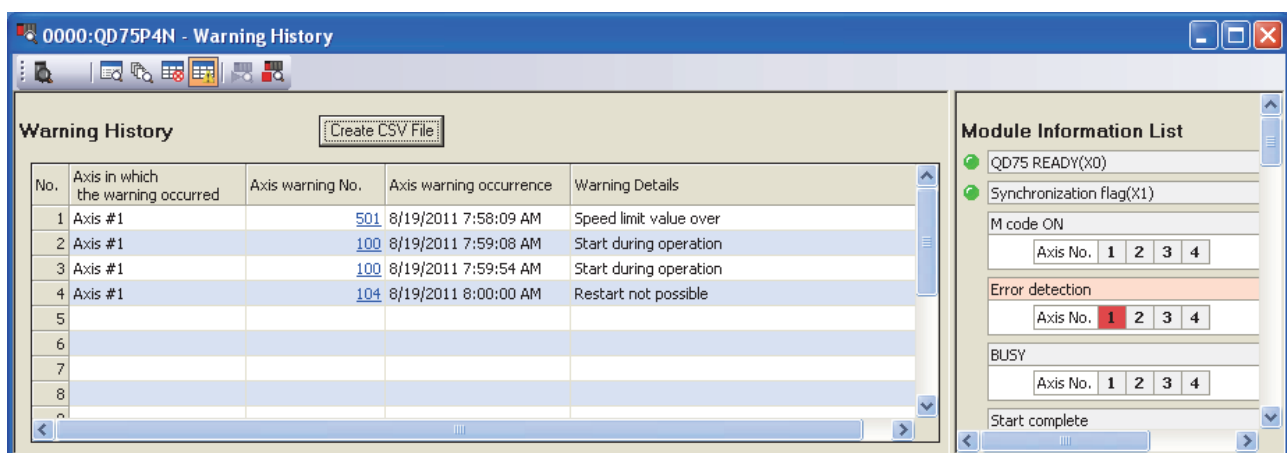
[Tool] → [Intelligent Function Module Tool] → [QD75/LD75 Positioning Module] → [Positioning Monitor...]



#### 2. Switching to the "Warning History" window

Click the "Warning History" button (  ) on the toolbar.

The display switches to the "Warning History" window.



When the number of logs exceeds 16, the latest log overwrites the oldest log so that the latest 16 warning histories can be monitored all the time.

Also, the warning details can be confirmed by clicking the axis warning No.

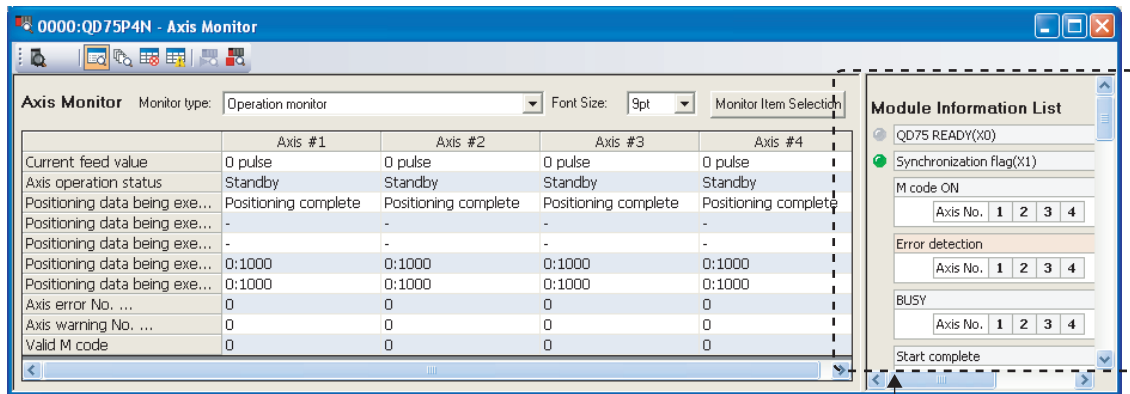
## ■ Module Information List

### (1) Operating procedure

#### 1. Starting the "Positioning Monitor" window

Display the "Positioning Monitor" window.

[Tool] → [Intelligent Function Module Tool] → [QD75/LD75 Positioning Module] → [Positioning Monitor...]



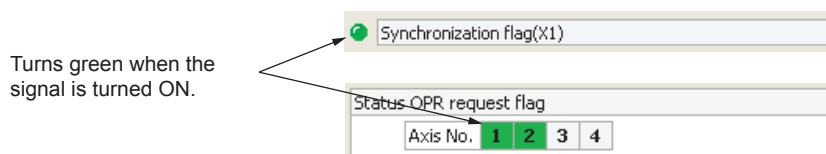
"Module Information List" window



The status by each axis of the following items is displayed in the "Module Information List" window.

| Monitor item                                | Device No. of the I/O signal device<br>or symbol of the buffer memory to<br>be referred |        |        |        | Reference     |
|---------------------------------------------|-----------------------------------------------------------------------------------------|--------|--------|--------|---------------|
|                                             | Axis 1                                                                                  | Axis 2 | Axis 3 | Axis 4 |               |
| QD75 READY (X0)                             | X0                                                                                      |        |        |        | Section 3.3.2 |
| Synchronization flag (X1)                   | X1                                                                                      |        |        |        |               |
| M code ON                                   | X4                                                                                      | X5     | X6     | X7     |               |
| Error detection                             | X8                                                                                      | X9     | XA     | XB     |               |
| BUSY                                        | XC                                                                                      | XD     | XE     | XF     |               |
| Start complete                              | X10                                                                                     | X11    | X12    | X13    |               |
| Positioning complete                        | X14                                                                                     | X15    | X16    | X17    |               |
| PLC READY (Y0)                              | Y0                                                                                      |        |        |        | Section 3.3.3 |
| Axis stop                                   | Y4                                                                                      | Y5     | Y6     | Y7     |               |
| Forward JOG start                           | Y8                                                                                      | YA     | YC     | YE     |               |
| Reverse JOG start                           | Y9                                                                                      | YB     | YD     | YF     |               |
| Positioning start                           | Y10                                                                                     | Y11    | Y12    | Y13    |               |
| Execution prohibition flag                  | Y14                                                                                     | Y15    | Y16    | Y17    |               |
| External I/O signal lower limit             | Md.30                                                                                   |        |        |        | Section 5.6.2 |
| External I/O signal upper limit             |                                                                                         |        |        |        |               |
| External I/O signal drive unit ready        |                                                                                         |        |        |        |               |
| External I/O signal stop signal             |                                                                                         |        |        |        |               |
| External I/O signal external command        |                                                                                         |        |        |        |               |
| External I/O signal zero signal             |                                                                                         |        |        |        |               |
| External I/O signal near-point dog signal   |                                                                                         |        |        |        |               |
| External I/O signal deviation counter clear |                                                                                         |        |        |        |               |
| External command valid                      | Cd.8                                                                                    |        |        |        | Section 5.7.2 |
| Status speed controlling flag               | Md.31                                                                                   |        |        |        | Section 5.6.2 |
| Status speed and position change latch flag |                                                                                         |        |        |        |               |
| Status command in-position flag             |                                                                                         |        |        |        |               |
| Status OPR request flag                     |                                                                                         |        |        |        |               |
| Status OPR complete flag                    |                                                                                         |        |        |        |               |
| Status position and speed change latch flag |                                                                                         |        |        |        |               |
| Status axis warning detection               |                                                                                         |        |        |        |               |
| Status speed change 0 flag                  |                                                                                         |        |        |        |               |

Statuses turned ON are displayed in green by each axis.



Also, at error occurrence, the axis in which the error occurred ("Error detection") is displayed in red.

At occurrence of warning, the axis in which the warning occurred ("Status axis warning detection") is displayed in orange.

## Appendix 10.5 Positioning test

This function allows users to perform the following tests while monitoring the current status of the QD75.

- Positioning start test
- JOG/manual pulse generator/OPR test
- Speed change test
- Current value change test

### ■ Positioning start test

Test operation is performed by specifying the positioning data No. or point No. of the block start data.

#### (1) Operating procedure

1. Open the "Positioning Test" dialog box.

[Tool] → [Intelligent Function Module Tool] → [QD75/LD75 Positioning Module] → [Positioning Test...]

Monitor current values such as the current feed value and feedrate.

Test each function while checking current values with the monitor.

**Positioning Test**

Target Module: QD75P4N I/O Address: 0000

| Monitor Item          | Axis #1   | Axis #2   | Axis #3   | Axis #4   |
|-----------------------|-----------|-----------|-----------|-----------|
| Current feed value    | 0 pulse   | 0 pulse   | 0 pulse   | 0 pulse   |
| Machine feed value    | 0 pulse   | 0 pulse   | 0 pulse   | 0 pulse   |
| Feedrate              | 0 pulse/s | 0 pulse/s | 0 pulse/s | 0 pulse/s |
| Axis error number     | 0         | 0         | 0         | 0         |
| Axis warning No.      | 0         | 0         | 0         | 0         |
| Valid M code          | 0         | 0         | 0         | 0         |
| Axis operation status | Standby   | Standby   | Standby   | Standby   |
| Current speed         | 0 pulse/s | 0 pulse/s | 0 pulse/s | 0 pulse/s |
| Axis feedrate         | 0 pulse/s | 0 pulse/s | 0 pulse/s | 0 pulse/s |

**Test**

Target Axis: Axis #1

Select Function: Positioning start signal Please set this function after stopping the positioning.

Start Type:
 

- ☒ Positioning Start Signal
- ☐ Block Start
- ☐ Multiple Axes Simultaneous Start

Positioning start data

Positioning Data No. (1 to 600): 1

Step:
 

- ☐ Start step
- Step Mode: Execute step operation by declaration

External Command:
 

- ☐ External Command Valid
- ☐ Speed-position Switching Enable Flag
- ☐ Position-speed Switching Enable Flag

Buttons: Starting, Skip, Stop Target Axis(J), Stop All Axis, Restart Stop Axis, Positioning Complete, Error/Warning Details Confirmation, Error/Warning Reset, M Code OFF Request, Close

## 2. Select the test target axis.

Select the test target axis from the pull-down menu of "Target Axis".

## 3. Select the function.

Select "Positioning start signal" from the pull-down menu of "Select Function".

The screenshot shows the 'Test' dialog box with the following settings:

- Target Axis:** Axis #1
- Select Function:** Positioning start signal
- Start Type:** Positioning Start Signal (selected), Block Start, Multiple Axes Simultaneous Start
- Positioning start data:** Positioning Data No. (1 to 600) set to 1
- Step:** Start step checkbox is checked, Step Mode is 'Execute step operation by declaration', Continue button is visible.
- External Command:** External Command Valid checkbox is checked, Speed-position Switching Enable Flag and Position-speed Switching Enable Flag are unchecked, Set button is visible.
- Buttons:** Starting, Stop, Stop Target Axis(J), Stop All Axis, Restart Stop Axis, Positioning Complete, Error/Warning Details Confirmation, Error/Warning Reset, M Code OFF Request, Close.

## 4. Perform positioning start test.

1) Select "Positioning Start Signal", "Block Start", or "Multiple Axes Simultaneous Start" for "Start Type".

2) Set positioning start data according to the setting for "Start Type".

- Positioning Start Signal: Set the positioning data No.
- Block Start: Set the block No. and point No.
- Multiple Axes Simultaneous Start: Set the multiple axes simultaneous start data No.

Set "Step" and "External Command" as needed.

## • Step

To perform test operation by step start, check the "Start step" checkbox, select a step mode from the pull-down list, and click the **Continue** button.

## • External Command

To enable external commands or control switch, select the corresponding item and click the **Set** button.

3) Click the **Starting** button to perform test operation.

## ■ JOG/manual pulse generator/OPR test

The following test can be performed when positioning control is debugged by the JOG operation test or manual pulse operation test.

- Direction check (forward run or reverse run)
- On/off status check of external input signals, such as upper/lower limit switches, zero signal, or near-point dog signal
- Operation test of speed and acceleration/deceleration
- Measurement of backlash compensation amount for forward/reverse rotation
- Accurate address check and movement amount measurement

In addition, the OP is established and the OPR basic and detailed parameters are corrected by the OPR test.

### (1) Operating procedure

1. Open the "Positioning Test" dialog box.

[Tool] → [Intelligent Function Module Tool] → [QD75/LD75 Positioning Module] → [Positioning Test]

Monitor current values such as the current feed value and feedrate.

Test each function while checking current values with the monitor.

**Positioning Test**

Target Module: QD75P4N I/O Address: 0000

| Monitor Item          | Axis #1   | Axis #2   | Axis #3   | Axis #4   |
|-----------------------|-----------|-----------|-----------|-----------|
| Current feed value    | 0 pulse   | 0 pulse   | 0 pulse   | 0 pulse   |
| Machine feed value    | 0 pulse   | 0 pulse   | 0 pulse   | 0 pulse   |
| Feedrate              | 0 pulse/s | 0 pulse/s | 0 pulse/s | 0 pulse/s |
| Axis error number     | 0         | 0         | 0         | 0         |
| Axis warning No.      | 0         | 0         | 0         | 0         |
| Valid M code          | 0         | 0         | 0         | 0         |
| Axis operation status | Standby   | Standby   | Standby   | Standby   |
| Current speed         | 0 pulse/s | 0 pulse/s | 0 pulse/s | 0 pulse/s |
| Axis feedrate         | 0 pulse/s | 0 pulse/s | 0 pulse/s | 0 pulse/s |

**Test**

Target Axis: Axis #1

Select Function: Positioning start signal Please set this function after stopping the positioning.

Start Type:
   
☒ Positioning Start Signal
   
☐ Block Start
   
☐ Multiple Axes Simultaneous Start

Positioning start data:
   
 Positioning Data No. (1 to 600): 1

Step:
   
☐ Start step
   
 Step Mode: Execute step operation by declaration

External Command:
   
☐ External Command Valid
   
☐ Speed-position Switching Enable Flag
   
☐ Position-speed Switching Enable Flag

Buttons: Starting, Skip, Stop Target Axis(1), Stop All Axis, Restart Stop Axis, Positioning Complete, Error/Warning Details Confirmation, Error/Warning Reset, M Code OFF Request, Close

2. Select the test target axis.

Select the test target axis from the pull-down menu of "Target Axis".

## 3. Select the function.

Select "JOG/Manual Pulse Generator/OPR" from the pull-down menu of "Select Function".

The screenshot shows the 'Test' window with the following settings:

- Target Axis: Axis #1
- Select Function: JOG/Manual Pulse Generator/OPR (Note: Please set this function after stopping the positioning.)
- JOG Speed: 1 pulse/s (1 to 1000000)
- Inching Movement Amount: 0 pulse (0 to 65535)
- Manual Pulse Generator:
  - Manual pulse generator enable flag: ☐
  - Manual Pulse 1 Pulse Generator Input Magnification: 1 x (1 to 100)
- OPR Operation:
  - OPR Method: Machine OPR
  - OPR button

Buttons at the bottom: Starting, Skip, Stop Target Axis(J), Stop All Axis, Restart Stop Axis, Positioning Complete, Error/Warning Details Confirmation, Error/Warning Reset, M Code OFF Request, Close.

## 1) Perform each operation.

- JOG operation

Set "JOG speed" to "1" or more, "Inching Movement Amount" to "0" and click the **Forward RUN** or **Reverse RUN** button.

- Manual Pulse Generator operation

Set "Manual pulse generator 1 pulse input magnification", select "Manual pulse generator enable flag" and use the manual pulse generator.

- OPR operation

Select the OPR method from "Machine OPR" and "Fast OPR", and click the **OPR** button.

## ■ Speed change test

The appropriate speed and acceleration/deceleration time can be checked by changing speed or acceleration/deceleration time or by performing override to the axis where the positioning start test, OPR test, or JOG operation test is performed.

### (1) Operating procedure

1. Open the "Positioning Test" dialog box.

Display the "Positioning Test" window.

[Tool] → [Intelligent Function Module Tool] → [QD75/LD75 Positioning Module] → [Positioning Test]

Monitor current values such as the current feed value and feedrate.

Test each function while checking current values with the monitor.

**Positioning Test**

Target Module: QD75P4N I/O Address: 0000

| Monitor Item          | Axis #1   | Axis #2   | Axis #3   | Axis #4   |
|-----------------------|-----------|-----------|-----------|-----------|
| Current feed value    | 0 pulse   | 0 pulse   | 0 pulse   | 0 pulse   |
| Machine feed value    | 0 pulse   | 0 pulse   | 0 pulse   | 0 pulse   |
| Feedrate              | 0 pulse/s | 0 pulse/s | 0 pulse/s | 0 pulse/s |
| Axis error number     | 0         | 0         | 0         | 0         |
| Axis warning No.      | 0         | 0         | 0         | 0         |
| Valid M code          | 0         | 0         | 0         | 0         |
| Axis operation status | Standby   | Standby   | Standby   | Standby   |
| Current speed         | 0 pulse/s | 0 pulse/s | 0 pulse/s | 0 pulse/s |
| Axis feedrate         | 0 pulse/s | 0 pulse/s | 0 pulse/s | 0 pulse/s |

**Test**

Target Axis: Axis #1

Select Function: Positioning start signal Please set this function after stopping the positioning.

Start Type: ☒ Positioning Start Signal ☐ Block Start ☐ Multiple Axes Simultaneous Start

Positioning start data: Positioning Data No. (1 to 600): 1

Step: ☐ Start step  Step Mode: Execute step operation by declaration

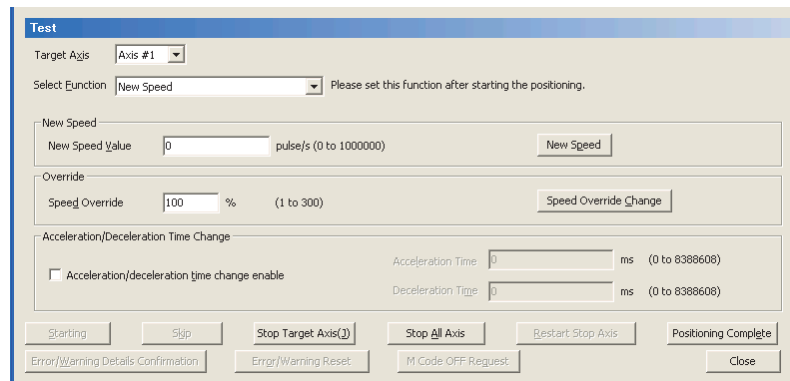
External Command: ☐ External Command Valid ☐ Speed-position Switching Enable Flag ☐ Position-speed Switching Enable Flag

2. Select the test target axis.

Select the test target axis from the pull-down menu of "Target Axis".

## 3. Select the function.

Select "New Speed" from the pull-down menu of "Select Function".



## 4. Execute the speed change test.

## 1) Execute the speed change function.

## • New Speed

Enter the new speed value during the startup of positioning start test or OPR test, and click the **New Speed** button.

## • Override

Enter the speed override value and click the **Speed Override Change** button.

## • Acceleration/Deceleration Time Change

Check the "Acceleration/deceleration time change enable" checkbox, set the acceleration time and deceleration time values, and click the **New Speed** button.

## ■ Current value change test

The current feed value of the QD75 can be changed to the specified address.

### (1) Operating procedure

1. Open the "Positioning Test" dialog box.

[Tool] → [Intelligent Function Module Tool] → [QD75/LD75 Positioning Module] → [Positioning Test]

Monitor current values such as the current feed value and feedrate.

Test each function while checking current values with the monitor.

**Positioning Test**

Target Module: QD75P4N I/O Address: 0000

| Monitor Item          | Axis #1   | Axis #2   | Axis #3   | Axis #4   |
|-----------------------|-----------|-----------|-----------|-----------|
| Current feed value    | 0 pulse   | 0 pulse   | 0 pulse   | 0 pulse   |
| Machine feed value    | 0 pulse   | 0 pulse   | 0 pulse   | 0 pulse   |
| Feedrate              | 0 pulse/s | 0 pulse/s | 0 pulse/s | 0 pulse/s |
| Axis error number     | 0         | 0         | 0         | 0         |
| Axis warning No.      | 0         | 0         | 0         | 0         |
| Valid M code          | 0         | 0         | 0         | 0         |
| Axis operation status | Standby   | Standby   | Standby   | Standby   |
| Current speed         | 0 pulse/s | 0 pulse/s | 0 pulse/s | 0 pulse/s |
| Axis feedrate         | 0 pulse/s | 0 pulse/s | 0 pulse/s | 0 pulse/s |

**Test**

Target Axis: Axis #1

Select Function: Positioning start signal Please set this function after stopping the positioning.

Start Type: ☒ Positioning Start Signal ☐ Block Start ☐ Multiple Axes Simultaneous Start

Positioning start data: Positioning Data No. (1 to 600): 1

Step: ☐ Start step  Step Mode: Execute step operation by declaration

External Command: ☐ External Command Valid ☐ Speed-position Switching Enable Flag ☐ Position-speed Switching Enable Flag

2. Select the test target axis.

Select the test target axis from the pull-down menu of "Target Axis".



## 3. Select the function.

Select "Current Value Changing" from the pull-down menu of "Select Function".

The screenshot shows the 'Test' dialog box with the following elements:

- Target Axis:** Axis #1
- Select Function:** Current Value Changing (Please set this function after stopping the positioning.)
- Current Value Changing sub-dialog:**
  - New Current Value:** 0
  - Range:** pulse (-2147483648 to 2147483647)
  - Button:** Current Value Changing
- Bottom Buttons:** Starting, Skip, Stop Target Axis(J), Stop All Axis, Restart Stop Axis, Positioning Complete, Error/Warning Details Confirmation, Error/Warning Reset, M Code OFF Request, Close.

1) Enter the new current value and click the Current Value Changing button.

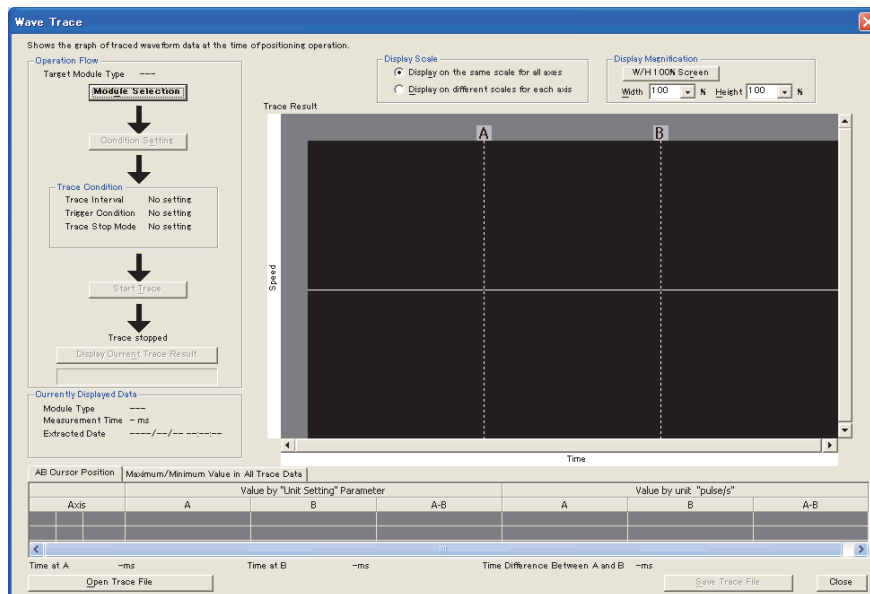
## Appendix 10.6 Wave trace

This function displays the speed command (axis speed) in positioning operation in waveform data.

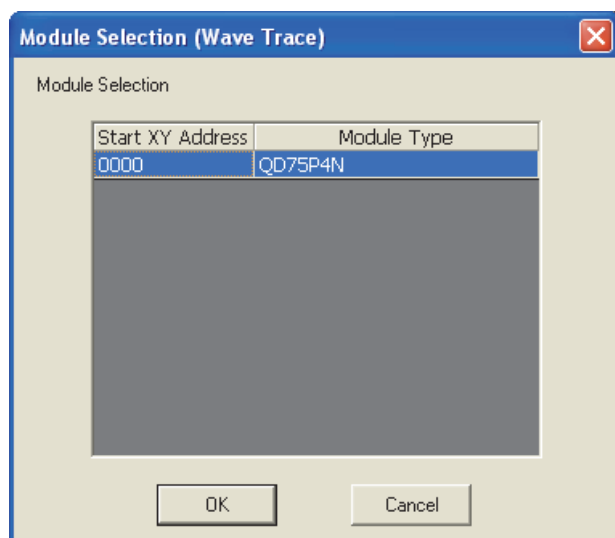
## (1) Operating procedure

1. Open the "Wave Trace" dialog box.

[Tool] → [Intelligent Function Module Tool] → [QD75/LD75 Positioning Module] → [Wave Trace]



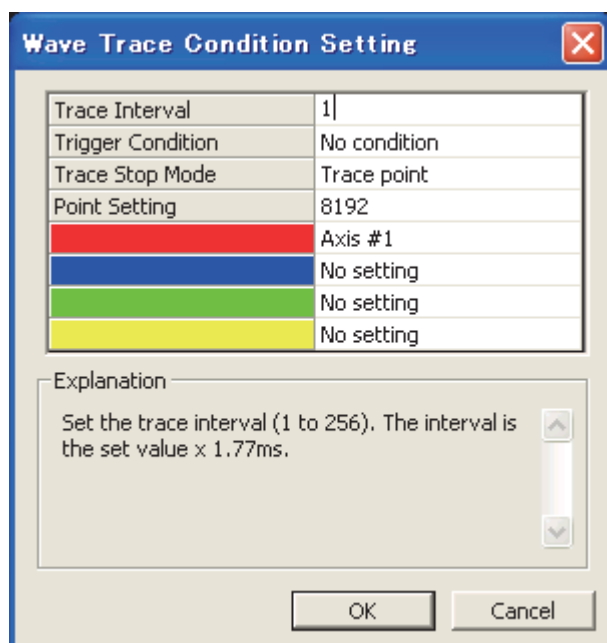
## 2. Select the module.



Click the **Module Selection** button.

Select the positioning module to trace waves and click the **OK** button.

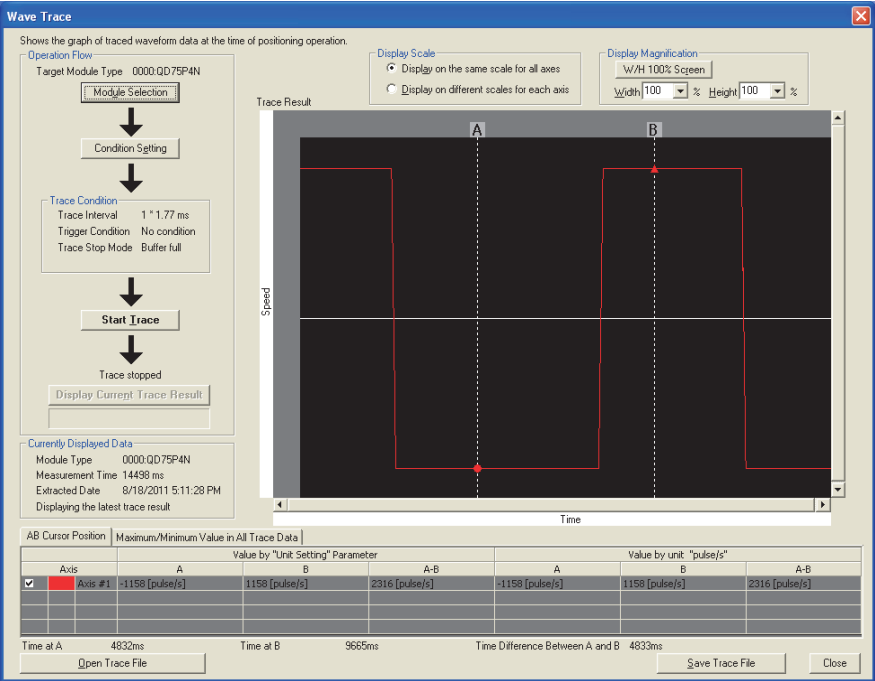
## 3. Set conditions.



Click the **Condition Setting** button.

Read the displayed description, set each condition, and click the **OK** button.

4. Start trace and display trace results.
- Click the **Start Trace** button to start trace.
- The trace result at that point can be displayed by clicking the **Display Current Trace Result** button during trace, even if the trace completion conditions are not met.
- When the completion conditions are met and the trace is completed, the trace result is displayed.



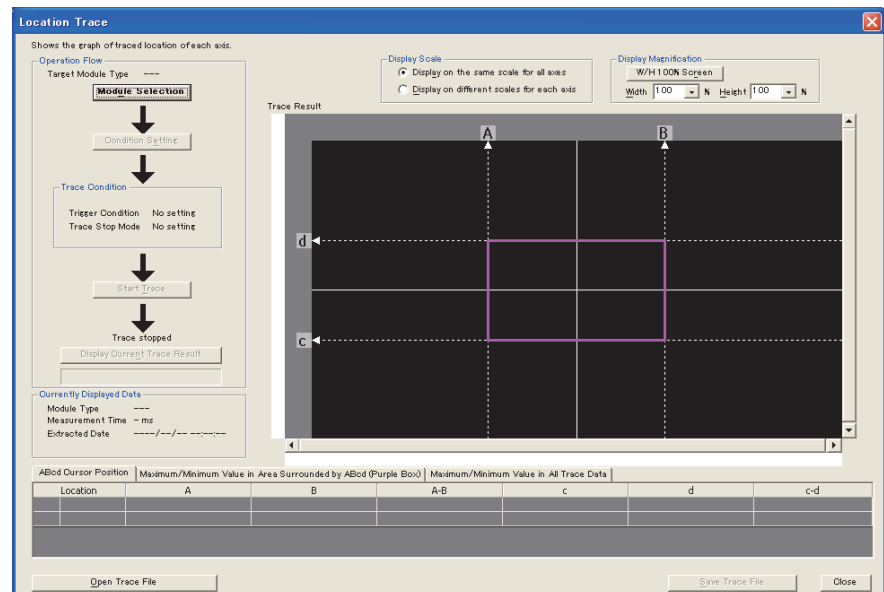
## Appendix 10.7 Location trace

This function displays 2-axis interpolation control and simultaneous start (2-axes) in locus data.

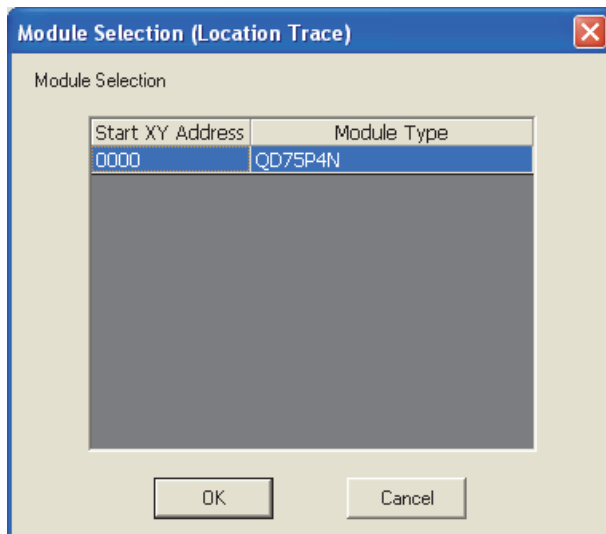
## (1) Operating procedure

1. Open the "Location Trace" dialog box.

[Tool] → [Intelligent Function Module Tool] → [QD75/LD75 Positioning Module] → [Location Trace]



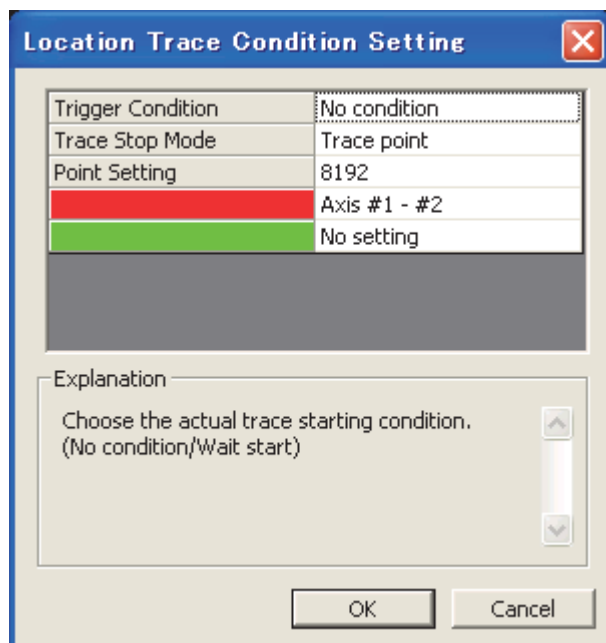
## 2. Select the module.



Click the **Module Selection** button.

Select the positioning module to trace locations and click the **OK** button.

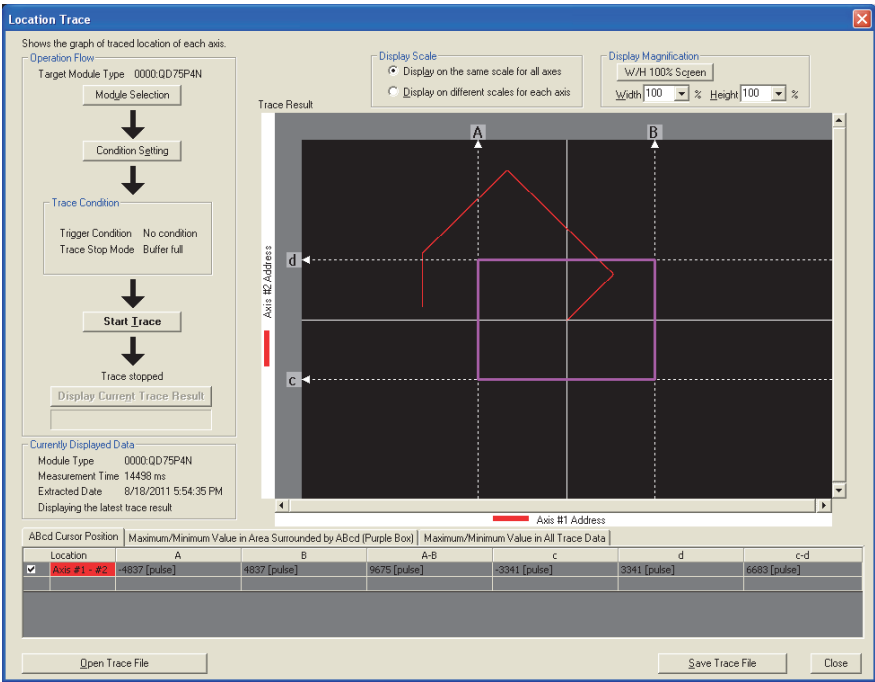
## 3. Set conditions.



Click the **Condition Setting** button.

Read the displayed description, set each condition, and click the **OK** button.

4. Start trace and display trace results.
- Click the **Start Trace** button to start trace.
- The trace result at that point can be displayed by clicking the **Display Current Trace Result** button during trace, even if the trace completion conditions are not met.
- When the completion conditions are met and the trace is completed, the trace result is displayed.



## Appendix 10.8 Parameter initialization function

The "parameter initialization function" is used to return the setting data set in the QD75 buffer memory and flash ROM to their factory-set initial values.

The details shown below explain about the "parameter initialization function".

- [1] Parameter initialization means
- [2] Control details
- [3] Control Precautions
- [4] Parameter initialization method

### [1] Parameter initialization means

- Initialization is executed with a program.
- Initialization is executed by GX Works2.

Refer to GX Works2 Version1 Operating Manual (Intelligent Function Module) for the execution method by GX Works2.

### [2] Control details

The following table shows the setting data initialized by the "parameter initialization function".

(The data initialized are "buffer memory" and " flash ROM " setting data.)

| Setting data                                                |
|-------------------------------------------------------------|
| Basic parameters ([Pr.1] to [Pr.10])                        |
| Detailed parameters ([Pr.11] to [Pr.42], [Pr.70], [Pr.150]) |
| OPR basic parameters ([Pr.43] to [Pr.48])                   |
| OPR detailed parameters ([Pr.49] to [Pr.57])                |
| Positioning data (No.1 to No.600)                           |
| Block start data (No.7000 to No.7004)                       |

### [3] Control Precautions

- (1) Parameter initialization is only executed when the positioning control is not carried out (when the PLC READY signal (Y0) is OFF).  
The warning "In PLC READY" (warning code: 111) will occur if executed when the PLC READY [Y0] is ON.
- (2) Writing to the flash ROM can be executed up to 100,000 times.  
If writing to the flash ROM exceeds 100,000 times, the writing may become impossible, and an error "Flash ROM write error" (error code: 801) will occur.
- (3) A "CPU module reset" or "CPU module power restart" must be carried out after the parameters are initialized.
- (4) If an error occurs on the parameter set in the QD75 when the PLC READY signal [Y0] is turned ON, the QD75 READY signal [X0] will not be turned ON and the control cannot be carried out.

#### Important

Parameter initialization takes about 10 seconds. (Up to 30 seconds are sometimes required.)  
Do not turn the power ON/OFF; reset the CPU module, etc., during parameter initialization. If the power is turned OFF or the CPU module is reset to forcibly end the process, the data backed up in the flash ROM will be lost.



#### [4] Parameter initialization method

- (1) Parameter initialization is carried out using the dedicated instruction "ZP.PINIT".  
(Refer to CHAPTER 14 "DEDICATED INSTRUCTIONS" for details.)

- (2) Parameter initialization can also be carried out by the writing of the data shown in the table below to the buffer memory using the TO instruction /intelligent function device.

The initialization of the parameter is executed at the time point the data is written to the QD75 buffer memory.

| Setting item                                                                                        | Setting value | Setting details                             | Buffer memory address |        |        |        |
|-----------------------------------------------------------------------------------------------------|---------------|---------------------------------------------|-----------------------|--------|--------|--------|
|                                                                                                     |               |                                             | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <span style="border: 1px solid black; padding: 0 2px;">Cd.2</span> Parameter initialization request | 1             | Set "1" (parameter initialization request). | 1901                  |        |        |        |

Refer to Section 5.7 "List of control data" for details on the setting details.

When the initialization is complete, "0" will be set in " Cd.2 Parameter initialization request" by the QD75 automatically.

# Appendix 10.9 Execution data backup function

When the QD75 buffer memory data is rewritten from the CPU module, "the data backed up in the QD75 flash ROM" may differ from "the data (buffer memory data) for which control is being executed". In cases like these, the data being executed will be lost when the programmable controller power is turned OFF.  
(Refer to CHAPTER 7.)

In cases like these, the "execution data backup function" backs up the data being executed by writing it to the flash ROM. The data that was backed up is then written to the buffer memory when the power is turned ON next.

The details shown below explain about the "execution data backup function".

- [1] Execution data backup means
- [2] Control details
- [3] Control Precautions
- [4] Execution data backup method

## [1] Execution data backup (written to flash ROM) means

- The backup is executed with a program.
- The backup is executed by GX Works2.

Refer to GX Works2 Version1 Operating Manual (Common) for execution data backup method by GX Works2.

## [2] Control details

The following shows the data that can be written to the flash ROM using the "execution data backup function".

| Buffer memory                                |   | Flash ROM                                    |
|----------------------------------------------|---|----------------------------------------------|
| Parameters (Pr.1 to Pr.57,<br>Pr.70, Pr.150) | → | Parameters (Pr.1 to Pr.57,<br>Pr.70, Pr.150) |
| Positioning data (No. 1 to No. 600)          |   | Positioning data (No. 1 to No. 600)          |
| Block start data (No. 7000 to 7004)          |   | Block start data (No. 7000 to 7004)          |

## [3] Control Precautions

- (1) Parameter initialization is only executed when the positioning control is not carried out (when the PLC READY signal (Y0) is OFF). The warning "In PLC READY" (warning code: 111) will occur if executed when the PLC READY [Y0] is ON.
- (2) Writing to the flash ROM can be executed up to 100,000 times.  
If writing to the flash ROM exceeds 100,000 times, the writing may become impossible, and an error "Flash ROM write error" (error code: 801) will occur.
- (3) After one power ON/CPU module reset operation, writing to the flash ROM using a program is limited to up to 25 times.  
If the 26th writing is executed, an error "Flash ROM write number error" (error code: 801) will occur. If this error occurs, carry out the power OFF ON/CPU module reset operation again.  
Refer to Md.19 of Section 5.1.7 "Types and roles of monitor data" for details.

### Important

Do not turn the power ON/OFF, reset the CPU module, during writing to the flash ROM. If the power is turned OFF or the CPU module is reset to forcibly end the process, the data backed up in the flash ROM will be lost.

#### [4] Execution data backup method

- (1) Execution data backup (writing to the flash ROM) is carried out using the dedicated instruction "ZP.PFWRT". (Refer to CHAPTER 14 "DEDICATED INSTRUCTIONS" for details.)
- (2) Refer to Section 7.2 "Data transmission process" for the data transmission processing at the backup of the execution data.
- (3) Execution data backup can also be carried out by the writing of the data shown in the table below to the QD75 buffer memory using the TO instruction /intelligent function device.

The writing to the flash ROM is executed at the time point the data is written to the QD75 buffer memory.

| Setting item                                                                               | Setting value | Setting details                    | Buffer memory address |        |        |        |
|--------------------------------------------------------------------------------------------|---------------|------------------------------------|-----------------------|--------|--------|--------|
|                                                                                            |               |                                    | Axis 1                | Axis 2 | Axis 3 | Axis 4 |
| <span style="border: 1px solid black; padding: 0 2px;">Cd.1</span> Flash ROM write request | 1             | Set "1" (flash ROM write request). | 1900                  |        |        |        |

Refer to Section 5.7 "List of control data" for details on the setting details.

When the writing to the flash ROM is complete, "0" will be set in "Cd.1 Flash ROM write request" by the QD75 automatically.

## Appendix 10.10 External I/O signal logic switching function

This function switches the signal logic according to the external equipment connected to the QD75.

For the system in which drive unit READY with b-contact, upper limit switch, and lower limit switch are not used, the parameter logic setting can be controlled without wiring if it is changed to a "positive logic".

When the drive unit READY, upper limit switch, and lower limit switch are used, ensure to use them with b-contact.

The details shown below explain about the "external I/O signal logic switching function".

[1] Parameter setting details

[2] Precautions on parameter setting

### [1] Parameter setting details

To use the "external I/O signal logic switching function", set the parameters shown in the following table.

| Setting item |                               | Setting details                                                     |                                | Factory-set initial value | Buffer memory address |        |        |        |                                         |                                         |
|--------------|-------------------------------|---------------------------------------------------------------------|--------------------------------|---------------------------|-----------------------|--------|--------|--------|-----------------------------------------|-----------------------------------------|
|              |                               |                                                                     |                                |                           | Axis 1                | Axis 2 | Axis 3 | Axis 4 |                                         |                                         |
| Pr.22        | Input signal logic selection  | • Selection of logic of signals input from external source to QD75  |                                | 0                         | 31                    | 181    | 331    | 481    |                                         |                                         |
|              |                               | b0                                                                  | Lower limit                    |                           |                       |        |        |        | 0: Negative logic,<br>1: Positive logic |                                         |
|              |                               | b1                                                                  | Upper limit                    |                           |                       |        |        |        |                                         |                                         |
|              |                               | b2                                                                  | Drive unit READY               |                           |                       |        |        |        |                                         |                                         |
|              |                               | b3                                                                  | Stop signal                    |                           |                       |        |        |        |                                         |                                         |
|              |                               | b4                                                                  | External command signal        |                           |                       |        |        |        |                                         |                                         |
|              |                               | b5                                                                  | Zero signal                    |                           |                       |        |        |        |                                         |                                         |
|              |                               | b6                                                                  | Near-point dog signal          |                           |                       |        |        |        |                                         |                                         |
|              |                               | b7                                                                  | Unused                         |                           |                       |        |        |        |                                         | Set "0".                                |
|              |                               | b8                                                                  | Manual pulse generator input   |                           |                       |        |        |        |                                         | 0: Negative logic,<br>1: Positive logic |
| b9 to b15    | Unused                        | Set "0".                                                            |                                |                           |                       |        |        |        |                                         |                                         |
| Pr.23        | Output signal logic selection | • Selection of logic of signals output from QD75 to external source |                                | 0                         | 32                    | 182    | 332    | 482    |                                         |                                         |
|              |                               | b0                                                                  | Command pulse signal           |                           |                       |        |        |        | 0: Negative logic,<br>1: Positive logic |                                         |
|              |                               | b1 to b3                                                            | Unused                         |                           |                       |        |        |        | Set "0".                                |                                         |
|              |                               | b4                                                                  | Deviation counter clear signal |                           |                       |        |        |        | 0: Negative logic,<br>1: Positive logic |                                         |
|              |                               | b5 to b15                                                           | Unused                         |                           |                       |        |        |        | Set "0".                                |                                         |

\* Refer to Section 5.2 "List of parameters" for the information on detail settings.

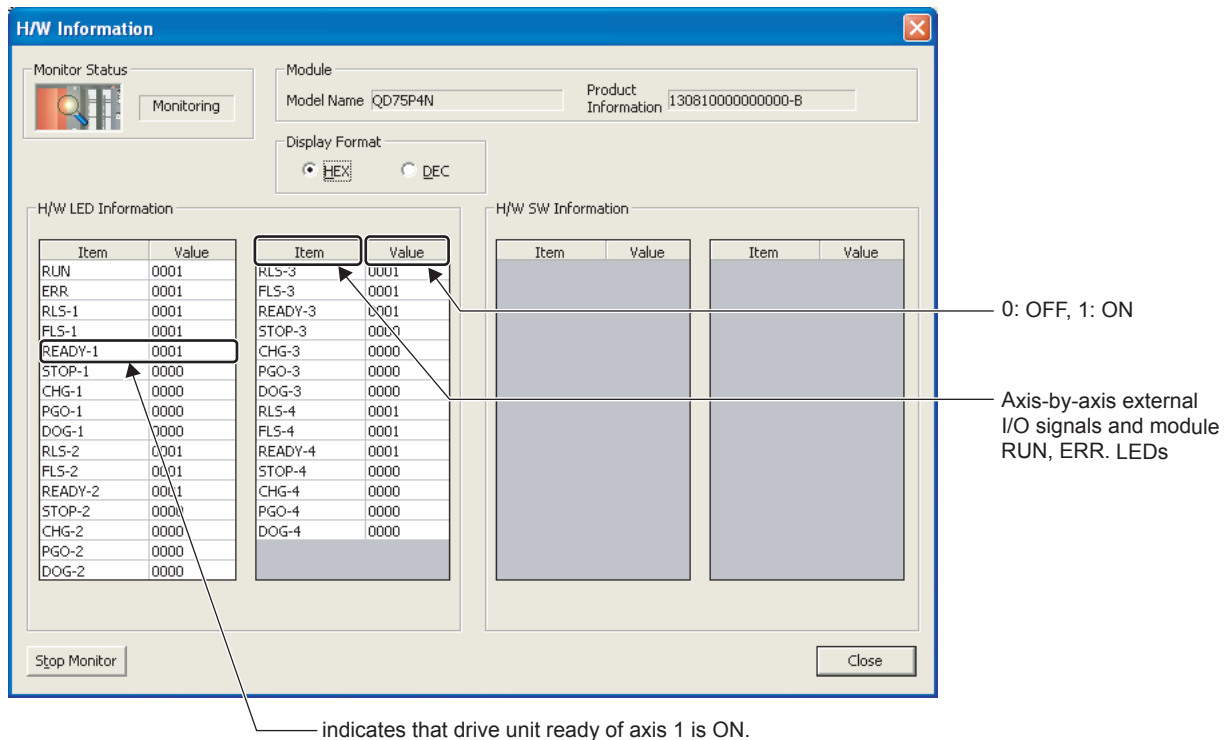
### [2] Precautions on parameter setting

- (1) The external I/O signal logic switching parameters are validated when the PLC READY signal [Y0] is turned OFF to ON. (The logic is negative right after power-on.)
- (2) If each signal logic is set erroneously, the operation may not be carried out correctly.  
Before setting, check the specifications of the equipment to be used.

## Appendix 10.11 External I/O signal monitor function

The "external I/O signal monitor function" monitors the module's information and external I/O signal statuses in the module's detailed information which can be displayed on the system monitor of GX Works2.

The information that can be monitored are the module's information (same as the QD75 front "RUN", "ERR" LED indicators) and the following external I/O signals. Set the logic of the external I/O signals in " **Pr.22** Input signal logic selection" and " **Pr.23** Output signal logic selection".



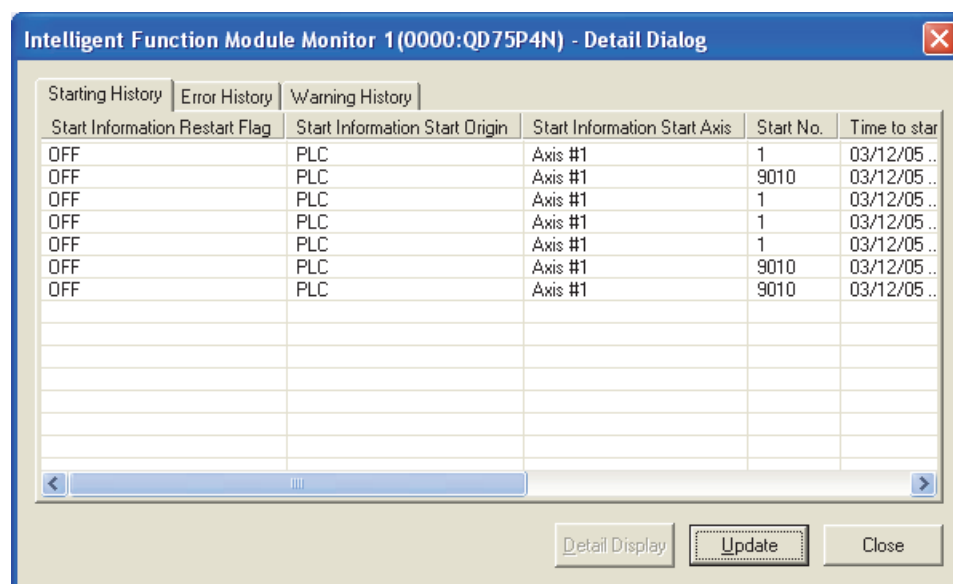
Refer to the following manual for the system monitor of GX Works2.  
GX Works2 Version1 Operating Manual (Common)

## Appendix 10.12 History monitor function

This function monitors starting history, error history, and warning history stored in the buffer memory of the QD75 during operation.

## [1] Starting history

Sixteen starting history logs of operations such as positioning operation, JOG operation, and manual pulse generator operation can be monitored. When the number of logs exceeds 16, the latest log overwrites the oldest log so that the latest 16 logs can be monitored all the time. This function allows users to check the operation sequence (whether the operations have been started in a predetermined sequence) at system start-up.

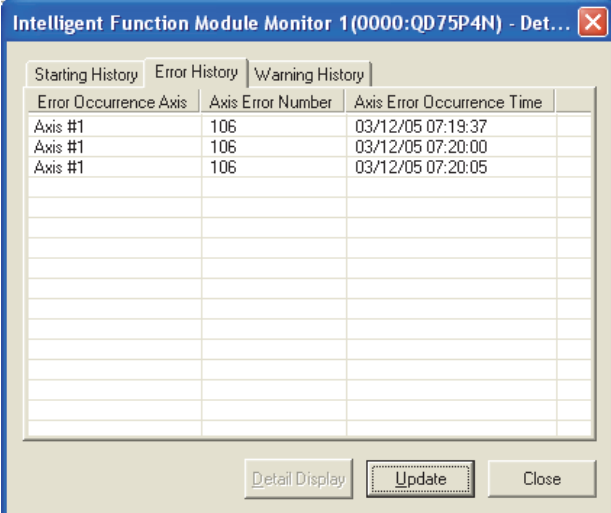


To monitor the starting history, register the QD75 to the "Intelligent Function Module Monitor Window".

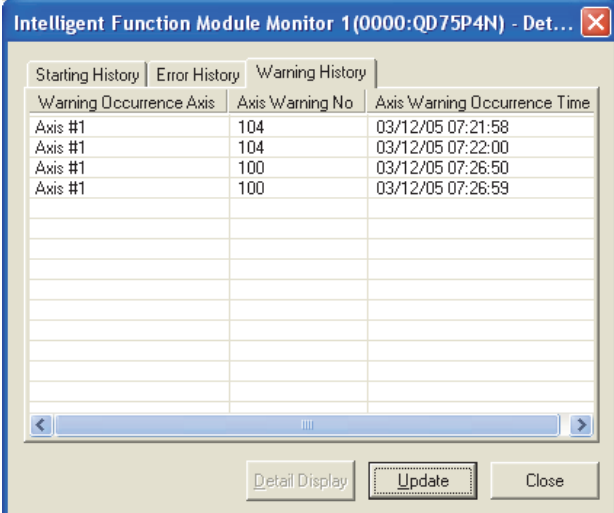
For details on the registration method and monitoring method, refer to the following.  
GX Works2 Version1 Operating Manual (Intelligent Function Module)

## [2] Error history, warning history

Sixteen error history logs and sixteen warning history logs can be monitored. When the number of logs exceeds 16, the latest log overwrites the oldest log so that the latest 16 logs can be monitored all the time.



| Error Occurrence Axis | Axis Error Number | Axis Error Occurrence Time |
|-----------------------|-------------------|----------------------------|
| Axis #1               | 106               | 03/12/05 07:19:37          |
| Axis #1               | 106               | 03/12/05 07:20:00          |
| Axis #1               | 106               | 03/12/05 07:20:05          |
|                       |                   |                            |
|                       |                   |                            |
|                       |                   |                            |
|                       |                   |                            |
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|                       |                   |                            |
|                       |                   |                            |
|                       |                   |                            |
|                       |                   |                            |



| Warning Occurrence Axis | Axis Warning No | Axis Warning Occurrence Time |
|-------------------------|-----------------|------------------------------|
| Axis #1                 | 104             | 03/12/05 07:21:58            |
| Axis #1                 | 104             | 03/12/05 07:22:00            |
| Axis #1                 | 100             | 03/12/05 07:26:50            |
| Axis #1                 | 100             | 03/12/05 07:26:59            |
|                         |                 |                              |
|                         |                 |                              |
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|                         |                 |                              |
|                         |                 |                              |

To monitor the error history and warning history, register the QD75 to the "Intelligent Function Module Monitor Window".

For details on the registration method and monitoring method, refer to the following.  
GX Works2 Version1 Operating Manual (Intelligent Function Module)

## Appendix 10.13 Checking errors

Error codes corresponding to the errors occurred in the QD75 can be checked on the following screen of GX Works2.

Select the screen according to the purpose and usage.

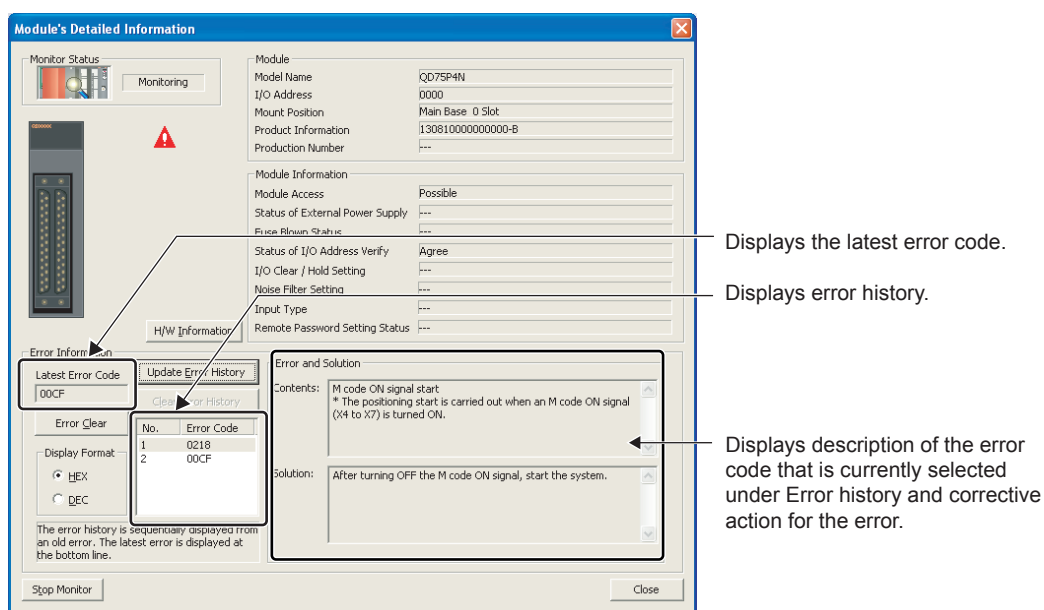
- "Module's Detailed Information" screen
- "Error History" screen (for the QD75P□N/QD75D□N only)

(1) Checking errors on the "Module's Detailed Information" screen

Select [Diagnostics] → [System Monitor...] on GX Works2.

Select QD75 for "Main Base" and click the **Detailed Information** button.

The "Module's Detailed Information" screen for the QD75 appears and the error code, error details, and corrective actions can be checked.

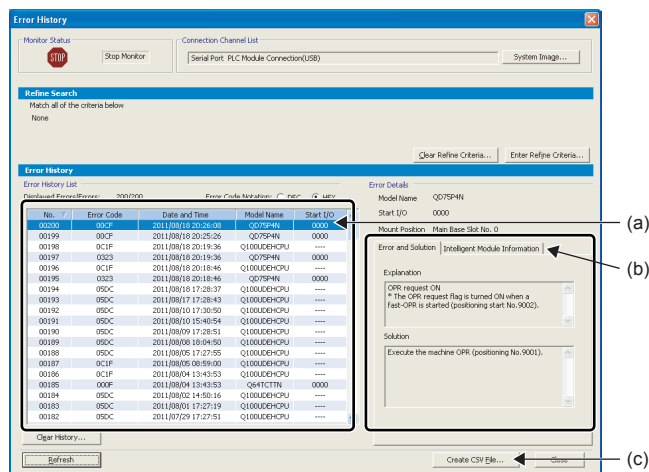




- (2) Checking errors on the "Error History" screen. (for the QD75P□N/QD75D□N only)

On the "Error History" screen, the error logs of the QD75 are displayed in a list together with the error logs of other modules. The logs can be output to a CSV format file. The error codes and the time of error occurrence can be checked even after the CPU module is powered off and then on or reset.

Select [Diagnostics] → [System Monitor...] → **System Error History** button on GX Works2.



- (a) Error History List  
Module error logs are displayed in a list.

## (b) Error and Solution, Intelligent Module Information

## • Error and Solution

Details of the selected in the "Error History List" and its corrective action are displayed.

## • Intelligent Module Information

The QD75P□N/QD75D□N status when the error selected in the "Error History List" occurred is displayed.

For the QD75P□N/QD75D□N, the following contents are displayed.

| Item                                                                                                                                                                                                                                                                                                       | Description                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Start axis                                                                                                                                                                                                                                                                                                 | The axis No. requested to start is stored.                                                   |
| Positioning start No.                                                                                                                                                                                                                                                                                      | The positioning start No. is stored.                                                         |
| Axis in which the error occurred                                                                                                                                                                                                                                                                           | The axis No. in which the error occurred is stored.                                          |
| Axis error occurrence (Data No.)                                                                                                                                                                                                                                                                           | The data No. in which the error occurred is stored.                                          |
| Current feed value                                                                                                                                                                                                                                                                                         | The current feed value (at the time of error) of the error axis is stored.                   |
| Input signals with PLC CPU(X0 -- XF)                                                                                                                                                                                                                                                                       | The status of input signals (X0 to XF) (at the time of error) is stored (in hexadecimal).    |
| Input signals with PLC CPU(X10 -- X1F)                                                                                                                                                                                                                                                                     | The status of input signals (X10 to X1F) (at the time of error) is stored (in hexadecimal).  |
| Output signals with PLC CPU(Y0 -- YF)                                                                                                                                                                                                                                                                      | The status of output signals (Y0 to YF) (at the time of error) is stored (in hexadecimal).   |
| Output signals with PLC CPU(Y10 -- Y1F)                                                                                                                                                                                                                                                                    | The status of output signals (Y10 to Y1F) (at the time of error) is stored (in hexadecimal). |
| <ul style="list-style-type: none"> <li>• Axis 1 drive unit READY signal</li> <li>• Axis 1 upper limit signal</li> <li>• Axis 1 lower limit signal</li> <li>• Axis 1 stop signal</li> <li>• Axis 1 external command signal</li> <li>• Axis 1 near-point dog signal</li> <li>• Axis 1 zero signal</li> </ul> | The status of axis 1 external input signals (at the time of error) is stored.                |
| <ul style="list-style-type: none"> <li>• Axis 2 drive unit READY signal</li> <li>• Axis 2 upper limit signal</li> <li>• Axis 2 lower limit signal</li> <li>• Axis 2 stop signal</li> <li>• Axis 2 external command signal</li> <li>• Axis 2 near-point dog signal</li> <li>• Axis 2 zero signal</li> </ul> | The status of axis 2 external input signals (at the time of error) is stored.                |
| <ul style="list-style-type: none"> <li>• Axis 3 drive unit READY signal</li> <li>• Axis 3 upper limit signal</li> <li>• Axis 3 lower limit signal</li> <li>• Axis 3 stop signal</li> <li>• Axis 3 external command signal</li> <li>• Axis 3 near-point dog signal</li> <li>• Axis 3 zero signal</li> </ul> | The status of axis 3 external input signals (at the time of error) is stored.                |
| <ul style="list-style-type: none"> <li>• Axis 4 drive unit READY signal</li> <li>• Axis 4 upper limit signal</li> <li>• Axis 4 lower limit signal</li> <li>• Axis 4 stop signal</li> <li>• Axis 4 external command signal</li> <li>• Axis 4 near-point dog signal</li> <li>• Axis 4 zero signal</li> </ul> | The status of axis 4 external input signals (at the time of error) is stored.                |

- (c) Create CSV File button

The module error logs are output to a CSV format file.

#### POINT

- (1) If errors frequently occur in the QD75P□N/QD75D□N, "\*\*HST.LOSS\*\*" (instead of an actual error code) may be displayed in the Error Code column.

(Display example)

| No. ▾ | Error Code        | Date and Time       | Model Name | Start I/O | ▲ |
|-------|-------------------|---------------------|------------|-----------|---|
| 00200 | 00CF              | 2011/06/28 02:52:50 | QD75P4N    | 0000      |   |
| 00199 | 020C              | 2011/06/28 02:51:24 | QD75P4N    | 0000      |   |
| 00198 | <b>'HST.LOSS'</b> | 2011/06/28 02:50:09 | QD75P4N    | 0000      |   |
| 00197 | 020C              | 2011/06/28 02:49:28 | QD75P4N    | 0000      |   |
| 00196 | 020C              | 2011/06/28 02:47:58 | QD75P4N    | 0000      |   |

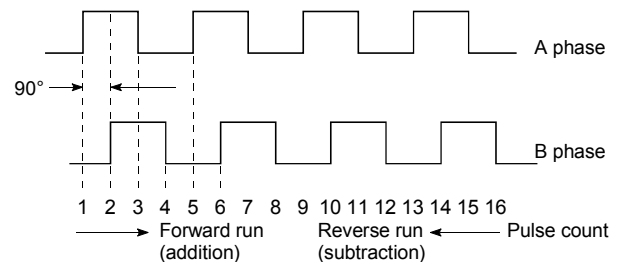
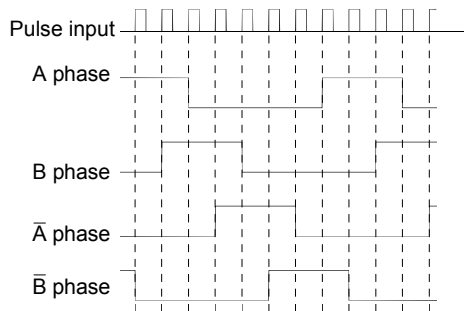
If "\*\*HST.LOSS\*\*" is frequently displayed, set a larger value for the number of errors collected per scan in the PLC RAS tab of the PLC Parameter dialog box. For the setting, refer to the QnUCPU User's Manual (Function Explanation, Program Fundamentals).

- (2) If the error occurred at the simultaneous start, the axis No. in which the error is detected is stored in the "Starting axis" in Error History.

## Appendix 11 MELSEC Explanation of positioning terms

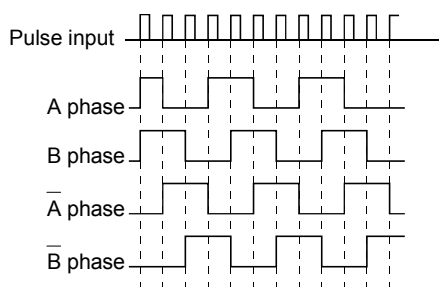
### 1-2 PHASE EXCITATION SYSTEM

This is one system for exciting each stepping motor coil in a determined order. In this system, one phase and two phases are alternately excited.



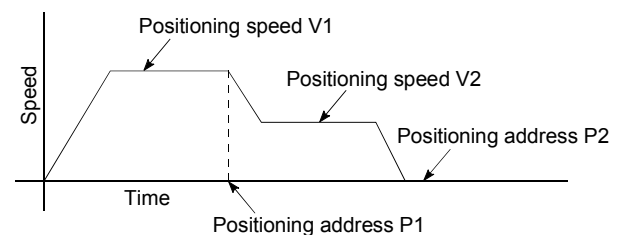
### 2-PHASE EXCITATION SYSTEM

This is one system for exciting each stepping motor coil in a determined order. In this system, a current constantly flows to 2 phases to carry out step feed.



### 2-SPEED TRAPEZOIDAL CONTROL

In this positioning control method, the positioning pattern, positioning addresses (P1, P2), and positioning speeds (V1, V2) are set in the sequence program. Positioning is carried out to positioning address P1 by issuing the 1st positioning start command. When P1 is reached, the positioning then automatically changes to positioning at the V2 speed.



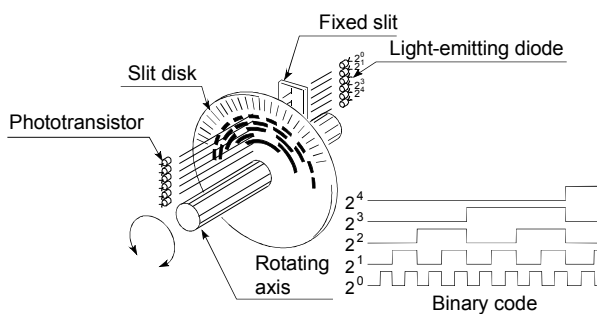
### 2-PHASE PULSE

An A phase and B phase double pulse. There is a phase difference between the two phases, so that difference can be automatically added and subtracted in the pulse count. The standard phase difference is a 90° electrical angle.

If the B phase were to lag behind the A phase in a forward run (B phase turns ON after the A phase), the A phase would lag behind the B phase in a reverse run (A phase turns ON after the B phase). In this way the forward and reverse run (addition and subtraction) can be automatically carried out.

## ABSOLUTE ENCODER

This is a detector that enables the angle data within 1 motor rotation to be output to an external destination. Absolute encoders are generally able to output  $360^\circ$  in 8 to 12 bits. Incremental encoders have a disadvantage in that the axis position is lost when a power failure occurs. However, with absolute encoders, the axis position is not lost even when a power failure occurs. Various codes such as a binary code and BCD code can be output. Absolute encoders are more expensive, more accurate, and larger than incremental encoders. Refer to "ENCODER".



## ABSOLUTE POSITION DETECTION SYSTEM

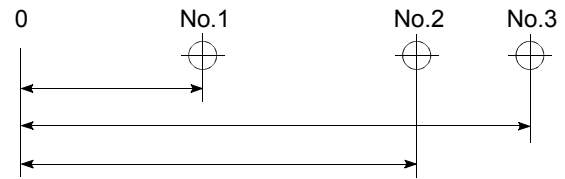
In the absolute position detection system, once an OPR is carried out at the system startup, the system stores the machine position in the memory and retains the current position even when the power is turned OFF. Mechanical deviation will be compensated, so that the OPR is not required after the power is turned ON next time. Configuring this system requires a motor with an absolute position detector and a servo amplifier and positioning module compatible with an absolute position detection system.

## ABSOLUTE SYSTEM

This is one system for expressing a positioning address.

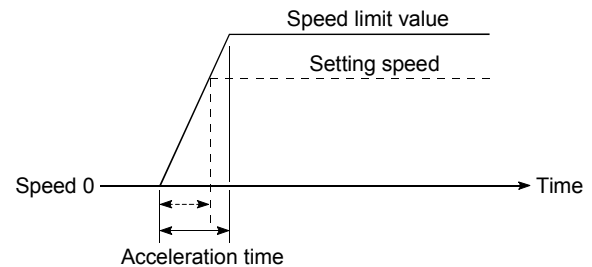
Absolute address system.

This system uses 0 as a reference, and expresses the address as the distance from 0. The direction is automatically determined, even when it is not designated. The other address system is the increment system.



## ACCELERATION TIME

The parameter acceleration time refers to the time from a stopped state to the time the speed limit value is reached, so it becomes proportionally shorter as the setting speed decreases. The acceleration time is determined by factors such as machine inertia, motor torque, and load resistance torque.

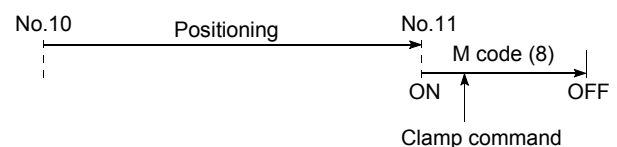


## ADDRESS

- 1) This is a numerical value to express the positioning position, designated in mm, inch, angle, or No. of pulse units.
- 2) The memory address. Many addresses are stored in the memory. An address is read or written after it is designated.

## AFTER mode

This is the mode that outputs the M code after positioning is complete (after stopping). Clamping can be commanded, drilling dimensions can be selected, etc., with this mode.



## AUTO TUNING (Automatic Tuning)

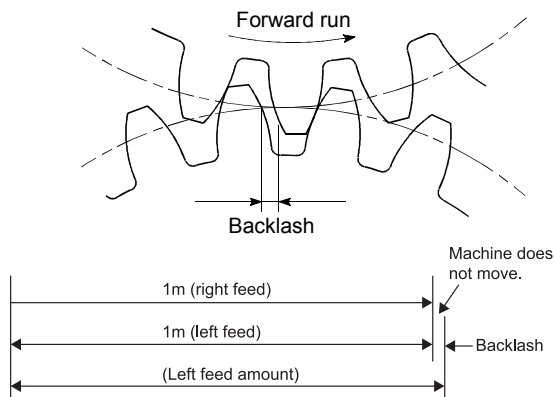
Properties such as responsiveness and stability of machines driven with a servomotor are affected by changes in the inertia moment and rigidity due to changes in the machine load, etc.

This function automatically adjusts the speed loop gain and position loop gain to match the machine state, so the machine's performance can be maintained at its optimum state.

A real time automatic tuning function should be used for machines having large load fluctuations.

## BACKLASH COMPENSATION

When a forward run operation changes to a reverse run operation, there is sometimes play (backlash) in the mesh of the toothed gears. This also occurs when using a worm gear. Because of this backlash, a left feed of 1m carried out after a right feed of 1m will not be sufficient to return the machine to its original position. The machine cannot be positioned to its original position without an extra feed equivalent to the backlash amount. This function compensates for that backlash amount.



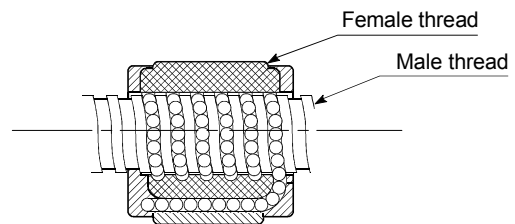
## BACKUP FUNCTION

Backup functions consist of the following.

- 1) Functions for storing the sequence program and device statuses stored in the RAM memory of the CPU module, so that they are not lost during power failures, etc.
- 2) Functions for storing the current value in absolute position compatible systems so that it is not lost during power failures, etc.
- 3) Functions for reading the CPU module data (programmable controller programs, parameters, positioning data, etc.) by a peripheral device when the old CPU module is replaced, and then write it to the new CPU module after the replacement is completed.

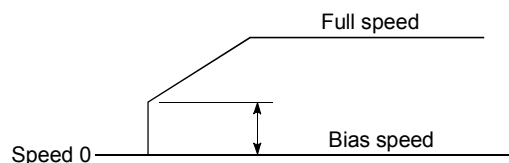
## BALL SCREW

This is a type of screw, with balls lined up in the threads like ball bearings. This reduces backlash, and enables rotation with little force.



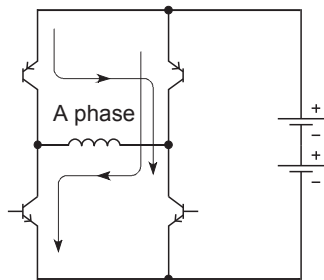
## BIAS SPEED AT START

A large amount of torque is required when the machine starts moving, but the torque may be unstable at speed 0 with stepping motors. Therefore, movement can be smoothly carried out by starting the movement at a given speed from the beginning. The bias speed at start is the speed set at that start.



## BIPOLAR DRIVE CONSTANT-CURRENT SYSTEM

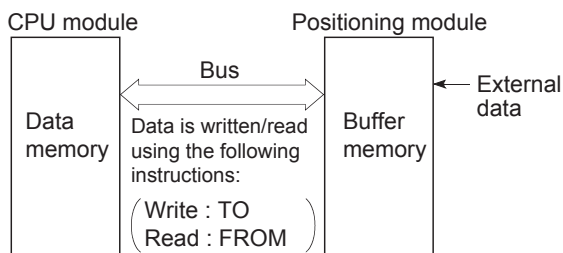
This is one system for driving a stepping motor. In this method, the orientation of the excitation current flowing to the stator coil is reversed, and the excitation current direction is in both the positive and negative direction. This enables the motor coil to be used effectively, and a large output torque can be obtained at low speeds.



Bipolar drive basic circuit (bridge method)

## BUFFER MEMORY

Memory used to temporarily store data. Before writing external data to the CPU module data memory, it is first temporarily stored in the buffer memory to be used for operation by the program. The buffer memory is used by the positioning module because the latest data can be read and written.



## BUSY

The device is doing some other work. It is in a positioning operation or in dwell time.

## CCW (Counterclockwise)

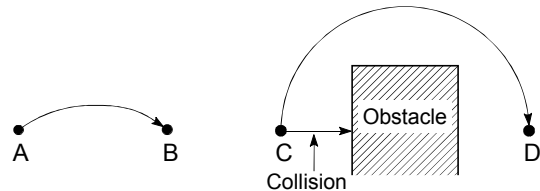
Rotation in the counterclockwise direction. In the motor, this is determined looking from the shaft end side. Also refer to "CW".

## CHANGE signal

The CHANGE signal is an external signal used to switch the speed-position control from the speed control being executed to position control.

## CIRCULAR INTERPOLATION

Automatic operation in which the machine path makes a circle when positioning is carried out by simultaneously operating both the longitudinal feed and latitudinal feed motors. The normal unit is 90°. Round shapes can be created with this type of interpolation, and obstacles in the machine path can also be avoided. Refer to the terms "INTERPOLATION OPERATION" and "LINEAR INTERPOLATION"

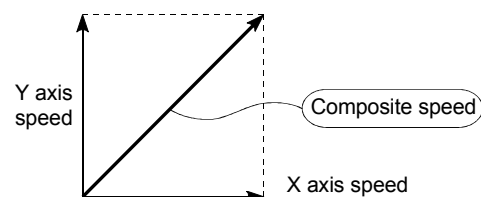


## COMMAND PULSE

Refer to term "FEEDBACK PULSE".

## COMPOSITE SPEED

The movement speed for the target control during interpolation operations.



## CONTROL UNIT

This is one type of positioning reference data. The unit to be used is designated as mm, inch, degree, or pulse.

## CP CONTROL (Continuous Path Control)

Continuous path is a control method in which a path is followed without interrupting such as in uniform speed control.

**CREEP SPEED**

A speed at which the machine moves very slowly.  
It is difficult for the machine to stop accurately when running at high speed, so the movement must first be changed to the creep speed before stopping.  
Refer to the term "NEAR-POINT DOG".

**CURRENT FEED VALUE**

The OP address at the completion of the machine OPR is stored.  
The position currently being executed is stored.  
This value changes when the current value is changed.

**CURRENT LOOP MODE**

One of the servo control modes in positioning.  
Mode which uses current for torque control.  
Also called the torque loop mode.  
Refer to the section of "position loop mode".

**CURRENT VALUE**

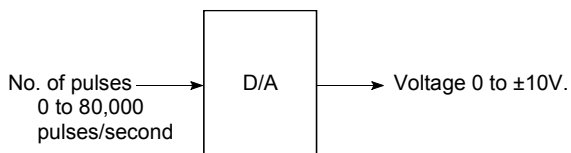
This is the current address (position) when stopped or during positioning.

**CW (Clockwise)**

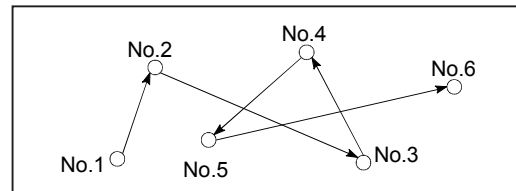
Rotation in the clockwise direction. Rotation in the clockwise direction looking from the motor shaft end side.

**D/A CONVERTER (Digital-to-Analog converter)**

A device having a function to convert the digital value expressing the No. of pulses to an analog value expressing the voltage (or current).

**DATA NO.**

To carry out positioning to 2 or more addresses, each position is assigned a sequence No. such as No. 1, No. 2, No. 3, etc. The positioning is then carried out following this sequence. The QD75 is capable of positioning up to No. 600.

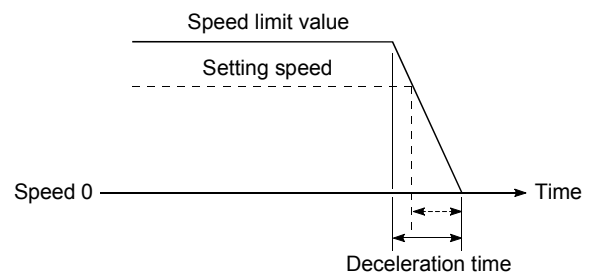
**DECELERATION RATIO**

A ratio used when the machine is decelerated using a toothed gear. This ratio is a numeral larger than 1.

$$\text{Deceleration ratio} = \frac{\text{Input gear speed}}{\text{Output gear speed}}$$

**DECELERATION TIME**

The parameter deceleration time is the same value as the acceleration time. Deceleration time refers to the time from the speed limit value to a stopped state, so it becomes proportionally shorter as the setting speed decreases.

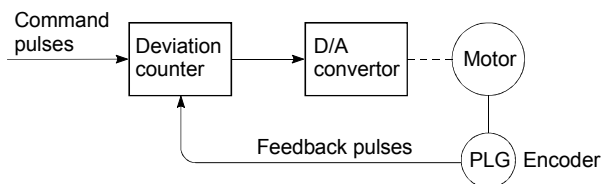




## DEVIATION COUNTER

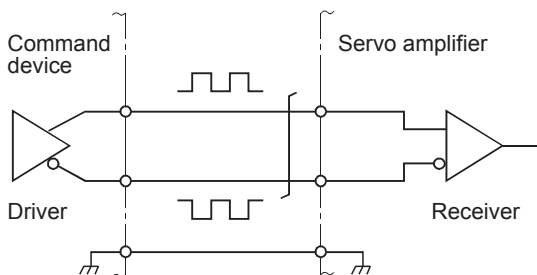
Deviation counters have the following two functions.

- 1) To count the command pulses issued from the QD75, and transmit the count value to the D/A converter.
- 2) To subtract the feedback pulses from the command pulses, and run the motor by the deviation value (droop pulse) of the command pulses and feedback pulses until the command pulses reaches 0.



## DIFFERENTIAL OUTPUT TYPE

This is one type of encoder feedback pulse output. When one signal is output with this method, a companion signal having the reverse polarity is simultaneously output. This method enables high-frequency transfer, and is resistant to noise, etc., so it is also used in high-speed signal transfer such as inputting and outputting of pulse trains. In general, the transmission side is called the driver, the reception side is called the receiver, and a dedicated IC is used.



## DIGITAL BUS CONNECTION

Commands are generally output from the positioning module to the servo amplifier as a pulse train. Recently, however, devices are being digitalized. Accompanying that, a connection method has appeared in which the bus lines of both the positioning module and the servo amplifier CPUs are connected. This has enabled the construction of higher-accuracy, higher-speed systems.

The MELSEC A70D, AD774M, A171SCPU, A273UCPU etc., models employ this digital bus connection.

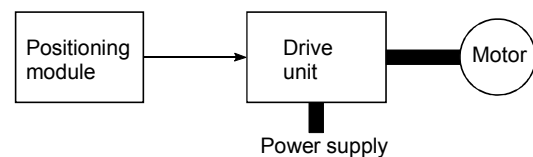
## DOG SIGNAL

The near-point dog of the OPR.

## DRIVE UNIT

The commands (such as pulses) output from the positioning module are low-voltage, low-current commands with insufficient energy to run the motor.

The drive unit increases the width of these commands so the motor can be run. It is an accessory on servomotors and stepping motors. Also called a servo amplifier.



## DRIVE UNIT READY

This signal is output when the drive unit for the motor is in a READY state.

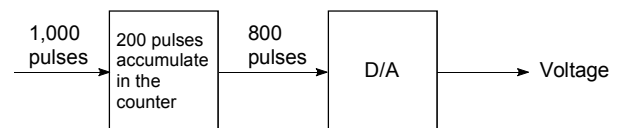
This signal remains OFF when the drive unit power is OFF, or during faults, etc.

## DROOP PULSE

Because of inertia ( $GD^2$ ) in the machine, it will lag behind and not be able to track if the positioning module speed commands are issued in their normal state.

Thus, for a servomotor, a method is used in which the speed command pulses are delayed by accumulation in a deviation counter. These accumulated pulses are called the droop pulse.

The deviation counter emits all pulses and returns to 0 when the machine stops.



## DWELL TIME

This is the time taken immediately after the positioning is completed to adjust for the droop pulses in the deviation counter. The positioning will not be accurate if this time is too short.

## DYNAMIC BRAKE

When protection circuits operate due to power failures, emergency stops (EMG signal) etc., this function is used to short-circuit between servomotor terminals via a resistor, thermally consume the rotation energy, and cause a sudden stop without allowing coasting of the motor.

Braking power is generated by electromagnetic brakes only when running motors with which a large brake torque can be obtained. Because electromagnetic brakes have no holding power, they are used in combination with mechanical brakes to prevent dropping of the vertical axis.

## ELECTROMAGNETIC BRAKE

This function is supplied on motors with electromagnetic brakes. Electromagnetic brakes are used to prevent slipping during power failures and faults when driving a vertical axis, or as a protective function when the machine is stopped.

These brakes are activated when not excited.

## ELECTRONIC GEAR

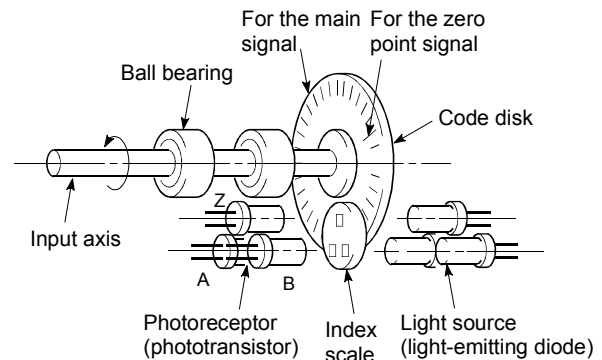
This function electrically increases/decreases the command pulses from the pulse command module by 1/50 to 50-fold. Thus, the positioning speed and movement amount can be controlled by the electronic gear ratio magnification.

## EMERGENCY STOP

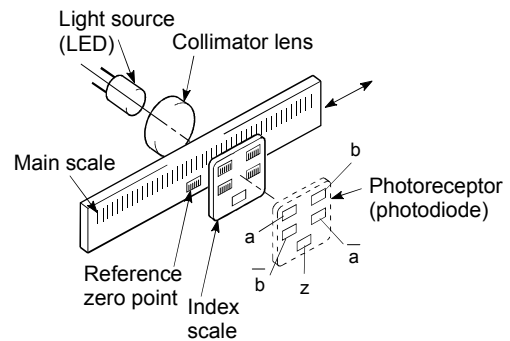
Emergency stops cannot be carried out by the QD75, so a method of shutting OFF the servo side power supply from outside the programmable controller, etc., must be considered.

## ENCODER

This device turns the input data into a binary code of 1 (ON) and 0 (OFF). A type of pulse generator.



Rotary encoder



Linear encoder

## ERROR CORRECTION

If a dimension error occurs in the machine, and that error is actually smaller or larger than 1m in spite of a 1m command being issued from the QD75, that error amount will be compensated. For example, when the error is actually smaller than 1m, the remaining distance to 1m is fed, and the correct 1m of positioning is carried out.

## ERROR RESET

This resets error of axis. Note that if the cause of the error is not eliminated at that time, the error will occur again.

## EXTERNAL REGENERATIVE BRAKE RESISTOR

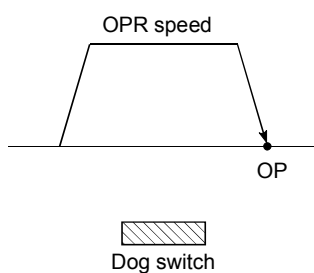
This is also called the regenerative brake. When a machine is moved with a motor, power is normally supplied to the motor from an amplifier. However, the rotation energy in the motor and machine counterflows (regenerates) to the amplifier when the motor is decelerating or when driving a descending load. The external regenerative resistor consumes this regeneration energy with resistance, obtains the regenerative brake torque, and enables the full capacity of the regeneration system during stopping. It is used when carrying out highly repetitive acceleration/deceleration.

## F

In the QD75, F is a status where the module itself has a fault.  
[HOLD]

## FAST OPR

The axis returns to the machine OP at the OPR speed without detecting the near-point dog.  
(This is not validated unless a machine OPR has been carried out first.)

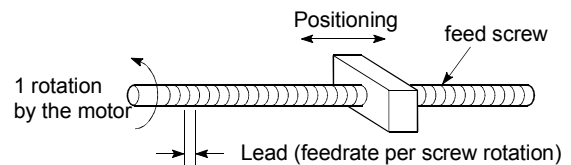


## FEED PULSE

This is a pulse issued from the positioning module to a servomotor or stepping motor. Also called a command pulse.

## FEED SCREW

This is the basic screw in mechanisms that position using screw rotation. Ball screws are often used to reduce backlash and dimension error.



## FEEDBACK PULSE

This is a method of using a returning pulse train to confirm whether the machine faithfully operated according to the commands issued in automatic control. If the machine did not faithfully operate according to the commands, a correction command is issued. For example, if a command is issued for 10,000 pulses, and a feedback pulse of 10,000 pulses is returned, then the balance becomes 0 and it can be judged that the command was faithfully followed.

Refer to the term "DEVIATION COUNTER".

## FIXED-FEED

This is the feeding of a set dimension for cutting sheet and bar workpieces into the designated dimensions. Incremental system positioning is often used. The current value is not incremented, even when the feed operation is repeated.

## FLASH MEMORY

This battery-less memory can be used to store parameters and positioning data for backup. Because it is battery-less, battery maintenance is not required.

## FLAT TYPE MOTOR (PANCAKE MOTOR)

About 100mm shorter in axial dimension than the standard motor. Used when the servomotor is installed in a small space.

**FLS SIGNAL (forward limit signal)**

This is the input signal that notifies the user that the limit switch (b contact configuration, normally ON) installed at the upper limit of the positioning control enabled range has been activated.

The positioning operation stops when the FLS signal turns OFF (non-continuity).

**G CODE**

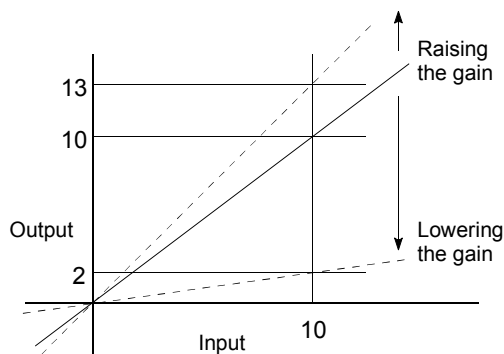
These are standardized (coded) 2-digit numerical values (00 to 99) designating various control functions of the NC module. Also called G functions.

Example :

- G01 Linear interpolation
- G02 Circular interpolation CW (clockwise)
- G04 Dwell
- G28 OPR
- G50 Max. spindle speed setting

**GAIN**

The changing of the ratio between two values having a proportional relation. Seen on a graph, the changing of the incline of the characteristics.



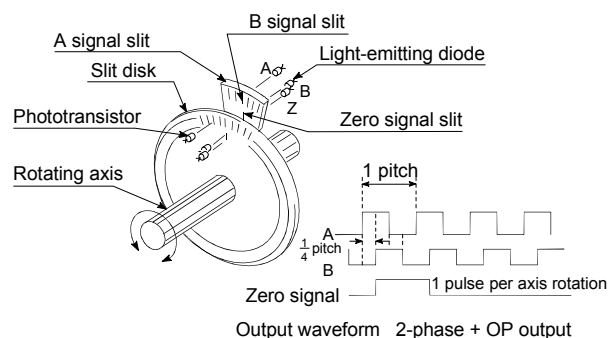
For example, when 10 is output for an input of 10, the output can be changed to 12, 5, etc., by changing the gain.

**GD<sup>2</sup>**

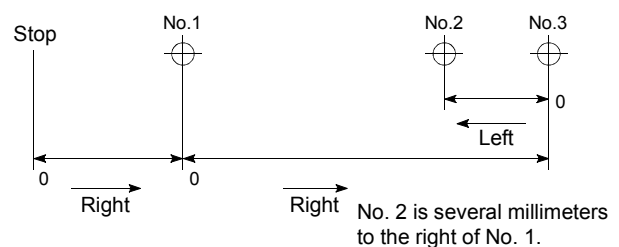
The inertia moment. The sum total of the mass (dm) of each small area configuring an object multiplied by the square of the distance (r) of each of those areas from a given straight line. The relation with  $I = \int r^2 dm$  GD<sup>2</sup> is given by  $4gl$ , with "g" being gravitational acceleration.

**INCREMENTAL ENCODER**

A device that simply outputs ON/OFF pulses by the rotation of the axis. 1-phase types output only A pulses, and do not indicate the axis rotation direction. 2-phase types output both A and B pulse trains, and can judge the rotation direction. The direction is judged to be forward if the B pulse train turns ON when A is ON, and judged to be reverse if A turns ON when B is ON. There is also another type of incremental encoder with a zero signal. The most commonly used incremental encoders output between 100 and 10,000 pulses per axis rotation. Refer to "ENCODER".

**INCREMENTAL SYSTEM**

The current value is 0 in this system. Positions are expressed by the designated direction and distance of travel. Also called the relative address system. This system is used in fixed-feed, etc. Compare ABSOLUTE SYSTEM.

**INERTIA**

The property of an object, when not being affected by external forces, where it tries to maintain its current condition. The inertia moment.

## INPUT TERMINAL

This is a pin connector wired by the user for inputting data to the QD75 from an external source. It is connected to the motor drive unit or machine side.

This terminal is used to output the following.

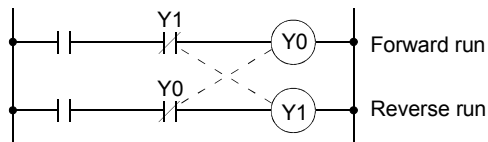
- DRIVE UNIT READY signal
- START signal
- STOP signal

, etc.

The input No. Xn is not directly related to the program, so it is not used.

## INTERLOCK

In this condition, the machine is blocked from moving to the next operation until the operation in progress is complete. This function is used to prevent damage to devices and malfunctioning.



## INTERPOLATION OPERATION

The simultaneous operation of multiple motors to carry out a composite operation. Each motor can be freely set with the positioning distance, acceleration/deceleration time, speed, and other factors, which are combined to move a target in a straight line, circle, etc. Linear interpolation and circular interpolation are available. (The circular interpolation uses two motors.)

## INVERTER

This refers to a device to change a direct current (DC) to an alternating current (AC). The device actually changes the motor speed by changing 50Hz or 60Hz of commercial frequency to direct current once, then changing it again to a 5 to 120Hz alternating current and controlling the motor speed.

## JOG

Jog. This refers to moving the tool little by little. Inching.  
Parameter setting is required to carry out JOG operation.

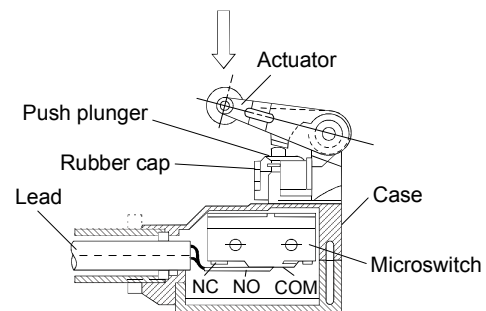
## kPPS

This is the abbreviation for "kilopulses per second". 80kPPS equals 80,000 pulses per second.

## LIMIT SWITCH

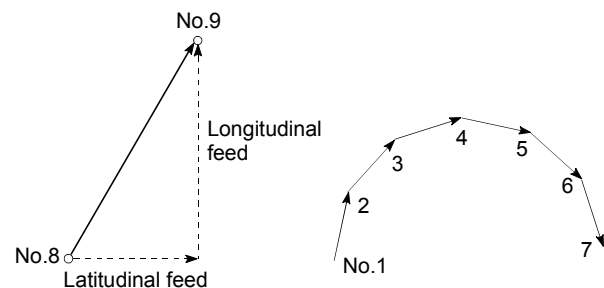
This is a switch set to stop a moving object at both ends, etc., of a movement device for safety reasons.

A circuit is created in which the moving body itself presses against the switch to activate the contact and forcibly shut the power OFF. For example, pressing on the actuator in the drawing below activates the internal microswitch. There are various other types.



## LINEAR INTERPOLATION

This automatic operation simultaneously operates two motors for the latitudinal (X) feed and longitudinal (Y) feed to move a target in a diagonal line for positioning. Three or four motors can also be operated simultaneously. The QD75 combines the operation of axis 1 through 4 for the linear interpolation. The same positioning data No. must be used for the setting. Refer to the term "INTERPOLATION OPERATION".



## LOAD INERTIA RATIO

$GDL^2/GDM^2$   
Refer to "GD<sup>2</sup>".

## LOW-INERTIA MOTOR

This is a motor used when frequent acceleration/deceleration is repeated. Low-inertia motors are longitudinally longer, to decrease the rotor diameter and cover the torque. This enables their inertia moment to be reduced up to 1/3 that of standard motors. The ideal load inertia ratio is 1 or less.

## M CODE (Machine Code)

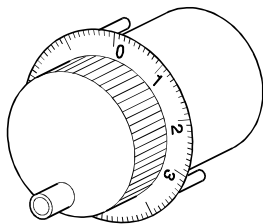
These are sub functions that interlock with the positioning operation to replace drills, tighten and loosen clamps, raise and lower welding electrodes, display various data, etc. Either of two modes can be entered when the machine code turns ON: AFTER or WITH. The machine does not move to the next positioning when the machine code is ON. M codes are turned OFF by the programmable controller program. Code Nos. from 1 to 65535 assigned by the user and used (1: Clamp, 2: Loosen, etc.). Comments can be written after 50 of the M codes, and they can be monitored using a peripheral device or displayed on an external display. Refer to "AFTER MODE" and "WITH MODE".

## MACHINE FEED VALUE

The OP address at the completion of the machine OPR is stored. The current position of the machine coordinates determined by a machine having the OP address as a reference. Even if the current value is changed, this value will not change.

## MANUAL PULSE GENERATOR

The handle of this device is manually rotated to generate pulses. This device is used when manually carrying out accurate positioning.



Made by Mitsubishi Electric Corp.  
(model: MR-HDP01)

## MASTER AXIS

When carrying out interpolation operations, this is the side on which the positioning data is executed in priority. For example, when positioning with the X axis and Y axis, the side with the largest movement distance will become the master axis, and the speed will follow that axis. The slave axis speed will be ignored.

## MOVEMENT AMOUNT PER PULSE

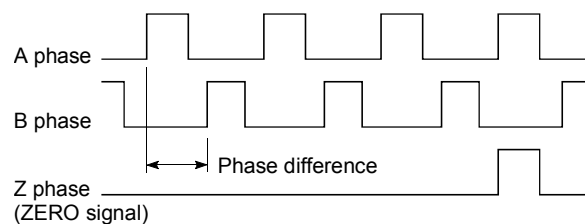
When using mm, inch, or degree units, the movement amount is calculated and output from the machine side showing how much the motor shaft moves per pulse. Positioning accuracy in smaller units is not possible. On the motor side, the movement amount per axis rotation is normally designed as a reference, so it is calculated as follows.

Movement amount per pulse =

$$\frac{\text{P rate}}{\text{No. of pulses per encoder rotation}} \times \text{Movement amount per rotation}$$

## MULTI-PHASE PULSE

A combination of pulses in which 2 or more phases differ. 2-phase pulses, etc.



## MULTIPLYING RATE SETTING

The P rate. Refer to the term "P RATE".

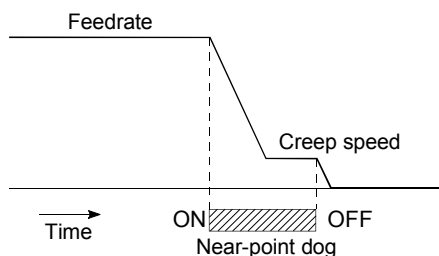
### NC LANGUAGE (Numerical Control Language)

This is the language punched into the paper tape that instructs the machining to the NC module.

The NC language consists of EIA codes (EIA language), ISO codes (ISO standards), and JIS codes (JIS standards).

### NEAR-POINT DOG

This is a switch placed before the OP. When this switch turns ON, the feedrate is changed to the creep speed. Because of that, the time that this switch is ON must be long enough to allow for the time required for deceleration from the feedrate to the creep speed.



### NEW CURRENT VALUE (CURRENT VALUE CHANGING)

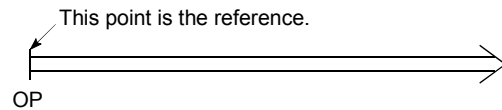
The QD75 has no way of knowing the current value when the machine is assembled and the positioning module is connected, so this function is used to teach it a temporary approximate value as the current value. This function can also be used to write a temporary current value when the current value has been lost due to accidents, etc. If an OPR is carried out after that, the positioning module will recognize the zero point.

To prevent the accumulated value from exceeding the stroke limit in fixed-feed, etc., rewrite the current value to 0 after the fixed-feed. The current value can be changed during a positioning stop.

### OP

This is the reference position for positioning. Positioning cannot start without a reference point.

The OP is normally set to the upper or lower stroke limit.



### OP SHIFT FUNCTION

The OP position can be shifted in the positive or negative direction by executing a machine OPR and determining the shift amount from the machine OPR complete position.

An OP can be set at a position besides the OP position, or outside the dog switch.

### OPERATION PATTERN

The kind of operation to be carried out after executing the positioning data is determined.

- 1) If "POSITIONING COMPLETE" is selected, the operation will stop after the positioning is complete.
- 2) If "CONTINUOUS POSITIONING CONTROL" is selected, the next data No. will be automatically executed after the positioning is complete.
- 3) If "CONTINUOUS PATH CONTROL" is selected, the positioning will not be completed. Only the speed will be automatically changed, and the next data No. will be executed.

### OPR METHOD

The OPR methods are shown below. The method used depends on the machine structure, stopping accuracy, etc.

OPR can be carried out when the OPR parameters are written.

- 1) Near-point dog method
- 2) Stopper method
- 3) Count method



## OPR PARAMETER

This parameter is required when returning to the OP. It is determined by the machine side design, so subsequent changes of this parameter must be accompanied by changes in the machine design.

The OP is the reference for positioning operations, so if the OP is lost due to a power failure during positioning, or because the power is turned OFF and the machine is moved manually, etc., it can be restored by carrying out an OPR. When a machine OPR command is issued, the machine will move in search of the near-point dog regardless of the current value, and will stop at the OP. At this time, the current value will be rewritten to the OP address. Data cannot be written during positioning. With the QD75, data is always written for all axes (from 1 to 4 axes). Refer to the term "NEAR-POINT DOG".

## OPR REQUEST

This signal turns ON when there is an error with the QD75. It will turn ON in the following situations.

- 1) When the power is turned ON.
- 2) When the PLC READY signal turns from OFF to ON.
- 3) When the machine OPR starts.
- 4) When the drive unit READY signal turns from ON to OFF.

The user judges whether to carry out a machine OPR in the above situations.

## OUTPUT TERMINAL

This is a pin connector for outputting data from the QD75 to an external source. It is connected to the motor drive unit.

This terminal is used to output the following.

- Feedback pulses for both forward run and reverse run
- Start
- Deviation counter clear

The terminal Nos. are determined for each axis.

The output No. Yn is not directly related to the program, so it is not used.

## OVERRIDE FUNCTION

With this function, the speed during positioning operations (current speed) can be varied between 1 and 300%.

The speed can also be changed by the same variable rate for continuous positioning with differing designated speeds.

## P RATE (Pulse Rate)

A coefficient that magnifies the feedback pulses per motor shaft rotation by 2-fold, 3-fold, 1/2 or 1/3.

It is the ratio of the feed pulses and feedback pulses.

For example, when the No. of pulses per motor shaft rotation is set to 2400 pulses, and the P rate is set to 2, the result will be equivalent to 1200 pulses.

The rotation per pulse is  $0.15^\circ$  when 2400 pulses per rotation are set, but this becomes  $0.3^\circ$  when 1200 pulses. The positioning accuracy drops as the P rate is increased.

## PARAMETER

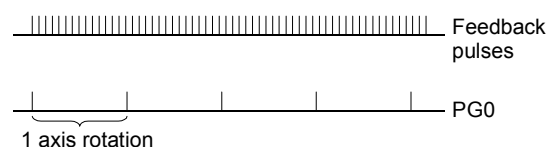
This is the basic data used in positioning.

Parameters are determined by the machine side design, so subsequent changes of parameters must be accompanied by changes in the machine design.

Data cannot be written during positioning. The initial parameter values are written by the maker.

## PG0 (Pulse Generator Zero)

Pronounced "pee-jee-zero". Refer to the term "ZERO SIGNAL".



## POSITION CONTROL

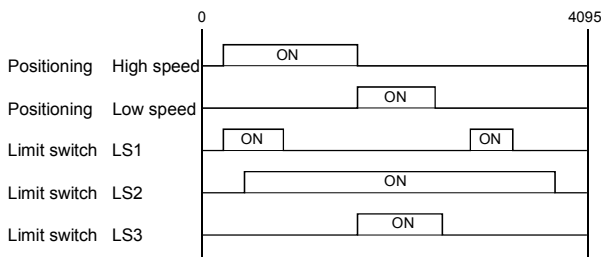
This is mainly the control of position and dimension, such as in fixed-feed, positioning, numerical control, etc. This is always controlled with feed pulses. There is also speed control.

Drive units may differ, even when the same motor is used.



## POSITION DETECTION MODULE

This is an abridged version of positioning. There are two types on MELSEC, the A1S62LS. This module has positioning and limit switch functions, and can use a total of 16 channels. The following drawing shows an example for 5 channels. A resolver is used in the positioning detection.



## POSITION LOOP GAIN

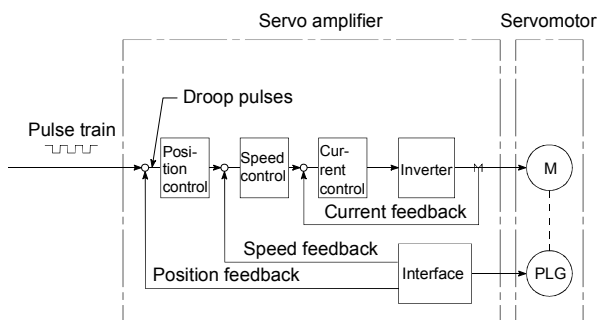
This is the ratio of the deviation counter droop pulses to the command pulse frequency.

$$\text{Position loop gain} = \frac{\text{Command pulse frequency}}{\text{Droop pulses}} (1/s)$$

The position loop gain can be set with the drive unit. Raise the gain to improve the stopping accuracy. However, overshooting will occur if the position loop gain is raised too far, and the operation will become unstable. If the position loop gain is lowered too far, the machine will stop more smoothly but the stopping error will increase.

## POSITION LOOP MODE

This is one servo control mode used in positioning. It is a mode for carrying out position control. The other servo control modes are the speed loop mode for carrying out speed control, and the torque loop mode for carrying out torque control (current control).



## POSITIONING

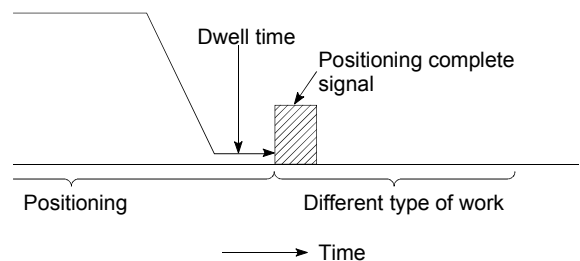
Accurately moving the machine from a point to a determined point. The distance, direction, speed, etc., for that movement are designated by the user. Positioning is used in operations such as cutting sheets, drilling holes in plates, mounting parts on a PCB, and welding. Positioning is also used with robots.

## POSITIONING COMPLETE

Refer to the term "OPERATION PATTERN".

## POSITIONING COMPLETE SIGNAL

This is a signal that occurs when the positioning is complete. A timer set beforehand starts when this signal is output, and the machine movement stops for that time. When this signal turns ON, the positioning start signal turns OFF.



## POSITIONING CONTINUED

Refer to the section of term "operation pattern".

## POSITIONING DATA

This is data for the user to carry out positioning. The No. of points to which positioning is carried out (the No. of addresses) is designated by the user. In the QD75, these are 600 points. As a principle, positioning is executed in the order of the data Nos.

## POSITIONING PARAMETER

This is basic data for carrying out positioning control. Types of data include control unit, movement amount per pulse, speed limit value, upper and lower stroke limit values, acceleration/deceleration time, positioning system, etc.

Parameters have an initial value, so that value is changed to match the control conditions.

## POSITIONING START

This refers the act of designating a target data No. and starting the positioning.

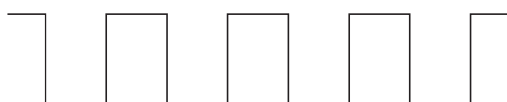
The operation after the positioning is complete for that data No. is determined by the data No.'s positioning pattern.

## PTP Control (Point To Point Control)

This is a type of positioning control. With this control method, the points to be passed are designated at random locations on the path. Movement only to a given target positioning is requested. Path control is not required during movement from a given point to the next value.

## PULSE

The turning ON and OFF of the current (voltage) for short periods. A pulse train is a series of pulses. The QD75 is the module that generates the pulses.



## PULSE GENERATOR

This is a device that generates pulses.

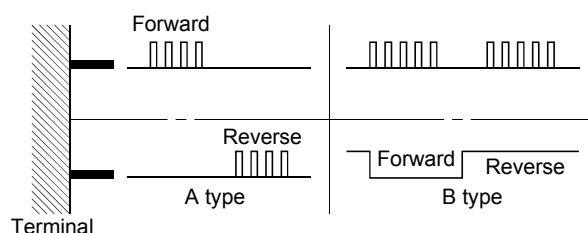
Examples include devices installed on the motor shaft that create pulses when the shaft rotates, and digital devices.

1-phase types output one pulse train. 2-phase types output two pulse trains with a phase difference. From 600 to 1,000,000 pulses can be output per shaft rotation. Generators with a ZERO signal function to output 1 or 2 pulses per shaft rotation. Abbreviated as PLG. Refer to the term "ENCODER".

## PULSE OUTPUT MODE

There are two methods used to issue forward run and reverse run commands to the servomotor. The type used differs according to the machine maker. In type A, the forward run pulses and reverse run pulses are output from separate terminals.

In type B, the forward run pulses and reverse run pulses are output from the same terminal, and a forward/reverse run identification signal is output from another terminal.



## READY

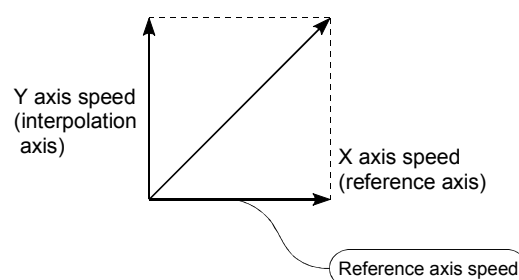
This means that preparation is complete.

## REAL-TIME AUTO TUNING (Real-time Automatic Tuning)

Refer to "AUTO TUNING".

## REFERENCE AXIS SPEED

This is the speed of the reference axis during interpolation operations.

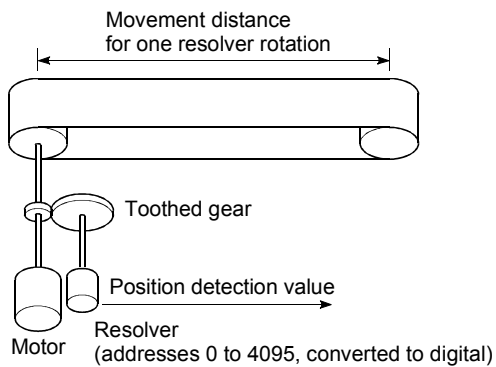


## REGENERATIVE BRAKE OPTION

This function is an option. It is used when carrying out highly repetitive acceleration/deceleration. Refer to "EXTERNAL REGENERATIVE RESISTOR".

## RESOLVER

This device detects the angle by resolving the two voltages of the analog input. Also called a 2-phase synchro. For a 1-phase voltage input, the axis rotation angle is converted into a perpendicular 2-phase voltage (analog voltage) and output.



## RLS SIGNAL (reverse limit signal)

This is the input signal that notifies the user that the limit switch (b contact configuration, normally ON) installed at the lower limit of the positioning control enabled range is activated. The positioning operation stops when the RLS signal turns OFF (non-continuity).

## ROTARY TABLE

A round table on which the workpiece is placed. Positioning control is carried out while rotating the workpiece in a 360° range.

## S-CURVE

### ACCELERATION/DECELERATION

In this pattern, the acceleration and deceleration follow a sine curve, and the movement is smooth. The S-curve ratio can be set from 1 to 100%.

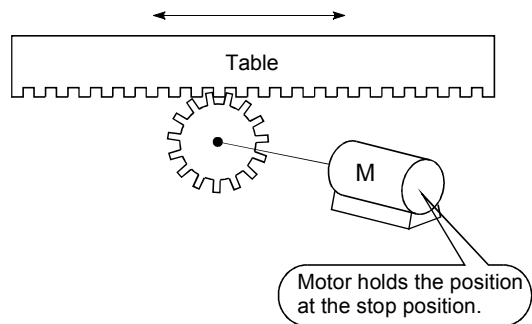


## SERVO AMPLIFIER

Refer to the term "DRIVE UNIT".

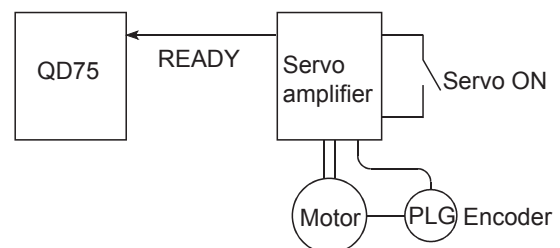
## SERVO LOCK

In positioning using a servomotor, stepping motor, etc., working power is required to hold the machine at the stop position. (The position will be lost if the machine is moved by external power.) This kind of state is called servo lock or servo lock torque.



## SERVO ON

The servo amplifier will not operate if the servo amplifier is in a normal state and this servo ON signal is OFF.



## SERVOMOTOR

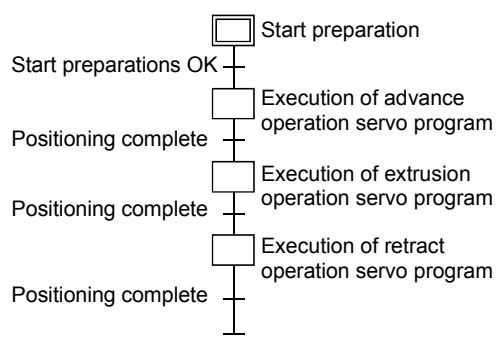
A motor that rotates true to the command. Servomotors are highly responsive, and can carry out frequent high-speed and high-accuracy starts and stops. DC and AC types are available, as well as large-capacity motors. A pulse generator accessory for speed detection is common, and feedback control is often carried out.

## SETTING UNIT

This is one setting item of the positioning reference parameters. The unit to be used is designated as mm, inch, degree, or pulse.

## SFC (Sequential Function Chart)

A sequential function chart is a programming method optimally structured for running a machine's automatic control in sequence with the programmable controller.



## SKIP FUNCTION

When a SKIP signal is input, the positioning being executed is interrupted, the motor is deceleration stopped, and the next positioning is automatically carried out.

## SLAVE AXIS

During interpolation operation, the positioning data is partially ignored on this side. This axis is moved by the master axis data.

## SPEED CONTROL

Speed control is mainly carried out with the servomotor. It is an application for grindstone rotation, welding speed, feedrate, etc. Speed control differs from position control in that the current position (address) is not controlled. Drive units may differ, even when the same motor is used.

## SPEED INTEGRAL COMPENSATION

This is one item in the parameters of the servo amplifier, and is used to raise the frequency response during speed control to improve transient characteristics.

When adjusting the speed loop gain, raising this value is effective if the overshooting during acceleration/deceleration remains large.

This compensation is set in ms units.

## SPEED LIMIT VALUE

This is the max. speed for positioning. Even if other data is mistakenly set to a higher speed than this, the positioning will be carried out at this speed limit value when it is set in the parameters. The acceleration time becomes the time to accelerate from a stopped state to the speed limit value, and the deceleration time becomes the time to decelerate from the speed limit value to a stopped state.

## SPEED LOOP GAIN

This is one item in the parameters of the servo amplifier, and expresses the speed of the control response during speed control. When the load inertia moment ratio increases, the control system speed response decreases and the operation may become unstable. If this happens, the operation can be improved by raising this setting value.

The overshoot will become larger if the speed loop gain is raised too far, and motor vibration noise will occur during operation and when stopped.

## SPEED LOOP MODE

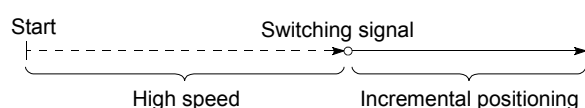
This is one servo control mode used in positioning. It is a mode for carrying out speed control. Refer to "POSITION LOOP MODE".

## SPEED-POSITION SWITCHING CONTROL

With this control, positioning is carried out to the end point of the movement amount while changing the speed at the speed switching point during positioning control.

## SPEED-POSITION SWITCHING CONTROL MODE

This is one method used for positioning. It is an application for operations such as high-speed movement to a point unrelated to positioning, then set dimension movement from the switching signal activation point.



## START COMPLETE

This signal gives an immediate response notifying the user that the QD75 that was started is now in a normal state and can start positioning.

## STARTING AXIS

One of the QD75 axis system axes (axis 1, axis 2, axis 3, or axis 4) or the reference axis for the interpolation operation is designated as the starting axis.

## STATUS

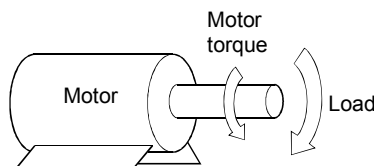
Data showing the state of the machine. Collectively refers to signals that turn ON when the battery voltage drops, during OPR requests, during dwell time, etc.

## STEP FUNCTION

When the operation is designed so that several positioning data Nos. are consecutively run, this function can be used to carry out a test operation for 1 data item at a time.

## STEP OUT

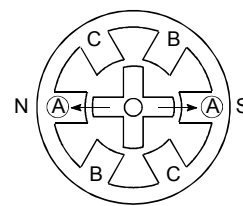
Stepping motors rotate in proportion to the No. of pulses (frequency), but the motor's rotation will deviate if the load is too large for the motor. This is called step out. If step out occurs, the motor must be replaced by one with a larger torque. Step out causes the positioning error to increase.



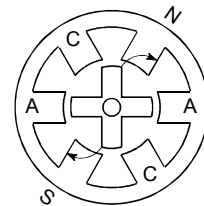
## STEPPING MOTOR

A motor that rotates a given angle (example:  $0.15^\circ$ ) when 1 pulse is generated.

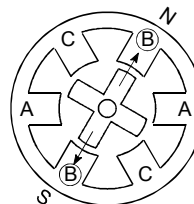
For that reason, a rotation proportional to the No. of pulses can be obtained. 2-phase to 5-phase stepping motors are available. In the 3-phase type, the rotor rotates in order from A to C when a voltage is applied. Often found in compact motors, stepping motors rotate accurately without feedback. Be careful of step out when overloaded.



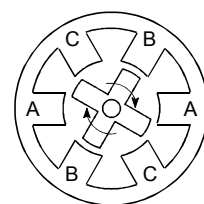
1) First, the A phase is excited by a pulse.



2) When the B phase is then excited, the force works in the direction shown by the arrows.



3) The nearest tooth to the B phase is attracted, and the rotation stops.



4) As the excitation phase is continuously changed, the rotor rotates in a clockwise direction.

## STOP SETTLING TIME

Refer to the term "DWEELL TIME".

## STOP SIGNAL

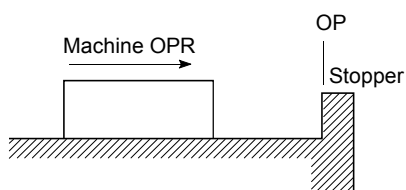
In positioning control, this is the input signal that directly stops the operation from an external source.

The operation stops when the external stop signal (a-contact) turns ON (continuity).

## STOP WITH STOPPER

This is one machine OPR method. With this method, a stopper is established at the OP, and the operation is stopped when the machine presses against it.

Motor burning would occur or the stopper would be damaged if the machine were left in that state. There are two methods to prevent this; a timer can be used to shut OFF the motor after a fixed time, or the motor can be stopped by limiting sudden increase in the motor torque when the machine presses against the stopper.



## STROKE

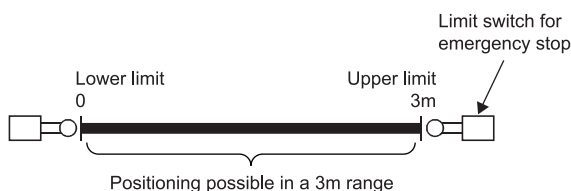
The stroke is the variation in the operation by the distance from a stopped state to the next stopped state after a movement.

## STROKE LIMIT

This is the range in which a positioning operation is possible, or the range in which the machine can be moved without damage occurring.

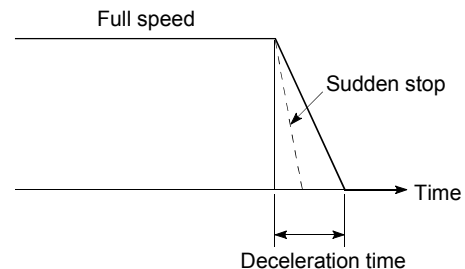
(Movement outside this range is possible in the manual operation.) For operations using a worm gear, the stroke limit is determined by the length of the screw. For operations using a fixed-feed, it is determined by the max. dimension to be cut.

The upper and lower limits are set in the parameters, but a separate limit switch should be established and an emergency stop circuit outside the programmable controller should be created. Refer to the term "LIMIT SWITCH".



## SUDDEN STOP

A stop carried out in a shorter time than the deceleration time designated in the parameters.



## TEACHING

When the positioning address is uncertain, or gauging is required, this function is used by the user to search for and teach the position to the machine.

For example, complex addresses such as drawings can be taught by tracing a model, and the positioning operation can be reproduced.

## TORQUE CONTROL

In this function, a limit is established for the resistance torque applied to the motor used for positioning. The power is turned OFF if torque exceeding that value is applied to the motor. When excessive torque is applied to a motor, it causes the current to suddenly increase. Motor burning and other stress on the motor occurs, and the life of the motor is shortened.

This function utilizes the sudden increase in the torque when the machine OPR to issue a command to stop the motor.

## TORQUE LOOP MODE

Also called the current loop mode.  
Refer to "POSITIONING LOOP MODE".

## TORQUE RIPPLE

Torque width variations, deviations in the torque.

## TRACKING FUNCTION

In this function, positioning is carried out at a speed relative to a moving target object by inputting the movement amount from an external encoder and adding it to the servo command value.

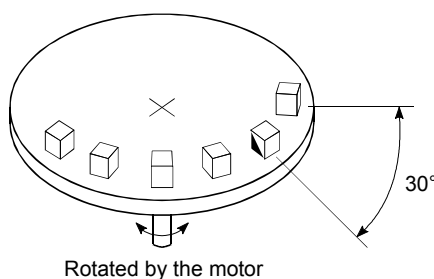
## TRAPEZOIDAL ACCELERATION/DECELERATION

An operation in which a graph of the time and speed takes a trapezoidal shape.



## TURNTABLE

A rotating table, which is turned using power. The table is used divided from one 360° rotation into the required locations for work. The positioning control unit is "degree".



## UNIT SETTING

This is the setting of the unit for the actual address to which positioning is required, or for the movement amount.

The following units can be set: mm, inch, degree and pulse. The initial value in the parameters is a pulse unit.

## WARNING

A warning is output as a warning code in when an abnormality is detected that is not serious enough to require cancellation or stoppage of the positioning operation. Warnings are handled differently than errors.

## WINDOW

These are the selection menus that appear on the screen when the QD75 is started.

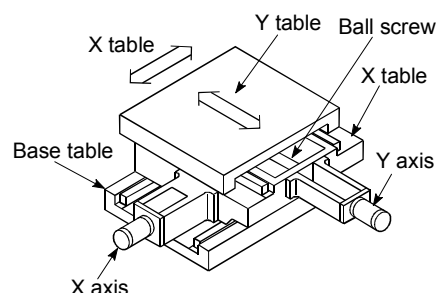
- Menu selection window
- Mode function selection window
- Sub-function selection window
- Execution/setting selection window

## WITH MODE

This is the mode that outputs the M code before the start of the positioning. This mode turns ON at the positioning start, enabling voltage to be applied to the welding electrodes, display of positioning speeds, etc. Refer to the term "AFTER MODE".

## XY TABLE

This is a device that moves a table in the X (latitudinal) and Y (longitudinal) directions so that positioning can be carried out easily. There are also commercially available products.



## Z PHASE

Also called "PG zero". Refer to "ZERO SIGNAL".

## ZERO SIGNAL

This refers to PG0 of the pulse generator (encoder) (one detection per shaft rotation). It also called "Z phase". Refer to the term "PULSE GENERATOR".

## Appendix 12 Positioning control troubleshooting

| Trouble type              | Questions/Trouble                                                                                                                                         | Remedy                                                                                                                                                                                                                                                                                                                                                                                                                                             | No. |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Parameter                 | Display reads "FFFFH" when a parameter is read with GX Configurator-QP.                                                                                   | The CPU module power was turned OFF or the CPU module was reset, etc., during flash ROM writing, which deleted the data in the flash ROM. Initialize the parameters, and reset the required parameters. (Refer to Section 13.2 "Parameter initialization function" for details.)                                                                                                                                                                   | 1   |
|                           | How can the parameters be returned to their initial values (default values)?                                                                              | Set the " <a href="#">Cd.2</a> Parameter initialization request" to "1". (Refer to Section 13.2 "Parameter initialization function" for details.)                                                                                                                                                                                                                                                                                                  | 2   |
|                           | A parameter error occurred although the parameter was set correctly by GX Configurator-QP.                                                                | The parameter may have been overwritten in the sequence program. Review the sequence program.                                                                                                                                                                                                                                                                                                                                                      | 3   |
| Hardware stroke limit     | The machine overruns if operating at high speeds when the hardware stroke limit range is exceeded.                                                        | In the QD75, deceleration stops are executed after the machine exceeds hardware stroke limit range. Because of this, more time is required for the deceleration stop as the speed increases, and the overrun becomes larger. (The deceleration time becomes shorter at lower speeds, so the overrun becomes smaller.)                                                                                                                              | 4   |
|                           | When the machine exceeded the hardware stroke limit range, positioning toward inside the range was started, but the machine did not start.                | Use a "JOG operation", "Inching operation" or "Manual pulse generator operation" to return the machine to inside the hardware stroke limit range. (When the hardware stroke limit range is exceeded, positioning will not start toward inside the range even when so commanded. Once the range is exceeded, a return to inside the range can only be executed using a "JOG operation", "Inching operation" or "Manual pulse generator operation".) | 5   |
| Degree                    | Exactly one rotation is required, but the setting range for a "degree" unit setting is "0 to 359.999...". Won't the rotation deviate by "0.00...1"?       | Designate "360.000" in the INC control. The motor will make exactly one rotation.                                                                                                                                                                                                                                                                                                                                                                  | 6   |
| Movement amount per pulse | If the "movement amount per pulse" is calculated as written in the manual, settings smaller than the basic parameter setting range cannot be carried out. | Set "movement amount per pulse" in the QD75 using the three parameter values of <a href="#">Pr.2</a> to <a href="#">Pr.4</a> . Try setting the values following the explanations for each parameter.                                                                                                                                                                                                                                               | 7   |
| Override                  | Will an override setting value written before starting be valid?                                                                                          | It will be valid.                                                                                                                                                                                                                                                                                                                                                                                                                                  | 8   |
|                           | During path control, will the override still be valid after the point is passed?                                                                          | It will still be valid.                                                                                                                                                                                                                                                                                                                                                                                                                            | 9   |
|                           | How can the override be canceled?                                                                                                                         | Set the " <a href="#">Cd.13</a> Positioning operation speed override" to "100".                                                                                                                                                                                                                                                                                                                                                                    | 10  |



| Trouble type                                  | Questions/Trouble                                                                                                                                                                            | Remedy                                                                                                                                                                                                                                                                                                                                                                                                                     | No. |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Acceleration/<br>deceleration time            | How can the deceleration stop time during stopping be shortened using the hardware stroke limit?                                                                                             | Set "1: Sudden stop" in the " [Pr.37] Stop group 1 sudden stop selection", and reduce the setting value of " [Pr.36] Sudden stop deceleration time".                                                                                                                                                                                                                                                                       | 11  |
|                                               | The motor does not operate at "60000ms" although the acceleration/deceleration time is set to "60000ms".                                                                                     | The value set for the acceleration/deceleration time is the time required for the machine to accelerate from speed "0" to the value set in " [Pr.8] Speed limit value". Because of that, the acceleration/deceleration time will also be shorter than "60000ms" if the command speed value is smaller than the " [Pr.8] Speed limit value". (Refer to the explanation for [Pr.9] and [Pr.10] for details.)                 | 12  |
|                                               | Can each acceleration/deceleration time be individually set to trapezoidal or S-curve acceleration/deceleration?                                                                             | The trapezoidal and S-curve acceleration/deceleration processing is a common setting for all acceleration/deceleration times, so individual setting is not possible. (Refer to Section 12.7.6 "Acceleration/deceleration process function".)                                                                                                                                                                               | 13  |
|                                               | The machine starts and stops suddenly when carrying out JOG operations and positioning operations. (Using an MR-J2S servo amplifier.)                                                        | Review the parameter settings for acceleration/deceleration time, speed limit value, JOG speed limit value, JOG acceleration/deceleration time, etc.                                                                                                                                                                                                                                                                       | 14  |
| Simplified absolute position detection system | Are simplified absolute position detection system possible in the QD75 Positional deviation models?                                                                                          | They are possible if the models are used in combination with a Mitsubishi "AC Servo". (Refer to "AC servo User's Manual" for details.)                                                                                                                                                                                                                                                                                     | 15  |
| Positional deviation                          | The physical position position detection system from the commanded position, although the positioning is complete (and the monitored current position is correct).                           | If the deviation counter value is not "0", the servo side is still moving. Increase the torque.                                                                                                                                                                                                                                                                                                                            | 16  |
| Electronic gear                               | A setting of "1 $\mu$ m = 1pulse" is required in the following system. <ul style="list-style-type: none"> <li>Ball screw pitch = 10mm</li> <li>No. of feedback pulses = 8192pulse</li> </ul> | In this case, the following values will result. <ul style="list-style-type: none"> <li>No. of pulses per rotation = 8192</li> <li>Movement amount per rotation = 10000</li> <li>Unit magnification = 10</li> </ul> Therefore, the "Movement amount per pulse" will become "1.2207 $\mu$ m". This value is fixed by the machine system, so it cannot be changed. Thus, the setting "1 $\mu$ m = 1pulse" cannot be achieved. | 17  |

| Trouble type       | Questions/Trouble                                                                                                                                                                                                                                                                                                                                            | Remedy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | No. |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Error compensation | <p>The machine only moves to "10081230", although positioning with a command value of "10081234" carried out.</p> <p>How can the error be compensated?</p> <p>The following values are currently set.</p> <ul style="list-style-type: none"> <li>• [Pr.2] No. of pulses per rotation = 8192</li> <li>• [Pr.3] Movement amount per rotation = 8000</li> </ul> | <p>Reset [Pr.3] and [Pr.2] in the following order.</p> <ol style="list-style-type: none"> <li>1) Calculate <math>8192/8000 \times 10081230/10081234</math>.</li> <li>2) Obtain the reduced value.</li> <li>3) Set the numerator in "[Pr.3] Movement amount per rotation", and the denominator in "[Pr.2] No. of pulses per rotation".</li> </ol>                                                                                                                                                                                                                                                  | 18  |
| OPR                | <p>When carrying out a count method machine OPR, the message "Leave Sufficient Distance From The OP Position To The Near-Point Dog OFF." appears. Is there a problem if the distance is short?</p>                                                                                                                                                           | <p>The near-point dog must be set to turn OFF at a position after the OP is passed.<br/>(When the machine OPR is started on the near-point dog on in a count-method machine OPR, the machine enters a normal machine OPR operation after returning to the near-point dog OFF region.)<br/>(If the near-point dog is turned OFF before the OP, and the machine OPR is started between the near-point dog OFF position and the OP, the machine will mistakenly interpret that its current position is before the near-point dog ON position, and it will pass over the OP and continue moving.)</p> | 19  |
|                    | <p>In the near-point dog method machine OPR, the stop positions are not uniform.</p>                                                                                                                                                                                                                                                                         | <p>Carry out the following measures.</p> <ol style="list-style-type: none"> <li>1) Separate the near-point dog signal and zero signal detection positions.</li> <li>2) Lower the values in "[Pr.46] OPR speed" and "[Pr.47] Creep speed".</li> <li>3) Confirm whether the zero signal and near-point dog signal turn ON normally.</li> <li>4) Check that there is no play (backlash) in the machine system.</li> </ol>                                                                                                                                                                            | 20  |
|                    | <p>Can the machine OPR be carried out with the OPR retry function when it is started with the near-point dog ON and the upper/lower limit OFF?</p>                                                                                                                                                                                                           | <p>A "Hardware stroke limit error" will occur and the operation will not be carried out.<br/>(The machine will interpret any position where the near-point dog is ON as being within the working range, and that the upper/lower limit is ON.)</p>                                                                                                                                                                                                                                                                                                                                                | 21  |
|                    | <p>Are ABS and INC positioning possible without carrying out an OPR?</p>                                                                                                                                                                                                                                                                                     | <p>They are possible.<br/>In this case, the position where the power is turned ON is handled as the current feed value "0".</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 22  |
|                    | <p>After an OPR, the OPR request flag sometimes turns ON for no apparent reason.</p>                                                                                                                                                                                                                                                                         | <p>The OPR request flag turns ON in the following cases.</p> <ol style="list-style-type: none"> <li>1) When the power is turned ON.</li> <li>2) When the PLC READY signal [YO] turns from OFF to ON.</li> <li>3) When an OPR is carried out.</li> <li>4) When the servo amplifier power is turned OFF.</li> </ol> <p>If no problem is found when the above are checked, then it is possible that the communication is being interrupted by "a fault in the bus cable", "noise influence", etc.</p>                                                                                                | 23  |
|                    | <p>The OPR complete flag ([Md.31] Status: b4) sometimes turns ON when it shouldn't be ON.</p>                                                                                                                                                                                                                                                                | <p>Check whether the drive unit READY signal is weak or the power supply is unstable.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 24  |

| Trouble type                     | Questions/Trouble                                                                                                                                                                                                                                                                                | Remedy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | No. |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Start                            | The positioning start signal [Y10] is kept ON until the BUSY signal is OFF, but is there any problem with turning it OFF before the BUSY signal turns OFF?                                                                                                                                       | After the BUSY signal turns ON, there is no problem with turning [Y10] OFF before the BUSY signal turns OFF. (The QD75 detects the rising edge (OFF → ON) of the positioning start signal [Y10].)                                                                                                                                                                                                                                                                                                                                      | 25  |
|                                  | The operation will not start even when the start signal is turned ON.                                                                                                                                                                                                                            | Check the " [Md.26] Axis operation status" and " [Md.23] Axis error No.".                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 26  |
| Stop                             | How many milliseconds should the axis stop signal [Y4] be turned ON for?                                                                                                                                                                                                                         | The signal should be turned ON at 4ms or more. (If possible, set the signal so it does not turn ON only momentarily, but instead stays ON until the BUSY signal turns OFF. This will keep the stop signal from skipping.)                                                                                                                                                                                                                                                                                                              | 27  |
|                                  | How can a sudden stop be selected?                                                                                                                                                                                                                                                               | Set "1: Sudden stop" in the parameter from [Pr.37] to [Pr.39] corresponding to the stop group, and reduce the setting value of " [Pr.36] Sudden stop deceleration time".                                                                                                                                                                                                                                                                                                                                                               | 28  |
|                                  | "Normal deceleration stop" was selected in " [Pr.39] Stop group 3 sudden stop", and Y stop was turned ON. If the [Pr.39] setting is changed to "sudden stop" during a deceleration stop, and the Y stop signal turns from OFF to ON, will the operation change to a sudden stop from that point? | The operation will not change. Even if the same stop factor is input again during the deceleration stop, it will be ignored. The same deceleration stop process used when the stop signal was first input will be continued. (This also applies for [Pr.37] and [Pr.38].)                                                                                                                                                                                                                                                              | 29  |
| Circular interpolation           | ABS system circular interpolation operates normally, but a vertically oblong circle results when INC system circular interpolation is carried out.                                                                                                                                               | The address designation may be incorrect. When carrying out INC system circular interpolation, designate the relative addresses from the starting point of both the center point and end point.                                                                                                                                                                                                                                                                                                                                        | 30  |
| Speed-position switching control | Can the speed be changed during speed control and position control by speed-position switching control?                                                                                                                                                                                          | No. The speed for the speed control and position control cannot be set differently.                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 31  |
| JOG operation                    | Even if the JOG start signal is turned ON, the response until it turns ON is sometimes slow.                                                                                                                                                                                                     | Either of the following is possible.<br>1) The sequence program may be incorrect. Check by creating a test program in which the JOG start signal is turned ON only.<br>2) If the machine is hitting something when the torque setting is low, it may be trying to move by JOG operation in the opposite direction. In this case, the machine will start moving only after the internal droop pulses have been reached 0 in the counter, even if the JOG start signal has been turned ON. This makes it seem that the response is slow. | 32  |
|                                  | The operation is not carried out at the set JOG speed, although the speed limit value has not been reached.                                                                                                                                                                                      | Either of the following is possible.<br>1) The JOG start signal may be chattering. Monitor the JOG start signal to confirm whether it is chattering. (When using the "BUSY signal" in the JOG operation start circuit, check the position of the BUSY signal.)<br>2) The " [Pr.31] JOG speed limit value" may not be appropriate. Review the setting value and carry out the JOG operation again.                                                                                                                                      | 33  |

| Trouble type                     | Questions/Trouble                                                                                                       | Remedy                                                                                                                                                                                                                                                                                                                                  | No. |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| JOG operation                    | When a JOG operation is attempted, an error results and the machine does not move.                                      | The " [Pr.31] JOG speed limit value" may be larger than the " [Pr.8] Speed limit value".<br>Review the parameters and carry out the JOG operation again.                                                                                                                                                                                | 34  |
|                                  | Why does the positioning complete signal turns ON during the JOG operation?                                             | If a value other than "0" is set for " [Cd.16] Inching movement amount", the inching operation is carried out and the positioning complete signal turns ON.<br>Confirm that the " [Cd.16] Inching movement amount" is set to "0".                                                                                                       | 35  |
| Manual pulse generator operation | Is it possible to count the pulses when the B phase is set to "1", and only A phase pulses are input?                   | Not possible.<br>(The QD75 counts 1, 0, 1, 0.)                                                                                                                                                                                                                                                                                          | 36  |
|                                  | Can a manual pulse generator other than the Mitsubishi Electric MR-HDP01 be used?                                       | Other manual pulse generators can be used if they conform to Section 3.4 "Specifications for input/output interfaces with external devices."                                                                                                                                                                                            | 37  |
|                                  | Can one manual pulse generator be operated connected to several QD75 modules?                                           | This is possible if the system conforms to the electrical specifications.                                                                                                                                                                                                                                                               | 38  |
| Current value changing           | The BUSY signal is not canceled by the current value changing. How can it be canceled?                                  | The BUSY signal may remain if the scan time is long.<br>Use a complete signal to check whether the current value changing has been executed.                                                                                                                                                                                            | 39  |
| QD75 READY signal                | The QD75 READY signal does not turn ON even when the PLC READY signal [Y0] is turned ON.                                | "A parameter error" has occurred.<br>Confirm the error No. in the error history, and correct the parameter.                                                                                                                                                                                                                             | 40  |
| M code ON signal                 | Is there any problem with setting an M code ON signal OFF request in the next scan after the M code ON signal ON?       | The QD75 checks the M code ON signal OFF request every "0.9ms" (QD75P□N/QD75D□N) or "1.8ms" (QD75P□/QD75D□), so there is a possibility that the M code ON signal OFF may be delayed by a maximum of "0.9ms" (QD75P□N/QD75D□N) or "1.8ms" (QD75P□/QD75D□) after the M code ON signal ON, even if an M code ON signal OFF request is set. | 41  |
| Deviation counter clear          | How long is the output time for the deviation counter clear signal?                                                     | The time set in " [Pr.55] Deviation counter clear signal output time". (Initial value: 11ms)                                                                                                                                                                                                                                            | 42  |
|                                  | Is a deviation counter clear signal output when the positioning is complete?                                            | A signal is not output.<br>The only time the QD75 outputs a deviation counter clear signal is for a machine OPR.                                                                                                                                                                                                                        | 43  |
|                                  | How can a deviation counter clear signal be output?                                                                     | The QD75 does not output a deviation counter clear signal except for machine OPR.                                                                                                                                                                                                                                                       | 44  |
| Module                           | An error "PLC READY OFF start" (error code: 537) occurs after the QD75 is replaced. (The sequence program is the same.) | The internal parameters of the QD75 may be different.<br>Check if the QD75 READY signal [X0] turns ON when the PLC READY signal [Y0] turns ON.<br>When the PLC READY signal is ON but the QD75 READY signal is OFF, the parameter error has occurred.<br>Check the error code and modify the parameter with the error.                  | 45  |
| Motor                            | The motor only rotates in one direction.                                                                                | The parameter settings on the QD75 side may not match those on the servo side.<br>Check the parameter settings.                                                                                                                                                                                                                         | 46  |
|                                  | Can the current motor speed be monitored?                                                                               | The speed shown on the QD75 monitor is calculated from the number of pulses output from the module. Thus, the actual motor speed cannot be monitored.<br>(" [Md.22] Feedrate" monitors the commanded speed. It does not show the actual motor speed.)                                                                                   | 47  |

| Trouble type                | Questions/Trouble                                                                                                                                                                                                                            | Remedy                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | No. |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Error/warning               | An error "Backlash compensation amount error" (error code: 920) occurs even when the backlash compensation value is set to "1".                                                                                                              | $0 \leq \frac{\text{Backlash compensation value}}{\text{Movement amount per pulse}} \leq 255$ <p>Setting is not possible if the above equation is not satisfied.<br/>Adjust by setting " [Pr.4] Unit magnification" to 10-fold (or 100-fold, or 1000-fold), and setting " [Pr.2] Movement amount per rotation" to 1/10 (or 1/100, or 1/1000).</p>                                                                                                                      | 48  |
|                             | When a JOG operation is attempted, errors such as "Hardware stroke limit (+)" or "Hardware stroke limit (-)" (error code: 104 or 105) occur and the machine does not move.                                                                   | The hardware stroke limit wiring has probably not been carried out.<br>Refer to Section 12.4.4 "Hardware stroke limit function" for details, and wire accordingly.                                                                                                                                                                                                                                                                                                     | 49  |
|                             | An error "Speed designation during OP shift error" (error code: 997) appears when the PLC READY signal [Y0] turns from OFF to ON.                                                                                                            | A value besides "0" or "1" may be set in the " [Pr.56] Speed designation during OP shift".<br>Review the set sequence program, and reset the correct parameters.                                                                                                                                                                                                                                                                                                       | 50  |
|                             | When the start signal was turned ON immediately after the stop signal ON, a warning "Start during operation" (warning code: 100) was detected, and the start was ignored.                                                                    | The QD75 starts the deceleration stop process when the stop signal ON is detected. Thus, the machine interprets that "positioning is still being executed" immediately after the stop signal ON. Even if the start signal is turned ON at that time, the start request will be ignored and a warning "Start during operation" (warning code: 100) will occur.                                                                                                          | 51  |
|                             | Does a warning "Deceleration/stop speed change" (warning code: 500) occur only during "stop deceleration" and "automatically deceleration"?<br>Is there any problem if the operation is continued in that state without resetting the error? | The warning occurs only at those times mentioned at the left. Because this is a warning, there is no problem if the operation can be continued without resetting the error.<br>(When the speed is changed using the override, the new value will not be reflected on the data being executed, but will be reflected from the next start.)                                                                                                                              | 52  |
| Positioning complete signal | Position control was executed but the positioning complete signal does not turn ON.                                                                                                                                                          | Depending on the stop occurrence factor, positioning may have not been completed normally.<br>Check the axis monitor " [Md.26] Axis operation status" after the BUSY signal has turned OFF.<br>Stop : The stop signal has turned ON during positioning.<br>Check the condition under which the stop signal (Y stop, external stop) turns ON.<br>Error : An error has occurred during positioning.<br>Check the error occurrence factor from " [Md.23] Axis error No.". | 53  |
|                             |                                                                                                                                                                                                                                              | The setting value of the detailed parameter 2 "Positioning complete signal output time" is 0 or shorter than the scan time.<br>Using the sequence program, set the time when the signal can be detected securely.                                                                                                                                                                                                                                                      | 54  |

## Appendix 13 List of buffer memory addresses

The following shows the relation between the buffer memory addresses and the various items.

(Do not use any address other than listed below. If used, the system may not operate correctly.)

| Buffer memory address |        |        |        | Item                                                              | Memory area           |                        |
|-----------------------|--------|--------|--------|-------------------------------------------------------------------|-----------------------|------------------------|
| Axis 1                | Axis 2 | Axis 3 | Axis 4 |                                                                   |                       |                        |
| 0                     | 150    | 300    | 450    | <a href="#">Pr.1</a> Unit setting                                 | Basic parameters 1    | Positioning parameters |
| 1                     | 151    | 301    | 451    | <a href="#">Pr.2</a> No. of pulses per rotation (Ap)              |                       |                        |
| 2                     | 152    | 302    | 452    | <a href="#">Pr.3</a> Movement amount per rotation (Al)            |                       |                        |
| 3                     | 153    | 303    | 453    | <a href="#">Pr.4</a> Unit magnification (Am)                      |                       |                        |
| 4                     | 154    | 304    | 454    | <a href="#">Pr.5</a> Pulse output mode                            |                       |                        |
| 5                     | 155    | 305    | 455    | <a href="#">Pr.6</a> Rotation direction setting                   |                       |                        |
| 6                     | 156    | 306    | 456    | <a href="#">Pr.7</a> Bias speed at start                          |                       |                        |
| 7                     | 157    | 307    | 457    |                                                                   |                       |                        |
| 8                     | 158    | 308    | 458    | Not used                                                          | Basic parameters 2    |                        |
| 9                     | 159    | 309    | 459    |                                                                   |                       |                        |
| 10                    | 160    | 310    | 460    | <a href="#">Pr.8</a> Speed limit value                            |                       |                        |
| 11                    | 161    | 311    | 461    | <a href="#">Pr.9</a> Acceleration time 0                          |                       |                        |
| 12                    | 162    | 312    | 462    |                                                                   |                       |                        |
| 13                    | 163    | 313    | 463    | <a href="#">Pr.10</a> Deceleration time 0                         |                       |                        |
| 14                    | 164    | 314    | 464    |                                                                   |                       |                        |
| 15                    | 165    | 315    | 465    |                                                                   |                       |                        |
| 17                    | 167    | 317    | 467    | <a href="#">Pr.11</a> Backlash compensation amount                | Detailed parameters 1 |                        |
| 18                    | 168    | 318    | 468    | <a href="#">Pr.12</a> Software stroke limit upper limit value     |                       |                        |
| 19                    | 169    | 319    | 469    |                                                                   |                       |                        |
| 20                    | 170    | 320    | 470    | <a href="#">Pr.13</a> Software stroke limit lower limit value     |                       |                        |
| 21                    | 171    | 321    | 471    |                                                                   |                       |                        |
| 22                    | 172    | 322    | 472    | <a href="#">Pr.14</a> Software stroke limit selection             |                       |                        |
| 23                    | 173    | 323    | 473    | <a href="#">Pr.15</a> Software stroke limit valid/invalid setting |                       |                        |
| 24                    | 174    | 324    | 474    | <a href="#">Pr.16</a> Command in-position width                   |                       |                        |
| 25                    | 175    | 325    | 475    |                                                                   |                       |                        |
| 26                    | 176    | 326    | 476    | <a href="#">Pr.17</a> Torque limit setting value                  |                       |                        |
| 27                    | 177    | 327    | 477    | <a href="#">Pr.18</a> M code ON signal output timing              |                       |                        |
| 28                    | 178    | 328    | 478    | <a href="#">Pr.19</a> Speed switching mode                        |                       |                        |
| 29                    | 179    | 329    | 479    | <a href="#">Pr.20</a> Interpolation speed designation method      |                       |                        |
| 30                    | 180    | 330    | 480    | <a href="#">Pr.21</a> Current feed value during speed control     |                       |                        |
| 31                    | 181    | 331    | 481    | <a href="#">Pr.22</a> Input signal logic selection                |                       |                        |
| 32                    | 182    | 332    | 482    | <a href="#">Pr.23</a> Output signal logic selection               |                       |                        |
| 33                    | —      | —      | —      | <a href="#">Pr.24</a> Manual pulse generator input selection      |                       |                        |
| 34                    | 184    | 334    | 484    | <a href="#">Pr.150</a> Speed-position function selection          | Detailed parameters 2 |                        |
| 35                    | 185    | 335    | 485    | Not used                                                          |                       |                        |
| 36                    | 186    | 336    | 486    | <a href="#">Pr.25</a> Acceleration time 1                         |                       |                        |
| 37                    | 187    | 337    | 487    |                                                                   |                       |                        |
| 38                    | 188    | 338    | 488    | <a href="#">Pr.26</a> Acceleration time 2                         |                       |                        |
| 39                    | 189    | 339    | 489    |                                                                   |                       |                        |

| Buffer memory address |            |            |            | Item                                                                          | Memory area             |                        |                      |                |
|-----------------------|------------|------------|------------|-------------------------------------------------------------------------------|-------------------------|------------------------|----------------------|----------------|
| Axis 1                | Axis 2     | Axis 3     | Axis 4     |                                                                               |                         |                        |                      |                |
| 40<br>41              | 190<br>191 | 340<br>341 | 490<br>491 | <a href="#">Pr.27</a> Acceleration time 3                                     | Detailed parameters 2   | Positioning parameters |                      |                |
| 42<br>43              | 192<br>193 | 342<br>343 | 492<br>493 | <a href="#">Pr.28</a> Deceleration time 1                                     |                         |                        |                      |                |
| 44<br>45              | 194<br>195 | 344<br>345 | 494<br>495 | <a href="#">Pr.29</a> Deceleration time 2                                     |                         |                        |                      |                |
| 46<br>47              | 196<br>197 | 346<br>347 | 496<br>497 | <a href="#">Pr.30</a> Deceleration time 3                                     |                         |                        |                      |                |
| 48<br>49              | 198<br>199 | 348<br>349 | 498<br>499 | <a href="#">Pr.31</a> JOG speed limit value                                   |                         |                        |                      |                |
| 50                    | 200        | 350        | 500        | <a href="#">Pr.32</a> JOG operation acceleration time selection               |                         |                        |                      |                |
| 51                    | 201        | 351        | 501        | <a href="#">Pr.33</a> JOG operation deceleration time selection               |                         |                        |                      |                |
| 52                    | 202        | 352        | 502        | <a href="#">Pr.34</a> Acceleration/deceleration process selection             |                         |                        |                      |                |
| 53                    | 203        | 353        | 503        | <a href="#">Pr.35</a> S-curve ratio                                           |                         |                        |                      |                |
| 54<br>55              | 204<br>205 | 354<br>355 | 504<br>505 | <a href="#">Pr.36</a> Sudden stop deceleration time                           |                         |                        |                      |                |
| 56                    | 206        | 356        | 506        | <a href="#">Pr.37</a> Stop group 1 sudden stop selection                      |                         |                        |                      |                |
| 57                    | 207        | 357        | 507        | <a href="#">Pr.38</a> Stop group 2 sudden stop selection                      |                         |                        |                      |                |
| 58                    | 208        | 358        | 508        | <a href="#">Pr.39</a> Stop group 3 sudden stop selection                      |                         |                        |                      |                |
| 59                    | 209        | 359        | 509        | <a href="#">Pr.40</a> Positioning complete signal output time                 |                         |                        |                      |                |
| 60<br>61              | 210<br>211 | 360<br>361 | 510<br>511 | <a href="#">Pr.41</a> Allowable circular interpolation error width            |                         |                        |                      |                |
| 62                    | 212        | 362        | 512        | <a href="#">Pr.42</a> External command function selection                     |                         |                        |                      |                |
| 70                    | 220        | 370        | 520        | <a href="#">Pr.43</a> OPR method                                              |                         |                        | OPR basic parameters | OPR parameters |
| 71                    | 221        | 371        | 521        | <a href="#">Pr.44</a> OPR direction                                           |                         |                        |                      |                |
| 72<br>73              | 222<br>223 | 372<br>373 | 522<br>523 | <a href="#">Pr.45</a> OP address                                              |                         |                        |                      |                |
| 74<br>75              | 224<br>225 | 374<br>375 | 524<br>525 | <a href="#">Pr.46</a> OPR speed                                               |                         |                        |                      |                |
| 76<br>77              | 226<br>227 | 376<br>377 | 526<br>527 | <a href="#">Pr.47</a> Creep speed                                             |                         |                        |                      |                |
| 78                    | 228        | 378        | 528        | <a href="#">Pr.48</a> OPR retry                                               |                         |                        |                      |                |
| 79                    | 229        | 379        | 529        | <a href="#">Pr.49</a> OPR dwell time                                          |                         |                        |                      |                |
| 80<br>81              | 230<br>231 | 380<br>381 | 530<br>531 | <a href="#">Pr.50</a> Setting for the movement amount after near-point dog ON | OPR detailed parameters |                        |                      |                |
| 82                    | 232        | 382        | 532        | <a href="#">Pr.51</a> OPR acceleration time selection                         |                         |                        |                      |                |
| 83                    | 233        | 383        | 533        | <a href="#">Pr.52</a> OPR deceleration time selection                         |                         |                        |                      |                |
| 84<br>85              | 234<br>235 | 384<br>385 | 534<br>535 | <a href="#">Pr.53</a> OP shift amount                                         |                         |                        |                      |                |
| 86                    | 236        | 386        | 536        | <a href="#">Pr.54</a> OPR torque limit value                                  |                         |                        |                      |                |
| 87                    | 237        | 387        | 537        | <a href="#">Pr.55</a> Deviation counter clear signal output time              |                         |                        |                      |                |
| 88                    | 238        | 388        | 538        | <a href="#">Pr.56</a> Speed designation during OP shift                       |                         |                        |                      |                |
| 89                    | 239        | 389        | 539        | <a href="#">Pr.57</a> Dwell time during OPR retry                             |                         |                        |                      |                |
| 140                   | —          | —          | —          | <a href="#">Pr.70</a> Positioning option valid/invalid setting                | Detailed parameters 1   | Positioning parameters |                      |                |

| Buffer memory address                                                                |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | Item                               |  | Memory area                                                  |                                     |                                         |
|--------------------------------------------------------------------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------------------------------------|--|--------------------------------------------------------------|-------------------------------------|-----------------------------------------|
| Common to axes 1, 2, 3, and 4                                                        |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |                                    |  |                                                              |                                     |                                         |
| 1200                                                                                 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | Md.1                               |  | In test mode flag                                            | System monitor data<br>Monitor data |                                         |
| 1201<br>1202<br>1203<br>1204<br>1205<br>1206<br>1207<br>1208<br>1209<br>1210<br>1211 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | Not used                           |  |                                                              |                                     |                                         |
| (0)                                                                                  | (1)  | (2)  | (3)  | (4)  | (5)  | (6)  | (7)  | (8)  | (9)  | (10) | (11) | (12) | (13) | (14) | (15) | (Pointer No.)                      |  | Start History                                                |                                     |                                         |
| 1212                                                                                 | 1217 | 1222 | 1227 | 1232 | 1237 | 1242 | 1247 | 1252 | 1257 | 1262 | 1267 | 1272 | 1277 | 1282 | 1287 | Md.3                               |  |                                                              |                                     | Start information                       |
| 1213                                                                                 | 1218 | 1223 | 1228 | 1233 | 1238 | 1243 | 1248 | 1253 | 1258 | 1263 | 1268 | 1273 | 1278 | 1283 | 1288 | Md.4                               |  |                                                              |                                     | Start No.                               |
| 1440                                                                                 | 1441 | 1442 | 1443 | 1444 | 1445 | 1446 | 1447 | 1448 | 1449 | 1450 | 1451 | 1452 | 1453 | 1454 | 1455 | Md.50                              |  |                                                              |                                     | Start (Year:month)<br>(QD75P□N/QD75D□N) |
| 1214                                                                                 | 1219 | 1224 | 1229 | 1234 | 1239 | 1244 | 1249 | 1254 | 1259 | 1264 | 1269 | 1274 | 1279 | 1284 | 1289 | Md.5                               |  |                                                              |                                     | Start (Day:hour)                        |
| 1215                                                                                 | 1220 | 1225 | 1230 | 1235 | 1240 | 1245 | 1250 | 1255 | 1260 | 1265 | 1270 | 1275 | 1280 | 1285 | 1290 | Md.6                               |  |                                                              |                                     | Start (Minute: second)                  |
| 1216                                                                                 | 1221 | 1226 | 1231 | 1236 | 1241 | 1246 | 1251 | 1256 | 1261 | 1266 | 1271 | 1276 | 1281 | 1286 | 1291 | Md.7                               |  |                                                              |                                     | Error judgment                          |
| 1292                                                                                 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | Md.8                               |  | Start history pointer                                        |                                     | Error History                           |
| (0)                                                                                  | (1)  | (2)  | (3)  | (4)  | (5)  | (6)  | (7)  | (8)  | (9)  | (10) | (11) | (12) | (13) | (14) | (15) | (Pointer No.)                      |  |                                                              |                                     |                                         |
| 1293                                                                                 | 1297 | 1301 | 1305 | 1309 | 1313 | 1317 | 1321 | 1325 | 1329 | 1333 | 1337 | 1341 | 1345 | 1349 | 1353 | Md.9                               |  | Axis in which the error occurred                             |                                     |                                         |
| 1294                                                                                 | 1298 | 1302 | 1306 | 1310 | 1314 | 1318 | 1322 | 1326 | 1330 | 1334 | 1338 | 1342 | 1346 | 1350 | 1354 | Md.10                              |  | Axis error No.                                               |                                     |                                         |
| 1456                                                                                 | 1457 | 1458 | 1459 | 1460 | 1461 | 1462 | 1463 | 1464 | 1465 | 1466 | 1467 | 1468 | 1469 | 1470 | 1471 | Md.51                              |  | Axis error occurrence<br>(Year:month)<br>(QD75P□N/QD75D□N)   |                                     |                                         |
| 1295                                                                                 | 1299 | 1303 | 1307 | 1311 | 1315 | 1319 | 1323 | 1327 | 1331 | 1335 | 1339 | 1343 | 1347 | 1351 | 1355 | Md.11                              |  | Axis error occurrence<br>(Day:hour)                          |                                     |                                         |
| 1296                                                                                 | 1300 | 1304 | 1308 | 1312 | 1316 | 1320 | 1324 | 1328 | 1332 | 1336 | 1340 | 1344 | 1348 | 1352 | 1356 | Md.12                              |  | Axis error occurrence<br>(Minute: second)                    |                                     |                                         |
| 1357                                                                                 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | Md.13                              |  | Error history pointer                                        |                                     | Warning history                         |
| (0)                                                                                  | (1)  | (2)  | (3)  | (4)  | (5)  | (6)  | (7)  | (8)  | (9)  | (10) | (11) | (12) | (13) | (14) | (15) | (Pointer No.)                      |  |                                                              |                                     |                                         |
| 1358                                                                                 | 1362 | 1366 | 1370 | 1374 | 1378 | 1382 | 1386 | 1390 | 1394 | 1398 | 1402 | 1406 | 1410 | 1414 | 1418 | Md.14                              |  | Axis in which the warning occurred                           |                                     |                                         |
| 1359                                                                                 | 1363 | 1367 | 1371 | 1375 | 1379 | 1383 | 1387 | 1391 | 1395 | 1399 | 1403 | 1407 | 1411 | 1415 | 1419 | Md.15                              |  | Axis warning No.                                             |                                     |                                         |
| 1472                                                                                 | 1473 | 1474 | 1475 | 1476 | 1477 | 1478 | 1479 | 1480 | 1481 | 1482 | 1483 | 1484 | 1485 | 1486 | 1487 | Md.52                              |  | Axis warning occurrence<br>(Year:month)<br>(QD75P□N/QD75D□N) |                                     |                                         |
| 1360                                                                                 | 1364 | 1368 | 1372 | 1376 | 1380 | 1384 | 1388 | 1392 | 1396 | 1400 | 1404 | 1408 | 1412 | 1416 | 1420 | Md.16                              |  | Axis warning occurrence (Day:hour)                           |                                     |                                         |
| 1361                                                                                 | 1365 | 1369 | 1373 | 1377 | 1381 | 1385 | 1389 | 1393 | 1397 | 1401 | 1405 | 1409 | 1413 | 1417 | 1421 | Md.17                              |  | Axis warning occurrence<br>(Minute: second)                  |                                     |                                         |
| 1422                                                                                 |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | Md.18                              |  | Warning history pointer                                      | Md.19                               |                                         |
| 1424<br>1425                                                                         |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | No. of write accesses to flash ROM |  |                                                              |                                     |                                         |



| Buffer memory address |            |              |              | Item  | Memory area                                               |                   |              |
|-----------------------|------------|--------------|--------------|-------|-----------------------------------------------------------|-------------------|--------------|
| Axis 1                | Axis 2     | Axis 3       | Axis 4       |       |                                                           |                   |              |
| 800<br>801            | 900<br>901 | 1000<br>1001 | 1100<br>1101 | Md.20 | Current feed value                                        | Axis monitor data | Monitor data |
| 802<br>803            | 902<br>903 | 1002<br>1003 | 1102<br>1103 | Md.21 | Machine feed value                                        |                   |              |
| 804<br>805            | 904<br>905 | 1004<br>1005 | 1104<br>1105 | Md.22 | Feedrate                                                  |                   |              |
| 806                   | 906        | 1006         | 1106         | Md.23 | Axis error No.                                            |                   |              |
| 807                   | 907        | 1007         | 1107         | Md.24 | Axis warning No.                                          |                   |              |
| 808                   | 908        | 1008         | 1108         | Md.25 | Valid M code                                              |                   |              |
| 809                   | 909        | 1009         | 1109         | Md.26 | Axis operation status                                     |                   |              |
| 810<br>811            | 910<br>911 | 1010<br>1011 | 1110<br>1111 | Md.27 | Current speed                                             |                   |              |
| 812<br>813            | 912<br>913 | 1012<br>1013 | 1112<br>1113 | Md.28 | Axis feedrate                                             |                   |              |
| 814<br>815            | 914<br>915 | 1014<br>1015 | 1114<br>1115 | Md.29 | Speed-position switching control<br>positioning amount    |                   |              |
| 816                   | 916        | 1016         | 1116         | Md.30 | External input/output signal                              |                   |              |
| 817                   | 917        | 1017         | 1117         | Md.31 | Status                                                    |                   |              |
| 818<br>819            | 918<br>919 | 1018<br>1019 | 1118<br>1119 | Md.32 | Target value                                              |                   |              |
| 820<br>821            | 920<br>921 | 1020<br>1021 | 1120<br>1121 | Md.33 | Target speed                                              |                   |              |
| 824<br>825            | 924<br>925 | 1024<br>1025 | 1124<br>1125 | Md.34 | Movement amount after near-point dog ON                   |                   |              |
| 826                   | 926        | 1026         | 1126         | Md.35 | Torque limit stored value                                 |                   |              |
| 827                   | 927        | 1027         | 1127         | Md.36 | Special start data instruction parameter<br>setting value |                   |              |
| 828                   | 928        | 1028         | 1128         | Md.37 | Special start data instruction parameter<br>setting value |                   |              |
| 829                   | 929        | 1029         | 1129         | Md.38 | Start positioning data No. setting value.                 |                   |              |
| 830                   | 930        | 1030         | 1130         | Md.39 | In speed control flag                                     |                   |              |
| 831                   | 931        | 1031         | 1131         | Md.40 | In speed change processing flag                           |                   |              |
| 832                   | 932        | 1032         | 1132         | Md.41 | Special start repetition counter                          |                   |              |
| 833                   | 933        | 1033         | 1133         | Md.42 | Control system repetition counter                         |                   |              |
| 834                   | 934        | 1034         | 1134         | Md.43 | Start data pointer being executed                         |                   |              |
| 835                   | 935        | 1035         | 1135         | Md.44 | Positioning data No. being executed                       |                   |              |
| 836                   | 936        | 1036         | 1136         | Md.45 | Block No. being executed                                  |                   |              |
| 837                   | 937        | 1037         | 1137         | Md.46 | Last executed positioning data No.                        |                   |              |
| 838 to 847            | 938 to 947 | 1038 to 1047 | 1138 to 1147 | Md.47 | Positioning data being executed                           |                   |              |
| 899                   | 999        | 1099         | 1199         | Md.48 | Deceleration start flag                                   |                   |              |

| Buffer memory address |        |        |        | Item     | Memory area                                                                         |                   |              |
|-----------------------|--------|--------|--------|----------|-------------------------------------------------------------------------------------|-------------------|--------------|
| Axis 1                | Axis 2 | Axis 3 | Axis 4 |          |                                                                                     |                   |              |
| 1500                  | 1600   | 1700   | 1800   | Cd.3     | Positioning start No.                                                               | Axis control data | Control data |
| 1501                  | 1601   | 1701   | 1801   | Cd.4     | Positioning starting point No.                                                      |                   |              |
| 1502                  | 1602   | 1702   | 1802   | Cd.5     | Axis error reset                                                                    |                   |              |
| 1503                  | 1603   | 1703   | 1803   | Cd.6     | Restart command                                                                     |                   |              |
| 1504                  | 1604   | 1704   | 1804   | Cd.7     | M code OFF request                                                                  |                   |              |
| 1505                  | 1605   | 1705   | 1805   | Cd.8     | External command valid                                                              |                   |              |
| 1506                  | 1606   | 1706   | 1806   | Cd.9     | New current value                                                                   |                   |              |
| 1507                  | 1607   | 1707   | 1807   |          |                                                                                     |                   |              |
| 1508                  | 1608   | 1708   | 1808   | Cd.10    | New acceleration time value                                                         |                   |              |
| 1509                  | 1609   | 1709   | 1809   |          |                                                                                     |                   |              |
| 1510                  | 1610   | 1710   | 1810   | Cd.11    | New deceleration time value                                                         |                   |              |
| 1511                  | 1611   | 1711   | 1811   |          |                                                                                     |                   |              |
| 1512                  | 1612   | 1712   | 1812   | Cd.12    | Acceleration/deceleration time change during speed change, enable/disable selection |                   |              |
| 1513                  | 1613   | 1713   | 1813   | Cd.13    | Positioning operation speed override                                                |                   |              |
| 1514                  | 1614   | 1714   | 1814   | Cd.14    | New speed value                                                                     |                   |              |
| 1515                  | 1615   | 1715   | 1815   |          |                                                                                     |                   |              |
| 1516                  | 1616   | 1716   | 1816   | Cd.15    | Speed change request                                                                |                   |              |
| 1517                  | 1617   | 1717   | 1817   | Cd.16    | Inching movement amount                                                             |                   |              |
| 1518                  | 1618   | 1718   | 1818   | Cd.17    | JOG speed                                                                           |                   |              |
| 1519                  | 1619   | 1719   | 1819   |          |                                                                                     |                   |              |
| 1520                  | 1620   | 1720   | 1820   | Cd.18    | Continuous operation interrupt request                                              |                   |              |
| 1521                  | 1621   | 1721   | 1821   | Cd.19    | OPR request flag OFF request                                                        |                   |              |
| 1522                  | 1622   | 1722   | 1822   | Cd.20    | Manual pulse generator 1 pulse input magnification                                  |                   |              |
| 1523                  | 1623   | 1723   | 1823   |          |                                                                                     |                   |              |
| 1524                  | 1624   | 1724   | 1824   | Cd.21    | Manual pulse generator enable flag                                                  |                   |              |
| 1525                  | 1625   | 1725   | 1825   | Cd.22    | New torque value                                                                    |                   |              |
| 1526                  | 1626   | 1726   | 1826   | Cd.23    | Speed-position switching control movement amount change register                    |                   |              |
| 1527                  | 1627   | 1727   | 1827   |          |                                                                                     |                   |              |
| 1528                  | 1628   | 1728   | 1828   | Cd.24    | Speed-position switching enable flag                                                |                   |              |
| 1529                  | 1629   | 1729   | 1829   | Not used |                                                                                     |                   |              |
| 1530                  | 1630   | 1730   | 1830   | Cd.25    | Position-speed switching control speed change register                              |                   |              |
| 1531                  | 1631   | 1731   | 1831   |          |                                                                                     |                   |              |
| 1532                  | 1632   | 1732   | 1832   | Cd.26    | Position-speed switching enable flag                                                |                   |              |
| 1533                  | 1633   | 1733   | 1833   | Not used |                                                                                     |                   |              |
| 1534                  | 1634   | 1734   | 1834   | Cd.27    | Target position change value (new address)                                          |                   |              |
| 1535                  | 1635   | 1735   | 1835   |          |                                                                                     |                   |              |
| 1536                  | 1636   | 1736   | 1836   | Cd.28    | Target position change value (new speed)                                            |                   |              |
| 1537                  | 1637   | 1737   | 1837   |          |                                                                                     |                   |              |
| 1538                  | 1638   | 1738   | 1838   | Cd.29    | Target position change request flag                                                 |                   |              |
| 1539                  | 1639   | 1739   | 1839   | Not used |                                                                                     |                   |              |
| 1540                  | 1640   | 1740   | 1840   | Cd.30    | Simultaneous starting axis start data No. (axis 1 start data No.)                   |                   |              |

| Buffer memory address |        |        |        | Item                                                                              | Memory area         |              |
|-----------------------|--------|--------|--------|-----------------------------------------------------------------------------------|---------------------|--------------|
| Axis 1                | Axis 2 | Axis 3 | Axis 4 |                                                                                   |                     |              |
| 1541                  | 1641   | 1741   | 1841   | <b>Cd.31</b> Simultaneous starting axis start data No.<br>(axis 2 start data No.) | Axis control data   | Control data |
| 1542                  | 1642   | 1742   | 1842   | <b>Cd.32</b> Simultaneous starting axis start data No.<br>(axis 3 start data No.) |                     |              |
| 1543                  | 1643   | 1743   | 1843   | <b>Cd.33</b> Simultaneous starting axis start data No.<br>(axis 4 start data No.) |                     |              |
| 1544                  | 1644   | 1744   | 1844   | <b>Cd.34</b> Step mode                                                            |                     |              |
| 1545                  | 1645   | 1745   | 1845   | <b>Cd.35</b> Step valid flag                                                      |                     |              |
| 1546                  | 1646   | 1746   | 1846   | <b>Cd.36</b> Step start information                                               |                     |              |
| 1547                  | 1647   | 1747   | 1847   | <b>Cd.37</b> Skip command                                                         |                     |              |
| 1548                  | 1648   | 1748   | 1848   | <b>Cd.38</b> Teaching data selection                                              |                     |              |
| 1549                  | 1649   | 1749   | 1849   | <b>Cd.39</b> Teaching positioning data No.                                        |                     |              |
| 1550                  | 1650   | 1750   | 1850   | <b>Cd.40</b> ABS direction in degrees                                             |                     |              |
| 1900                  |        |        |        | <b>Cd.1</b> Flash ROM write request                                               | System Control data | Control data |
| 1901                  |        |        |        | <b>Cd.2</b> Parameter initialization request                                      |                     |              |
| 1905                  |        |        |        | <b>Cd.41</b> Deceleration start flag valid                                        |                     |              |
| 1907                  |        |        |        | <b>Cd.42</b> Stop command processing for deceleration<br>stop selection           |                     |              |
| 1934                  |        |        |        | <b>Cd.43</b> Output timing selection of near pass control                         |                     |              |

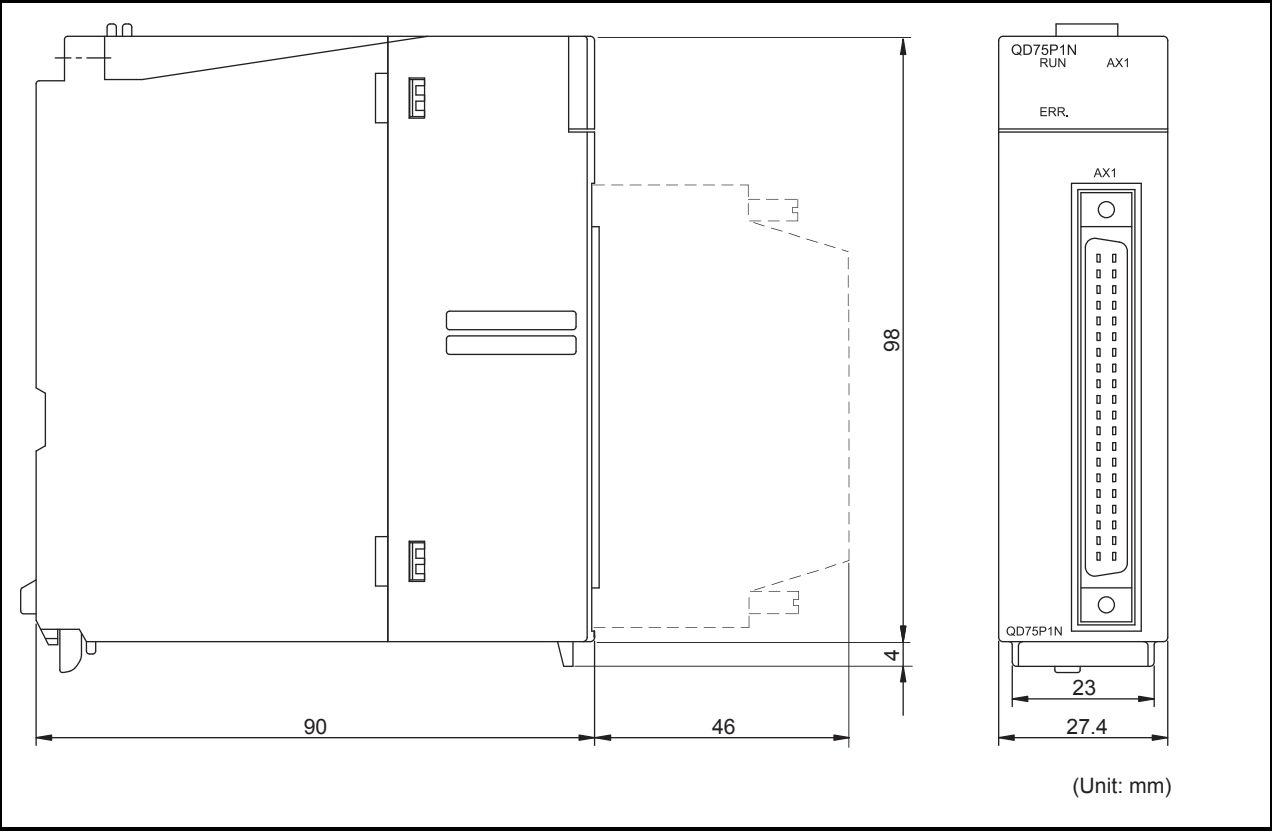
| Buffer memory address |                      |                      |                      | Item                                                                                                                                                                                             | Memory area |                  |                  |
|-----------------------|----------------------|----------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------|------------------|
| Axis 1                | Axis 2               | Axis 3               | Axis 4               |                                                                                                                                                                                                  |             |                  |                  |
| 2000                  | 8000                 | 14000                | 20000                | <div>Da.1</div> Operation pattern<br><div>Da.2</div> Control system<br><div>Da.3</div> Acceleration time No.<br><div>Da.4</div> Deceleration time No.<br><div>Da.5</div> Axis to be interpolated | No.1        | Positioning data | Positioning data |
| 2001                  | 8001                 | 14001                | 20001                | <div>Da.10</div> M code/condition data No.<br>/No. of LOOP to LEND repetitions                                                                                                                   |             |                  |                  |
| 2002                  | 8002                 | 14002                | 20002                | <div>Da.9</div> Dwell time (JUMP destination<br>positioning data No.)                                                                                                                            |             |                  |                  |
| 2003                  | 8003                 | 14003                | 20003                | <div>Da.27</div> M code ON signal output timing<br><div>Da.28</div> ABS direction in degrees<br><div>Da.29</div> Interpolation speed designation<br>method                                       |             |                  |                  |
| 2004<br>2005          | 8004<br>8005         | 14004<br>14005       | 20004<br>20005       | <div>Da.8</div> Command speed                                                                                                                                                                    |             |                  |                  |
| 2006<br>2007          | 8006<br>8007         | 14006<br>14007       | 20006<br>20007       | <div>Da.6</div> Positioning address/movement<br>amount                                                                                                                                           |             |                  |                  |
| 2008<br>2009          | 8008<br>8009         | 14008<br>14009       | 20008<br>20009       | <div>Da.7</div> Arc address                                                                                                                                                                      |             |                  |                  |
| 2010<br>to<br>2019    | 8010<br>to<br>8019   | 14010<br>to<br>14019 | 20010<br>to<br>20019 | No.2                                                                                                                                                                                             |             |                  |                  |
| 2020<br>to<br>2029    | 8020<br>to<br>8029   | 14020<br>to<br>14029 | 20020<br>to<br>20029 | No.3                                                                                                                                                                                             |             |                  |                  |
| to                    | to                   | to                   | to                   | to                                                                                                                                                                                               |             |                  |                  |
| 7990<br>to<br>7999    | 13990<br>to<br>13999 | 19990<br>to<br>19999 | 25990<br>to<br>25999 | No.600                                                                                                                                                                                           |             |                  |                  |

| Buffer memory address |       |                |       |                |             |                |       | Item             |                           | Memory area      |                  |                  |                                        |
|-----------------------|-------|----------------|-------|----------------|-------------|----------------|-------|------------------|---------------------------|------------------|------------------|------------------|----------------------------------------|
| Axis 1                |       | Axis 2         |       | Axis 3         |             | Axis 4         |       |                  |                           |                  |                  |                  |                                        |
| 26000                 | 26050 | 27000          | 27050 | 28000          | 28050       | 29000          | 29050 | Da.11            | Shape                     | 1st point        | Block start data | Starting block 0 | Positioning data (Starting block data) |
|                       |       |                |       |                |             |                |       | Da.12            | Start data No.            |                  |                  |                  |                                        |
|                       |       |                |       |                |             |                |       | Da.13            | Special start instruction |                  |                  |                  |                                        |
|                       |       |                |       |                |             |                |       | Da.14            | Parameter                 |                  |                  |                  |                                        |
| 26001                 | 26051 | 27001          | 27051 | 28001          | 28051       | 29001          | 29051 | 2nd point        |                           |                  |                  |                  |                                        |
| 26002                 | 26052 | 27002          | 27052 | 28002          | 28052       | 29002          | 29052 | 3rd point        |                           |                  |                  |                  |                                        |
| to                    |       | to             |       | to             |             | to             |       | to               |                           |                  |                  |                  |                                        |
| 26049                 | 26099 | 27049          | 27099 | 28049          | 28099       | 29049          | 29099 | 50th point       |                           |                  |                  |                  |                                        |
| 26100                 |       | 27100          |       | 28100          |             | 29100          |       | Da.15            | Condition target          | No. 1            | Condition data   |                  |                                        |
|                       |       |                |       |                |             |                |       | Da.16            | Condition operator        |                  |                  |                  |                                        |
| 26102                 | 26103 | 27102          | 27103 | 28102          | 28103       | 29102          | 29103 | Da.17            | Address                   |                  |                  |                  |                                        |
| 26104                 |       | 27104          |       | 28104          |             | 29104          |       | Da.18            | Parameter 1               |                  |                  |                  |                                        |
| 26105                 | 27105 | 28105          | 29105 | Da.19          | Parameter 2 |                |       |                  |                           |                  |                  |                  |                                        |
| 26110 to 26119        |       | 27110 to 27119 |       | 28110 to 28119 |             | 29110 to 29119 |       | No. 2            |                           |                  |                  |                  |                                        |
| 26120 to 26129        |       | 27120 to 27129 |       | 28120 to 28129 |             | 29120 to 29129 |       | No. 3            |                           |                  |                  |                  |                                        |
| to                    |       | to             |       | to             |             | to             |       | to               |                           |                  |                  |                  |                                        |
| 26190 to 26199        |       | 27190 to 27199 |       | 28190 to 28199 |             | 29190 to 29199 |       | No. 10           |                           |                  |                  |                  |                                        |
| 26200 to 26299        |       | 27200 to 27299 |       | 28200 to 28299 |             | 29200 to 29299 |       | Block start data |                           | Starting block 1 |                  |                  |                                        |
| 26300 to 26399        |       | 27300 to 27399 |       | 28300 to 28399 |             | 29300 to 29399 |       | Condition data   |                           |                  |                  |                  |                                        |
| 26400 to 26499        |       | 27400 to 27499 |       | 28400 to 28499 |             | 29400 to 29499 |       | Block start data |                           | Starting block 2 |                  |                  |                                        |
| 26500 to 26599        |       | 27500 to 27599 |       | 28500 to 28599 |             | 29500 to 29599 |       | Condition data   |                           |                  |                  |                  |                                        |
| 26600 to 26699        |       | 27600 to 27699 |       | 28600 to 28699 |             | 29600 to 29699 |       | Block start data |                           | Starting block 3 |                  |                  |                                        |
| 26700 to 26799        |       | 27700 to 27799 |       | 28700 to 28799 |             | 29700 to 29799 |       | Condition data   |                           |                  |                  |                  |                                        |
| 26800 to 26899        |       | 27800 to 27899 |       | 28800 to 28899 |             | 29800 to 29899 |       | Block start data |                           | Starting block 4 |                  |                  |                                        |
| 26900 to 26999        |       | 27900 to 27999 |       | 28900 to 28999 |             | 29900 to 29999 |       | Condition data   |                           |                  |                  |                  |                                        |

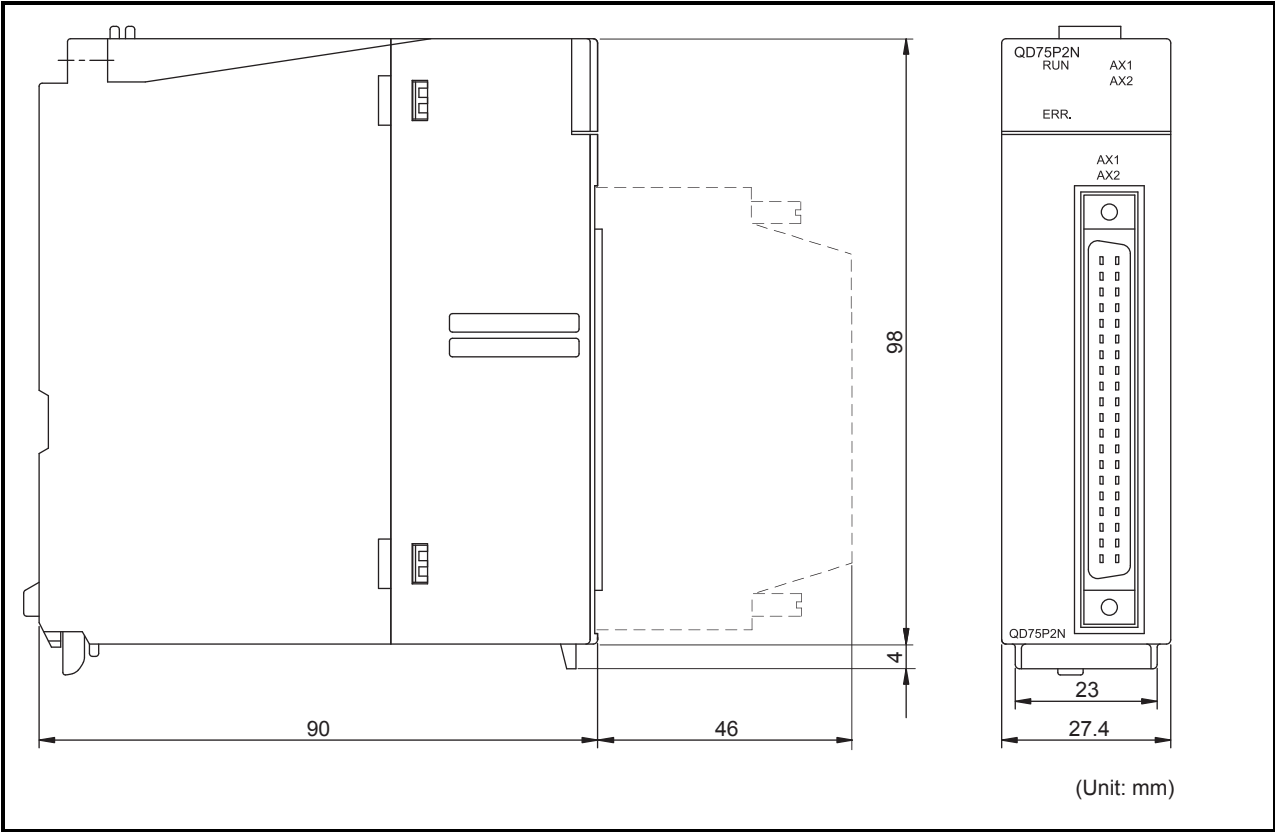
| Buffer memory address |        |        |        | Item                                                  | Memory area            |                  |
|-----------------------|--------|--------|--------|-------------------------------------------------------|------------------------|------------------|
| Axis 1                | Axis 2 | Axis 3 | Axis 4 |                                                       | PLC CPU<br>memory area | Positioning data |
| 30000                 |        |        |        | Condition judgement target data of the condition data |                        |                  |
| to                    |        |        |        |                                                       |                        |                  |
| 30099                 |        |        |        |                                                       |                        |                  |

Appendix 14 External dimension drawing

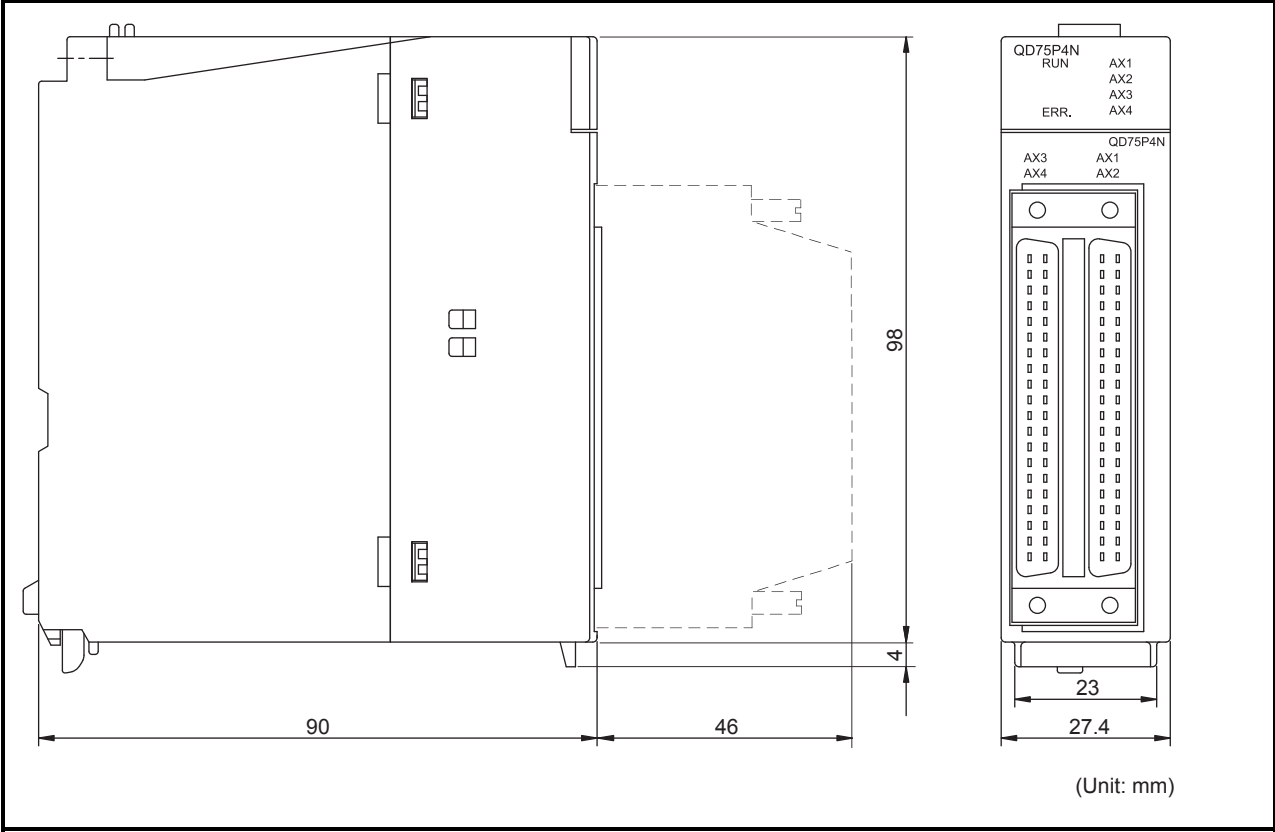
[1] QD75P1N



[2] QD75P2N

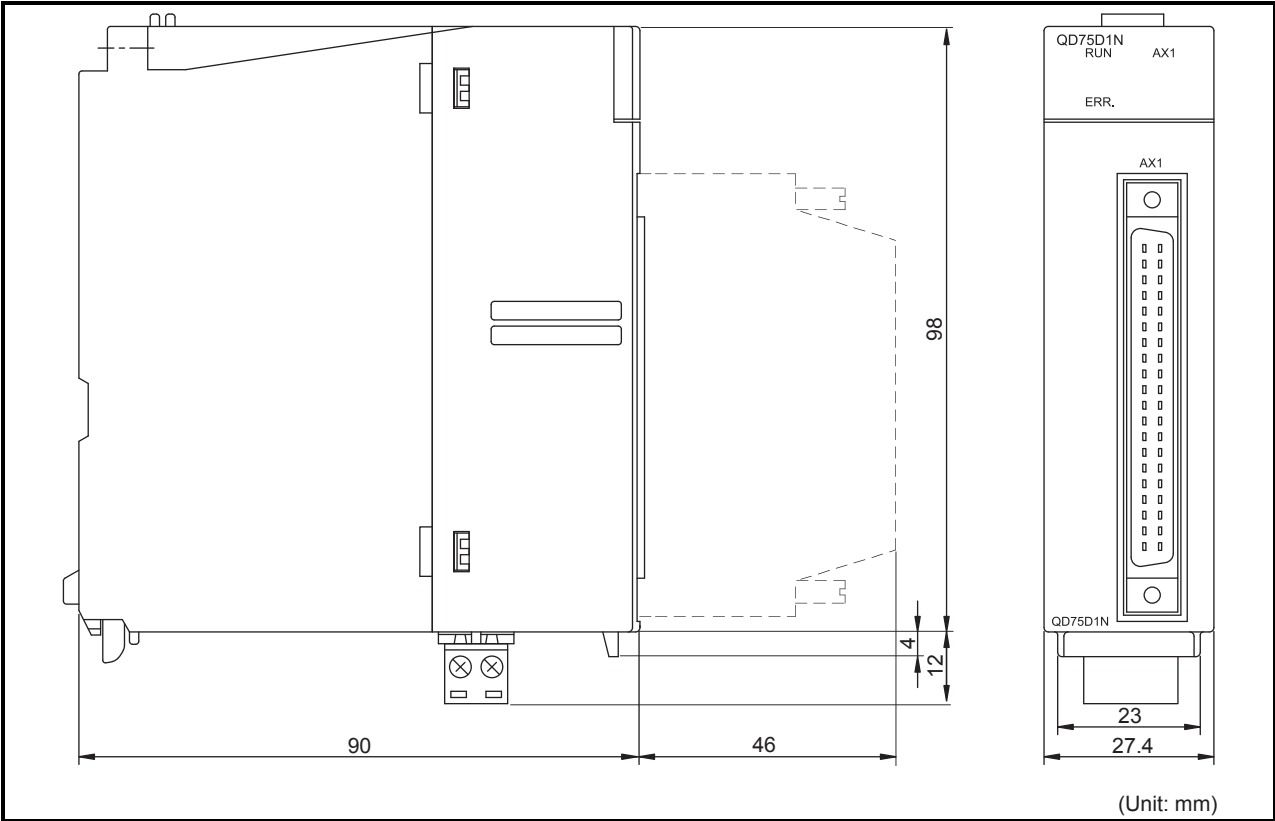


[3] QD75P4N

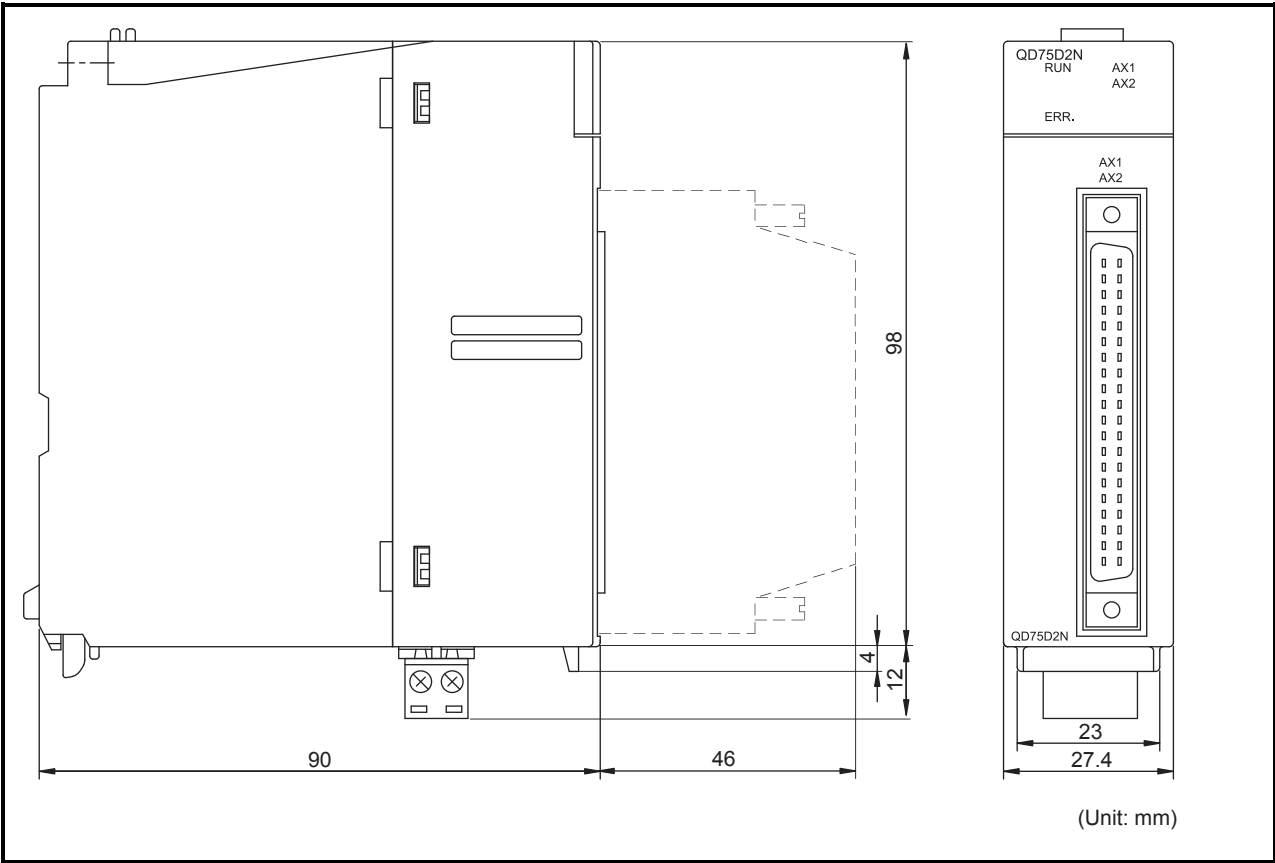




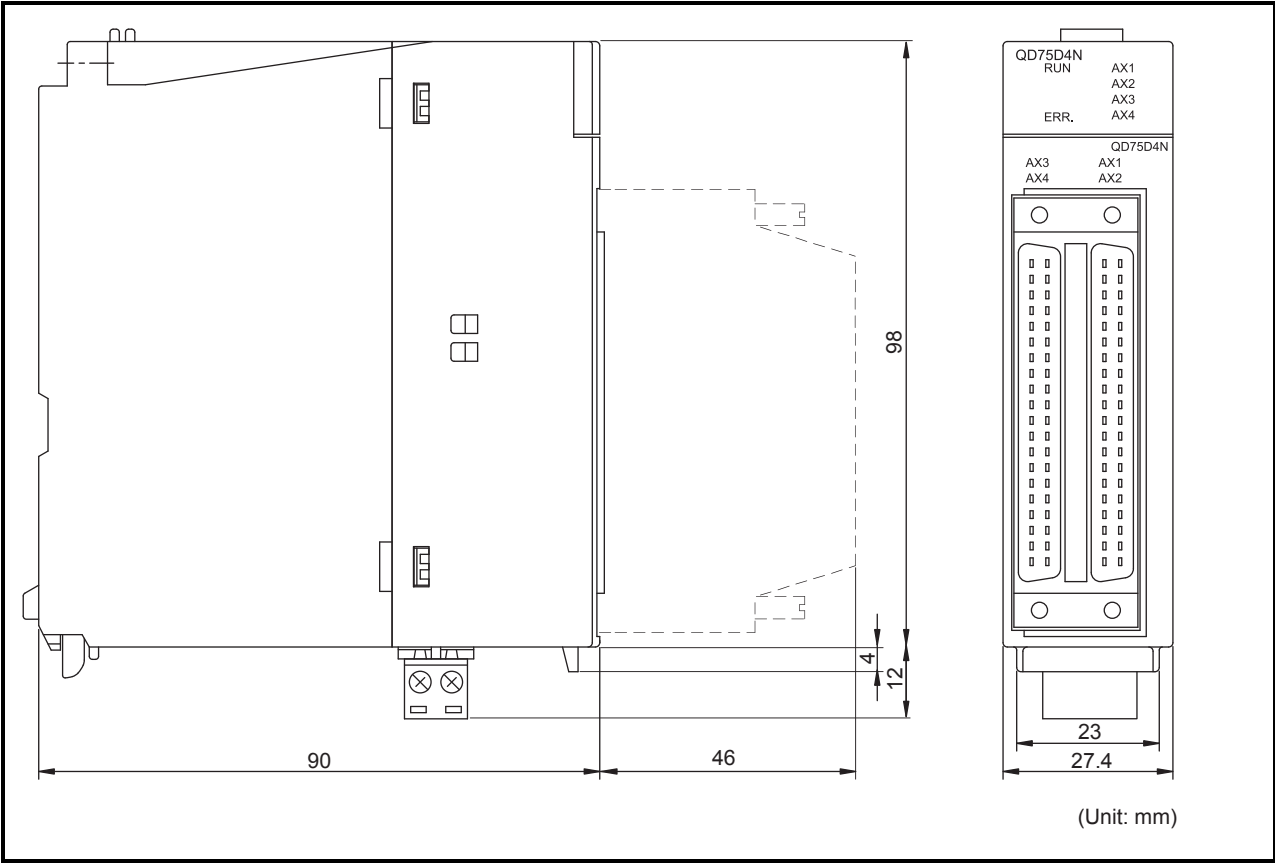
[4] QD75D1N



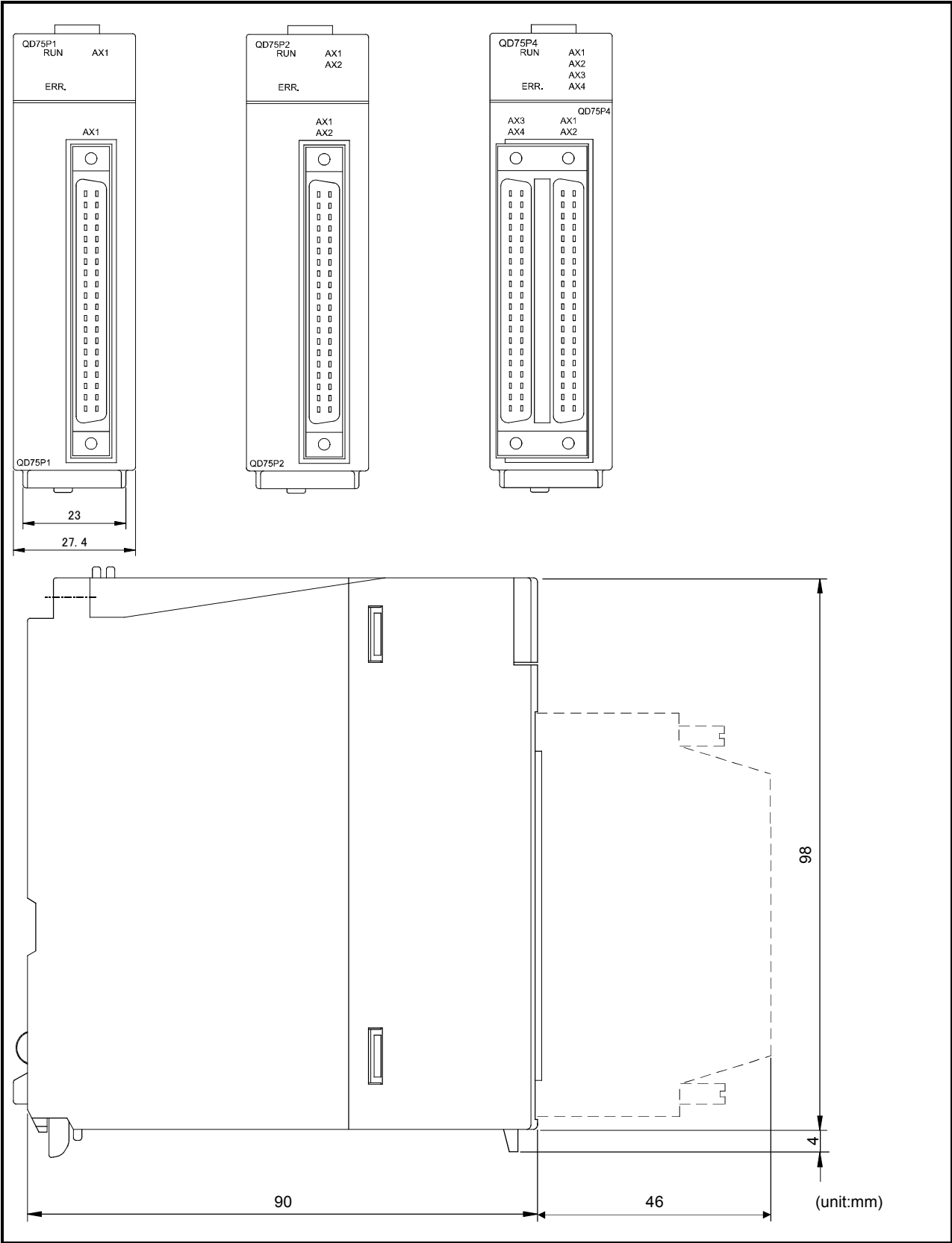
[5] QD75D2N



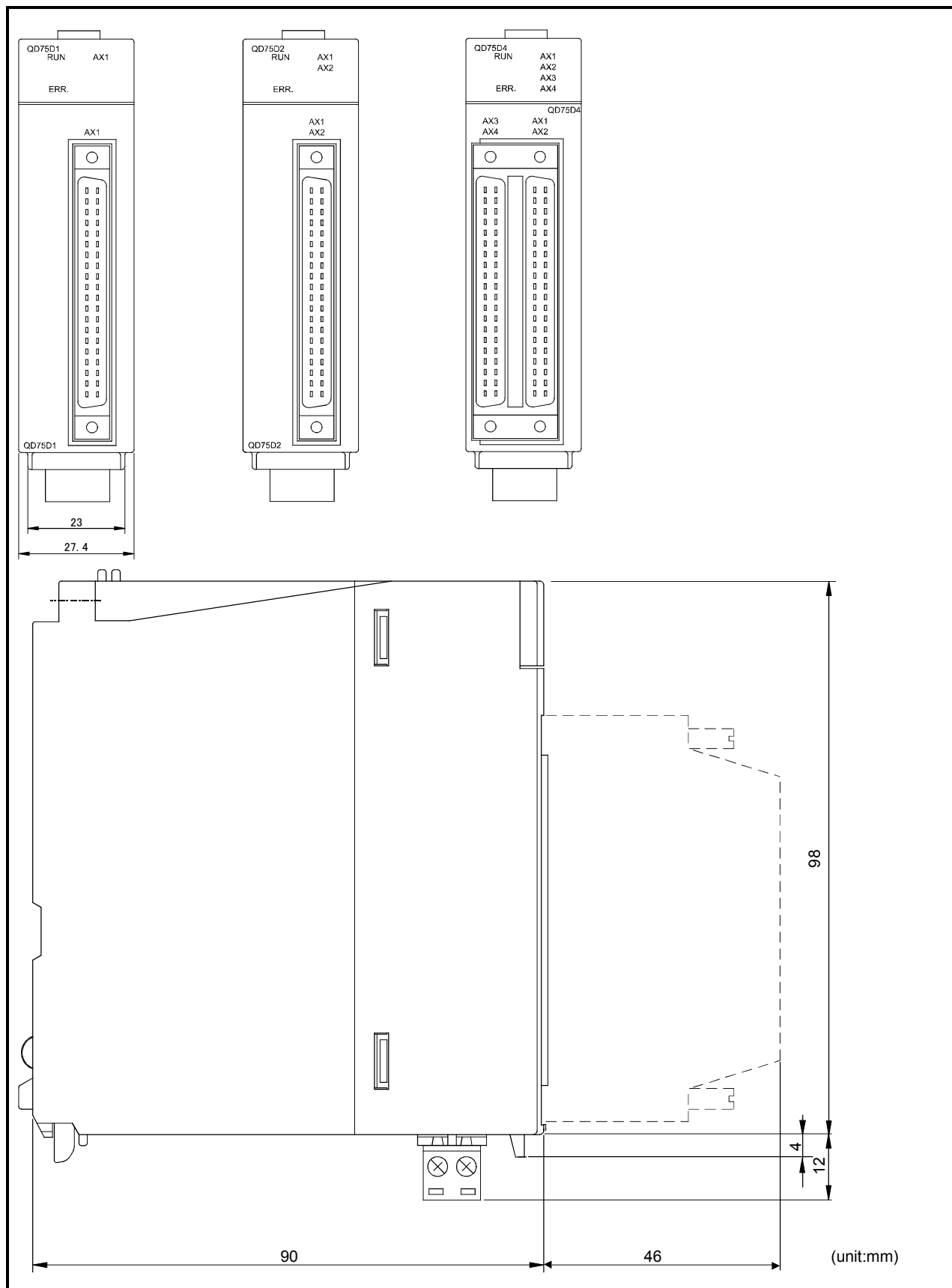
[6] QD75D4N



[7] QD75P1/QD75P2/QD75P4



## [8] QD75D1/QD75D2/QD75D4



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MODEL CODE: 13JR09

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